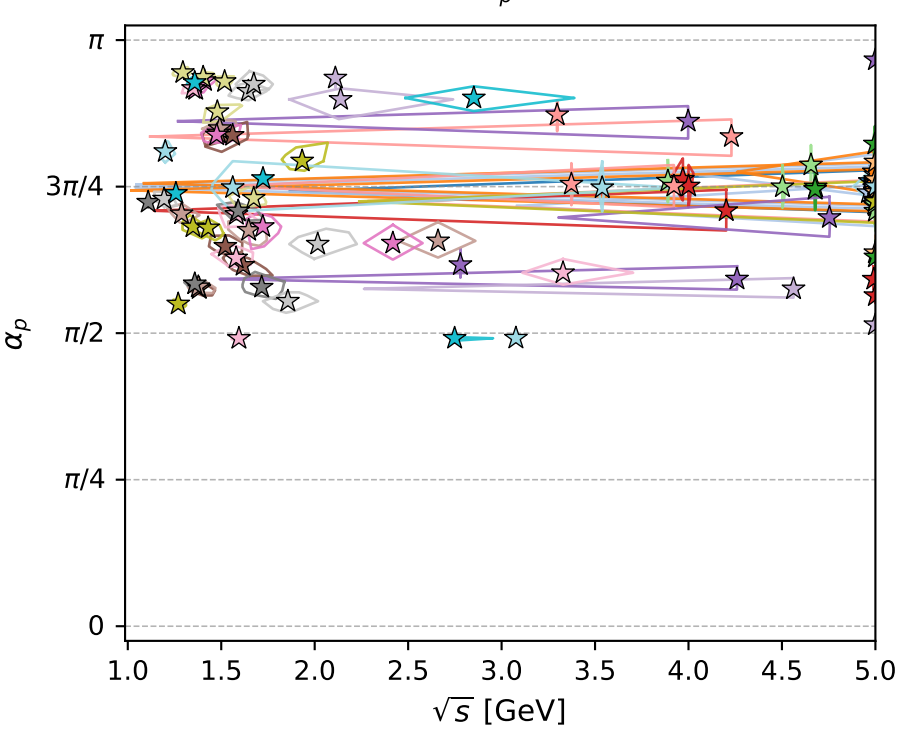
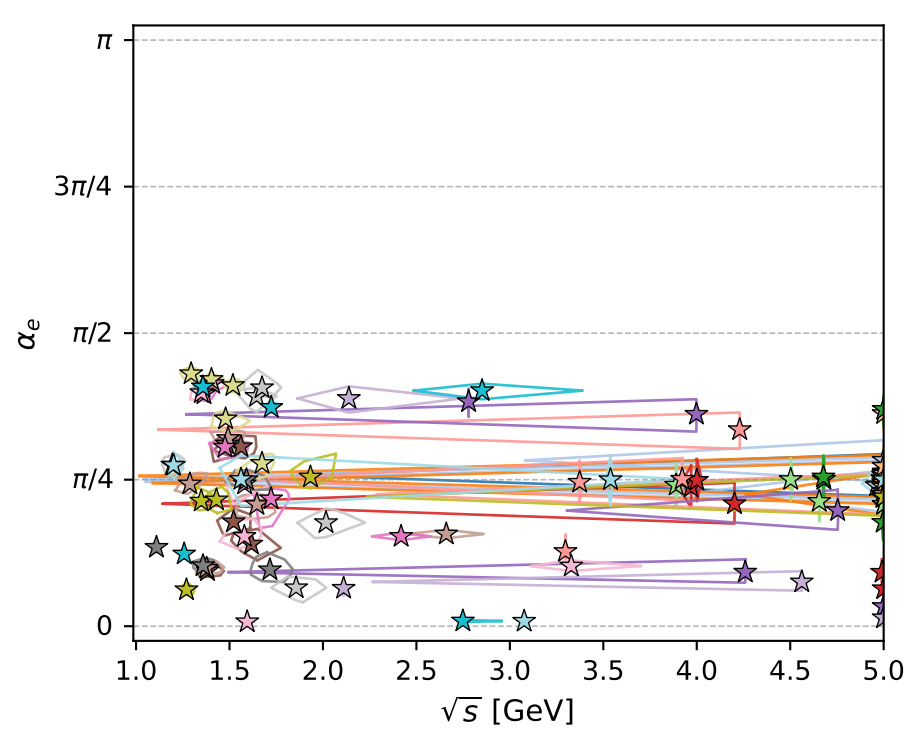
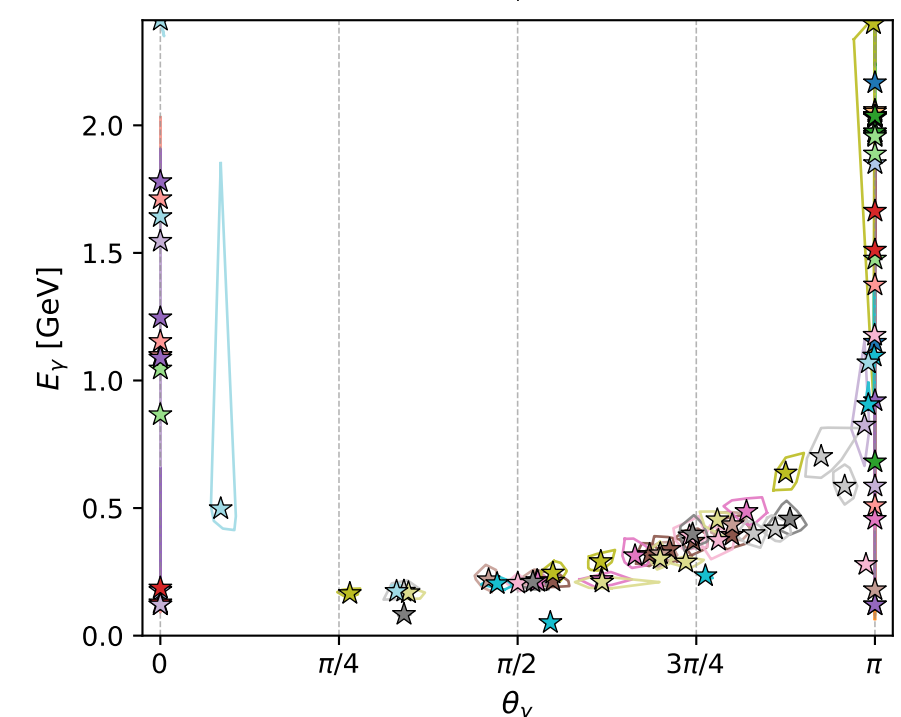
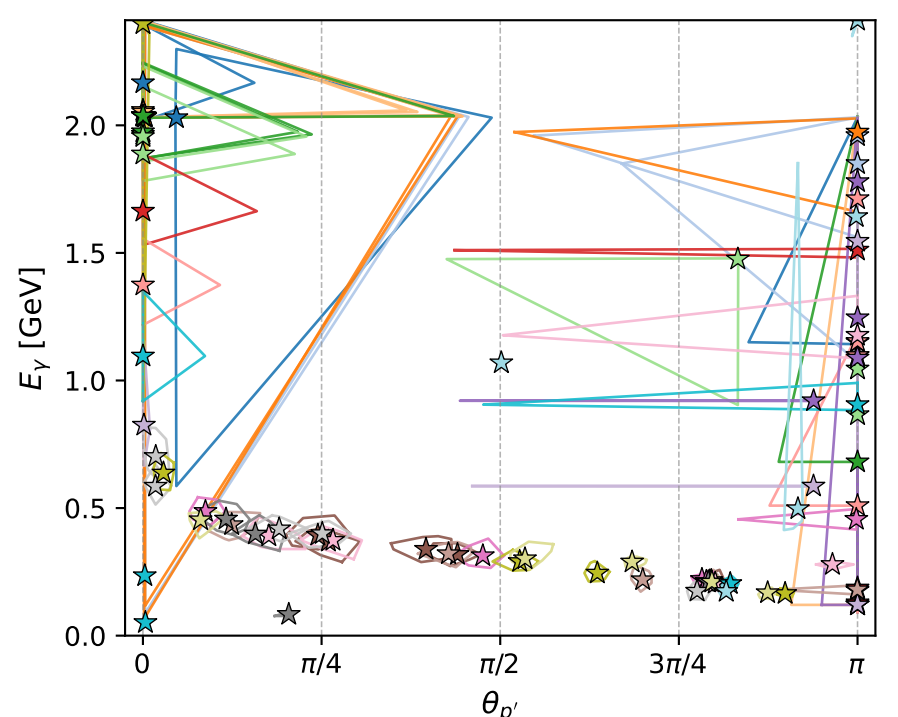
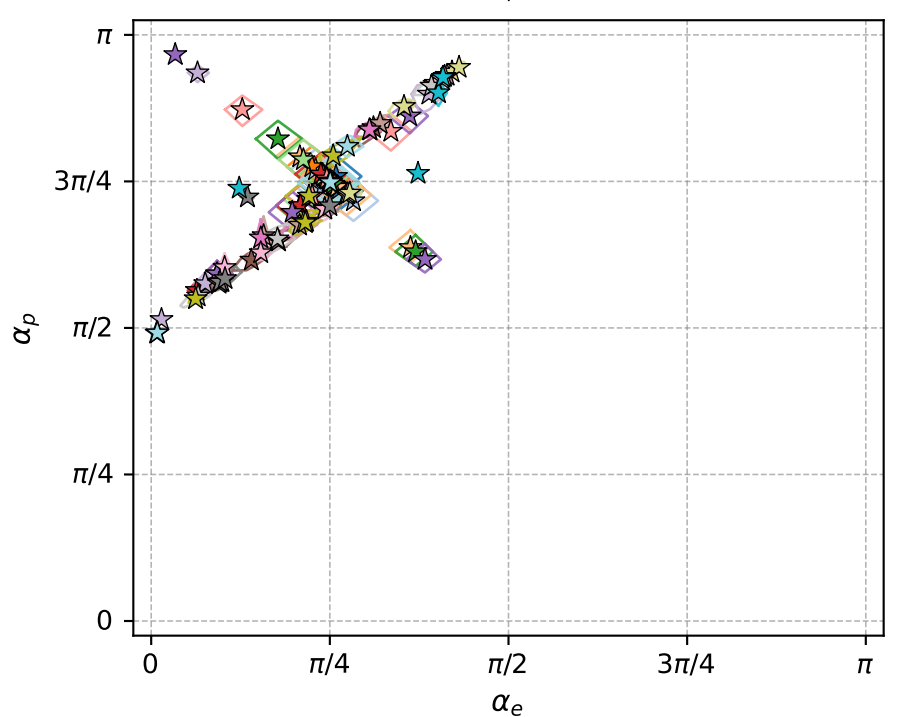
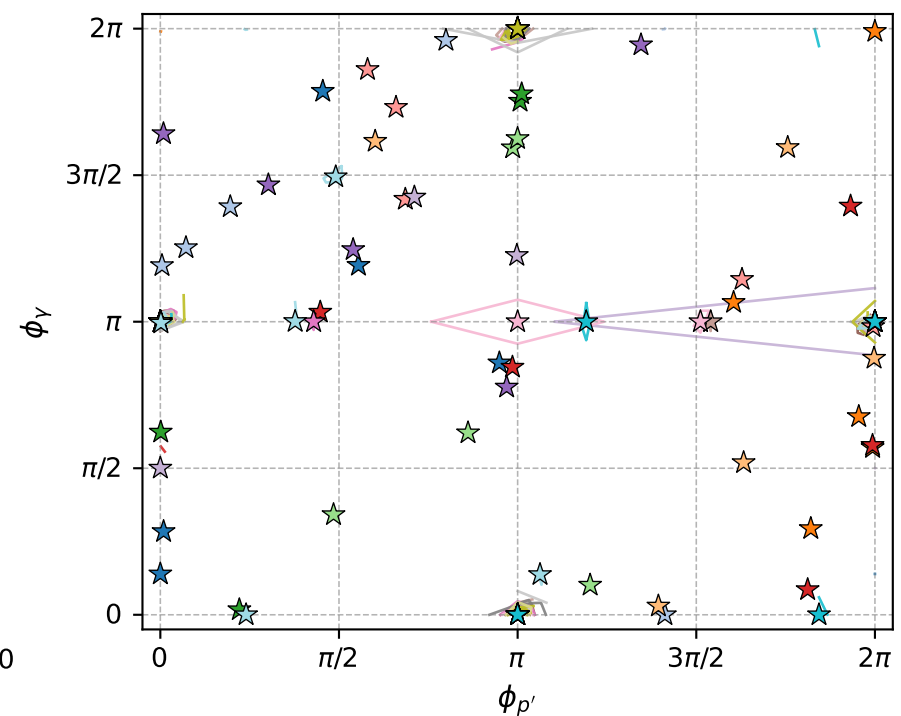
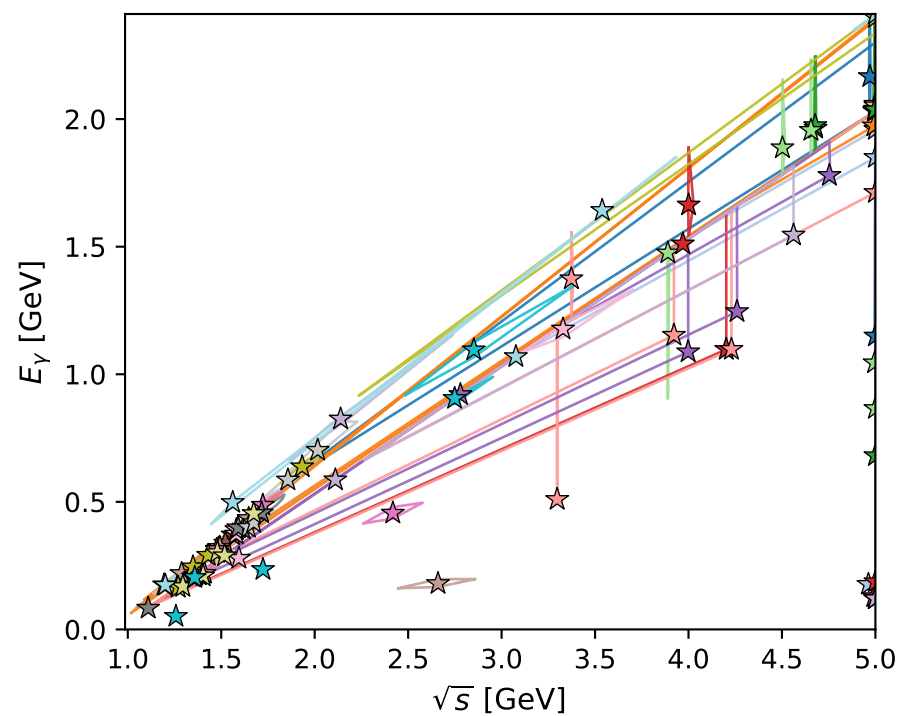
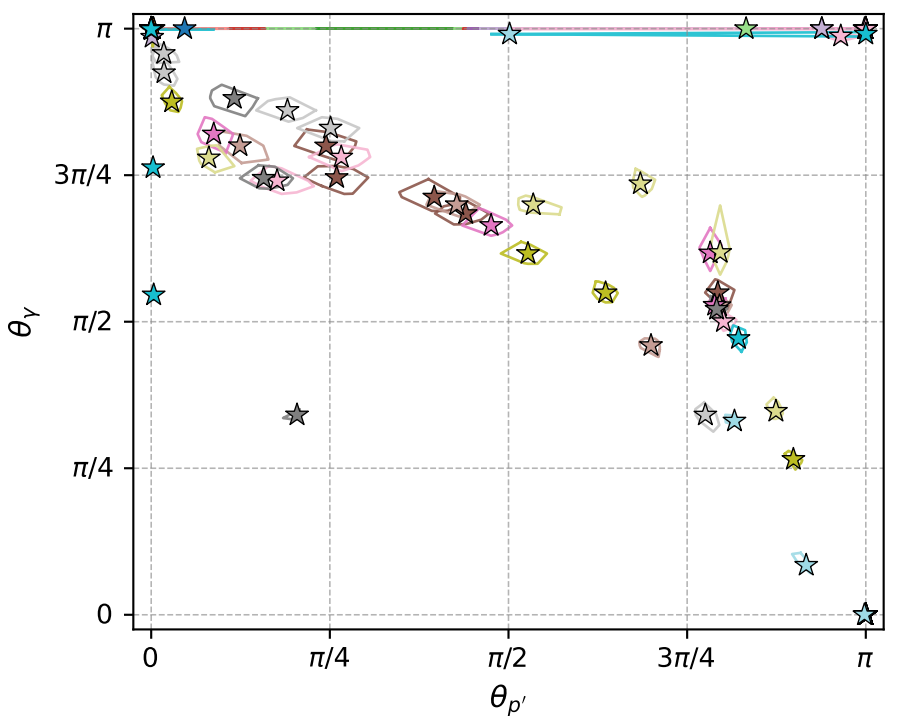


electron: polarization cluster P2; all local-minimum configurations and pairwise projections of their 8D contours



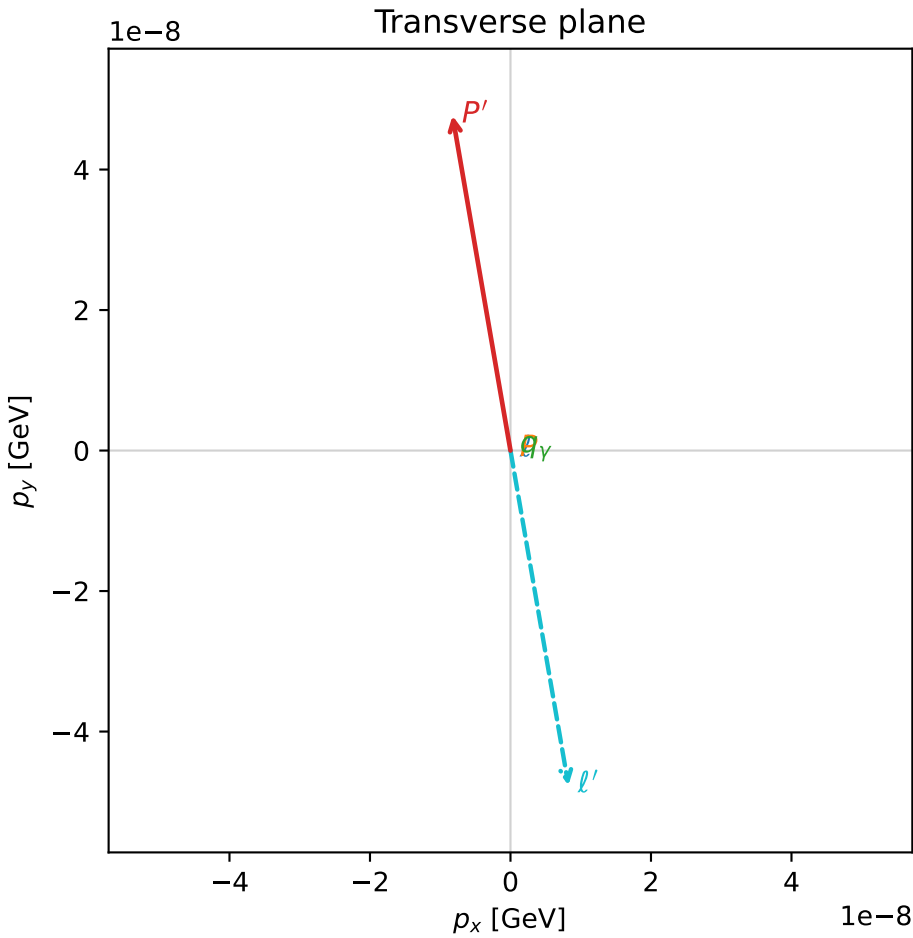
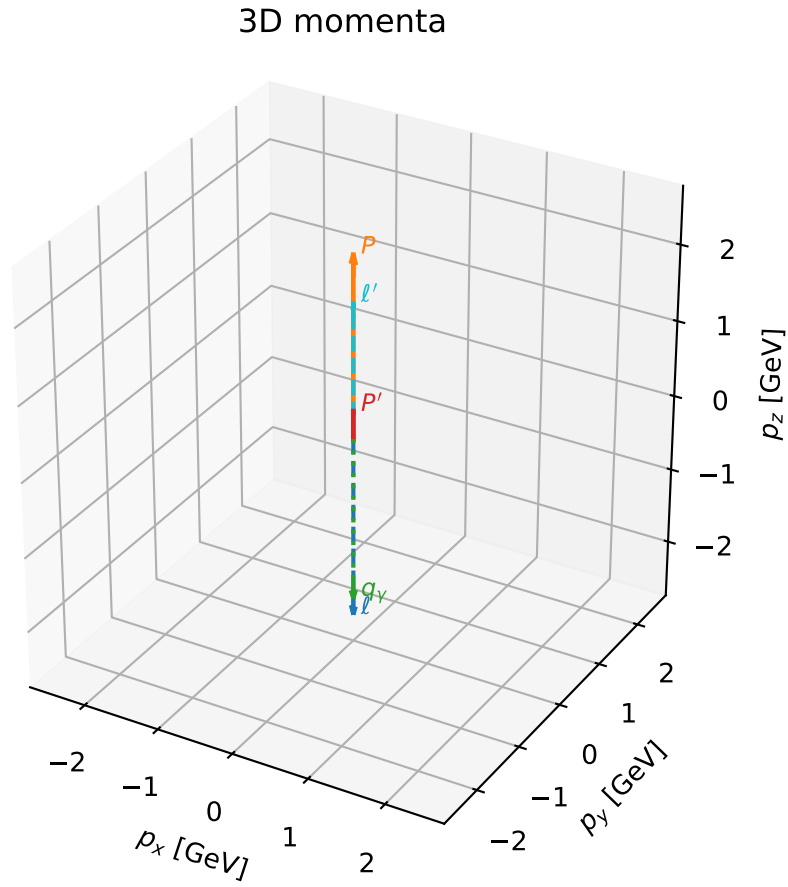
polarization cluster P2: α_e in $[0, \pi/2)$, $\alpha_p \sim 3\pi/4$
 97 local minima in this polarization cluster
 contour delta=0.05
 configured 8D samples/minimum=256
 dGHZ range= 2.228346×10^{-8} to 0.03557047
 global-minimum offsets=0 to 0.03557045

coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

dghz_mixing_angles_polarization_02_local_minimum_0 $d_{\{\text{rm GHZ}\}}=2.22835\text{e-}08$
 region: polarization_cluster_02_local_minimum_0
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=0.7579928$ rad
 incoming lepton: $0.726217|+\rangle +0.687465|-\rangle$
 proton mixing angle: $\alpha_p=2.3835959$ rad
 incoming proton: $-0.726215|+\rangle +0.687468|-\rangle$

$s=24.7062$, $\sqrt{s}=4.97053$, $|\vec{P}|=2.39676$, $|\vec{P}'|=0.371823$
 $\theta_{P'}=1.30111\text{e-}07$, $\theta_Y=3.14159$, $\phi_{P'}=1.74218$, $\phi_Y=3.74627$
 $E_Y=2.16667$, $\theta_{\ell'}=2.58096\text{e-}08$, $\phi_{\ell'}=4.88378$
 $Q^2=17.2073$, $x_B=0.741984$, $t=-1.65188$, $W^2=6.86346$, $y=0.973331$

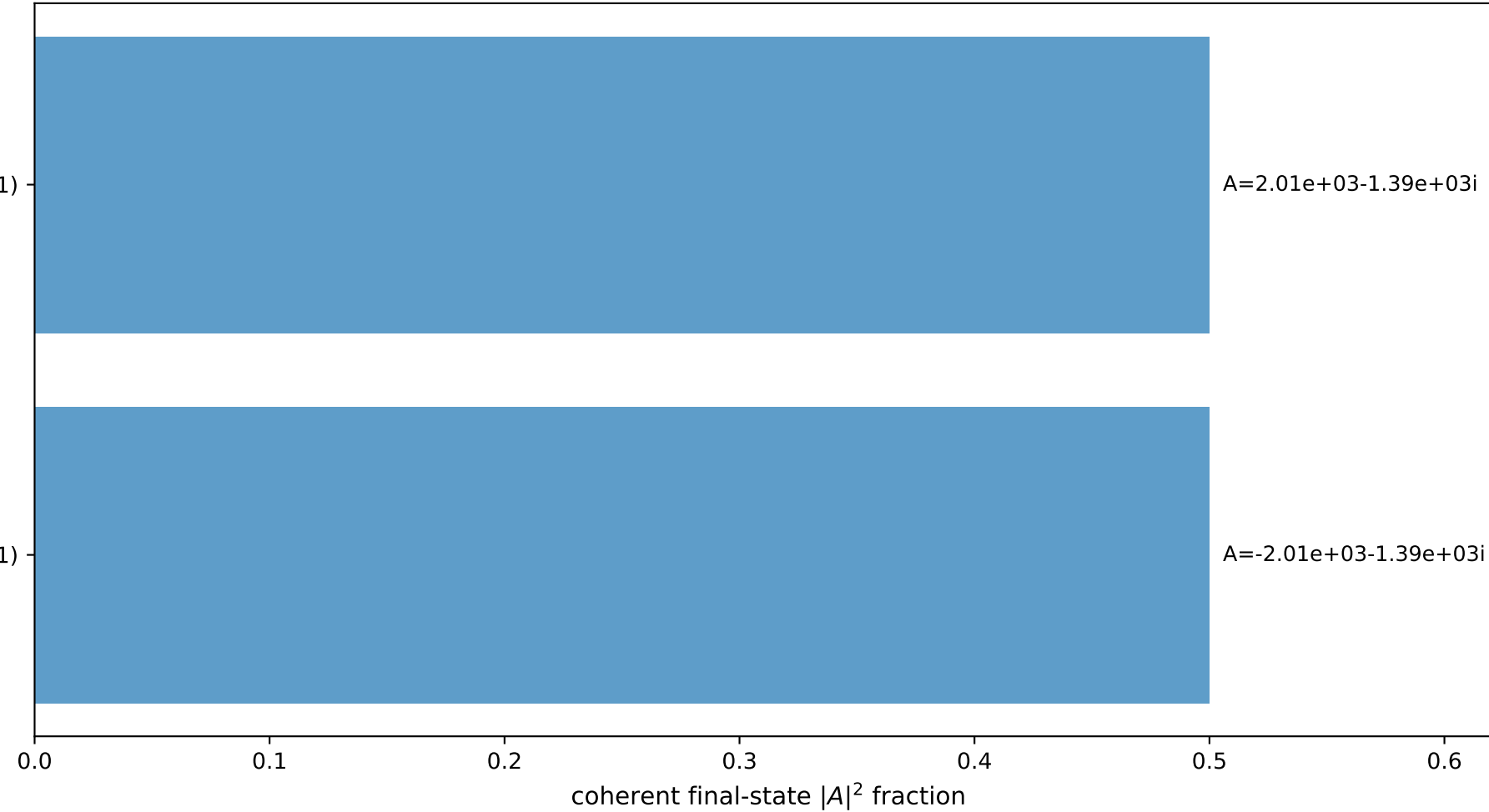
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.3968$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.3968$
 P $E= 2.5738$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.3968$
 ℓ' $E= 1.7948$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 1.7948$
 P' $E= 1.0090$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 0.3718$
 q_Y $E= 2.1667$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= -2.1667$



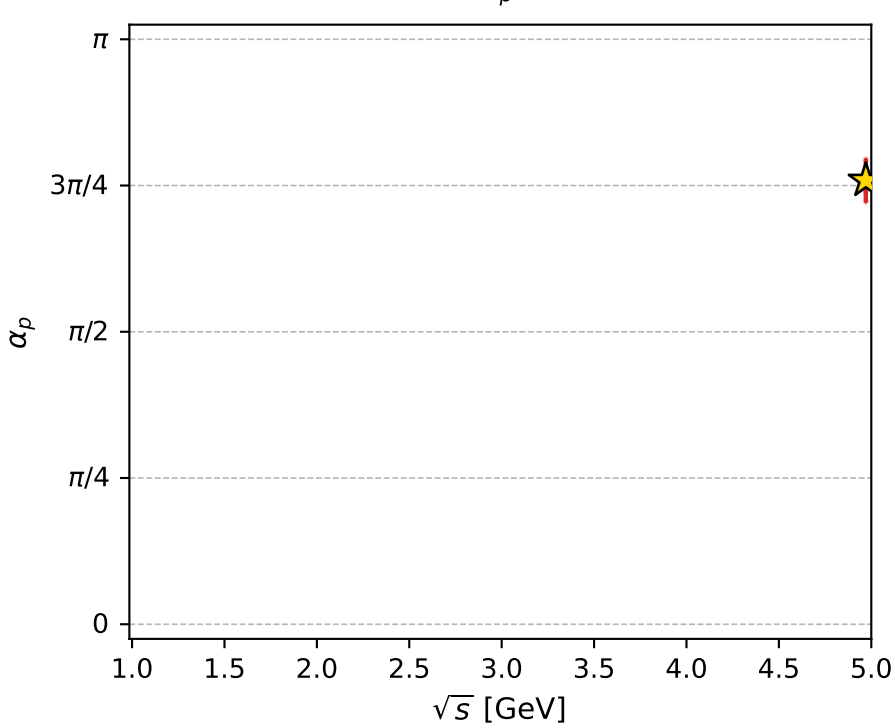
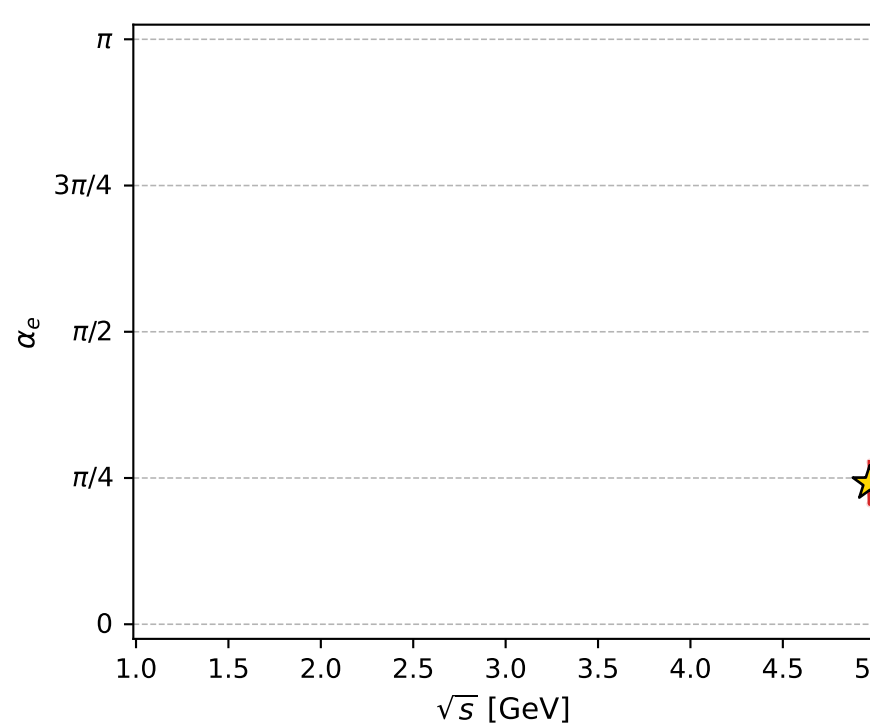
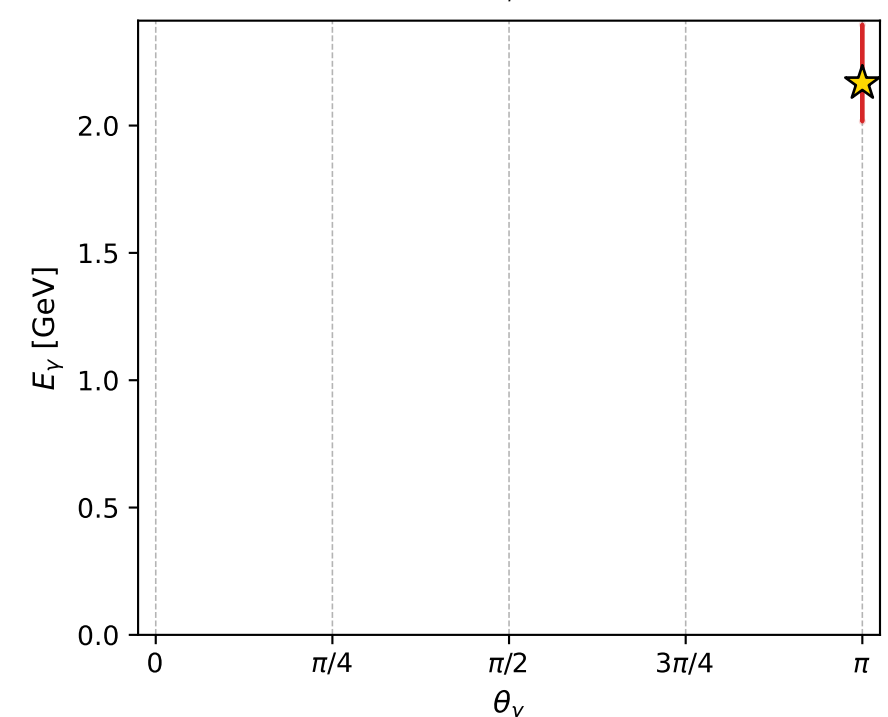
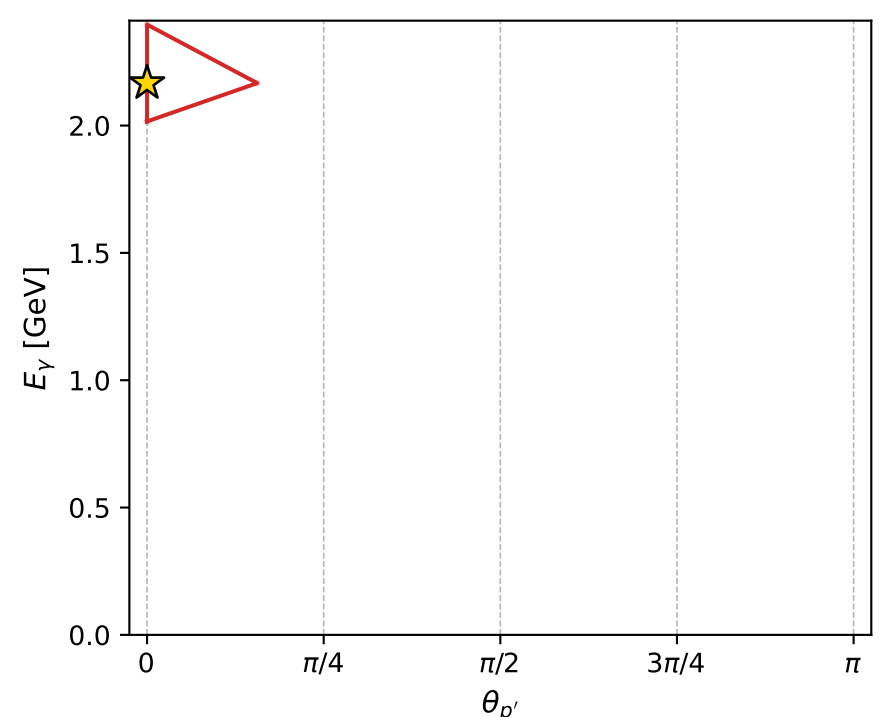
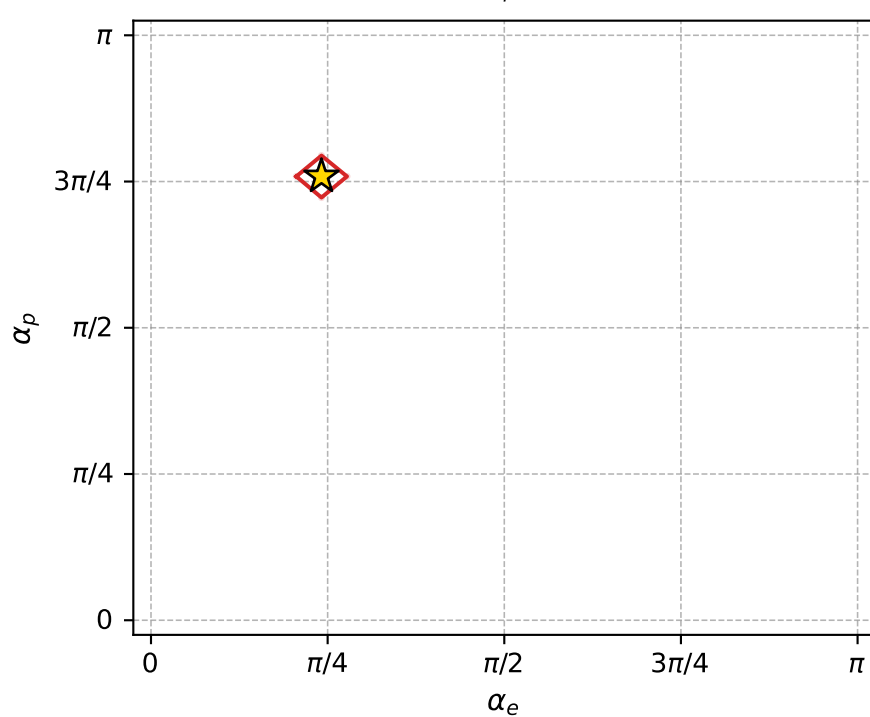
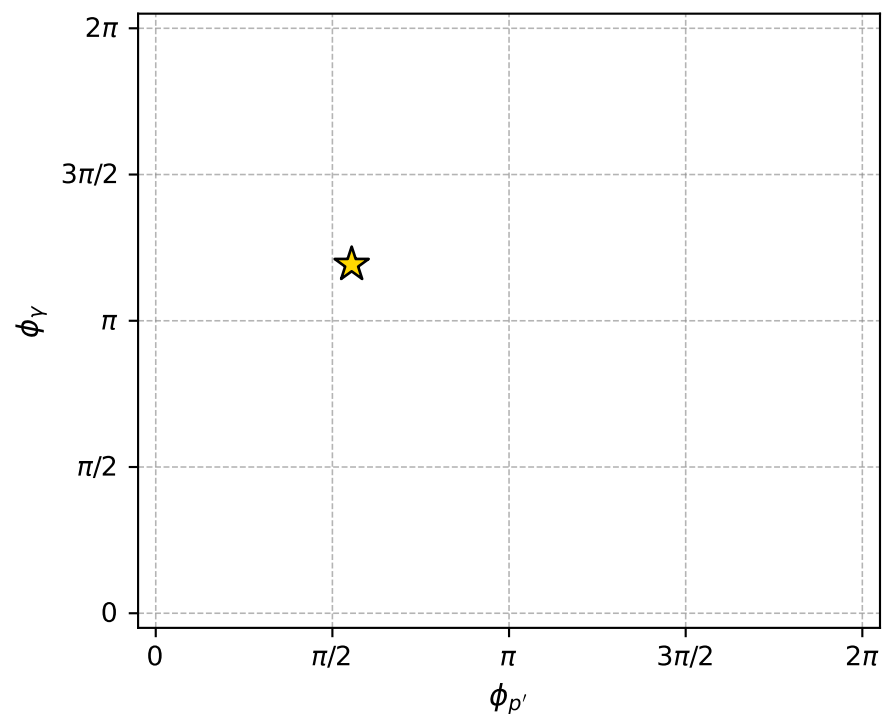
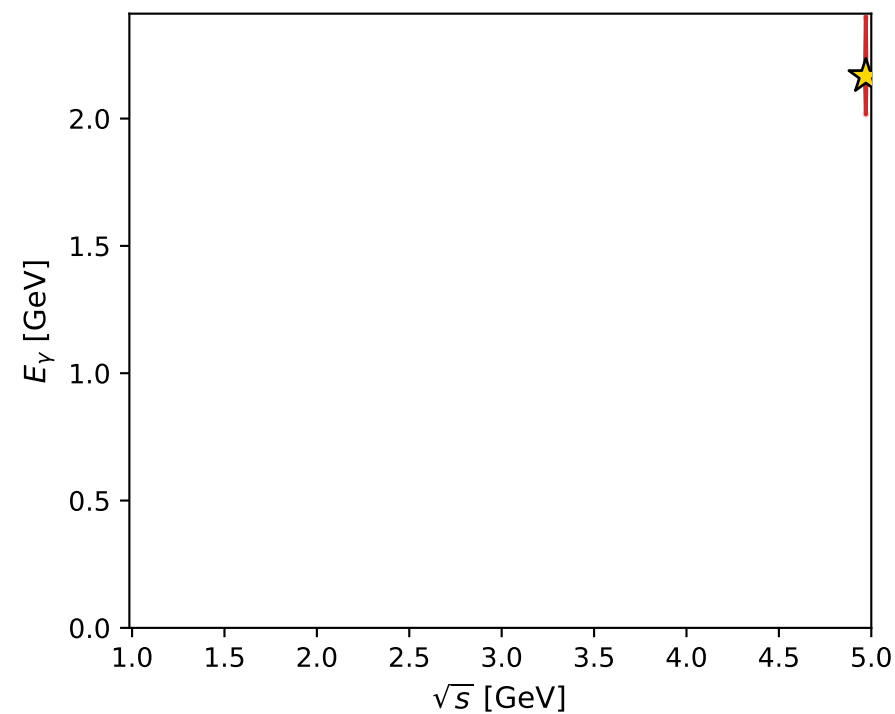
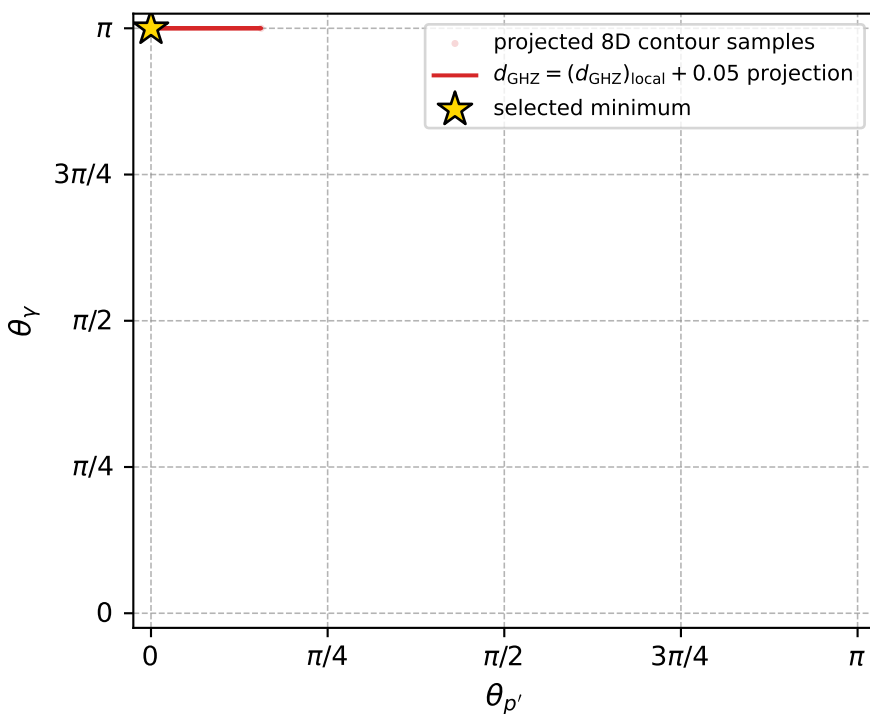
$(h_p, h_Y, h_{\ell}) = (+1, +1, -1)$

$(h_p, h_Y, h_{\ell}) = (-1, -1, +1)$

Leading final-state amplitudes (N=2, retained=100.0%)



electron: polarization cluster P2, local minimum 0; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 2.2283462\text{e-}08$
 $d_{\text{GHZ}} \text{ contour} = 0.050000022$
 above global minimum = 0
 8D contour samples = 81

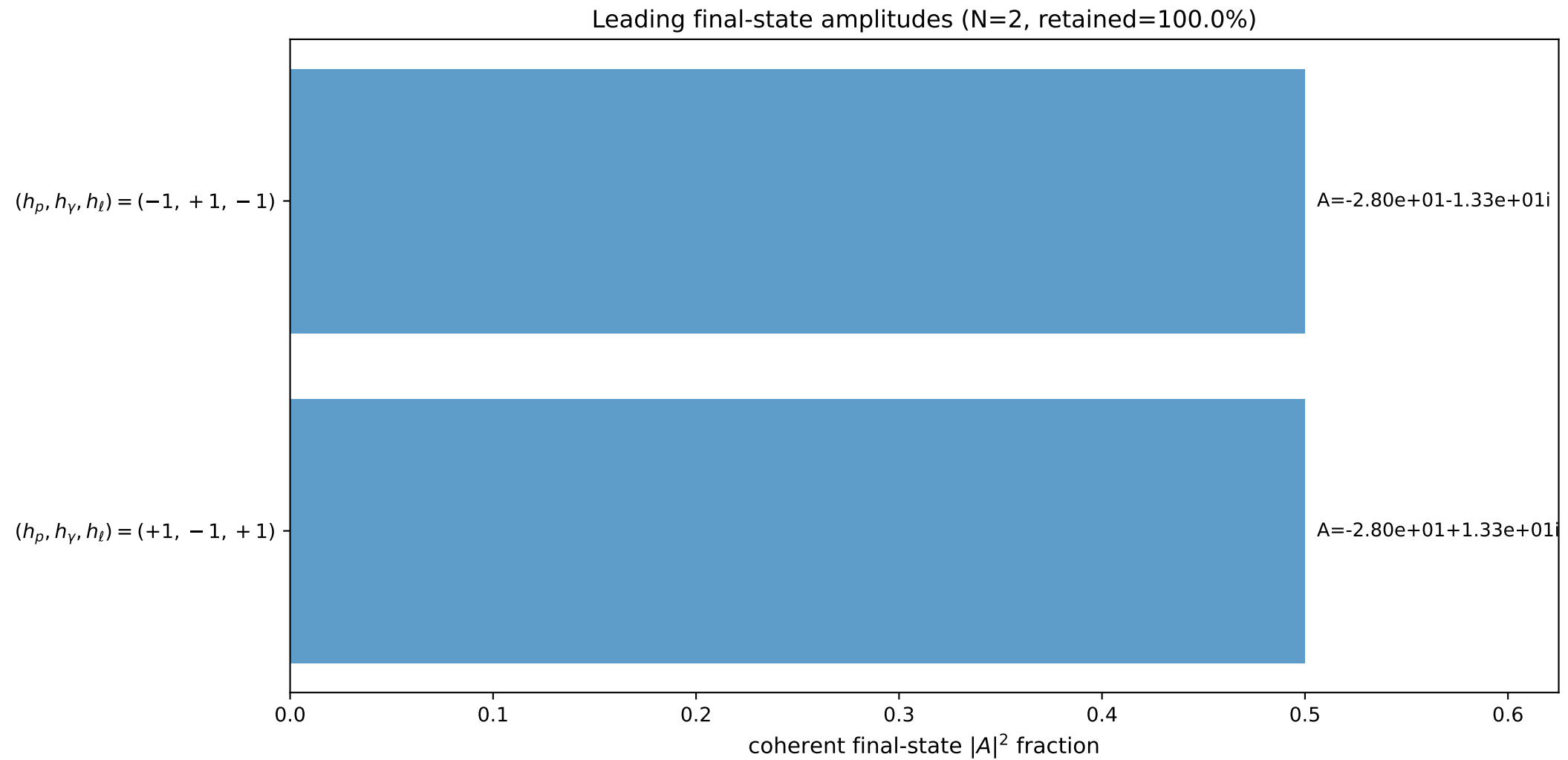
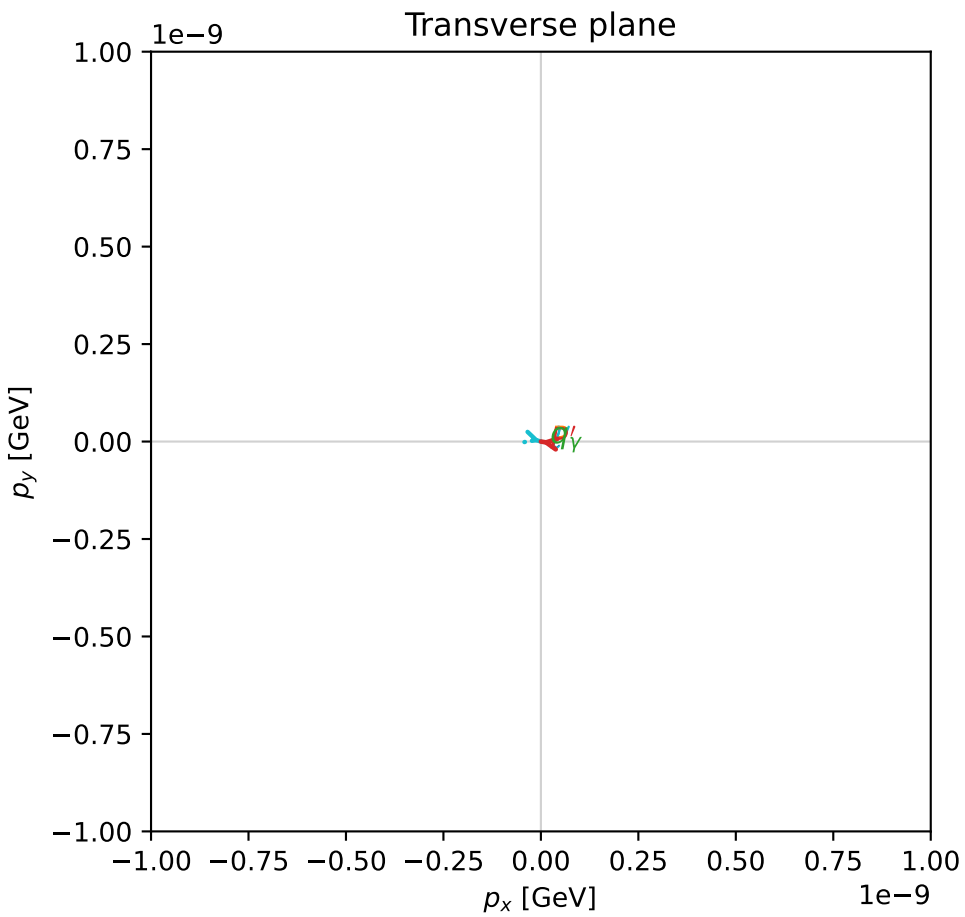
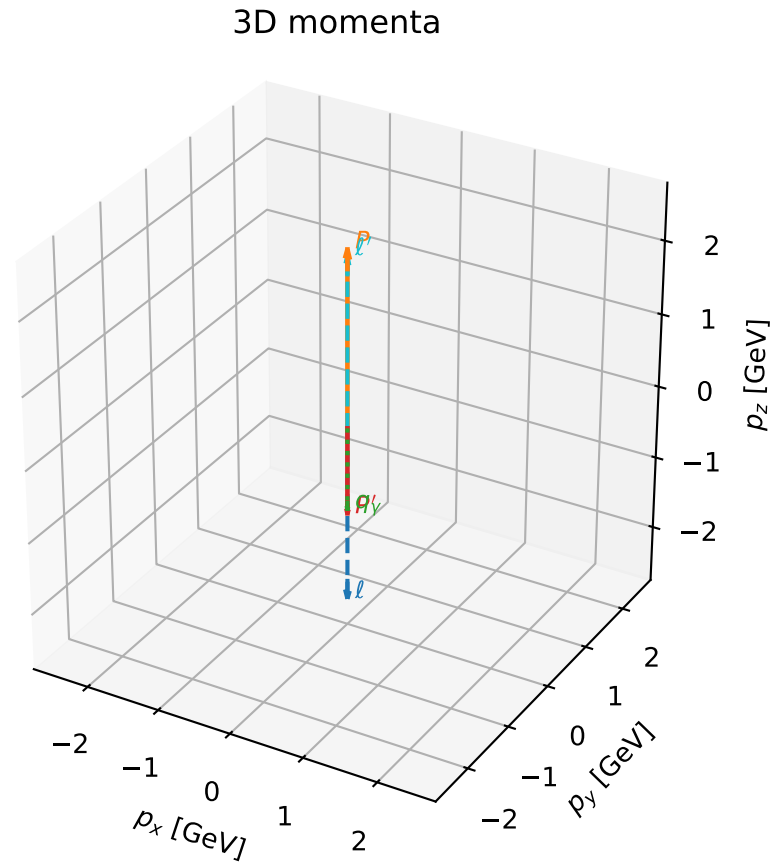
selected phase-space point:
 $\text{sqrt_s} = 4.970528$
 $\text{theta_p_out} = 1.301108\text{e-}07$
 $\text{theta_gamma_out} = 3.141593$
 $\text{q0ut} = 2.166672$
 $\text{phi_p_out} = 1.742183$
 $\text{phi_gamma_out} = 3.74627$
 $\text{alpha_e} = 0.7579928$
 $\text{alpha_p} = 2.383596$

coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

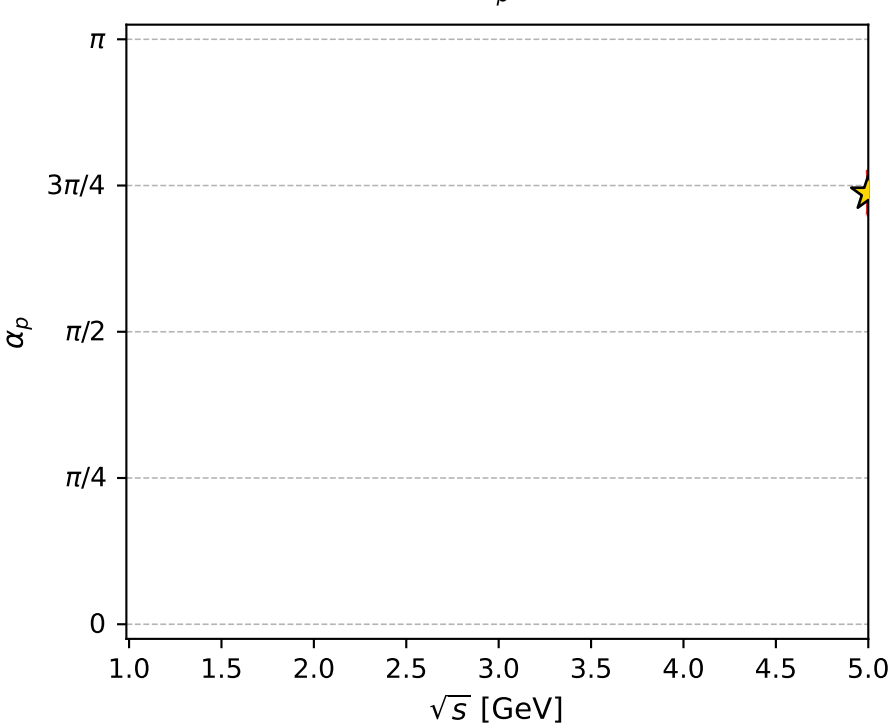
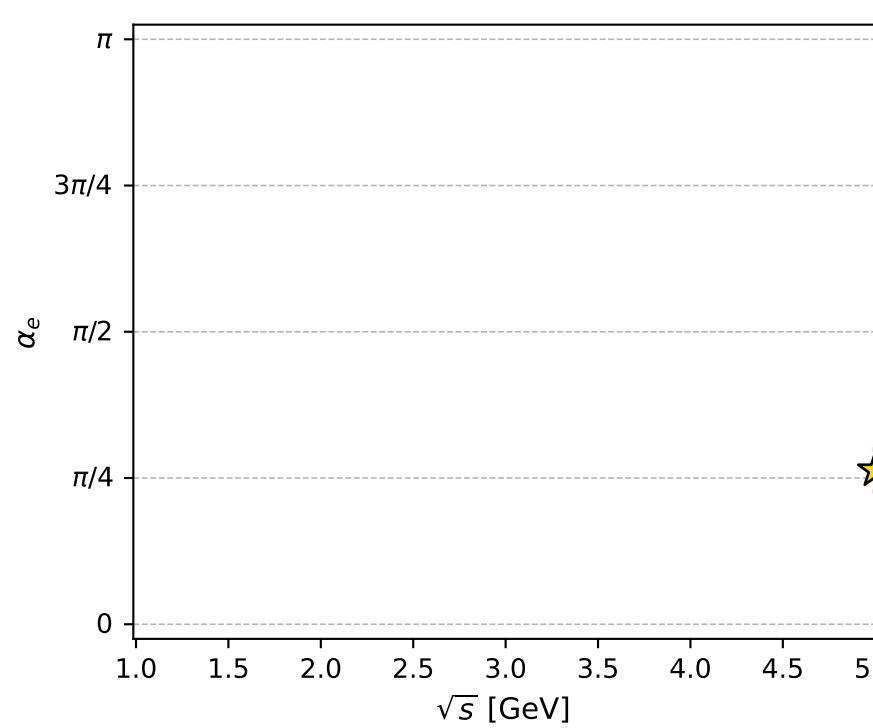
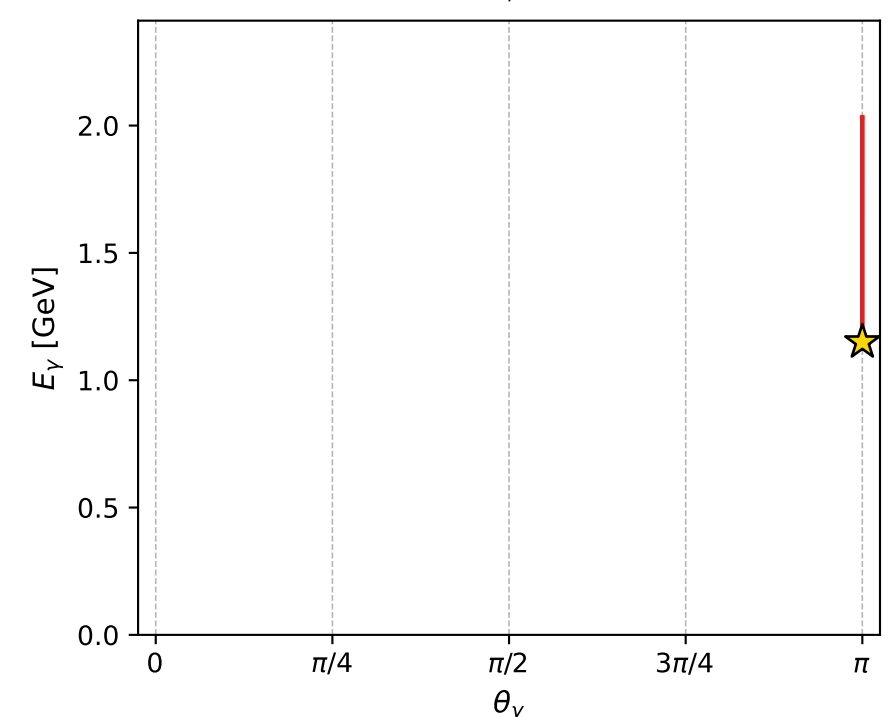
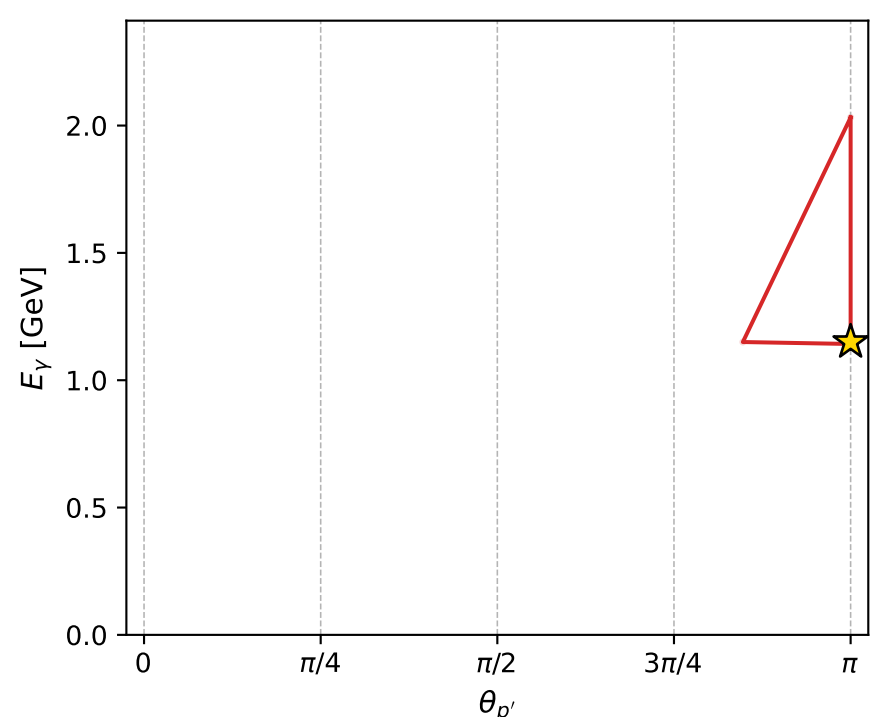
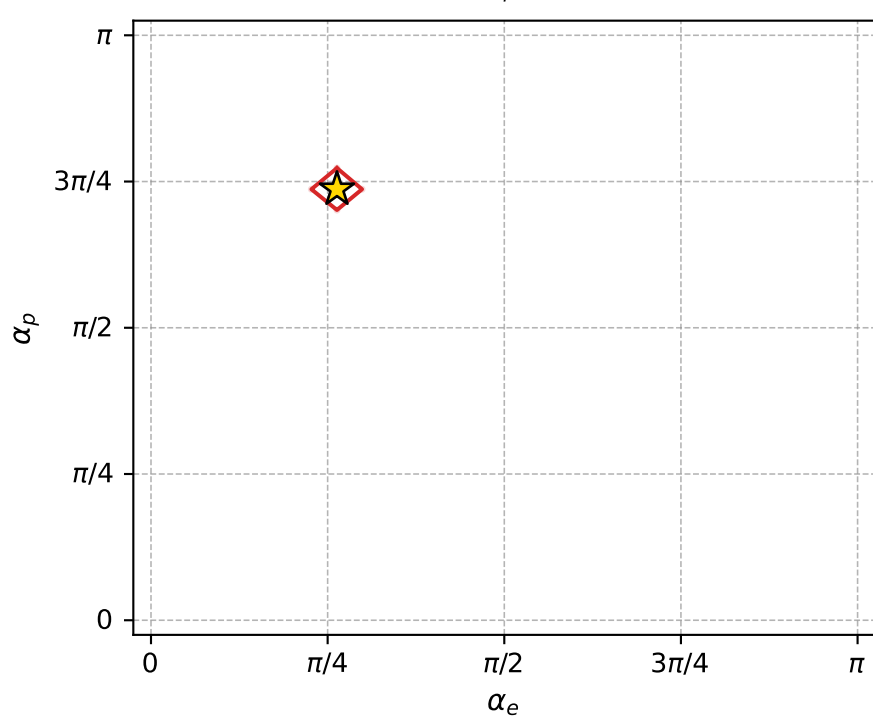
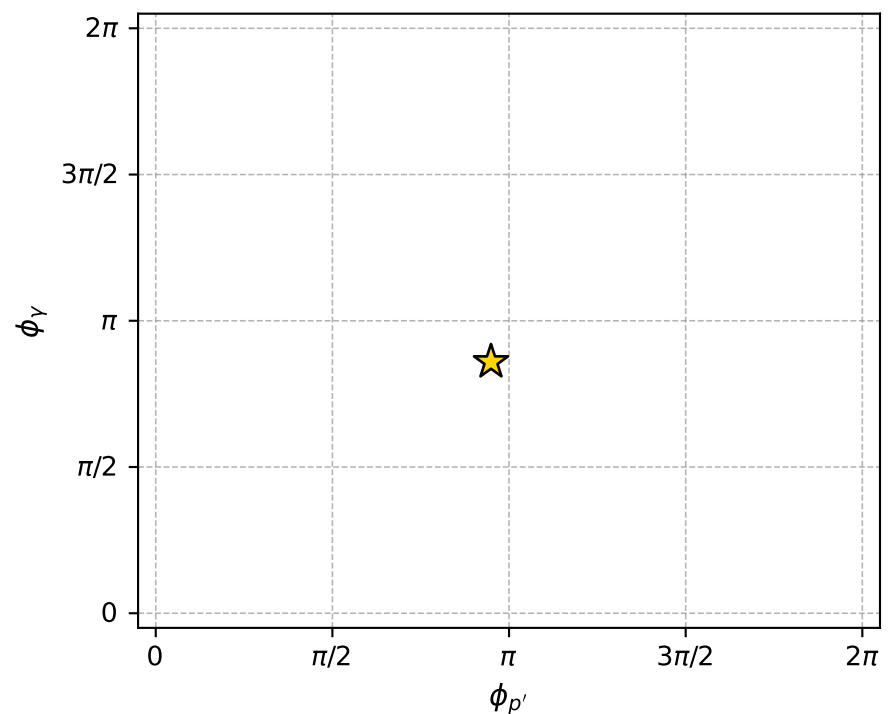
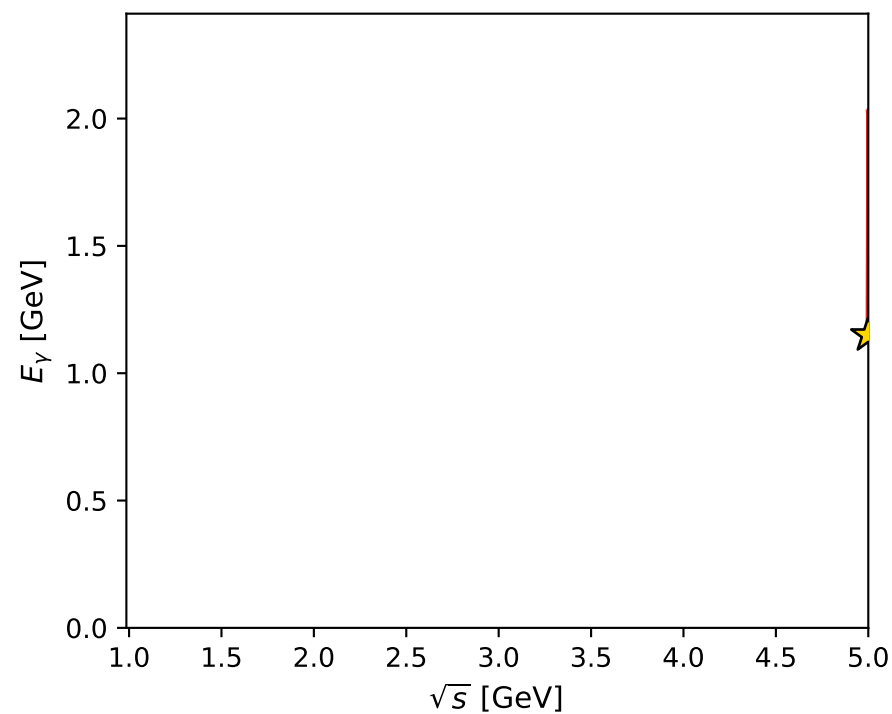
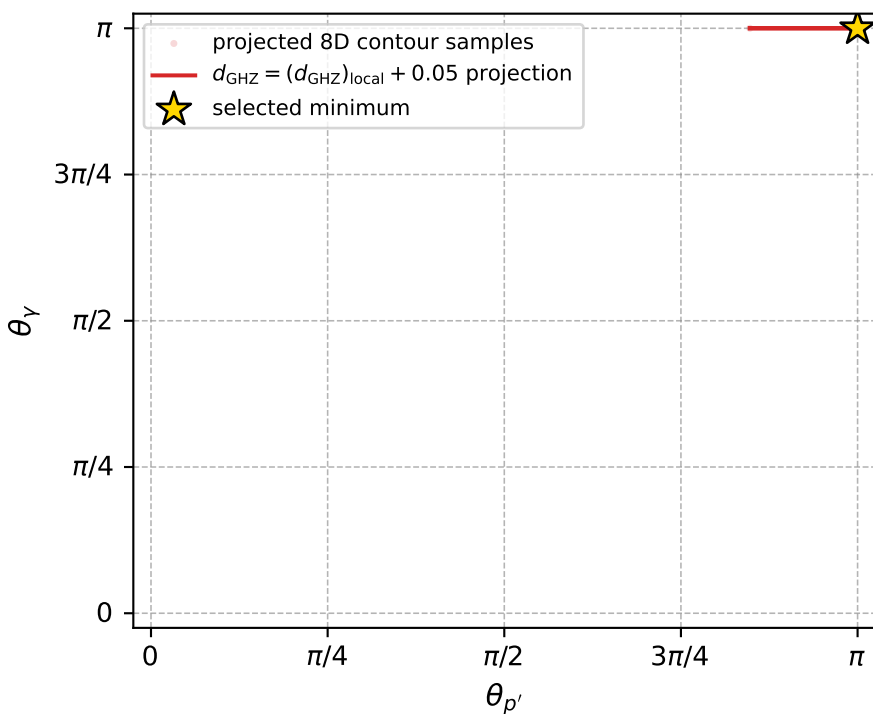
dghz_mixing_angles_polarization_02_local_minimum_3 $d_{\{\text{rm GHZ}\}}=2.32433\text{e-}08$
 region: polarization_cluster_02_local_minimum_3
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=0.82685667$ rad
 incoming lepton: $0.677192|+\rangle +0.735806|-\rangle$
 proton mixing angle: $\alpha_p=2.3147292$ rad
 incoming proton: $-0.677187|+\rangle +0.735811|-\rangle$

$s=24.9996$, $\sqrt{s}=4.99996$, $|\vec{P}|=2.412$, $|\vec{P}'|=1.18707$
 $\theta_{p'}=3.14159$, $\theta_Y=3.14159$, $\phi_{p'}=2.98106$, $\phi_Y=2.69995$
 $E_Y=1.14997$, $\theta_{\ell'}=0$, $\phi_{\ell'}=5.98435$
 $Q^2=22.5478$, $x_B=0.96783$, $t=-11.7976$, $W^2=1.62932$, $y=0.965899$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4120$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 P $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4120$
 ℓ' $E= 2.3370$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 2.3370$
 P' $E= 1.5129$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= -1.1871$
 q_Y $E= 1.1500$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= -1.1500$



electron: polarization cluster P2, local minimum 3; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 2.3243342\text{e-}08$
 $d_{\text{GHZ}} \text{ contour} = 0.050000023$
 above global minimum = $9.5987982\text{e-}10$
 8D contour samples = 57

selected phase-space point:

$\text{sqrt_s} = 4.999963$
 $\text{theta_p_out} = 3.141593$
 $\text{theta_gamma_out} = 3.141593$
 $\text{q0out} = 1.149975$
 $\text{phi_p_out} = 2.981065$
 $\text{phi_gamma_out} = 2.699954$
 $\text{alpha_e} = 0.8268567$
 $\text{alpha_p} = 2.314729$

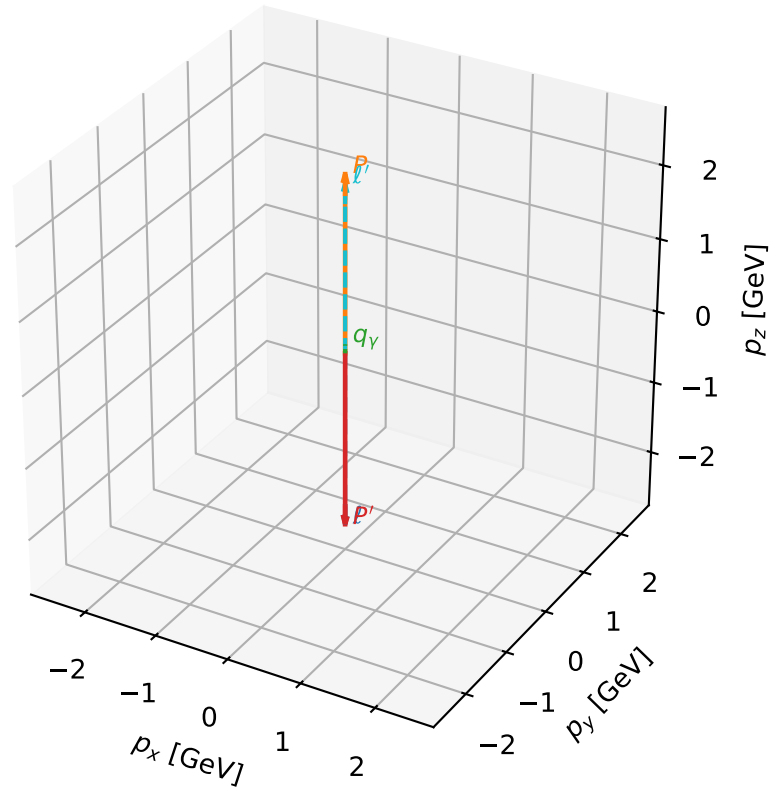
coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

dghz_mixing_angles_polarization_02_local_minimum_4 d_{\rm GHZ}=2.36714e-08
region: polarization_cluster_02_local_minimum_4
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: $\alpha_e=0.81232036$ rad
incoming lepton: $0.687816|+\rangle +0.725885|-\rangle$
proton mixing angle: $\alpha_p=2.3831047$ rad
incoming proton: $-0.725877|+\rangle +0.687825|-\rangle$

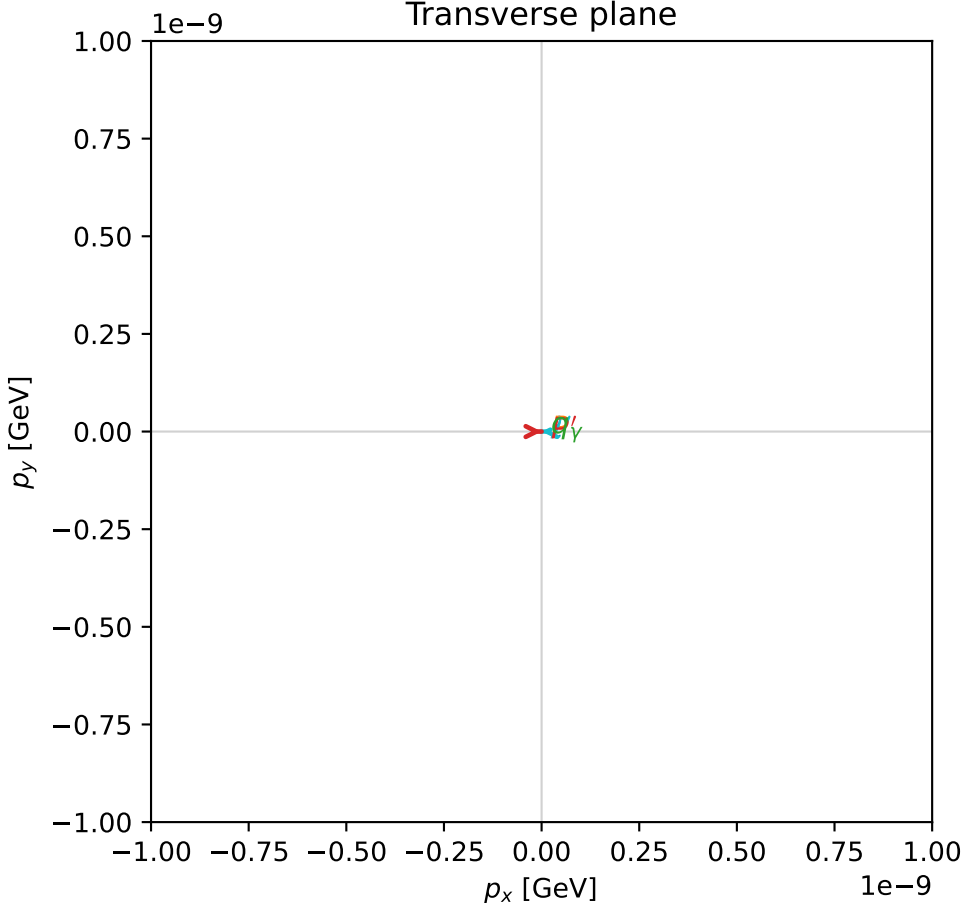
$s=25$, $\sqrt{s}=5$, $|\vec{P}|=2.41202$, $|\vec{P}'|=2.41202$
 $\theta_{P'}=3.14159$, $\theta_Y=0$, $\phi_{P'}=0$, $\phi_Y=0.437635$
 $E_Y=0.120596$, $\theta_{\ell'}=0$, $\phi_{\ell'}=3.14159$
 $Q^2=22.1078$, $x_B=0.948273$, $t=-23.2713$, $W^2=2.0858$, $y=0.966566$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4120$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 P $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4120$
 ℓ' $E= 2.2914$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 2.2914$
 P' $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 q_Y $E= 0.1206$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1206$

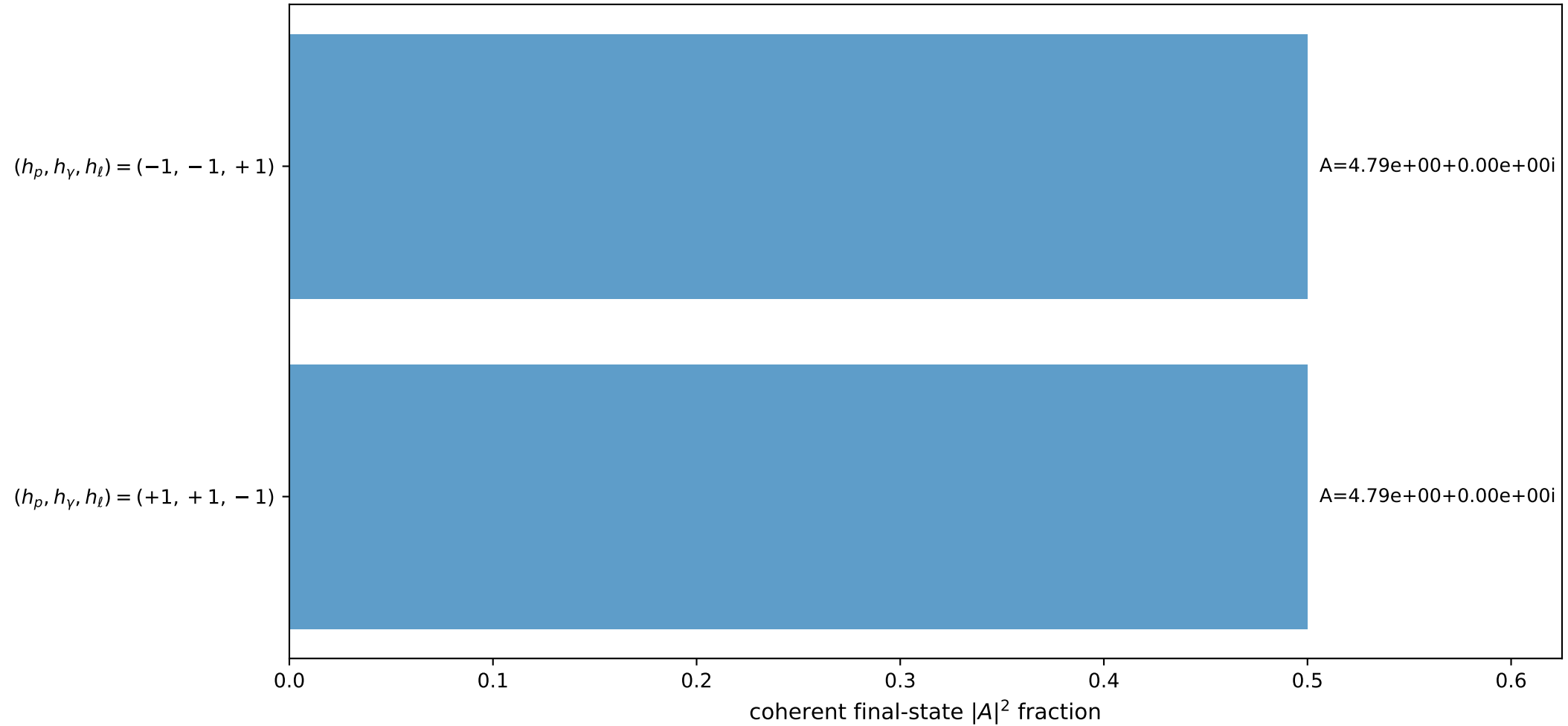
3D momenta



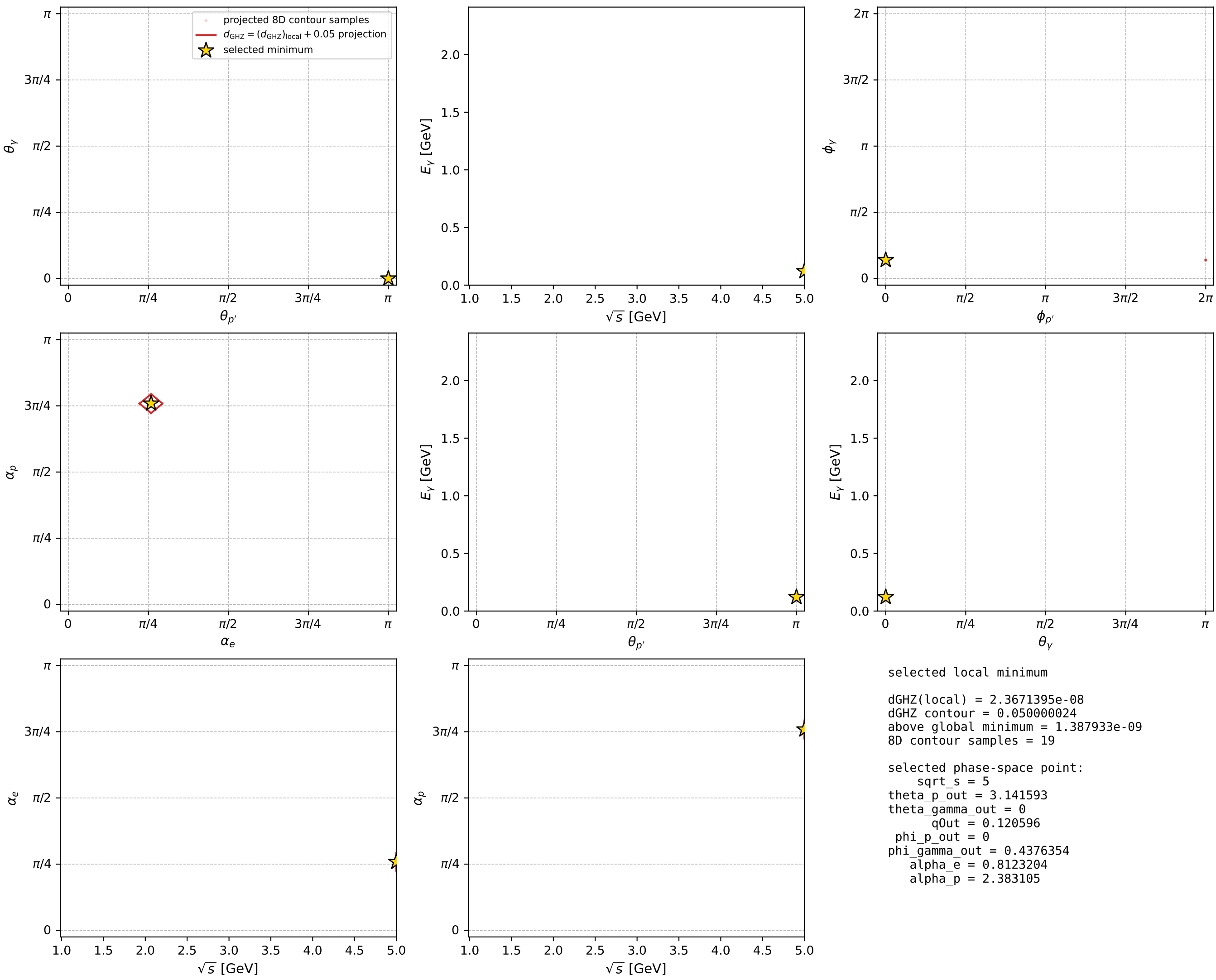
Transverse plane



Leading final-state amplitudes (N=2, retained=100.0%)



electron: polarization cluster P2, local minimum 4; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour

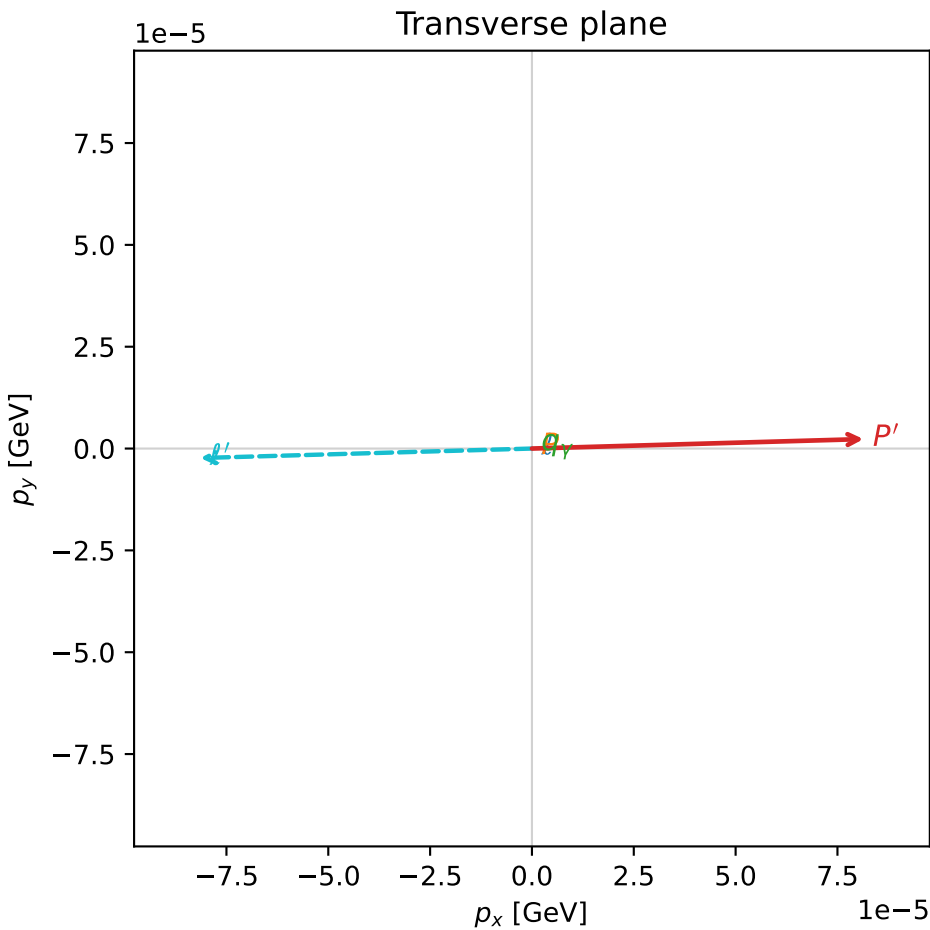
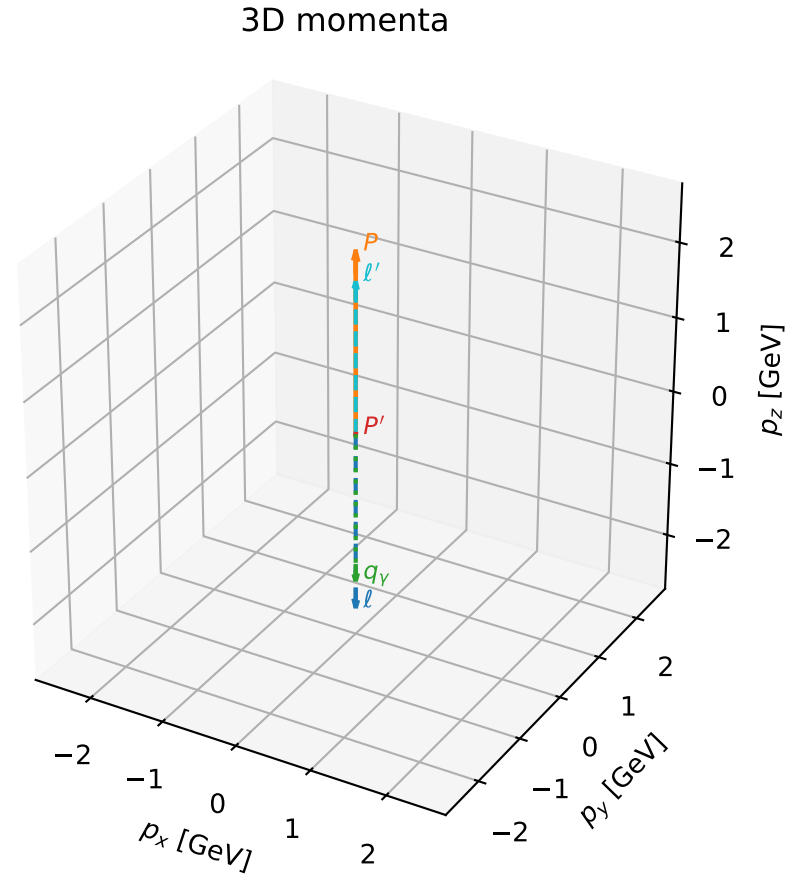


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

dghz_mixing_angles_polarization_02_local_minimum_6 $d_{\{\text{rm GHZ}\}}=2.39329\text{e-}08$
region: polarization_cluster_02_local_minimum_6
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: $\alpha_e=0.81040845$ rad
incoming lepton: $0.689203|+\rangle +0.724569|-\rangle$
proton mixing angle: $\alpha_p=2.331173$ rad
incoming proton: $-0.689194|+\rangle +0.724576|-\rangle$

$s=24.9736$, $\sqrt{s}=4.99736$, $|\vec{P}|=2.41065$, $|\vec{P}'|=0.000556214$
 $\theta_{P'}=0.146926$, $\theta_Y=3.14159$, $\phi_{P'}=0.0285838$, $\phi_Y=0.891756$
 $E_Y=2.02996$, $\theta_{\ell'}=4.01243\text{e-}05$, $\phi_{\ell'}=3.17018$
 $Q^2=19.5688$, $x_B=0.837016$, $t=-3.09033$, $W^2=4.69028$, $y=0.970341$

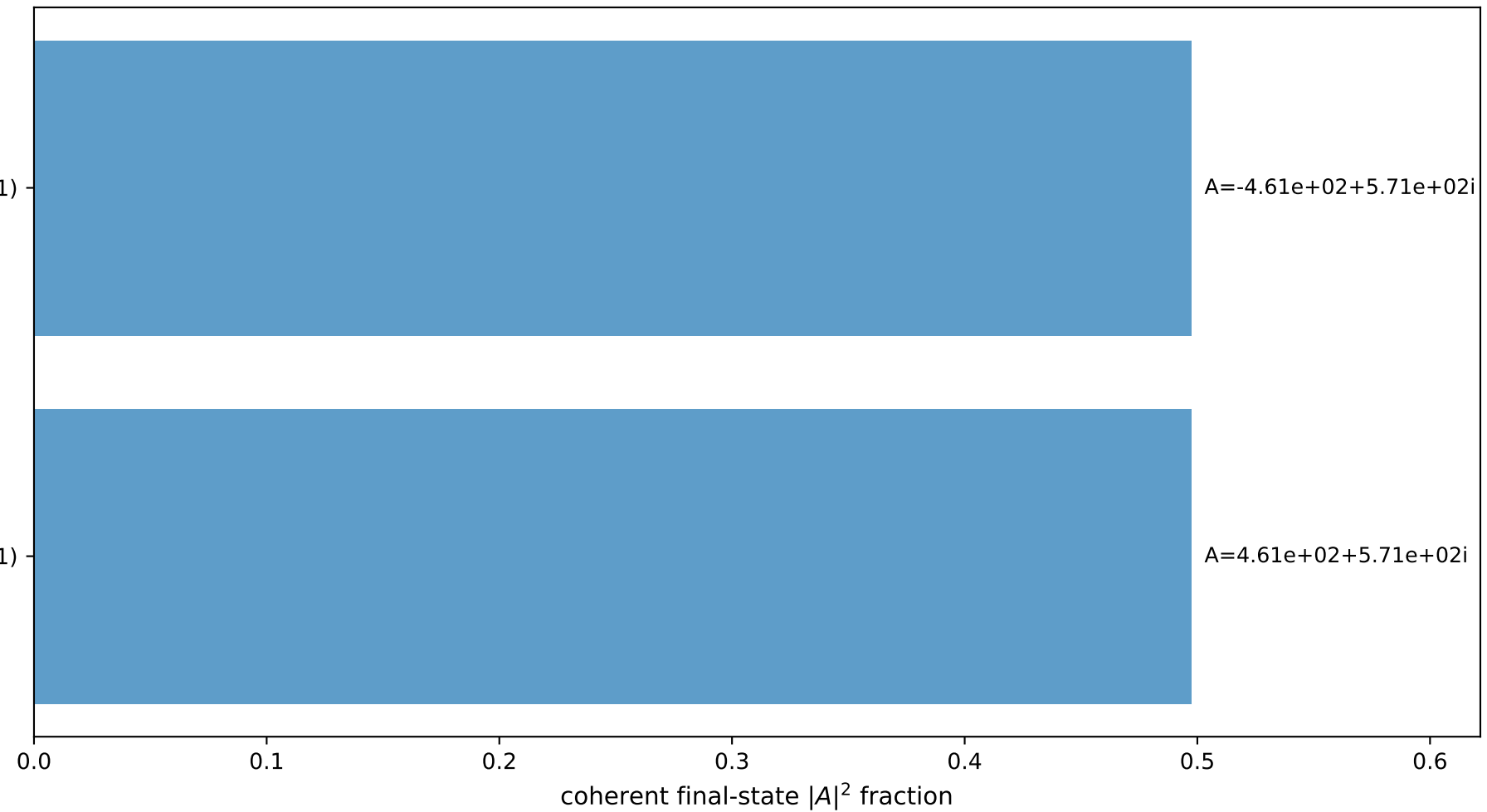
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4107$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4107$
 P $E= 2.5867$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4107$
 ℓ' $E= 2.0294$ $p_x= -0.0001$ $p_y= -0.0000$ $p_z= 2.0294$
 P' $E= 0.9380$ $p_x= 0.0001$ $p_y= 0.0000$ $p_z= 0.0006$
 q_Y $E= 2.0300$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.0300$



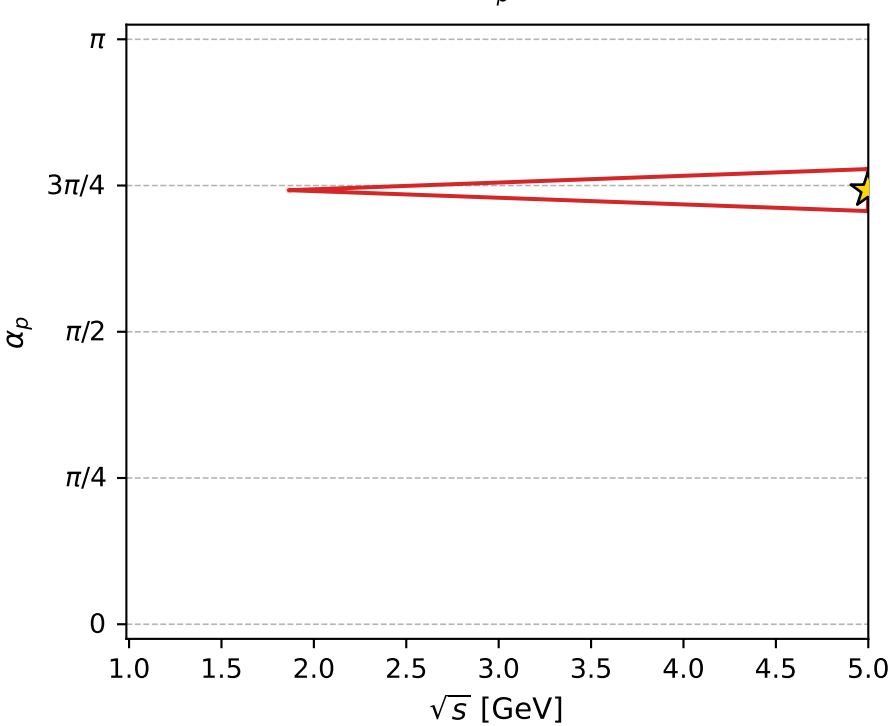
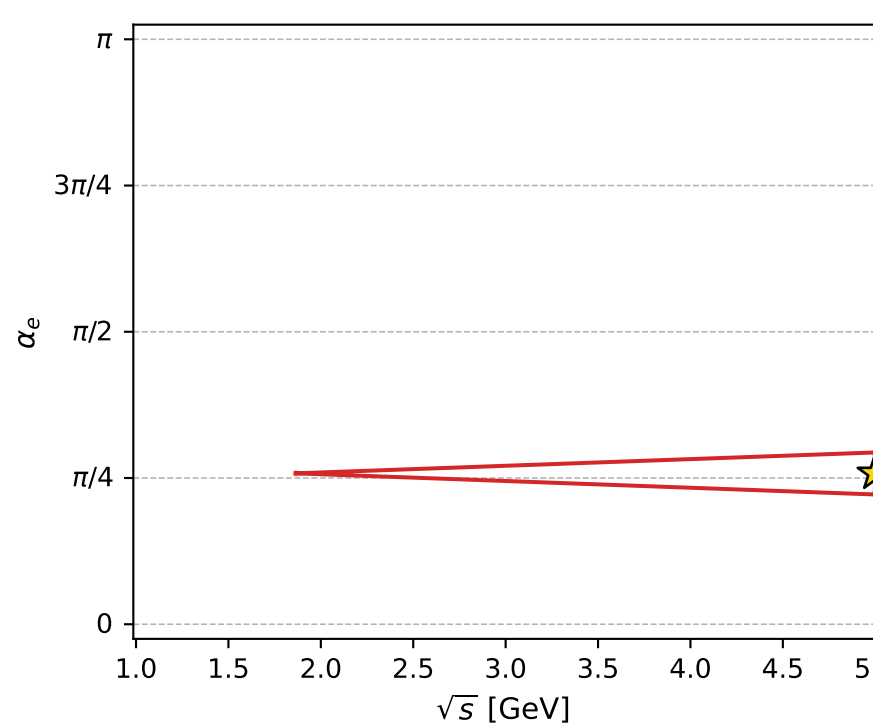
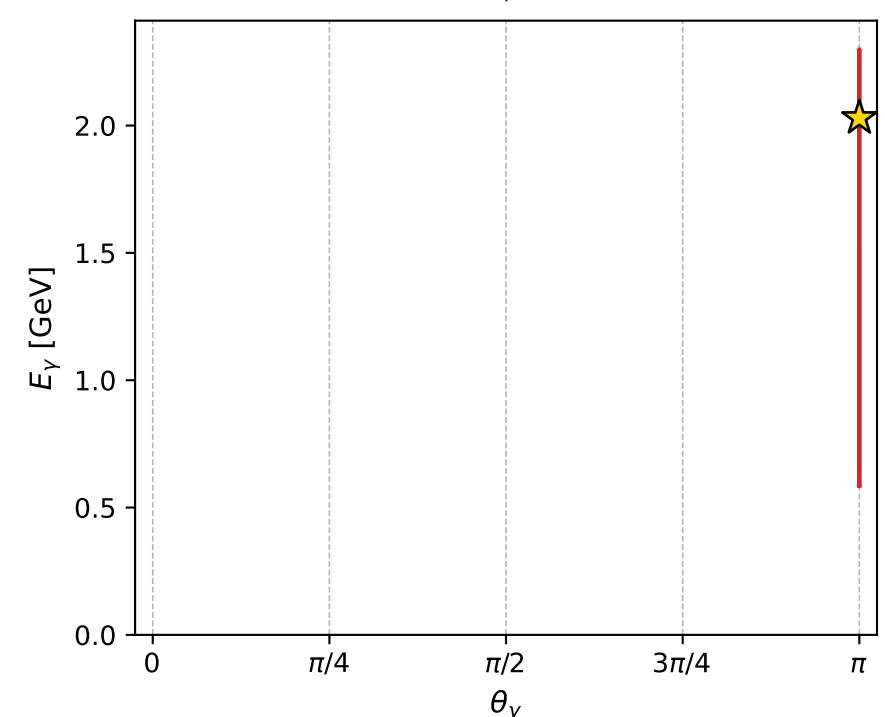
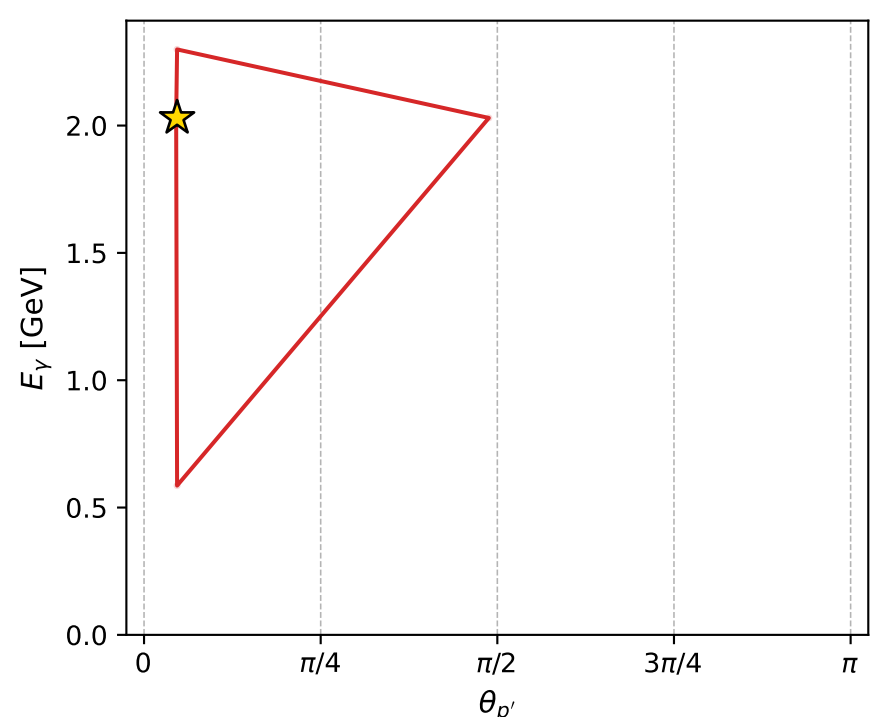
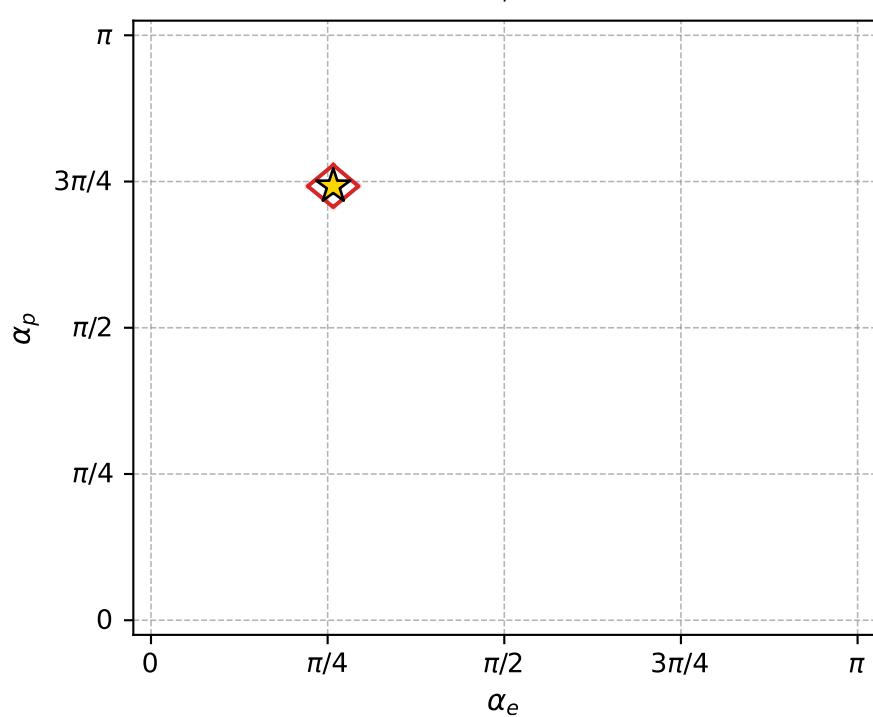
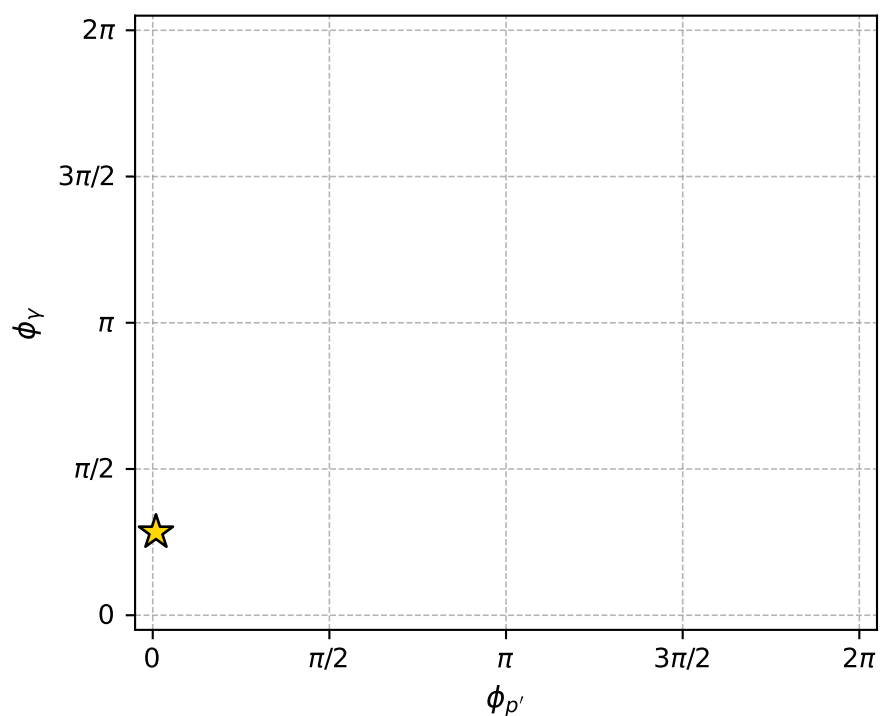
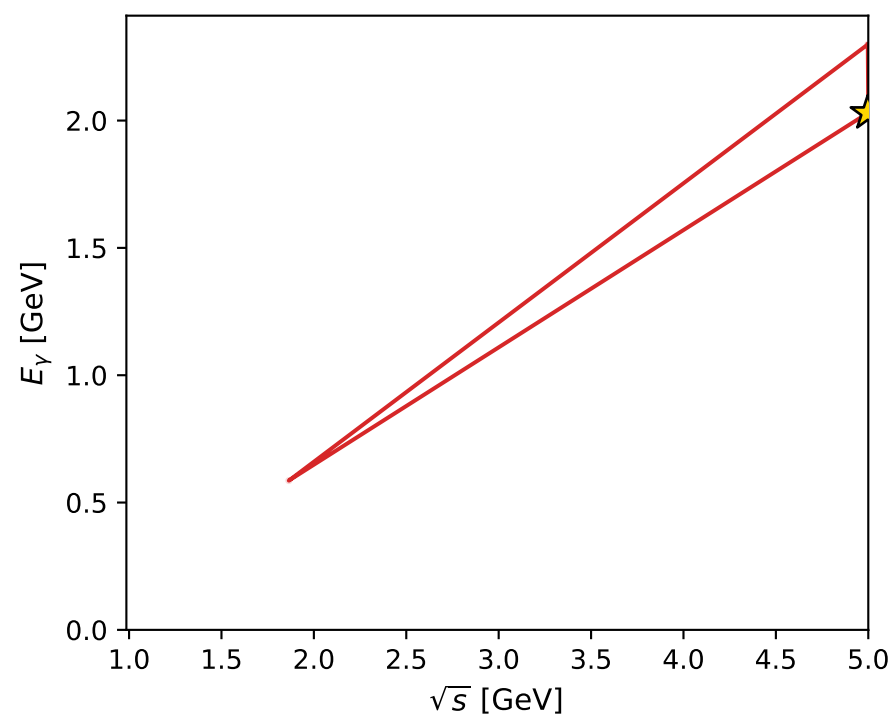
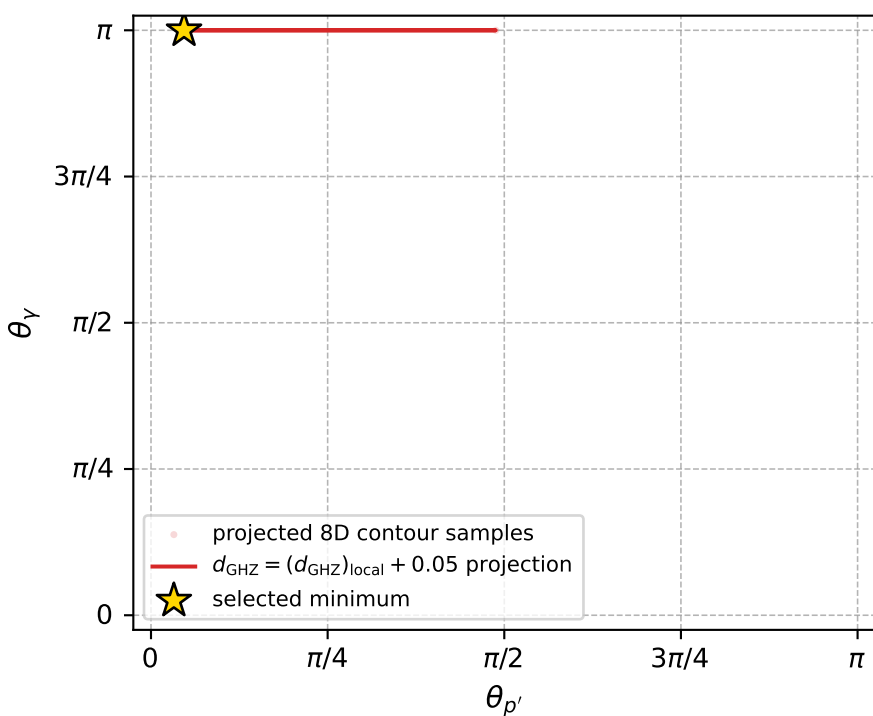
$(h_p, h_Y, h_\ell) = (+1, +1, -1)$

$(h_p, h_Y, h_\ell) = (-1, -1, +1)$

Leading final-state amplitudes (N=2, retained=99.5%)



electron: polarization cluster P2, local minimum 6; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 2.3932934\text{e-}08$
 $d_{\text{GHZ}} \text{ contour} = 0.050000024$
 above global minimum = $1.649472\text{e-}09$
 8D contour samples = 125

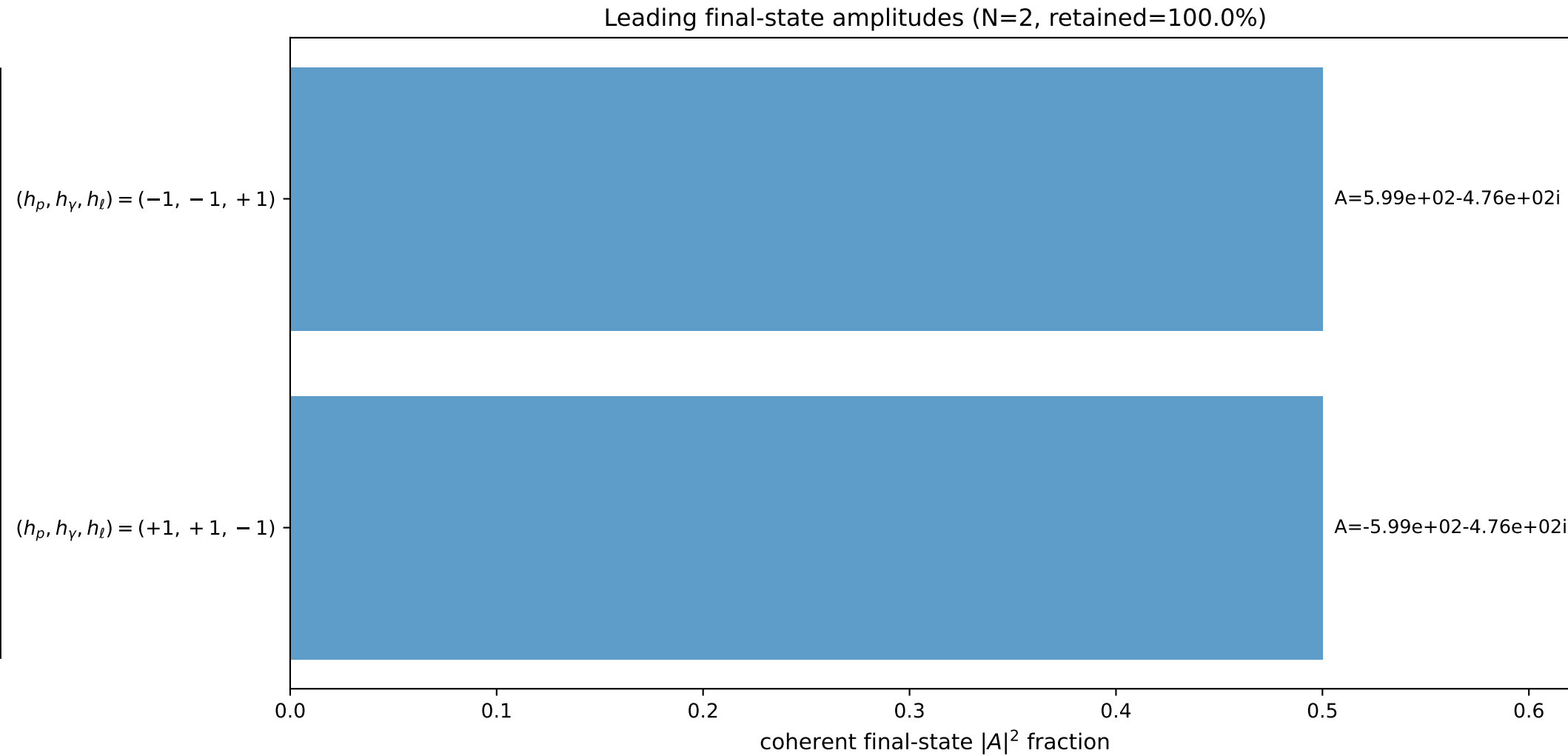
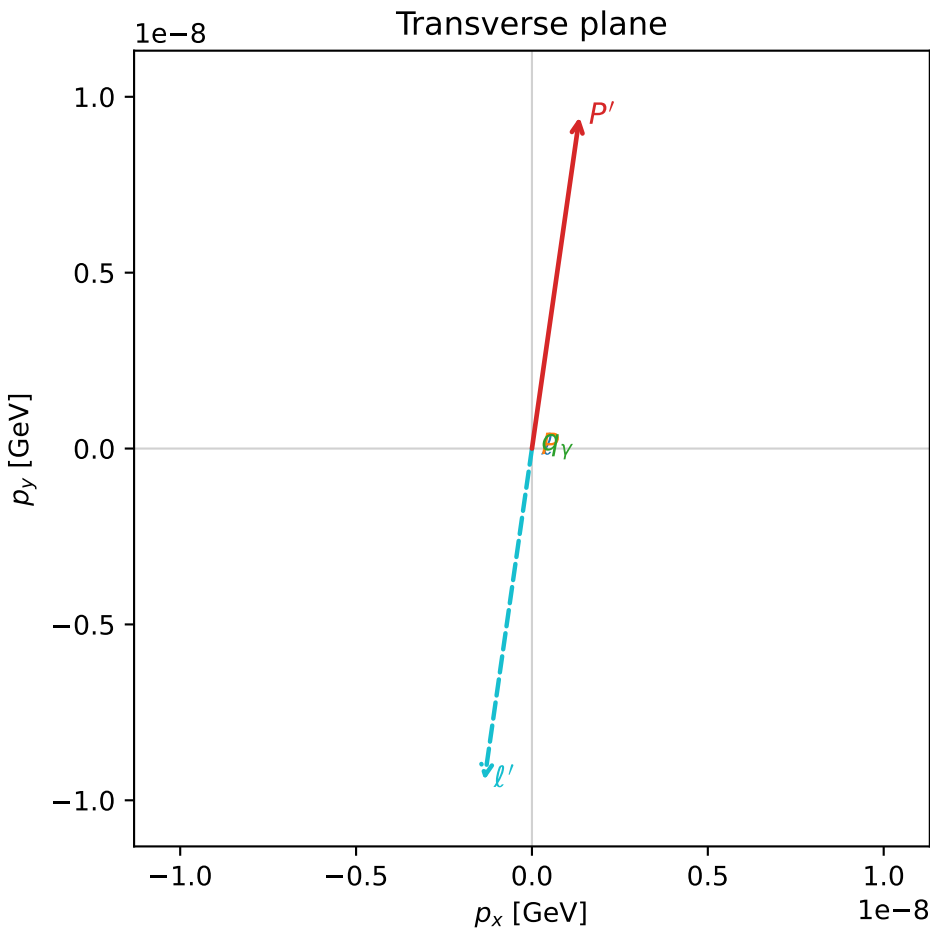
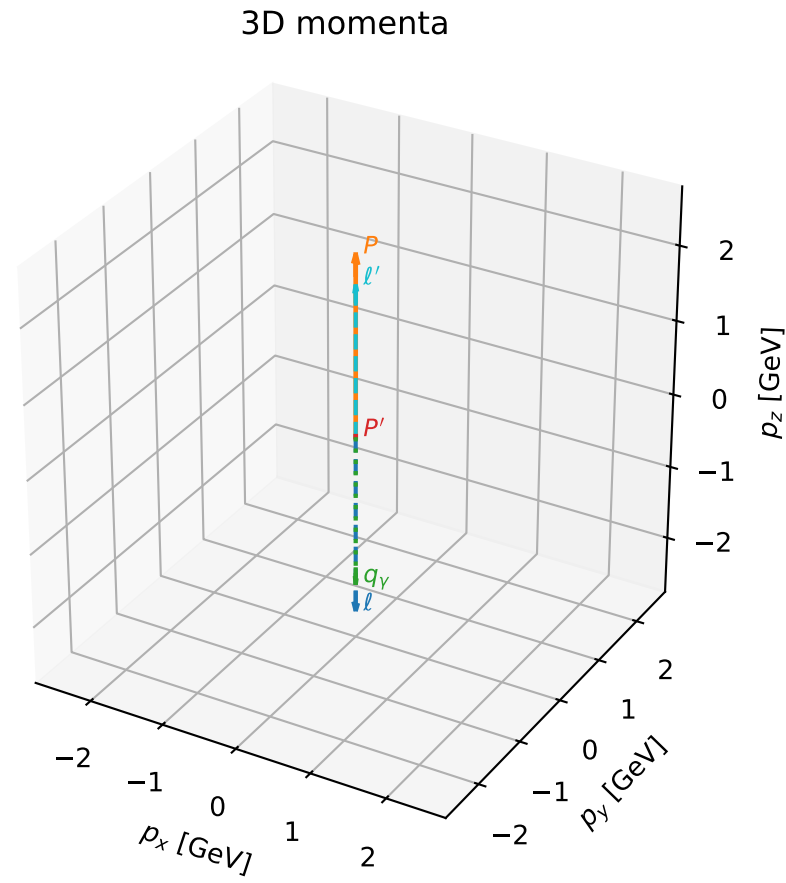
selected phase-space point:
 $\text{sqrt_s} = 4.997364$
 $\text{theta_p_out} = 0.1469258$
 $\text{theta_gamma_out} = 3.141593$
 $\text{q0ut} = 2.029957$
 $\text{phi_p_out} = 0.02858377$
 $\text{phi_gamma_out} = 0.8917557$
 $\text{alpha_e} = 0.8104085$
 $\text{alpha_p} = 2.331173$

coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

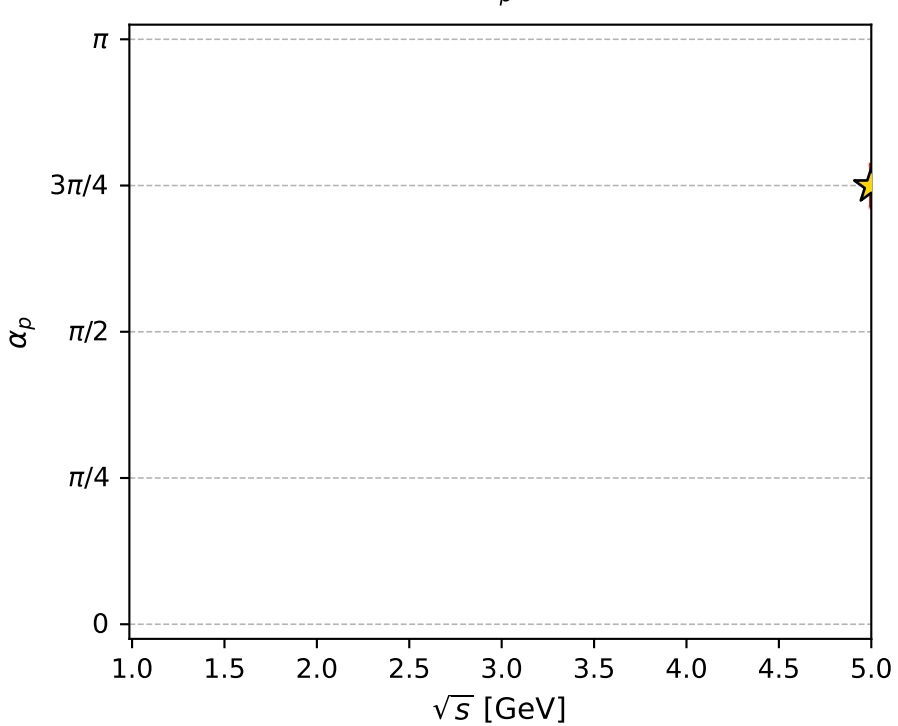
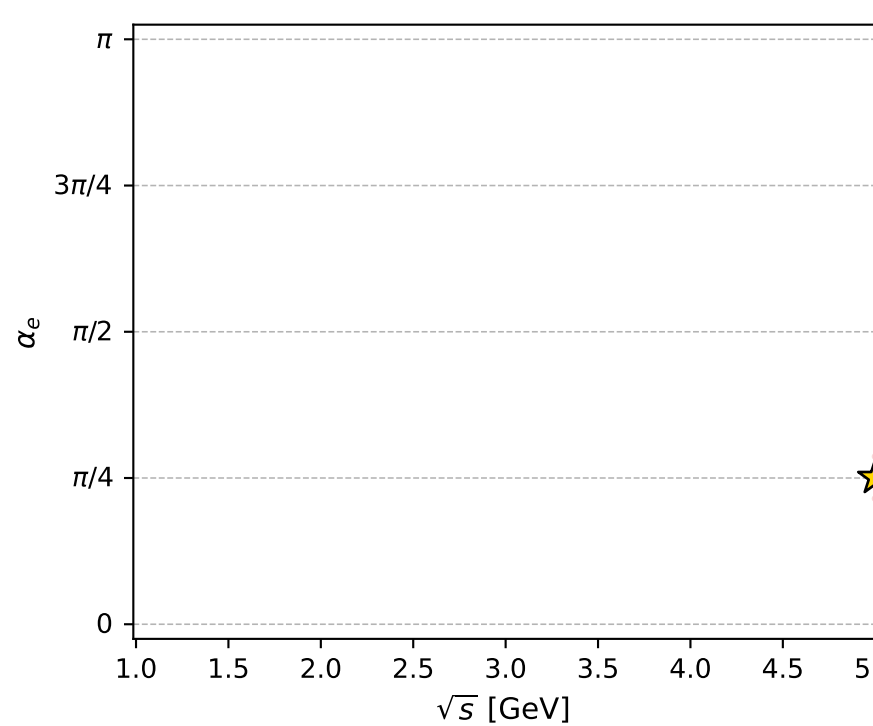
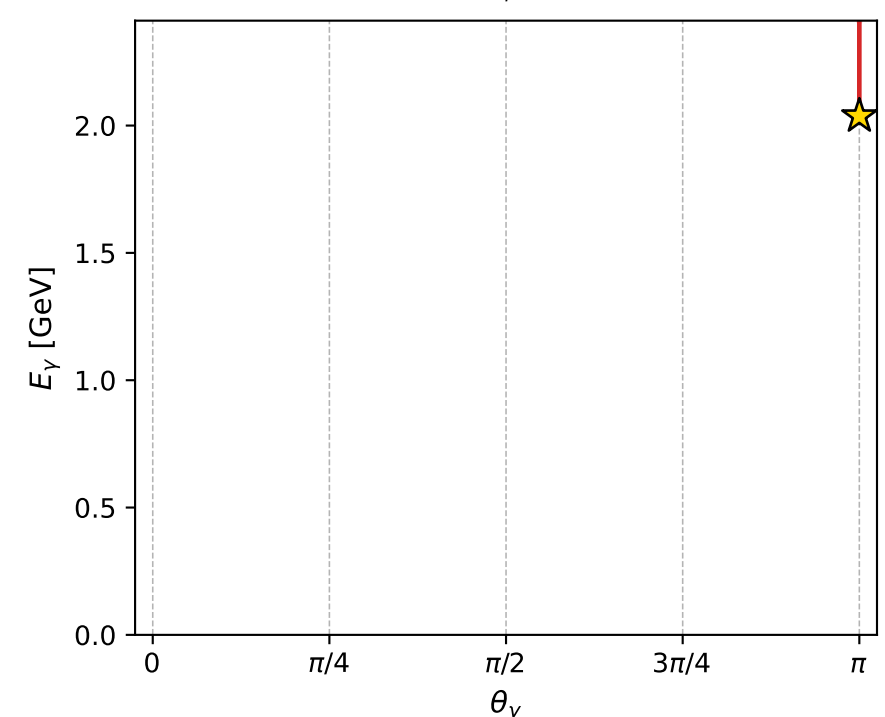
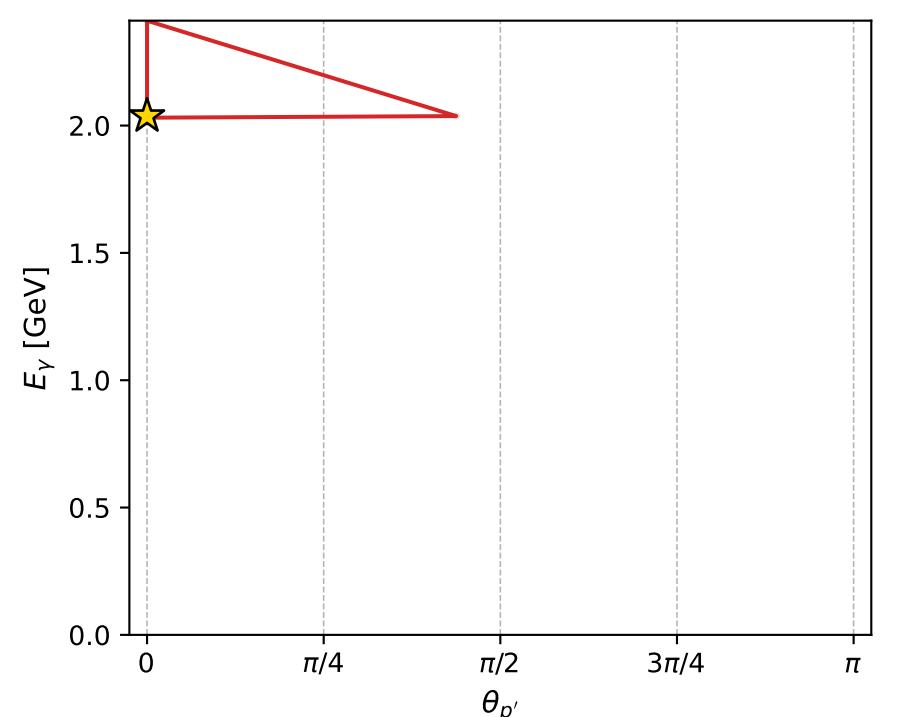
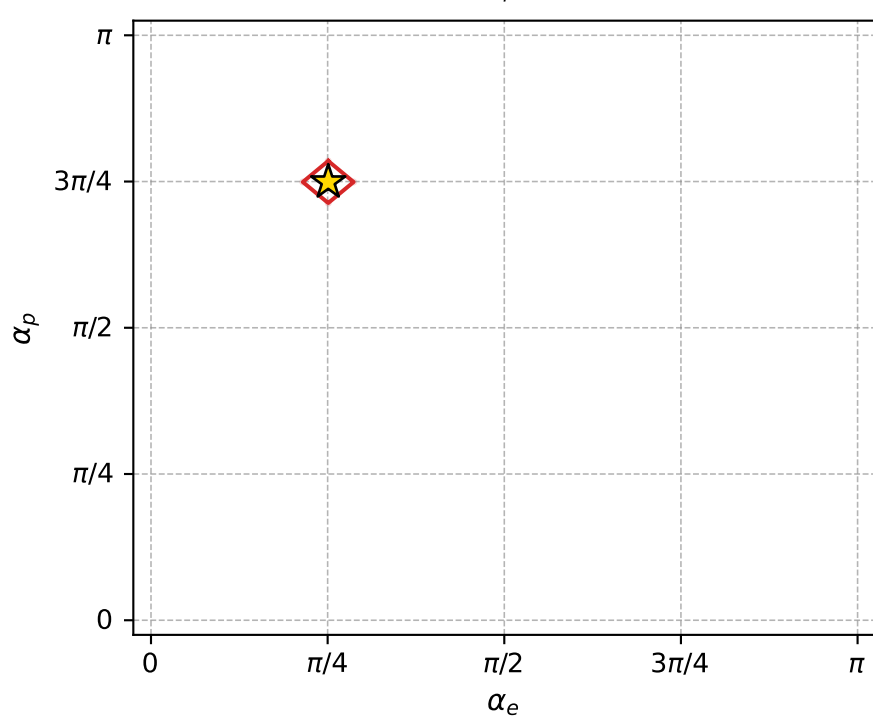
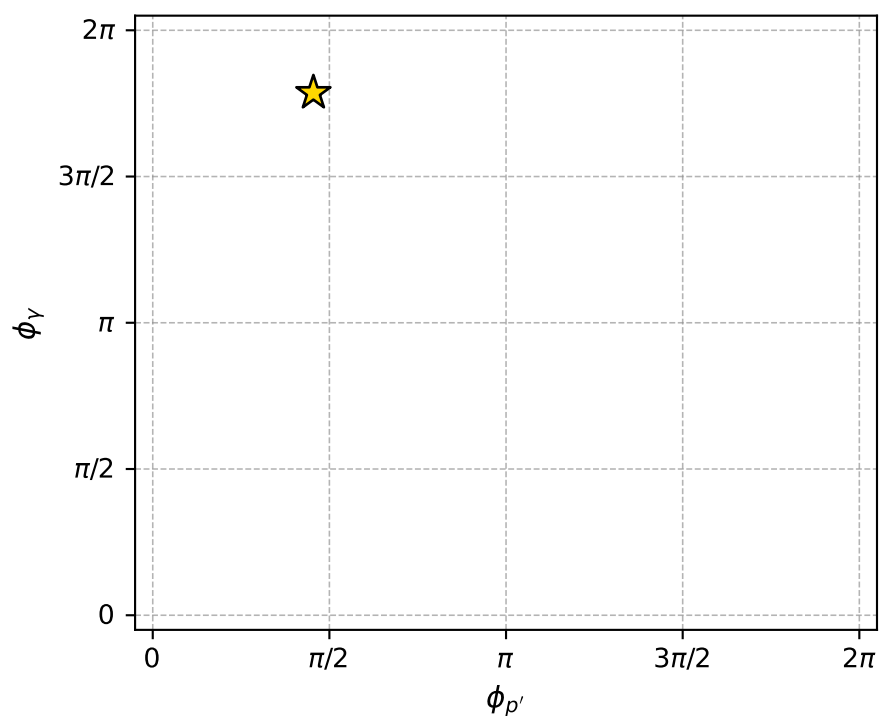
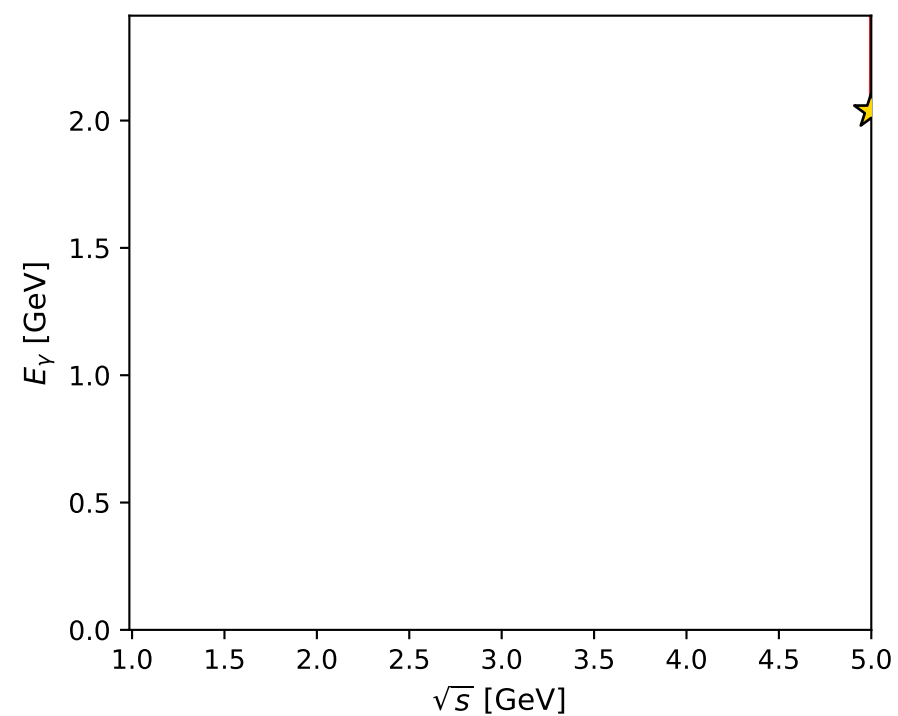
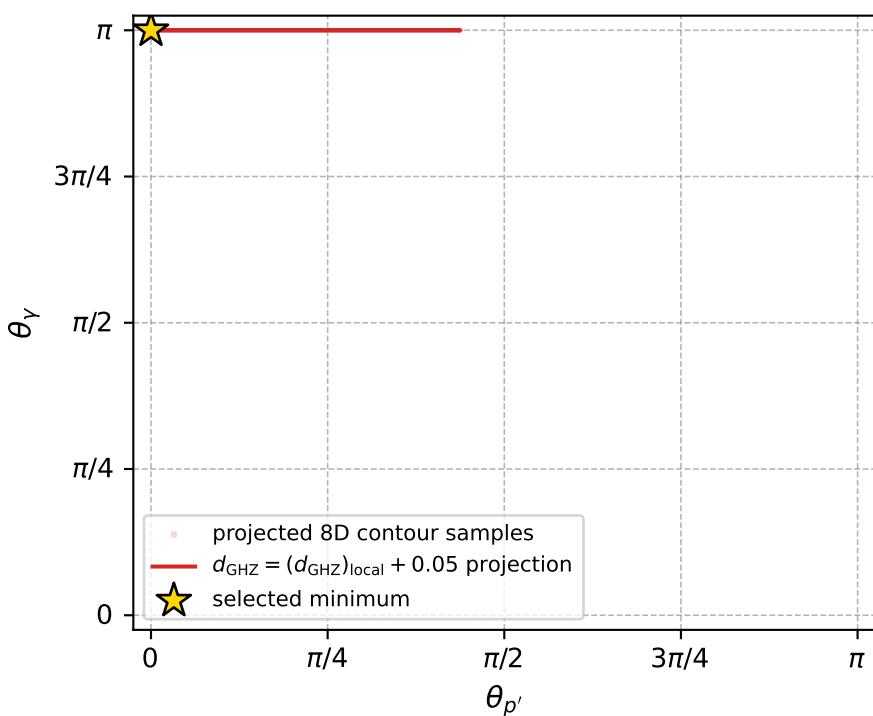
dghz_mixing_angles_polarization_02_local_minimum_7 $d_{\{\text{rm GHZ}\}}=2.41149\text{e-}08$
 region: polarization_cluster_02_local_minimum_7
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=0.78796497$ rad
 incoming lepton: $0.705289|+\rangle +0.708919|-\rangle$
 proton mixing angle: $\alpha_p=2.3536584$ rad
 incoming proton: $-0.705311|+\rangle +0.708898|-\rangle$

$s=24.9993$, $\sqrt{s}=4.99993$, $|\vec{P}|=2.41198$, $|\vec{P}'|=0.0126281$
 $\theta_{P'}=7.54042\text{e-}07$, $\theta_Y=3.14159$, $\phi_{P'}=1.42848$, $\phi_Y=5.61149$
 $E_Y=2.03724$, $\theta_{\ell'}=0$, $\phi_{\ell'}=4.57008$
 $Q^2=19.5333$, $x_B=0.834508$, $t=-3.03483$, $W^2=4.7535$, $y=0.970458$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4120$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 P $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4120$
 ℓ' $E= 2.0246$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 2.0246$
 P' $E= 0.9381$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.0126$
 q_Y $E= 2.0372$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= -2.0372$



electron: polarization cluster P2, local minimum 7; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 2.4114925\text{e-}08$
 $d_{\text{GHZ}} \text{ contour} = 0.050000024$
 above global minimum = $1.8314628\text{e-}09$
 8D contour samples = 51

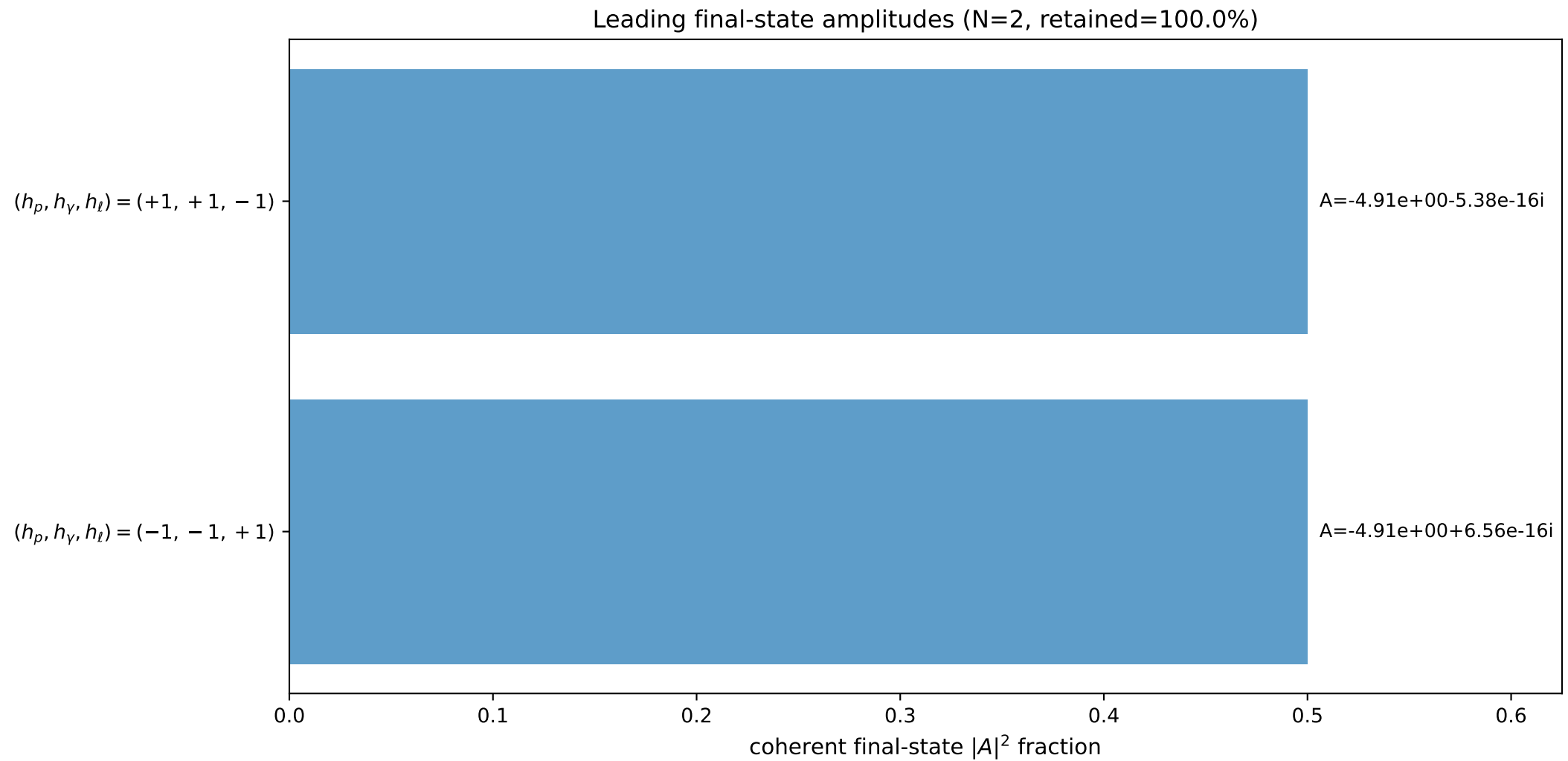
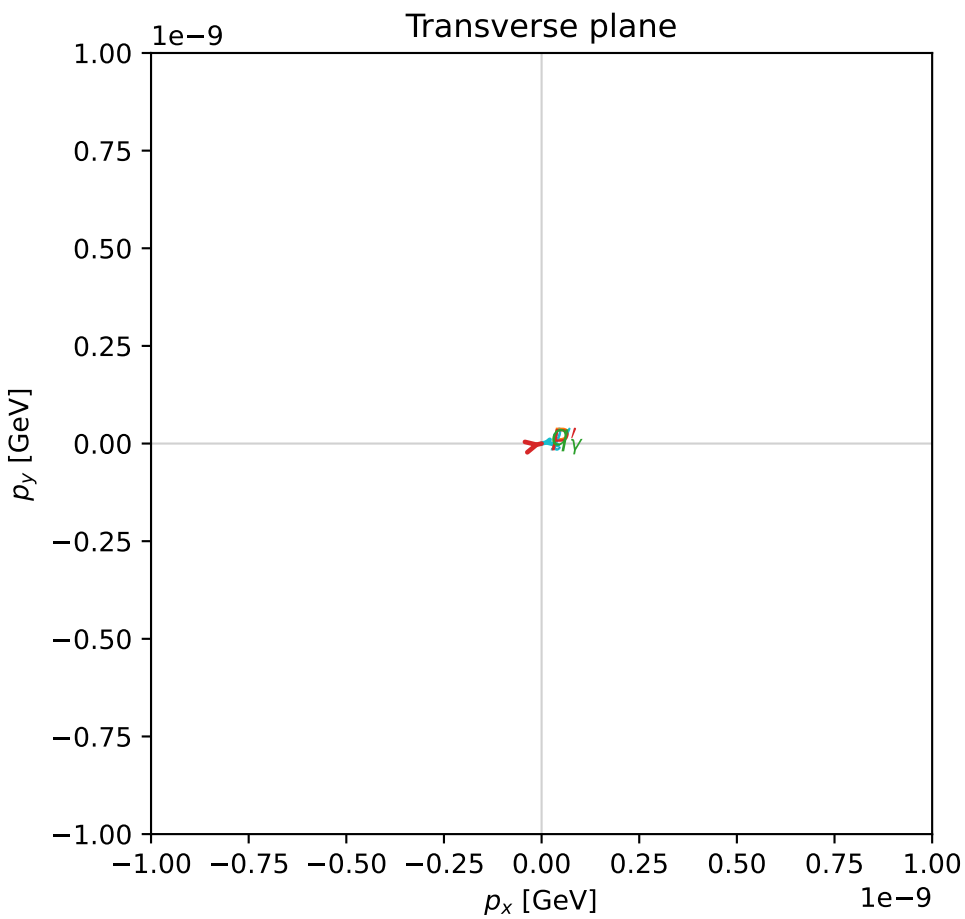
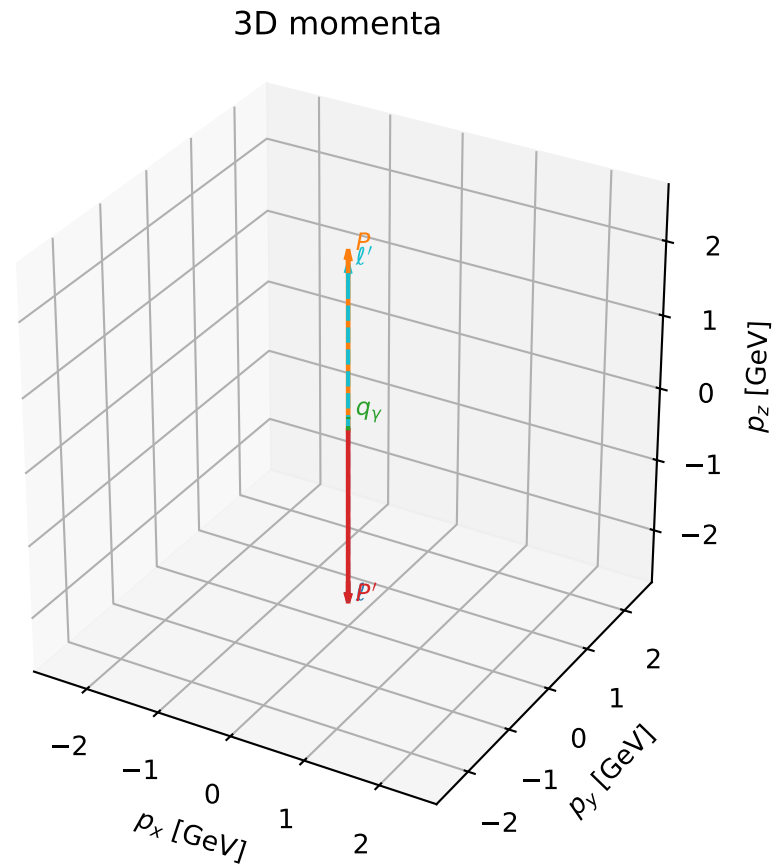
selected phase-space point:
 $\text{sqrt_s} = 4.999935$
 $\text{theta_p_out} = 7.54042\text{e-}07$
 $\text{theta_gamma_out} = 3.141593$
 $\text{q0out} = 2.037239$
 $\text{phi_p_out} = 1.428483$
 $\text{phi_gamma_out} = 5.611485$
 $\text{alpha_e} = 0.787965$
 $\text{alpha_p} = 2.353658$

coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

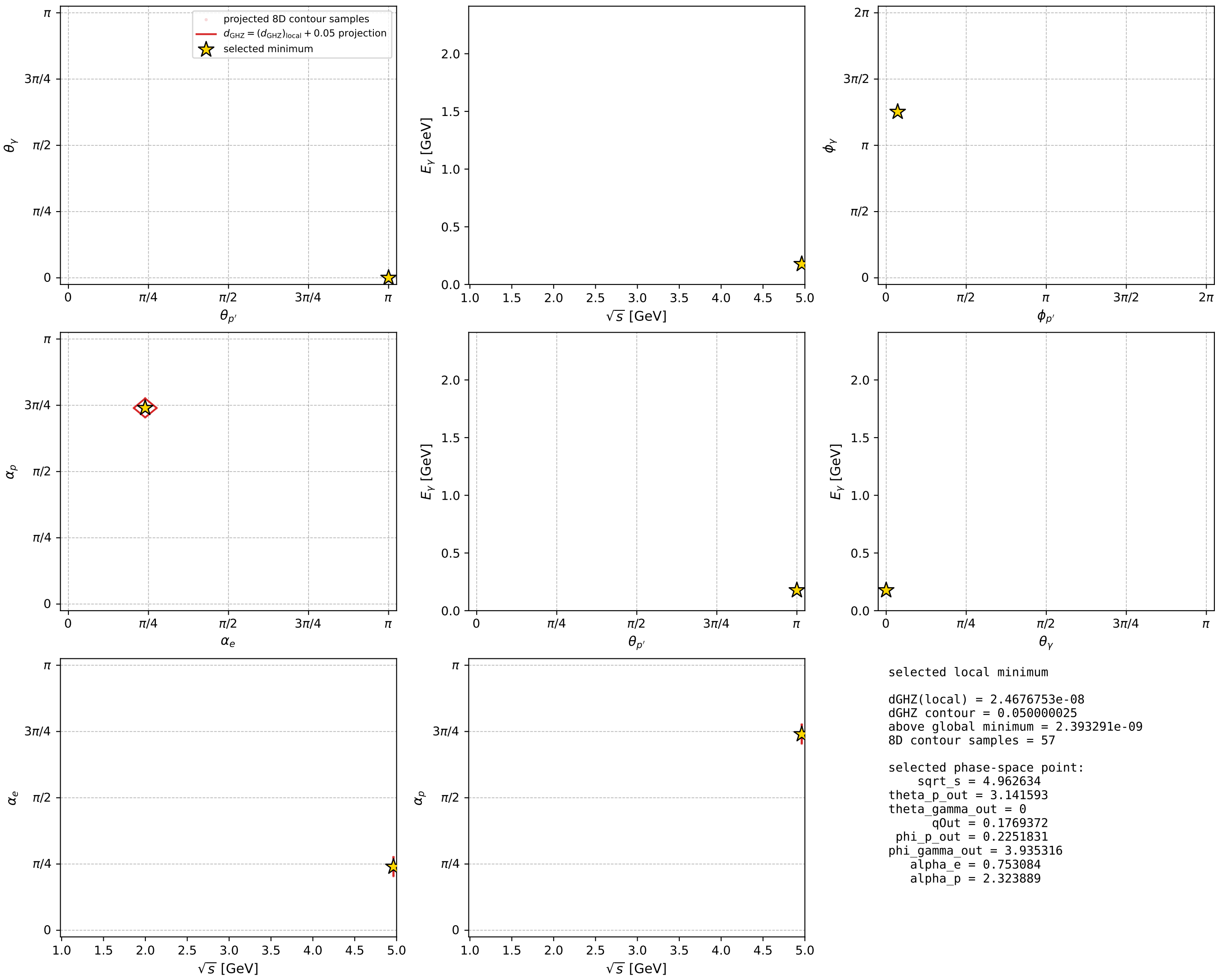
dghz_mixing_angles_polarization_02_local_minimum_14 $d_{\{\text{rm GHZ}\}}=2.46768\text{e-}08$
 region: polarization_cluster_02_local_minimum_14
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=0.75308403$ rad
 incoming lepton: $0.729583|+\rangle +0.683892|-\rangle$
 proton mixing angle: $\alpha_p=2.3238894$ rad
 incoming proton: $-0.683899|+\rangle +0.729577|-\rangle$

$s=24.6277$, $\sqrt{s}=4.96263$, $|\vec{P}|=2.39267$, $|\vec{P}'|=2.39267$
 $\theta_{P'}=3.14159$, $\theta_Y=0$, $\phi_{P'}=0.225183$, $\phi_Y=3.93532$
 $E_Y=0.176937$, $\theta_{\ell'}=0$, $\phi_{\ell'}=3.36678$
 $Q^2=21.2061$, $x_B=0.92352$, $t=-22.8995$, $W^2=2.63599$, $y=0.966916$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.3927$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.3927$
 P $E= 2.5700$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.3927$
 ℓ' $E= 2.2157$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 2.2157$
 P' $E= 2.5700$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.3927$
 q_Y $E= 0.1769$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 0.1769$



electron: polarization cluster P2, local minimum 14; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour

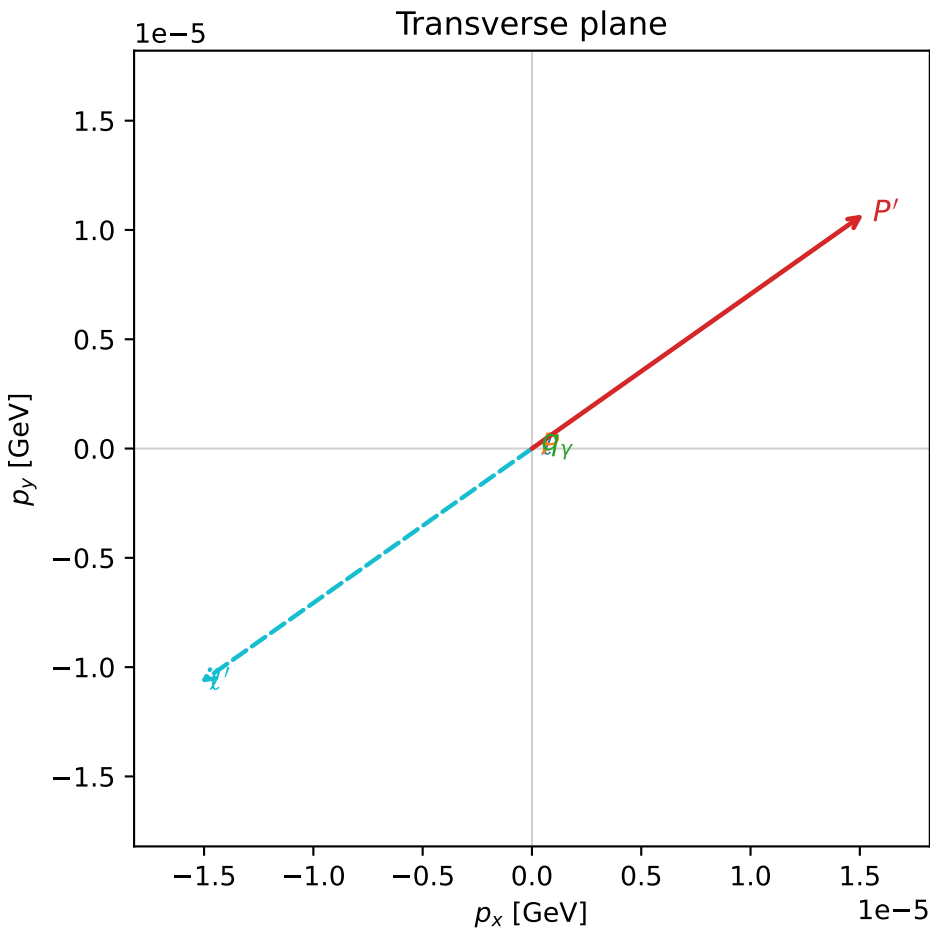
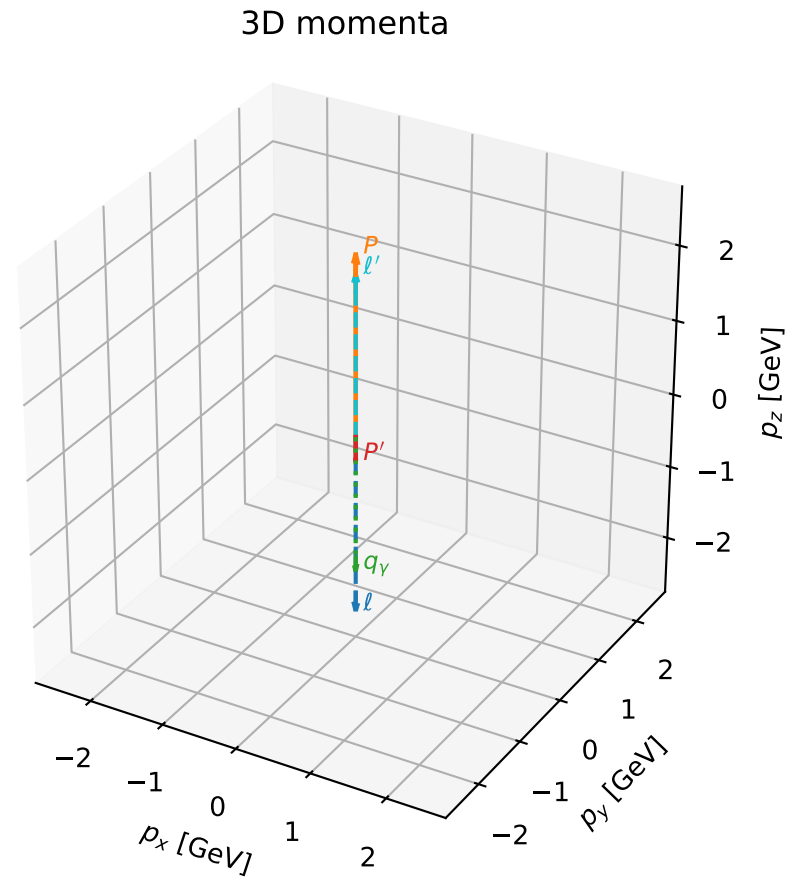


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

dghz_mixing_angles_polarization_02_local_minimum_21 $d_{\{\text{rm GHZ}\}}=2.55996\text{e-}08$
 region: polarization_cluster_02_local_minimum_21
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=0.88921427$ rad
 incoming lepton: $0.630022|+\rangle +0.776577|-\rangle$
 proton mixing angle: $\alpha_p=2.2523645$ rad
 incoming proton: $-0.630012|+\rangle +0.776586|-\rangle$

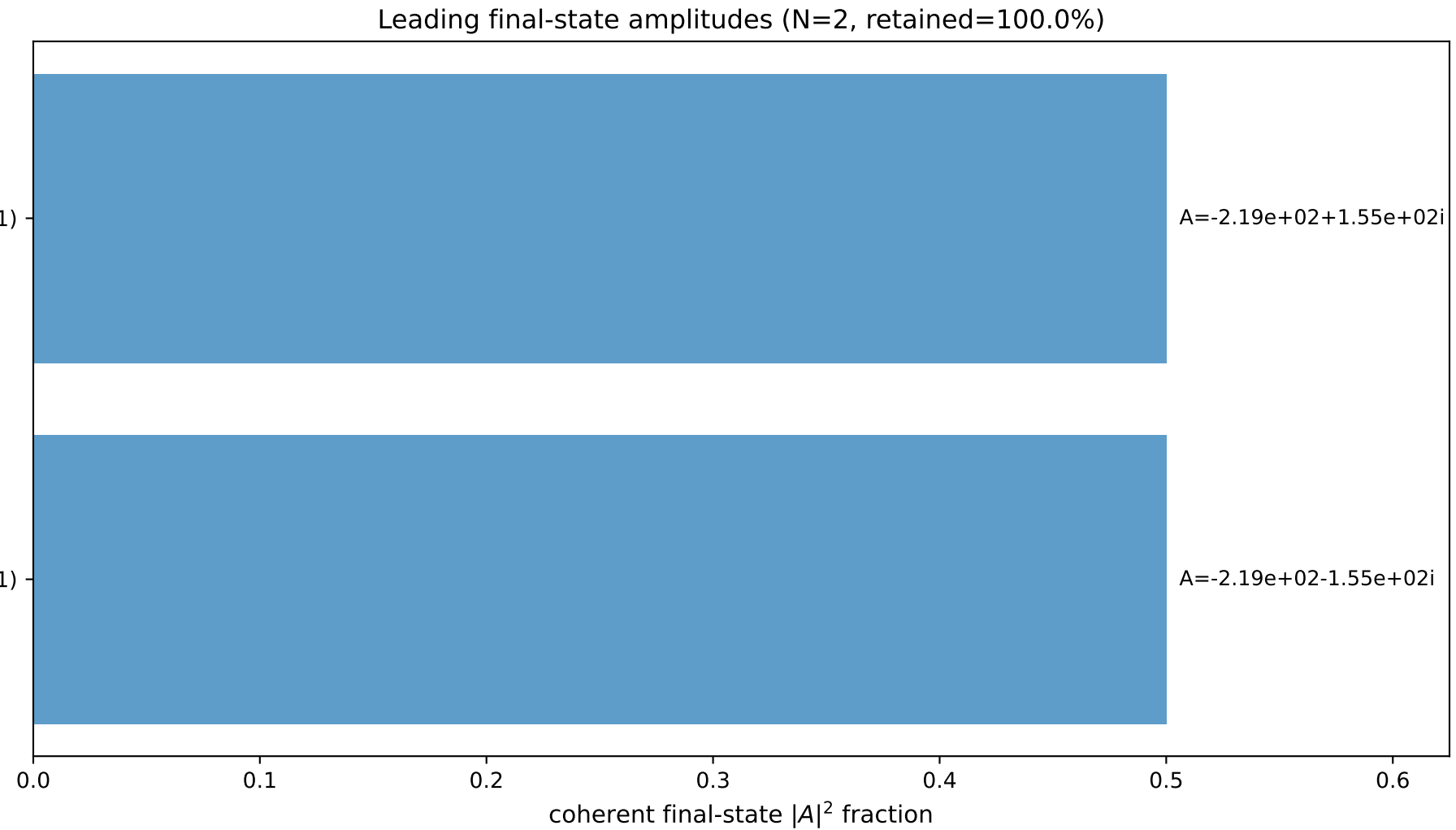
$s=25$, $\sqrt{s}=5$, $|\vec{P}|=2.41202$, $|\vec{P}'|=0.311681$
 $\theta_{P'}=3.14153$, $\theta_Y=3.14159$, $\phi_{P'}=0.615032$, $\phi_Y=4.37445$
 $E_Y=1.84995$, $\theta_{\ell'}=8.59128\text{e-}06$, $\phi_{\ell'}=3.75662$
 $Q^2=20.8555$, $x_B=0.89281$, $t=-4.85994$, $W^2=3.38373$, $y=0.96846$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4120$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 P $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4120$
 ℓ' $E= 2.1616$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 2.1616$
 P' $E= 0.9884$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.3117$
 q_Y $E= 1.8499$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= -1.8499$



$(h_p, h_Y, h_{\ell}) = (-1, +1, -1)$

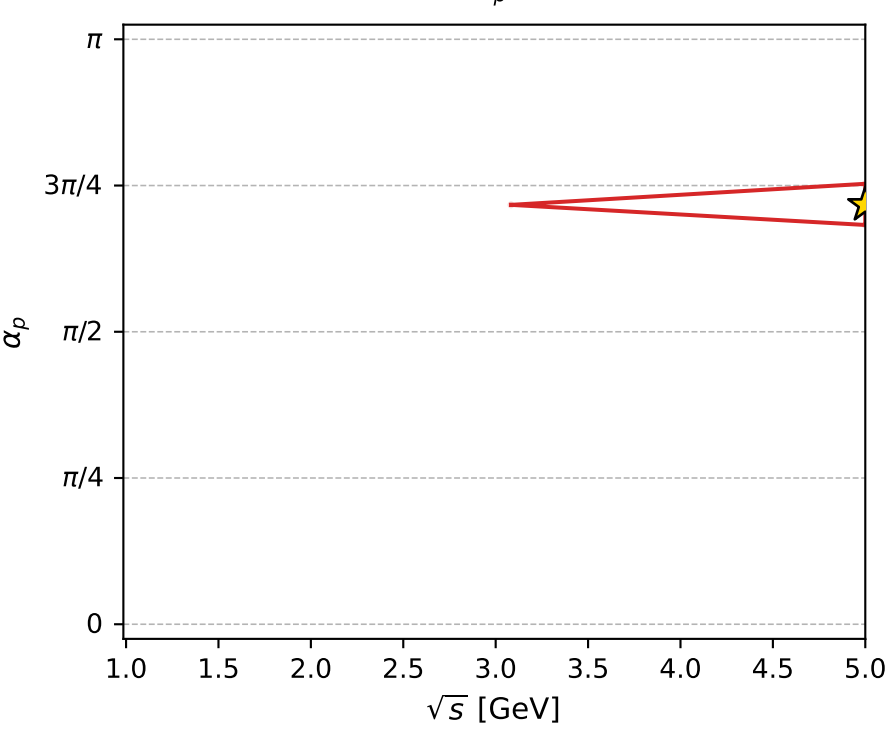
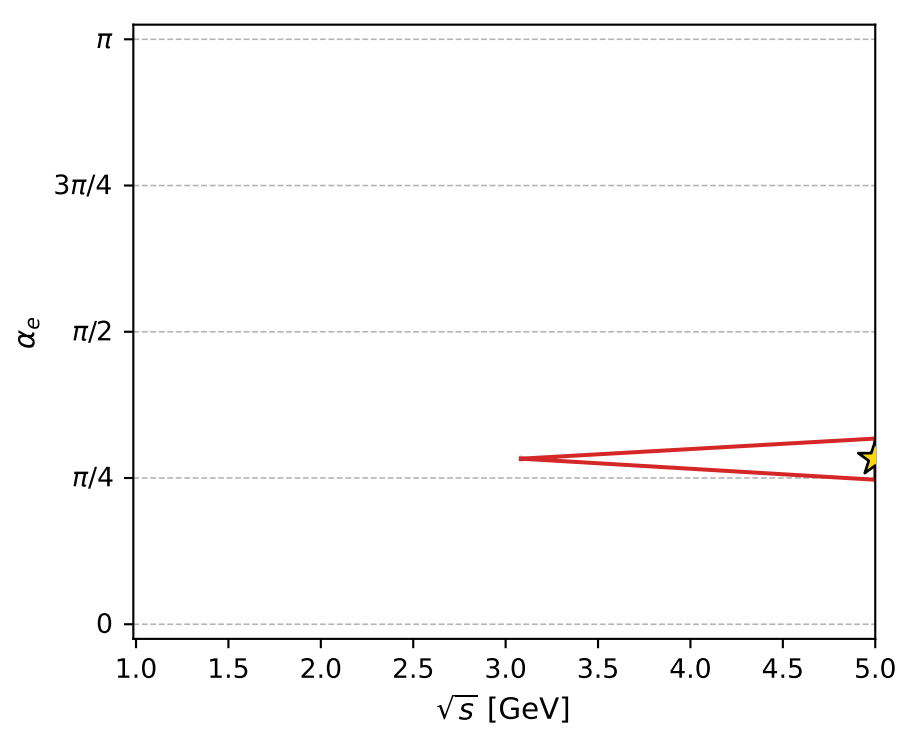
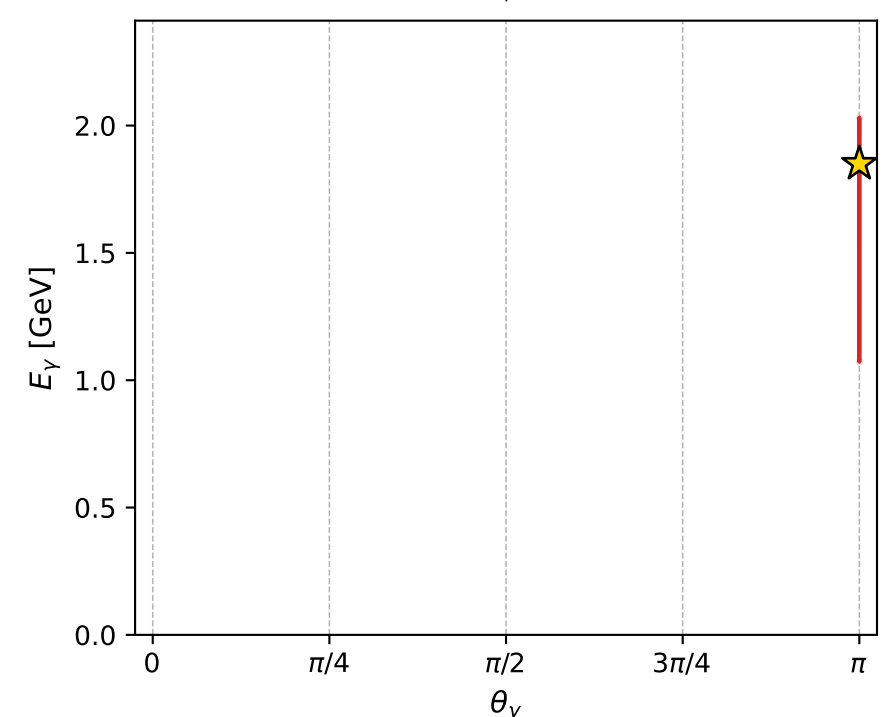
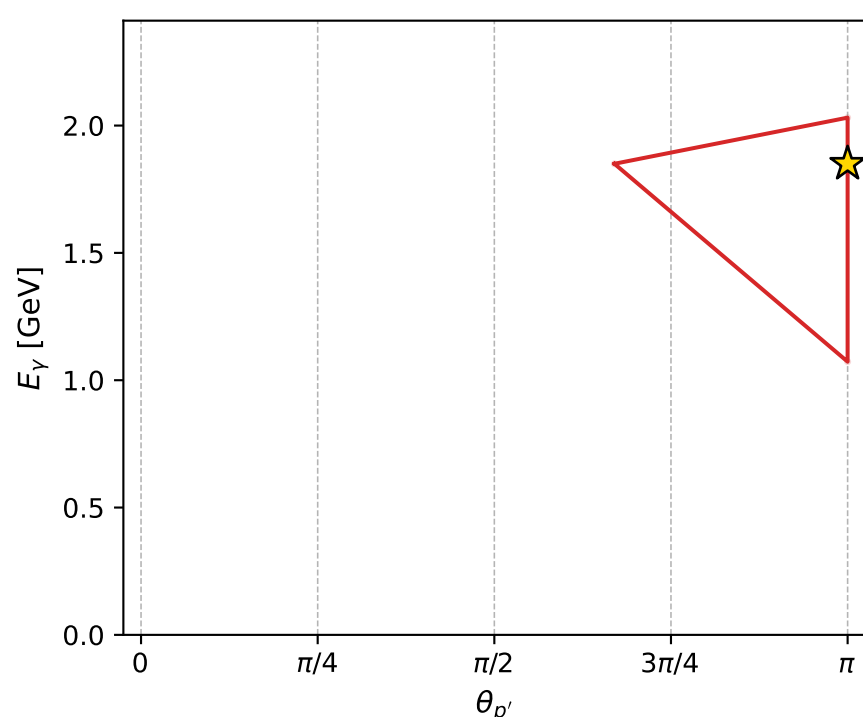
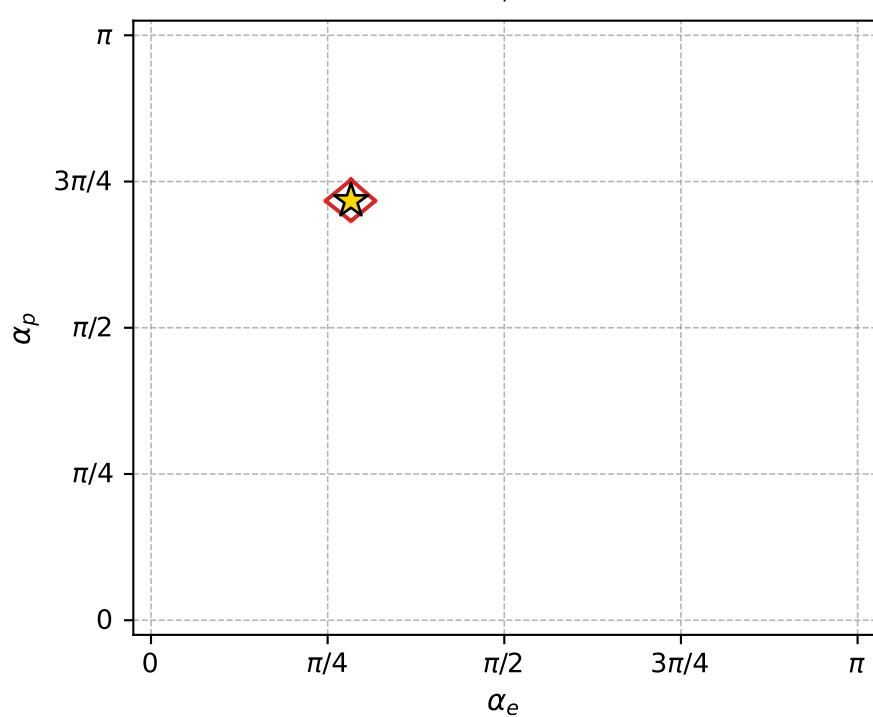
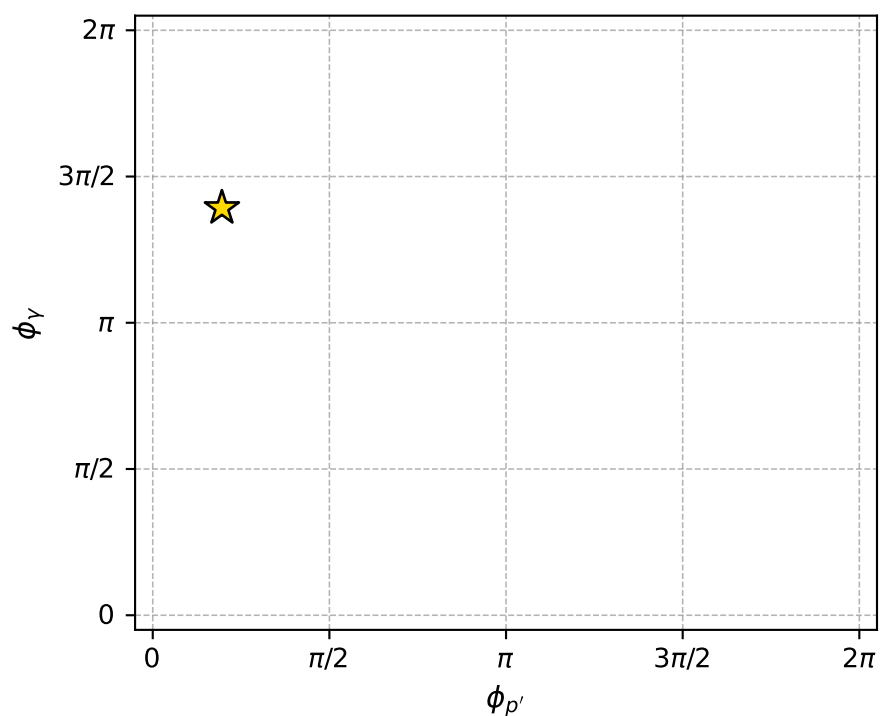
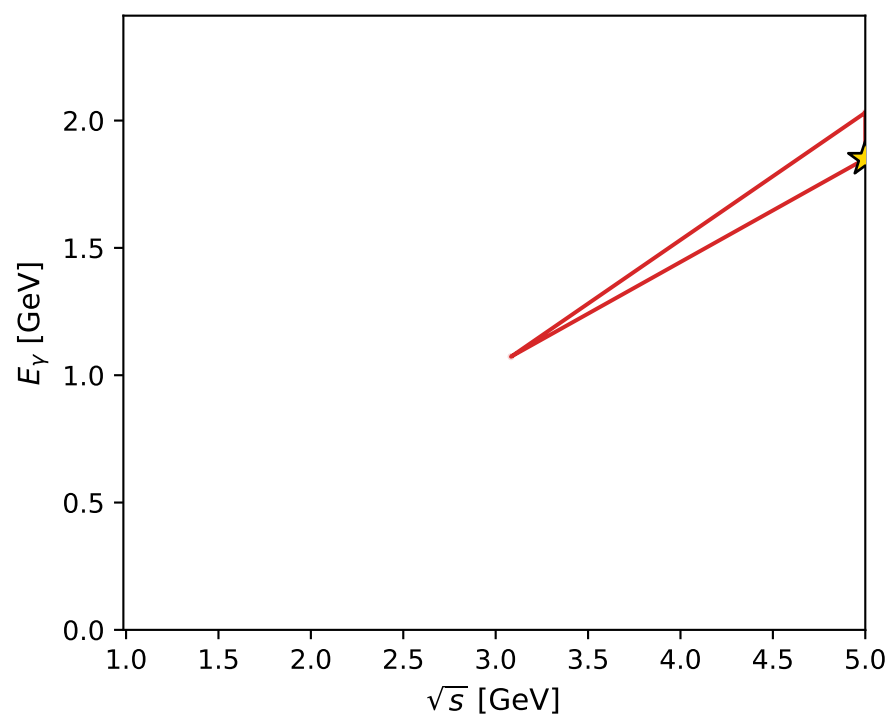
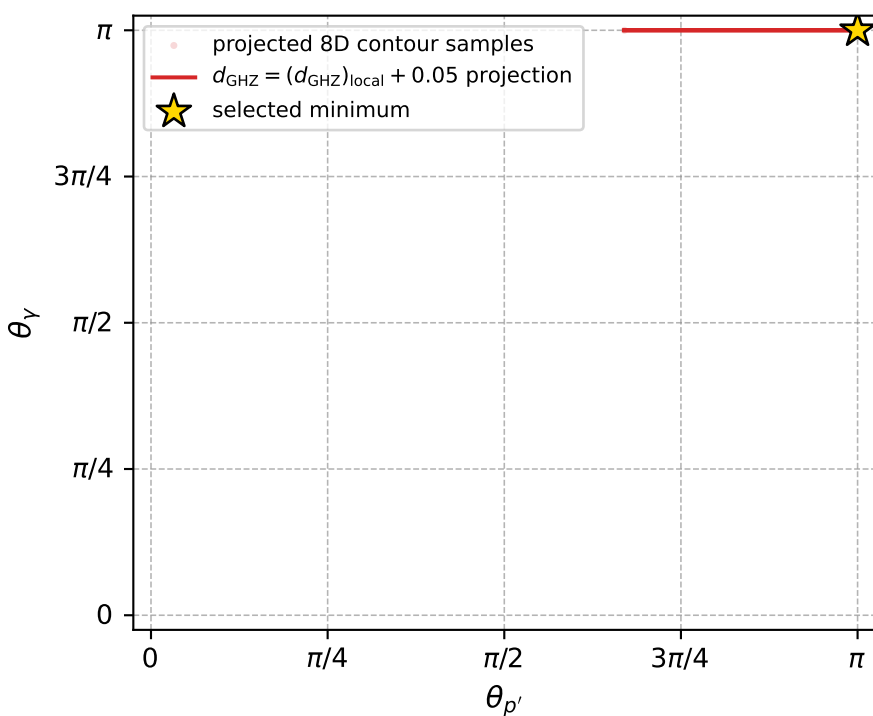
$(h_p, h_Y, h_{\ell}) = (+1, -1, +1)$



$A=-2.19\text{e}+02+1.55\text{e}+02i$

$A=-2.19\text{e}+02-1.55\text{e}+02i$

electron: polarization cluster P2, local minimum 21; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 2.5599618\text{e-}08$
 $d_{\text{GHZ}} \text{ contour} = 0.050000026$
 above global minimum = $3.3161559\text{e-}09$
 8D contour samples = 66

selected phase-space point:

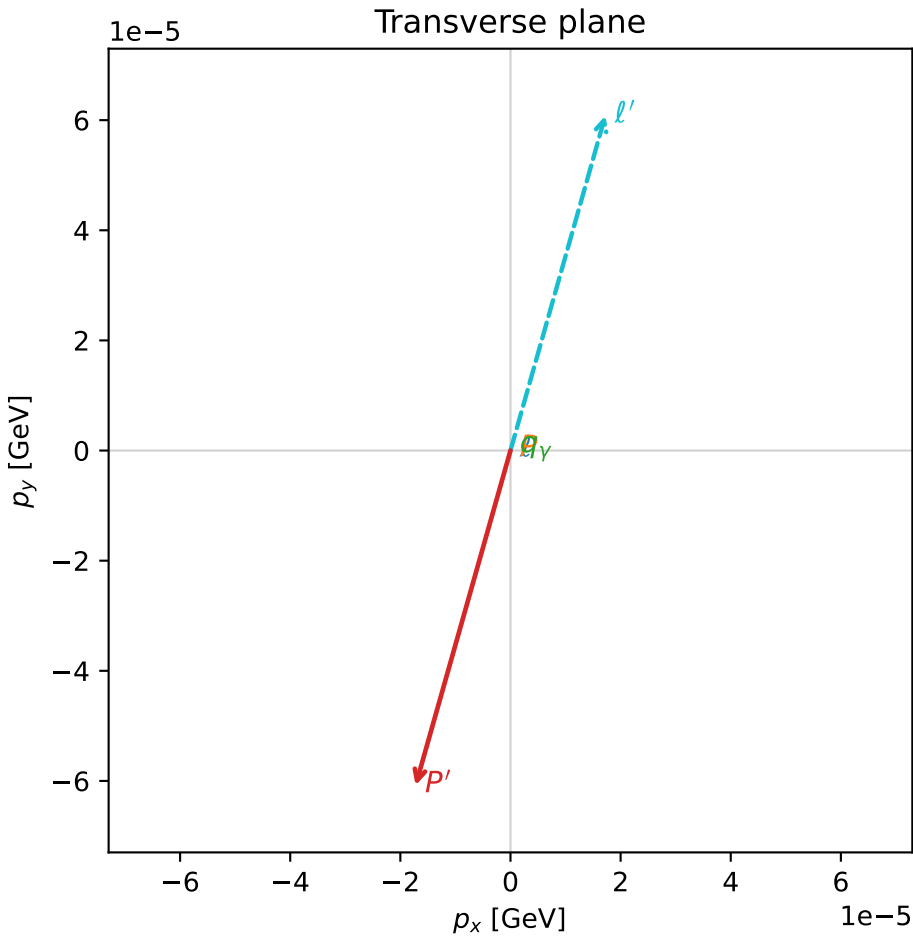
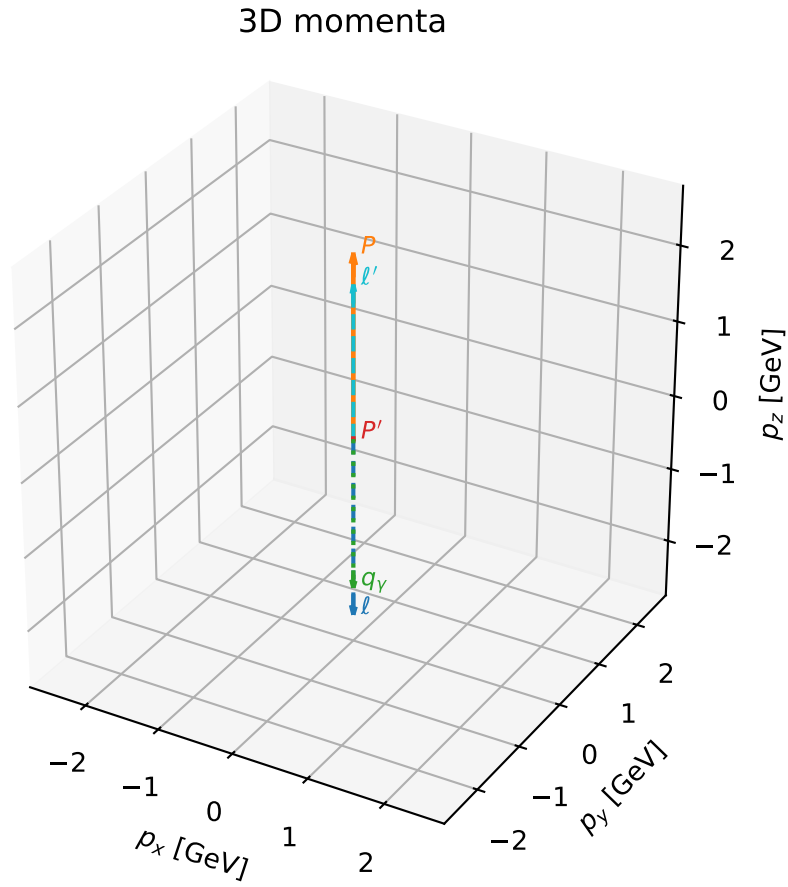
$\text{sqrt_s} = 5$
 $\text{theta_p_out} = 3.141533$
 $\text{theta_gamma_out} = 3.141593$
 $\text{q0out} = 1.849945$
 $\text{phi_p_out} = 0.6150317$
 $\text{phi_gamma_out} = 4.374448$
 $\text{alpha_e} = 0.8892143$
 $\text{alpha_p} = 2.252364$

coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

dghz_mixing_angles_polarization_02_local_minimum_26 $d_{\{\text{rm GHZ}\}}=2.59712\text{e-}08$
region: polarization_cluster_02_local_minimum_26
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: $\alpha_e=0.77423168$ rad
incoming lepton: $0.714958|+\rangle +0.699167|-\rangle$
proton mixing angle: $\alpha_p=2.3673692$ rad
incoming proton: $-0.714964|+\rangle +0.699161|-\rangle$

$s=24.9729$, $\sqrt{s}=4.99729$, $|\vec{P}|=2.41061$, $|\vec{P}'|=0.00645061$
 $\theta_{P'}=0.00979874$, $\theta_Y=3.14159$, $\phi_{P'}=4.43638$, $\phi_Y=6.38506\text{e-}05$
 $E_Y=2.03286$, $\theta_{\ell'}=3.11916\text{e-}05$, $\phi_{\ell'}=1.29479$
 $Q^2=19.5395$, $x_B=0.835755$, $t=-3.06193$, $W^2=4.7198$, $y=0.970383$

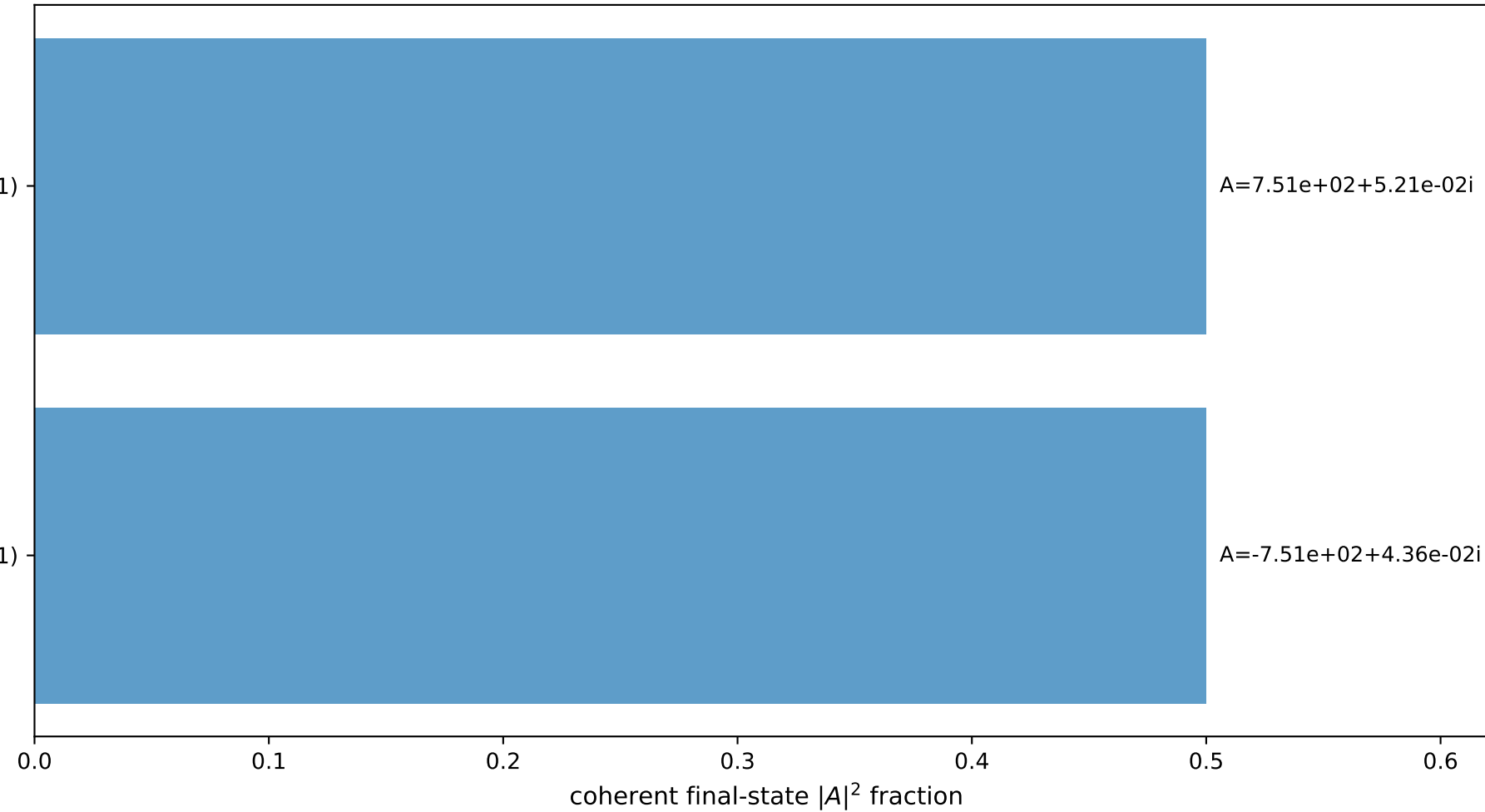
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4106$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4106$
 P $E= 2.5867$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4106$
 ℓ' $E= 2.0264$ $p_x= 0.0000$ $p_y= 0.0001$ $p_z= 2.0264$
 P' $E= 0.9380$ $p_x= -0.0000$ $p_y= -0.0001$ $p_z= 0.0065$
 q_Y $E= 2.0329$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.0329$



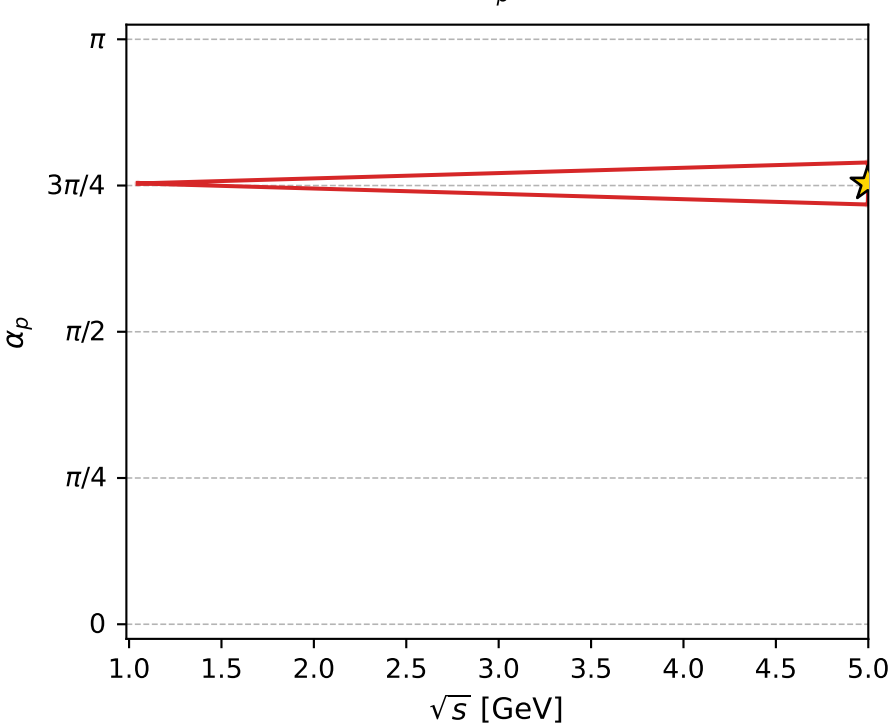
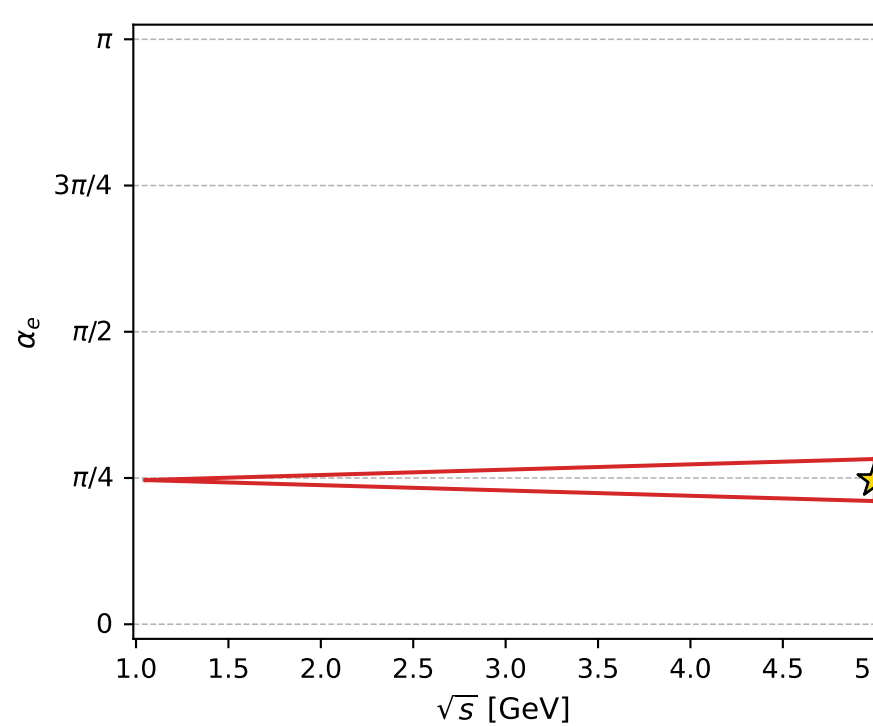
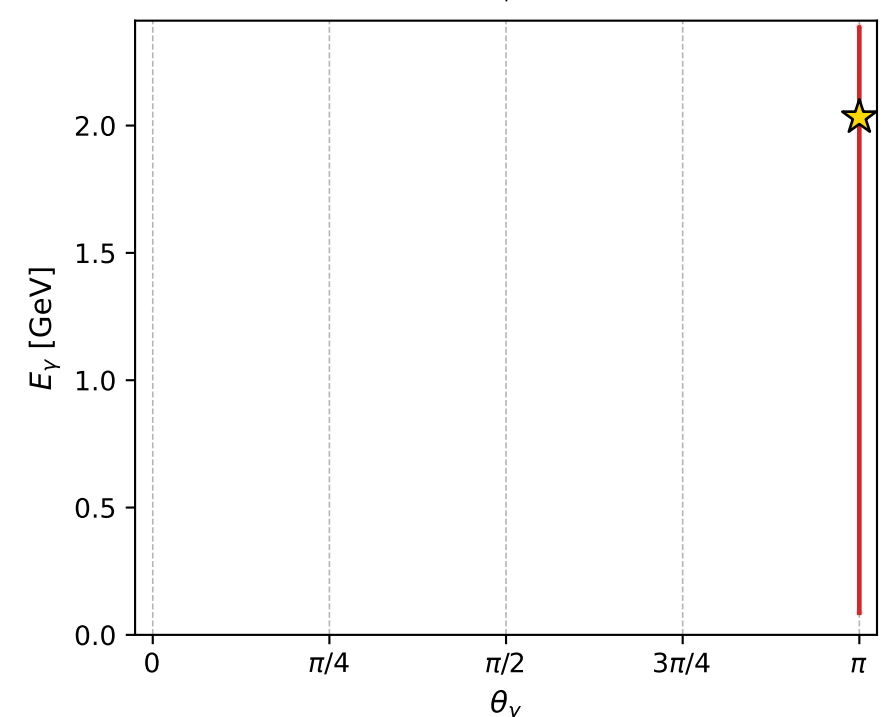
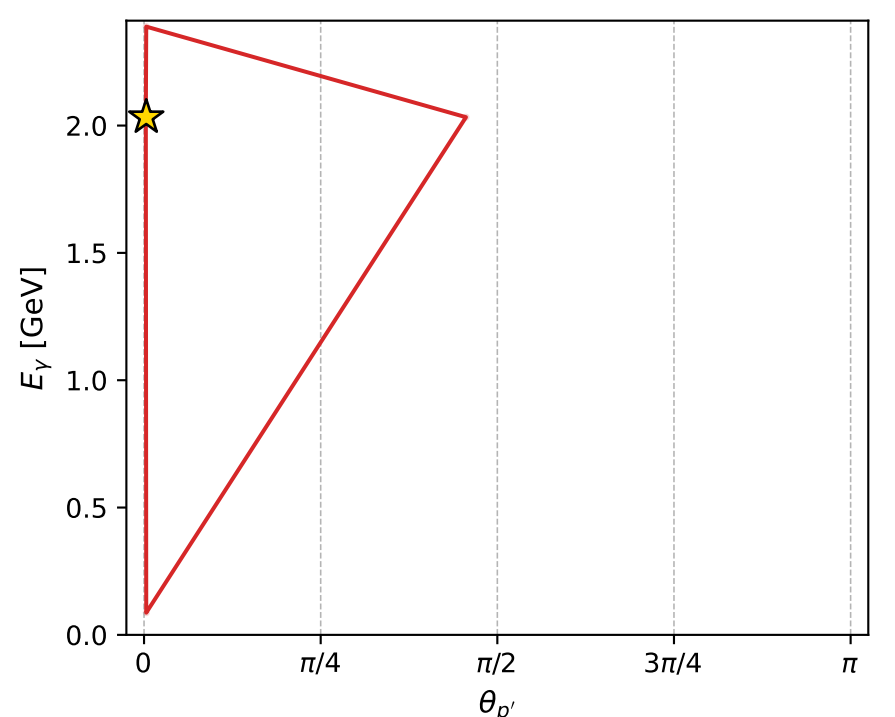
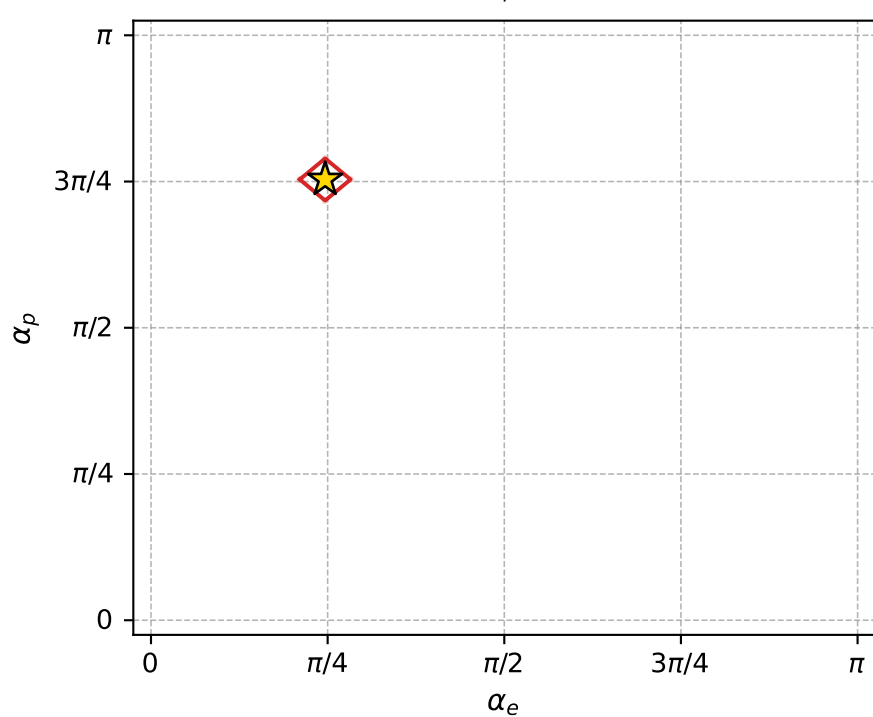
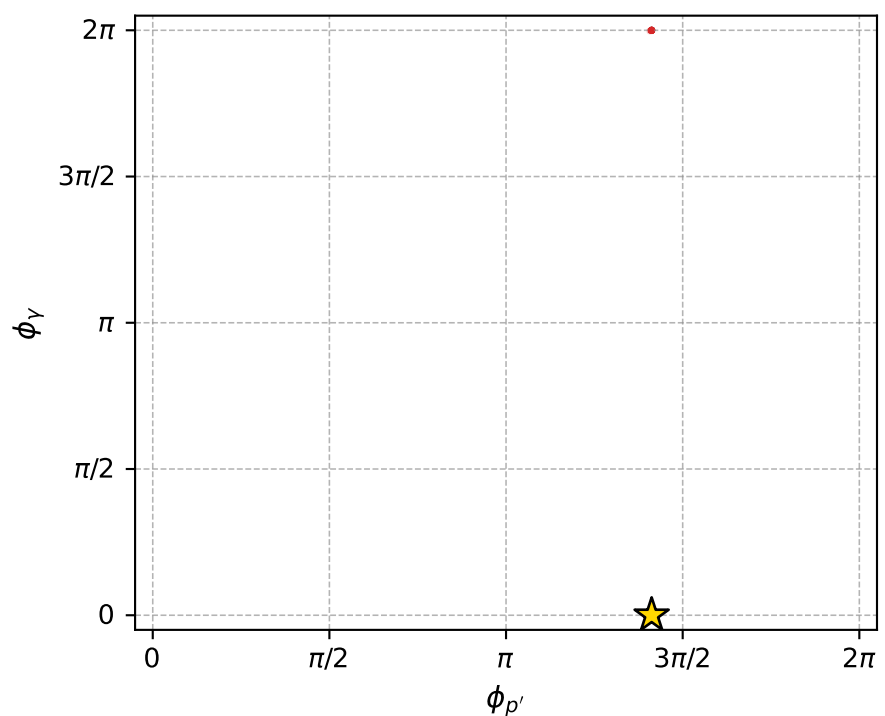
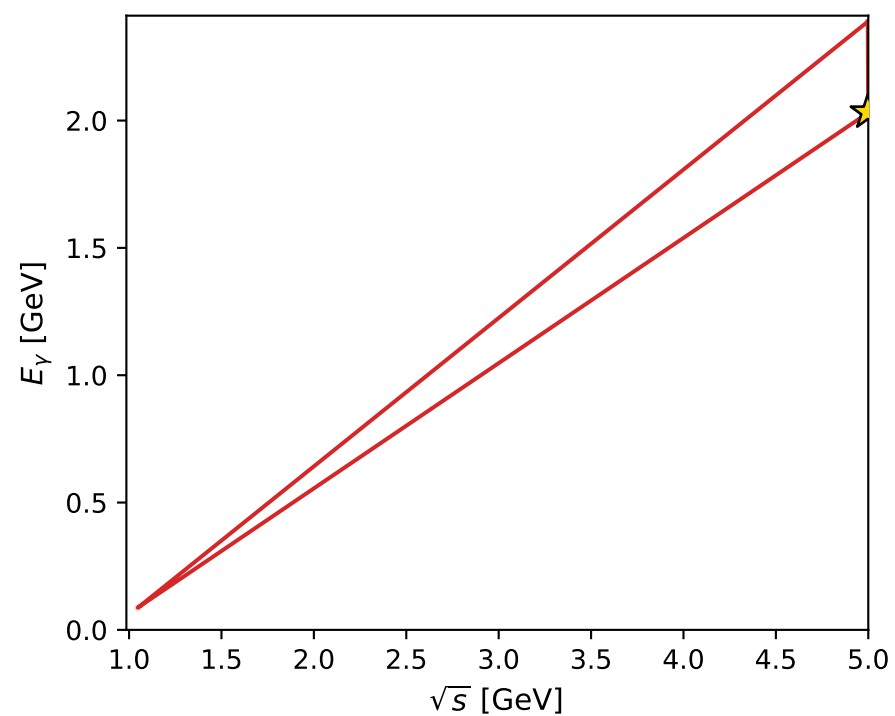
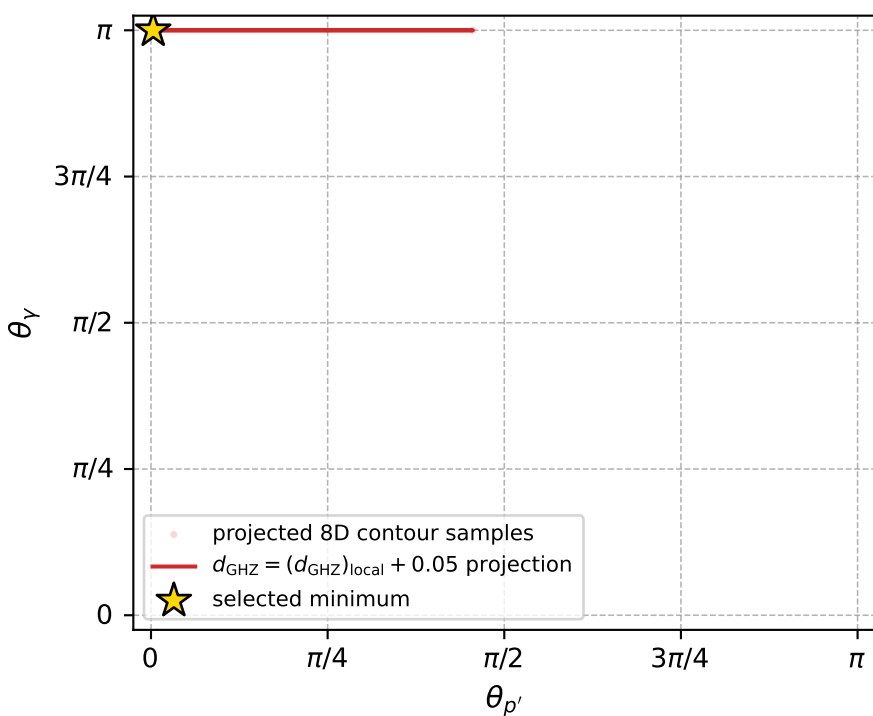
$(h_p, h_Y, h_\ell) = (-1, -1, +1)$

$(h_p, h_Y, h_\ell) = (+1, +1, -1)$

Leading final-state amplitudes (N=2, retained=100.0%)



electron: polarization cluster P2, local minimum 26; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 2.597121\text{e-}08$
 $d_{\text{GHZ}} \text{ contour} = 0.050000026$
 above global minimum = $3.6877476\text{e-}09$
 8D contour samples = 126

selected phase-space point:

$\text{sqrt_s} = 4.997287$
 $\text{theta_p_out} = 0.009798743$
 $\text{theta_gamma_out} = 3.141593$
 $\text{q0out} = 2.032857$
 $\text{phi_p_out} = 4.436384$
 $\text{phi_gamma_out} = 6.385065\text{e-}05$
 $\text{alpha_e} = 0.7742317$
 $\text{alpha_p} = 2.367369$

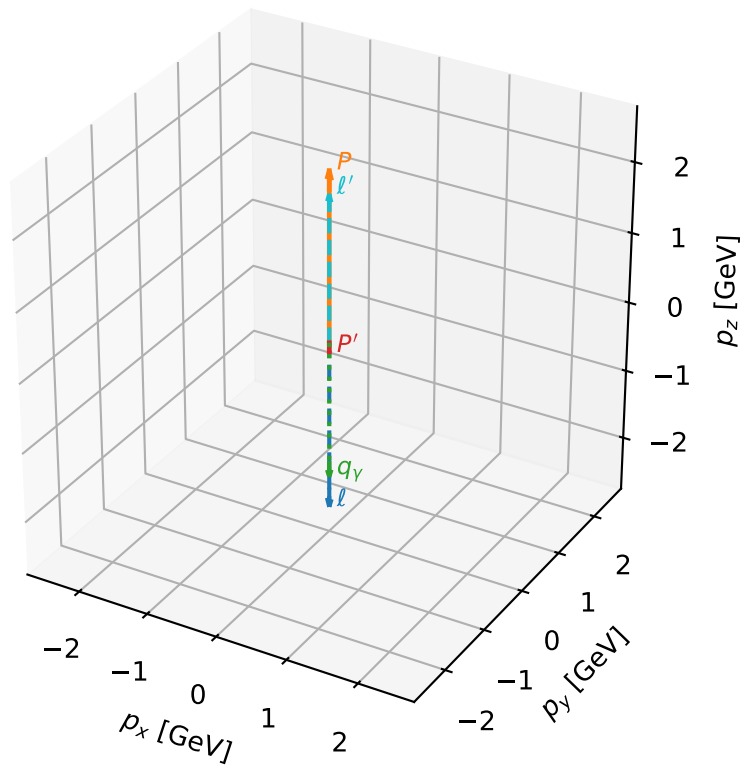
coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

dghz_mixing_angles_polarization_02_local_minimum_31 $d_{\{\text{rm GHZ}\}}=2.62572\text{e-}08$
 region: polarization_cluster_02_local_minimum_31
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=0.72294087$ rad
 incoming lepton: $0.749863|+\rangle +0.661593|-\rangle$
 proton mixing angle: $\alpha_p=2.4186407$ rad
 incoming proton: $-0.749856|+\rangle +0.661601|-\rangle$

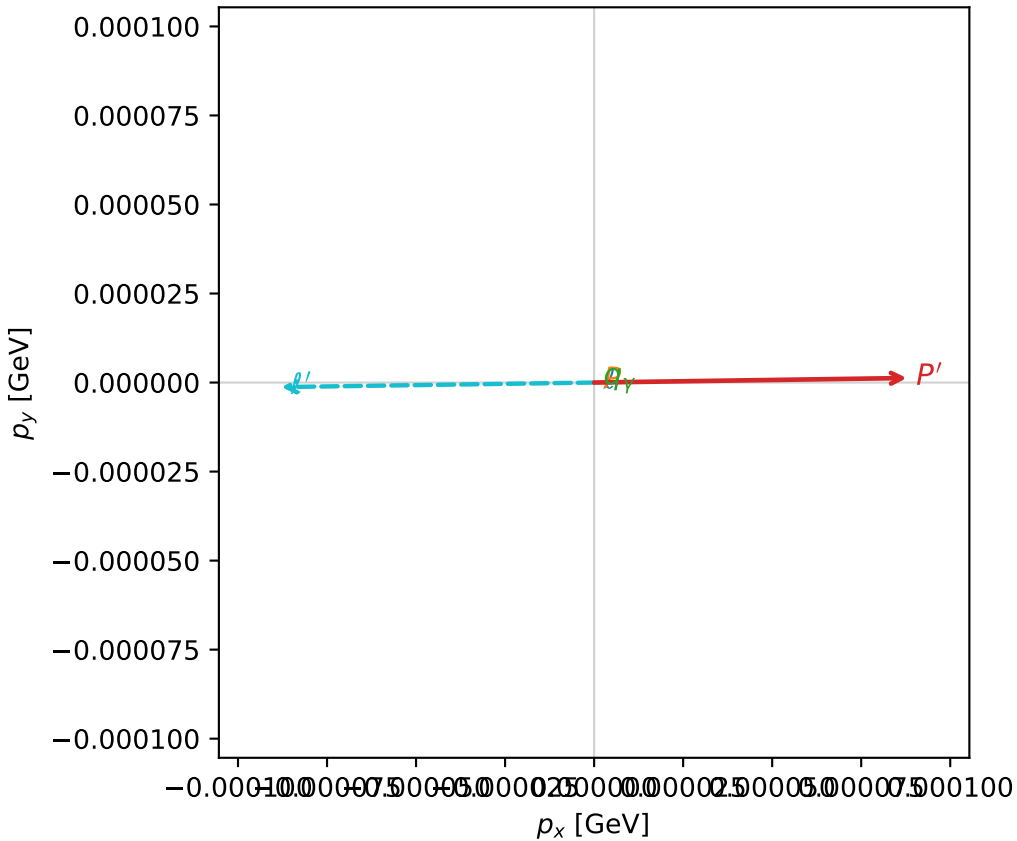
$s=24.9982$, $\sqrt{s}=4.99982$, $|\vec{P}|=2.41192$, $|\vec{P}'|=0.133473$
 $\theta_{p'}=3.14093$, $\theta_\gamma=3.14159$, $\phi_{p'}=0.0144773$, $\phi_\gamma=3.74298$
 $E_\gamma=1.95945$, $\theta_{l'}=4.19601\text{e-}05$, $\phi_{l'}=3.15607$
 $Q^2=20.1919$, $x_B=0.863574$, $t=-3.78797$, $W^2=4.06974$, $y=0.969459$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 2.4119$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4119$
 P $E= 2.5879$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4119$
 l' $E= 2.0929$ $p_x= -0.0001$ $p_y= -0.0000$ $p_z= 2.0929$
 P' $E= 0.9474$ $p_x= 0.0001$ $p_y= 0.0000$ $p_z= -0.1335$
 q_γ $E= 1.9595$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= -1.9595$

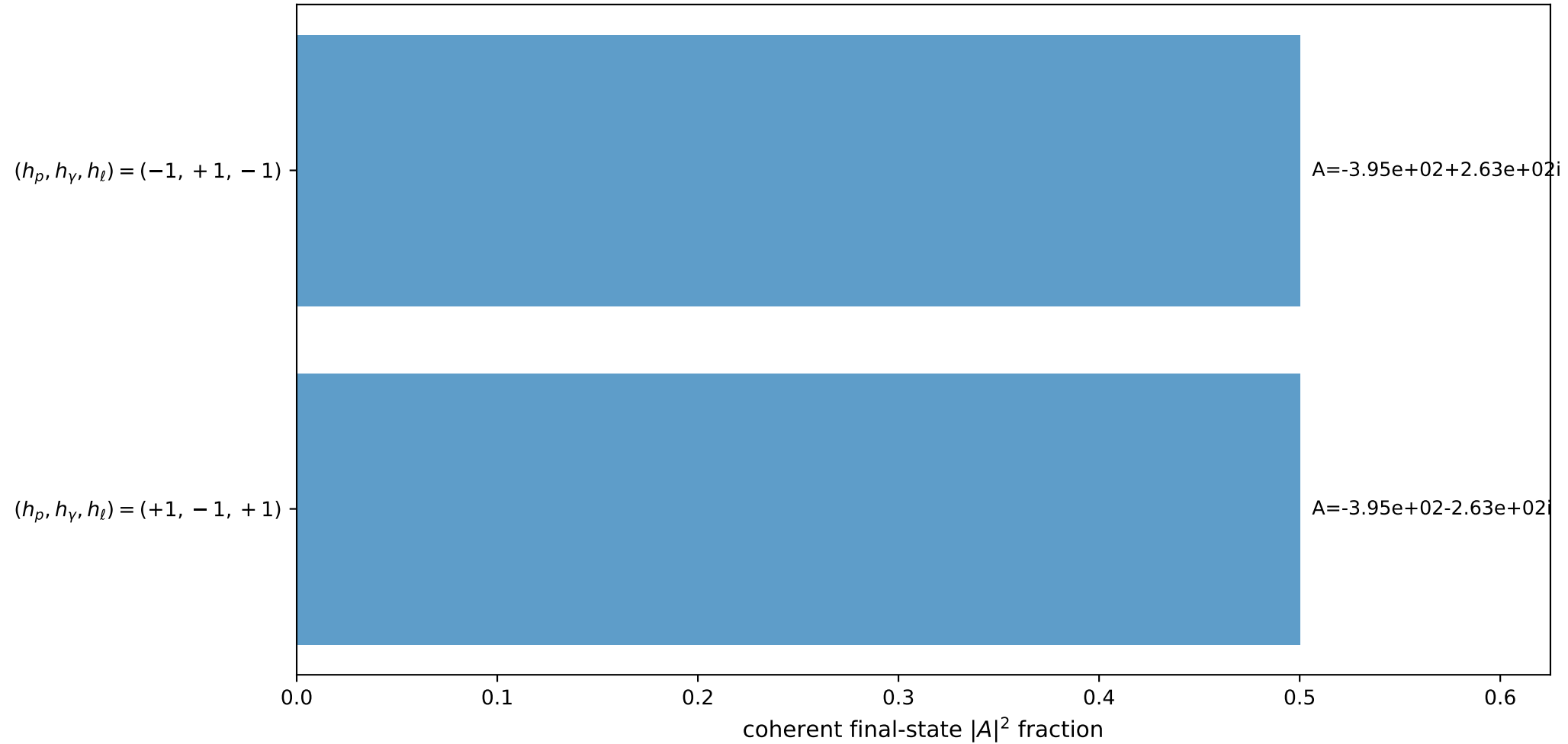
3D momenta



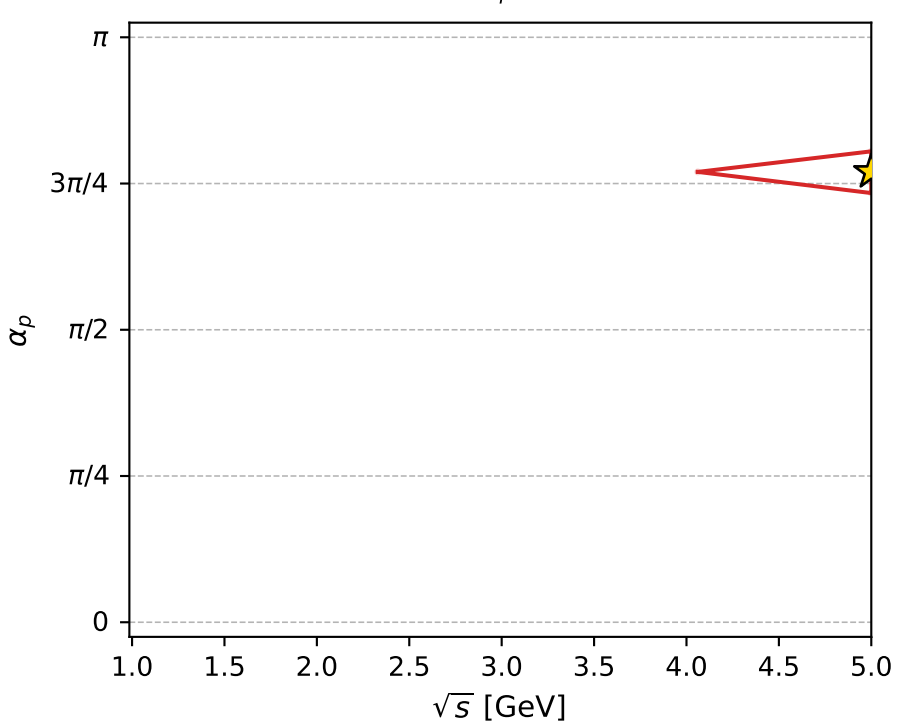
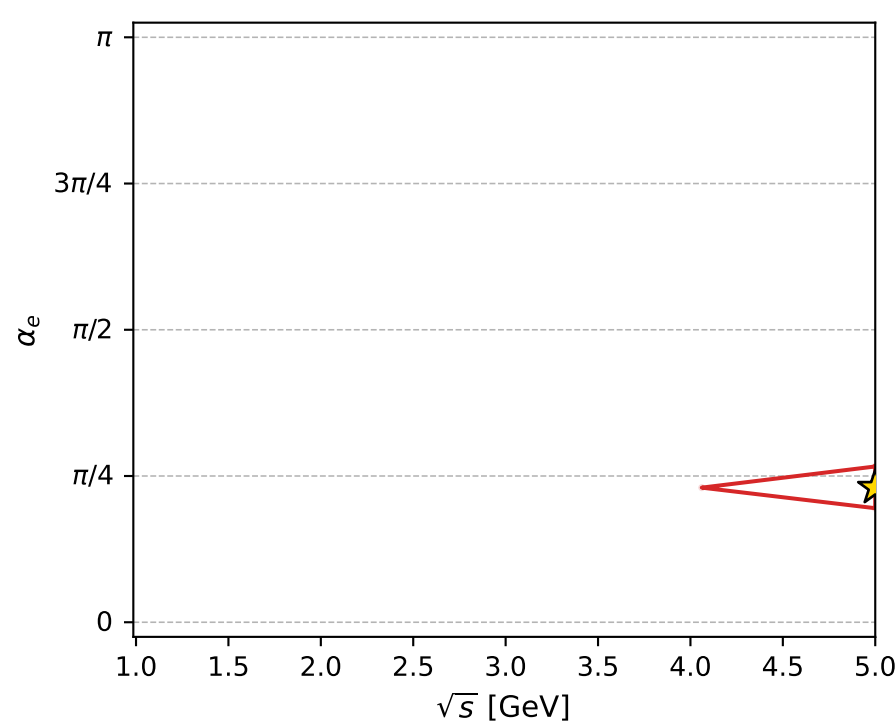
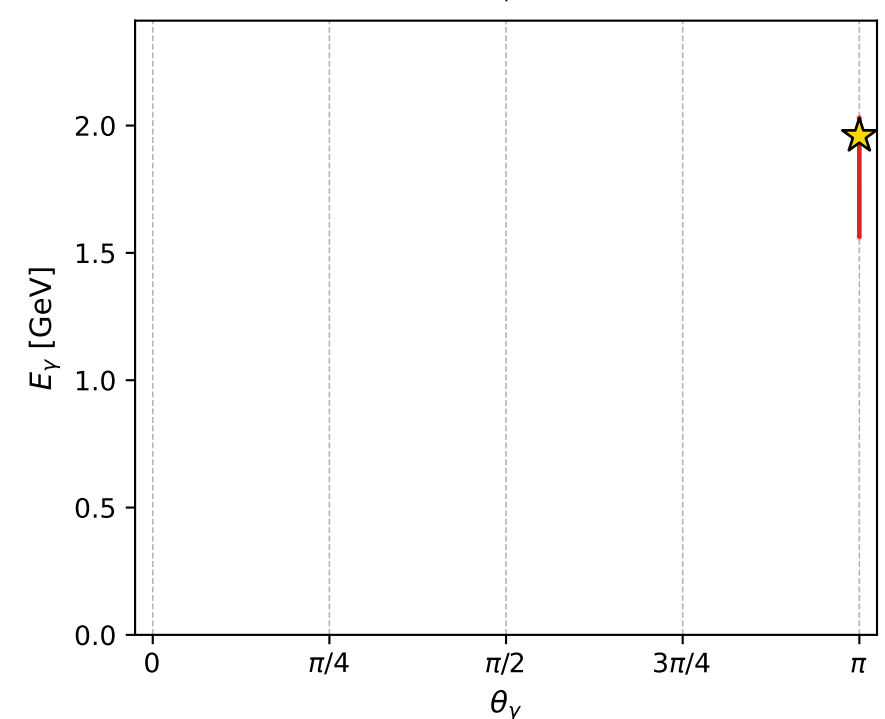
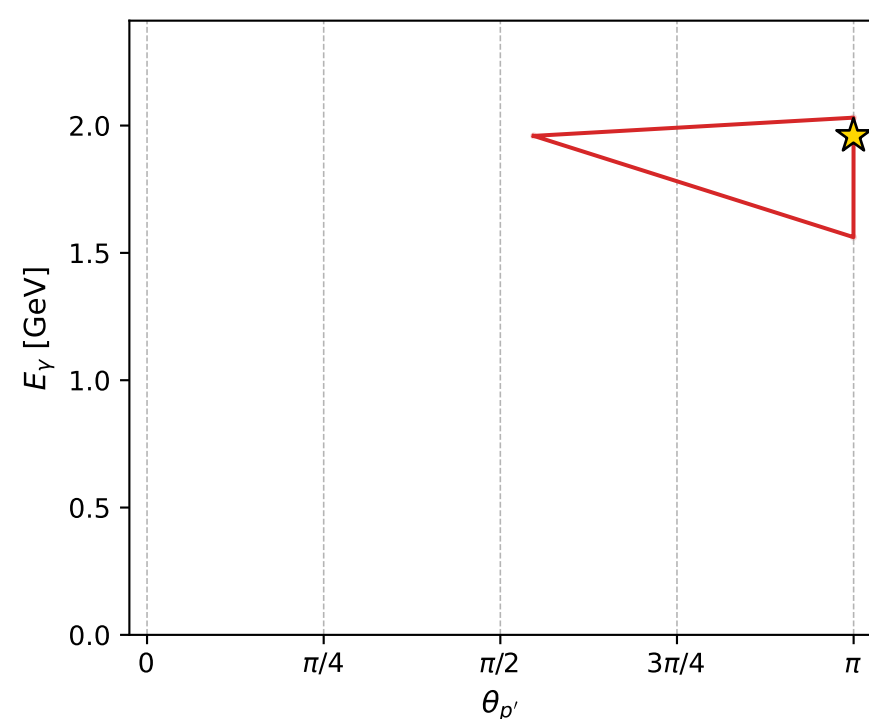
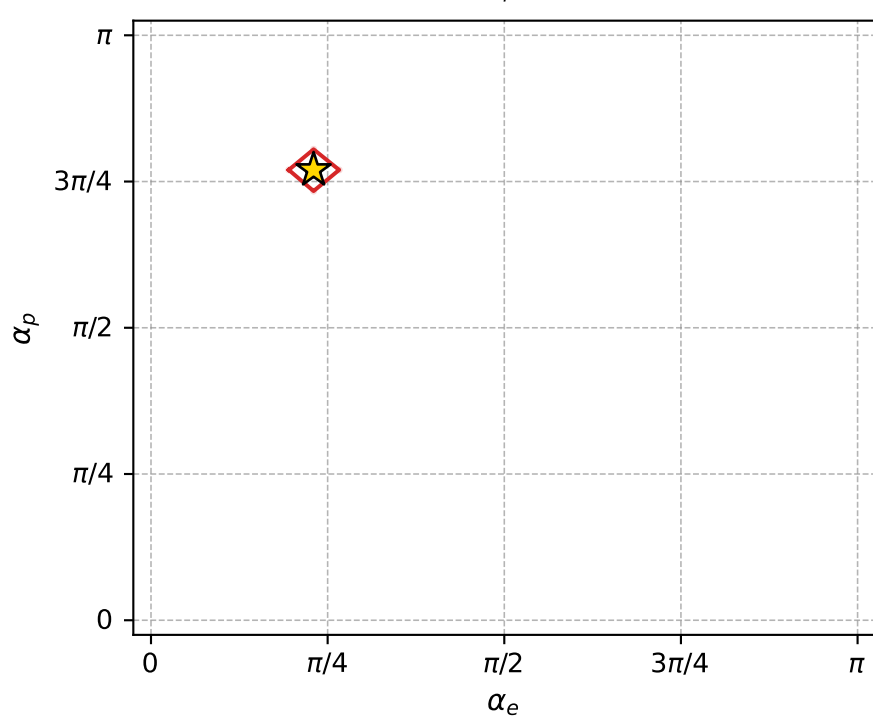
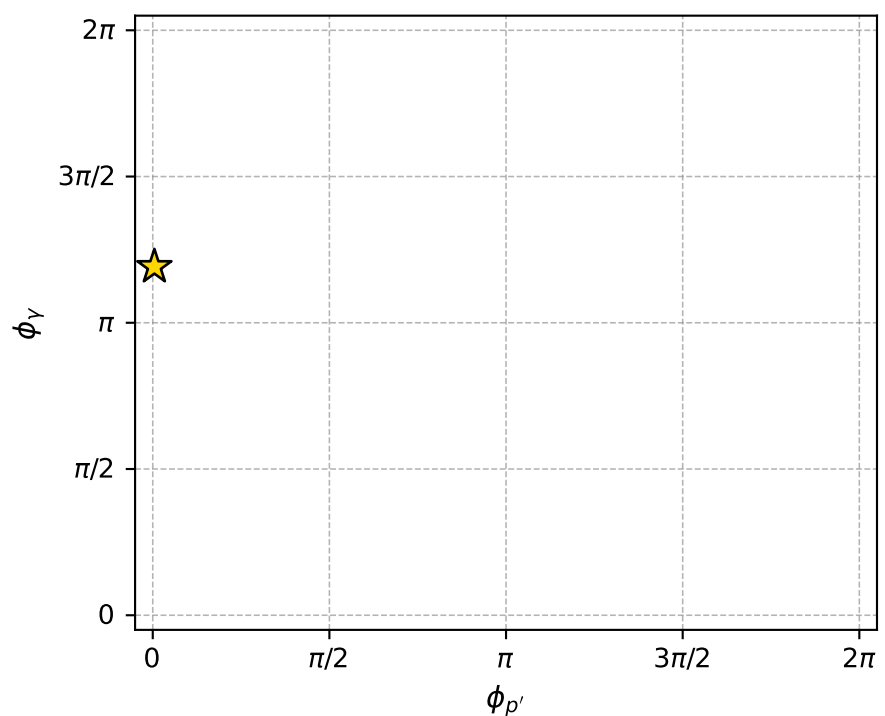
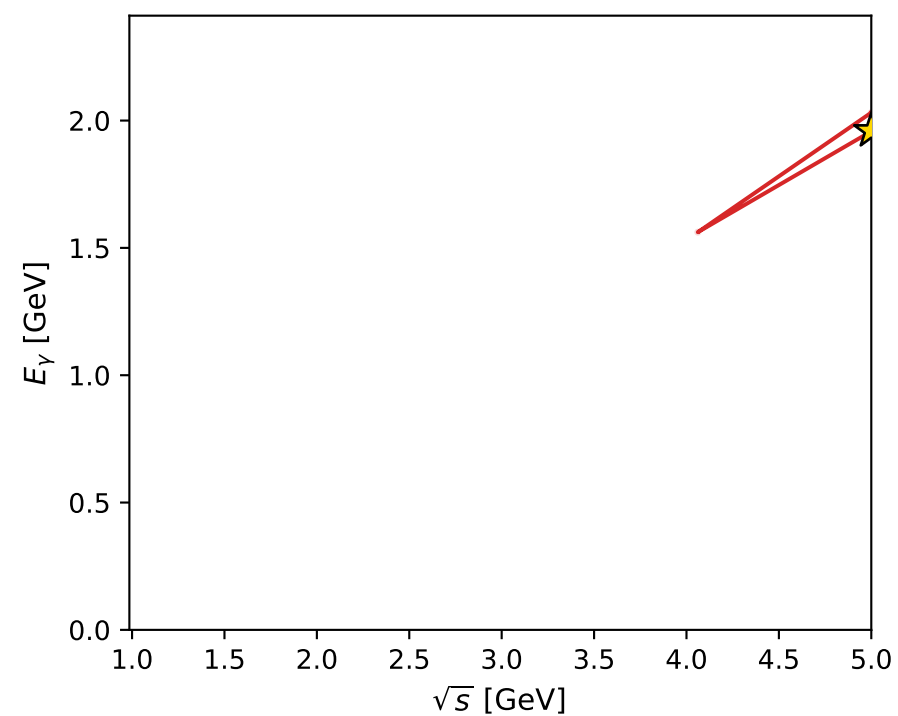
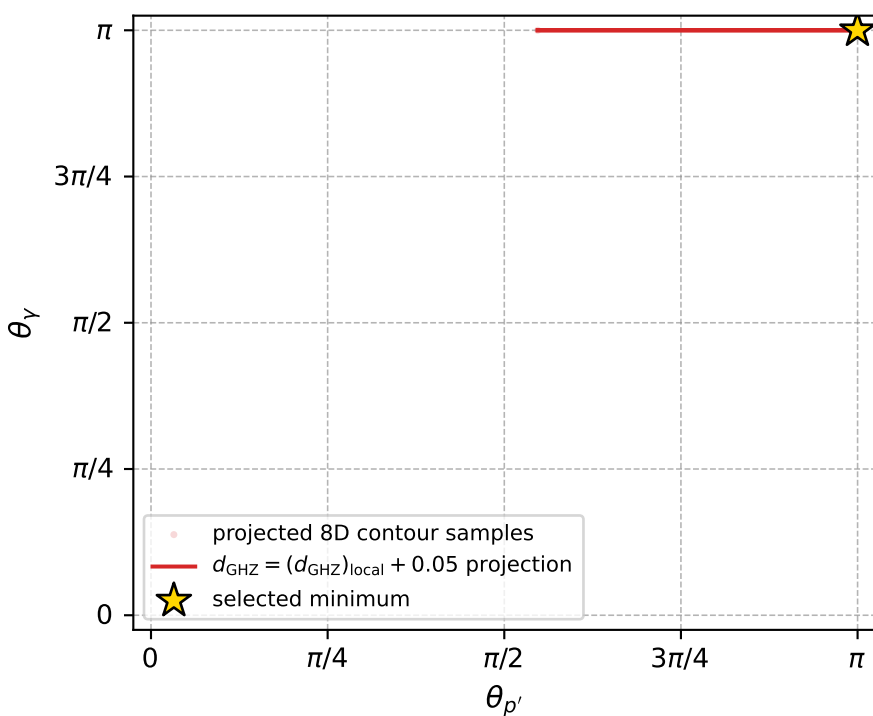
Transverse plane



Leading final-state amplitudes (N=2, retained=100.0%)



electron: polarization cluster P2, local minimum 31; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 2.6257246\text{e-}08$
 $d_{\text{GHZ}} \text{ contour} = 0.050000026$
 above global minimum = $3.9737839\text{e-}09$
 8D contour samples = 104

selected phase-space point:

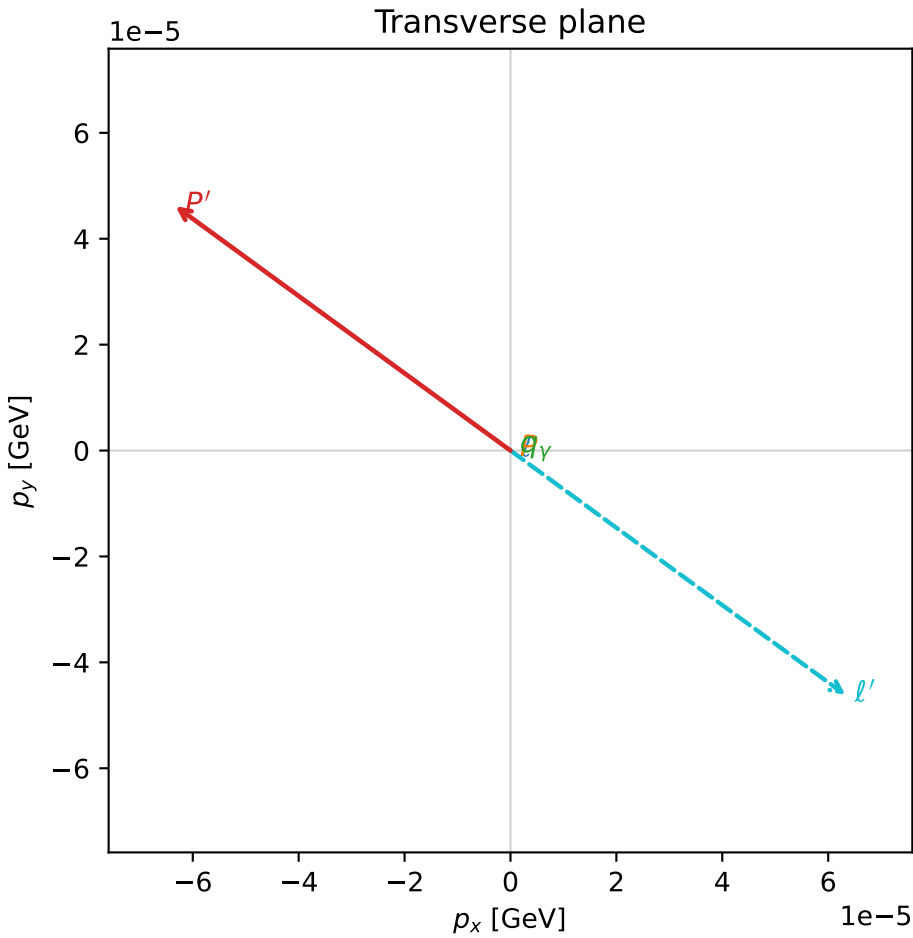
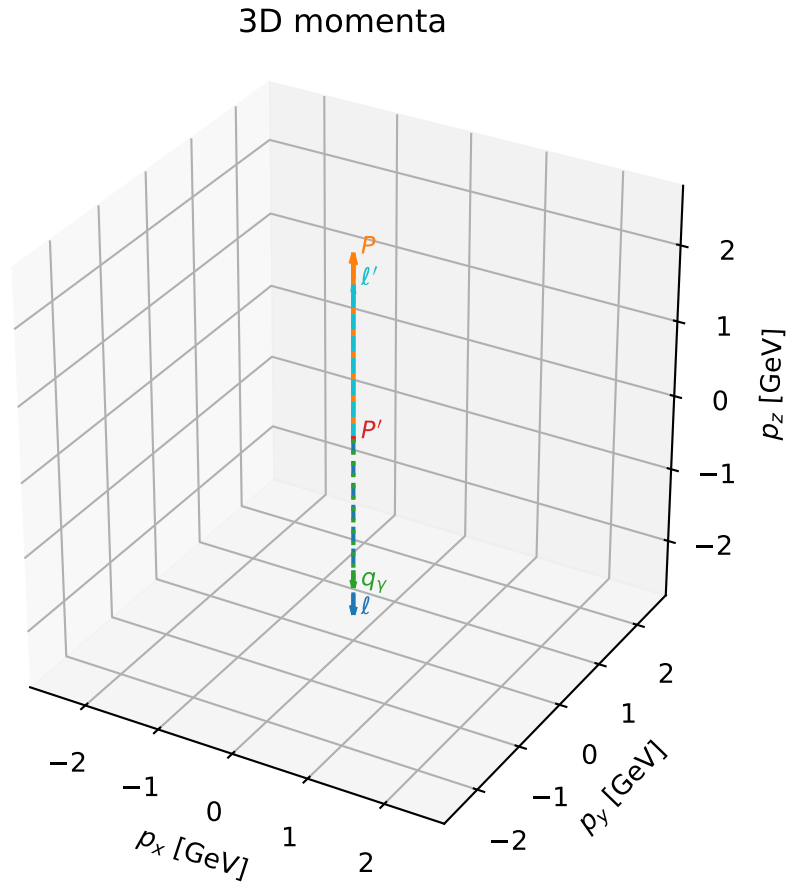
- $\text{sqrt_s} = 4.999824$
- $\text{theta_p_out} = 3.140935$
- $\text{theta_gamma_out} = 3.141593$
- $\text{q0ut} = 1.959451$
- $\text{phi_p_out} = 0.01447733$
- $\text{phi_gamma_out} = 3.74298$
- $\text{alpha_e} = 0.7229409$
- $\text{alpha_p} = 2.418641$

coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

dghz_mixing_angles_polarization_02_local_minimum_33 $d_{\{\text{rm GHZ}\}}=2.63319\text{e-}08$
region: polarization_cluster_02_local_minimum_33
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: $\alpha_e=0.79116621$ rad
incoming lepton: $0.703016|+\rangle +0.711174|-\rangle$
proton mixing angle: $\alpha_p=2.3504057$ rad
incoming proton: $-0.703002|+\rangle +0.711188|-\rangle$

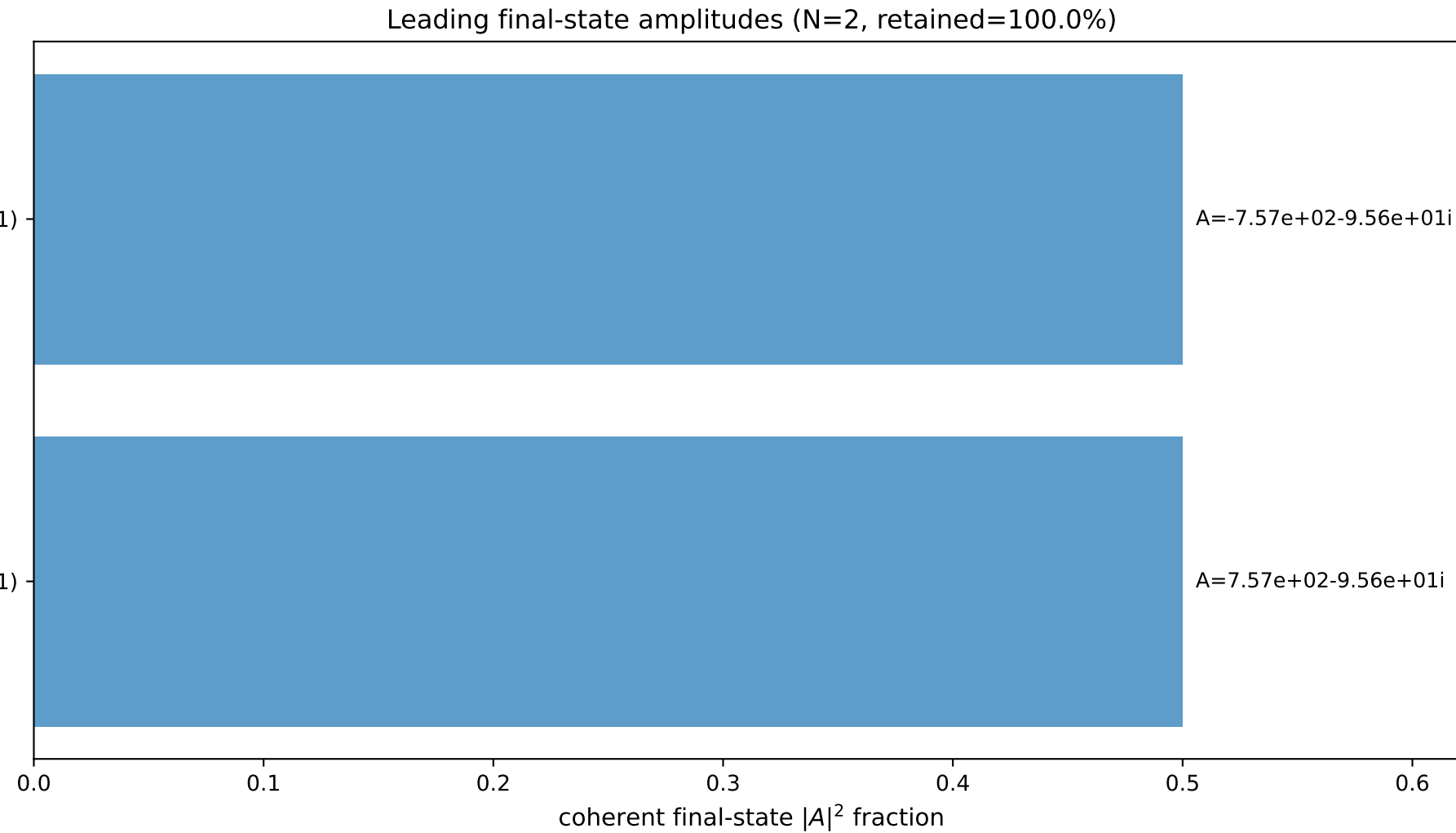
$s=24.934$, $\sqrt{s}=4.9934$, $|\vec{P}|=2.4086$, $|\vec{P}'|=0.0106855$
 $\theta_{P'}=0.0073264$, $\theta_Y=3.14159$, $\phi_{P'}=2.51128$, $\phi_Y=6.15755$
 $E_Y=2.03301$, $\theta_{\ell'}=3.87108\text{e-}05$, $\phi_{\ell'}=5.65288$
 $Q^2=19.4839$, $x_B=0.834731$, $t=-3.03824$, $W^2=4.73747$, $y=0.970372$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4086$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4086$
 P $E= 2.5848$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4086$
 ℓ' $E= 2.0223$ $p_x= 0.0001$ $p_y= -0.0000$ $p_z= 2.0223$
 P' $E= 0.9381$ $p_x= -0.0001$ $p_y= 0.0000$ $p_z= 0.0107$
 q_Y $E= 2.0330$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= -2.0330$

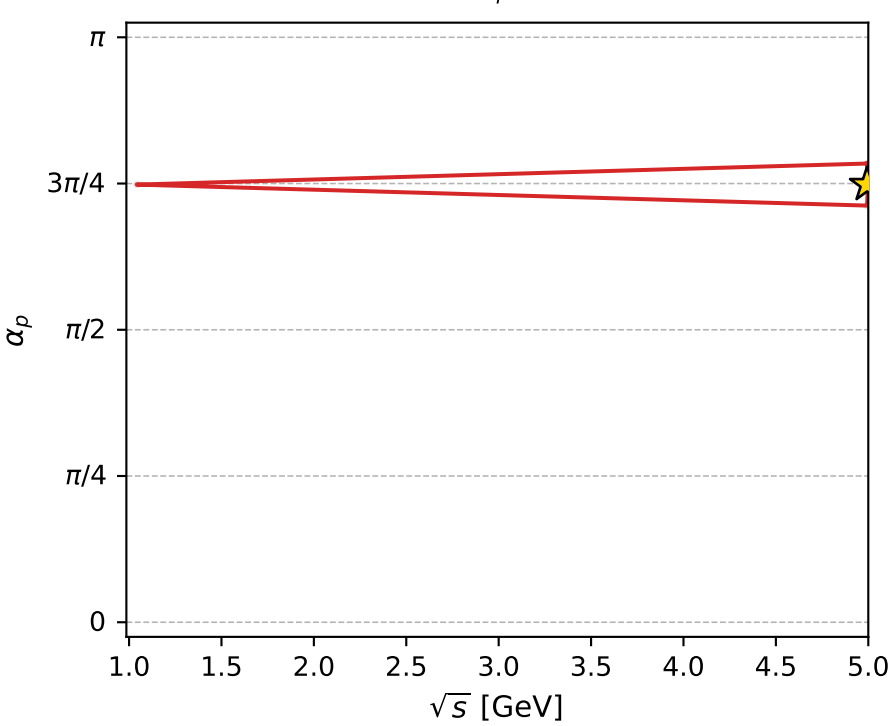
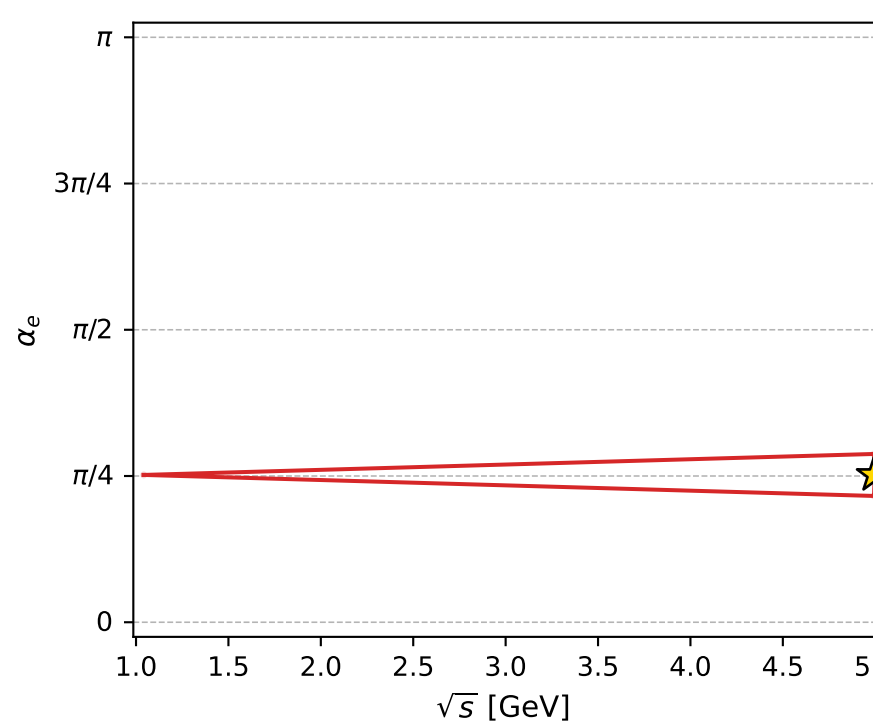
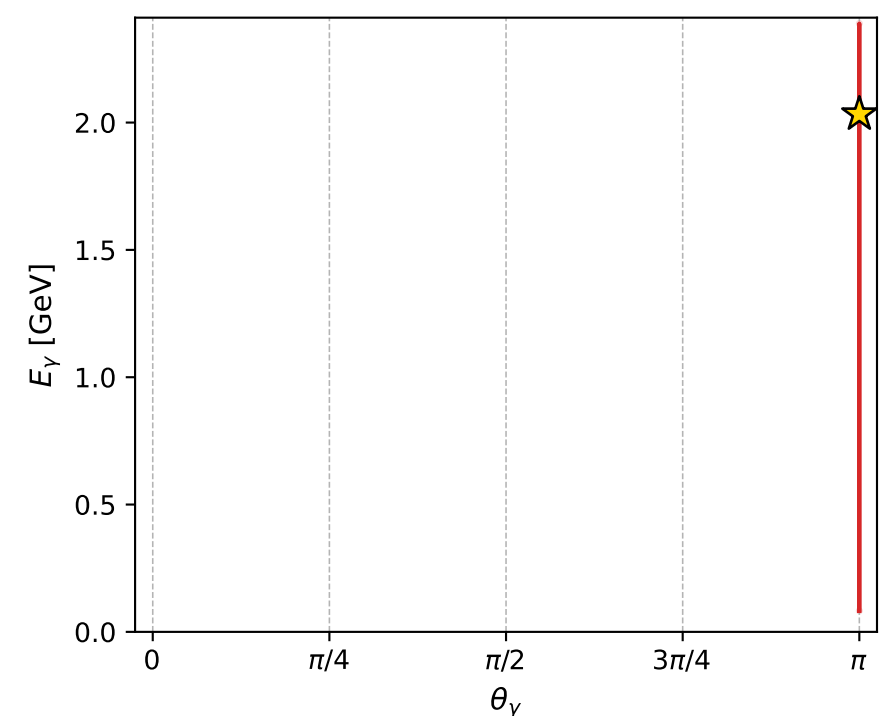
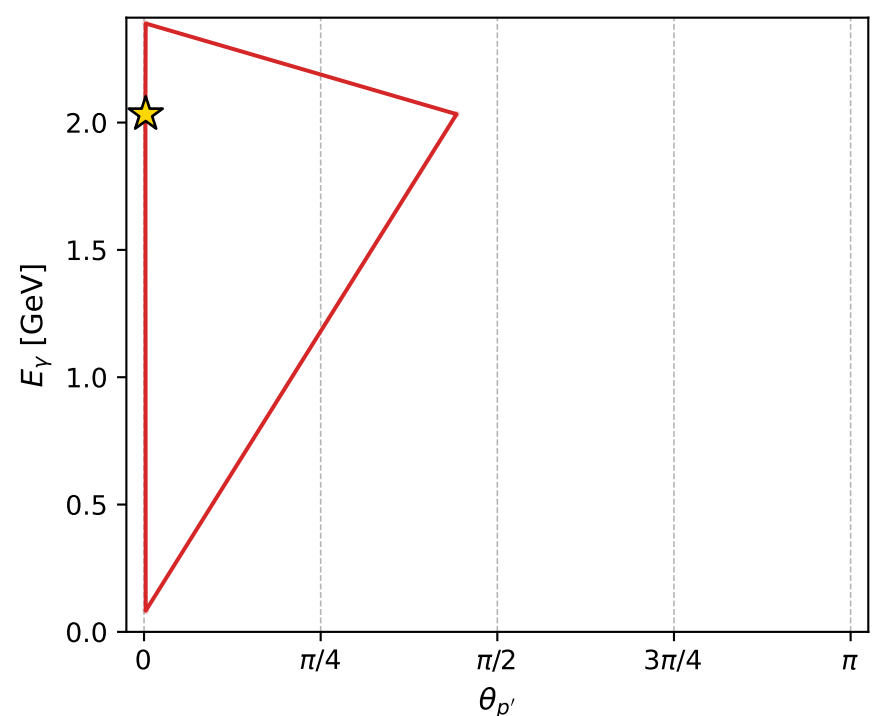
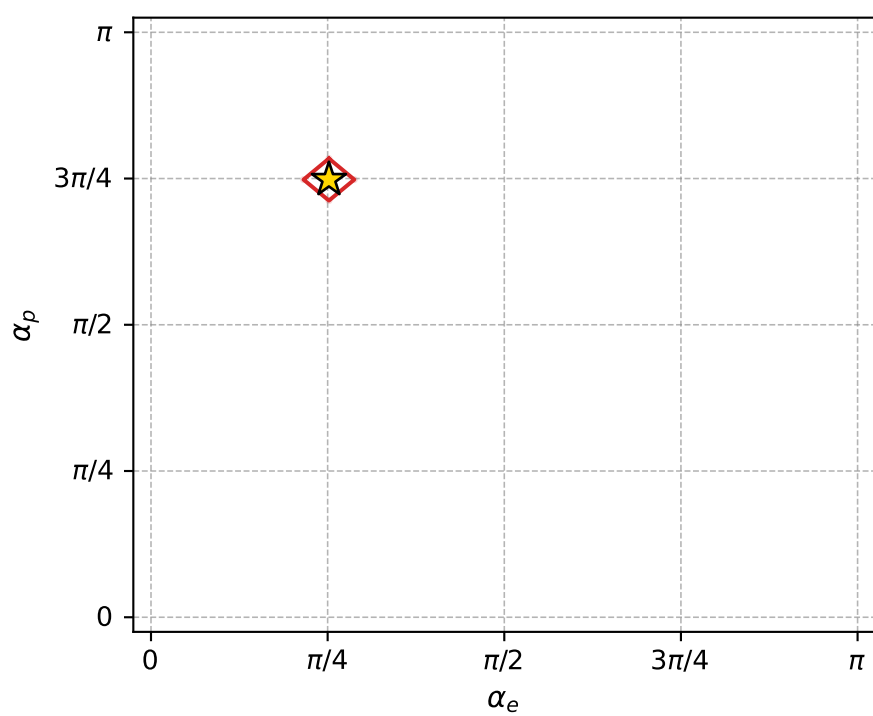
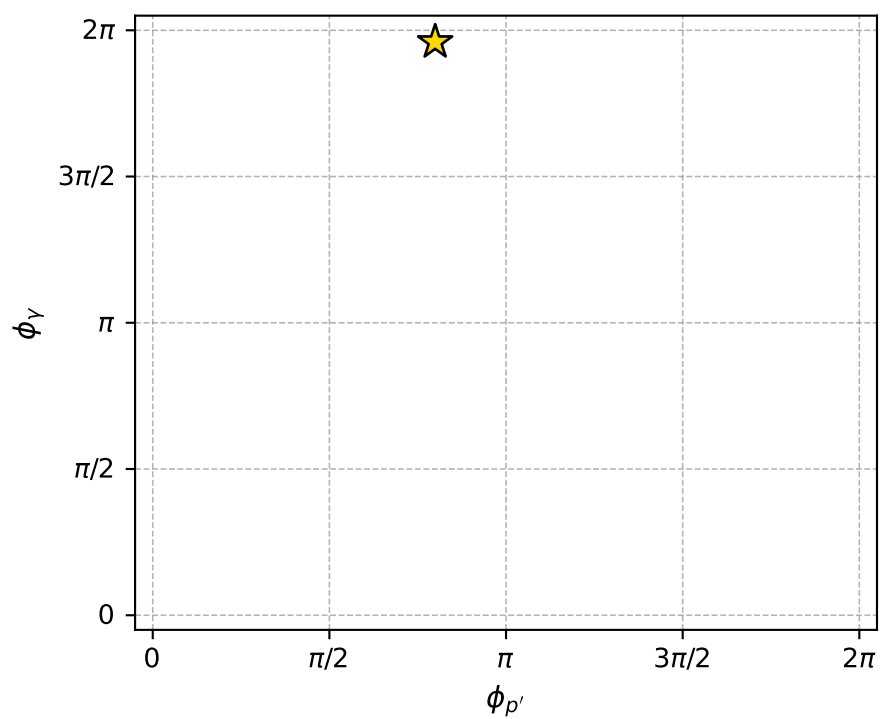
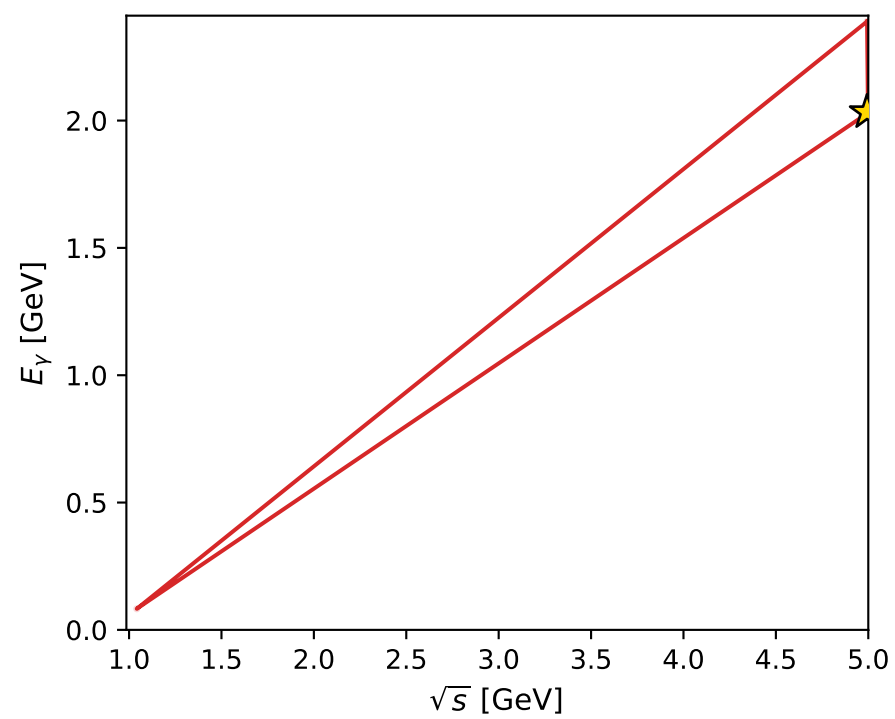
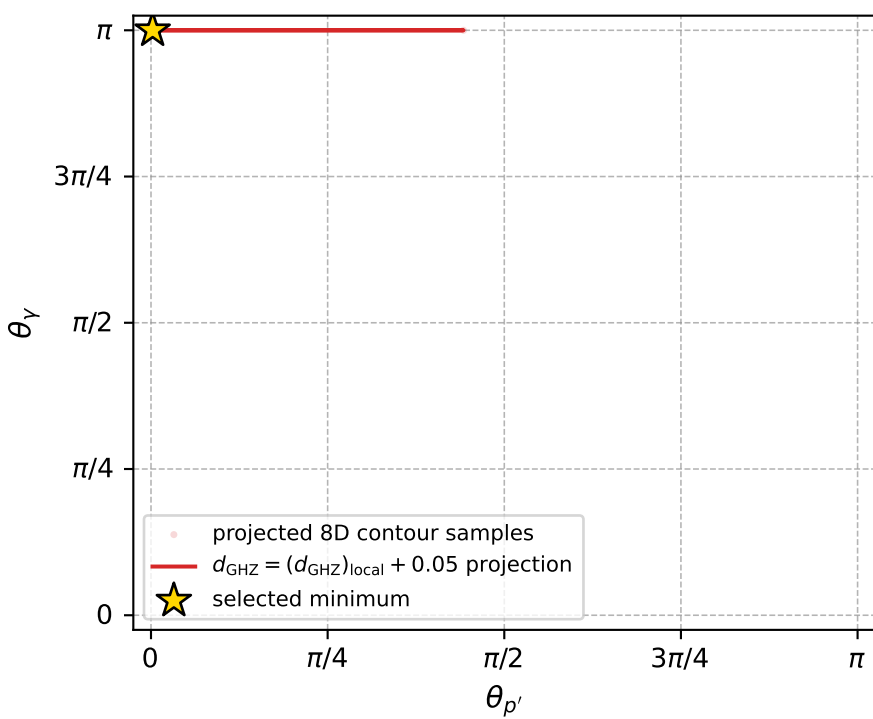


$(h_p, h_Y, h_\ell) = (+1, +1, -1)$

$(h_p, h_Y, h_\ell) = (-1, -1, +1)$



electron: polarization cluster P2, local minimum 33; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 2.6331935\text{e-}08$
 $d_{\text{GHZ}} \text{ contour} = 0.050000026$
 above global minimum = $4.0484731\text{e-}09$
 8D contour samples = 119

selected phase-space point:

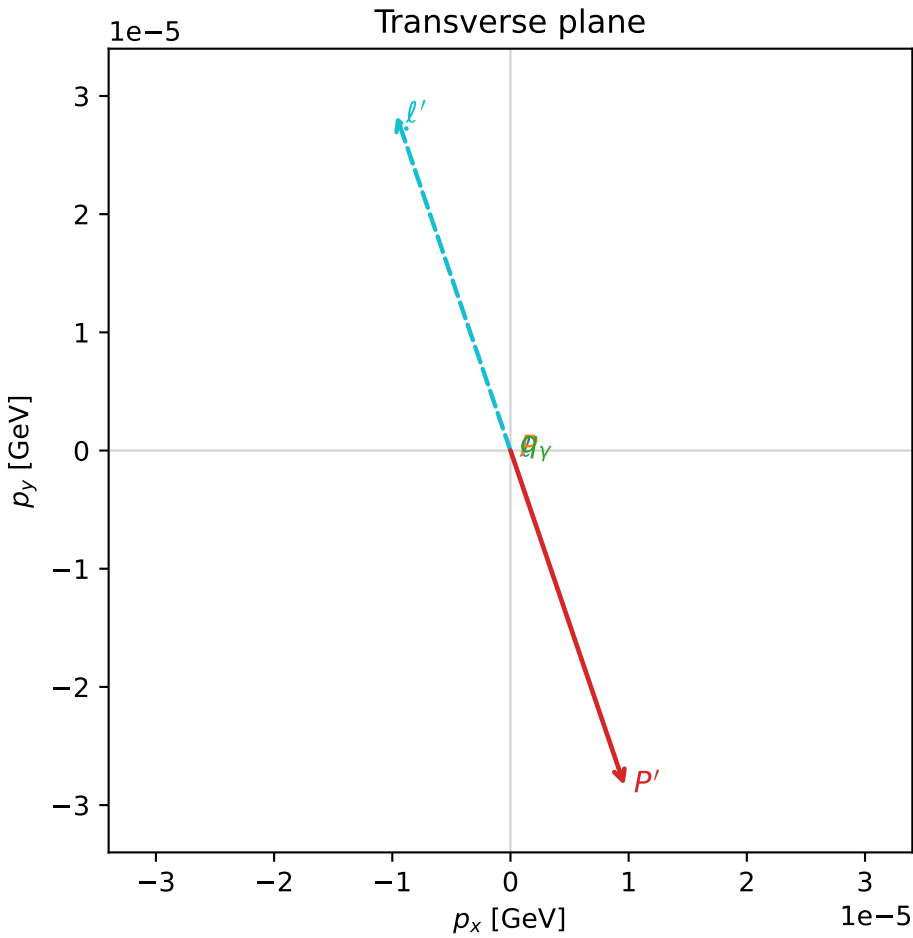
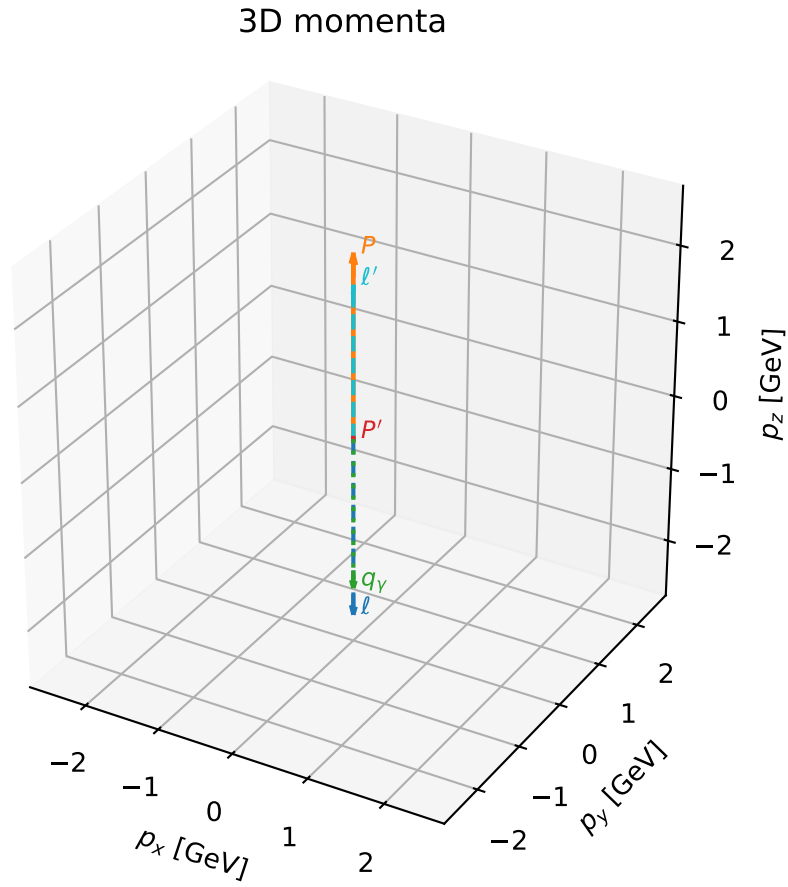
$\text{sqrt_s} = 4.993397$
 $\text{theta_p_out} = 0.007326399$
 $\text{theta_gamma_out} = 3.141593$
 $\text{q0ut} = 2.03301$
 $\text{phi_p_out} = 2.511284$
 $\text{phi_gamma_out} = 6.157547$
 $\text{alpha_e} = 0.7911662$
 $\text{alpha_p} = 2.350406$

coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

dghz_mixing_angles_polarization_02_local_minimum_34 $d_{\{\text{rm GHZ}\}}=2.63343\text{e-}08$
region: polarization_cluster_02_local_minimum_34
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: $\alpha_e=0.76007598$ rad
incoming lepton: $0.724784|+\rangle +0.688977|-\rangle$
proton mixing angle: $\alpha_p=2.3815115$ rad
incoming proton: $-0.72478|+\rangle +0.68898|-\rangle$

$s=25$, $\sqrt{s}=5$, $|\vec{P}|=2.41202$, $|\vec{P}'|=0.0121954$
 $\theta_{P'}=0.00245437$, $\theta_Y=3.14159$, $\phi_{P'}=5.04048$, $\phi_Y=3.34859$
 $E_Y=2.03706$, $\theta_{\ell'}=1.47823\text{e-}05$, $\phi_{\ell'}=1.89889$
 $Q^2=19.536$, $x_B=0.834603$, $t=-3.03695$, $W^2=4.75137$, $y=0.970455$

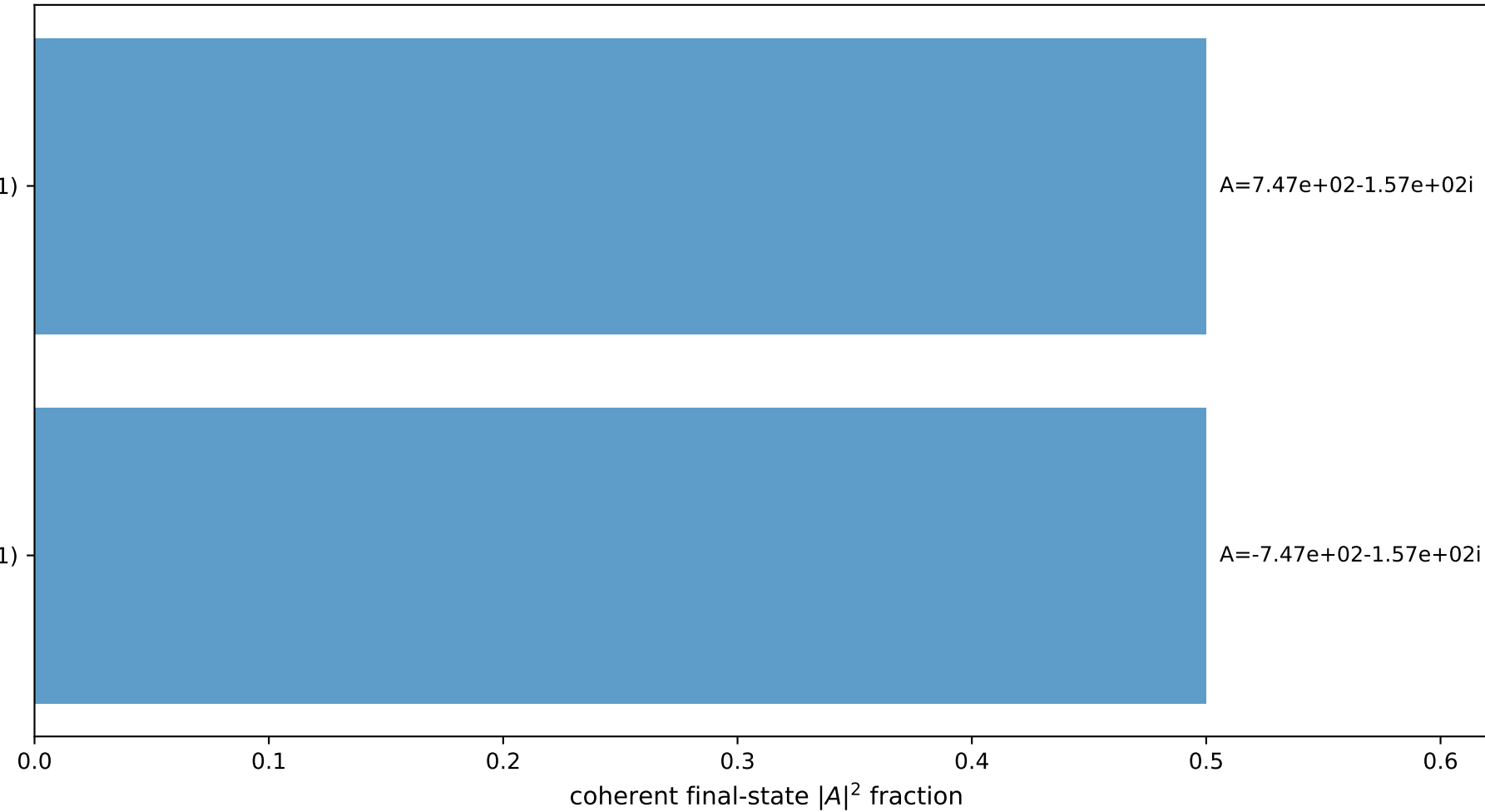
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4120$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 P $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4120$
 ℓ' $E= 2.0249$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 2.0249$
 P' $E= 0.9381$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 0.0122$
 q_Y $E= 2.0371$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= -2.0371$



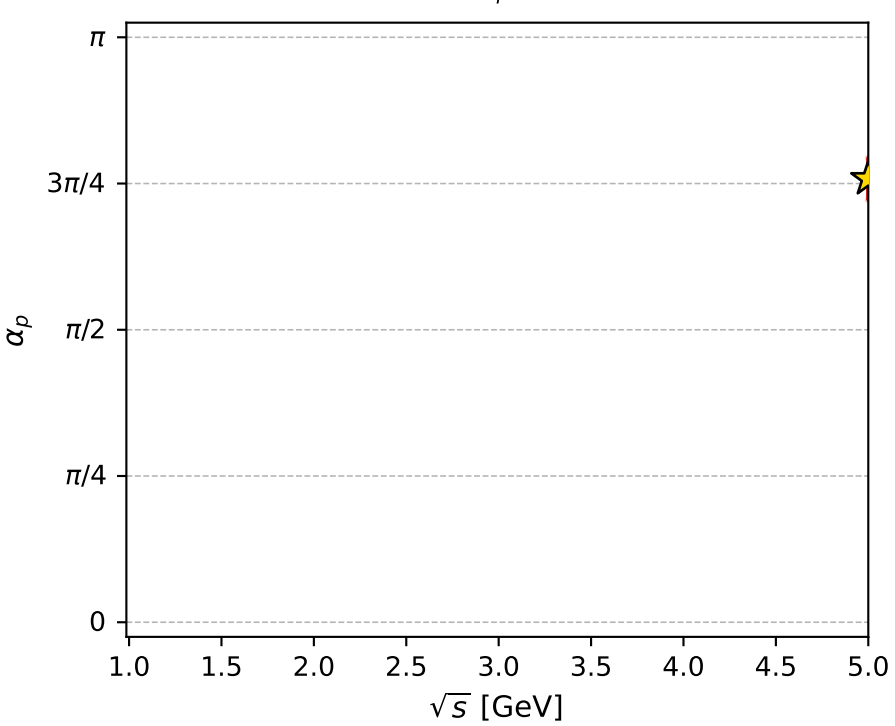
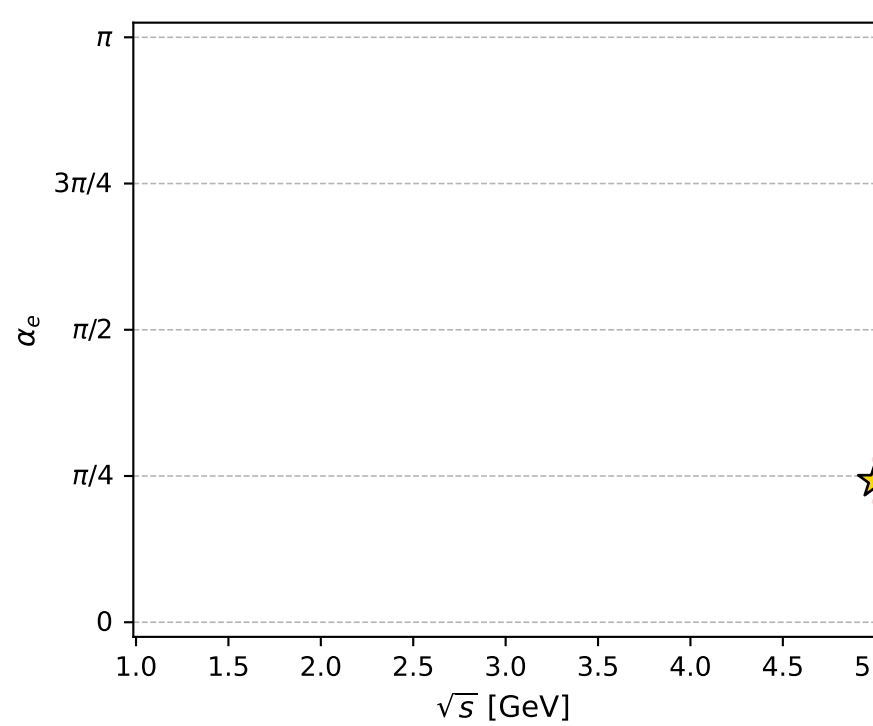
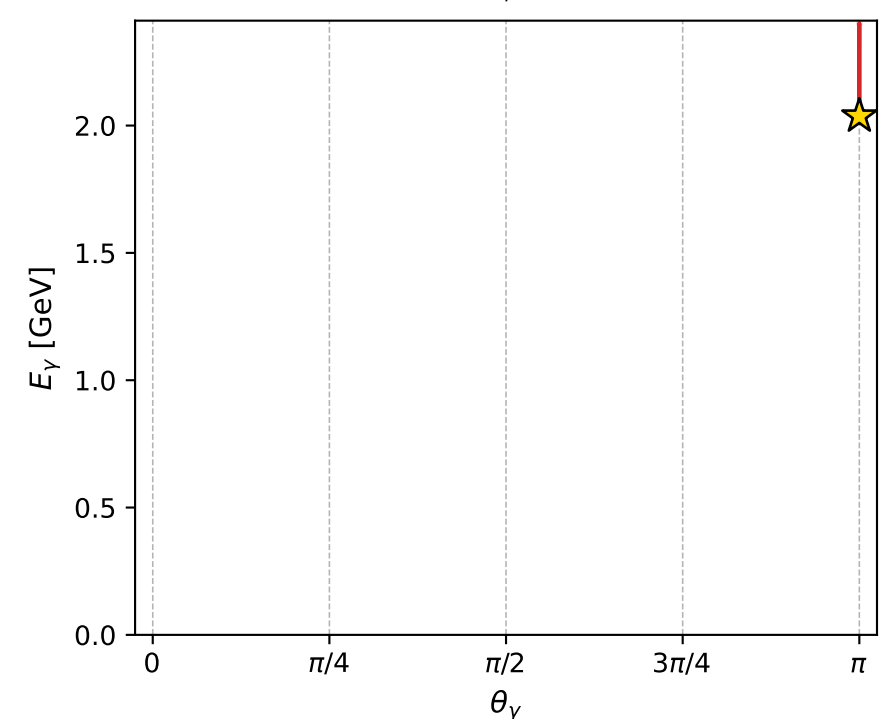
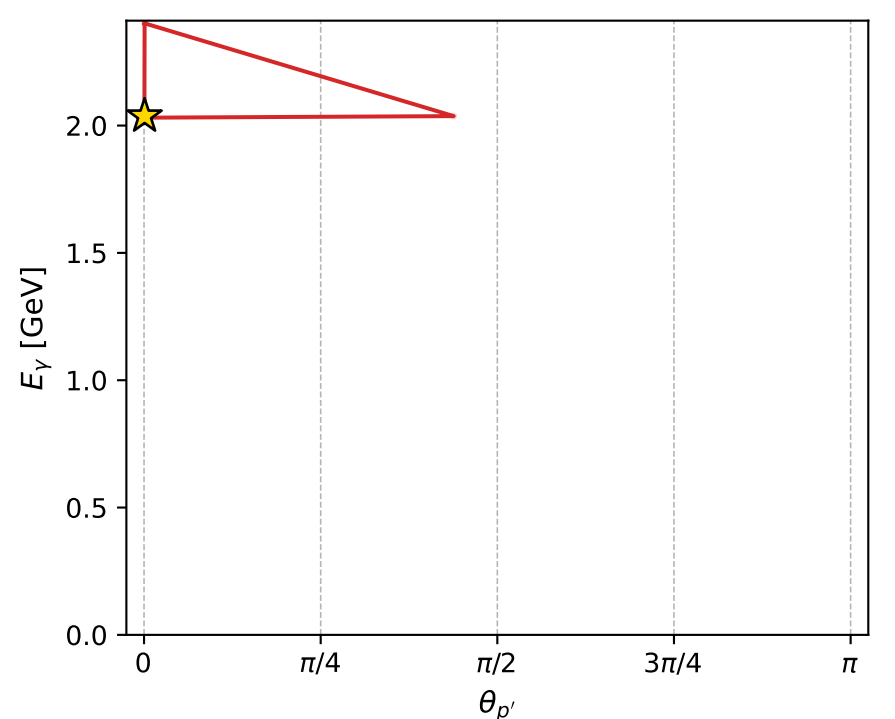
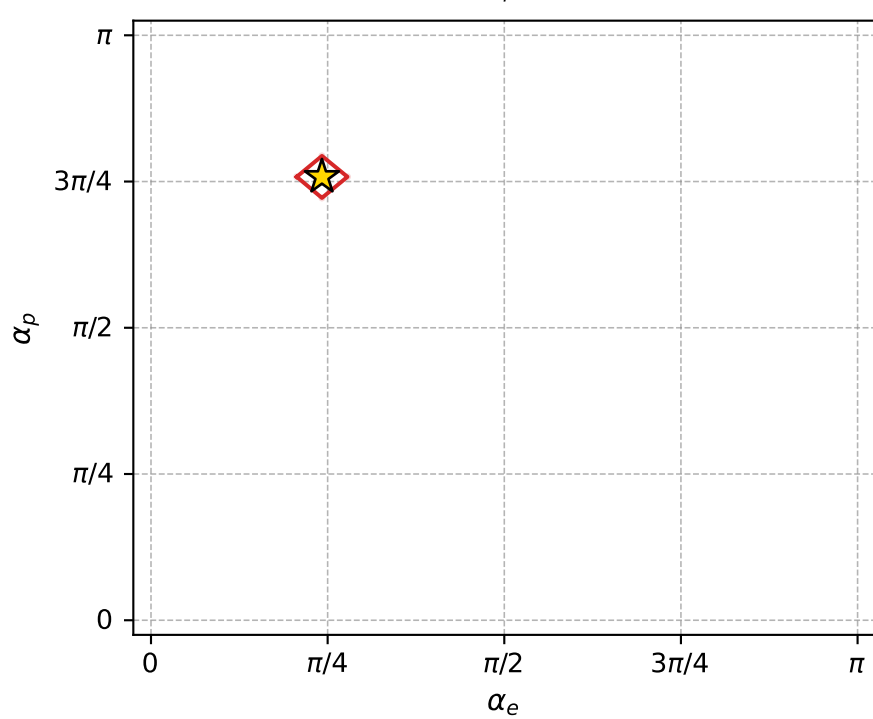
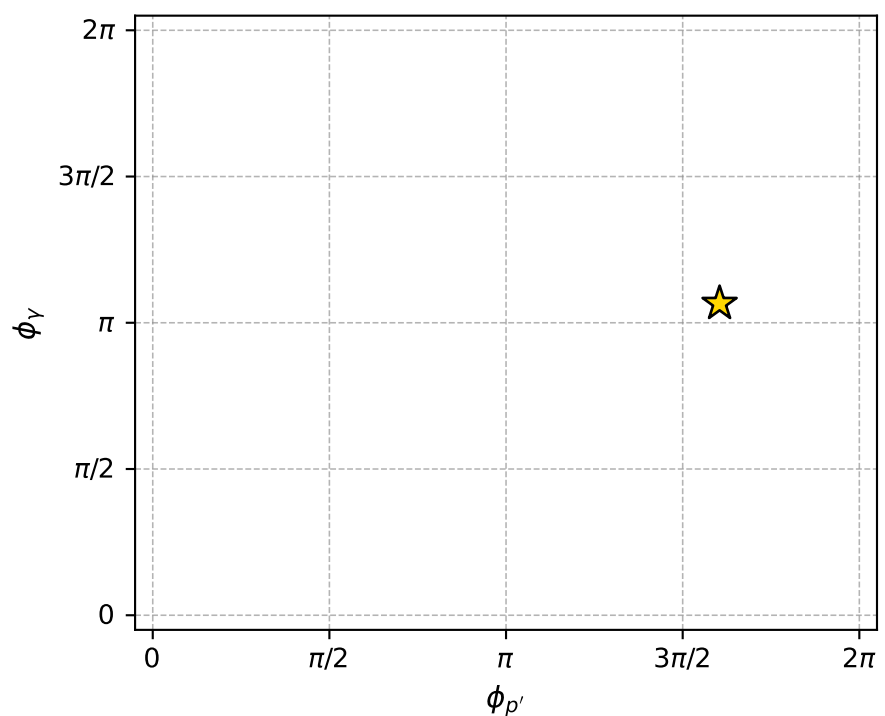
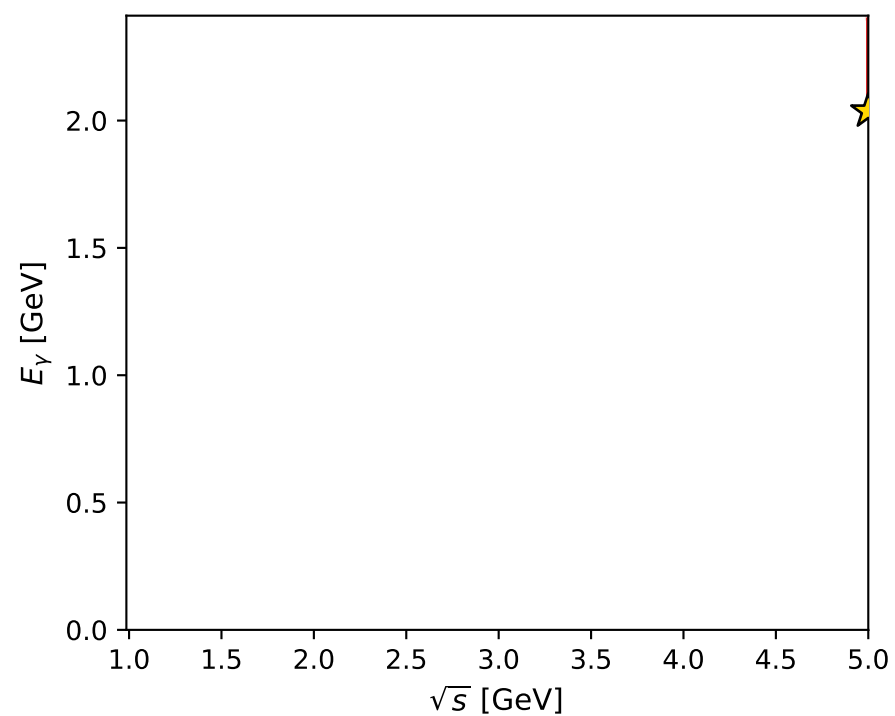
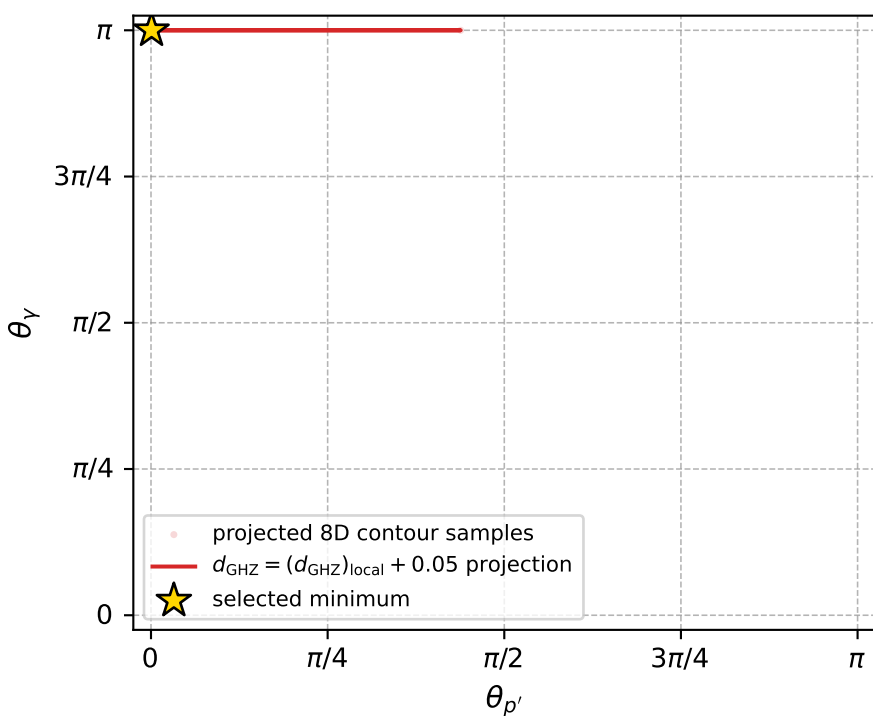
$(h_p, h_Y, h_{\ell}) = (+1, +1, -1)$

$(h_p, h_Y, h_{\ell}) = (-1, -1, +1)$

Leading final-state amplitudes (N=2, retained=100.0%)



electron: polarization cluster P2, local minimum 34; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 2.633433\text{e-}08$
 $d_{\text{GHZ}} \text{ contour} = 0.050000026$
 above global minimum = $4.0508684\text{e-}09$
 8D contour samples = 76

selected phase-space point:

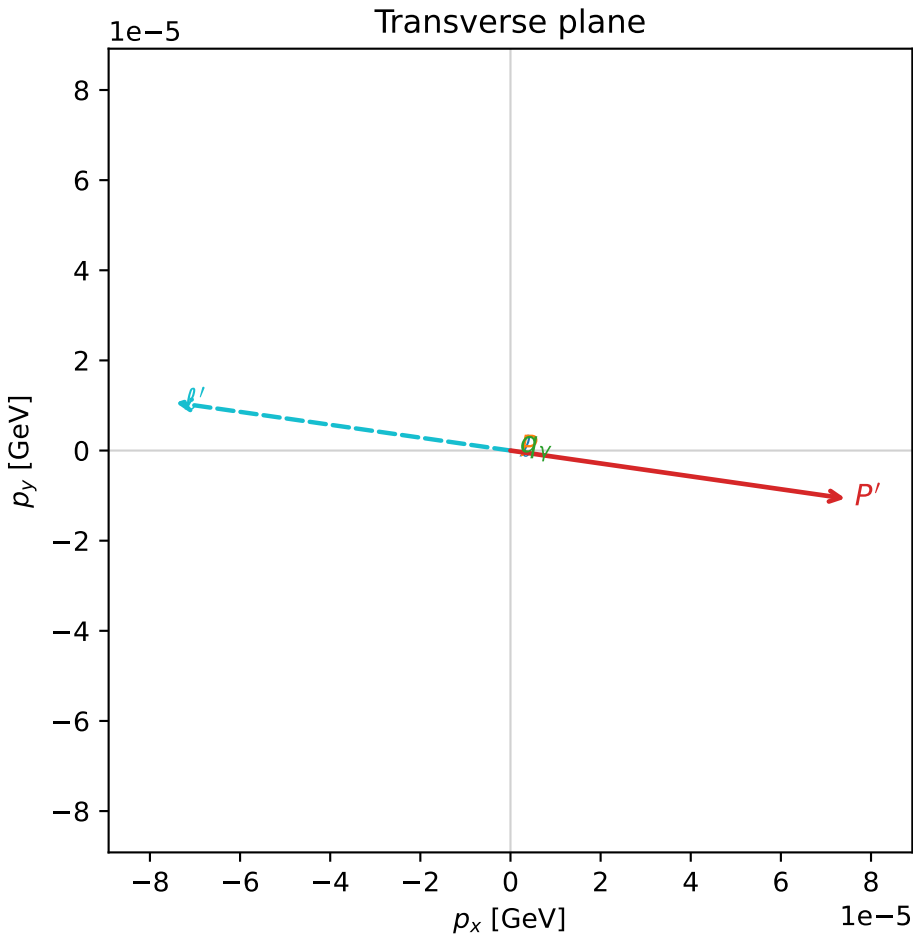
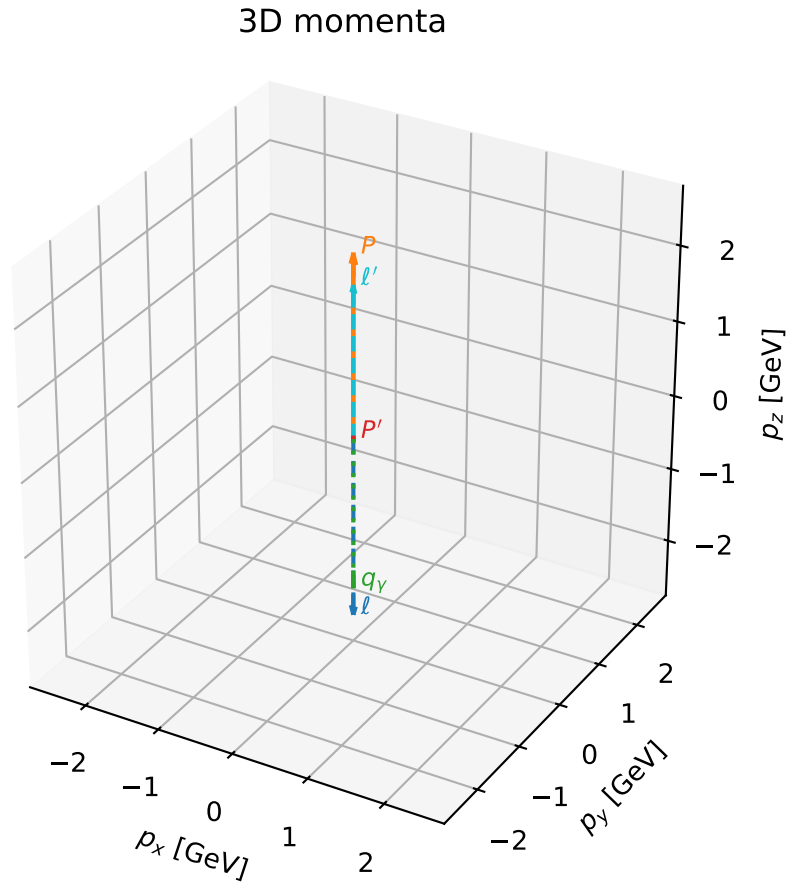
$\text{sqrt_s} = 5$
 $\text{theta_p_out} = 0.002454369$
 $\text{theta_gamma_out} = 3.141593$
 $\text{q0out} = 2.037058$
 $\text{phi_p_out} = 5.040482$
 $\text{phi_gamma_out} = 3.348585$
 $\text{alpha_e} = 0.760076$
 $\text{alpha_p} = 2.381512$

coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

dghz_mixing_angles_polarization_02_local_minimum_36 $d_{\{\text{rm GHZ}\}}=2.63908\text{e-}08$
 region: polarization_cluster_02_local_minimum_36
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=0.80546213$ rad
 incoming lepton: $0.692778|+\rangle +0.721151|-\rangle$
 proton mixing angle: $\alpha_p=2.3361295$ rad
 incoming proton: $-0.692777|+\rangle +0.721152|-\rangle$

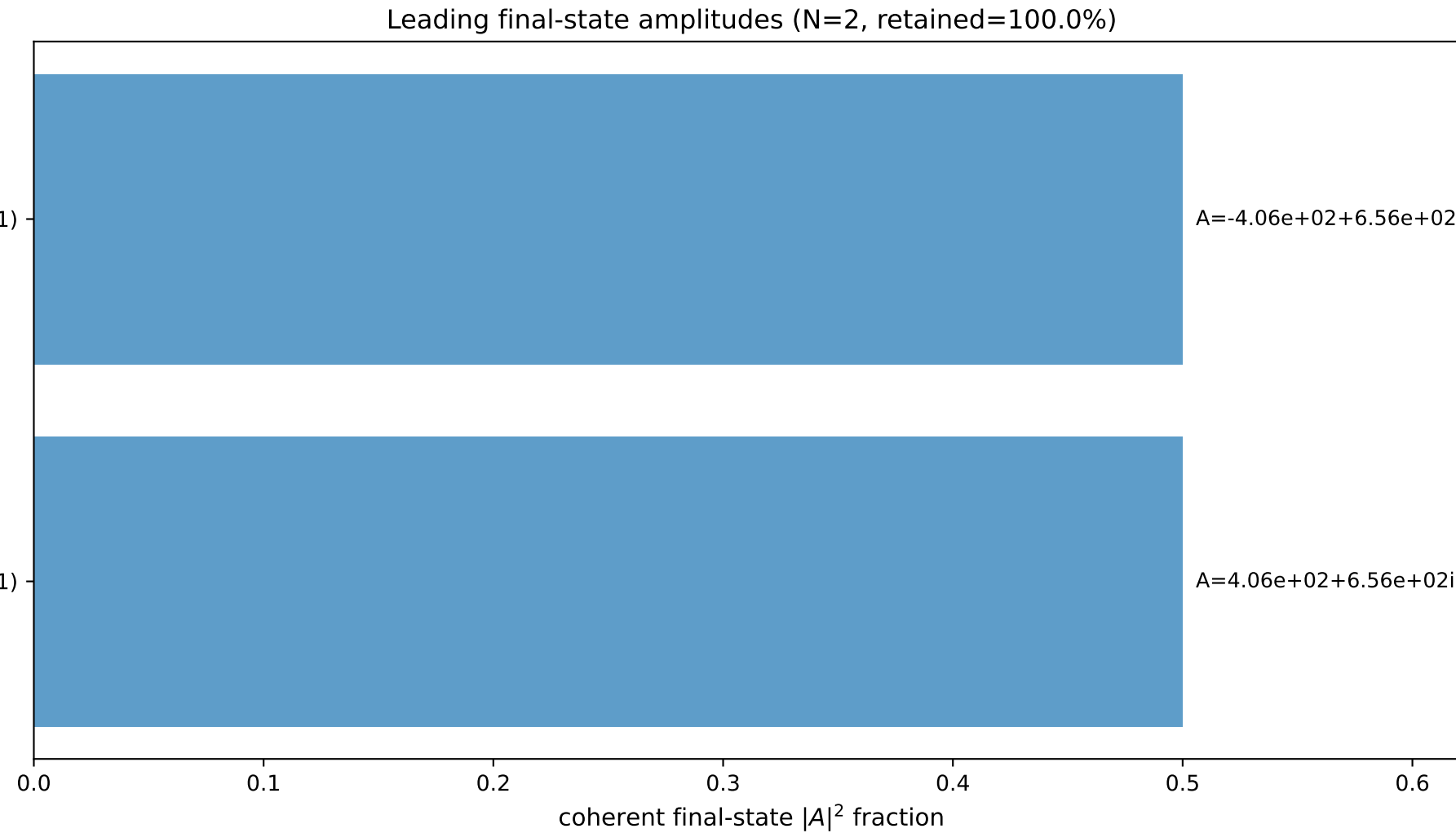
$s=25$, $\sqrt{s}=5$, $|\vec{P}|=2.41202$, $|\vec{P}'|=0.0152919$
 $\theta_{P'}=0.00490874$, $\theta_Y=3.14159$, $\phi_{P'}=6.14104$, $\phi_Y=2.12467$
 $E_Y=2.03858$, $\theta_{\ell'}=3.70997\text{e-}05$, $\phi_{\ell'}=2.99944$
 $Q^2=19.5208$, $x_B=0.833936$, $t=-3.02225$, $W^2=4.76708$, $y=0.970478$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4120$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 P $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4120$
 ℓ' $E= 2.0233$ $p_x= -0.0001$ $p_y= 0.0000$ $p_z= 2.0233$
 P' $E= 0.9381$ $p_x= 0.0001$ $p_y= -0.0000$ $p_z= 0.0153$
 q_Y $E= 2.0386$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= -2.0386$

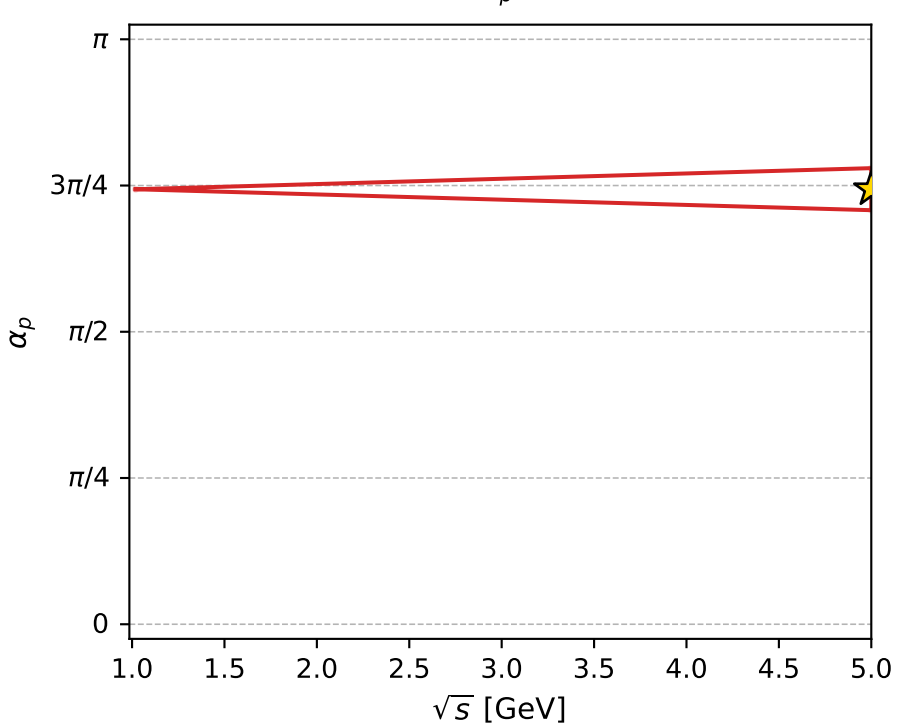
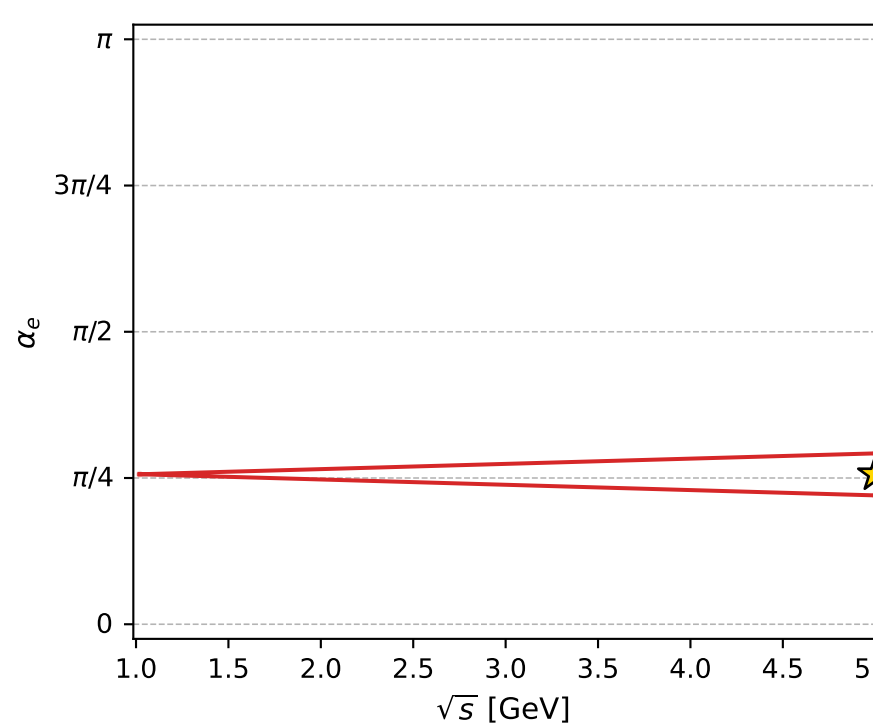
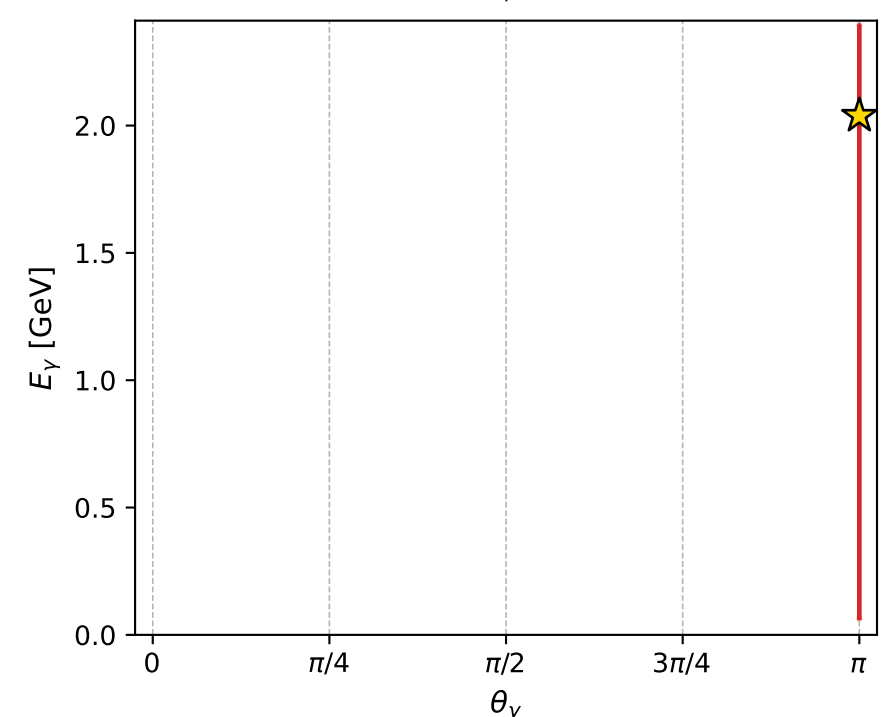
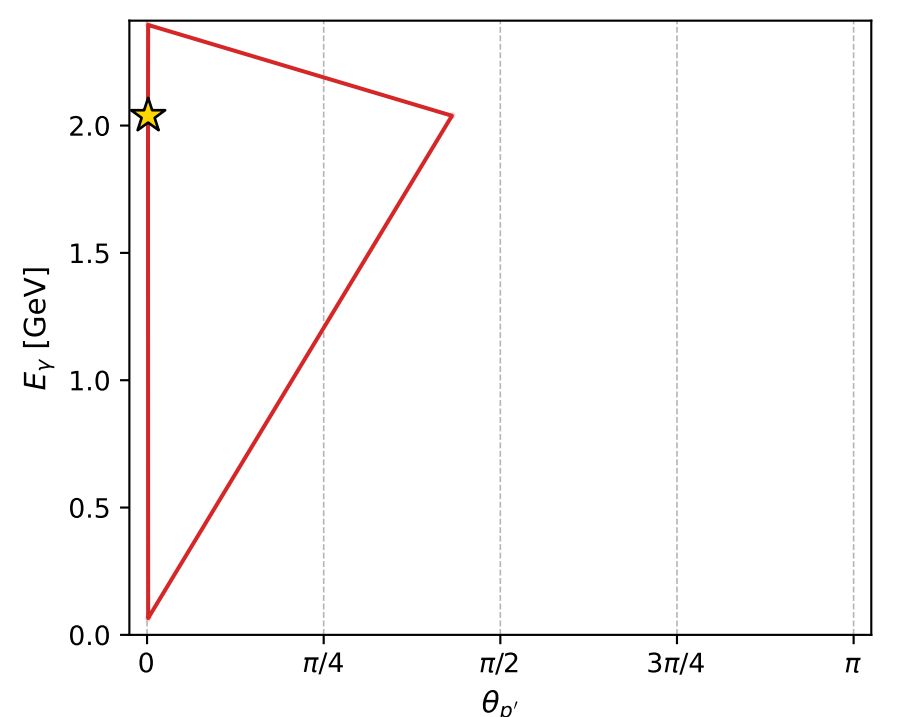
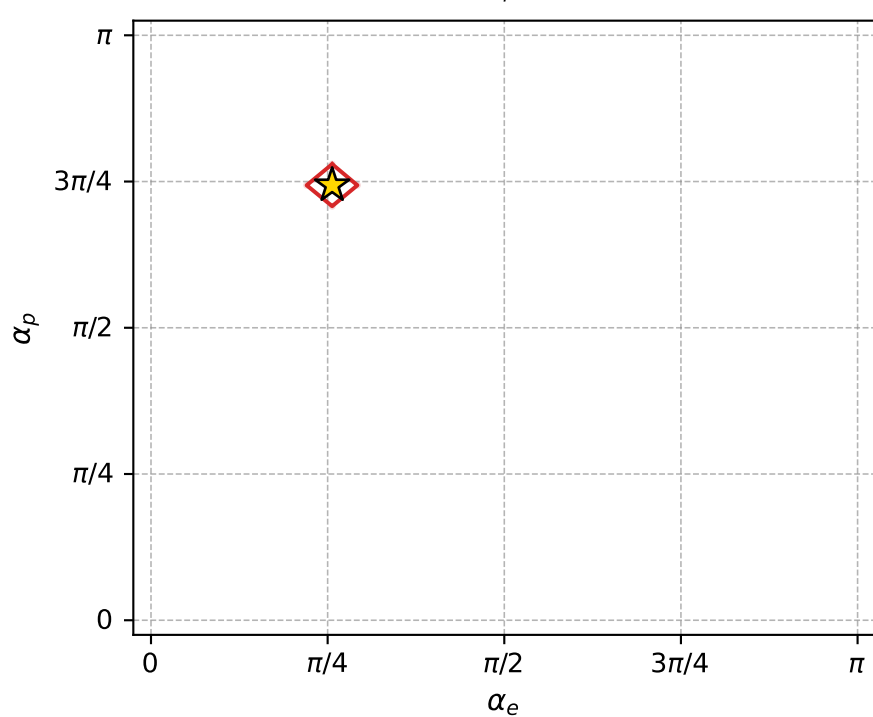
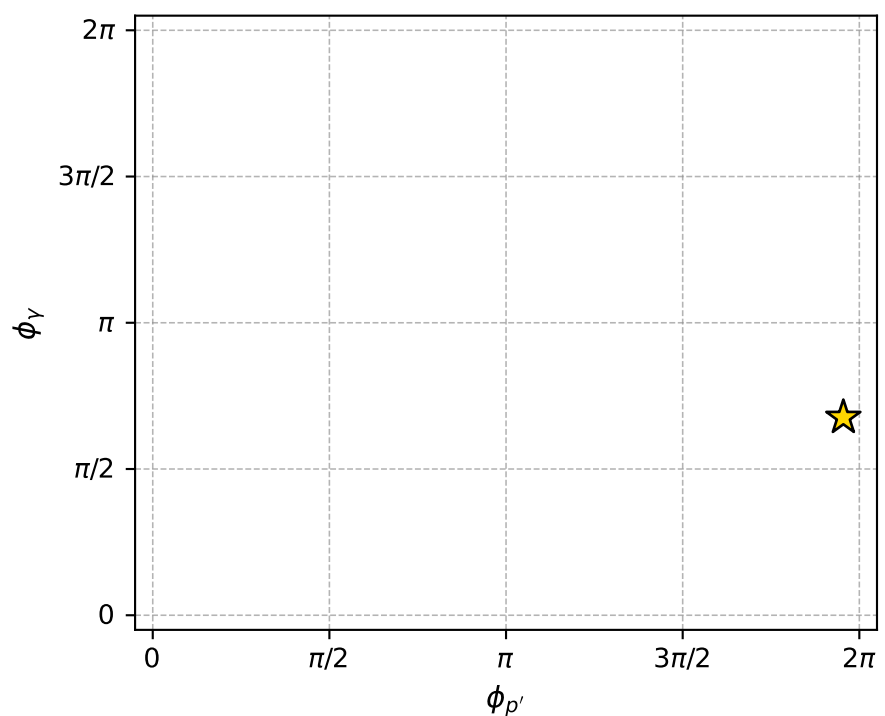
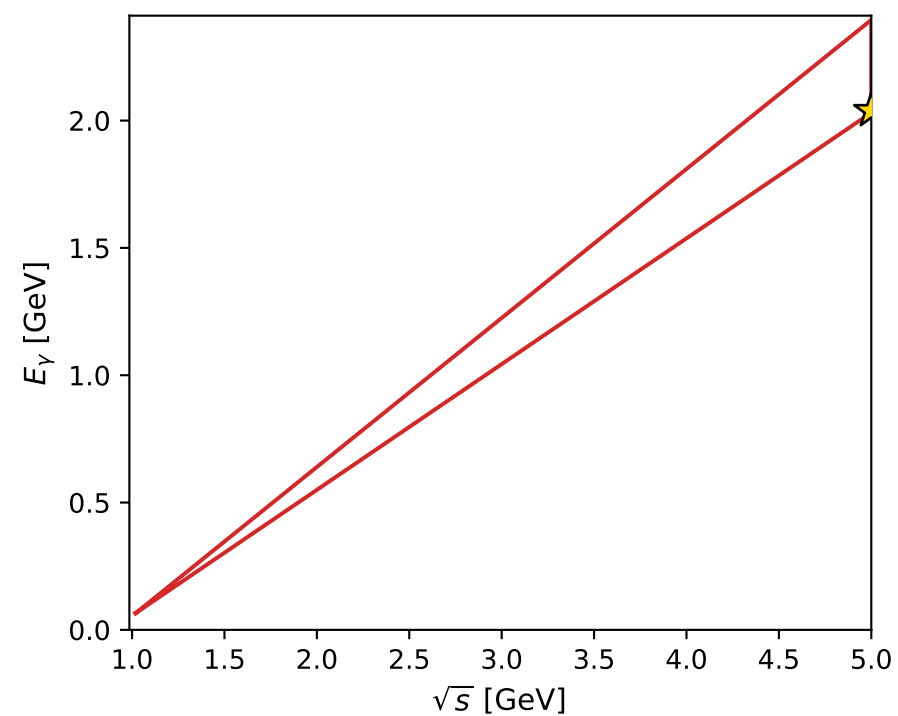
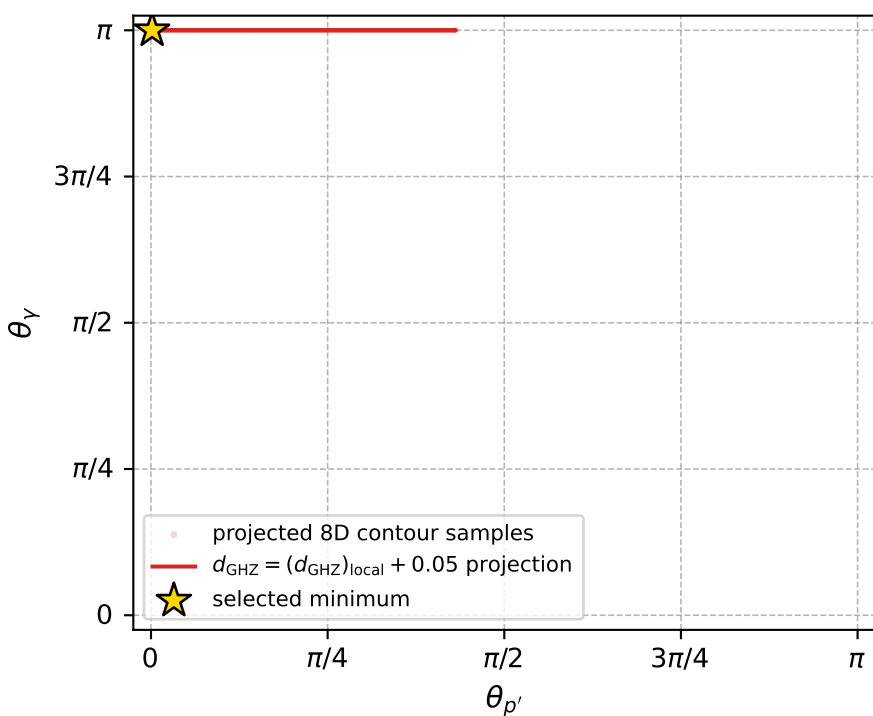


$(h_p, h_Y, h_\ell) = (-1, -1, +1)$

$(h_p, h_Y, h_\ell) = (+1, +1, -1)$



electron: polarization cluster P2, local minimum 36; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 2.6390795\text{e-}08$
 $d_{\text{GHZ}} \text{ contour} = 0.050000026$
 above global minimum = $4.1073327\text{e-}09$
 8D contour samples = 76

selected phase-space point:

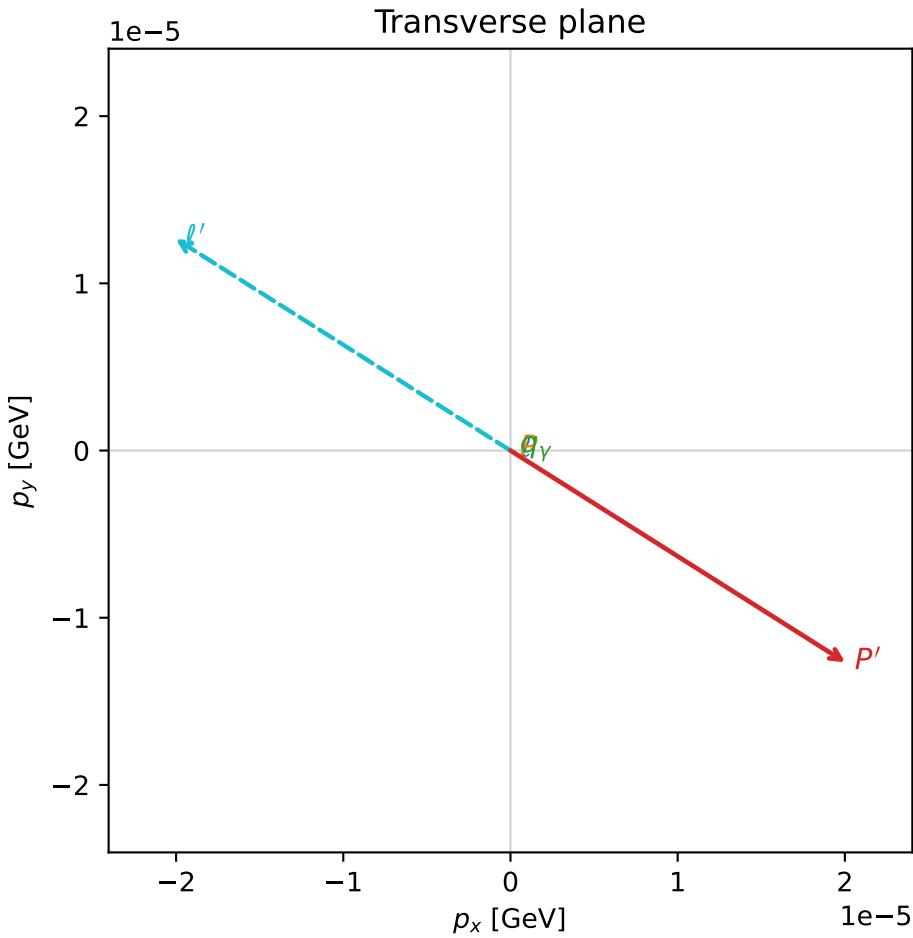
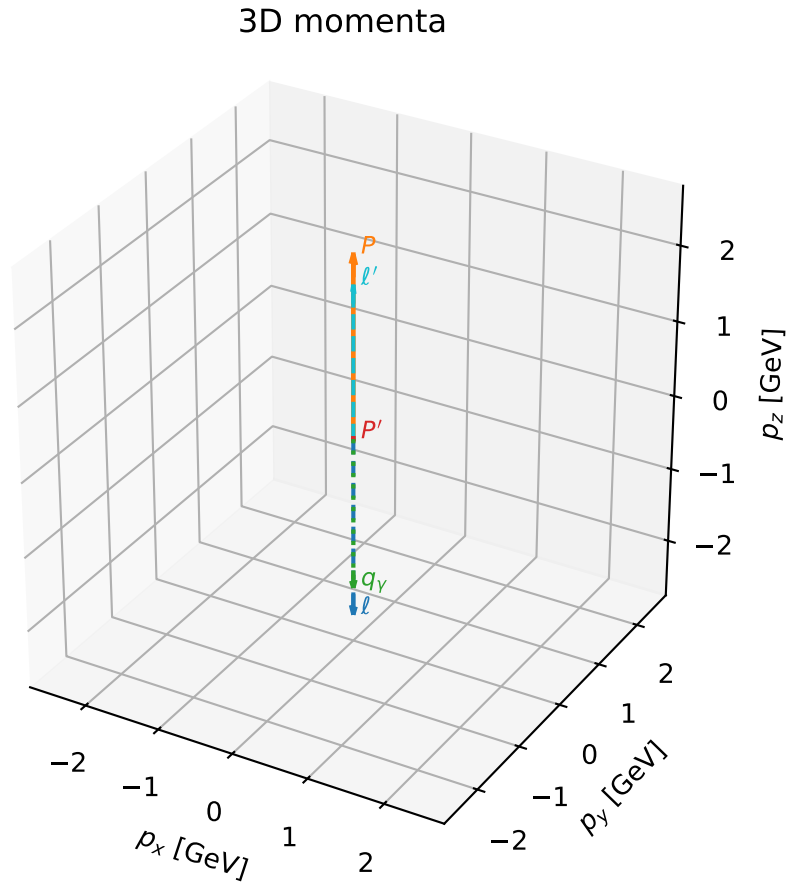
$\text{sqrt_s} = 5$
 $\text{theta_p_out} = 0.004908739$
 $\text{theta_gamma_out} = 3.141593$
 $\text{q0out} = 2.038583$
 $\text{phi_p_out} = 6.141035$
 $\text{phi_gamma_out} = 2.124665$
 $\text{alpha_e} = 0.8054621$
 $\text{alpha_p} = 2.336129$

coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

dghz_mixing_angles_polarization_02_local_minimum_37 $d_{\{\text{rm GHZ}\}}=2.64099\text{e-}08$
region: polarization_cluster_02_local_minimum_37
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: $\alpha_e=0.81766347$ rad
incoming lepton: $0.683928|+\rangle +0.72955|-\rangle$
proton mixing angle: $\alpha_p=2.3239276$ rad
incoming proton: $-0.683927|+\rangle +0.729551|-\rangle$

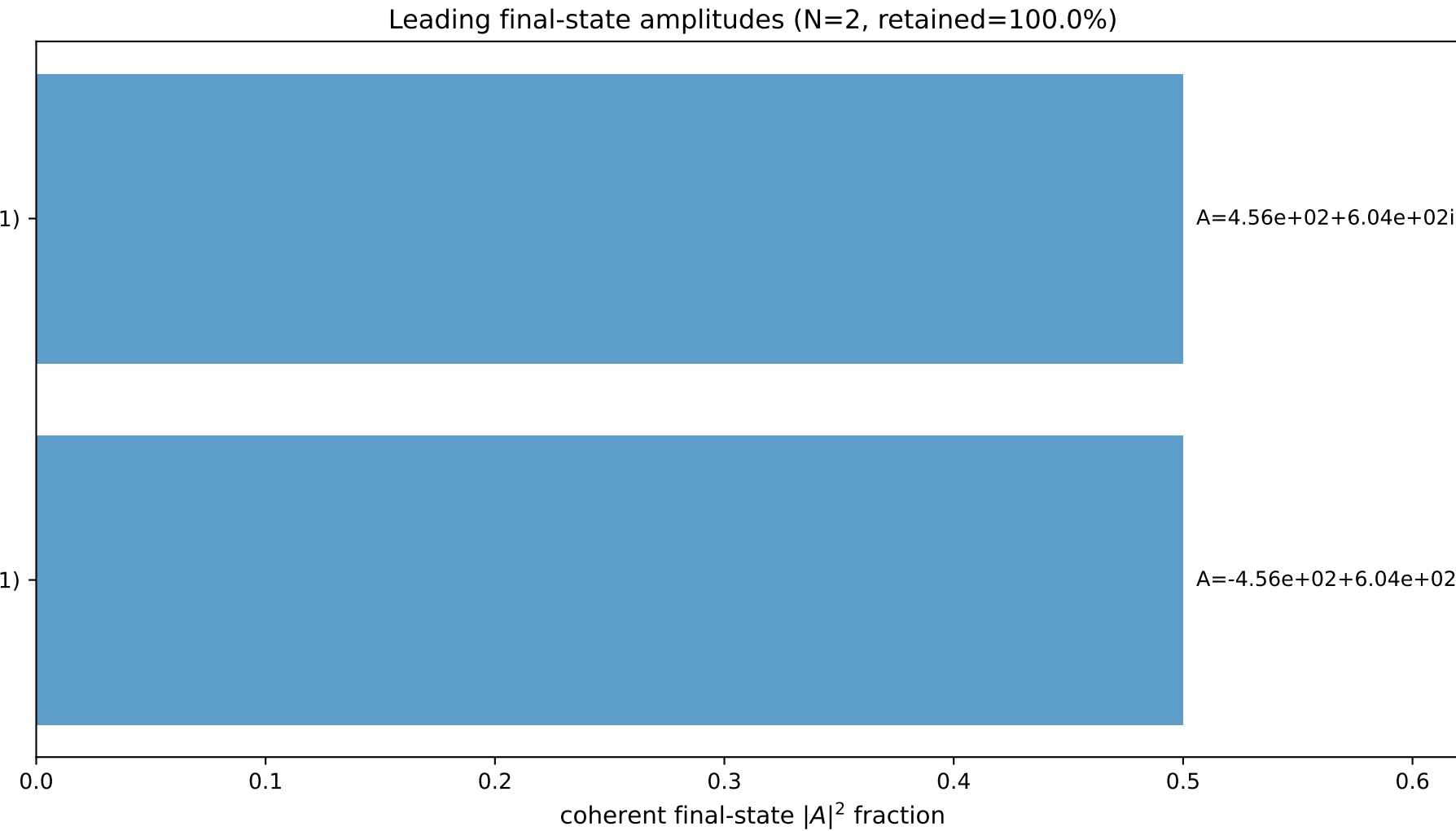
$s=25$, $\sqrt{s}=5$, $|\vec{P}|=2.41202$, $|\vec{P}'|=0.0096556$
 $\theta_{P'}=0.00245437$, $\theta_Y=3.14159$, $\phi_{P'}=5.7193$, $\phi_Y=0.92409$
 $E_Y=2.0358$, $\theta_{\ell'}=1.16963\text{e-}05$, $\phi_{\ell'}=2.5777$
 $Q^2=19.5484$, $x_B=0.835149$, $t=-3.04905$, $W^2=4.73853$, $y=0.970436$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4120$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 P $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4120$
 ℓ' $E= 2.0261$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 2.0261$
 P' $E= 0.9380$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 0.0097$
 q_Y $E= 2.0358$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.0358$

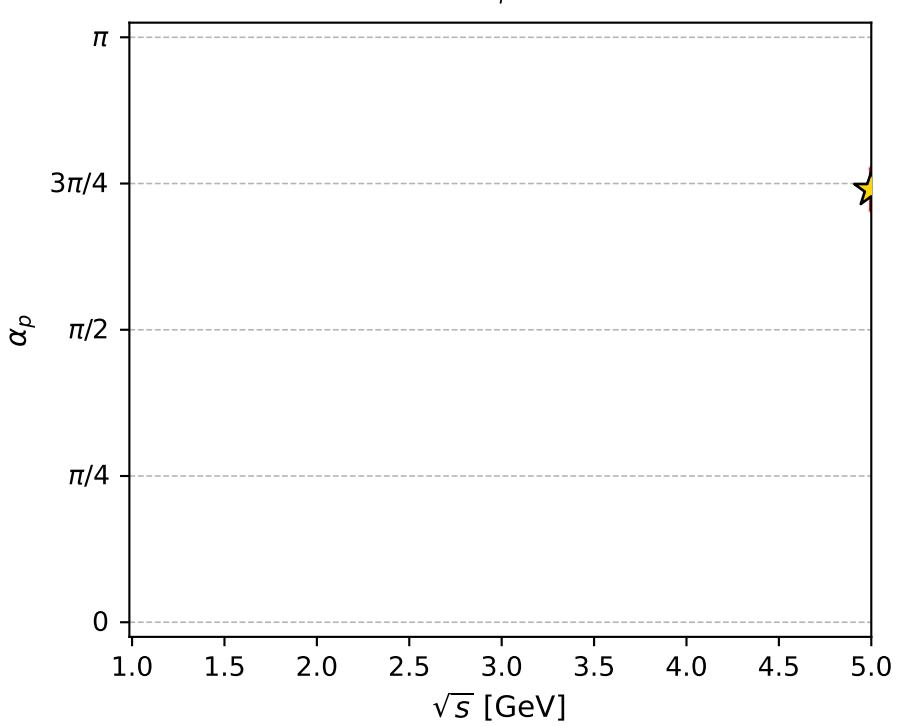
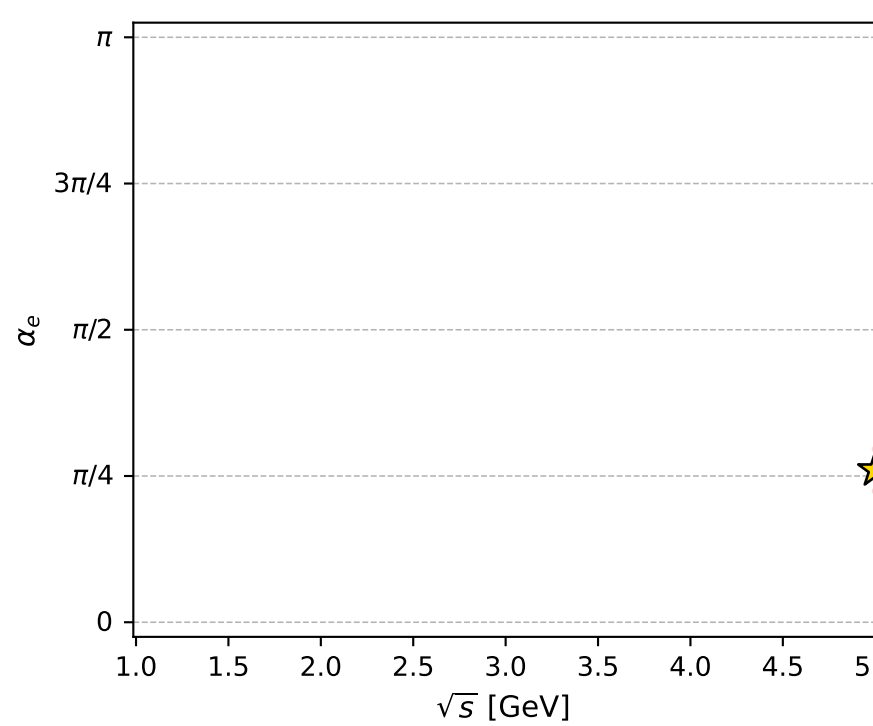
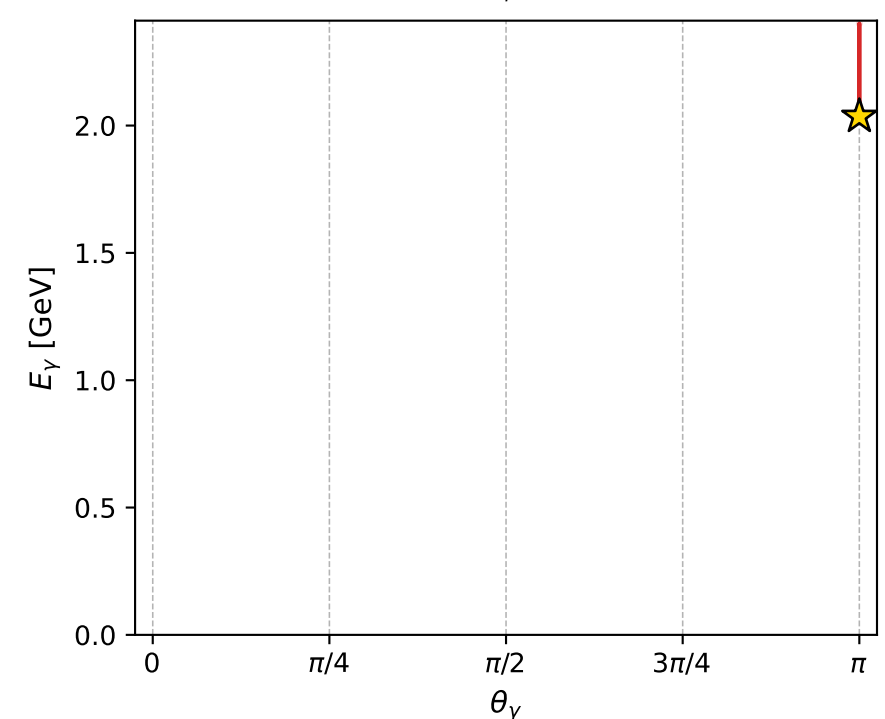
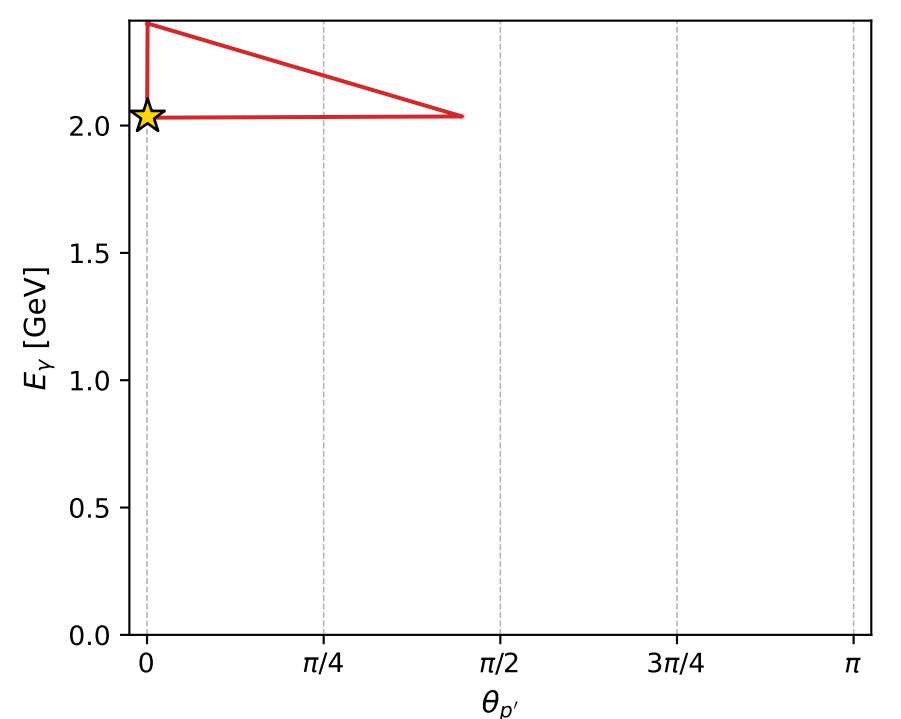
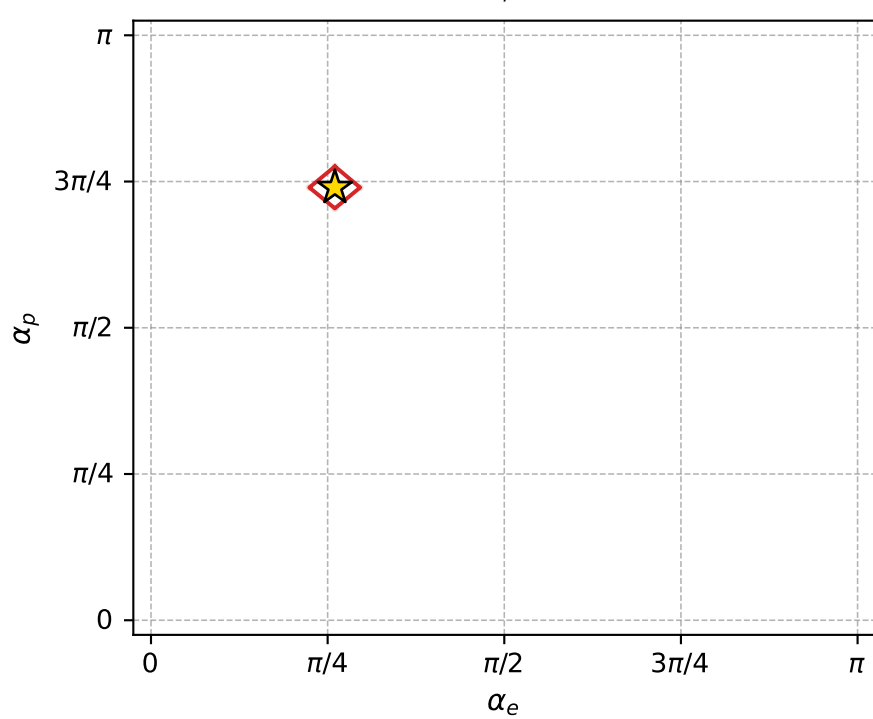
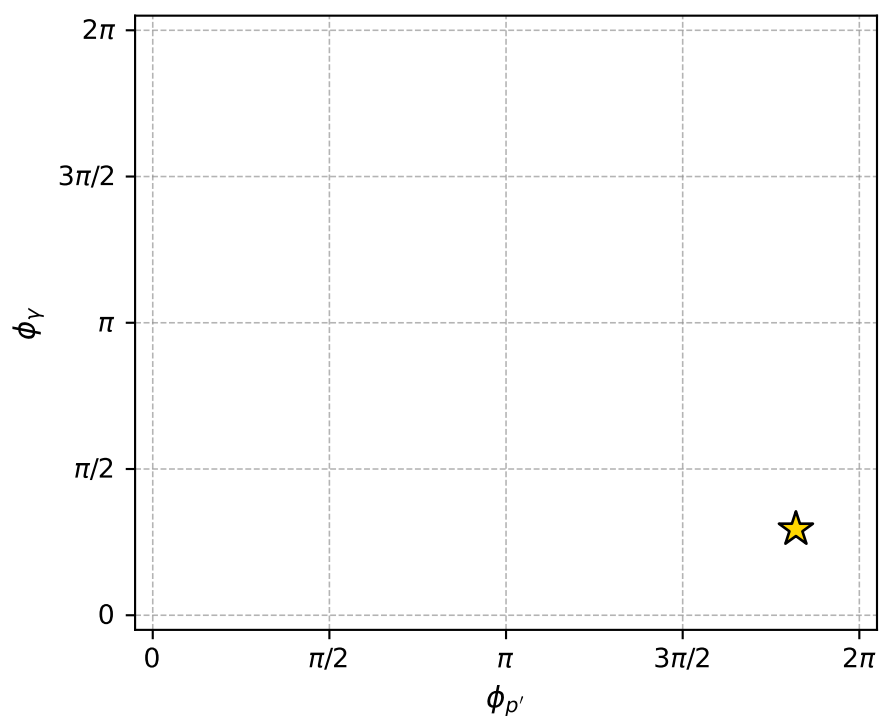
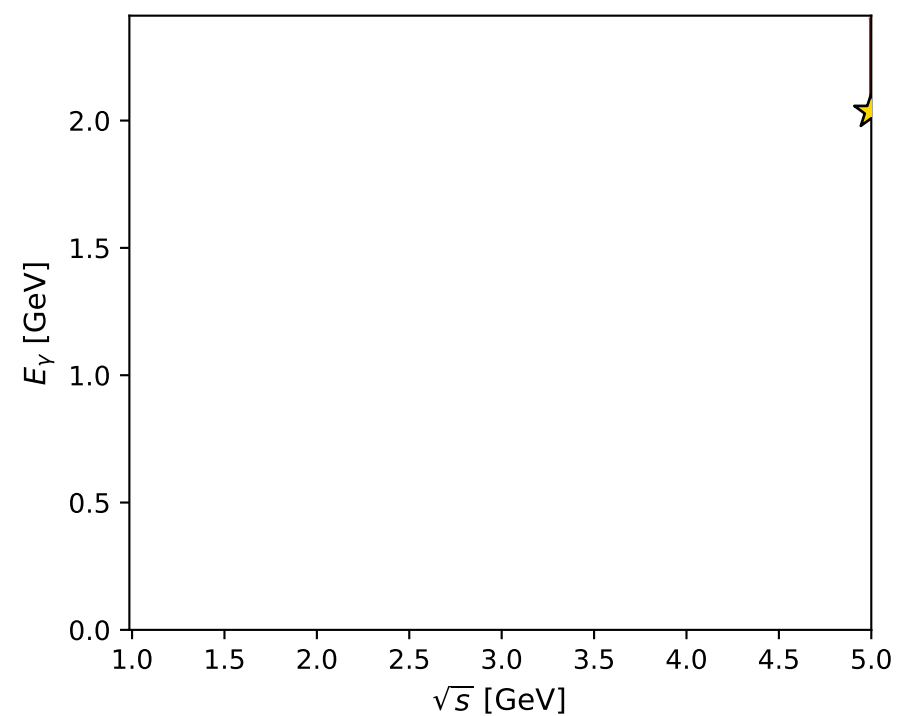
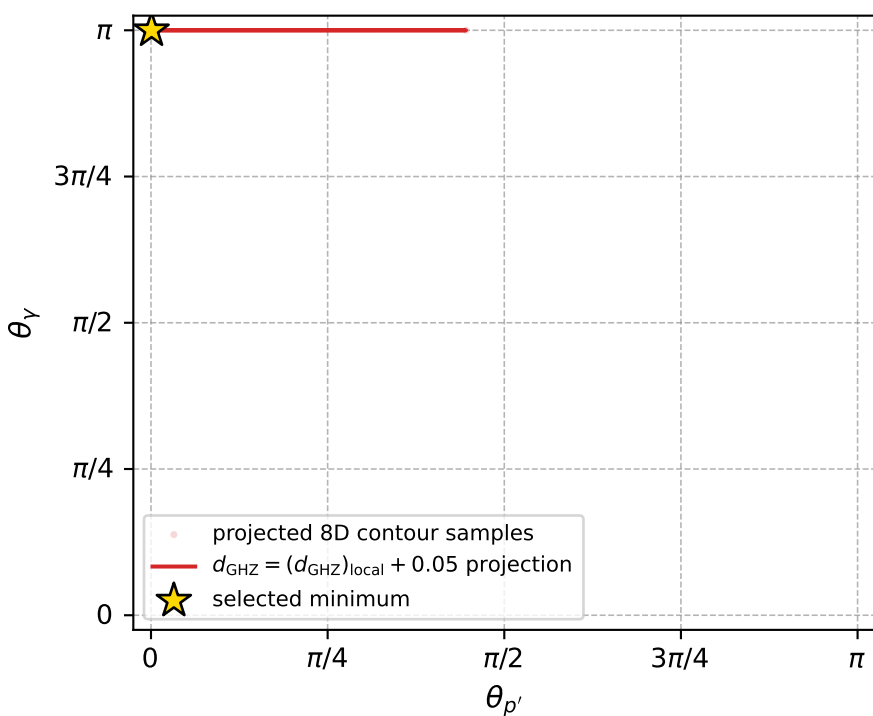


$(h_p, h_Y, h_{\ell}) = (-1, -1, +1)$

$(h_p, h_Y, h_{\ell}) = (+1, +1, -1)$



electron: polarization cluster P2, local minimum 37; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 2.6409884\text{e-}08$
 $d_{\text{GHZ}} \text{ contour} = 0.050000026$
 above global minimum = $4.1264218\text{e-}09$
 8D contour samples = 67

selected phase-space point:

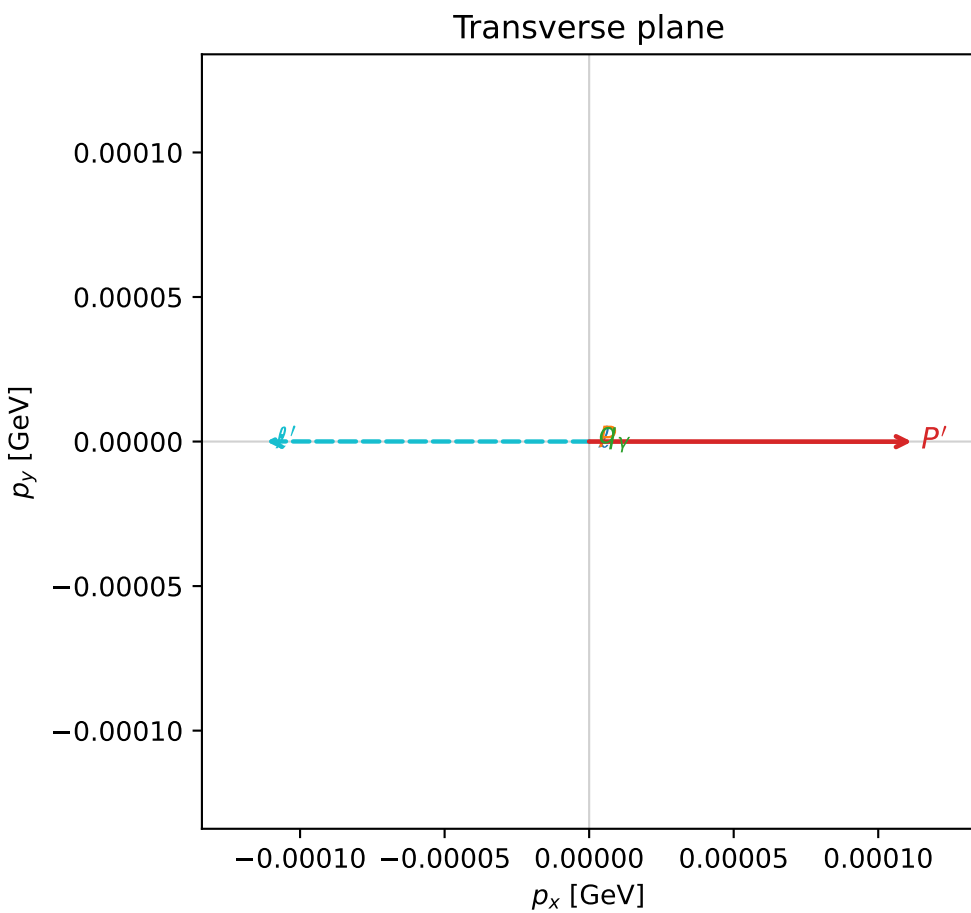
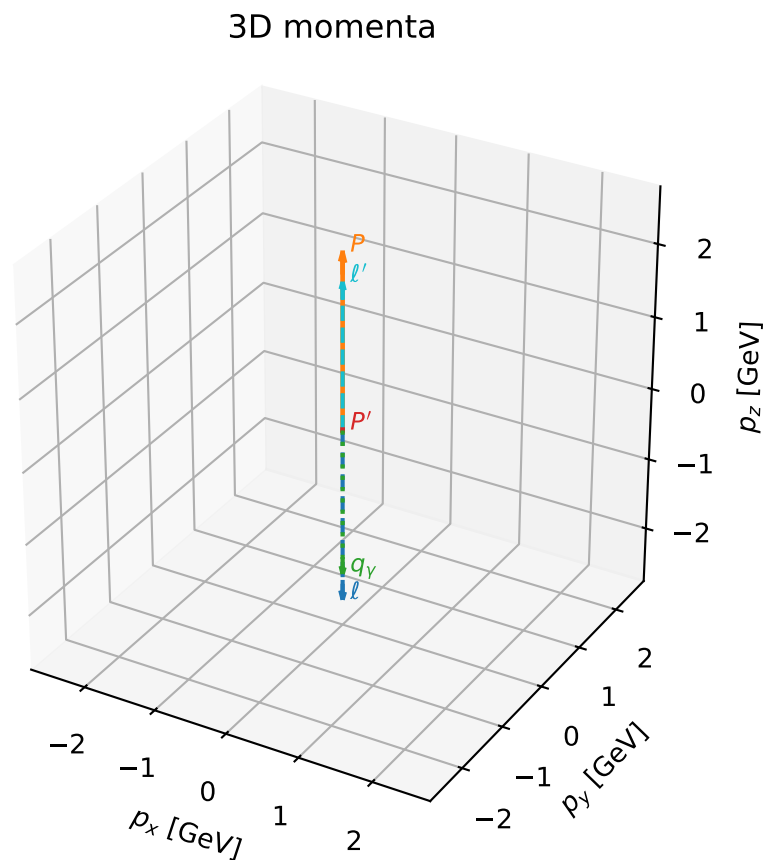
$\text{sqrt_s} = 5$
 $\text{theta_p_out} = 0.002454369$
 $\text{theta_gamma_out} = 3.141593$
 $\text{q0out} = 2.035803$
 $\text{phi_p_out} = 5.719296$
 $\text{phi_gamma_out} = 0.9240905$
 $\text{alpha_e} = 0.8176635$
 $\text{alpha_p} = 2.323928$

coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

dghz_mixing_angles_polarization_02_local_minimum_40 d_{\rm GHZ}=2.64788e-08
region: polarization_cluster_02_local_minimum_40
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: $\alpha_e=0.76726665$ rad
incoming lepton: $0.719811|+\rangle +0.69417|-\rangle$
proton mixing angle: $\alpha_p=2.3743383$ rad
incoming proton: $-0.719819|+\rangle +0.694161|-\rangle$

$s=24.9961$, $\sqrt{s}=4.99961$, $|\vec{P}|=2.41181$, $|\vec{P}'|=0.0116328$
 $\theta_{p'}=0.00959617$, $\theta_\gamma=3.14159$, $\phi_{p'}=6.28284$, $\phi_\gamma=6.2538$
 $E_\gamma=2.03658$, $\theta_{l'}=5.51265e-05$, $\phi_{l'}=3.14125$
 $Q^2=19.5352$, $x_B=0.834713$, $t=-3.03928$, $W^2=4.74815$, $y=0.970447$

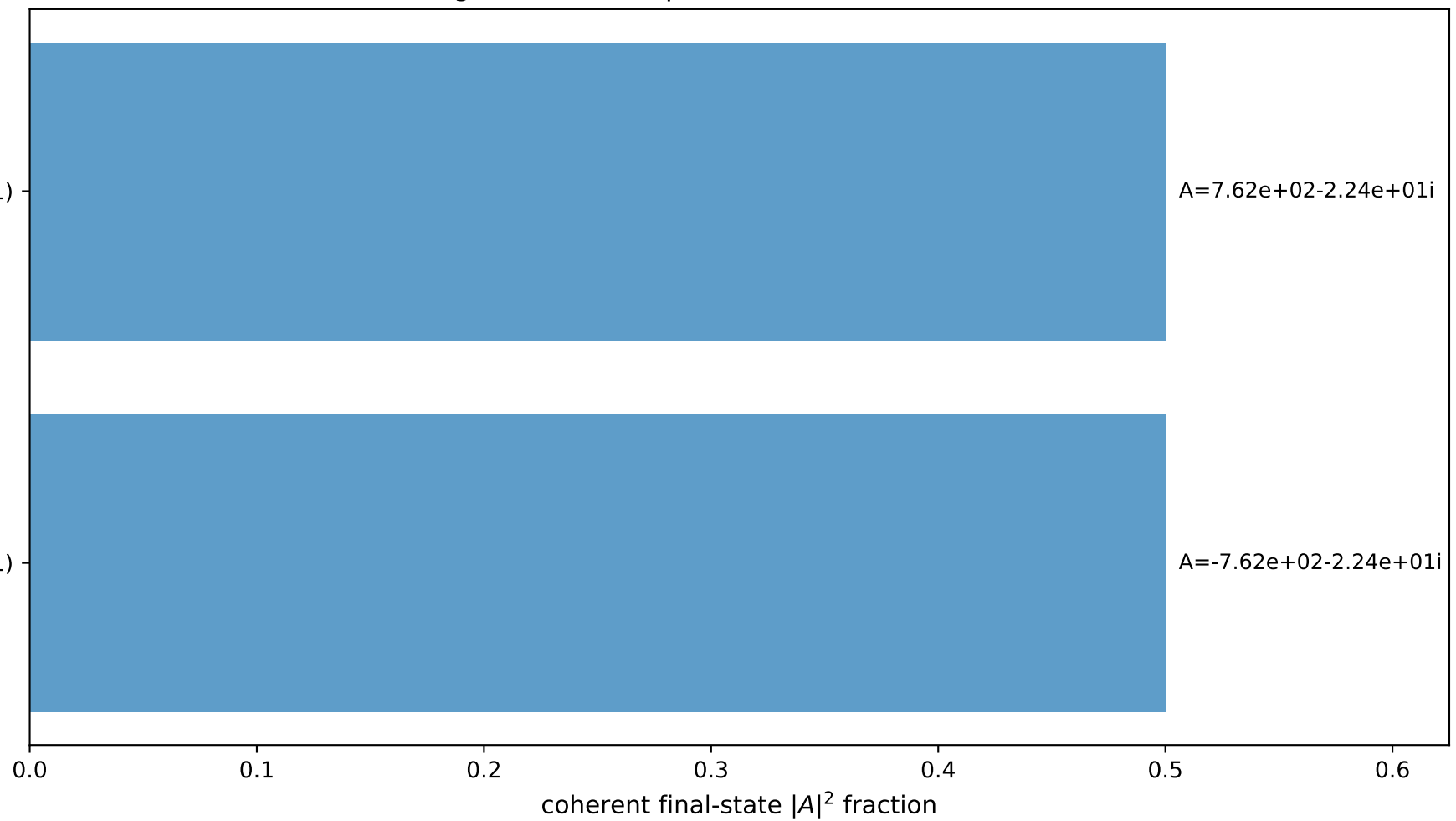
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 2.4118$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4118$
 P $E= 2.5878$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4118$
 l' $E= 2.0250$ $p_x= -0.0001$ $p_y= 0.0000$ $p_z= 2.0250$
 P' $E= 0.9381$ $p_x= 0.0001$ $p_y= -0.0000$ $p_z= 0.0116$
 q_γ $E= 2.0366$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= -2.0366$



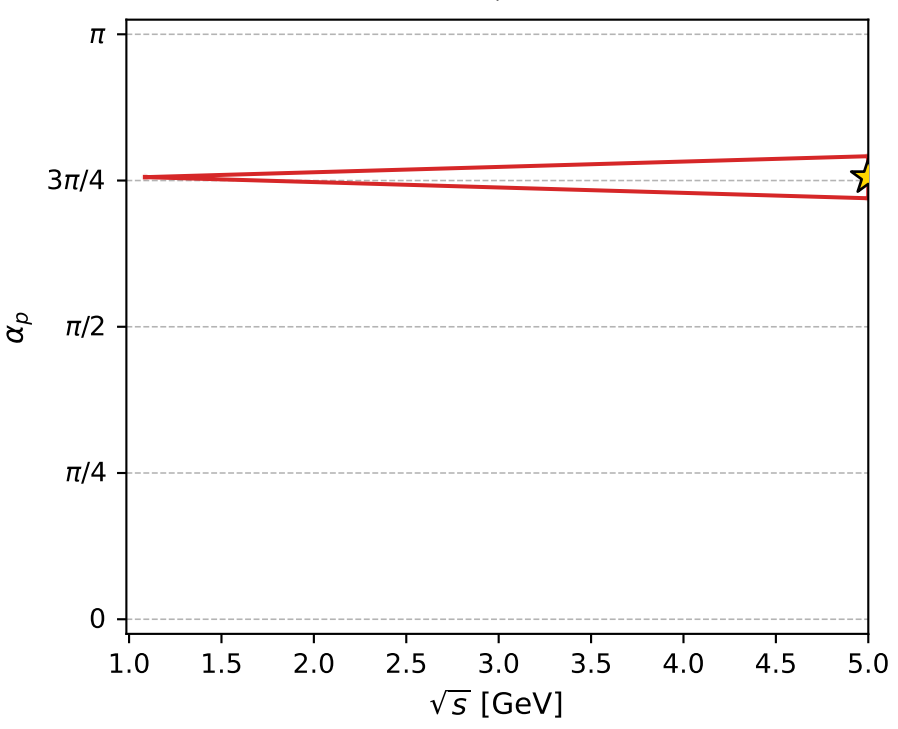
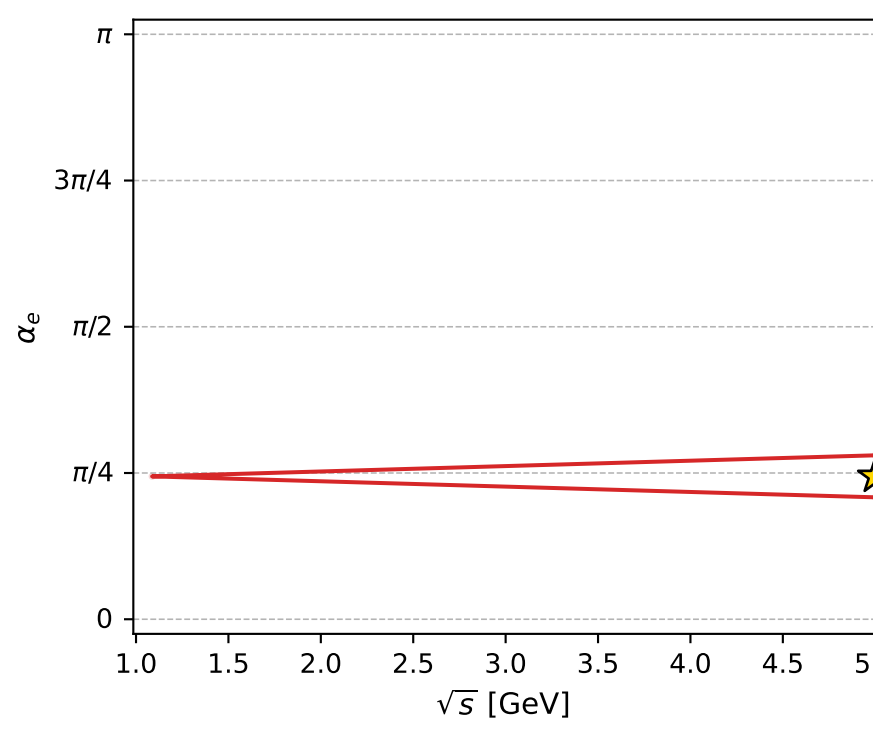
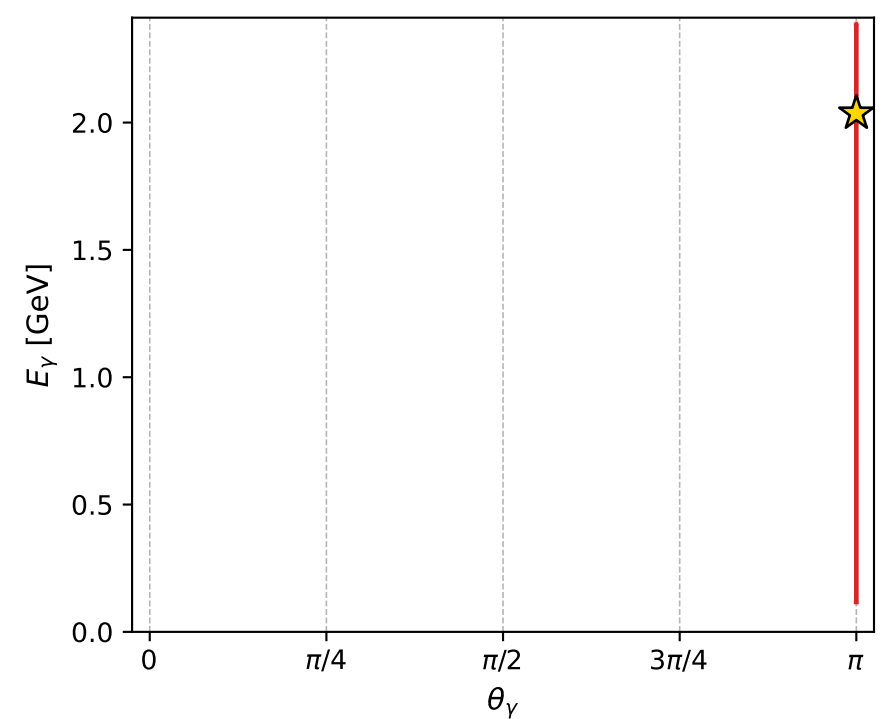
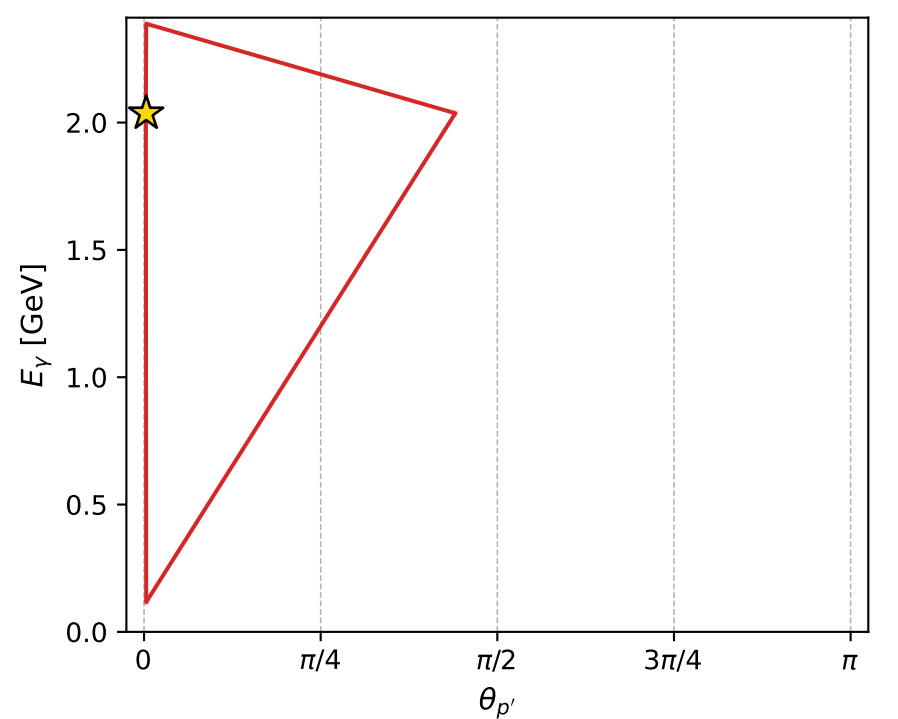
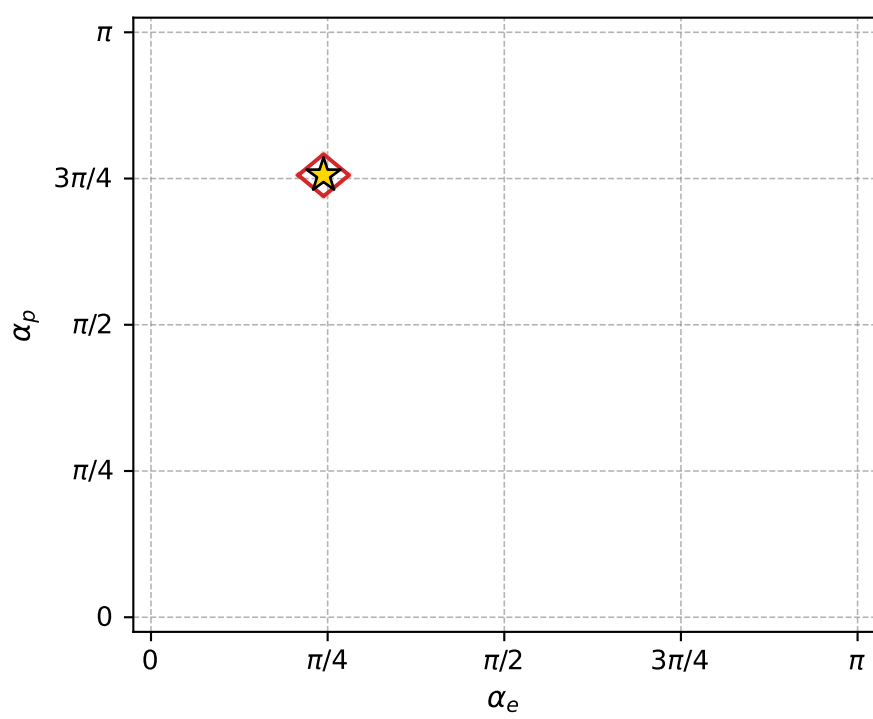
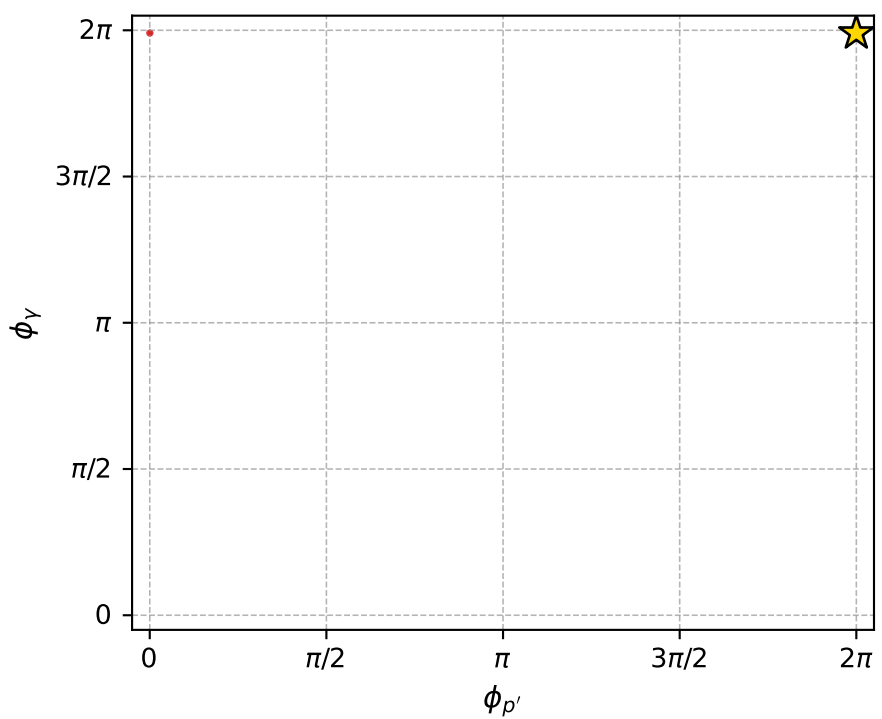
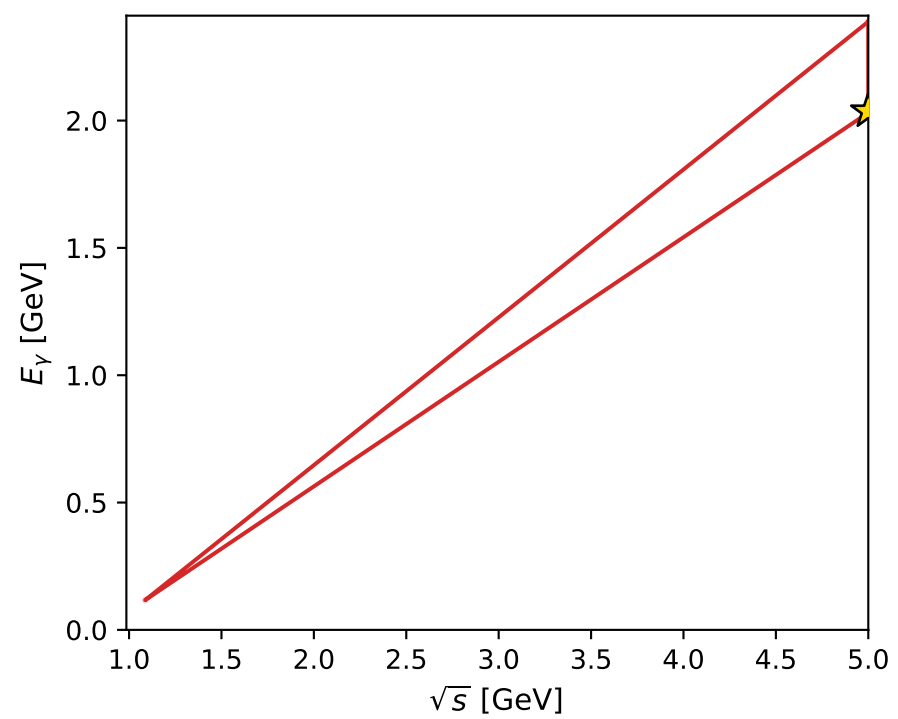
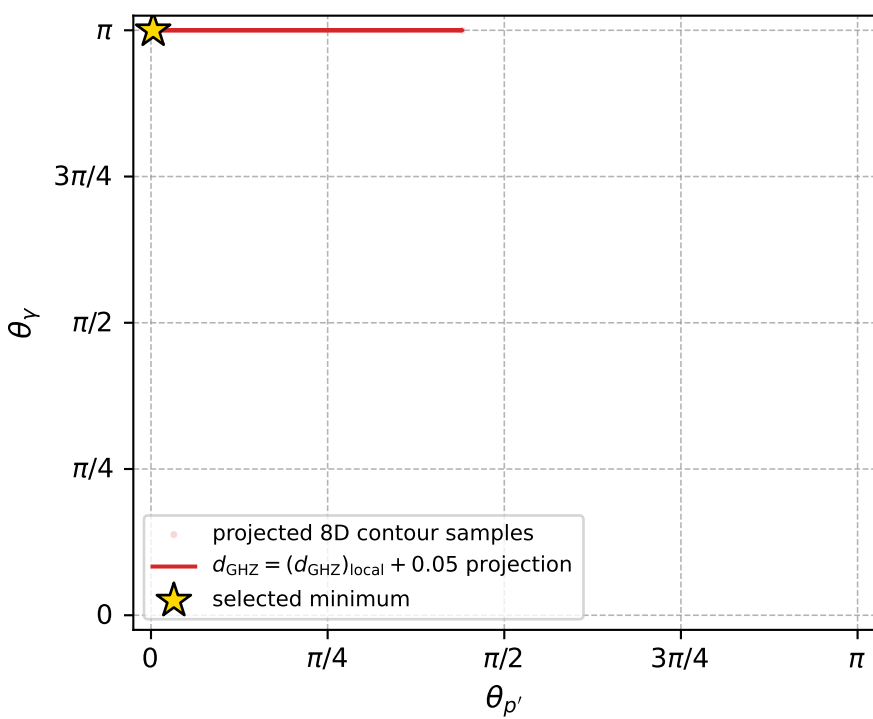
$(h_p, h_\gamma, h_l) = (-1, -1, +1)$

$(h_p, h_\gamma, h_l) = (+1, +1, -1)$

Leading final-state amplitudes (N=2, retained=100.0%)



electron: polarization cluster P2, local minimum 40; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 2.6478834\text{e-}08$
 $d_{\text{GHZ}} \text{ contour} = 0.050000026$
 above global minimum = $4.1953723\text{e-}09$
 8D contour samples = 112

selected phase-space point:

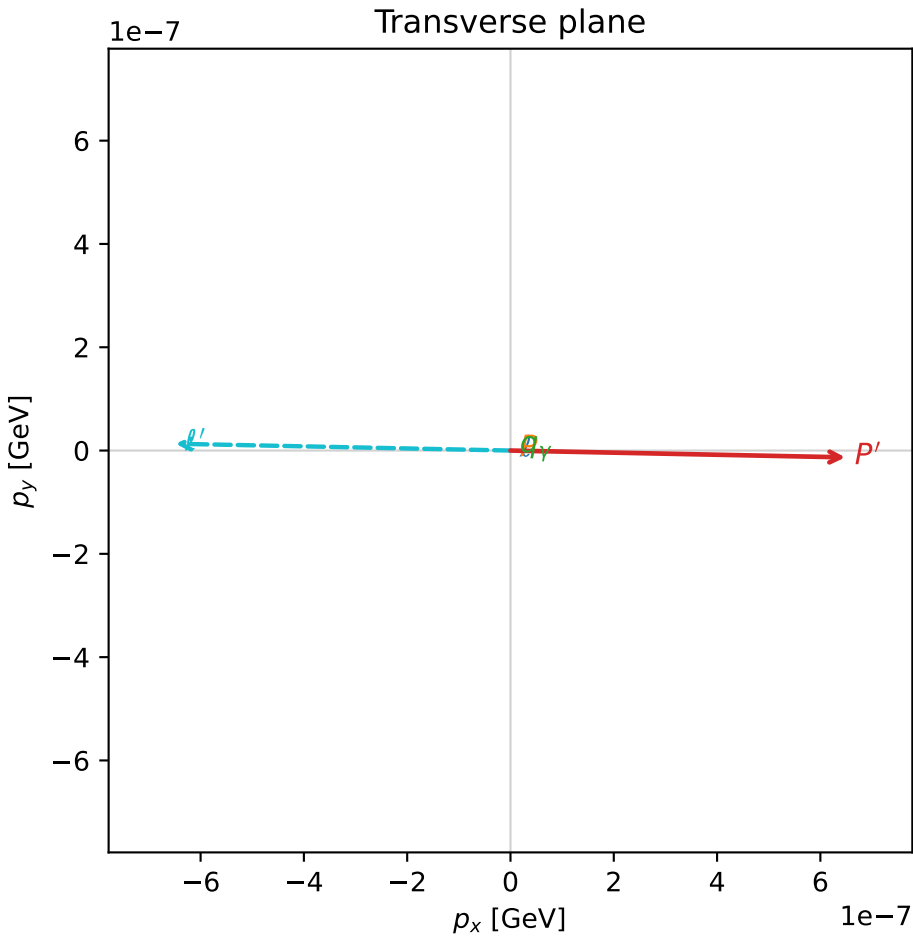
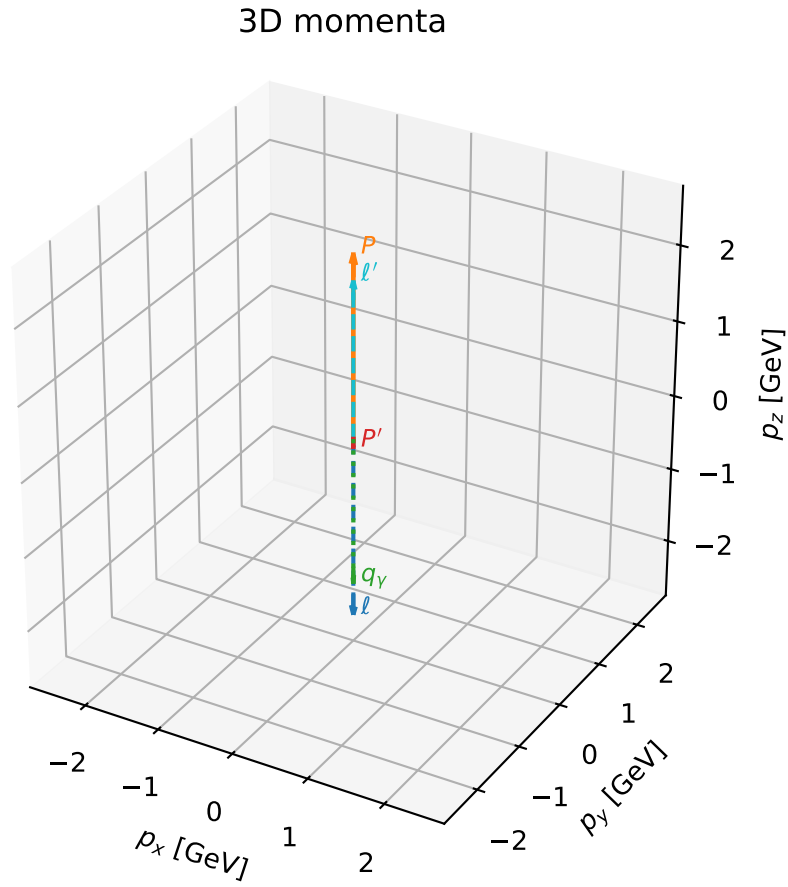
$\text{sqrt_s} = 4.999608$
 $\text{theta_p_out} = 0.00959617$
 $\text{theta_gamma_out} = 3.141593$
 $\text{qOut} = 2.036584$
 $\text{phi_p_out} = 6.282844$
 $\text{phi_gamma_out} = 6.253795$
 $\text{alpha_e} = 0.7672666$
 $\text{alpha_p} = 2.374338$

coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

dghz_mixing_angles_polarization_02_local_minimum_41 $d_{\{\text{rm GHZ}\}}=2.65018\text{e-}08$
region: polarization_cluster_02_local_minimum_41
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: $\alpha_e=0.70648598$ rad
incoming lepton: $0.760648|+\rangle +0.649165|-\rangle$
proton mixing angle: $\alpha_p=2.4350875$ rad
incoming proton: $-0.760635|+\rangle +0.649179|-\rangle$

$s=24.9206$, $\sqrt{s}=4.99205$, $|\vec{P}|=2.4079$, $|\vec{P}'|=0.101522$
 $\theta_{P'}=3.14159$, $\theta_Y=3.14159$, $\phi_{P'}=6.26287$, $\phi_Y=1.79274$
 $E_Y=1.97352$, $\theta_{\ell'}=3.12924\text{e-}07$, $\phi_{\ell'}=3.12128$
 $Q^2=19.986$, $x_B=0.857428$, $t=-3.6054$, $W^2=4.20309$, $y=0.969575$

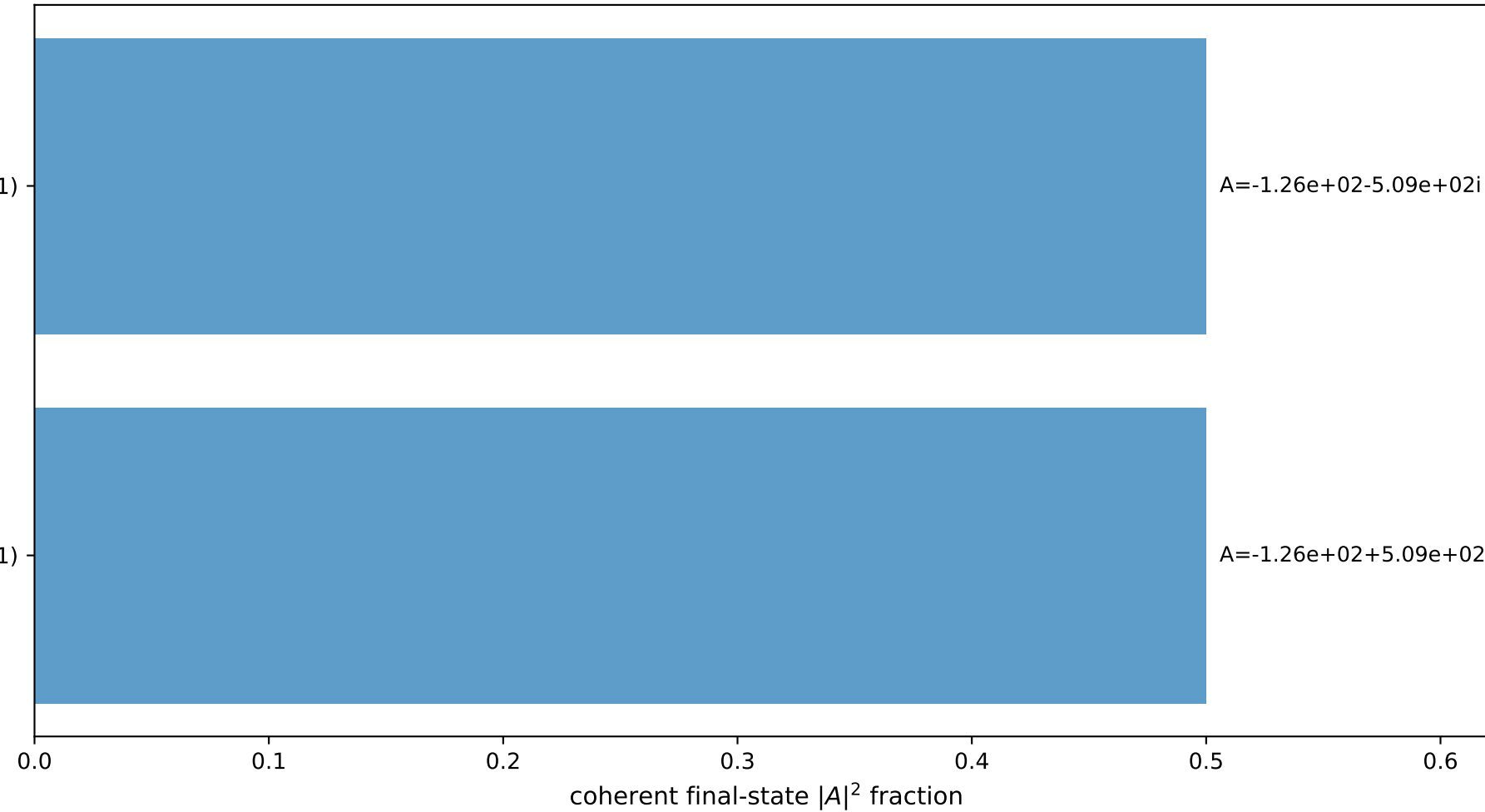
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4079$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4079$
 P $E= 2.5841$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4079$
 ℓ' $E= 2.0750$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 2.0750$
 P' $E= 0.9435$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= -0.1015$
 q_Y $E= 1.9735$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= -1.9735$



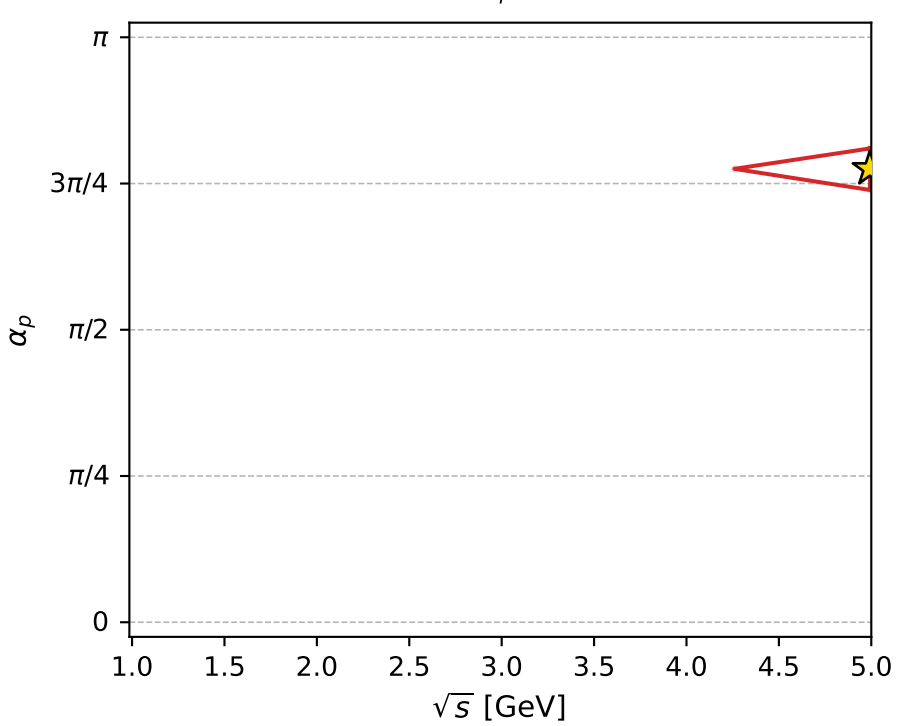
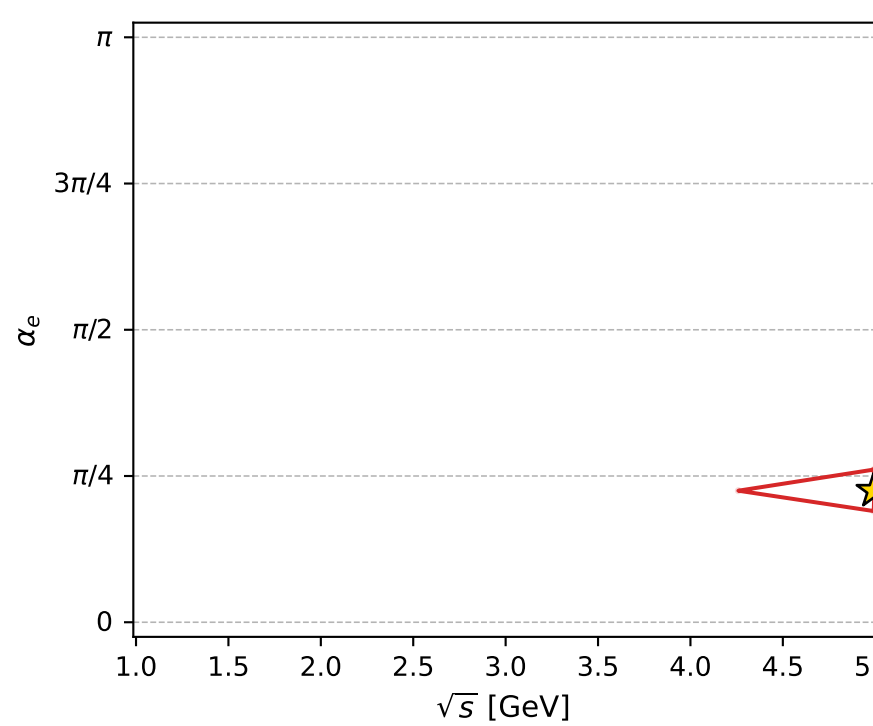
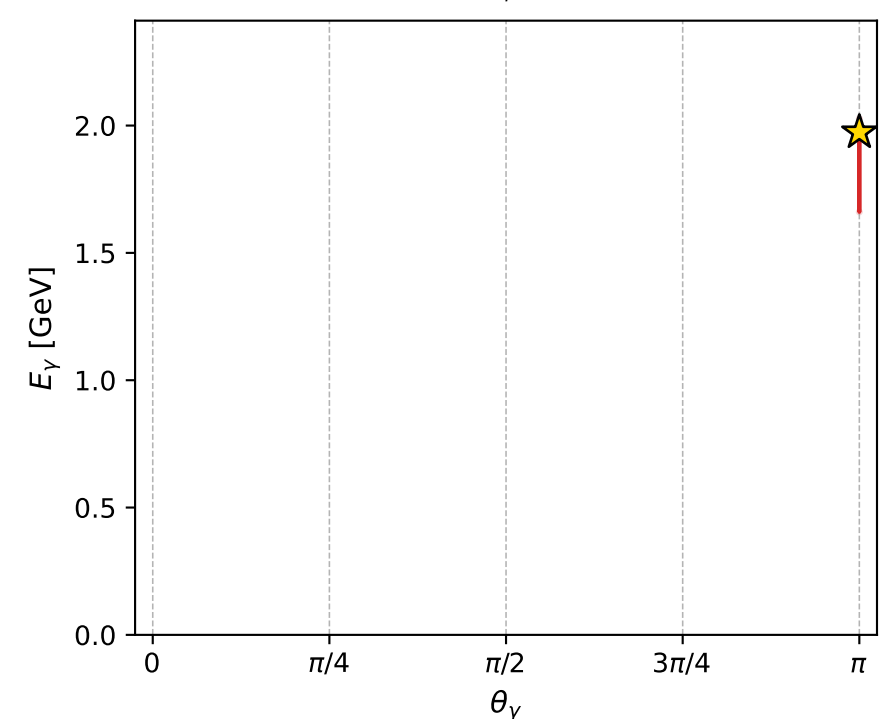
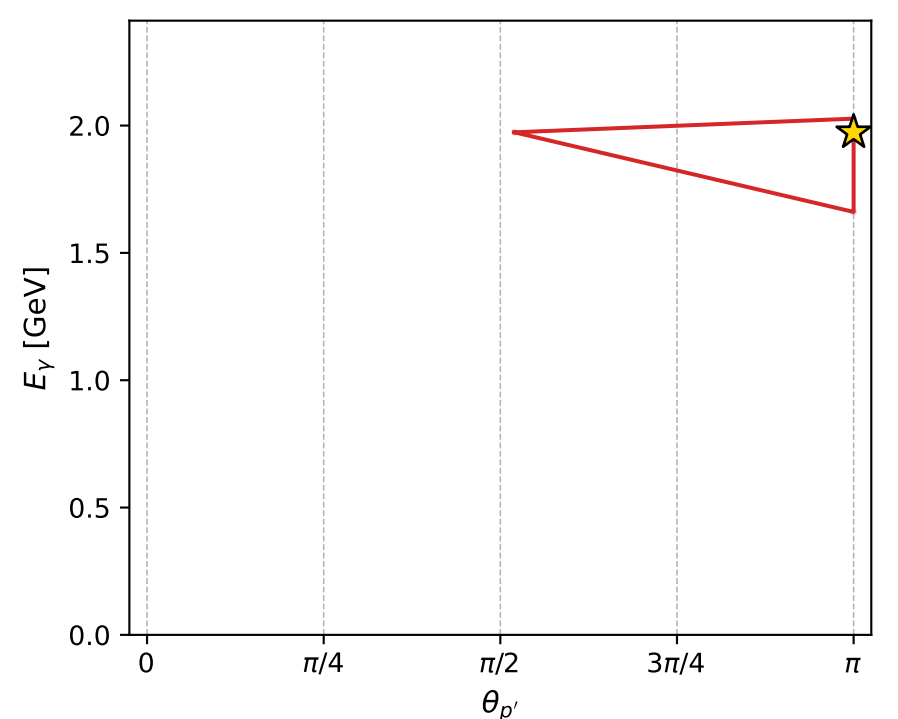
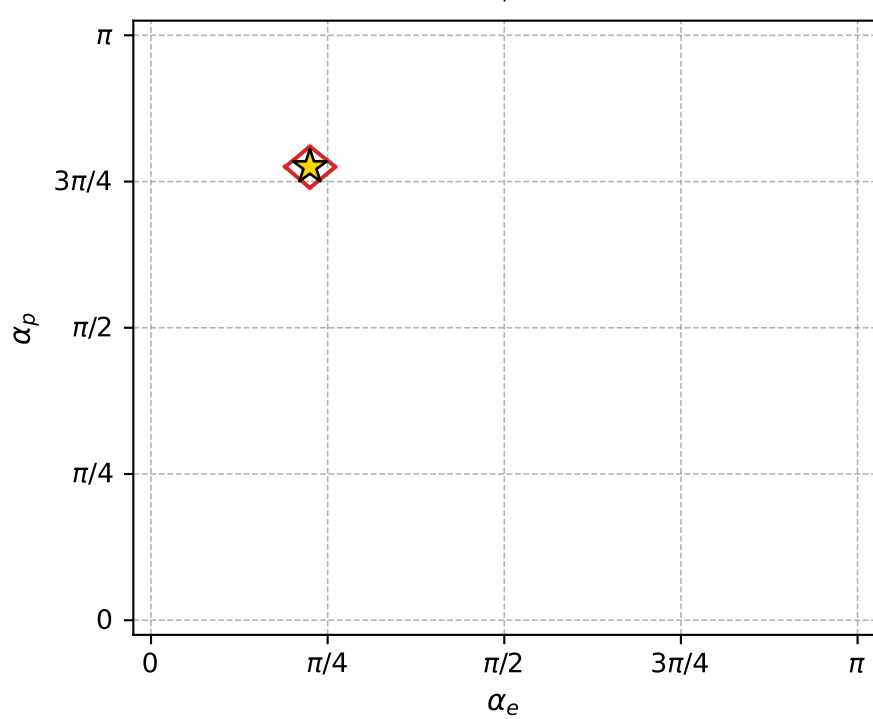
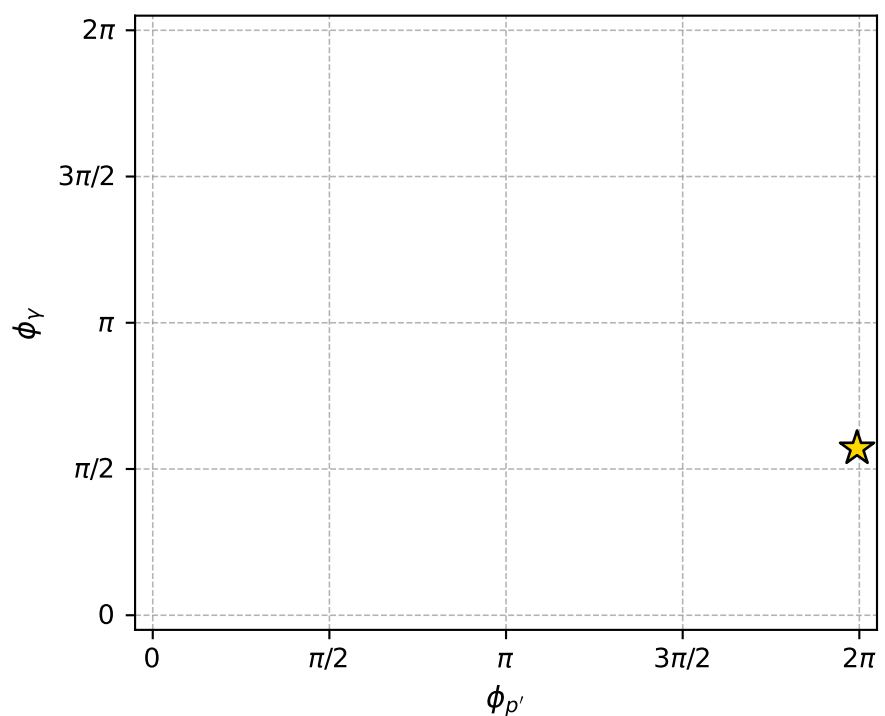
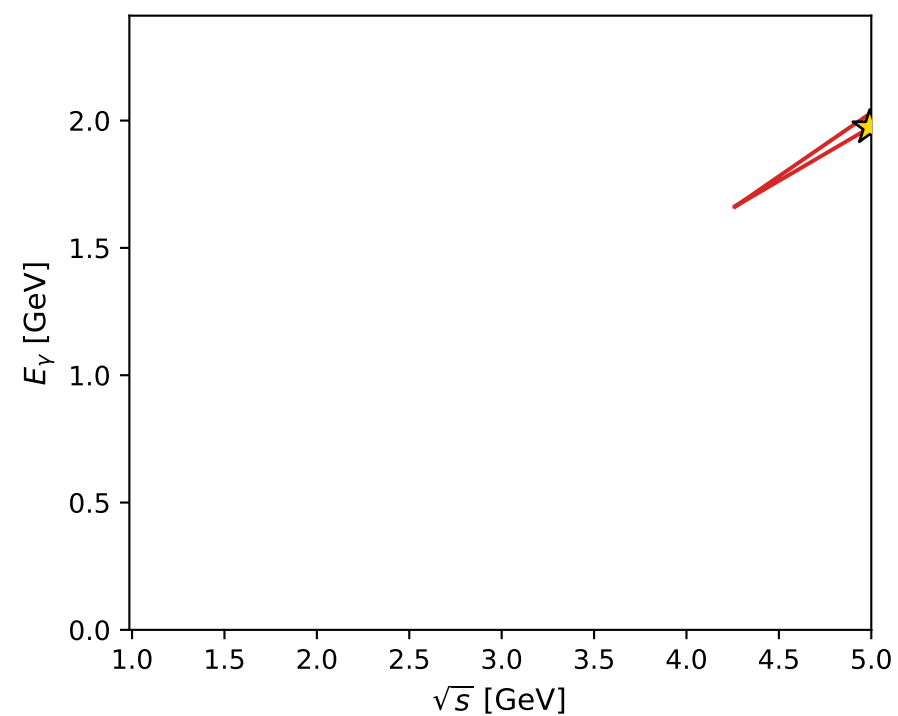
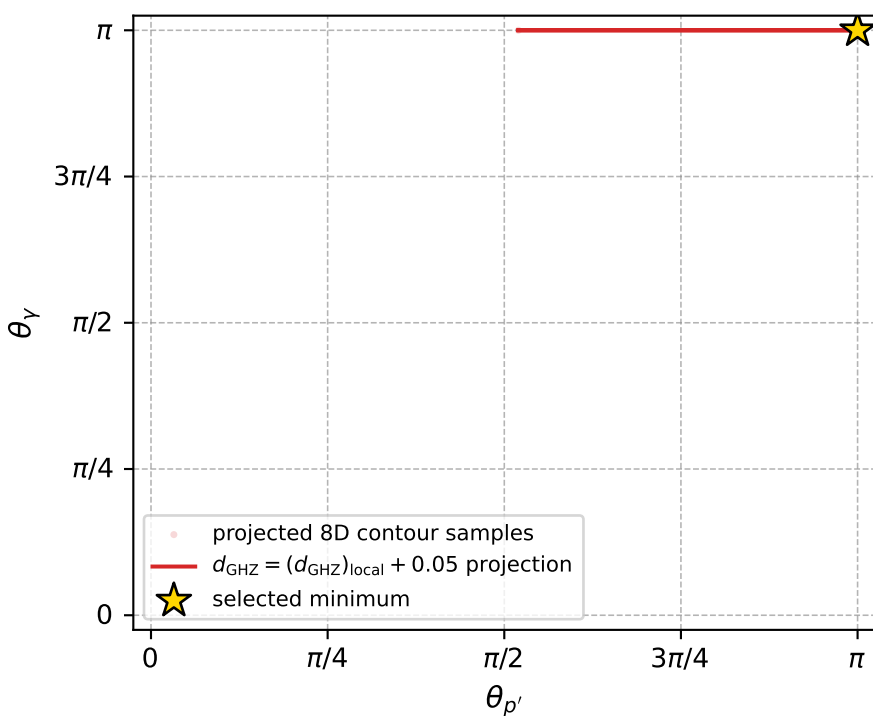
$(h_p, h_Y, h_\ell) = (-1, +1, -1)$

$(h_p, h_Y, h_\ell) = (+1, -1, +1)$

Leading final-state amplitudes (N=2, retained=100.0%)



electron: polarization cluster P2, local minimum 41; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 2.6501783\text{e-}08$
 $d_{\text{GHZ}} \text{ contour} = 0.050000027$
 above global minimum = $4.2183209\text{e-}09$
 8D contour samples = 79

selected phase-space point:

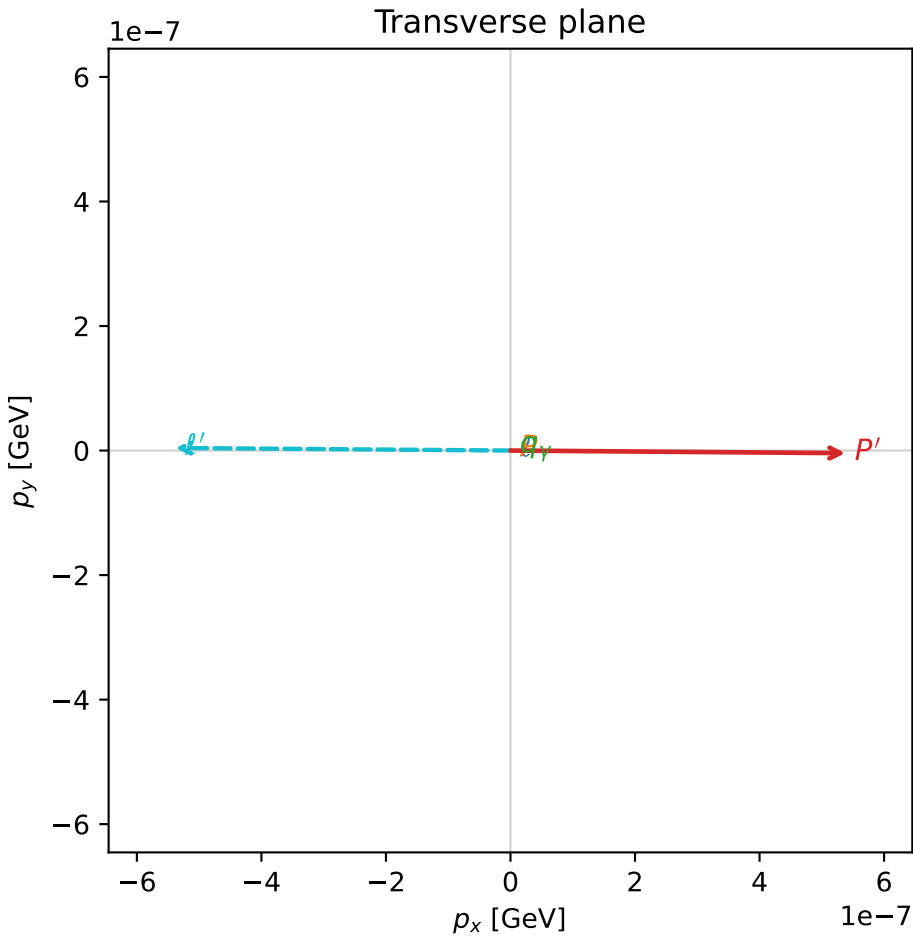
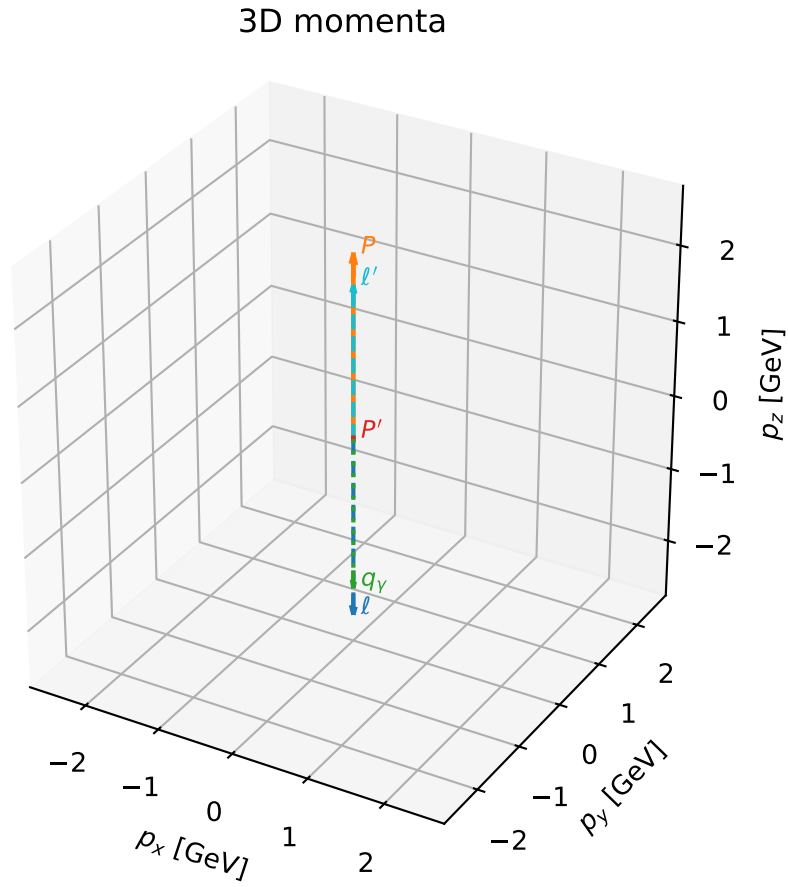
$\text{sqrt_s} = 4.99205$
 $\text{theta_p_out} = 3.141586$
 $\text{theta_gamma_out} = 3.141593$
 $\text{q0ut} = 1.973525$
 $\text{phi_p_out} = 6.262871$
 $\text{phi_gamma_out} = 1.792742$
 $\alpha_e = 0.706486$
 $\alpha_p = 2.435087$

coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

dghz_mixing_angles_polarization_02_local_minimum_42 $d_{\{\text{rm GHZ}\}}=2.65173\text{e-}08$
region: polarization_cluster_02_local_minimum_42
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: $\alpha_e=0.7472291$ rad
incoming lepton: $0.733575|+\rangle +0.679609|-\rangle$
proton mixing angle: $\alpha_p=2.394356$ rad
incoming proton: $-0.73357|+\rangle +0.679614|-\rangle$

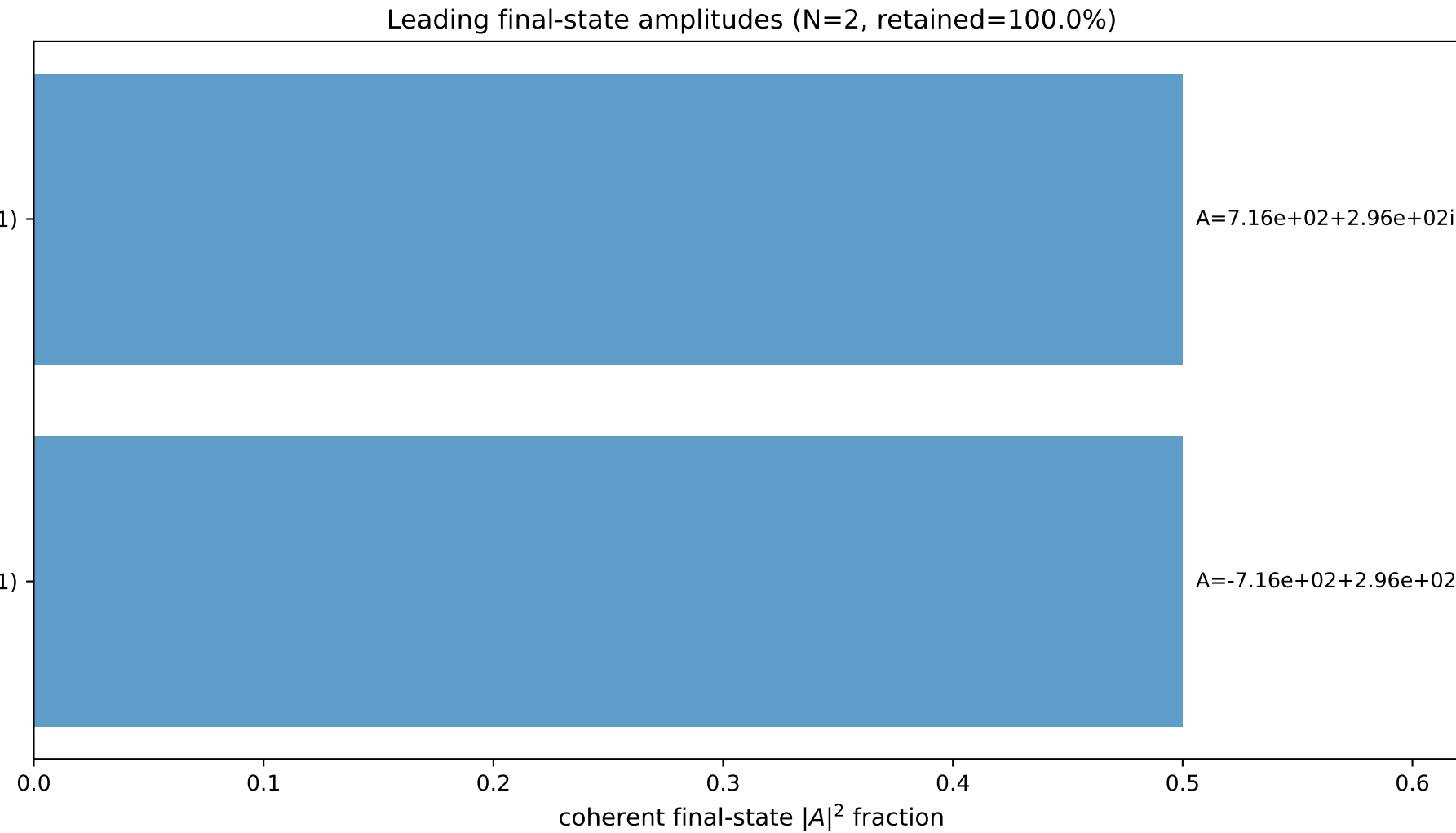
$s=24.9854$, $\sqrt{s}=4.99854$, $|\vec{P}|=2.41126$, $|\vec{P}'|=0.0171201$
 $\theta_{P'}=3.14159\text{e-}05$, $\theta_Y=3.14159$, $\phi_{P'}=6.27545$, $\phi_Y=2.74997$
 $E_Y=2.03875$, $\theta_{\ell'}=2.66143\text{e-}07$, $\phi_{\ell'}=3.13386$
 $Q^2=19.4987$, $x_B=0.833497$, $t=-3.01229$, $W^2=4.77498$, $y=0.970476$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4113$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4113$
 P $E= 2.5873$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4113$
 ℓ' $E= 2.0216$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 2.0216$
 P' $E= 0.9382$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 0.0171$
 q_Y $E= 2.0388$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= -2.0388$



$(h_p, h_Y, h_\ell) = (+1, +1, -1)$

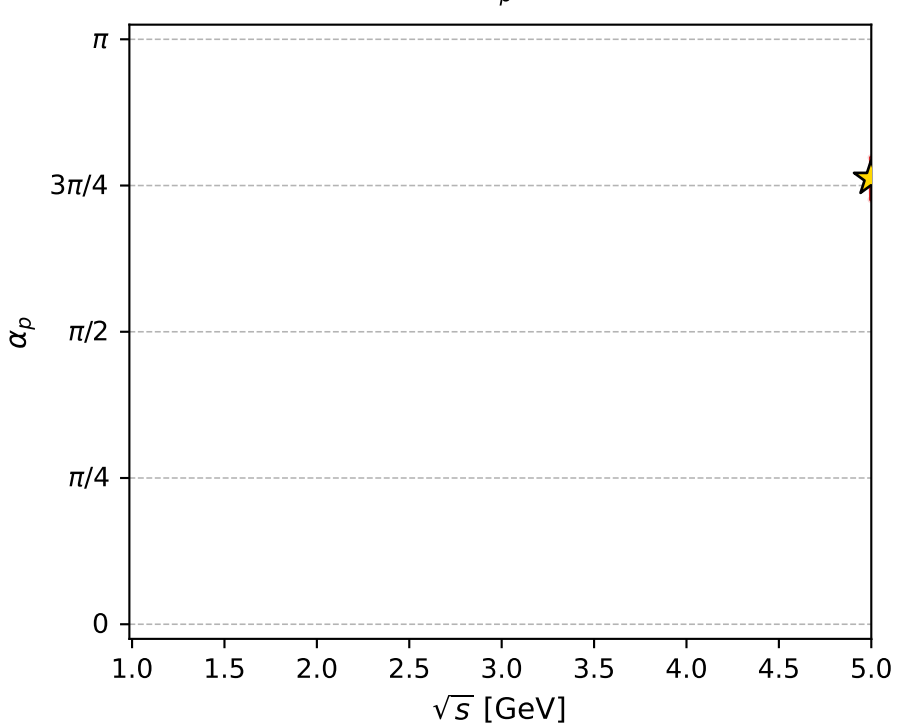
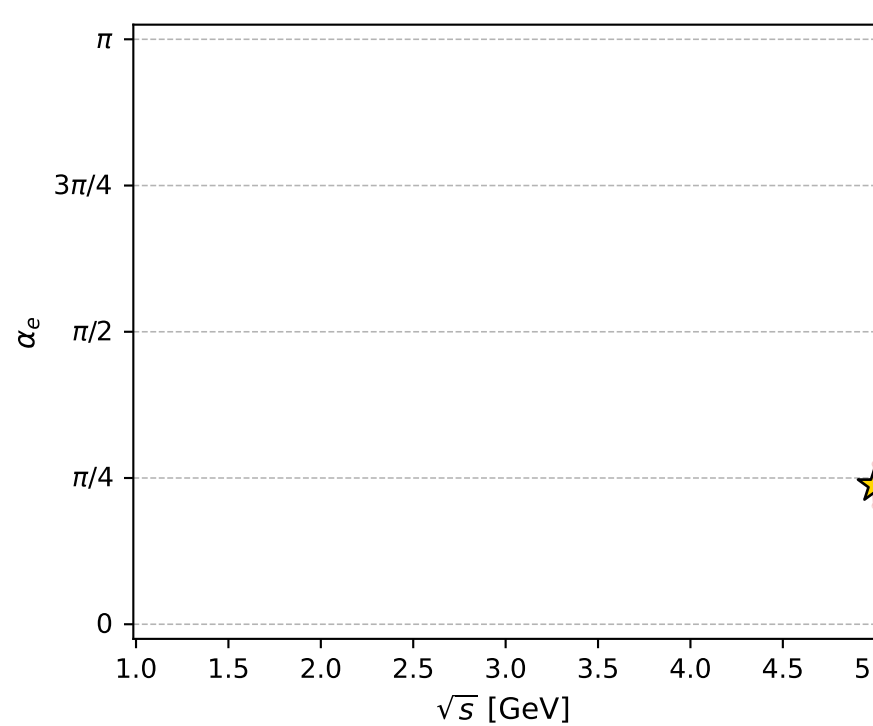
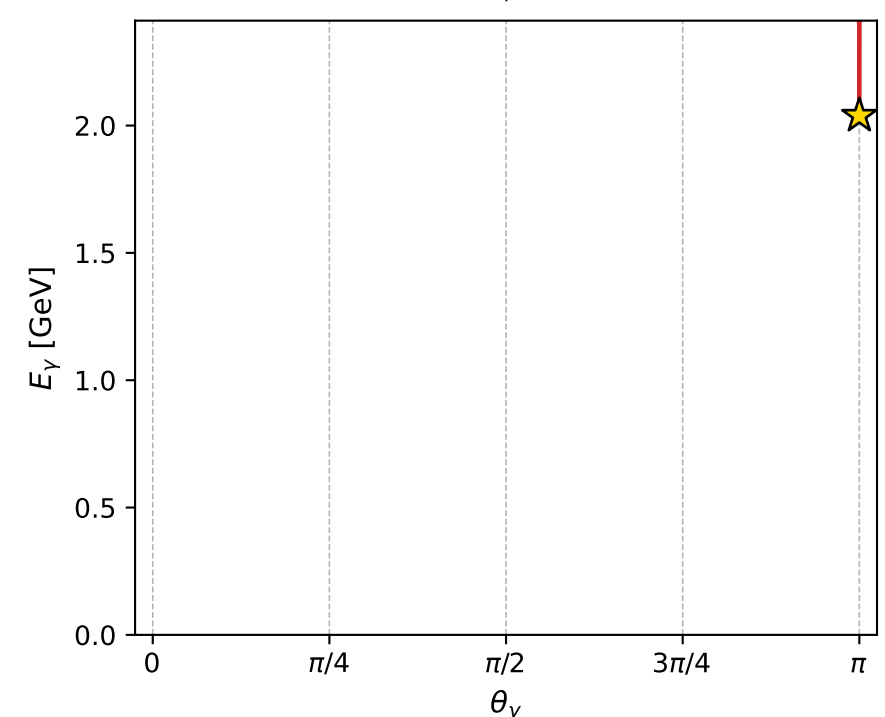
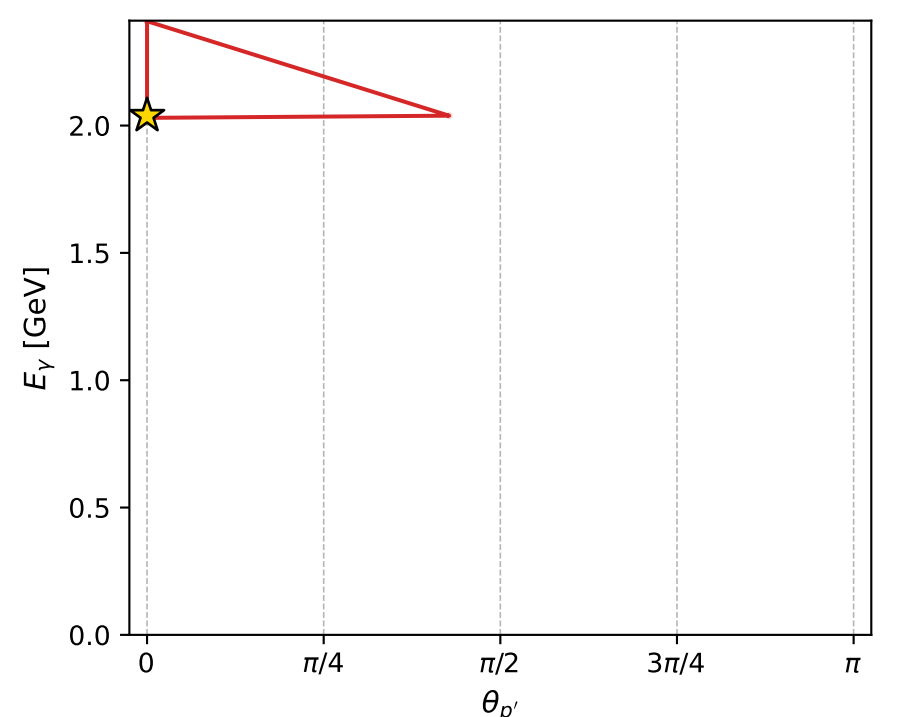
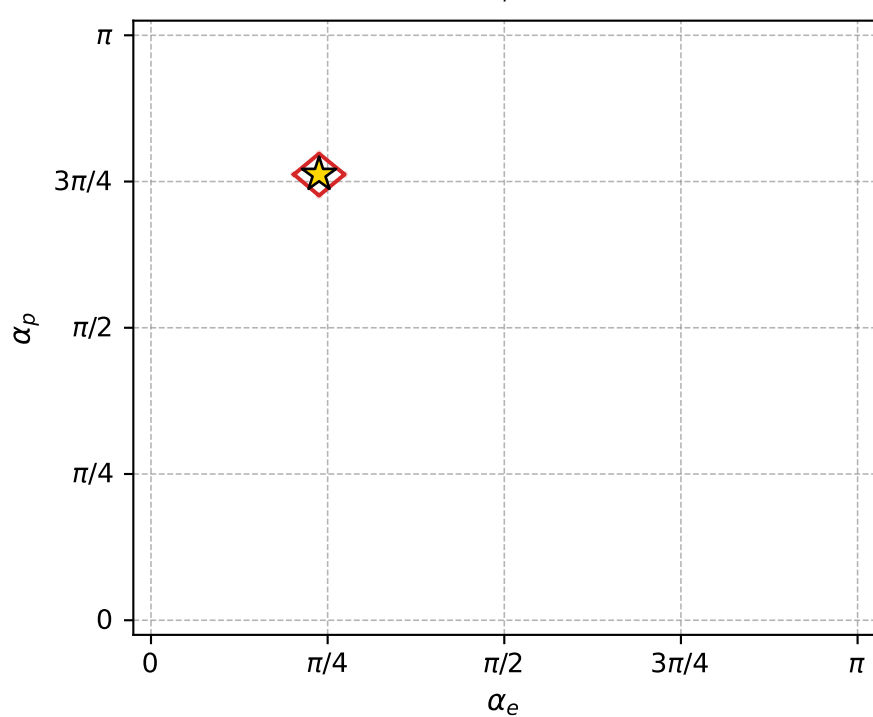
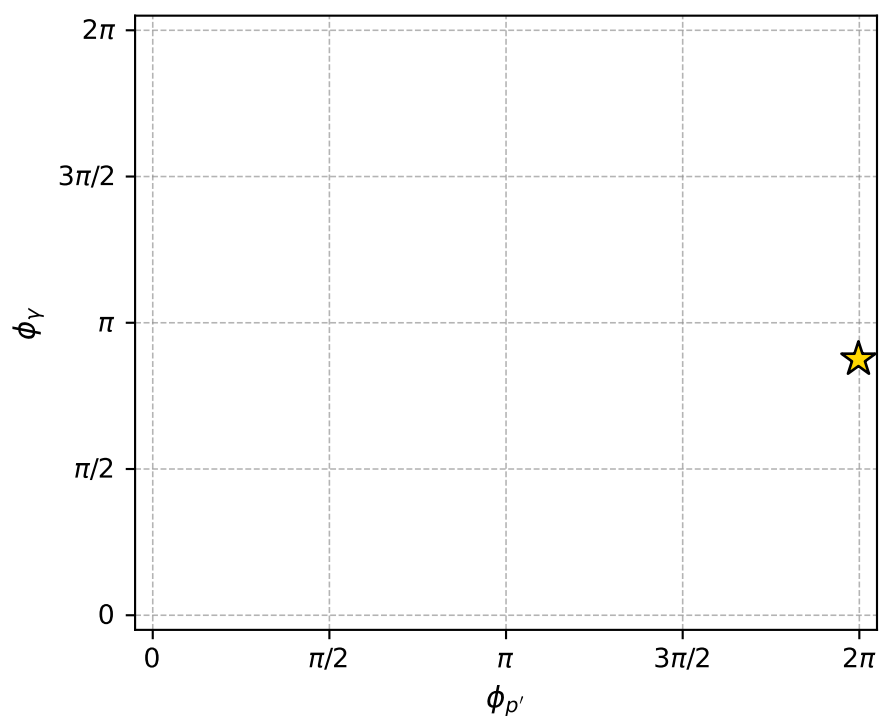
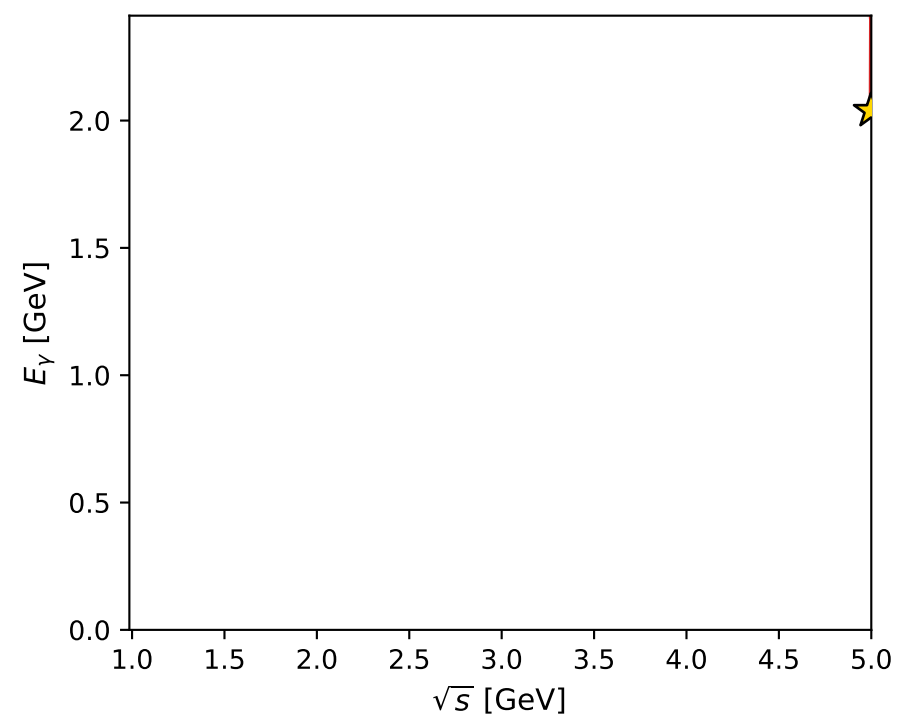
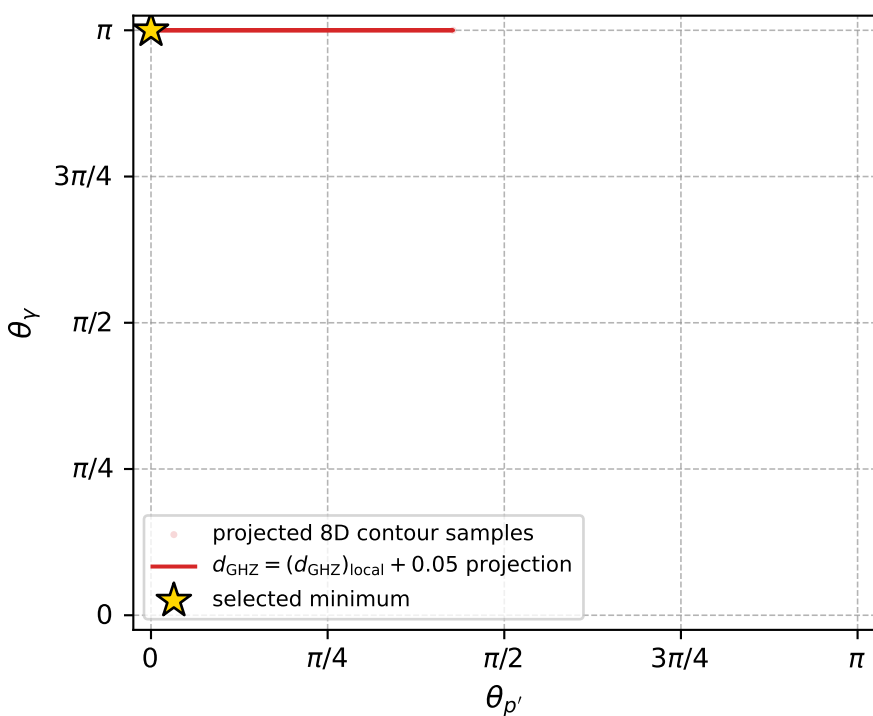
$(h_p, h_Y, h_\ell) = (-1, -1, +1)$



$A=7.16\text{e+}02+2.96\text{e+}02i$

$A=-7.16\text{e+}02+2.96\text{e+}02i$

electron: polarization cluster P2, local minimum 42; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 2.6517304\text{e-}08$
 $d_{\text{GHZ}} \text{ contour} = 0.050000027$
 above global minimum = $4.2338426\text{e-}09$
 8D contour samples = 65

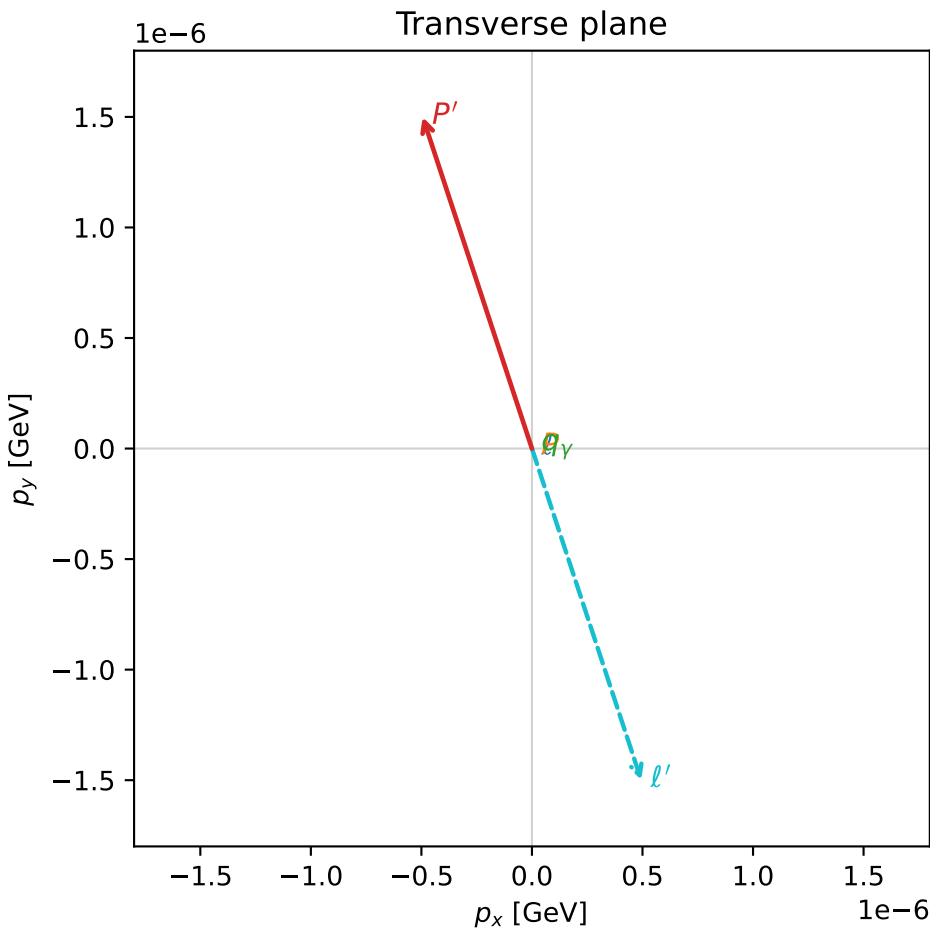
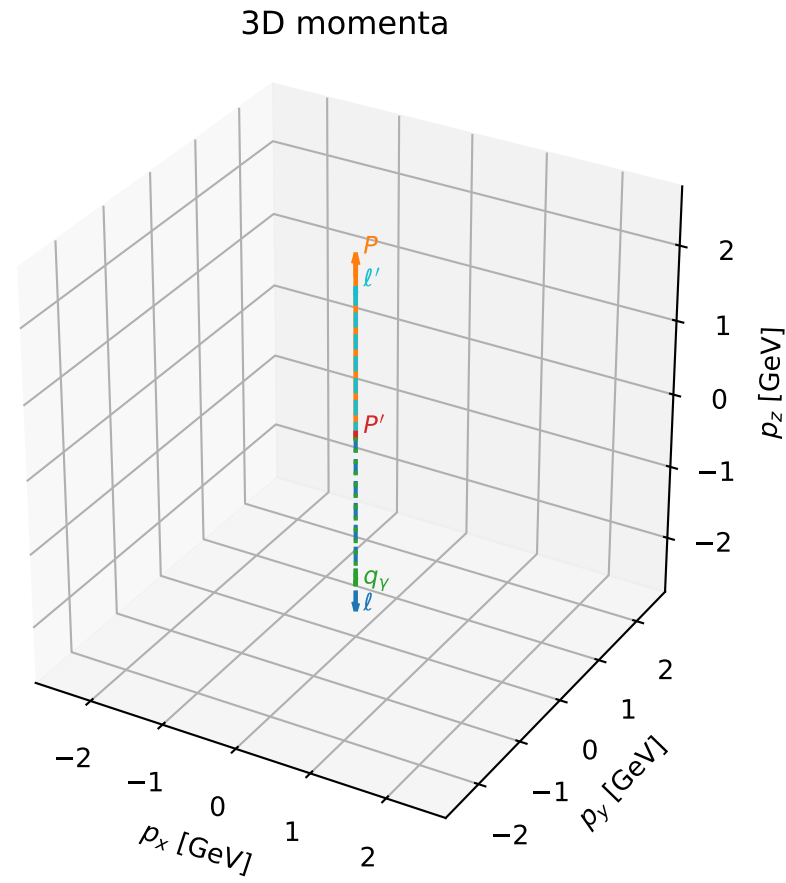
selected phase-space point:
 $\text{sqrt_s} = 4.998537$
 $\text{theta_p_out} = 3.141593\text{e-}05$
 $\text{theta_gamma_out} = 3.141593$
 $\text{q0out} = 2.038751$
 $\text{phi_p_out} = 6.275449$
 $\text{phi_gamma_out} = 2.749972$
 $\text{alpha_e} = 0.7472291$
 $\text{alpha_p} = 2.394356$

coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

dghz_mixing_angles_polarization_02_local_minimum_49 d_{\rm GHZ}=2.67844e-08
region: polarization_cluster_02_local_minimum_49
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: $\alpha_e=0.81607917$ rad
incoming lepton: $0.685083|+\rangle +0.728465|-\rangle$
proton mixing angle: $\alpha_p=2.3255025$ rad
incoming proton: $-0.685075|+\rangle +0.728473|-\rangle$

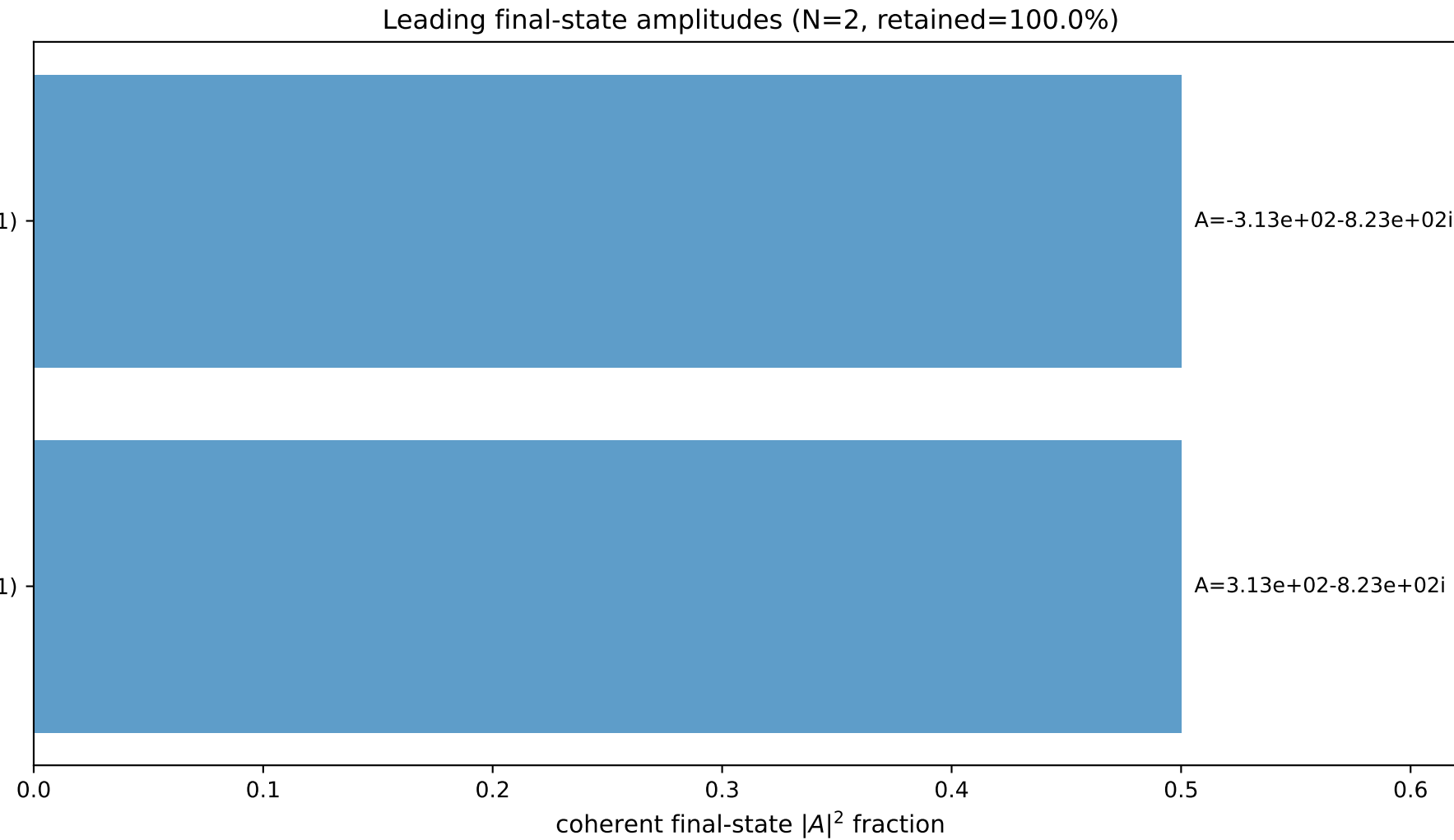
$s=24.9993$, $\sqrt{s}=4.99993$, $|\vec{P}|=2.41198$, $|\vec{P}'|=0.0563059$
 $\theta_{P'}=2.80434e-05$, $\theta_Y=3.14159$, $\phi_{P'}=1.88915$, $\phi_Y=5.07566$
 $E_Y=2.05828$, $\theta_{\ell'}=7.88636e-07$, $\phi_{\ell'}=5.03075$
 $Q^2=19.3148$, $x_B=0.824895$, $t=-2.83243$, $W^2=4.9799$, $y=0.970788$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4120$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 P $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4120$
 ℓ' $E= 2.0020$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 2.0020$
 P' $E= 0.9397$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 0.0563$
 q_Y $E= 2.0583$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= -2.0583$

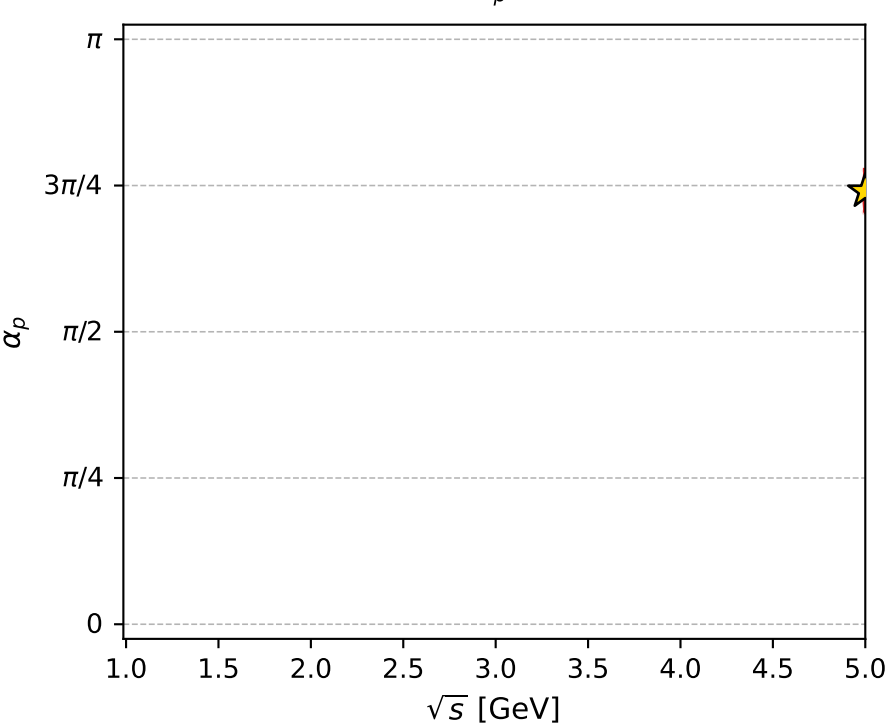
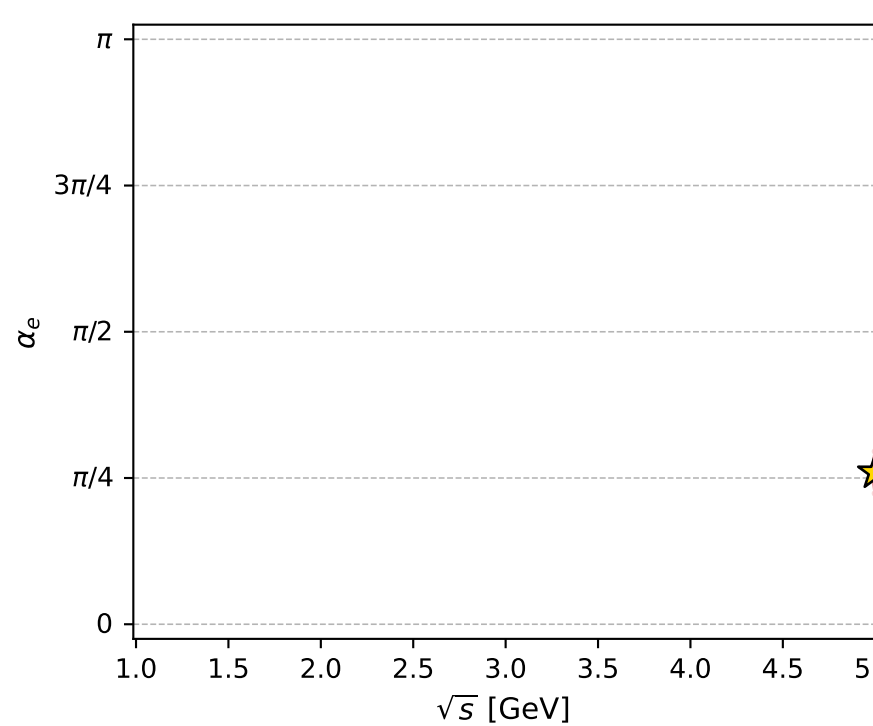
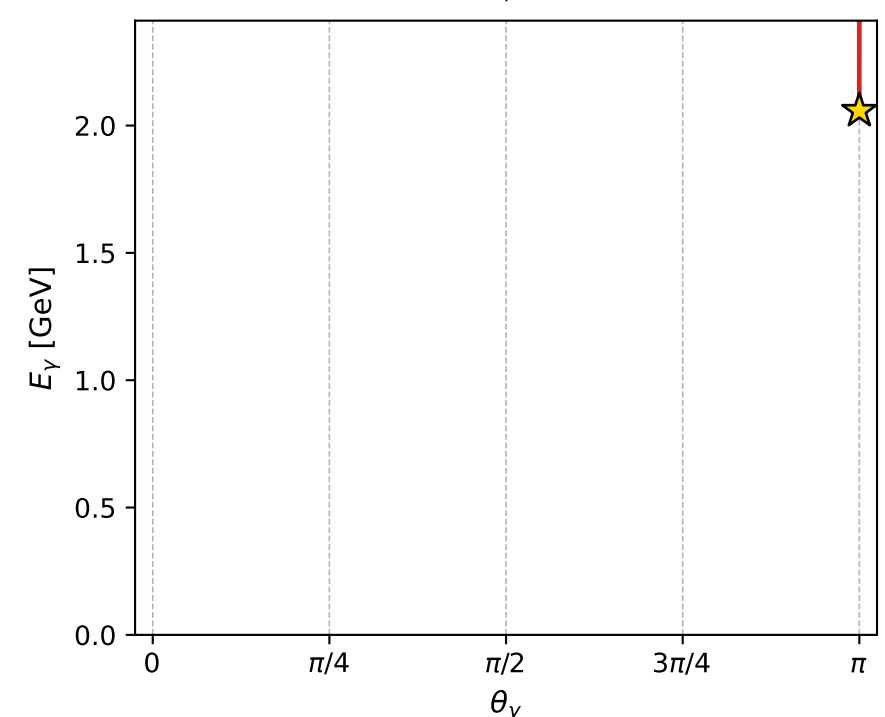
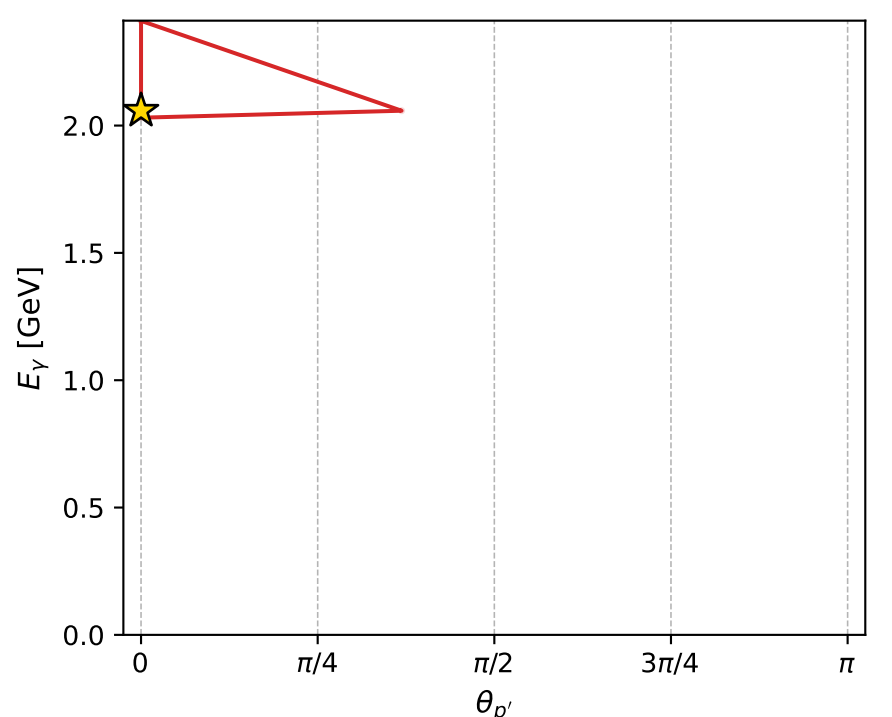
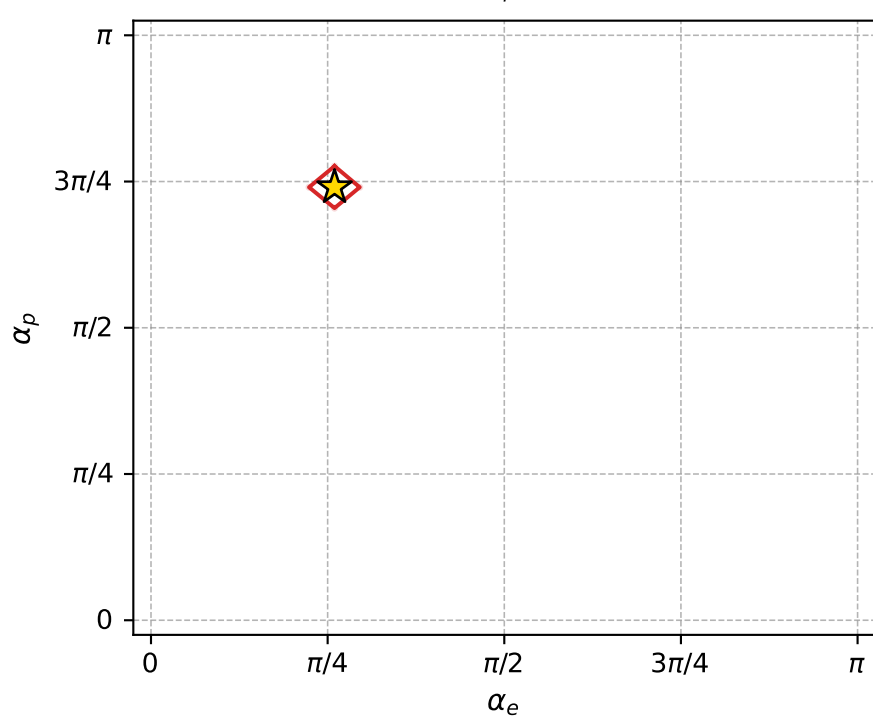
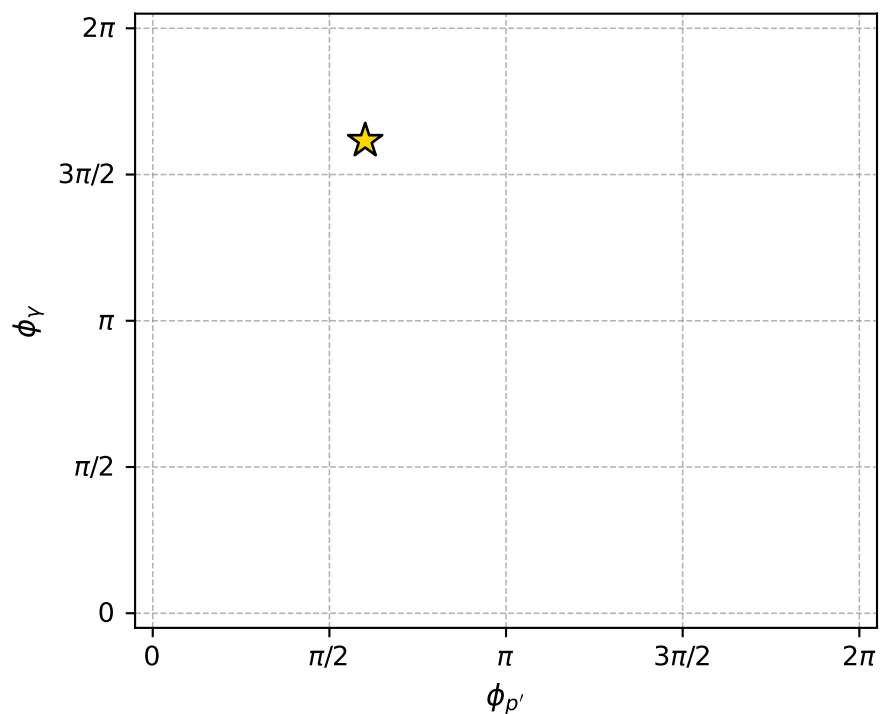
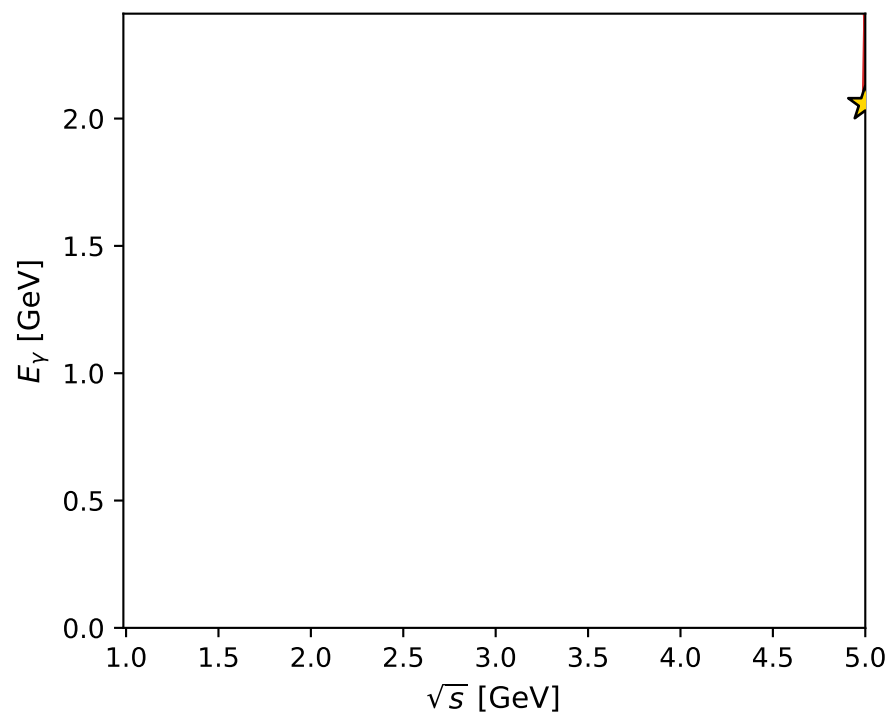
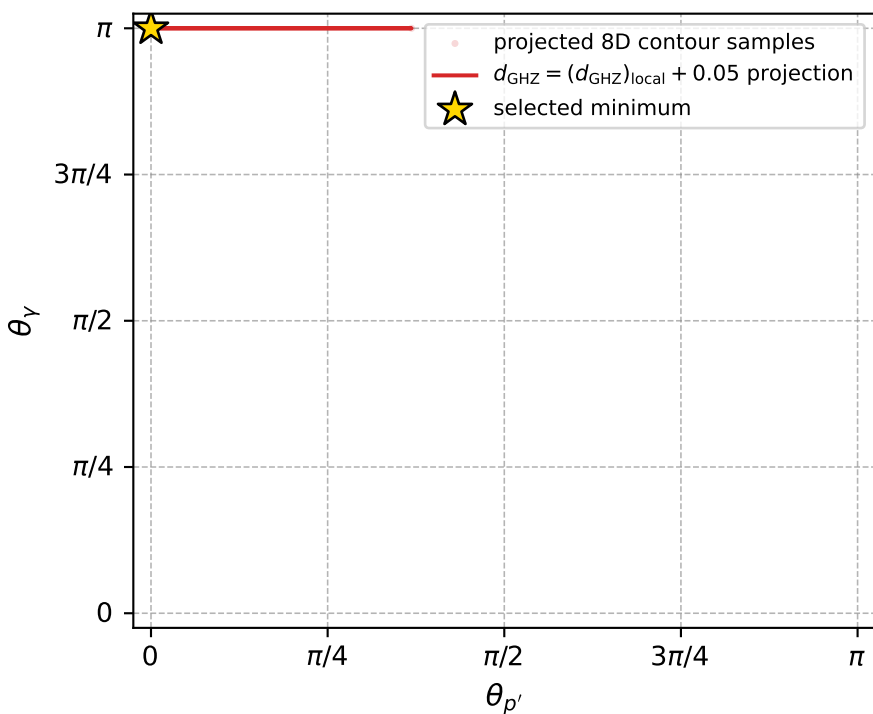


$(h_p, h_Y, h_\ell) = (+1, +1, -1)$

$(h_p, h_Y, h_\ell) = (-1, -1, +1)$



electron: polarization cluster P2, local minimum 49; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 2.6784355\text{e-}08$
 $d_{\text{GHZ}} \text{ contour} = 0.050000027$
 above global minimum = $4.5008934\text{e-}09$
 8D contour samples = 61

selected phase-space point:

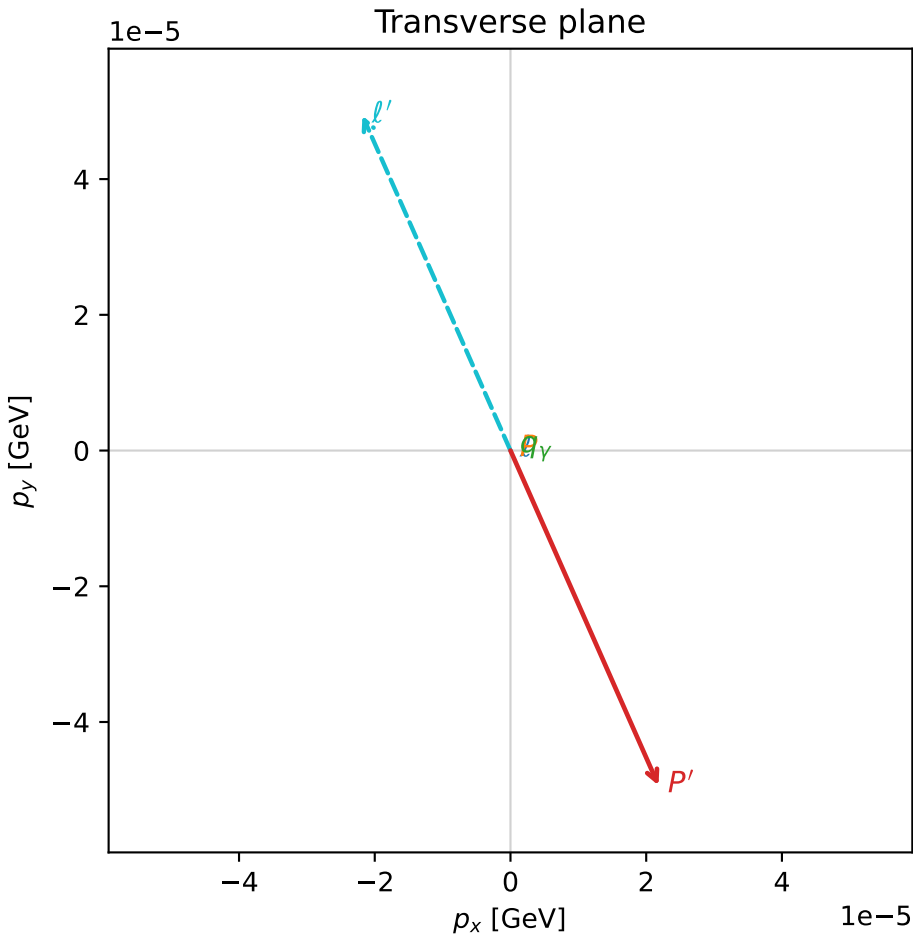
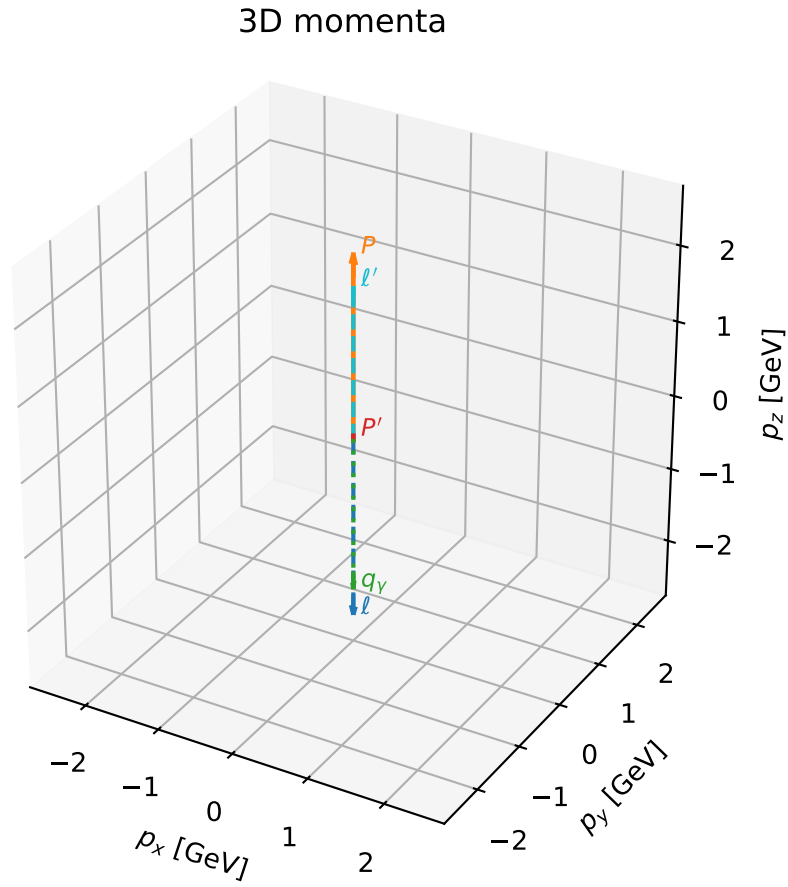
$\text{sqrt_s} = 4.999933$
 $\text{theta_p_out} = 2.804344\text{e-}05$
 $\text{theta_gamma_out} = 3.141593$
 $\text{q0out} = 2.058275$
 $\text{phi_p_out} = 1.889155$
 $\text{phi_gamma_out} = 5.075655$
 $\text{alpha_e} = 0.8160792$
 $\text{alpha_p} = 2.325503$

coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

dghz_mixing_angles_polarization_02_local_minimum_54 $d_{\{\text{rm GHZ}\}}=2.70595\text{e-}08$
region: polarization_cluster_02_local_minimum_54
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: $\alpha_e=0.85741662$ rad
incoming lepton: $0.654393|+\rangle +0.756155|-\rangle$
proton mixing angle: $\alpha_p=2.2841843$ rad
incoming proton: $-0.654399|+\rangle +0.756149|-\rangle$

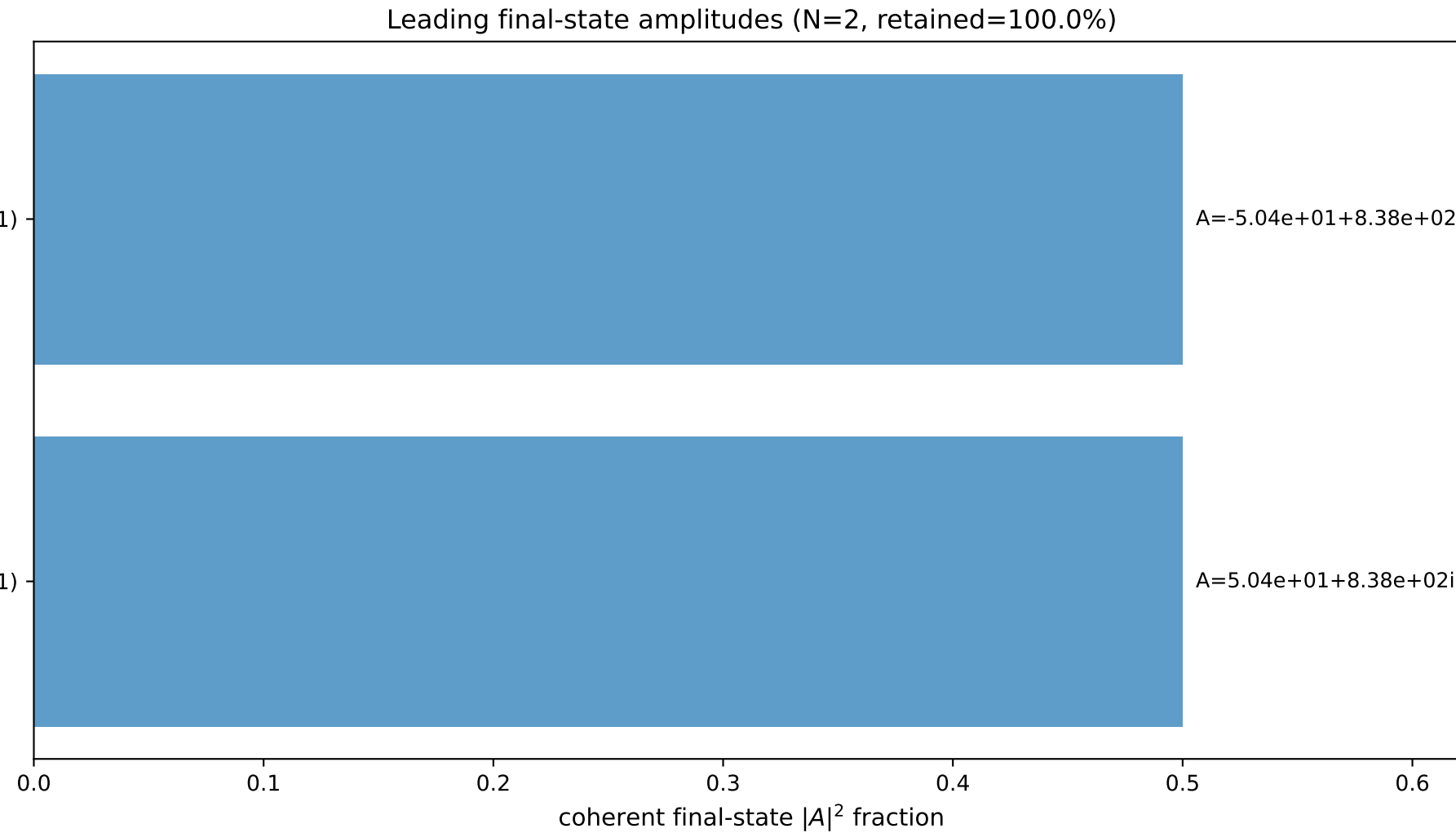
$s=24.9965$, $\sqrt{s}=4.99965$, $|\vec{P}|=2.41184$, $|\vec{P}'|=0.0441066$
 $\theta_{P'}=0.00122325$, $\theta_Y=3.14159$, $\phi_{P'}=5.12862$, $\phi_Y=1.63085$
 $E_Y=2.05236$, $\theta_{\ell'}=2.68658\text{e-}05$, $\phi_{\ell'}=1.98702$
 $Q^2=19.3743$, $x_B=0.827614$, $t=-2.88766$, $W^2=4.91537$, $y=0.970691$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4118$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4118$
 P $E= 2.5878$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4118$
 ℓ' $E= 2.0083$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 2.0083$
 P' $E= 0.9390$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 0.0441$
 q_Y $E= 2.0524$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= -2.0524$

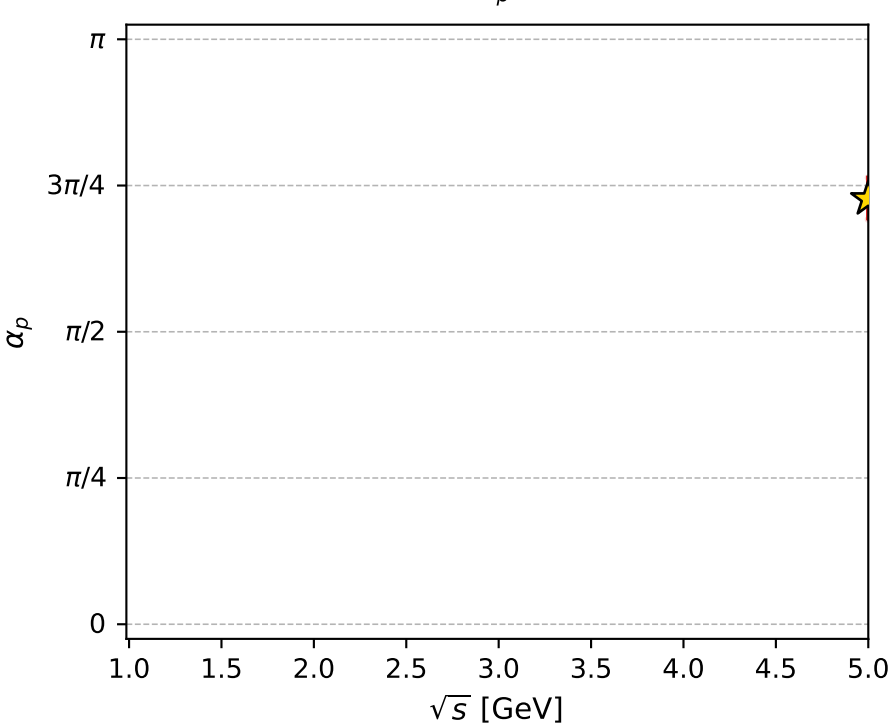
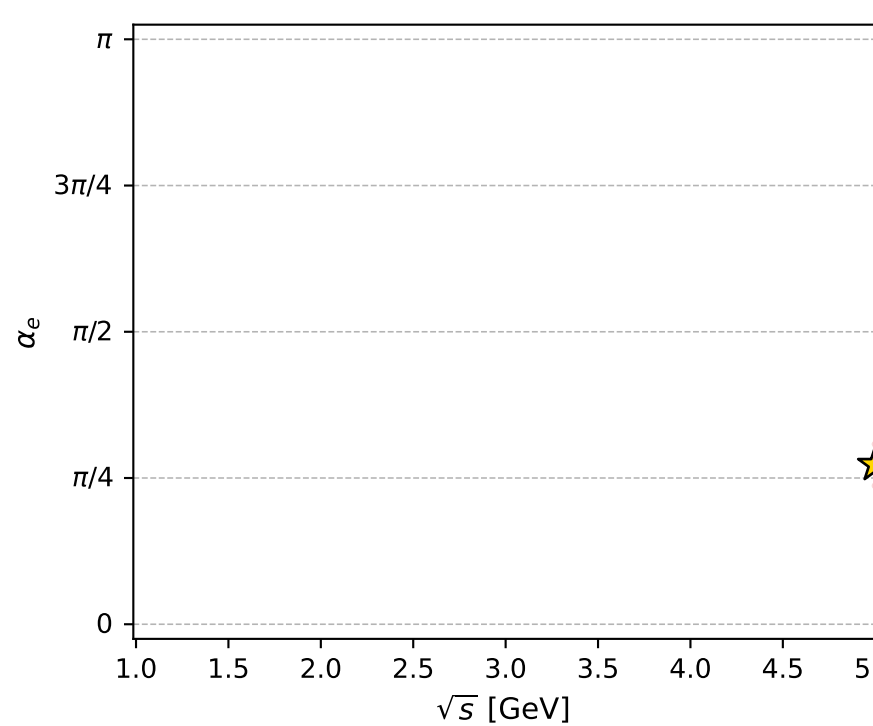
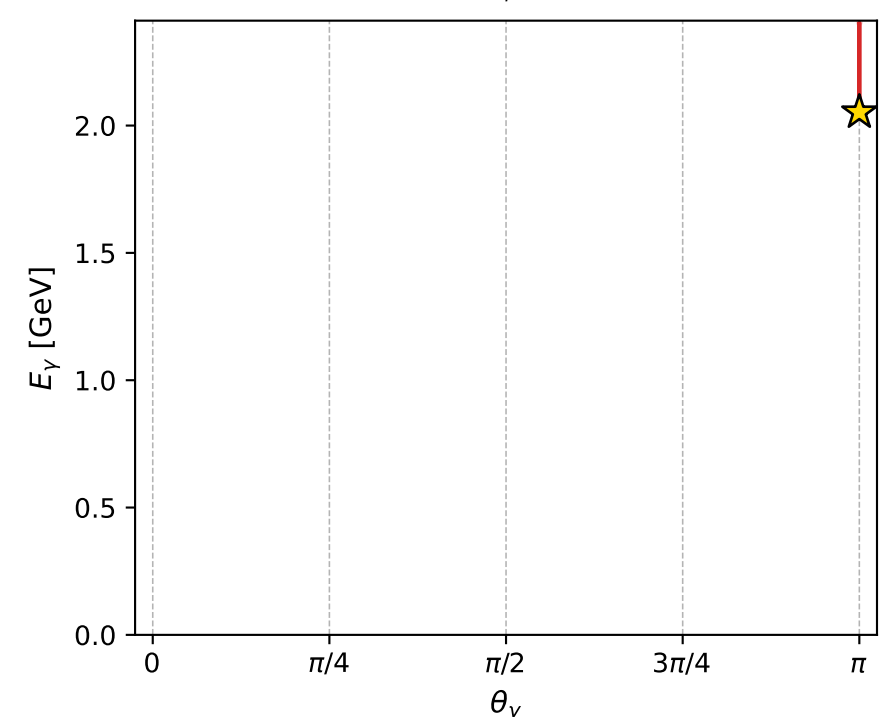
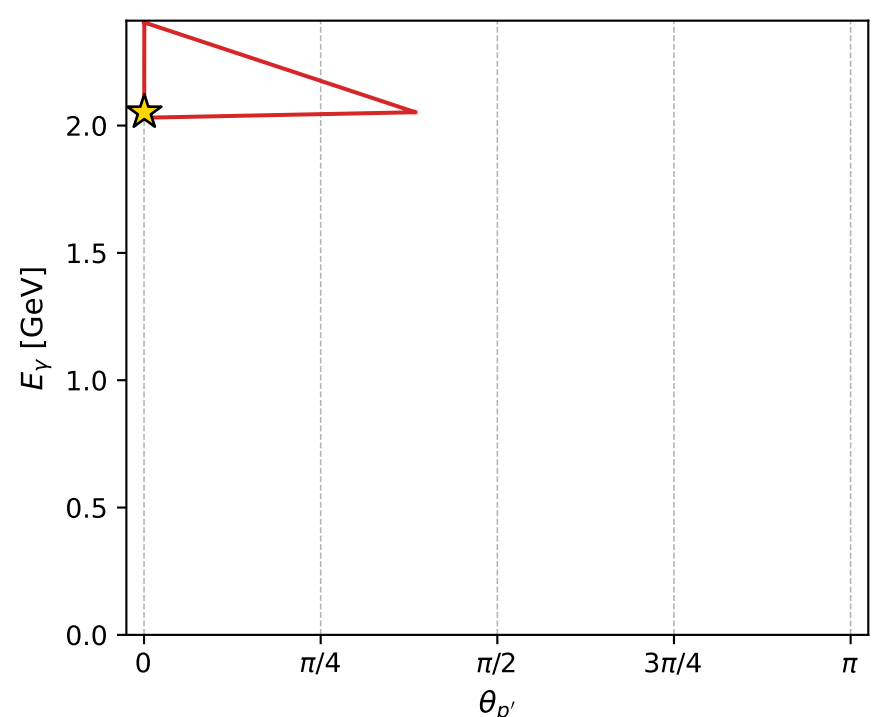
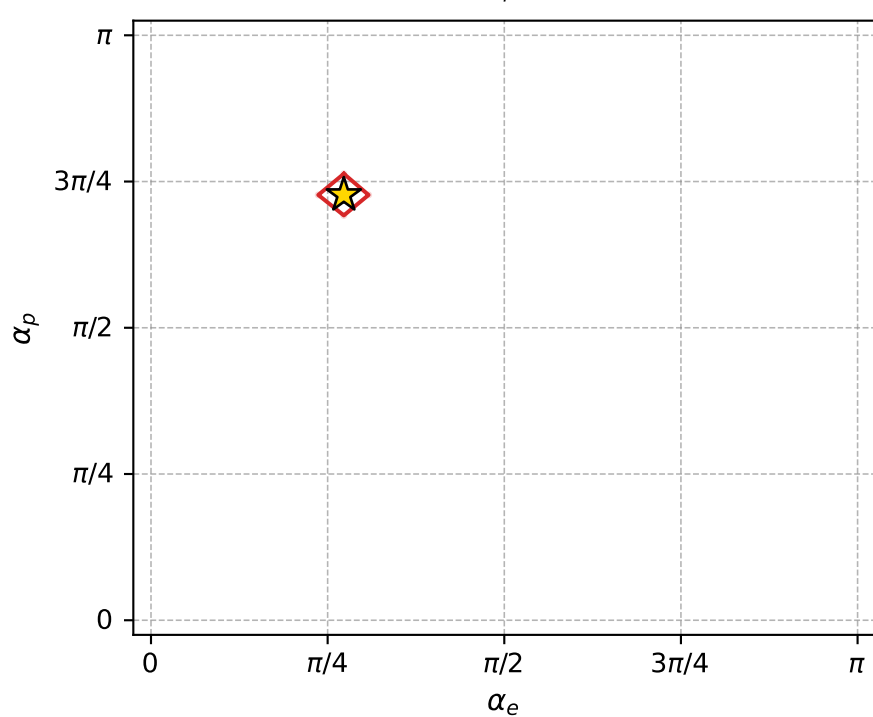
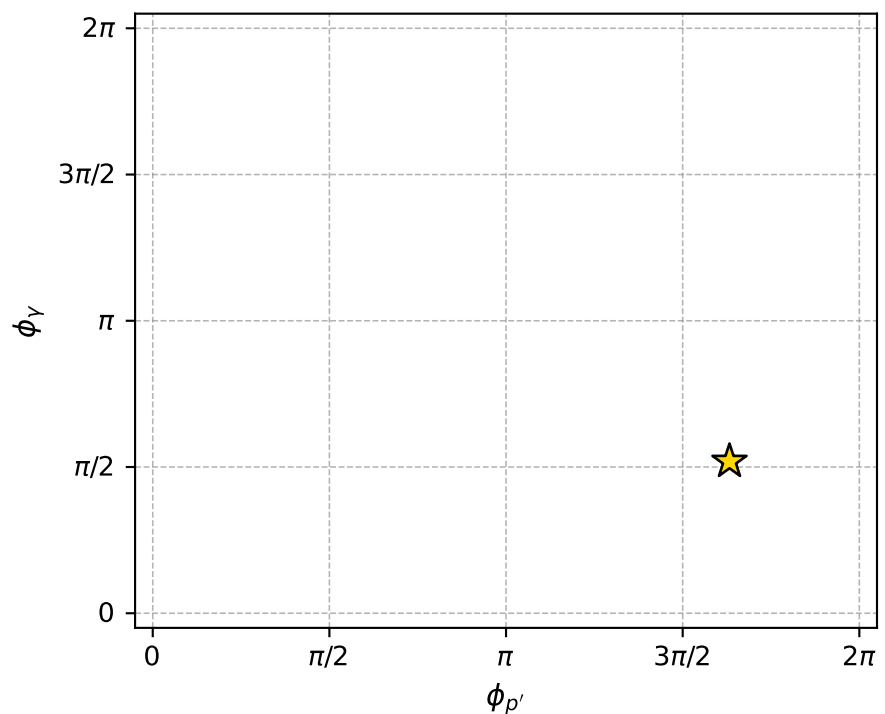
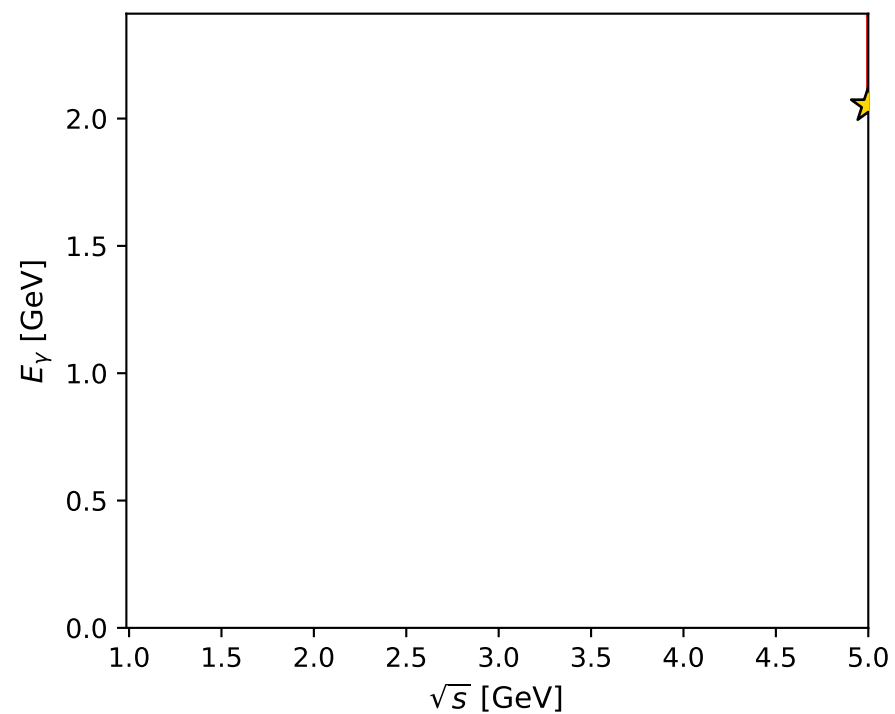
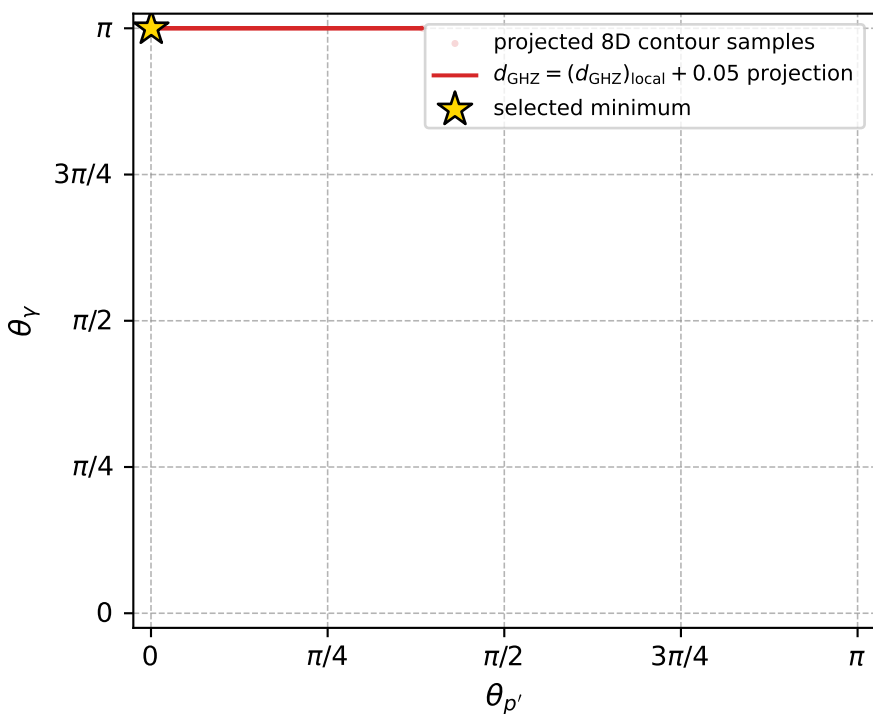


$(h_p, h_Y, h_\ell) = (-1, -1, +1)$

$(h_p, h_Y, h_\ell) = (+1, +1, -1)$



electron: polarization cluster P2, local minimum 54; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 2.7059466\text{e-}08$
 $d_{\text{GHZ}} \text{ contour} = 0.050000027$
 above global minimum = $4.7760038\text{e-}09$
 8D contour samples = 116

selected phase-space point:

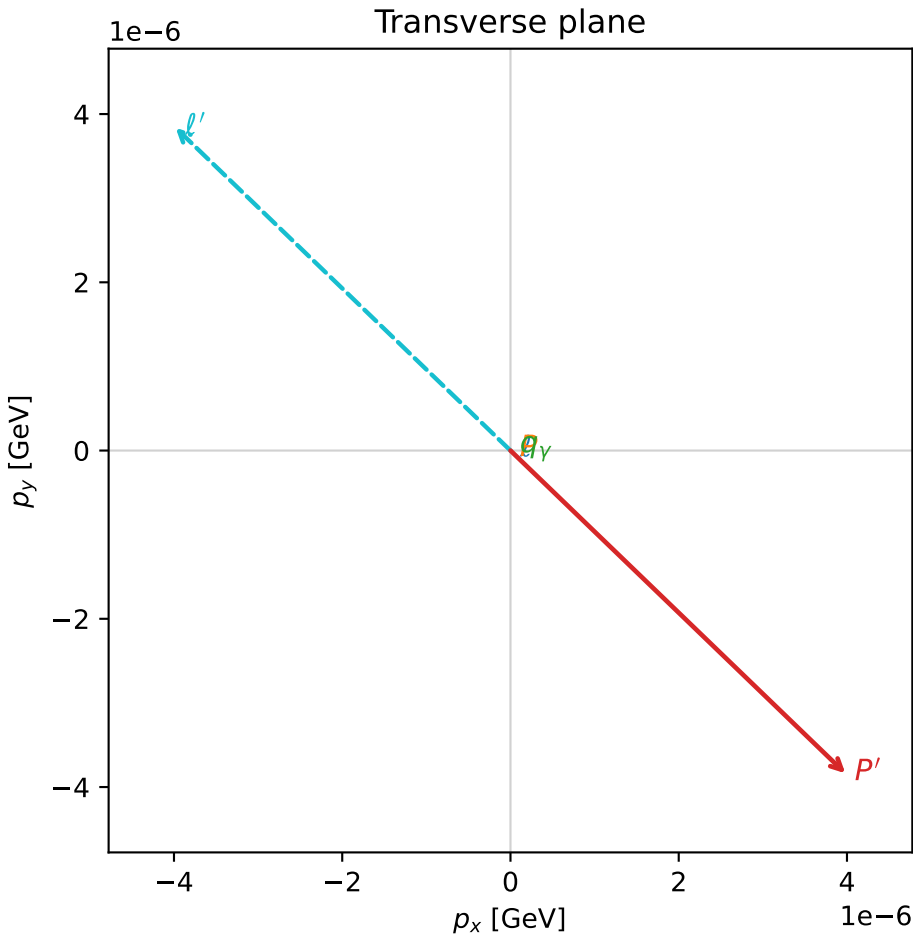
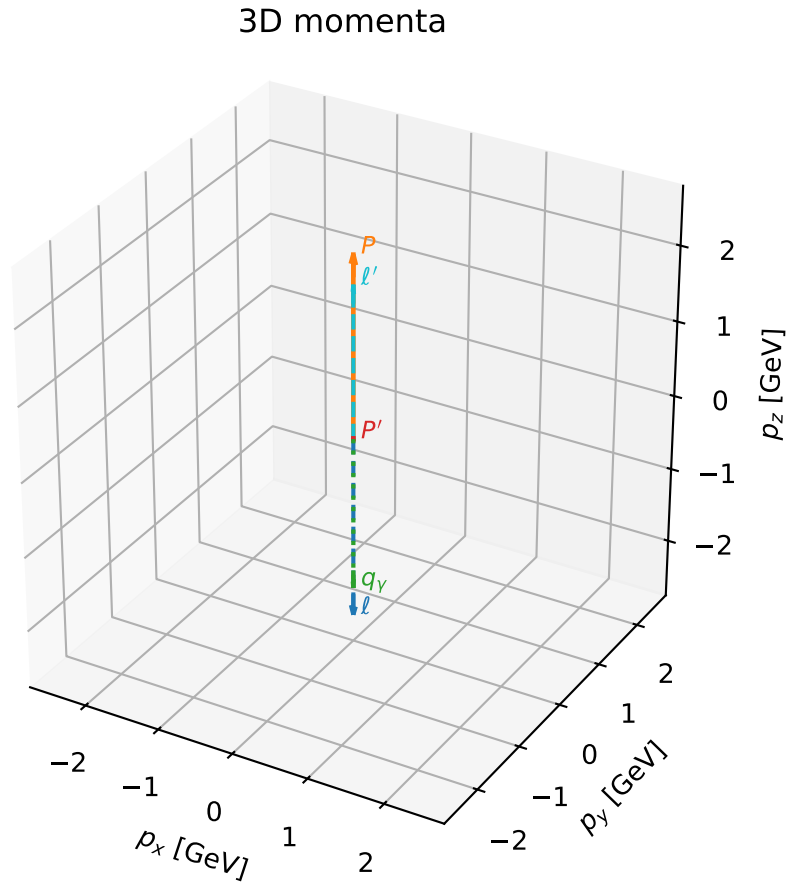
$\text{sqrt_s} = 4.999653$
 $\text{theta_p_out} = 0.001223249$
 $\text{theta_gamma_out} = 3.141593$
 $\text{q0ut} = 2.052361$
 $\text{phi_p_out} = 5.128616$
 $\text{phi_gamma_out} = 1.630851$
 $\text{alpha_e} = 0.8574166$
 $\text{alpha_p} = 2.284184$

coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

dghz_mixing_angles_polarization_02_local_minimum_62 $d_{\{\text{rm GHZ}\}}=2.75303\text{e-}08$
region: polarization_cluster_02_local_minimum_62
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: $\alpha_e=0.65340217$ rad
incoming lepton: $0.79402|+\rangle +0.607891|-\rangle$
proton mixing angle: $\alpha_p=2.4881891$ rad
incoming proton: $-0.794019|+\rangle +0.607892|-\rangle$

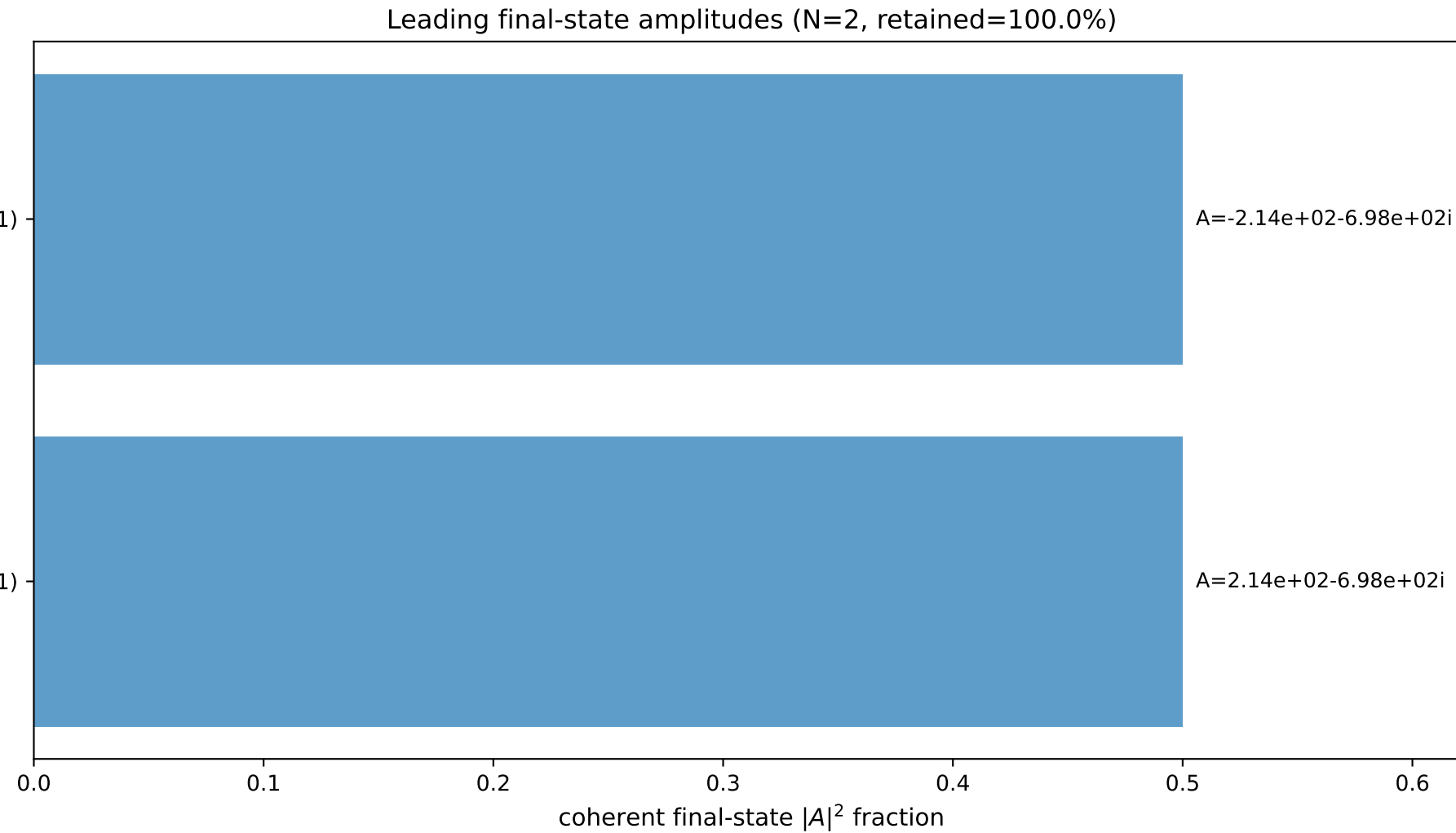
$s=25$, $\sqrt{s}=5$, $|\vec{P}|=2.41202$, $|\vec{P}'|=0.00901078$
 $\theta_{P'}=0.000613592$, $\theta_Y=3.14159$, $\phi_{P'}=5.51618$, $\phi_Y=5.00919$
 $E_Y=2.03548$, $\theta_{\ell'}=2.72834\text{e-}06$, $\phi_{\ell'}=2.37459$
 $Q^2=19.5515$, $x_B=0.835287$, $t=-3.05213$, $W^2=4.73527$, $y=0.970432$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4120$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 P $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4120$
 ℓ' $E= 2.0265$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 2.0265$
 P' $E= 0.9380$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 0.0090$
 q_Y $E= 2.0355$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= -2.0355$

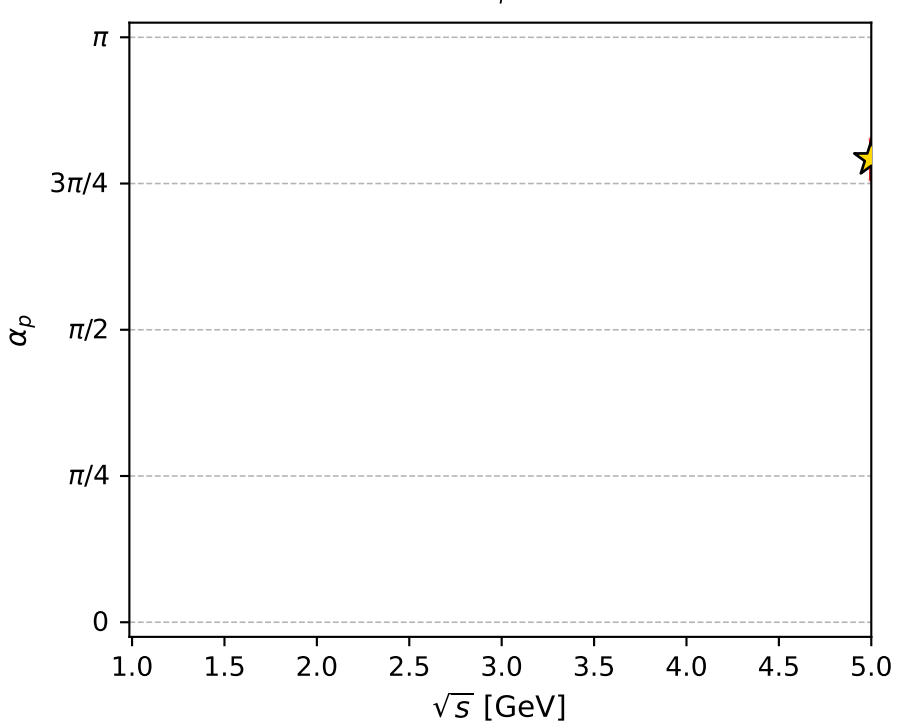
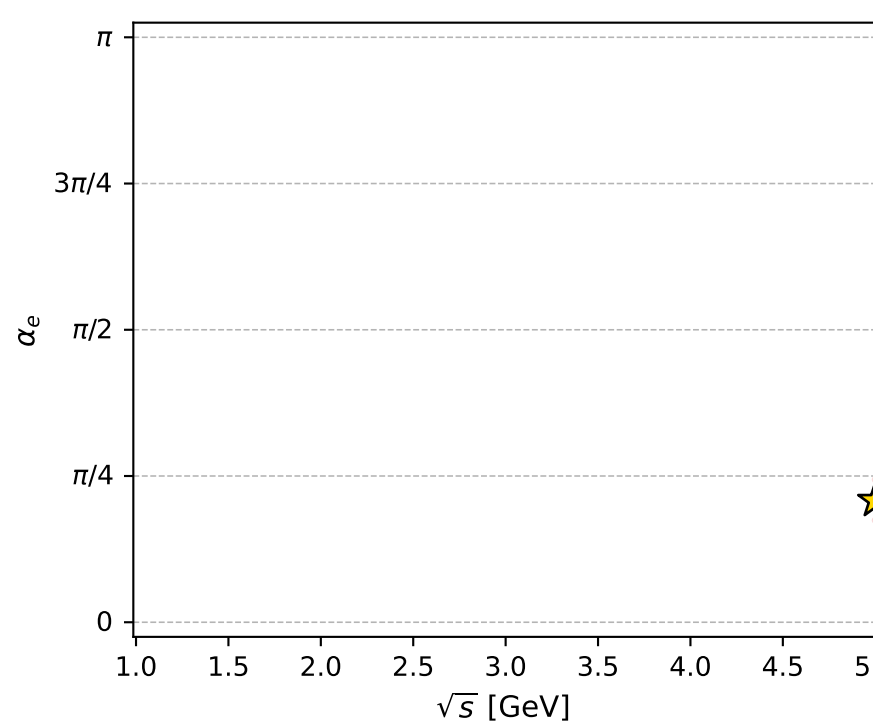
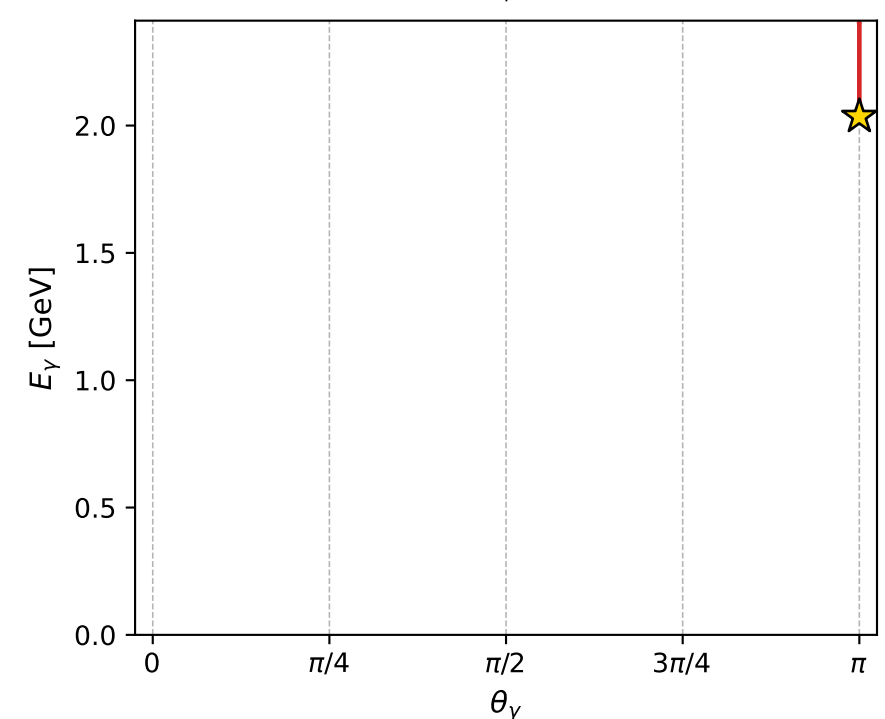
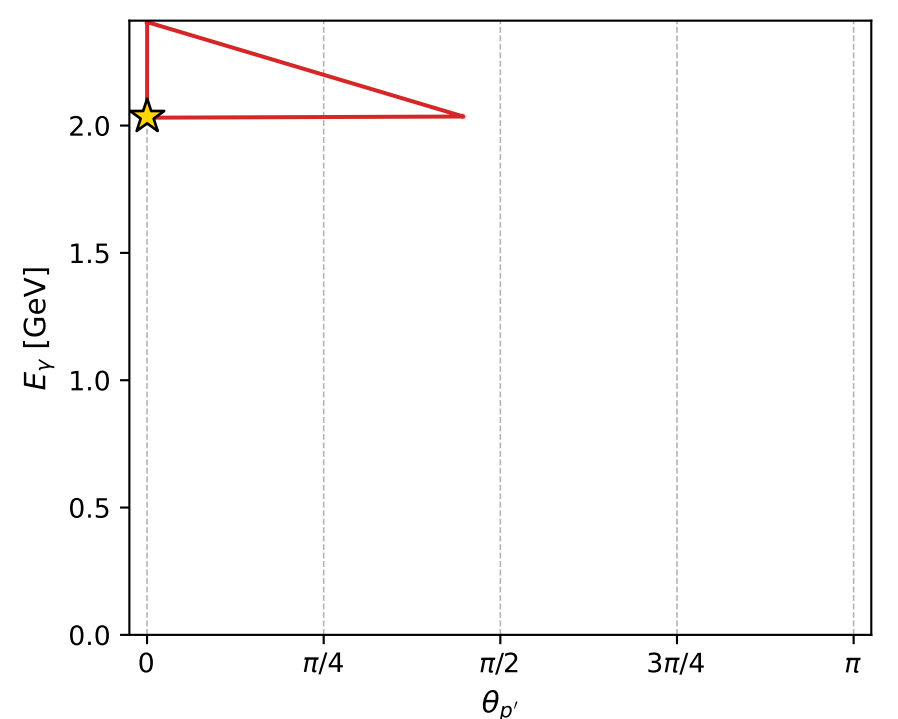
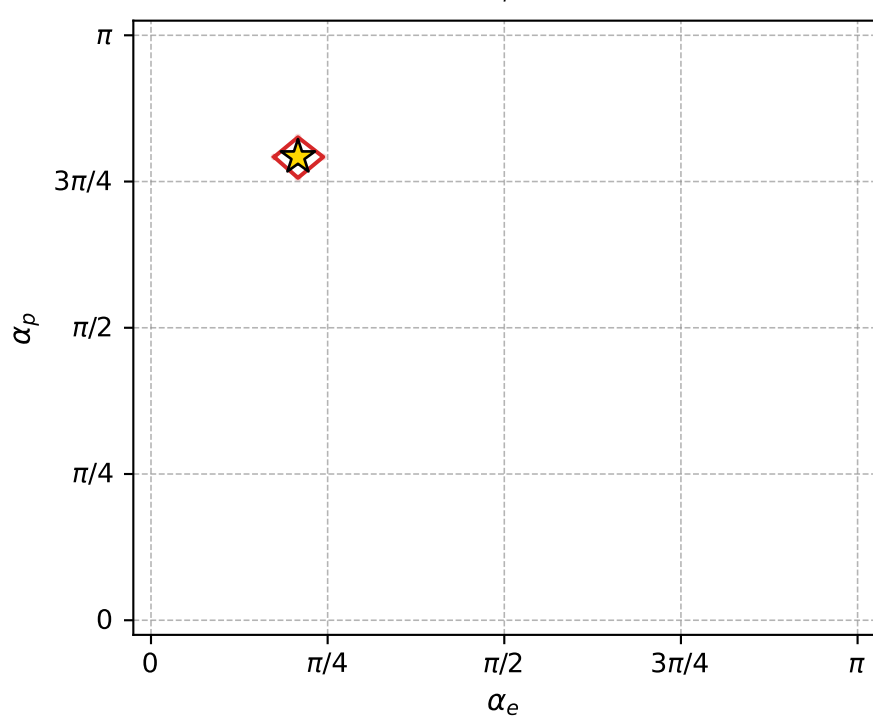
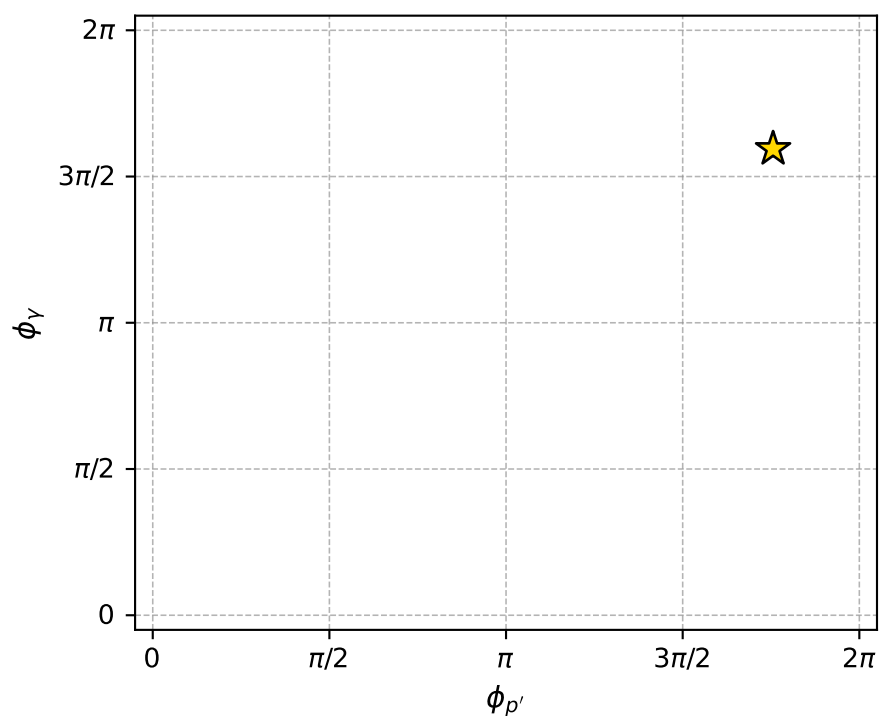
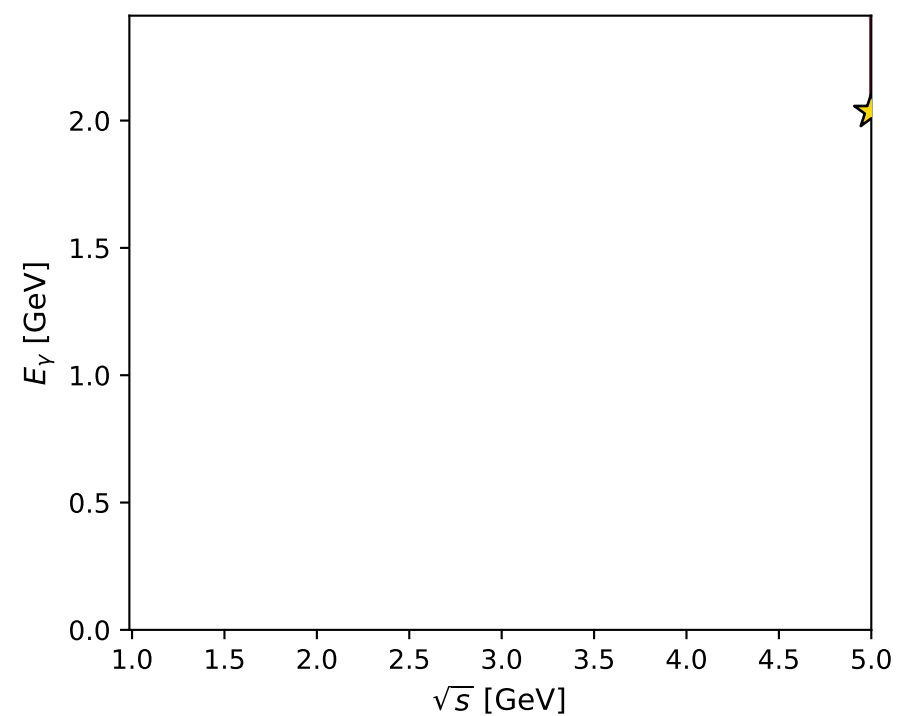
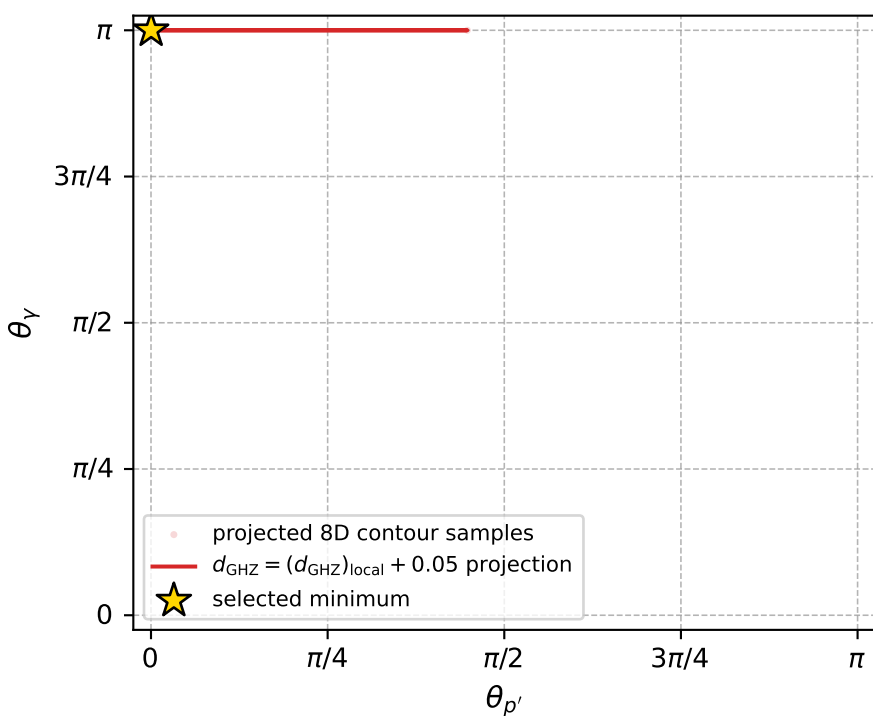


$(h_p, h_Y, h_{\ell}) = (+1, +1, -1)$

$(h_p, h_Y, h_{\ell}) = (-1, -1, +1)$



electron: polarization cluster P2, local minimum 62; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 2.7530269\text{e-}08$
 $d_{\text{GHZ}} \text{ contour} = 0.050000028$
 above global minimum = $5.2468074\text{e-}09$
 8D contour samples = 63

selected phase-space point:

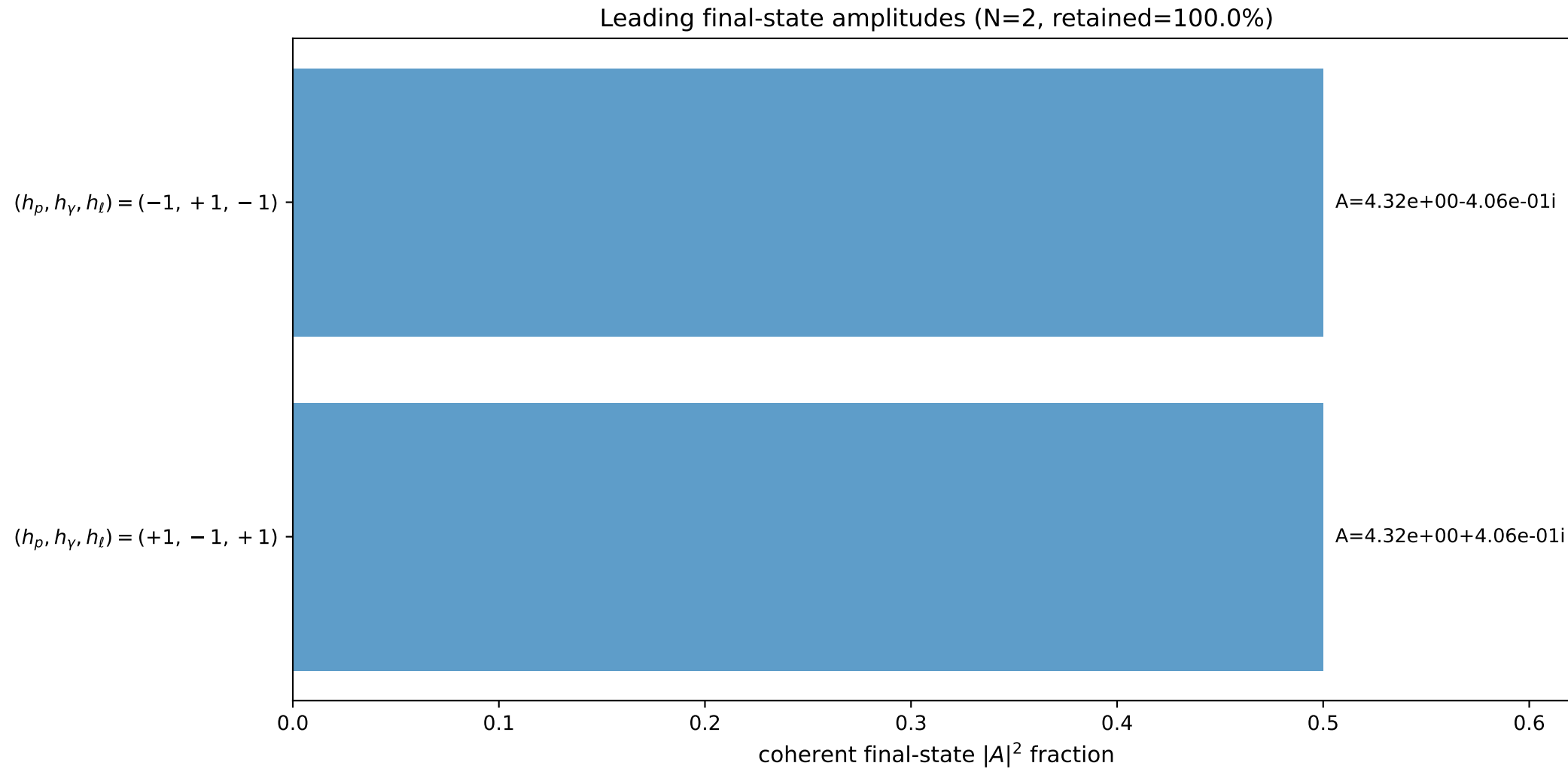
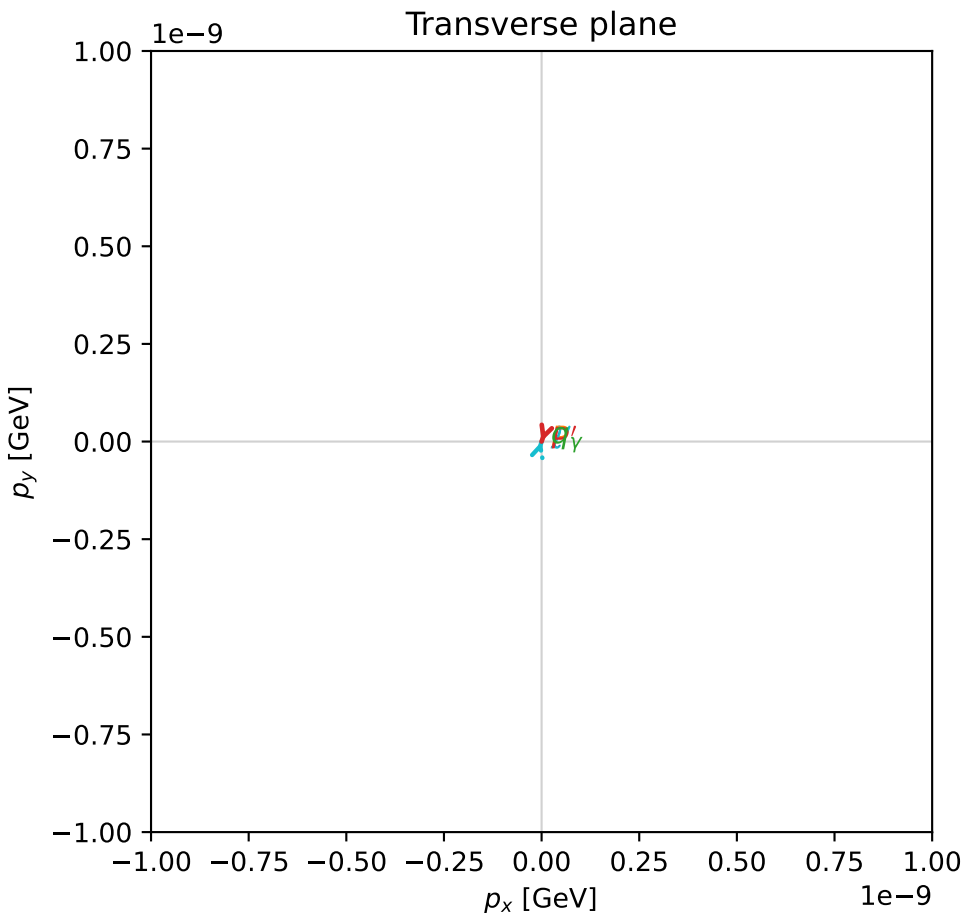
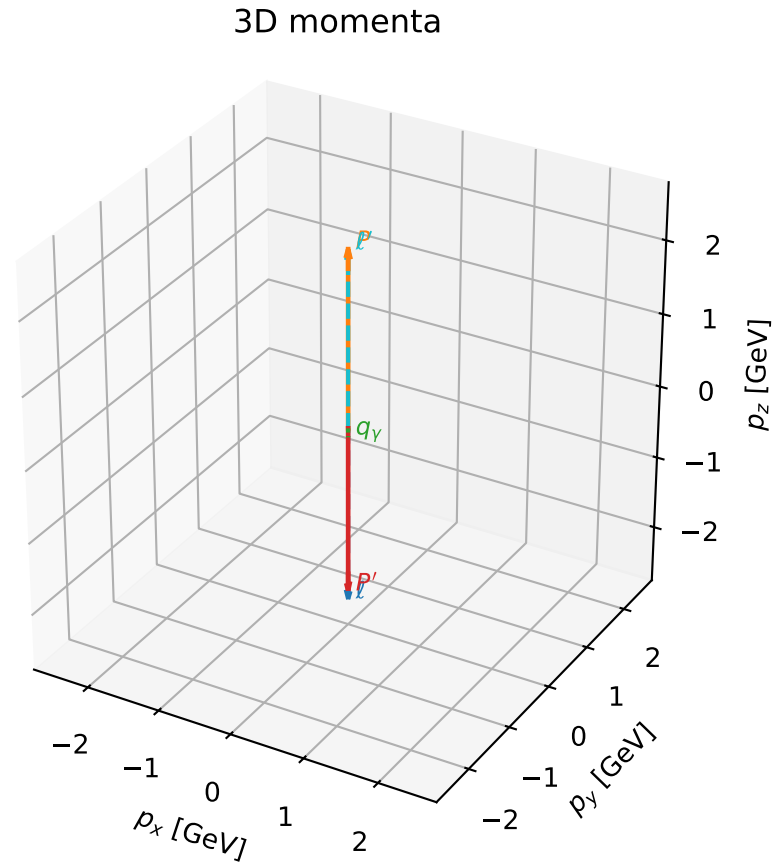
$\text{sqrt_s} = 5$
 $\text{theta_p_out} = 0.0006135923$
 $\text{theta_gamma_out} = 3.141593$
 $\text{qOut} = 2.035484$
 $\text{phi_p_out} = 5.516182$
 $\text{phi_gamma_out} = 5.009187$
 $\text{alpha_e} = 0.6534022$
 $\text{alpha_p} = 2.488189$

coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

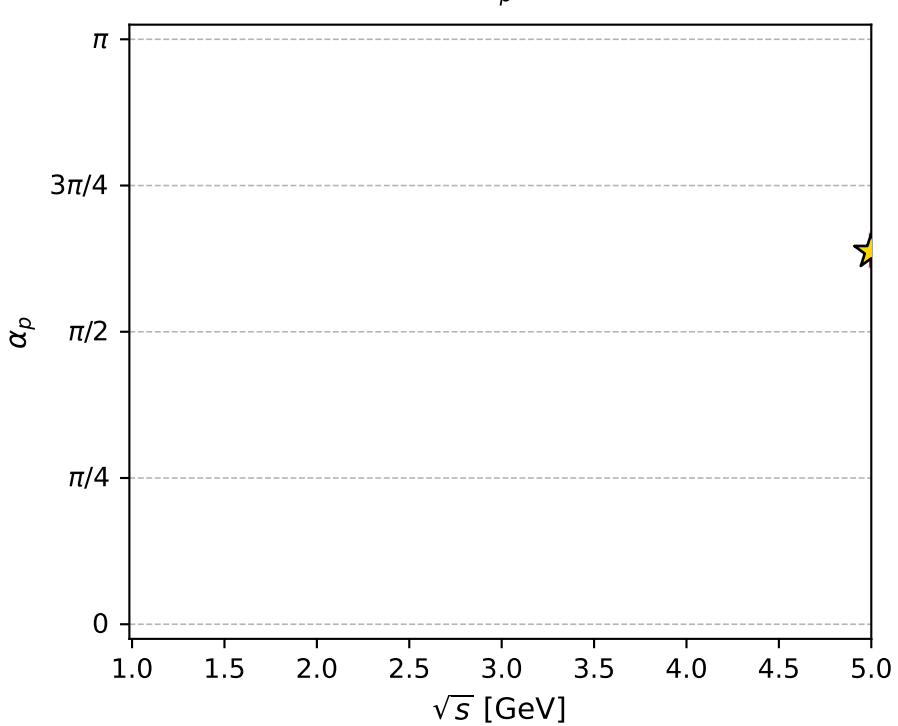
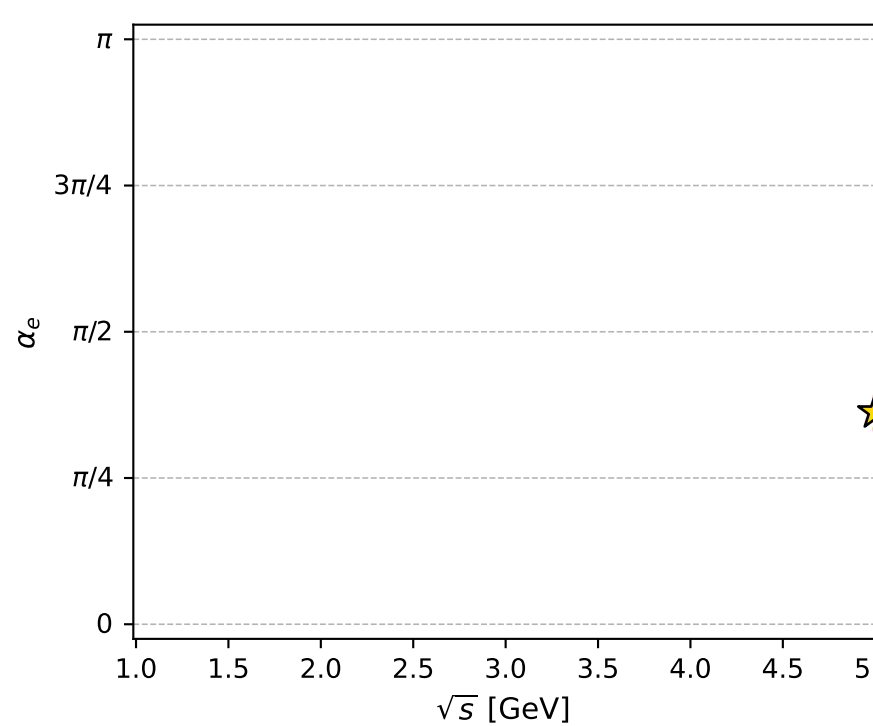
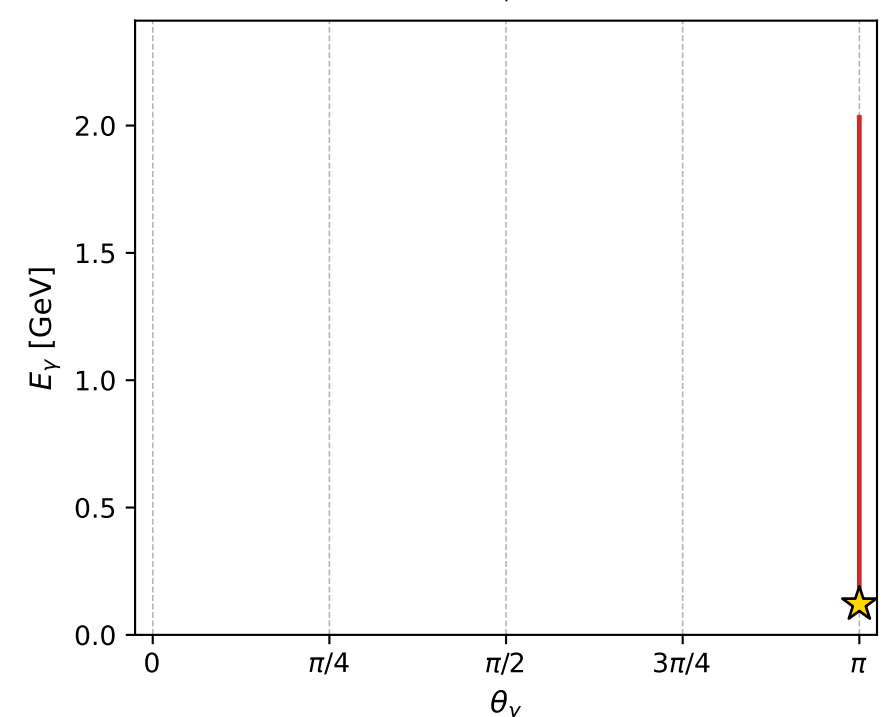
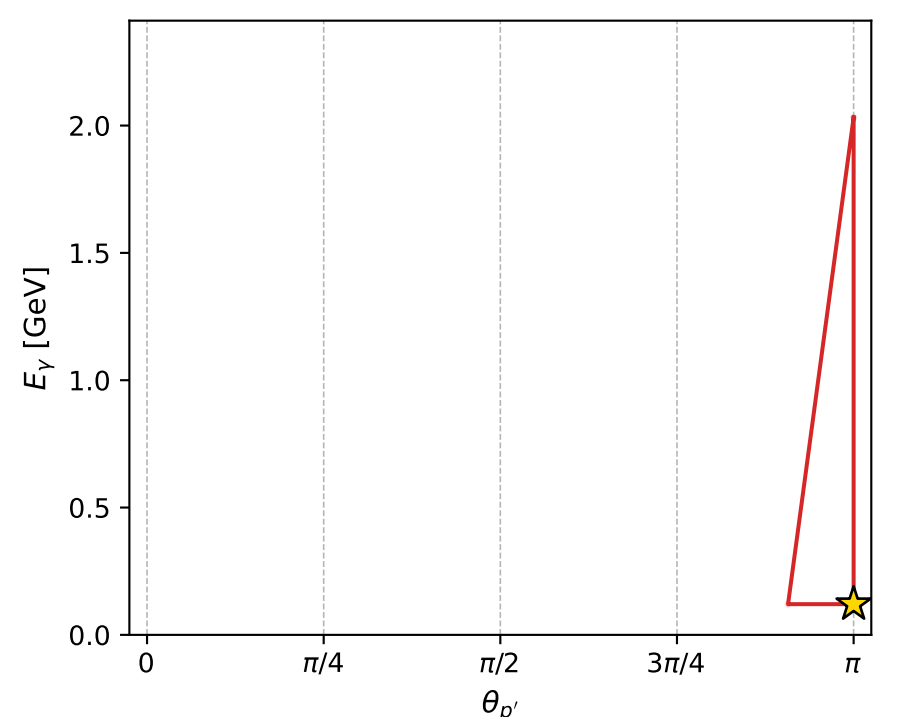
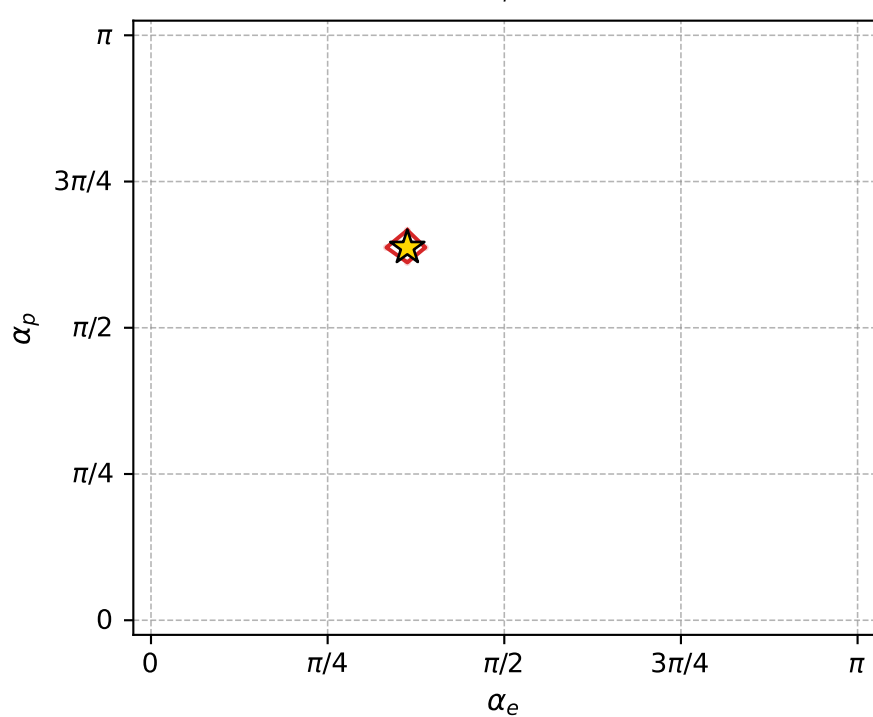
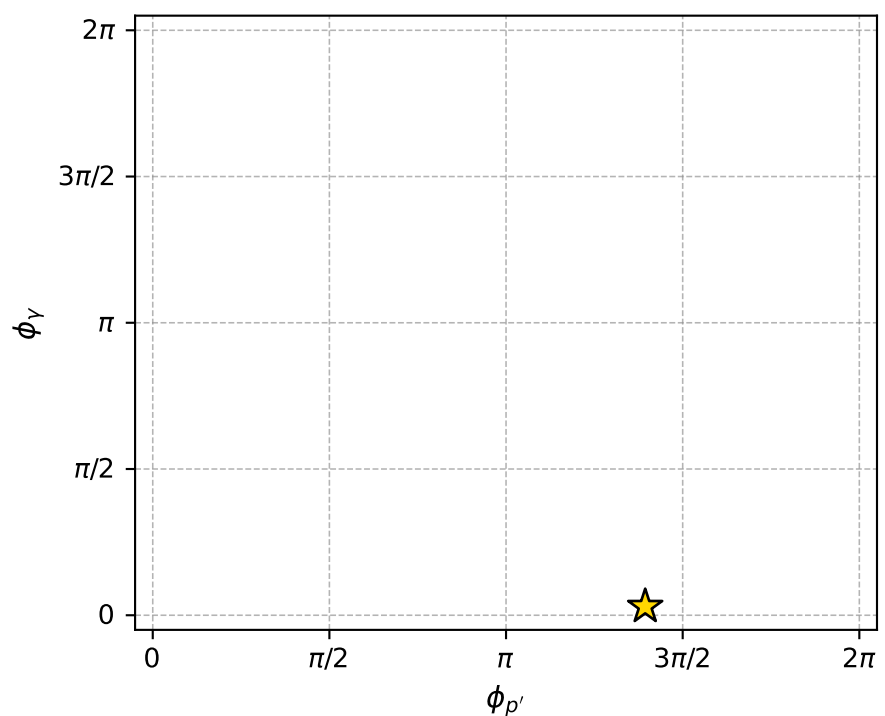
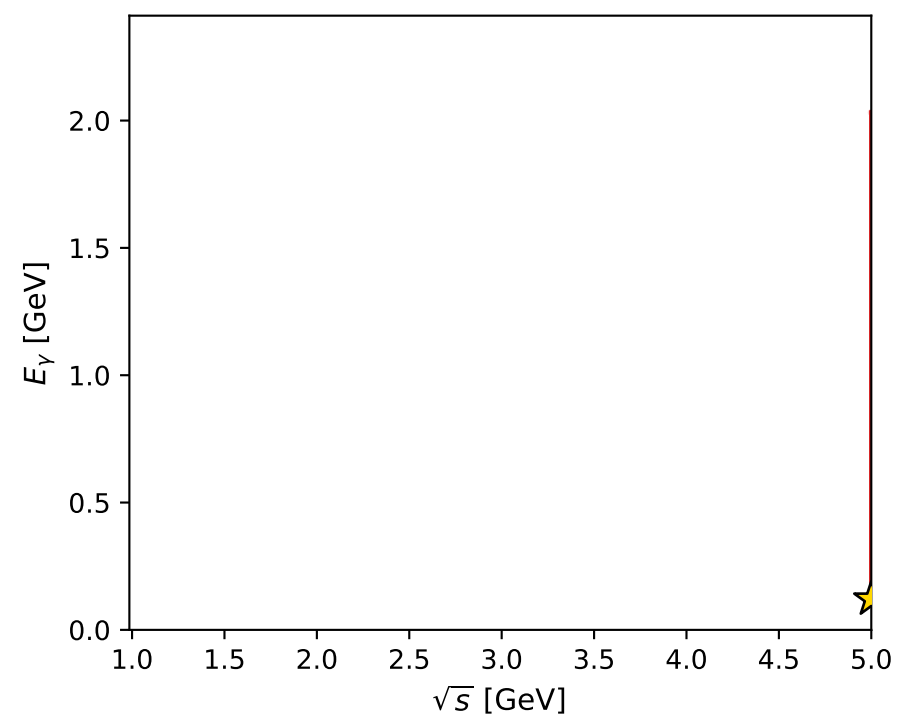
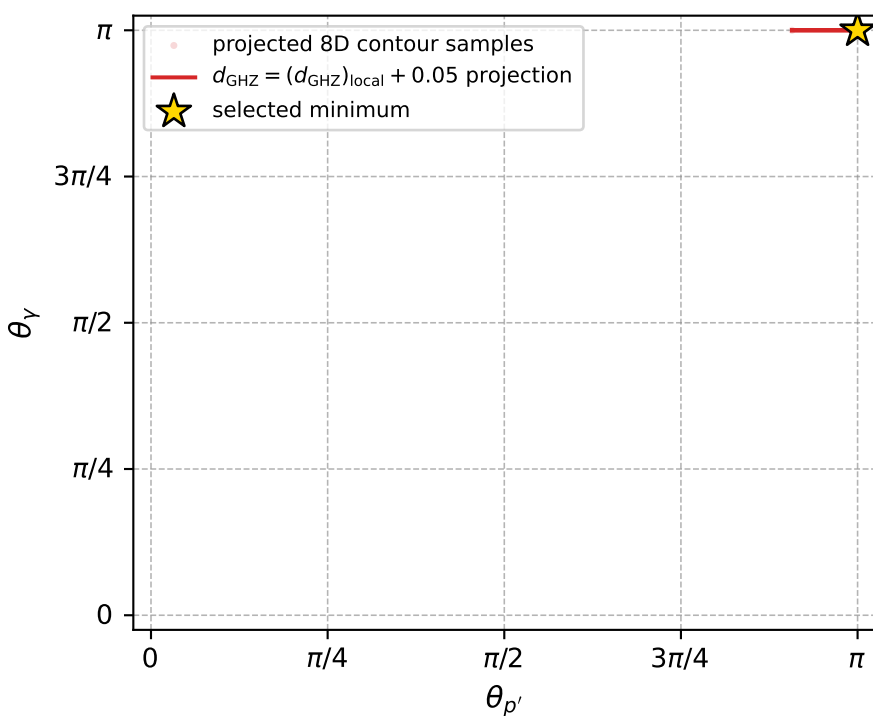
dghz_mixing_angles_polarization_02_local_minimum_77 $d_{\{\text{rm GHZ}\}}=2.96256\text{e-}08$
 region: polarization_cluster_02_local_minimum_77
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=1.1396606$ rad
 incoming lepton: $0.417903|+\rangle +0.908492|-\rangle$
 proton mixing angle: $\alpha_p=2.0019206$ rad
 incoming proton: $-0.417892|+\rangle +0.908497|-\rangle$

$s=25$, $\sqrt{s}=5$, $|\vec{P}|=2.41202$, $|\vec{P}'|=2.28693$
 $\theta_{P'}=3.14159$, $\theta_Y=3.14159$, $\phi_{P'}=4.37912$, $\phi_Y=0.0937313$
 $E_Y=0.120627$, $\theta_{\ell'}=0$, $\phi_{\ell'}=1.28657$
 $Q^2=23.2282$, $x_B=0.998083$, $t=-22.0666$, $W^2=0.924449$, $y=0.964871$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4120$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 P $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4120$
 ℓ' $E= 2.4076$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4076$
 P' $E= 2.4718$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= -2.2869$
 q_Y $E= 0.1206$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1206$



electron: polarization cluster P2, local minimum 77; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 2.9625613\text{e-}08$
 $d_{\text{GHZ}} \text{ contour} = 0.05000003$
 above global minimum = $7.342151\text{e-}09$
 8D contour samples = 41

selected phase-space point:

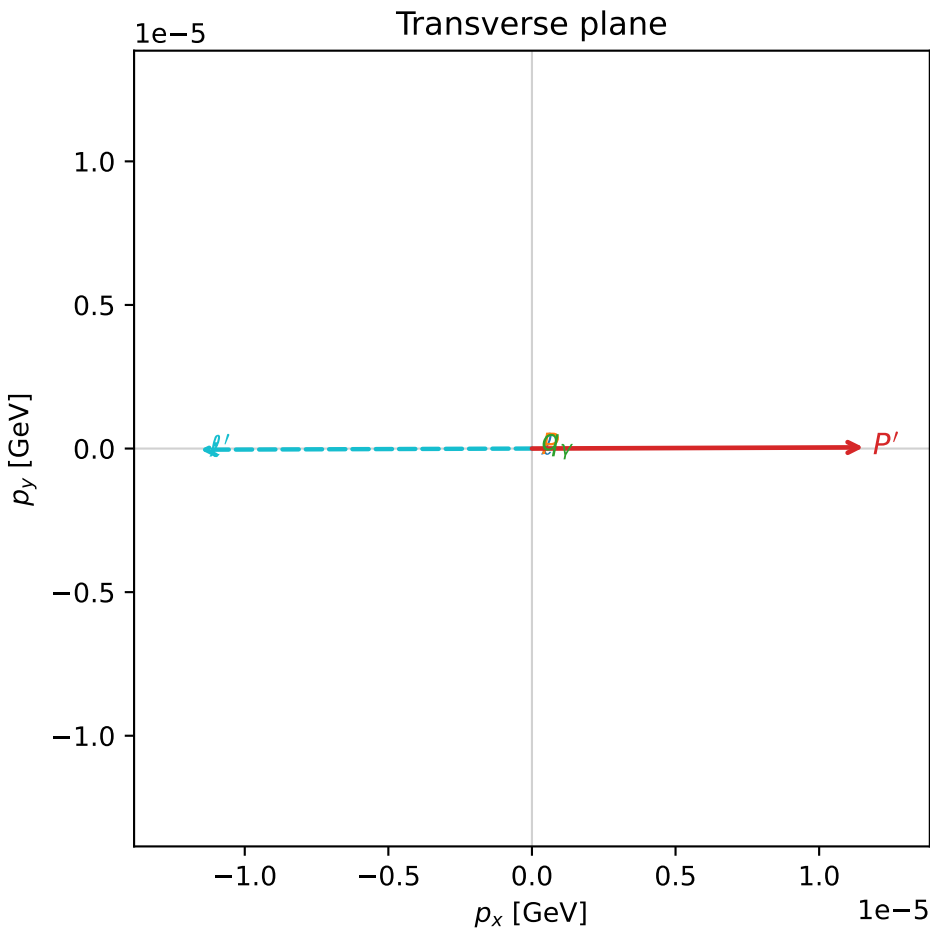
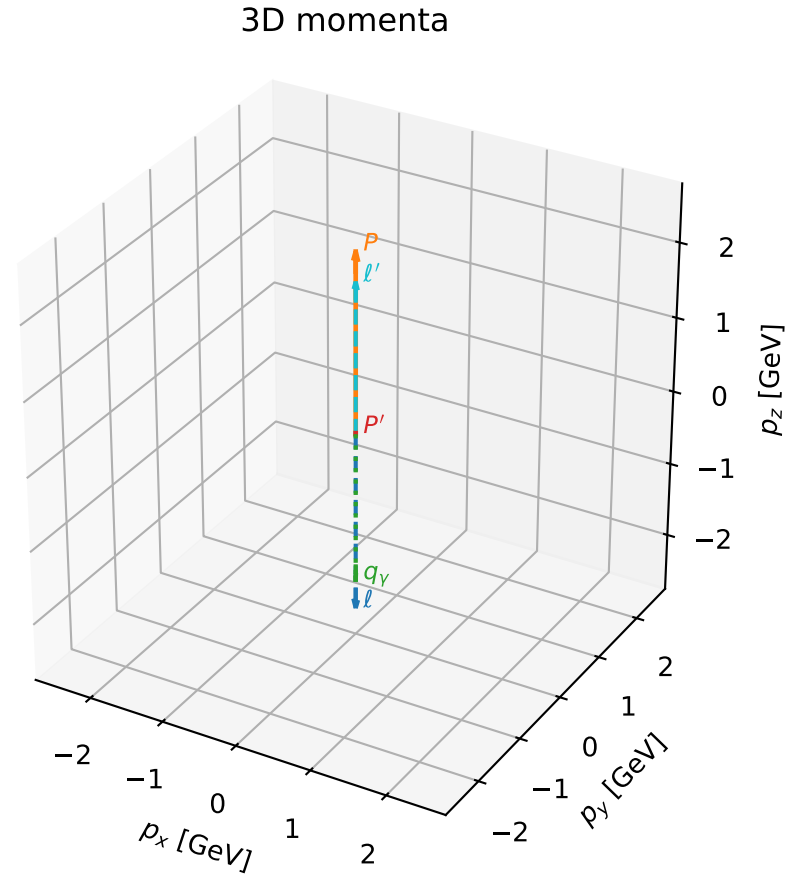
$\text{sqrt_s} = 5$
 $\text{theta_p_out} = 3.141593$
 $\text{theta_gamma_out} = 3.141593$
 $\text{q0ut} = 0.1206268$
 $\text{phi_p_out} = 4.379124$
 $\text{phi_gamma_out} = 0.09373129$
 $\text{alpha_e} = 1.139661$
 $\text{alpha_p} = 2.001921$

coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

dghz_mixing_angles_polarization_02_local_minimum_78 $d_{\{\text{rm GHZ}\}}=2.97007\text{e-}08$
region: polarization_cluster_02_local_minimum_78
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: $\alpha_e=0.55713074$ rad
incoming lepton: $0.848776|+\rangle +0.528753|-\rangle$
proton mixing angle: $\alpha_p=2.5844424$ rad
incoming proton: $-0.848765|+\rangle +0.52877|-\rangle$

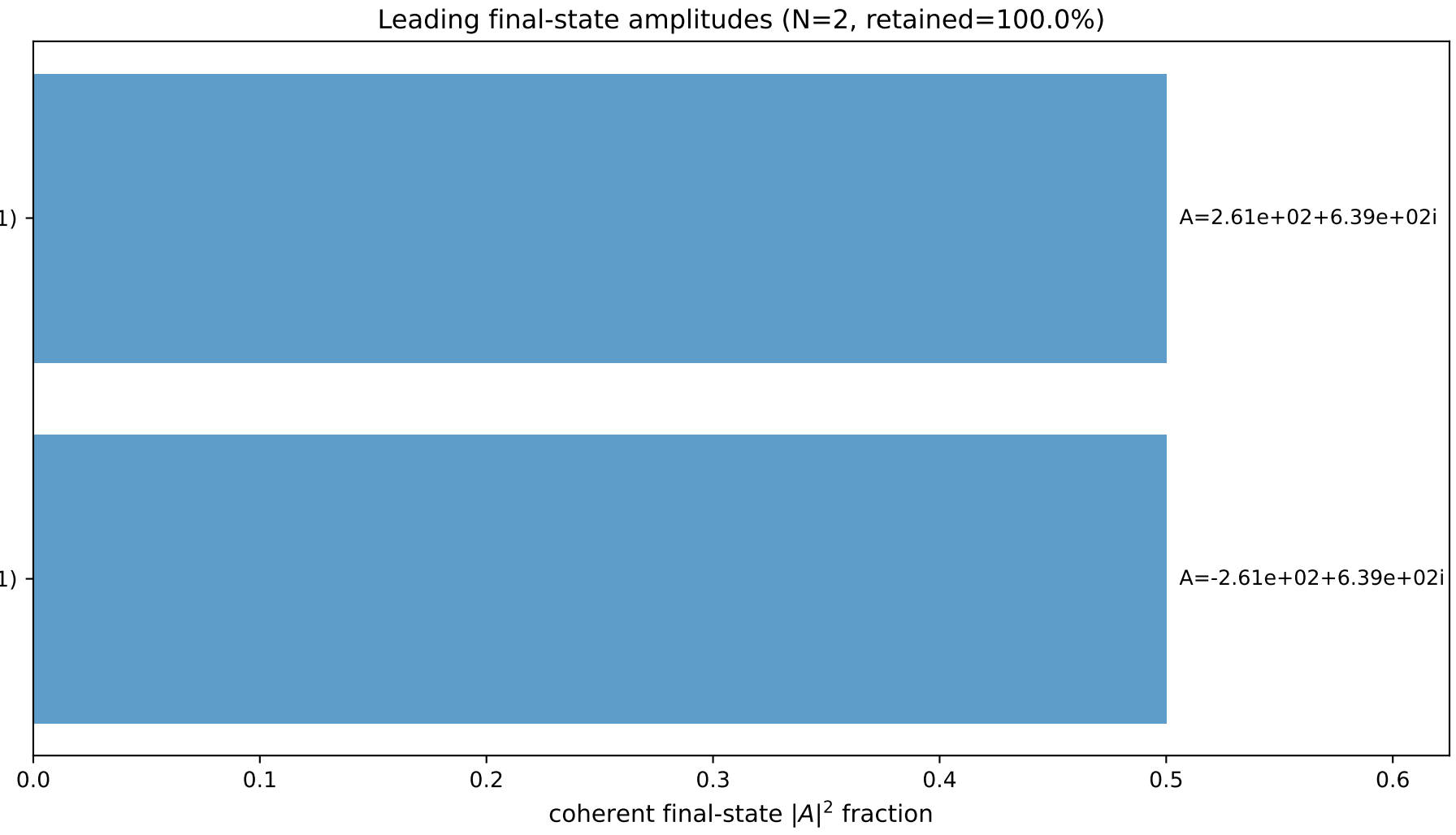
$s=24.9739$, $\sqrt{s}=4.99739$, $|\vec{P}|=2.41067$, $|\vec{P}'|=0.0137879$
 $\theta_{P'}=0.000837323$, $\theta_Y=3.14159$, $\phi_{P'}=0.00351513$, $\phi_Y=1.9584$
 $E_Y=2.03654$, $\theta_{\ell'}=5.70749\text{e-}06$, $\phi_{\ell'}=3.14511$
 $Q^2=19.5047$, $x_B=0.834182$, $t=-3.02706$, $W^2=4.75696$, $y=0.970439$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4107$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4107$
 P $E= 2.5867$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4107$
 ℓ' $E= 2.0228$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 2.0228$
 P' $E= 0.9381$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.0138$
 q_Y $E= 2.0365$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= -2.0365$

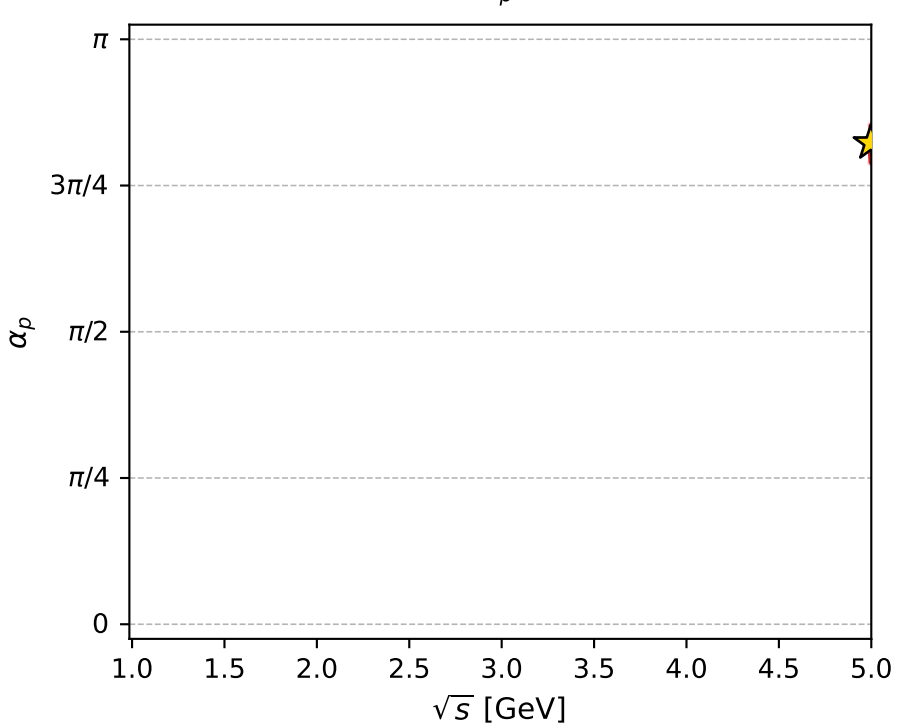
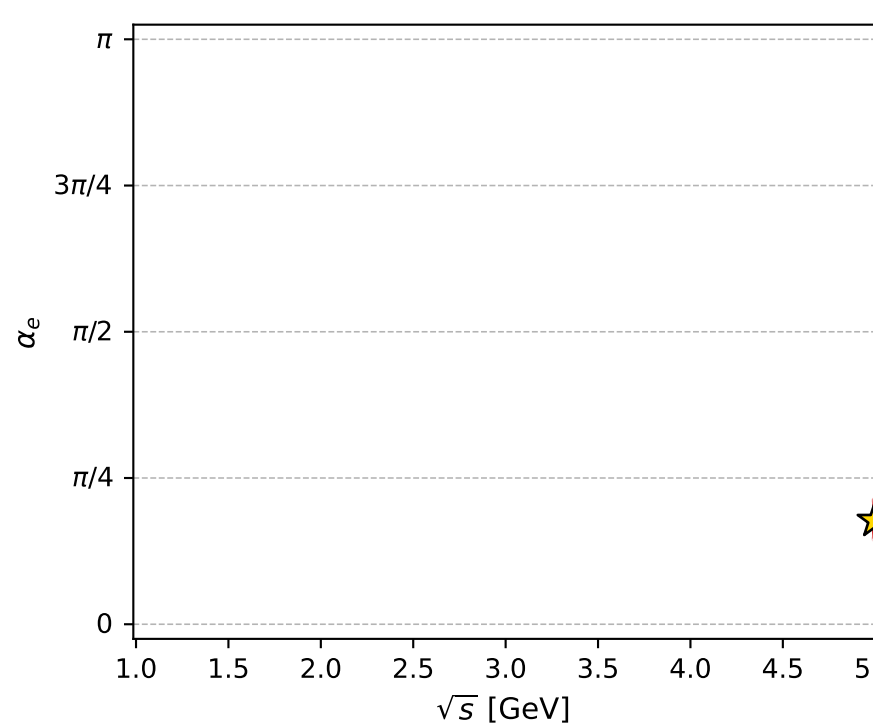
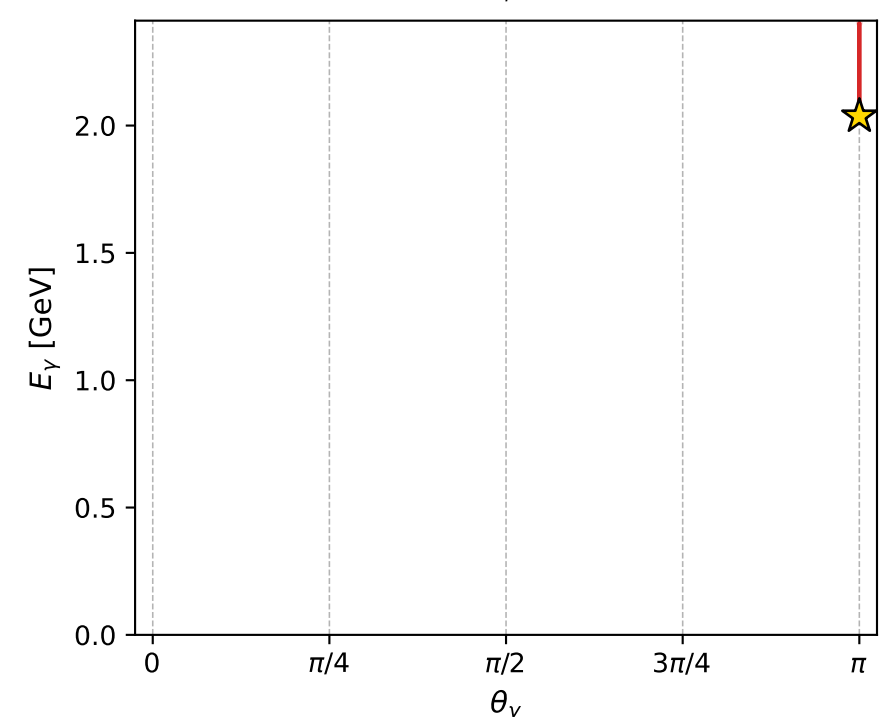
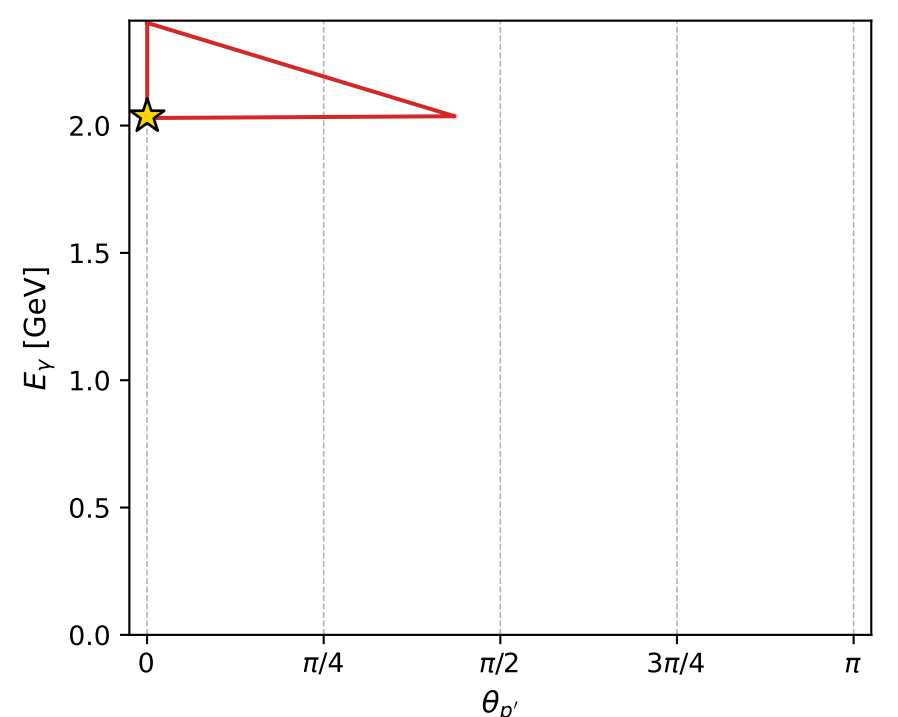
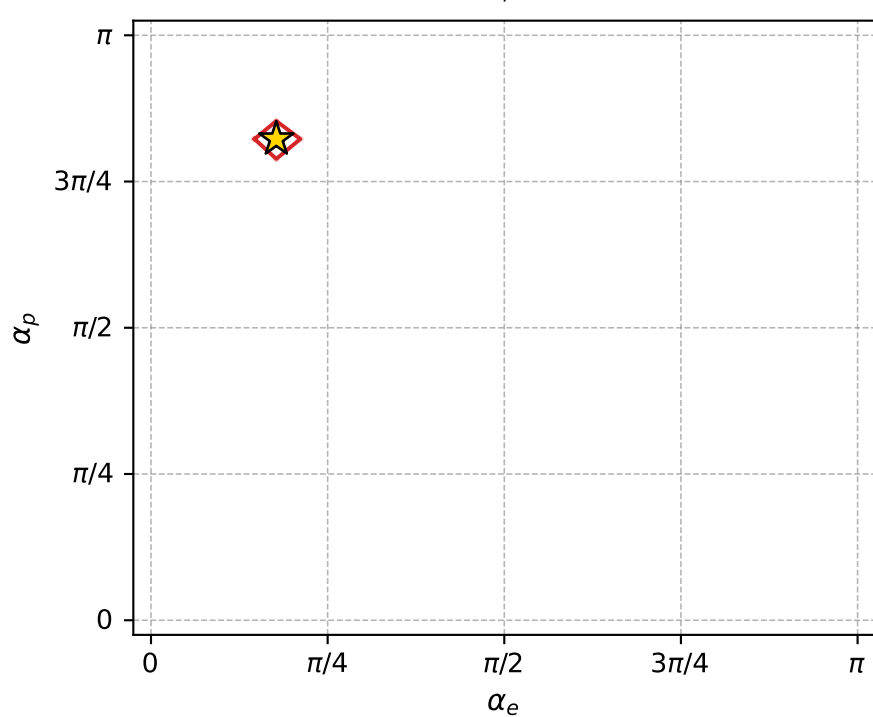
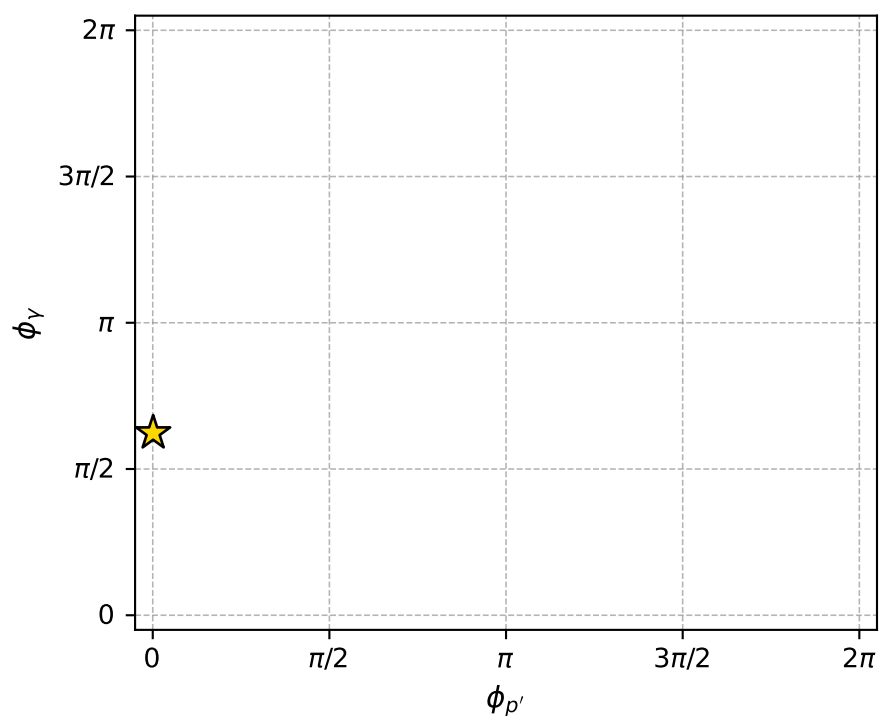
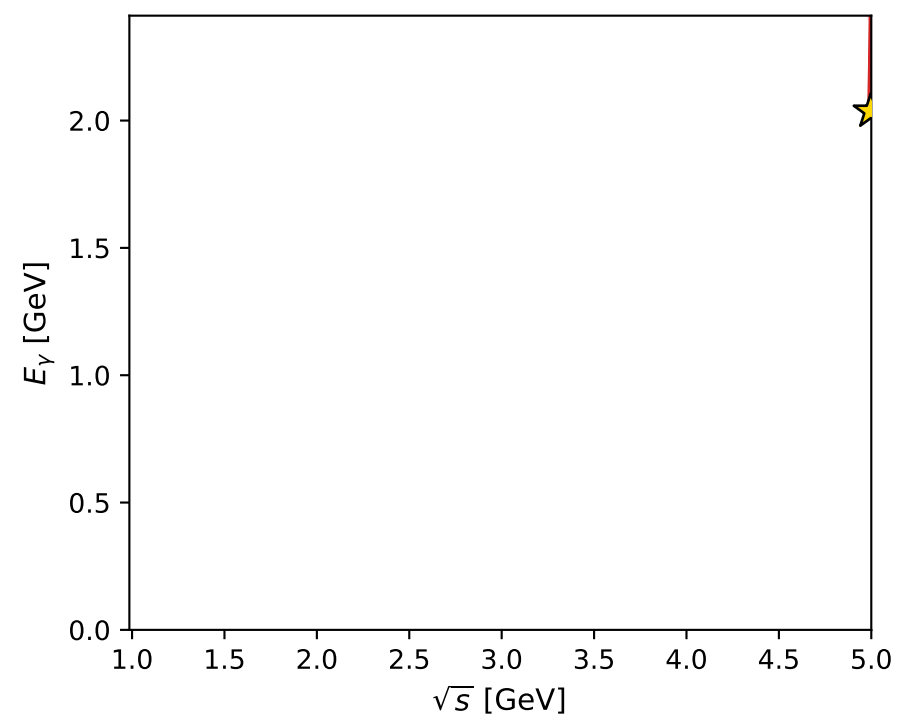
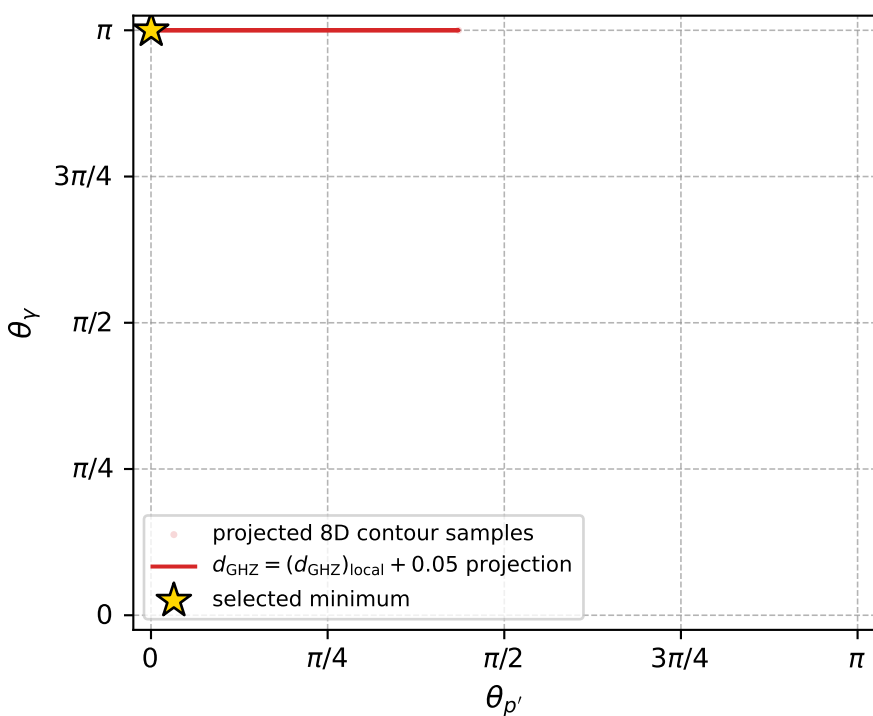


$(h_p, h_Y, h_{\ell}) = (+1, +1, -1)$

$(h_p, h_Y, h_{\ell}) = (-1, -1, +1)$



electron: polarization cluster P2, local minimum 78; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 2.9700713\text{e-}08$
 $d_{\text{GHZ}} \text{ contour} = 0.05000003$
 above global minimum = $7.4172516\text{e-}09$
 8D contour samples = 117

selected phase-space point:

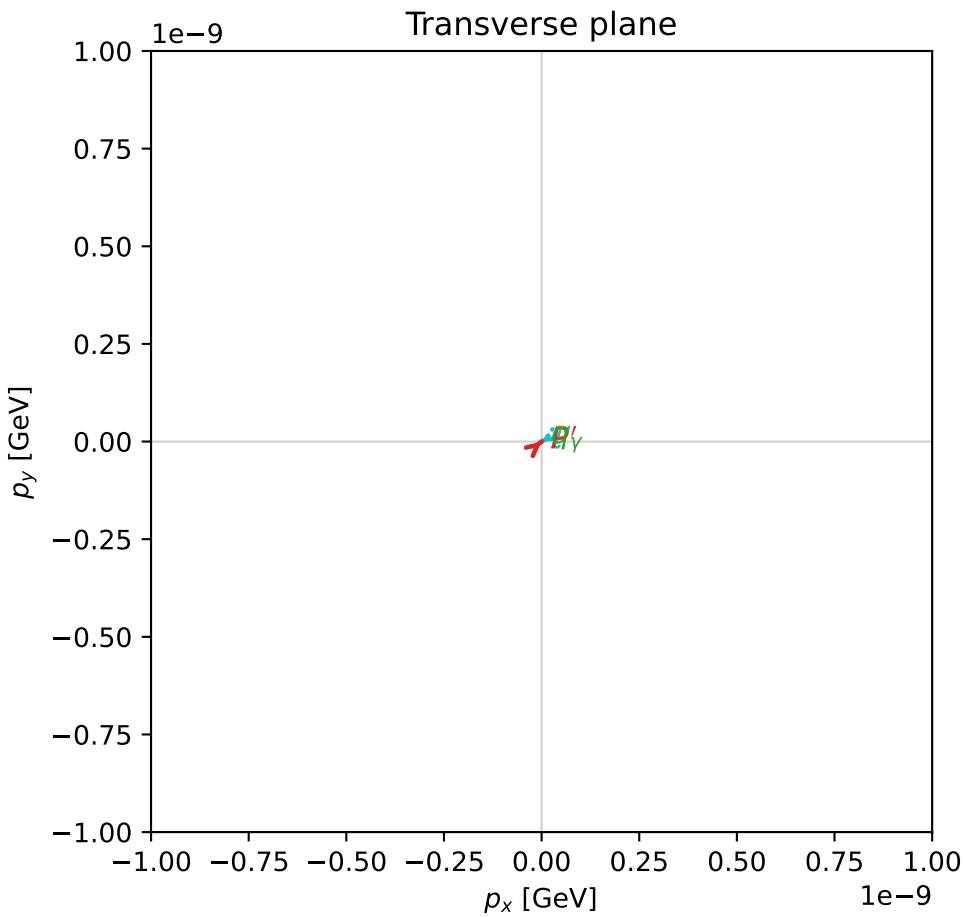
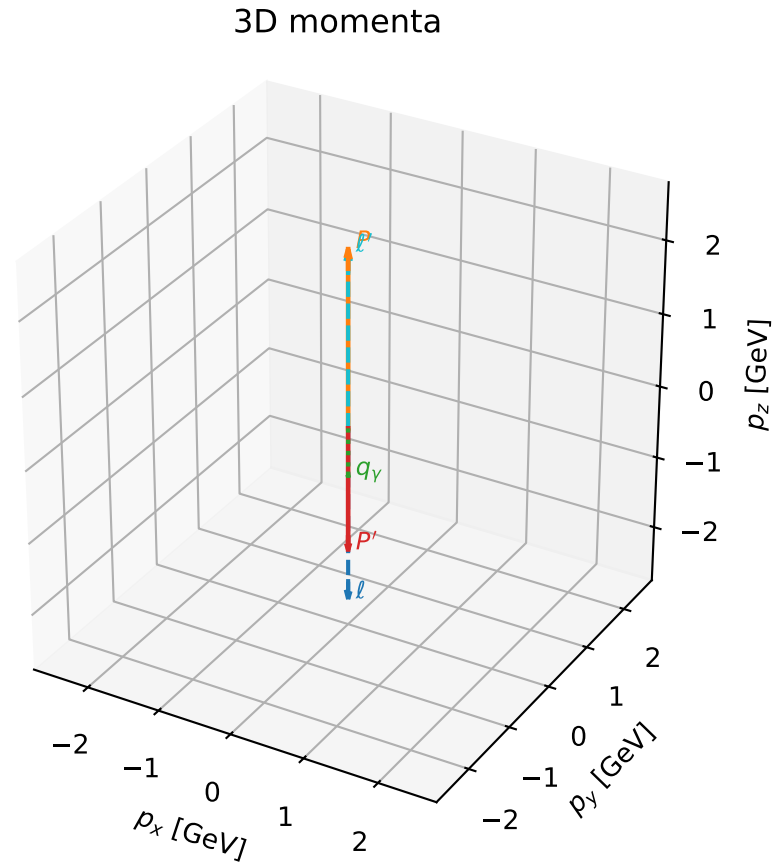
$\text{sqrt_s} = 4.997391$
 $\text{theta_p_out} = 0.000837323$
 $\text{theta_gamma_out} = 3.141593$
 $\text{q0ut} = 2.036539$
 $\text{phi_p_out} = 0.003515132$
 $\text{phi_gamma_out} = 1.958396$
 $\text{alpha_e} = 0.5571307$
 $\text{alpha_p} = 2.584442$

coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

dghz_mixing_angles_polarization_02_local_minimum_86 d_{\rm GHZ}=3.12213e-08
 region: polarization_cluster_02_local_minimum_86
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=1.1624483$ rad
 incoming lepton: $0.397094|+\rangle +0.917778|-\rangle$
 proton mixing angle: $\alpha_p=1.9791355$ rad
 incoming proton: $-0.397086|+\rangle +0.917782|-\rangle$

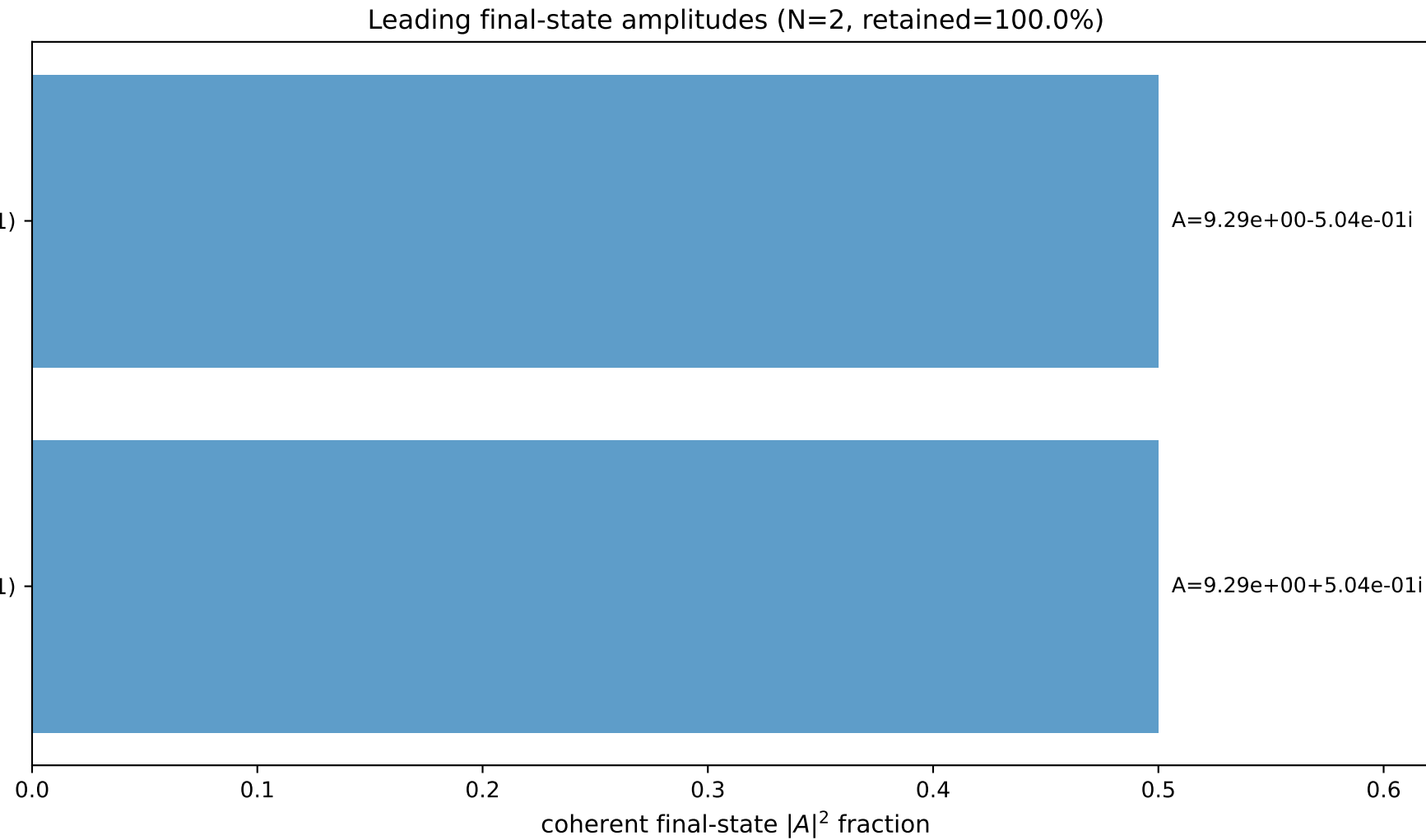
$s=25$, $\sqrt{s}=5$, $|\vec{P}|=2.41202$, $|\vec{P}'|=1.69806$
 $\theta_{P'}=3.14159$, $\theta_Y=3.14159$, $\phi_{P'}=0.694775$, $\phi_Y=0.0541782$
 $E_Y=0.681015$, $\theta_{\ell'}=0$, $\phi_{\ell'}=3.65694$
 $Q^2=22.9535$, $x_B=0.985852$, $t=-16.4727$, $W^2=1.20925$, $y=0.965287$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4120$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 P $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4120$
 ℓ' $E= 2.3791$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 2.3791$
 P' $E= 1.9399$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.6981$
 q_Y $E= 0.6810$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.6810$

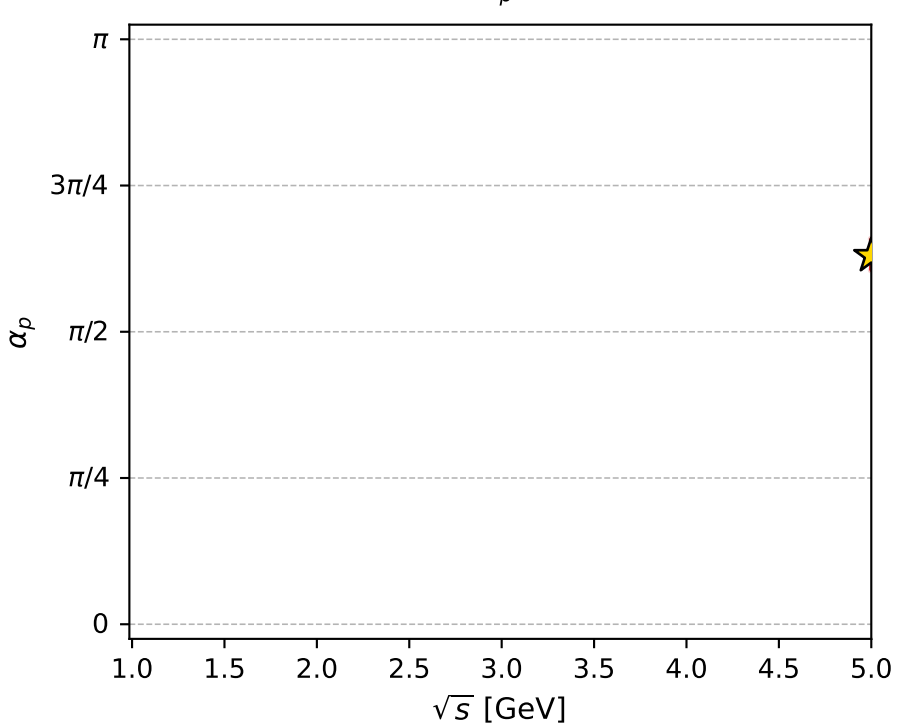
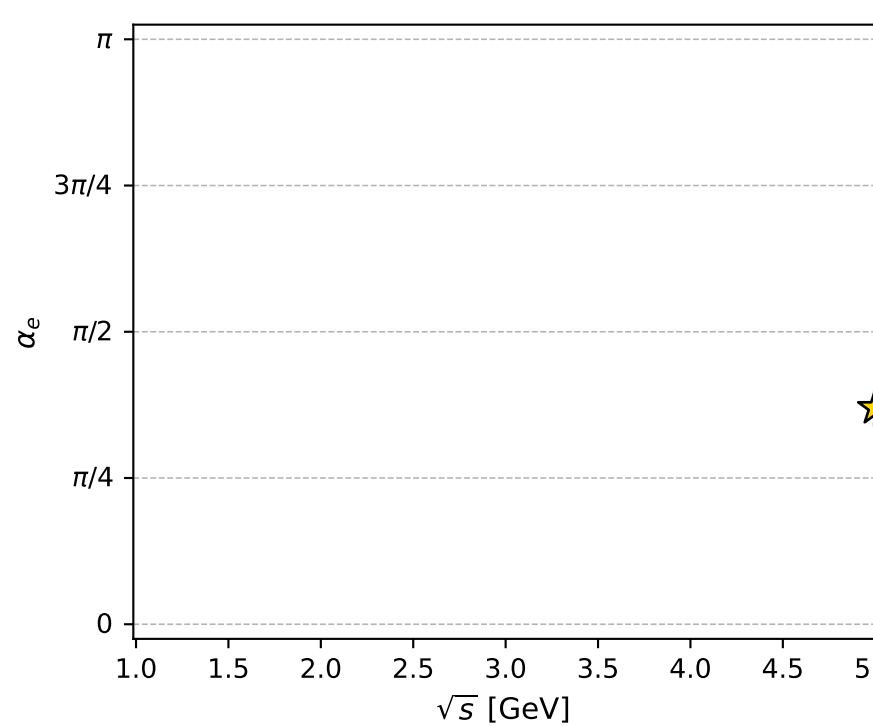
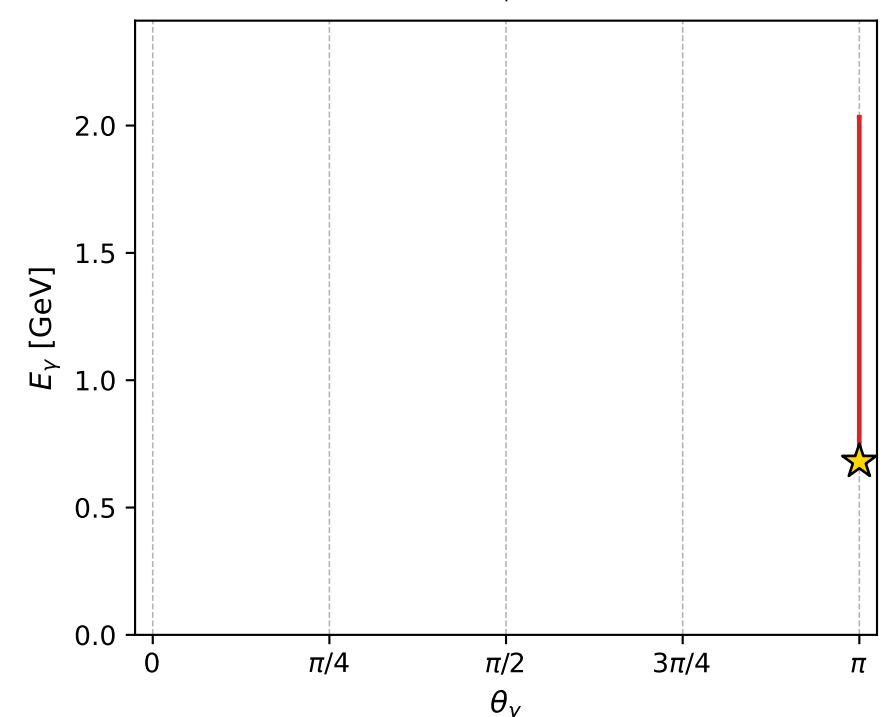
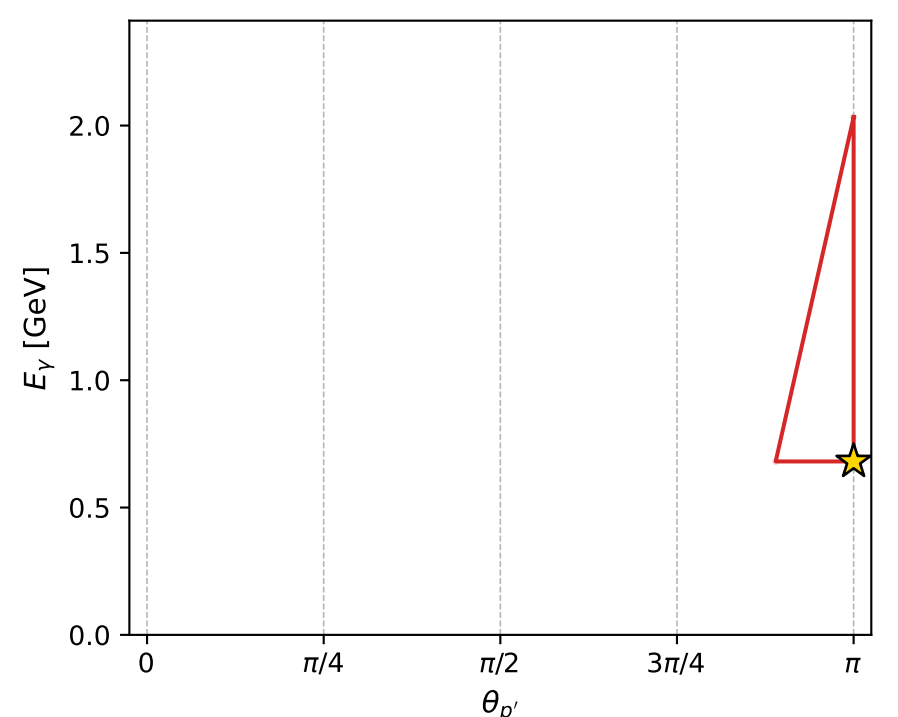
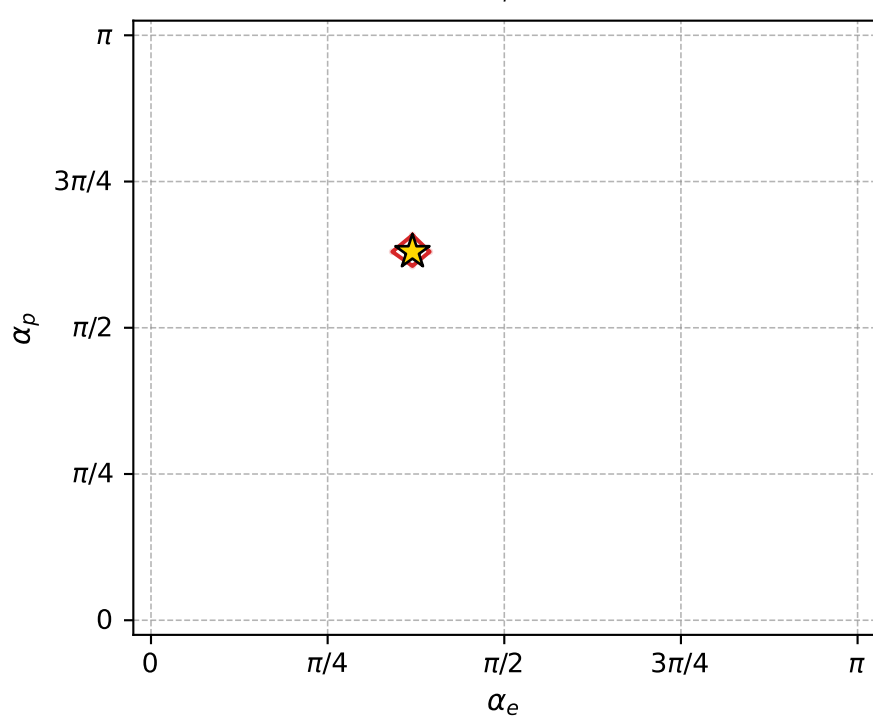
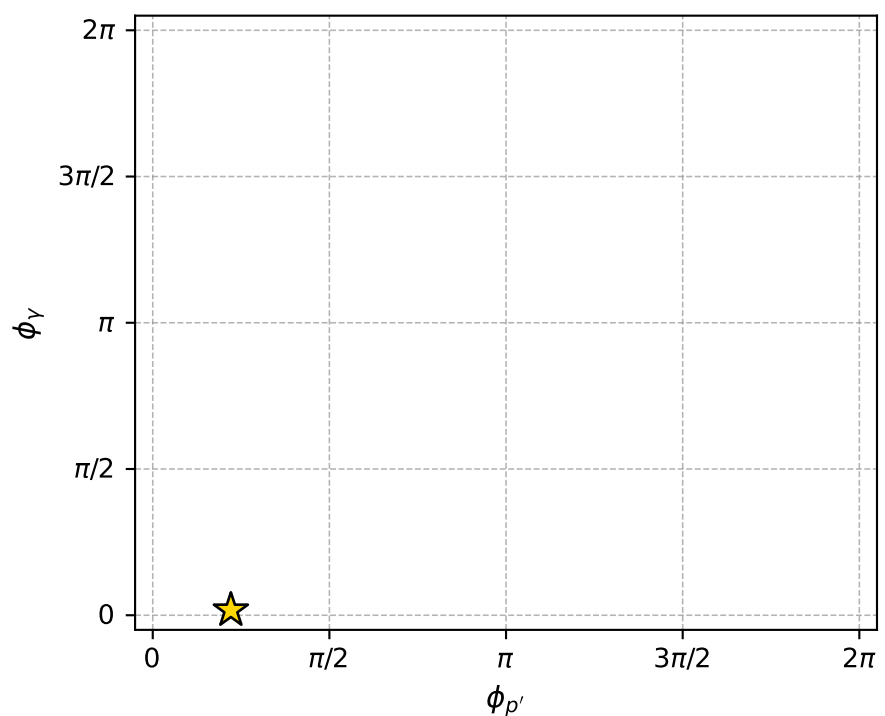
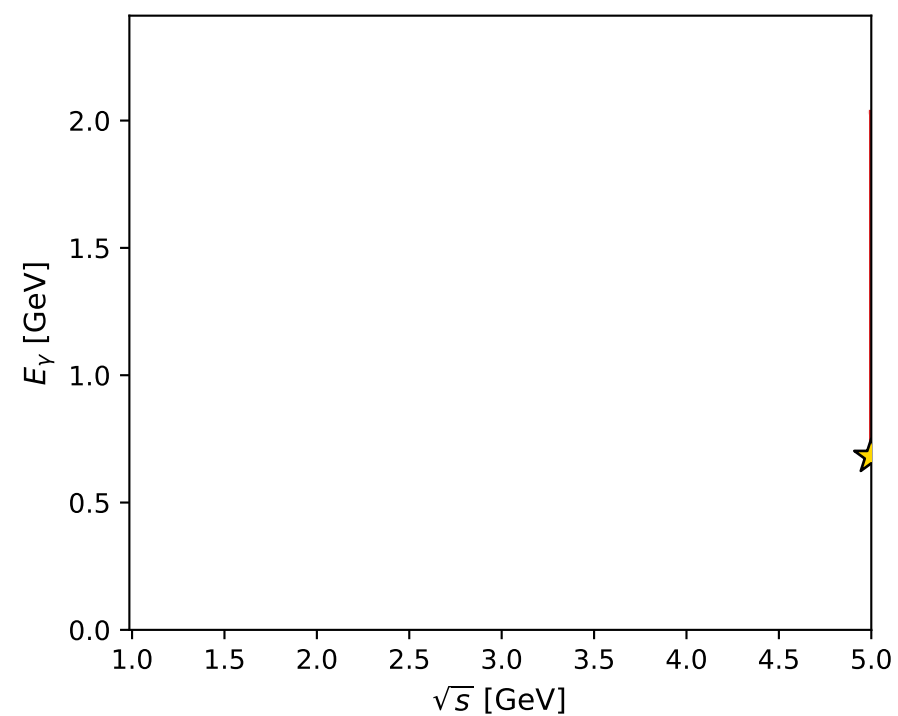
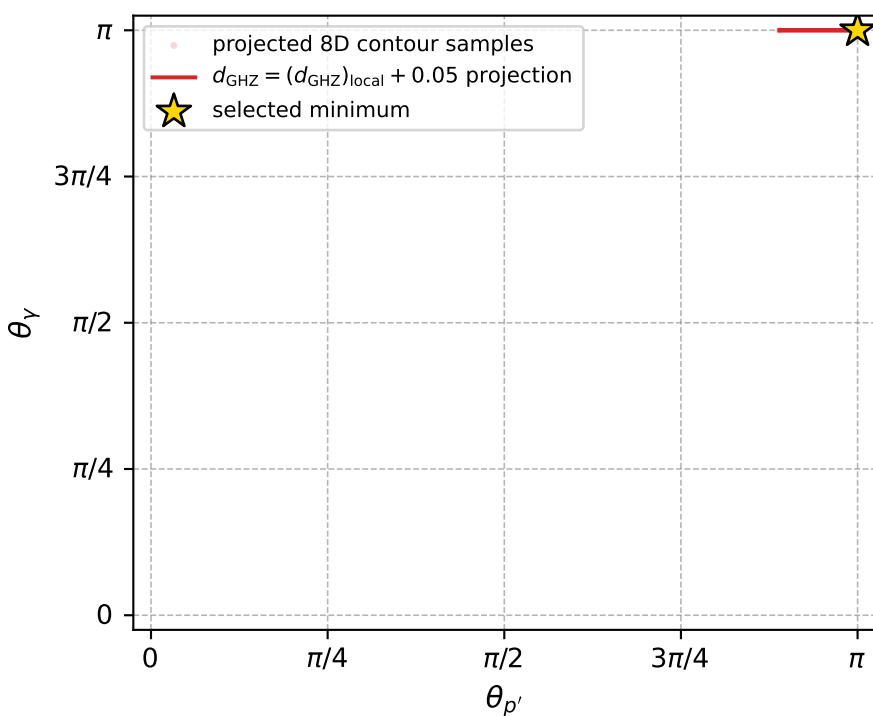


$(h_p, h_Y, h_{\ell}) = (-1, +1, -1)$

$(h_p, h_Y, h_{\ell}) = (+1, -1, +1)$



electron: polarization cluster P2, local minimum 86; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 3.1221313\text{e-}08$
 $d_{\text{GHZ}} \text{ contour} = 0.050000031$
 above global minimum = $8.9378508\text{e-}09$
 8D contour samples = 39

selected phase-space point:

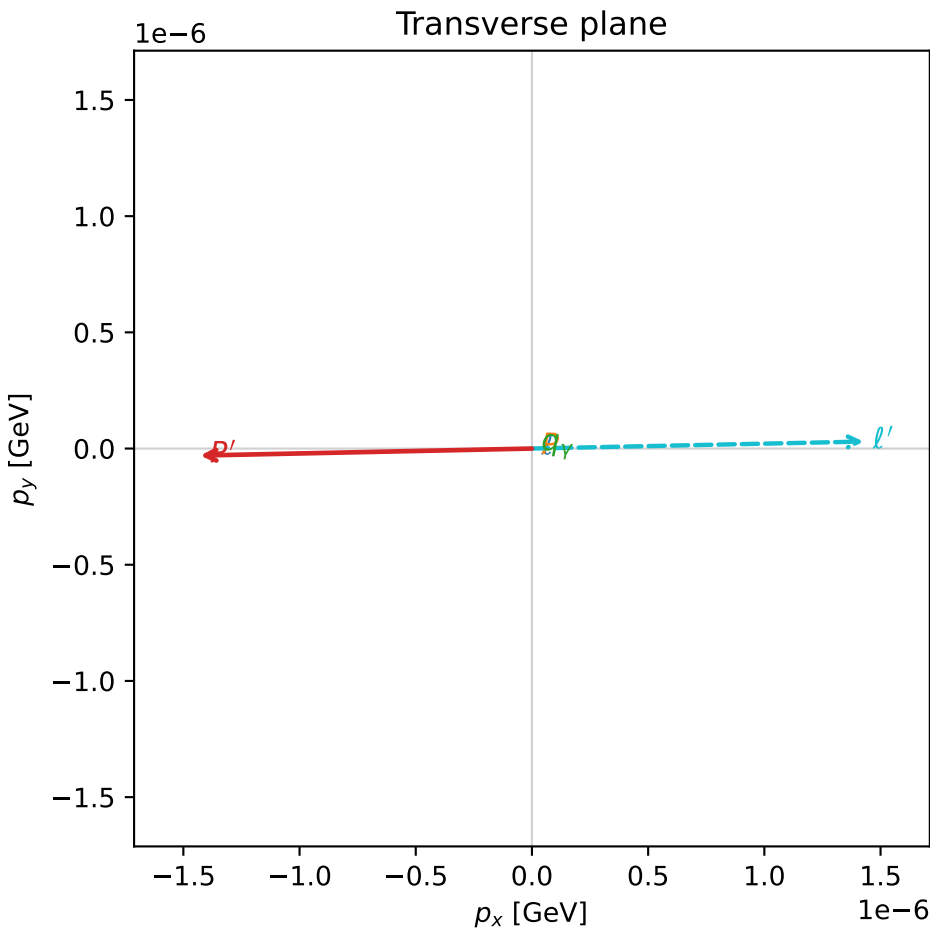
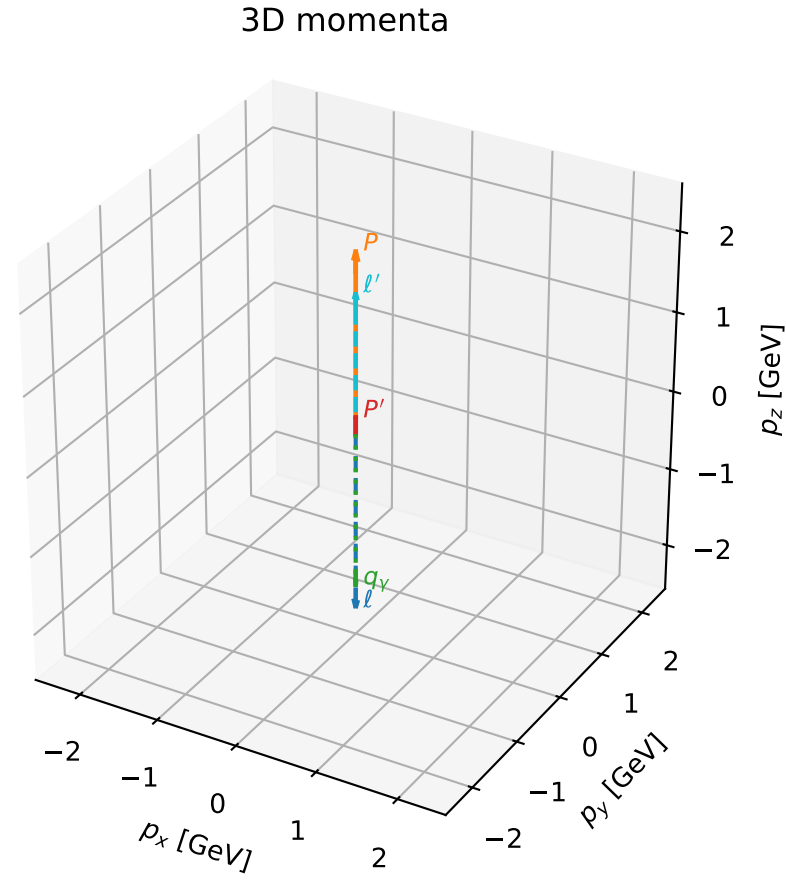
$\text{sqrt_s} = 5$
 $\text{theta_p_out} = 3.141593$
 $\text{theta_gamma_out} = 3.141593$
 $\text{q0out} = 0.6810148$
 $\text{phi_p_out} = 0.6947754$
 $\text{phi_gamma_out} = 0.05417817$
 $\text{alpha_e} = 1.162448$
 $\text{alpha_p} = 1.979136$

coherent mixing angles: minimum d_{GHz} configuration (gradient_local_minimum)

dghz_mixing_angles_polarization_02_local_minimum_90 $d_{\{\text{rm GHz}\}}=3.23249\text{e-}08$
 region: polarization_cluster_02_local_minimum_90
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=0.78715457$ rad
 incoming lepton: $0.705864|+\rangle +0.708348|-\rangle$
 proton mixing angle: $\alpha_p=2.3544342$ rad
 incoming proton: $-0.705861|+\rangle +0.70835|-\rangle$

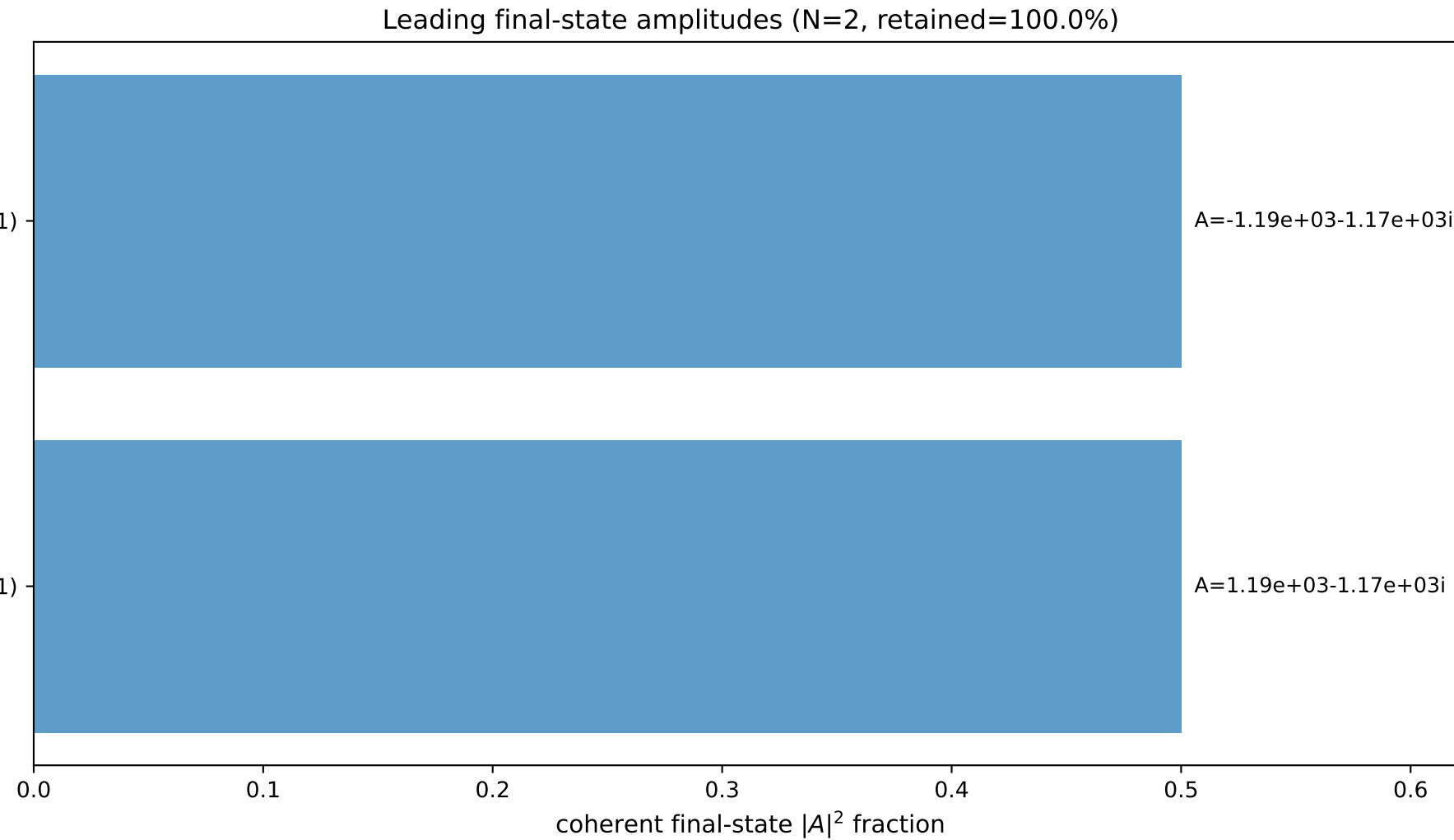
$s=21.9038$, $\sqrt{s}=4.68015$, $|\vec{P}|=2.24608$, $|\vec{P}'|=0.210445$
 $\theta_{P'}=6.7809\text{e-}06$, $\theta_Y=3.14159$, $\phi_{P'}=3.16268$, $\phi_Y=5.50499$
 $E_Y=1.96464$, $\theta_{\ell'}=8.13445\text{e-}07$, $\phi_{\ell'}=0.0210885$
 $Q^2=15.7602$, $x_B=0.773911$, $t=-1.97479$, $W^2=5.48402$, $y=0.968628$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.2461$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.2461$
 P $E= 2.4341$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.2461$
 ℓ' $E= 1.7542$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.7542$
 P' $E= 0.9613$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 0.2104$
 q_Y $E= 1.9646$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= -1.9646$

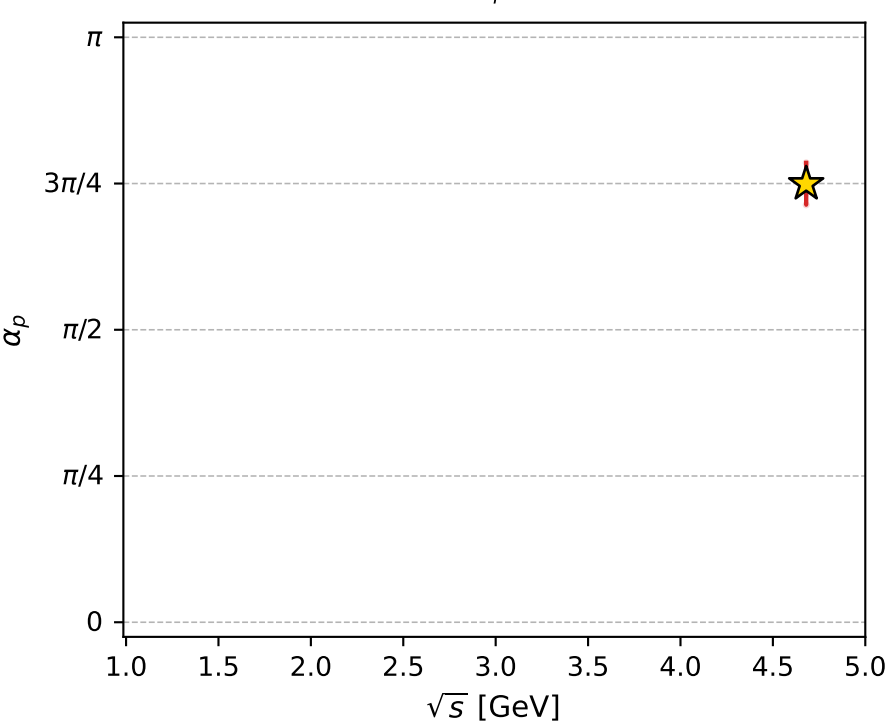
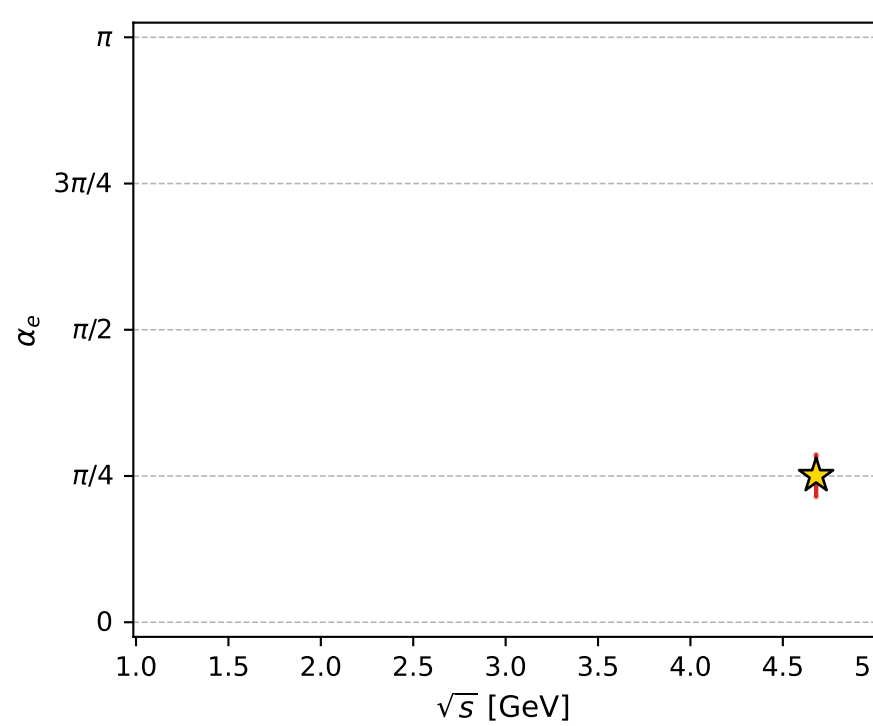
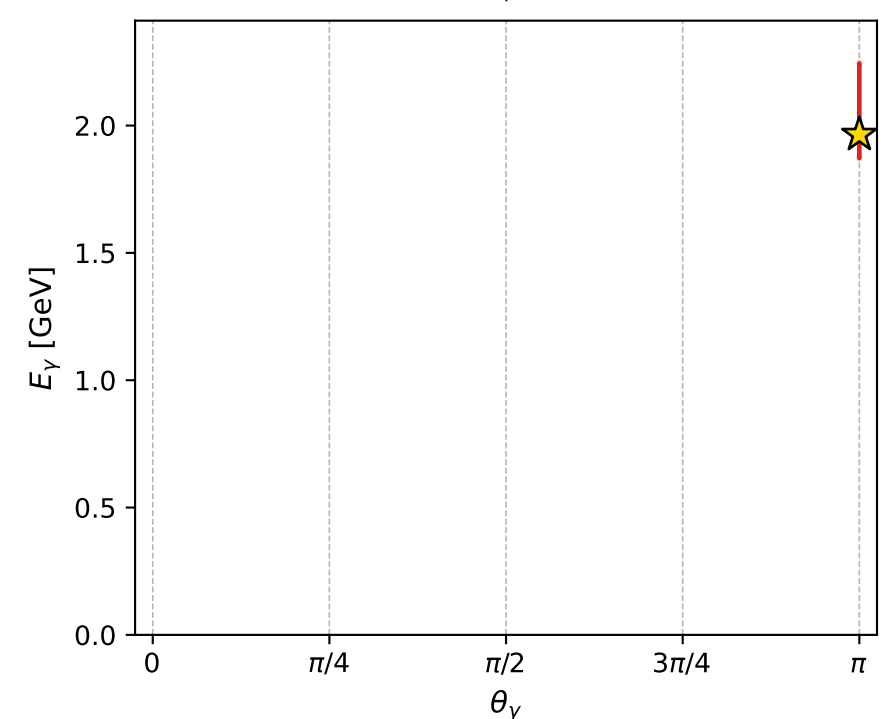
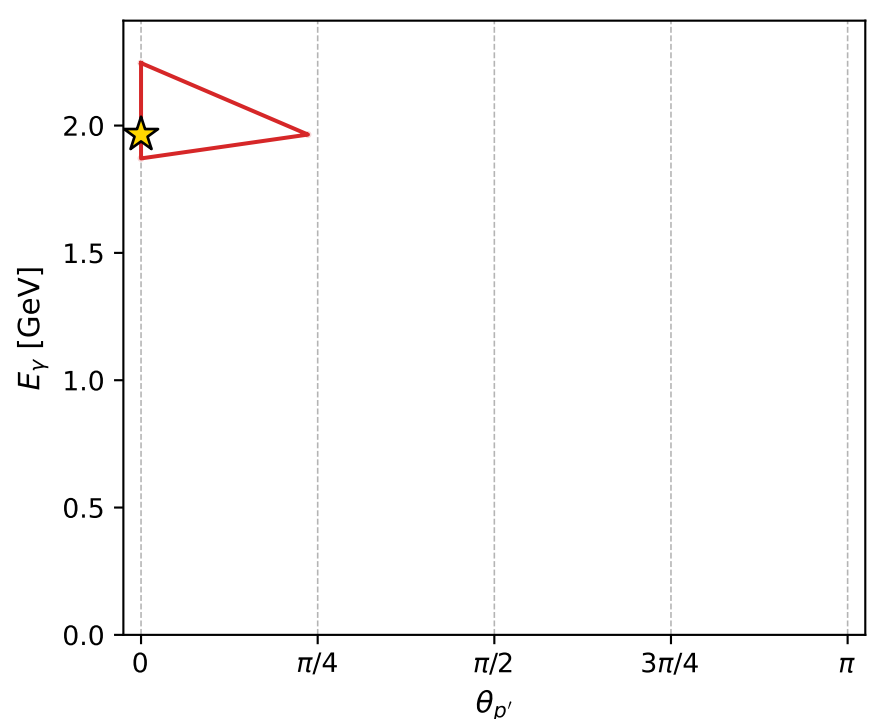
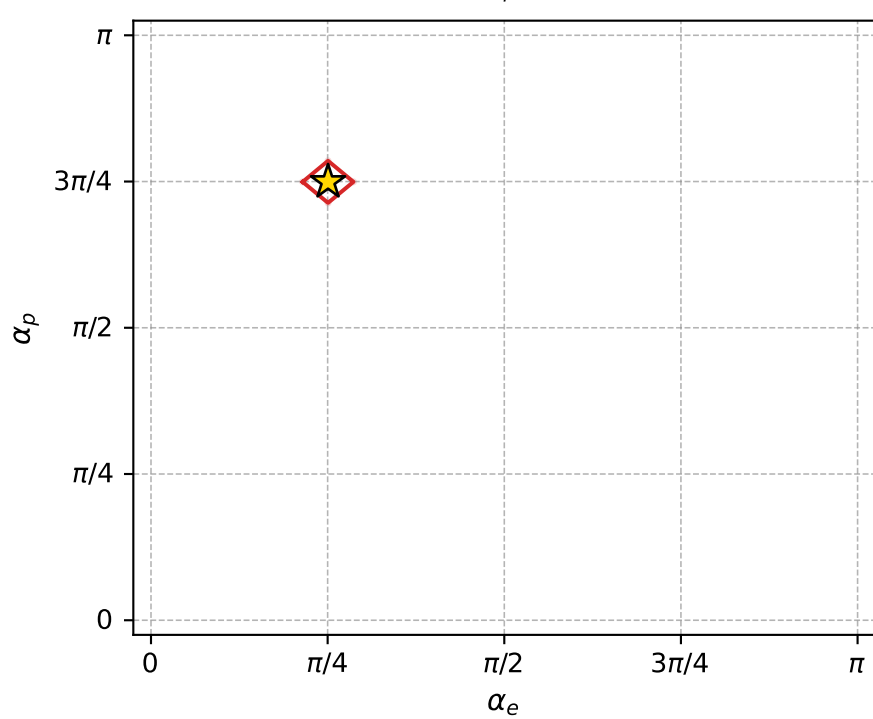
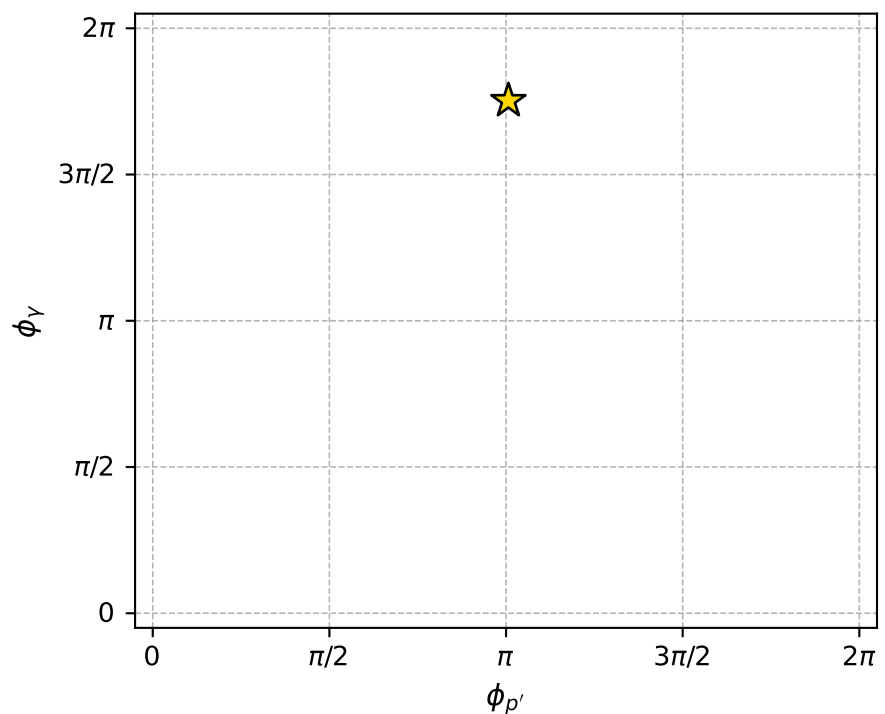
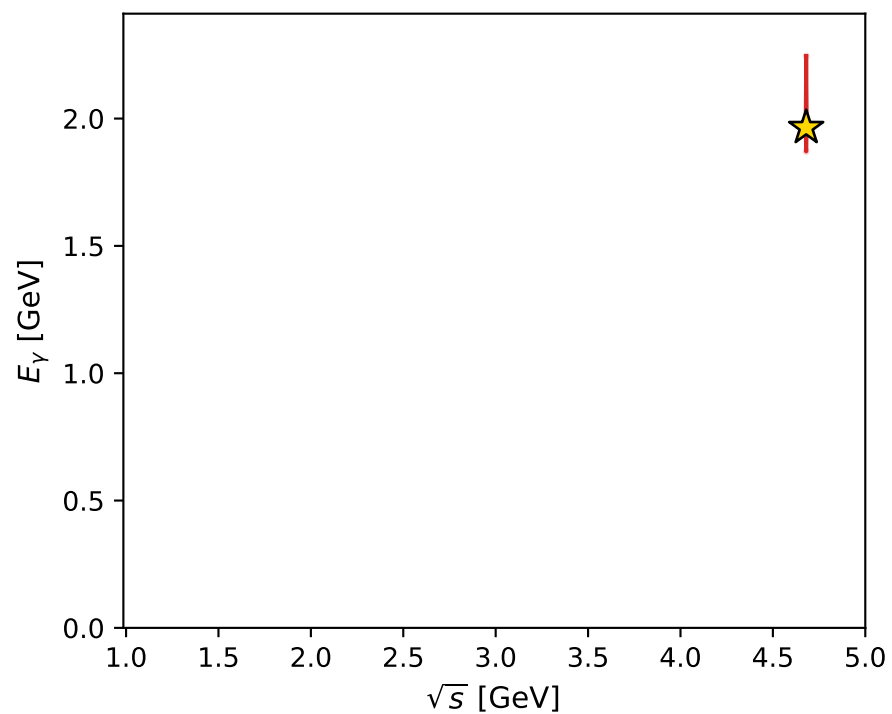
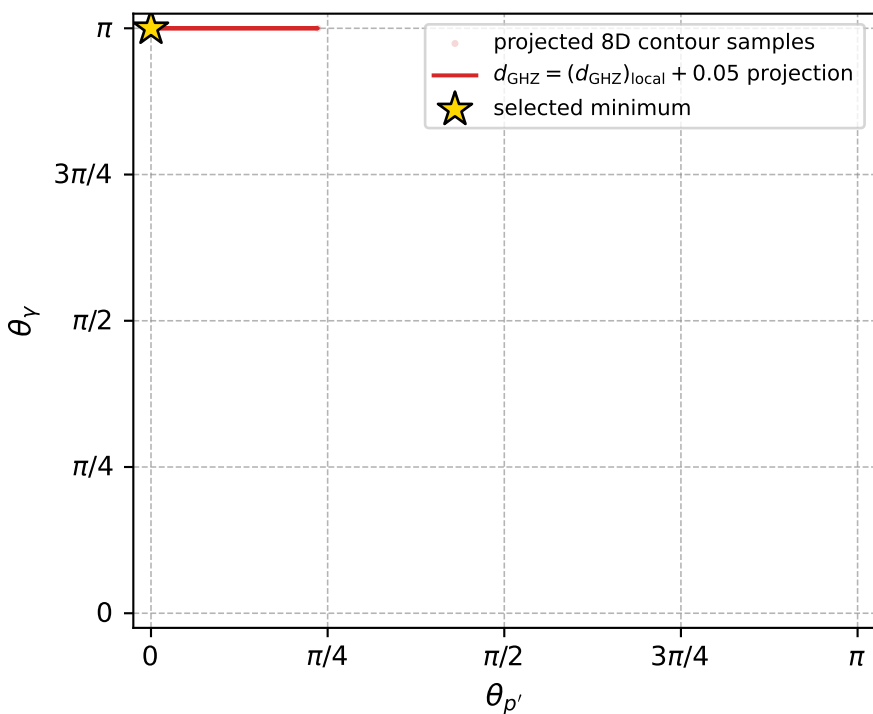


$(h_p, h_Y, h_{\ell}) = (+1, +1, -1)$

$(h_p, h_Y, h_{\ell}) = (-1, -1, +1)$



electron: polarization cluster P2, local minimum 90; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 3.2324919\text{e-}08$
 $d_{\text{GHZ}} \text{ contour} = 0.050000032$
 above global minimum = $1.0041457\text{e-}08$
 8D contour samples = 75

selected phase-space point:

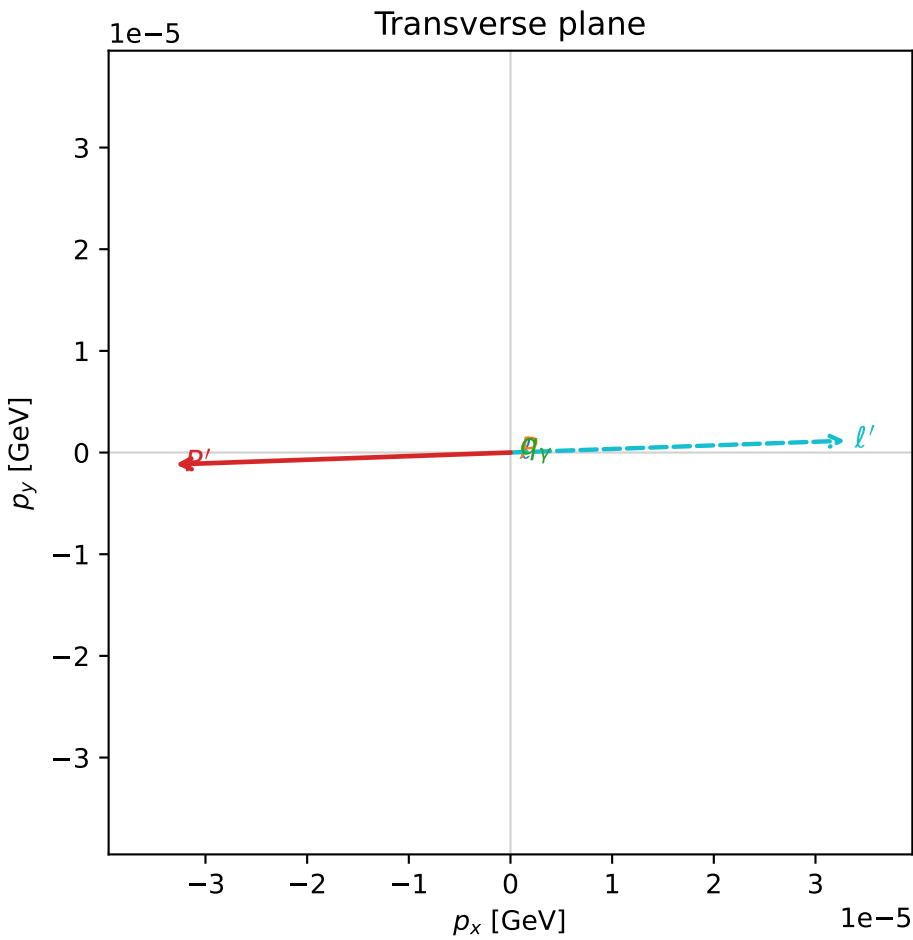
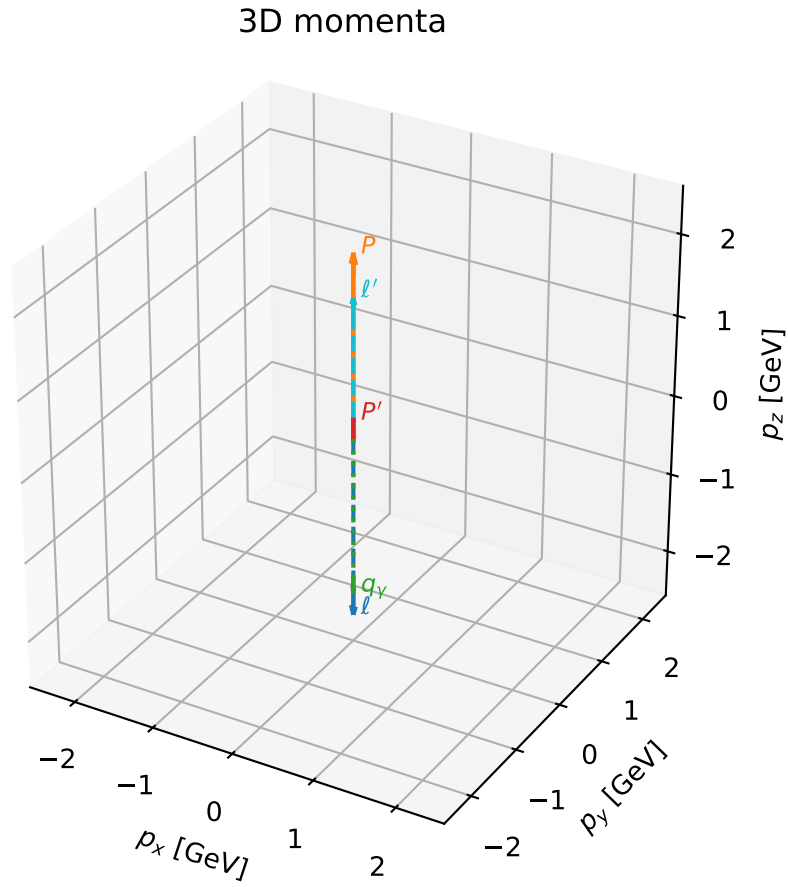
$\text{sqrt_s} = 4.68015$
 $\text{theta_p_out} = 6.780901\text{e-}06$
 $\text{theta_gamma_out} = 3.141593$
 $\text{q0out} = 1.964639$
 $\text{phi_p_out} = 3.162681$
 $\text{phi_gamma_out} = 5.504987$
 $\text{alpha_e} = 0.7871546$
 $\text{alpha_p} = 2.354434$

coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

dghz_mixing_angles_polarization_02_local_minimum_94 $d_{\{\text{rm GHZ}\}}=3.33046\text{e-}08$
region: polarization_cluster_02_local_minimum_94
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: $\alpha_e=0.79992921$ rad
incoming lepton: $0.696757|+\rangle +0.717307|-\rangle$
proton mixing angle: $\alpha_p=2.3416514$ rad
incoming proton: $-0.696749|+\rangle +0.717315|-\rangle$

$s=21.8786$, $\sqrt{s}=4.67746$, $|\vec{P}|=2.24468$, $|\vec{P}'|=0.239827$
 $\theta_{P'}=0.000137432$, $\theta_Y=3.14159$, $\phi_{P'}=3.17695$, $\phi_Y=5.58436$
 $E_Y=1.97455$, $\theta_{\ell'}=1.9\text{e-}05$, $\phi_{\ell'}=0.0353563$
 $Q^2=15.5756$, $x_B=0.765531$, $t=-1.87435$, $W^2=5.65037$, $y=0.968921$

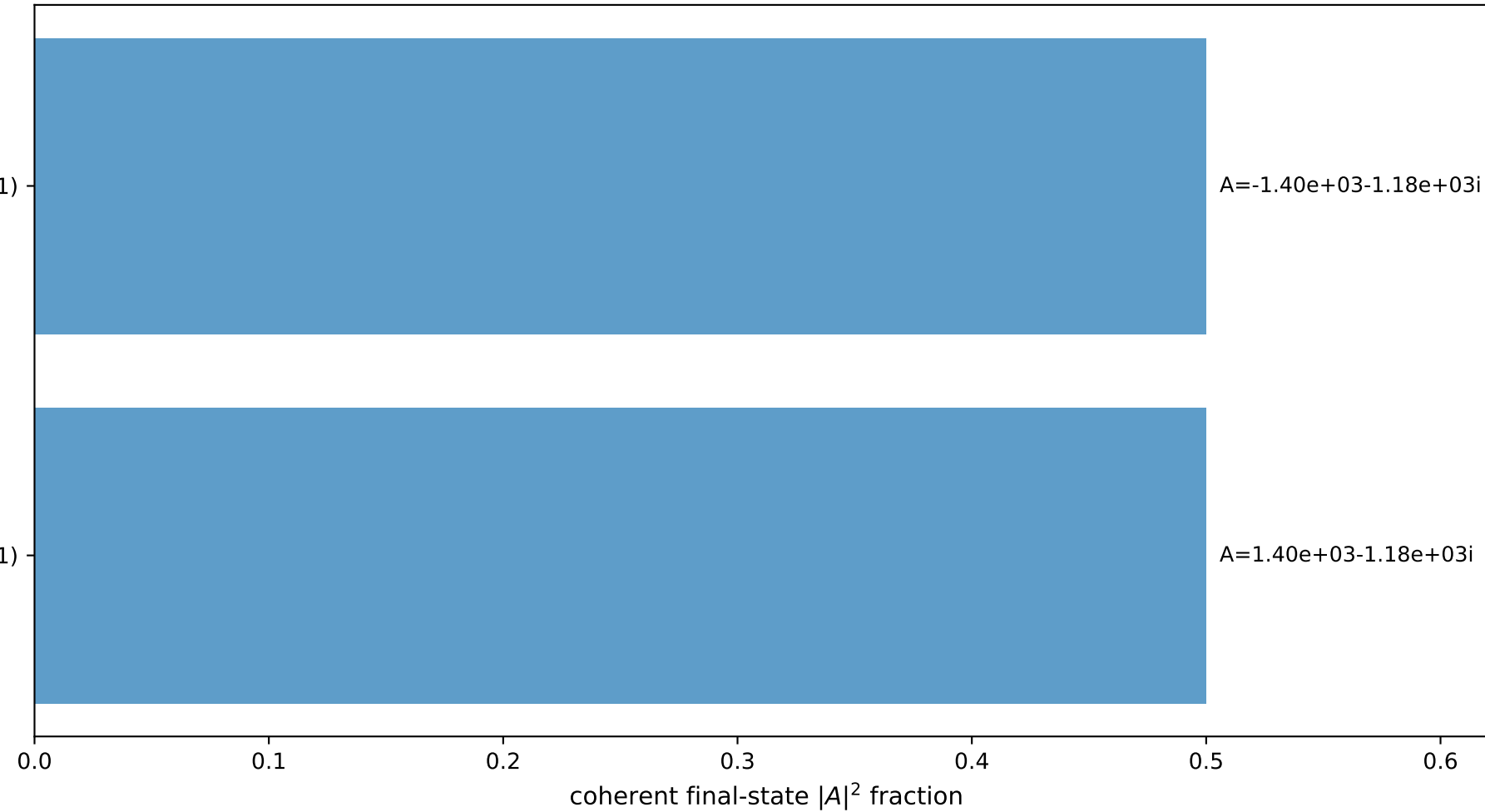
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.2447$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.2447$
 P $E= 2.4328$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.2447$
 ℓ' $E= 1.7347$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.7347$
 P' $E= 0.9682$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 0.2398$
 q_Y $E= 1.9746$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= -1.9746$



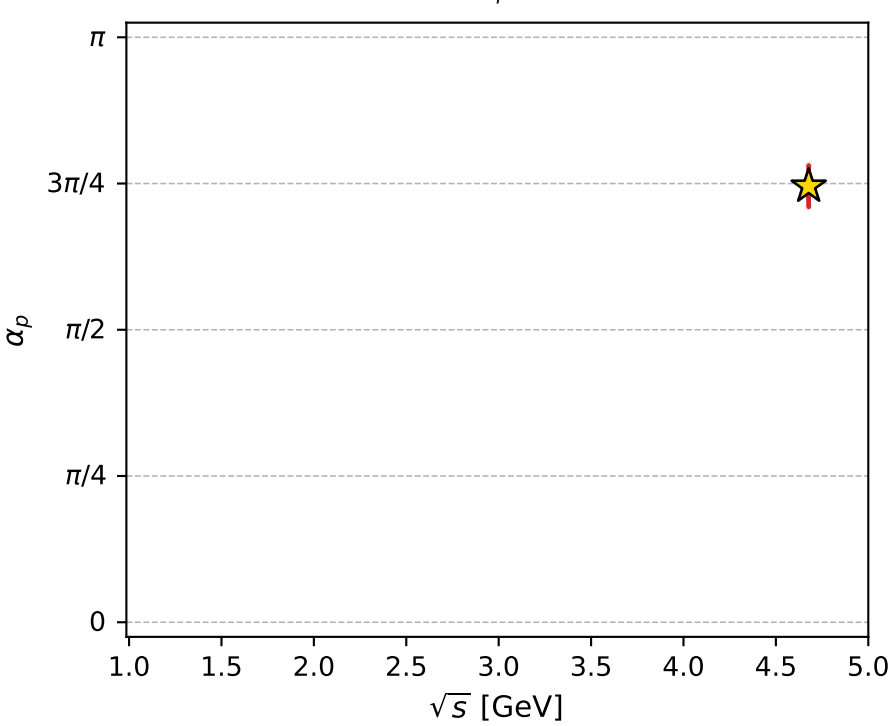
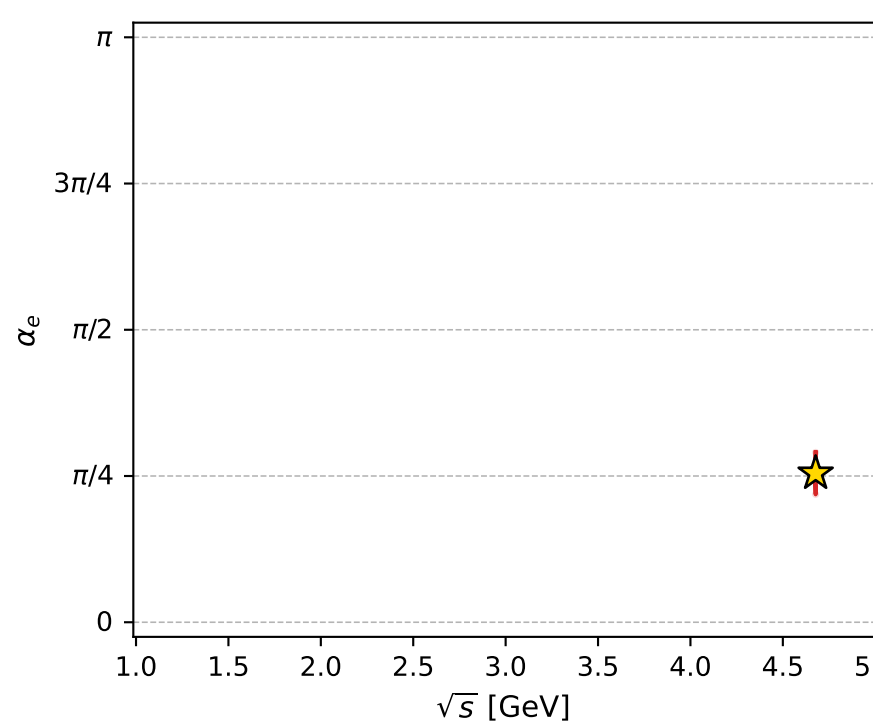
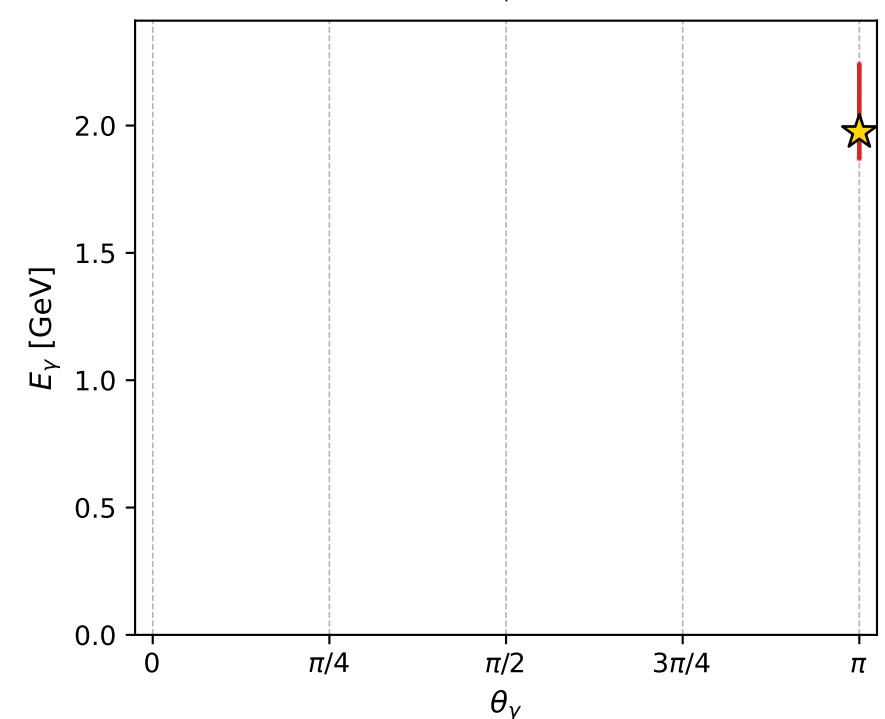
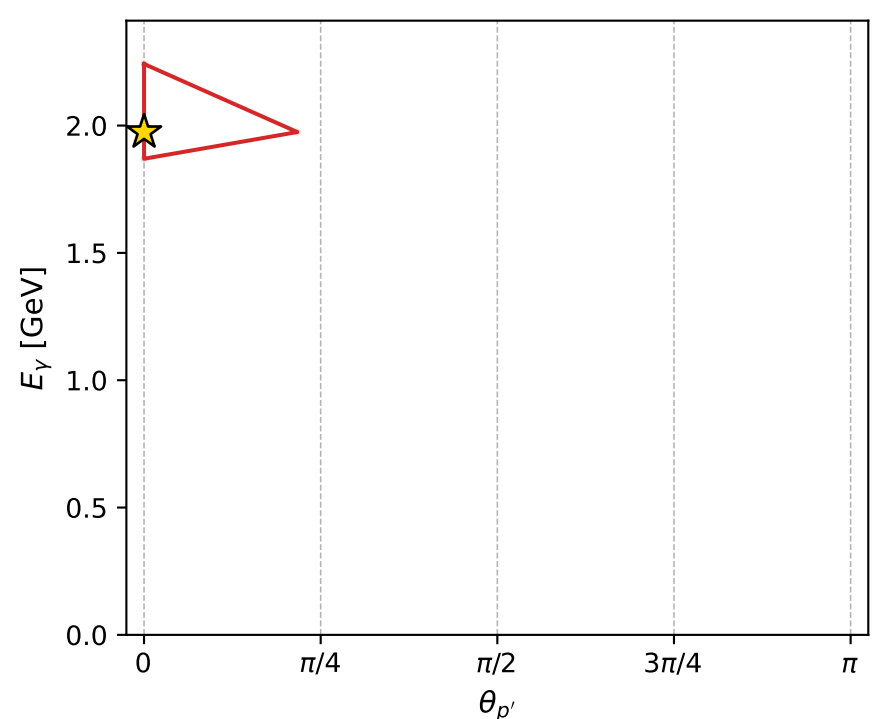
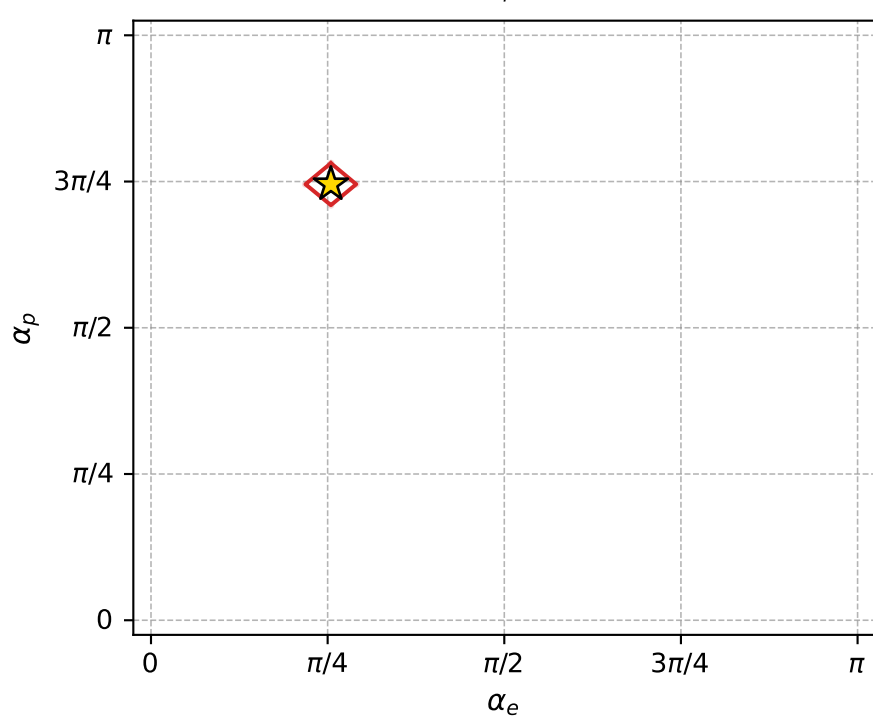
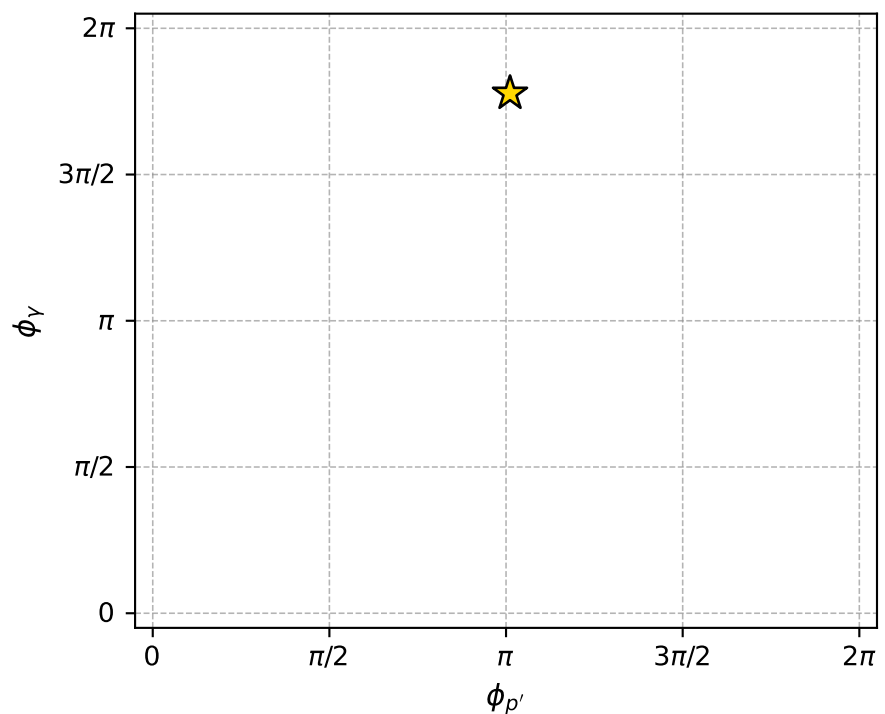
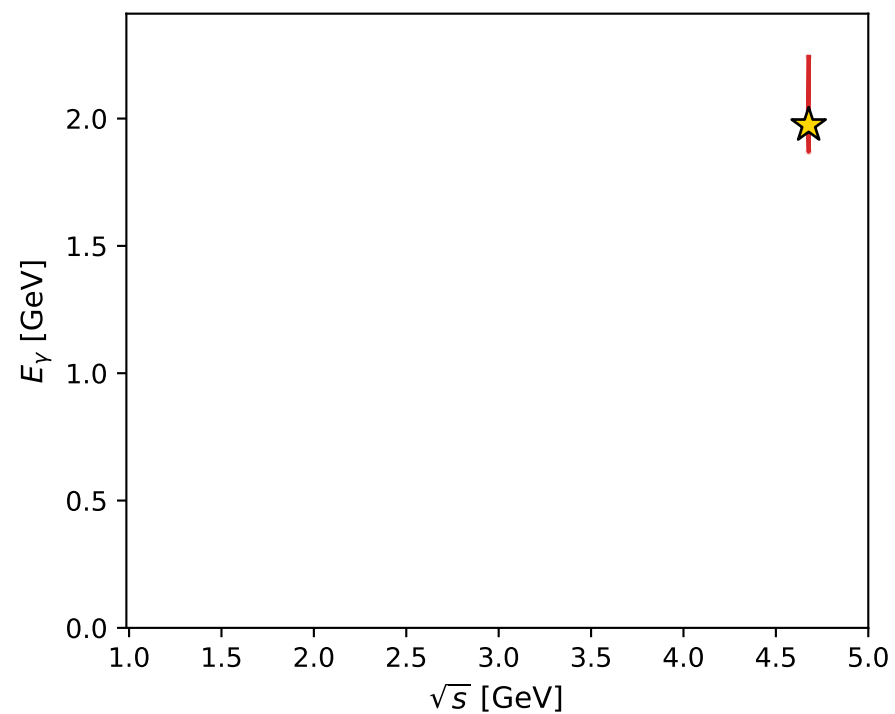
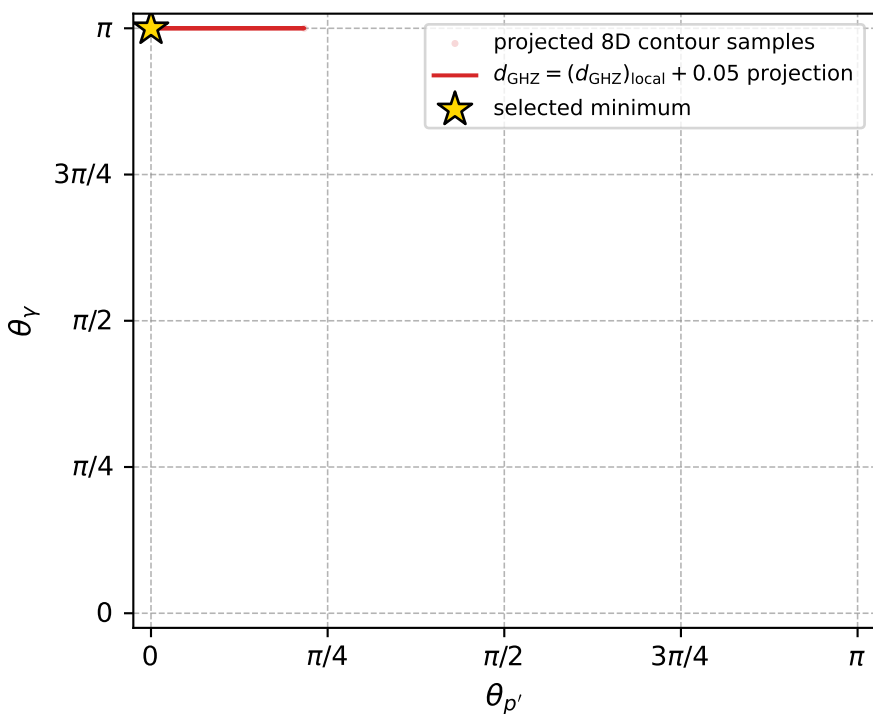
$(h_p, h_Y, h_{\ell}) = (+1, +1, -1)$

$(h_p, h_Y, h_{\ell}) = (-1, -1, +1)$

Leading final-state amplitudes (N=2, retained=100.0%)



electron: polarization cluster P2, local minimum 94; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 3.3304589\text{e-}08$
 $d_{\text{GHZ}} \text{ contour} = 0.050000033$
 above global minimum = $1.1021127\text{e-}08$
 8D contour samples = 105

selected phase-space point:

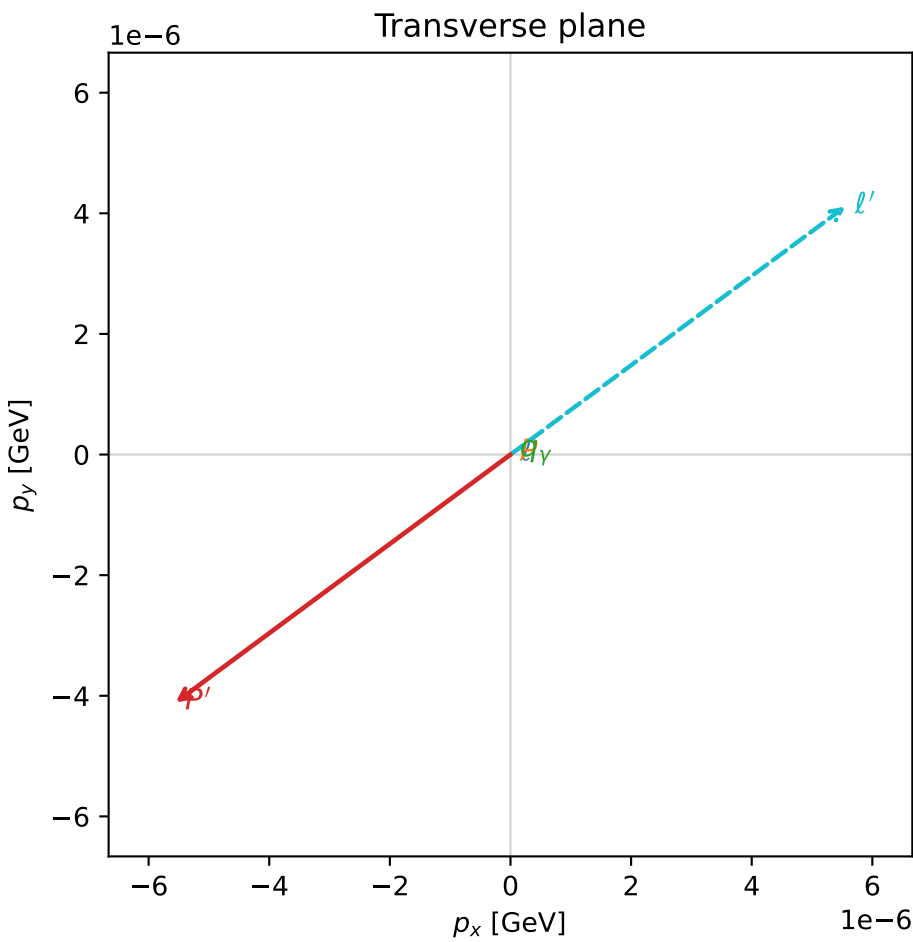
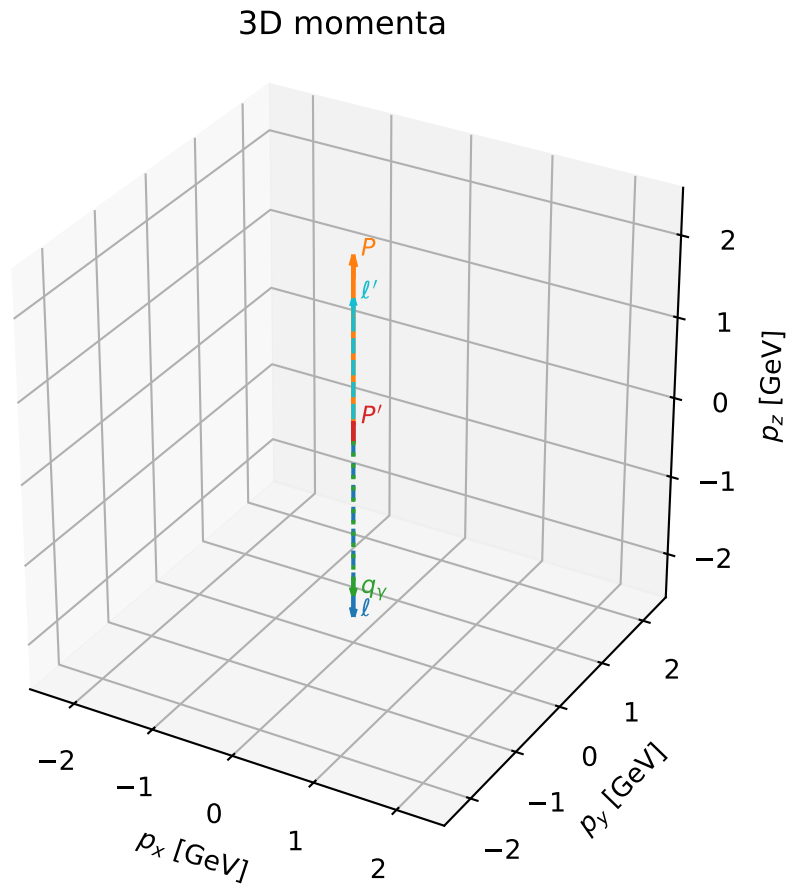
$\text{sqrt_s} = 4.677457$
 $\text{theta_p_out} = 0.0001374315$
 $\text{theta_gamma_out} = 3.141593$
 $\text{q0ut} = 1.974555$
 $\text{phi_p_out} = 3.176949$
 $\text{phi_gamma_out} = 5.584363$
 $\text{alpha_e} = 0.7999292$
 $\text{alpha_p} = 2.341651$

coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

dghz_mixing_angles_polarization_02_local_minimum_96 $d_{\{\text{rm GHZ}\}}=3.42638\text{e-}08$
region: polarization_cluster_02_local_minimum_96
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: $\alpha_e=0.66918829$ rad
incoming lepton: $0.784325|+\rangle +0.62035|-\rangle$
proton mixing angle: $\alpha_p=2.4723833$ rad
incoming proton: $-0.784312|+\rangle +0.620366|-\rangle$

$s=21.6714$, $\sqrt{s}=4.65526$, $|\vec{P}|=2.23313$, $|\vec{P}'|=0.219926$
 $\theta_{P'}=3.14159\text{e-}05$, $\theta_Y=3.14159$, $\phi_{P'}=3.77898$, $\phi_Y=0.317472$
 $E_Y=1.95587$, $\theta_{\ell'}=3.98006\text{e-}06$, $\phi_{\ell'}=0.637385$
 $Q^2=15.5064$, $x_B=0.770105$, $t=-1.9252$, $W^2=5.50887$, $y=0.96844$

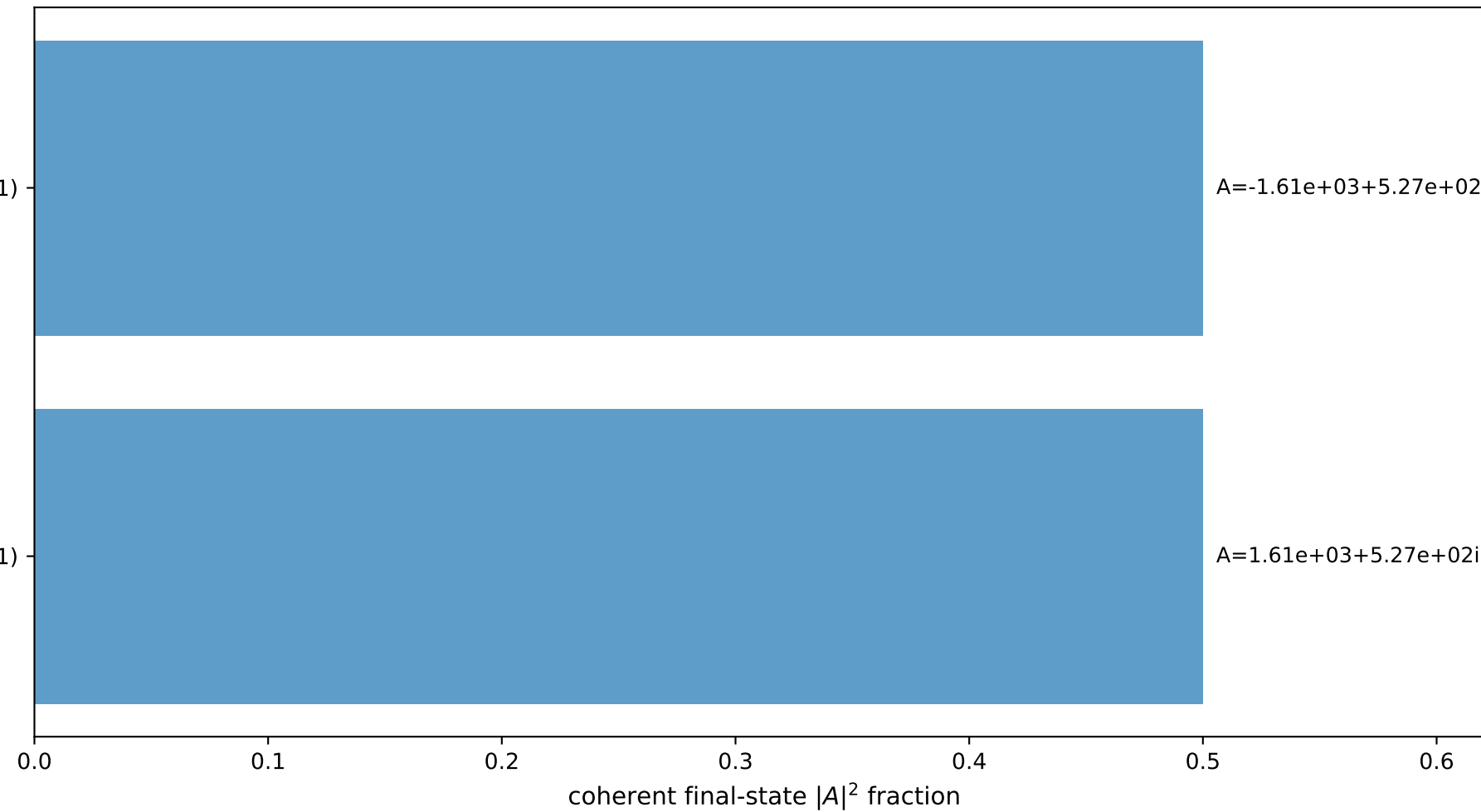
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.2331$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.2331$
 P $E= 2.4221$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.2331$
 ℓ' $E= 1.7359$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.7359$
 P' $E= 0.9634$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 0.2199$
 q_Y $E= 1.9559$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.9559$



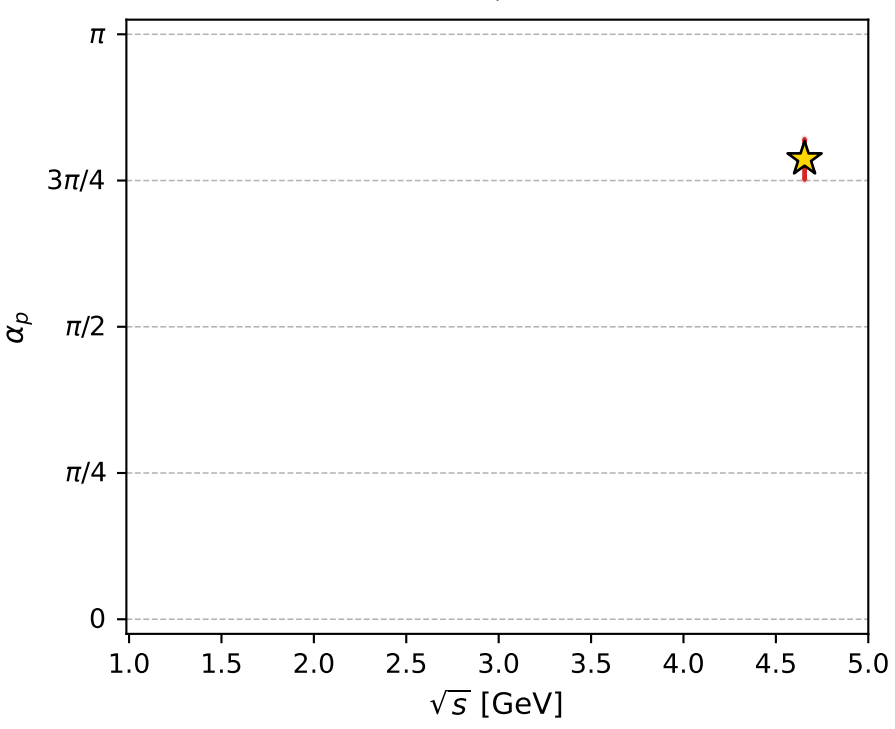
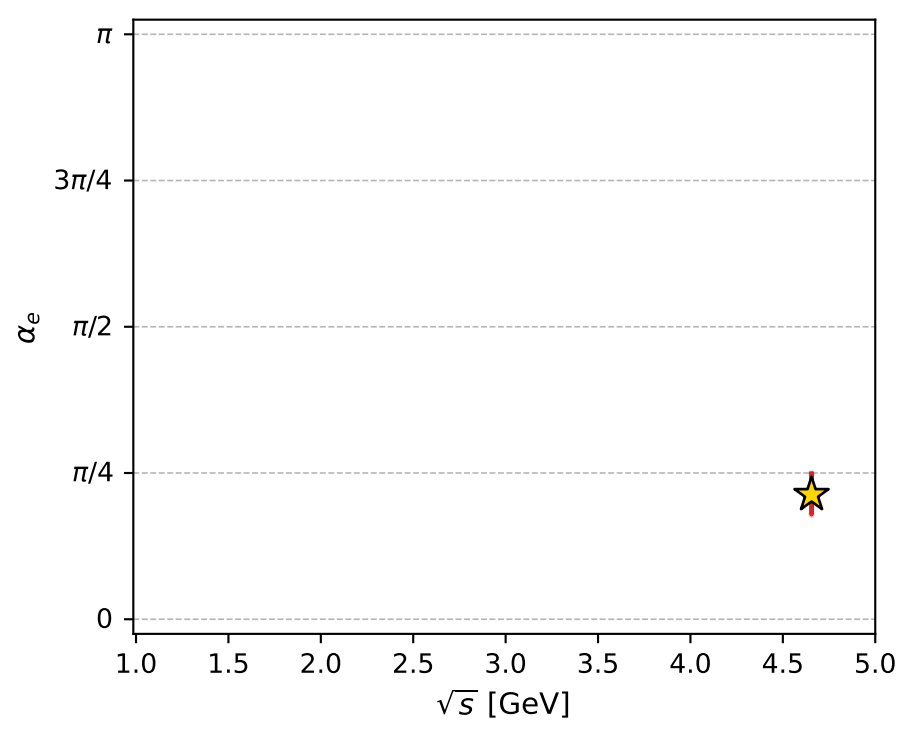
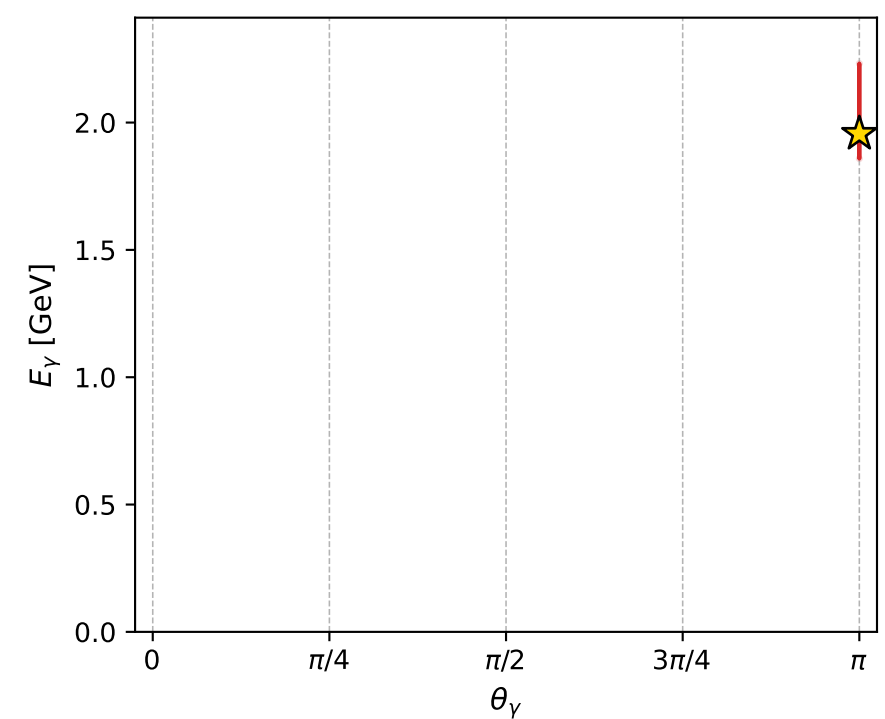
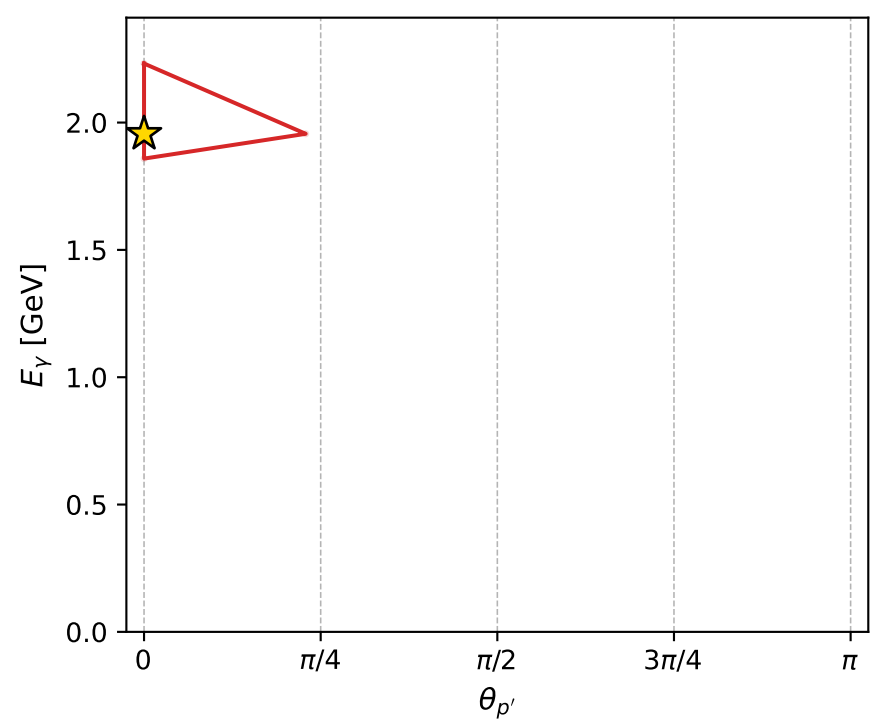
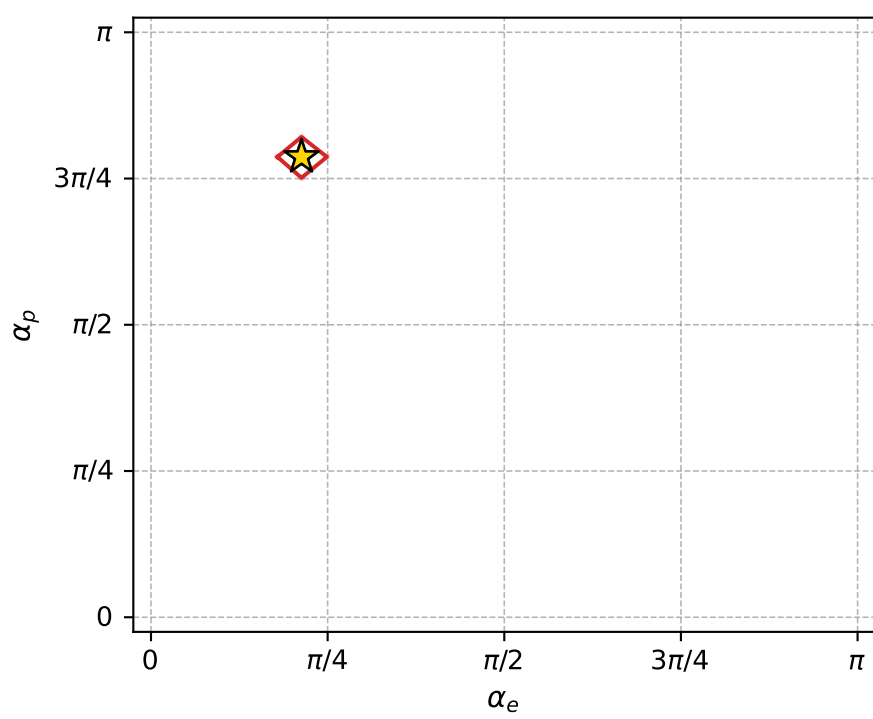
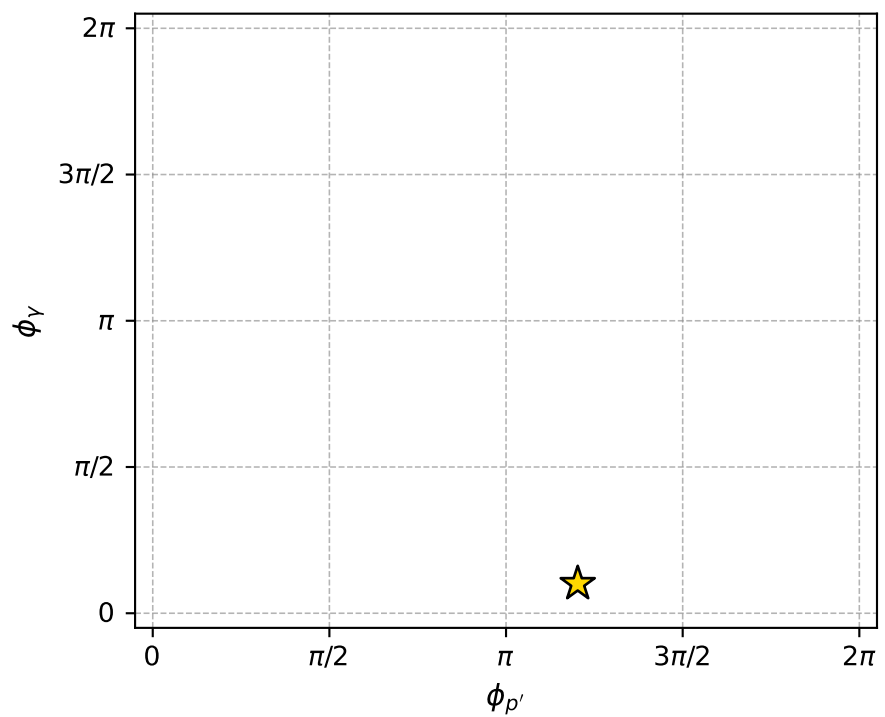
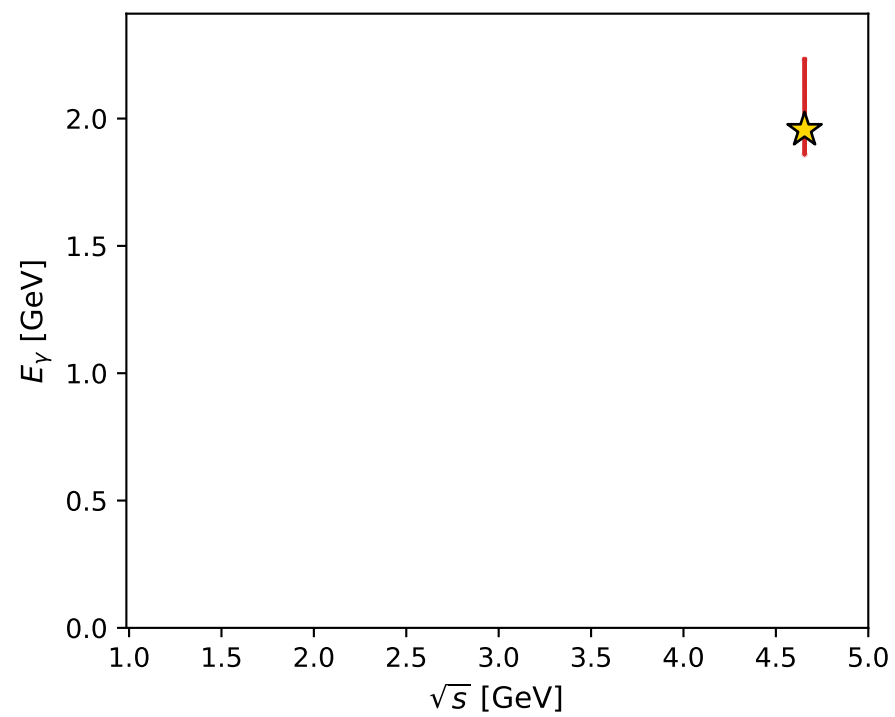
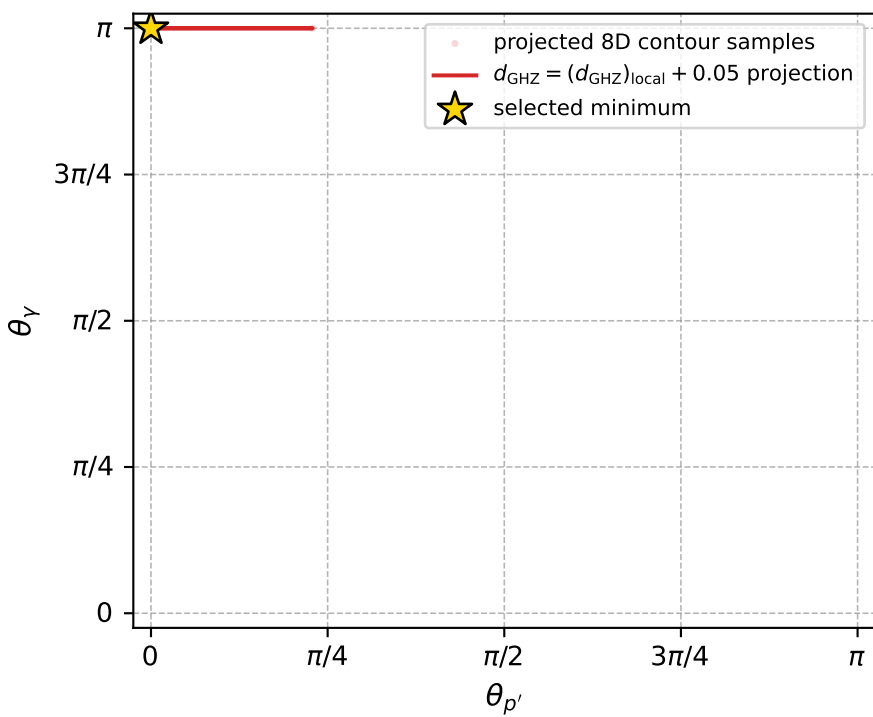
$(h_p, h_Y, h_\ell) = (+1, +1, -1)$

$(h_p, h_Y, h_\ell) = (-1, -1, +1)$

Leading final-state amplitudes (N=2, retained=100.0%)



electron: polarization cluster P2, local minimum 96; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 3.4263849\text{e-}08$
 $d_{\text{GHZ}} \text{ contour} = 0.050000034$
 above global minimum = $1.1980387\text{e-}08$
 8D contour samples = 102

selected phase-space point:

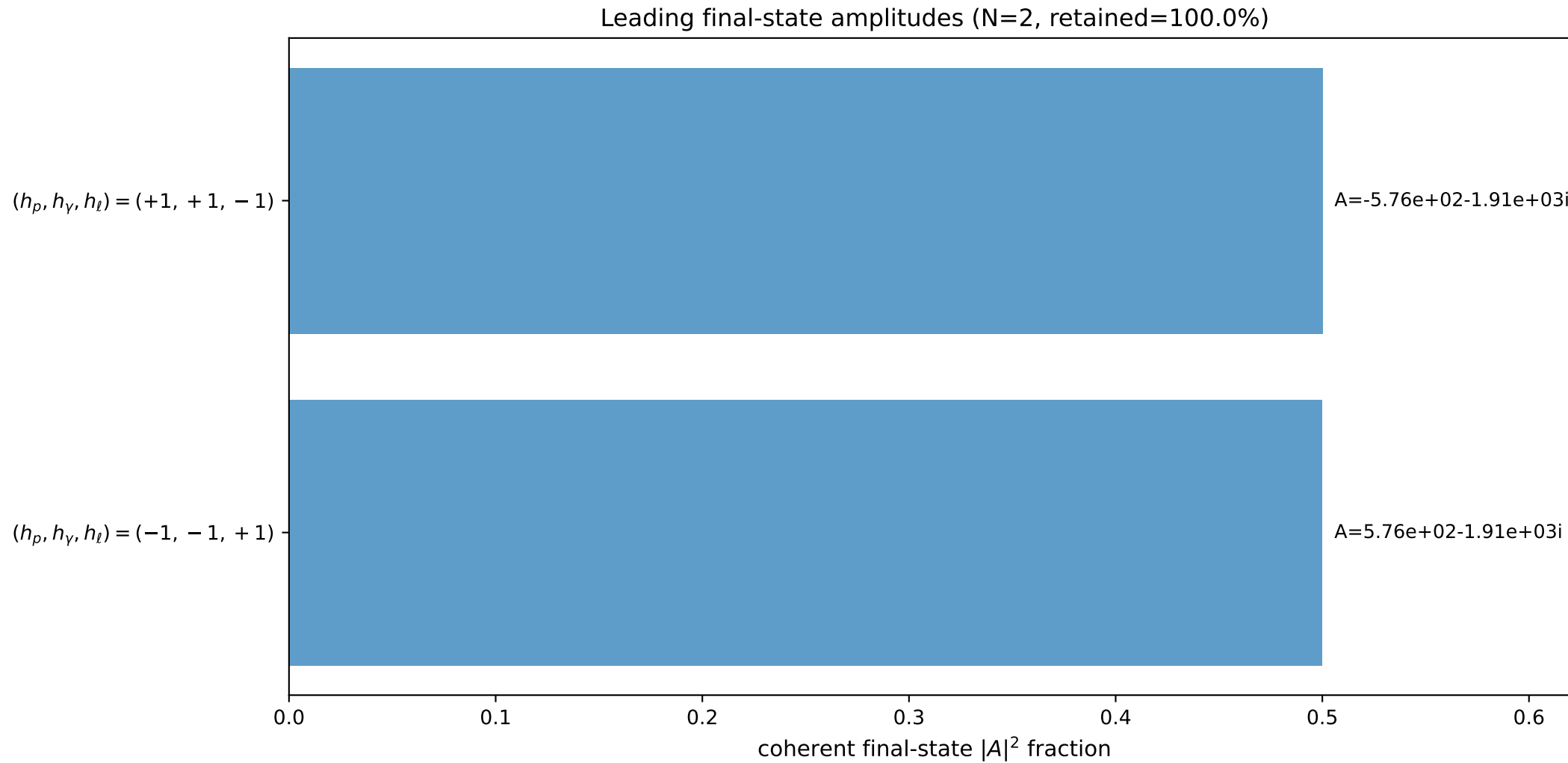
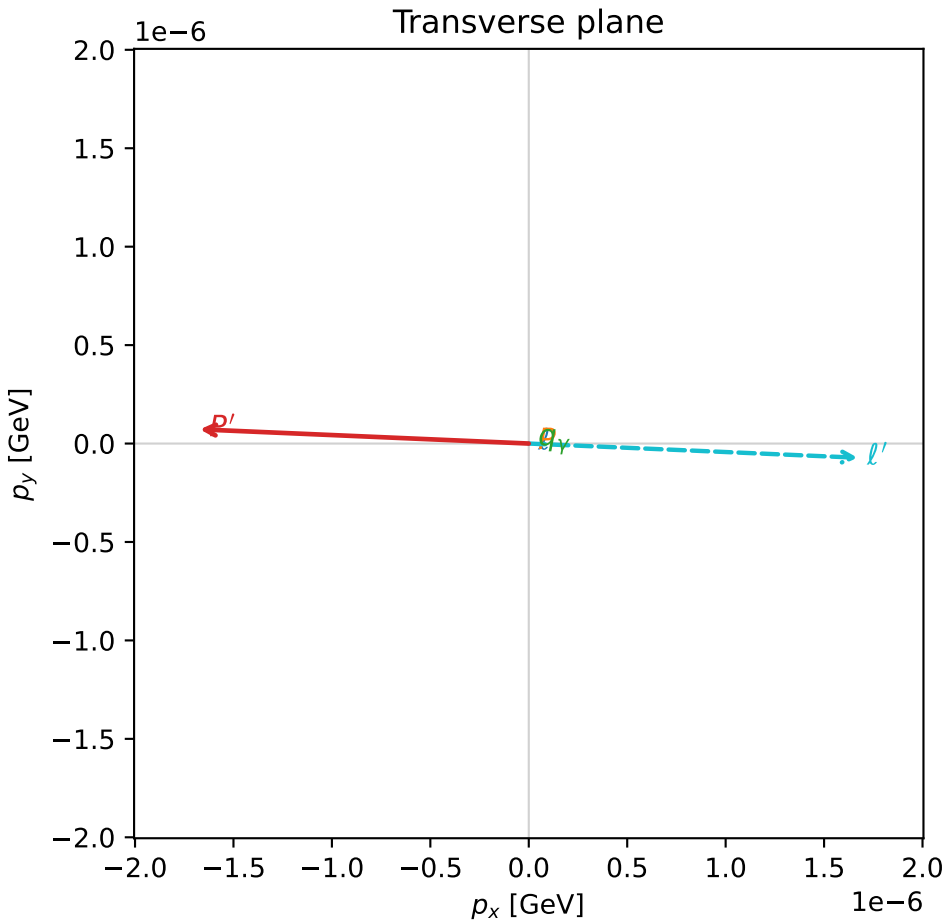
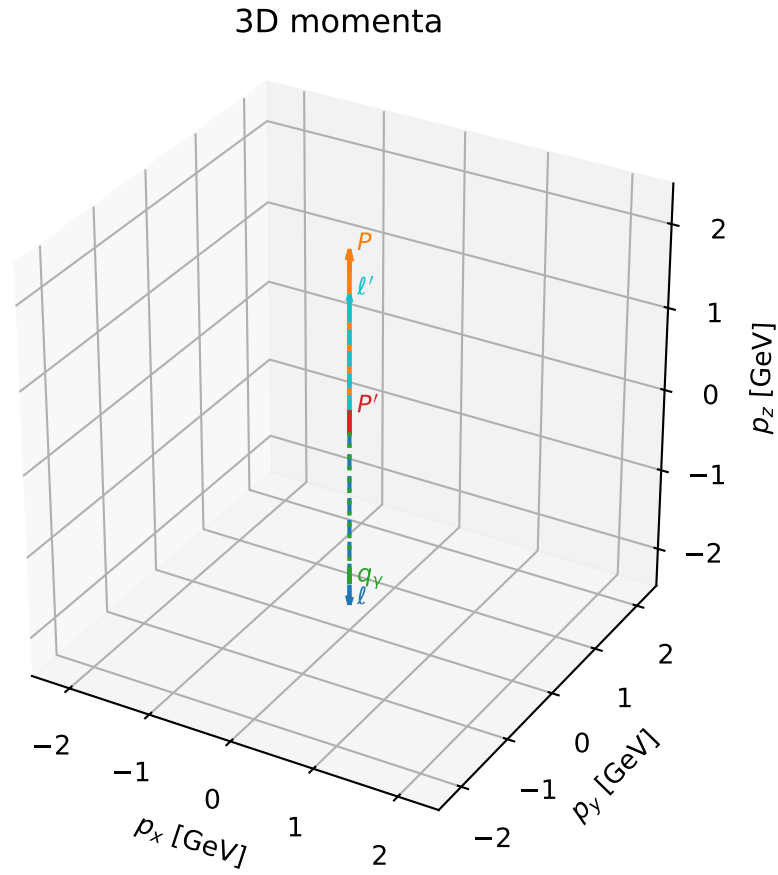
$\text{sqrt_s} = 4.65526$
 $\text{theta_p_out} = 3.141593\text{e-}05$
 $\text{theta_gamma_out} = 3.141593$
 $\text{q0out} = 1.955874$
 $\text{phi_p_out} = 3.778978$
 $\text{phi_gamma_out} = 0.3174718$
 $\text{alpha_e} = 0.6691883$
 $\text{alpha_p} = 2.472383$

coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

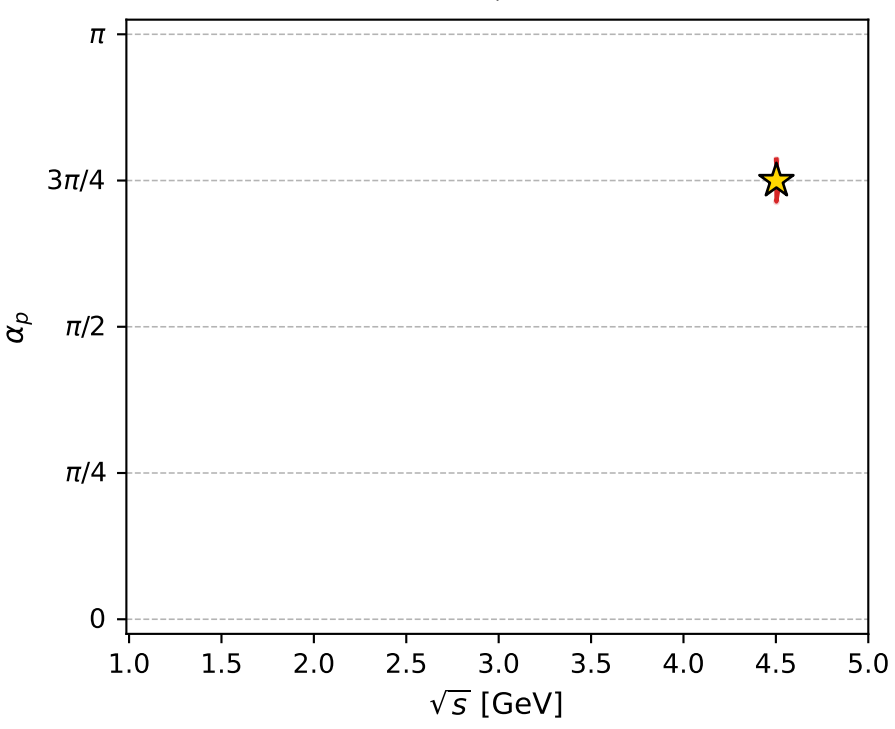
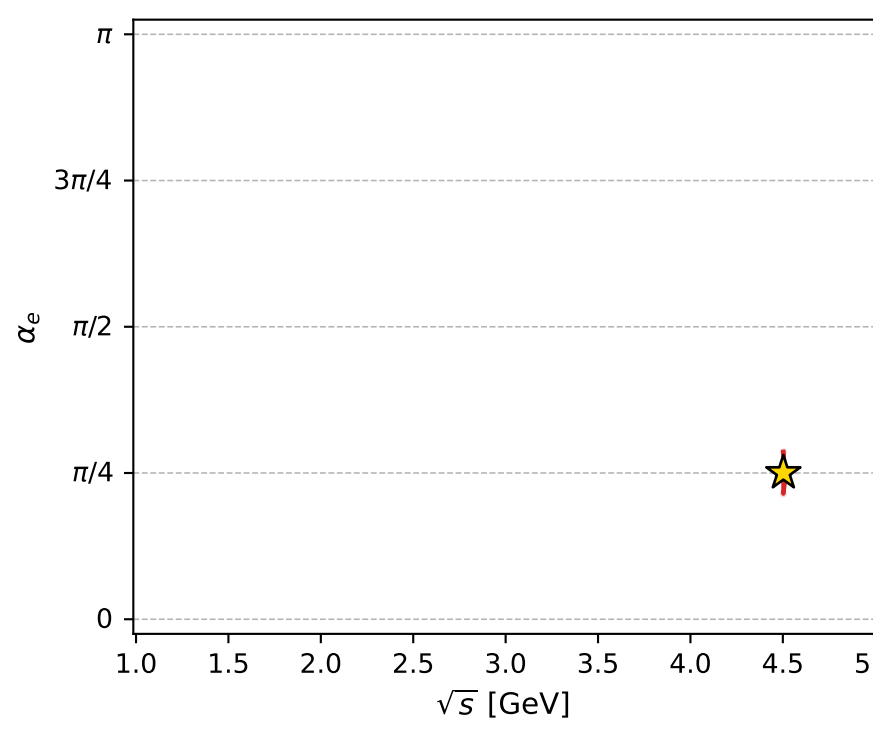
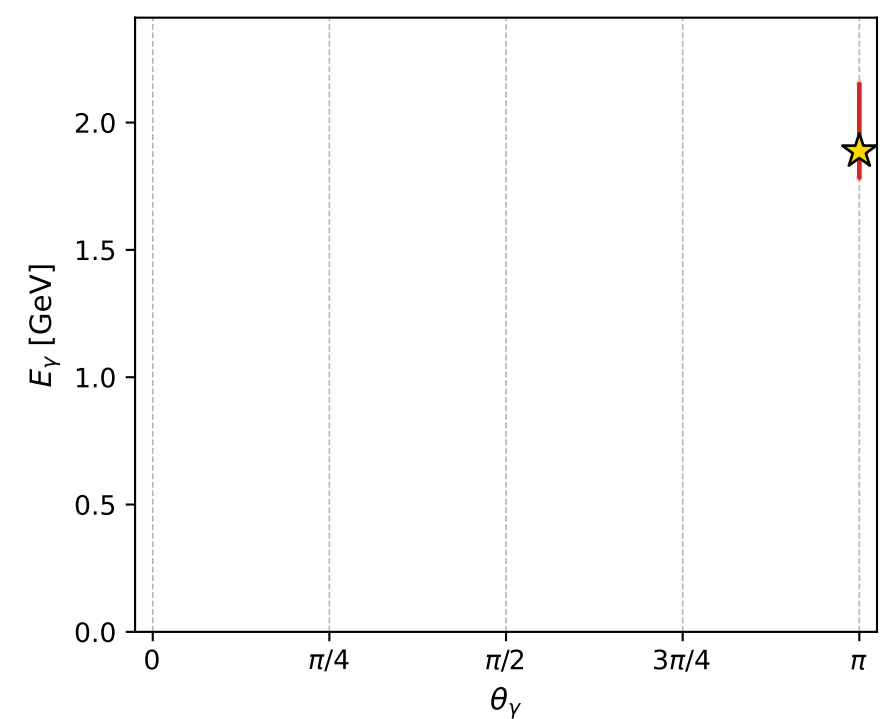
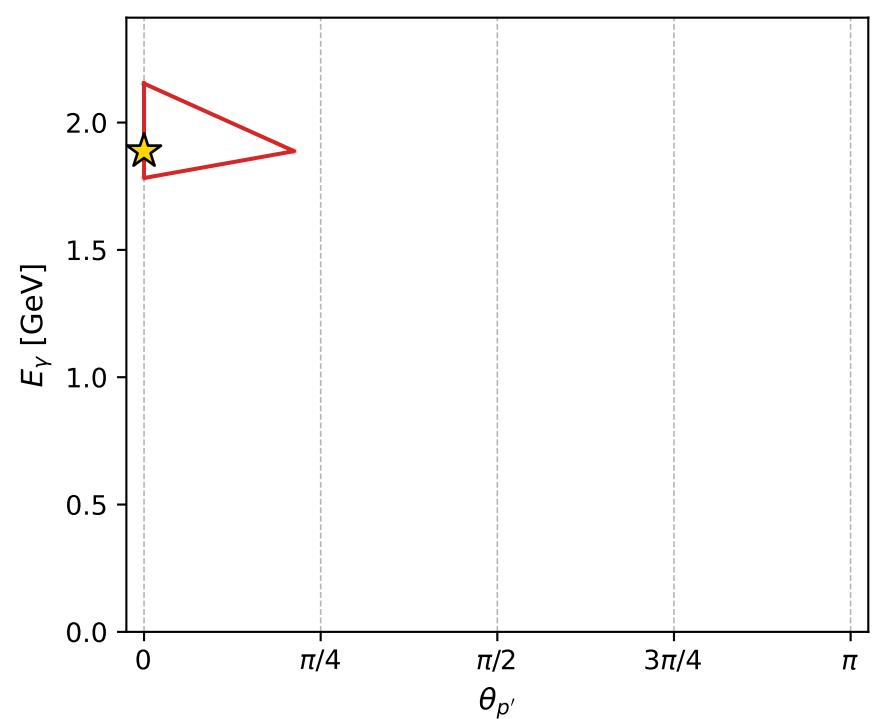
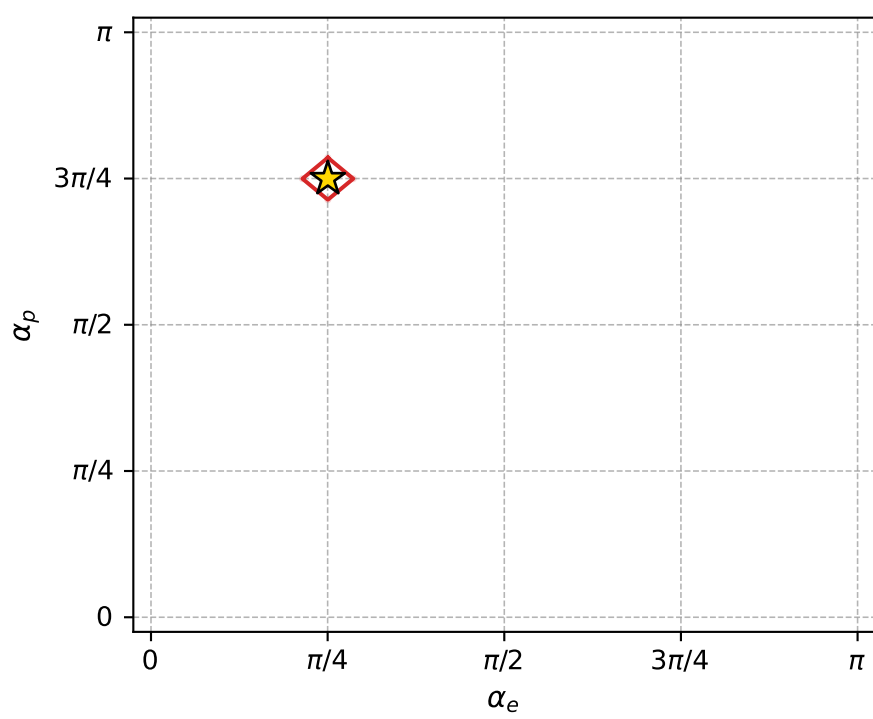
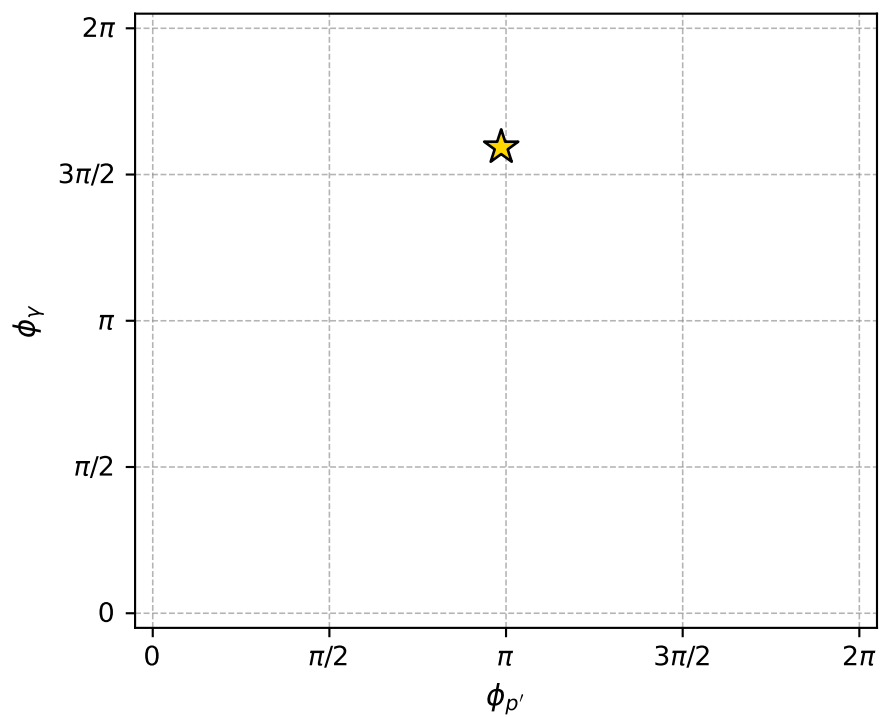
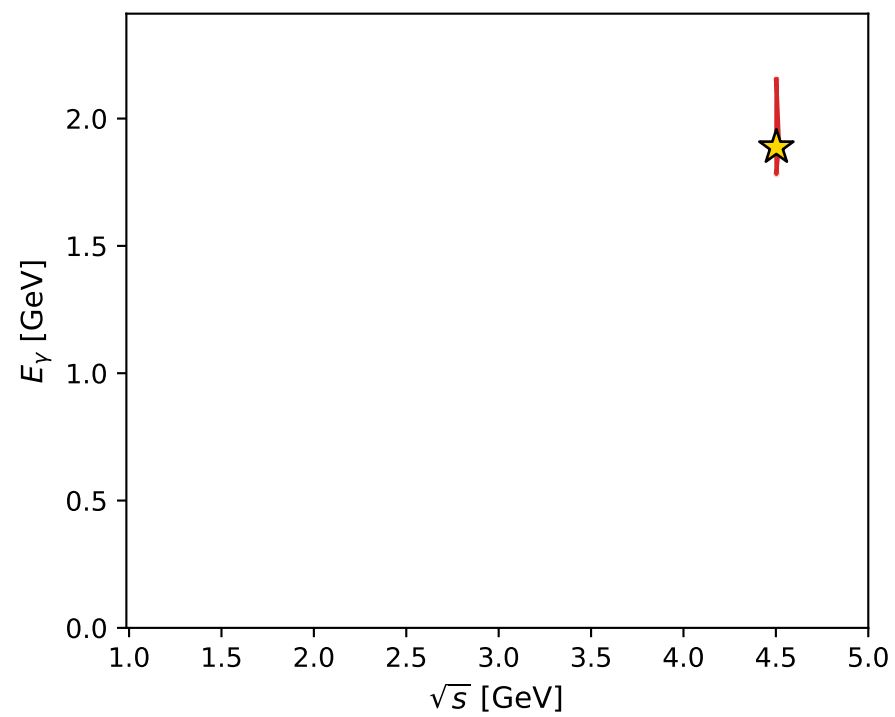
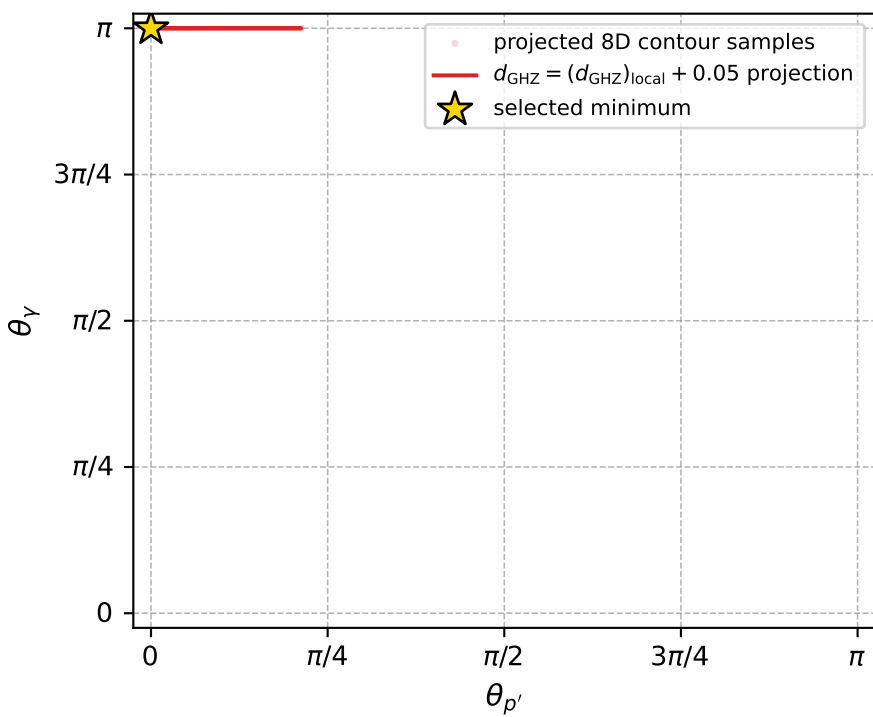
dghz_mixing_angles_polarization_02_local_minimum_99 d_{\rm GHZ}=3.58348e-08
region: polarization_cluster_02_local_minimum_99
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: $\alpha_e=0.78633215$ rad
incoming lepton: $0.706446|+\rangle +0.707767|-\rangle$
proton mixing angle: $\alpha_p=2.3552466$ rad
incoming proton: $-0.706436|+\rangle +0.707777|-\rangle$

$s=20.2744$, $\sqrt{s}=4.5027$, $|\vec{P}|=2.15365$, $|\vec{P}'|=0.24118$
 $\theta_{p'}=6.93359\text{e-}06$, $\theta_\gamma=3.14159$, $\phi_{p'}=3.0985$, $\phi_\gamma=5.00556$
 $E_\gamma=1.88769$, $\theta_{l'}=1.01558\text{e-}06$, $\phi_{l'}=6.2401$
 $Q^2=14.184$, $x_B=0.756439$, $t=-1.75164$, $W^2=5.44687$, $y=0.966822$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 2.1537$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.1537$
 P $E= 2.3491$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.1537$
 l' $E= 1.6465$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 1.6465$
 P' $E= 0.9685$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 0.2412$
 q_γ $E= 1.8877$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= -1.8877$



electron: polarization cluster P2, local minimum 99; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 3.5834754\text{e-}08$
 $d_{\text{GHZ}} \text{ contour} = 0.050000036$
 above global minimum = $1.3551293\text{e-}08$
 8D contour samples = 68

selected phase-space point:

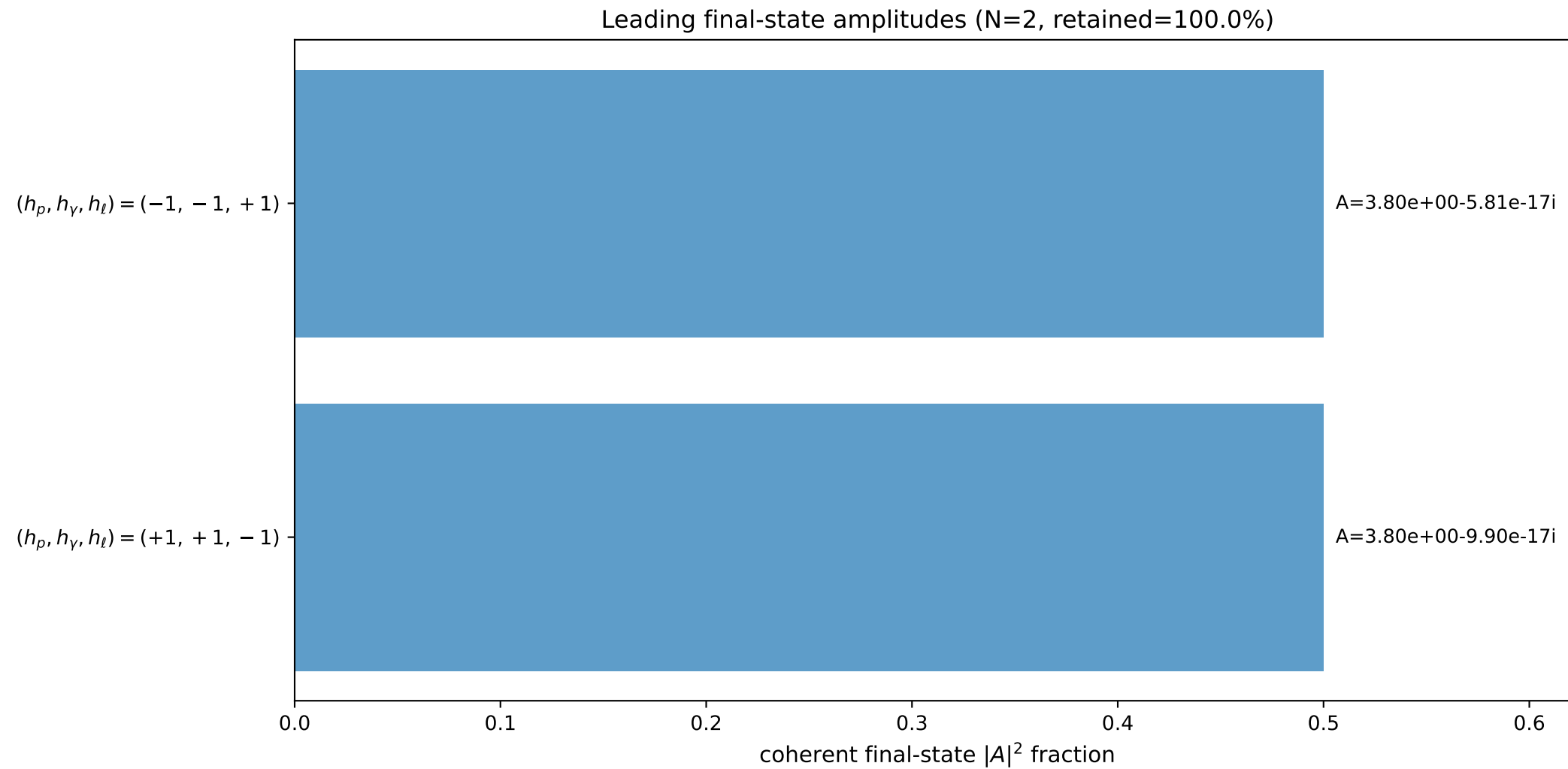
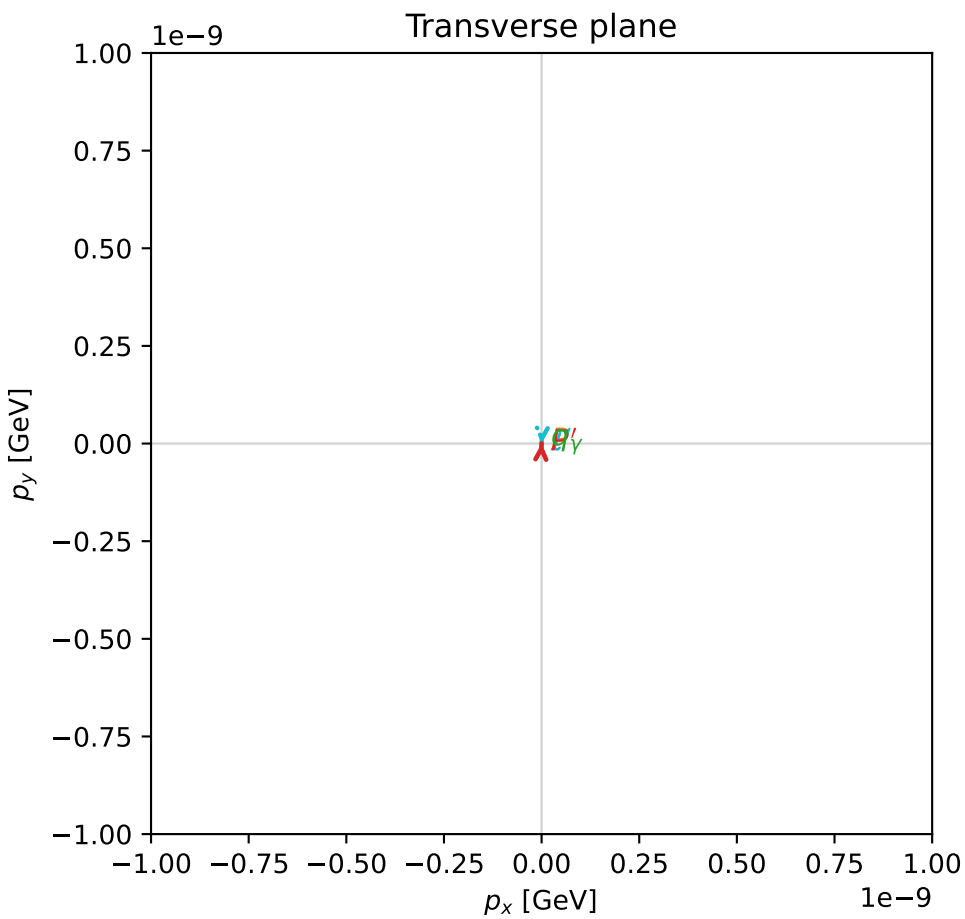
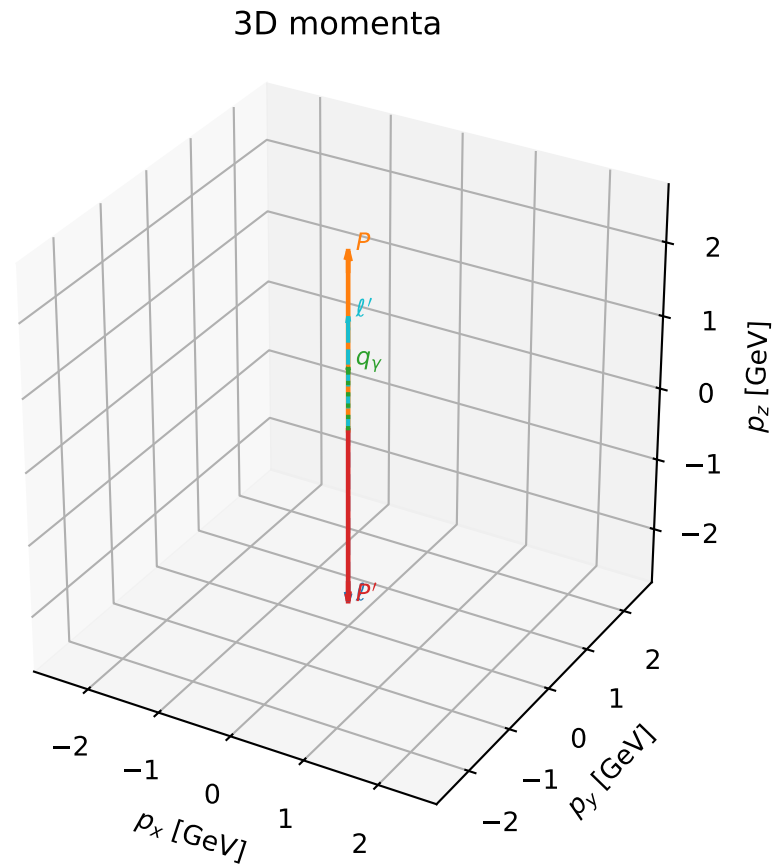
$\text{sqrt_s} = 4.502705$
 $\text{theta_p_out} = 6.933593\text{e-}06$
 $\text{theta_gamma_out} = 3.141593$
 $\text{q0ut} = 1.887687$
 $\text{phi_p_out} = 3.098503$
 $\text{phi_gamma_out} = 5.005563$
 $\text{alpha_e} = 0.7863321$
 $\text{alpha_p} = 2.355247$

coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

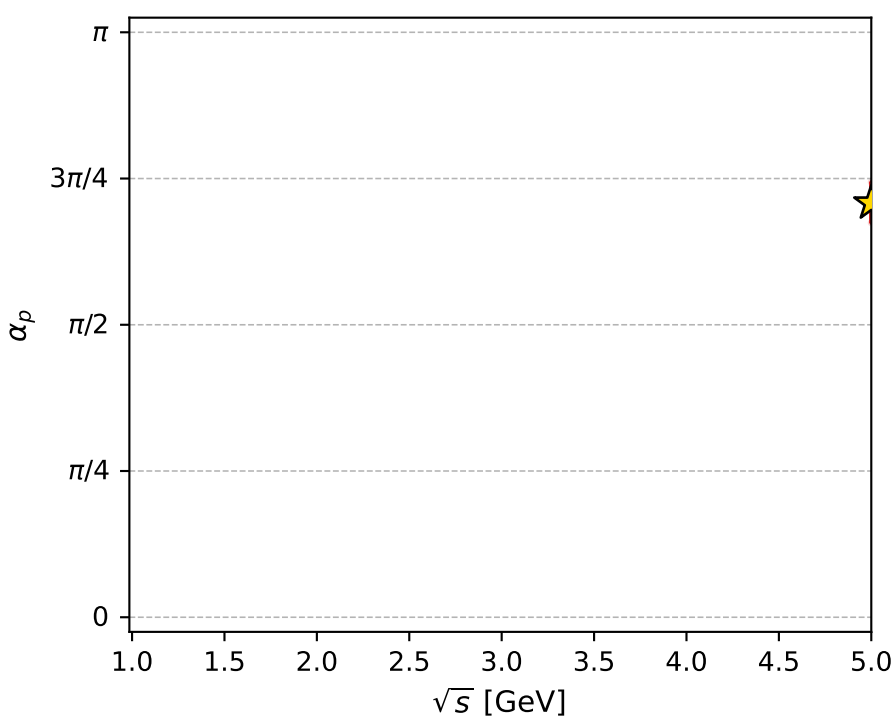
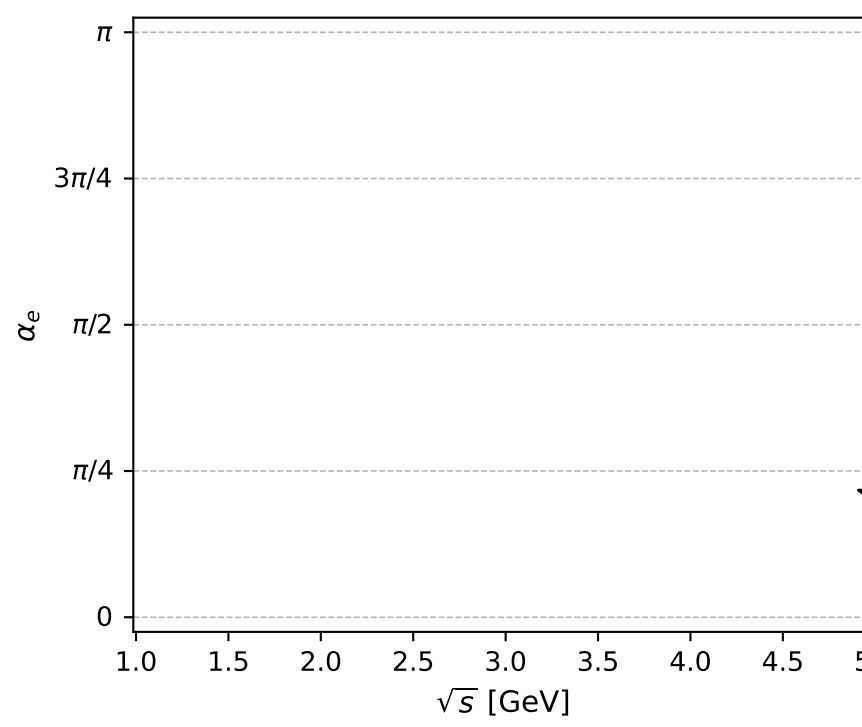
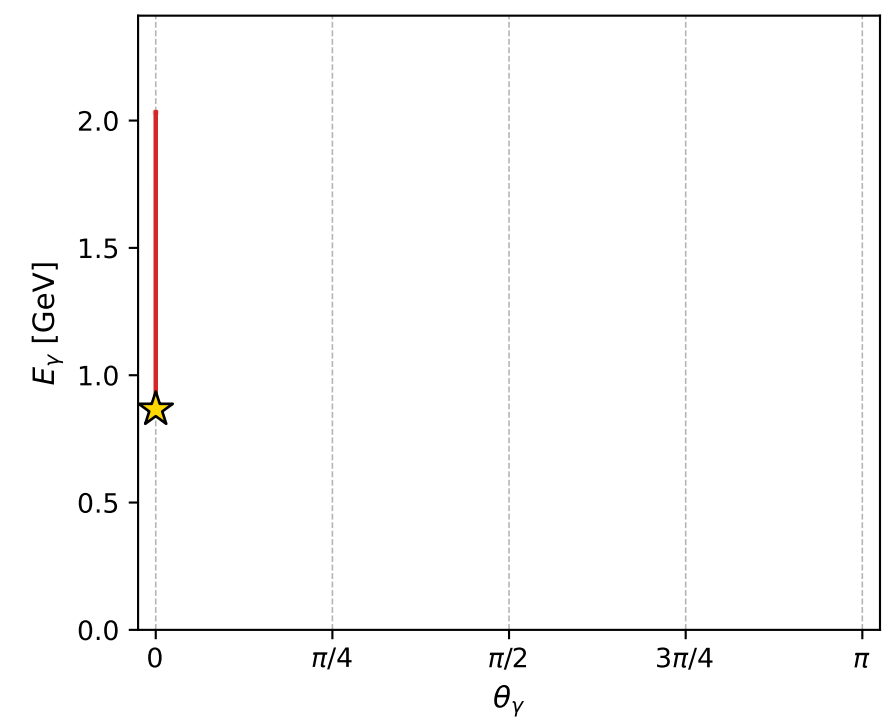
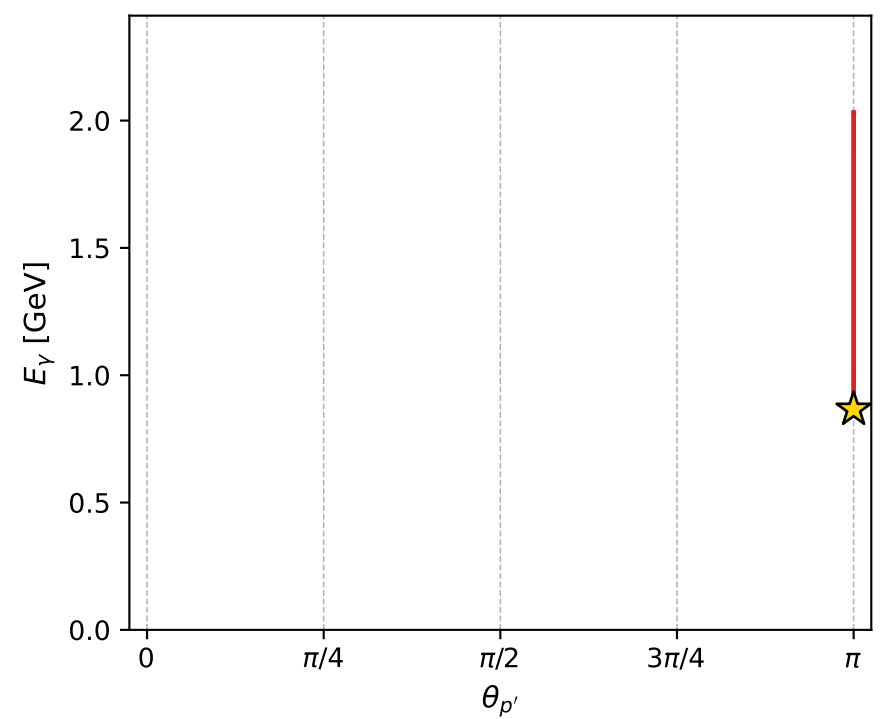
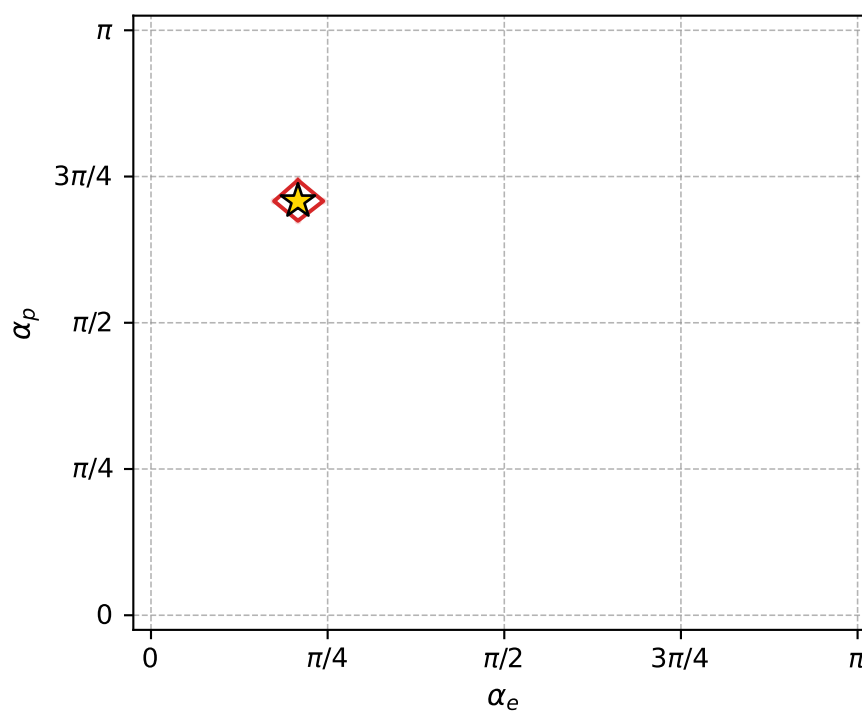
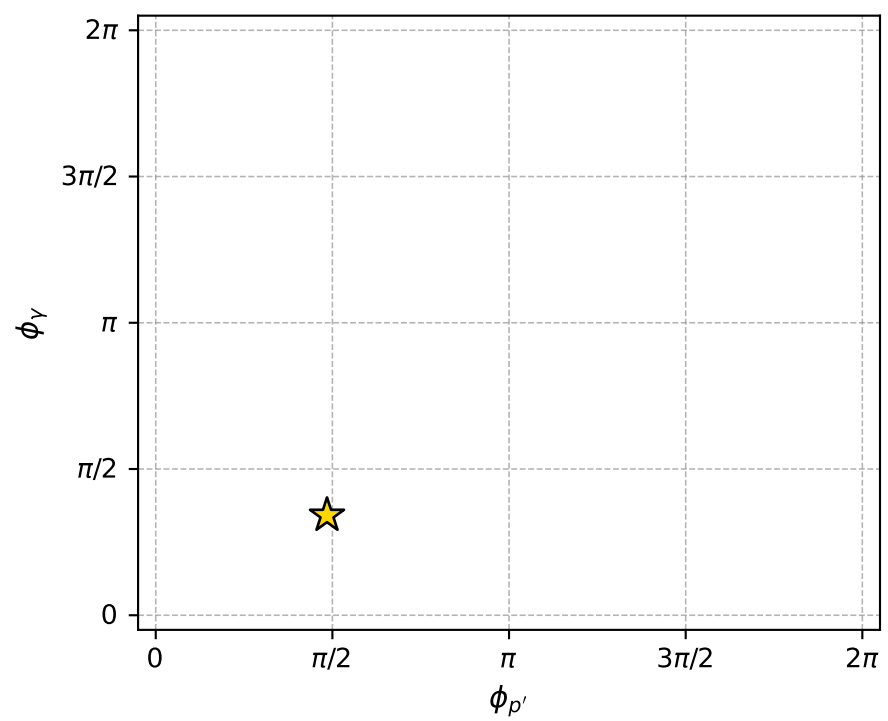
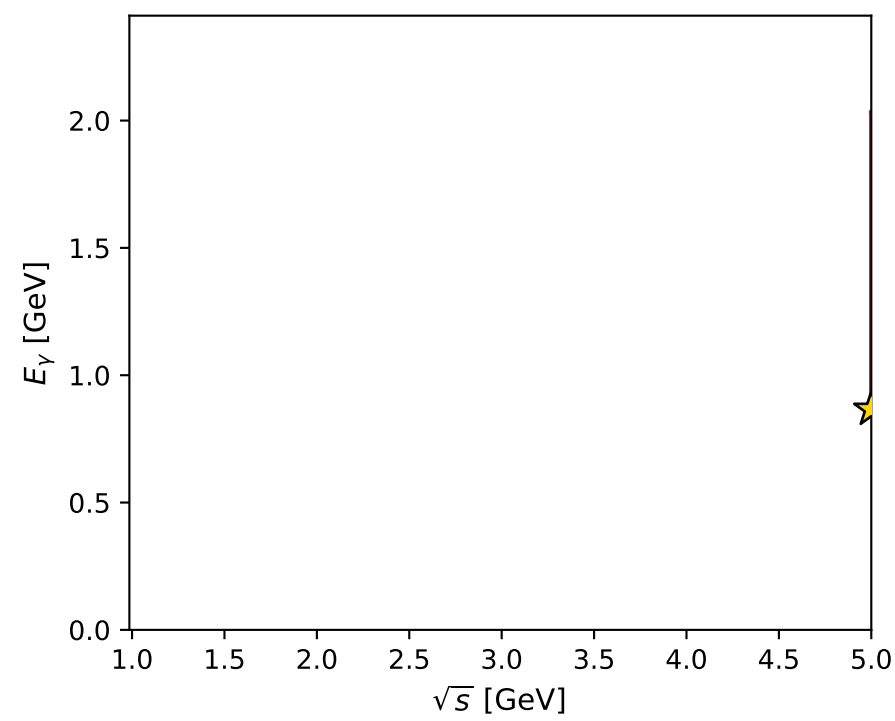
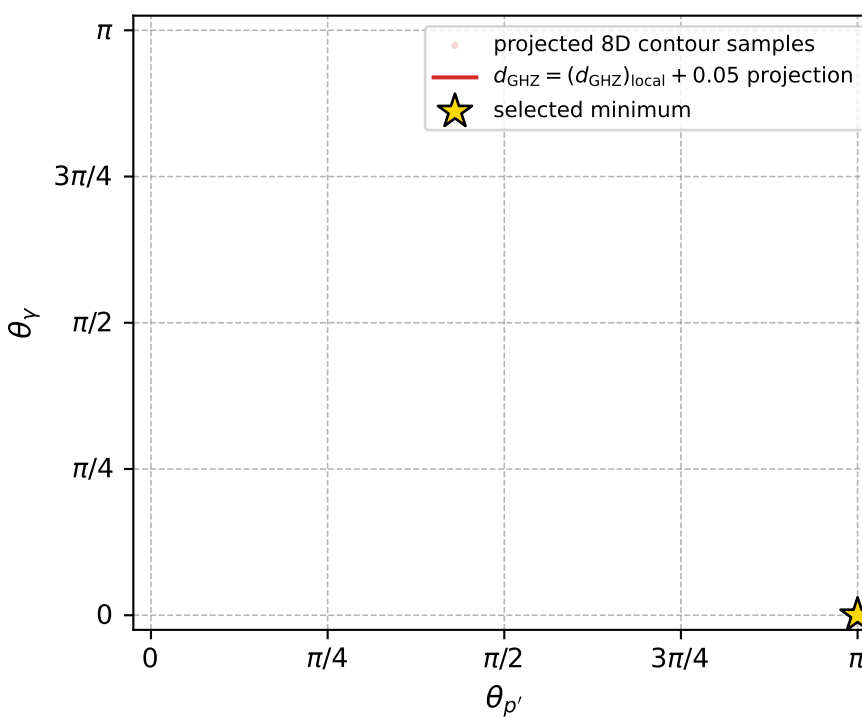
dghz_mixing_angles_polarization_02_local_minimum_100 $d_{\{\text{rm GHZ}\}}=3.62681\text{e-}08$
 region: polarization_cluster_02_local_minimum_100
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=0.65339387$ rad
 incoming lepton: $0.794025|+\rangle +0.607885|-\rangle$
 proton mixing angle: $\alpha_p=2.224186$ rad
 incoming proton: $-0.607881|+\rangle +0.794028|-\rangle$

$s=25$, $\sqrt{s}=5$, $|\vec{P}|=2.41202$, $|\vec{P}'|=2.41202$
 $\theta_{P'}=3.14159$, $\theta_Y=0$, $\phi_{P'}=1.52284$, $\phi_Y=1.07341$
 $E_Y=0.865987$, $\theta_{\ell'}=0$, $\phi_{\ell'}=4.66443$
 $Q^2=14.9162$, $x_B=0.632684$, $t=-23.2713$, $W^2=9.53971$, $y=0.977442$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4120$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 P $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4120$
 ℓ' $E= 1.5460$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 1.5460$
 P' $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 q_Y $E= 0.8660$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.8660$



electron: polarization cluster P2, local minimum 100; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 3.6268124\text{e-}08$
 $d_{\text{GHZ}} \text{ contour} = 0.050000036$
 above global minimum = $1.3984662\text{e-}08$
 8D contour samples = 40

selected phase-space point:

$\text{sqrt_s} = 5$
 $\text{theta_p_out} = 3.141593$
 $\text{theta_gamma_out} = 0$
 $\text{q0out} = 0.8659869$
 $\text{phi_p_out} = 1.522838$
 $\text{phi_gamma_out} = 1.073415$
 $\text{alpha_e} = 0.6533939$
 $\text{alpha_p} = 2.224186$

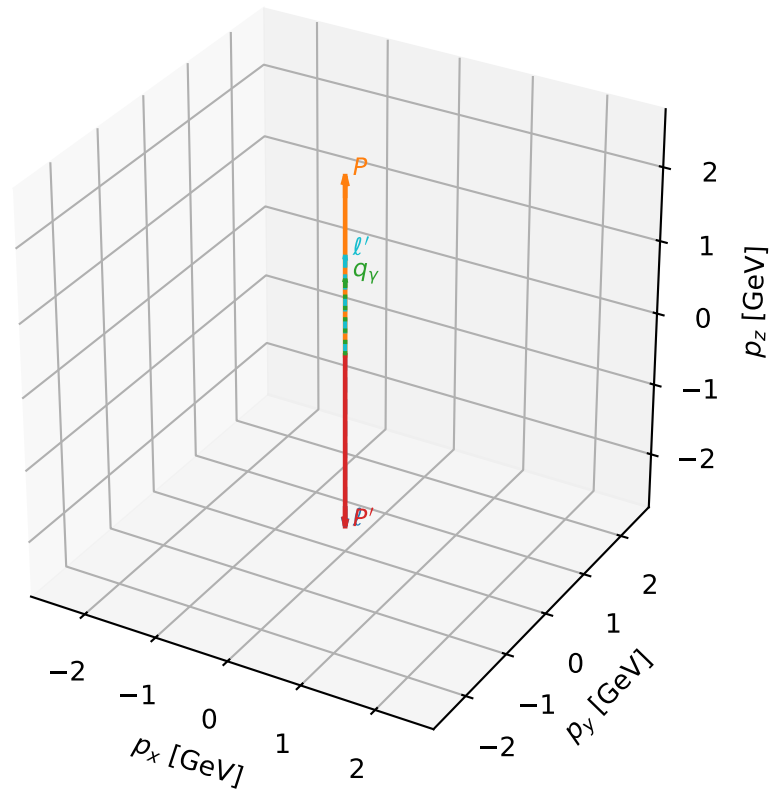
coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

dghz_mixing_angles_polarization_02_local_minimum_110 $d_{\{\text{rm GHZ}\}}=4.05095\text{e-}08$
 region: polarization_cluster_02_local_minimum_110
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=0.68998436$ rad
 incoming lepton: $0.771256|+\rangle +0.636525|-\rangle$
 proton mixing angle: $\alpha_p=2.2608057$ rad
 incoming proton: $-0.636544|+\rangle +0.77124|-\rangle$

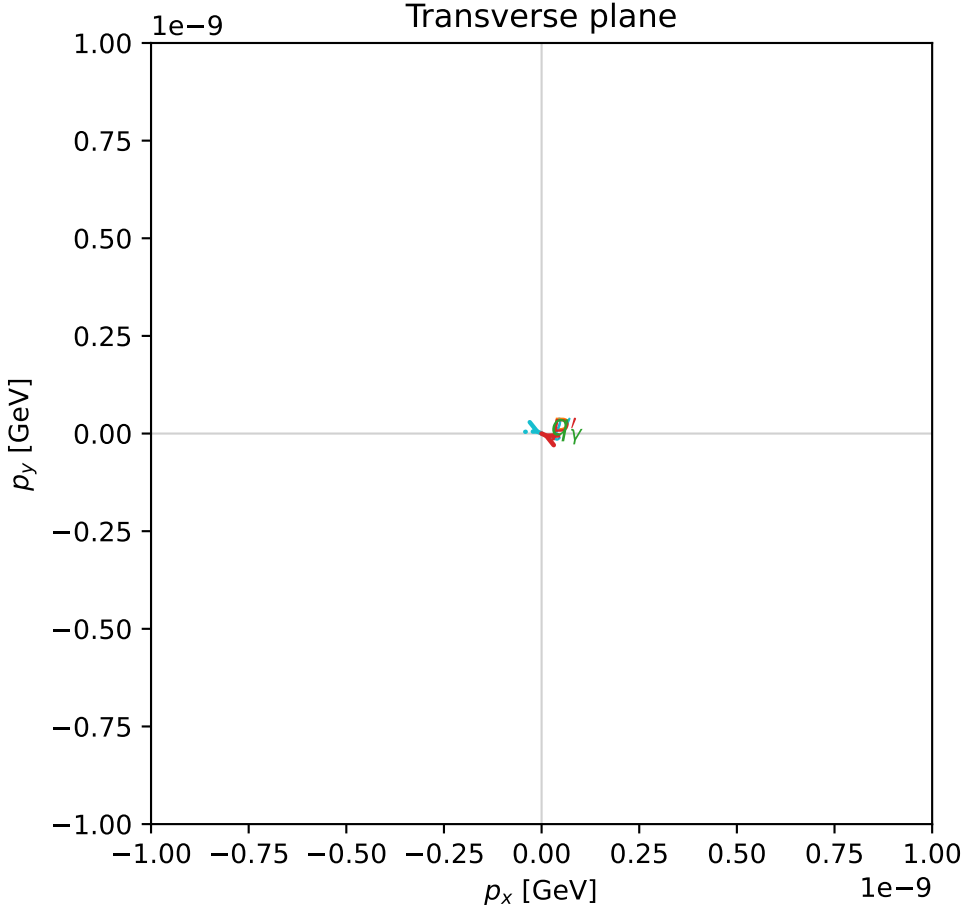
$s=24.9589$, $\sqrt{s}=4.99589$, $|\vec{P}|=2.40989$, $|\vec{P}'|=2.40989$
 $\theta_{P'}=3.14159$, $\theta_Y=0$, $\phi_{P'}=2.7054$, $\phi_Y=1.95022$
 $E_Y=1.04501$, $\theta_{\ell'}=0$, $\phi_{\ell'}=5.84699$
 $Q^2=13.1568$, $x_B=0.557533$, $t=-23.2303$, $W^2=11.3213$, $y=0.980035$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4099$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4099$
 P $E= 2.5860$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4099$
 ℓ' $E= 1.3649$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 1.3649$
 P' $E= 2.5860$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= -2.4099$
 q_Y $E= 1.0450$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 1.0450$

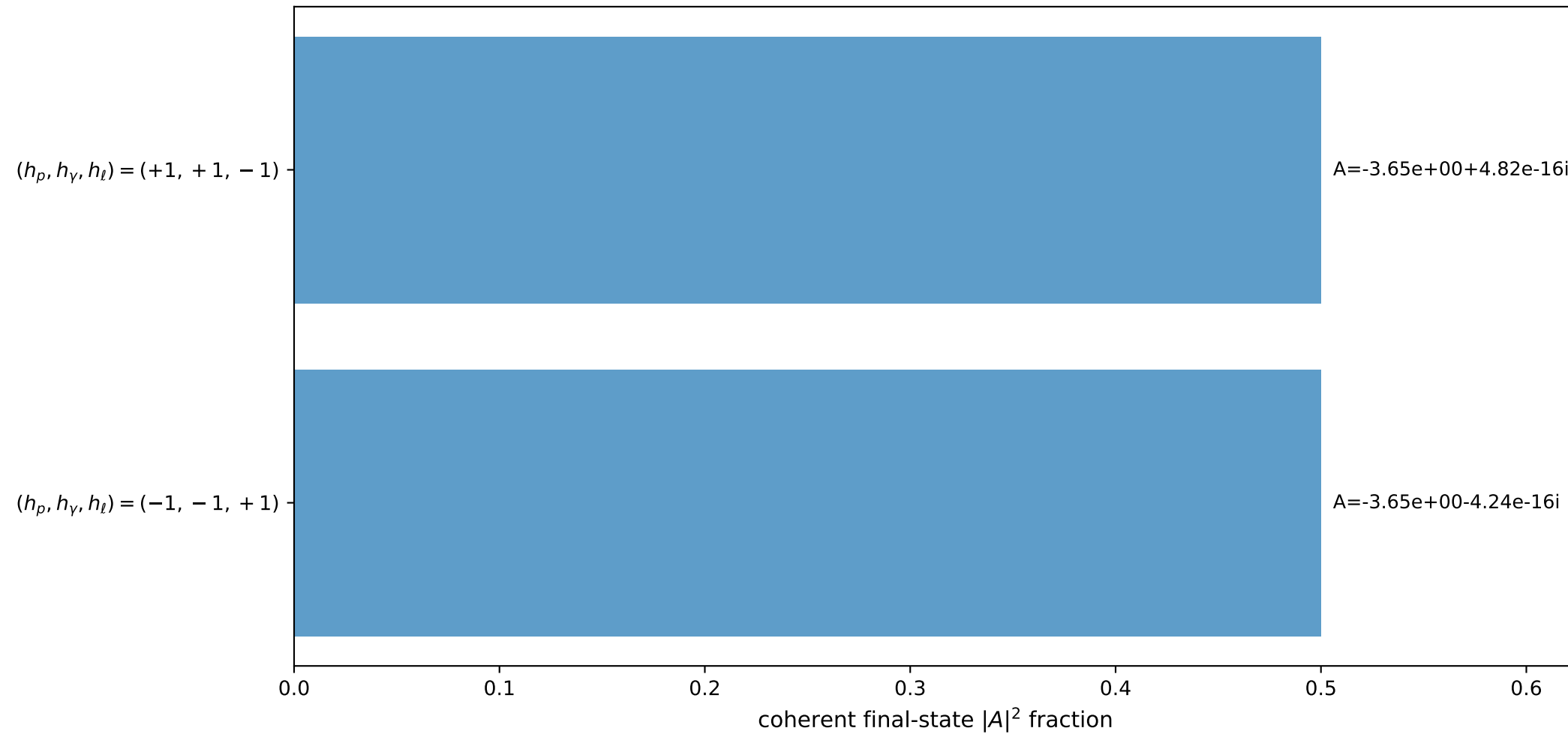
3D momenta



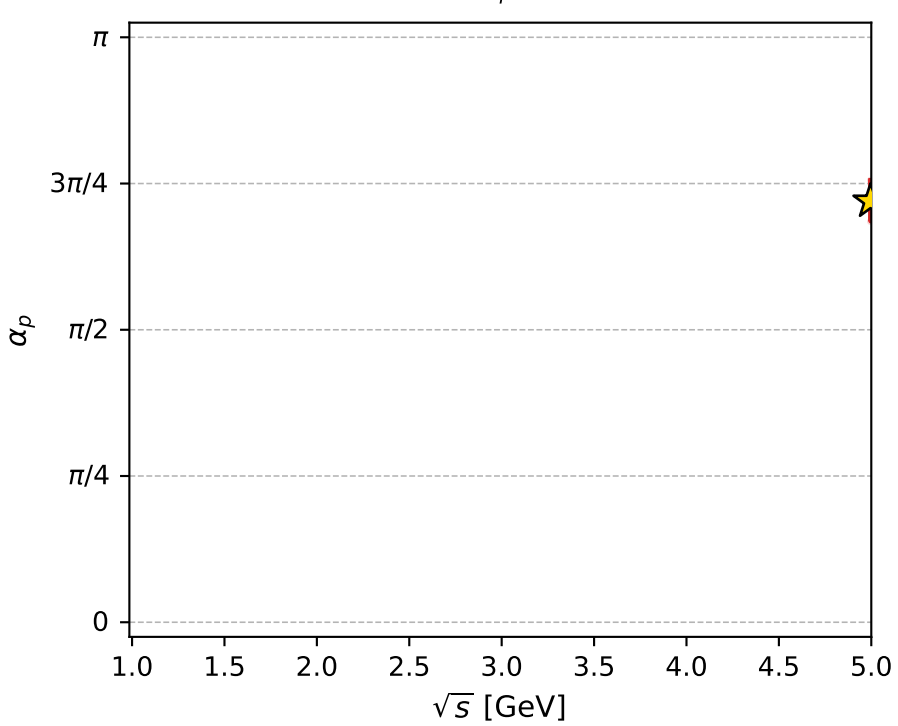
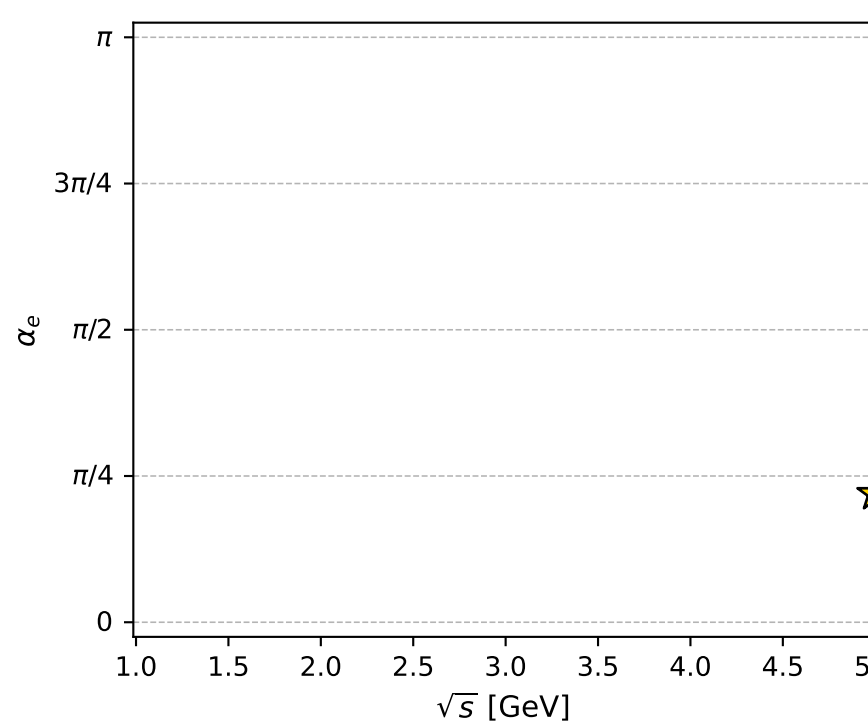
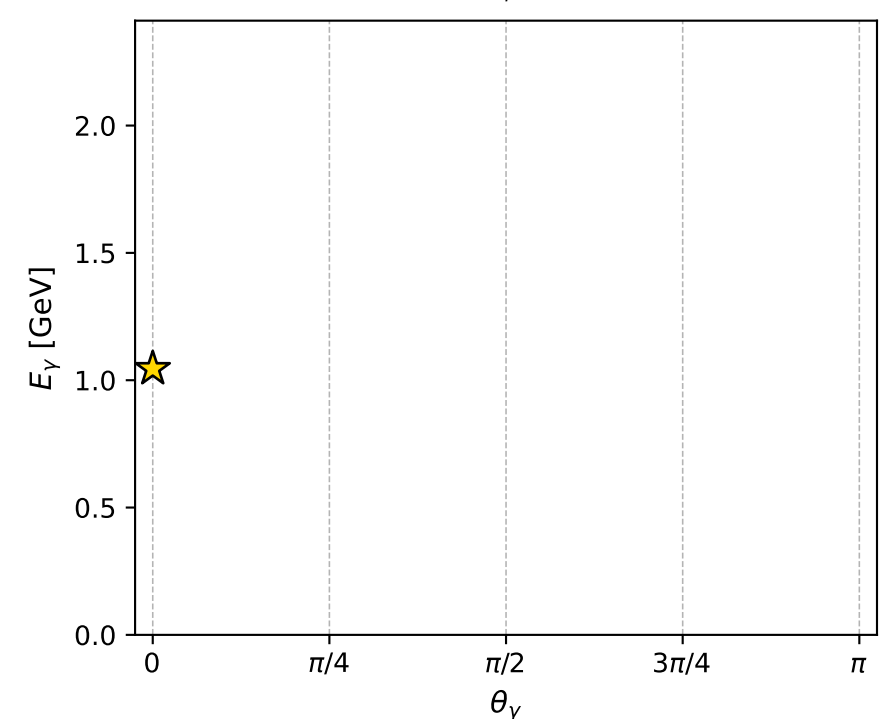
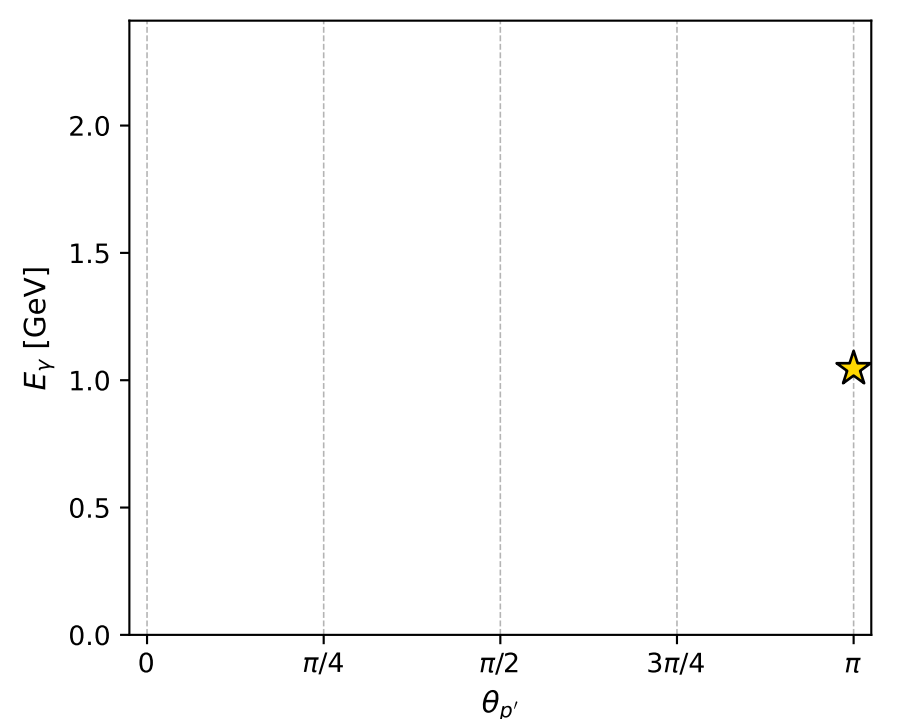
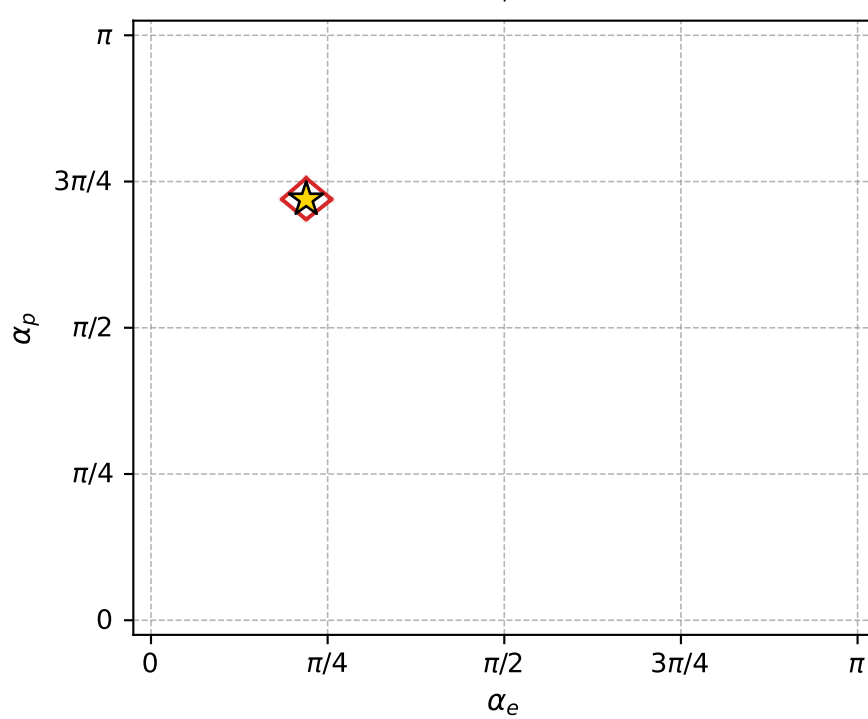
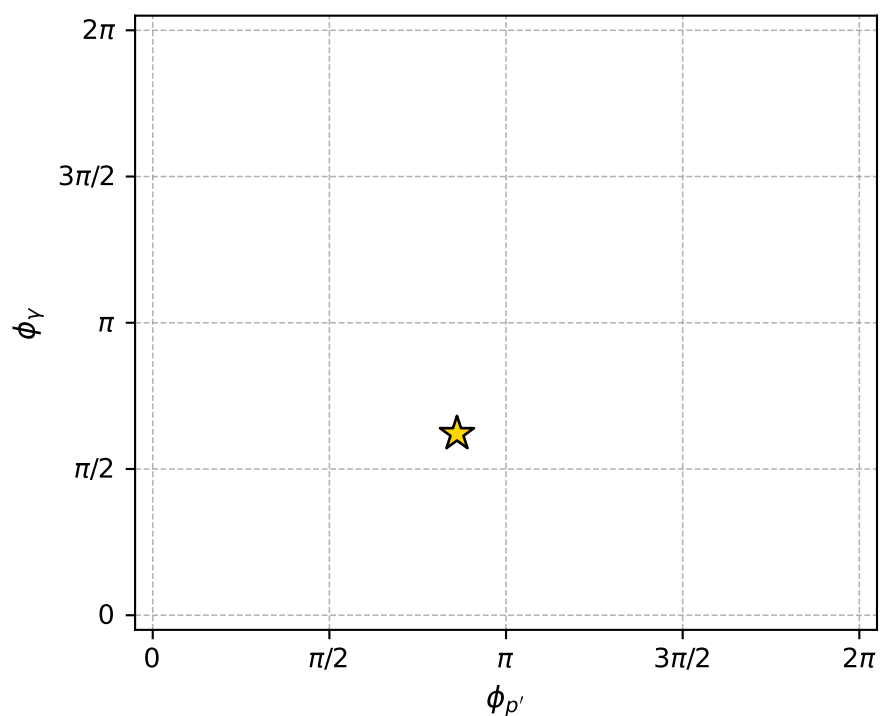
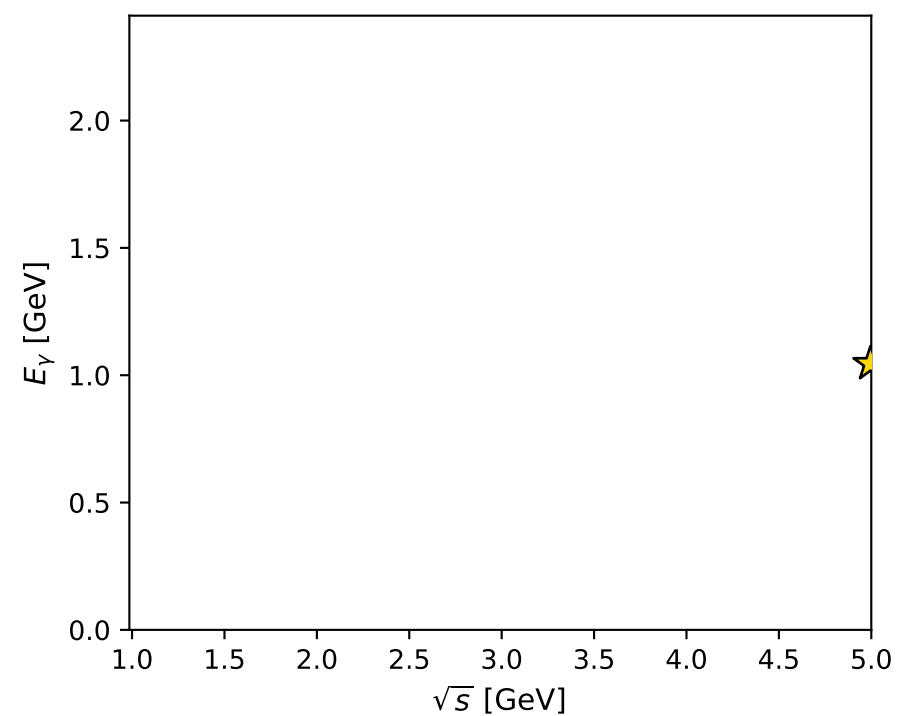
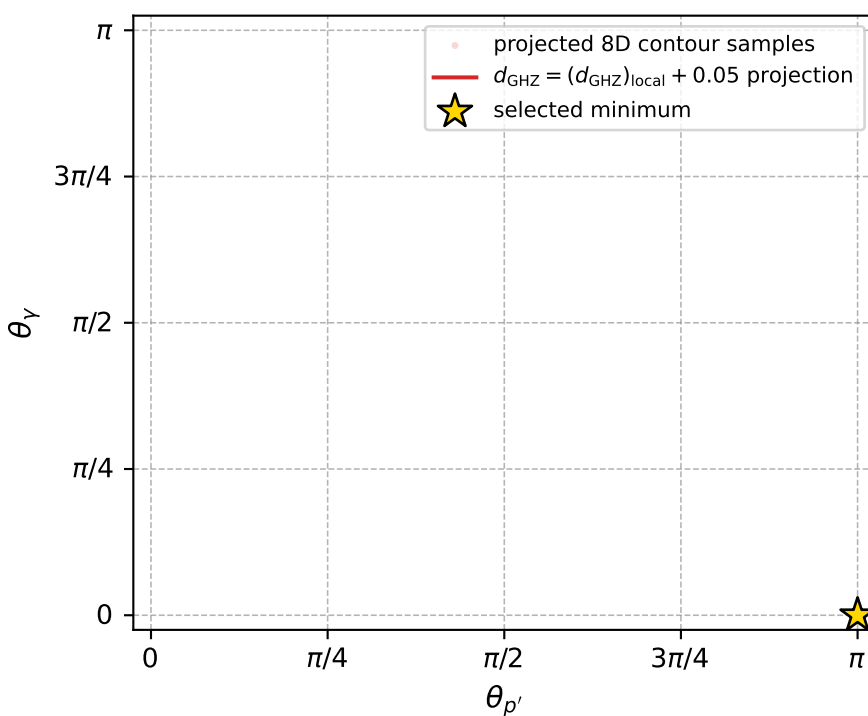
Transverse plane



Leading final-state amplitudes (N=2, retained=100.0%)



electron: polarization cluster P2, local minimum 110; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 4.0509472\text{e-}08$
 $d_{\text{GHZ}} \text{ contour} = 0.050000041$
 above global minimum = $1.822601\text{e-}08$
 8D contour samples = 66

selected phase-space point:

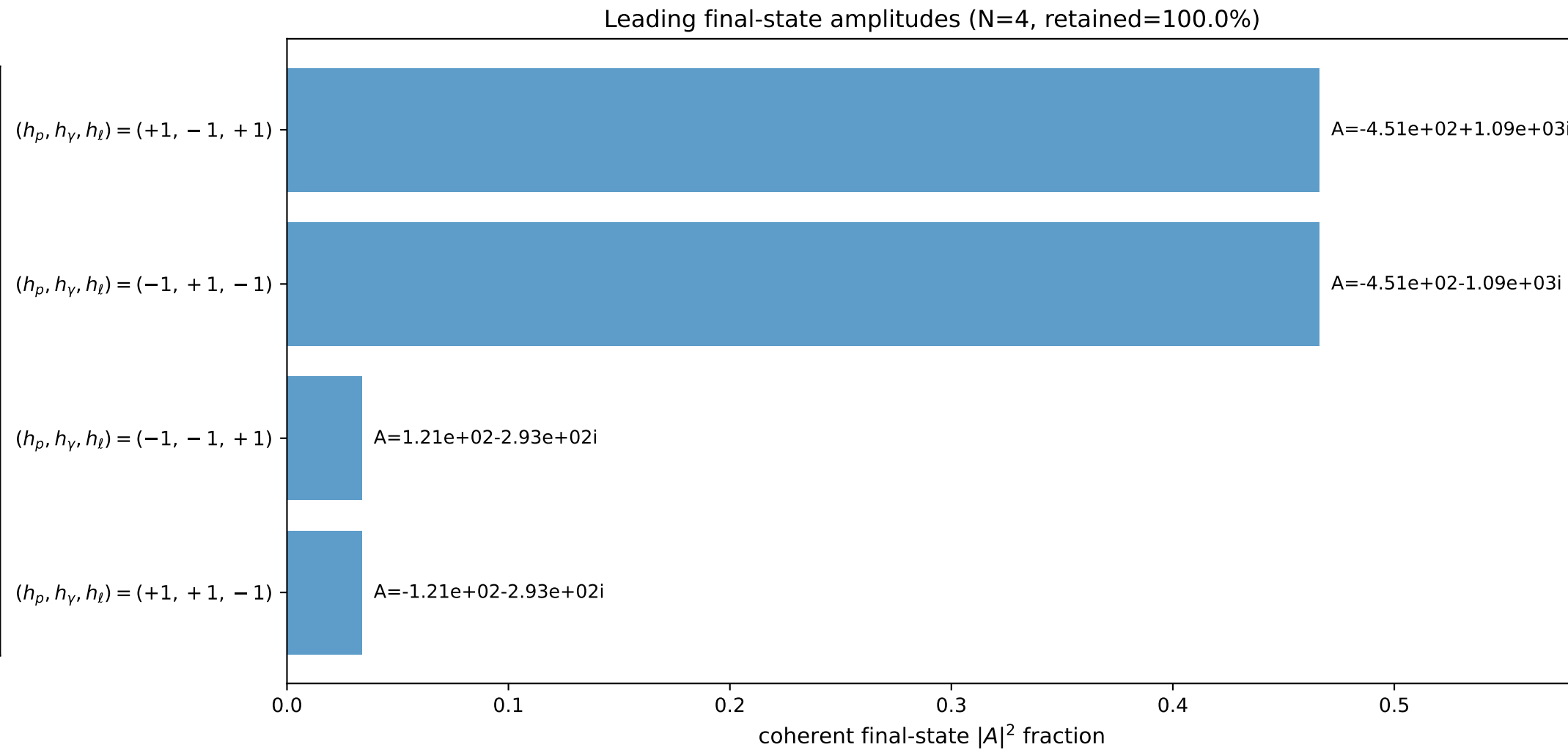
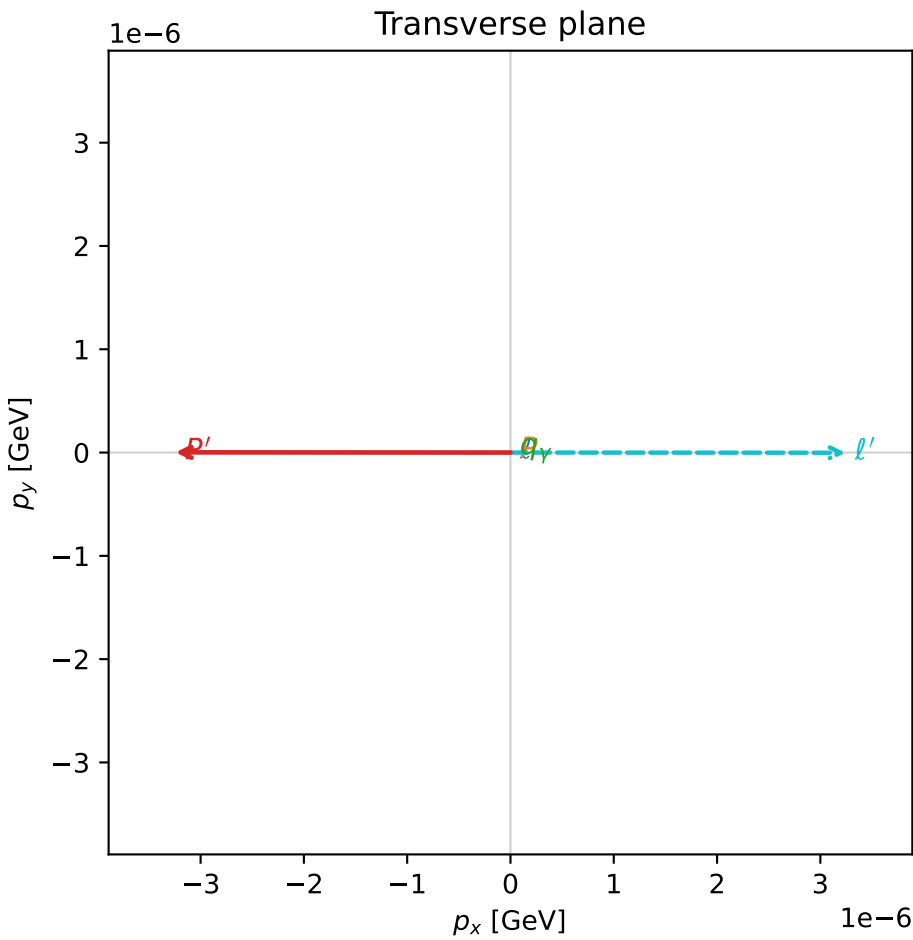
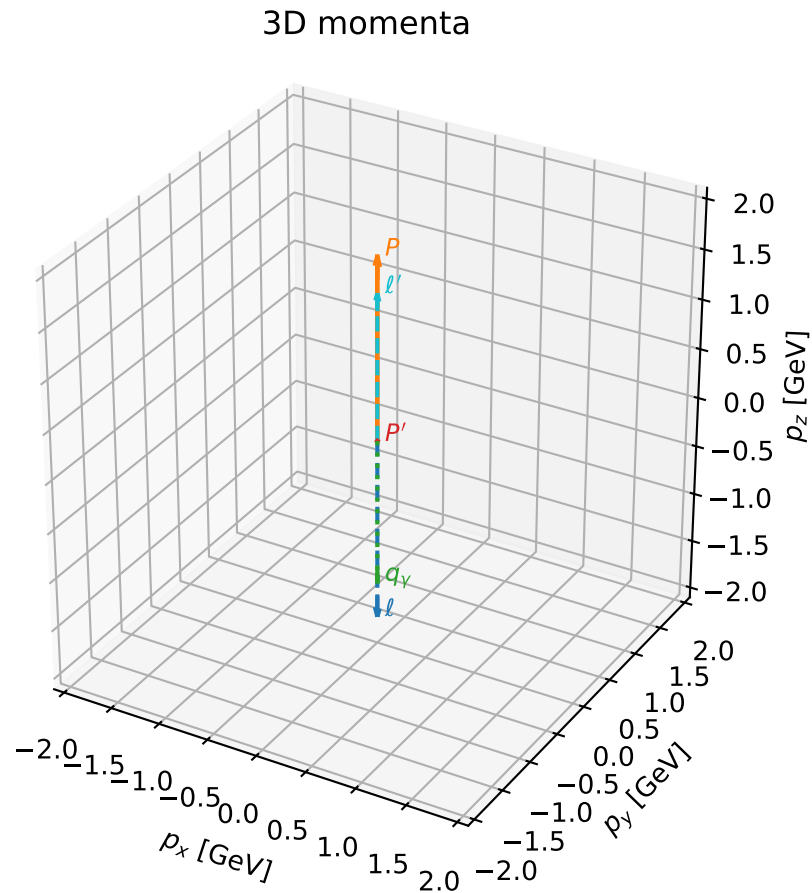
$\text{sqrt_s} = 4.995892$
 $\text{theta_p_out} = 3.141593$
 $\text{theta_gamma_out} = 0$
 $\text{qOut} = 1.045008$
 $\text{phi_p_out} = 2.705397$
 $\text{phi_gamma_out} = 1.950219$
 $\text{alpha_e} = 0.6899844$
 $\text{alpha_p} = 2.260806$

coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

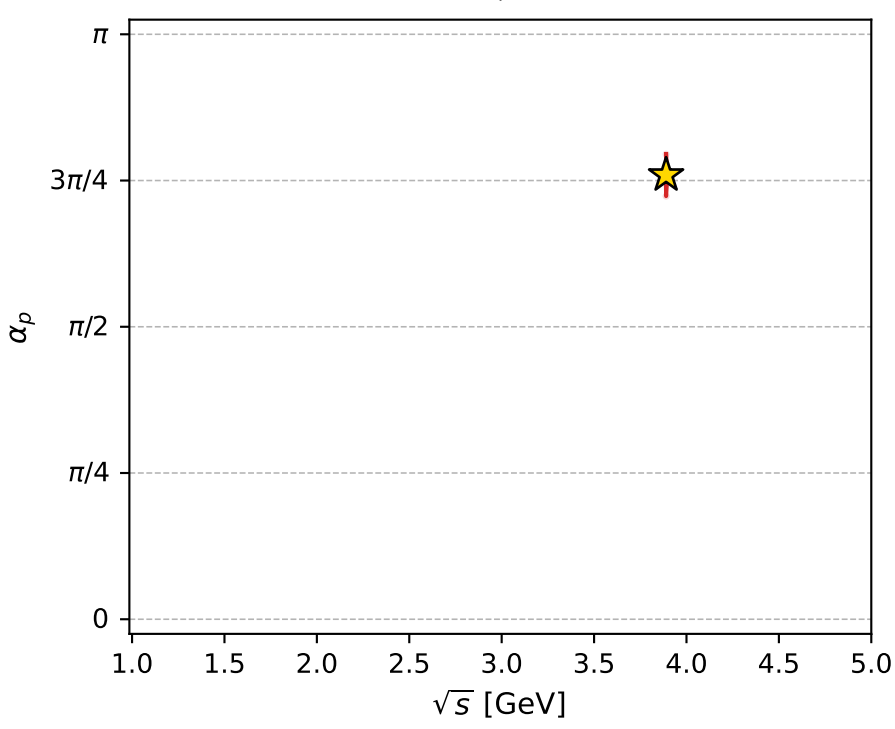
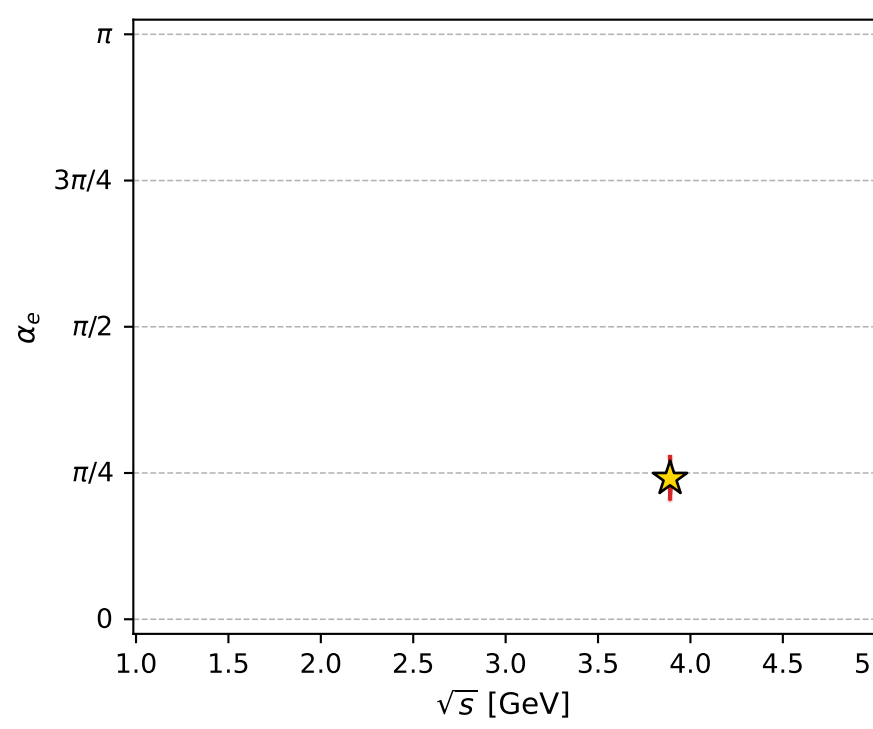
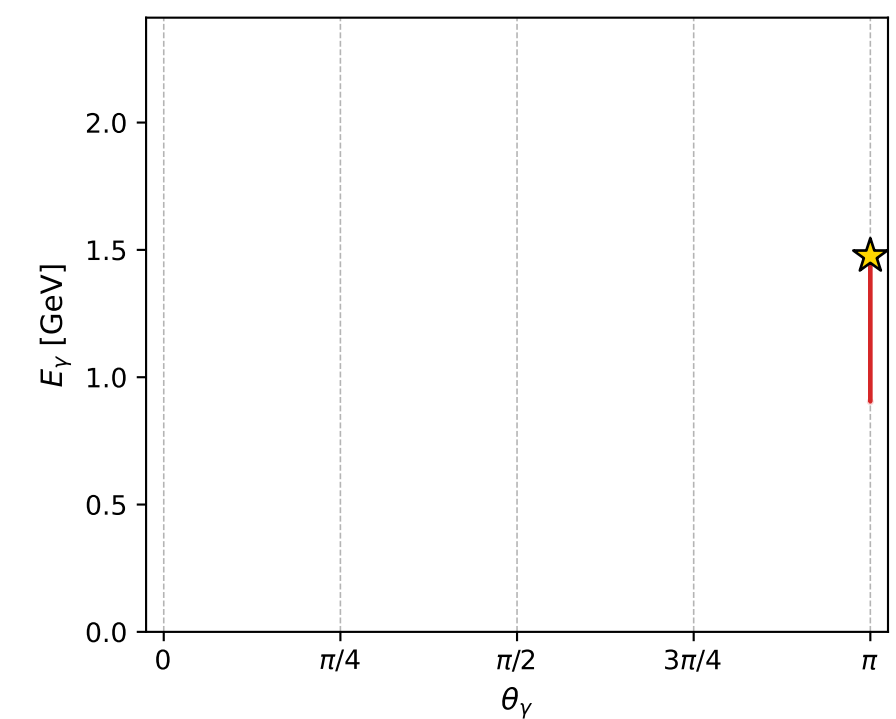
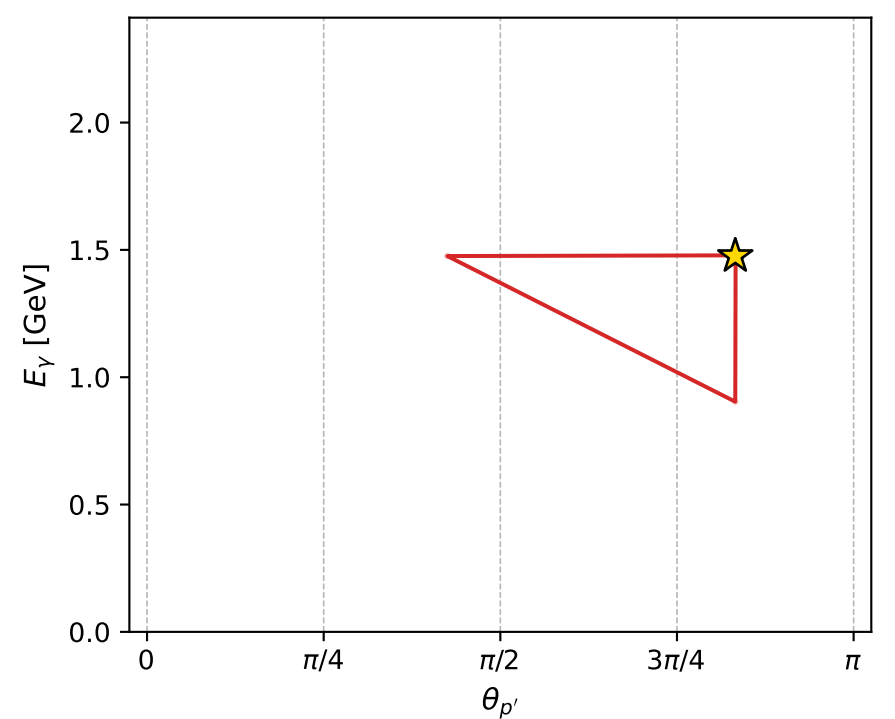
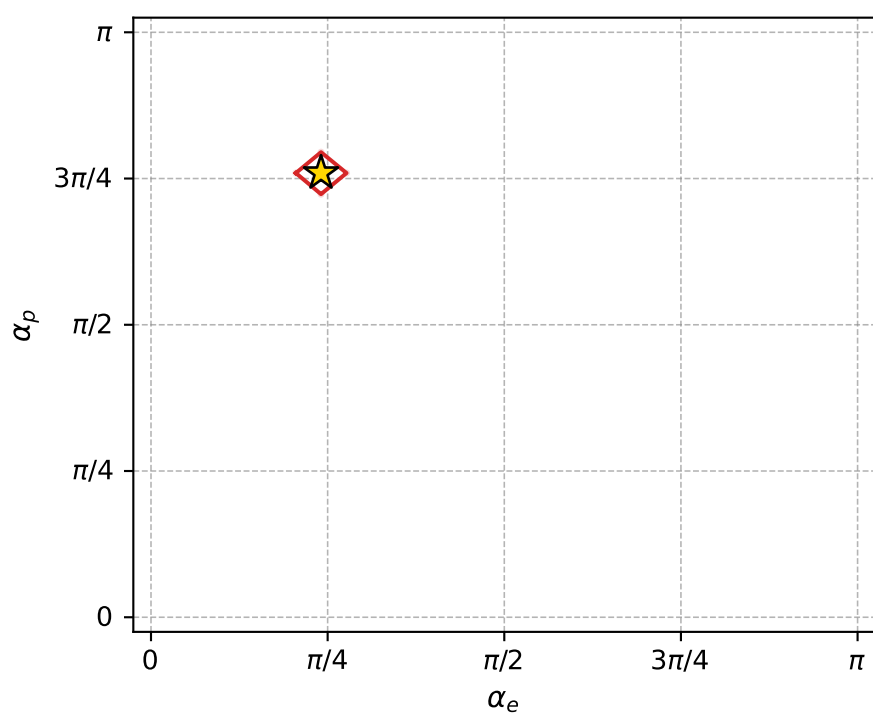
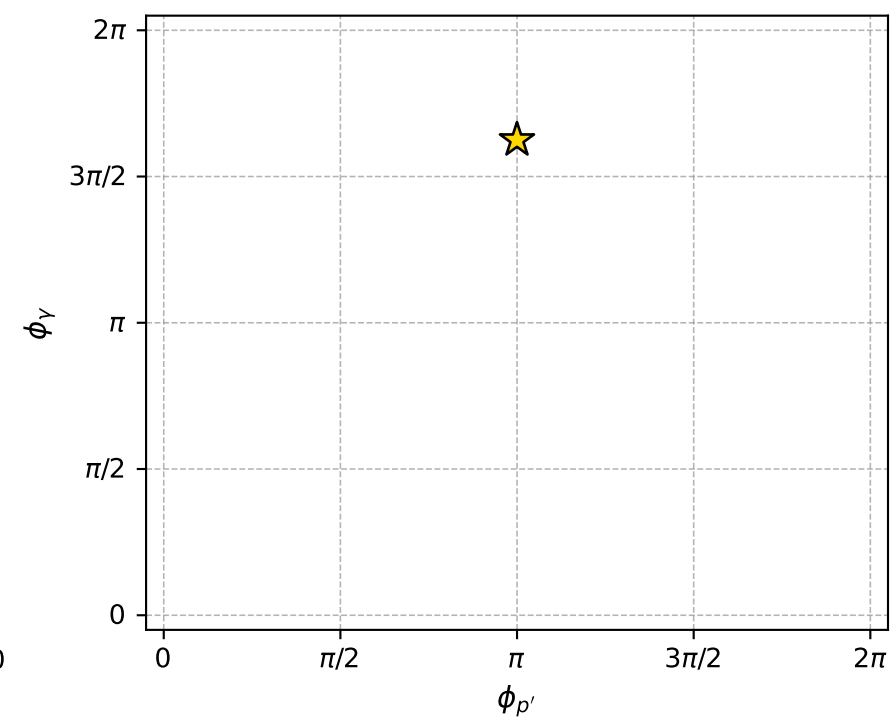
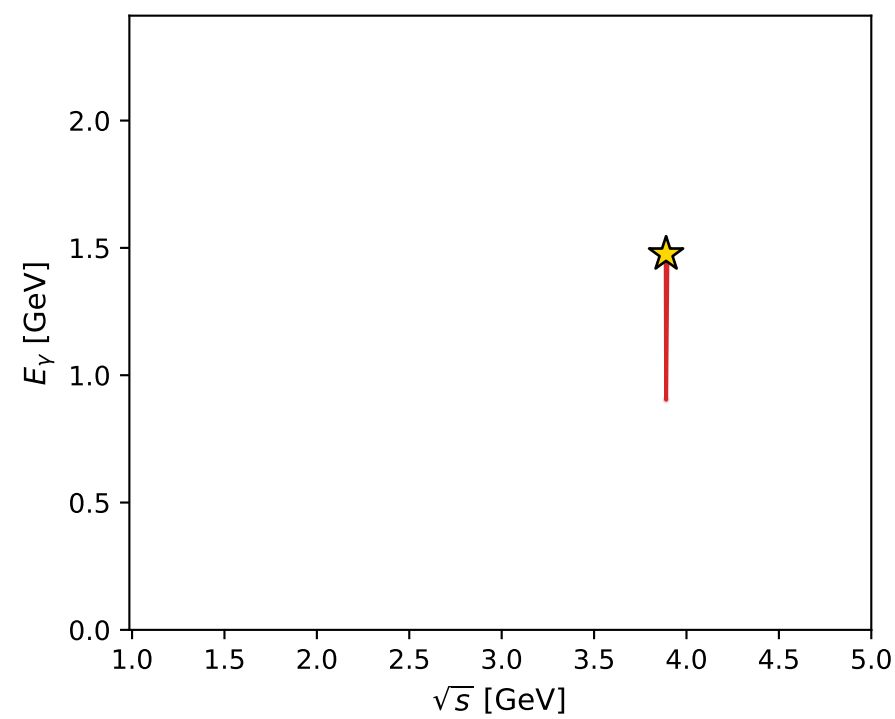
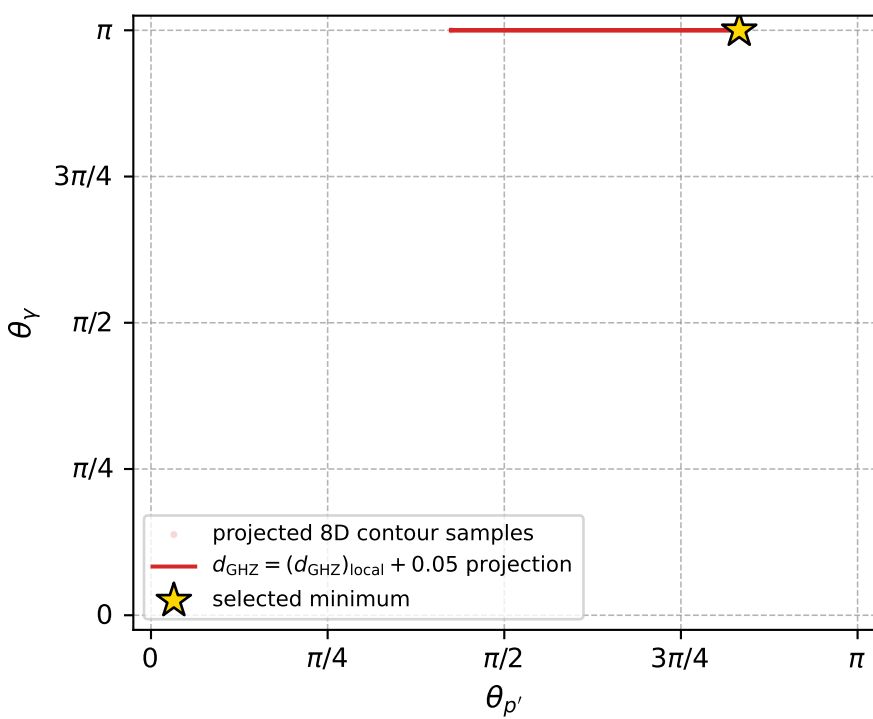
dghz_mixing_angles_polarization_02_local_minimum_111 d_{\rm GHZ}=4.10283e-08
region: polarization_cluster_02_local_minimum_111
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: $\alpha_e=0.75596649$ rad
incoming lepton: $0.727609|+\rangle +0.685992|-\rangle$
proton mixing angle: $\alpha_p=2.3856361$ rad
incoming proton: $-0.727616|+\rangle +0.685985|-\rangle$

$s=15.1296$, $\sqrt{s}=3.88967$, $|\vec{P}|=1.83174$, $|\vec{P}'|=6.45464\text{e-}06$
 $\theta_{p'}=2.61537$, $\theta_Y=3.14159$, $\phi_{p'}=3.141$, $\phi_Y=5.10493$
 $E_Y=1.47583$, $\theta_{\ell'}=2.19675\text{e-}06$, $\phi_{\ell'}=6.28259$
 $Q^2=10.8134$, $x_B=0.796154$, $t=-2.10102$, $W^2=3.64849$, $y=0.953145$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 1.8317$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.8317$
 P $E= 2.0579$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.8317$
 ℓ' $E= 1.4758$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 1.4758$
 P' $E= 0.9380$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= -0.0000$
 q_Y $E= 1.4758$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= -1.4758$



electron: polarization cluster P2, local minimum 111; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 4.1028322\text{e-}08$
 $d_{\text{GHZ}} \text{ contour} = 0.050000041$
 above global minimum = $1.874486\text{e-}08$
 8D contour samples = 129

selected phase-space point:

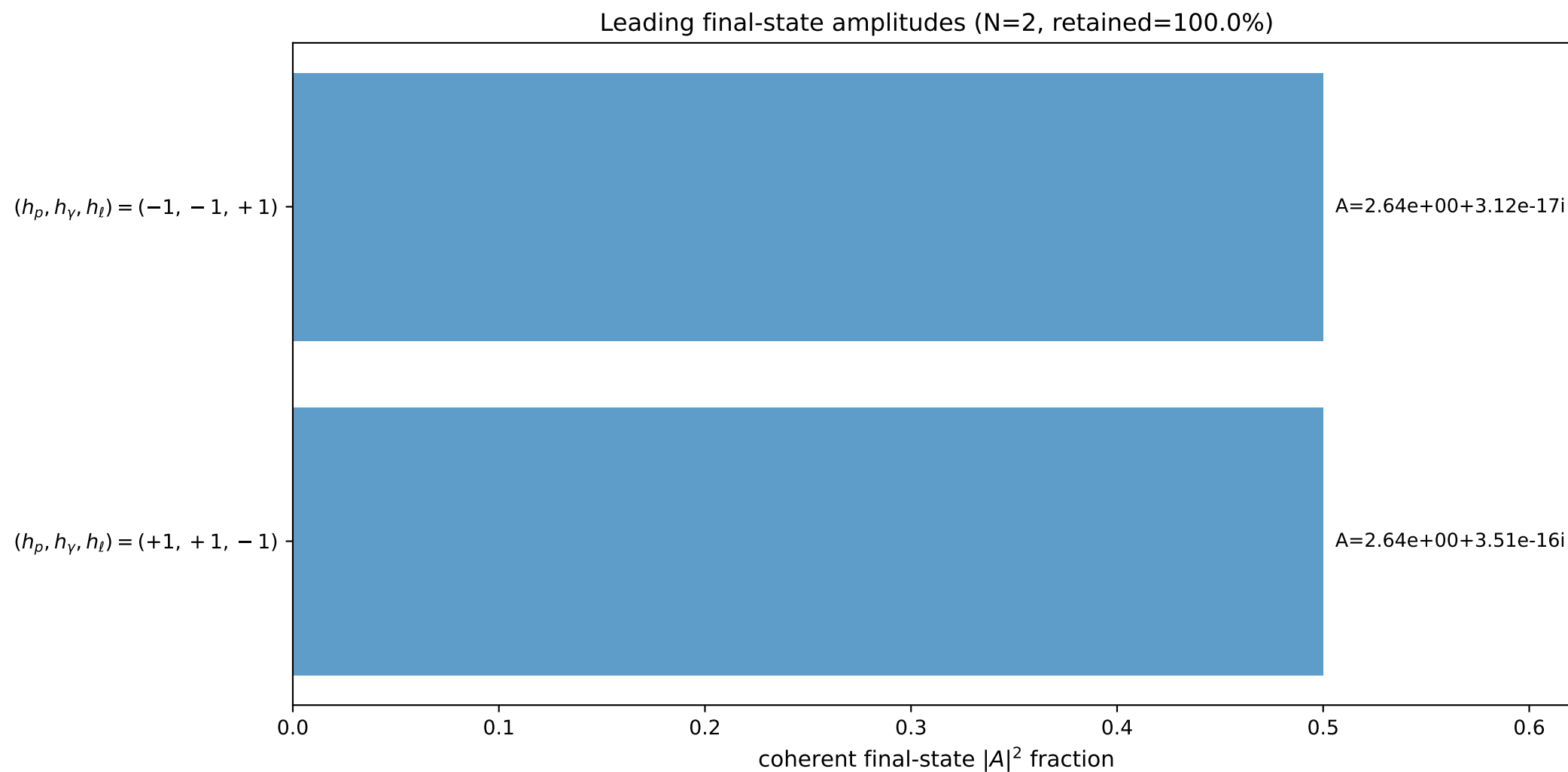
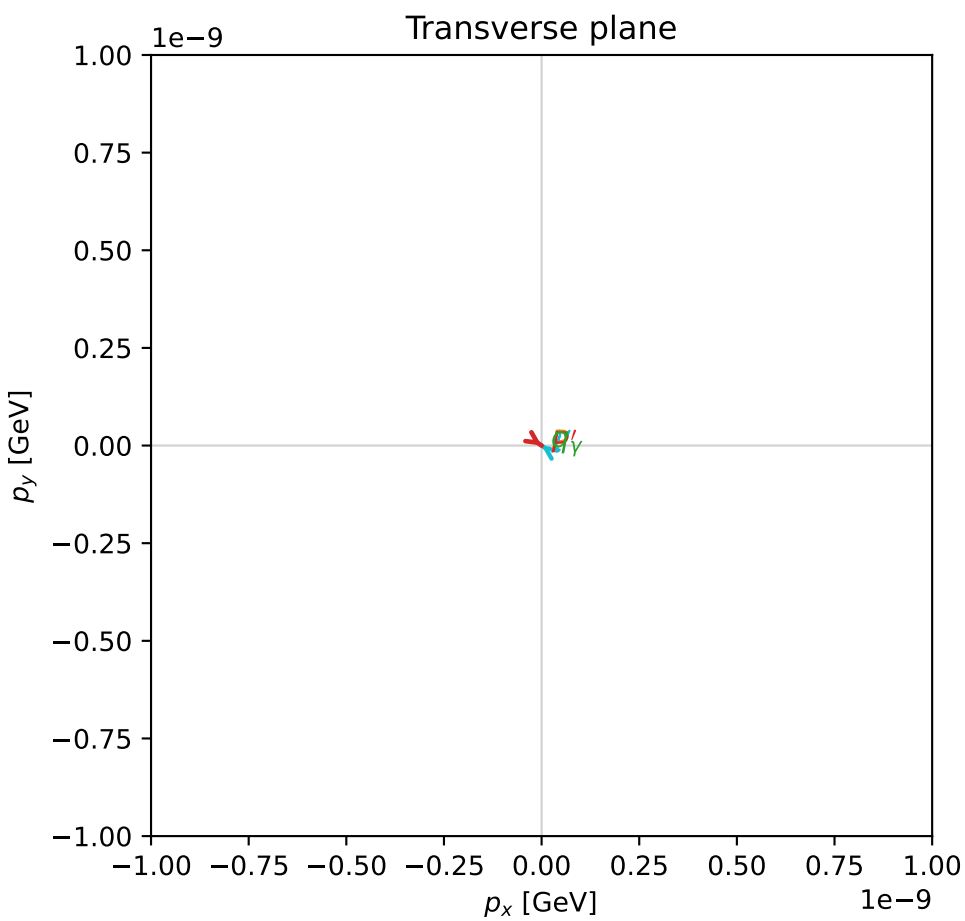
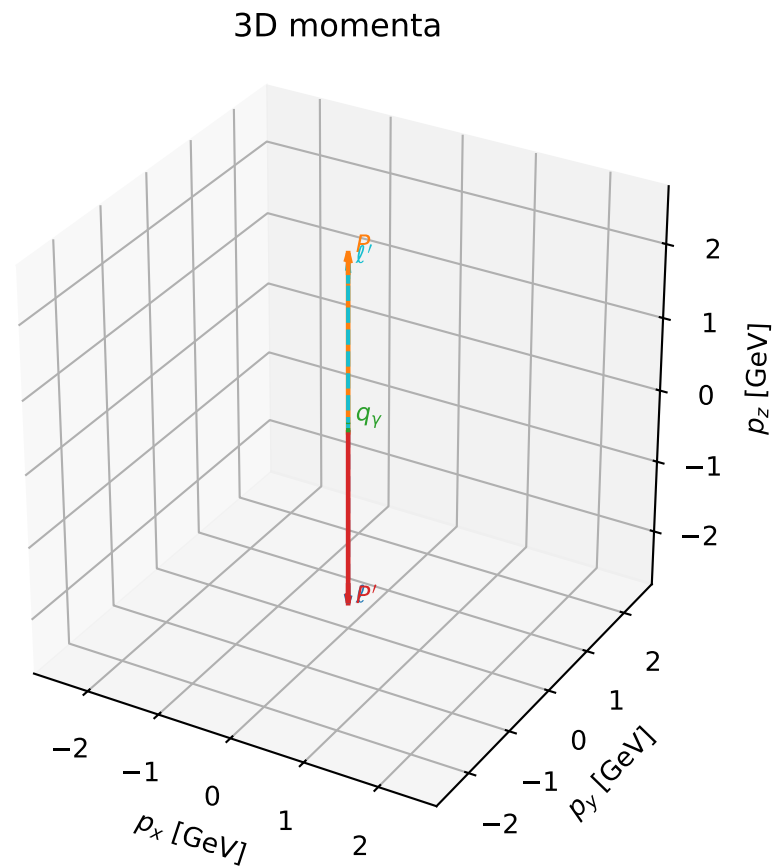
$\text{sqrt_s} = 3.889674$
 $\text{theta_p_out} = 2.615366$
 $\text{theta_gamma_out} = 3.141593$
 $\text{q0ut} = 1.475834$
 $\text{phi_p_out} = 3.140999$
 $\text{phi_gamma_out} = 5.104931$
 $\text{alpha_e} = 0.7559665$
 $\text{alpha_p} = 2.385636$

coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

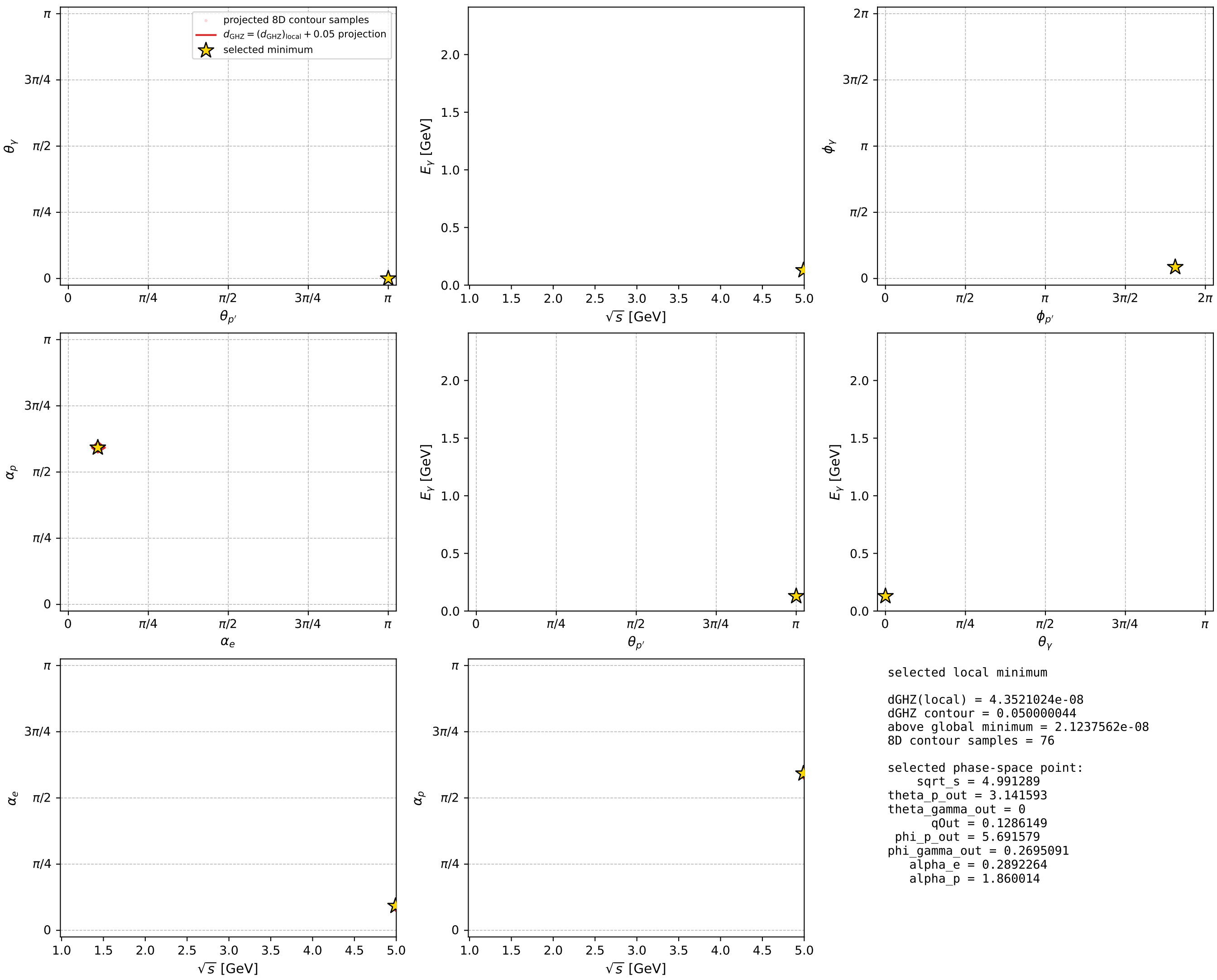
dghz_mixing_angles_polarization_02_local_minimum_116 $d_{\{\text{rm GHZ}\}}=4.3521\text{e-}08$
 region: polarization_cluster_02_local_minimum_116
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=0.28922643$ rad
 incoming lepton: $0.958465|+\rangle +0.285211|-\rangle$
 proton mixing angle: $\alpha_p=1.8600136$ rad
 incoming proton: $-0.285202|+\rangle +0.958467|-\rangle$

$s=24.913$, $\sqrt{s}=4.99129$, $|\vec{P}|=2.40751$, $|\vec{P}'|=2.40751$
 $\theta_{P'}=3.14159$, $\theta_Y=0$, $\phi_{P'}=5.69158$, $\phi_Y=0.269509$
 $E_Y=0.128615$, $\theta_{\ell'}=0$, $\phi_{\ell'}=2.54999$
 $Q^2=21.9458$, $x_B=0.94473$, $t=-23.1844$, $W^2=2.16375$, $y=0.96657$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4075$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4075$
 P $E= 2.5838$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4075$
 ℓ' $E= 2.2789$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 2.2789$
 P' $E= 2.5838$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= -2.4075$
 q_Y $E= 0.1286$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1286$



electron: polarization cluster P2, local minimum 116; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



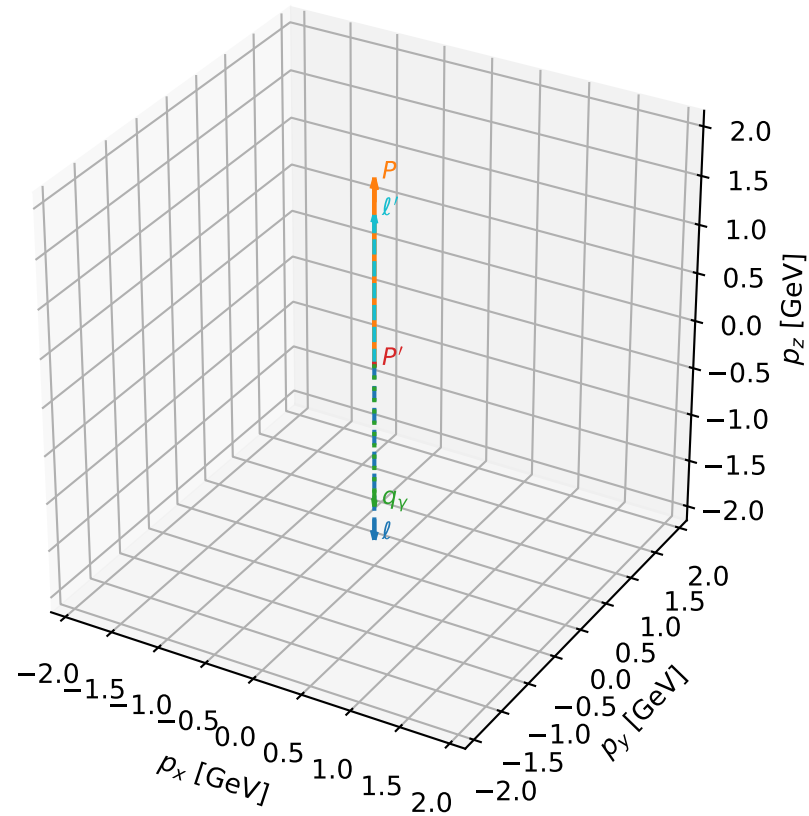
coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

dghz_mixing_angles_polarization_02_local_minimum_120 $d_{\{\text{rm GHZ}\}}=4.59538\text{e-}08$
region: polarization_cluster_02_local_minimum_120
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: $\alpha_e=0.74588136$ rad
incoming lepton: $0.73449|+\rangle +0.678619|-\rangle$
proton mixing angle: $\alpha_p=2.3956994$ rad
incoming proton: $-0.734482|+\rangle +0.678628|-\rangle$

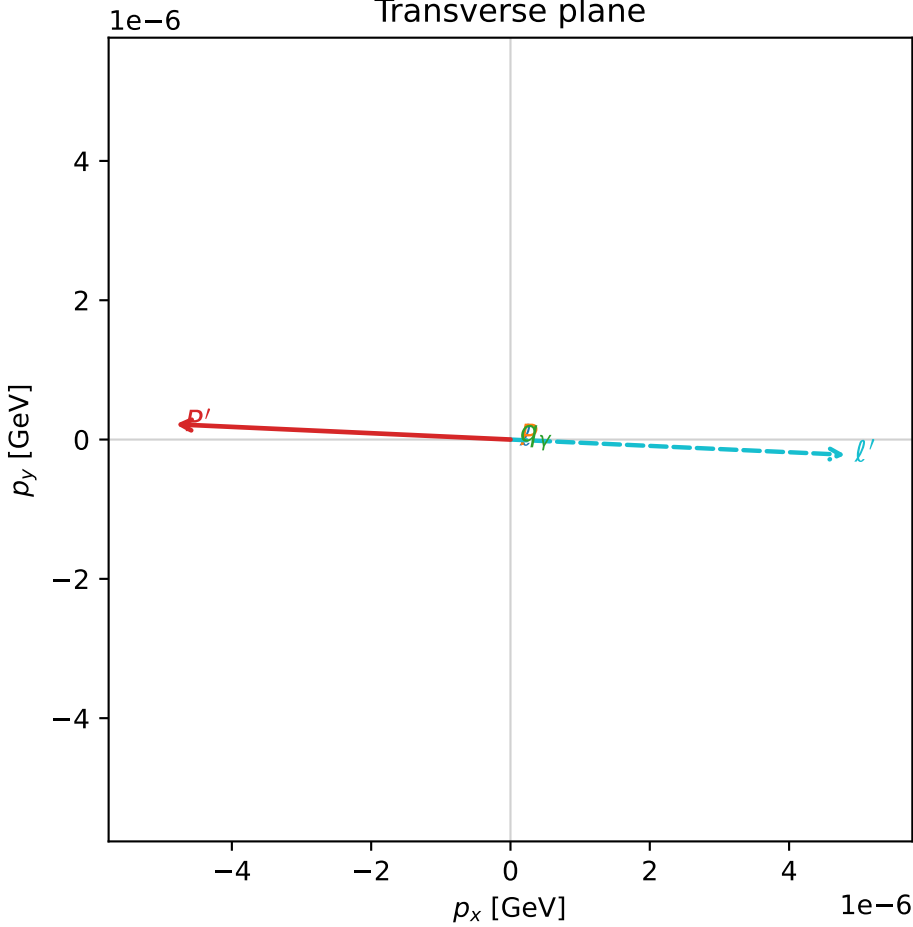
$s=15.7593$, $\sqrt{s}=3.9698$, $|\vec{P}|=1.87408$, $|\vec{P}'|=0.00992053$
 $\theta_{p'}=3.14111$, $\theta_Y=3.14159$, $\phi_{p'}=3.09611$, $\phi_Y=2.65443$
 $E_Y=1.51091$, $\theta_{\ell'}=3.16449\text{e-}06$, $\phi_{\ell'}=6.23771$
 $Q^2=11.4006$, $x_B=0.802563$, $t=-2.20928$, $W^2=3.68449$, $y=0.954693$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 1.8741$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.8741$
 P $E= 2.0957$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.8741$
 ℓ' $E= 1.5208$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 1.5208$
 P' $E= 0.9381$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= -0.0099$
 q_Y $E= 1.5109$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= -1.5109$

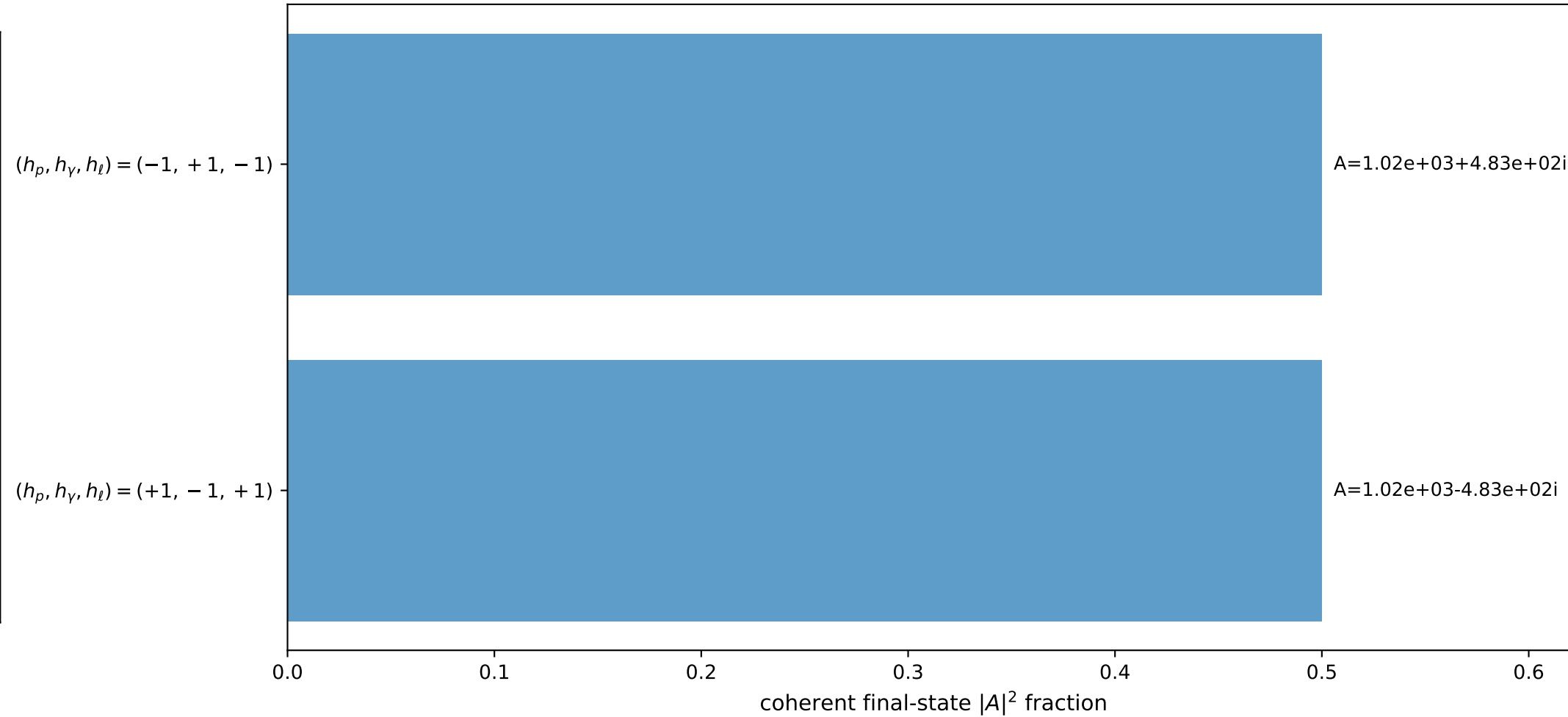
3D momenta



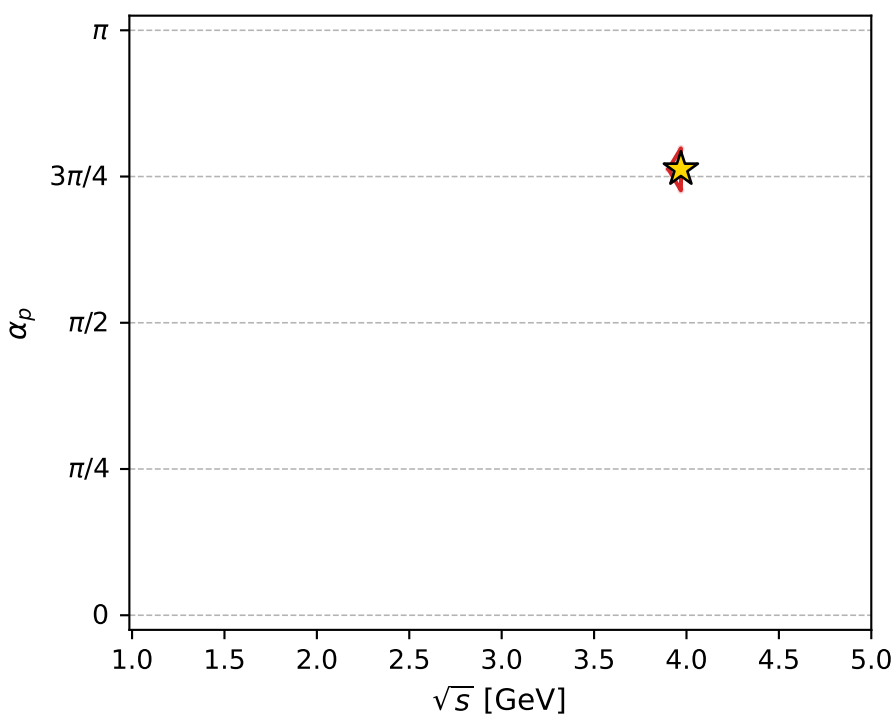
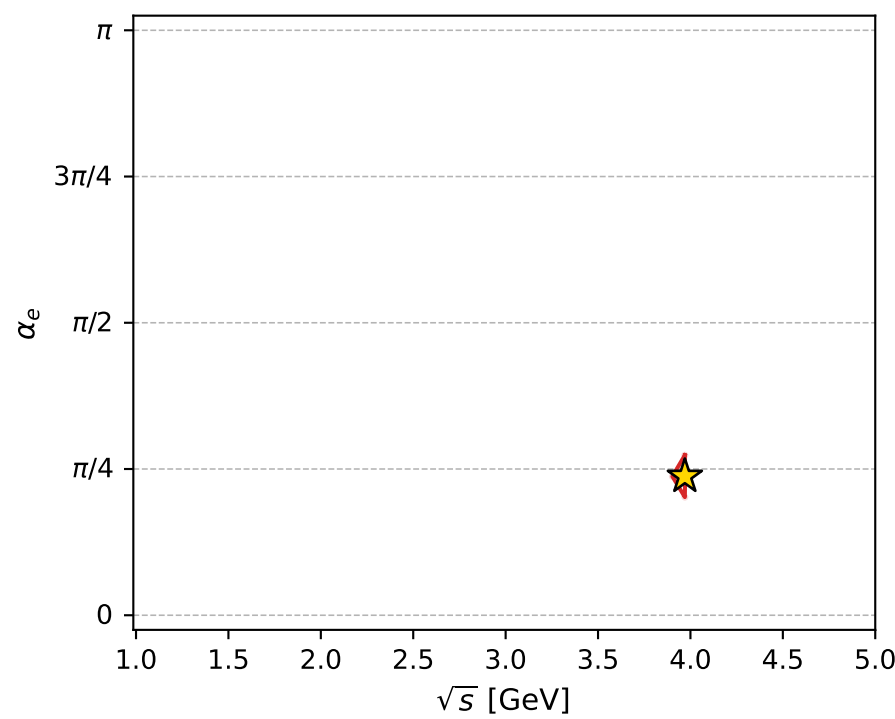
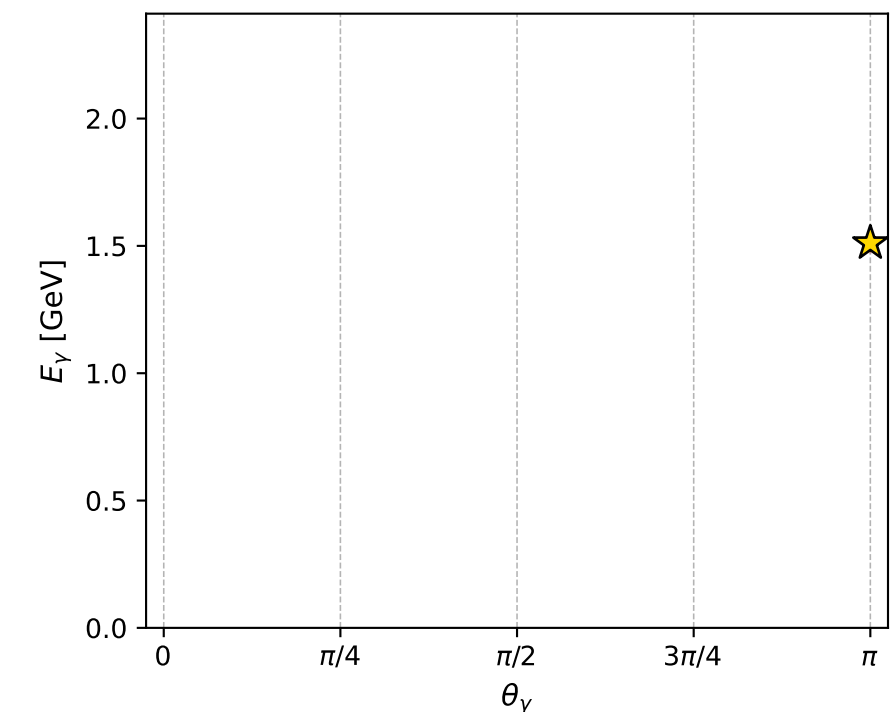
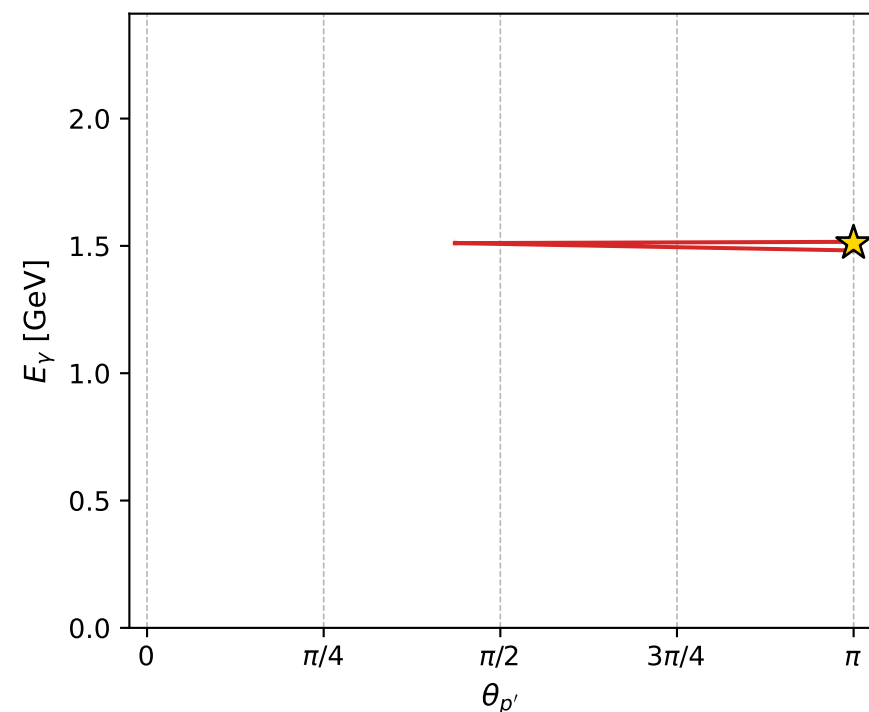
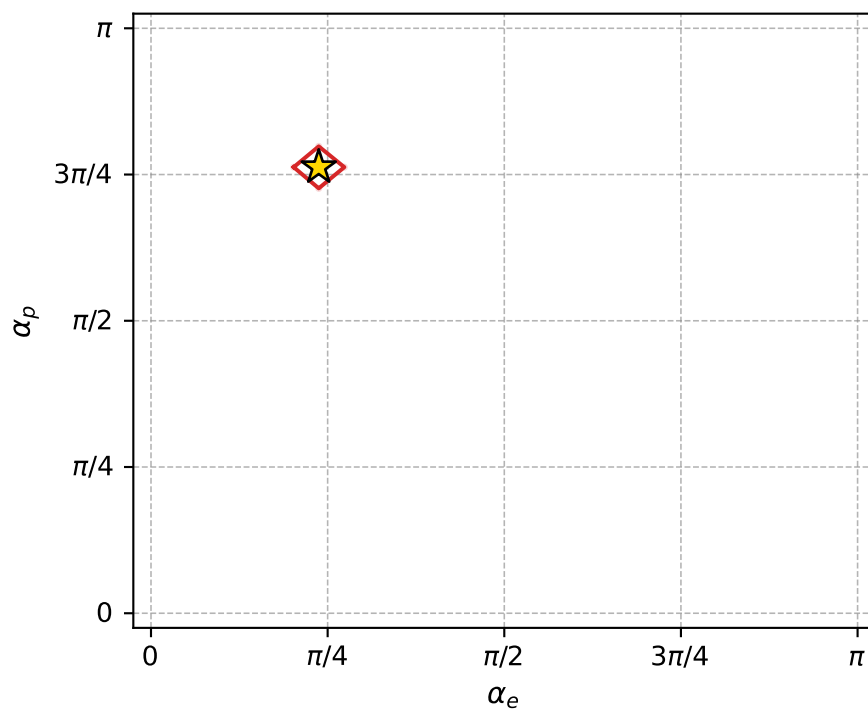
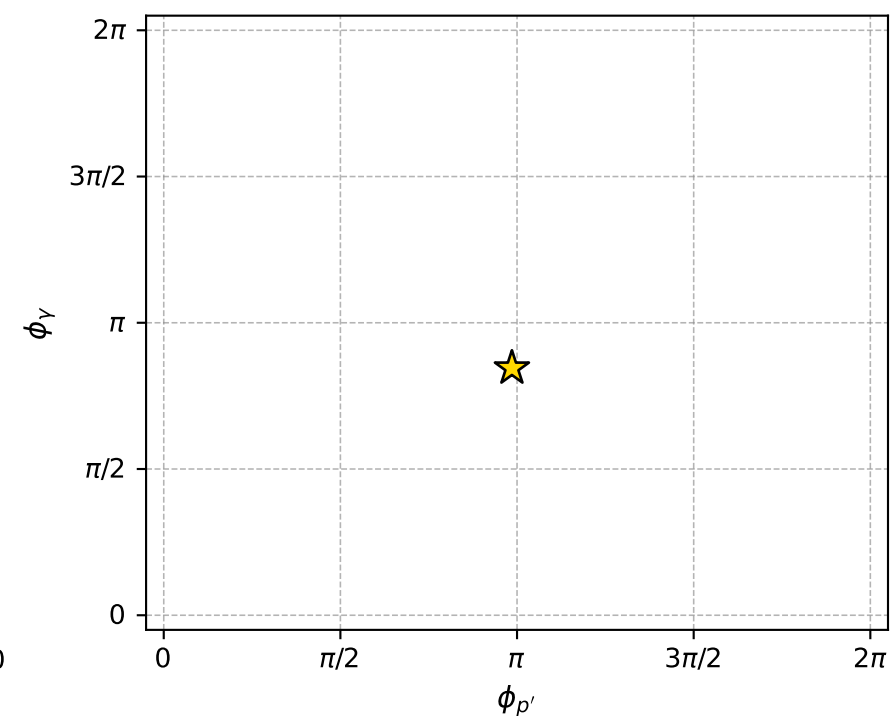
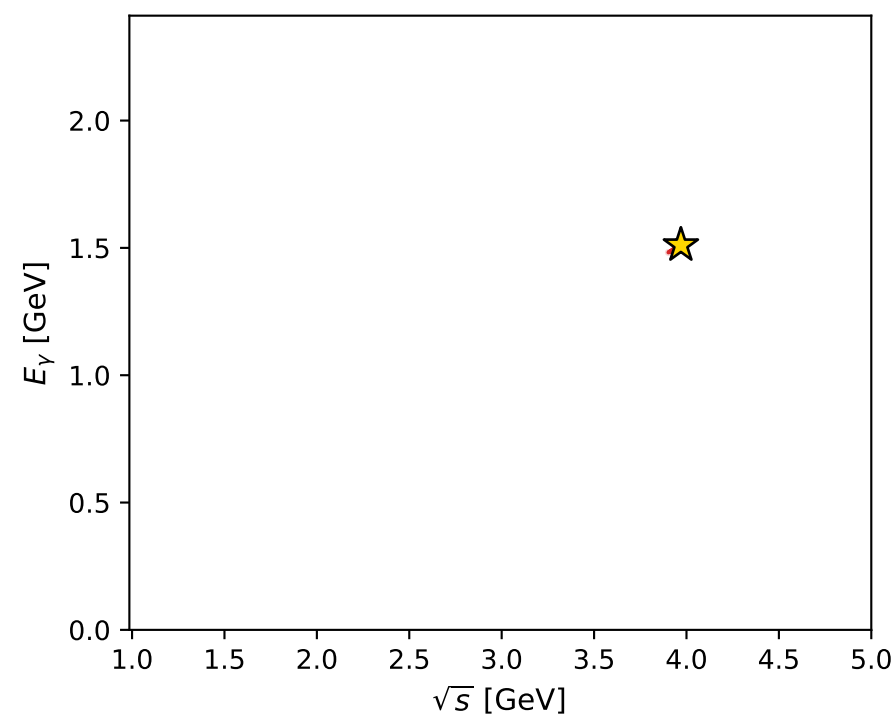
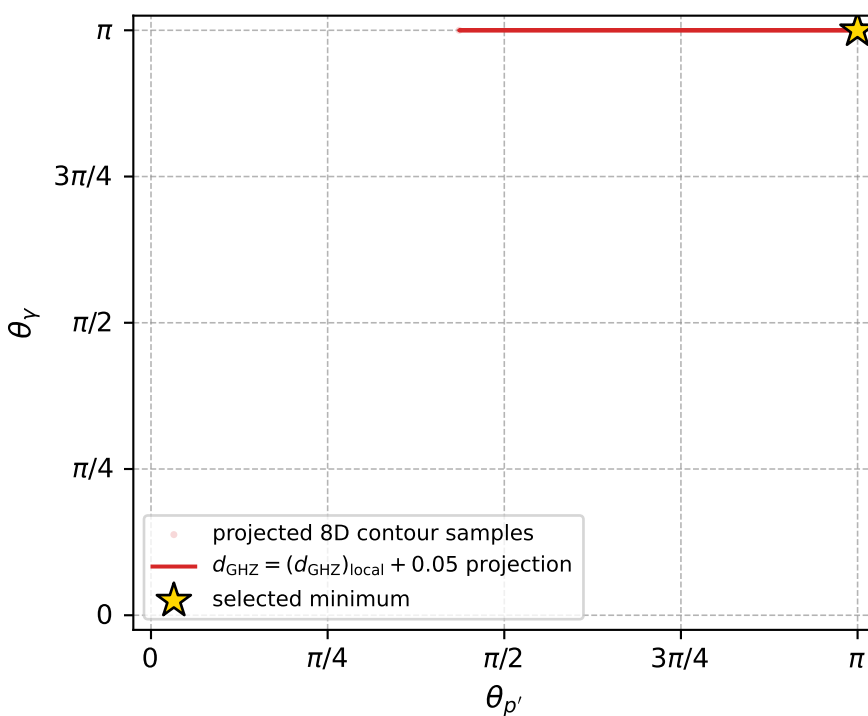
Transverse plane



Leading final-state amplitudes (N=2, retained=100.0%)



electron: polarization cluster P2, local minimum 120; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 4.5953813\text{e-}08$
 $d_{\text{GHZ}} \text{ contour} = 0.050000046$
 above global minimum = $2.3670351\text{e-}08$
 8D contour samples = 112

selected phase-space point:

$\text{sqrt_s} = 3.969796$
 $\text{theta_p_out} = 3.141108$
 $\text{theta_gamma_out} = 3.141593$
 $\text{q0out} = 1.510912$
 $\text{phi_p_out} = 3.096113$
 $\text{phi_gamma_out} = 2.654434$
 $\text{alpha_e} = 0.7458814$
 $\text{alpha_p} = 2.395699$

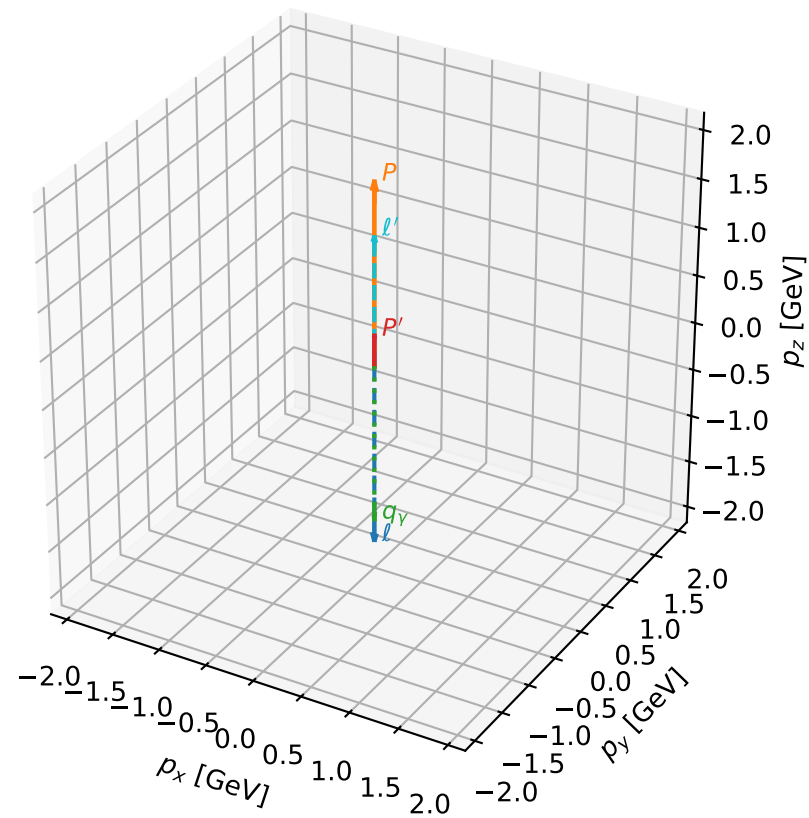
coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

dghz_mixing_angles_polarization_02_local_minimum_127 d_{\rm GHZ}=5.10304e-08
region: polarization_cluster_02_local_minimum_127
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: $\alpha_e=0.78216053$ rad
incoming lepton: $0.709392|+\rangle +0.704814|-\rangle$
proton mixing angle: $\alpha_p=2.3594272$ rad
incoming proton: $-0.709389|+\rangle +0.704817|-\rangle$

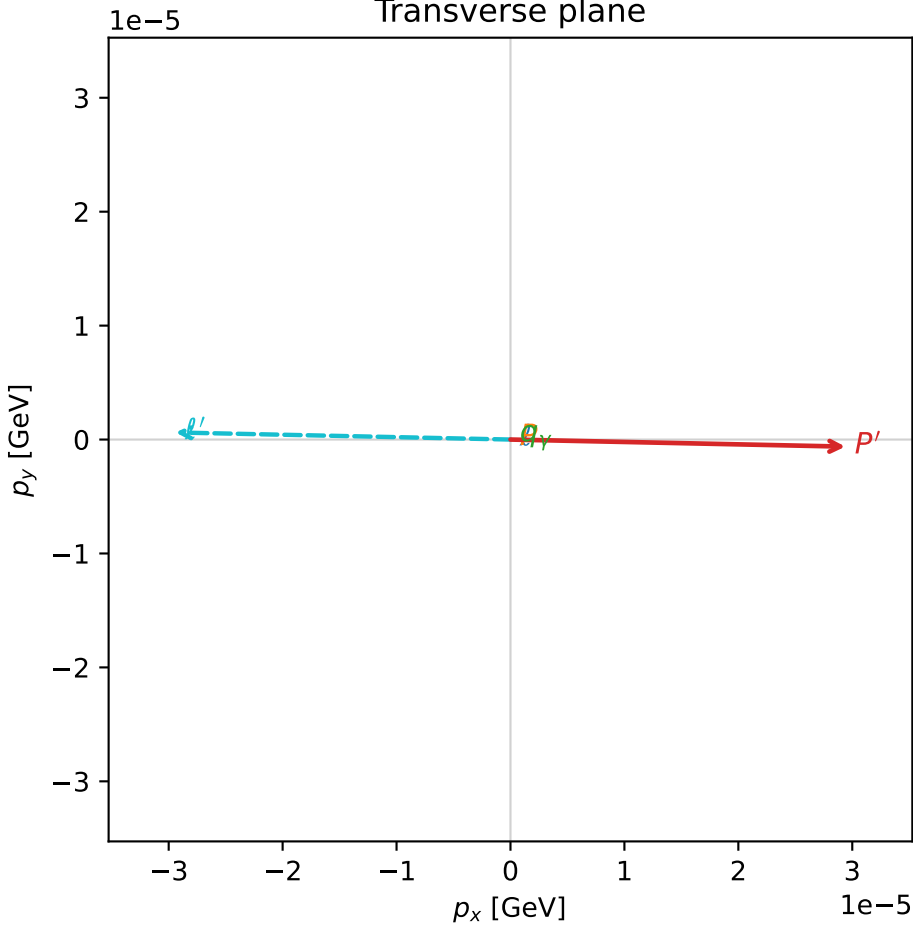
$s=16.0046$, $\sqrt{s}=4.00057$, $|\vec{P}|=1.89032$, $|\vec{P}'|=0.316348$
 $\theta_{P'}=9.29374e-05$, $\theta_Y=3.14159$, $\phi_{P'}=6.26205$, $\phi_Y=1.81607$
 $E_Y=1.6635$, $\theta_{\ell'}=2.18242e-05$, $\phi_{\ell'}=3.12046$
 $Q^2=10.1862$, $x_B=0.700944$, $t=-1.22222$, $W^2=5.22577$, $y=0.960822$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 1.8903$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.8903$
 P $E= 2.1102$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.8903$
 ℓ' $E= 1.3472$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 1.3472$
 P' $E= 0.9899$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 0.3163$
 q_Y $E= 1.6635$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= -1.6635$

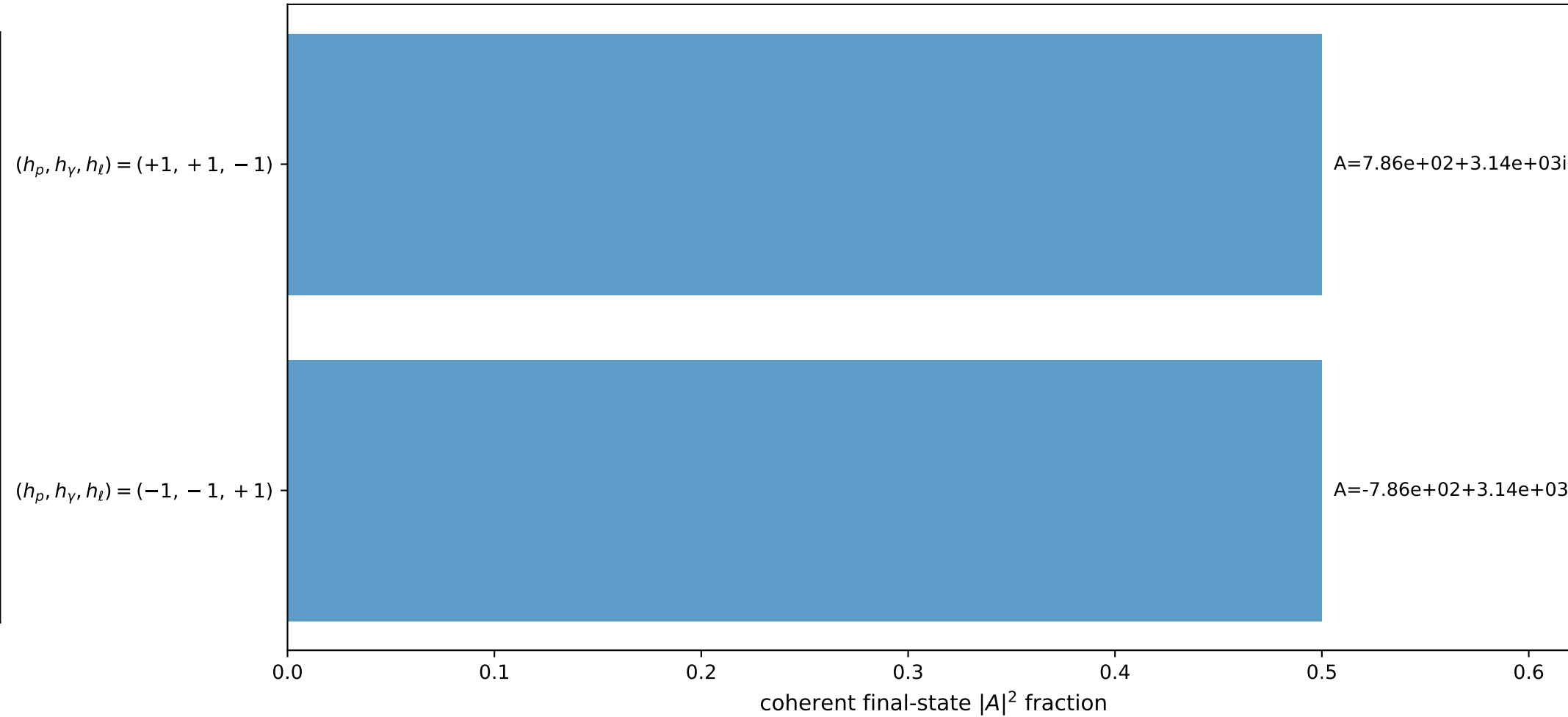
3D momenta



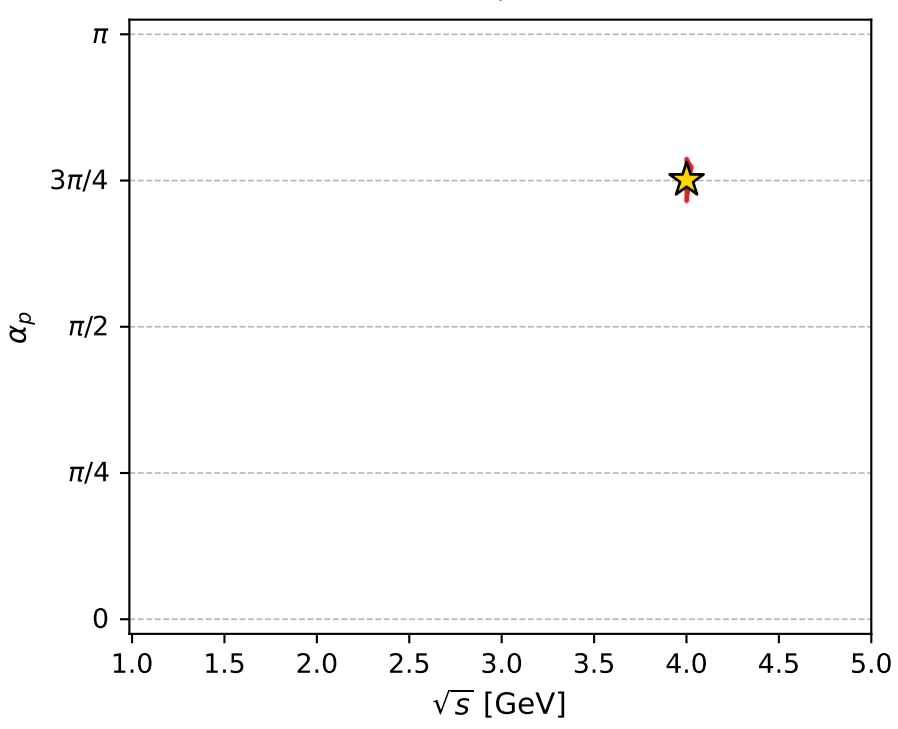
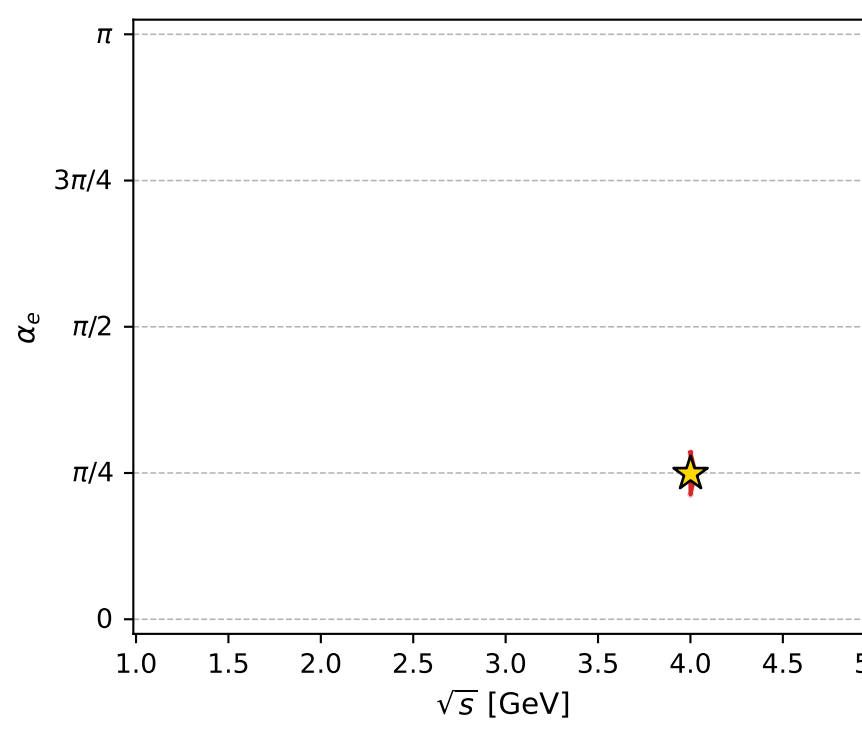
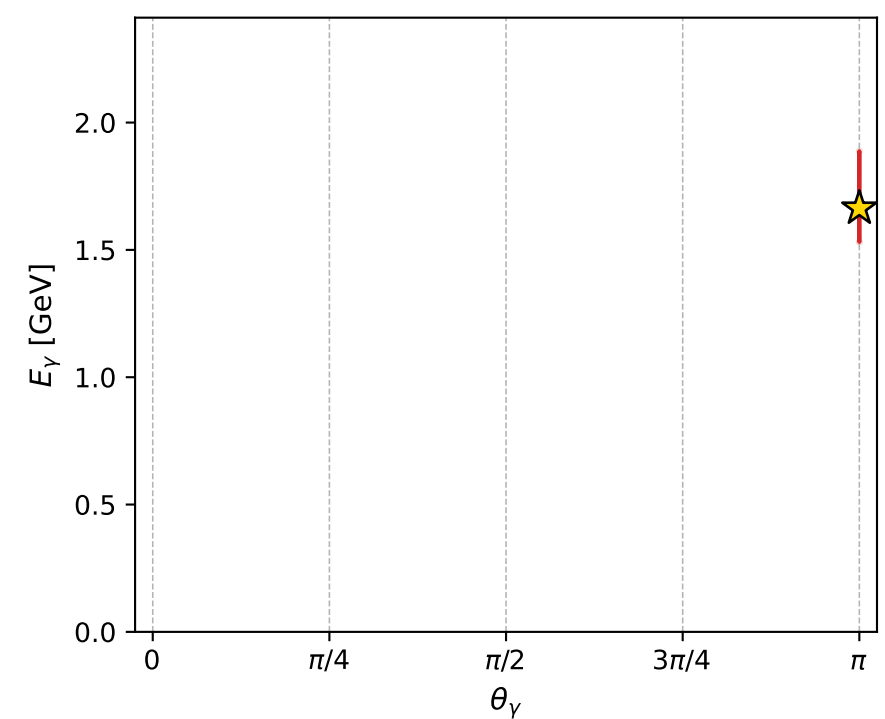
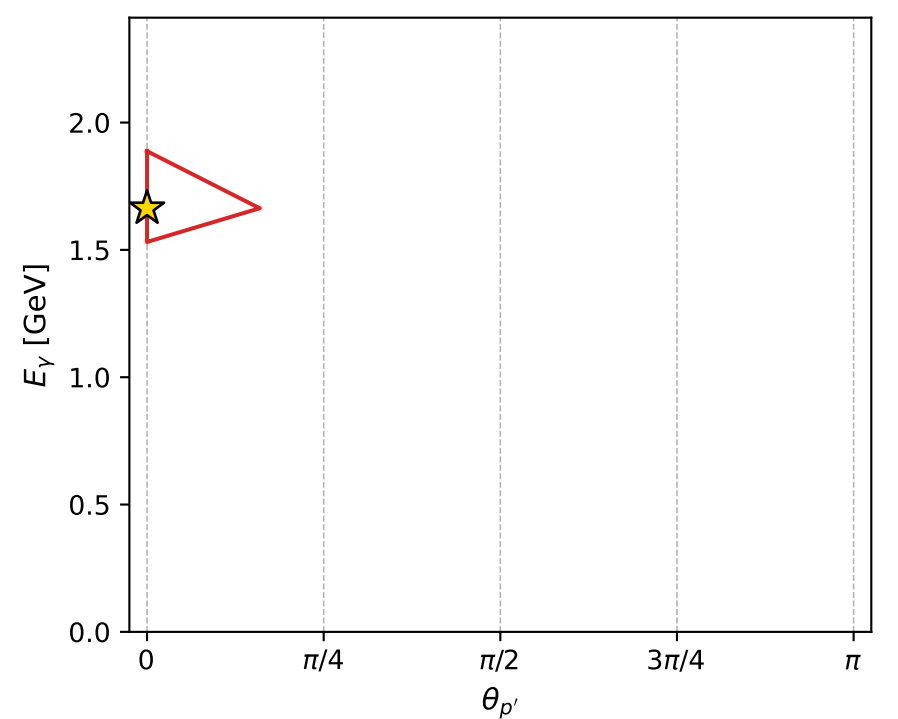
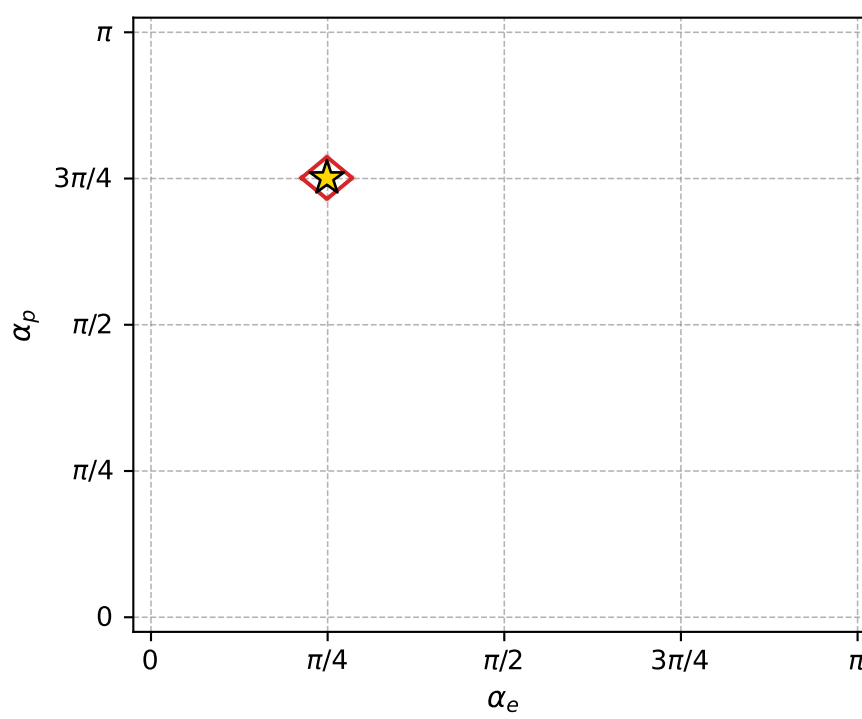
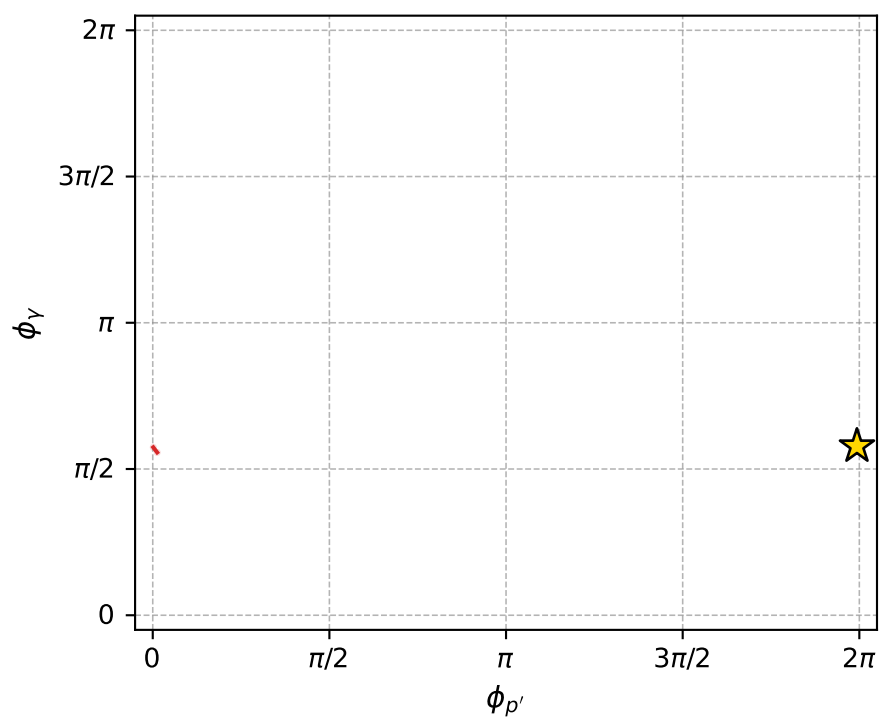
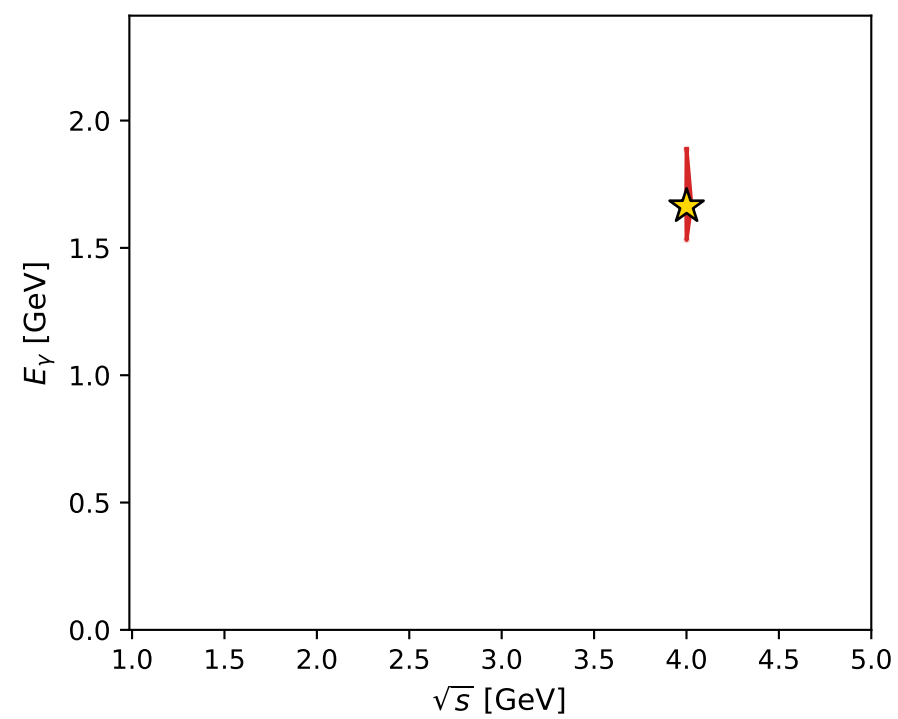
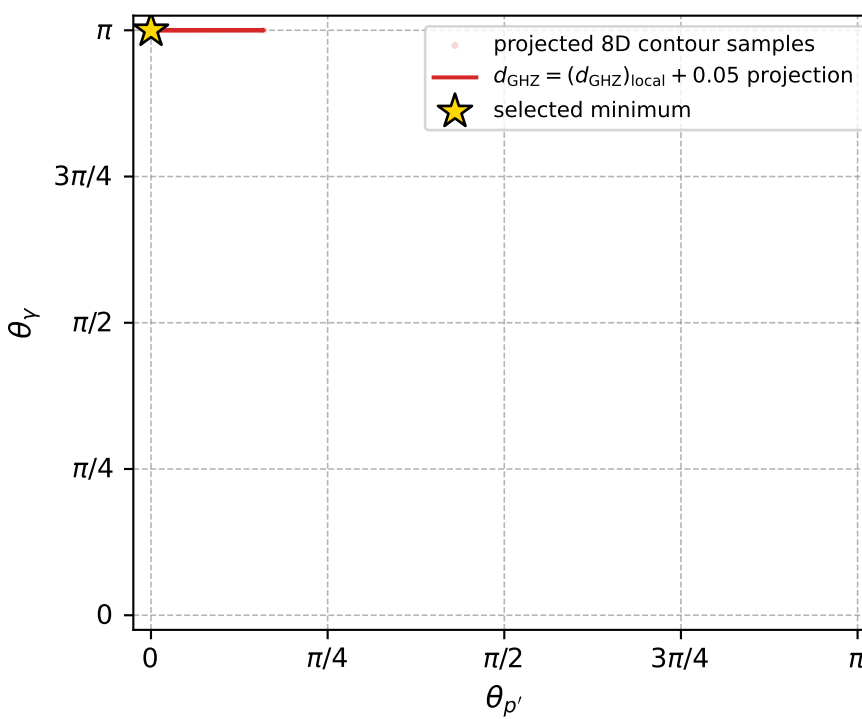
Transverse plane



Leading final-state amplitudes (N=2, retained=100.0%)



electron: polarization cluster P2, local minimum 127; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 5.1030393\text{e-}08$
 $d_{\text{GHZ}} \text{ contour} = 0.050000051$
 above global minimum = $2.8746931\text{e-}08$
 8D contour samples = 85

selected phase-space point:

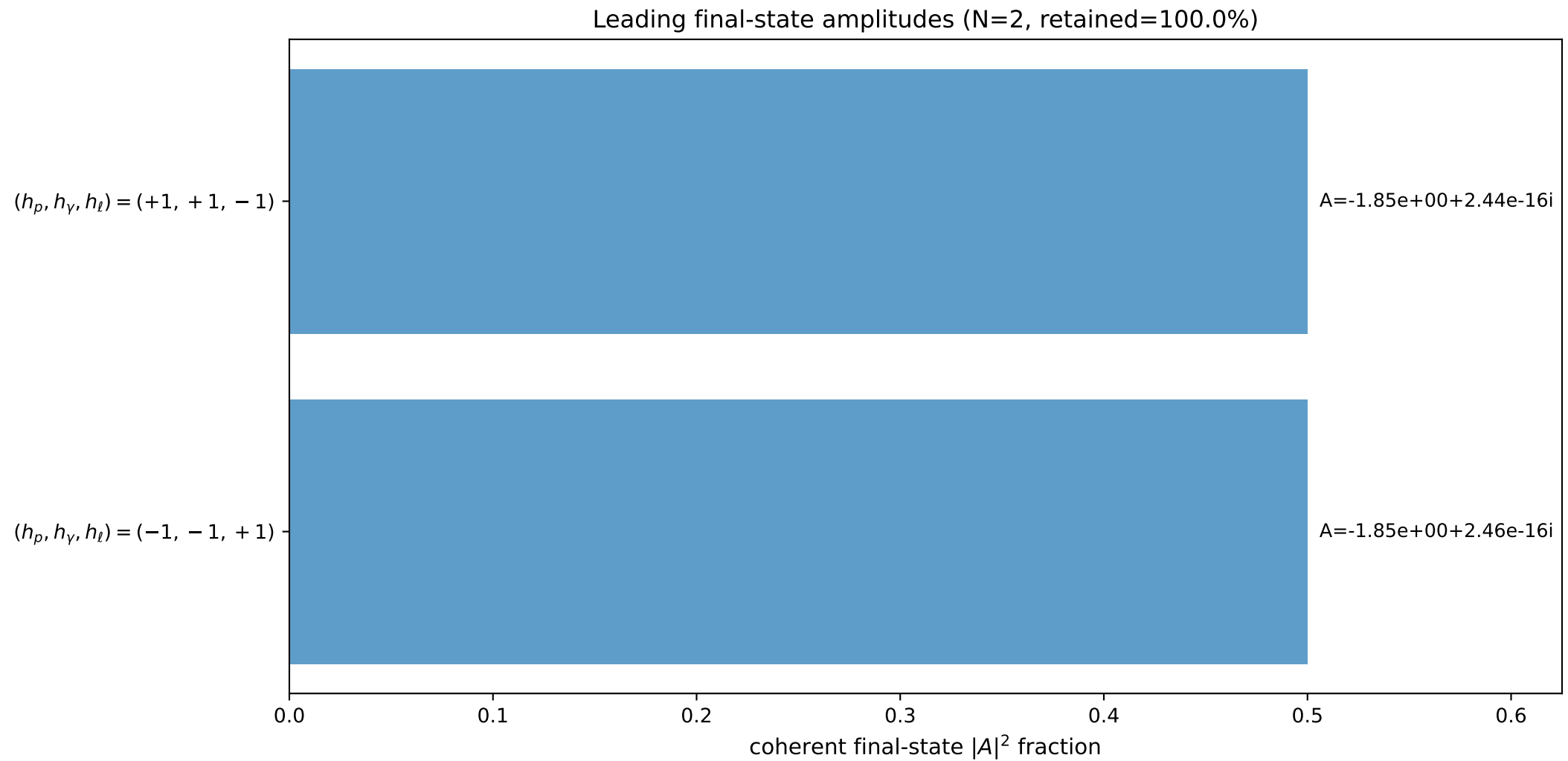
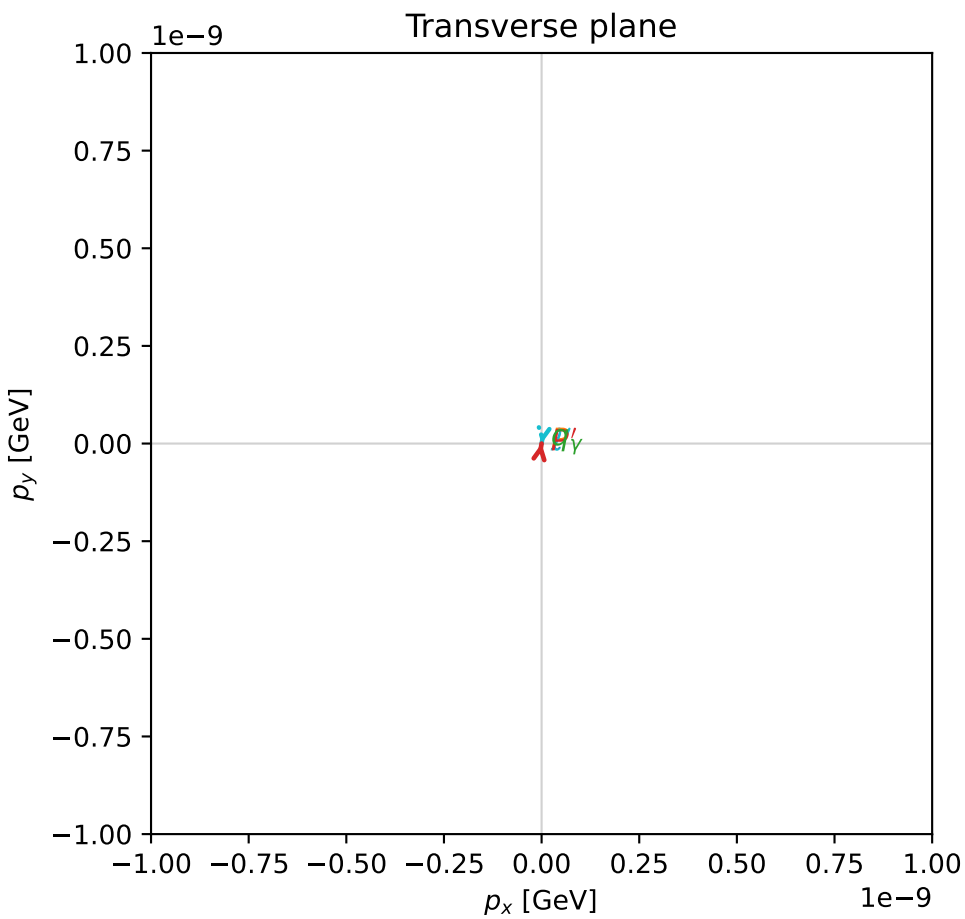
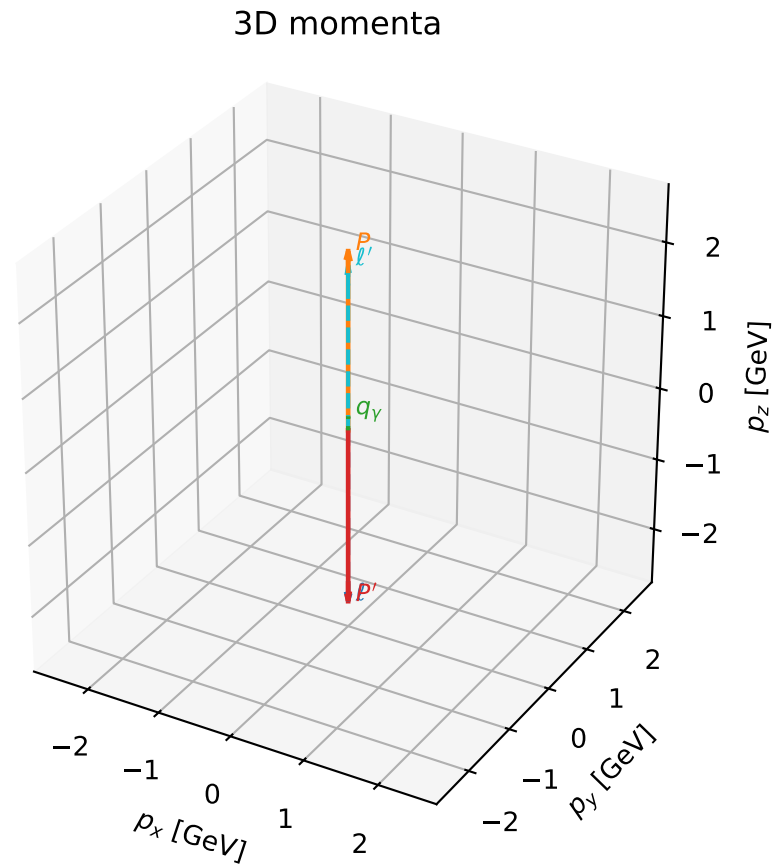
$\text{sqrt_s} = 4.00057$
 $\text{theta_p_out} = 9.29374\text{e-}05$
 $\text{theta_gamma_out} = 3.141593$
 $\text{q0ut} = 1.663505$
 $\text{phi_p_out} = 6.26205$
 $\text{phi_gamma_out} = 1.816075$
 $\text{alpha_e} = 0.7821605$
 $\text{alpha_p} = 2.359427$

coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

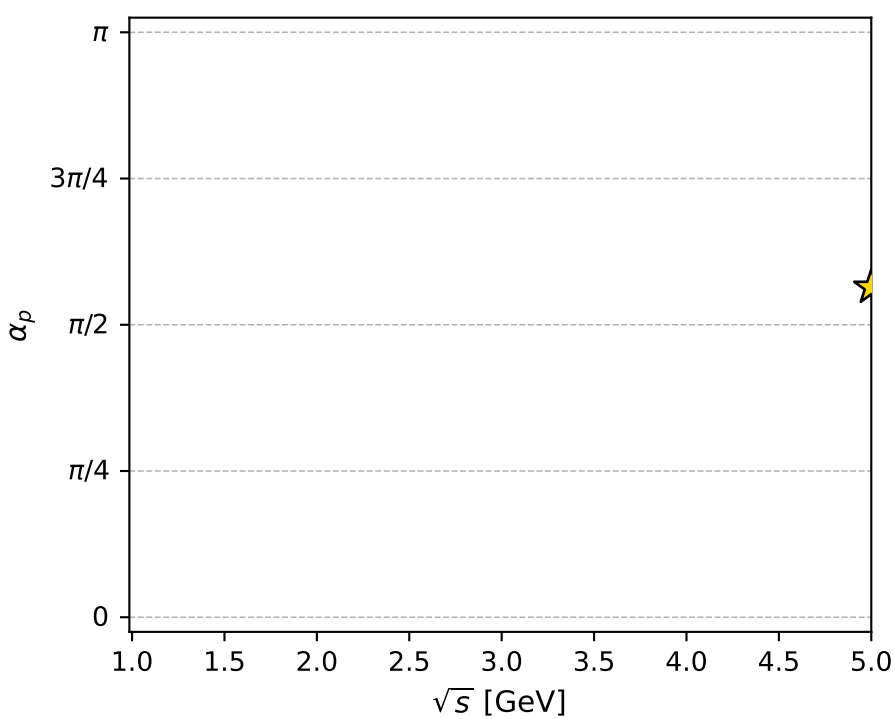
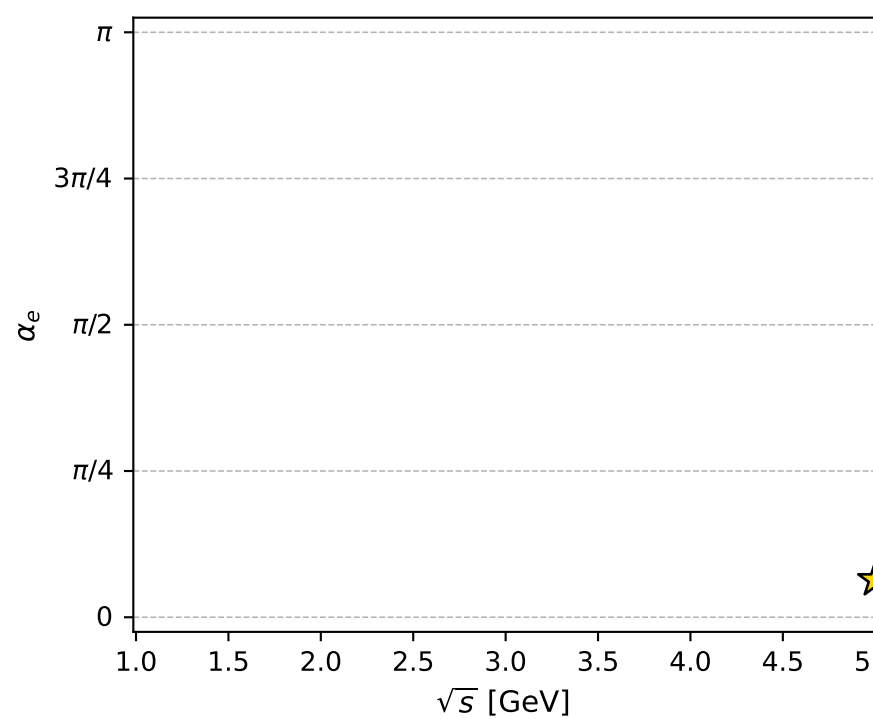
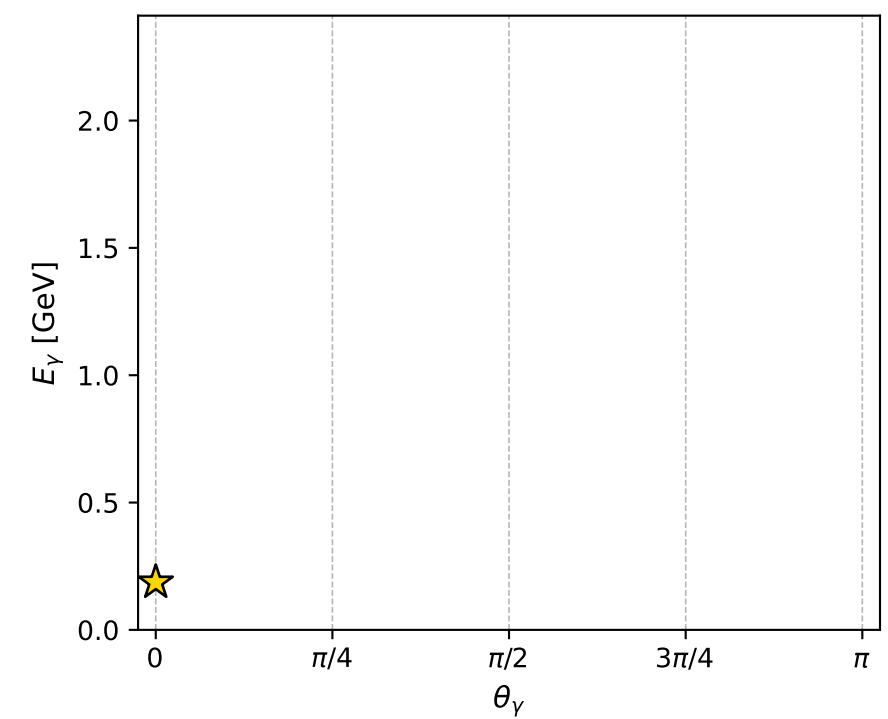
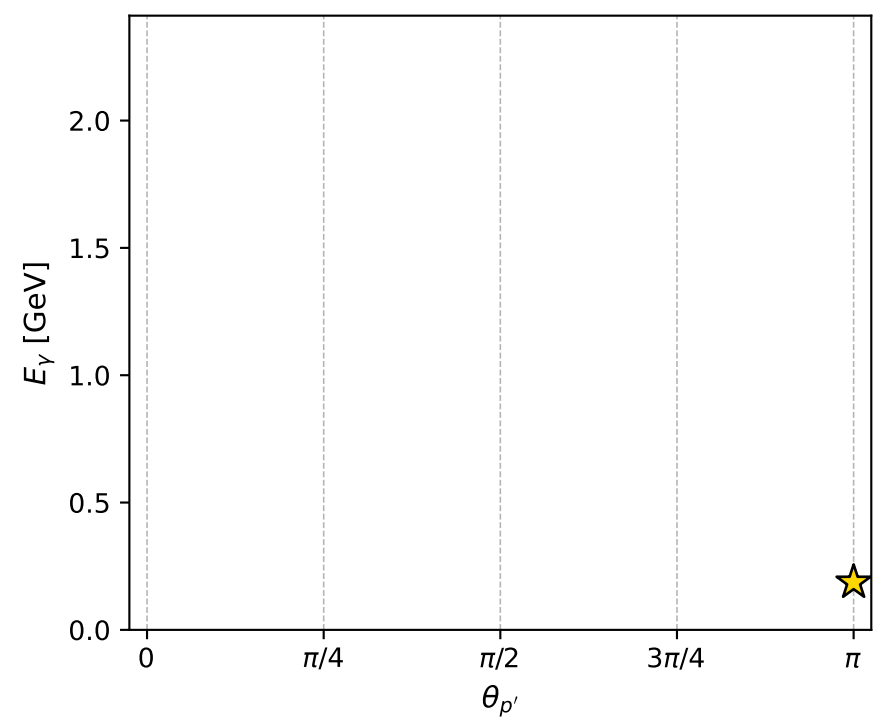
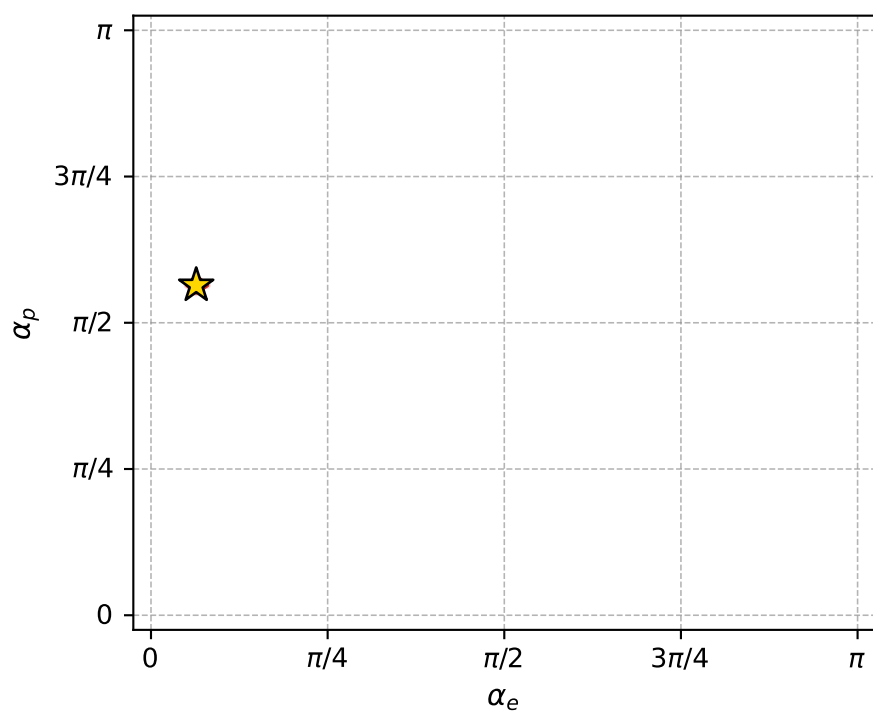
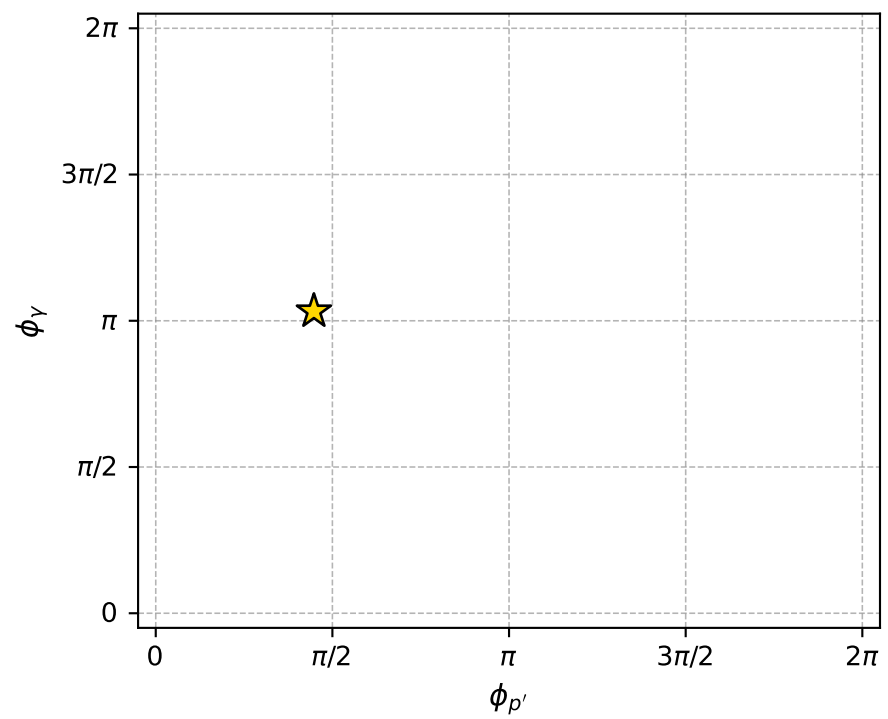
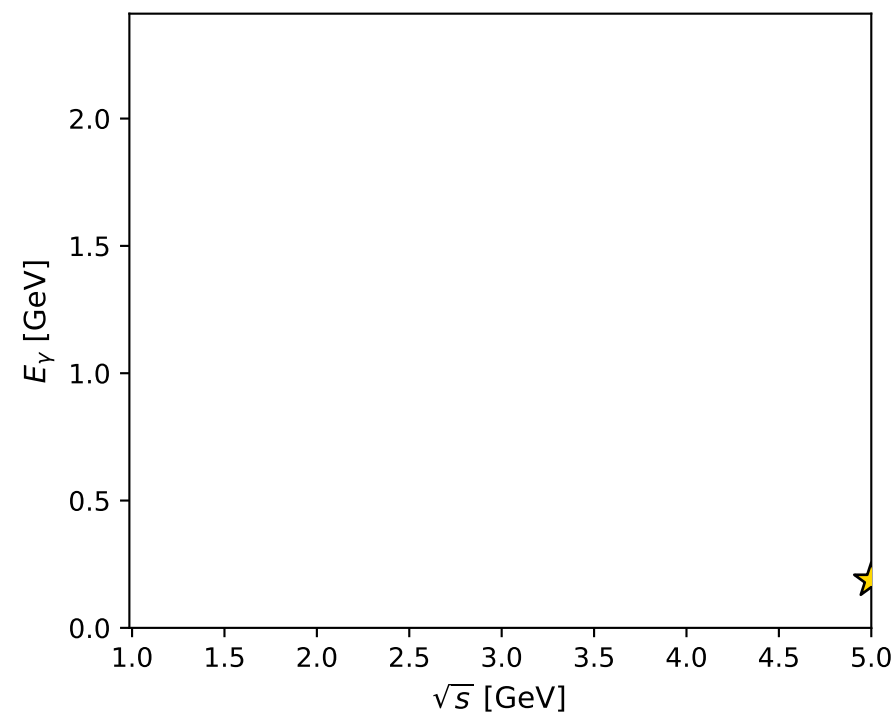
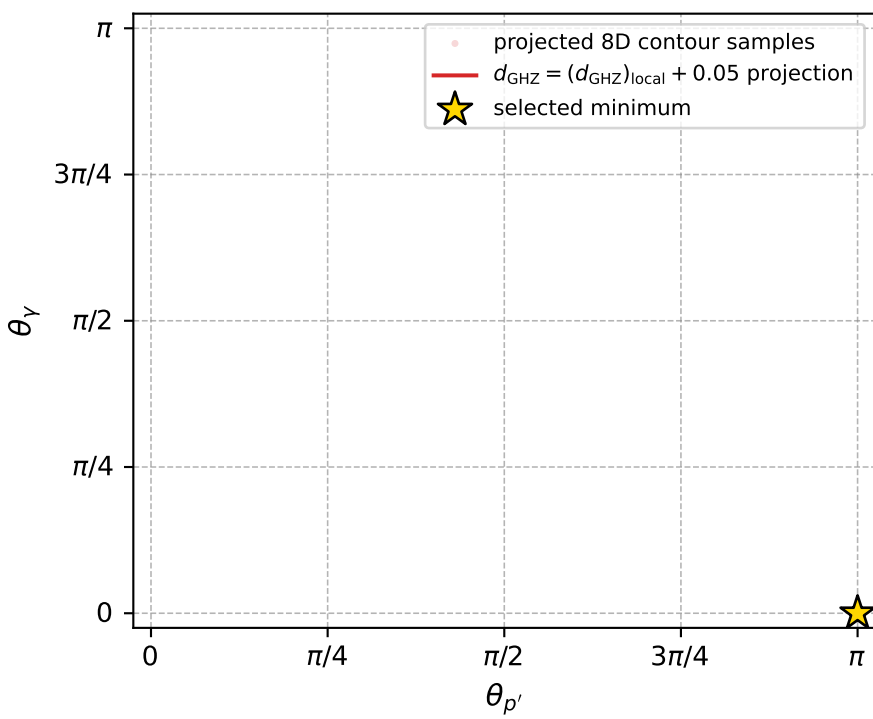
dghz_mixing_angles_polarization_02_local_minimum_136 $d_{\{\text{rm GHZ}\}}=6.21675\text{e-}08$
 region: polarization_cluster_02_local_minimum_136
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=0.20100578$ rad
 incoming lepton: $0.979866|+\rangle +0.199655|-\rangle$
 proton mixing angle: $\alpha_p=1.7718087$ rad
 incoming proton: $-0.199661|+\rangle +0.979865|-\rangle$

$s=24.9997$, $\sqrt{s}=4.99997$, $|\vec{P}|=2.412$, $|\vec{P}'|=2.412$
 $\theta_{P'}=3.14159$, $\theta_Y=0$, $\phi_{P'}=1.40635$, $\phi_Y=3.24397$
 $E_Y=0.186808$, $\theta_{\ell'}=0$, $\phi_{\ell'}=4.54794$
 $Q^2=21.4686$, $x_B=0.919952$, $t=-23.2709$, $W^2=2.74791$, $y=0.967532$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4120$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 P $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4120$
 ℓ' $E= 2.2252$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 2.2252$
 P' $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 q_Y $E= 0.1868$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 0.1868$



electron: polarization cluster P2, local minimum 136; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 6.216753\text{e-}08$
 $d_{\text{GHZ}} \text{ contour} = 0.050000062$
 above global minimum = $3.9884068\text{e-}08$
 8D contour samples = 68

selected phase-space point:

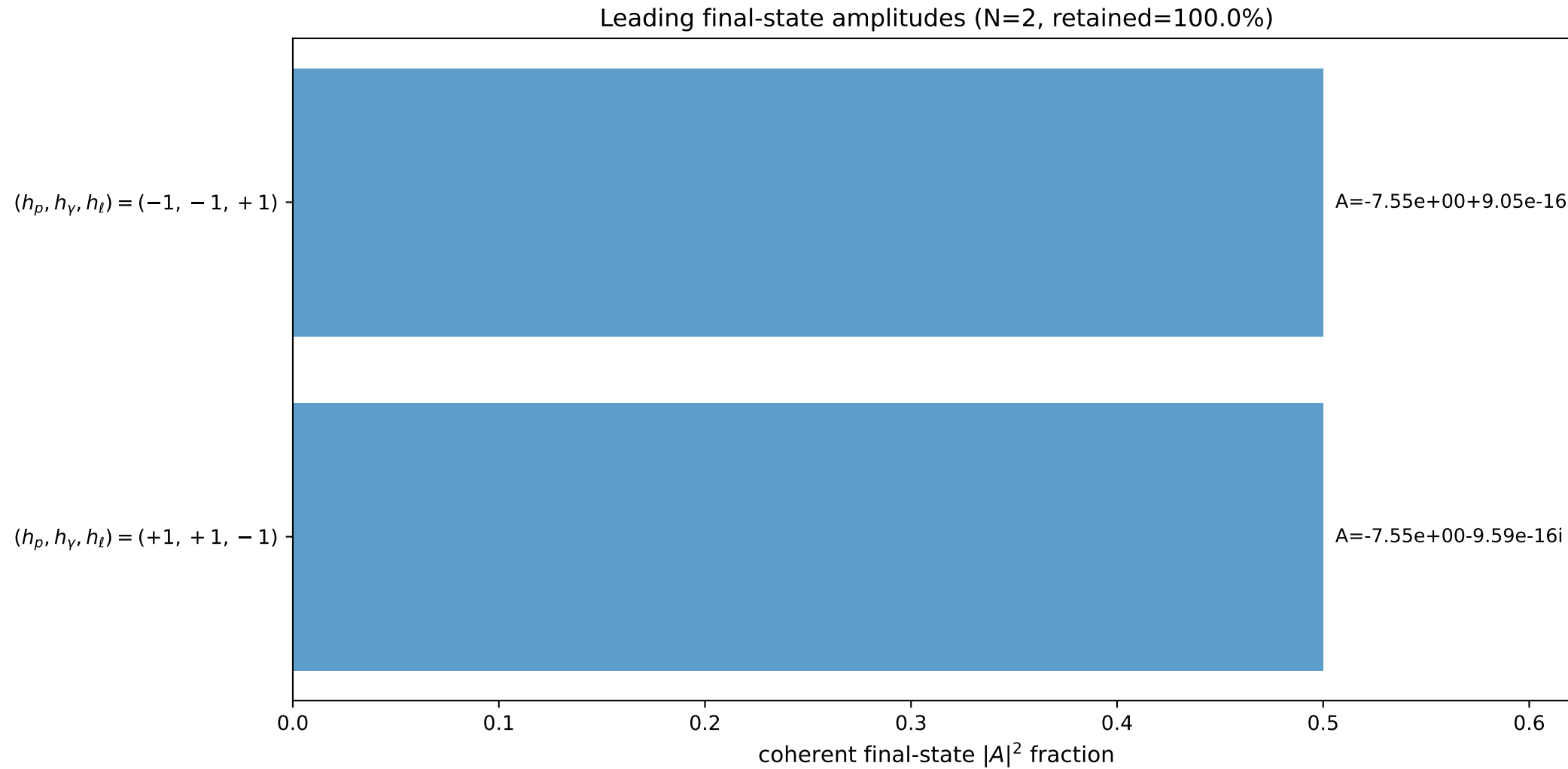
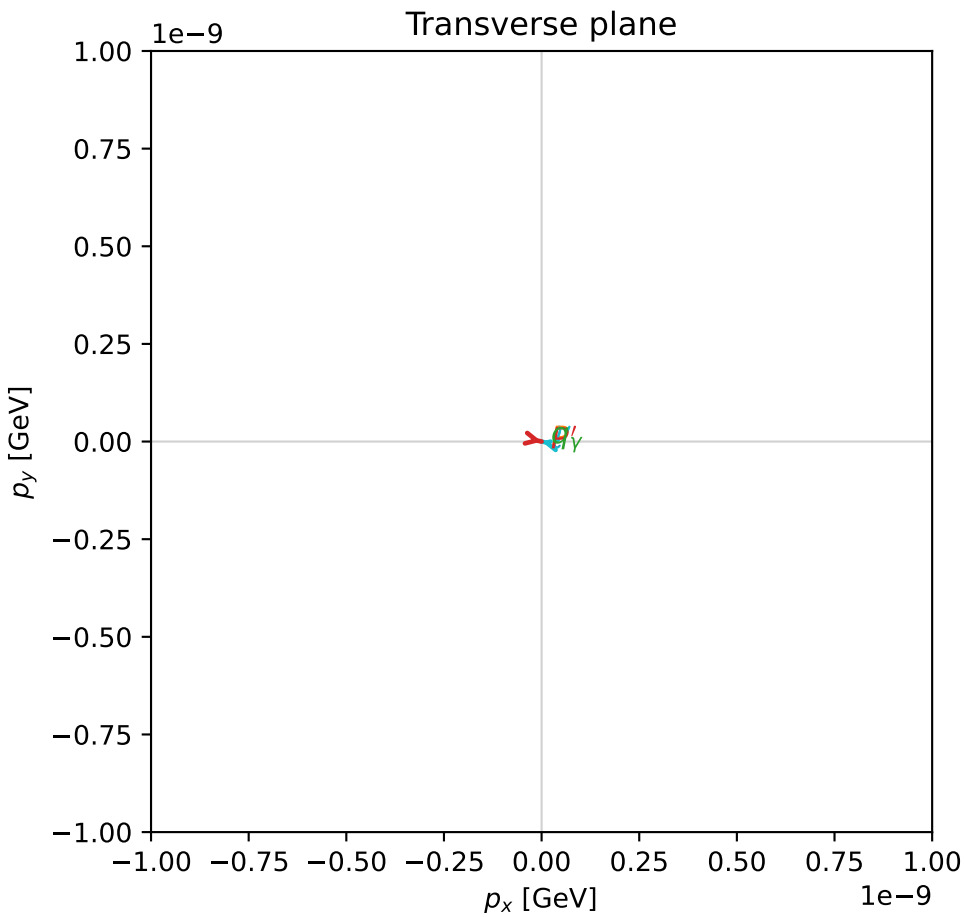
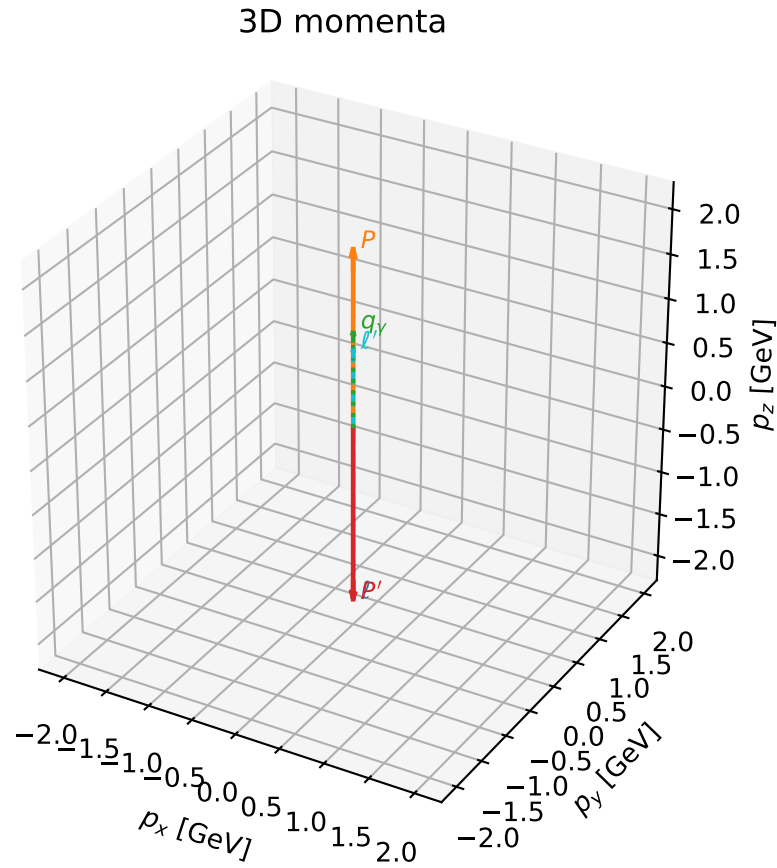
$\text{sqrt_s} = 4.999965$
 $\text{theta_p_out} = 3.141593$
 $\text{theta_gamma_out} = 0$
 $\text{qOut} = 0.1868075$
 $\text{phi_p_out} = 1.406351$
 $\text{phi_gamma_out} = 3.243974$
 $\text{alpha_e} = 0.2010058$
 $\text{alpha_p} = 1.771809$

coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

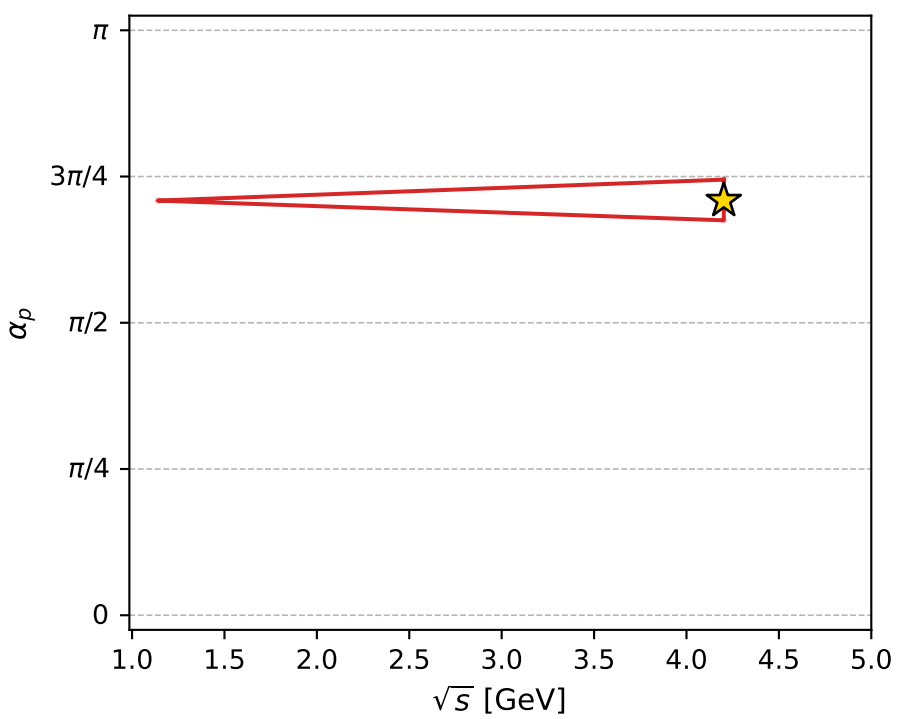
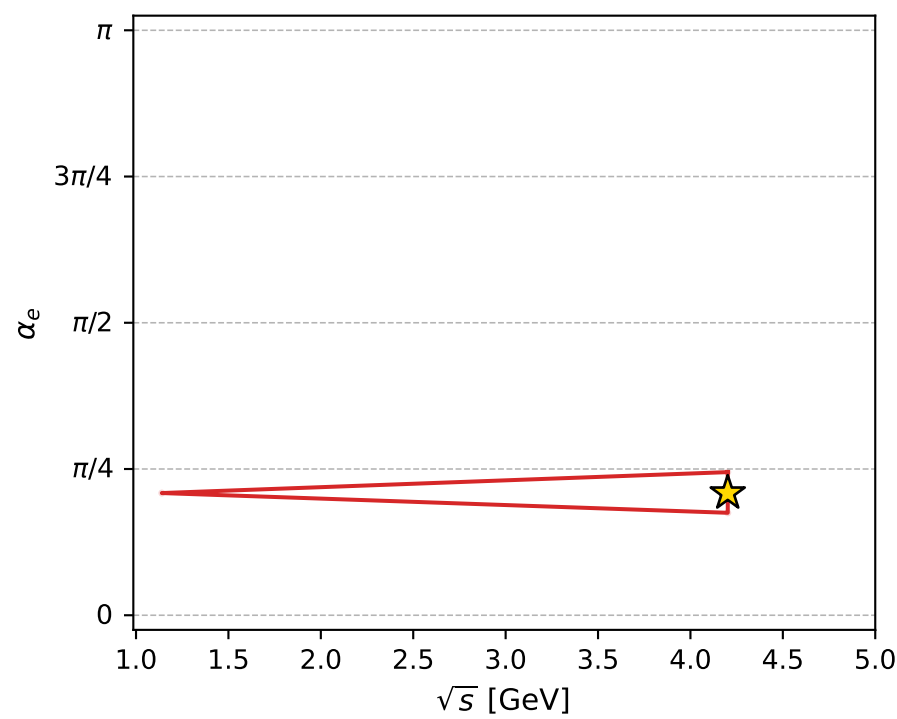
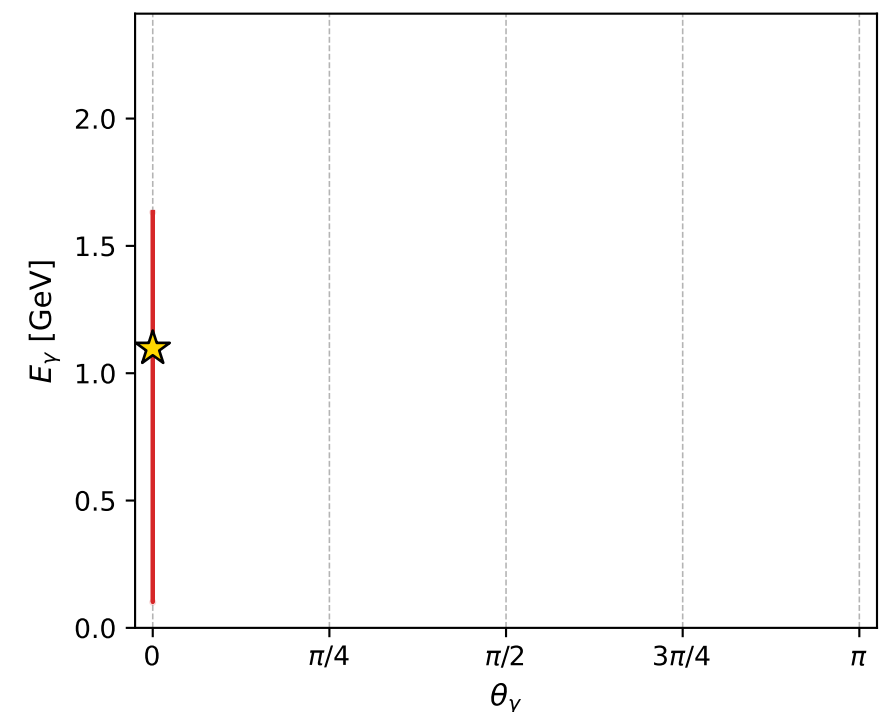
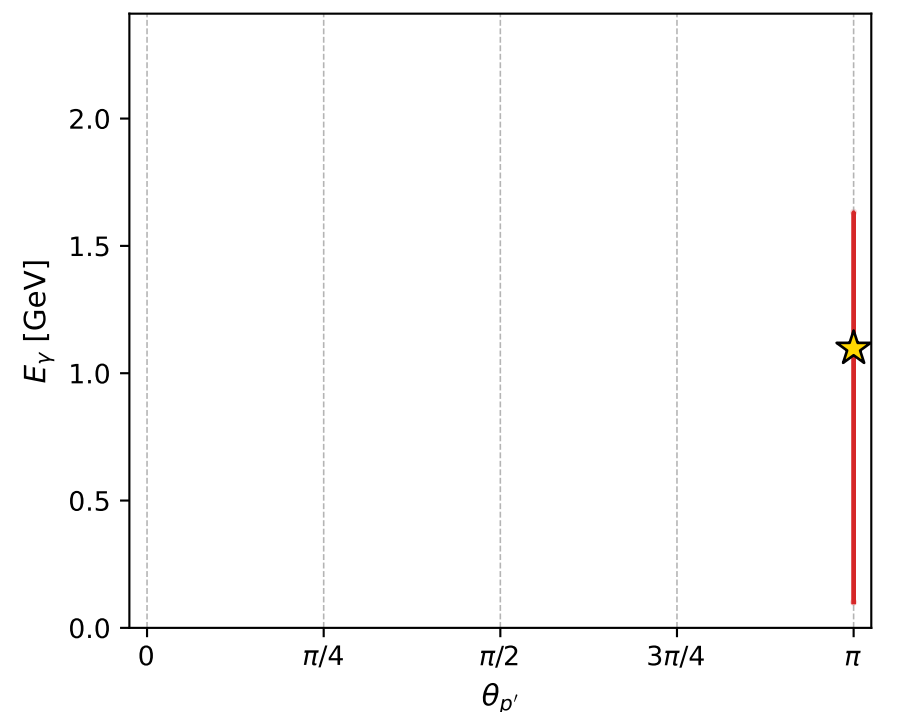
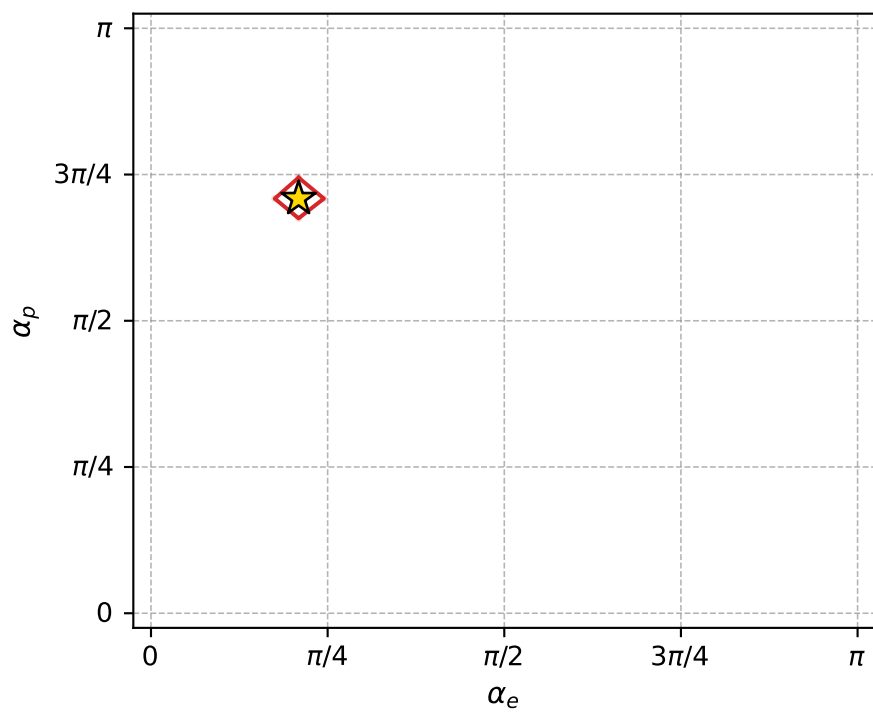
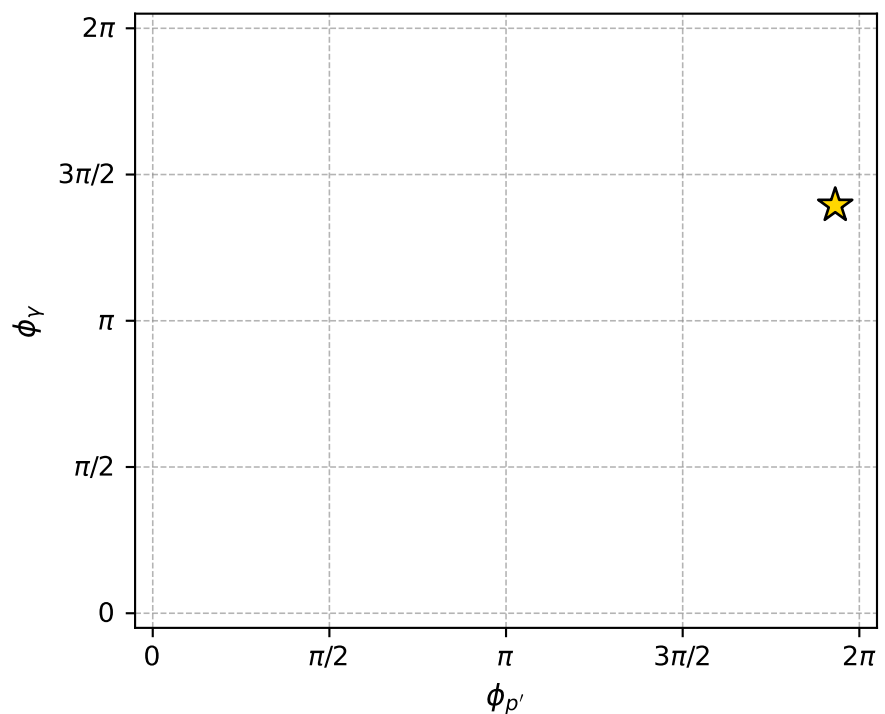
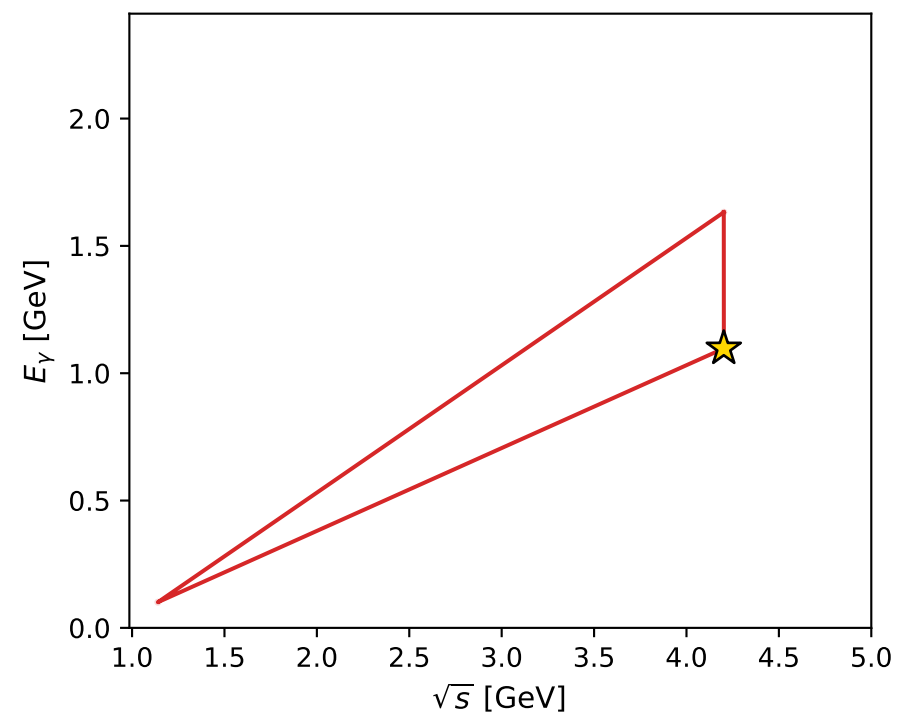
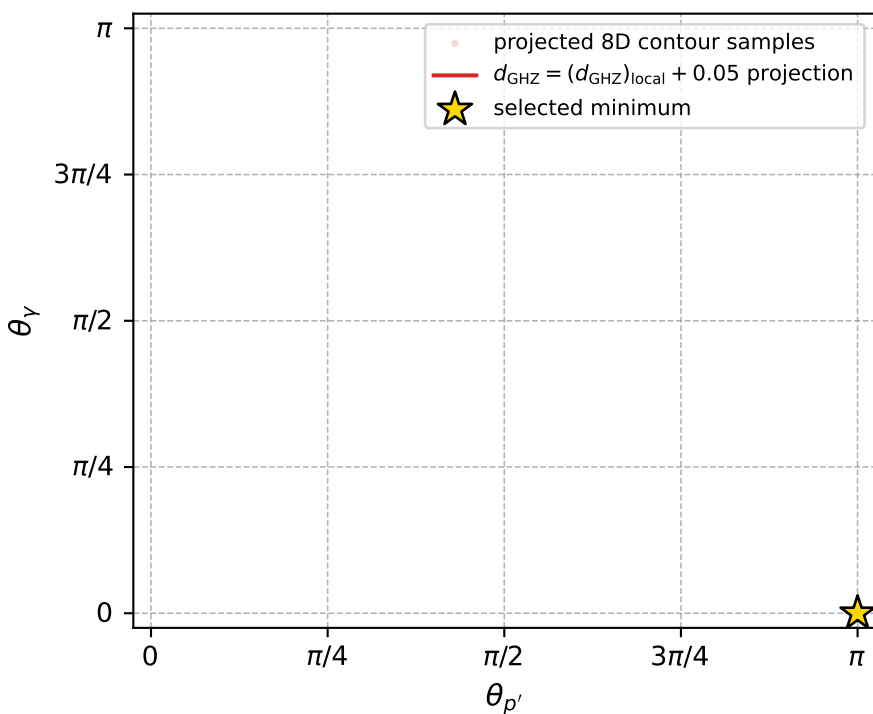
dghz_mixing_angles_polarization_02_local_minimum_142 $d_{\{\text{rm GHZ}\}}=7.52972\text{e-}08$
 region: polarization_cluster_02_local_minimum_142
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=0.65625654$ rad
 incoming lepton: $0.792282|+\rangle +0.610155|-\rangle$
 proton mixing angle: $\alpha_p=2.2270306$ rad
 incoming proton: $-0.610138|+\rangle +0.792295|-\rangle$

$s=17.6556$, $\sqrt{s}=4.20186$, $|\vec{P}|=1.99623$, $|\vec{P}'|=1.99623$
 $\theta_{P'}=3.14159$, $\theta_Y=0$, $\phi_{P'}=6.06891$, $\phi_Y=4.38119$
 $E_Y=1.09748$, $\theta_{\ell'}=0$, $\phi_{\ell'}=2.92732$
 $Q^2=7.17649$, $x_B=0.437607$, $t=-15.9398$, $W^2=10.1027$, $y=0.977564$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 1.9962$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.9962$
 P $E= 2.2056$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.9962$
 ℓ' $E= 0.8988$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 0.8988$
 P' $E= 2.2056$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= -1.9962$
 q_Y $E= 1.0975$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 1.0975$



electron: polarization cluster P2, local minimum 142; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 7.5297241\text{e-}08$
 $d_{\text{GHZ}} \text{ contour} = 0.050000075$
 above global minimum = $5.301378\text{e-}08$
 8D contour samples = 79

selected phase-space point:

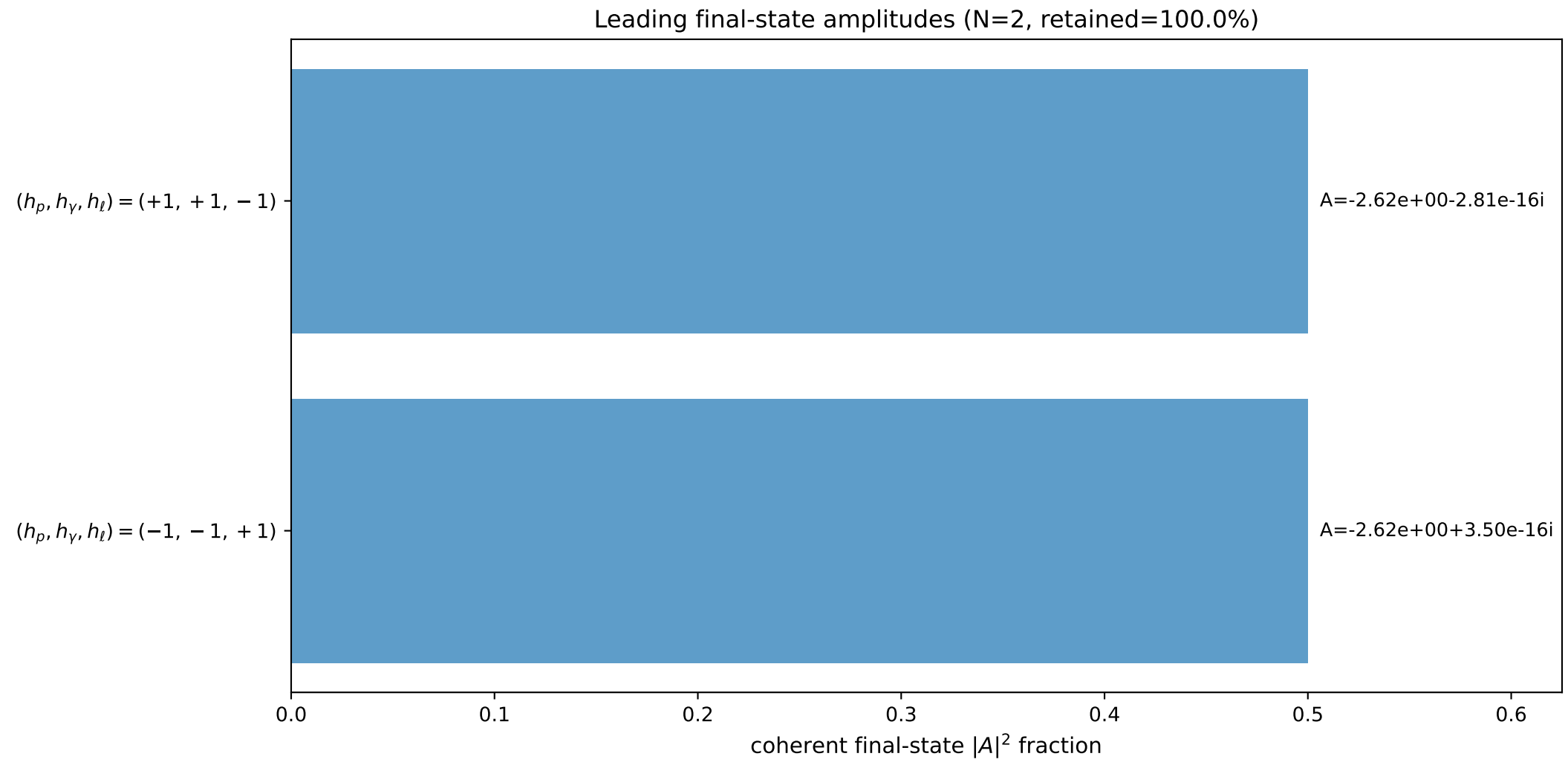
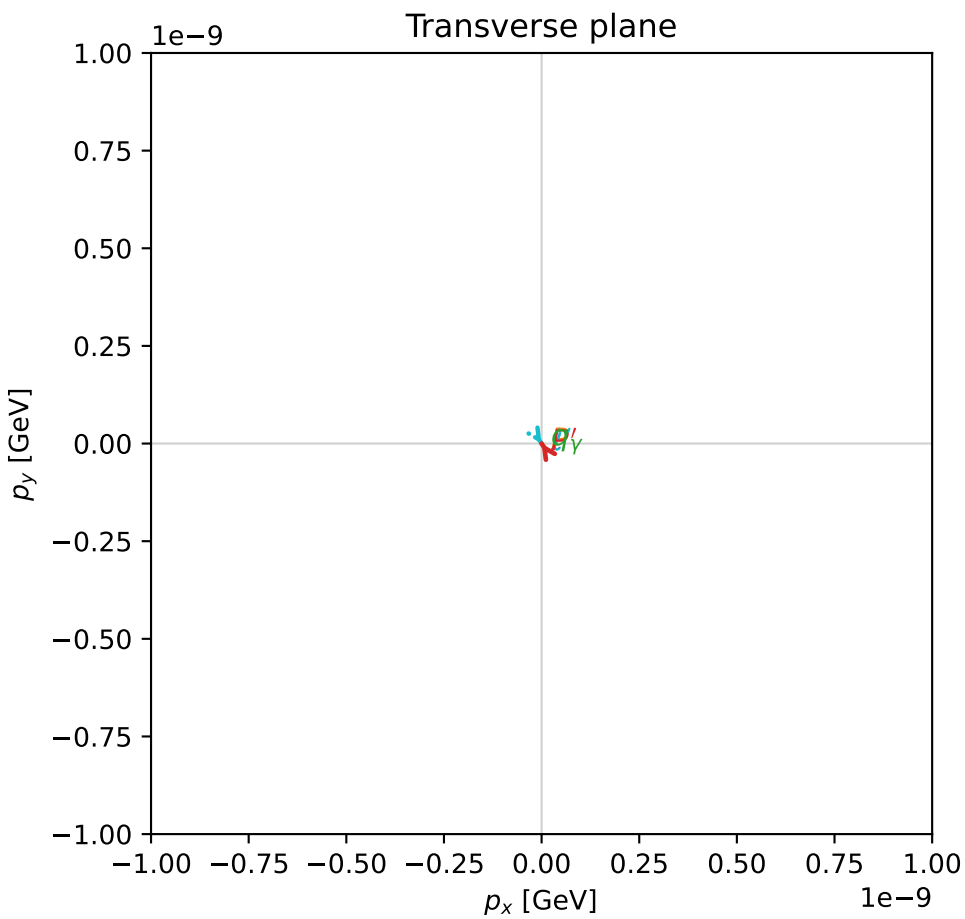
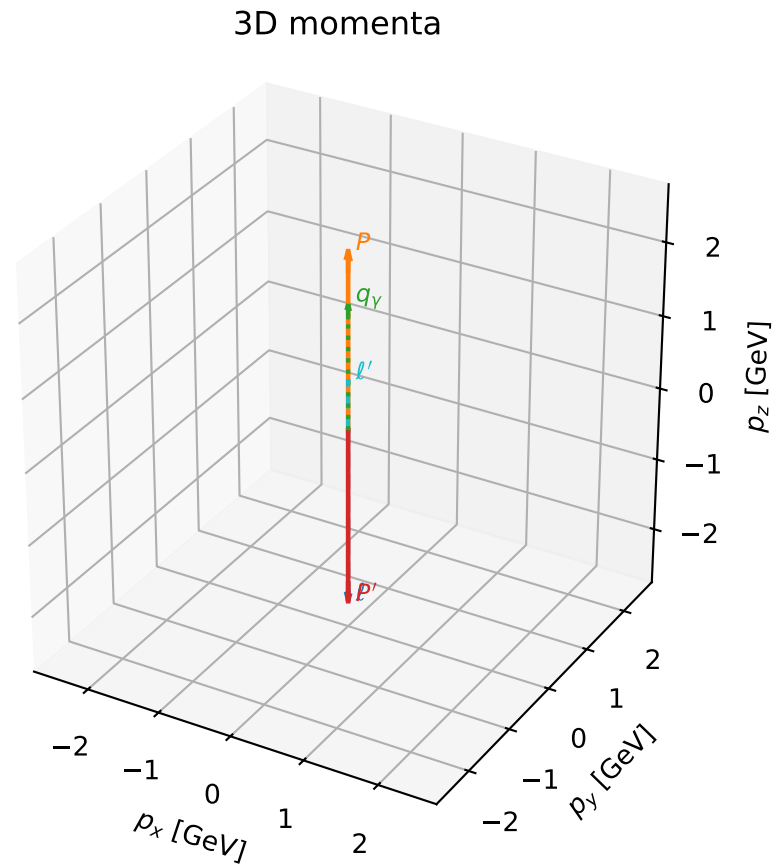
$\text{sqrt_s} = 4.201857$
 $\text{theta_p_out} = 3.141593$
 $\text{theta_gamma_out} = 0$
 $\text{q0out} = 1.097477$
 $\text{phi_p_out} = 6.068912$
 $\text{phi_gamma_out} = 4.381192$
 $\text{alpha_e} = 0.6562565$
 $\text{alpha_p} = 2.227031$

coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

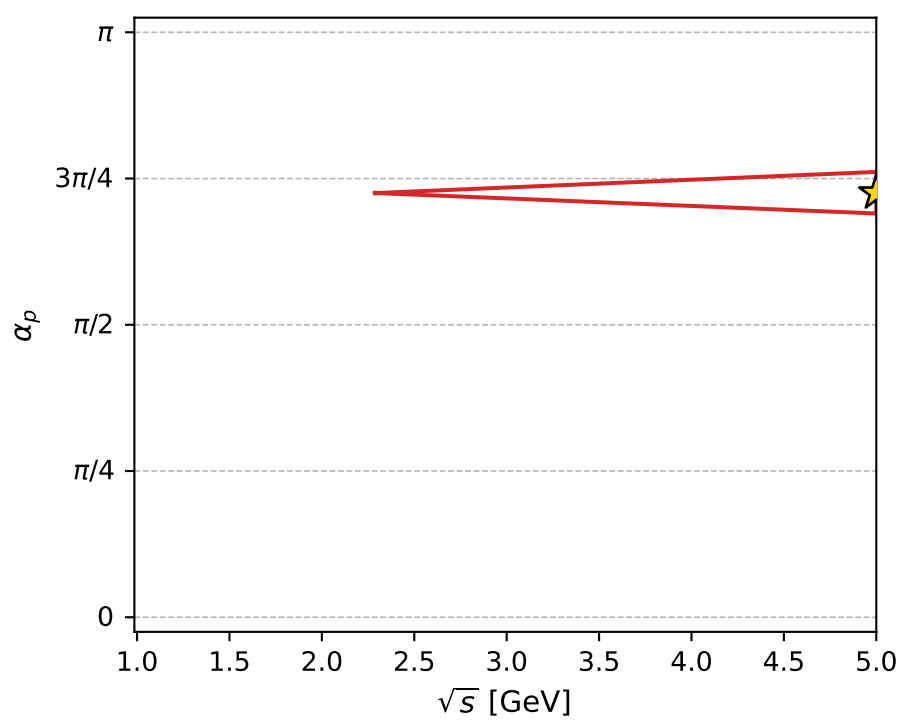
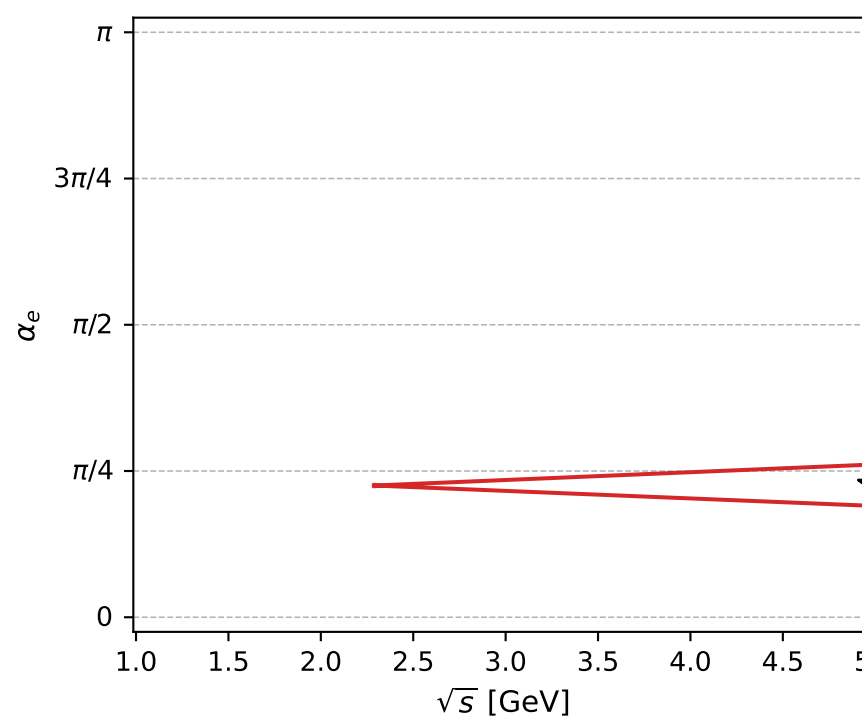
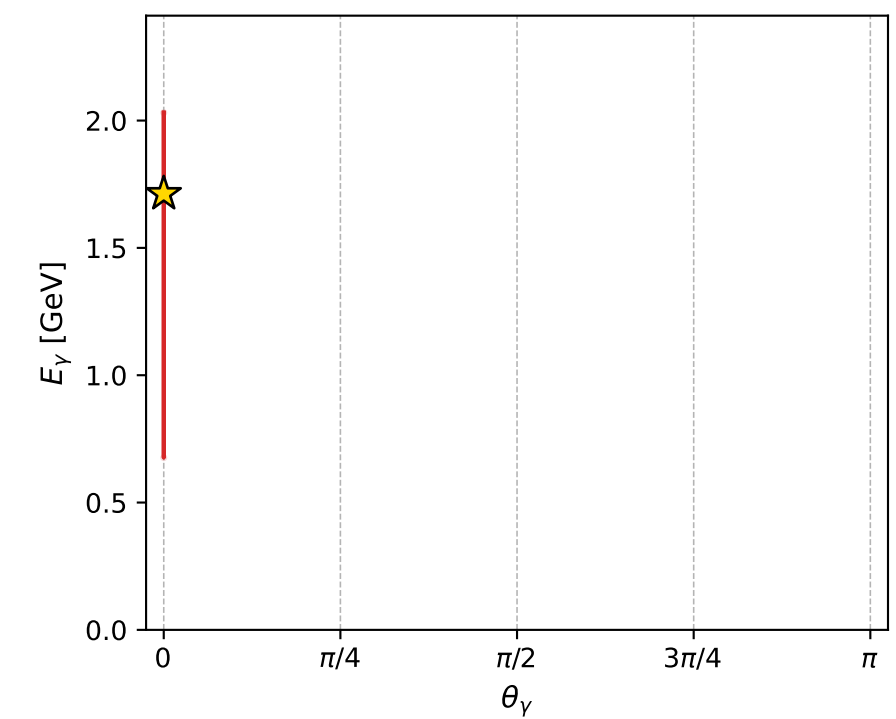
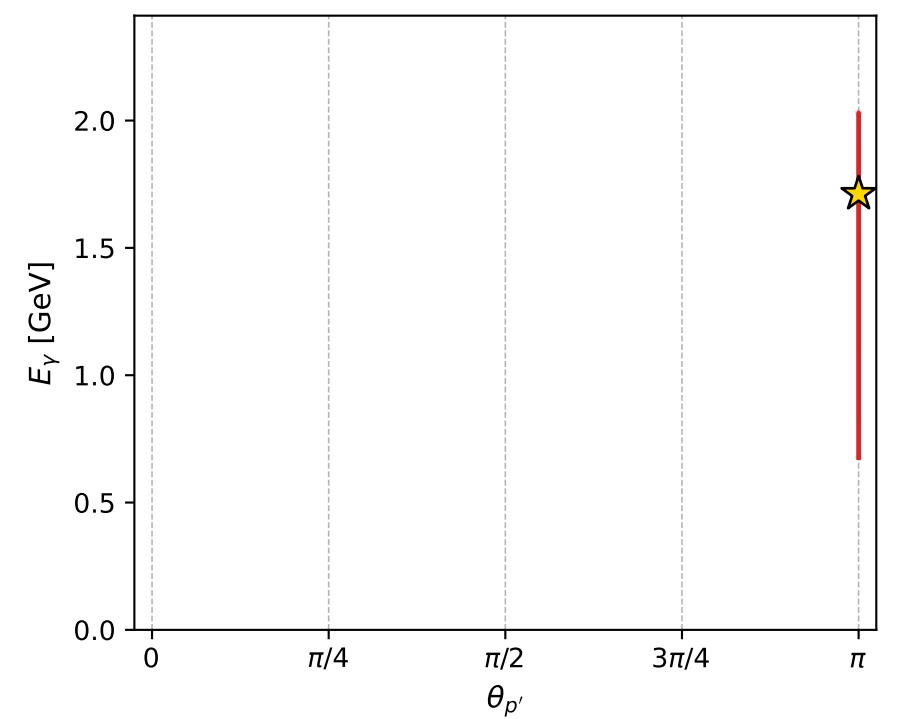
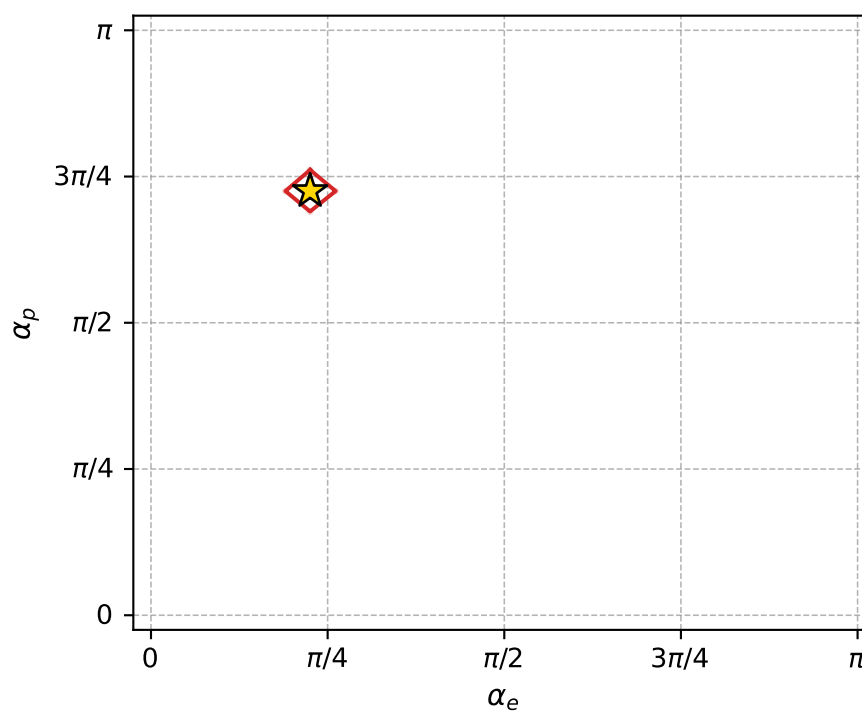
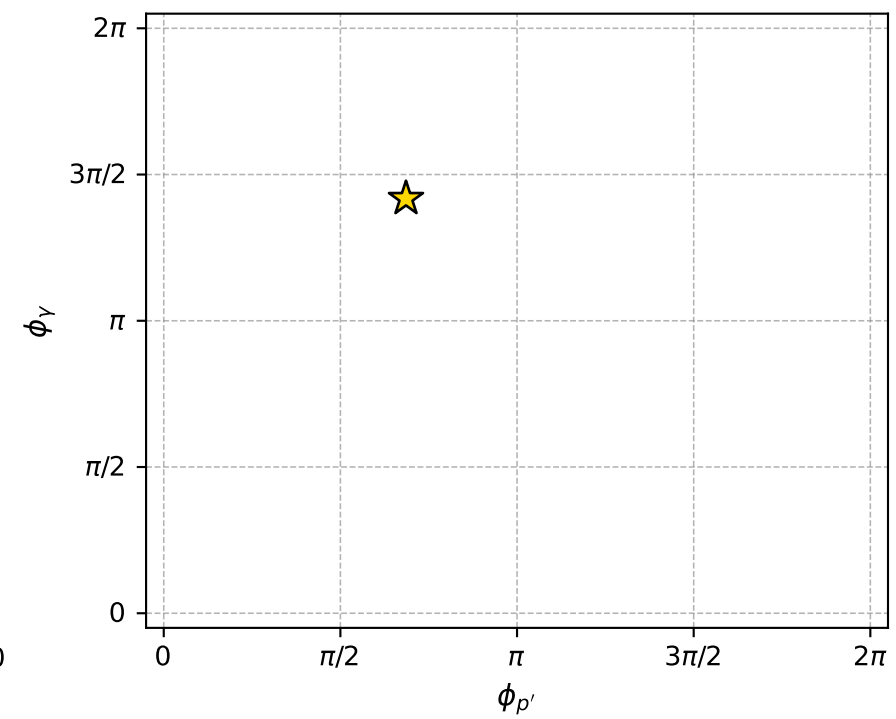
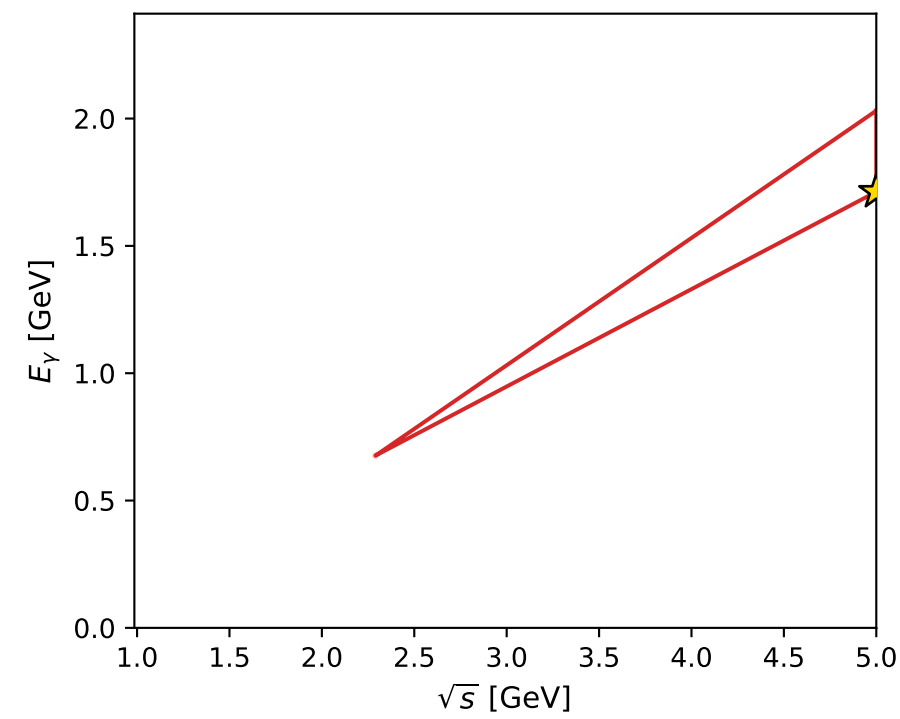
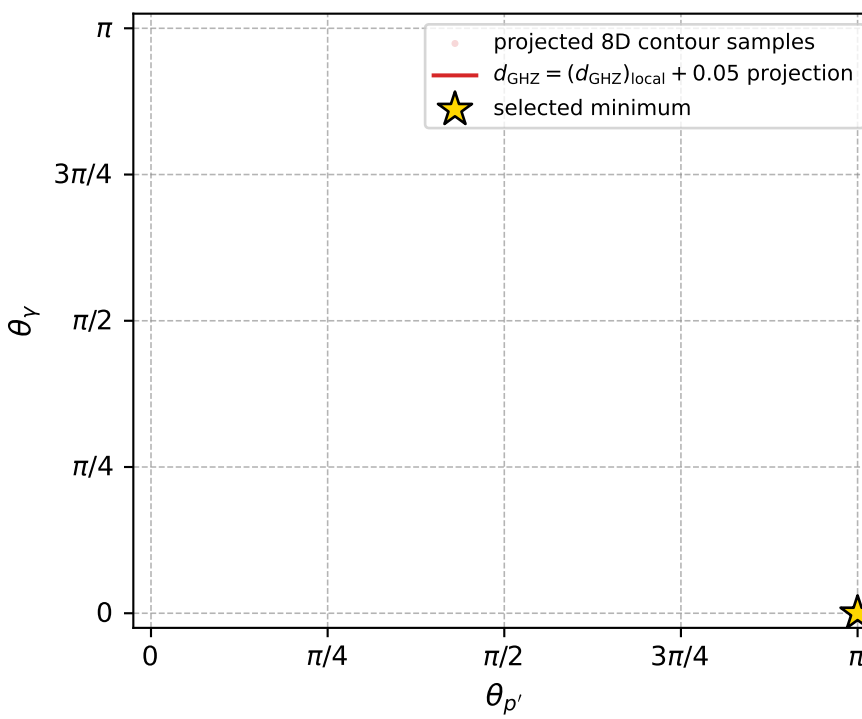
dghz_mixing_angles_polarization_02_local_minimum_144 $d_{\{\text{rm GHZ}\}}=7.83262\text{e-}08$
 region: polarization_cluster_02_local_minimum_144
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=0.70738913$ rad
 incoming lepton: $0.760061|+\rangle +0.649852|-\rangle$
 proton mixing angle: $\alpha_p=2.2781902$ rad
 incoming proton: $-0.649855|+\rangle +0.760058|-\rangle$

$s=25$, $\sqrt{s}=5$, $|\vec{P}|=2.41202$, $|\vec{P}'|=2.41202$
 $\theta_{P'}=3.14159$, $\theta_Y=0$, $\phi_{P'}=2.15414$, $\phi_Y=4.45412$
 $E_Y=1.71245$, $\theta_{\ell'}=0$, $\phi_{\ell'}=5.29573$
 $Q^2=6.74947$, $x_B=0.282713$, $t=-23.2713$, $W^2=18.0043$, $y=0.989793$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4120$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 P $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4120$
 ℓ' $E= 0.6996$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 0.6996$
 P' $E= 2.5880$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 q_Y $E= 1.7124$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 1.7124$



electron: polarization cluster P2, local minimum 144; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 7.8326232\text{e-}08$
 $d_{\text{GHZ}} \text{ contour} = 0.050000078$
 above global minimum = $5.604277\text{e-}08$
 8D contour samples = 40

selected phase-space point:

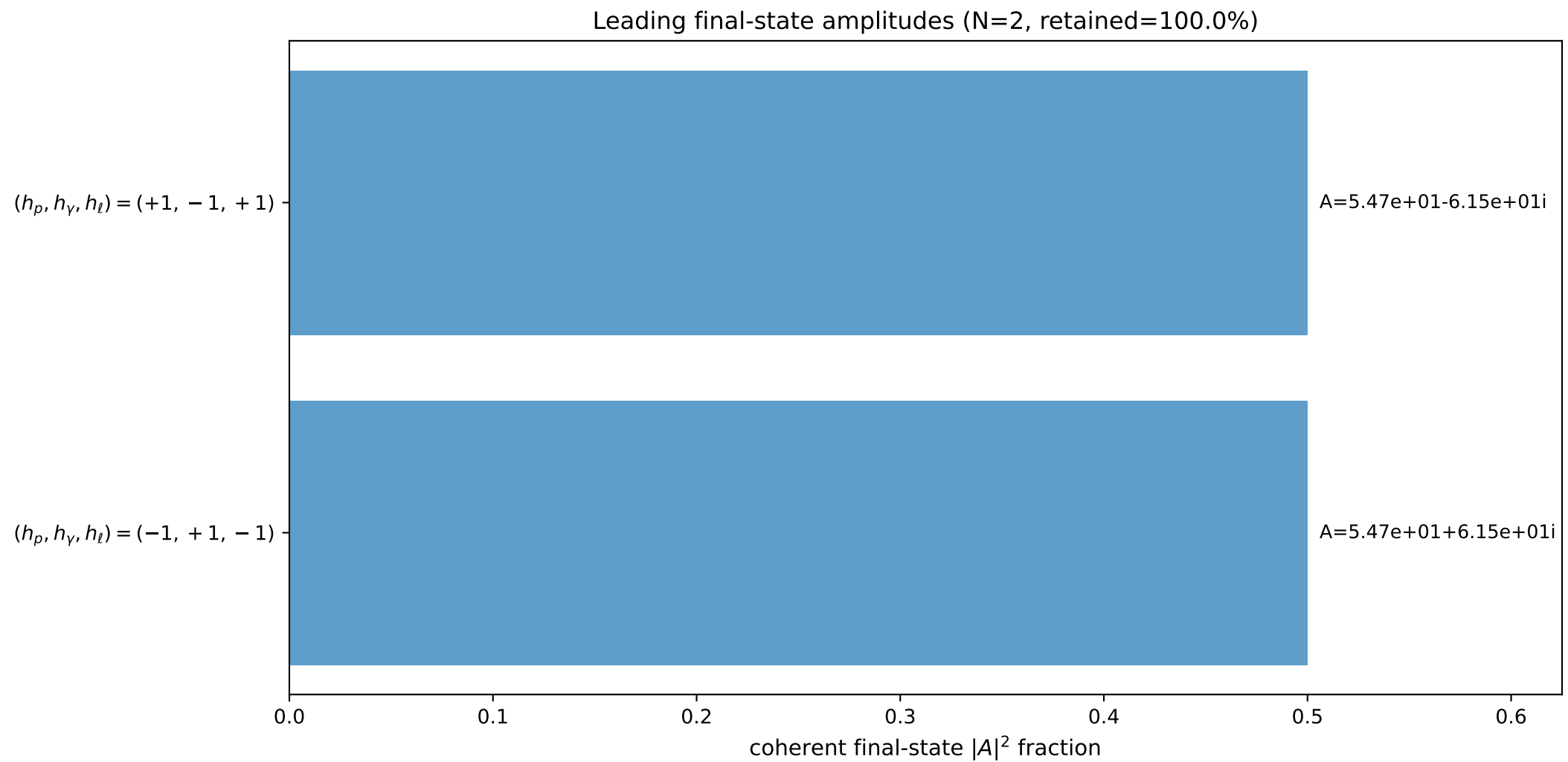
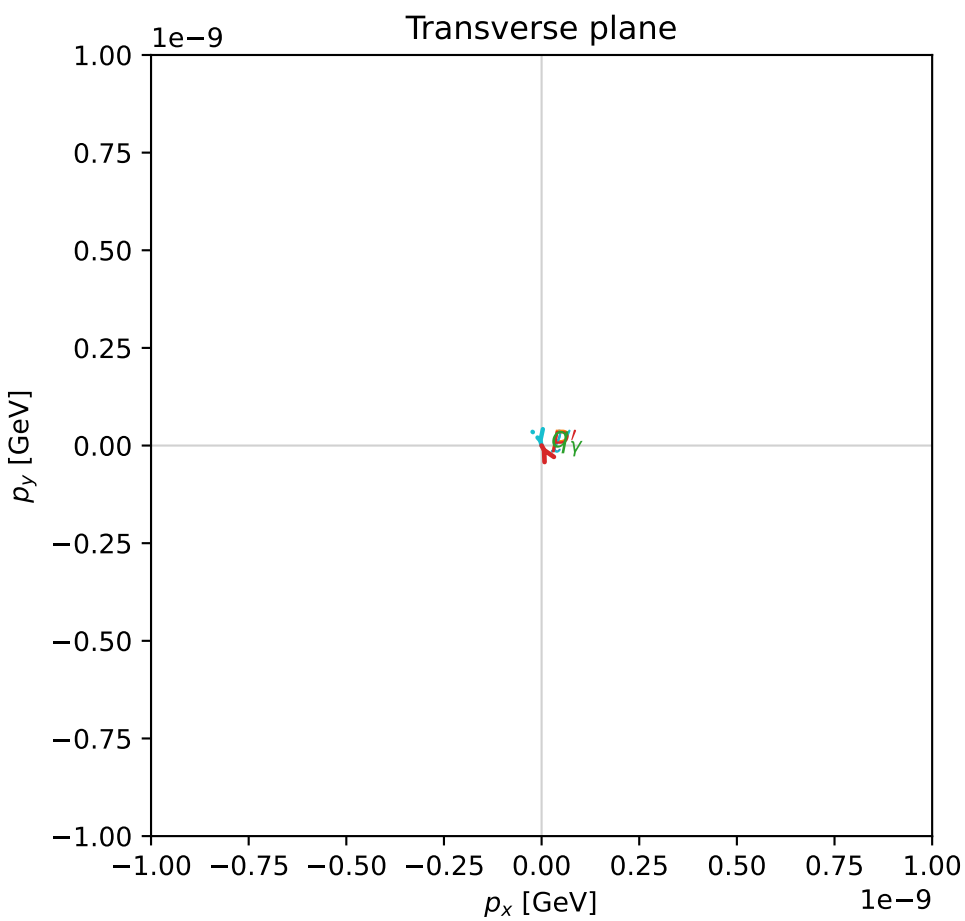
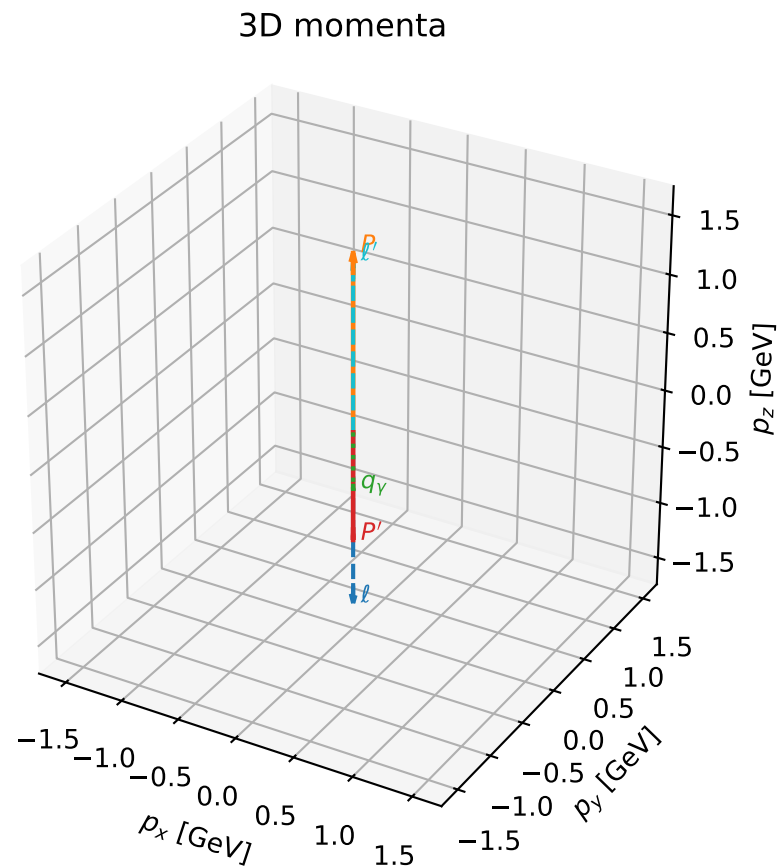
$\text{sqrt_s} = 5$
 $\text{theta_p_out} = 3.141593$
 $\text{theta_gamma_out} = 0$
 $\text{q0ut} = 1.712448$
 $\text{phi_p_out} = 2.154135$
 $\text{phi_gamma_out} = 4.454121$
 $\text{alpha_e} = 0.7073891$
 $\text{alpha_p} = 2.27819$

coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

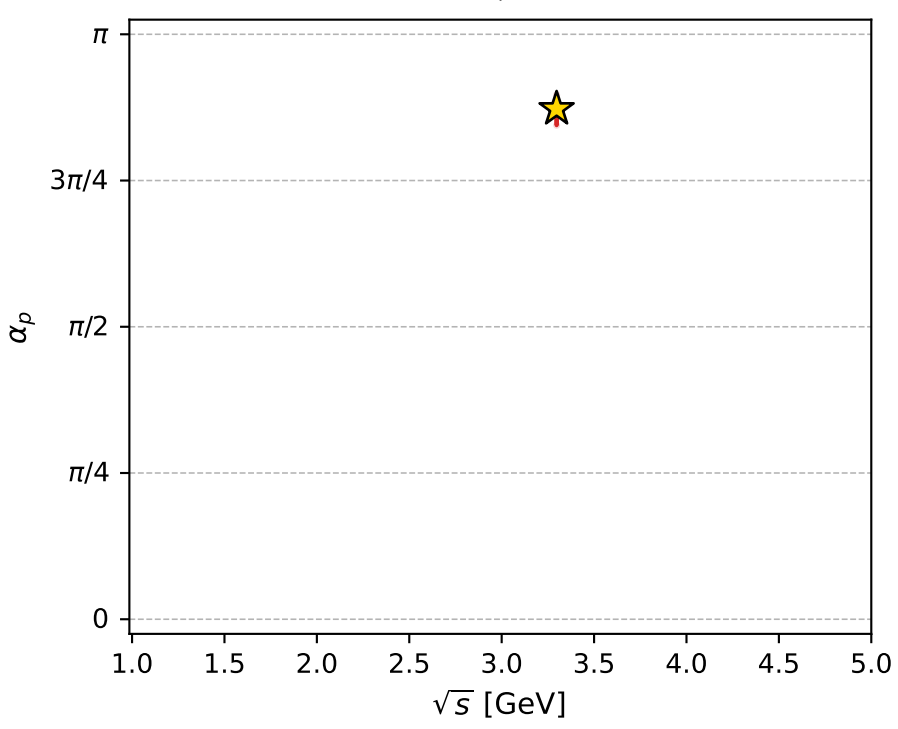
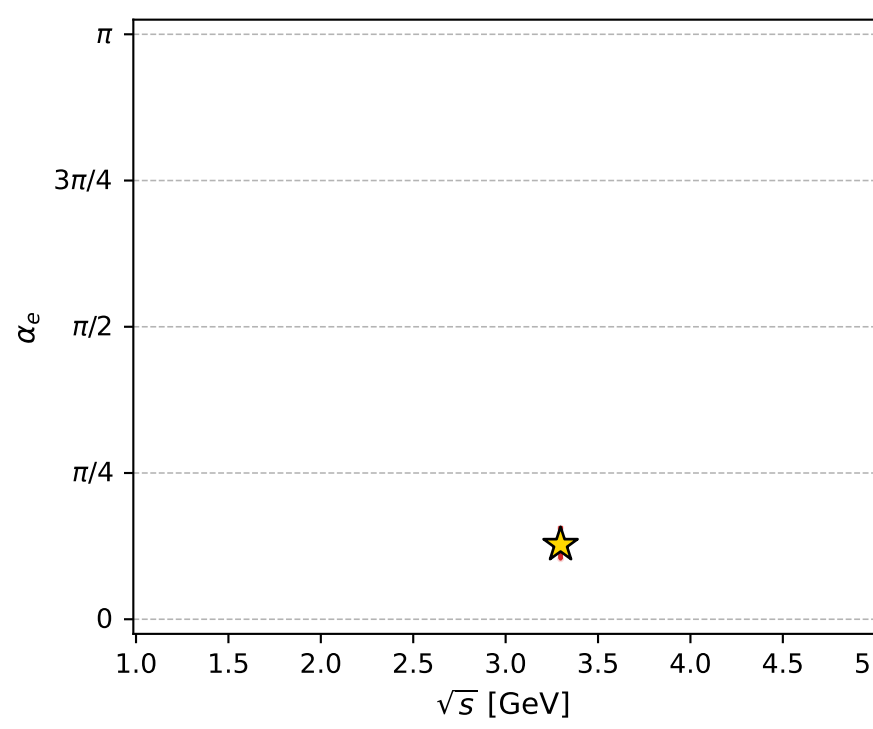
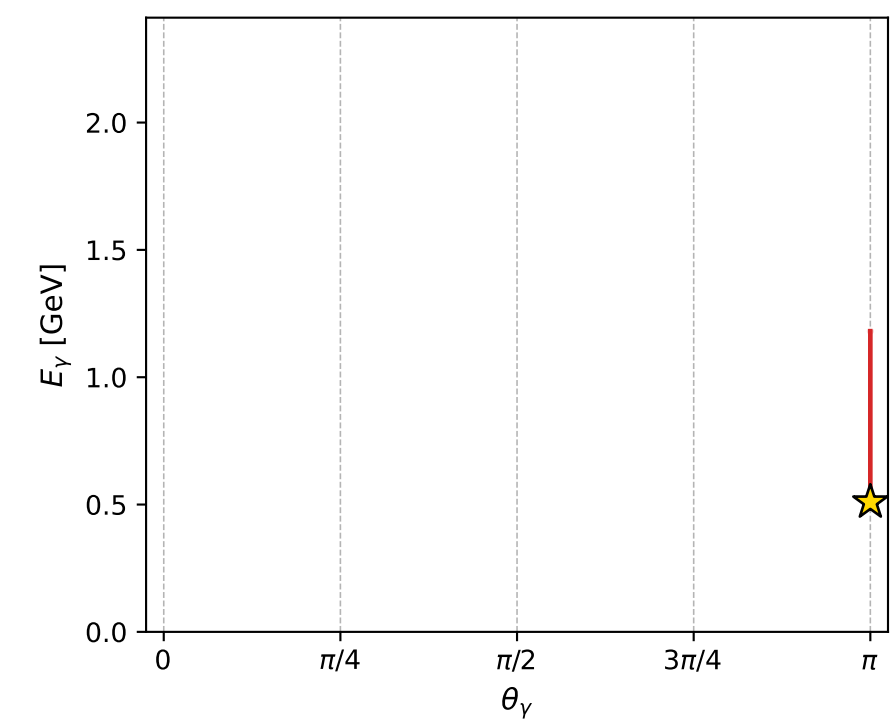
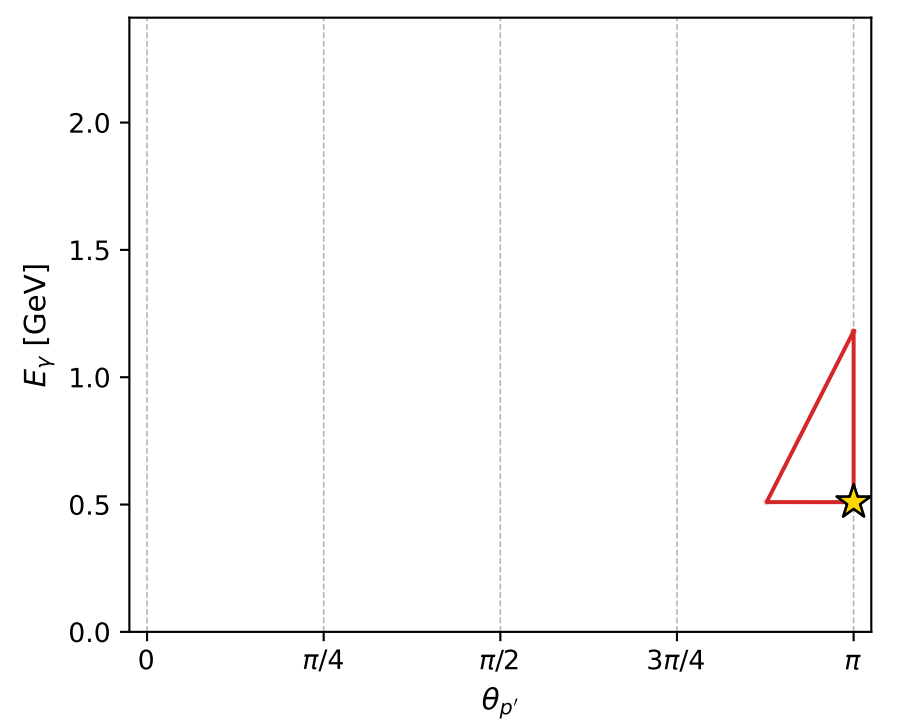
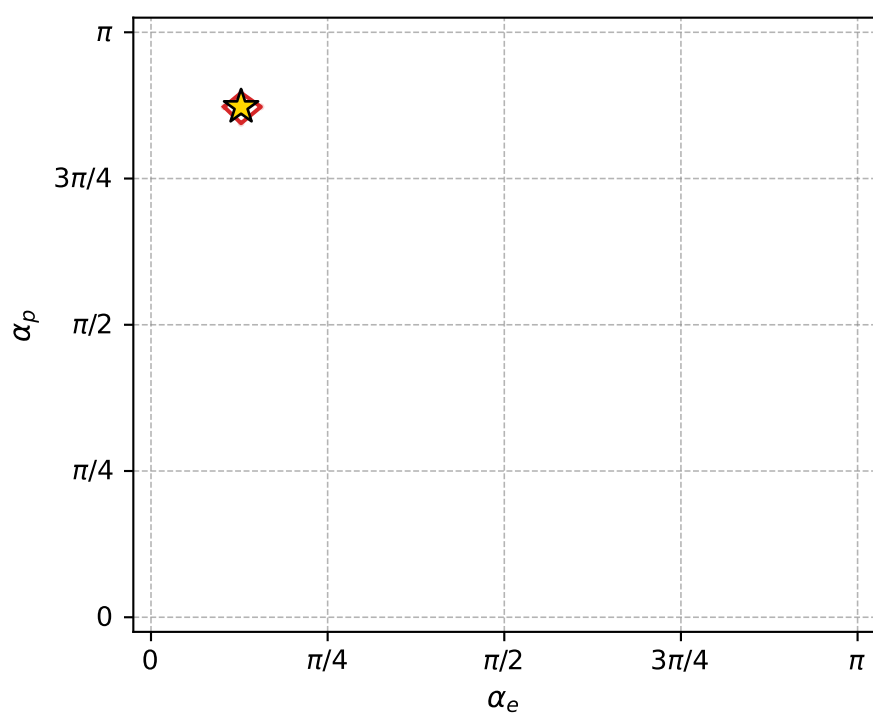
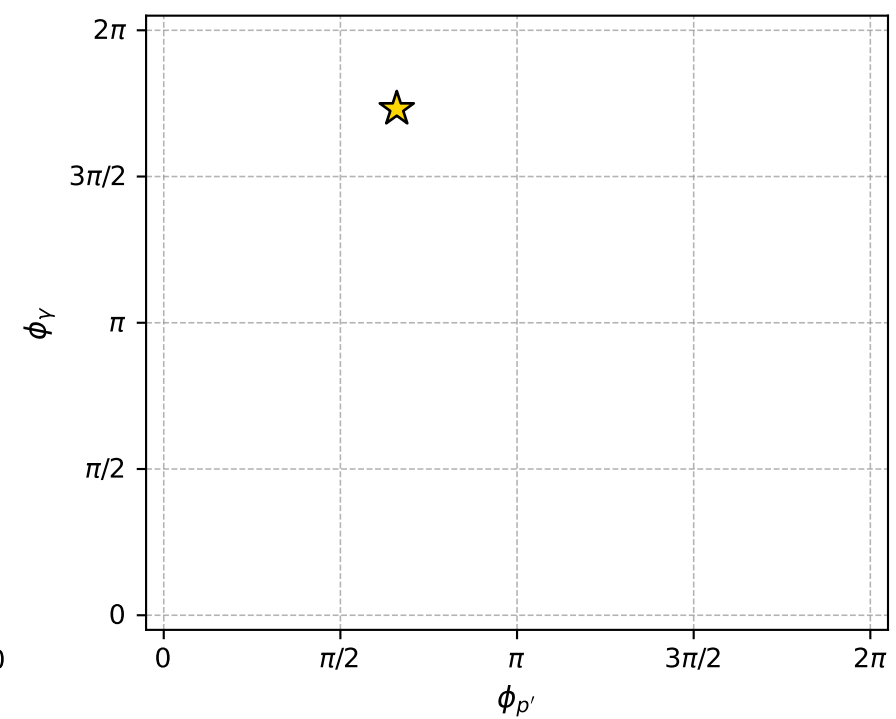
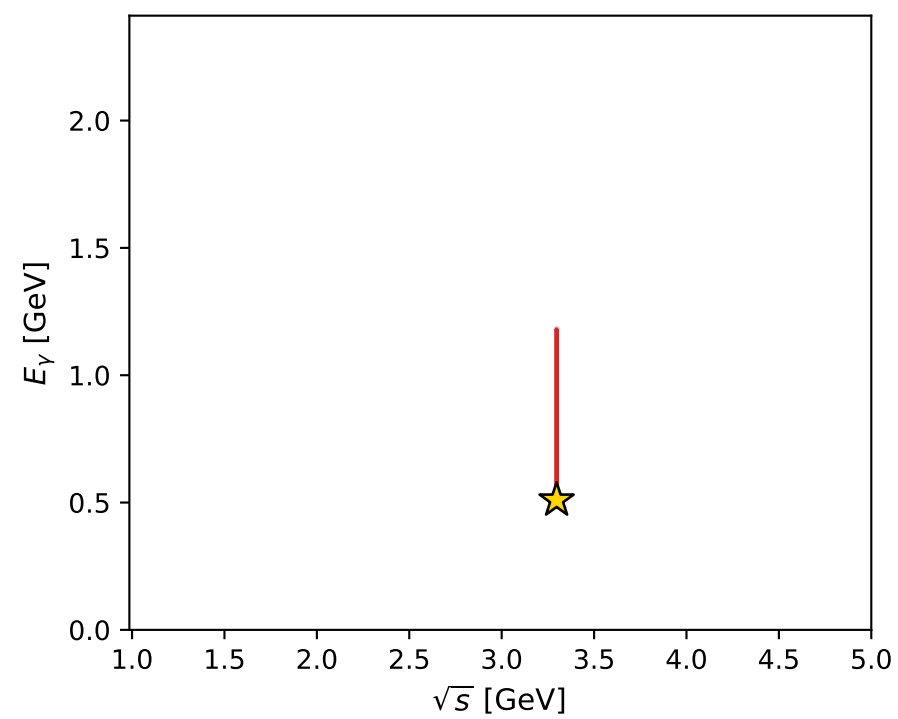
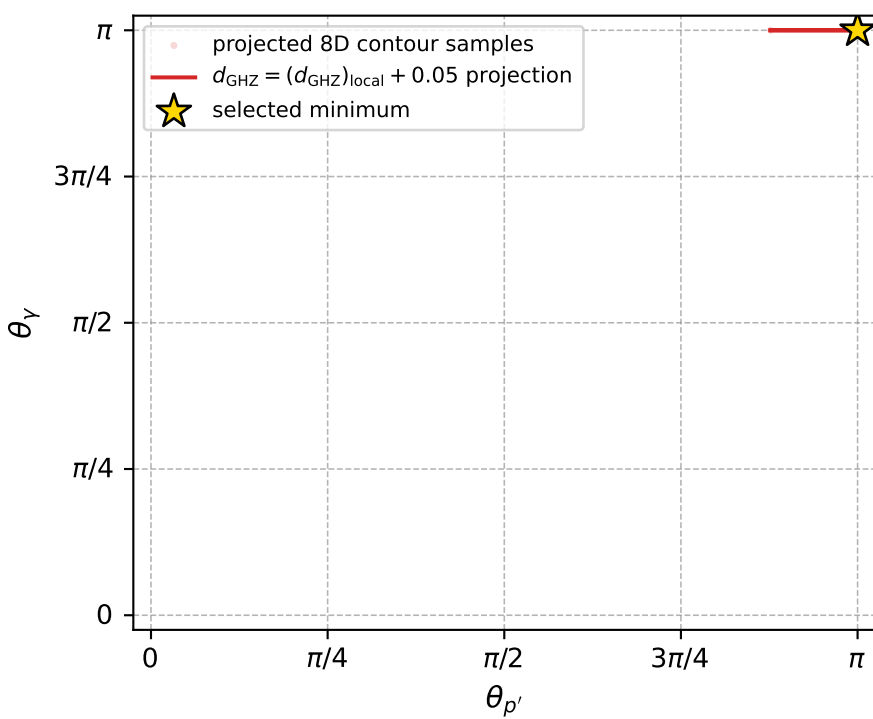
dghz_mixing_angles_polarization_02_local_minimum_147 $d_{\{\text{rm GHZ}\}}=8.24447\text{e-}08$
 region: polarization_cluster_02_local_minimum_147
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=0.40049047$ rad
 incoming lepton: $0.92087|+\rangle \rightarrow +0.38987|-\rangle$
 proton mixing angle: $\alpha_p=2.7411031$ rad
 incoming proton: $-0.92087|+\rangle \rightarrow +0.389869|-\rangle$

$s=10.8717$, $\sqrt{s}=3.29723$, $|\vec{P}|=1.51519$, $|\vec{P}'|=0.945903$
 $\theta_{P'}=3.14159$, $\theta_Y=3.14159$, $\phi_{P'}=2.07214$, $\phi_Y=5.43934$
 $E_Y=0.509598$, $\theta_{\ell'}=0$, $\phi_{\ell'}=4.96521$
 $Q^2=8.82147$, $x_B=0.957283$, $t=-5.85459$, $W^2=1.27349$, $y=0.922259$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 1.5152$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.5152$
 P $E= 1.7820$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.5152$
 ℓ' $E= 1.4555$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 1.4555$
 P' $E= 1.3321$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= -0.9459$
 q_Y $E= 0.5096$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= -0.5096$



electron: polarization cluster P2, local minimum 147; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 8.2444691\text{e-}08$
 $d_{\text{GHZ}} \text{ contour} = 0.050000082$
 above global minimum = $6.0161229\text{e-}08$
 8D contour samples = 55

selected phase-space point:

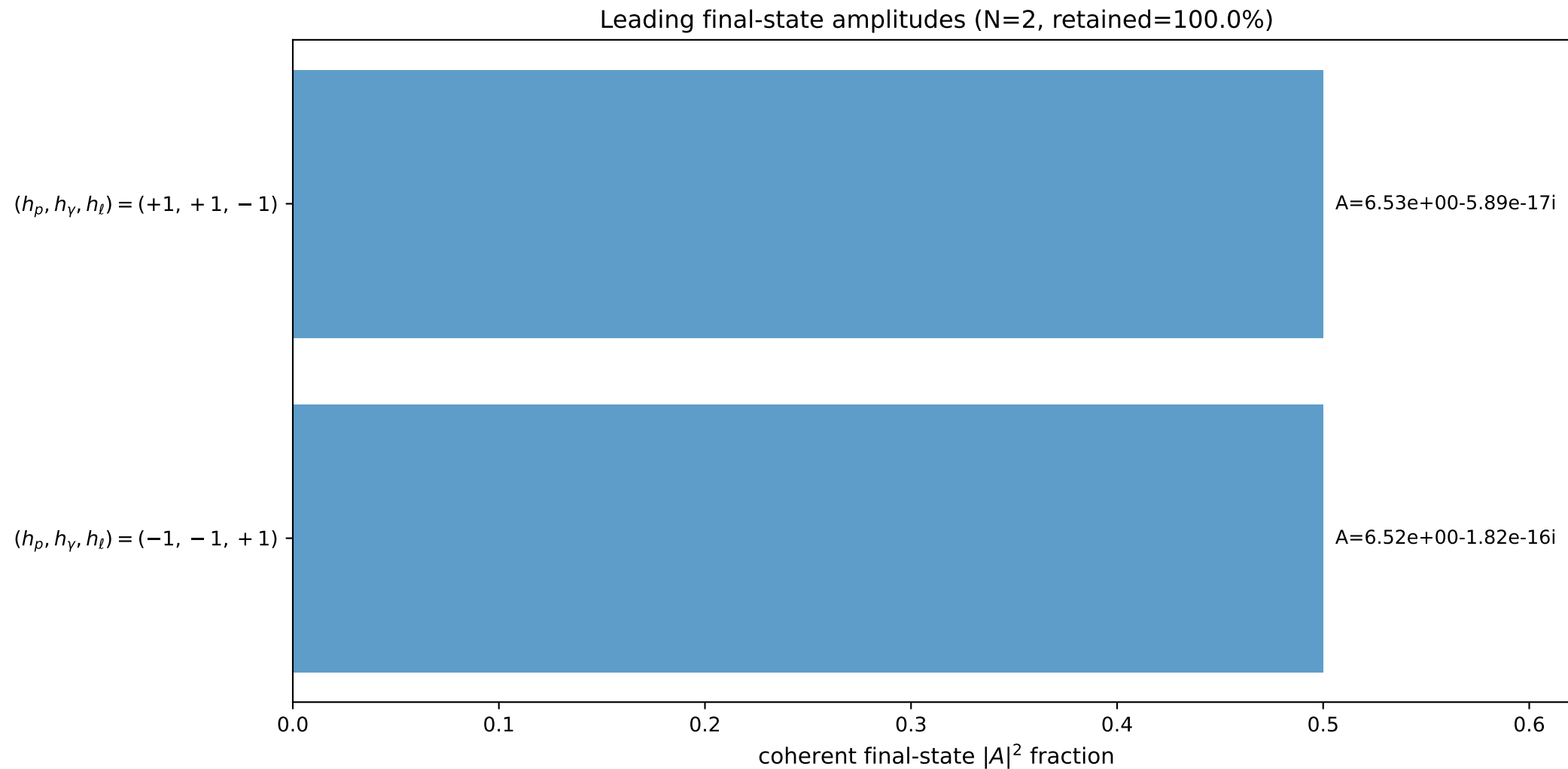
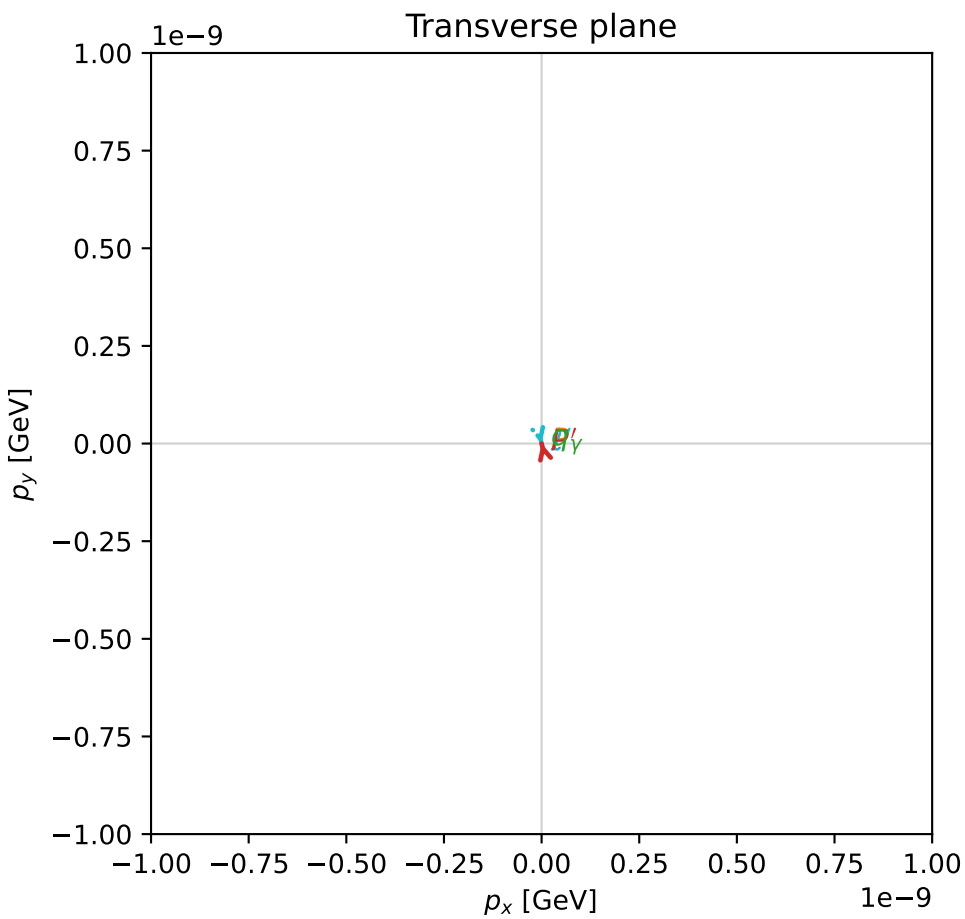
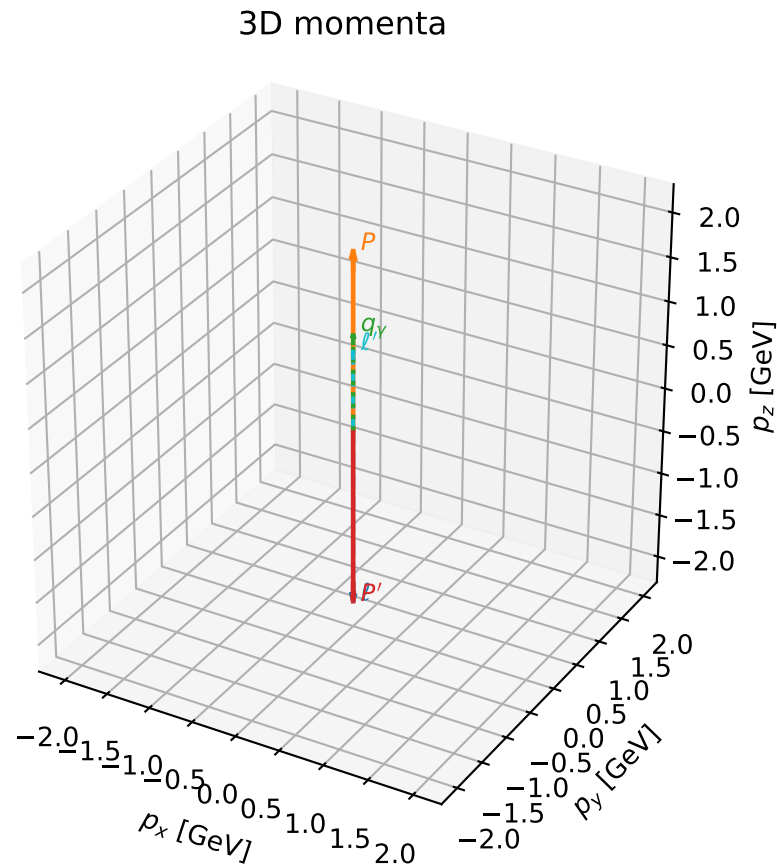
$\text{sqrt_s} = 3.297232$
 $\text{theta_p_out} = 3.141593$
 $\text{theta_gamma_out} = 3.141593$
 $\text{q0out} = 0.5095979$
 $\text{phi_p_out} = 2.07214$
 $\text{phi_gamma_out} = 5.439341$
 $\text{alpha_e} = 0.4004905$
 $\text{alpha_p} = 2.741103$

coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

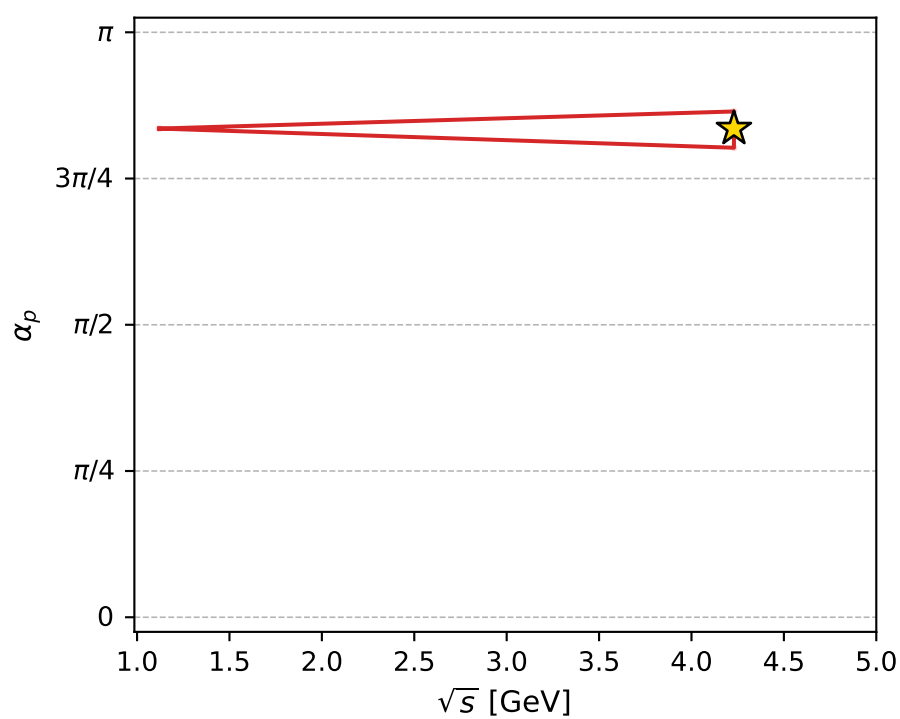
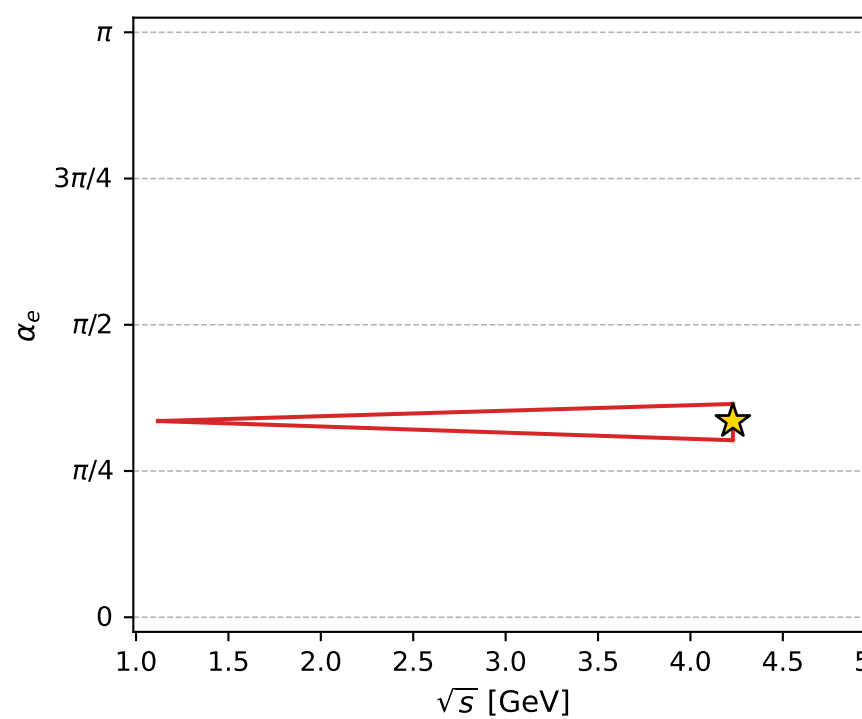
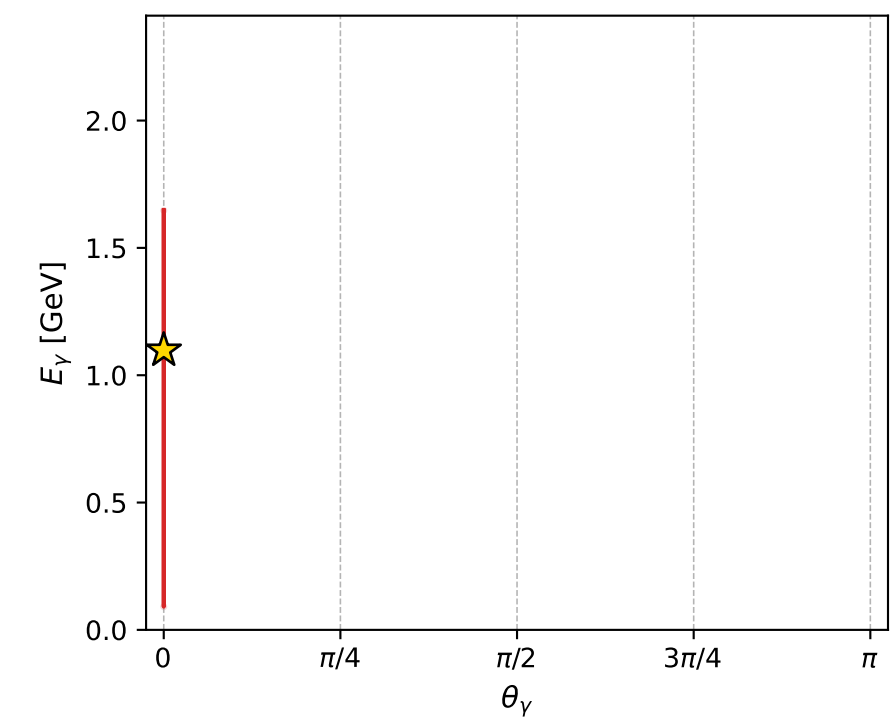
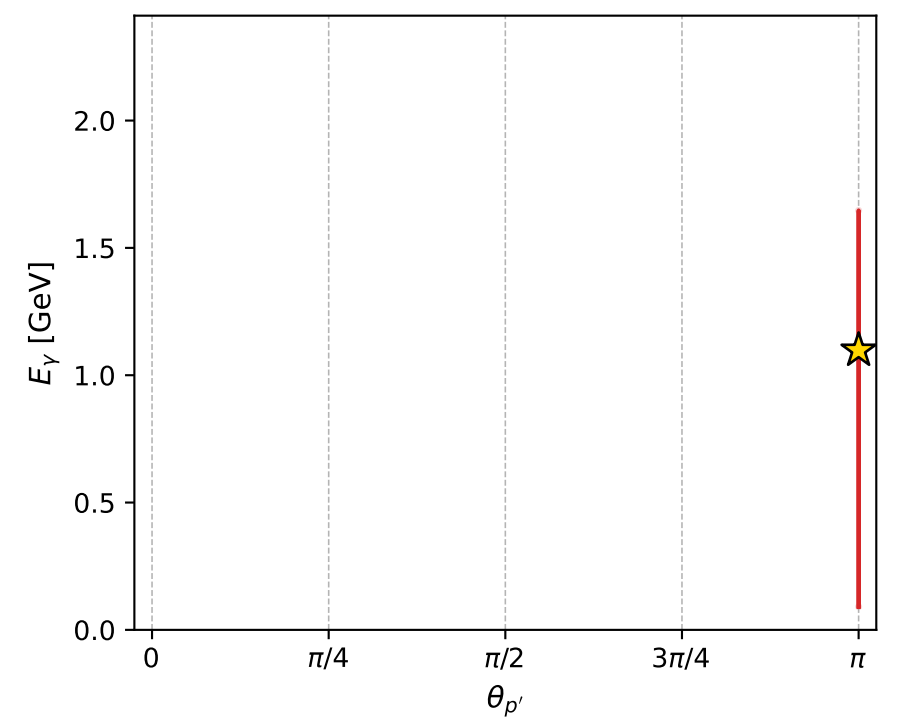
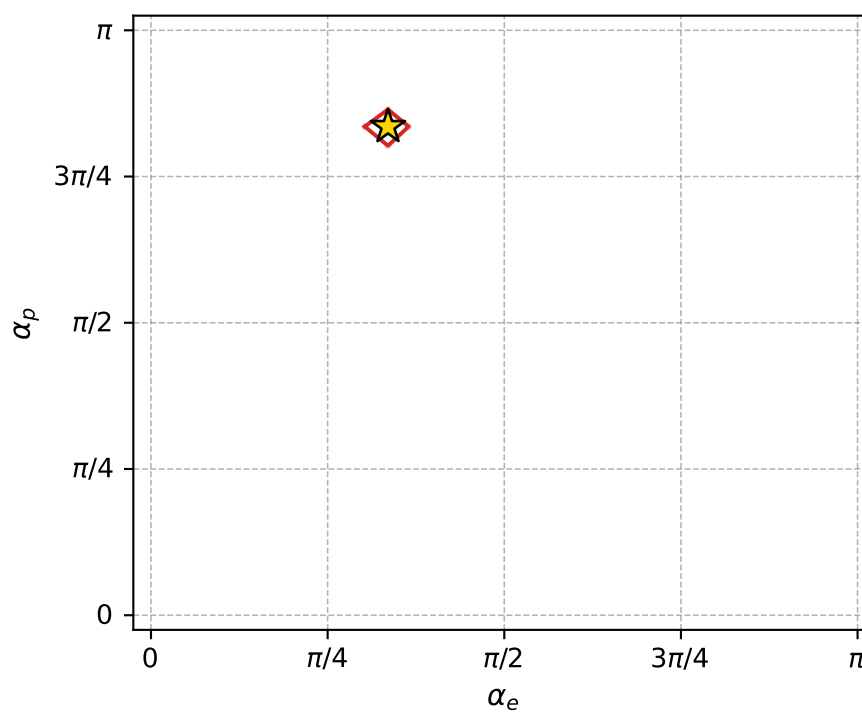
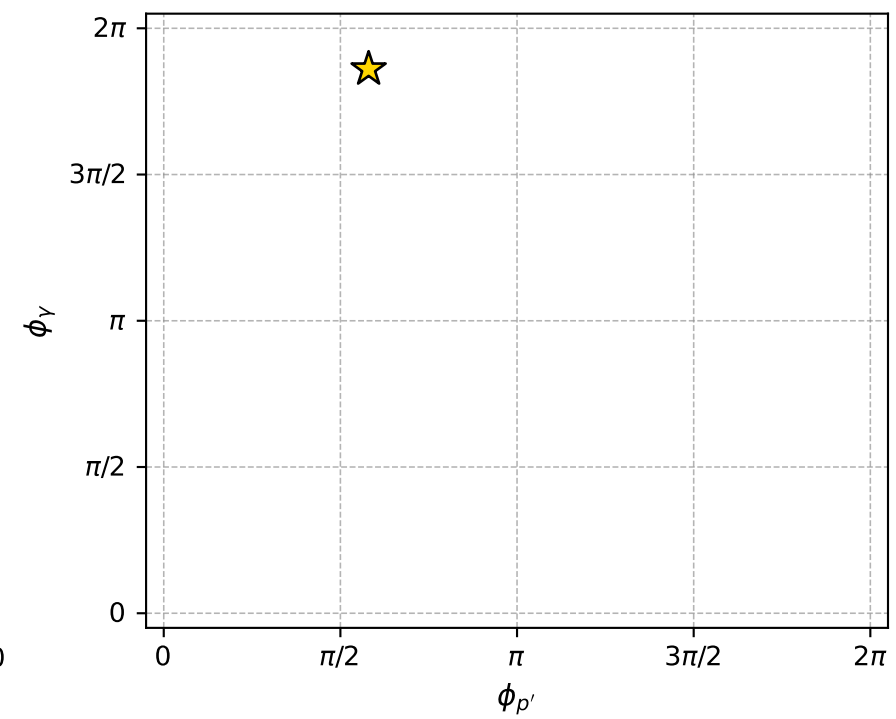
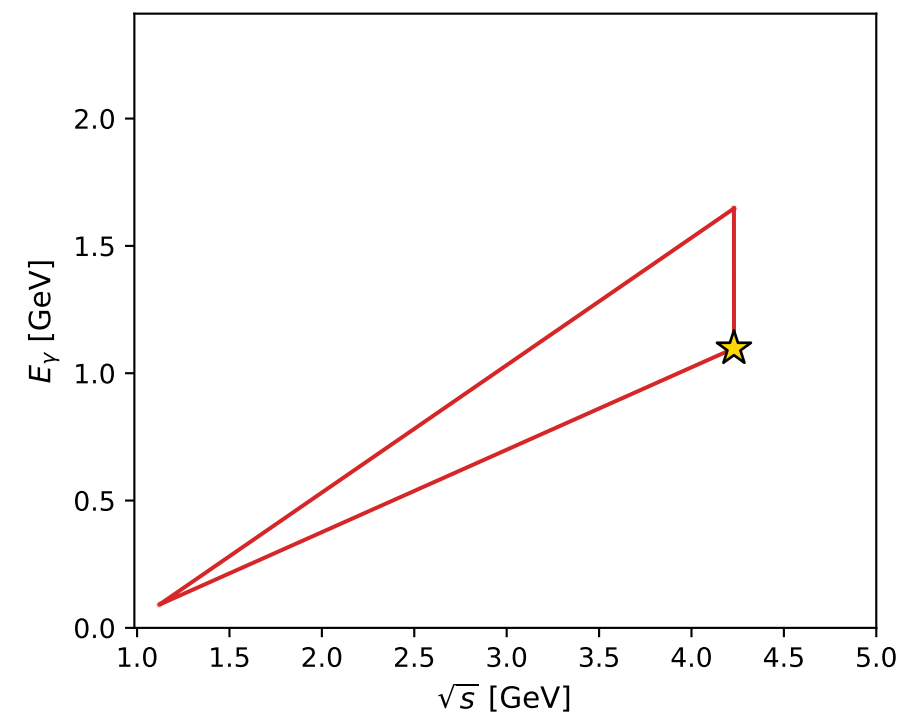
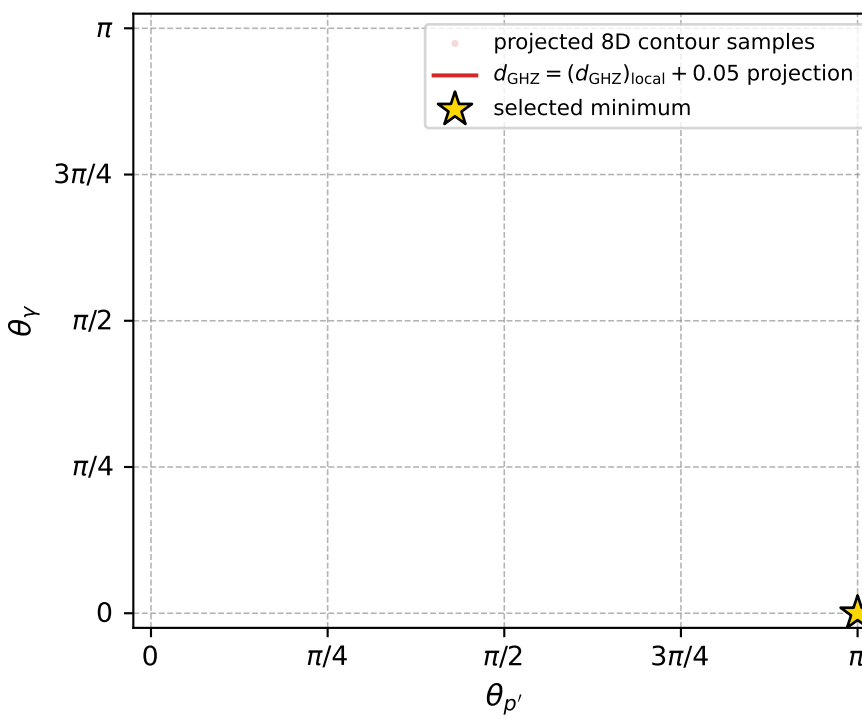
dghz_mixing_angles_polarization_02_local_minimum_148 $d_{\{\text{rm GHZ}\}}=8.27454\text{e-}08$
region: polarization_cluster_02_local_minimum_148
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: $\alpha_e=1.0537095$ rad
incoming lepton: $0.49435|+\rangle + 0.869263|-\rangle$
proton mixing angle: $\alpha_p=2.6245189$ rad
incoming proton: $-0.869269|+\rangle + 0.494339|-\rangle$

$s=17.8907$, $\sqrt{s}=4.22973$, $|\vec{P}|=2.01086$, $|\vec{P}'|=2.01086$
 $\theta_{P'}=3.14159$, $\theta_Y=0$, $\phi_{P'}=1.82192$, $\phi_Y=5.84502$
 $E_Y=1.09781$, $\theta_{\ell'}=0$, $\phi_{\ell'}=4.96351$
 $Q^2=7.34406$, $x_B=0.44159$, $t=-16.1742$, $W^2=10.1667$, $y=0.97767$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.0109$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.0109$
 P $E= 2.2189$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.0109$
 ℓ' $E= 0.9130$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 0.9130$
 P' $E= 2.2189$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= -2.0109$
 q_Y $E= 1.0978$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 1.0978$



electron: polarization cluster P2, local minimum 148; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 8.2745406\text{e-}08$
 $d_{\text{GHZ}} \text{ contour} = 0.050000083$
 above global minimum = $6.0461944\text{e-}08$
 8D contour samples = 58

selected phase-space point:

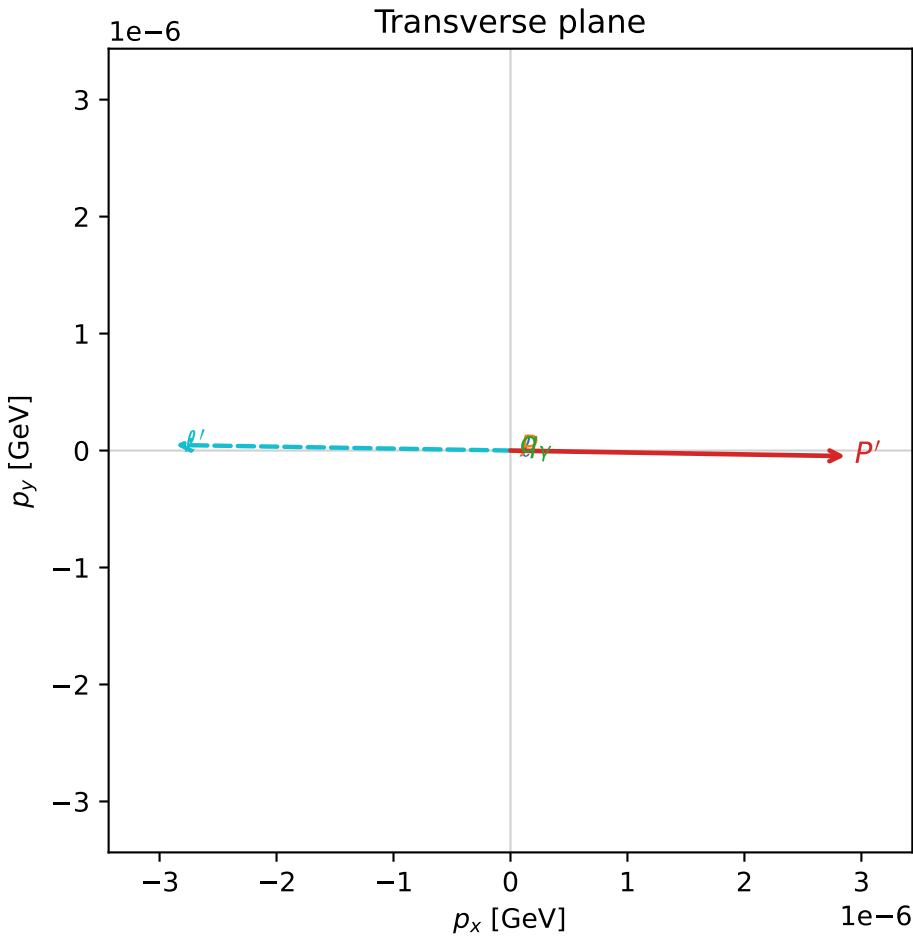
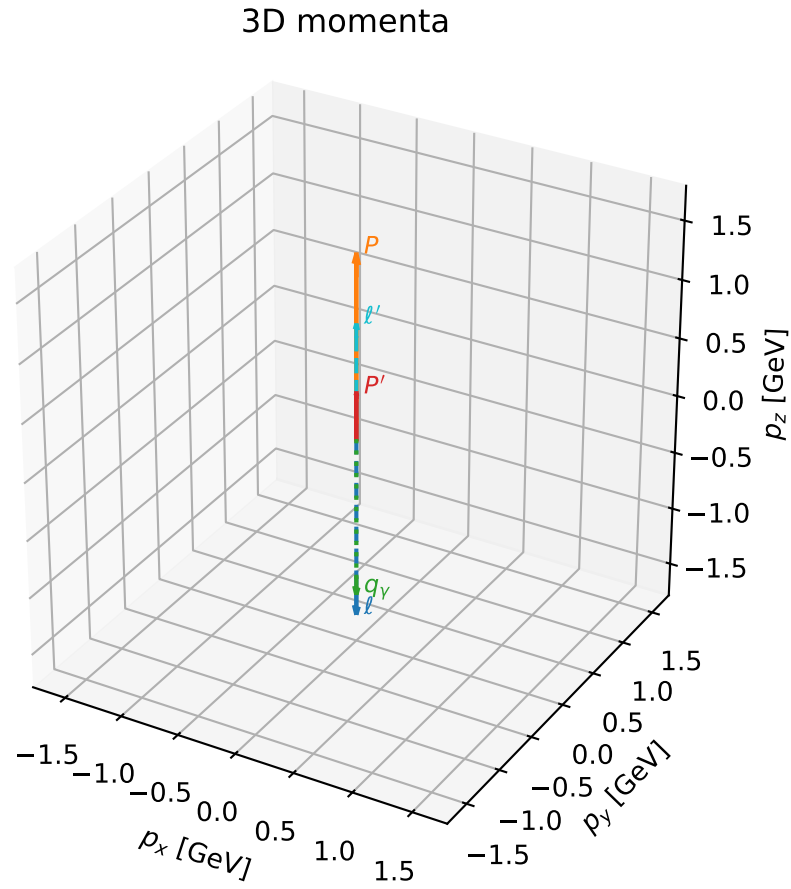
$\text{sqrt_s} = 4.229735$
 $\text{theta_p_out} = 3.141593$
 $\text{theta_gamma_out} = 0$
 $\text{q0ut} = 1.097811$
 $\text{phi_p_out} = 1.821921$
 $\text{phi_gamma_out} = 5.845022$
 $\text{alpha_e} = 1.053709$
 $\text{alpha_p} = 2.624519$

coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

dghz_mixing_angles_polarization_02_local_minimum_150 $d_{\{\text{rm GHZ}\}}=8.40814\text{e-}08$
region: polarization_cluster_02_local_minimum_150
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: $\alpha_e=0.77196915$ rad
incoming lepton: $0.716538|+\rangle +0.697548|-\rangle$
proton mixing angle: $\alpha_p=2.3696107$ rad
incoming proton: $-0.71653|+\rangle +0.697557|-\rangle$

$s=11.3797$, $\sqrt{s}=3.37339$, $|\vec{P}|=1.55628$, $|\vec{P}'|=0.391465$
 $\theta_{P'}=7.31453\text{e-}06$, $\theta_Y=3.14159$, $\phi_{P'}=6.26658$, $\phi_Y=3.08863$
 $E_Y=1.37422$, $\theta_{\ell'}=2.91362\text{e-}06$, $\phi_{\ell'}=3.12499$
 $Q^2=6.1178$, $x_B=0.61256$, $t=-0.715694$, $W^2=4.74931$, $y=0.951176$

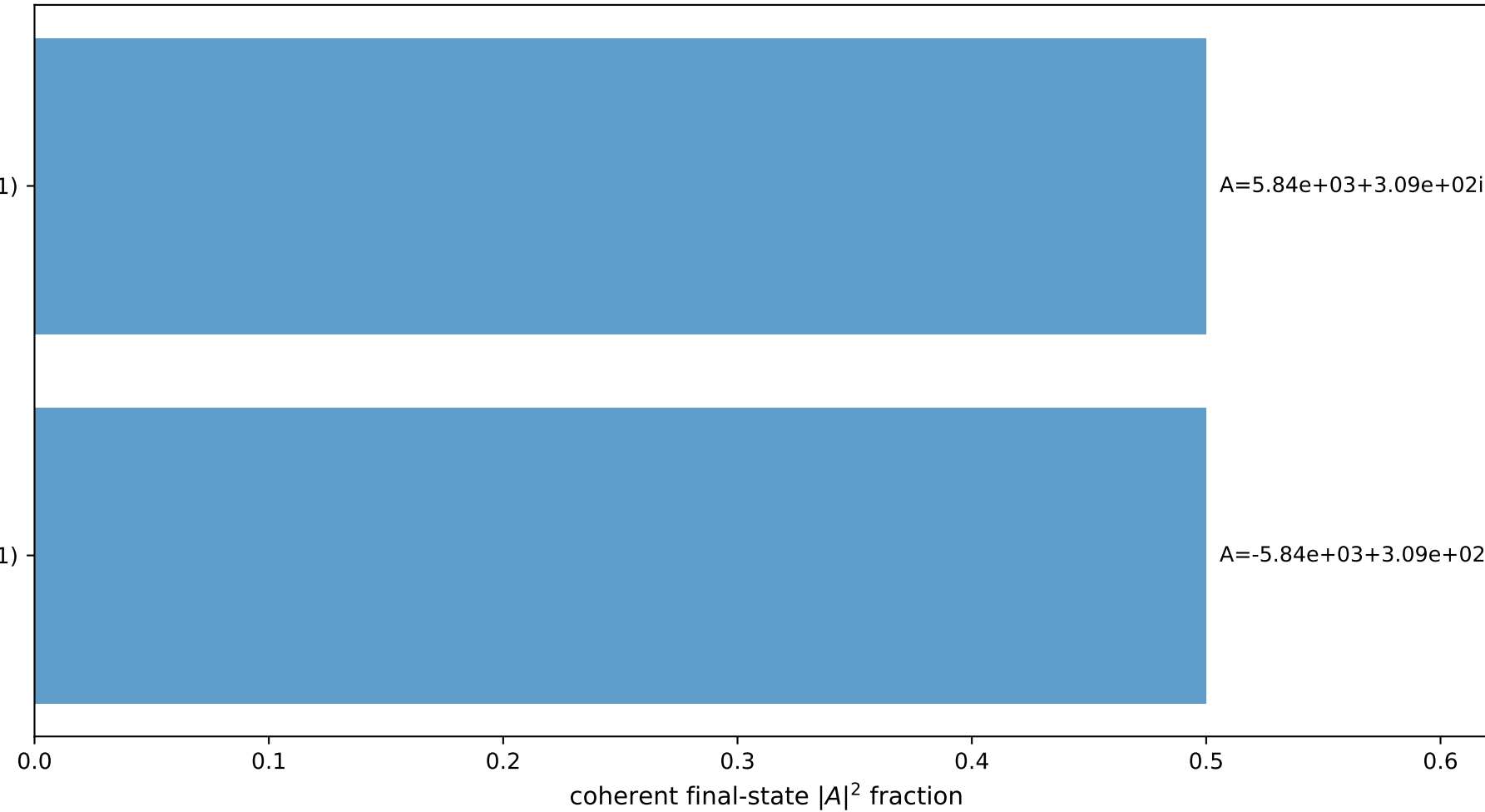
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 1.5563$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.5563$
 P $E= 1.8171$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.5563$
 ℓ' $E= 0.9828$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 0.9828$
 P' $E= 1.0164$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 0.3915$
 q_Y $E= 1.3742$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= -1.3742$



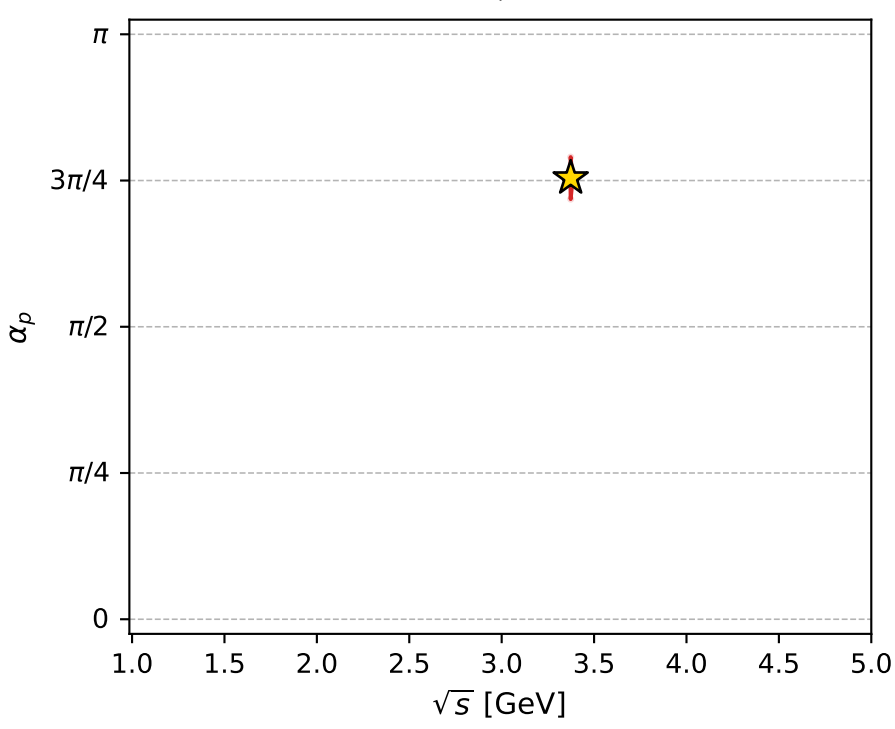
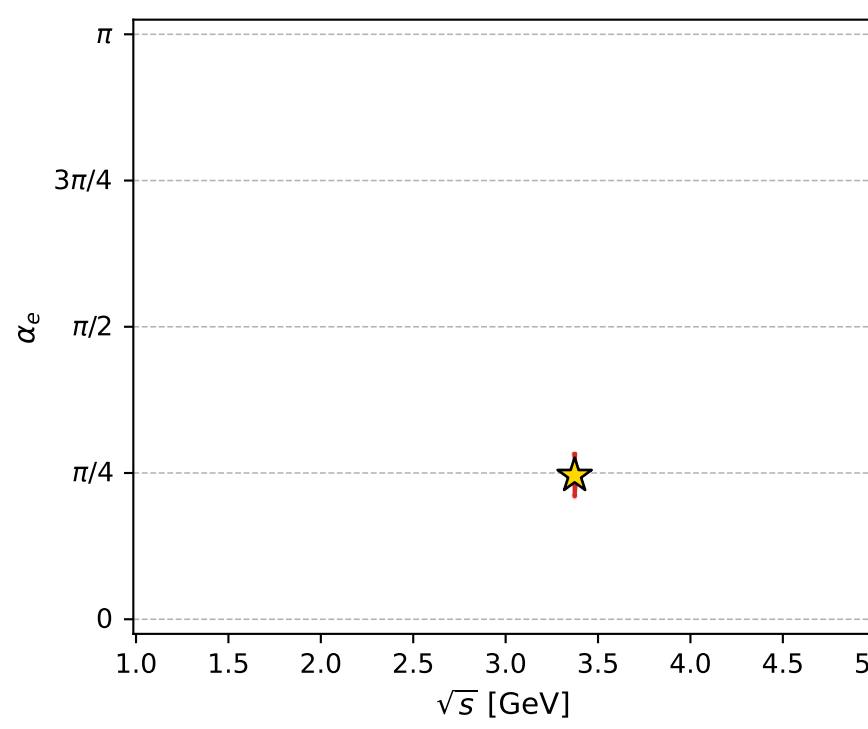
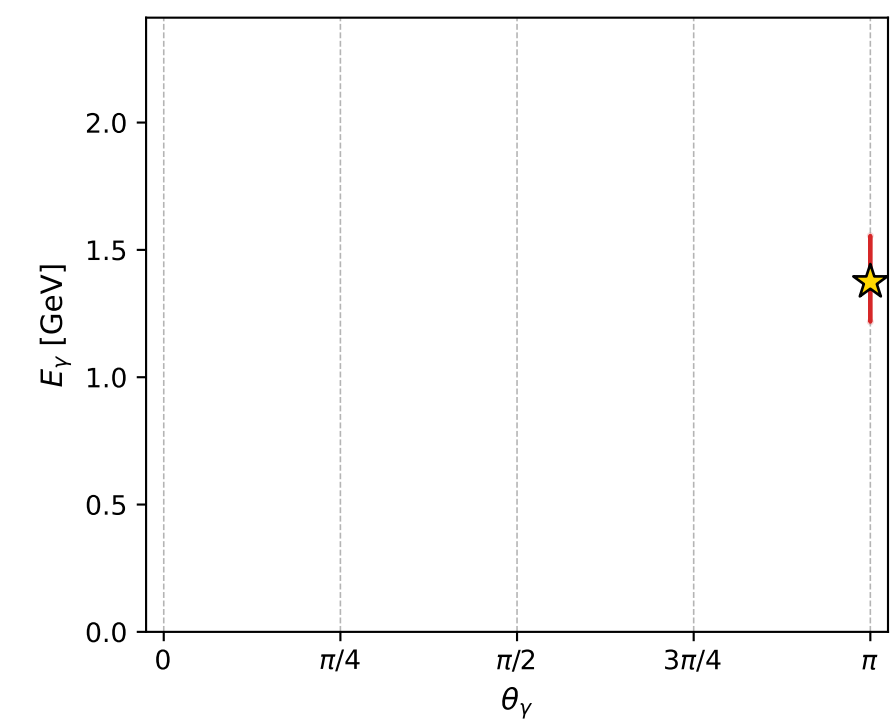
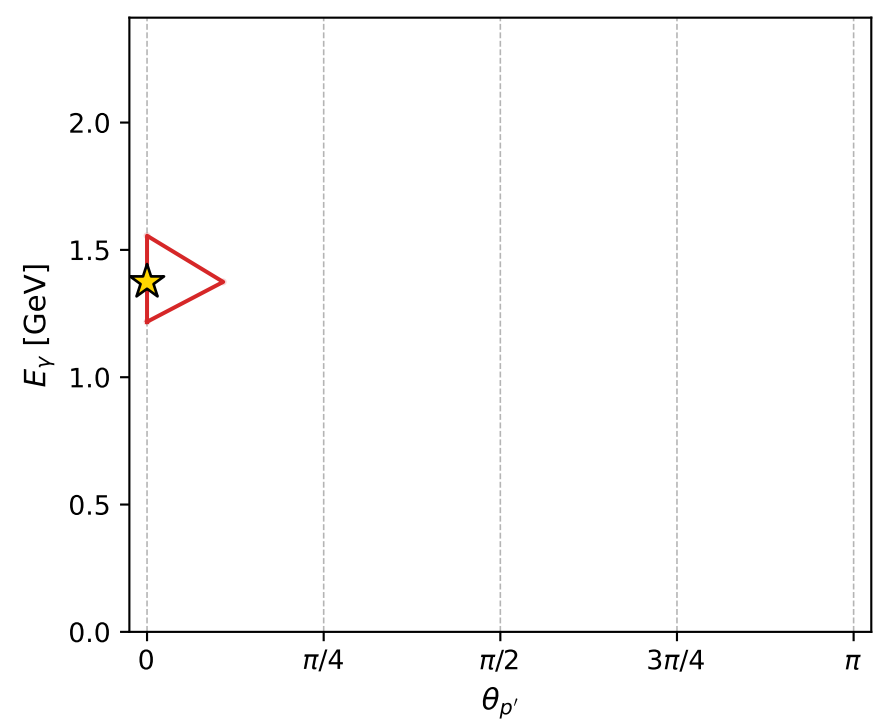
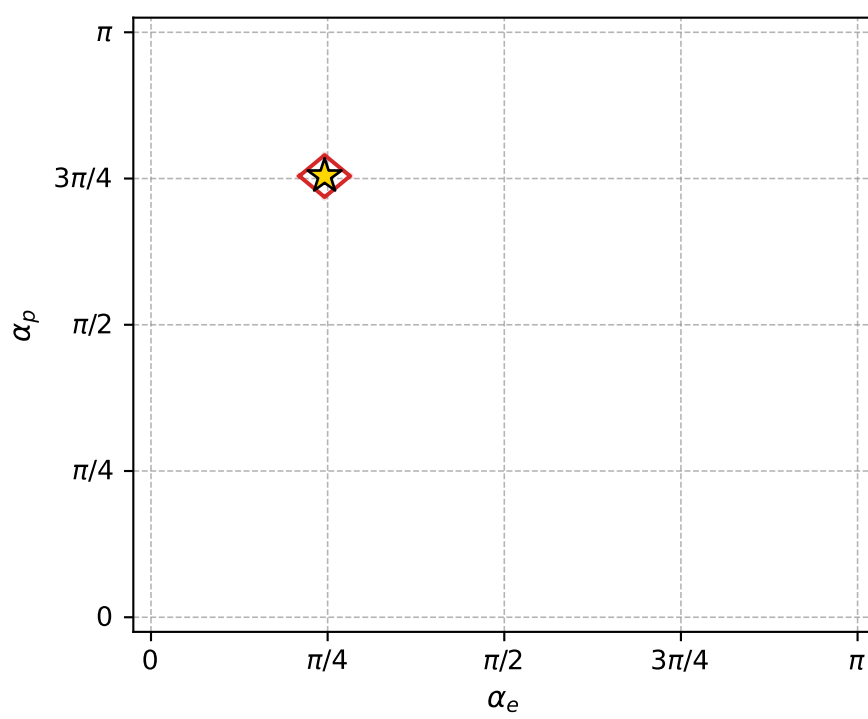
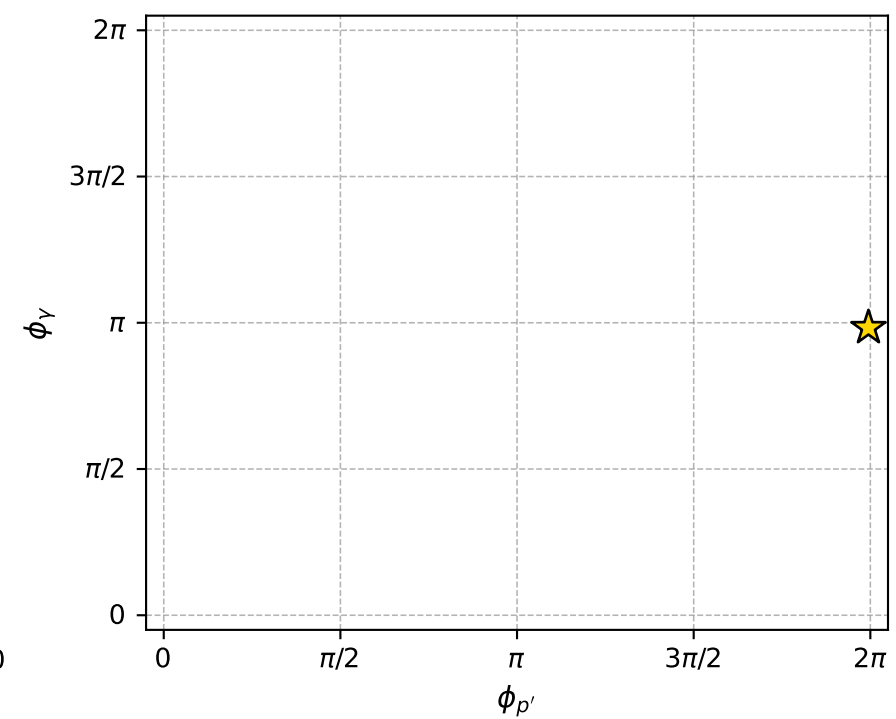
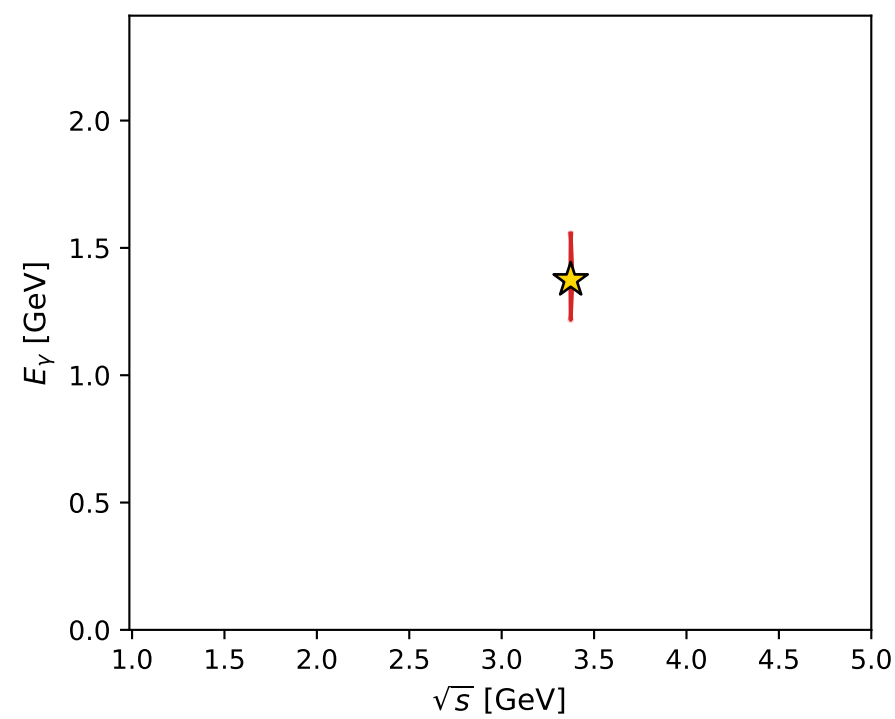
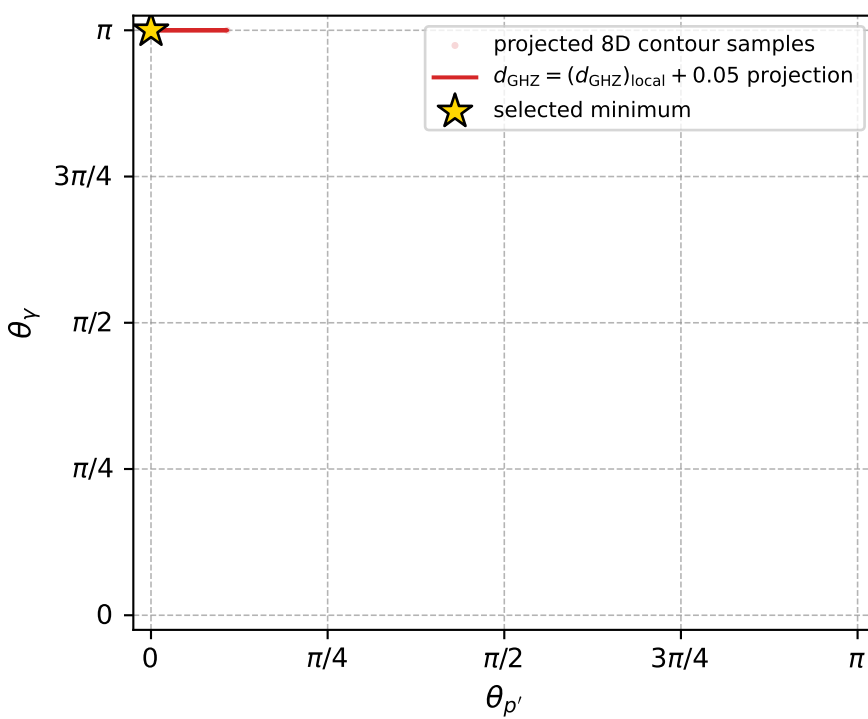
$(h_p, h_Y, h_\ell) = (+1, +1, -1)$

$(h_p, h_Y, h_\ell) = (-1, -1, +1)$

Leading final-state amplitudes (N=2, retained=100.0%)



electron: polarization cluster P2, local minimum 150; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 8.4081406\text{e-}08$
 $d_{\text{GHZ}} \text{ contour} = 0.050000084$
 above global minimum = $6.1797944\text{e-}08$
 8D contour samples = 60

selected phase-space point:

$\text{sqrt_s} = 3.373388$
 $\text{theta_p_out} = 7.314535\text{e-}06$
 $\text{theta_gamma_out} = 3.141593$
 $\text{q0ut} = 1.374221$
 $\text{phi_p_out} = 6.266583$
 $\text{phi_gamma_out} = 3.088627$
 $\text{alpha_e} = 0.7719692$
 $\text{alpha_p} = 2.369611$

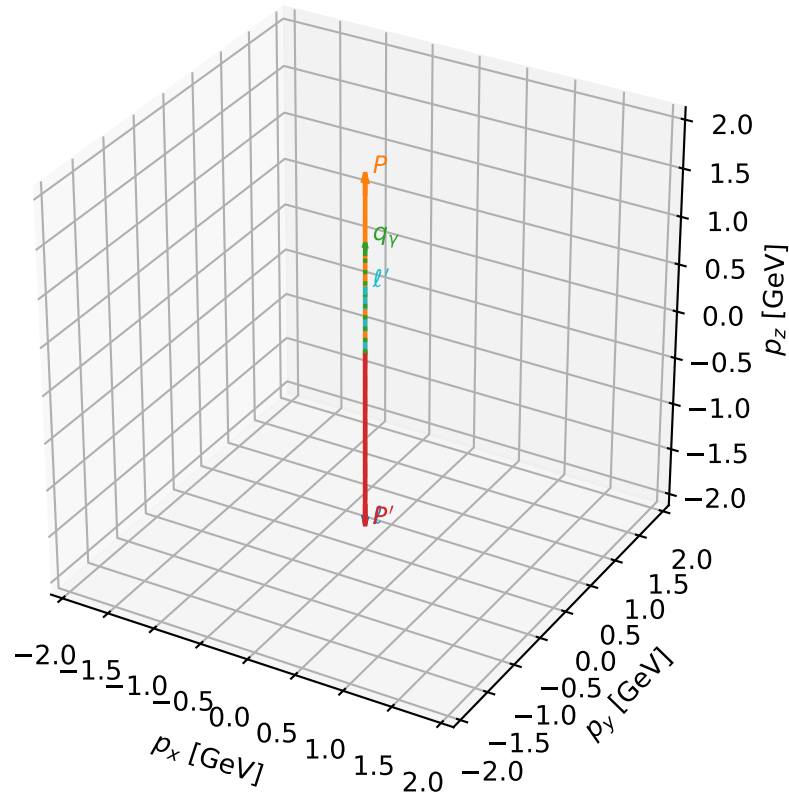
coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

dghz_mixing_angles_polarization_02_local_minimum_157 $d_{\{\text{rm GHZ}\}}=1.01492\text{e-}07$
 region: polarization_cluster_02_local_minimum_157
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=0.78801275$ rad
 incoming lepton: $0.705256|+\rangle +0.708953|-\rangle$
 proton mixing angle: $\alpha_p=2.3588075$ rad
 incoming proton: $-0.708952|+\rangle +0.705257|-\rangle$

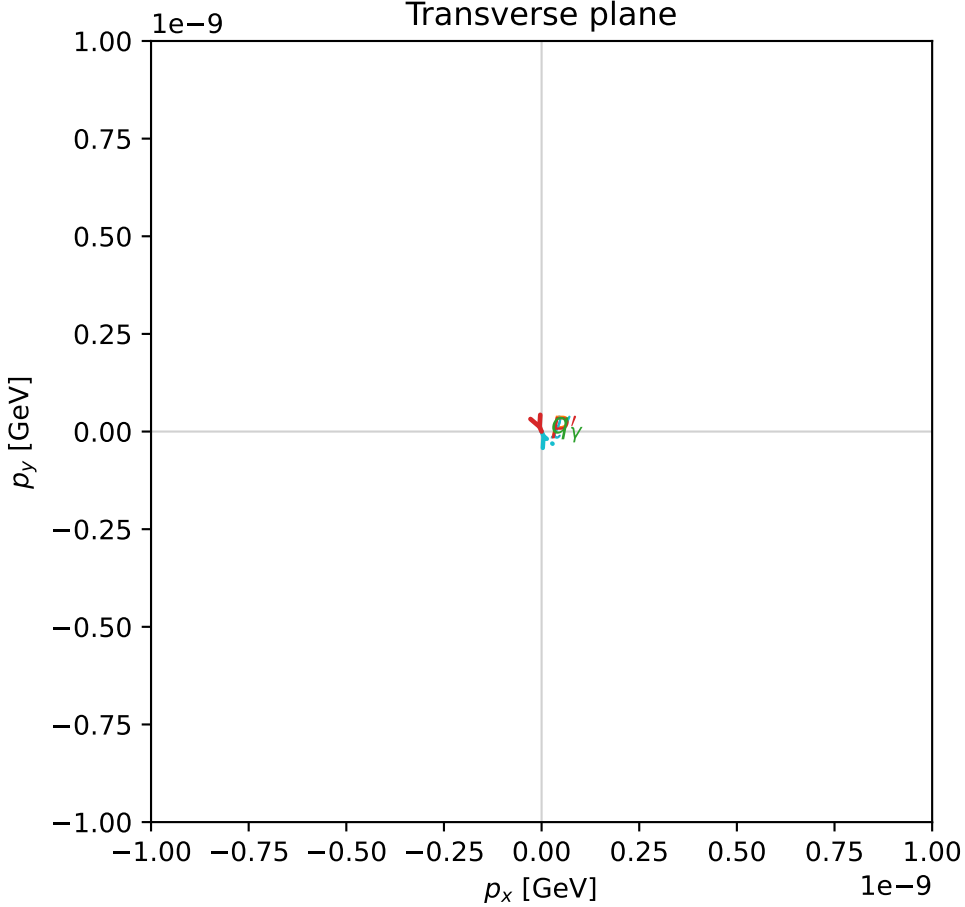
$s=15.3816$, $\sqrt{s}=3.92194$, $|\vec{P}|=1.8488$, $|\vec{P}'|=1.8488$
 $\theta_{P'}=3.14159$, $\theta_V=0$, $\phi_{P'}=5.11449$, $\phi_V=3.59494$
 $E_V=1.15299$, $\theta_{\ell'}=0$, $\phi_{\ell'}=1.97289$
 $Q^2=5.14568$, $x_B=0.362639$, $t=-13.6722$, $W^2=9.92371$, $y=0.978472$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 1.8488$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.8488$
 P $E= 2.0731$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.8488$
 ℓ' $E= 0.6958$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 0.6958$
 P' $E= 2.0731$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= -1.8488$
 q_V $E= 1.1530$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 1.1530$

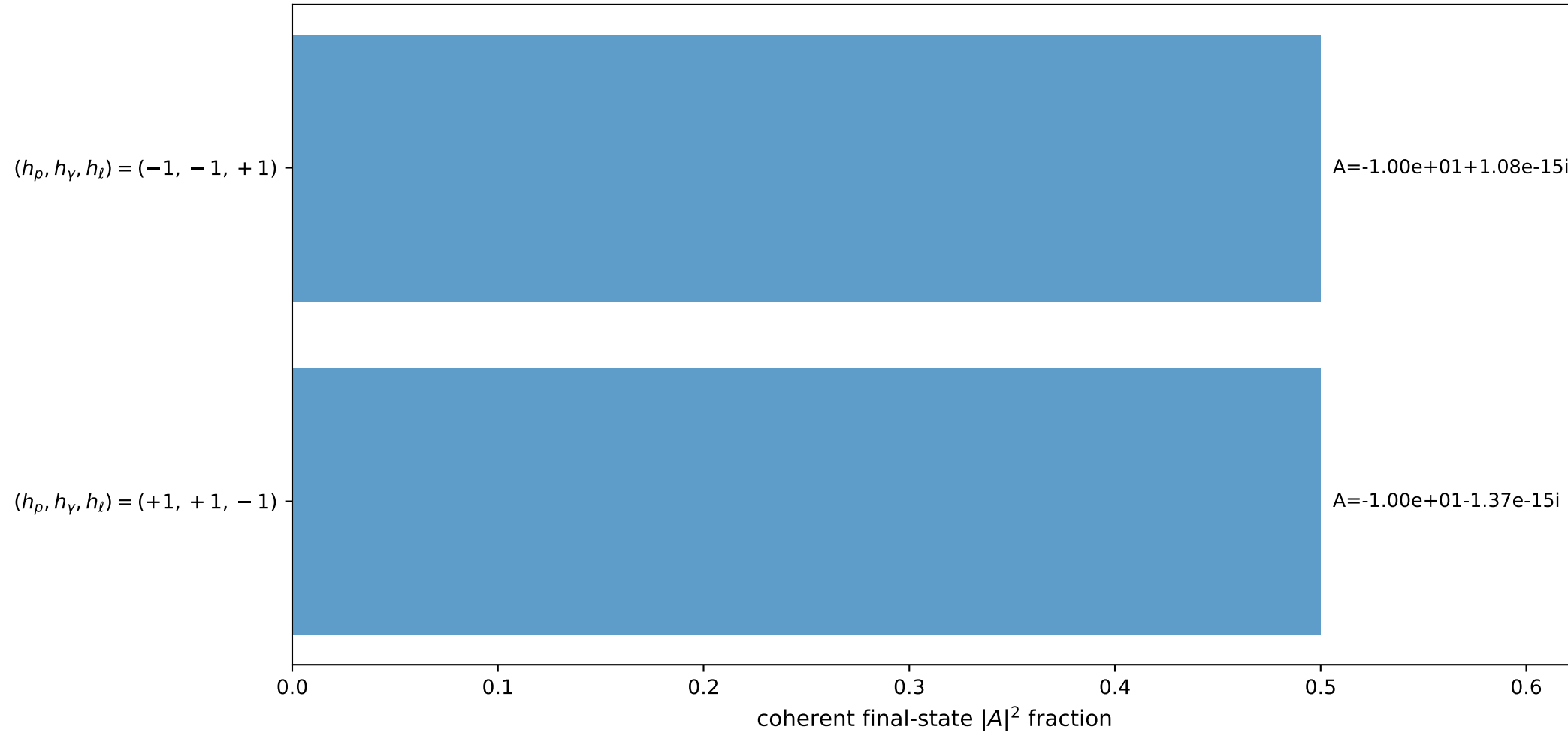
3D momenta



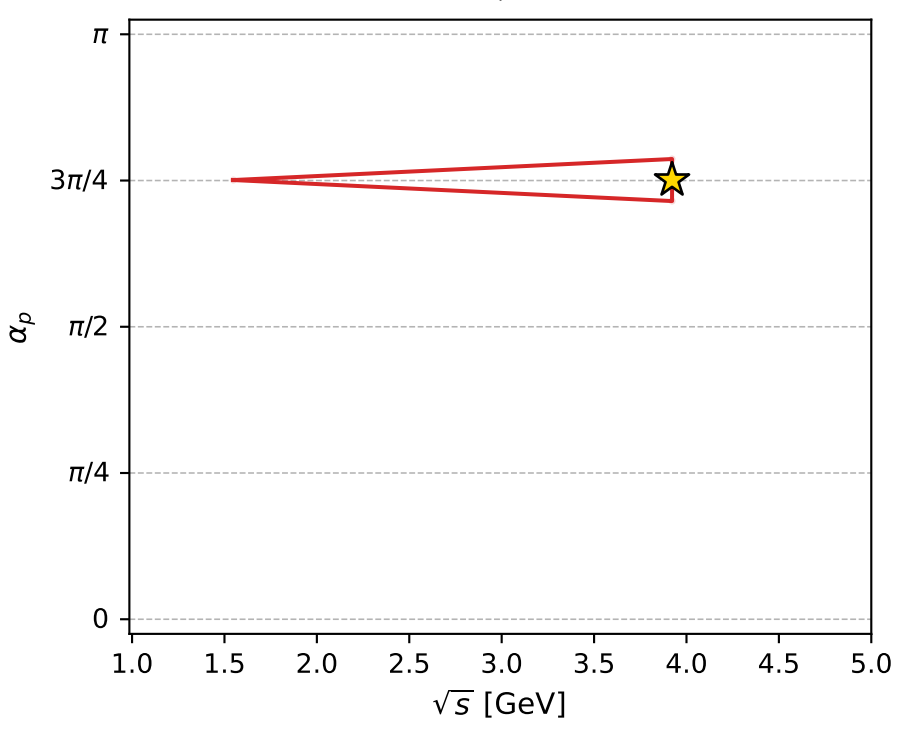
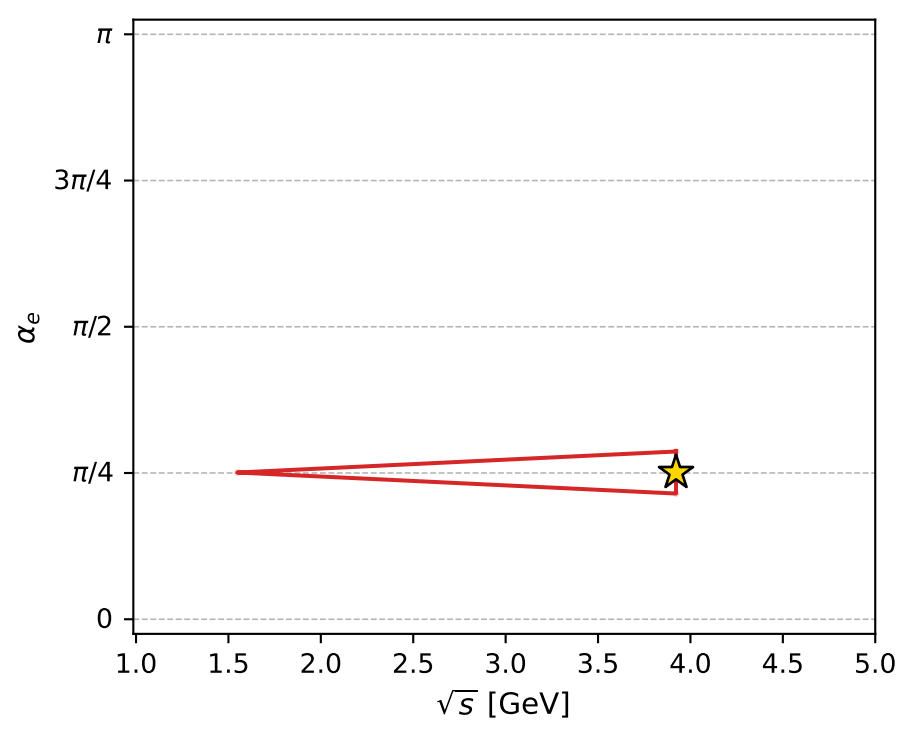
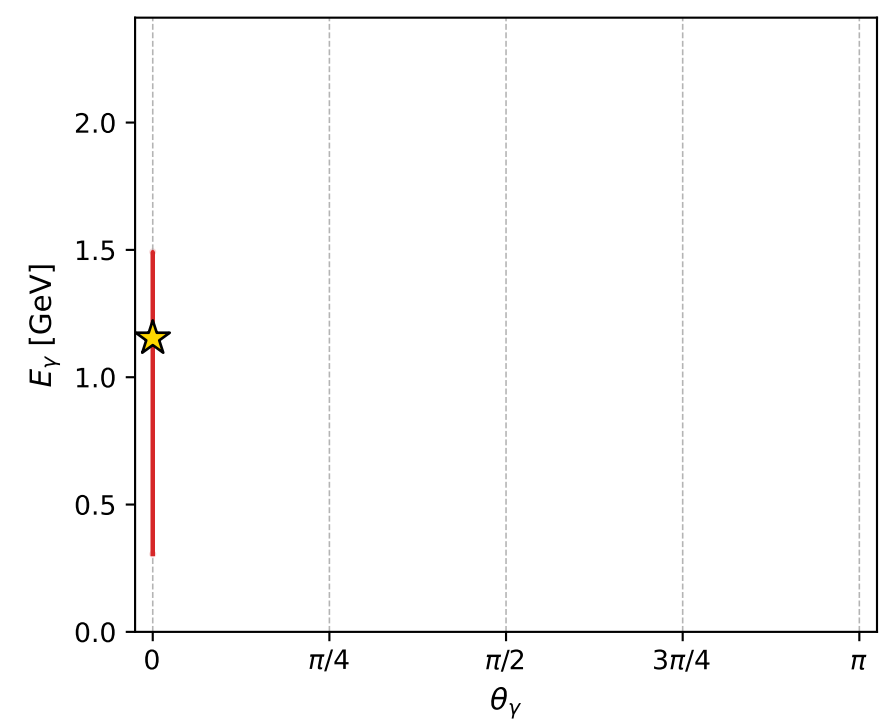
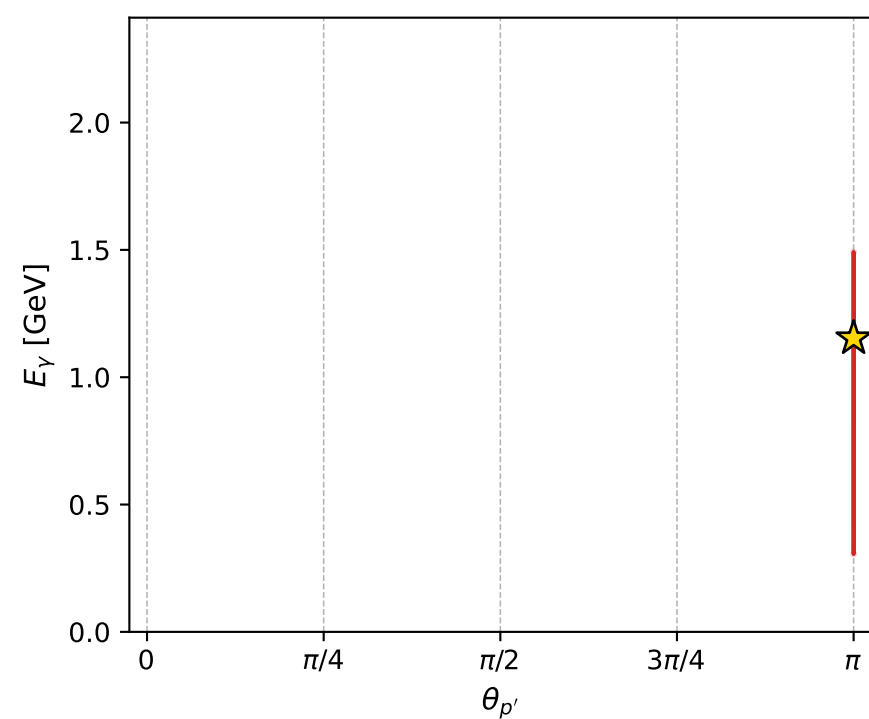
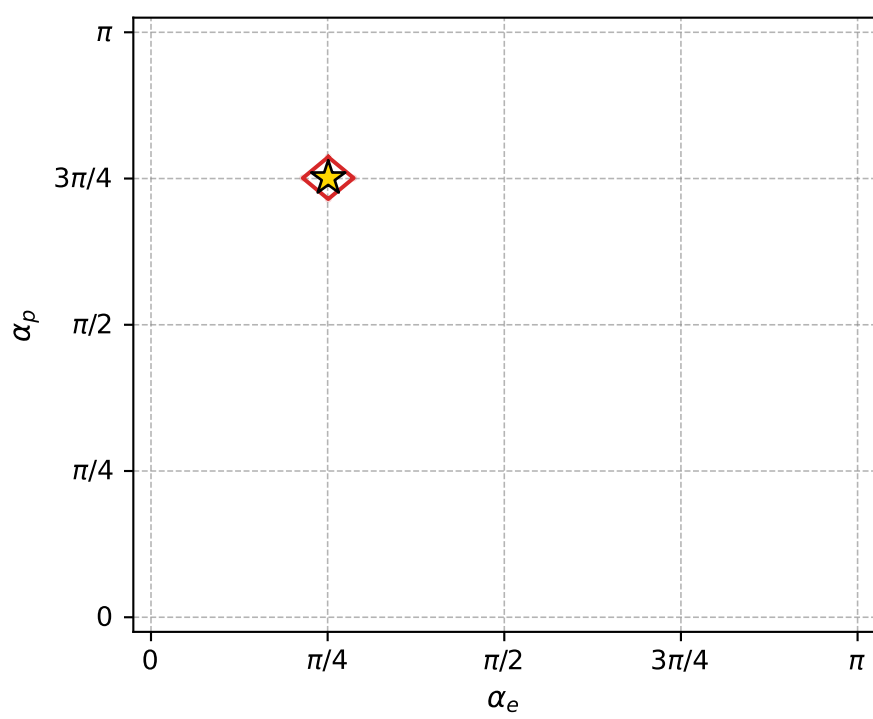
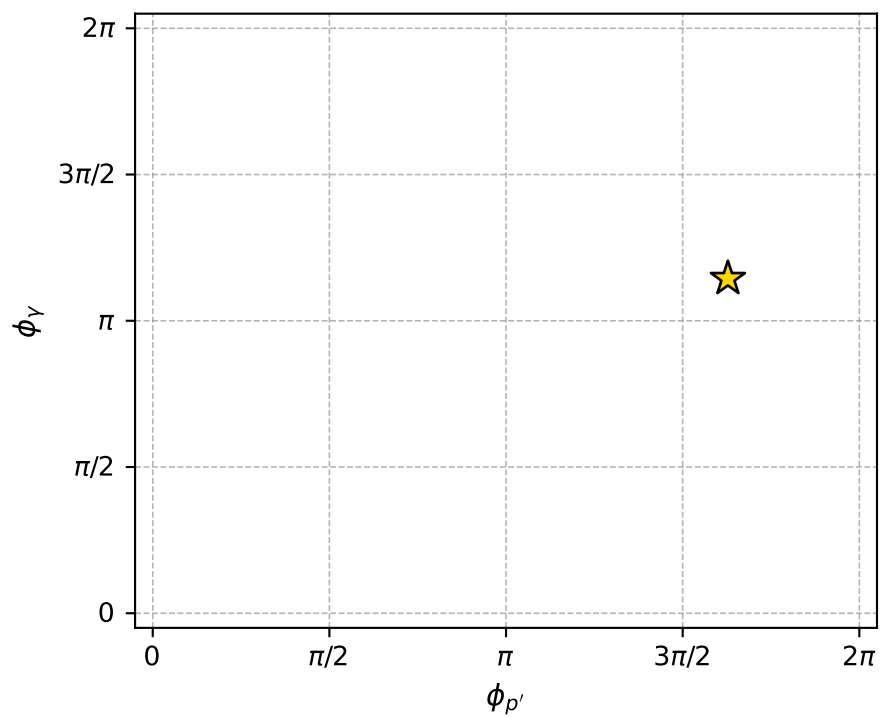
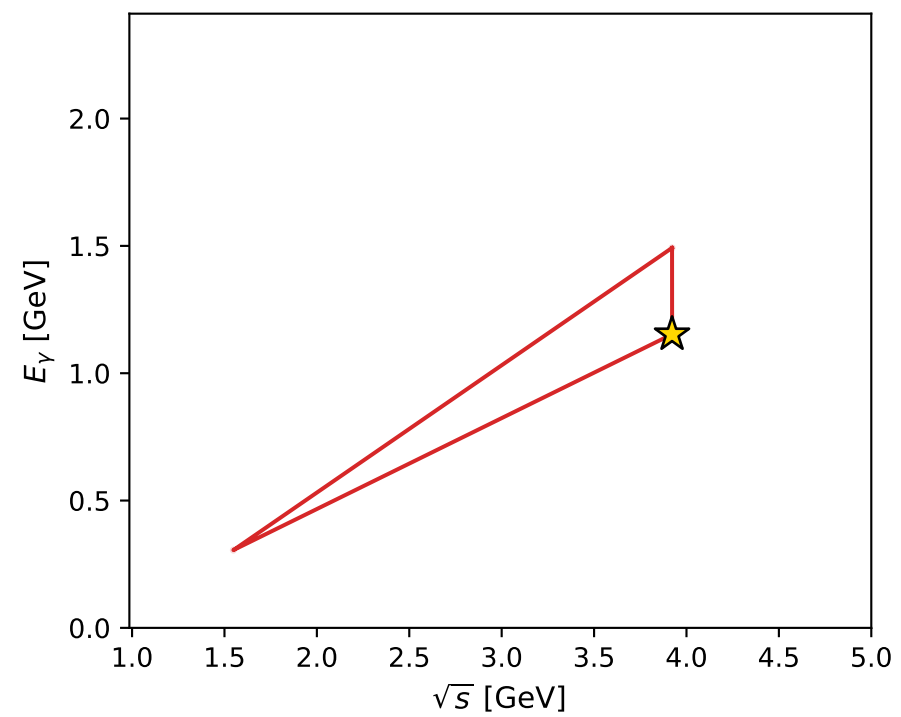
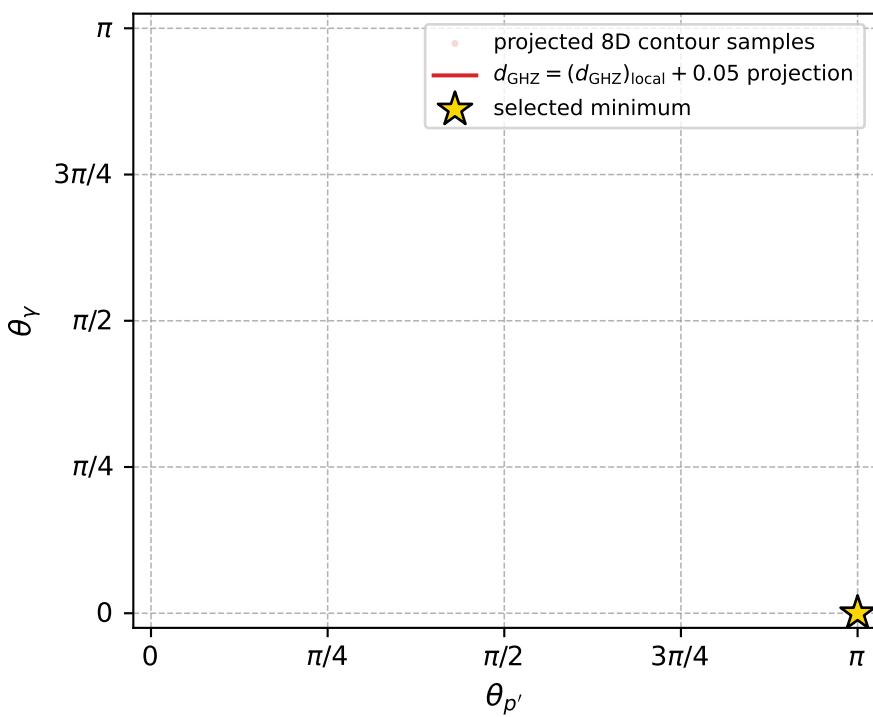
Transverse plane



Leading final-state amplitudes (N=2, retained=100.0%)



electron: polarization cluster P2, local minimum 157; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 1.0149213\text{e-}07$
 $d_{\text{GHZ}} \text{ contour} = 0.050000101$
 above global minimum = $7.920867\text{e-}08$
 8D contour samples = 60

selected phase-space point:

$\text{sqrt_s} = 3.921937$
 $\text{theta_p_out} = 3.141593$
 $\text{theta_gamma_out} = 0$
 $\text{q0out} = 1.152985$
 $\text{phi_p_out} = 5.114486$
 $\text{phi_gamma_out} = 3.59494$
 $\text{alpha_e} = 0.7880128$
 $\text{alpha_p} = 2.358807$

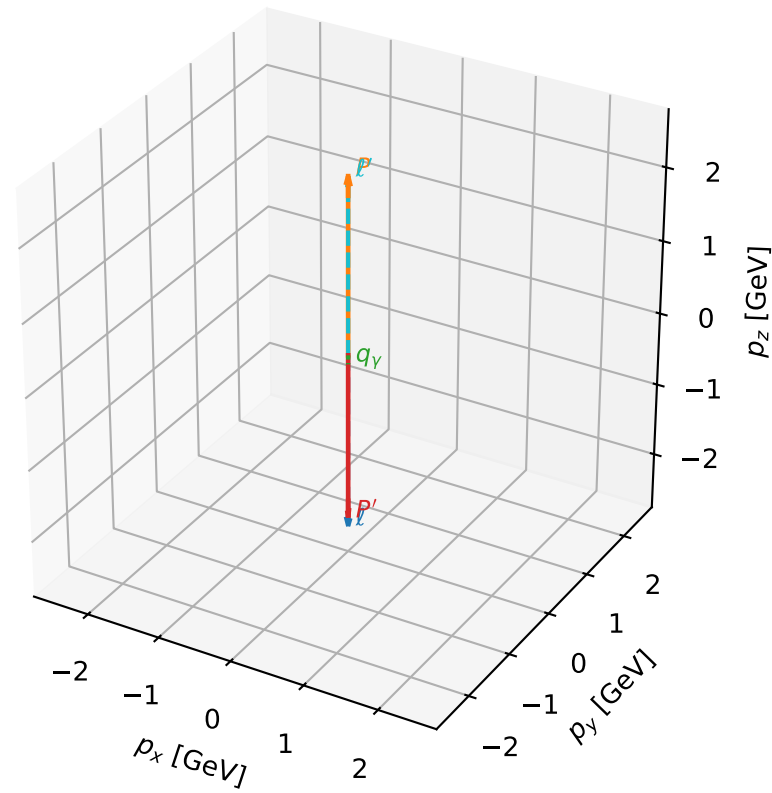
coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

dghz_mixing_angles_polarization_02_local_minimum_159 d_{\rm GHZ}=1.07274e-07
region: polarization_cluster_02_local_minimum_159
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: $\alpha_e=0.10562157$ rad
incoming lepton: $0.994427|+\rangle +0.105425|-\rangle$
proton mixing angle: $\alpha_p=3.0359631$ rad
incoming proton: $-0.994426|+\rangle +0.105433|-\rangle$

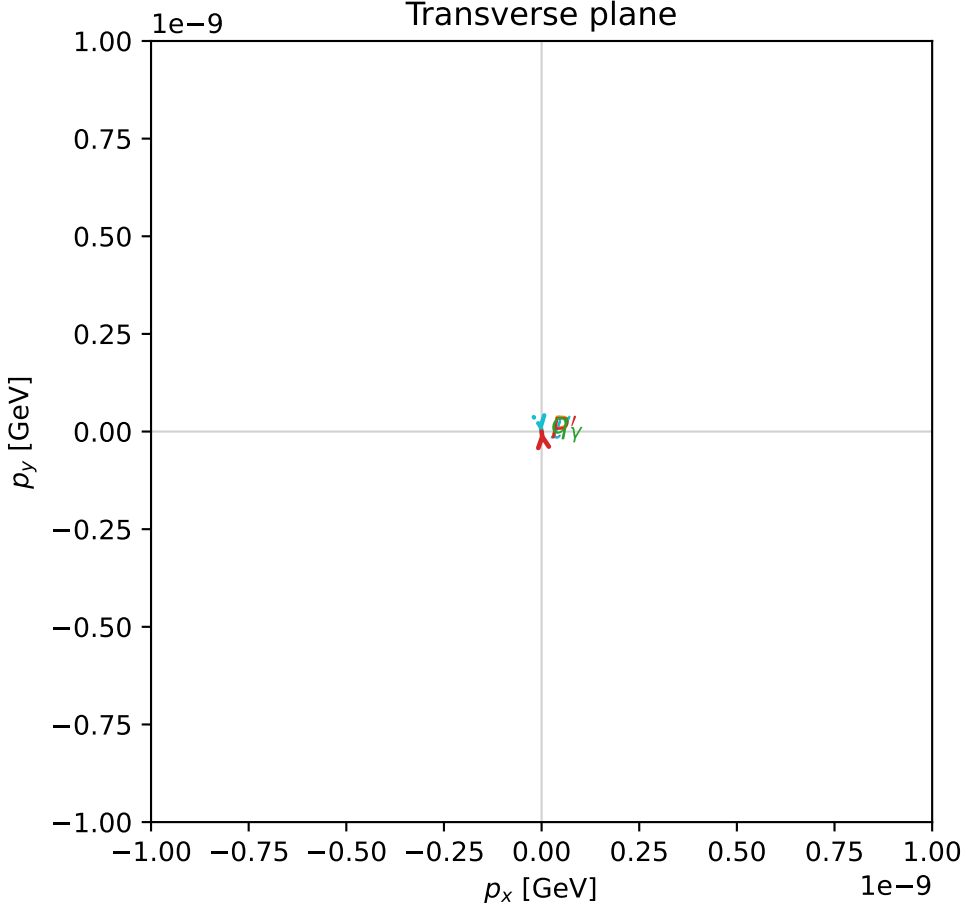
$s=25$, $\sqrt{s}=5$, $|\vec{P}|=2.41202$, $|\vec{P}'|=2.28696$
 $\theta_{P'}=3.14159$, $\theta_Y=3.14159$, $\phi_{P'}=1.69535$, $\phi_Y=3.91457$
 $E_Y=0.120596$, $\theta_{\ell'}=0$, $\phi_{\ell'}=4.88033$
 $Q^2=23.2283$, $x_B=0.998084$, $t=-22.0669$, $W^2=0.924437$, $y=0.964871$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4120$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 P $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4120$
 ℓ' $E= 2.4076$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 2.4076$
 P' $E= 2.4718$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= -2.2870$
 q_Y $E= 0.1206$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= -0.1206$

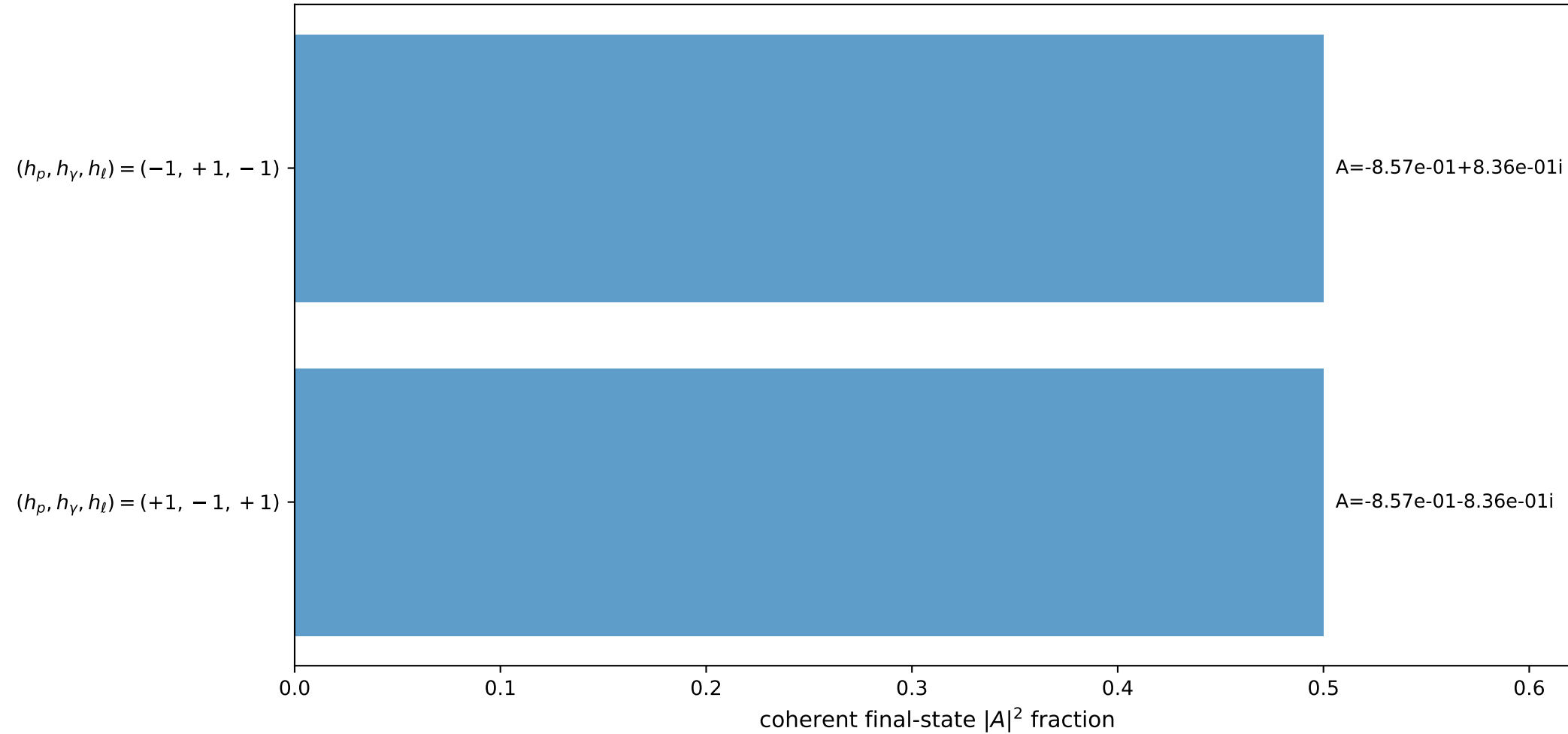
3D momenta



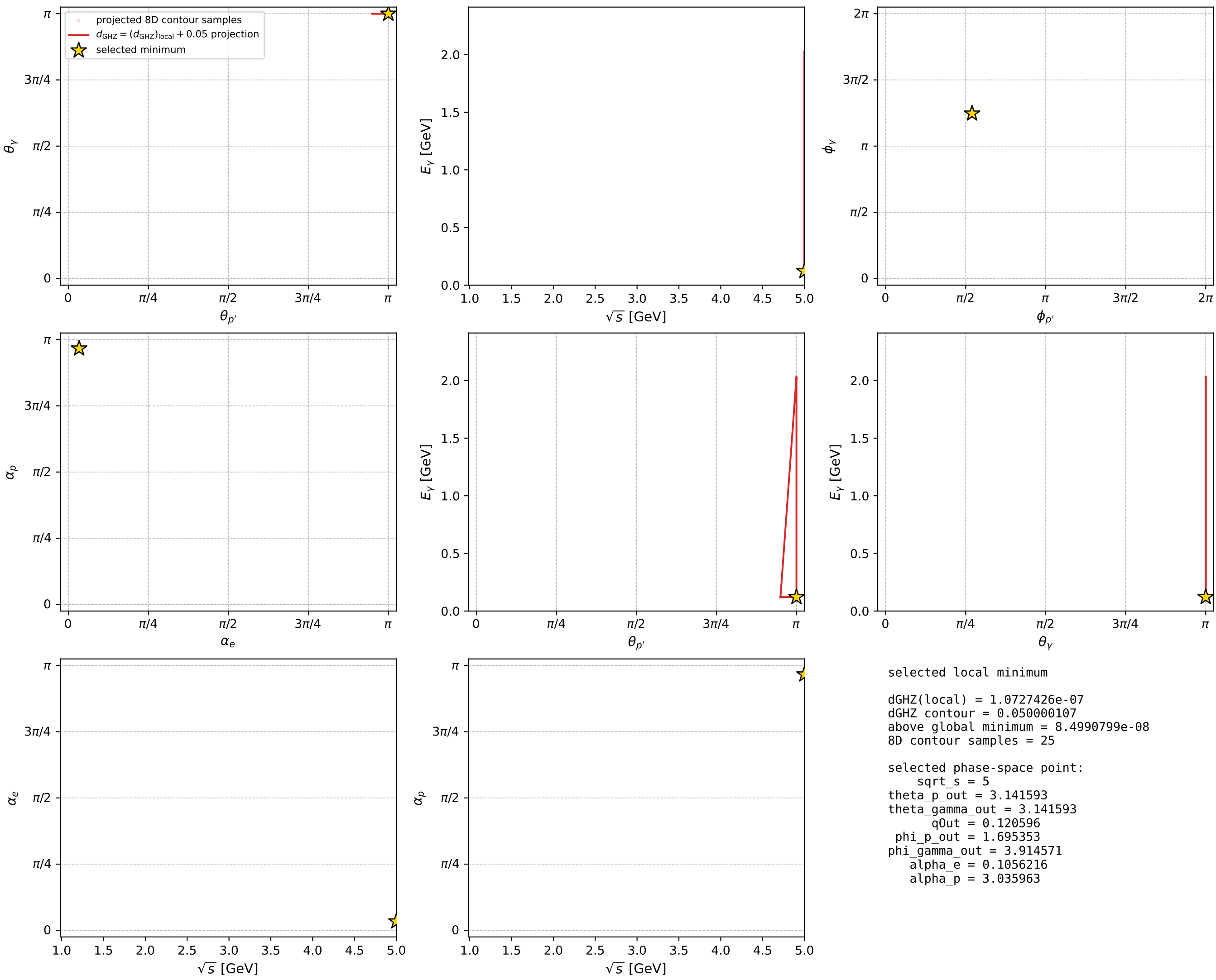
Transverse plane



Leading final-state amplitudes (N=2, retained=100.0%)



electron: polarization cluster P2, local minimum 159; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour

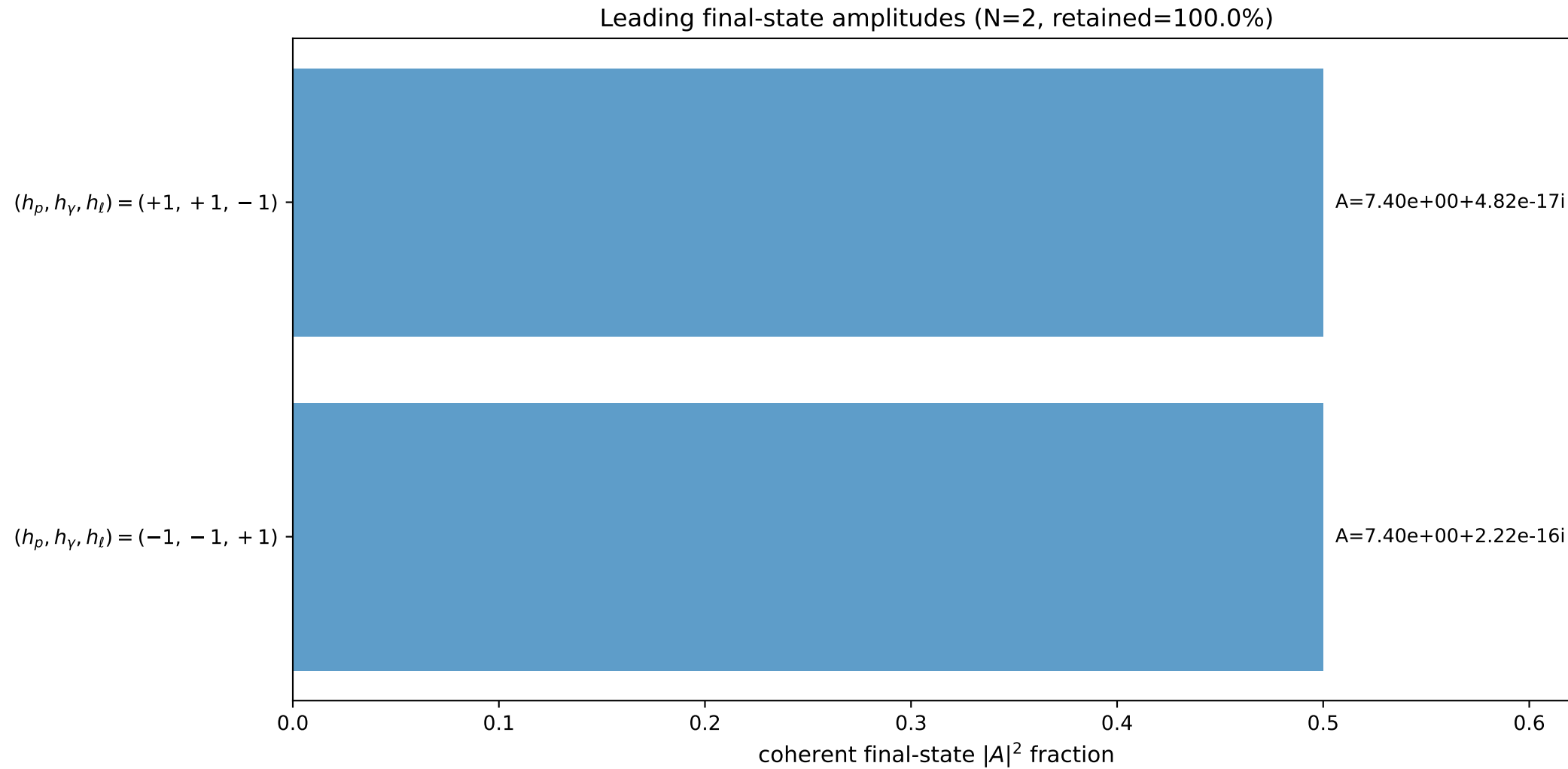
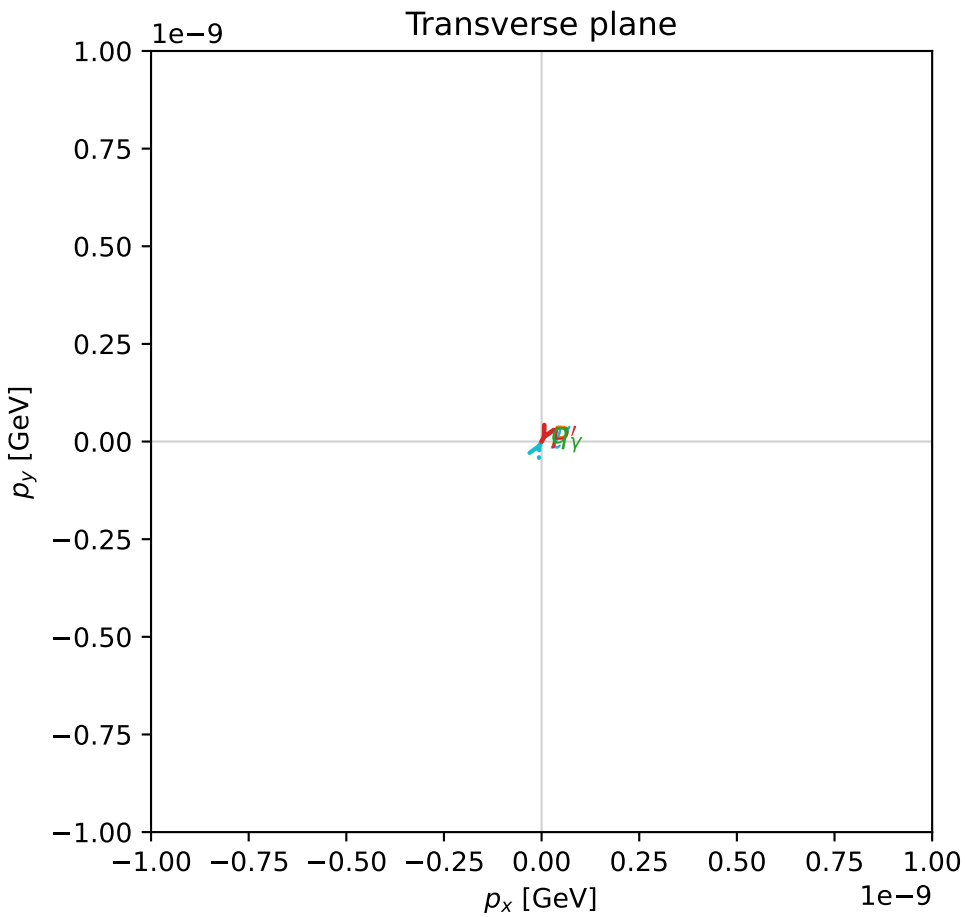
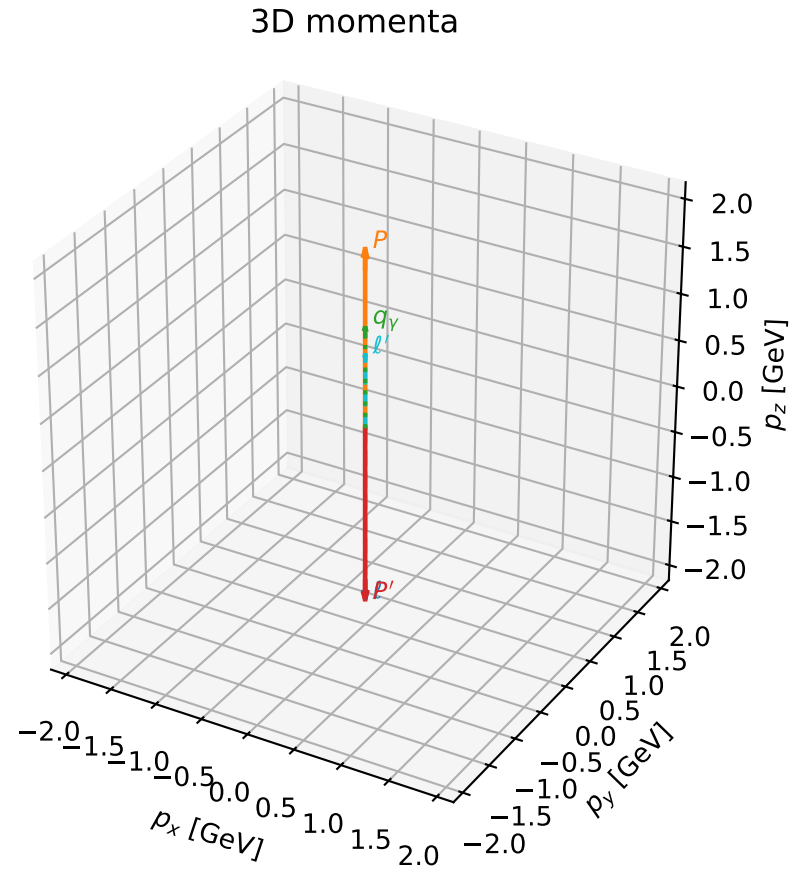


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

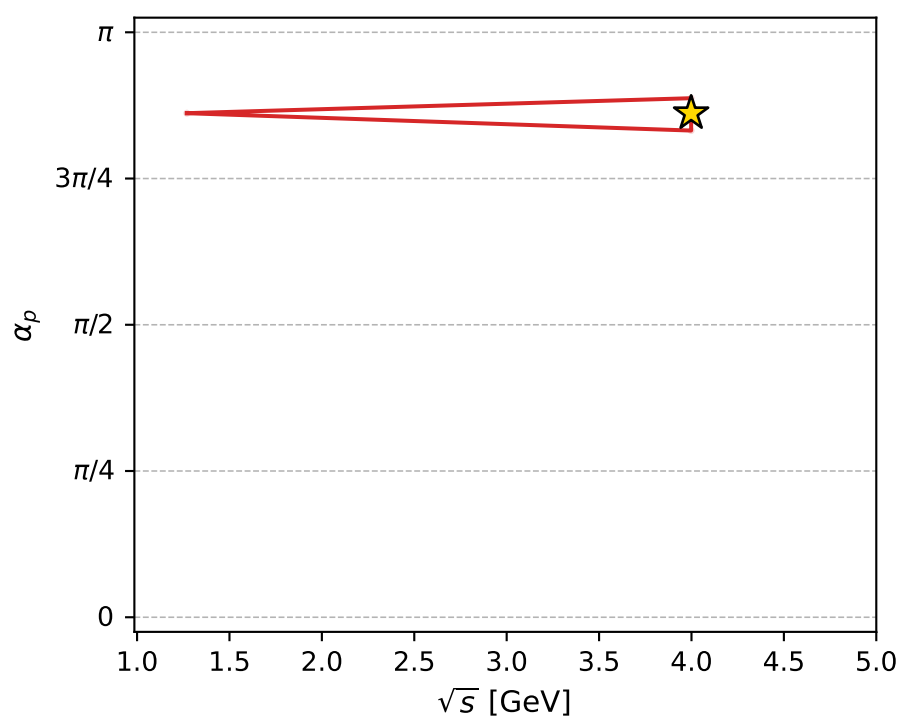
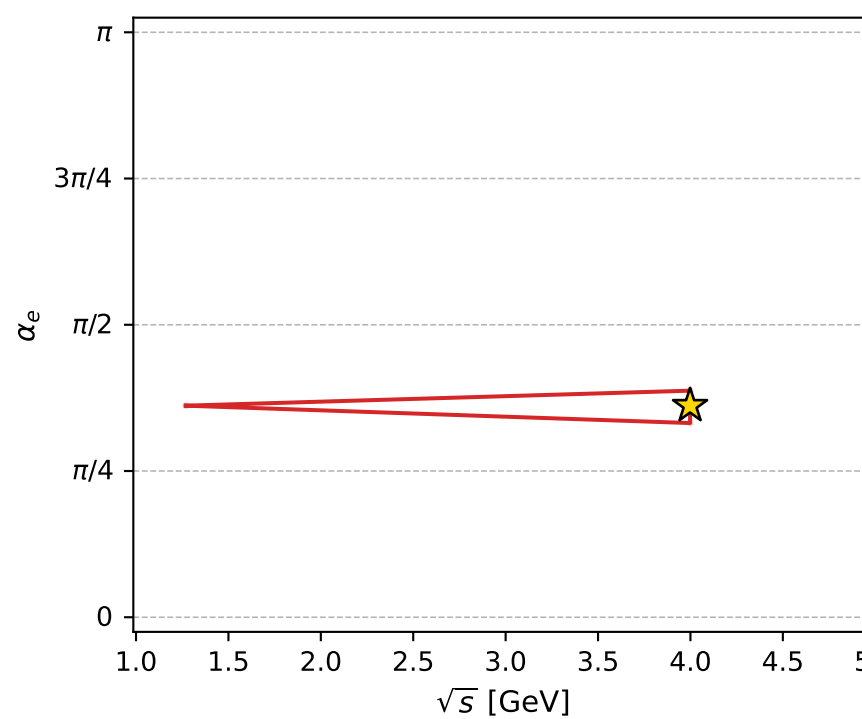
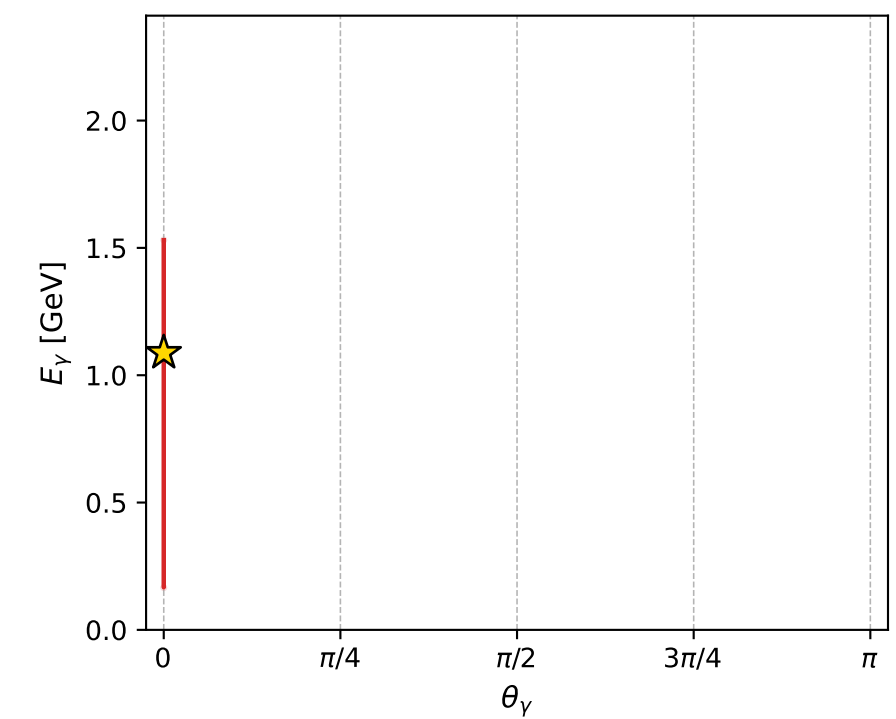
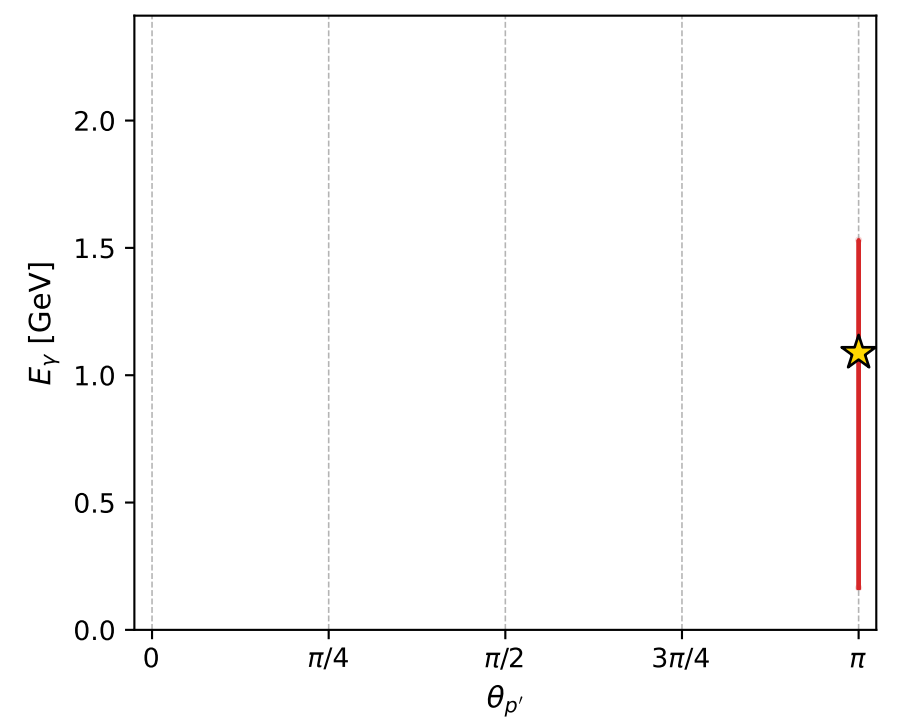
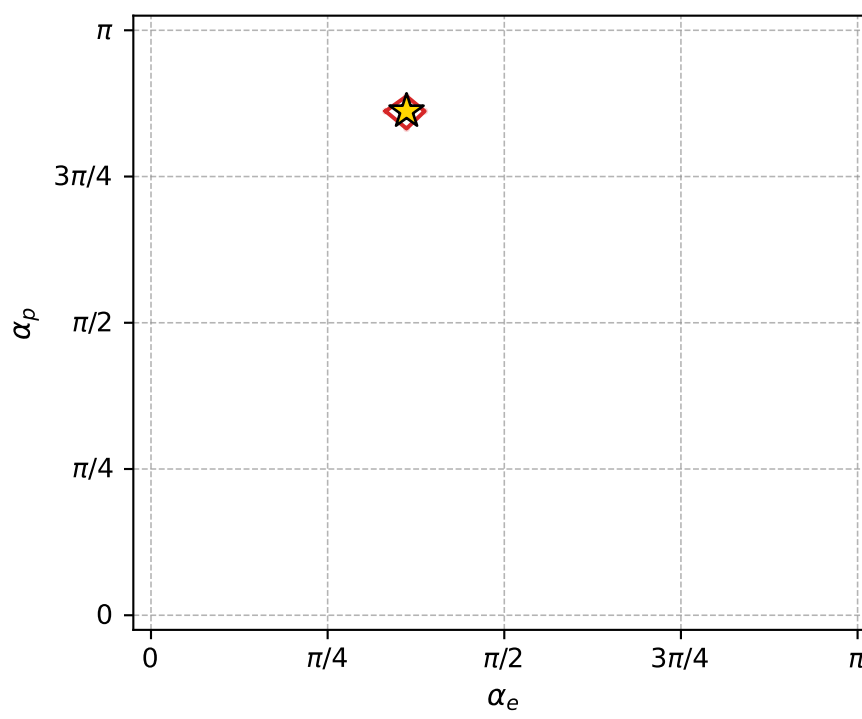
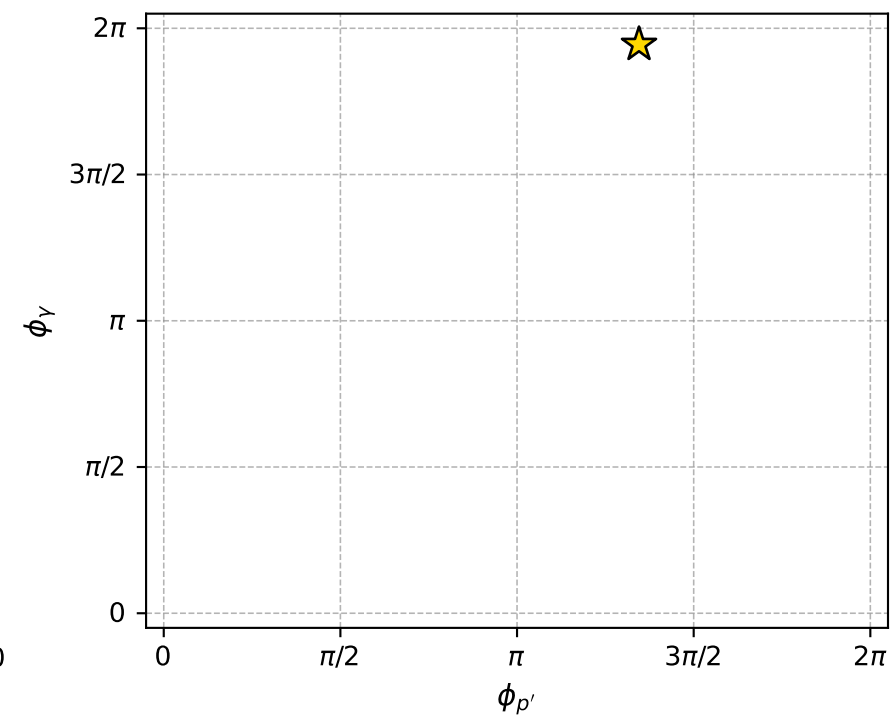
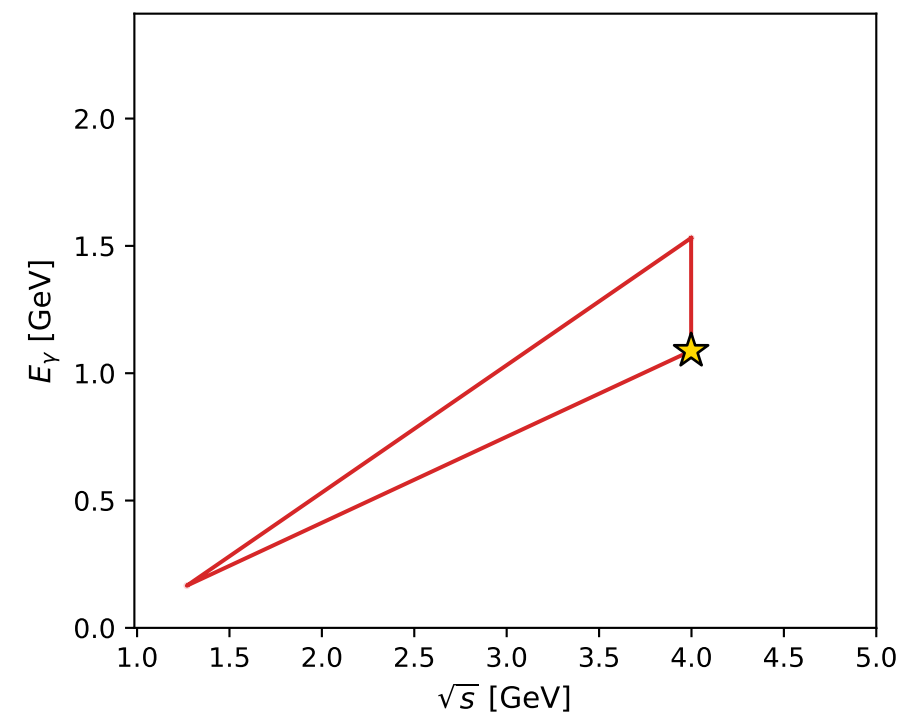
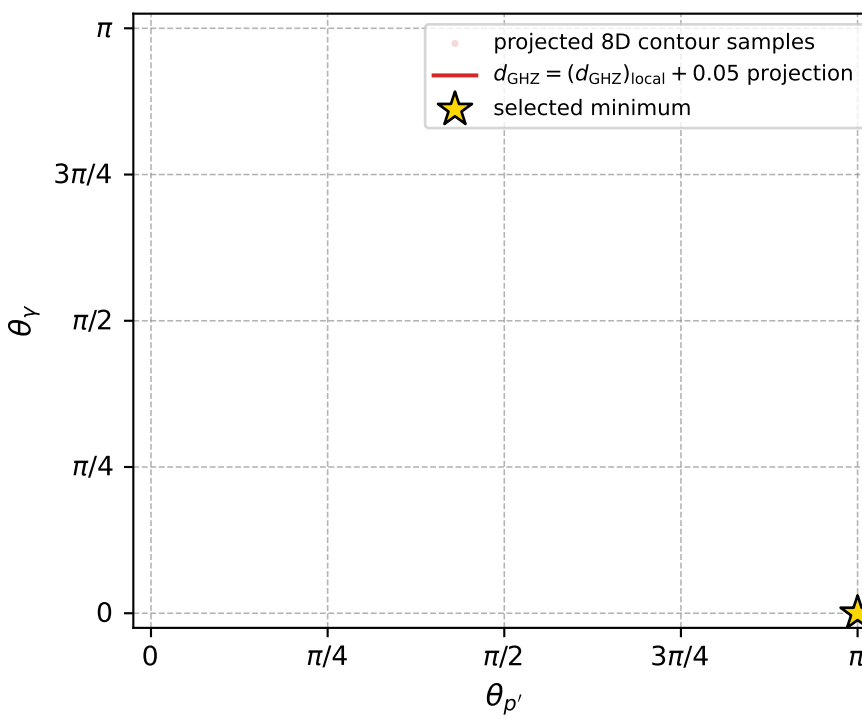
dghz_mixing_angles_polarization_02_local_minimum_160 d_{\rm GHZ}=1.12987e-07
 region: polarization_cluster_02_local_minimum_160
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=1.1362778$ rad
 incoming lepton: $0.420974|+\rangle +0.907073|-\rangle$
 proton mixing angle: $\alpha_p=2.7070795$ rad
 incoming proton: $-0.907075|+\rangle +0.420969|-\rangle$

$s=15.9848$, $\sqrt{s}=3.9981$, $|\vec{P}|=1.88901$, $|\vec{P}'|=1.88901$
 $\theta_{P'}=3.14159$, $\theta_Y=0$, $\phi_{P'}=4.2262$, $\phi_Y=6.10955$
 $E_Y=1.08804$, $\theta_{\ell'}=0$, $\phi_{\ell'}=1.08461$
 $Q^2=6.05225$, $x_B=0.410256$, $t=-14.2735$, $W^2=9.57998$, $y=0.976661$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 1.8890$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.8890$
 P $E= 2.1091$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.8890$
 ℓ' $E= 0.8010$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.8010$
 P' $E= 2.1091$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= -1.8890$
 q_Y $E= 1.0880$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 1.0880$



electron: polarization cluster P2, local minimum 160; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 1.1298674\text{e-}07$
 $d_{\text{GHZ}} \text{ contour} = 0.050000113$
 above global minimum = $9.0703281\text{e-}08$
 8D contour samples = 66

selected phase-space point:

$\text{sqrt_s} = 3.998095$
 $\text{theta_p_out} = 3.141593$
 $\text{theta_gamma_out} = 0$
 $\text{q0ut} = 1.088035$
 $\text{phi_p_out} = 4.226202$
 $\text{phi_gamma_out} = 6.10955$
 $\text{alpha_e} = 1.136278$
 $\text{alpha_p} = 2.70708$

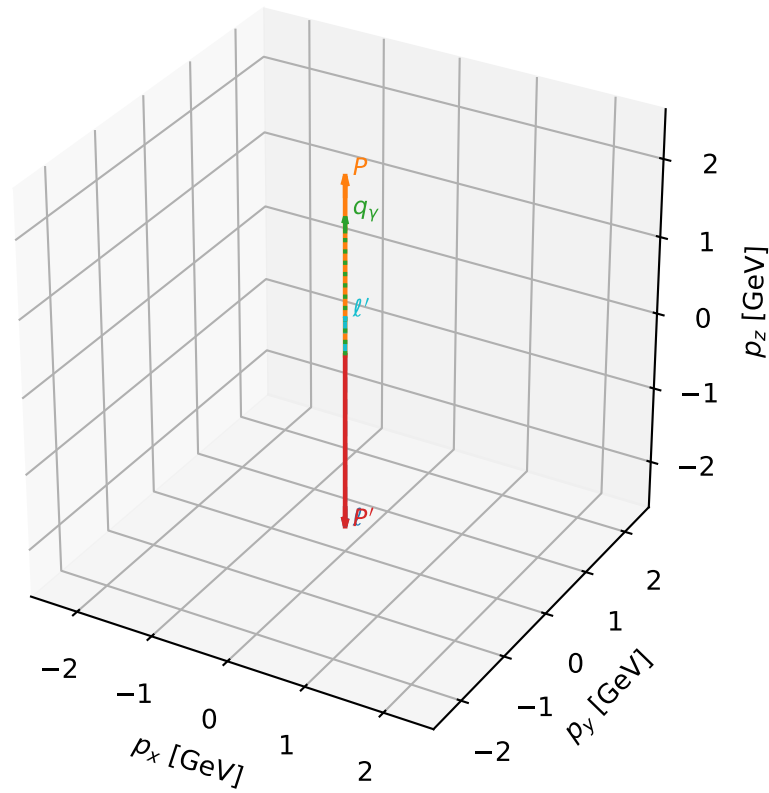
coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

dghz_mixing_angles_polarization_02_local_minimum_164 $d_{\{\text{rm GHZ}\}}=1.19686\text{e-}07$
region: polarization_cluster_02_local_minimum_164
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: $\alpha_e=0.61981444$ rad
incoming lepton: $0.813986|+\rangle +0.580884|-\rangle$
proton mixing angle: $\alpha_p=2.1905907$ rad
incoming proton: $-0.580868|+\rangle +0.813998|-\rangle$

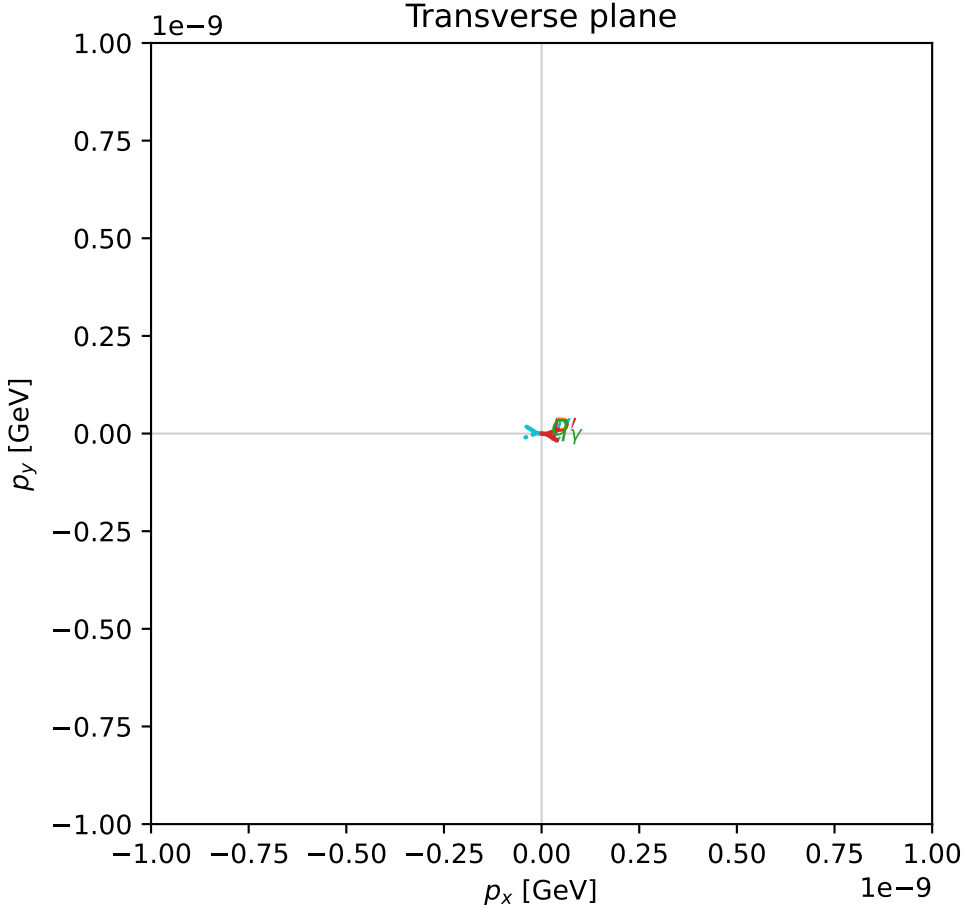
$s=22.5997$, $\sqrt{s}=4.75391$, $|\vec{P}|=2.28442$, $|\vec{P}'|=2.28442$
 $\theta_{P'}=3.14159$, $\theta_Y=0$, $\phi_{P'}=3.04512$, $\phi_Y=2.44102$
 $E_Y=1.7794$, $\theta_{\ell'}=0$, $\phi_{\ell'}=6.18671$
 $Q^2=4.61466$, $x_B=0.214308$, $t=-20.8742$, $W^2=17.7981$, $y=0.991393$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.2844$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.2844$
 P $E= 2.4695$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.2844$
 ℓ' $E= 0.5050$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 0.5050$
 P' $E= 2.4695$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= -2.2844$
 q_Y $E= 1.7794$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 1.7794$

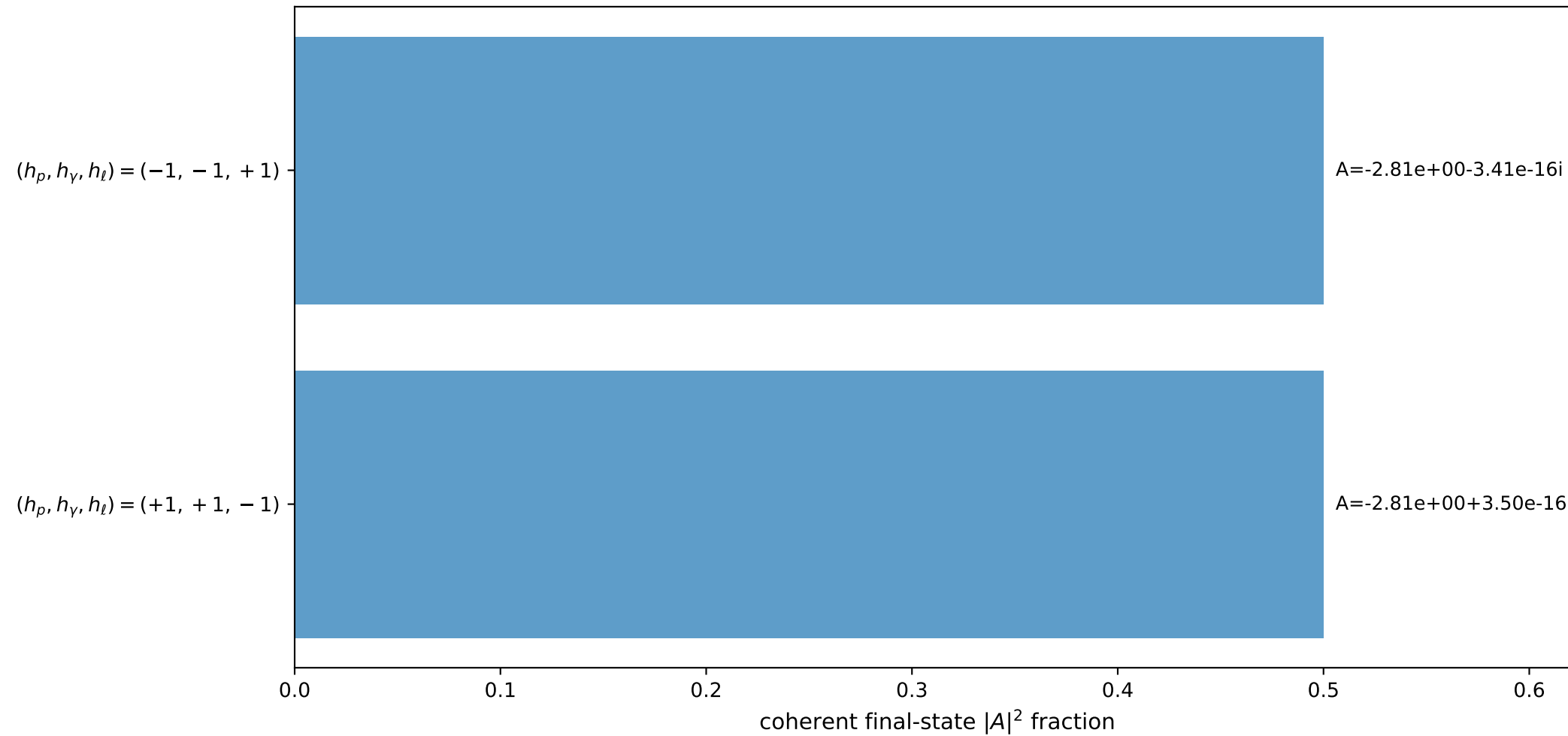
3D momenta



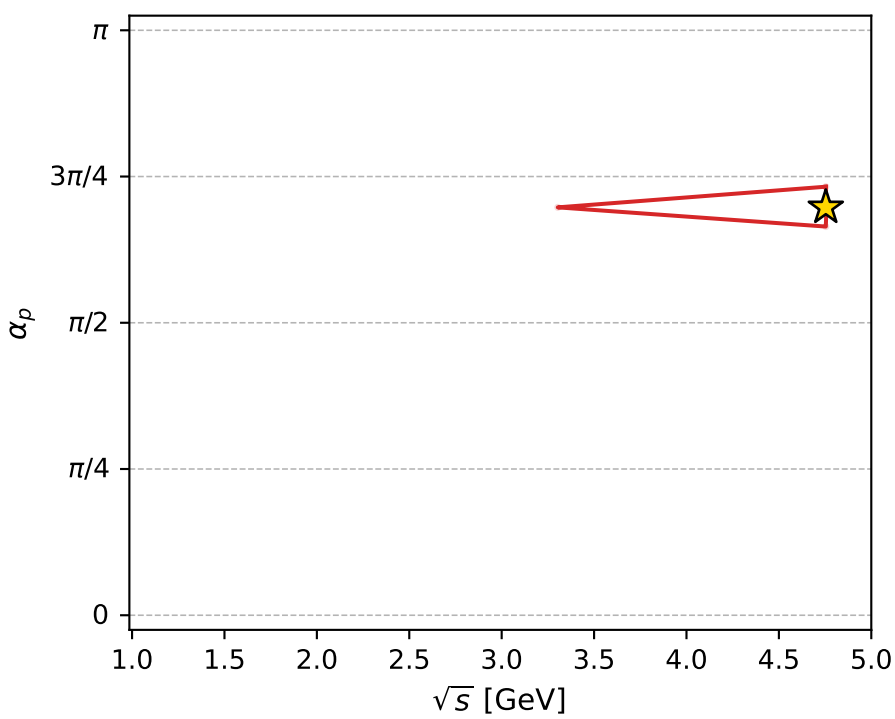
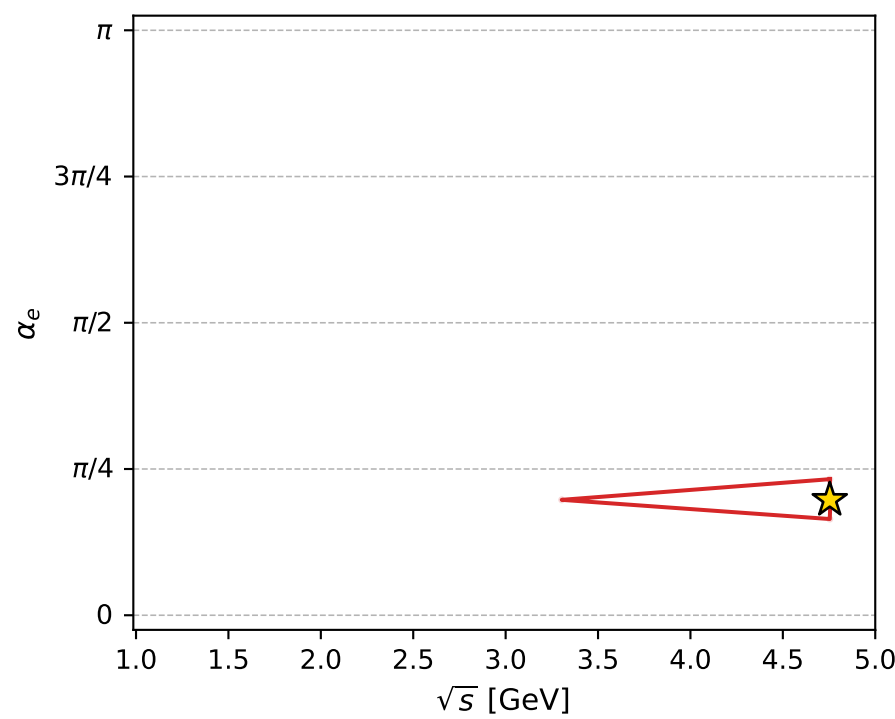
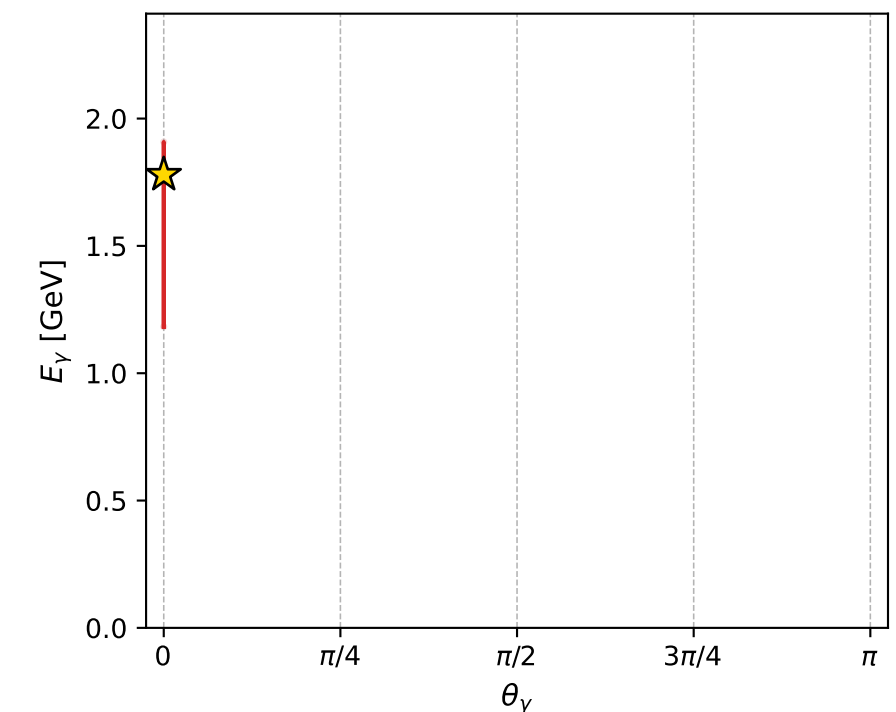
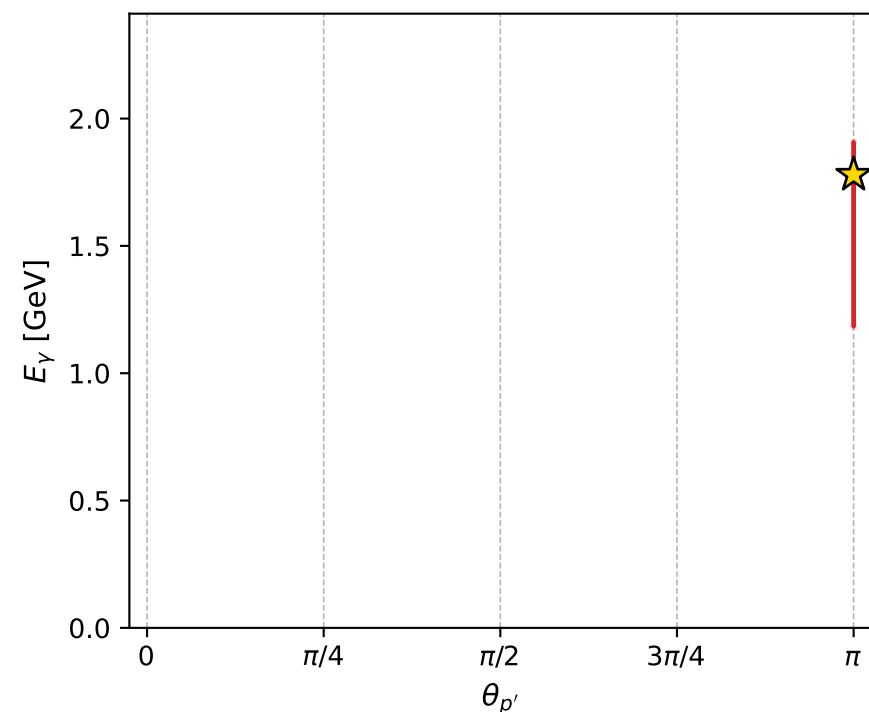
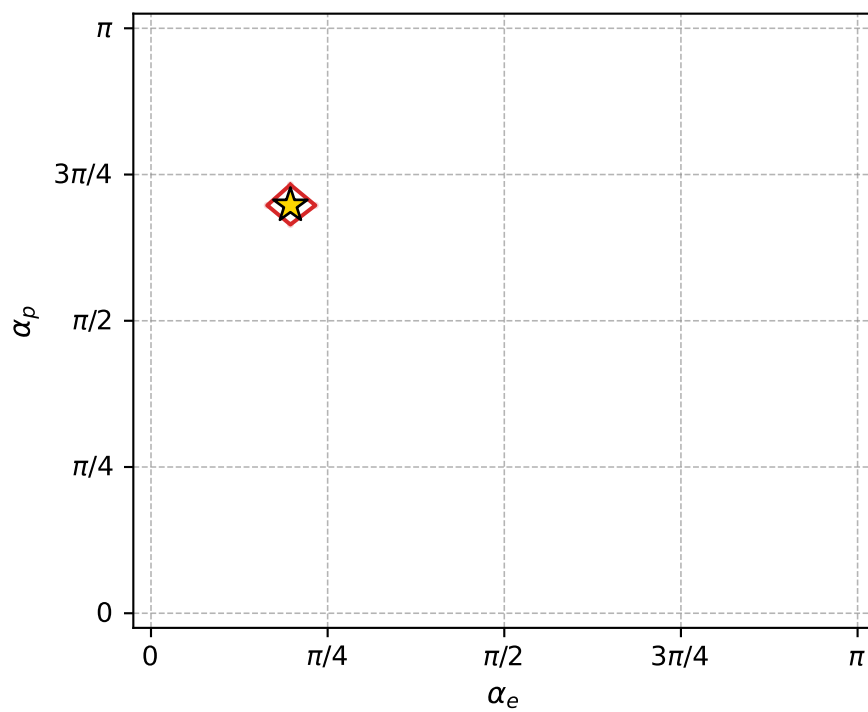
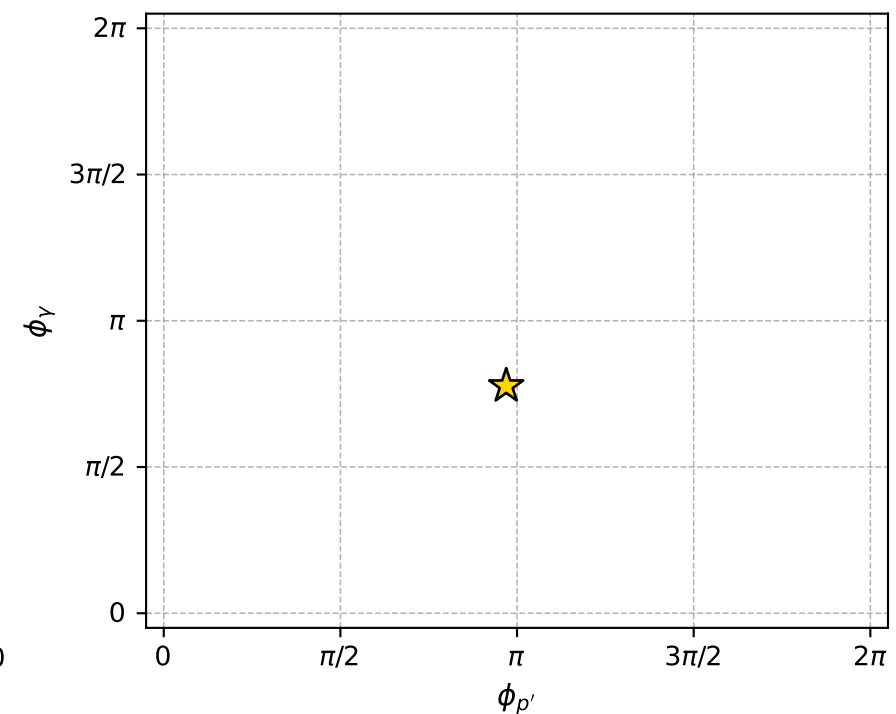
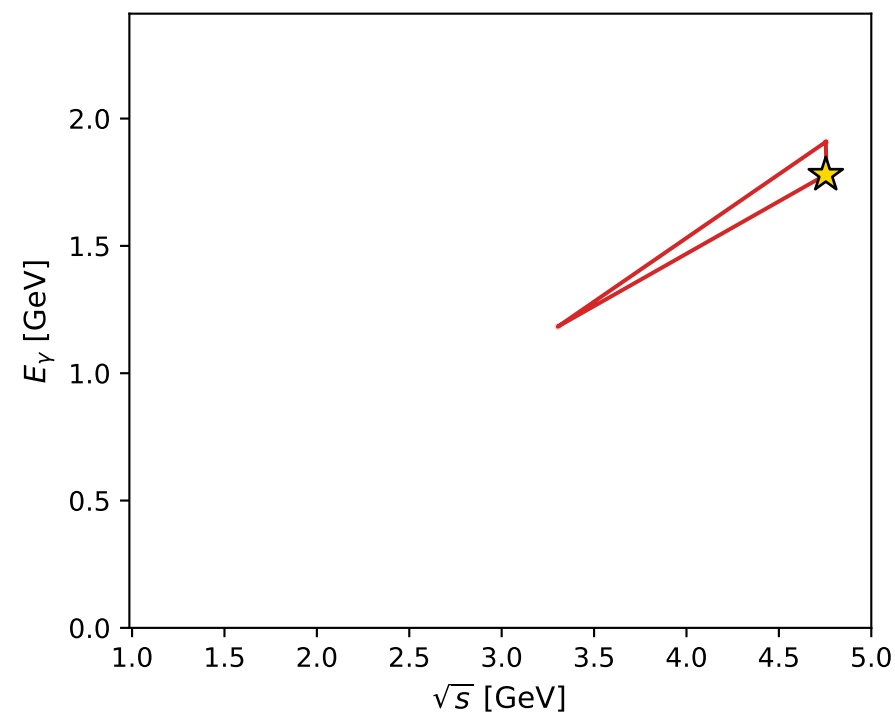
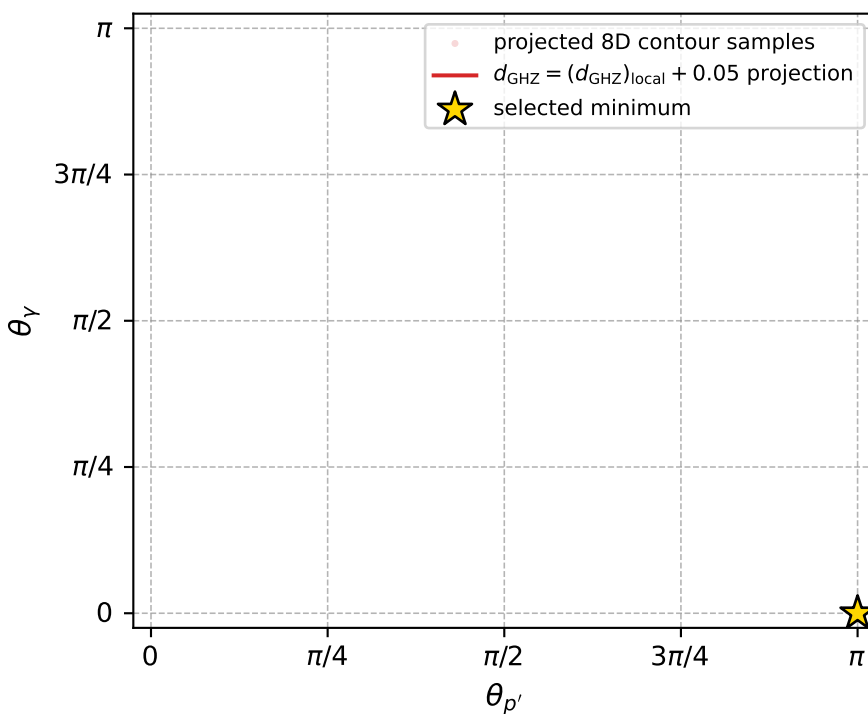
Transverse plane



Leading final-state amplitudes (N=2, retained=100.0%)



electron: polarization cluster P2, local minimum 164; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 1.1968572\text{e-}07$
 $d_{\text{GHZ}} \text{ contour} = 0.05000012$
 above global minimum = $9.7402259\text{e-}08$
 8D contour samples = 70

selected phase-space point:

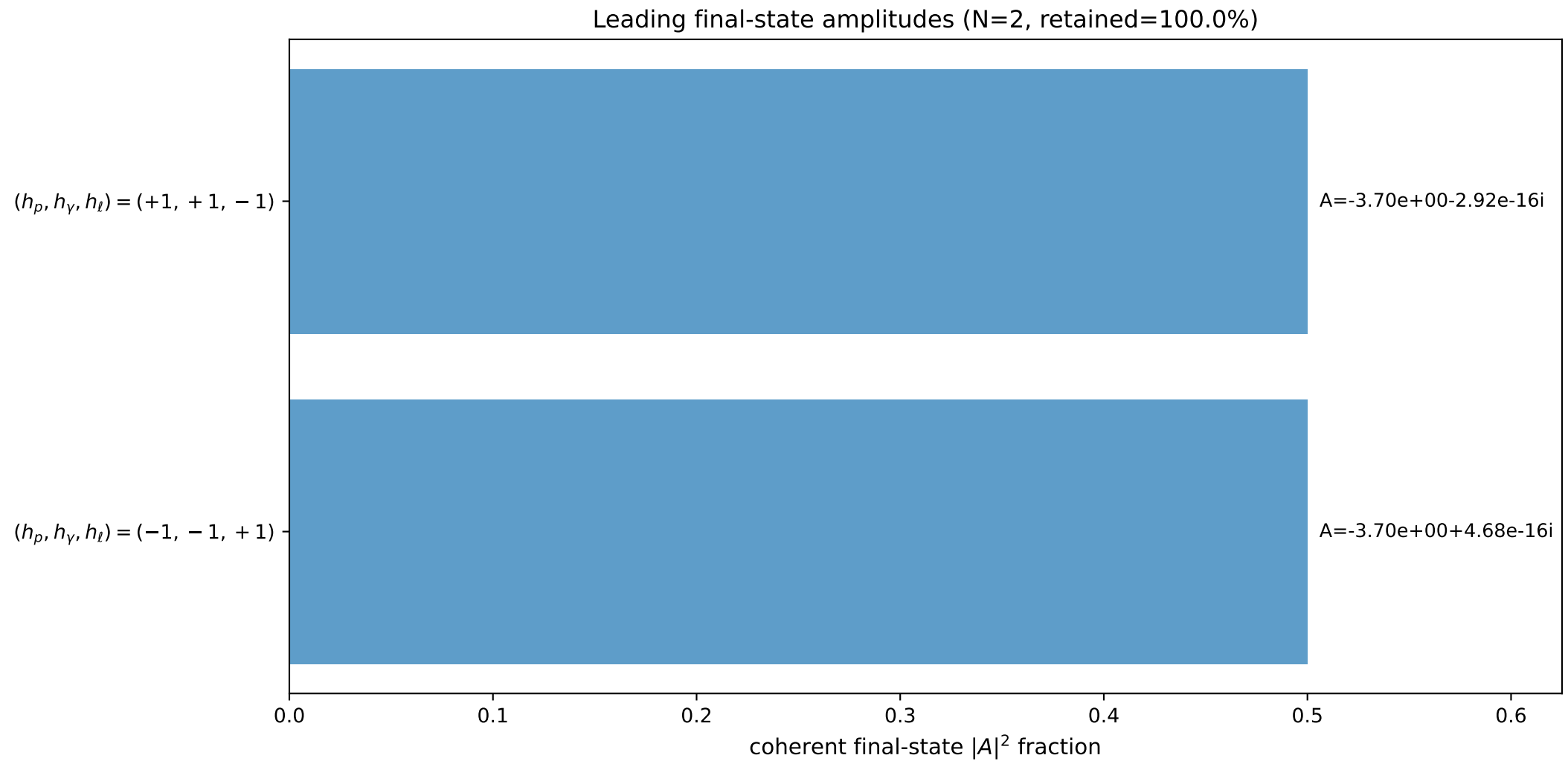
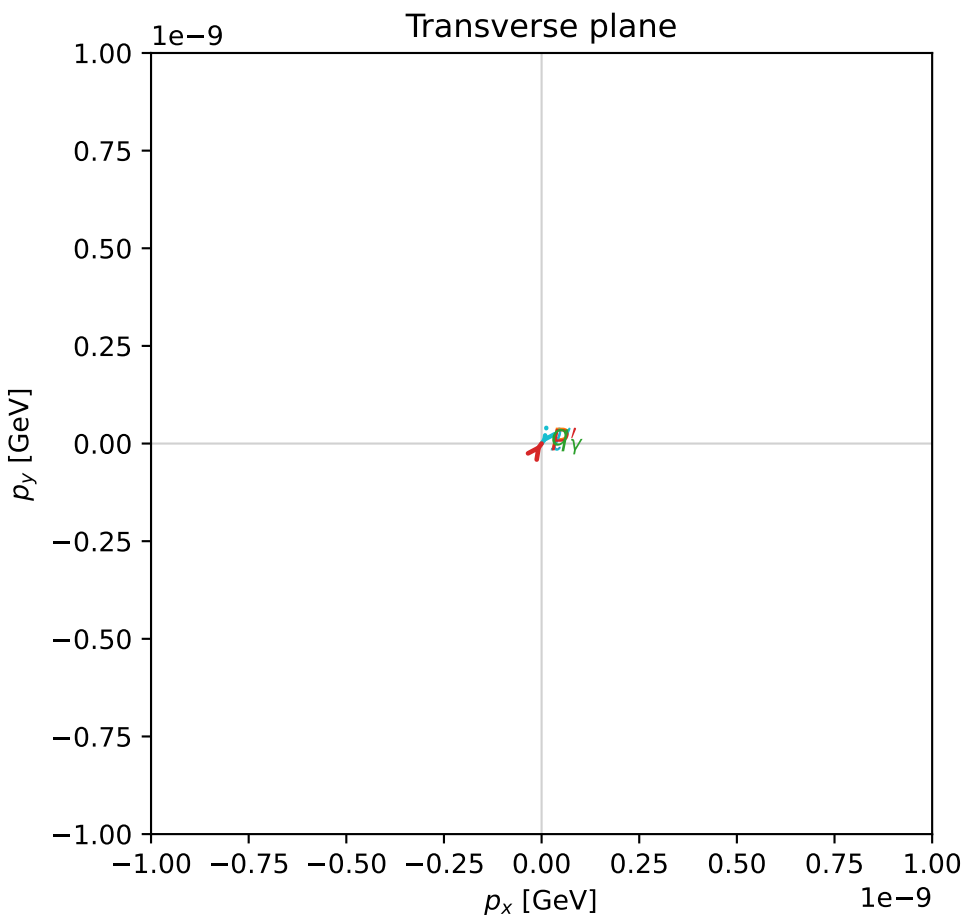
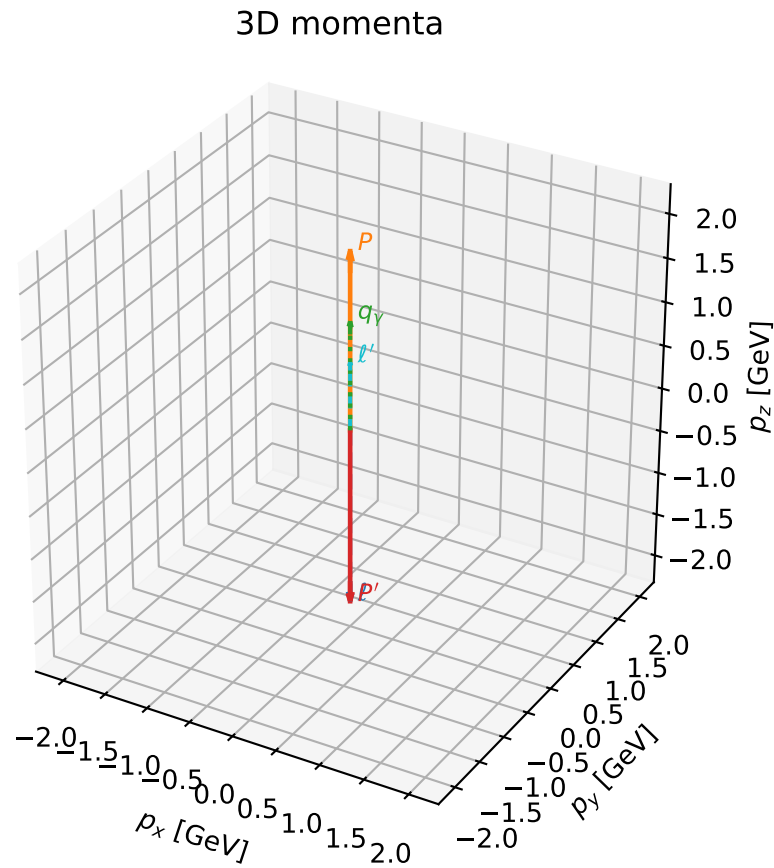
$\text{sqrt_s} = 4.753909$
 $\text{theta_p_out} = 3.141593$
 $\text{theta_gamma_out} = 0$
 $\text{q0ut} = 1.7794$
 $\text{phi_p_out} = 3.045116$
 $\text{phi_gamma_out} = 2.441022$
 $\text{alpha_e} = 0.6198144$
 $\text{alpha_p} = 2.190591$

coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

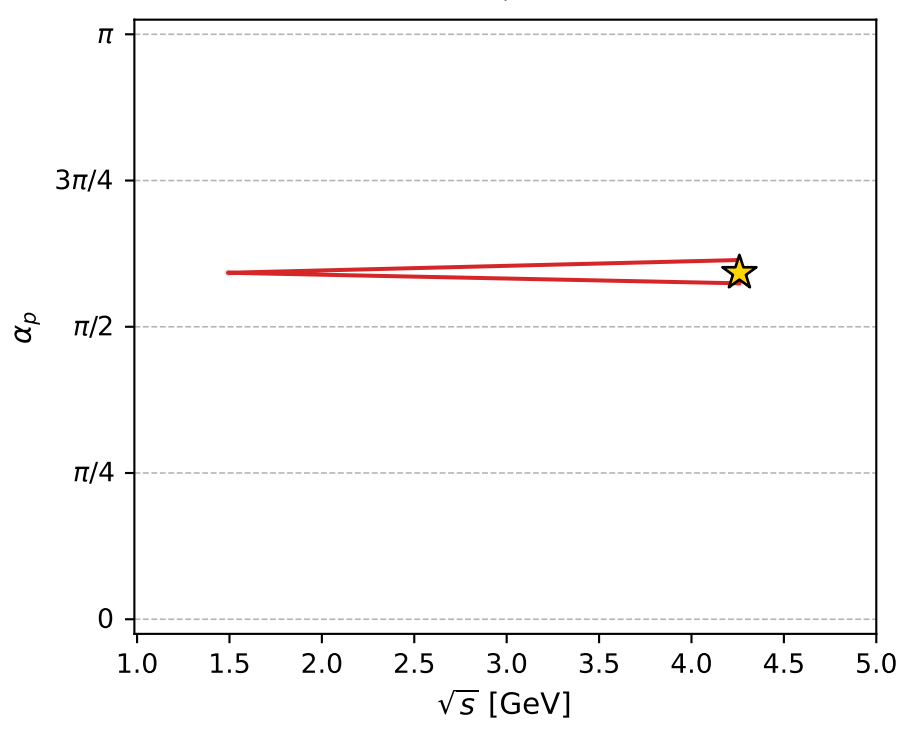
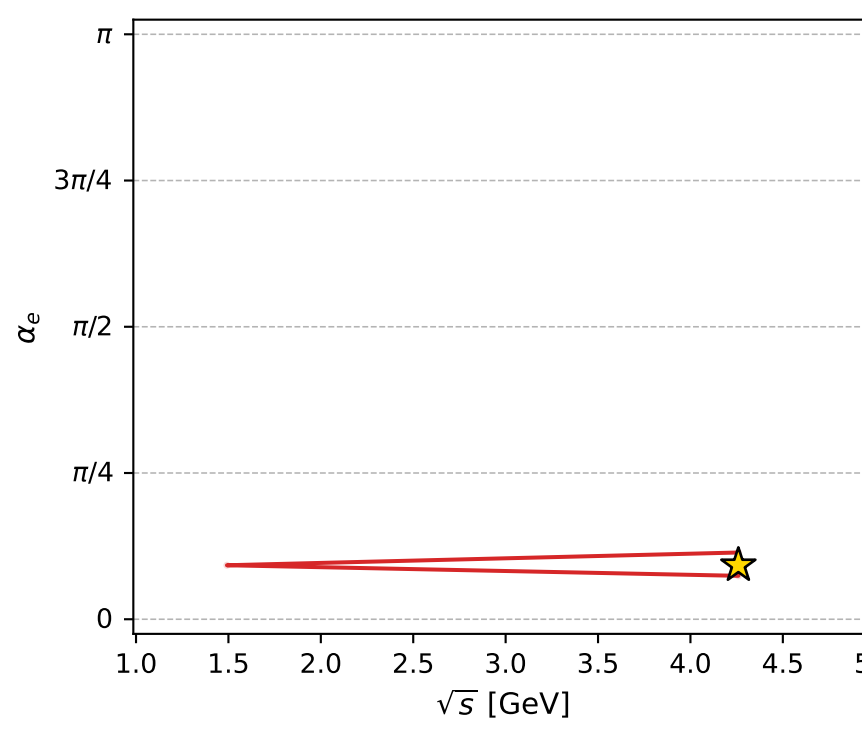
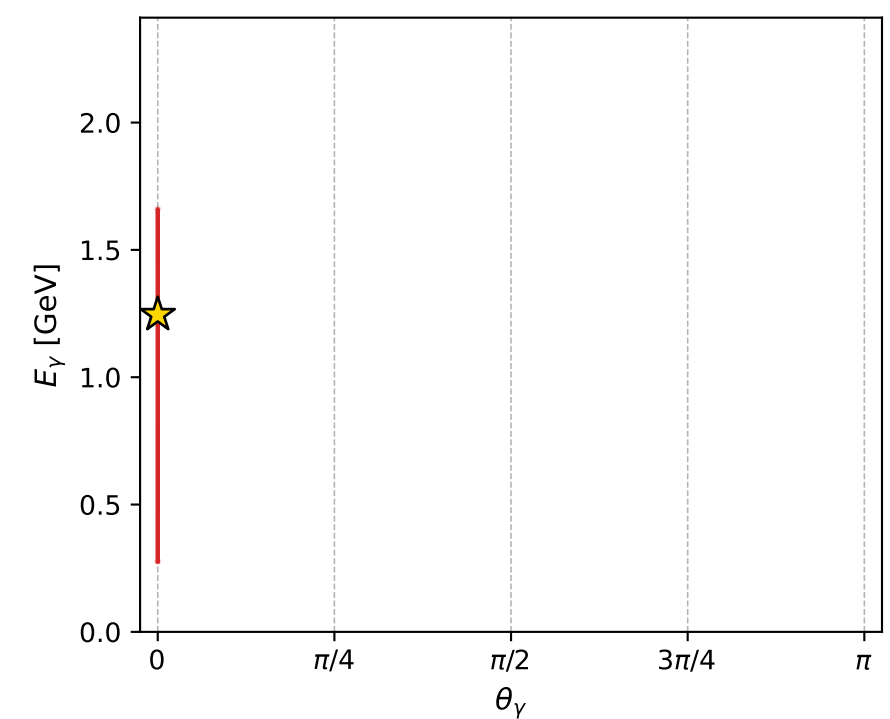
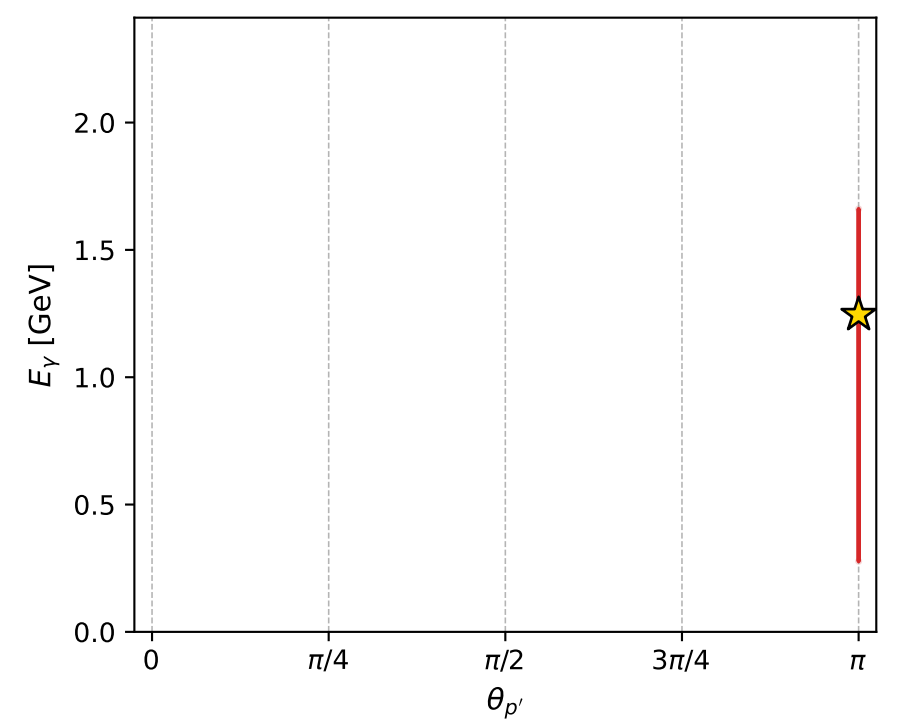
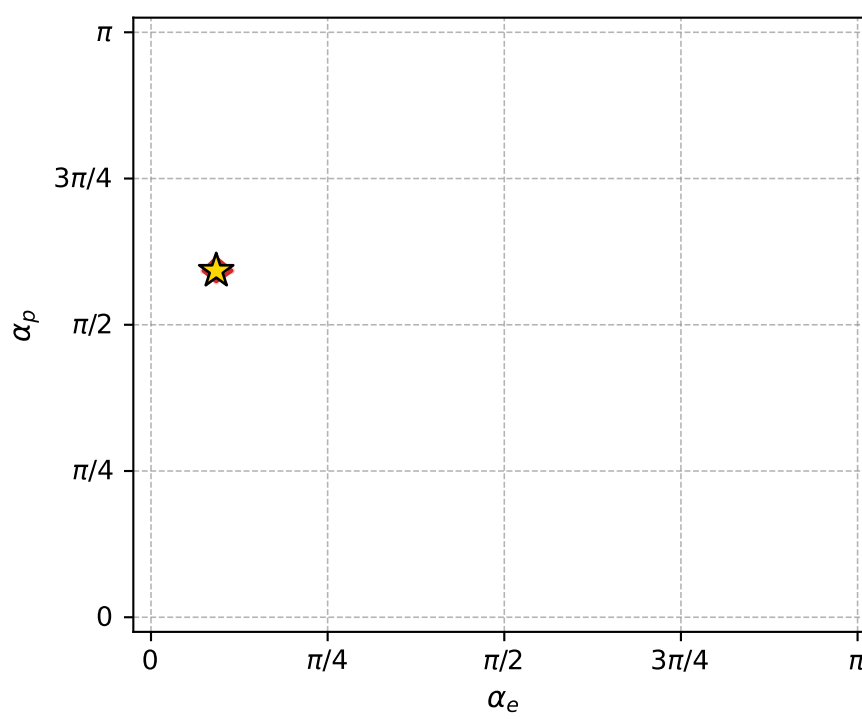
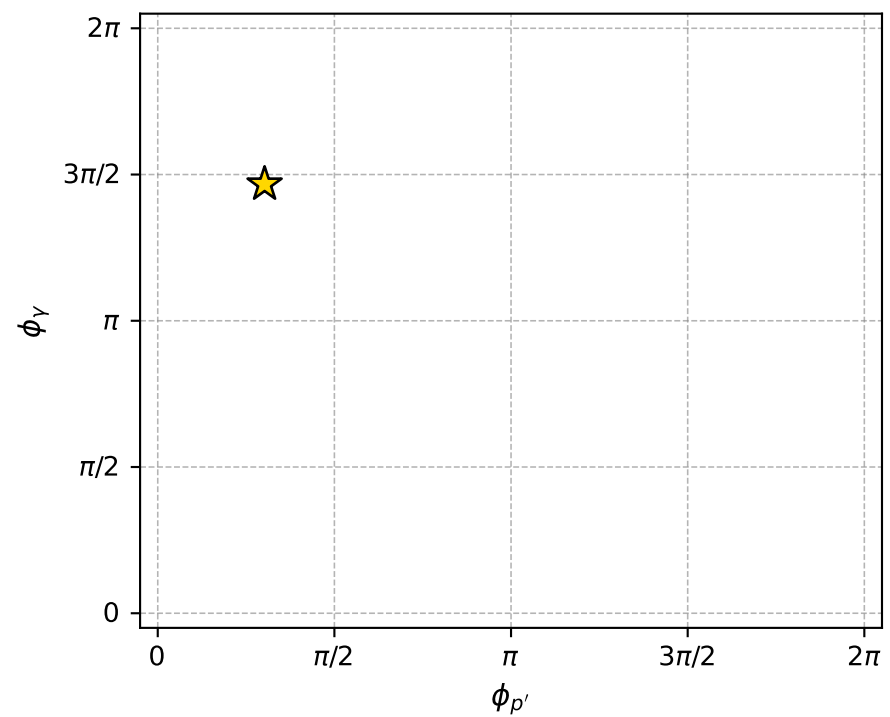
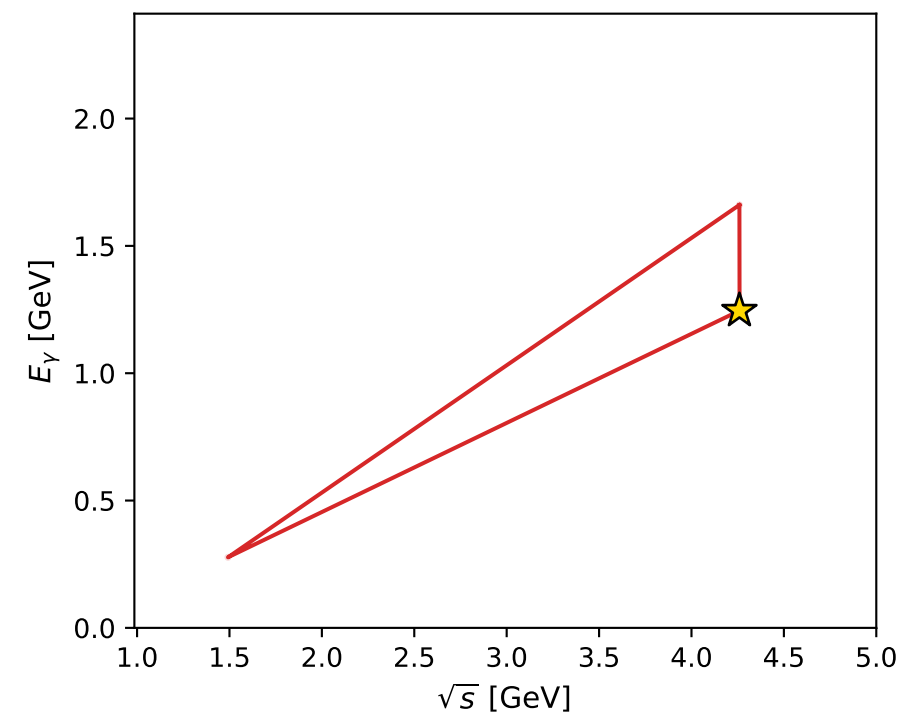
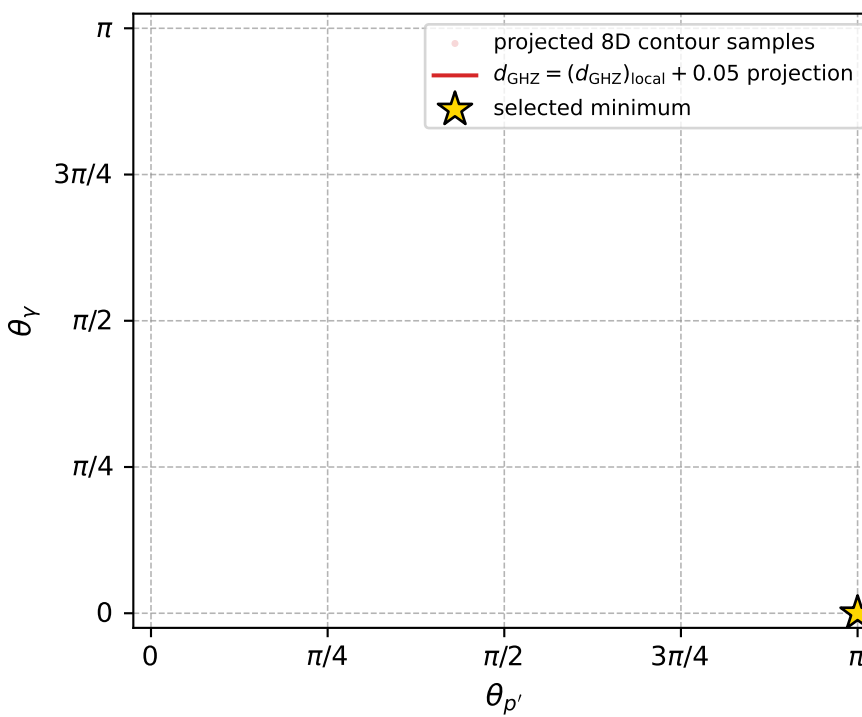
dghz_mixing_angles_polarization_02_local_minimum_167 $d_{\{\text{rm GHZ}\}}=1.5068\text{e-}07$
region: polarization_cluster_02_local_minimum_167
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: $\alpha_e=0.28996137$ rad
incoming lepton: $0.958255|+\rangle +0.285915|-\rangle$
proton mixing angle: $\alpha_p=1.8607761$ rad
incoming proton: $-0.285933|+\rangle +0.95825|-\rangle$

$s=18.1422$, $\sqrt{s}=4.25936$, $|\vec{P}|=2.0264$, $|\vec{P}'|=2.0264$
 $\theta_{P'}=3.14159$, $\theta_Y=0$, $\phi_{P'}=0.950105$, $\phi_Y=4.60864$
 $E_Y=1.24571$, $\theta_{\ell'}=0$, $\phi_{\ell'}=4.0917$
 $Q^2=6.32793$, $x_B=0.373554$, $t=-16.4252$, $W^2=11.4917$, $y=0.981316$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.0264$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.0264$
 P $E= 2.2330$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.0264$
 ℓ' $E= 0.7807$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 0.7807$
 P' $E= 2.2330$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.0264$
 q_Y $E= 1.2457$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 1.2457$



electron: polarization cluster P2, local minimum 167; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 1.5067991\text{e-}07$
 $d_{\text{GHZ}} \text{ contour} = 0.050000151$
 above global minimum = $1.2839645\text{e-}07$
 8D contour samples = 65

selected phase-space point:

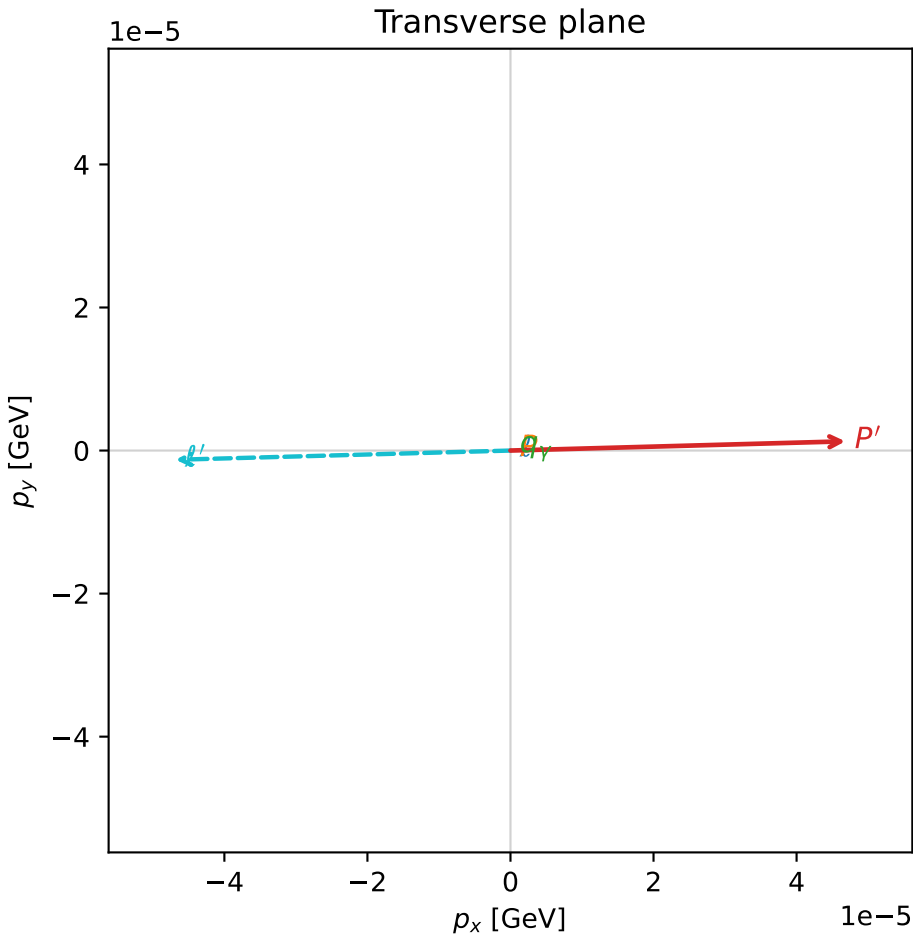
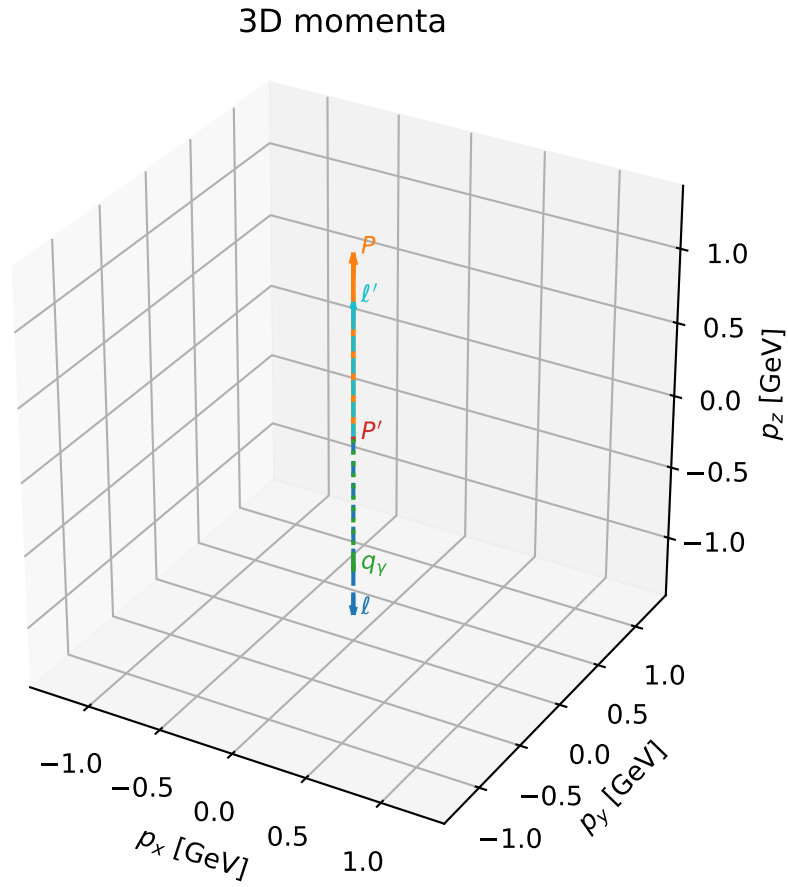
$\text{sqrt_s} = 4.25935$
 $\text{theta_p_out} = 3.141593$
 $\text{theta_gamma_out} = 0$
 $\text{q0ut} = 1.245712$
 $\text{phi_p_out} = 0.9501052$
 $\text{phi_gamma_out} = 4.60864$
 $\text{alpha_e} = 0.2899614$
 $\text{alpha_p} = 1.860776$

coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

dghz_mixing_angles_polarization_02_local_minimum_168 $d_{\{\text{rm GHZ}\}}=1.70137\text{e-}07$
region: polarization_cluster_02_local_minimum_168
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: $\alpha_e=1.2036147$ rad
incoming lepton: $0.358986|+\rangle +0.933343|-\rangle$
proton mixing angle: $\alpha_p=1.9380061$ rad
incoming proton: $-0.359013|+\rangle +0.933333|-\rangle$

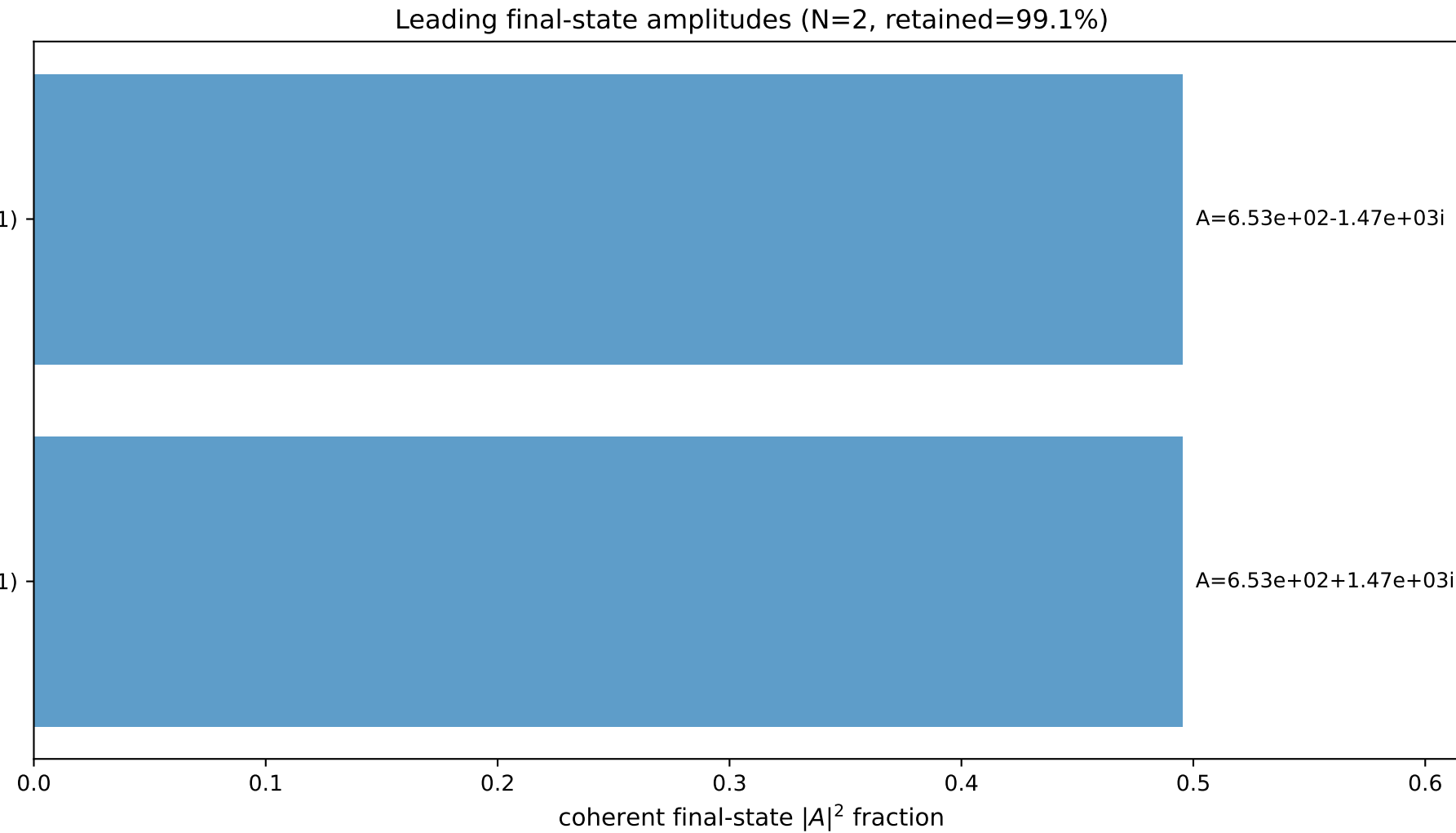
$s=7.72597$, $\sqrt{s}=2.77956$, $|\vec{P}|=1.23151$, $|\vec{P}'|=0.000244016$
 $\theta_{P'}=2.94845$, $\theta_Y=3.14159$, $\phi_{P'}=0.027548$, $\phi_Y=5.15721$
 $E_Y=0.920662$, $\theta_{\ell'}=5.08603\text{e-}05$, $\phi_{\ell'}=3.16914$
 $Q^2=4.5364$, $x_B=0.724304$, $t=-1.14505$, $W^2=2.60656$, $y=0.914842$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 1.2315$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.2315$
 P $E= 1.5481$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.2315$
 ℓ' $E= 0.9209$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 0.9209$
 P' $E= 0.9380$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.0002$
 q_Y $E= 0.9207$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= -0.9207$

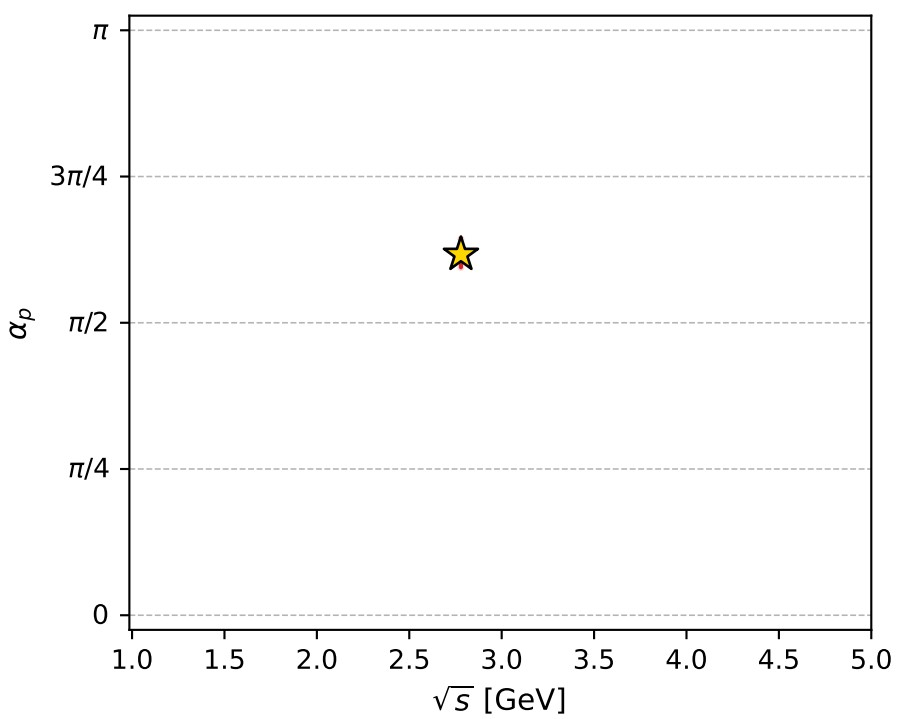
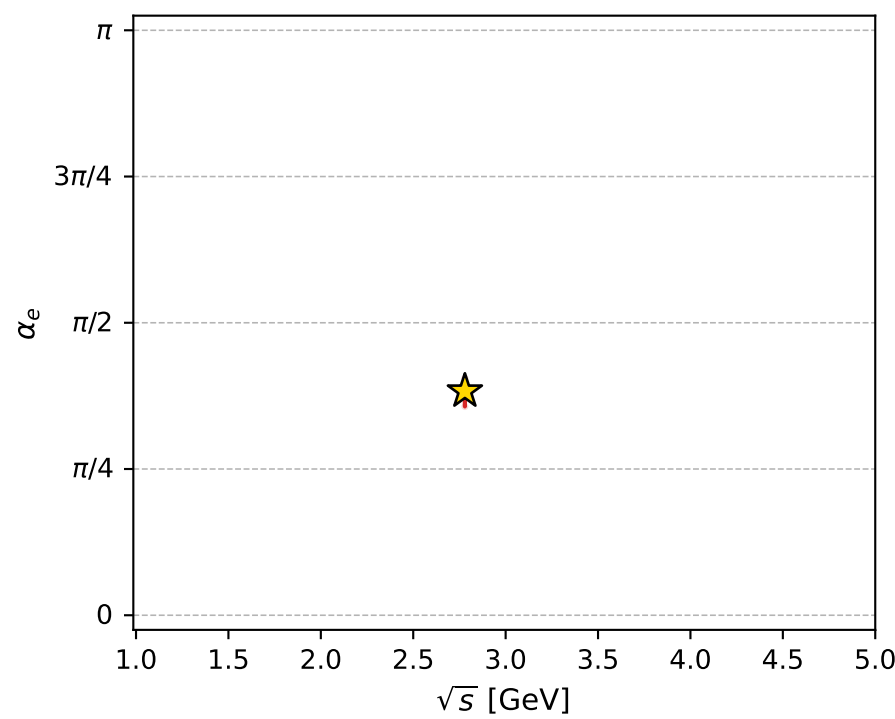
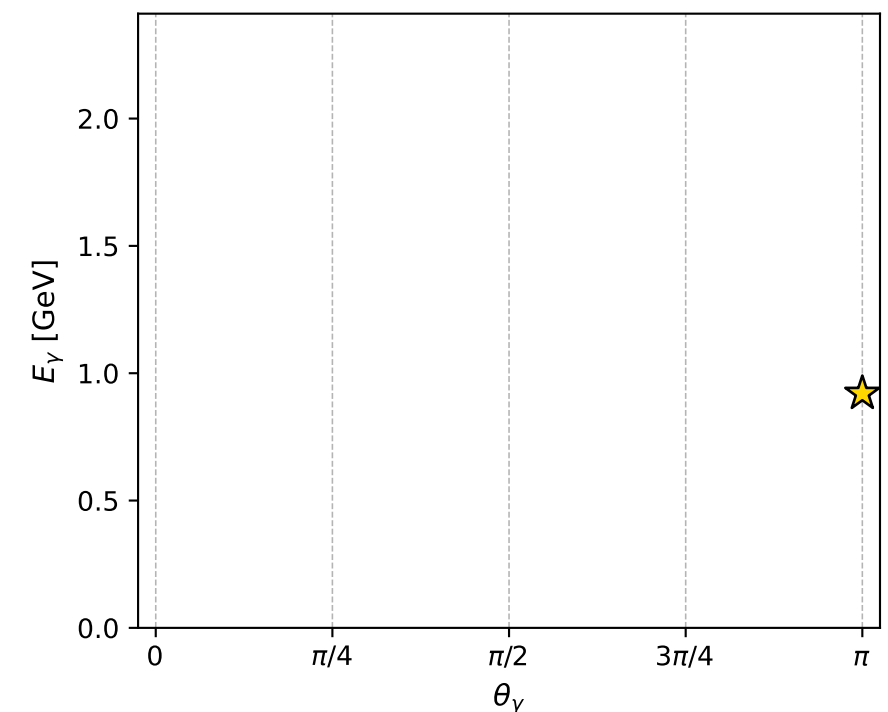
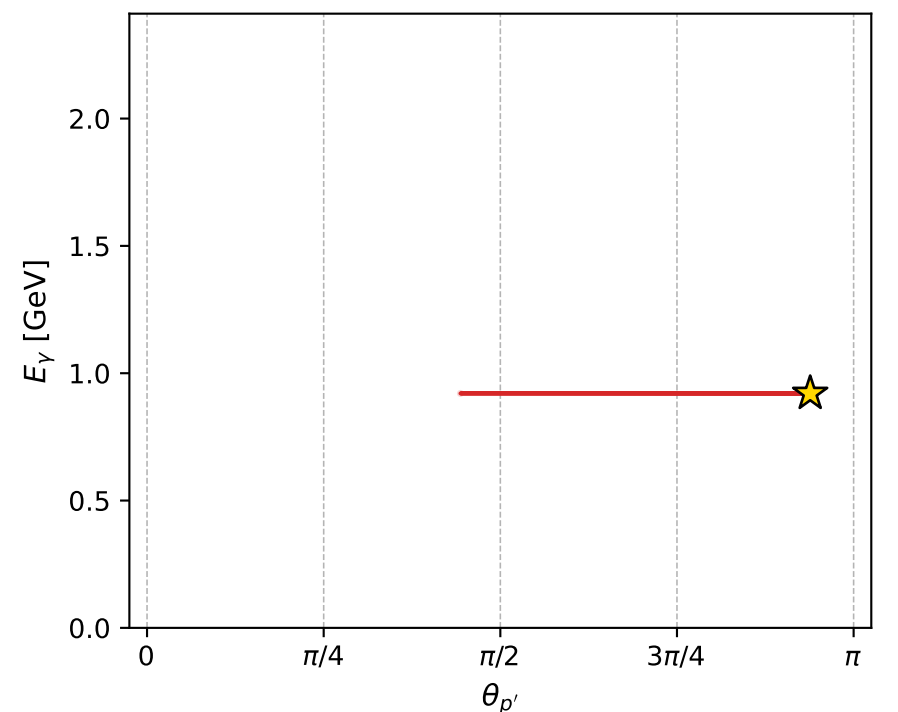
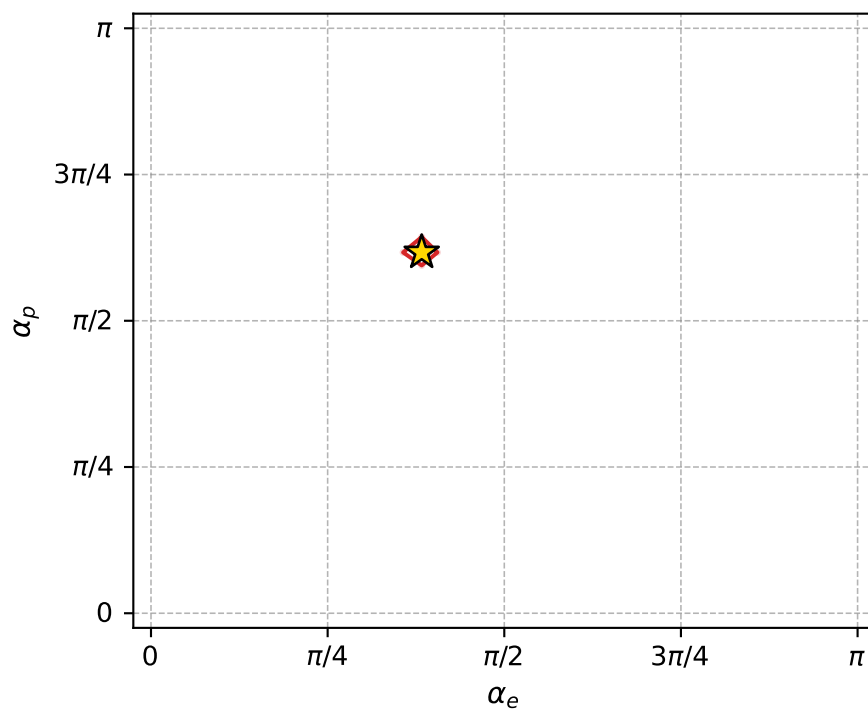
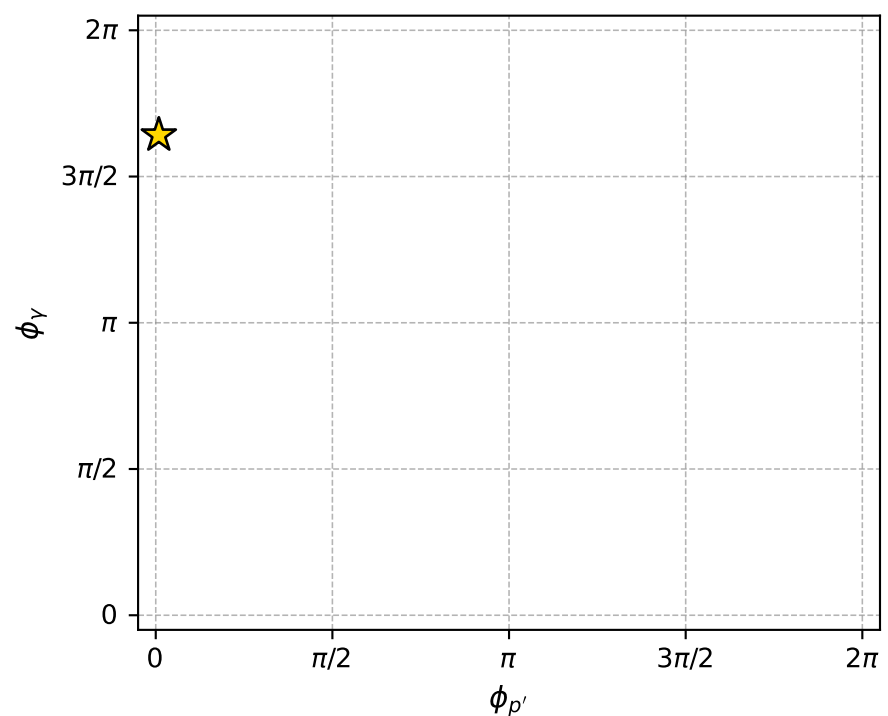
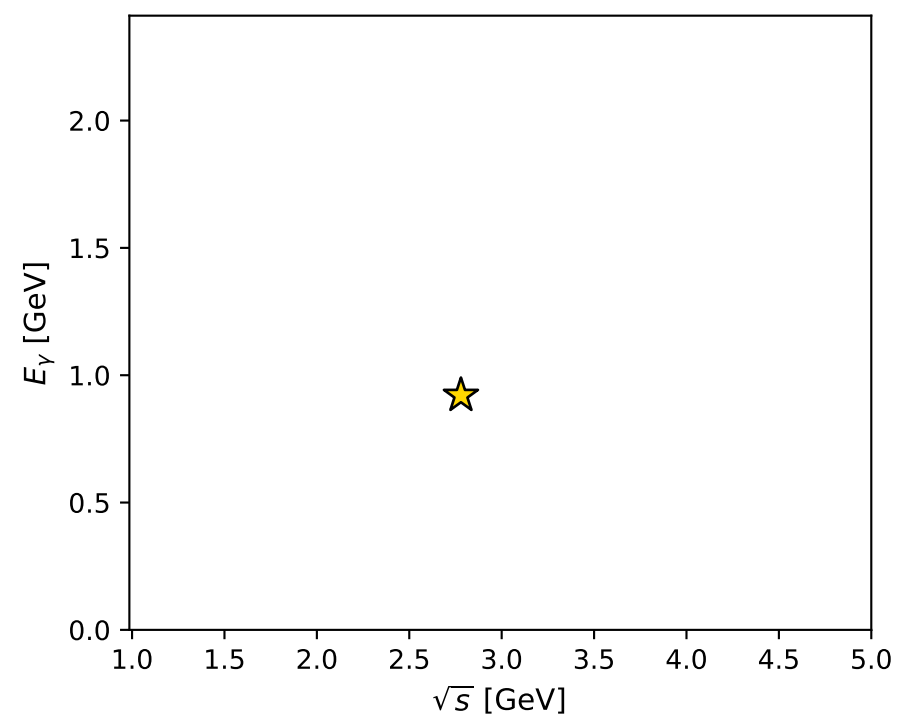
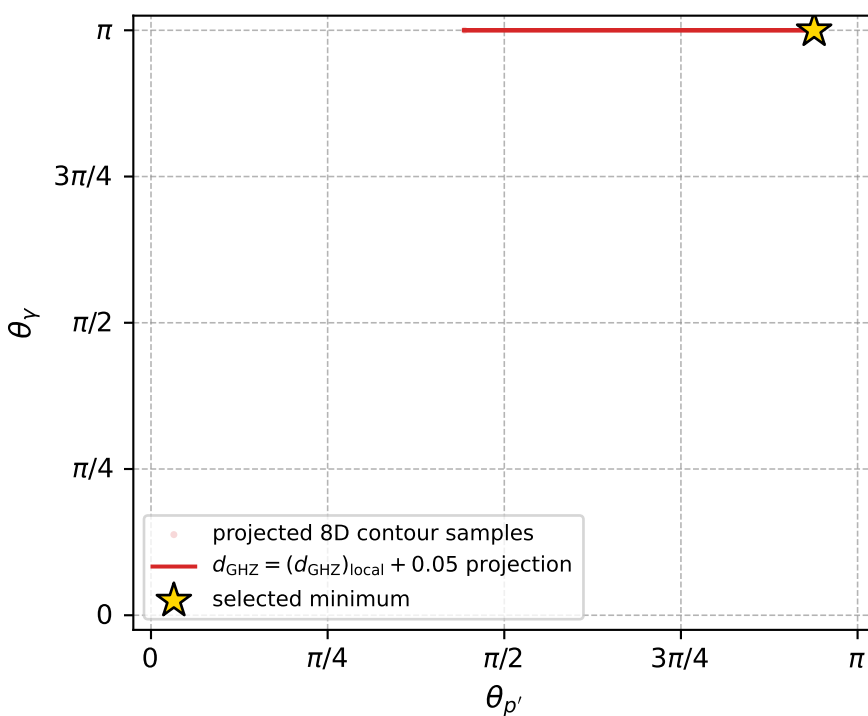


$(h_p, h_Y, h_{\ell}) = (+1, -1, +1)$

$(h_p, h_Y, h_{\ell}) = (-1, +1, -1)$



electron: polarization cluster P2, local minimum 168; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 1.7013674\text{e-}07$
 $d_{\text{GHZ}} \text{ contour} = 0.05000017$
 above global minimum = $1.4785328\text{e-}07$
 8D contour samples = 129

selected phase-space point:

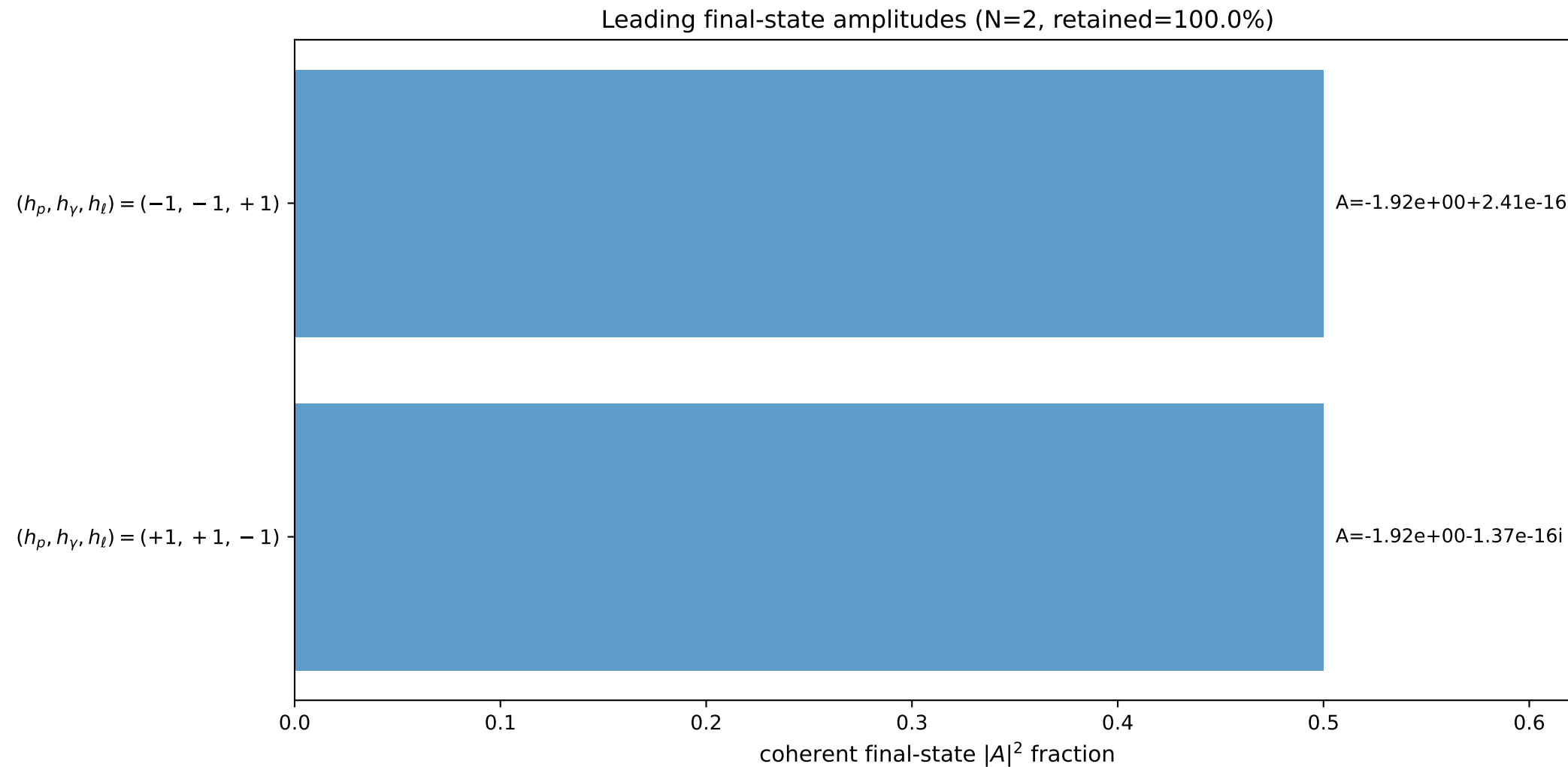
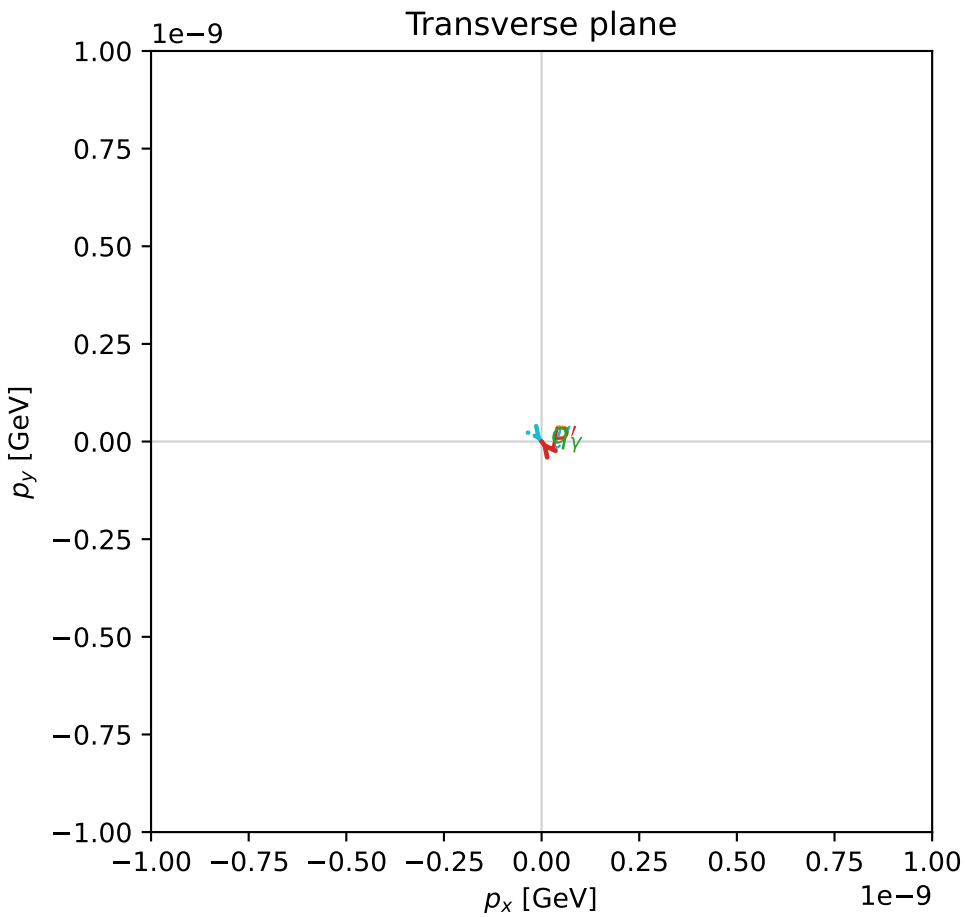
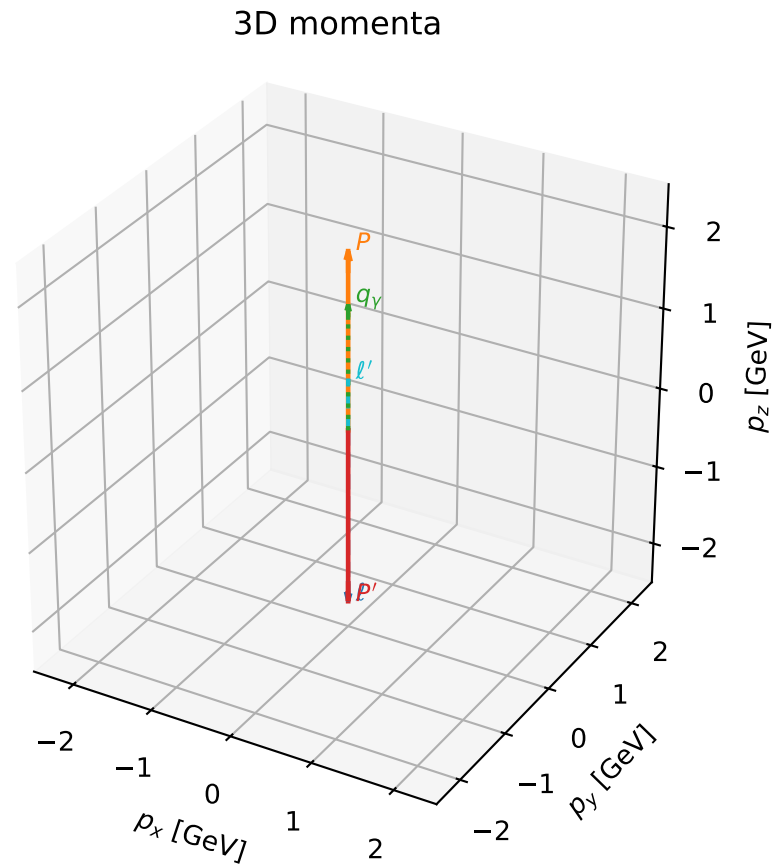
$\text{sqrt_s} = 2.779564$
 $\text{theta_p_out} = 2.948451$
 $\text{theta_gamma_out} = 3.141593$
 $\text{q0ut} = 0.920662$
 $\text{phi_p_out} = 0.02754796$
 $\text{phi_gamma_out} = 5.157206$
 $\text{alpha_e} = 1.203615$
 $\text{alpha_p} = 1.938006$

coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

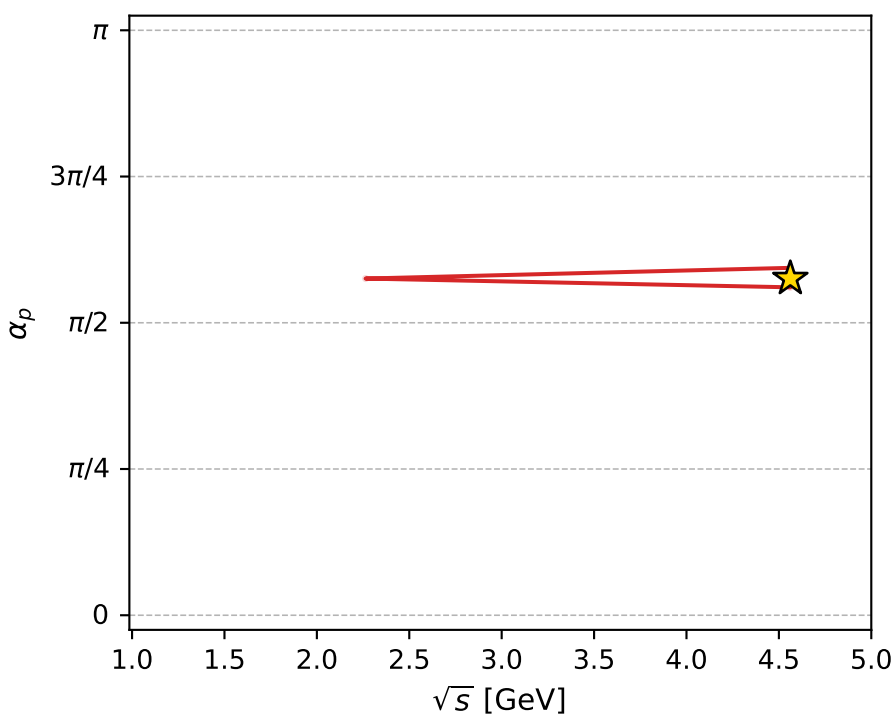
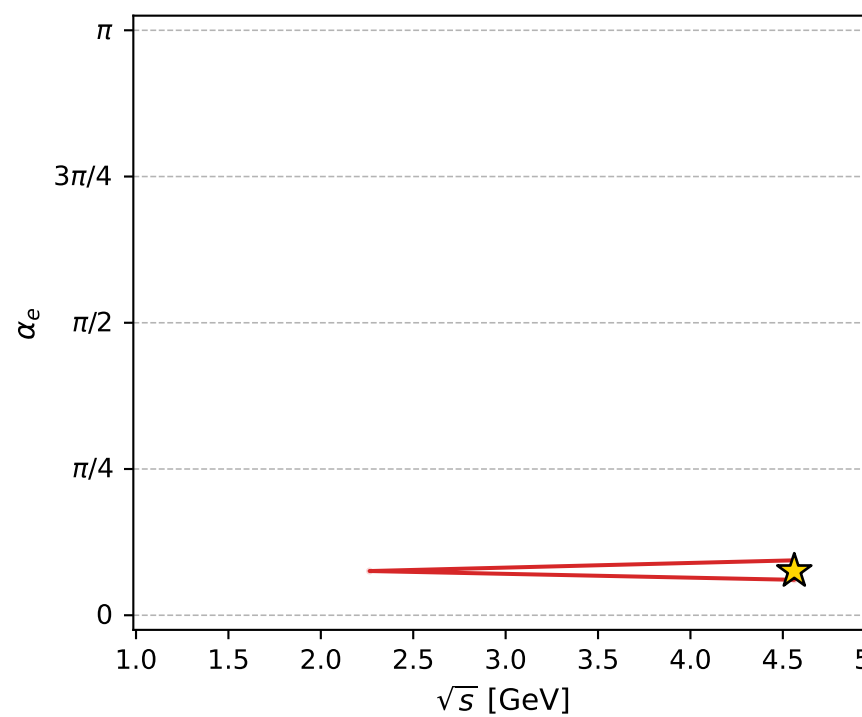
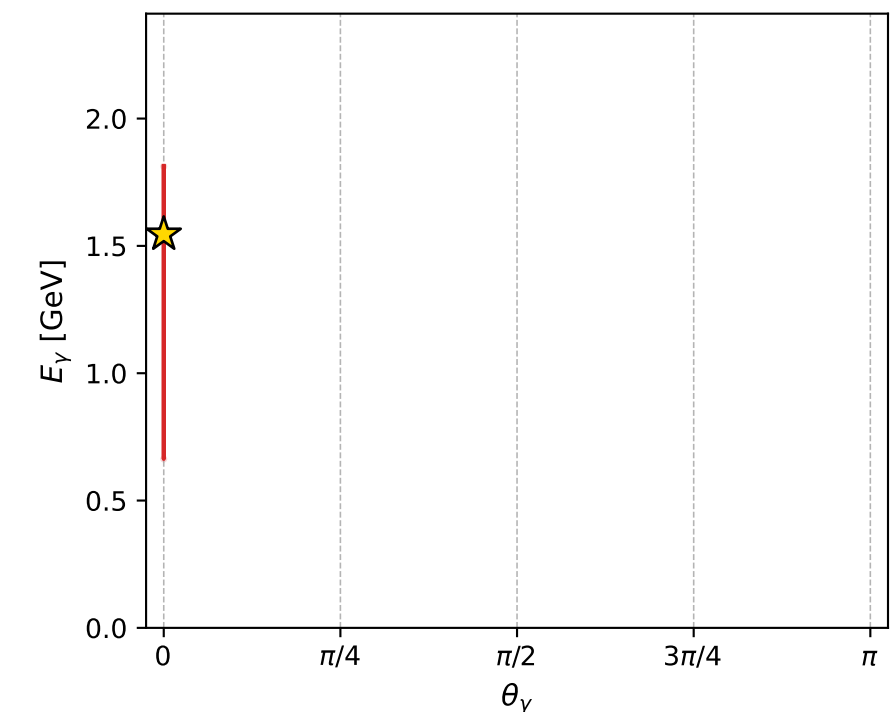
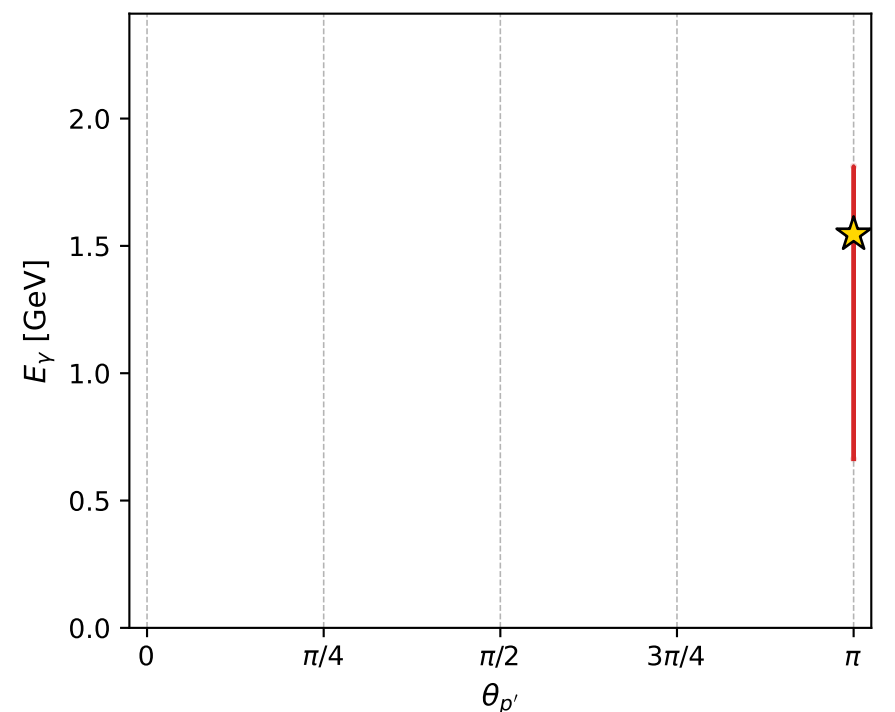
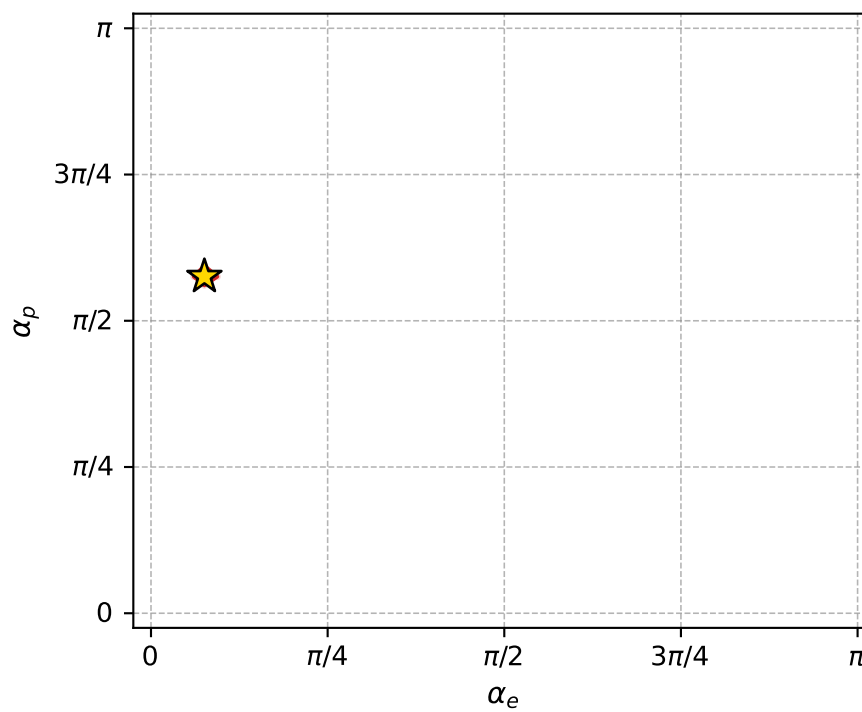
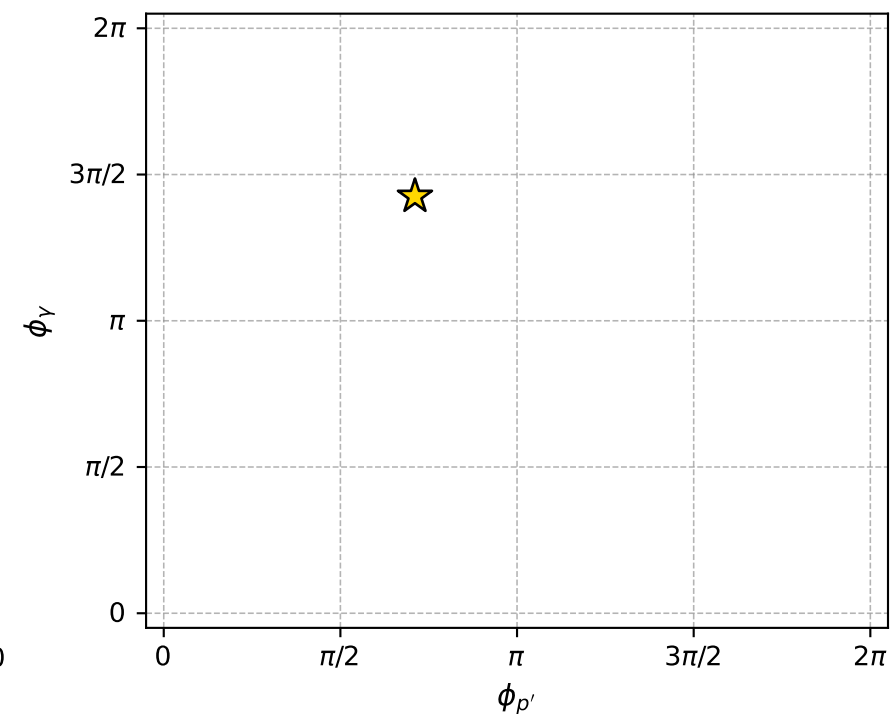
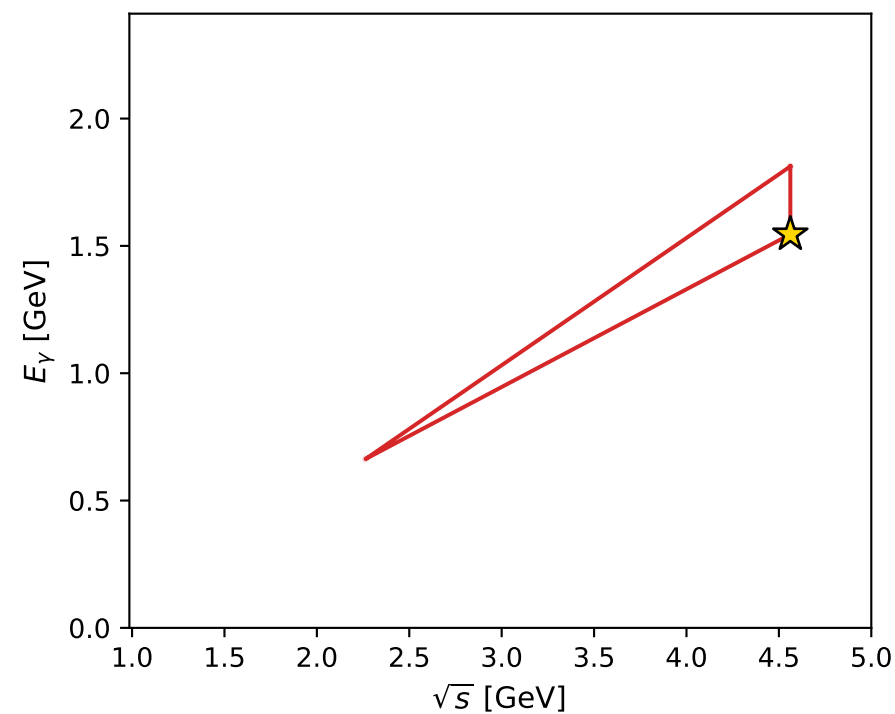
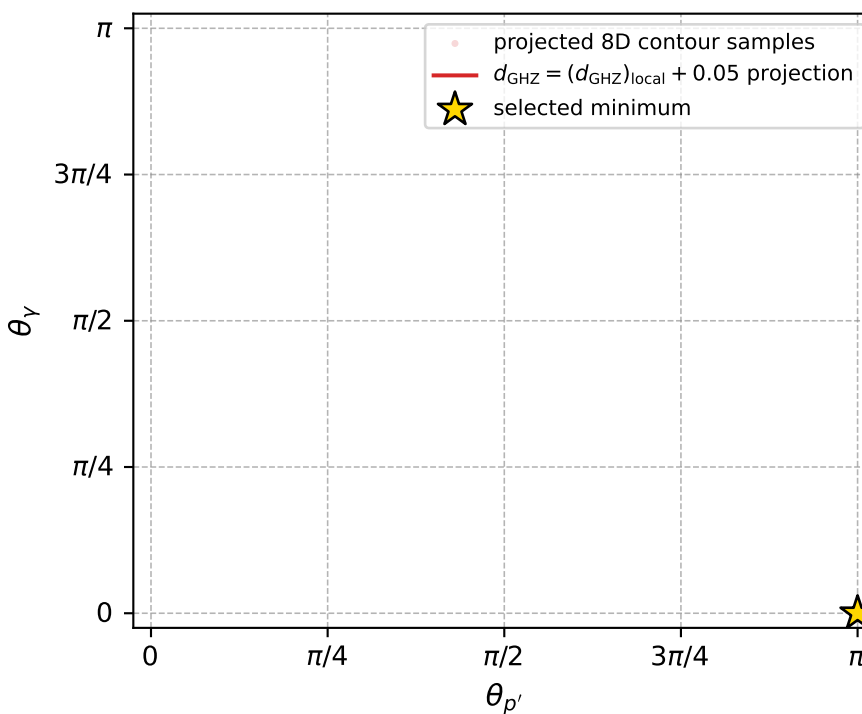
dghz_mixing_angles_polarization_02_local_minimum_171 $d_{\{\text{rm GHZ}\}}=2.04591\text{e-}07$
 region: polarization_cluster_02_local_minimum_171
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=0.23742392$ rad
 incoming lepton: $0.971947|+\rangle +0.2352|-\rangle$
 proton mixing angle: $\alpha_p=1.8082114$ rad
 incoming proton: $-0.235191|+\rangle +0.971949|-\rangle$

$s=20.8112$, $\sqrt{s}=4.56193$, $|\vec{P}|=2.18453$, $|\vec{P}'|=2.18453$
 $\theta_{P'}=3.14159$, $\theta_Y=0$, $\phi_{P'}=2.23383$, $\phi_Y=4.47775$
 $E_Y=1.54558$, $\theta_{\ell'}=0$, $\phi_{\ell'}=5.37543$
 $Q^2=5.58324$, $x_B=0.283631$, $t=-19.0887$, $W^2=14.9815$, $y=0.987634$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.1845$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.1845$
 P $E= 2.3774$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.1845$
 ℓ' $E= 0.6390$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 0.6390$
 P' $E= 2.3774$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= -2.1845$
 q_Y $E= 1.5456$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 1.5456$



electron: polarization cluster P2, local minimum 171; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 2.0459092\text{e-}07$
 $d_{\text{GHZ}} \text{ contour} = 0.050000205$
 above global minimum = $1.8230746\text{e-}07$
 8D contour samples = 63

selected phase-space point:

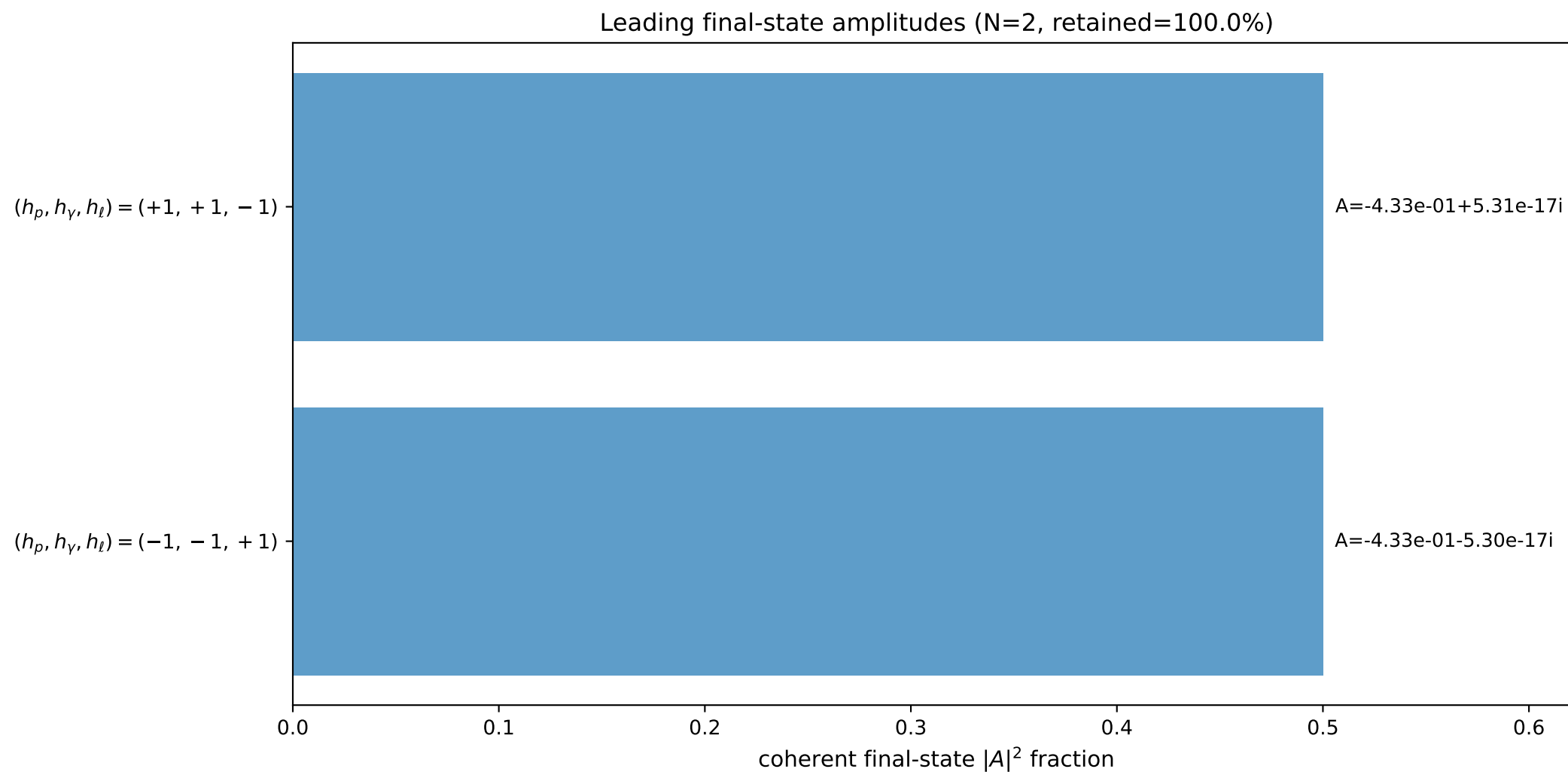
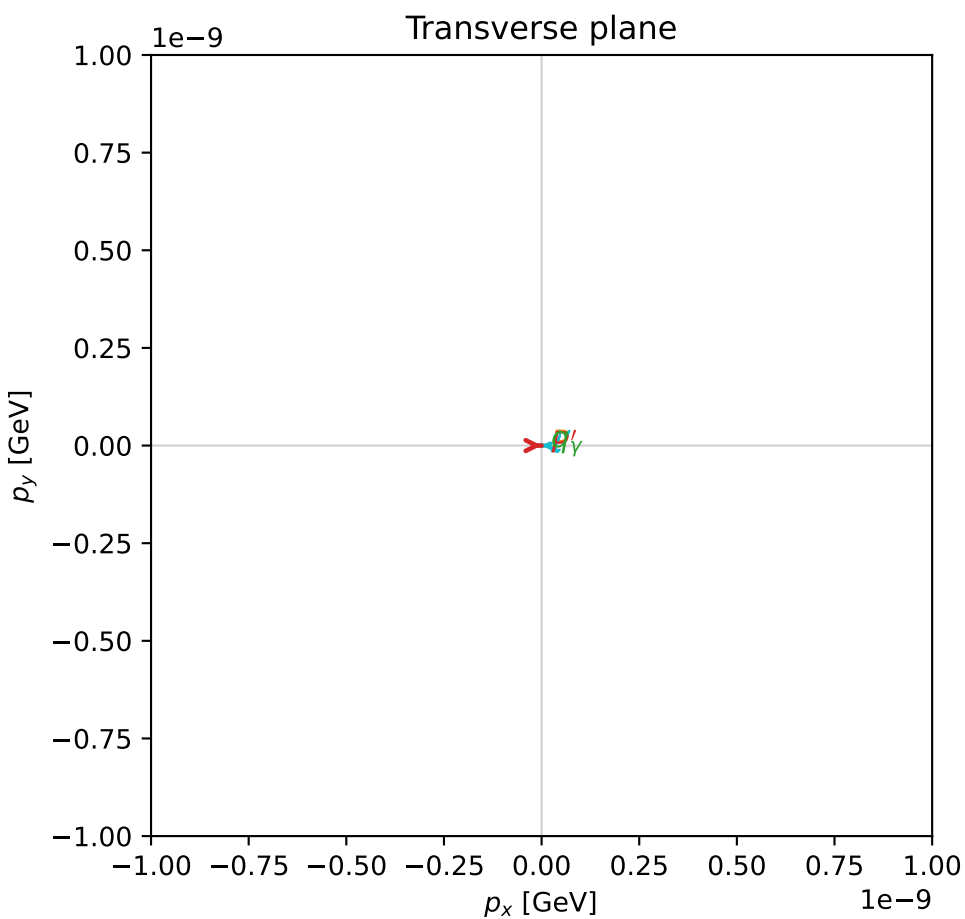
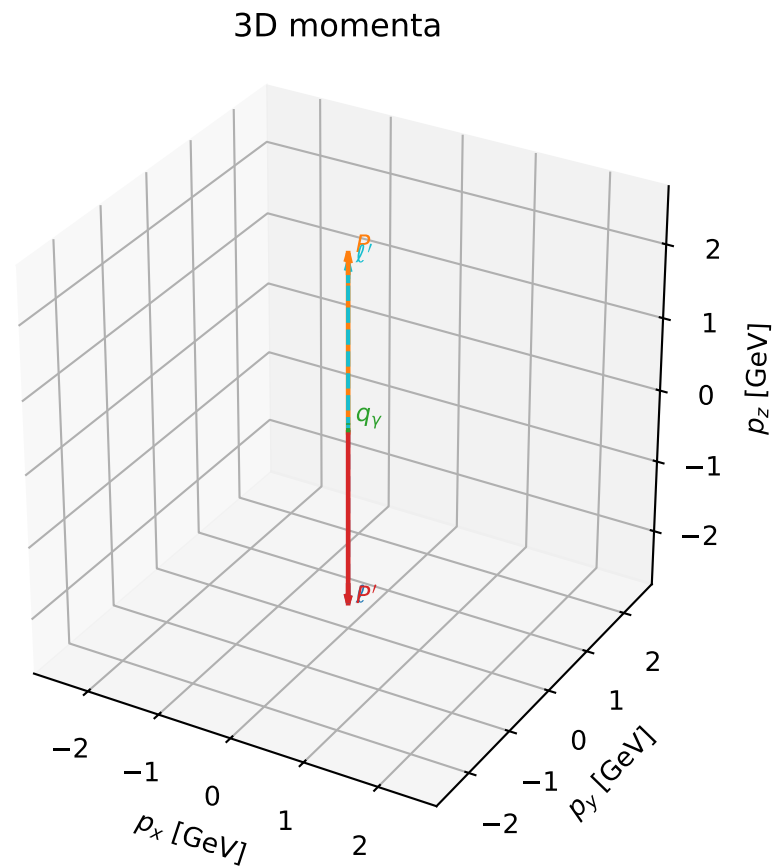
$\text{sqrt_s} = 4.561926$
 $\text{theta_p_out} = 3.141593$
 $\text{theta_gamma_out} = 0$
 $\text{q0ut} = 1.545577$
 $\text{phi_p_out} = 2.233832$
 $\text{phi_gamma_out} = 4.477746$
 $\text{alpha_e} = 0.2374239$
 $\text{alpha_p} = 1.808211$

coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

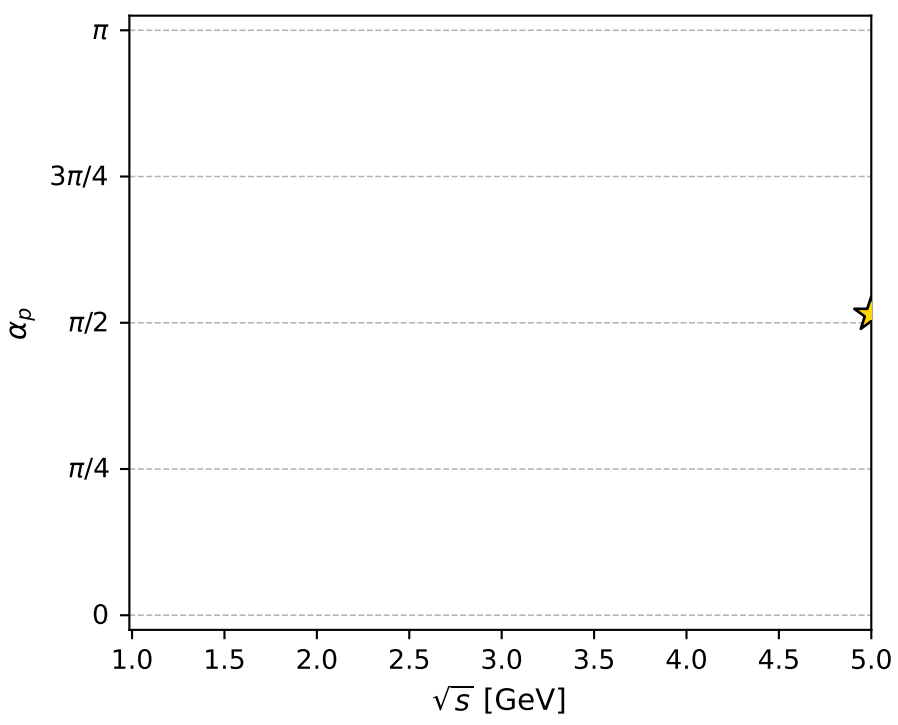
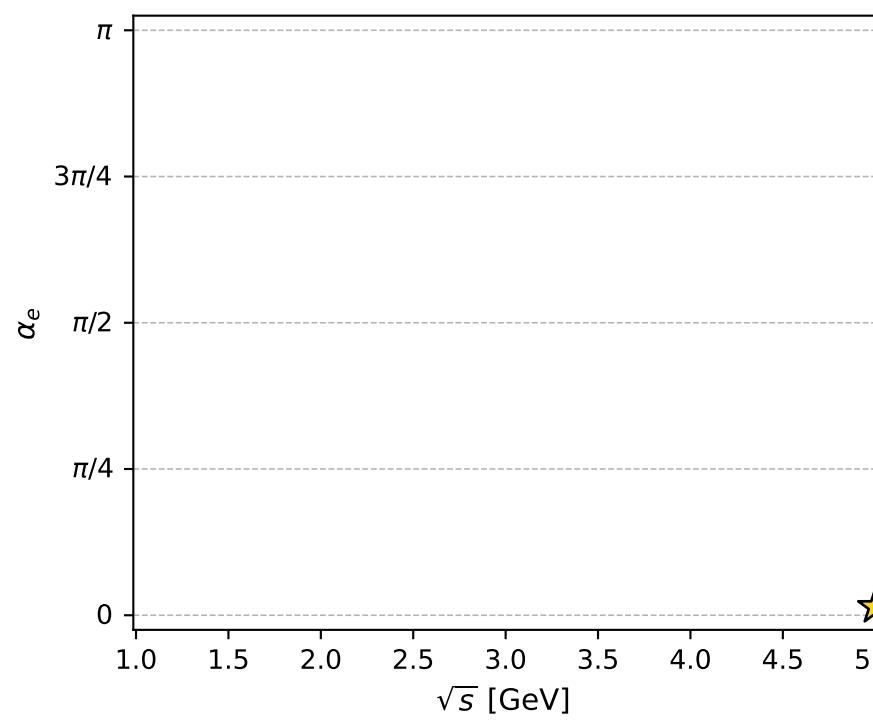
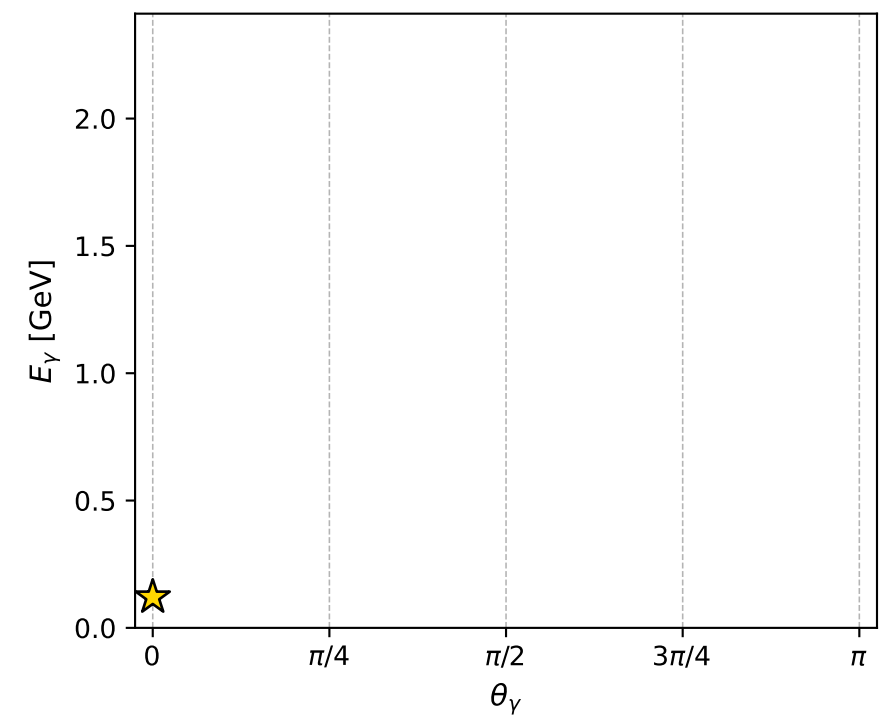
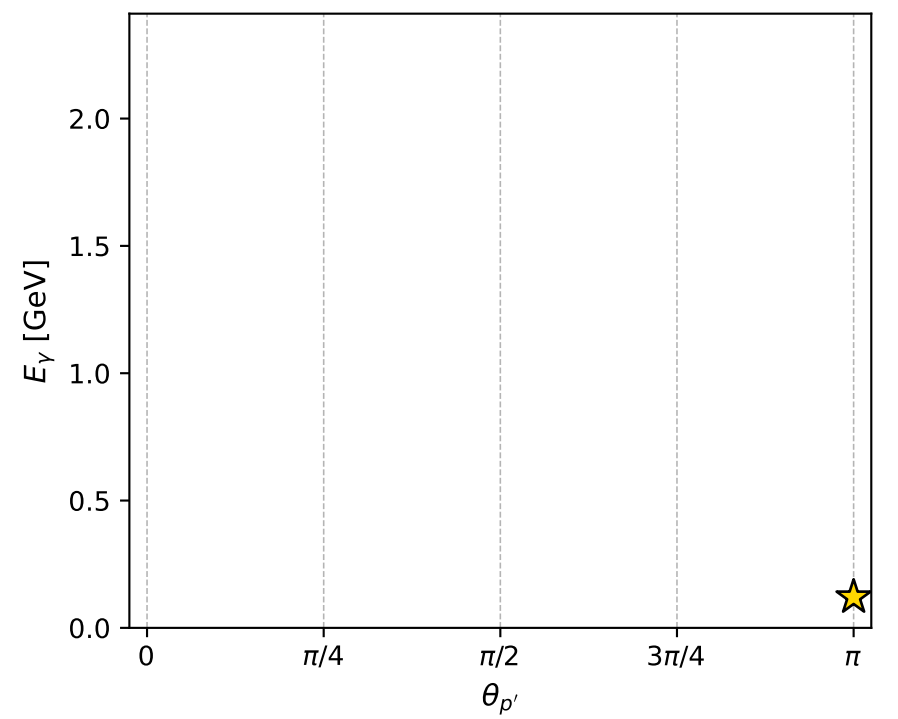
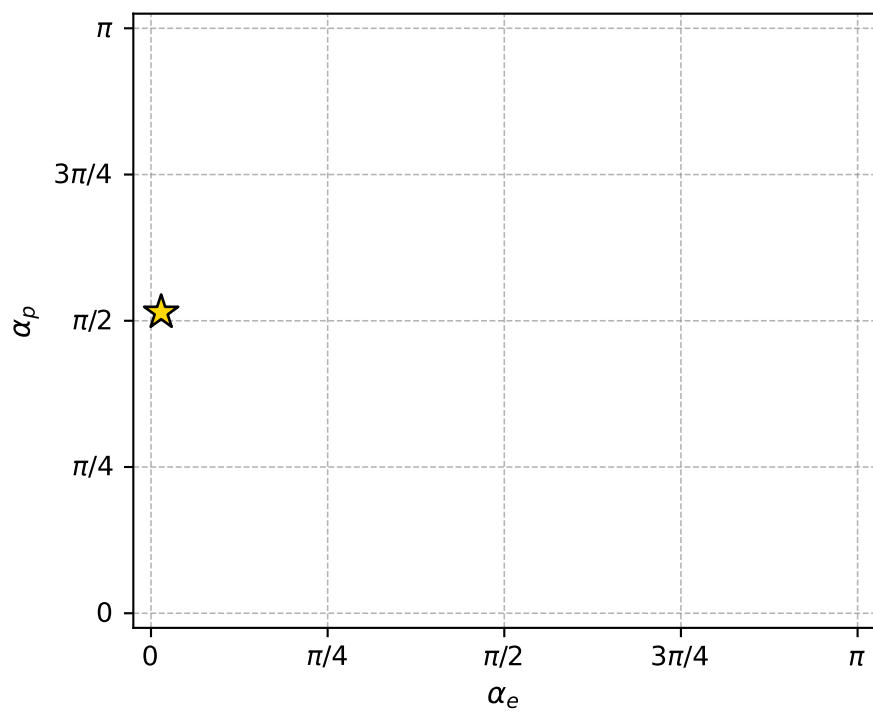
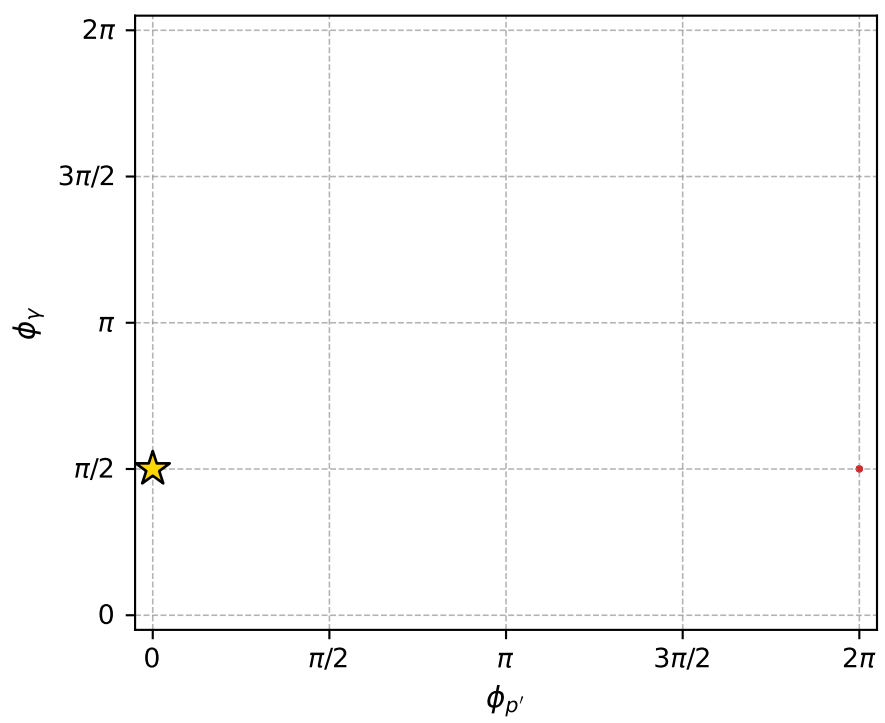
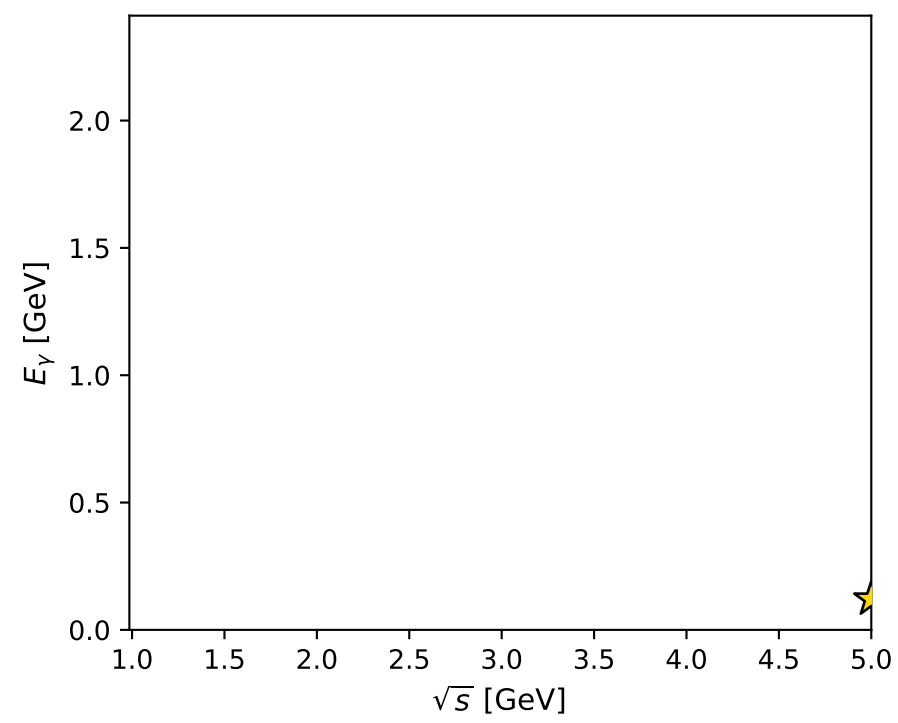
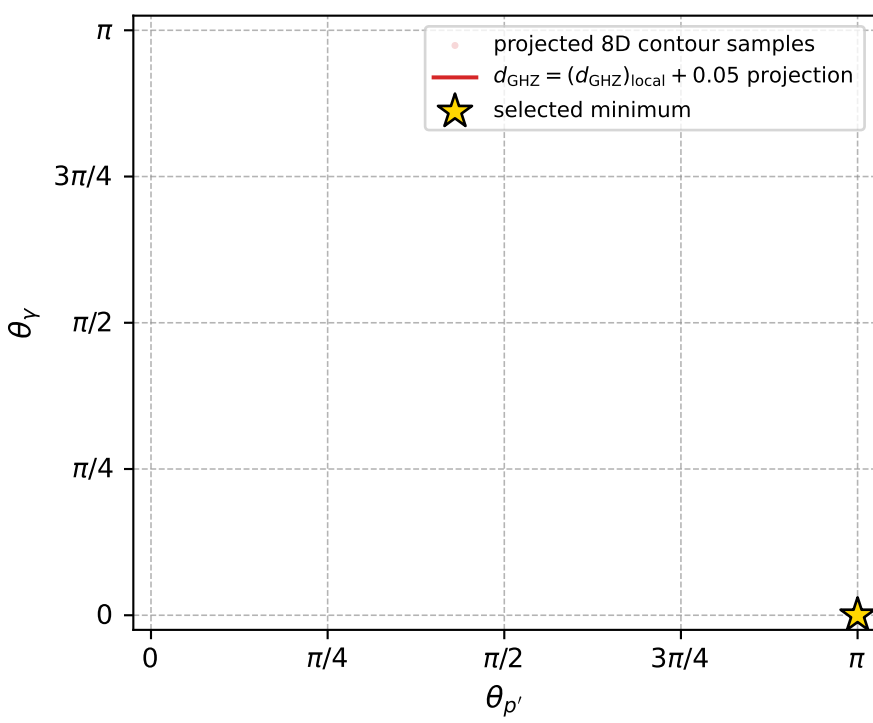
dghz_mixing_angles_polarization_02_local_minimum_175 $d_{\{\text{rm GHZ}\}}=2.71276\text{e-}07$
region: polarization_cluster_02_local_minimum_175
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: $\alpha_e=0.045198059$ rad
incoming lepton: $0.998979|+\rangle +0.0451827|-\rangle$
proton mixing angle: $\alpha_p=1.6160065$ rad
incoming proton: $-0.0451948|+\rangle +0.998978|-\rangle$

$s=25$, $\sqrt{s}=5$, $|\vec{P}|=2.41202$, $|\vec{P}'|=2.41202$
 $\theta_{P'}=3.14159$, $\theta_Y=0$, $\phi_{P'}=2.96908\text{e-}29$, $\phi_Y=1.572$
 $E_Y=0.120652$, $\theta_{\ell'}=0$, $\phi_{\ell'}=3.14159$
 $Q^2=22.1072$, $x_B=0.948249$, $t=-23.2713$, $W^2=2.08636$, $y=0.966567$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4120$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 P $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4120$
 ℓ' $E= 2.2914$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 2.2914$
 P' $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 q_Y $E= 0.1207$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 0.1207$



electron: polarization cluster P2, local minimum 175; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 2.7127631\text{e-}07$
 $d_{\text{GHZ}} \text{ contour} = 0.050000271$
 above global minimum = $2.4899285\text{e-}07$
 8D contour samples = 53

selected phase-space point:

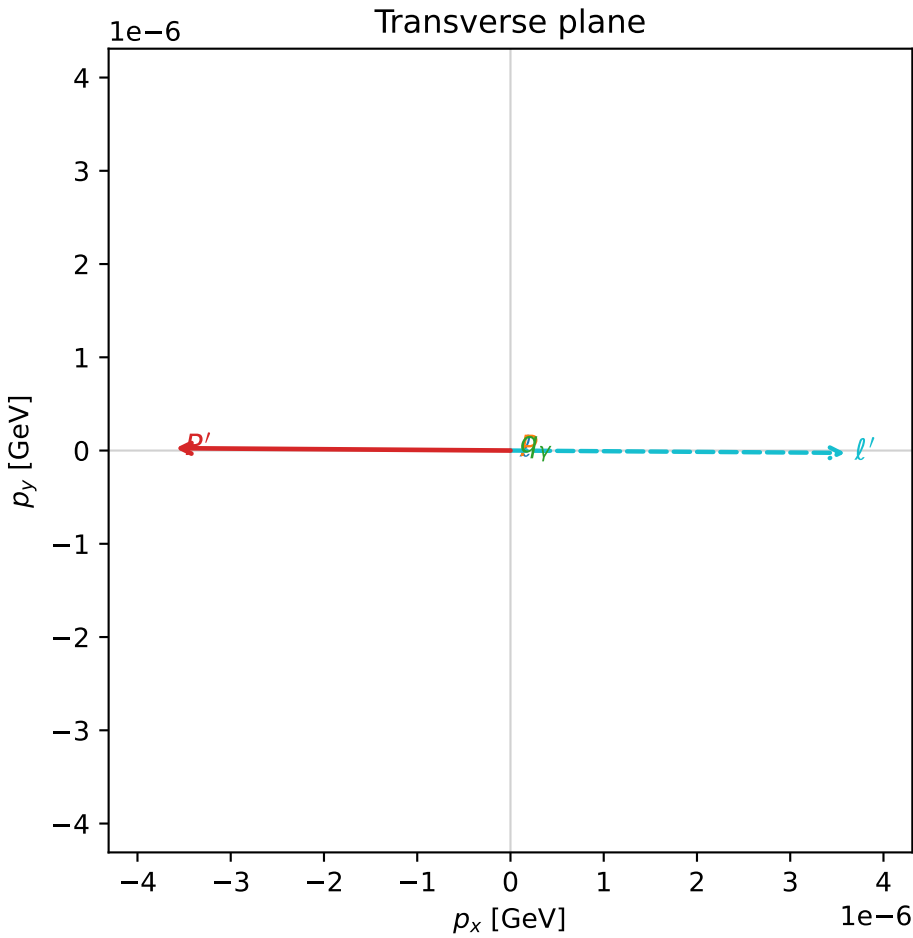
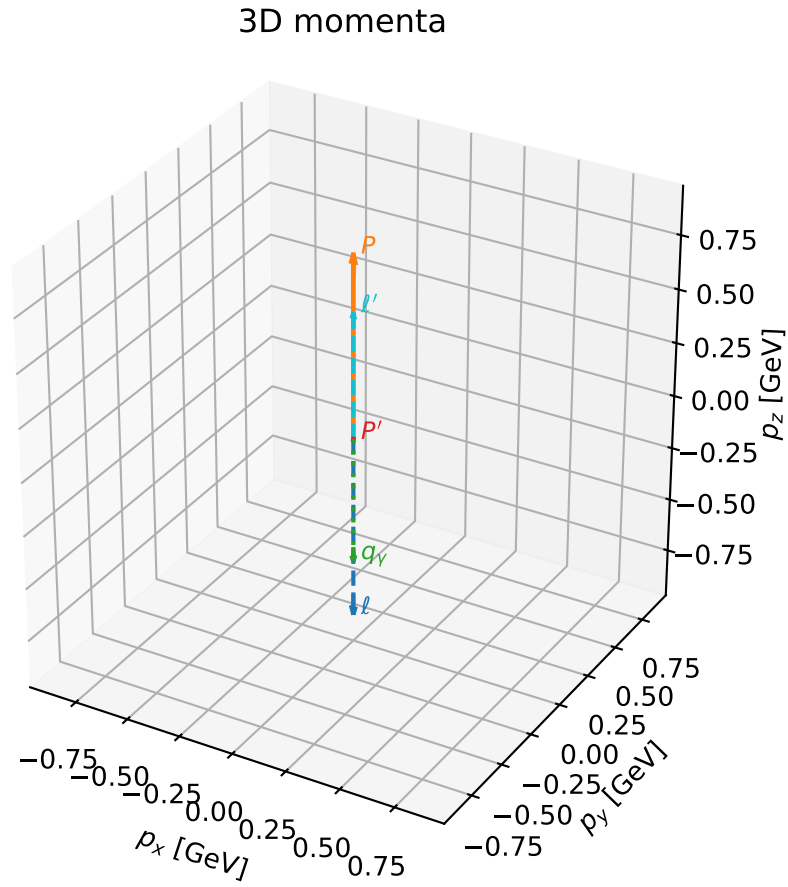
$\text{sqrt_s} = 5$
 $\text{theta_p_out} = 3.141593$
 $\text{theta_gamma_out} = 0$
 $\text{q0out} = 0.1206519$
 $\text{phi_p_out} = 2.969081\text{e-}29$
 $\text{phi_gamma_out} = 1.572$
 $\text{alpha_e} = 0.04519806$
 $\text{alpha_p} = 1.616007$

coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

dghz_mixing_angles_polarization_02_local_minimum_182 $d_{\{\text{rm GHZ}\}}=6.61545\text{e-}07$
region: polarization_cluster_02_local_minimum_182
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: $\alpha_e=0.2035518$ rad
incoming lepton: $0.979355|+\rangle +0.202149|-\rangle$
proton mixing angle: $\alpha_p=2.9380379$ rad
incoming proton: $-0.979354|+\rangle +0.202152|-\rangle$

$s=4.45467$, $\sqrt{s}=2.11061$, $|\vec{P}|=0.846871$, $|\vec{P}'|=1.86603\text{e-}05$
 $\theta_{P'}=2.94793$, $\theta_Y=3.14159$, $\phi_{P'}=3.13457$, $\phi_Y=3.85199$
 $E_Y=0.586295$, $\theta_{\ell'}=6.12523\text{e-}06$, $\phi_{\ell'}=6.27616$
 $Q^2=1.98613$, $x_B=0.643594$, $t=-0.611116$, $W^2=1.97971$, $y=0.863258$

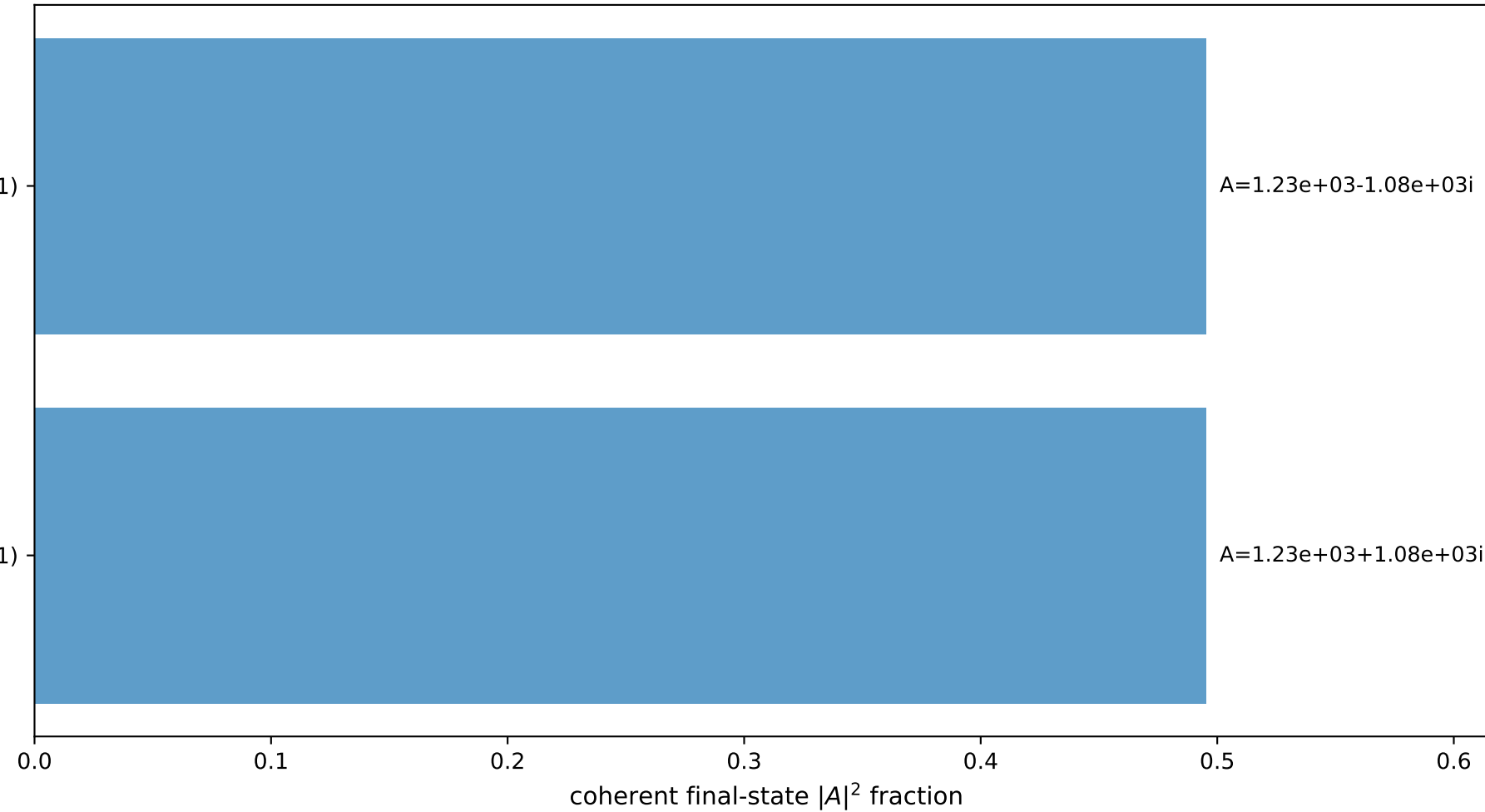
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.8469$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.8469$
 P $E= 1.2637$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.8469$
 ℓ' $E= 0.5863$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 0.5863$
 P' $E= 0.9380$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= -0.0000$
 q_Y $E= 0.5863$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= -0.5863$



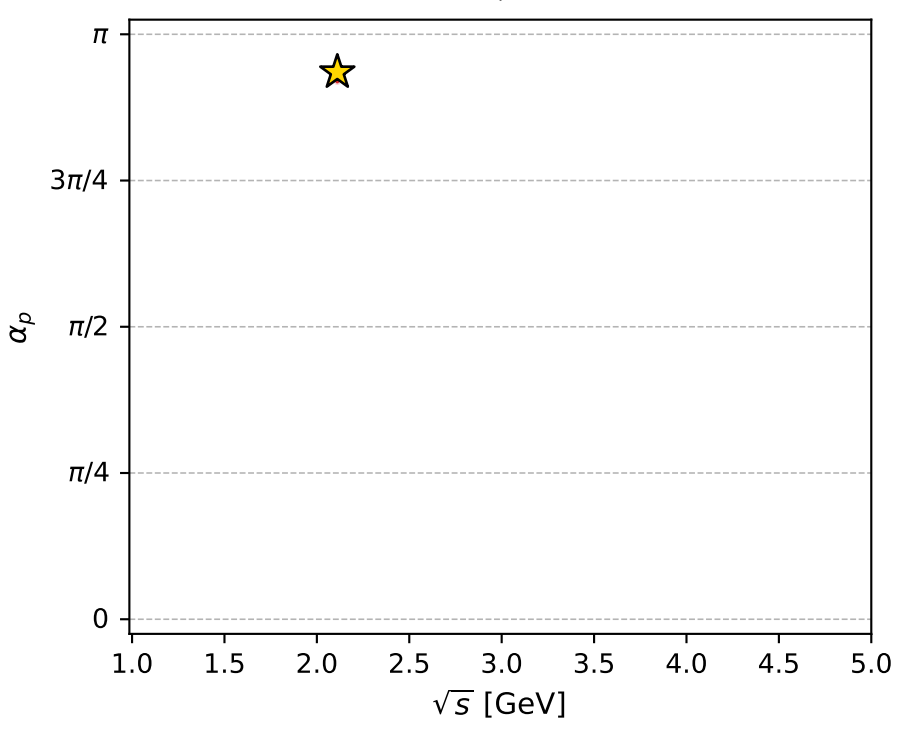
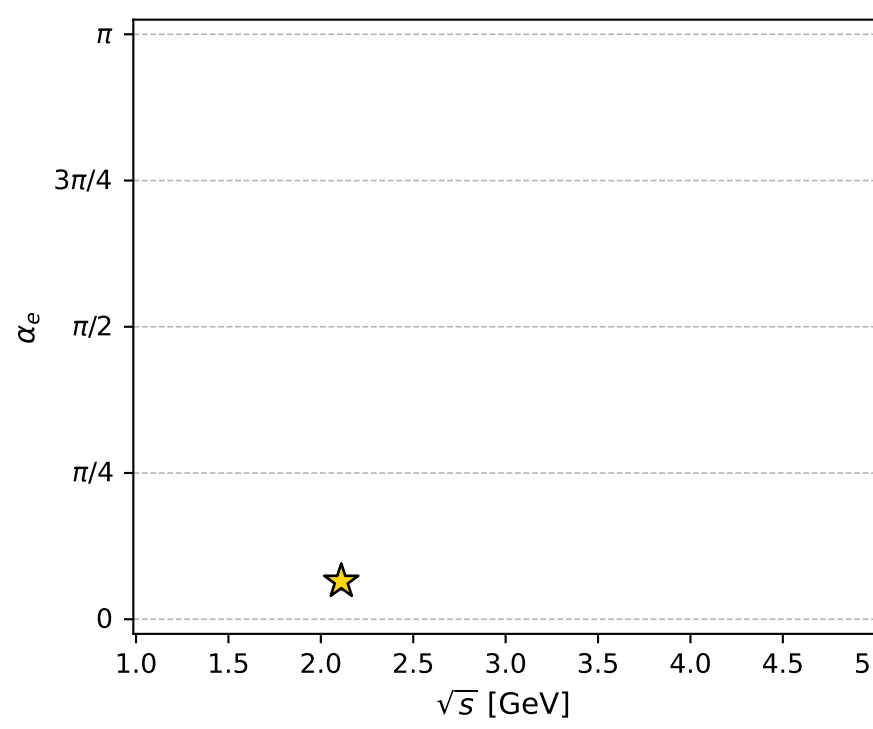
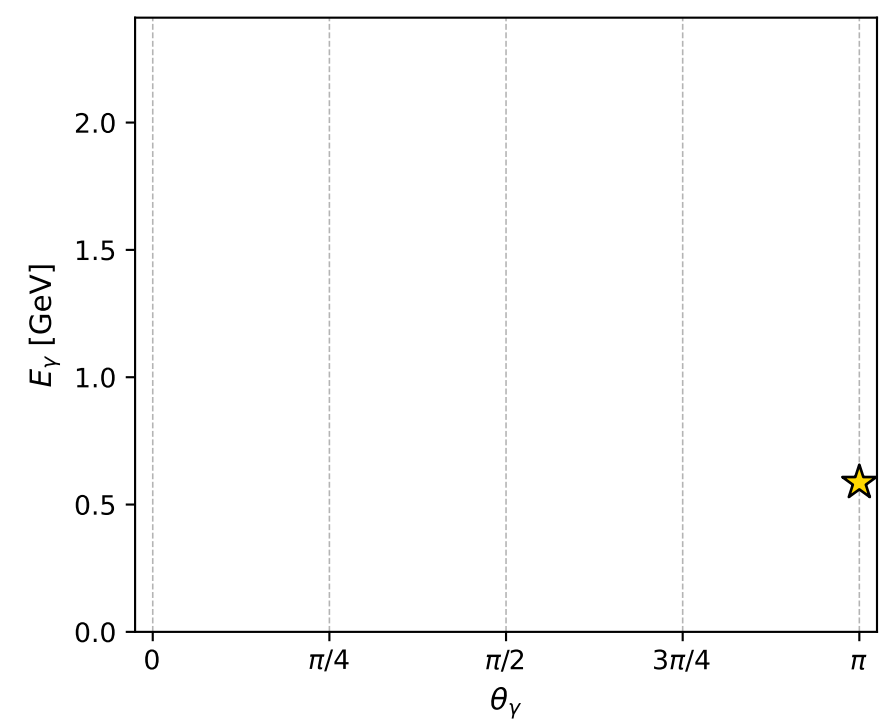
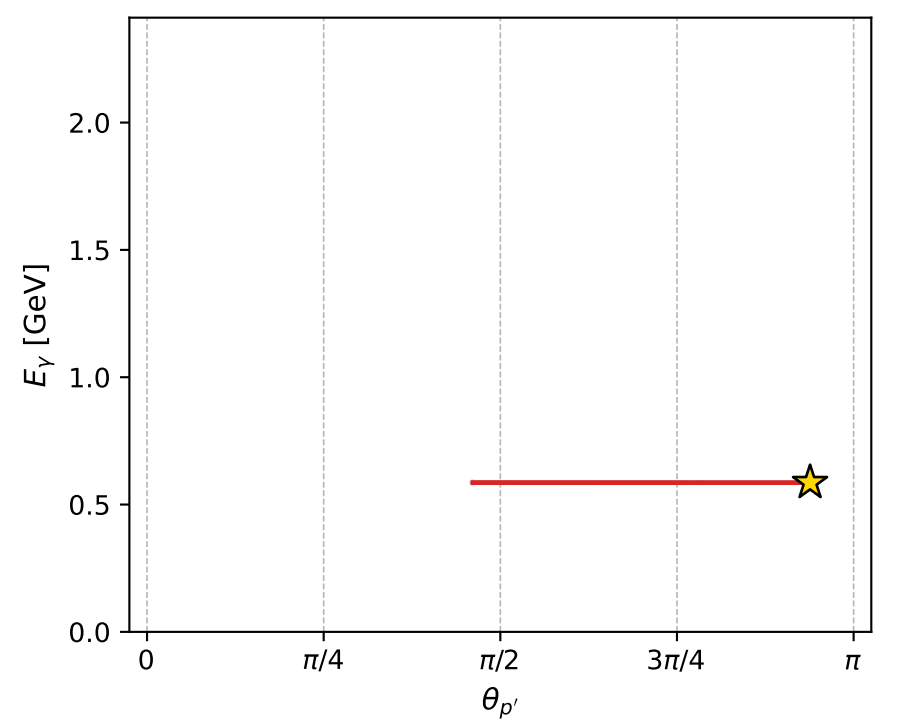
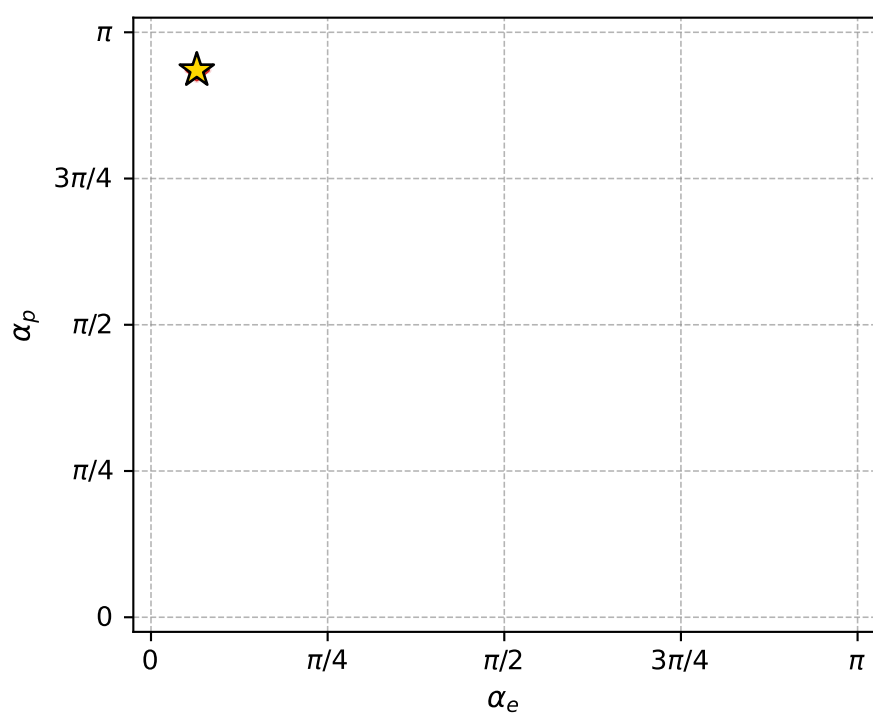
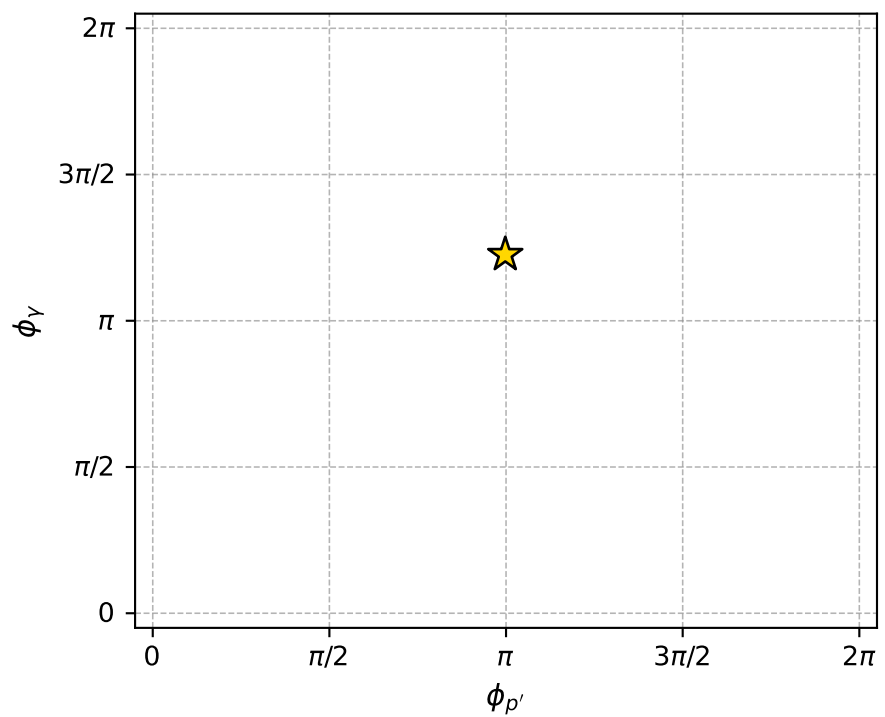
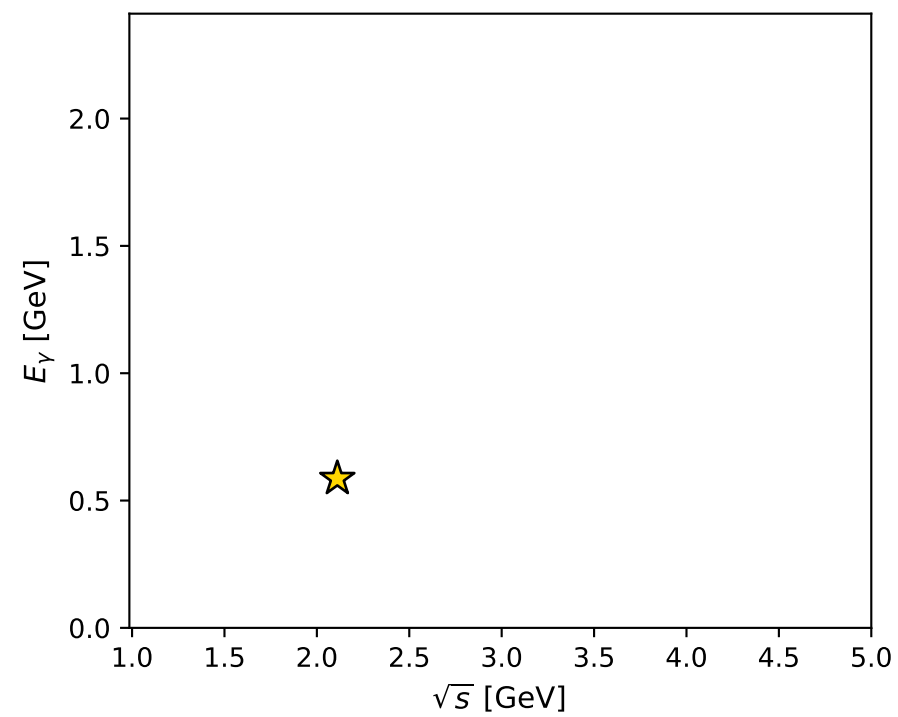
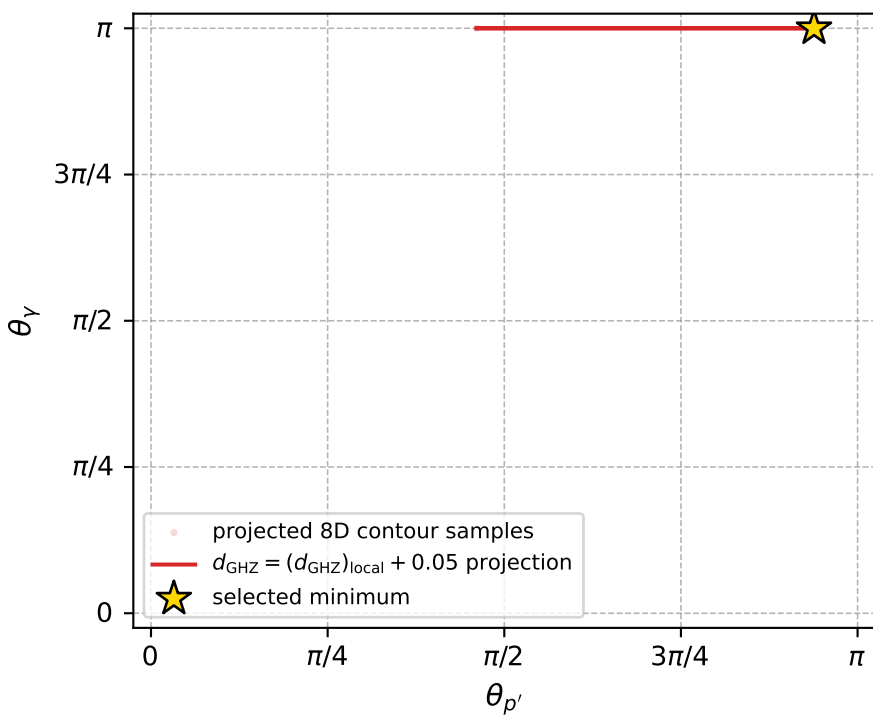
$(h_p, h_Y, h_\ell) = (-1, +1, -1)$

$(h_p, h_Y, h_\ell) = (+1, -1, +1)$

Leading final-state amplitudes (N=2, retained=99.1%)



electron: polarization cluster P2, local minimum 182; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 6.6154515\text{e-}07$
 $d_{\text{GHZ}} \text{ contour} = 0.050000662$
 above global minimum = $6.3926169\text{e-}07$
 8D contour samples = 133

selected phase-space point:

$\text{sqrt_s} = 2.110609$
 $\text{theta_p_out} = 2.947928$
 $\text{theta_gamma_out} = 3.141593$
 $\text{q0out} = 0.5862952$
 $\text{phi_p_out} = 3.134567$
 $\text{phi_gamma_out} = 3.851991$
 $\text{alpha_e} = 0.2035518$
 $\text{alpha_p} = 2.938038$

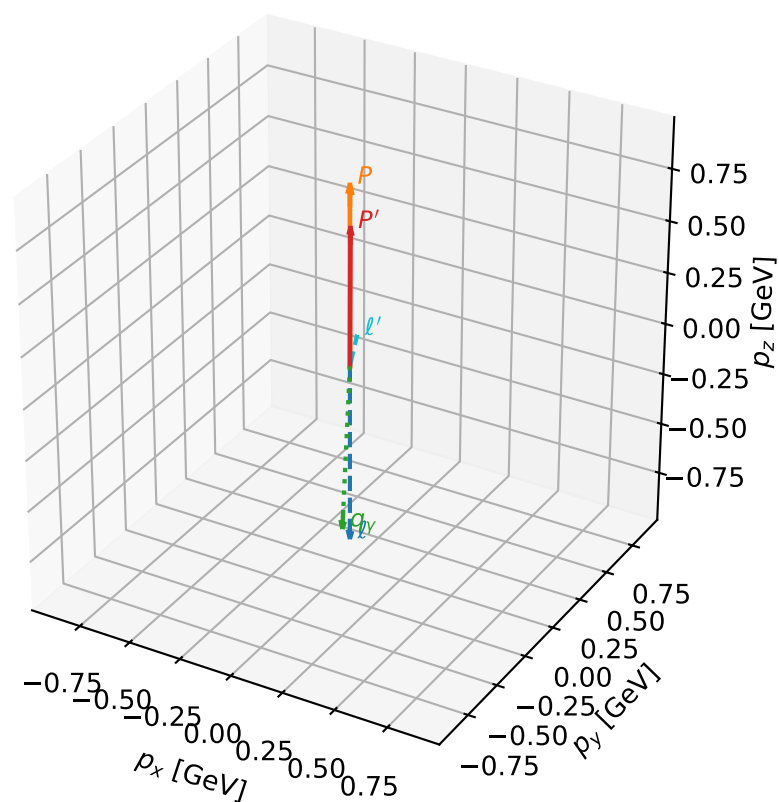
coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

dghz_mixing_angles_polarization_02_local_minimum_185 $d_{\{\text{rm GHZ}\}}=3.76832\text{e-}06$
 region: polarization_cluster_02_local_minimum_185
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=1.2208886$ rad
 incoming lepton: $0.342811|+\rangle +0.939404|-\rangle$
 proton mixing angle: $\alpha_p=2.8239017$ rad
 incoming proton: $-0.949959|+\rangle +0.312374|-\rangle$

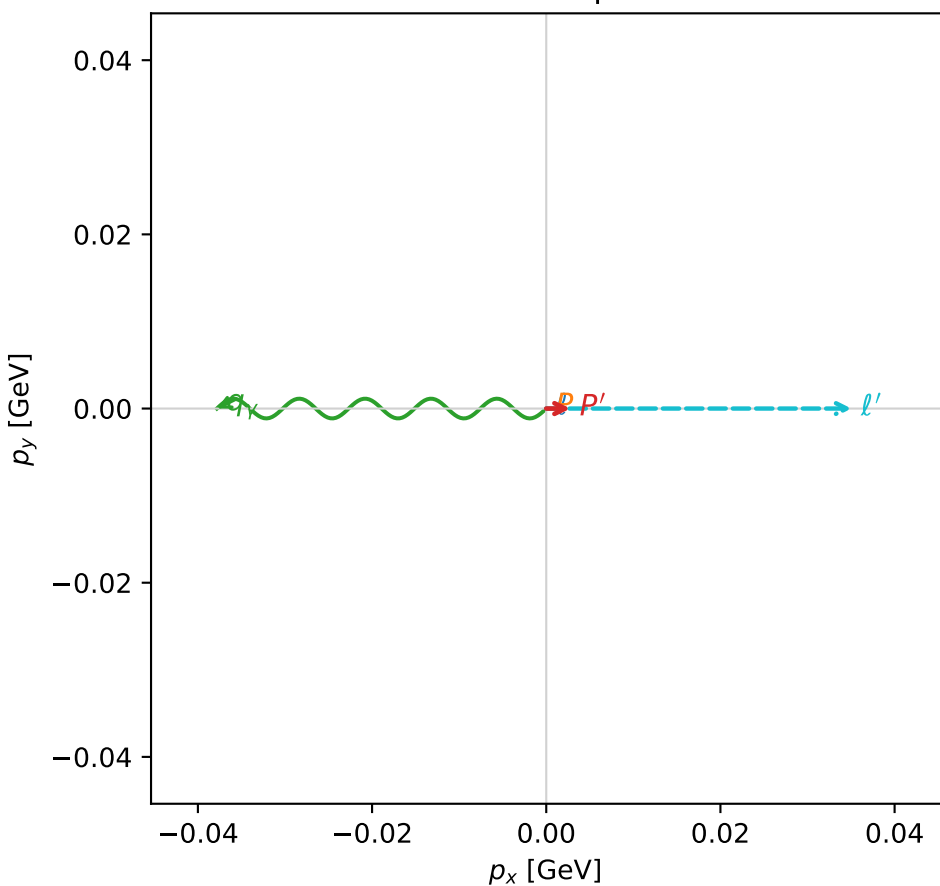
$s=4.57376$, $\sqrt{s}=2.13864$, $|\vec{P}|=0.863616$, $|\vec{P}'|=0.662371$
 $\theta_{P'}=0.00416983$, $\theta_Y=3.09573$, $\phi_{P'}=6.28319$, $\phi_Y=3.1416$
 $E_Y=0.824908$, $\theta_{\ell'}=0.213519$, $\phi_{\ell'}=6.85947\text{e-}06$
 $Q^2=0.564992$, $x_B=0.159094$, $t=-0.0244499$, $W^2=3.86616$, $y=0.961393$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.8636$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.8636$
 P $E= 1.2750$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.8636$
 ℓ' $E= 0.1654$ $p_x= 0.0351$ $p_y= 0.0000$ $p_z= 0.1617$
 P' $E= 1.1483$ $p_x= 0.0028$ $p_y= -0.0000$ $p_z= 0.6624$
 q_Y $E= 0.8249$ $p_x= -0.0378$ $p_y= -0.0000$ $p_z= -0.8240$

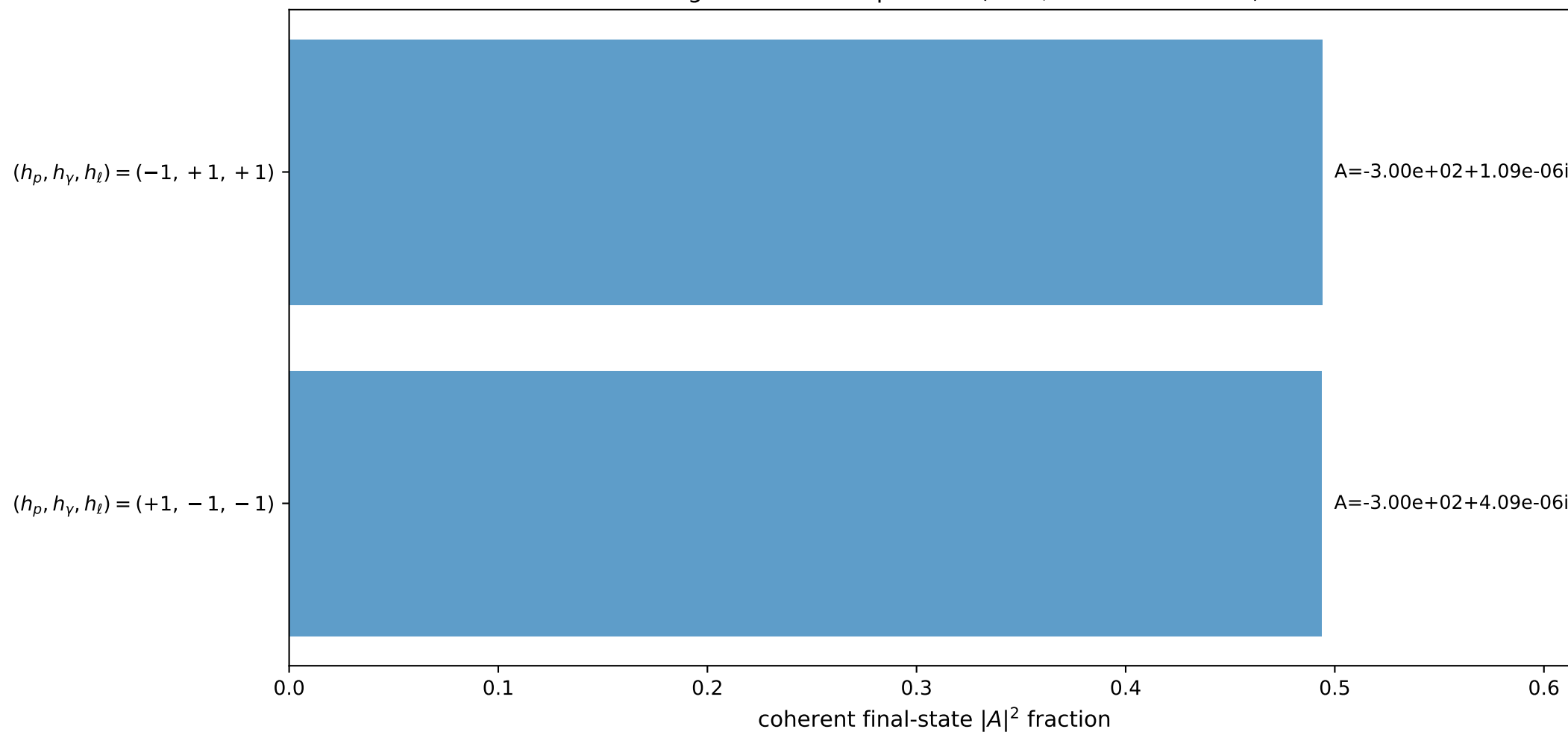
3D momenta



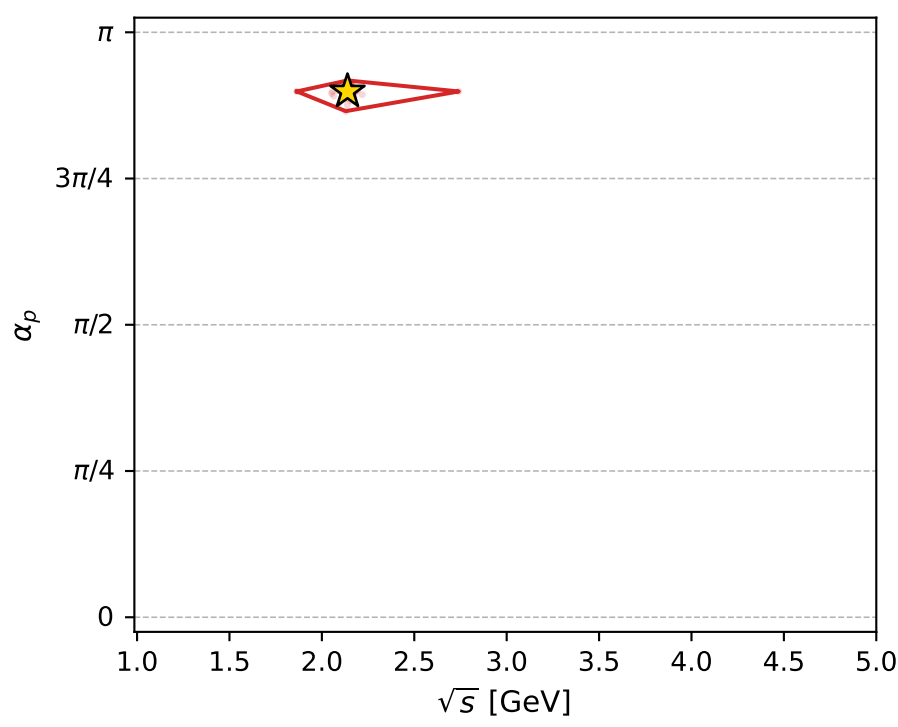
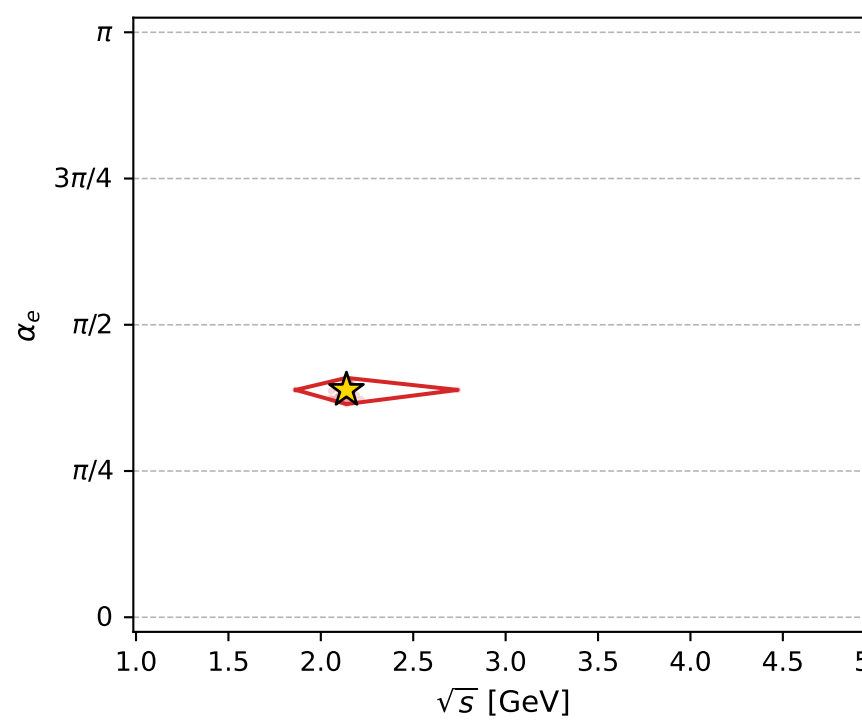
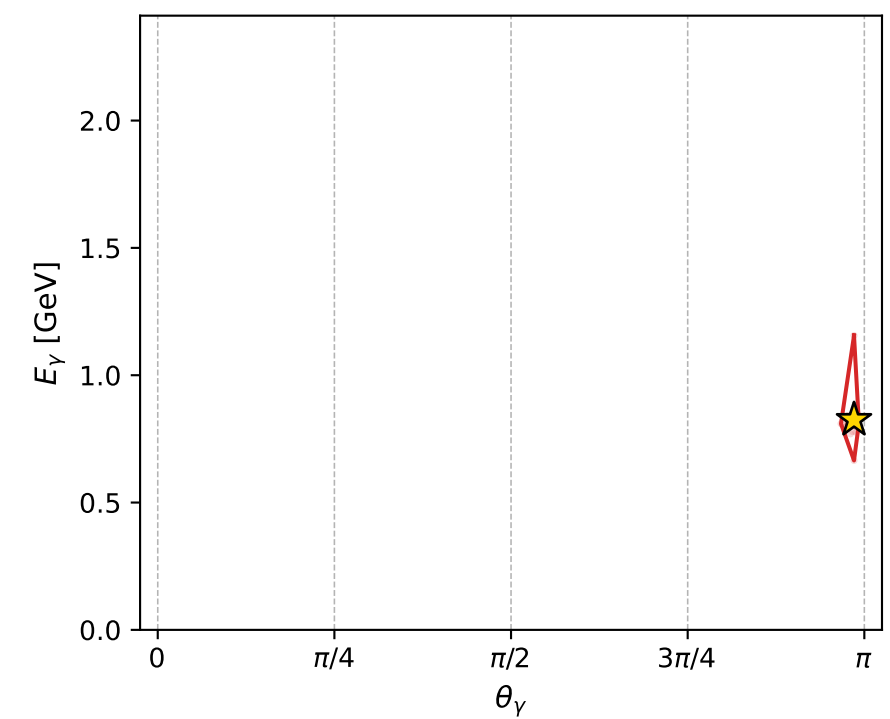
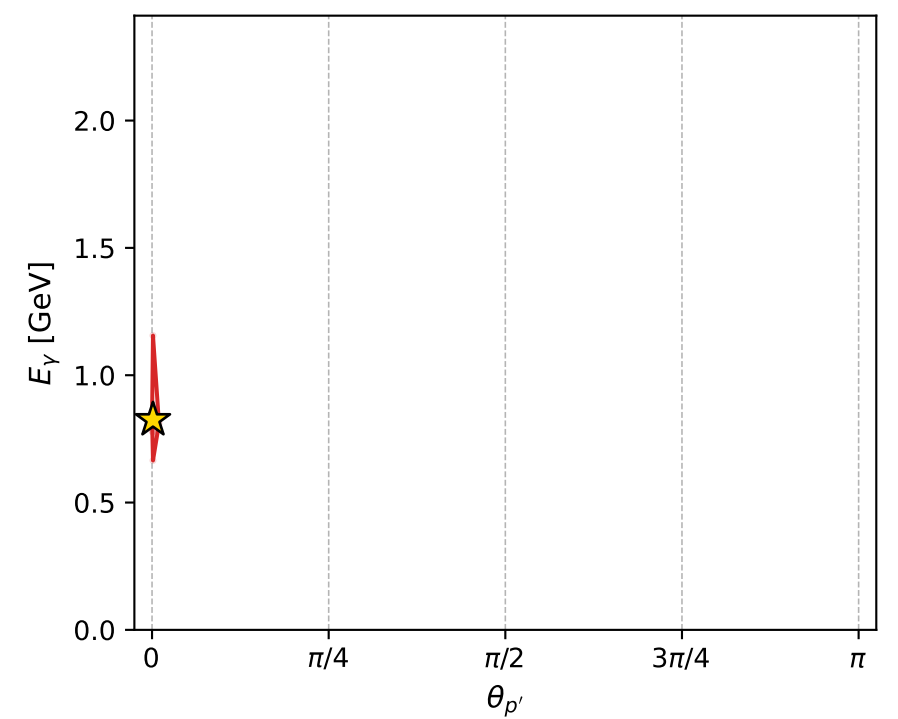
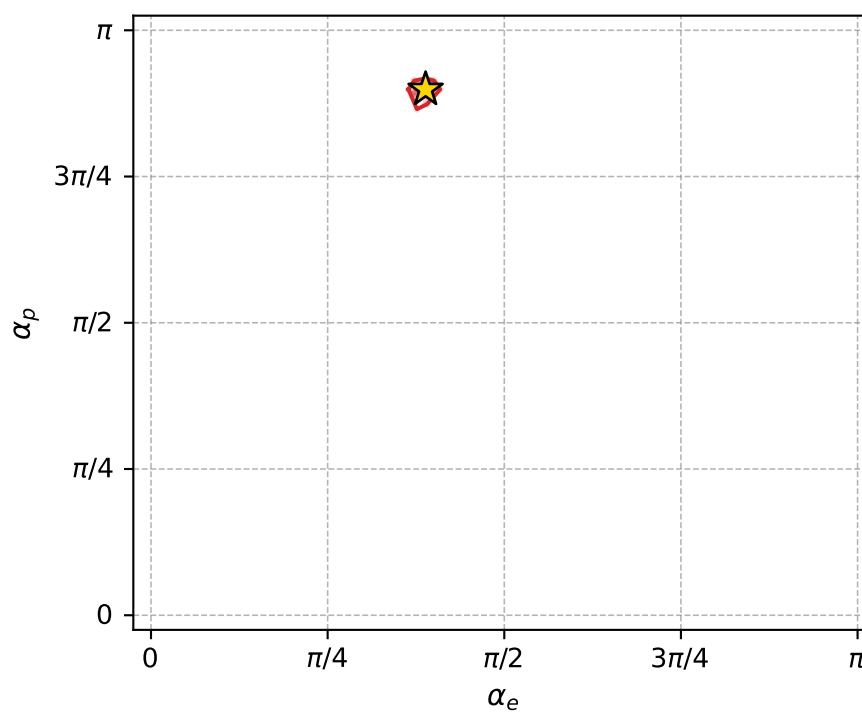
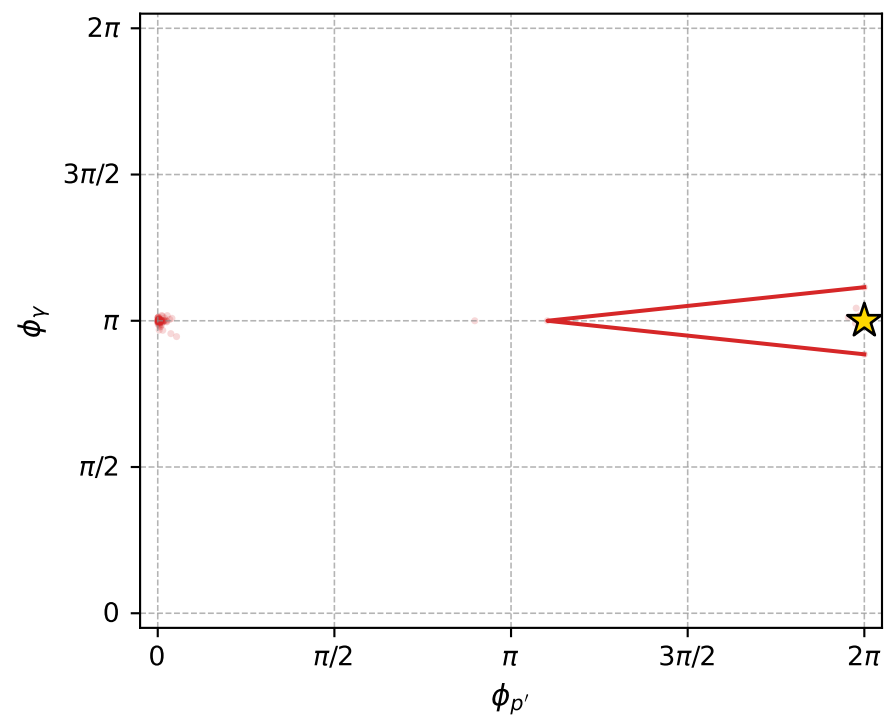
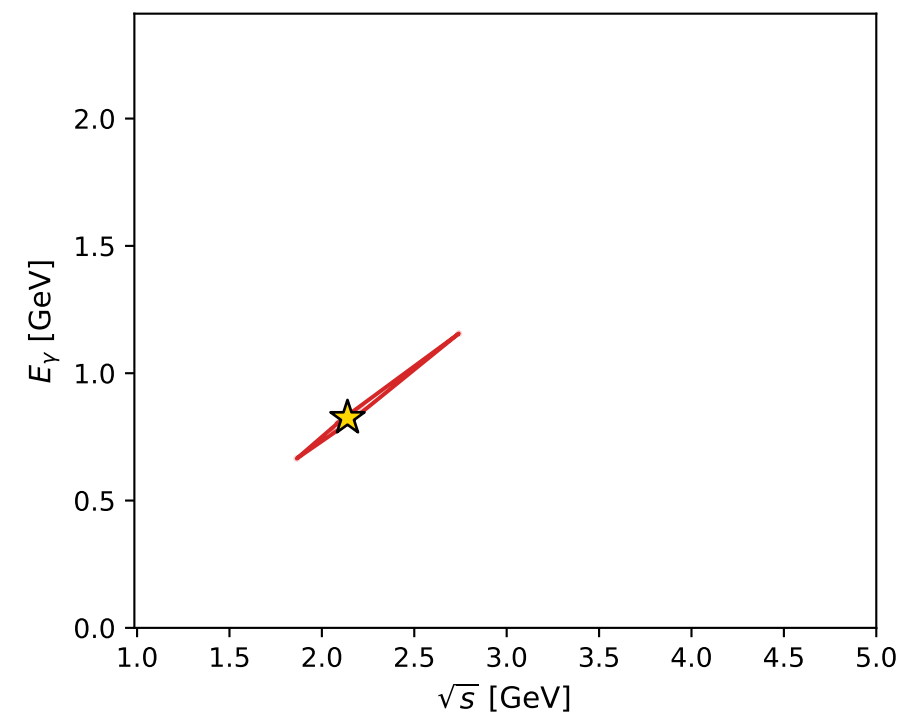
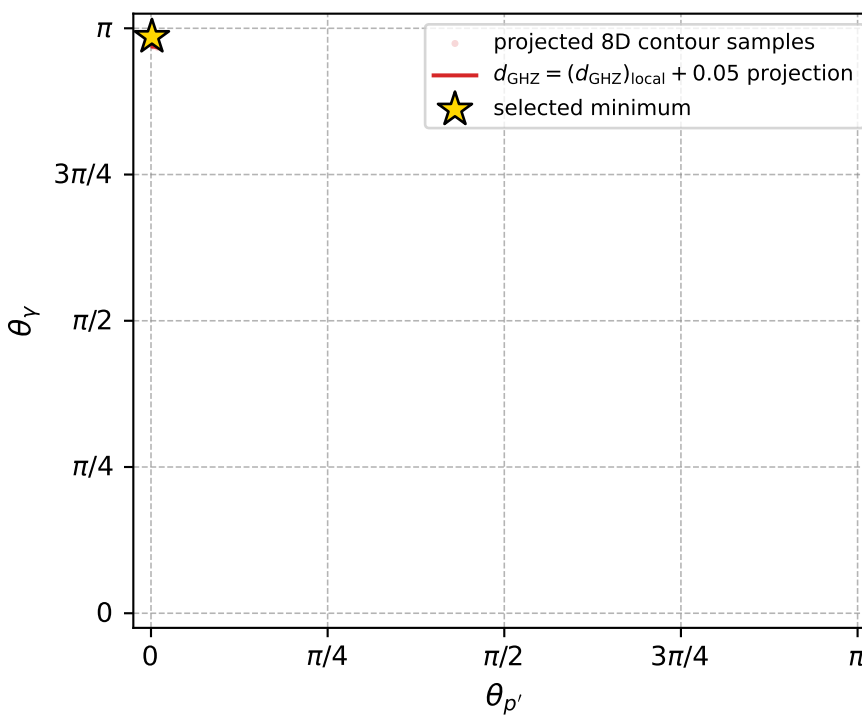
Transverse plane



Leading final-state amplitudes (N=2, retained=98.8%)



electron: polarization cluster P2, local minimum 185; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 3.7683168\text{e-}06$
 $d_{\text{GHZ}} \text{ contour} = 0.050003768$
 above global minimum = $3.7460334\text{e-}06$
 8D contour samples = 164

selected phase-space point:

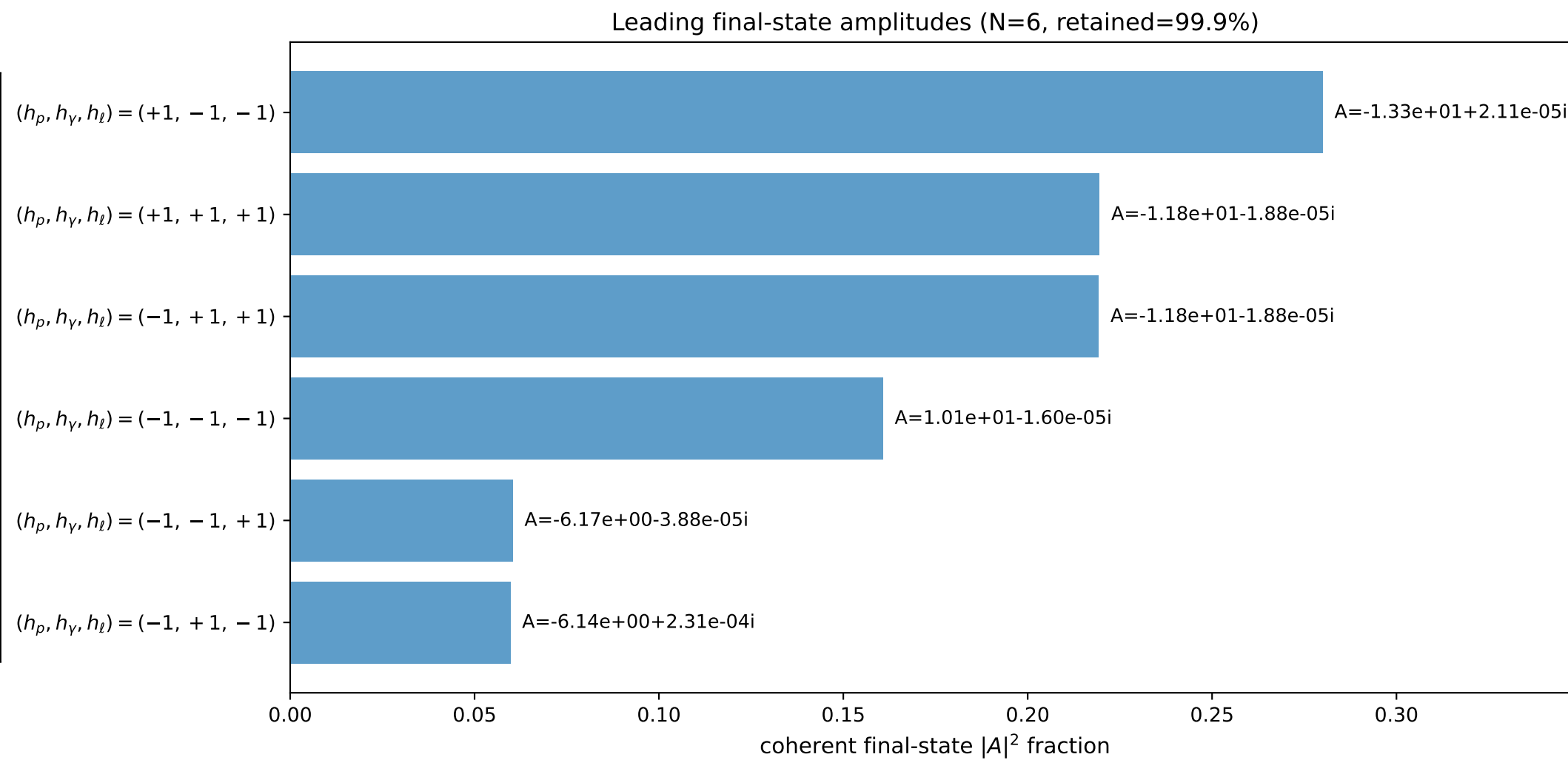
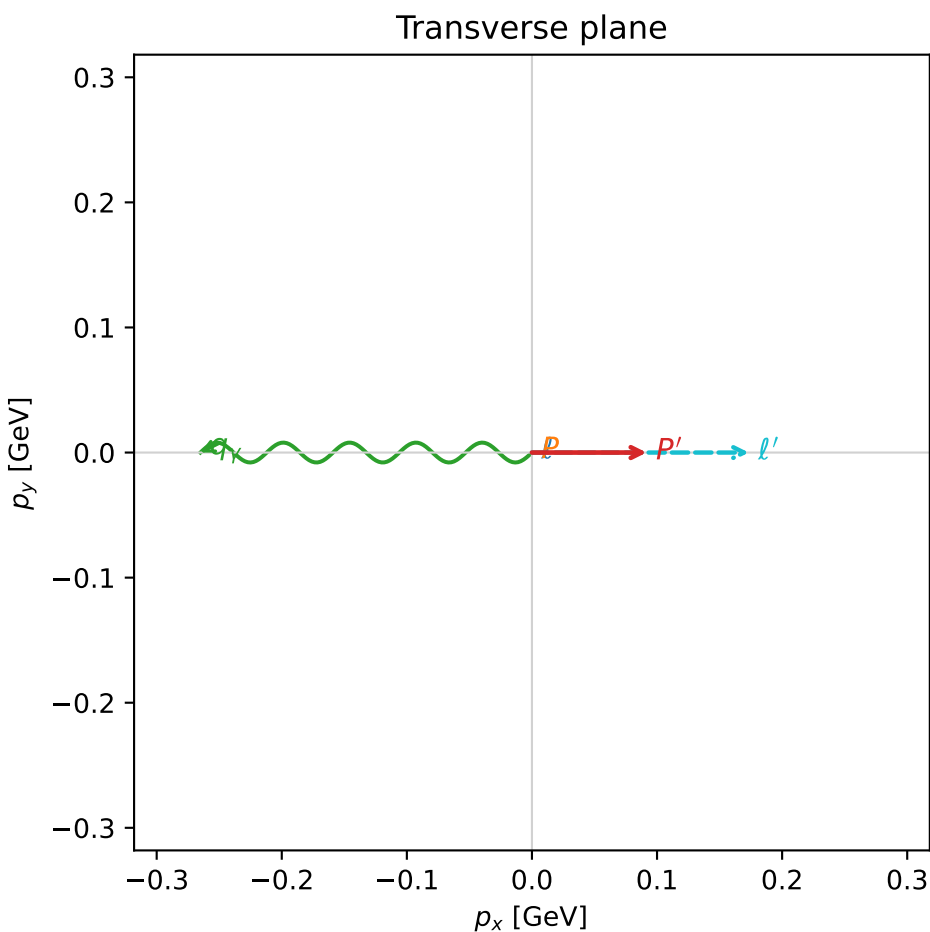
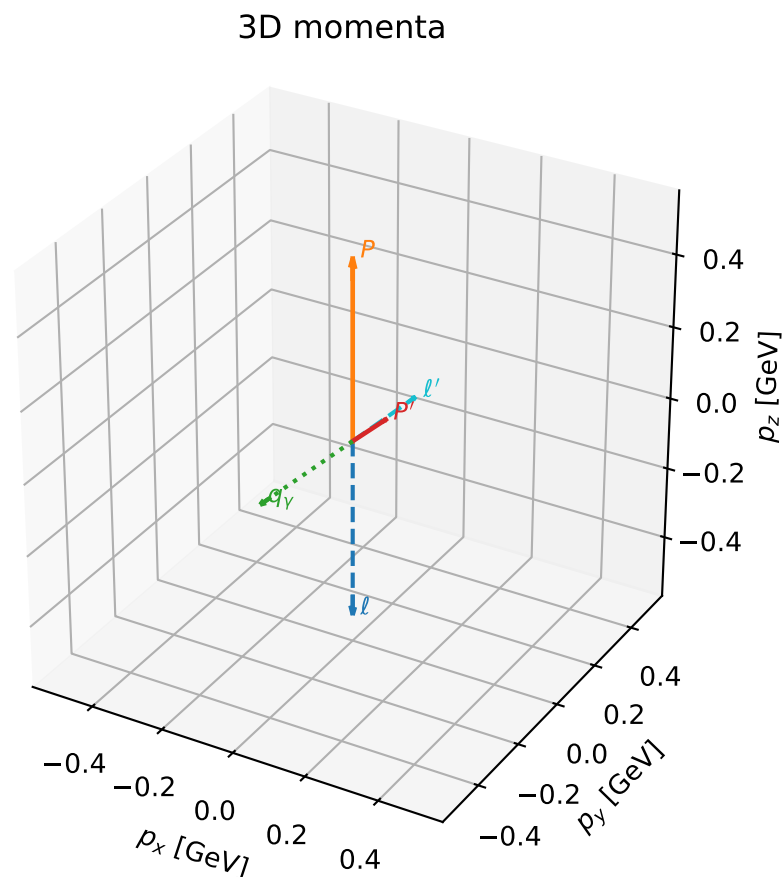
$\text{sqrt_s} = 2.138636$
 $\text{theta_p_out} = 0.004169826$
 $\text{theta_gamma_out} = 3.095733$
 $\text{q0out} = 0.8249084$
 $\text{phi_p_out} = 6.283185$
 $\text{phi_gamma_out} = 3.141599$
 $\text{alpha_e} = 1.220889$
 $\text{alpha_p} = 2.823902$

coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

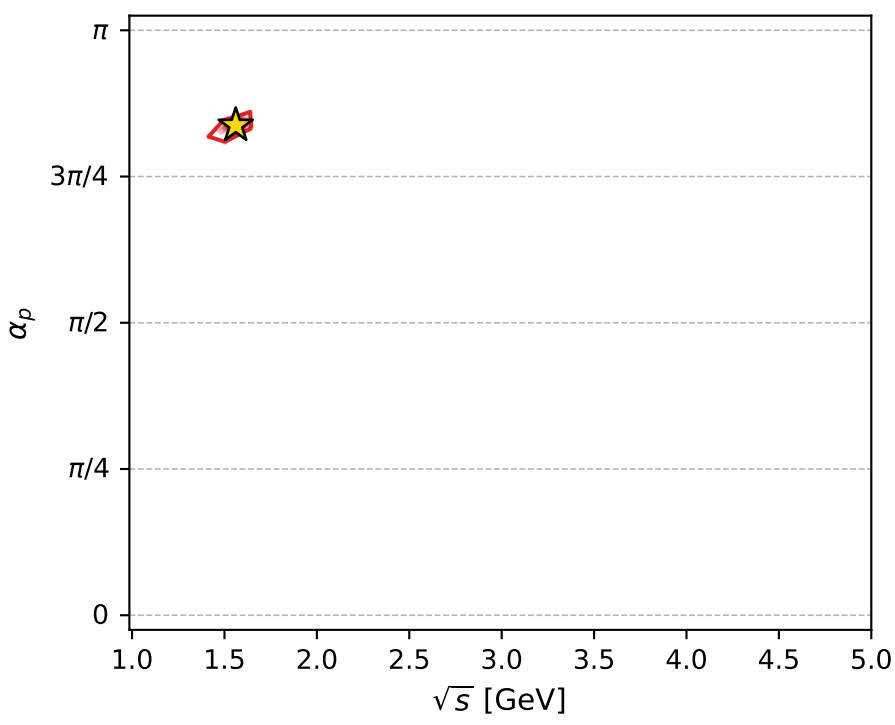
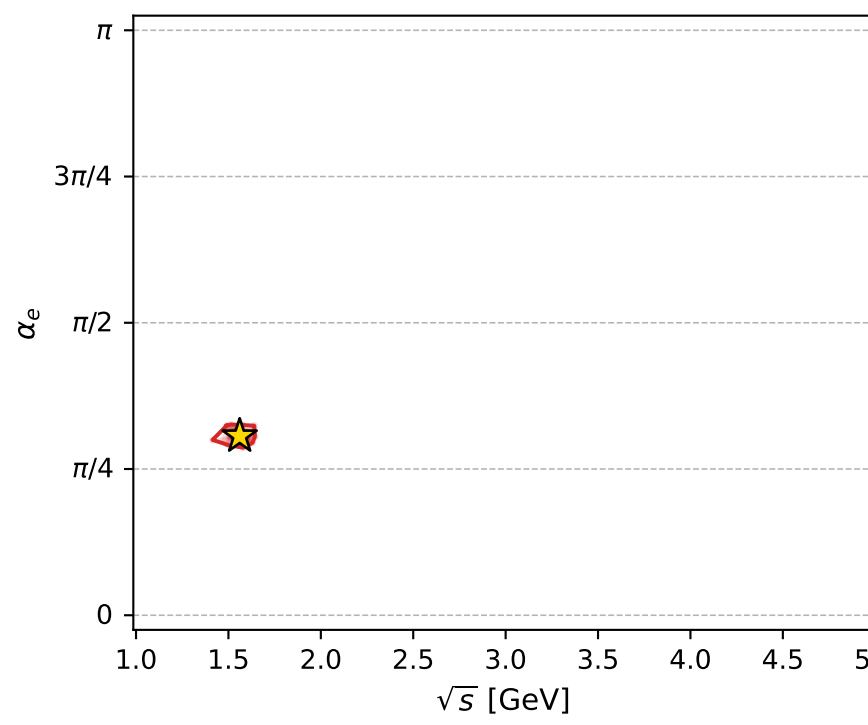
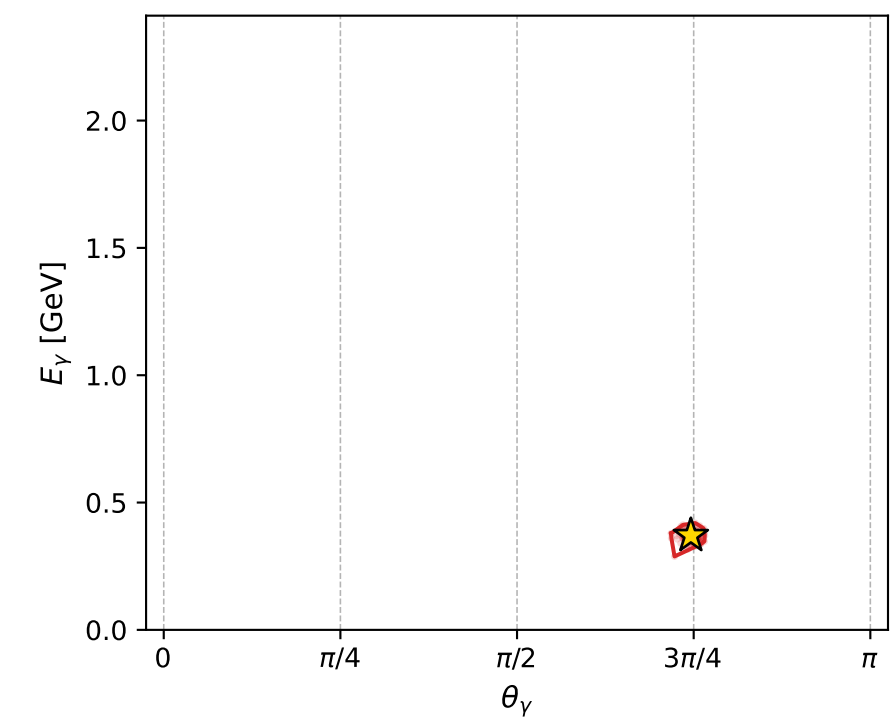
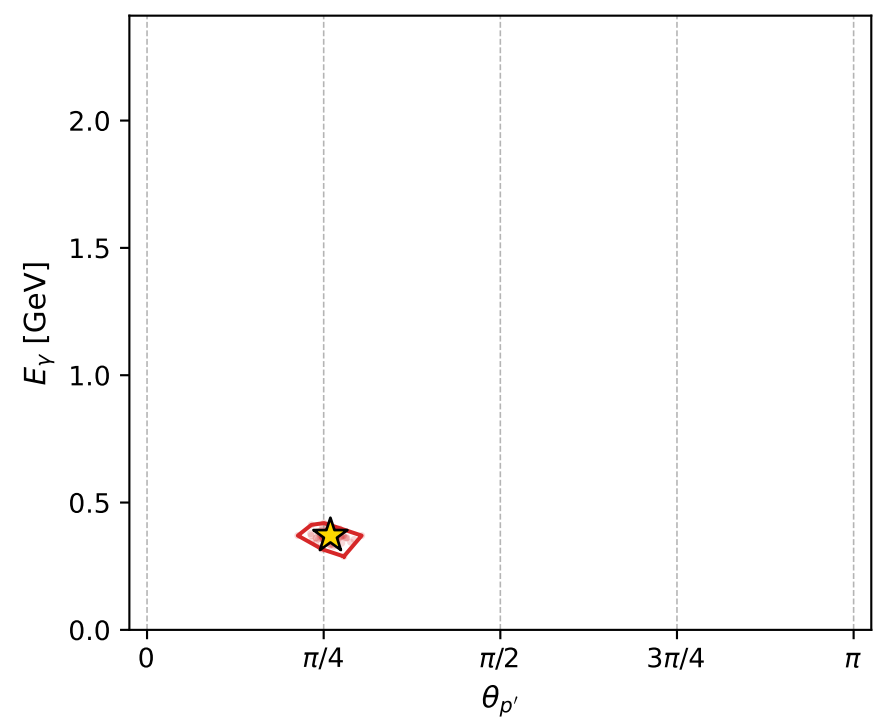
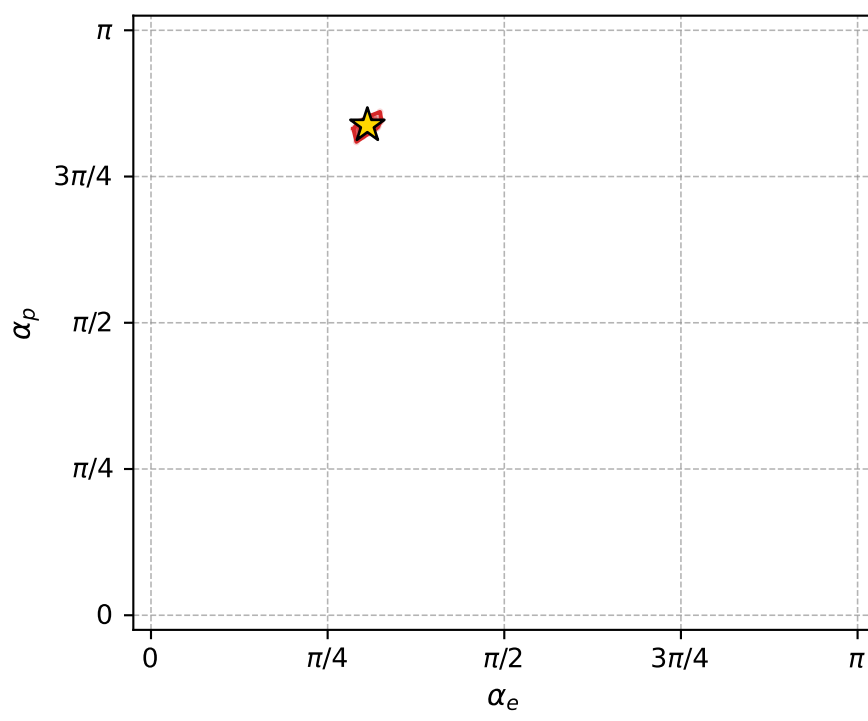
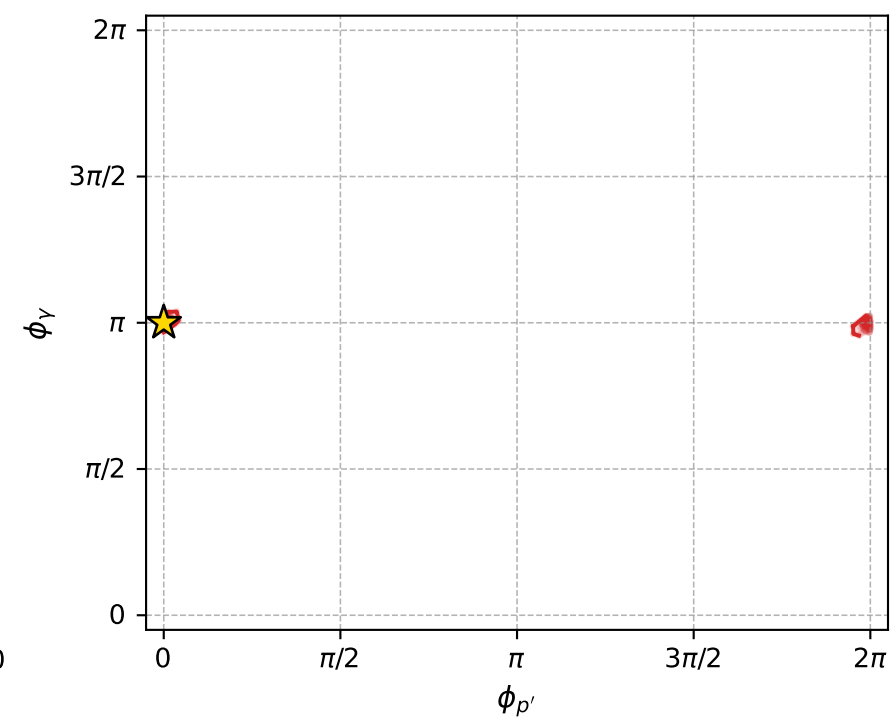
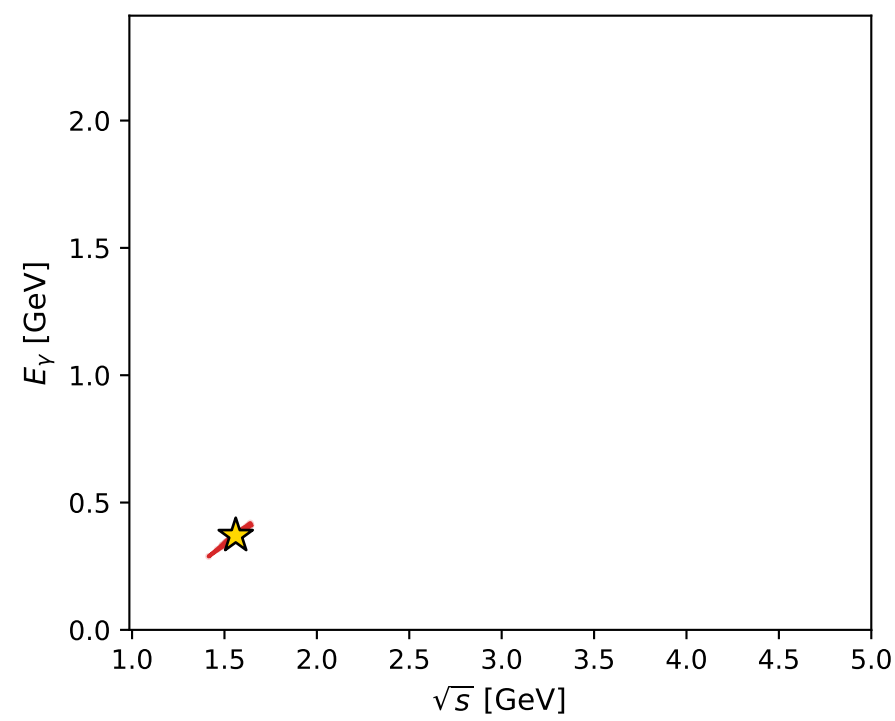
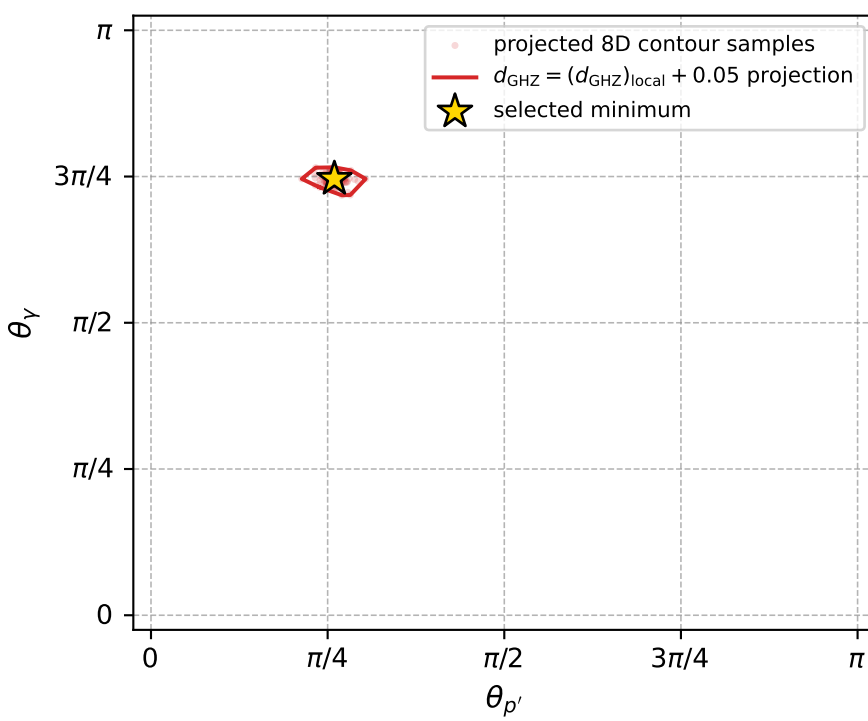
dghz_mixing_angles_polarization_02_local_minimum_191 d_{\rm GHZ}=1.65317e-05
 region: polarization_cluster_02_local_minimum_191
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=0.96200677$ rad
 incoming lepton: $0.571875|+\rangle +0.820341|-\rangle$
 proton mixing angle: $\alpha_p=2.6310204$ rad
 incoming proton: $-0.872465|+\rangle +0.488677|-\rangle$

$s=2.43664$, $\sqrt{s}=1.56098$, $|\vec{P}|=0.498662$, $|\vec{P}'|=0.125924$
 $\theta_{P'}=0.815321$, $\theta_Y=2.34363$, $\phi_{P'}=0$, $\phi_Y=3.14157$
 $E_Y=0.370228$, $\theta_{\ell'}=0.789011$, $\phi_{\ell'}=6.28315$
 $Q^2=0.415363$, $x_B=0.343455$, $t=-0.164982$, $W^2=1.67385$, $y=0.776829$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.4987$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.4987$
 P $E= 1.0623$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.4987$
 ℓ' $E= 0.2443$ $p_x= 0.1734$ $p_y= -0.0000$ $p_z= 0.1721$
 P' $E= 0.9464$ $p_x= 0.0917$ $p_y= 0.0000$ $p_z= 0.0863$
 q_Y $E= 0.3702$ $p_x= -0.2651$ $p_y= 0.0000$ $p_z= -0.2585$



electron: polarization cluster P2, local minimum 191; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 1.6531737\text{e-}05$
 $d_{\text{GHZ}} \text{ contour} = 0.050016532$
 above global minimum = $1.6509453\text{e-}05$
 8D contour samples = 256

selected phase-space point:

$\text{sqrt_s} = 1.560975$
 $\text{theta_p_out} = 0.8153207$
 $\text{theta_gamma_out} = 2.343634$
 $\text{q0ut} = 0.3702275$
 $\text{phi_p_out} = 0$
 $\text{phi_gamma_out} = 3.141573$
 $\text{alpha_e} = 0.9620068$
 $\text{alpha_p} = 2.63102$

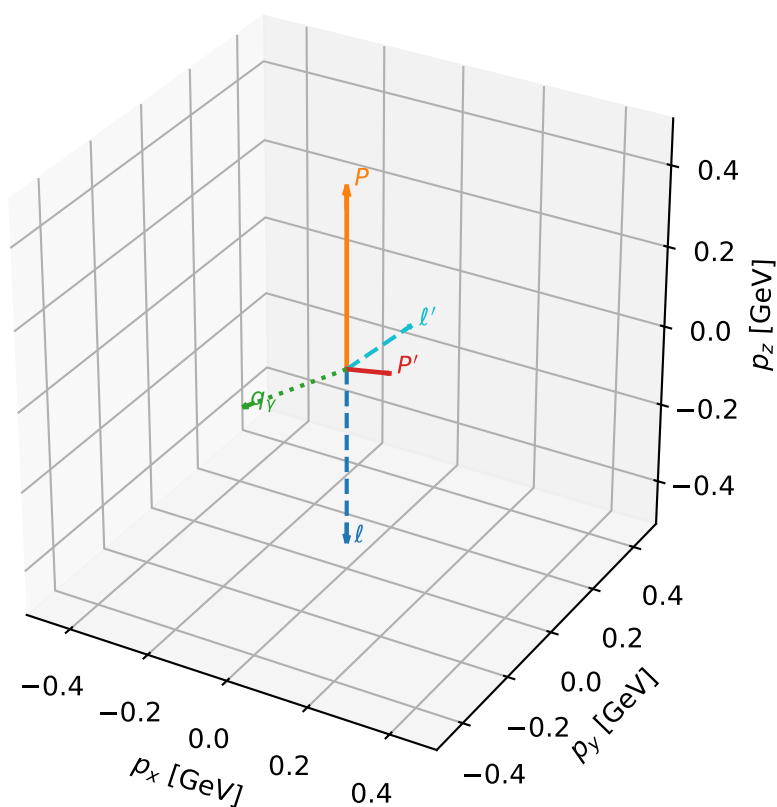
coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

dghz_mixing_angles_polarization_02_local_minimum_195 $d_{\{\text{rm GHZ}\}}=1.87442\text{e-}05$
 region: polarization_cluster_02_local_minimum_195
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=0.9771237$ rad
 incoming lepton: $0.559409|+\rangle +0.828892|-\rangle$
 proton mixing angle: $\alpha_p=2.6442362$ rad
 incoming proton: $-0.878847|+\rangle +0.477104|-\rangle$

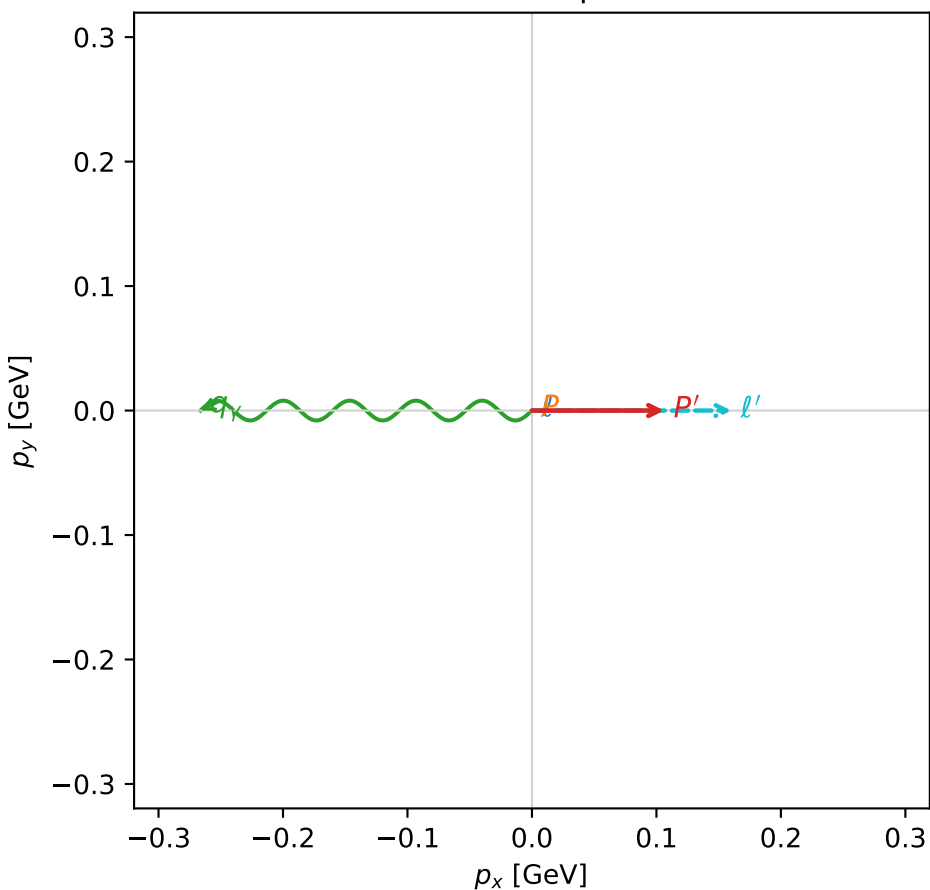
$s=2.20516$, $\sqrt{s}=1.48498$, $|\vec{P}|=0.446242$, $|\vec{P}'|=0.108153$
 $\theta_{P'}=1.38284$, $\theta_Y=2.15071$, $\phi_{P'}=2.53083\text{e-}05$, $\phi_Y=3.14163$
 $E_Y=0.318422$, $\theta_{\ell'}=0.803986$, $\phi_{\ell'}=4.9851\text{e-}05$
 $Q^2=0.336121$, $x_B=0.335754$, $t=-0.183858$, $W^2=1.54481$, $y=0.755358$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.4462$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.4462$
 P $E= 1.0387$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.4462$
 ℓ' $E= 0.2223$ $p_x= 0.1601$ $p_y= 0.0000$ $p_z= 0.1543$
 P' $E= 0.9442$ $p_x= 0.1062$ $p_y= 0.0000$ $p_z= 0.0202$
 q_Y $E= 0.3184$ $p_x= -0.2664$ $p_y= -0.0000$ $p_z= -0.1745$

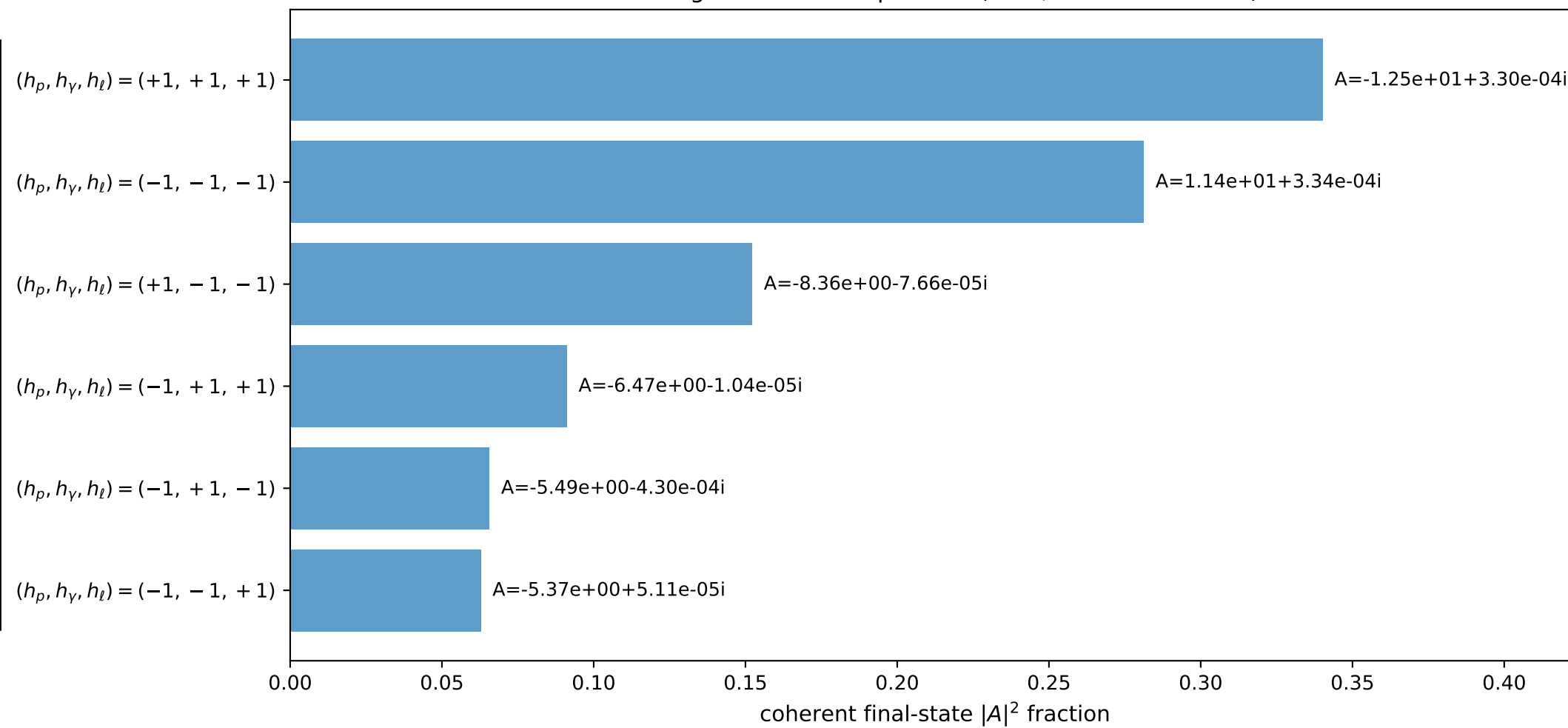
3D momenta



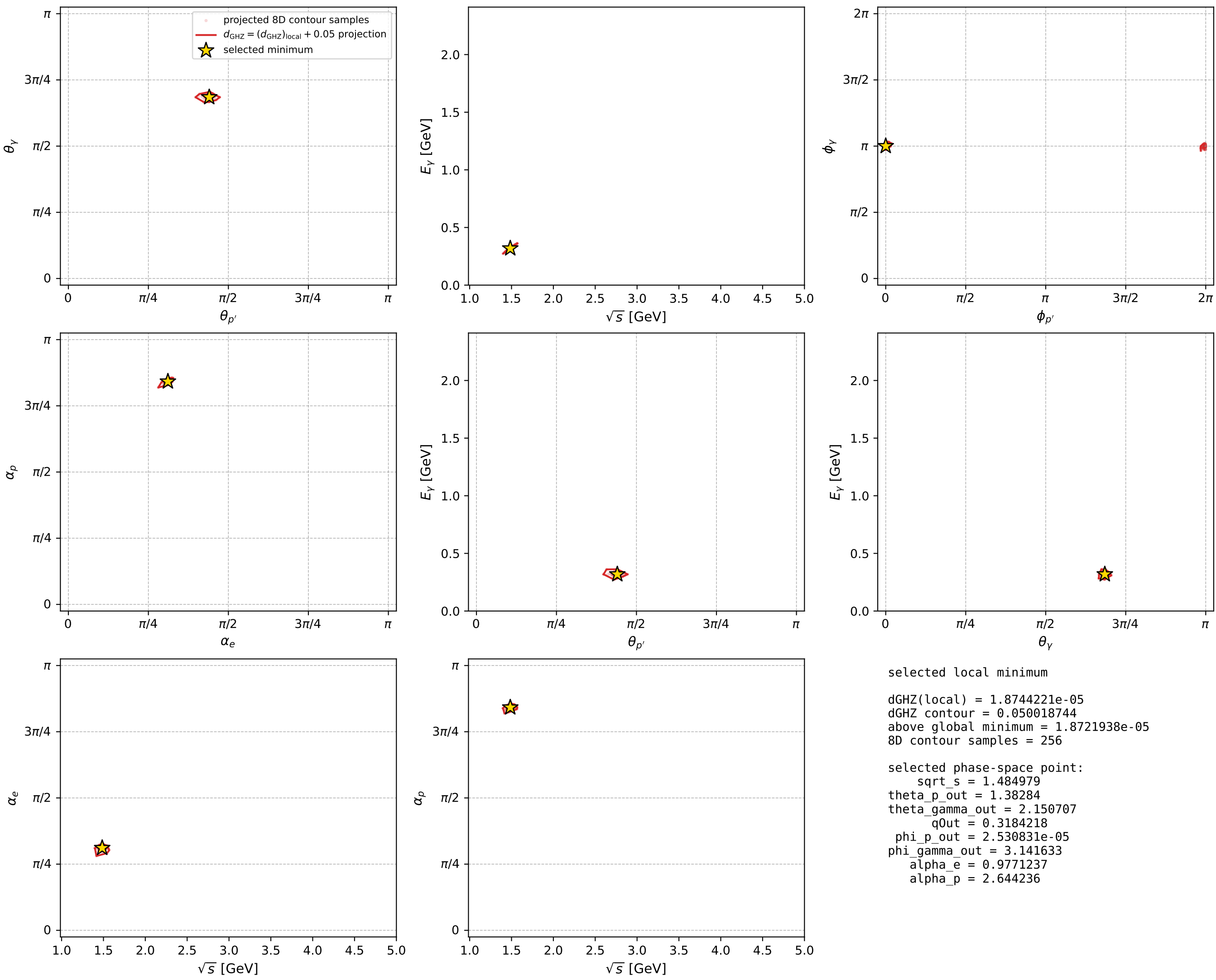
Transverse plane



Leading final-state amplitudes (N=6, retained=99.3%)



electron: polarization cluster P2, local minimum 195; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



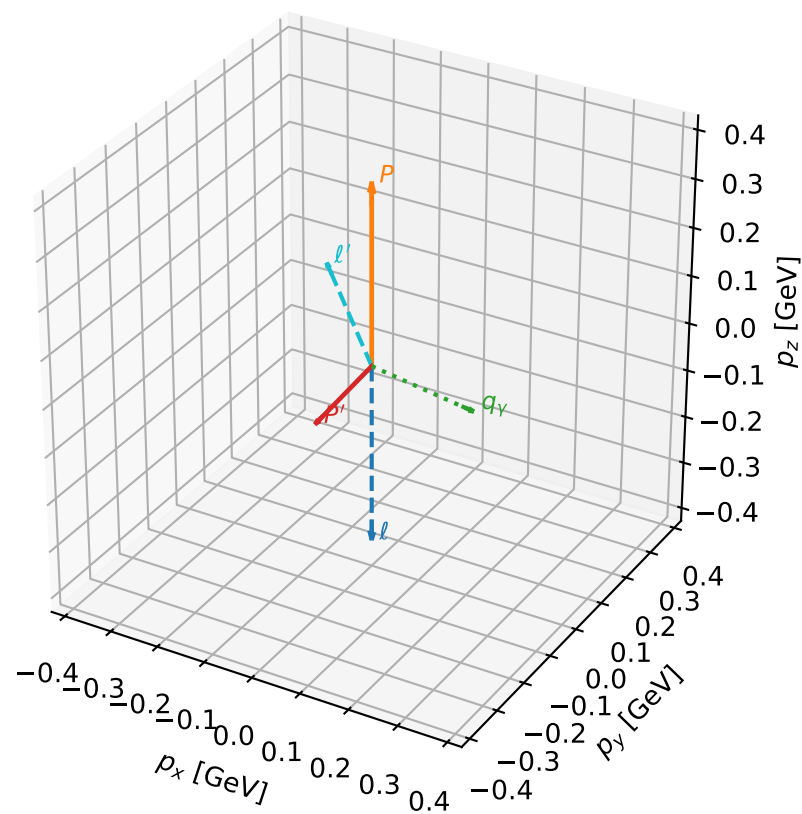
coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

dghz_mixing_angles_polarization_02_local_minimum_199 $d_{\{\text{rm GHZ}\}}=2.15468\text{e-}05$
 region: polarization_cluster_02_local_minimum_199
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=0.29778618$ rad
 incoming lepton: $0.955988|+\rangle +0.293405|-\rangle$
 proton mixing angle: $\alpha_p=1.8098026$ rad
 incoming proton: $-0.236737|+\rangle +0.971574|-\rangle$

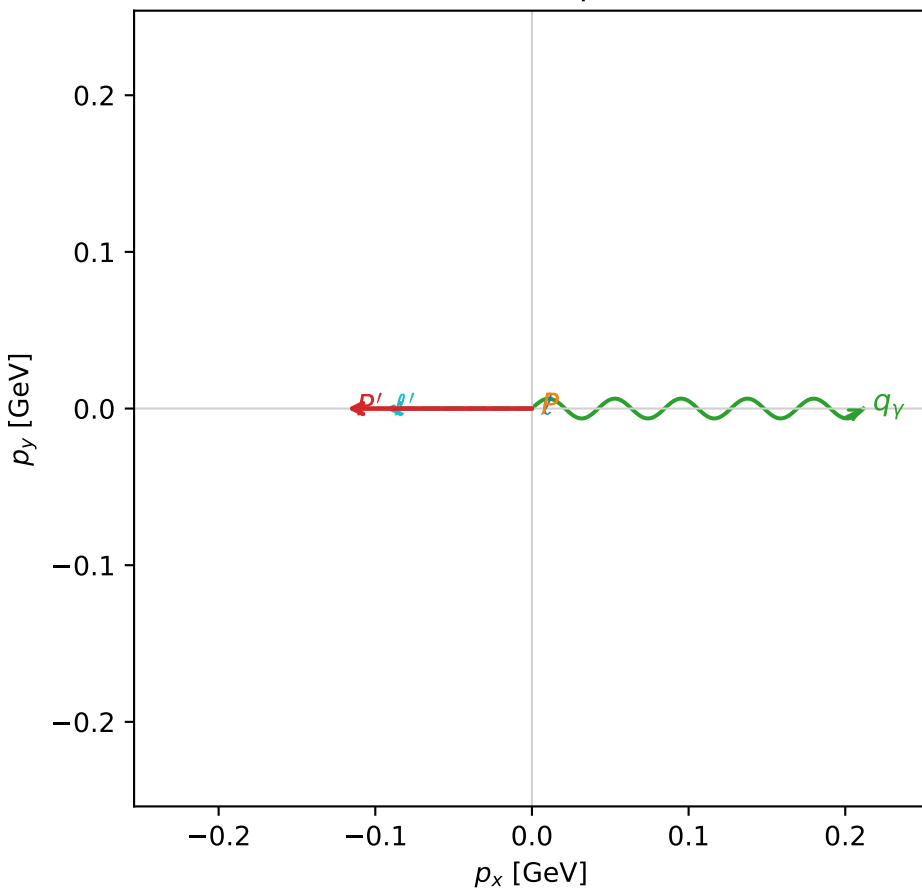
$s=1.91083$, $\sqrt{s}=1.38233$, $|\vec{P}|=0.372915$, $|\vec{P}'|=0.194713$
 $\theta_{P'}=2.49083$, $\theta_Y=1.72612$, $\phi_{P'}=3.14163$, $\phi_Y=1.92672\text{e-}05$
 $E_Y=0.214225$, $\theta_{\ell'}=0.462206$, $\phi_{\ell'}=3.14159$
 $Q^2=0.296962$, $x_B=0.397498$, $t=-0.289878$, $W^2=1.32996$, $y=0.724627$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.3729$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.3729$
 P $E= 1.0094$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.3729$
 ℓ' $E= 0.2101$ $p_x= -0.0937$ $p_y= 0.0000$ $p_z= 0.1881$
 P' $E= 0.9580$ $p_x= -0.1180$ $p_y= -0.0000$ $p_z= -0.1549$
 q_Y $E= 0.2142$ $p_x= 0.2116$ $p_y= 0.0000$ $p_z= -0.0331$

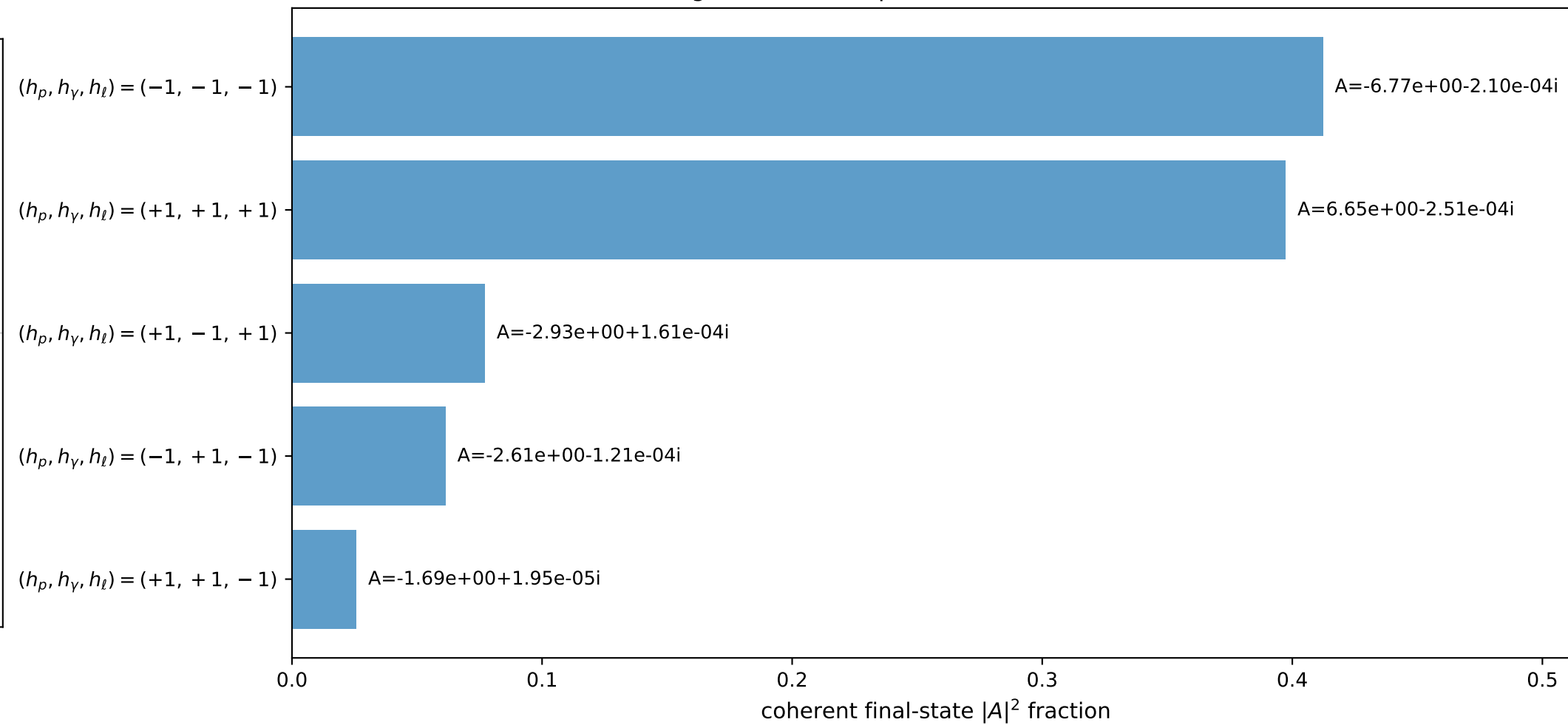
3D momenta



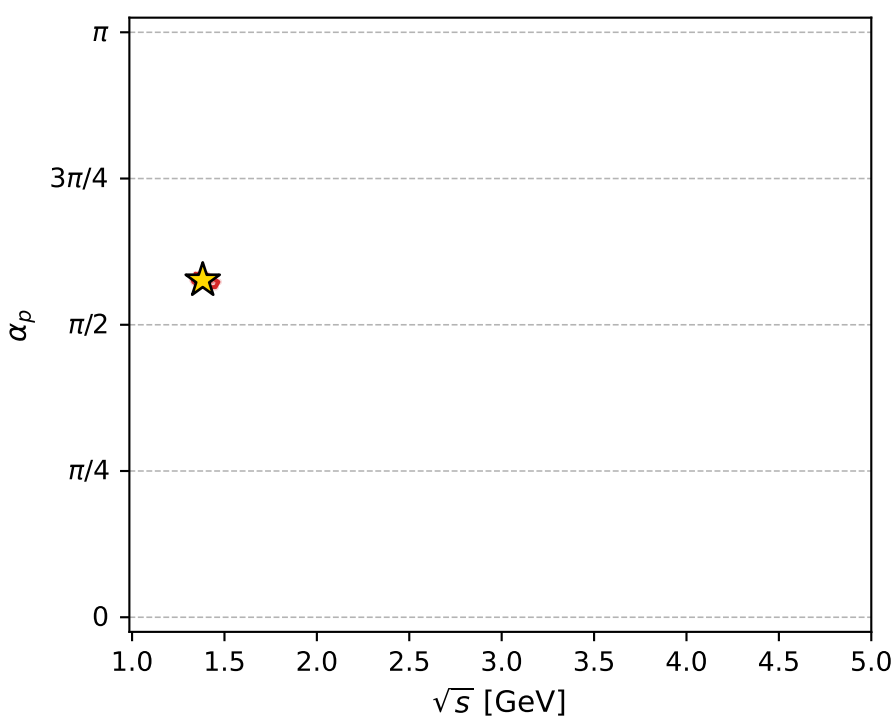
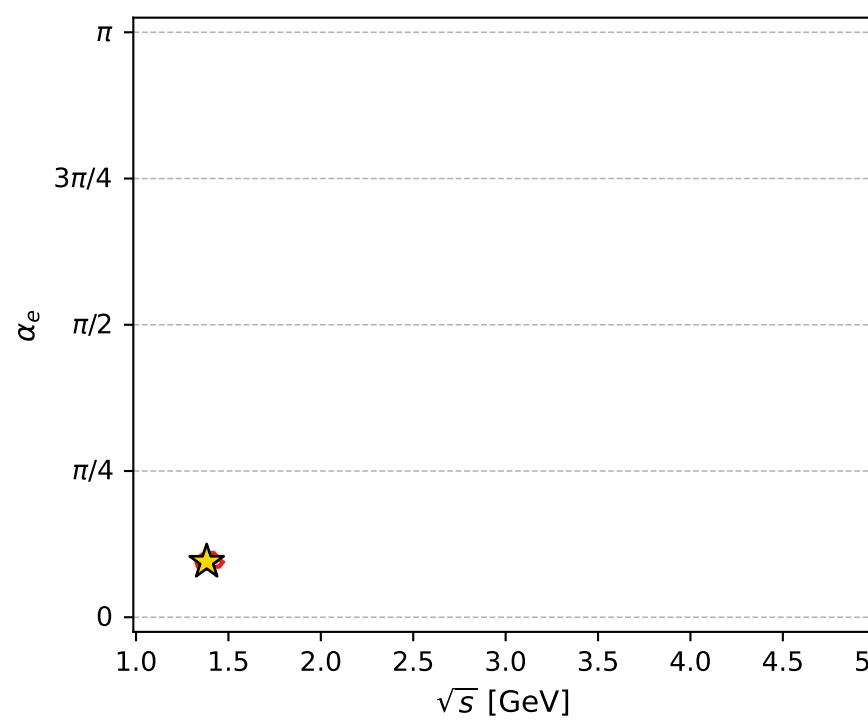
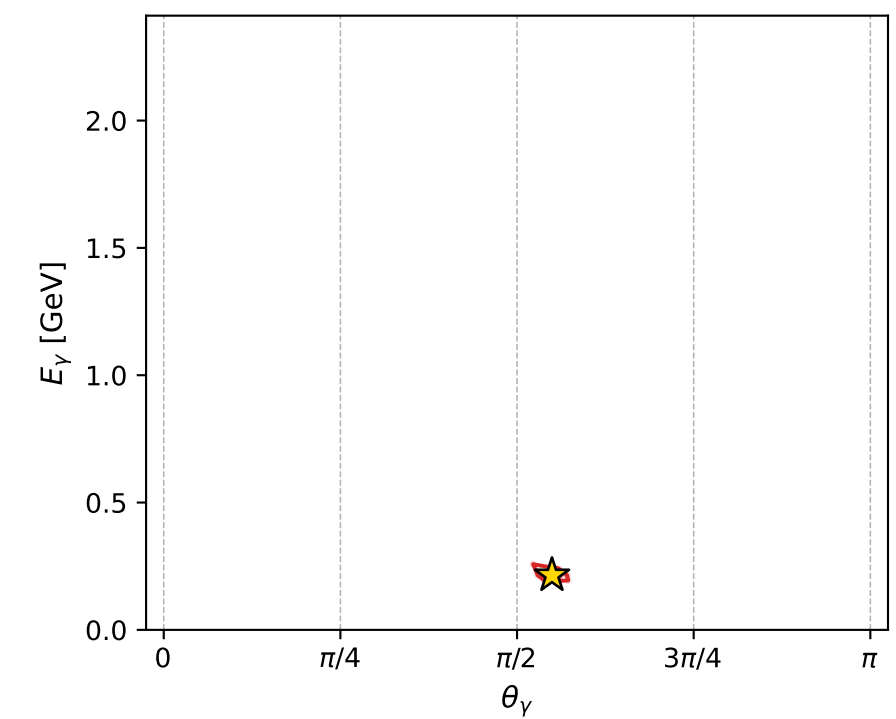
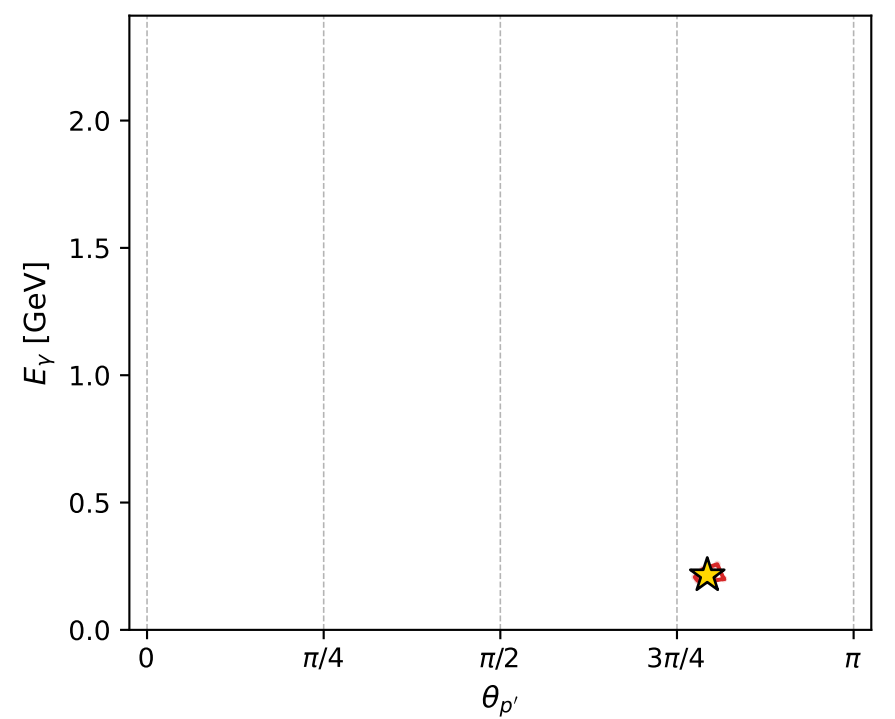
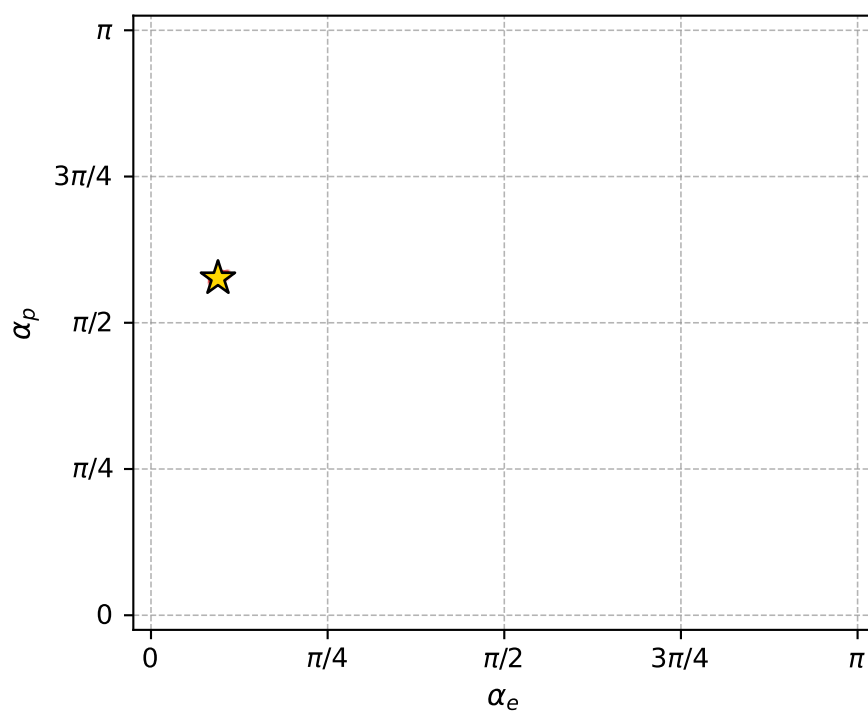
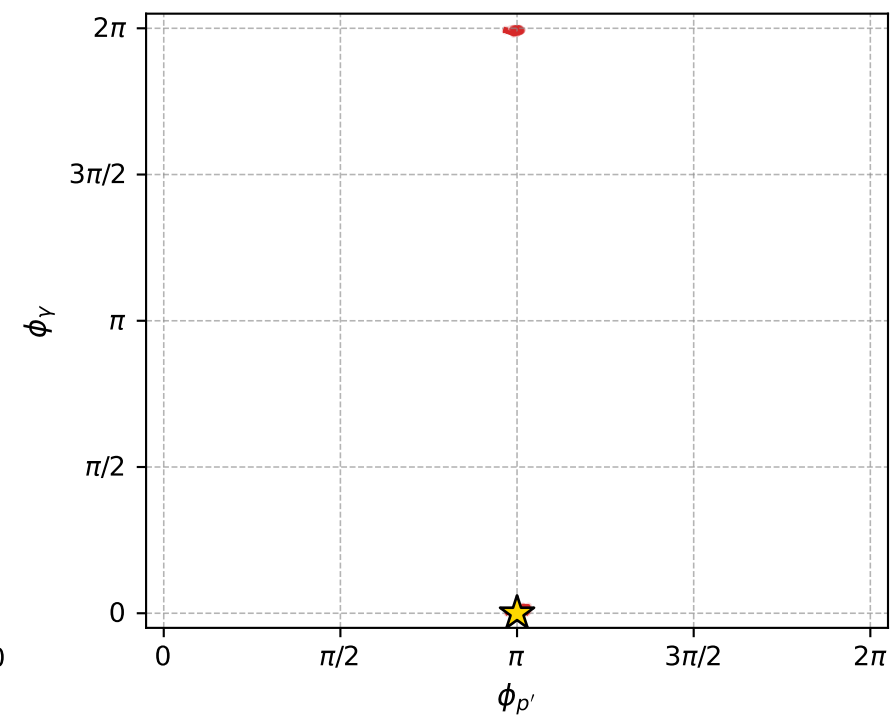
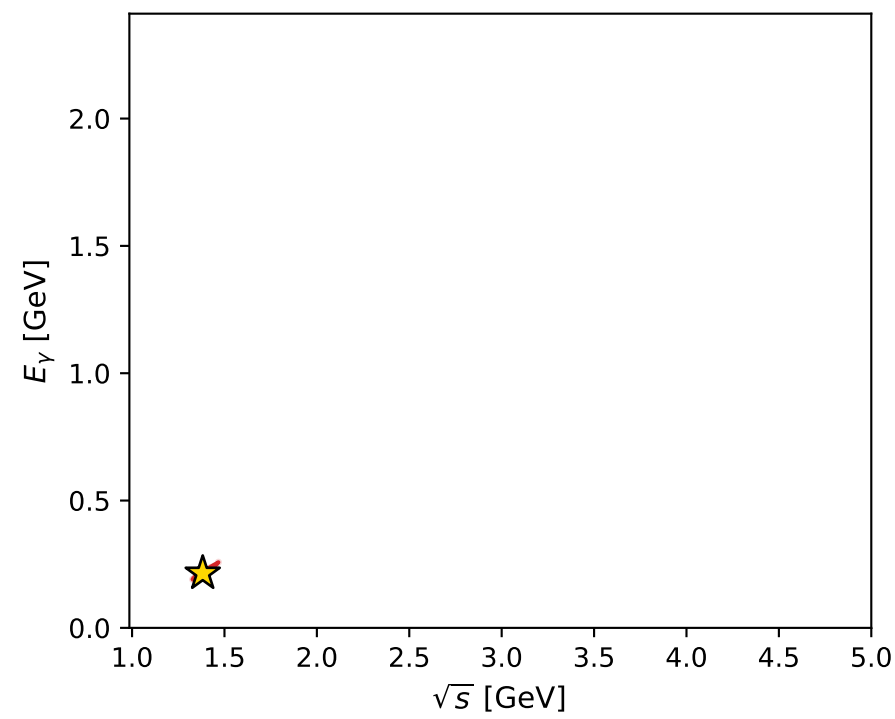
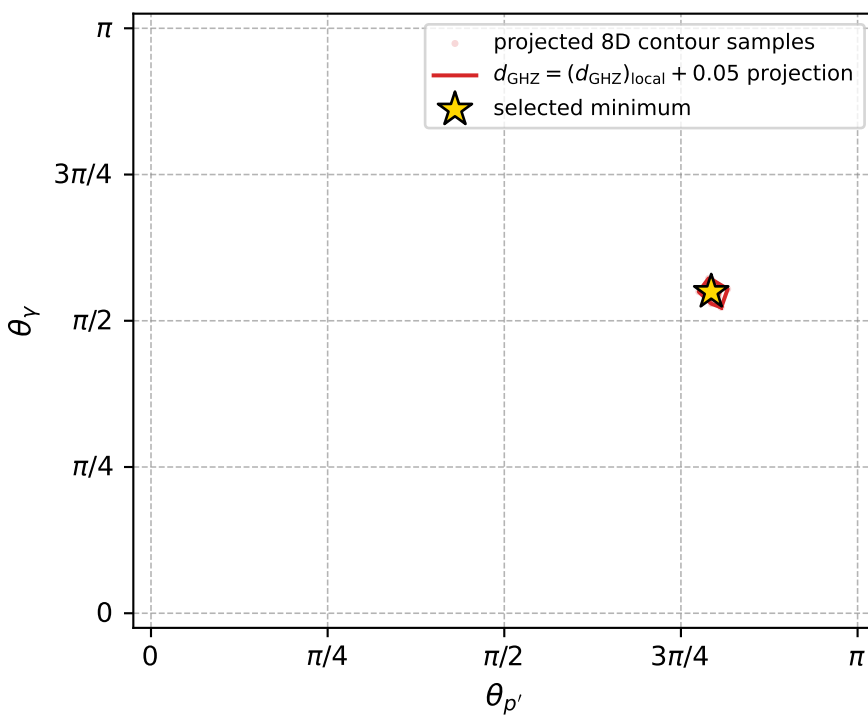
Transverse plane



Leading final-state amplitudes (N=5, retained=97.4%)



electron: polarization cluster P2, local minimum 199; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 2.1546757\text{e-}05$
 $d_{\text{GHZ}} \text{ contour} = 0.050021547$
 above global minimum = $2.1524474\text{e-}05$
 8D contour samples = 256

selected phase-space point:
 $\text{sqrt_s} = 1.382326$
 $\text{theta_p_out} = 2.490833$
 $\text{theta_gamma_out} = 1.726116$
 $\text{q0out} = 0.2142248$
 $\text{phi_p_out} = 3.141627$
 $\text{phi_gamma_out} = 1.926716\text{e-}05$
 $\text{alpha_e} = 0.2977862$
 $\text{alpha_p} = 1.809803$

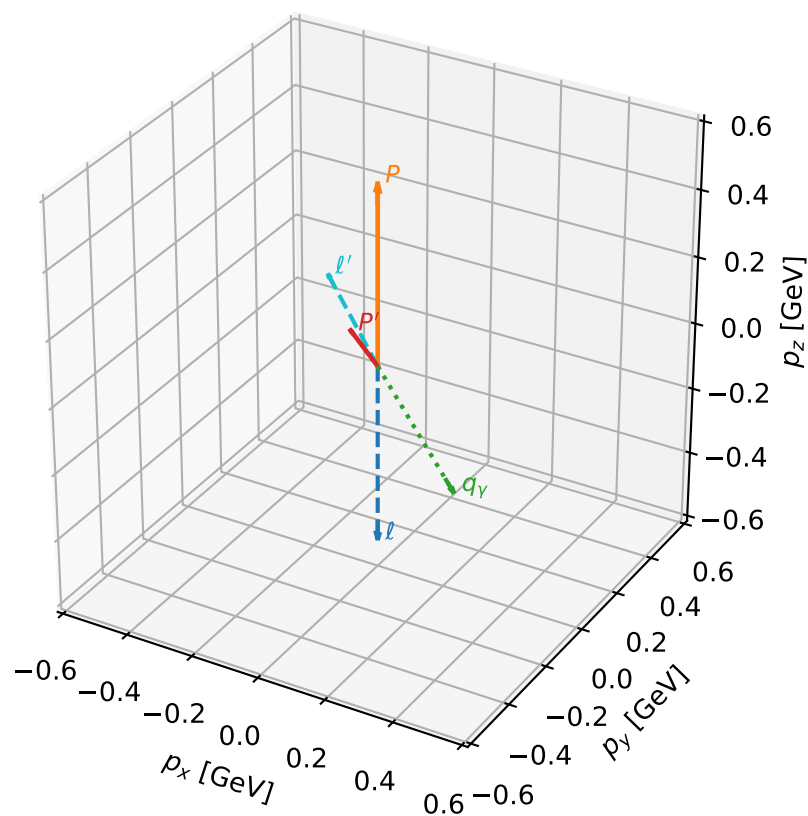
coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

dghz_mixing_angles_polarization_02_local_minimum_202 d_{\rm GHZ}=2.30639e-05
region: polarization_cluster_02_local_minimum_202
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: $\alpha_e=0.44035838$ rad
incoming lepton: $0.904599|+\rangle +0.426264|-\rangle$
proton mixing angle: $\alpha_p=1.9343705$ rad
incoming proton: $-0.355617|+\rangle +0.934632|-\rangle$

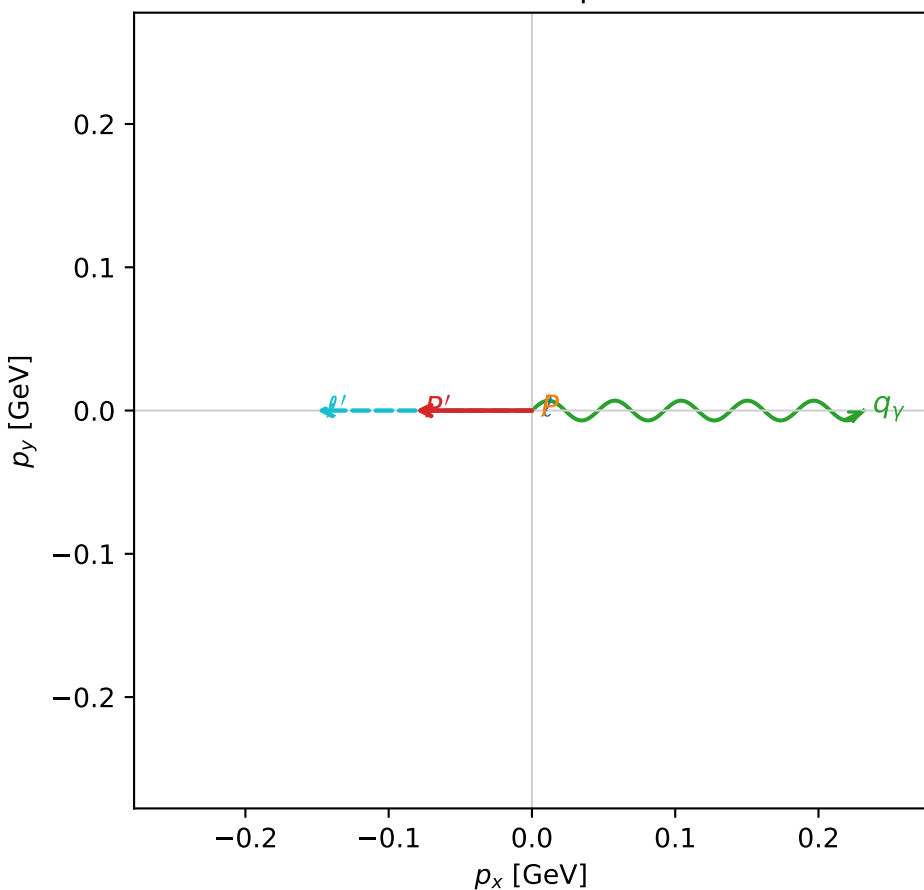
$s=2.61462$, $\sqrt{s}=1.61698$, $|\vec{P}|=0.536425$, $|\vec{P}'|=0.117035$
 $\theta_{P'}=0.768953$, $\theta_Y=2.51296$, $\phi_{P'}=3.14161$, $\phi_Y=0$
 $E_Y=0.393554$, $\theta_{\ell'}=0.569745$, $\phi_{\ell'}=3.14158$
 $Q^2=0.549692$, $x_B=0.396907$, $t=-0.192915$, $W^2=1.71509$, $y=0.798337$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.5364$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.5364$
 P $E= 1.0806$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.5364$
 ℓ' $E= 0.2782$ $p_x= -0.1500$ $p_y= 0.0000$ $p_z= 0.2342$
 P' $E= 0.9453$ $p_x= -0.0814$ $p_y= -0.0000$ $p_z= 0.0841$
 q_Y $E= 0.3936$ $p_x= 0.2314$ $p_y= 0.0000$ $p_z= -0.3183$

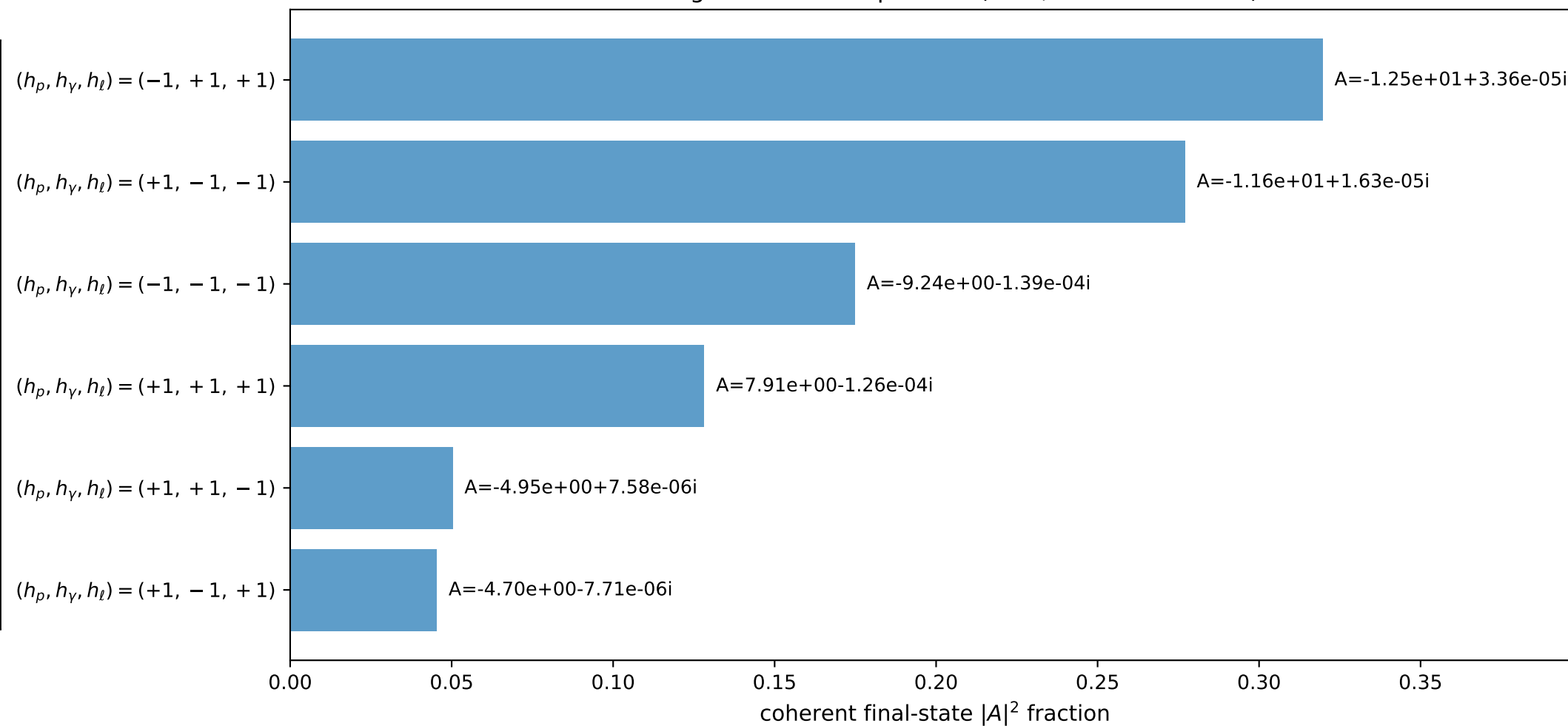
3D momenta



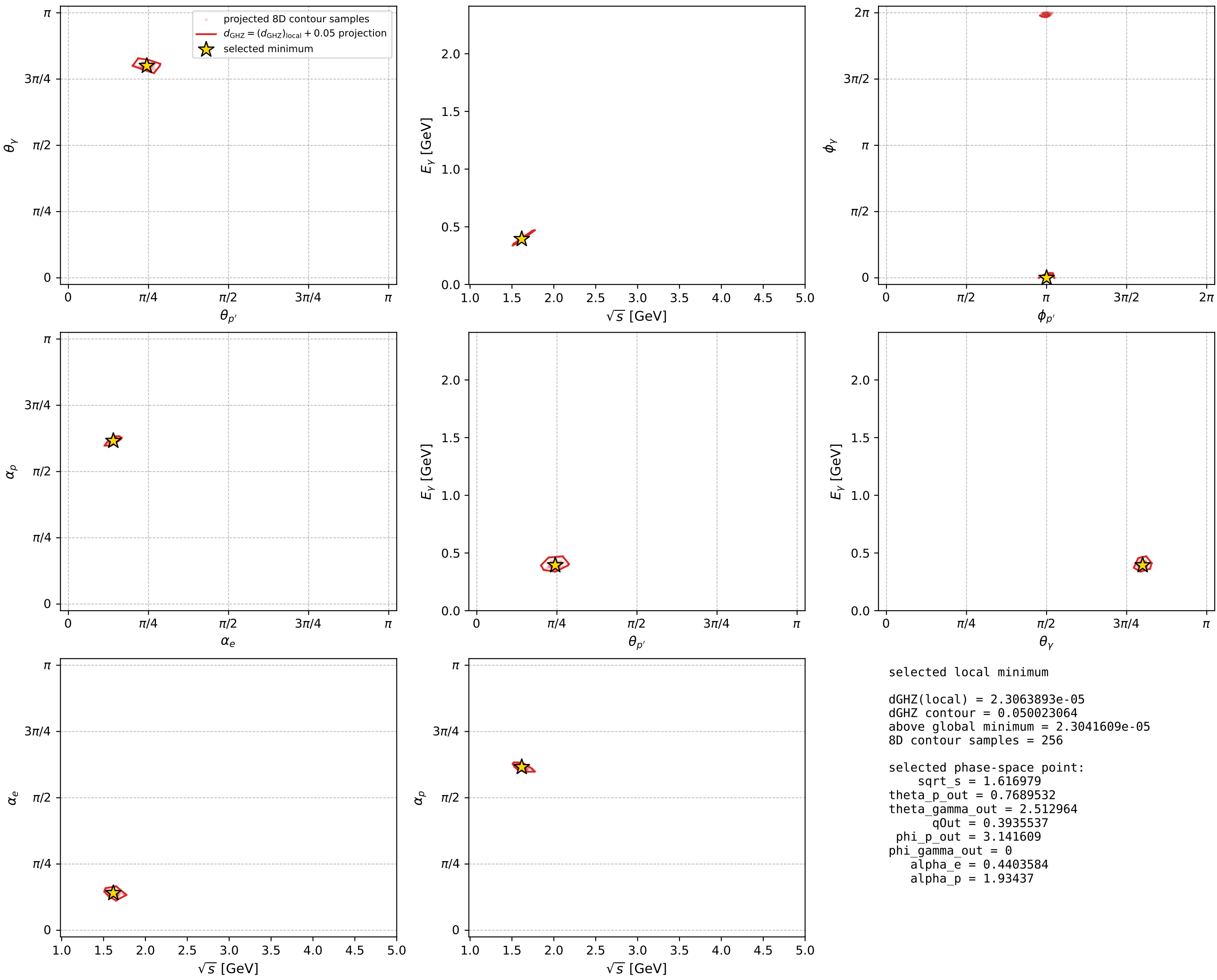
Transverse plane



Leading final-state amplitudes (N=6, retained=99.5%)



electron: polarization cluster P2, local minimum 202; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour

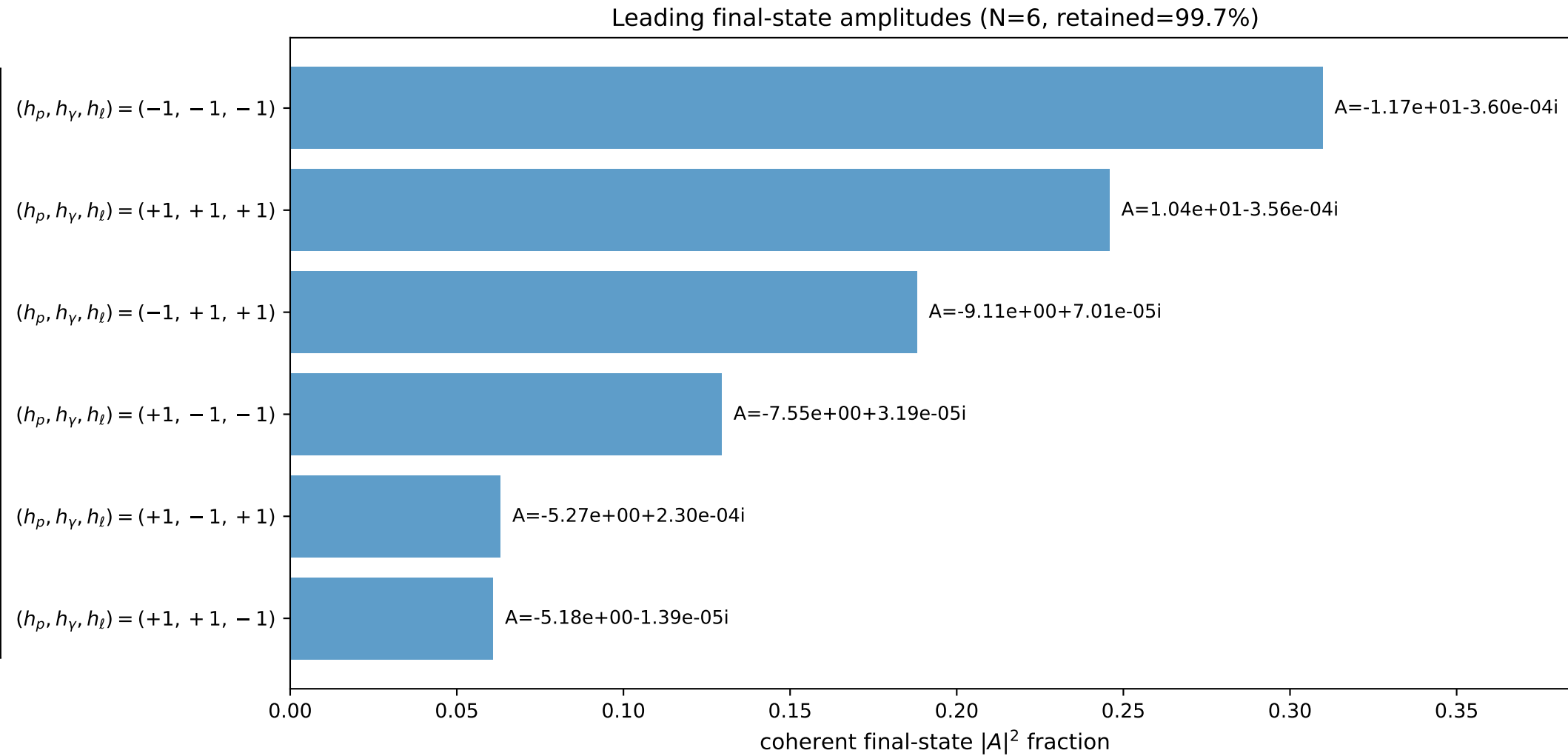
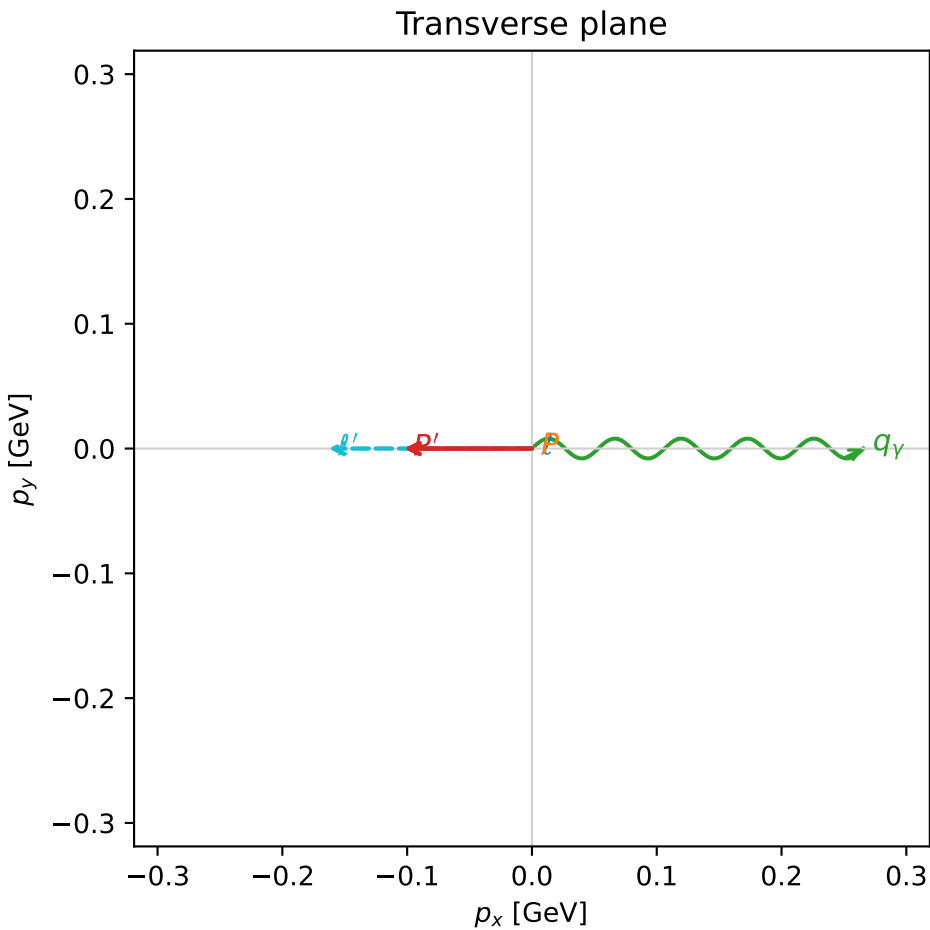
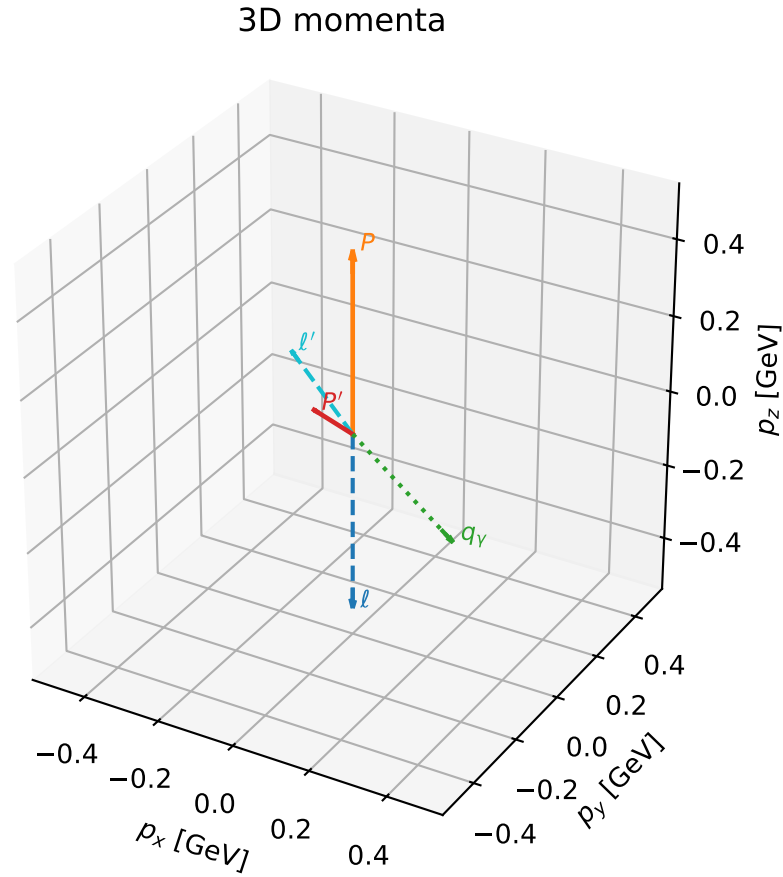


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

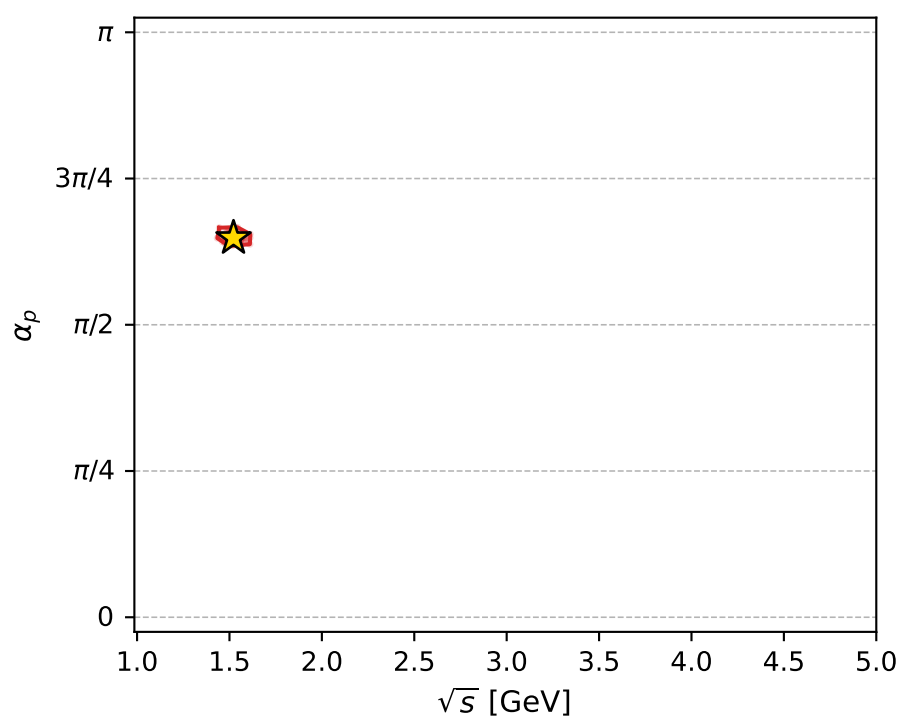
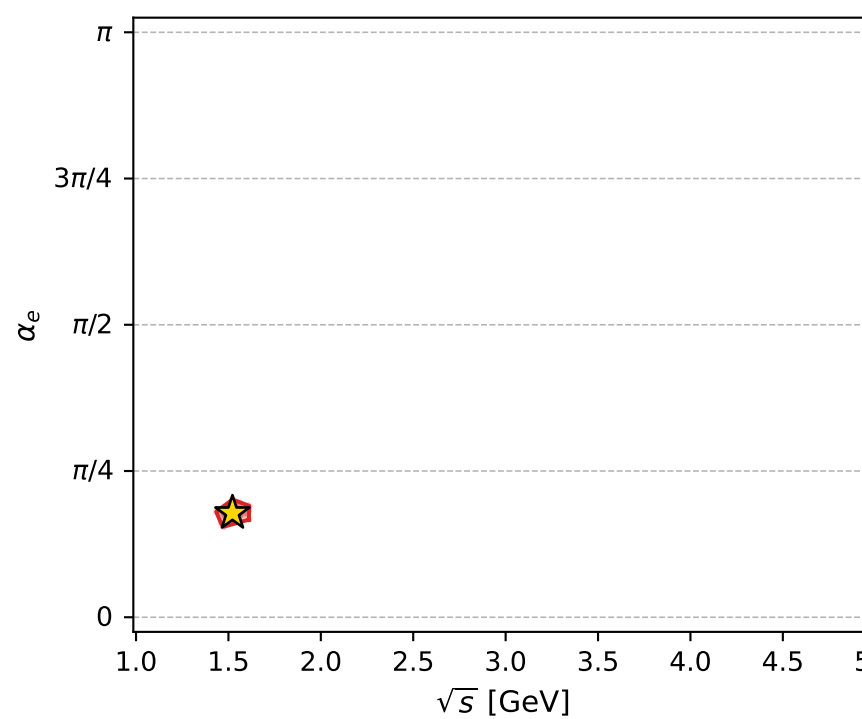
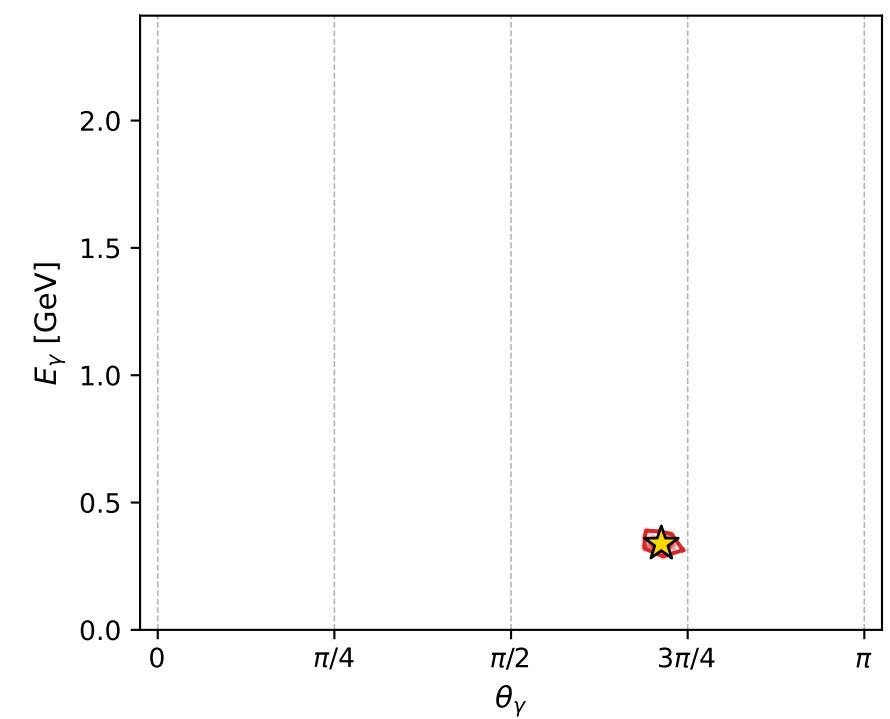
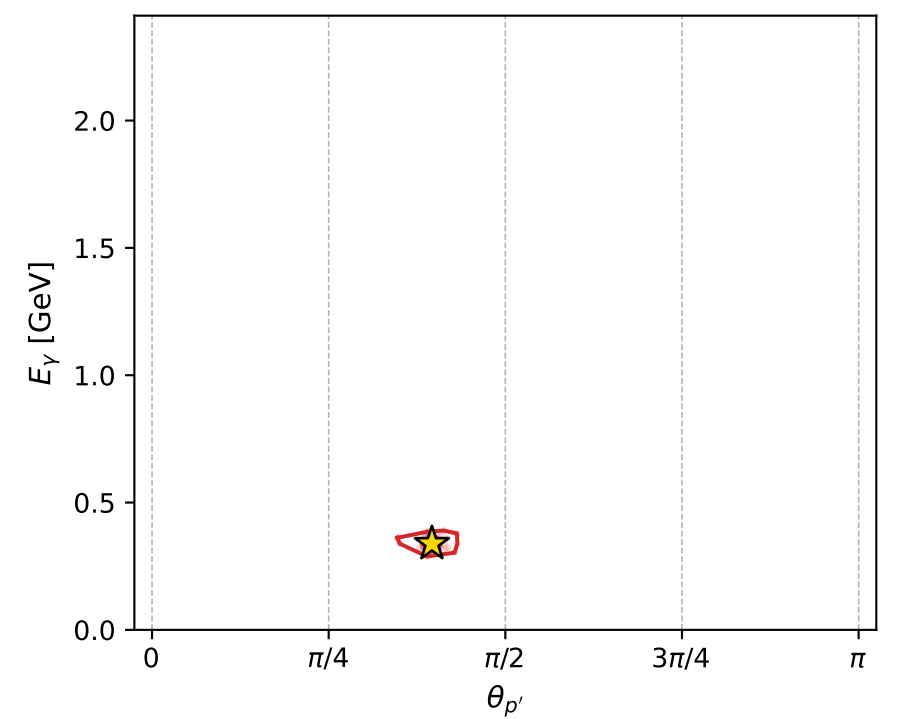
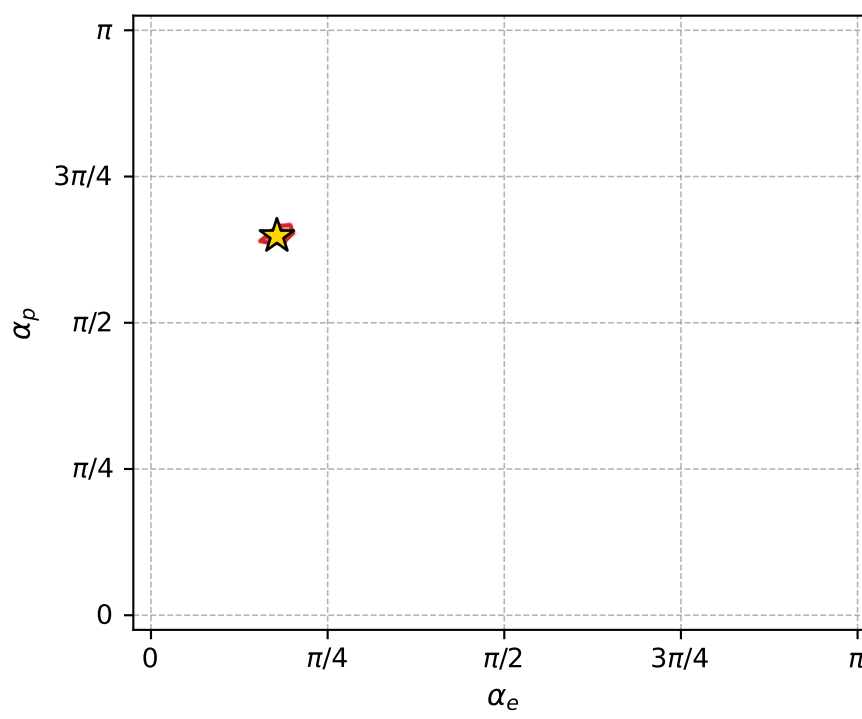
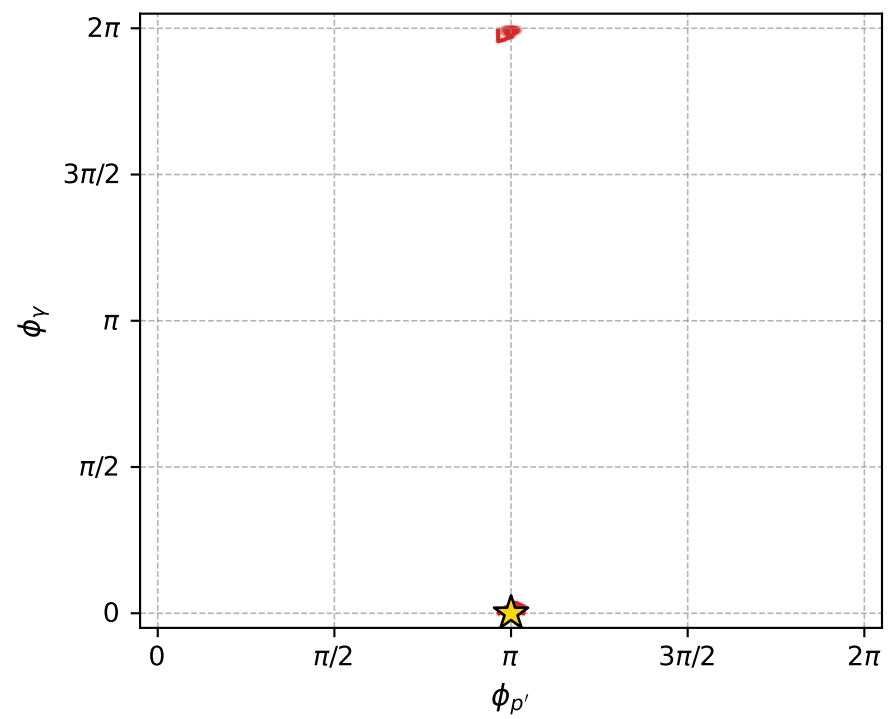
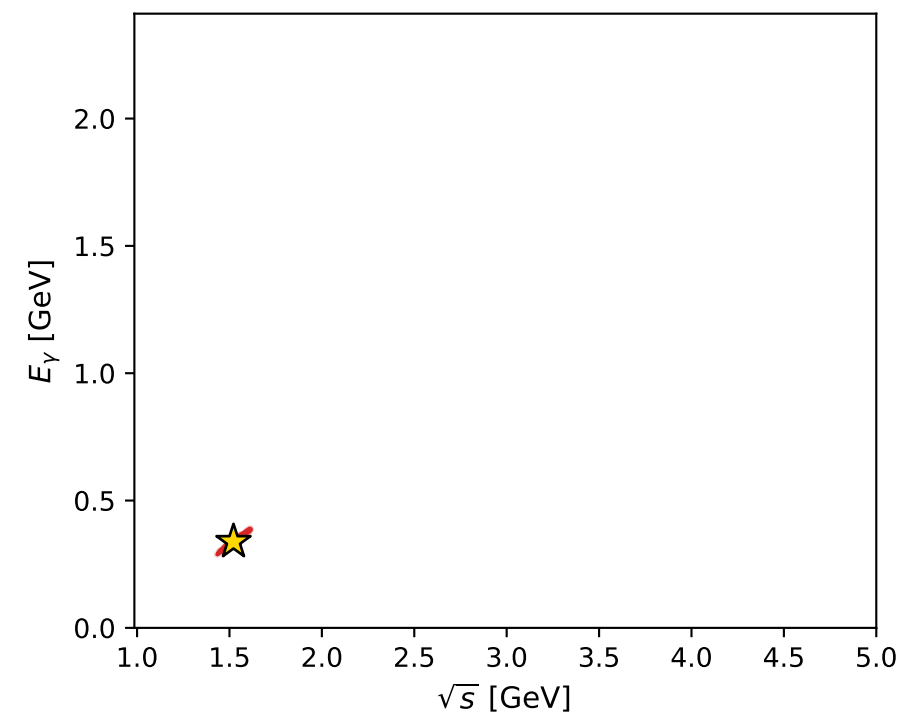
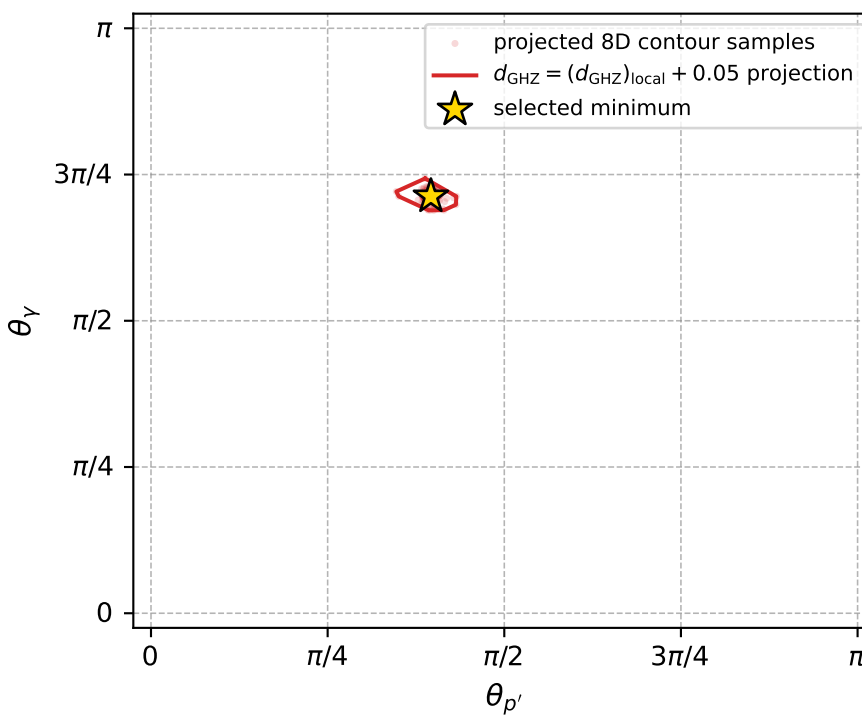
dghz_mixing_angles_polarization_02_local_minimum_217 $d_{\{\text{rm GHZ}\}}=4.09954\text{e-}05$
 region: polarization_cluster_02_local_minimum_217
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=0.55982921$ rad
 incoming lepton: $0.847346|+\rangle +0.531041|-\rangle$
 proton mixing angle: $\alpha_p=2.036484$ rad
 incoming proton: $-0.449037|+\rangle +0.893513|-\rangle$

$s=2.31584$, $\sqrt{s}=1.52179$, $|\vec{P}|=0.471811$, $|\vec{P}'|=0.108419$
 $\theta_{P'}=1.24447$, $\theta_Y=2.23894$, $\phi_{P'}=3.14163$, $\phi_Y=2.29619\text{e-}05$
 $E_Y=0.338456$, $\theta_{\ell'}=0.750064$, $\phi_{\ell'}=3.14161$
 $Q^2=0.390672$, $x_B=0.355484$, $t=-0.190385$, $W^2=1.58816$, $y=0.765315$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.4718$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.4718$
 P $E= 1.0500$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.4718$
 ℓ' $E= 0.2391$ $p_x= -0.1630$ $p_y= -0.0000$ $p_z= 0.1749$
 P' $E= 0.9442$ $p_x= -0.1027$ $p_y= -0.0000$ $p_z= 0.0348$
 q_Y $E= 0.3385$ $p_x= 0.2657$ $p_y= 0.0000$ $p_z= -0.2097$



electron: polarization cluster P2, local minimum 217; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 4.0995364\text{e-}05$
 $d_{\text{GHZ}} \text{ contour} = 0.050040995$
 above global minimum = $4.097308\text{e-}05$
 8D contour samples = 256

selected phase-space point:

$\text{sqrt_s} = 1.521788$
 $\text{theta_p_out} = 1.244467$
 $\text{theta_gamma_out} = 2.238936$
 $\text{q0out} = 0.3384558$
 $\text{phi_p_out} = 3.141625$
 $\text{phi_gamma_out} = 2.296195\text{e-}05$
 $\text{alpha_e} = 0.5598292$
 $\text{alpha_p} = 2.036484$

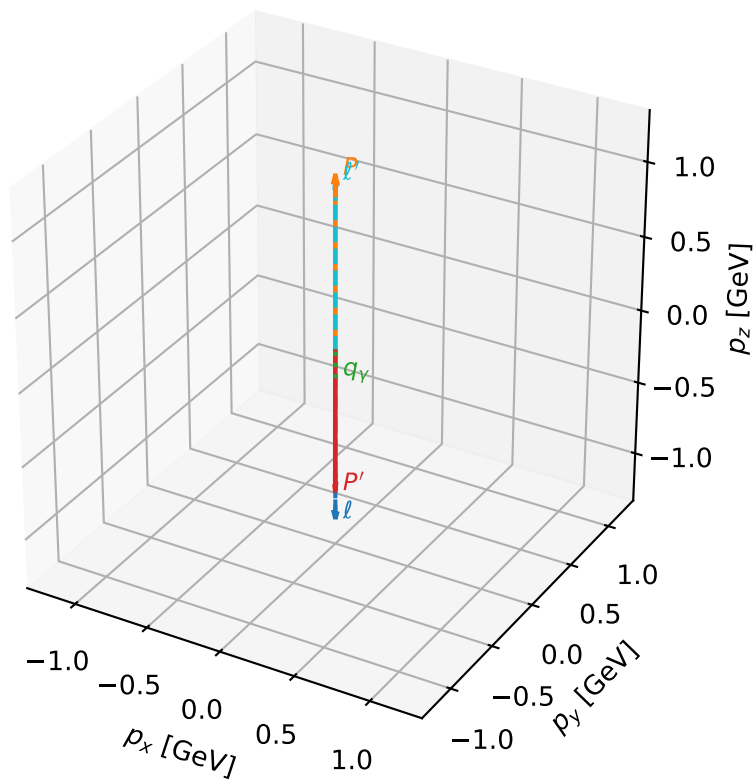
coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

dghz_mixing_angles_polarization_02_local_minimum_225 $d_{\{\text{rm GHZ}\}}=4.66004\text{e-}05$
 region: polarization_cluster_02_local_minimum_225
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=0.49415536$ rad
 incoming lepton: $0.88037|+\rangle +0.474288|-\rangle$
 proton mixing angle: $\alpha_p=2.0653512$ rad
 incoming proton: $-0.47464|+\rangle +0.88018|-\rangle$

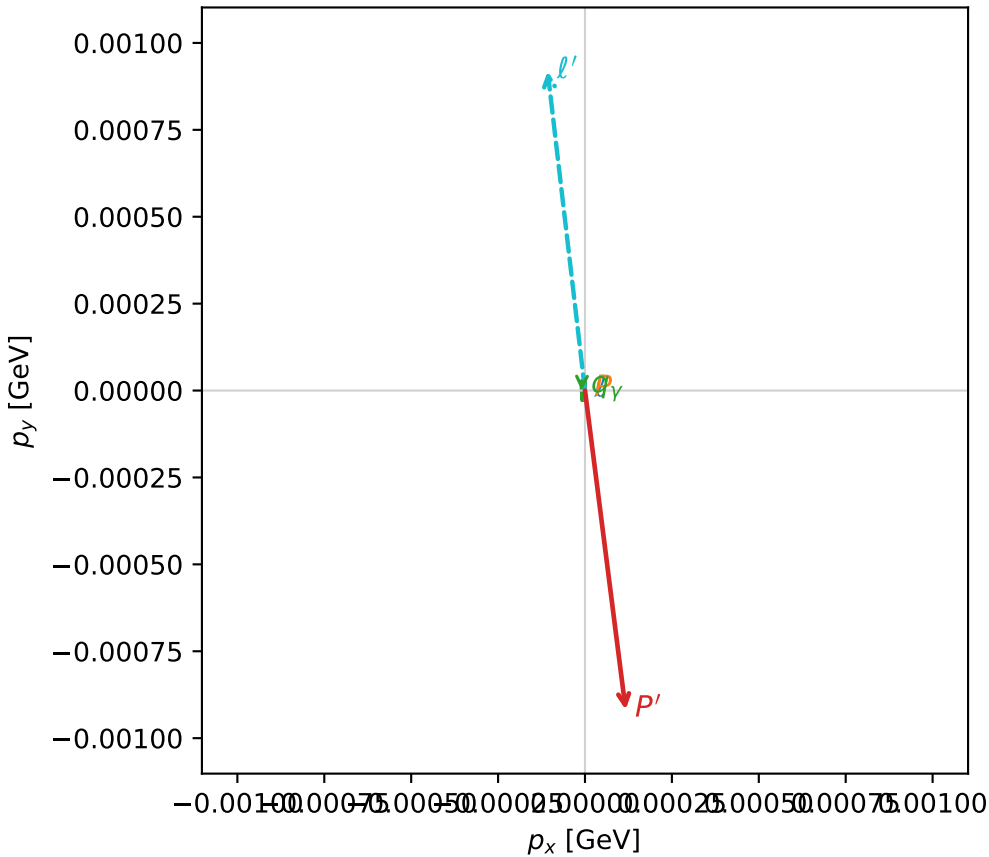
$s=7.0749$, $\sqrt{s}=2.65987$, $|\vec{P}|=1.16454$, $|\vec{P}'|=0.958683$
 $\theta_{P'}=3.14063$, $\theta_Y=3.14154$, $\phi_{P'}=4.83908$, $\phi_Y=3.14156$
 $E_Y=0.179974$, $\theta_{\ell'}=0.000812026$, $\phi_{\ell'}=1.68729$
 $Q^2=5.30405$, $x_B=0.974696$, $t=-4.48434$, $W^2=1.01754$, $y=0.878403$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 1.1645$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.1645$
 P $E= 1.4953$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.1645$
 ℓ' $E= 1.1387$ $p_x= -0.0001$ $p_y= 0.0009$ $p_z= 1.1387$
 P' $E= 1.3412$ $p_x= 0.0001$ $p_y= -0.0009$ $p_z= -0.9587$
 q_Y $E= 0.1800$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= -0.1800$

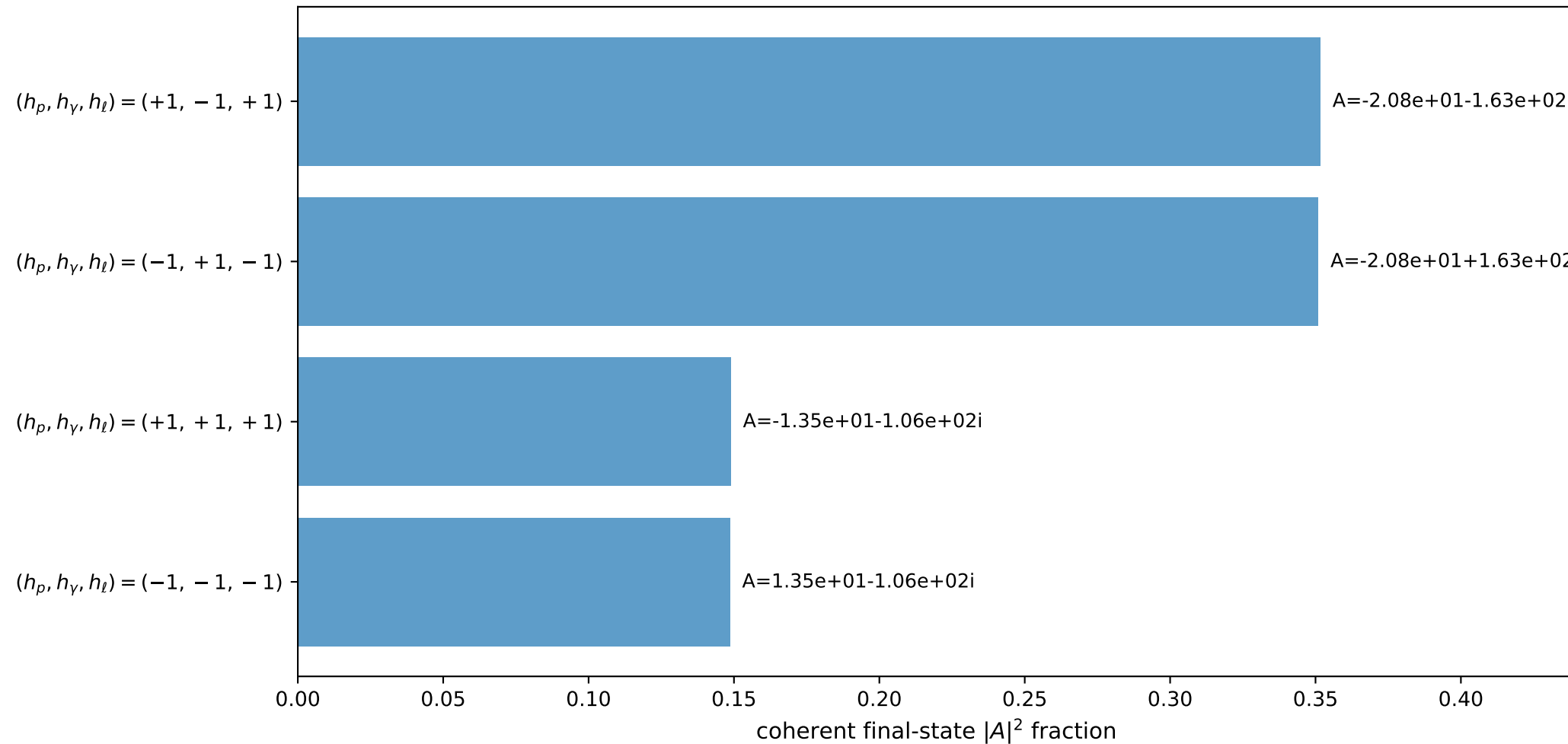
3D momenta



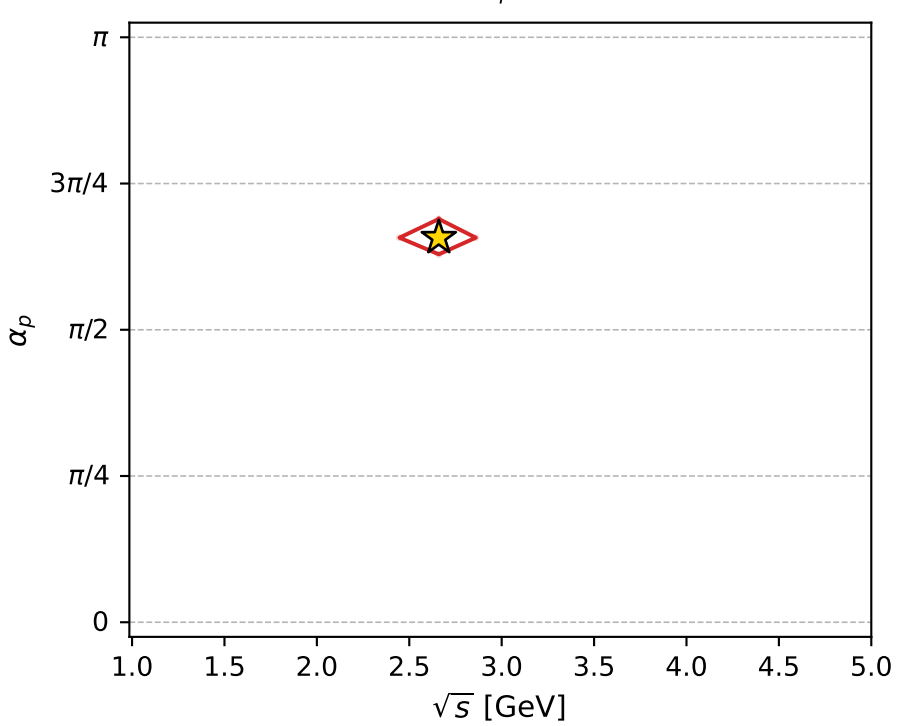
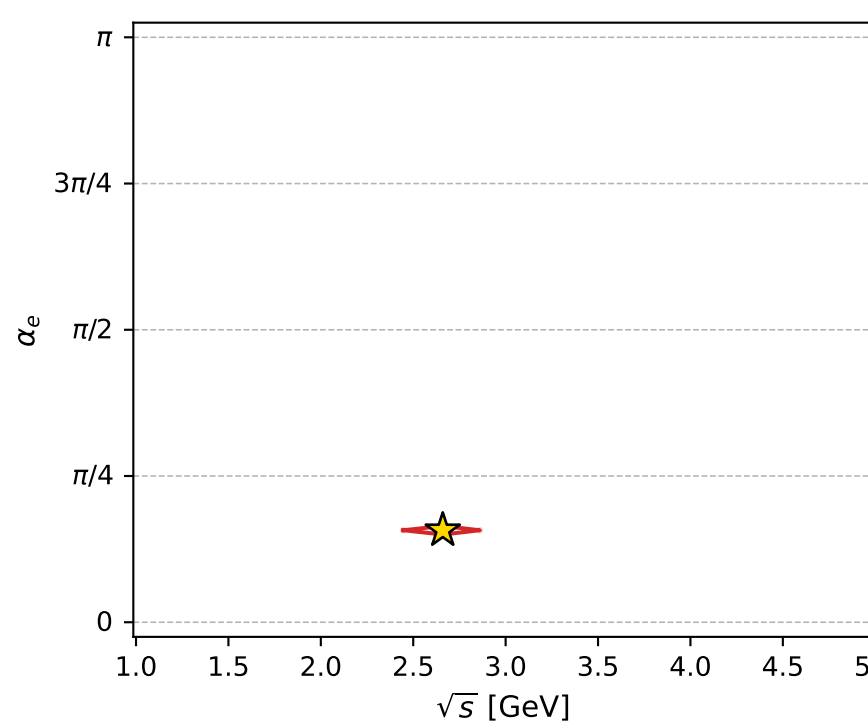
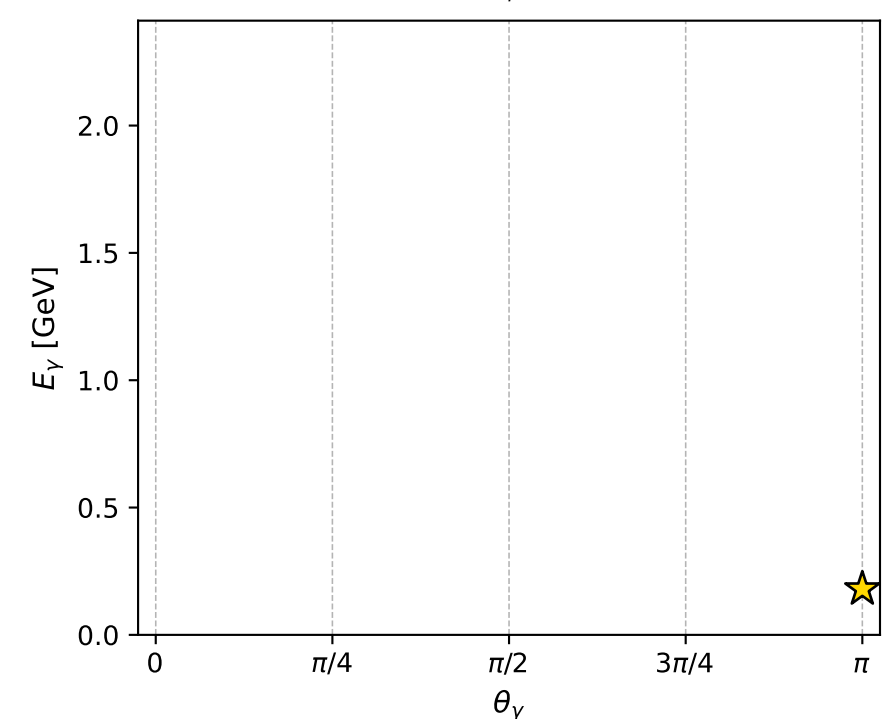
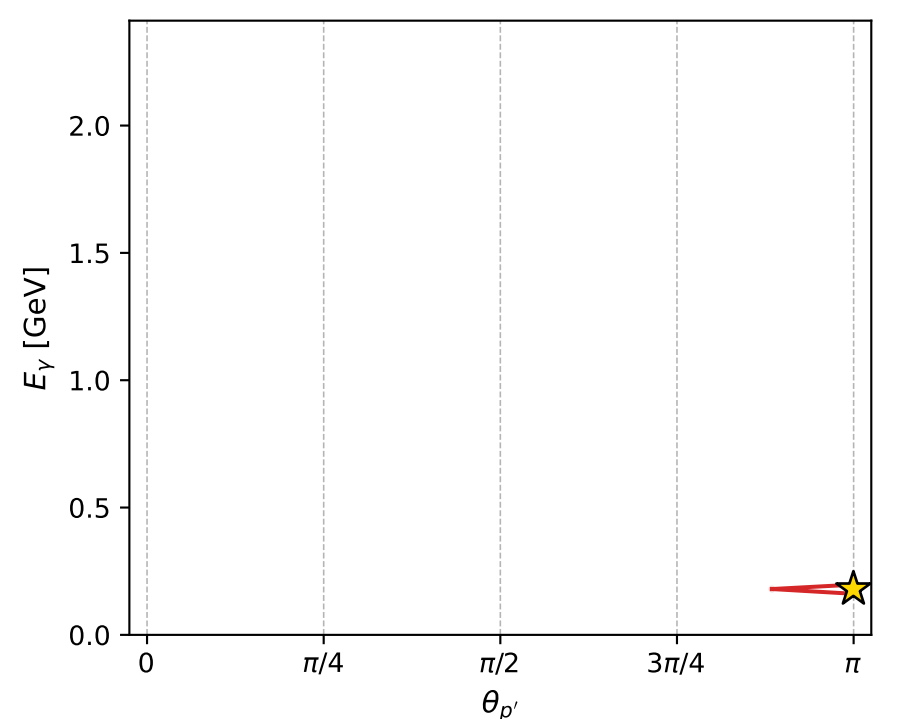
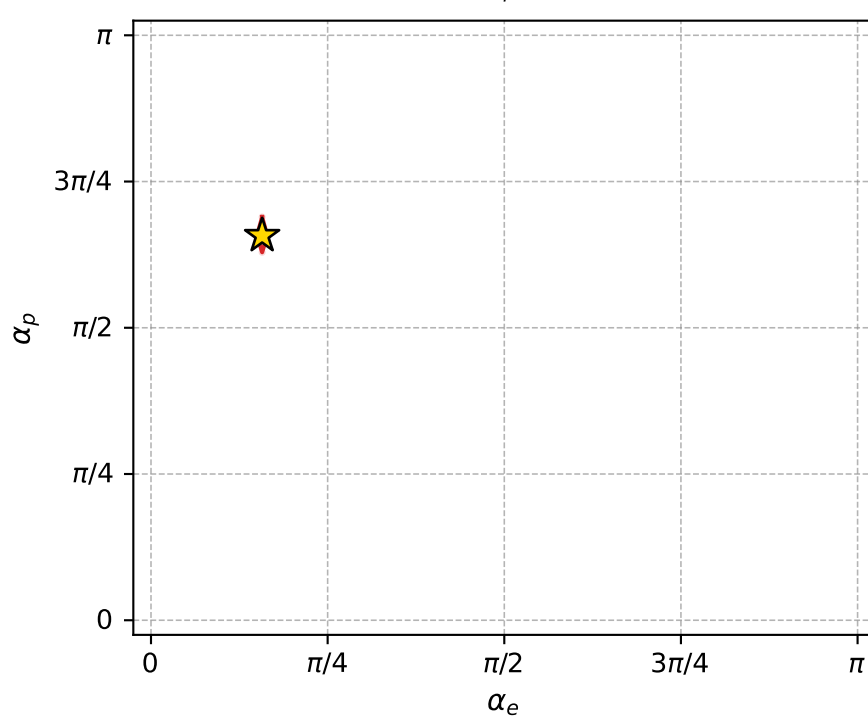
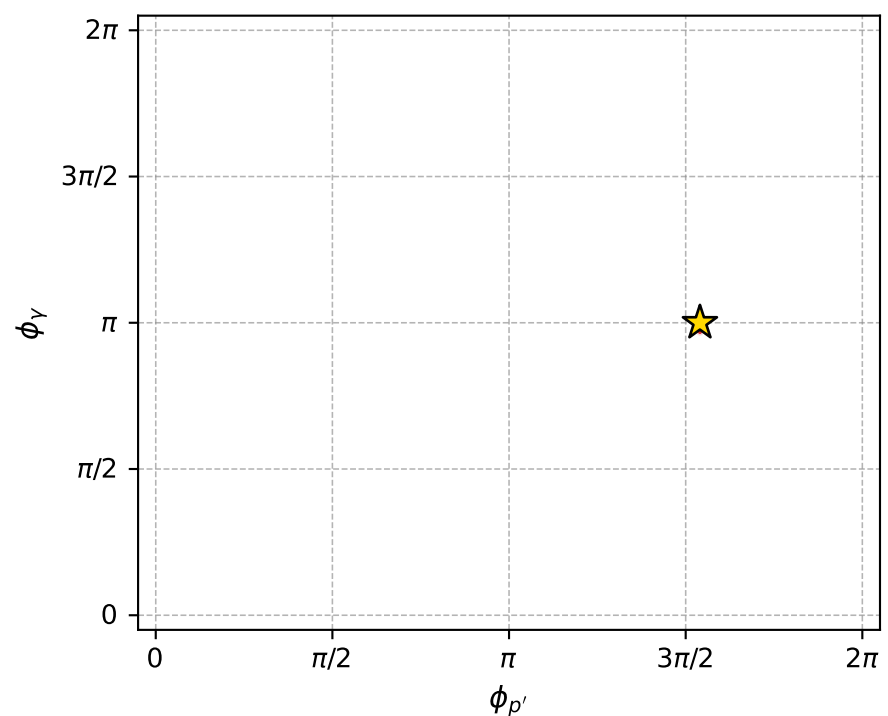
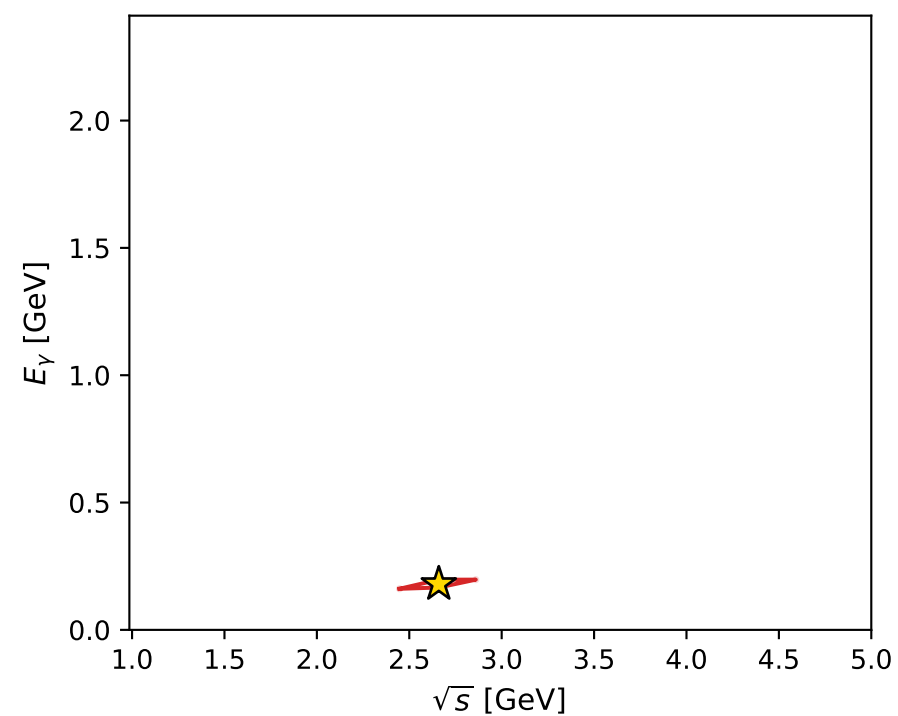
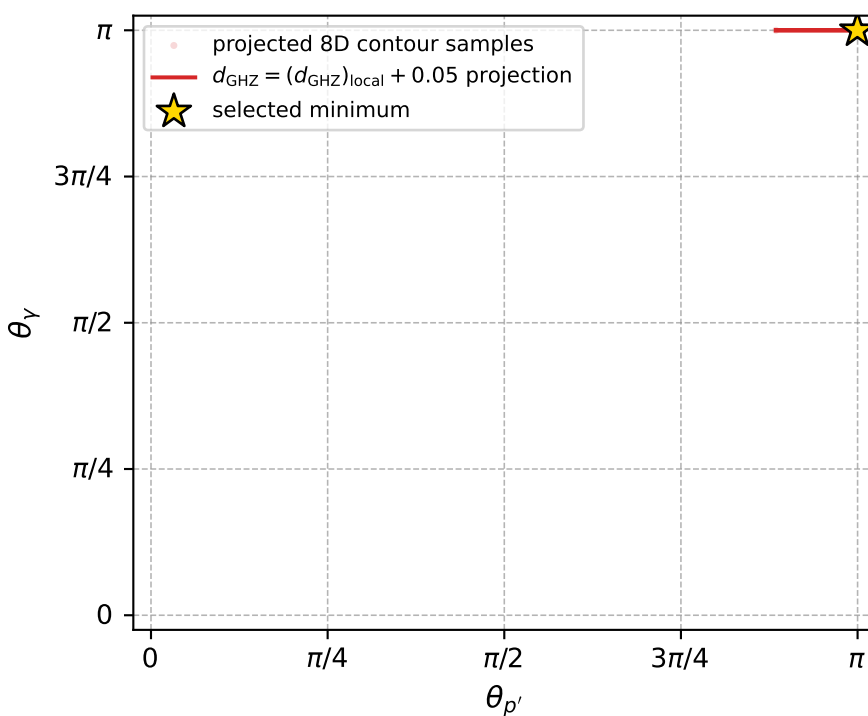
Transverse plane



Leading final-state amplitudes (N=4, retained=100.0%)



electron: polarization cluster P2, local minimum 225; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 4.6600448\text{e-}05$
 $d_{\text{GHZ}} \text{ contour} = 0.0500466$
 above global minimum = $4.6578164\text{e-}05$
 8D contour samples = 253

selected phase-space point:

$\text{sqrt_s} = 2.659868$
 $\text{theta_p_out} = 3.140627$
 $\text{theta_gamma_out} = 3.14154$
 $\text{q0ut} = 0.1799736$
 $\text{phi_p_out} = 4.839081$
 $\text{phi_gamma_out} = 3.141564$
 $\text{alpha_e} = 0.4941554$
 $\text{alpha_p} = 2.065351$

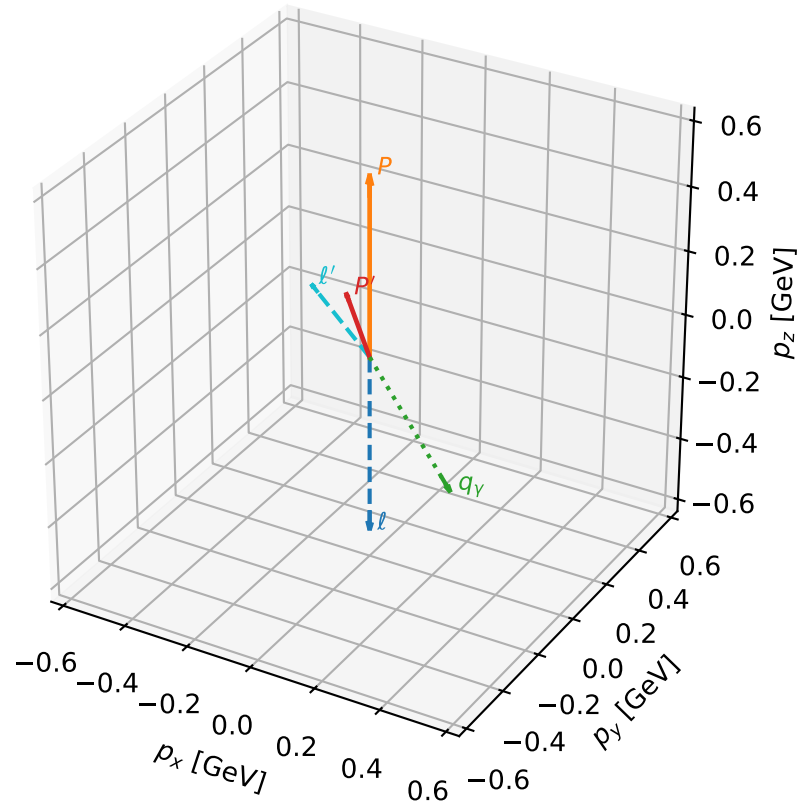
coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

dghz_mixing_angles_polarization_02_local_minimum_244 $d_{\{\text{r m GHZ}\}}=0.000103386$
 region: polarization_cluster_02_local_minimum_244
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=0.65215217$ rad
 incoming lepton: $0.794779|+\rangle +0.606898|-\rangle$
 proton mixing angle: $\alpha_p=2.1218088$ rad
 incoming proton: $-0.52355|+\rangle +0.851995|-\rangle$

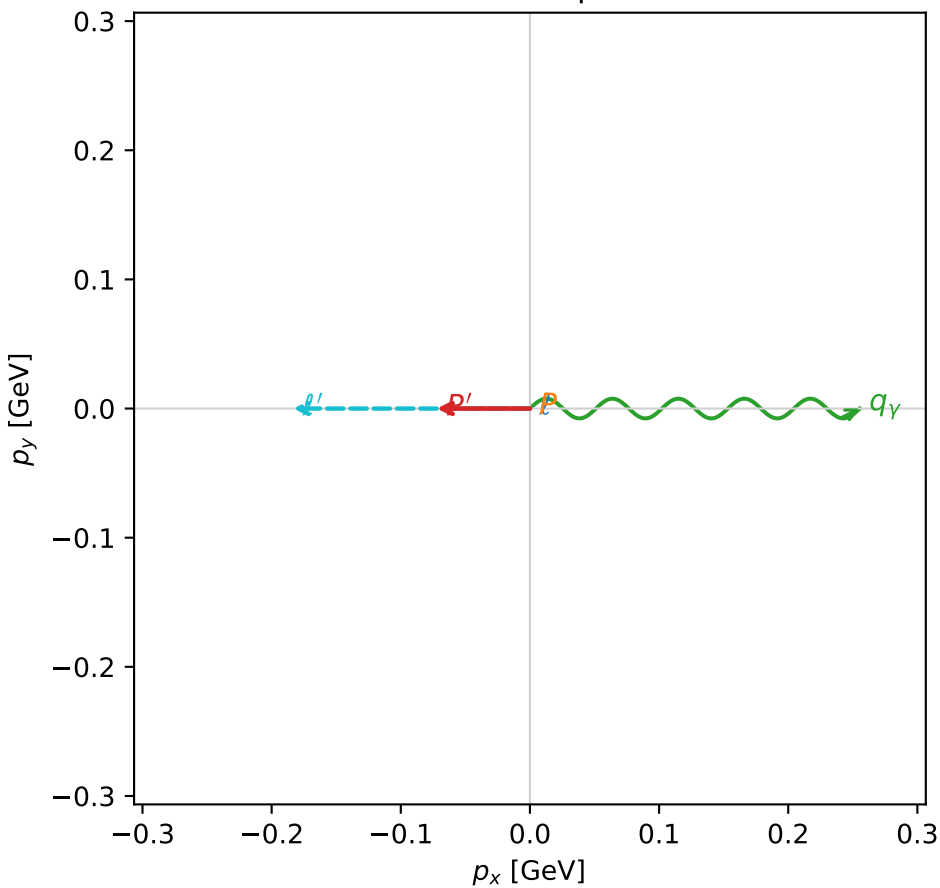
$s=2.71189$, $\sqrt{s}=1.64678$, $|\vec{P}|=0.55625$, $|\vec{P}'|=0.189636$
 $\theta_{P'}=0.390176$, $\theta_Y=2.51415$, $\phi_{P'}=3.14162$, $\phi_Y=6.28318$
 $E_Y=0.435098$, $\theta_{\ell'}=0.803346$, $\phi_{\ell'}=3.14158$
 $Q^2=0.480097$, $x_B=0.325876$, $t=-0.132425$, $W^2=1.873$, $y=0.804158$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.5563$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.5563$
 P $E= 1.0905$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.5563$
 ℓ' $E= 0.2547$ $p_x= -0.1833$ $p_y= 0.0000$ $p_z= 0.1768$
 P' $E= 0.9570$ $p_x= -0.0721$ $p_y= -0.0000$ $p_z= 0.1754$
 q_Y $E= 0.4351$ $p_x= 0.2554$ $p_y= -0.0000$ $p_z= -0.3522$

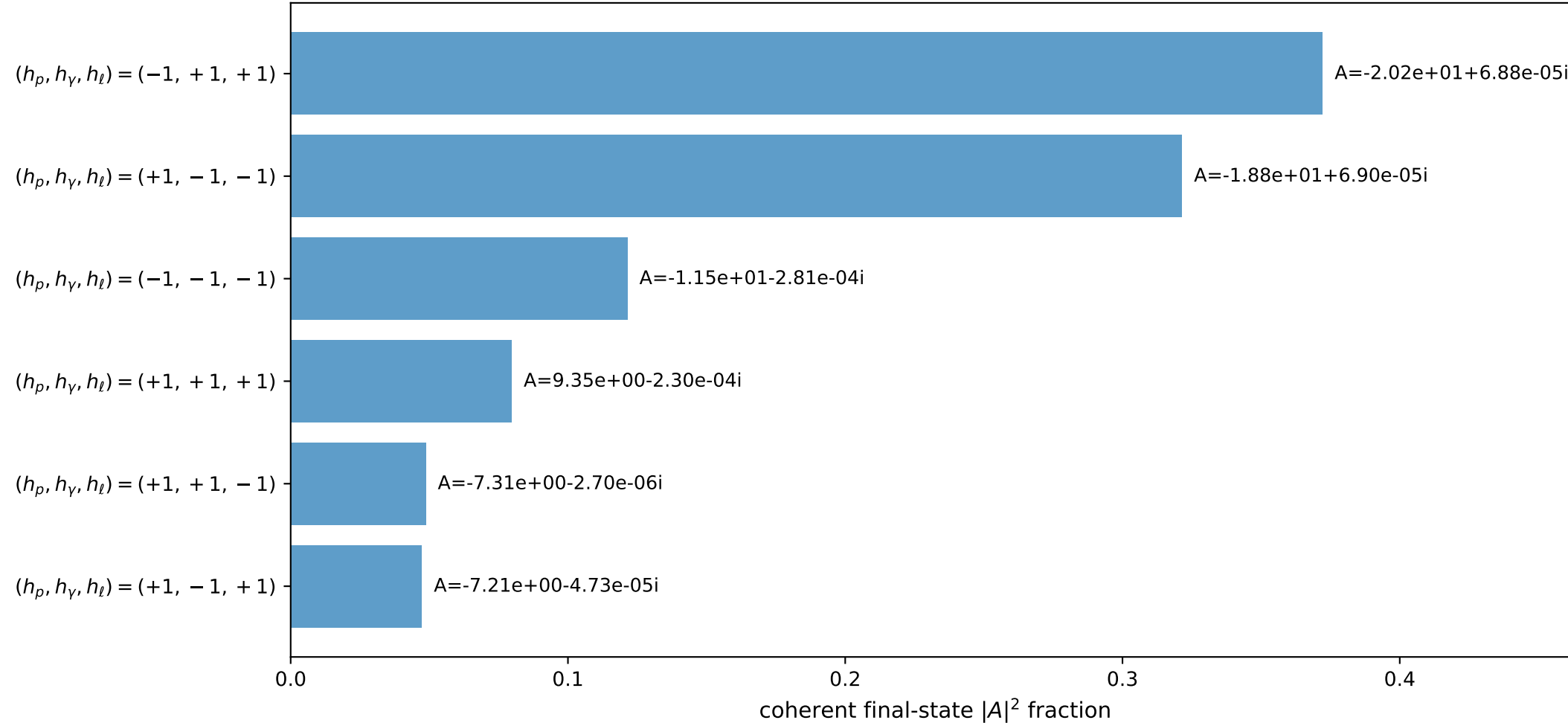
3D momenta



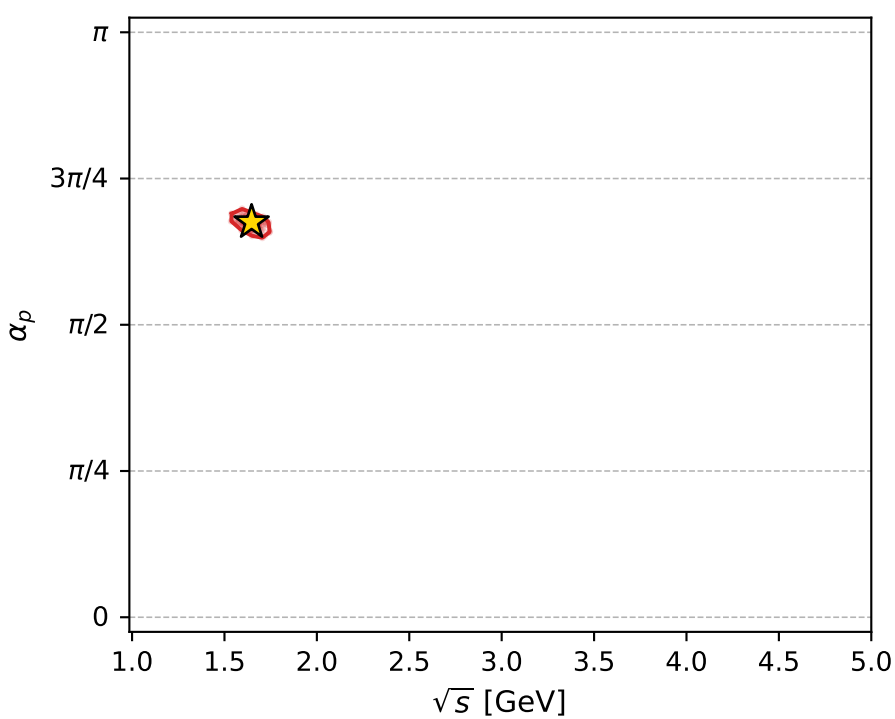
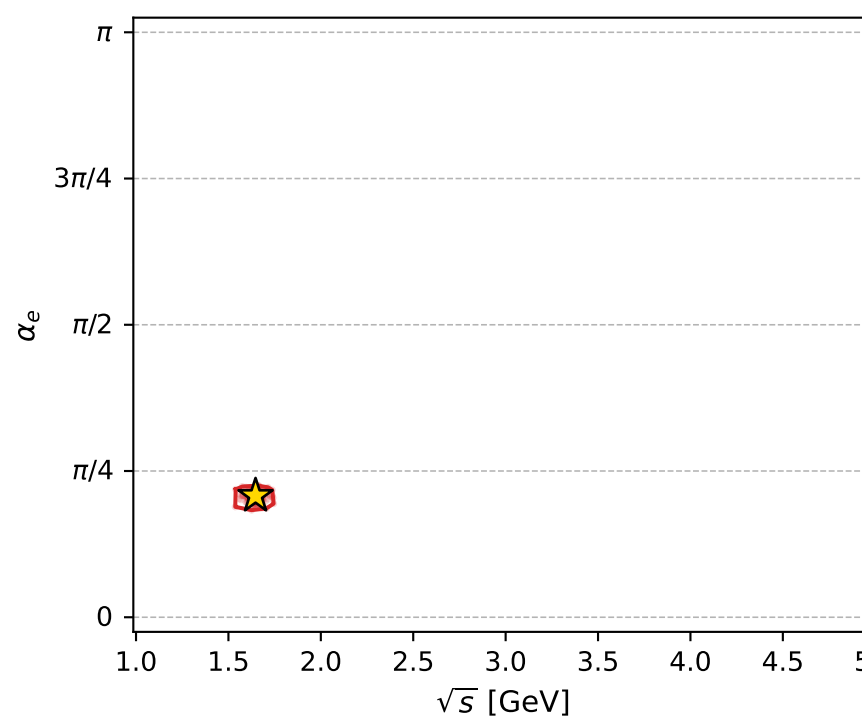
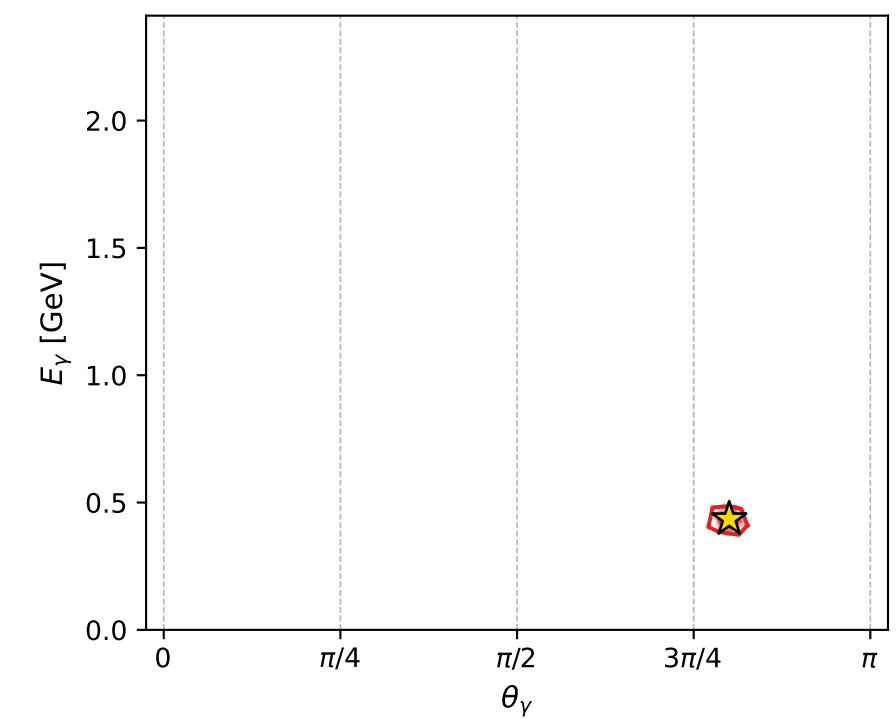
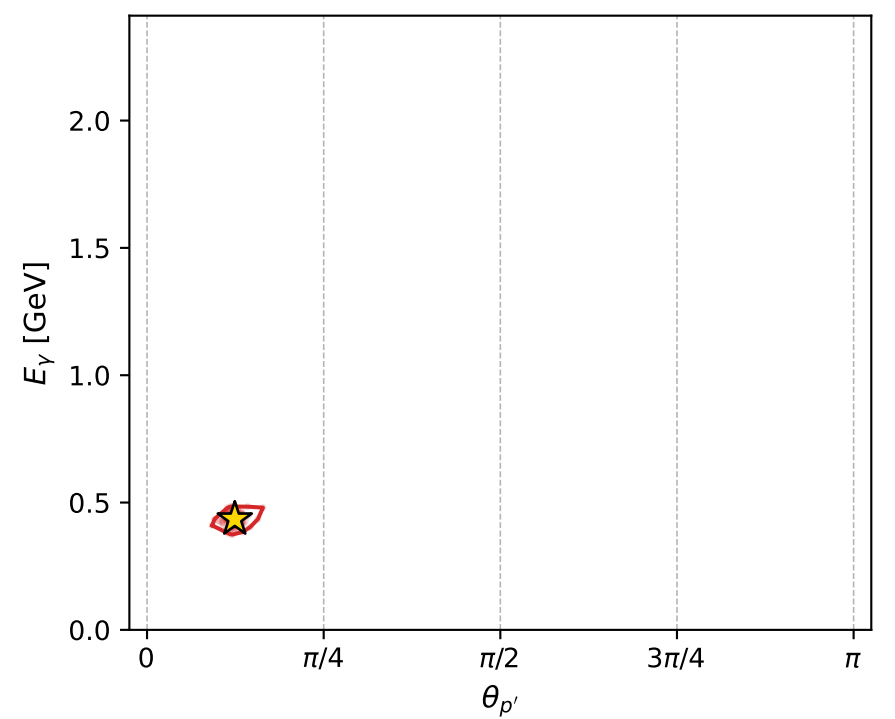
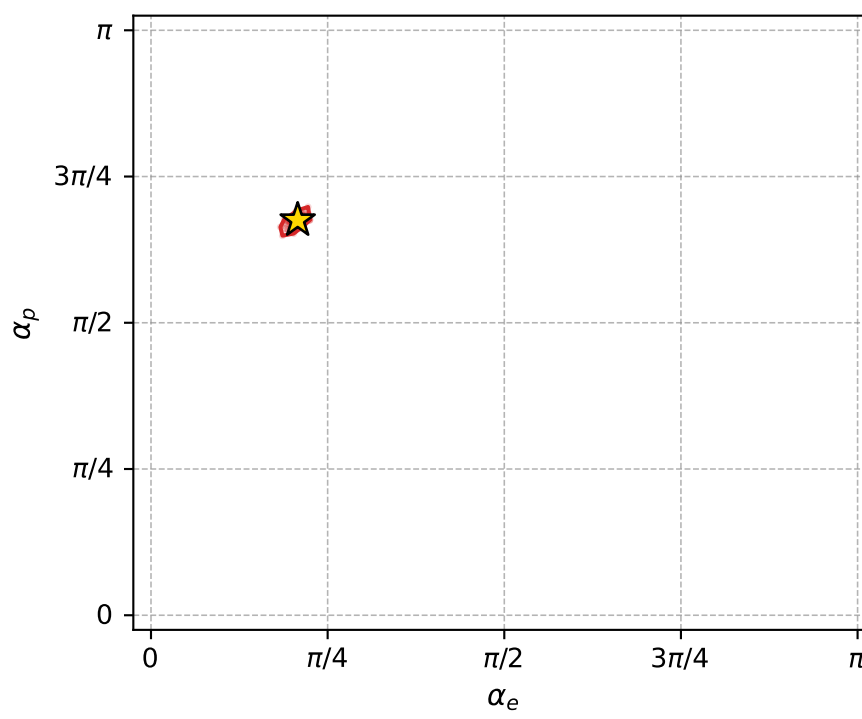
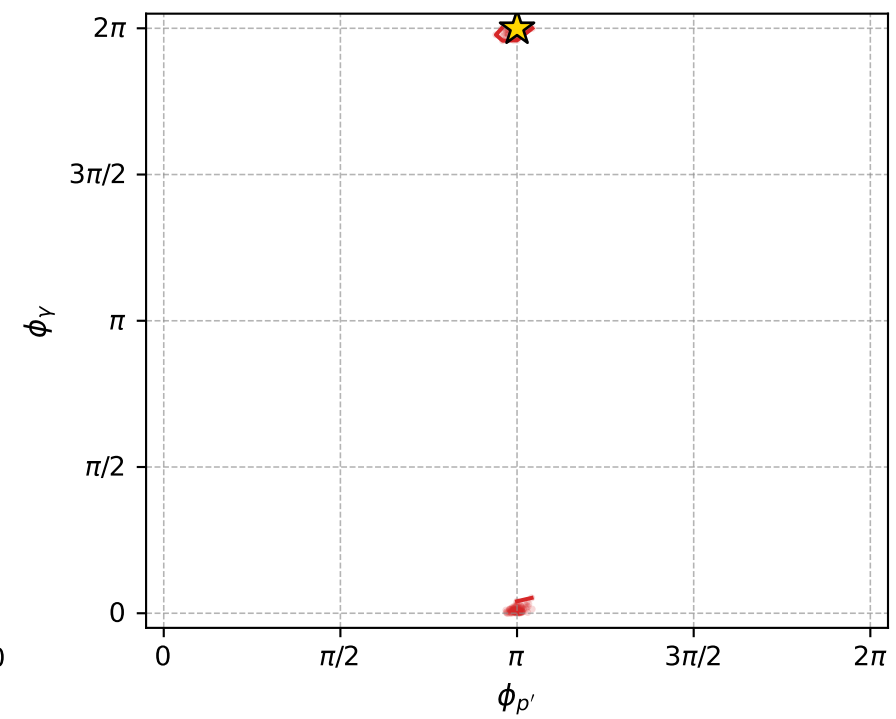
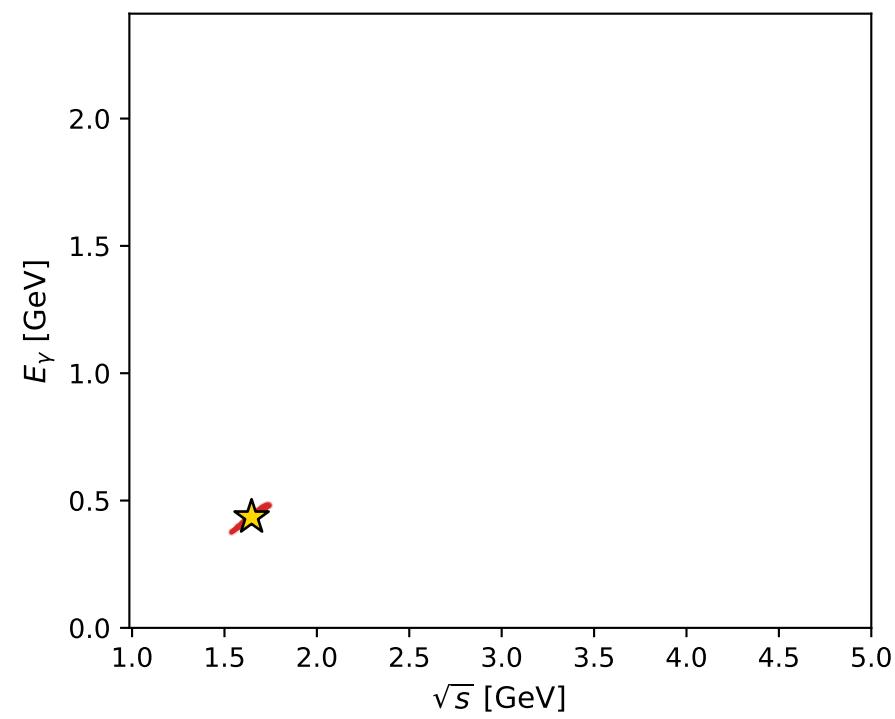
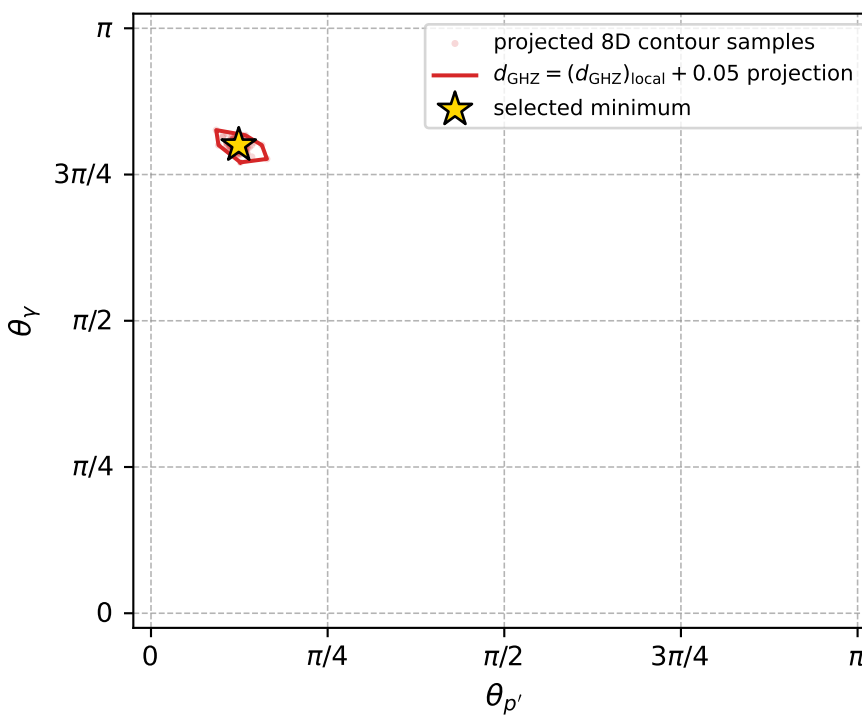
Transverse plane



Leading final-state amplitudes (N=6, retained=99.1%)



electron: polarization cluster P2, local minimum 244; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 0.00010338615$
 $d_{\text{GHZ}} \text{ contour} = 0.050103386$
 above global minimum = 0.00010336387
 8D contour samples = 256

selected phase-space point:

$\text{sqrt_s} = 1.646782$
 $\text{theta_p_out} = 0.3901755$
 $\text{theta_gamma_out} = 2.514151$
 $\text{q0ut} = 0.4350982$
 $\text{phi_p_out} = 3.141618$
 $\text{phi_gamma_out} = 6.283184$
 $\text{alpha_e} = 0.6521522$
 $\text{alpha_p} = 2.121809$

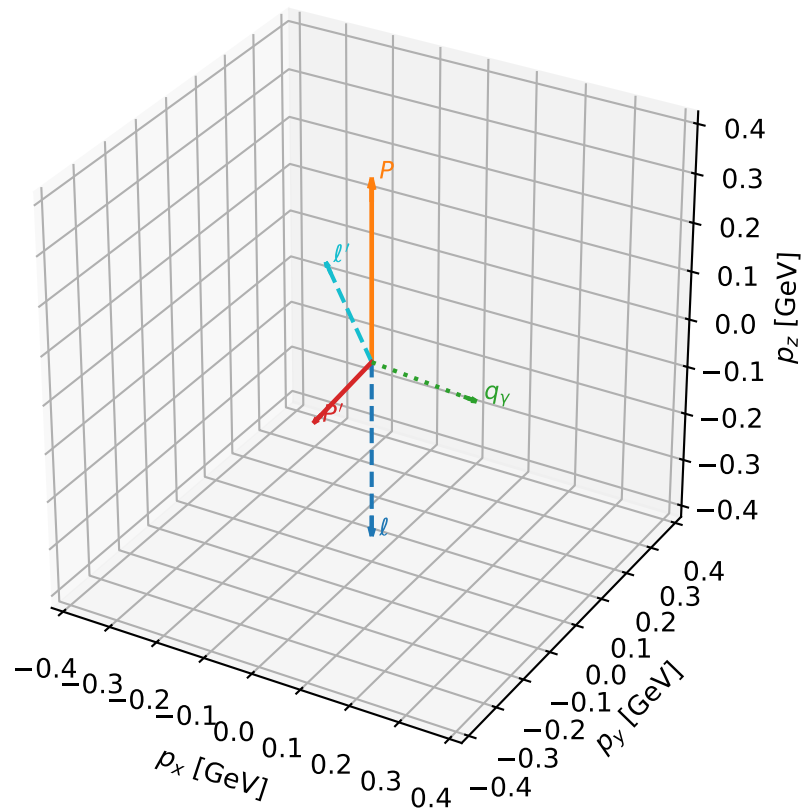
coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

dghz_mixing_angles_polarization_02_local_minimum_254 $d_{\{\text{rm GHZ}\}}=0.000146875$
 region: polarization_cluster_02_local_minimum_254
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=0.31073161$ rad
 incoming lepton: $0.95211|+\rangle + 0.305755|-\rangle$
 proton mixing angle: $\alpha_p=1.819137$ rad
 incoming proton: $-0.245796|+\rangle + 0.969322|-\rangle$

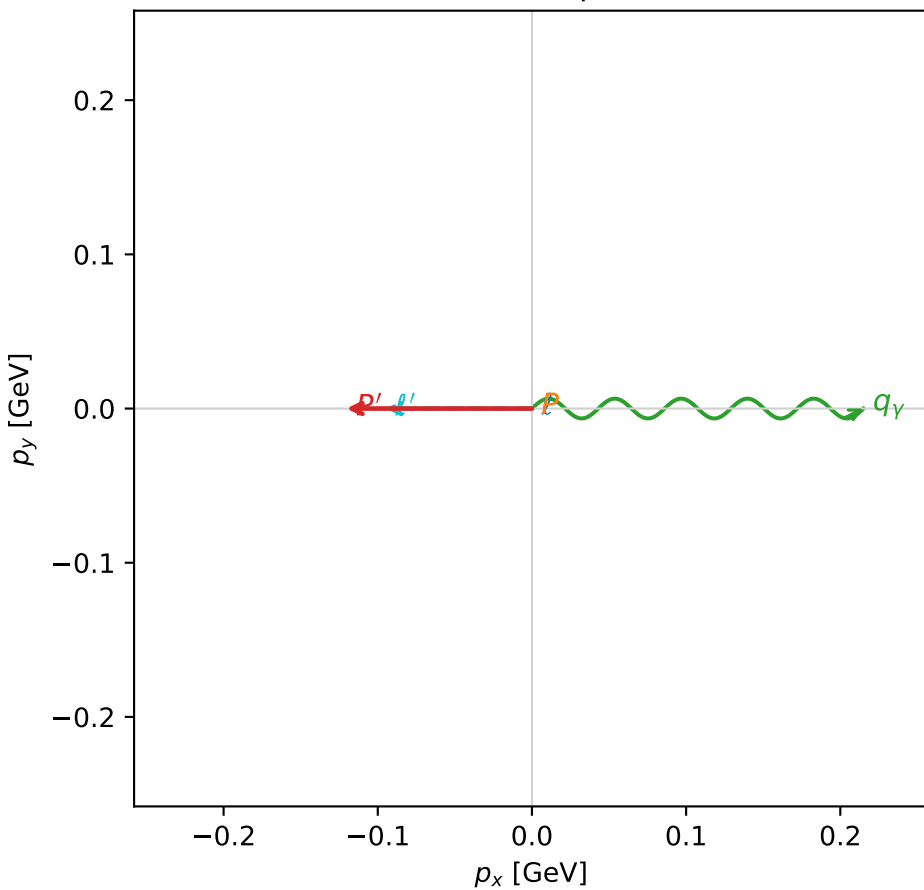
$s=1.89755$, $\sqrt{s}=1.37752$, $|\vec{P}|=0.369399$, $|\vec{P}'|=0.200503$
 $\theta_{P'}=2.49843$, $\theta_Y=1.6565$, $\phi_{P'}=3.14154$, $\phi_Y=6.28315$
 $E_Y=0.215843$, $\theta_{\ell'}=0.487251$, $\phi_{\ell'}=3.14157$
 $Q^2=0.281779$, $x_B=0.379942$, $t=-0.292798$, $W^2=1.3397$, $y=0.728734$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.3694$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.3694$
 P $E= 1.0081$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.3694$
 ℓ' $E= 0.2025$ $p_x= -0.0948$ $p_y= 0.0000$ $p_z= 0.1789$
 P' $E= 0.9592$ $p_x= -0.1202$ $p_y= 0.0000$ $p_z= -0.1604$
 q_Y $E= 0.2158$ $p_x= 0.2151$ $p_y= -0.0000$ $p_z= -0.0185$

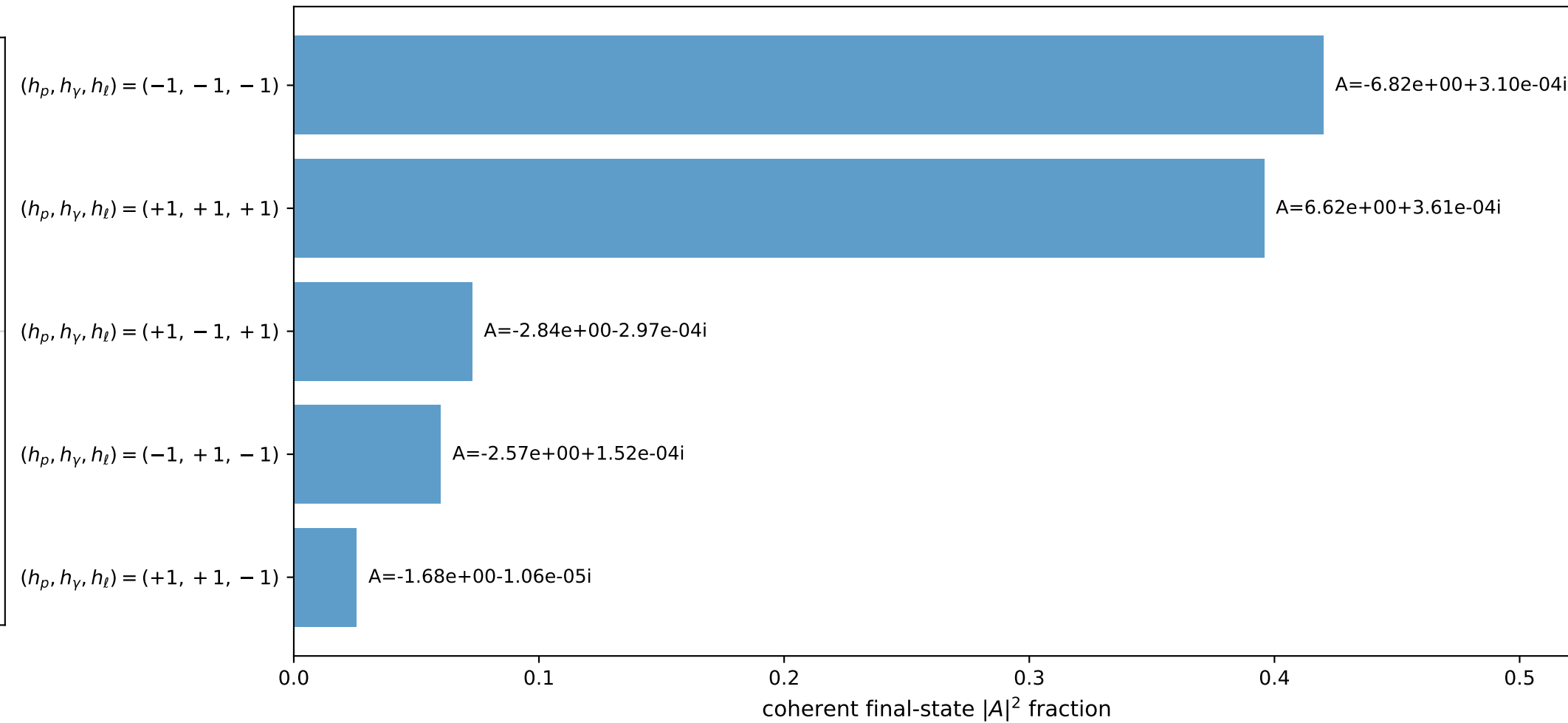
3D momenta



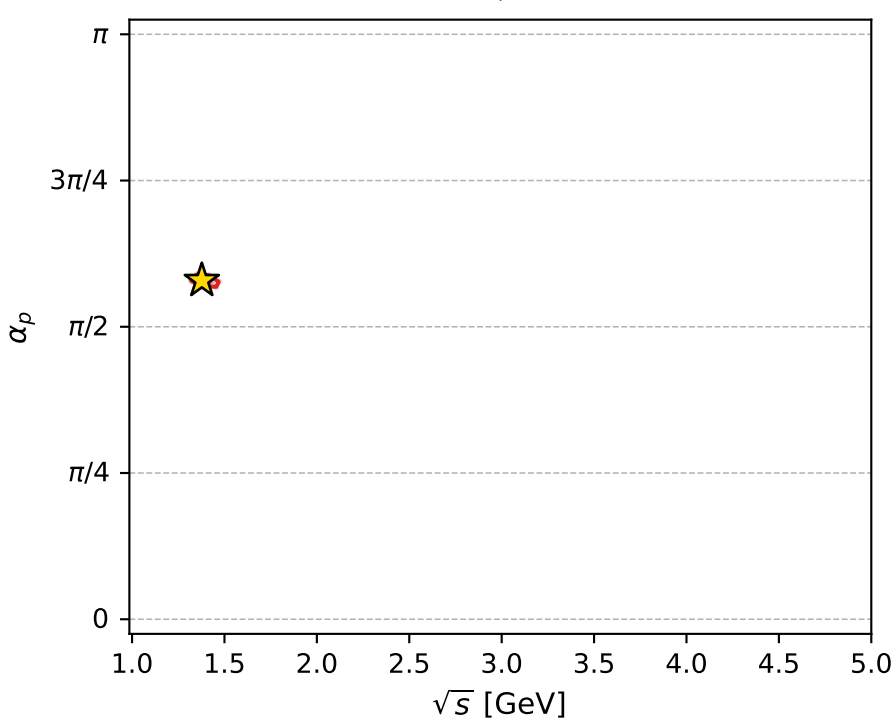
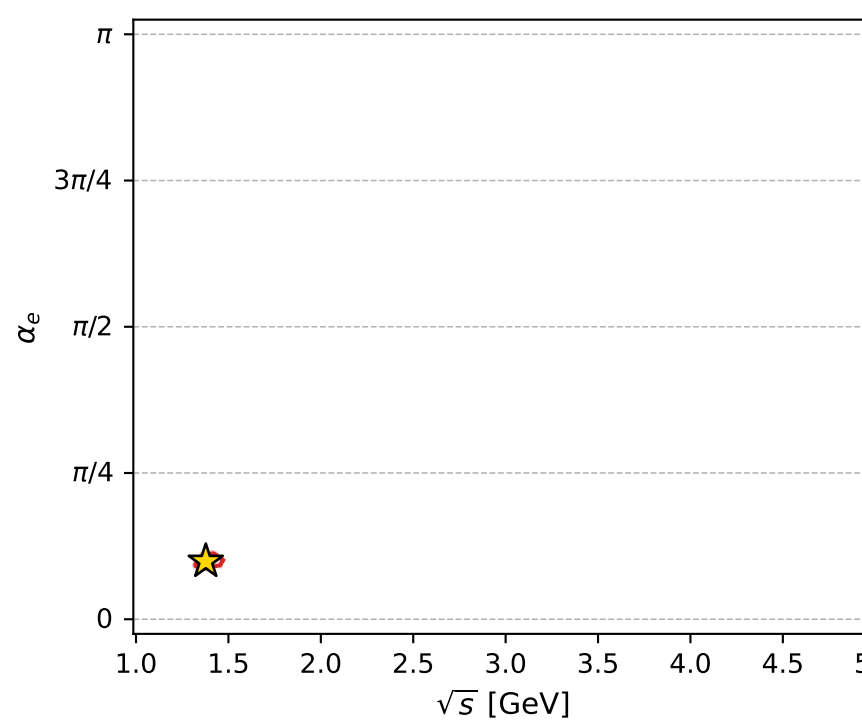
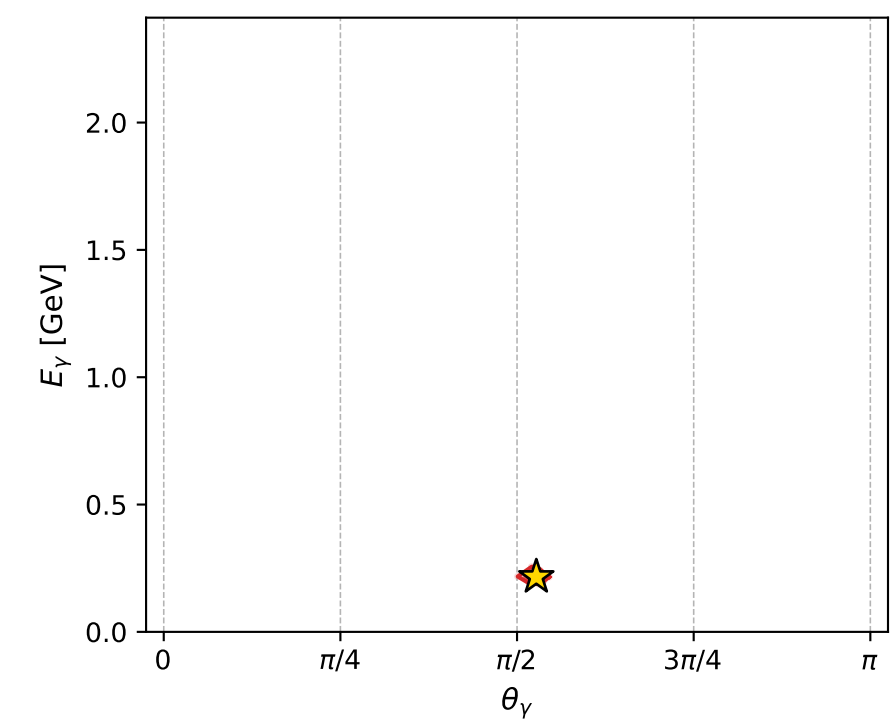
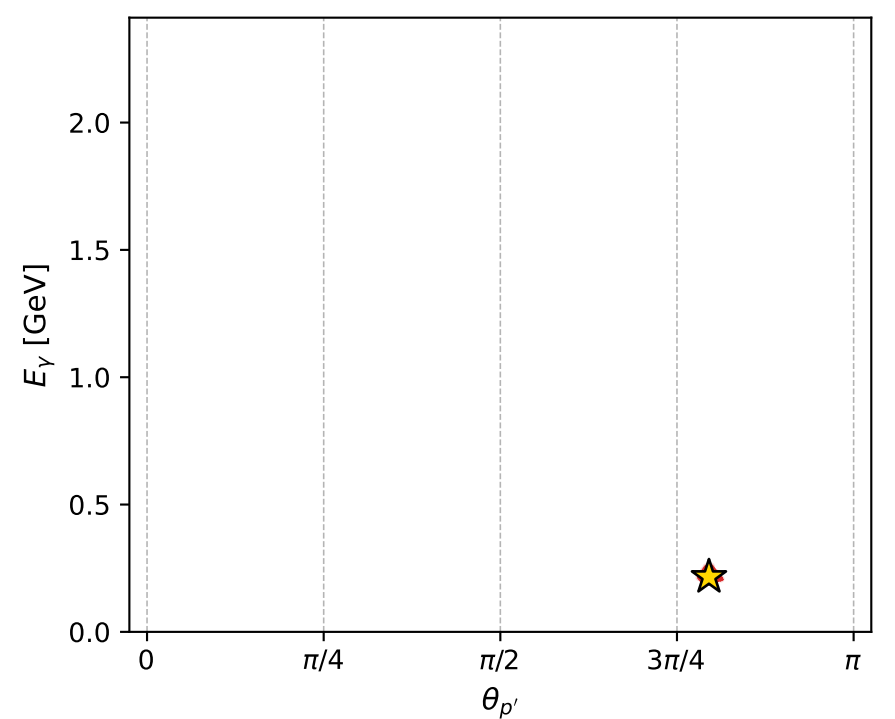
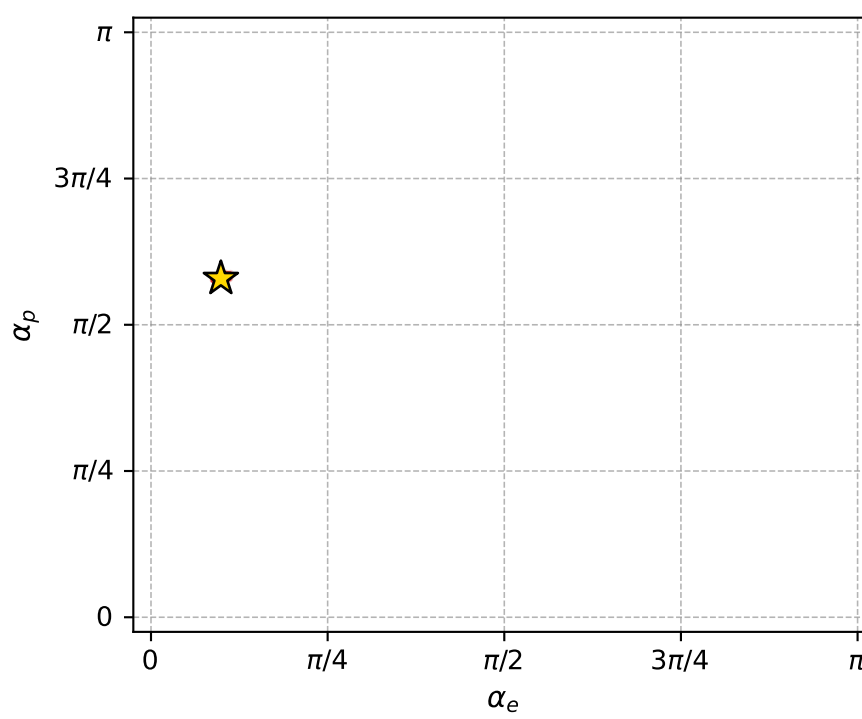
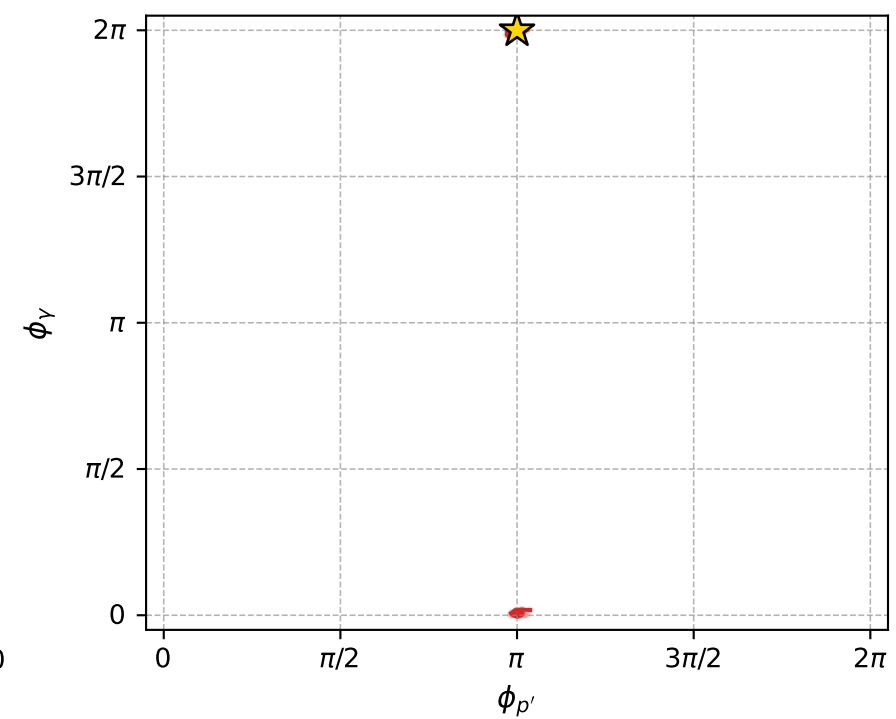
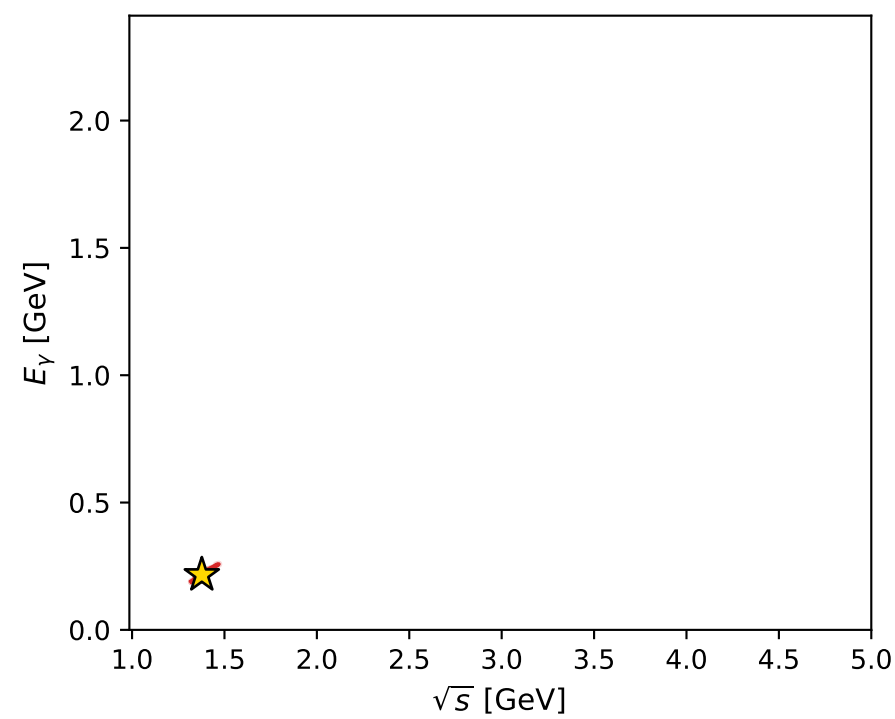
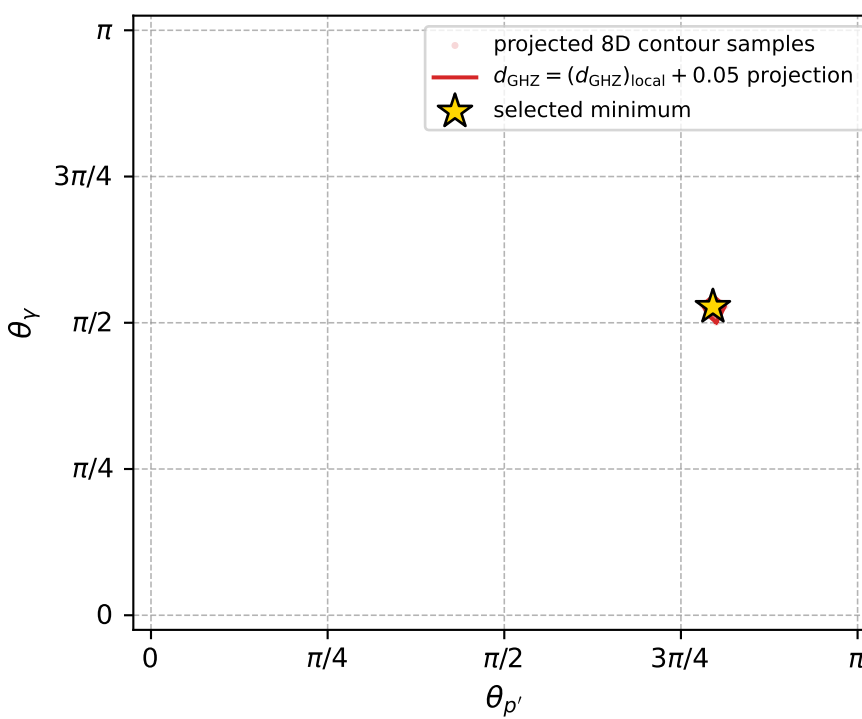
Transverse plane



Leading final-state amplitudes (N=5, retained=97.4%)



electron: polarization cluster P2, local minimum 254; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 0.00014687471$
 $d_{\text{GHZ}} \text{ contour} = 0.050146875$
 above global minimum = 0.00014685243
 8D contour samples = 256

selected phase-space point:

$\text{sqrt_s} = 1.377516$
 $\text{theta_p_out} = 2.498427$
 $\text{theta_gamma_out} = 1.656499$
 $\text{q0out} = 0.2158426$
 $\text{phi_p_out} = 3.141544$
 $\text{phi_gamma_out} = 6.283146$
 $\text{alpha_e} = 0.3107316$
 $\text{alpha_p} = 1.819137$

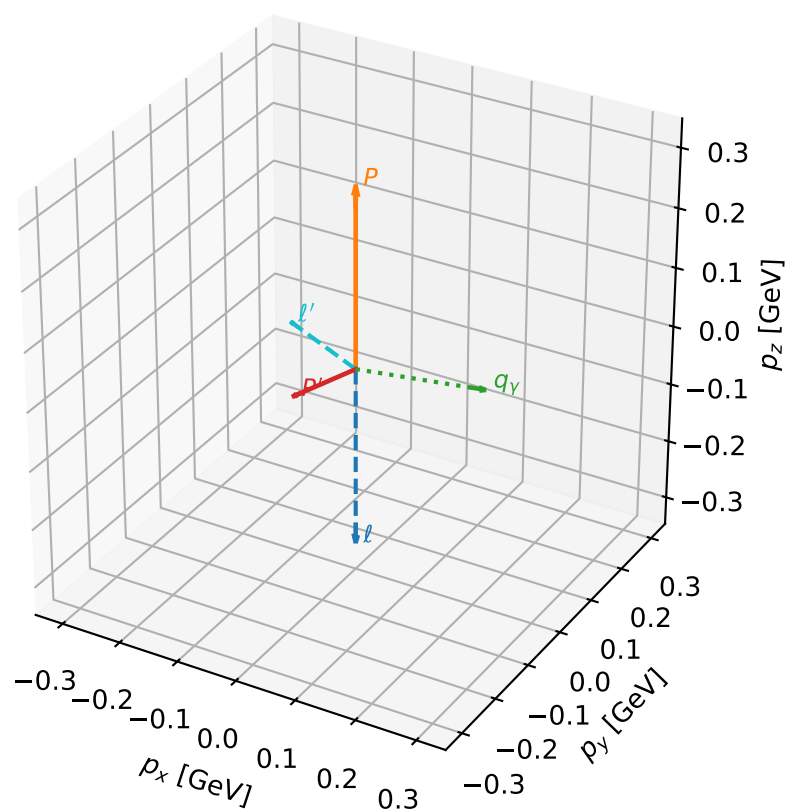
coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

dghz_mixing_angles_polarization_02_local_minimum_255 $d_{\{\text{rm GHZ}\}}=0.000148536$
 region: polarization_cluster_02_local_minimum_255
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=0.76282833$ rad
 incoming lepton: $0.722885|+\rangle +0.690969|-\rangle$
 proton mixing angle: $\alpha_p=2.2105728$ rad
 incoming proton: $-0.597016|+\rangle +0.802229|-\rangle$

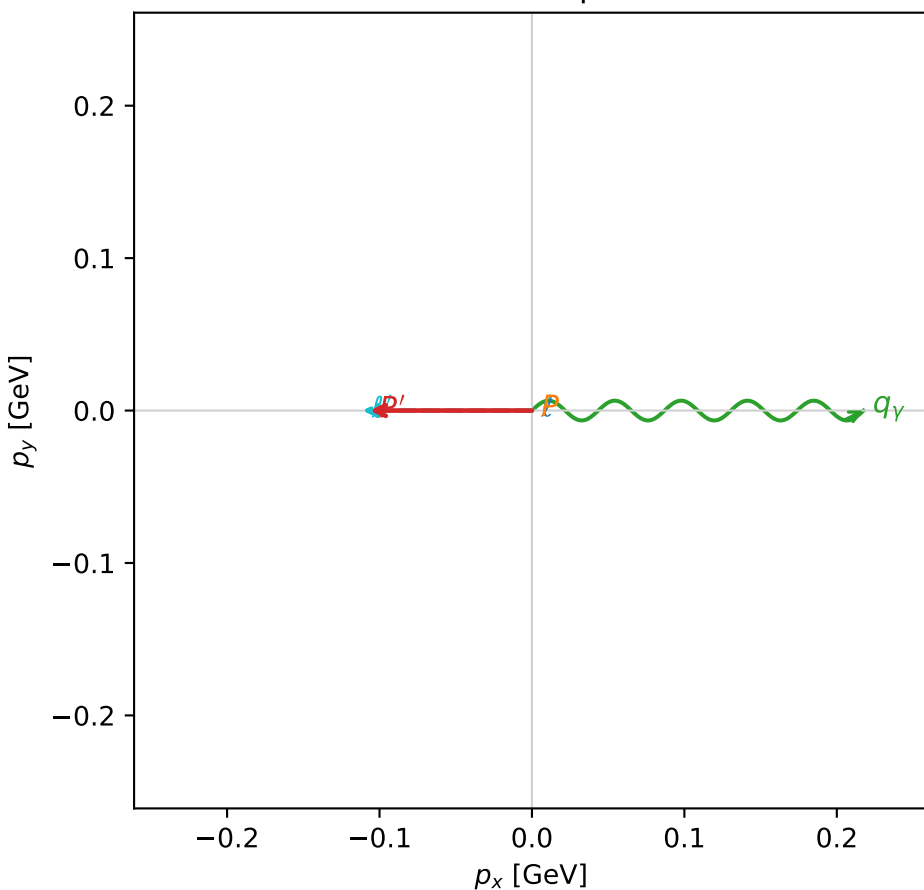
$s=1.65883$, $\sqrt{s}=1.28796$, $|\vec{P}|=0.302412$, $|\vec{P}'|=0.13123$
 $\theta_{p'}=2.19725$, $\theta_Y=1.44311$, $\phi_{p'}=3.14153$, $\phi_Y=6.2831$
 $E_Y=0.219298$, $\theta_{\ell'}=1.15568$, $\phi_{\ell'}=3.14148$
 $Q^2=0.103142$, $x_B=0.181239$, $t=-0.153732$, $W^2=1.3458$, $y=0.730559$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.3024$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.3024$
 P $E= 0.9855$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.3024$
 ℓ' $E= 0.1215$ $p_x= -0.1112$ $p_y= 0.0000$ $p_z= 0.0490$
 P' $E= 0.9471$ $p_x= -0.1063$ $p_y= 0.0000$ $p_z= -0.0769$
 q_Y $E= 0.2193$ $p_x= 0.2175$ $p_y= -0.0000$ $p_z= 0.0279$

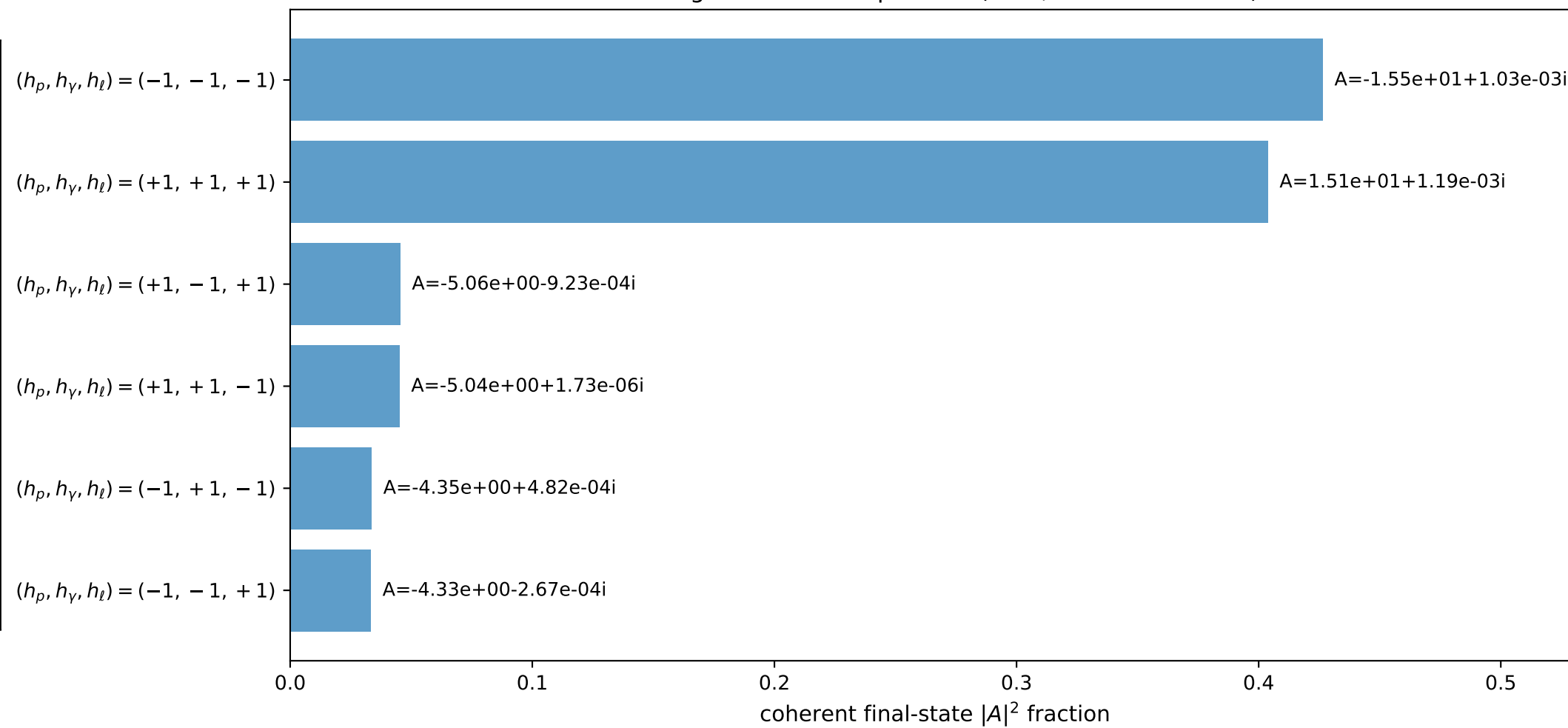
3D momenta



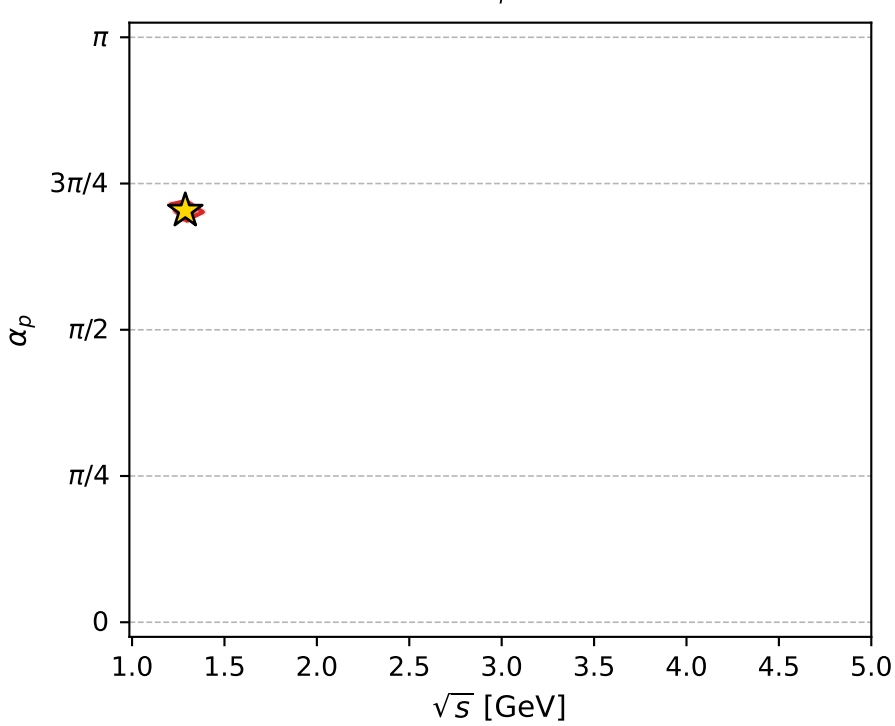
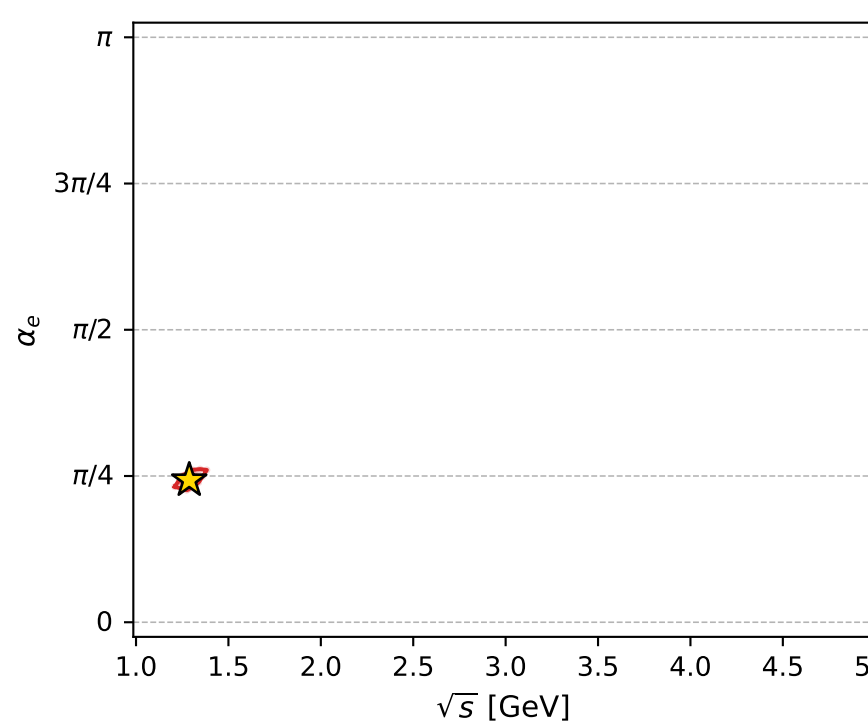
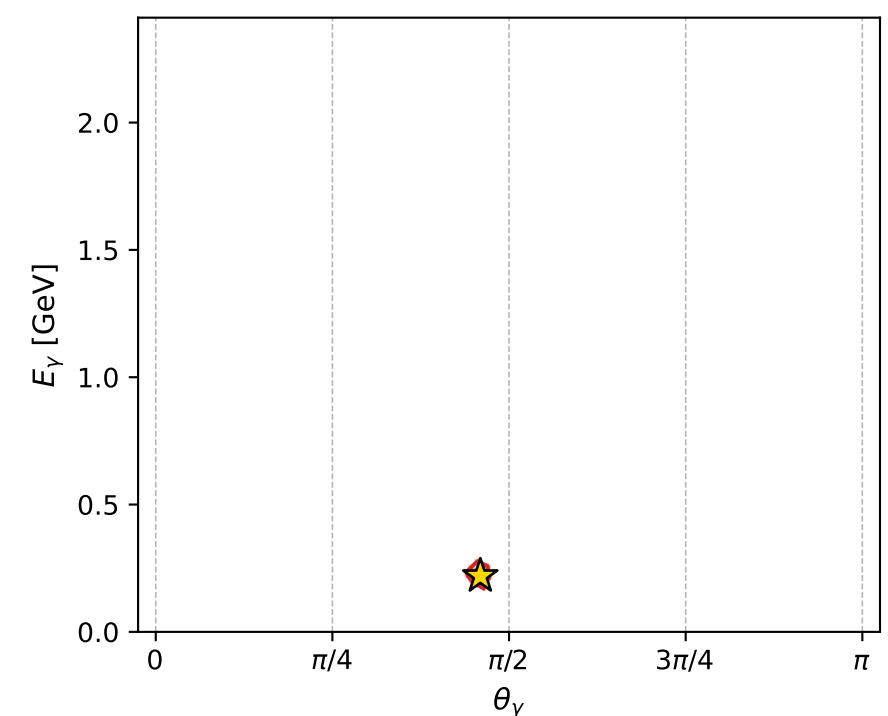
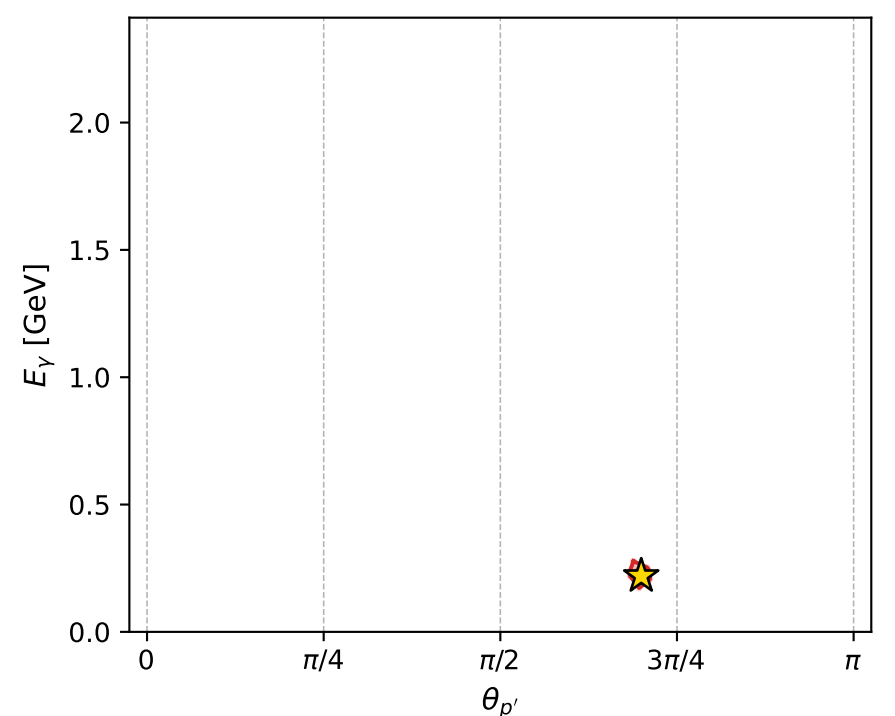
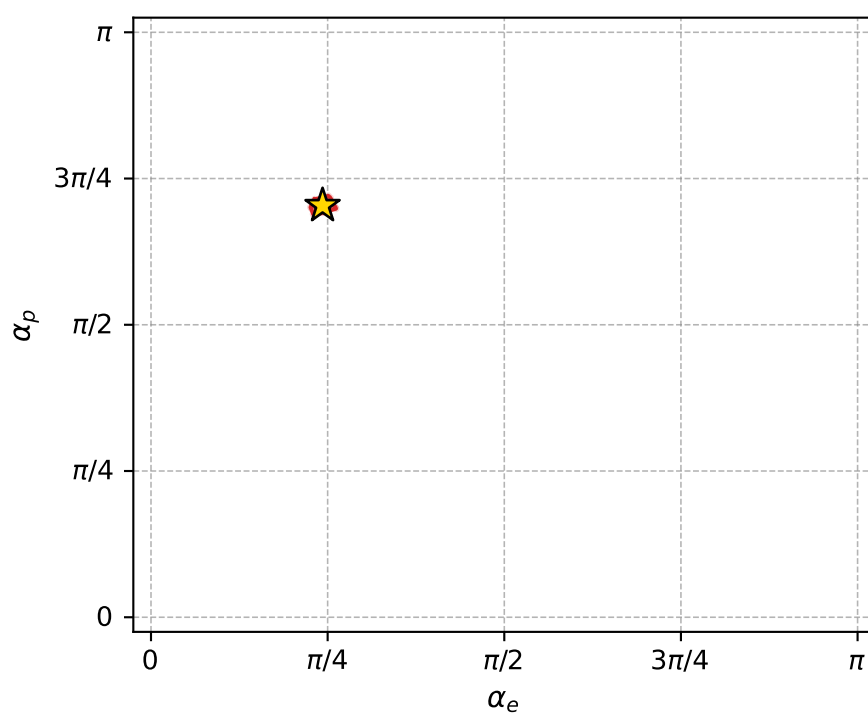
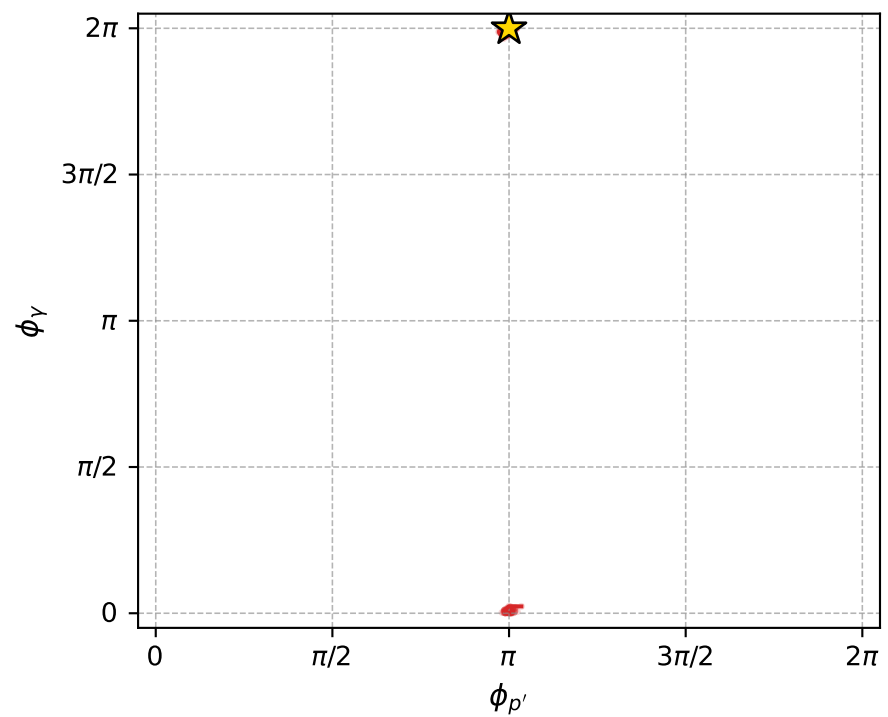
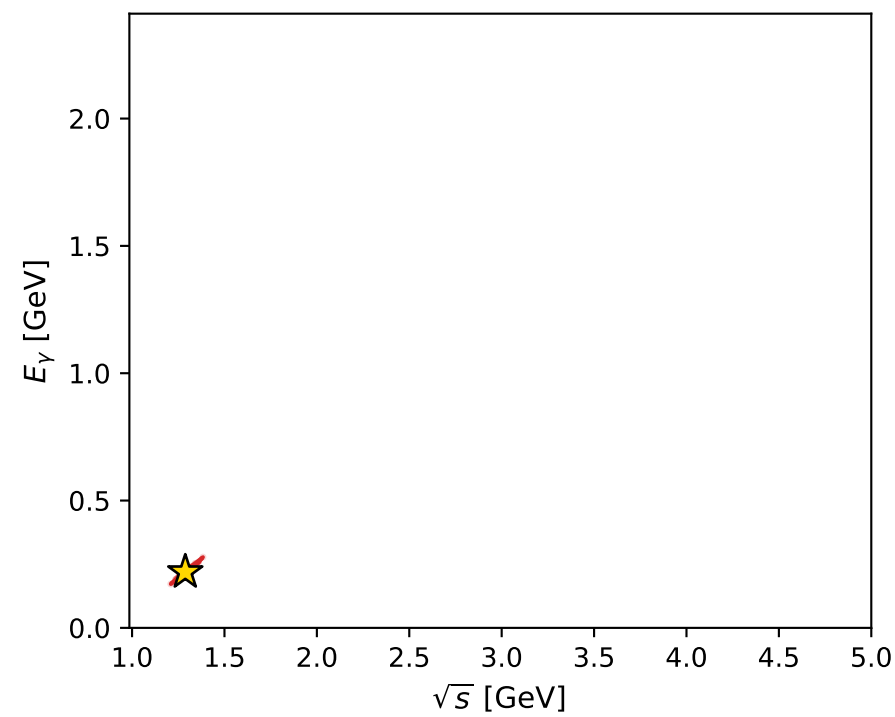
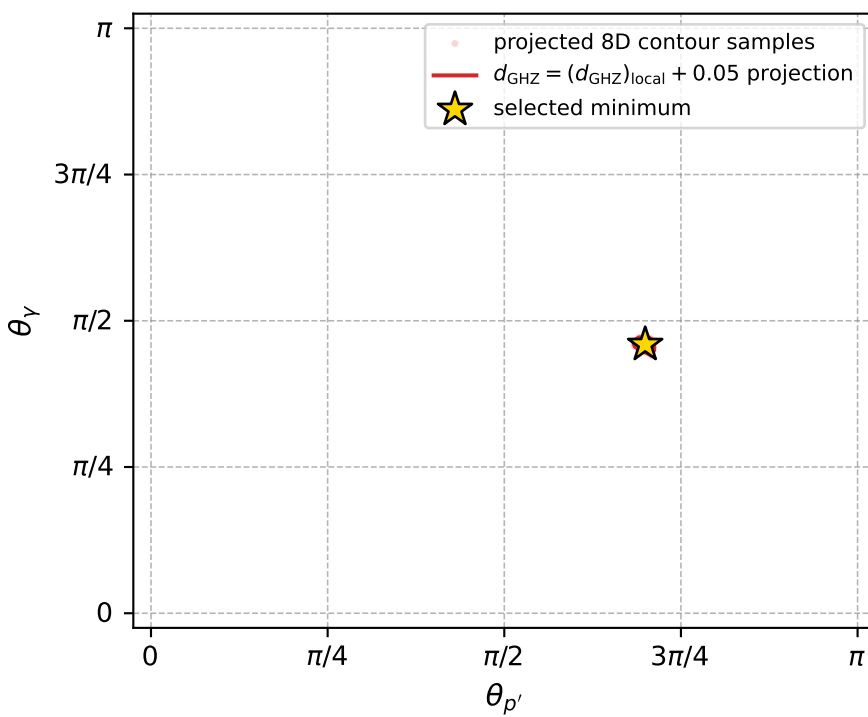
Transverse plane



Leading final-state amplitudes (N=6, retained=98.7%)



electron: polarization cluster P2, local minimum 255; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 0.00014853583$
 $d_{\text{GHZ}} \text{ contour} = 0.050148536$
 above global minimum = 0.00014851355
 8D contour samples = 256

selected phase-space point:

$\text{sqrt_s} = 1.287957$
 $\text{theta_p_out} = 2.197245$
 $\text{theta_gamma_out} = 1.443106$
 $\text{q0out} = 0.2192977$
 $\text{phi_p_out} = 3.141533$
 $\text{phi_gamma_out} = 6.283097$
 $\text{alpha_e} = 0.7628283$
 $\text{alpha_p} = 2.210573$

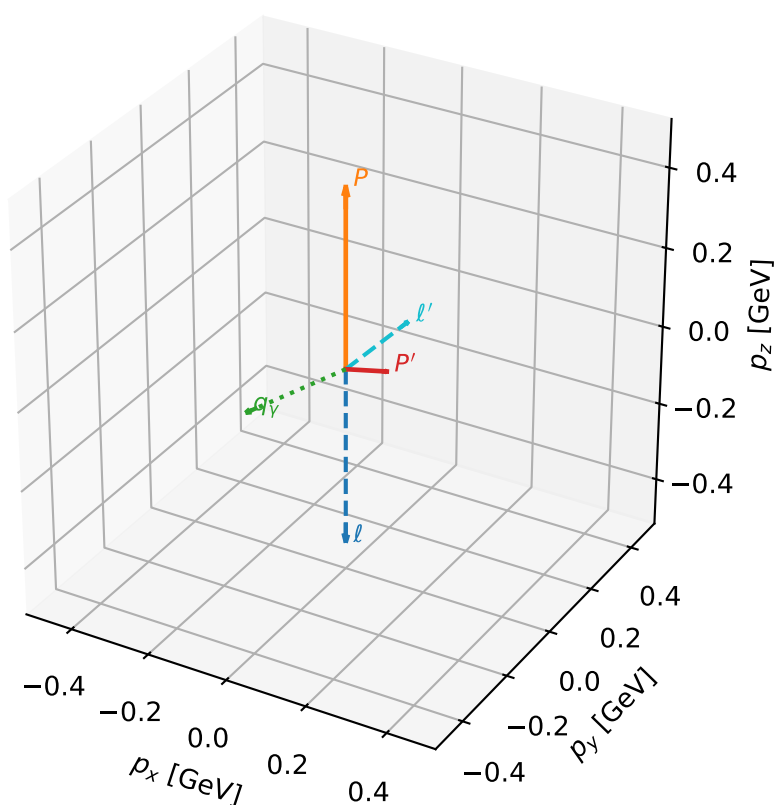
coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

dghz_mixing_angles_polarization_02_local_minimum_256 $d_{\{\text{rm GHZ}\}}=0.000156933$
 region: polarization_cluster_02_local_minimum_256
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=1.006269$ rad
 incoming lepton: $0.535017|+\rangle +0.844842|-\rangle$
 proton mixing angle: $\alpha_p=2.6672121$ rad
 incoming proton: $-0.889576|+\rangle +0.456787|-\rangle$

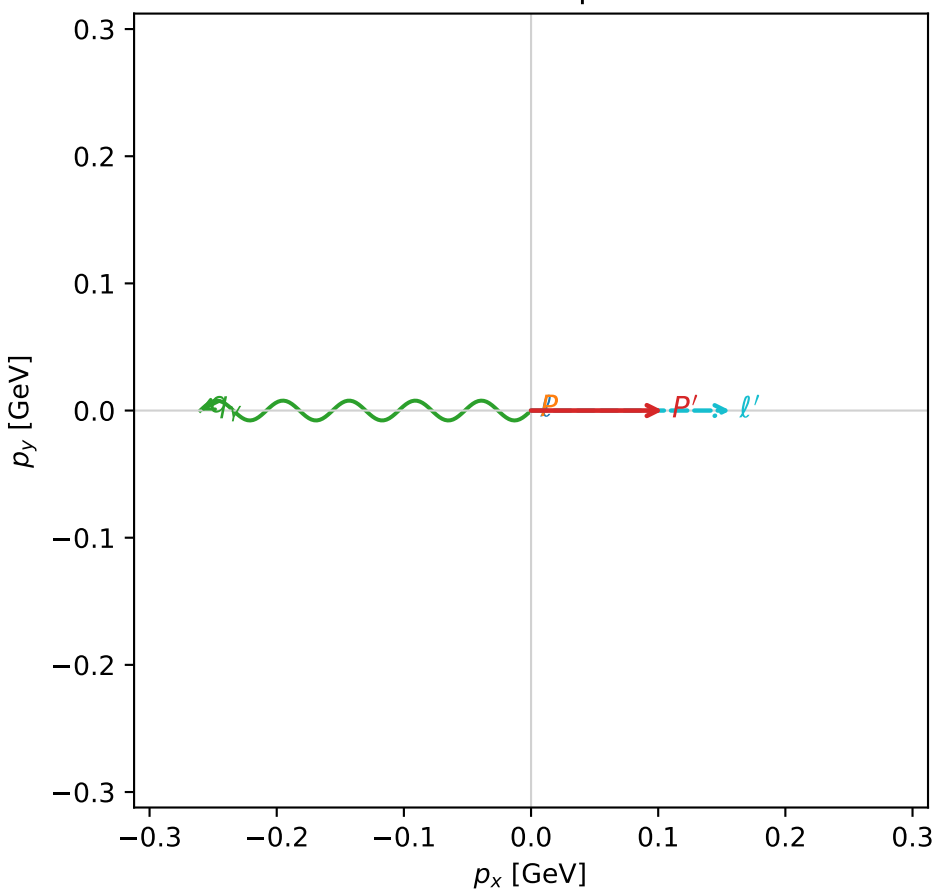
$s=2.22737$, $\sqrt{s}=1.49244$, $|\vec{P}|=0.451451$, $|\vec{P}'|=0.106265$
 $\theta_{p'}=1.3434$, $\theta_\gamma=2.19786$, $\phi_{p'}=6.28318$, $\phi_\gamma=3.14158$
 $E_\gamma=0.321266$, $\theta_{\ell'}=0.760694$, $\phi_{\ell'}=6.28316$
 $Q^2=0.353689$, $x_B=0.34569$, $t=-0.184064$, $W^2=1.54929$, $y=0.759271$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.4515$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.4515$
 P $E= 1.0410$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.4515$
 ℓ' $E= 0.2272$ $p_x= 0.1566$ $p_y= -0.0000$ $p_z= 0.1646$
 P' $E= 0.9440$ $p_x= 0.1035$ $p_y= -0.0000$ $p_z= 0.0240$
 q_γ $E= 0.3213$ $p_x= -0.2601$ $p_y= 0.0000$ $p_z= -0.1885$

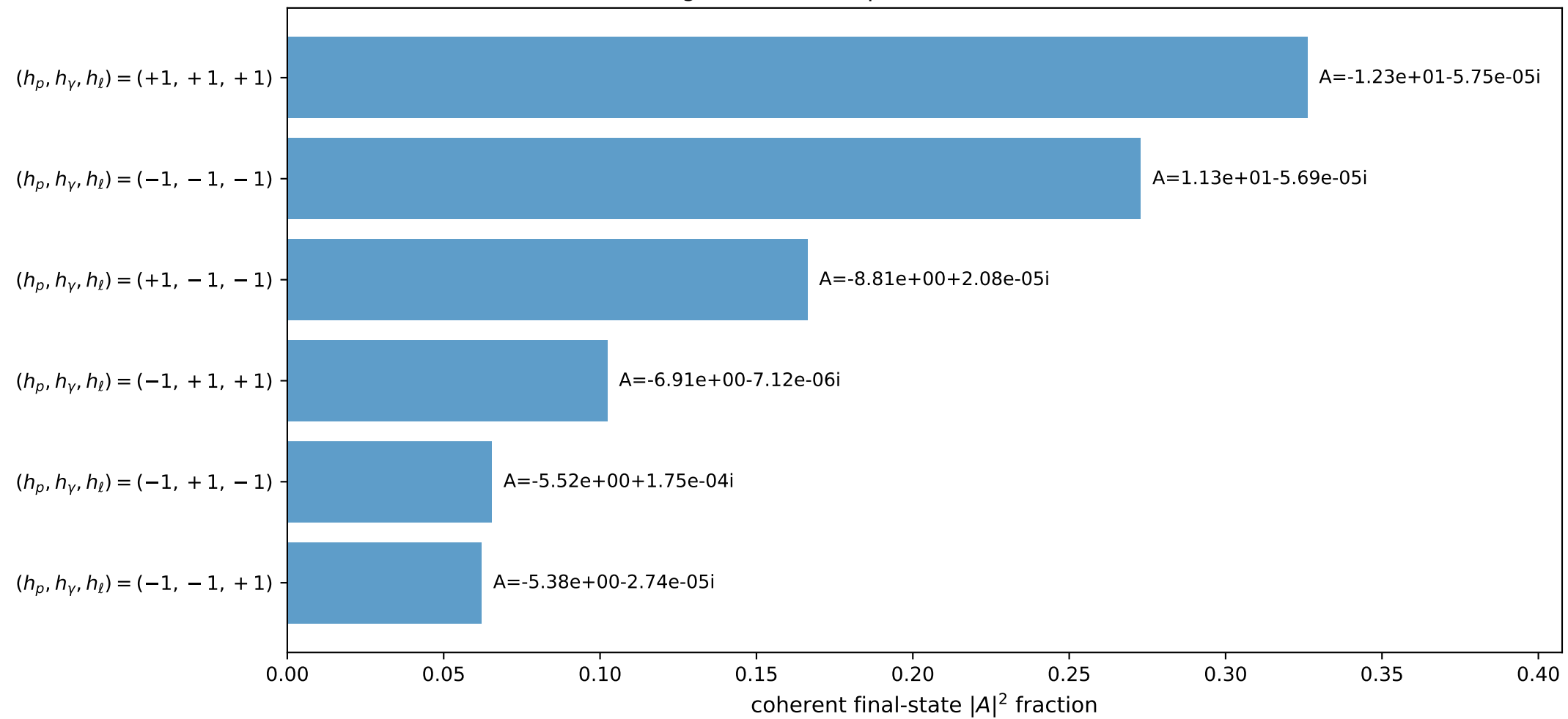
3D momenta



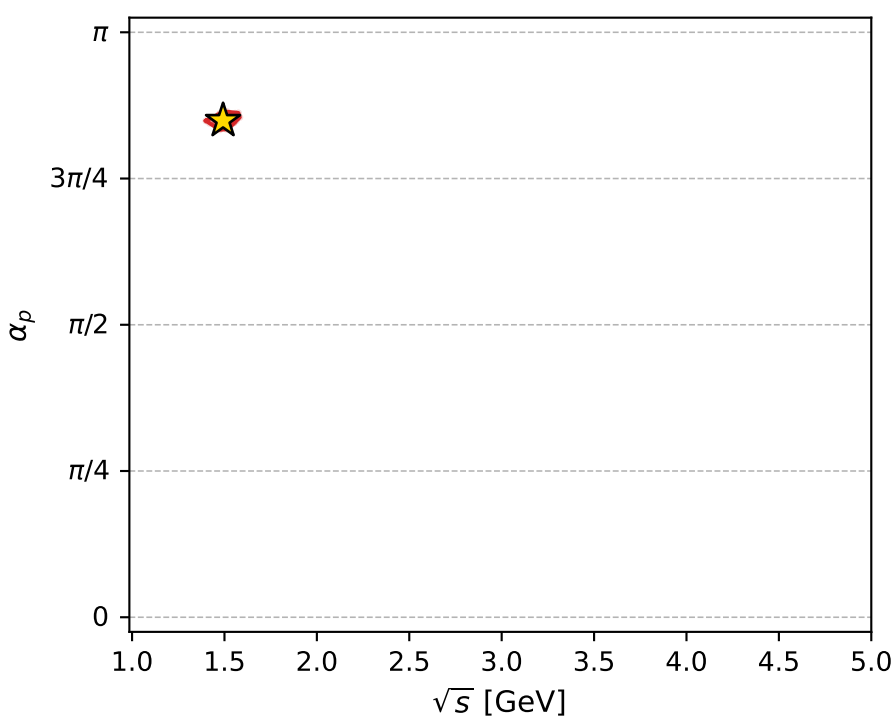
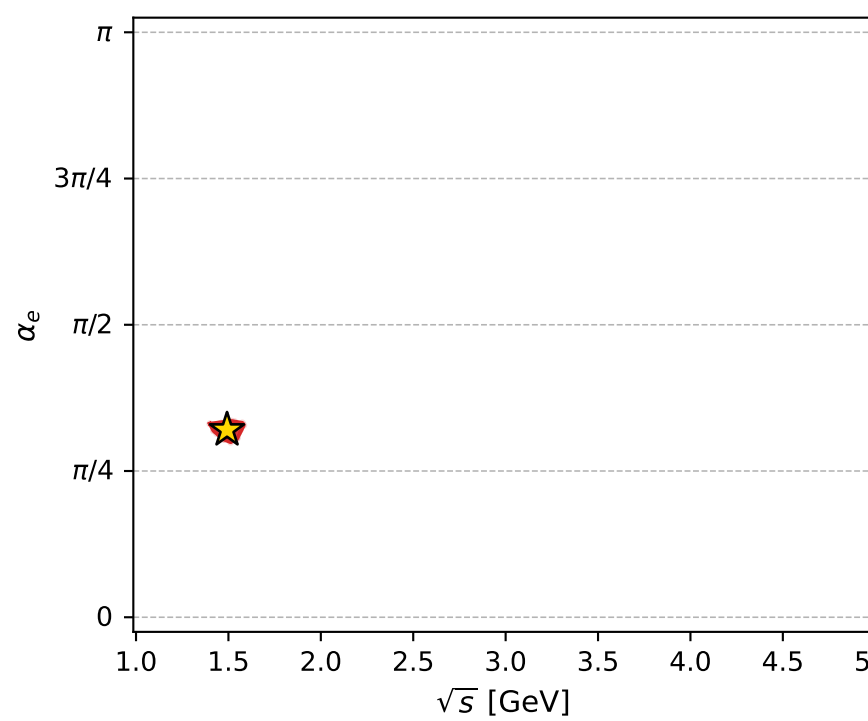
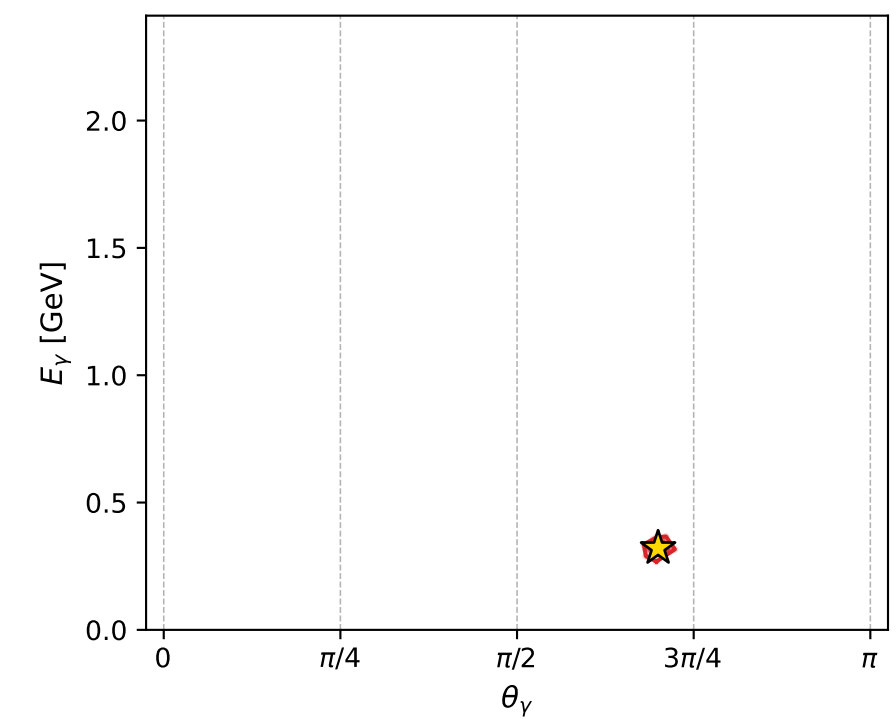
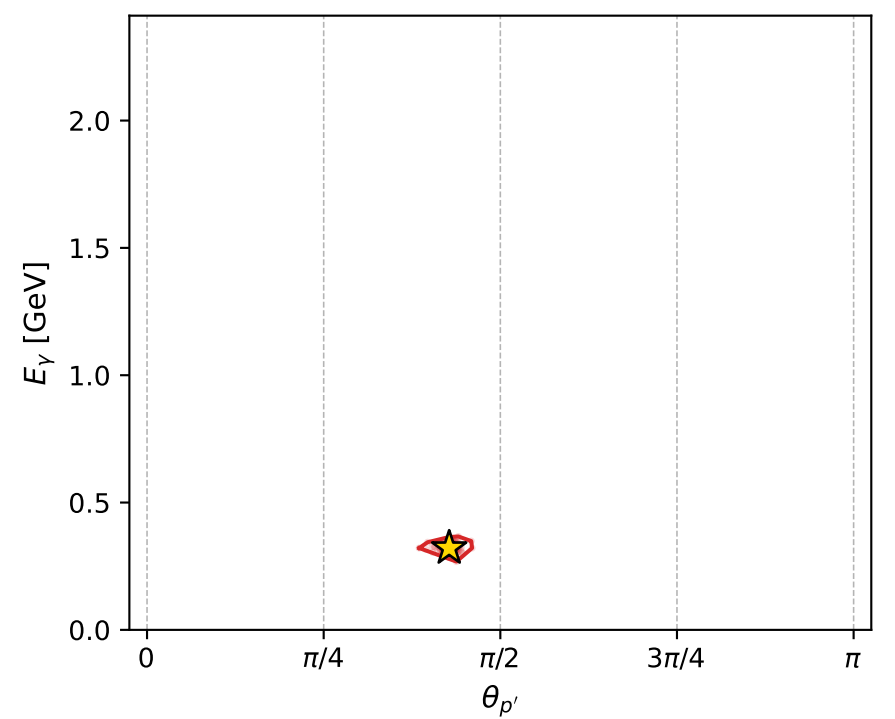
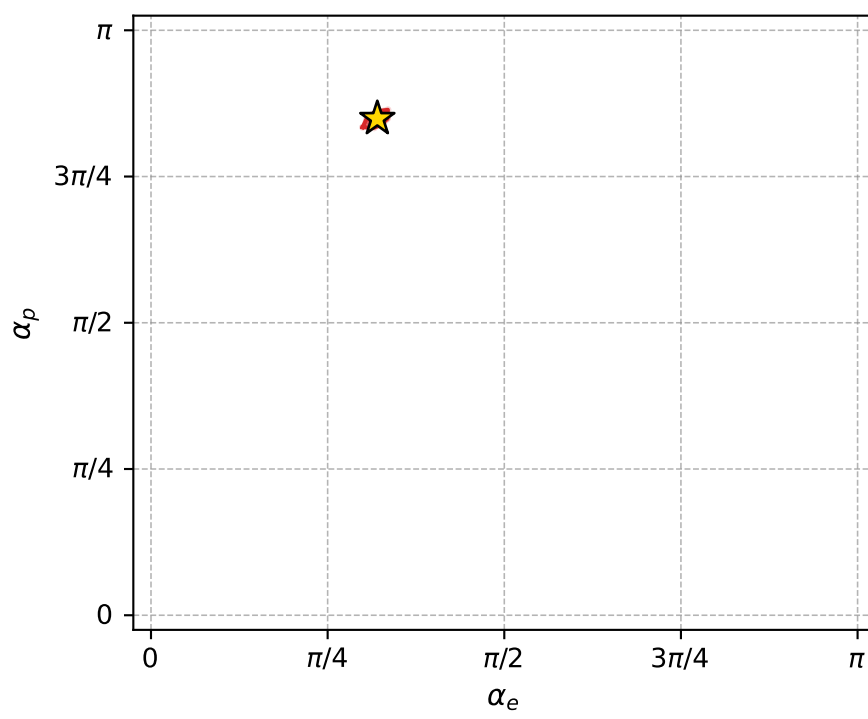
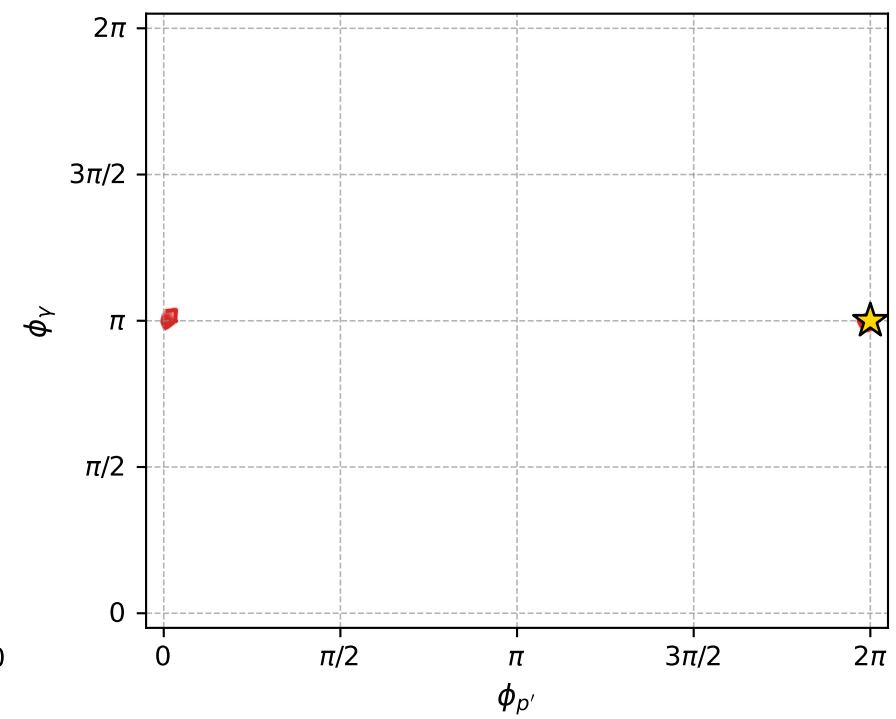
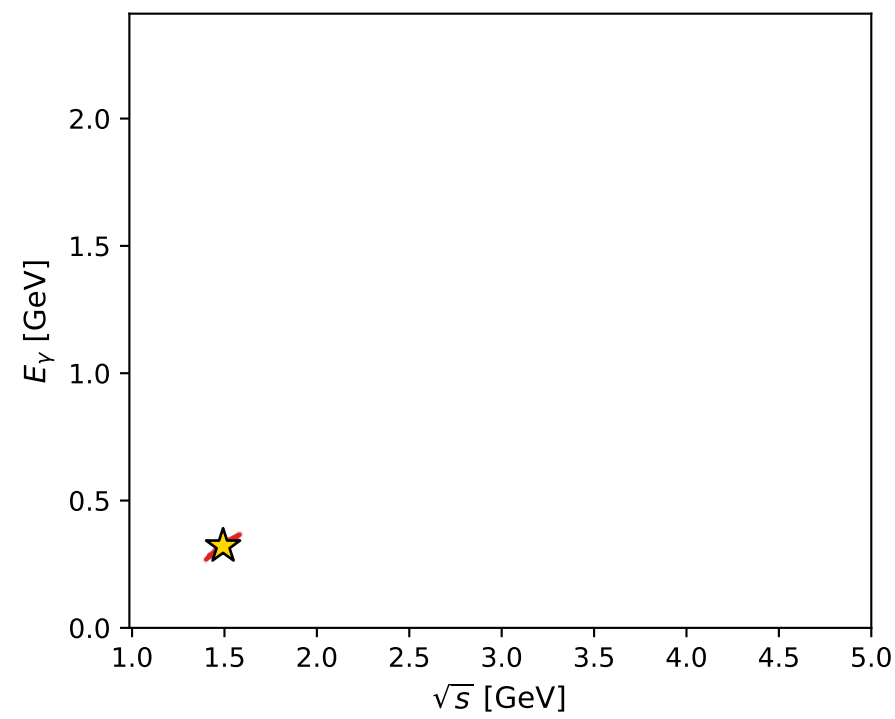
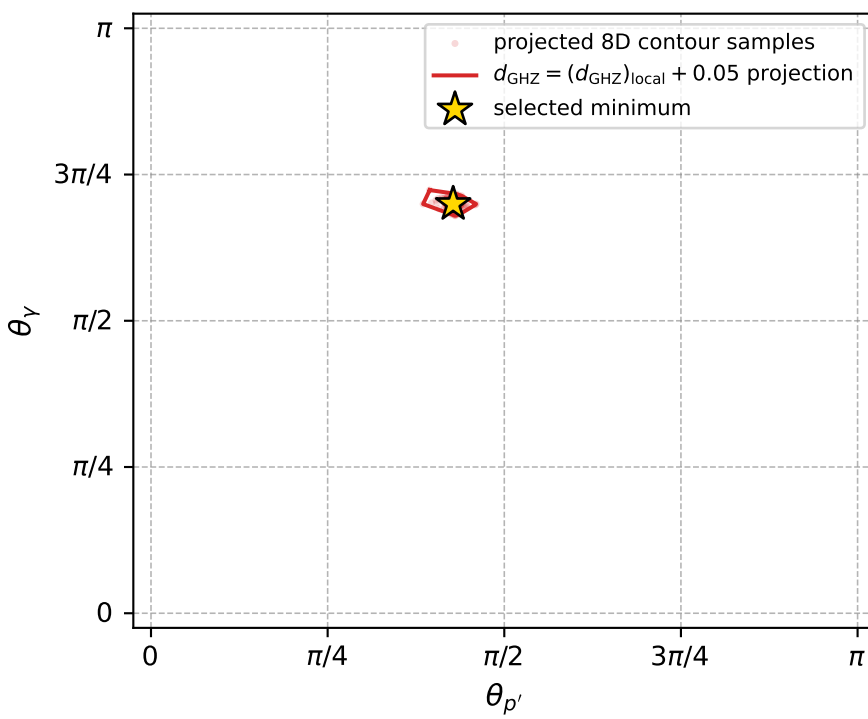
Transverse plane



Leading final-state amplitudes (N=6, retained=99.5%)



electron: polarization cluster P2, local minimum 256; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 0.0001569332$
 $d_{\text{GHZ}} \text{ contour} = 0.050156933$
 above global minimum = 0.00015691091
 8D contour samples = 256

selected phase-space point:

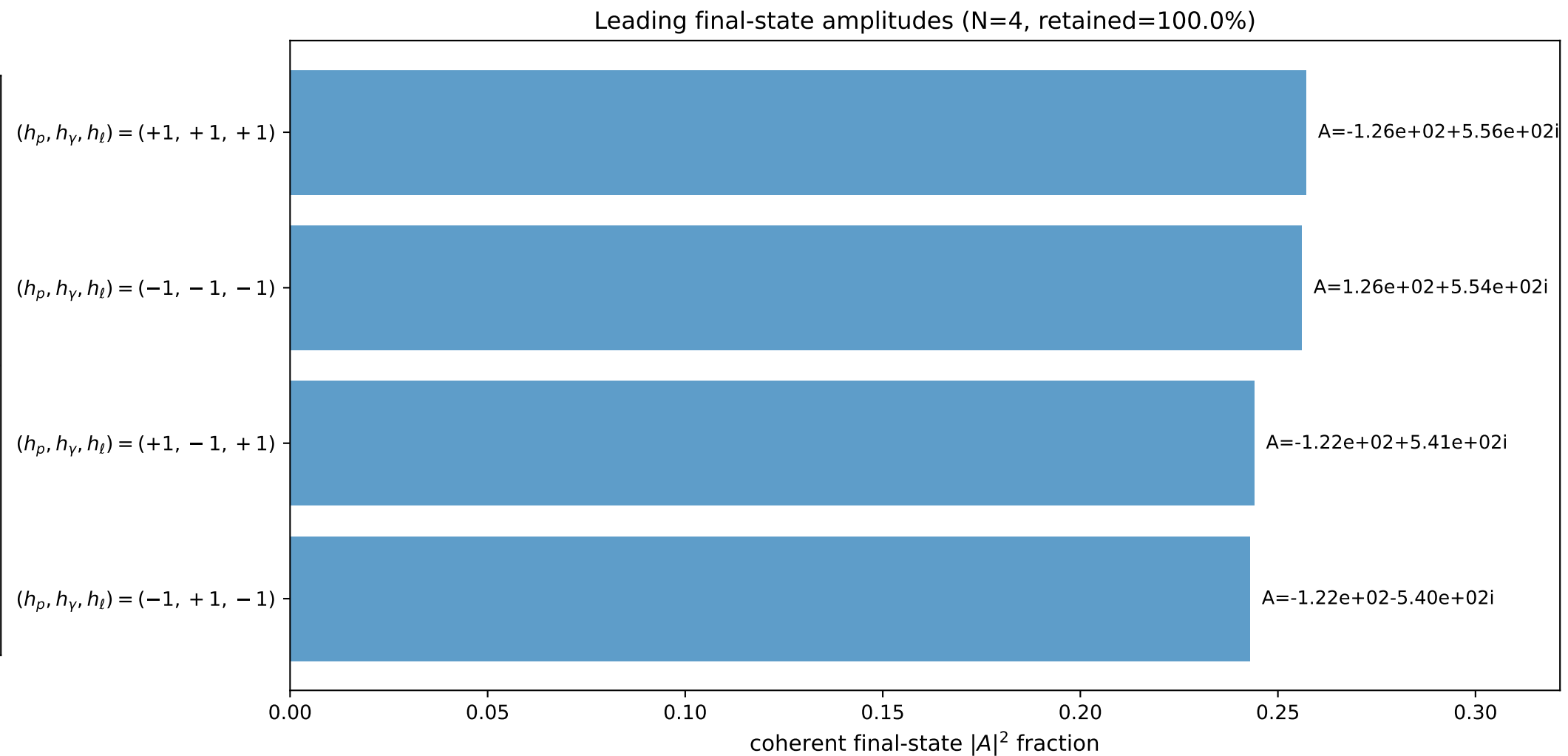
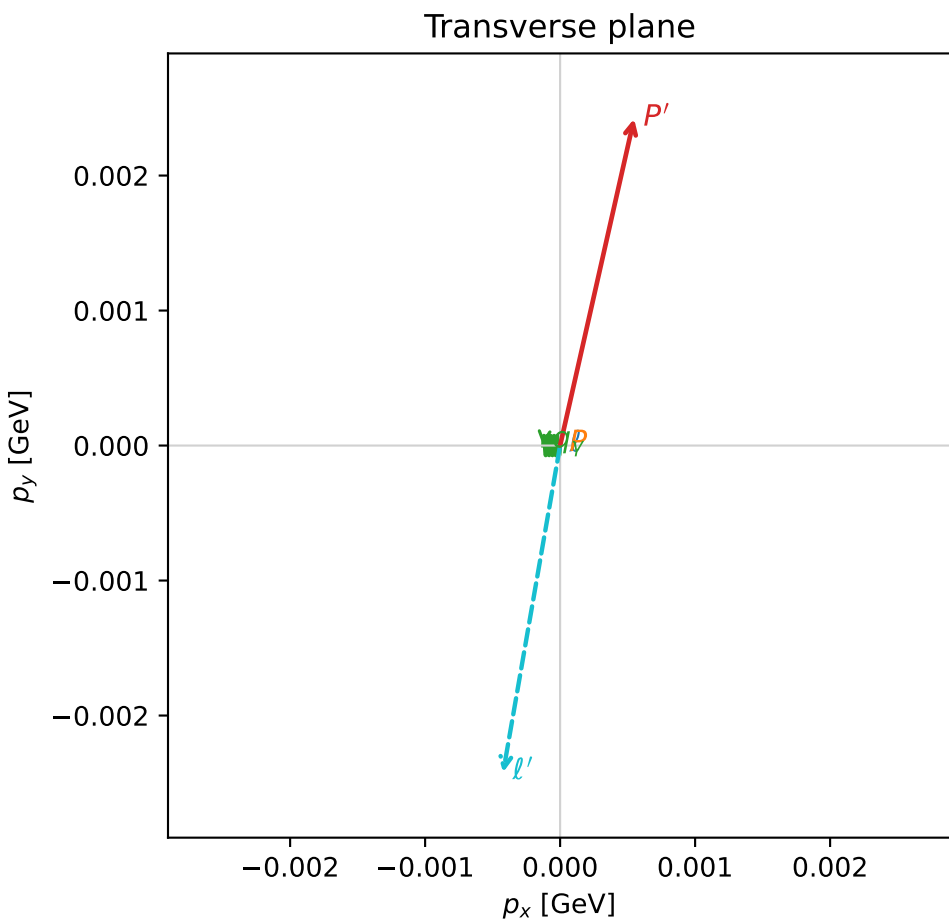
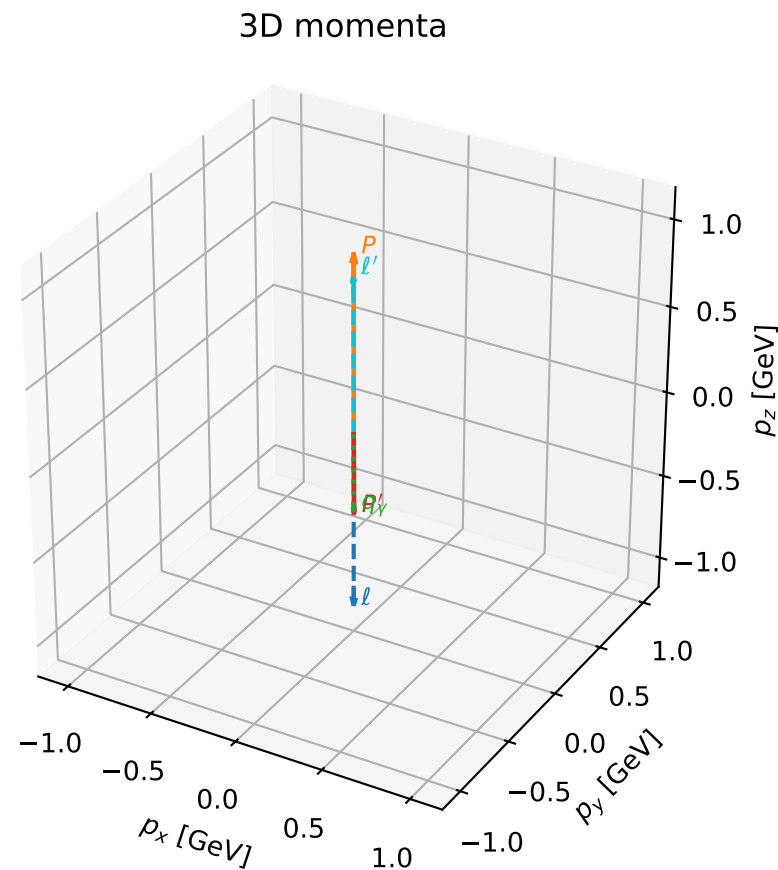
$\text{sqrt_s} = 1.492438$
 $\text{theta_p_out} = 1.3434$
 $\text{theta_gamma_out} = 2.19786$
 $\text{q0out} = 0.3212665$
 $\text{phi_p_out} = 6.283182$
 $\text{phi_gamma_out} = 3.141577$
 $\text{alpha_e} = 1.006269$
 $\text{alpha_p} = 2.667212$

coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

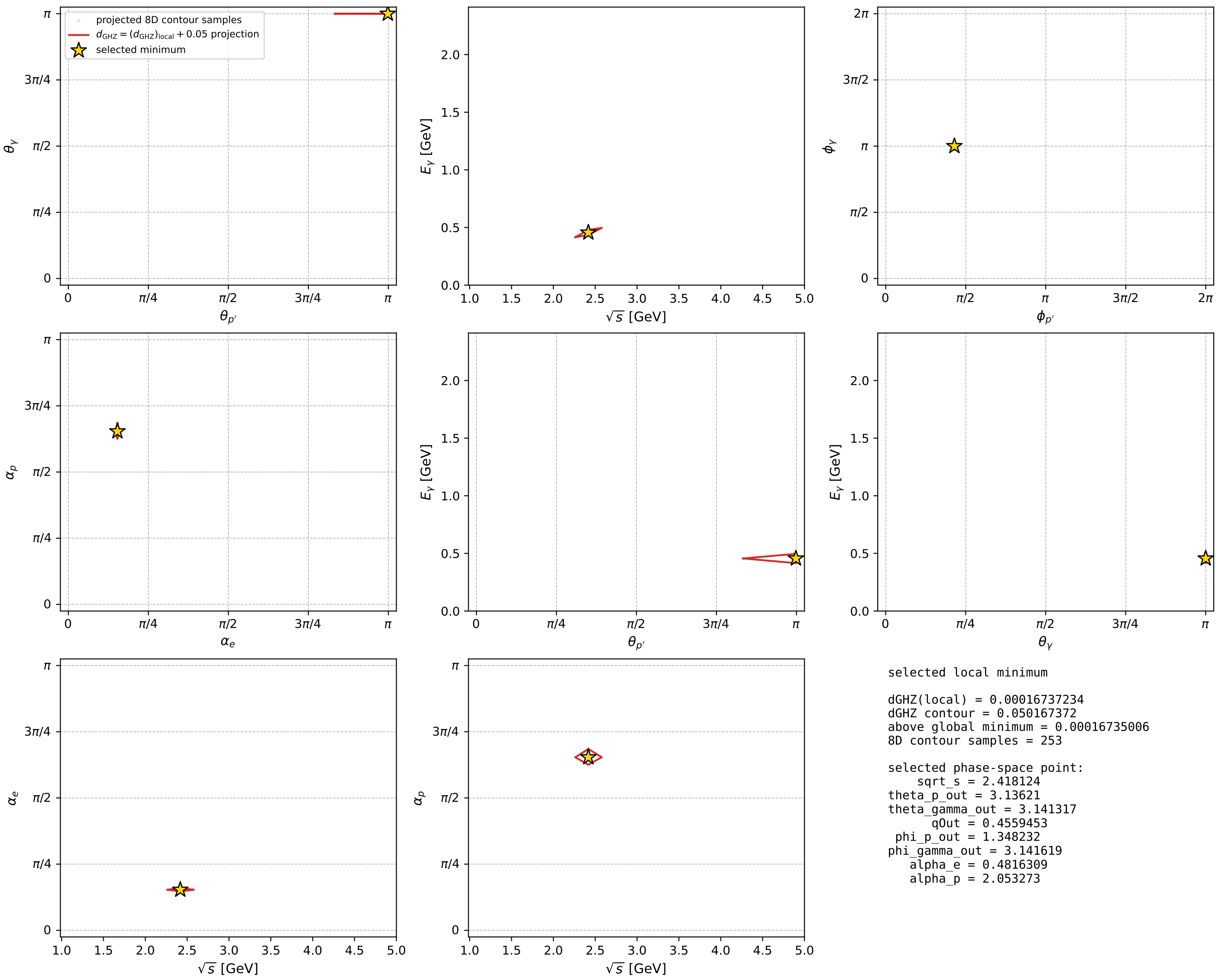
dghz_mixing_angles_polarization_02_local_minimum_259 $d_{\text{GHZ}}=0.000167372$
 region: polarization_cluster_02_local_minimum_259
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=0.48163088$ rad
 incoming lepton: $0.886241|+\rangle +0.463225|-\rangle$
 proton mixing angle: $\alpha_p=2.053273$ rad
 incoming proton: $-0.463975|+\rangle +0.885849|-\rangle$

$s=5.84733$, $\sqrt{s}=2.41812$, $|\vec{P}|=1.02714$, $|\vec{P}'|=0.461052$
 $\theta_{p'}=3.13621$, $\theta_Y=3.14132$, $\phi_{p'}=1.34823$, $\phi_Y=3.14162$
 $E_Y=0.455945$, $\theta_{\ell'}=0.00267942$, $\phi_{\ell'}=4.53983$
 $Q^2=3.7675$, $x_B=0.876128$, $t=-2.09511$, $W^2=1.41252$, $y=0.865664$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 1.0271$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.0271$
 P $E= 1.3910$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.0271$
 ℓ' $E= 0.9170$ $p_x= -0.0004$ $p_y= -0.0024$ $p_z= 0.9170$
 P' $E= 1.0452$ $p_x= 0.0005$ $p_y= 0.0024$ $p_z= -0.4610$
 q_Y $E= 0.4559$ $p_x= -0.0001$ $p_y= -0.0000$ $p_z= -0.4559$



electron: polarization cluster P2, local minimum 259; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



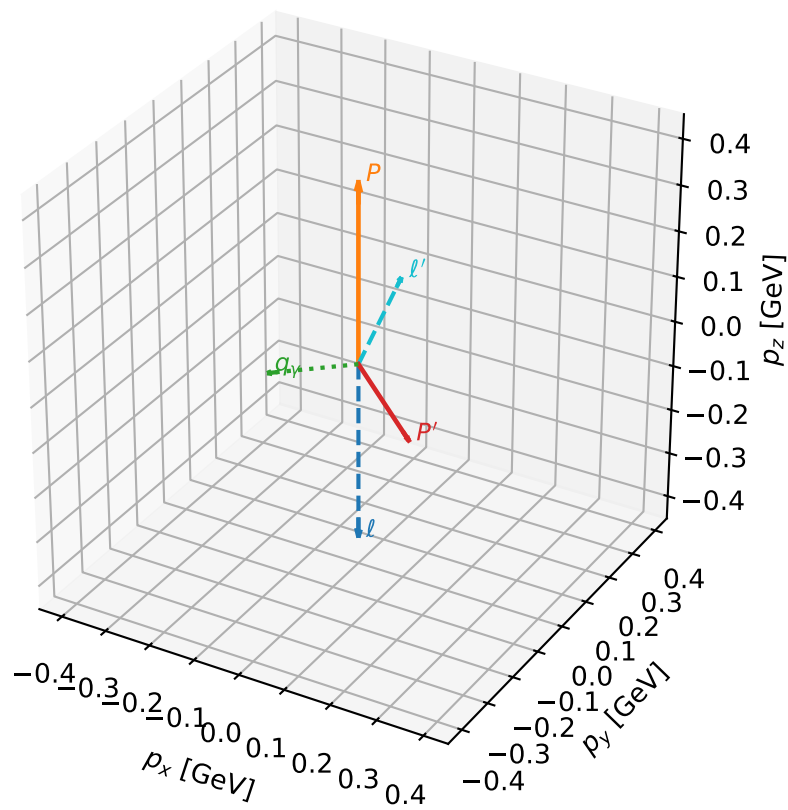
coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

dghz_mixing_angles_polarization_02_local_minimum_263 $d_{\text{GHZ}}=0.000178682$
 region: polarization_cluster_02_local_minimum_263
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=1.3007588$ rad
 incoming lepton: $0.266768|+\rangle +0.963761|-\rangle$
 proton mixing angle: $\alpha_p=2.9247007$ rad
 incoming proton: $-0.976571|+\rangle +0.215195|-\rangle$

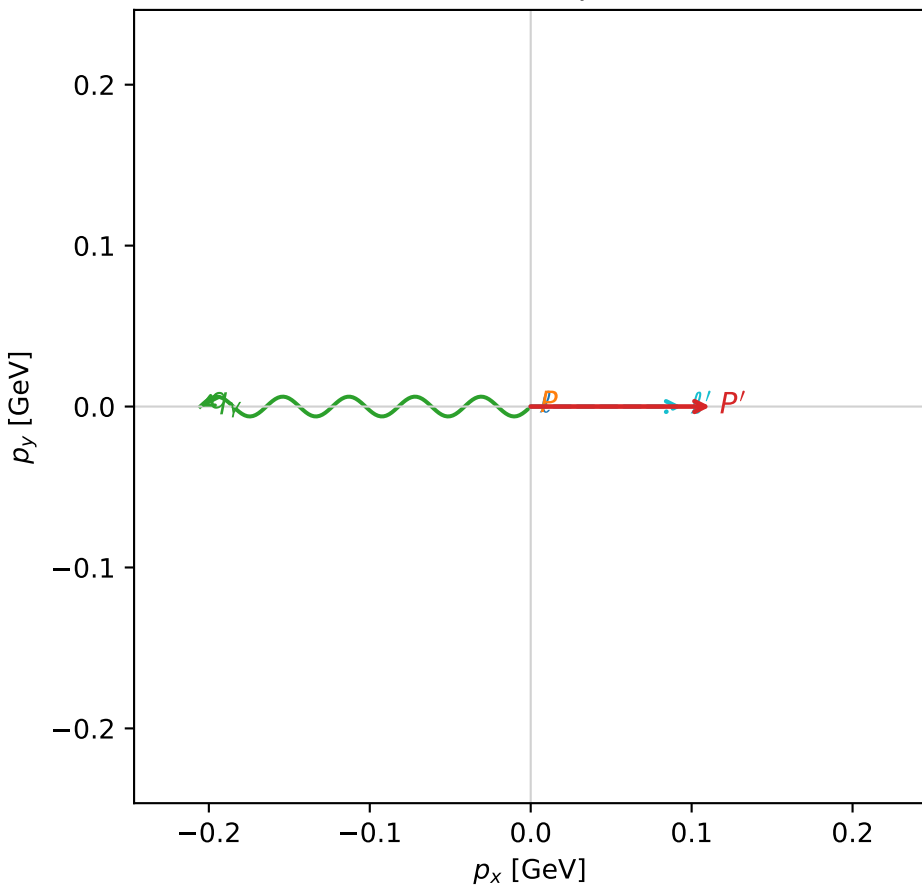
$s=1.98913$, $\sqrt{s}=1.41037$, $|\vec{P}|=0.393263$, $|\vec{P}'|=0.176607$
 $\theta_{P'}=2.45753$, $\theta_Y=1.93955$, $\phi_{P'}=6.79407\text{e-}05$, $\phi_Y=3.14162$
 $E_Y=0.220186$, $\theta_{\ell'}=0.409202$, $\phi_{\ell'}=6.28317$
 $Q^2=0.35546$, $x_B=0.444378$, $t=-0.289578$, $W^2=1.32429$, $y=0.721097$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.3933$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.3933$
 P $E= 1.0171$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.3933$
 ℓ' $E= 0.2357$ $p_x= 0.0938$ $p_y= -0.0000$ $p_z= 0.2162$
 P' $E= 0.9545$ $p_x= 0.1116$ $p_y= 0.0000$ $p_z= -0.1369$
 q_Y $E= 0.2202$ $p_x= -0.2054$ $p_y= -0.0000$ $p_z= -0.0794$

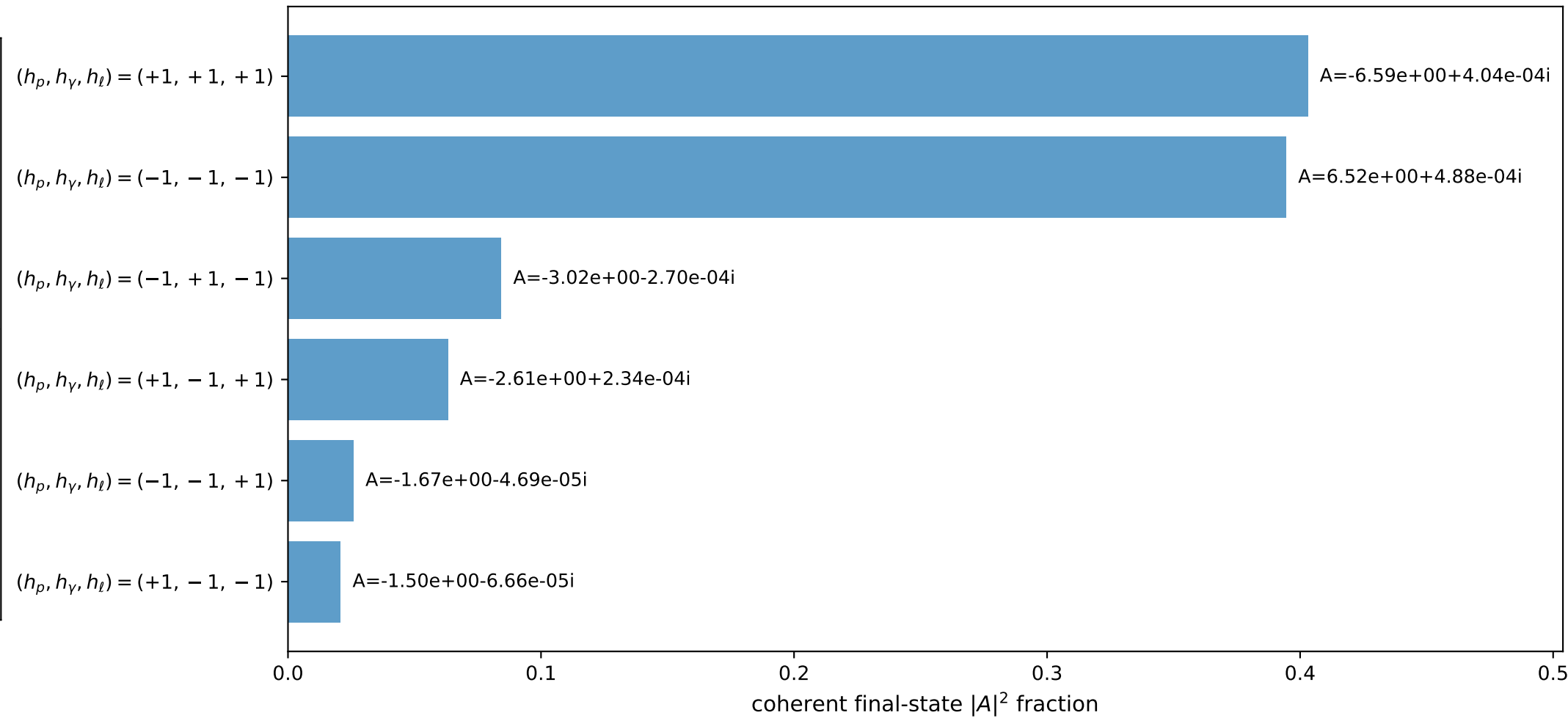
3D momenta



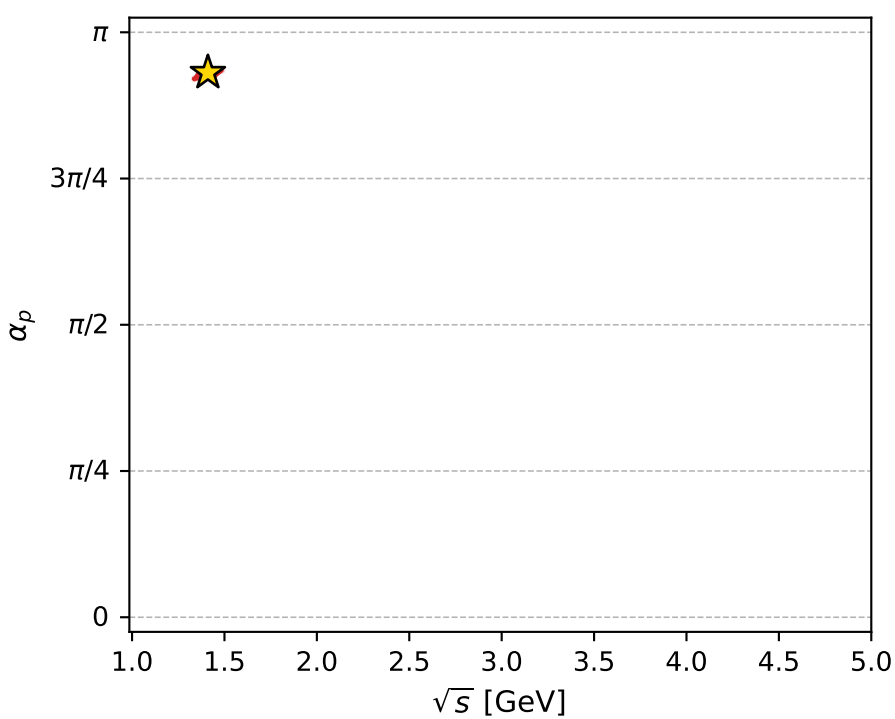
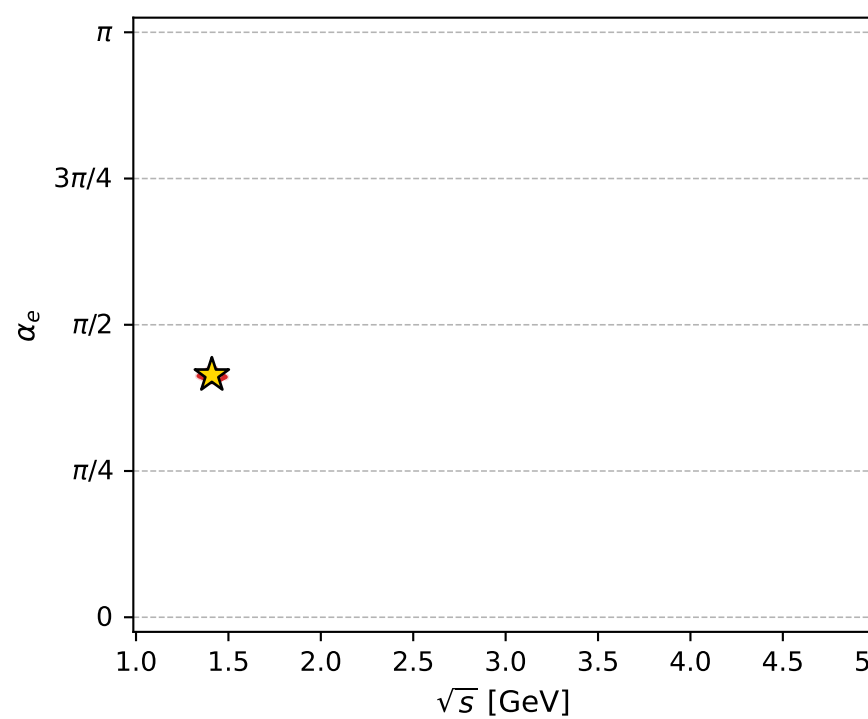
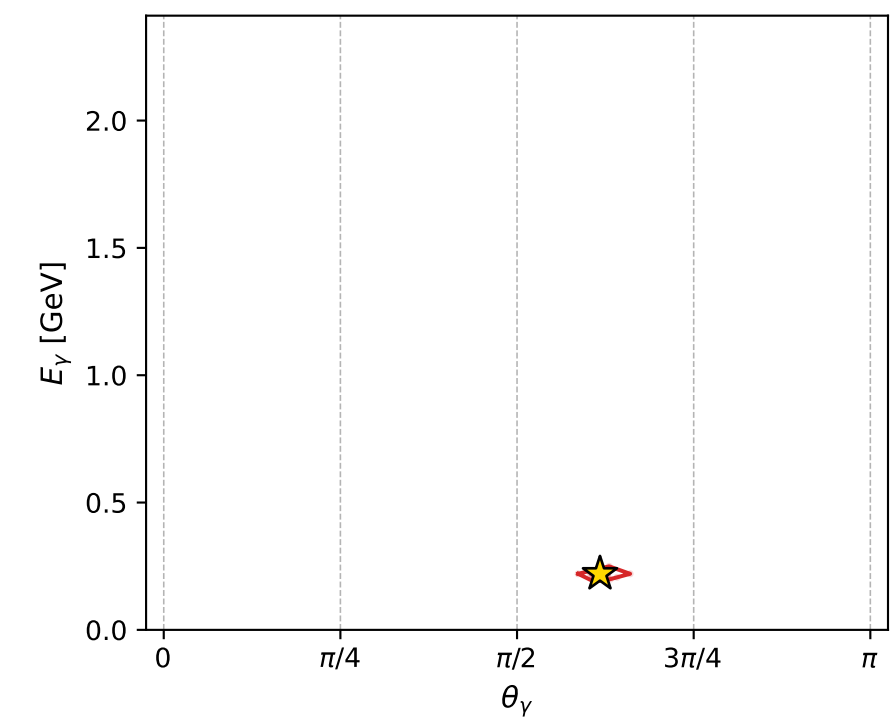
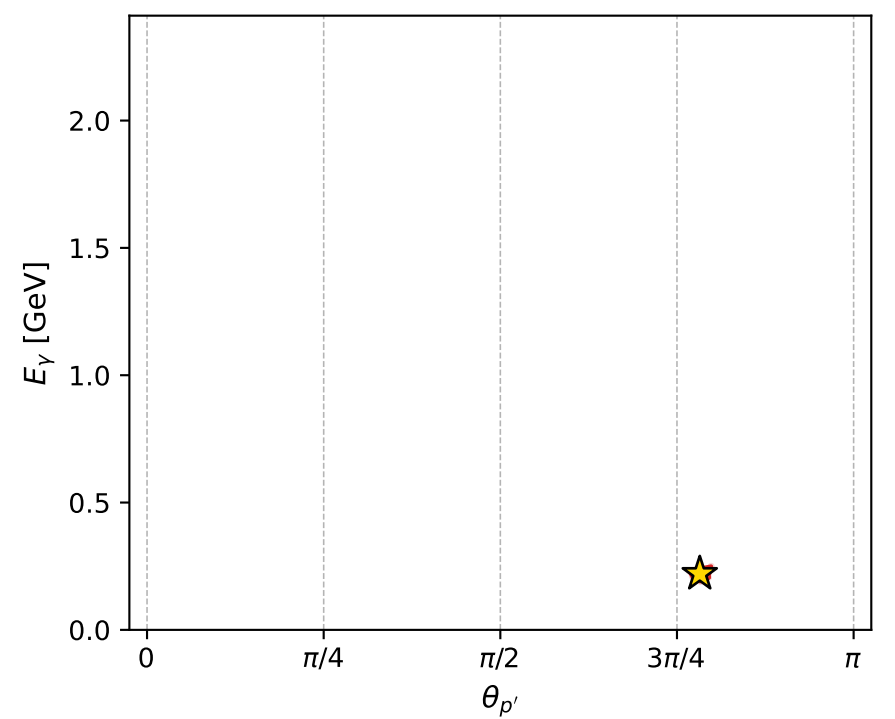
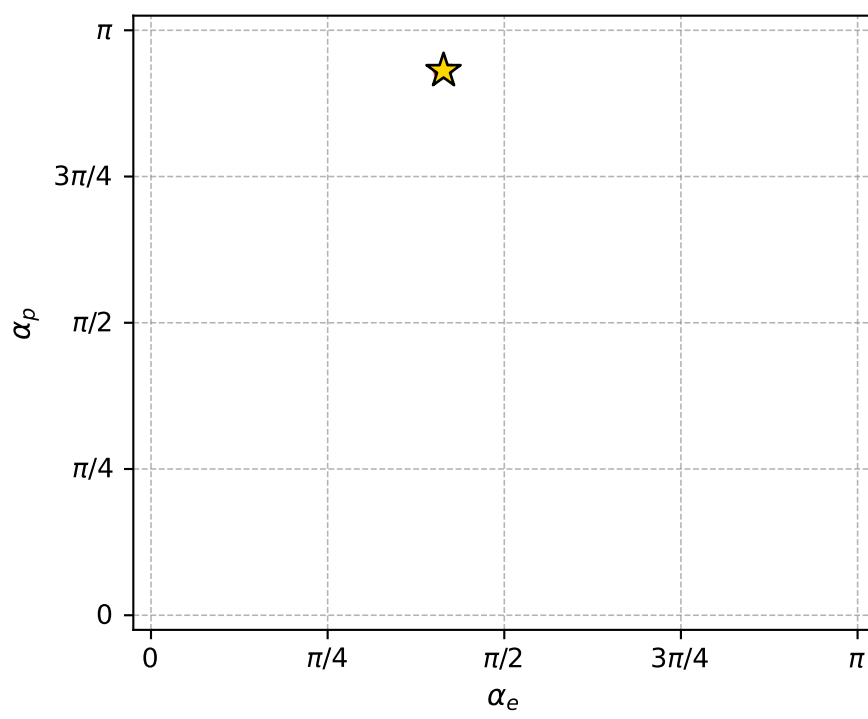
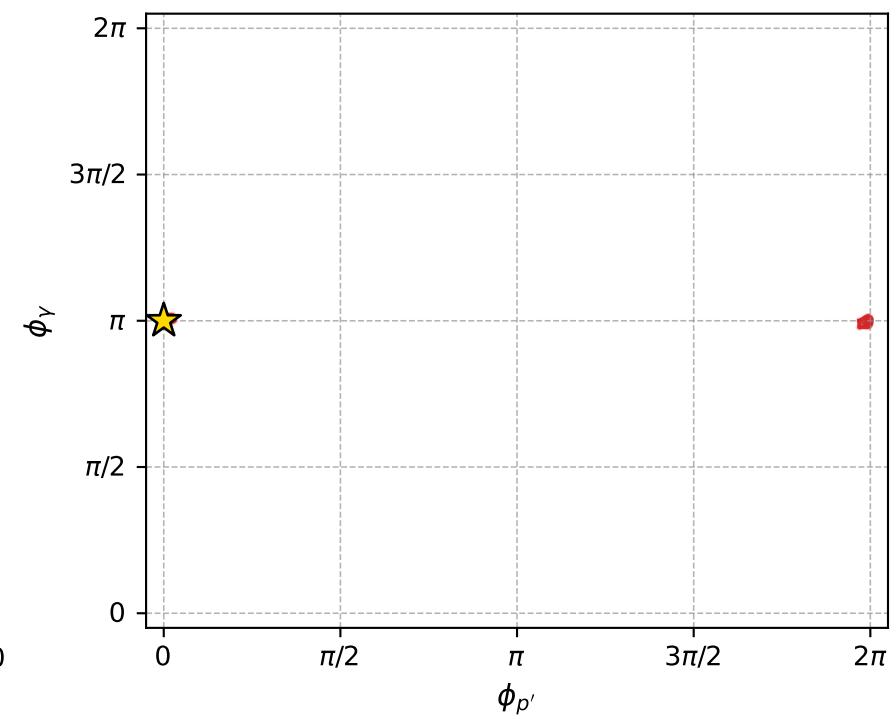
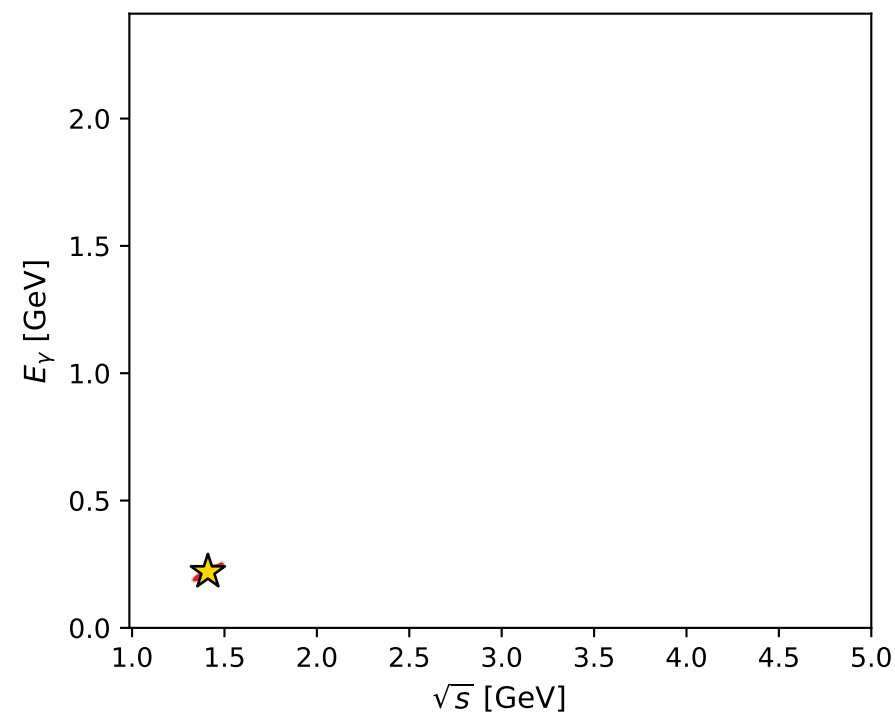
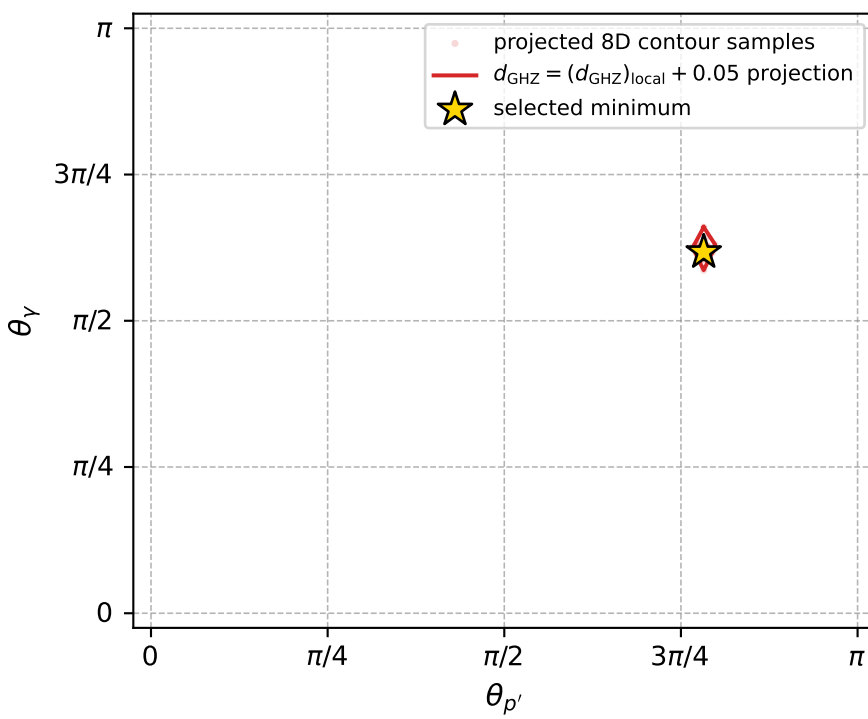
Transverse plane



Leading final-state amplitudes (N=6, retained=99.2%)



electron: polarization cluster P2, local minimum 263; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 0.00017868222$
 $d_{\text{GHZ}} \text{ contour} = 0.050178682$
 above global minimum = 0.00017865994
 8D contour samples = 256

selected phase-space point:

$\text{sqrt}_s = 1.410366$
 $\text{theta}_p \text{ out} = 2.457535$
 $\text{theta}_\gamma \text{ out} = 1.939547$
 $q_{\text{out}} = 0.2201861$
 $\text{phi}_p \text{ out} = 6.794067\text{e-}05$
 $\text{phi}_\gamma \text{ out} = 3.141621$
 $\alpha_e = 1.300759$
 $\alpha_p = 2.924701$

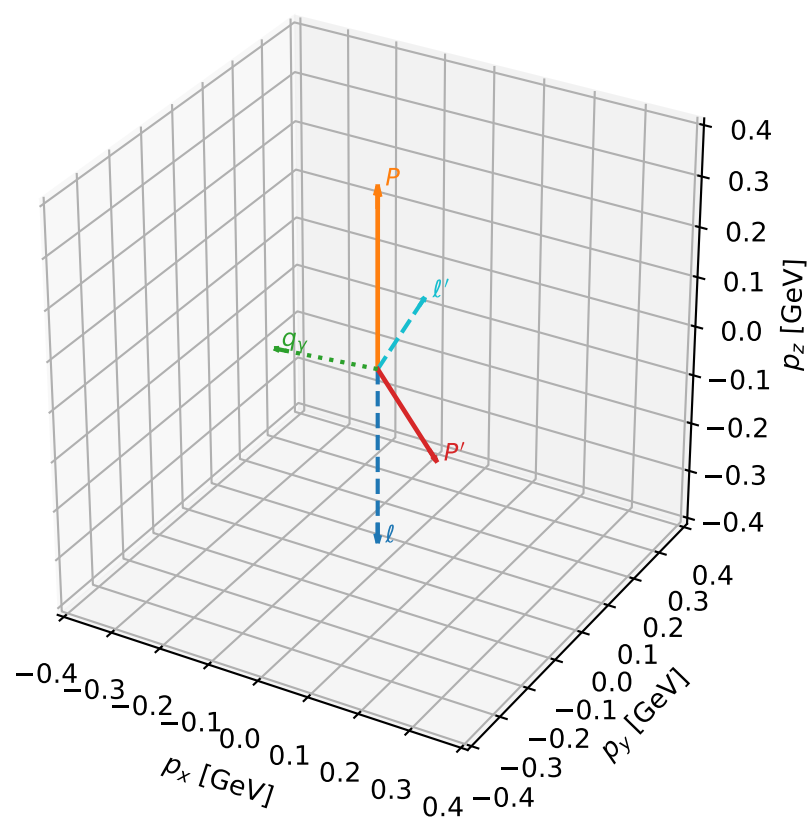
coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

dghz_mixing_angles_polarization_02_local_minimum_270 $d_{\{\text{rm GHZ}\}}=0.000198685$
 region: polarization_cluster_02_local_minimum_270
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=1.2456602$ rad
 incoming lepton: $0.319438|+> +0.947607|->$
 proton mixing angle: $\alpha_p=2.8771946$ rad
 incoming proton: $-0.96525|+> +0.261328|->$

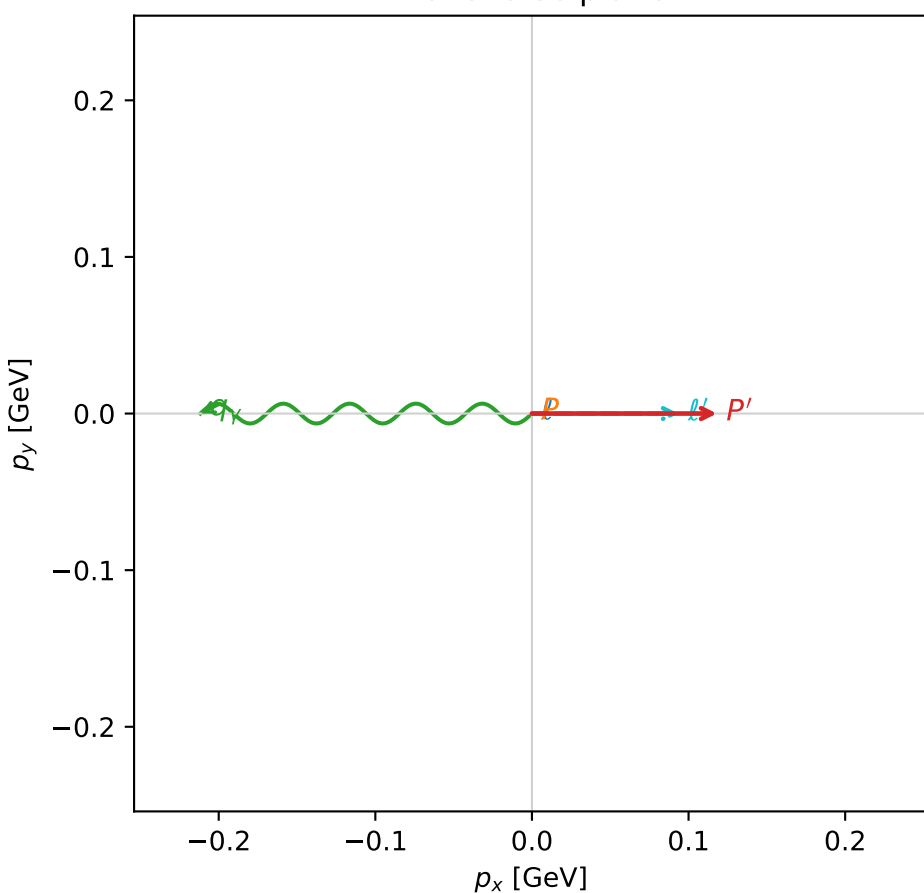
$s=1.859$, $\sqrt{s}=1.36345$, $|\vec{P}|=0.359073$, $|\vec{P}'|=0.191491$
 $\theta_{P'}=2.47702$, $\theta_Y=1.65891$, $\phi_{P'}=0.000108404$, $\phi_Y=3.14167$
 $E_Y=0.212527$, $\theta_{\ell'}=0.504713$, $\phi_{\ell'}=4.20313\text{e-}05$
 $Q^2=0.2607$, $x_B=0.366158$, $t=-0.271642$, $W^2=1.33113$, $y=0.727145$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.3591$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.3591$
 P $E= 1.0044$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.3591$
 ℓ' $E= 0.1936$ $p_x= 0.0936$ $p_y= 0.0000$ $p_z= 0.1694$
 P' $E= 0.9573$ $p_x= 0.1181$ $p_y= 0.0000$ $p_z= -0.1507$
 q_Y $E= 0.2125$ $p_x= -0.2117$ $p_y= -0.0000$ $p_z= -0.0187$

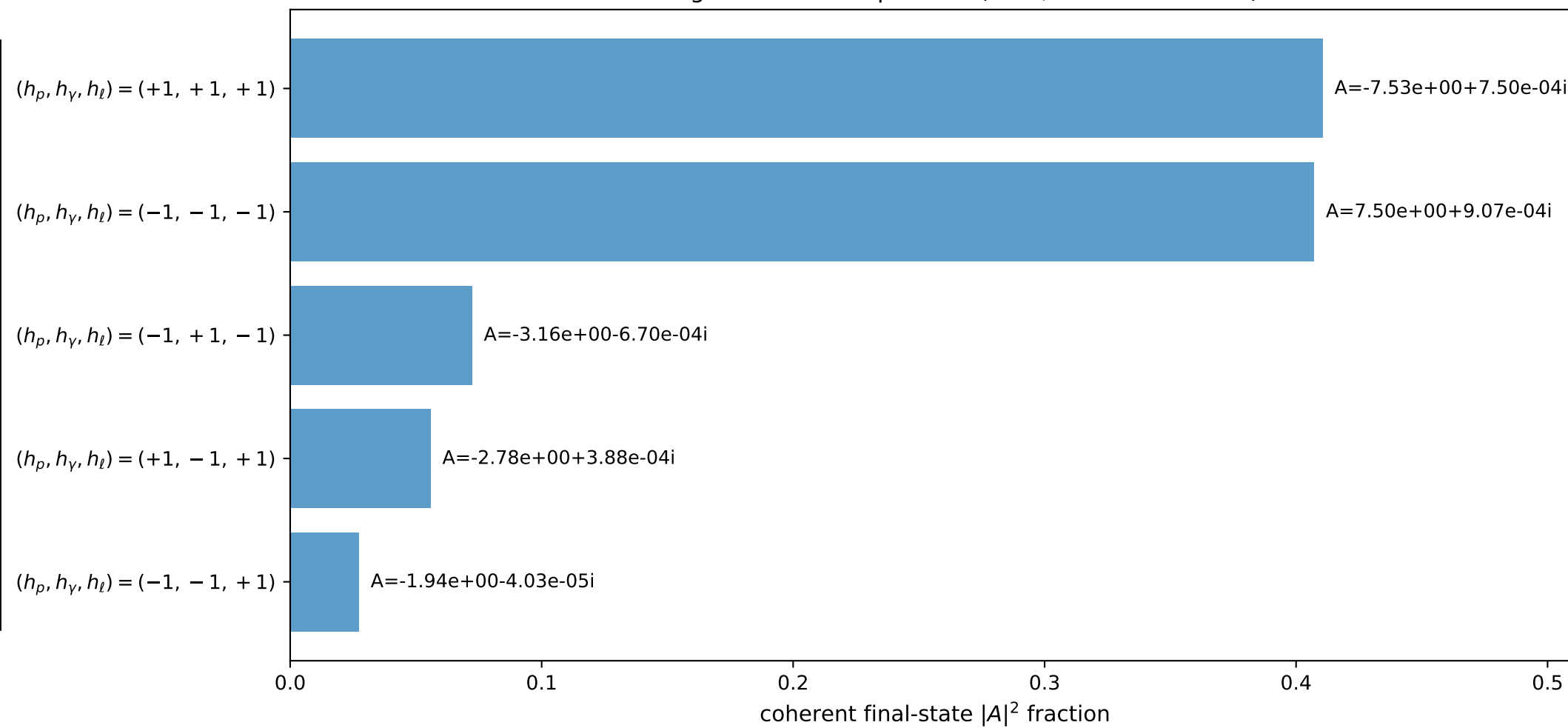
3D momenta



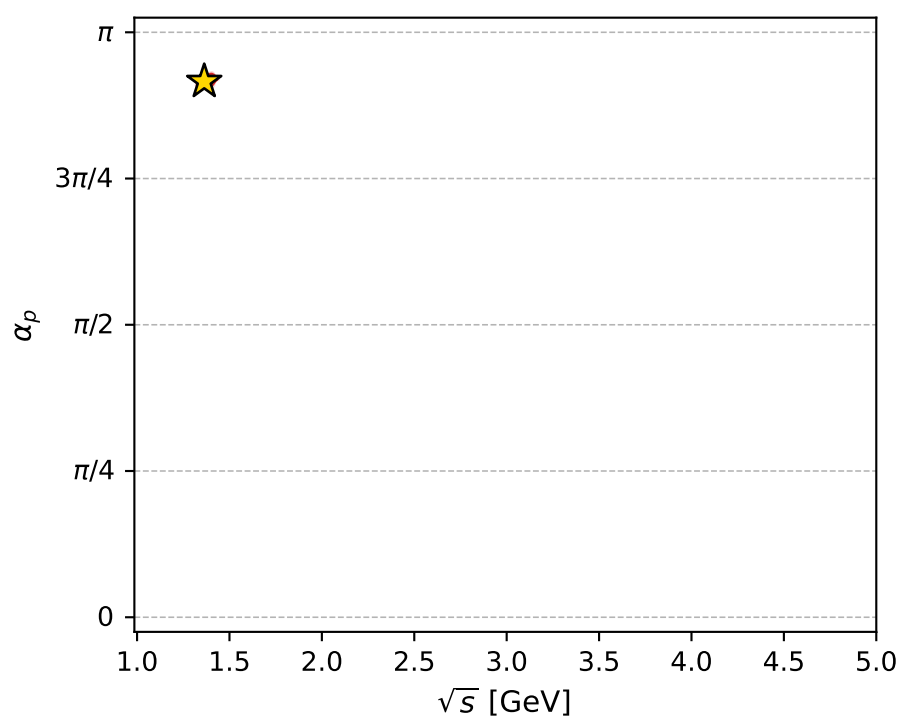
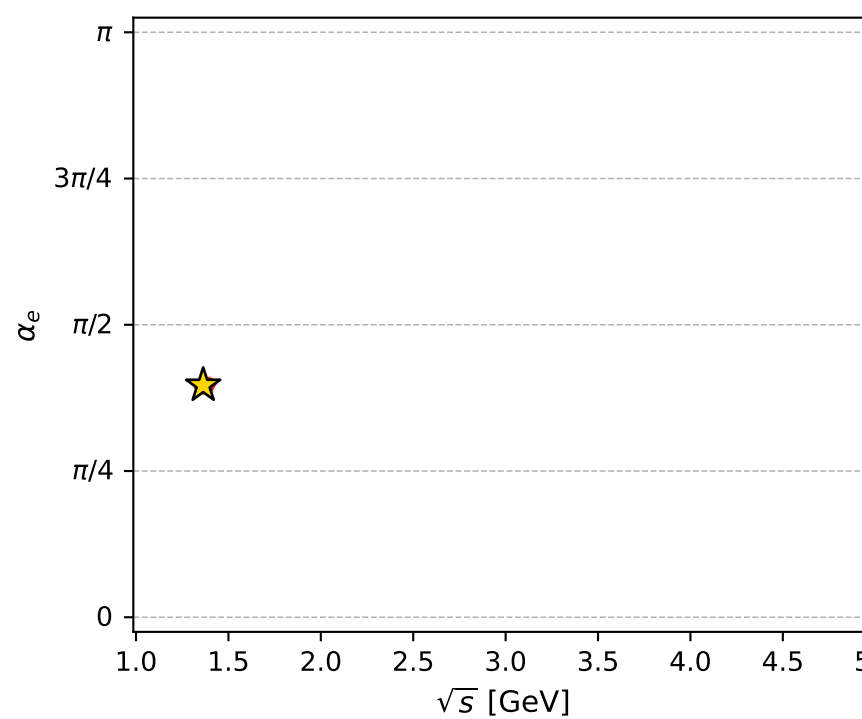
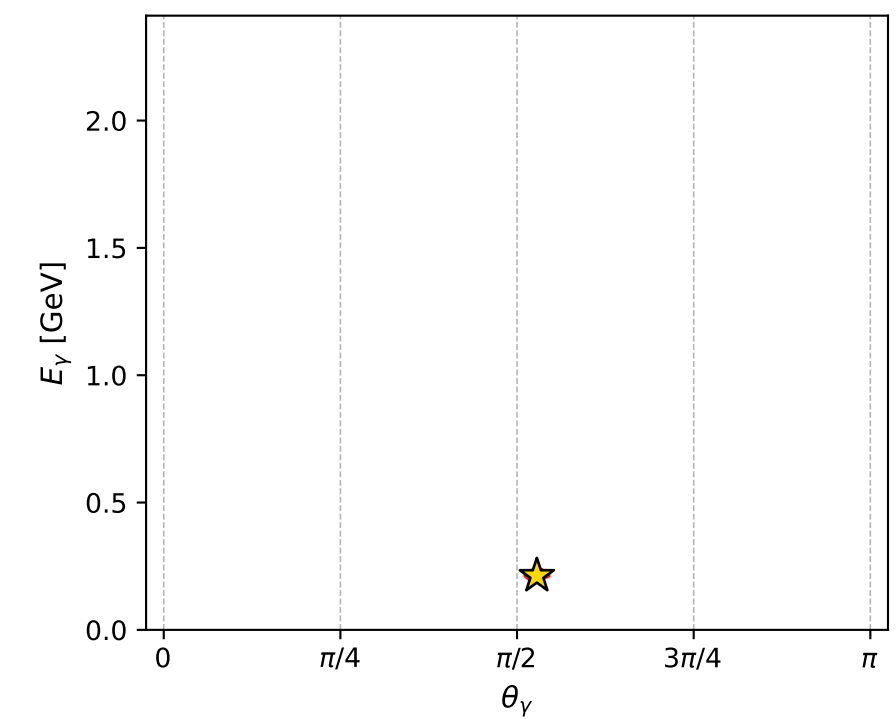
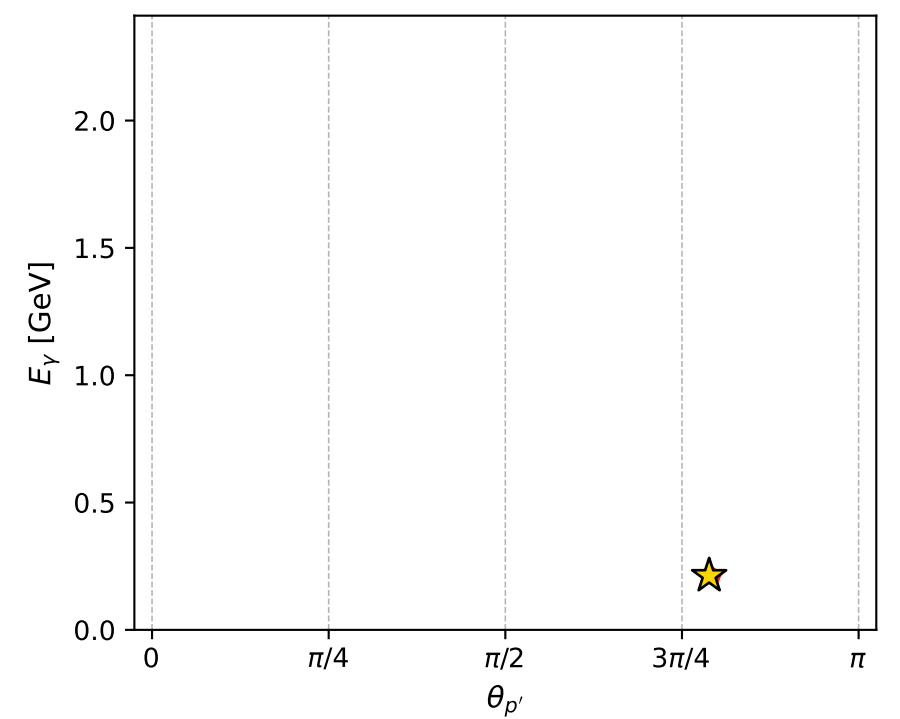
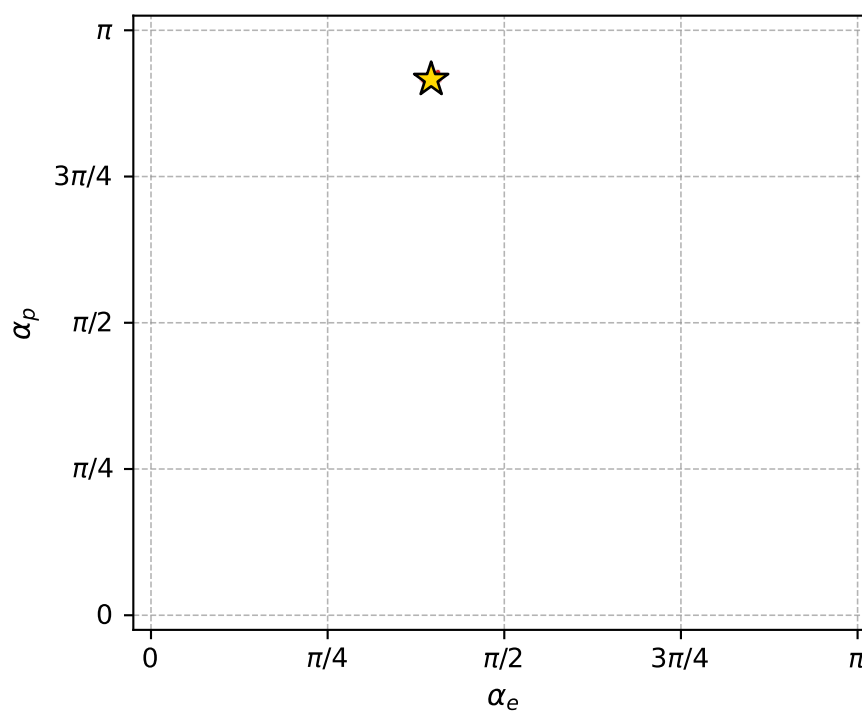
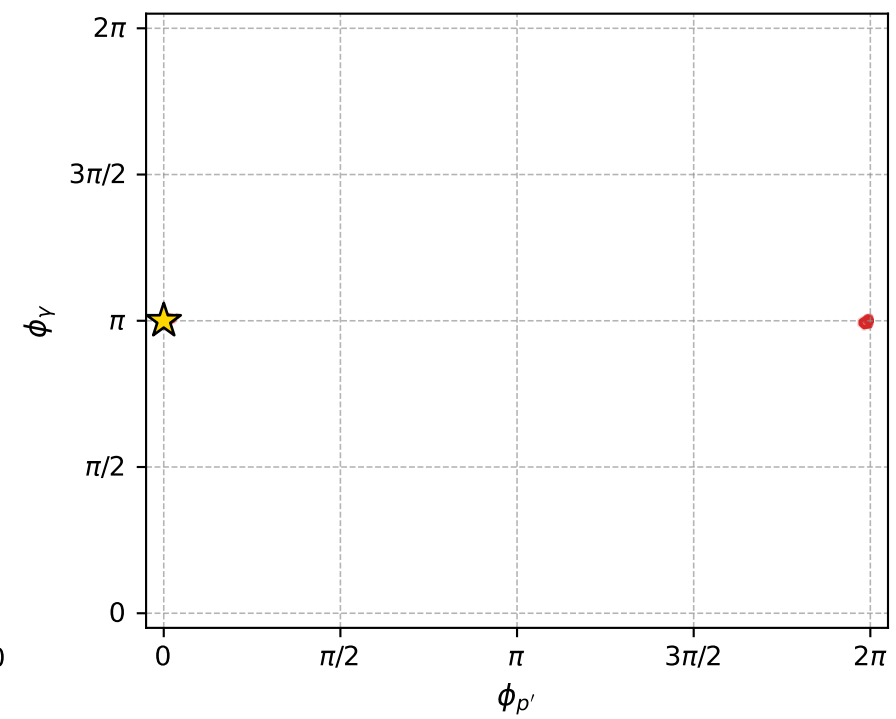
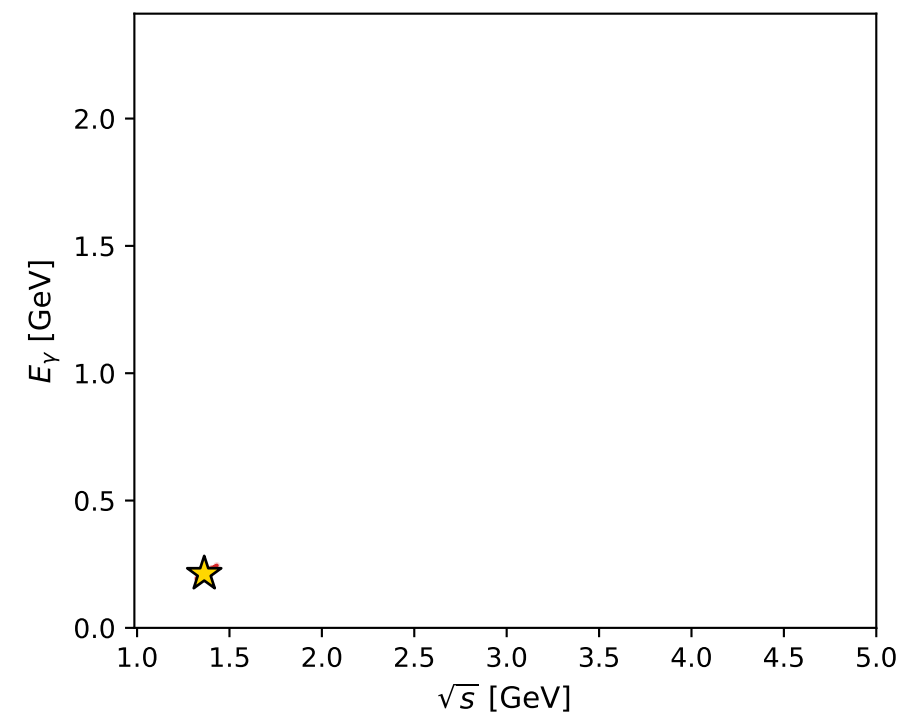
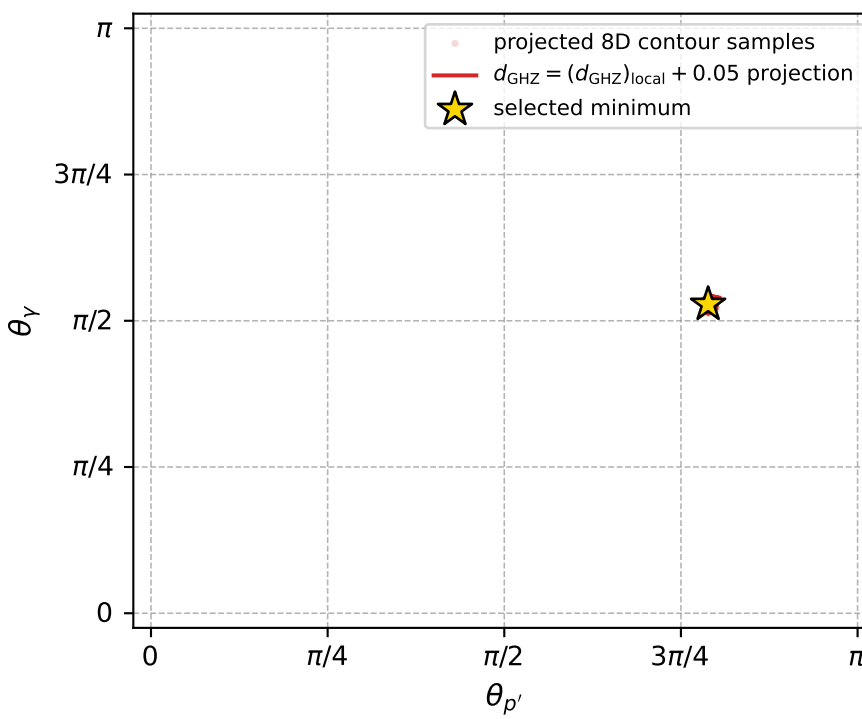
Transverse plane



Leading final-state amplitudes (N=5, retained=97.3%)



electron: polarization cluster P2, local minimum 270; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 0.00019868481$
 $d_{\text{GHZ}} \text{ contour} = 0.050198685$
 above global minimum = 0.00019866253
 8D contour samples = 256

selected phase-space point:

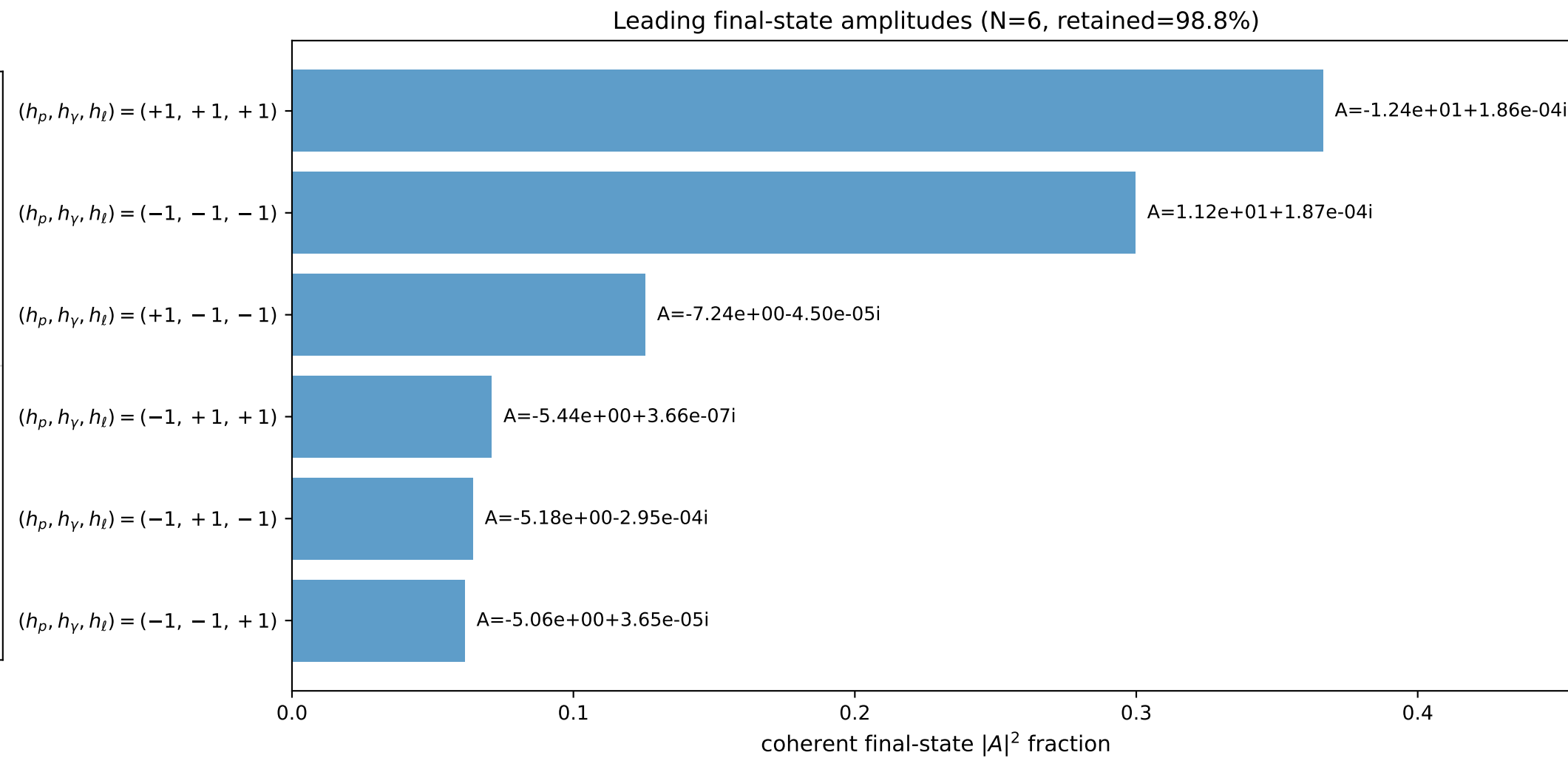
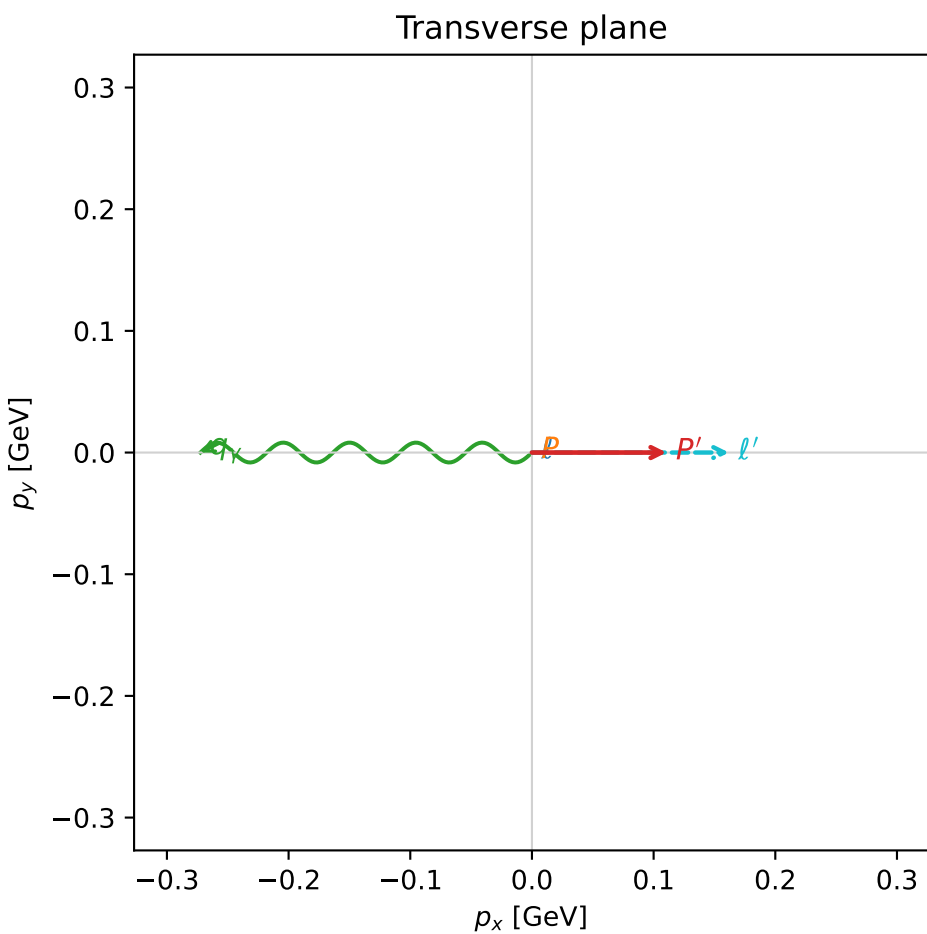
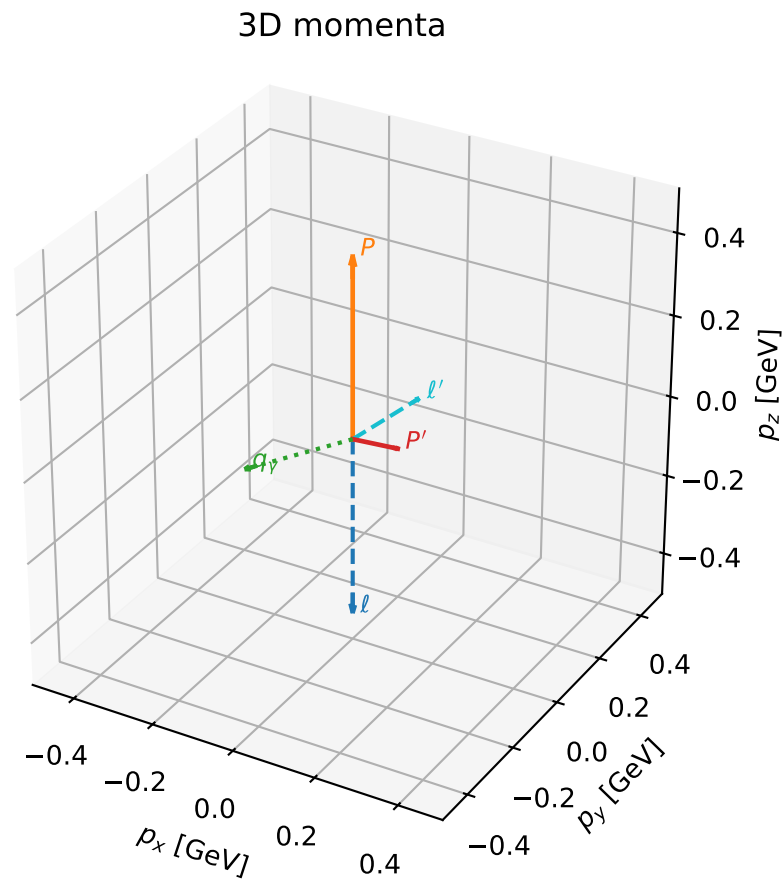
$\text{sqrt_s} = 1.363452$
 $\text{theta_p_out} = 2.47702$
 $\text{theta_gamma_out} = 1.658912$
 $\text{qOut} = 0.2125273$
 $\text{phi_p_out} = 0.0001084042$
 $\text{phi_gamma_out} = 3.141672$
 $\text{alpha_e} = 1.24566$
 $\text{alpha_p} = 2.877195$

coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

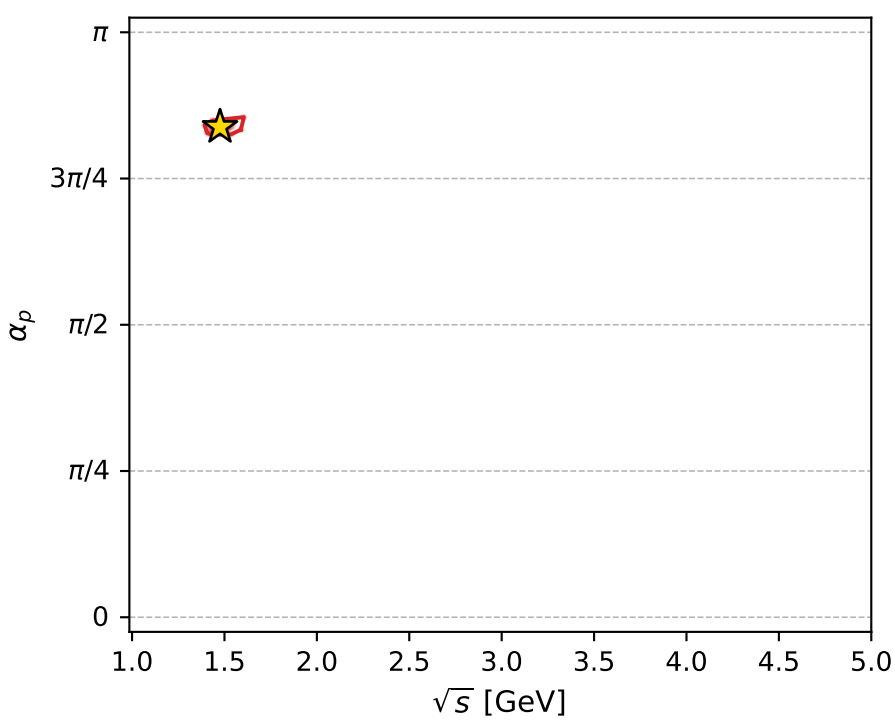
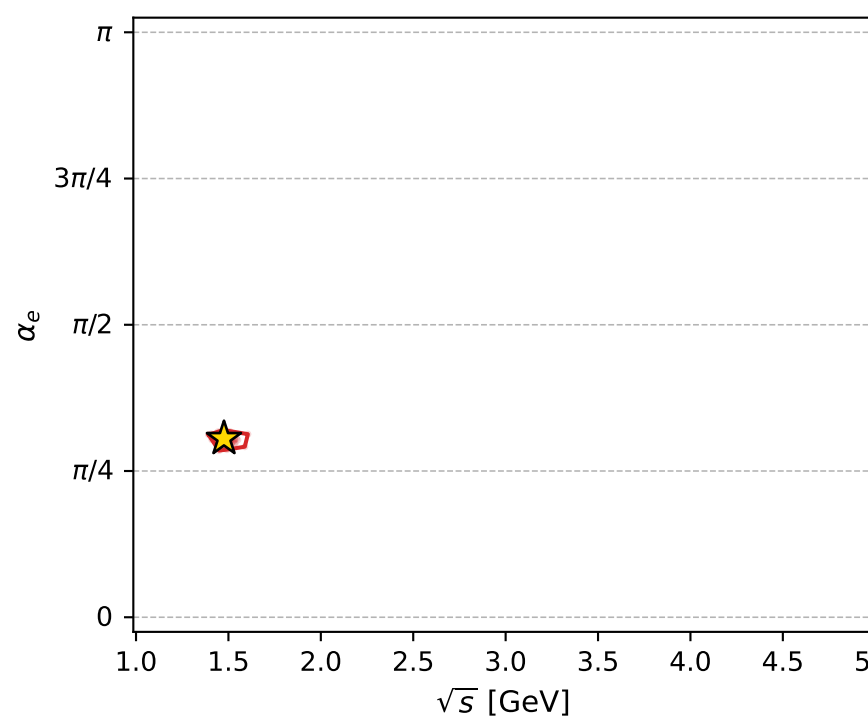
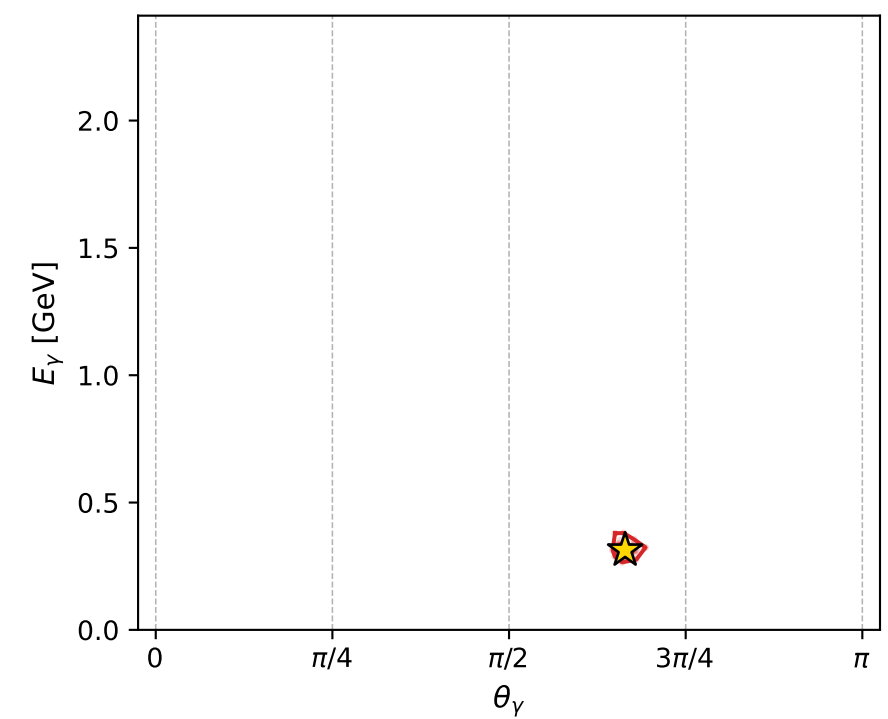
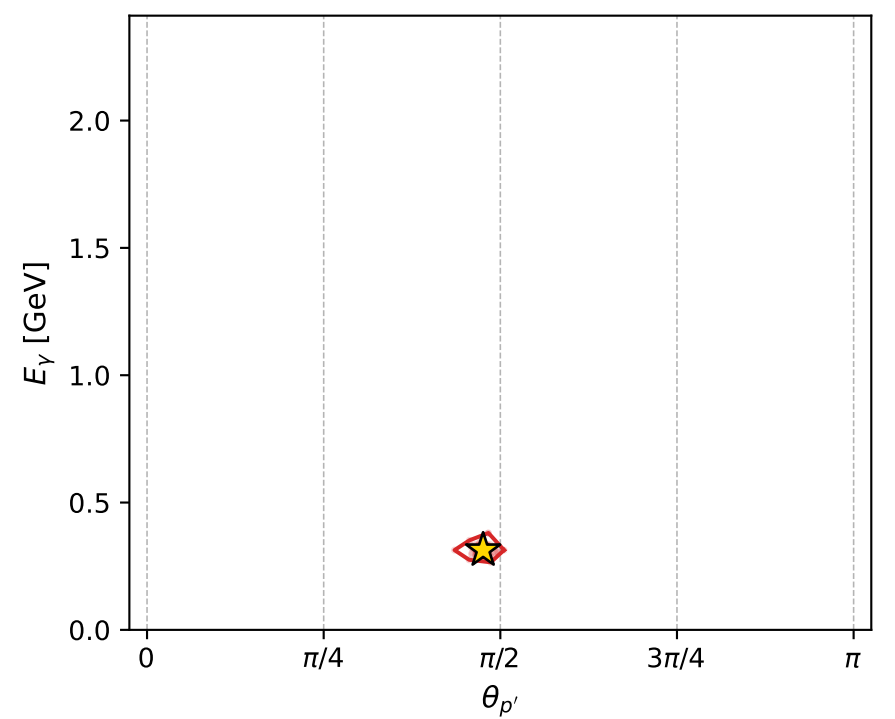
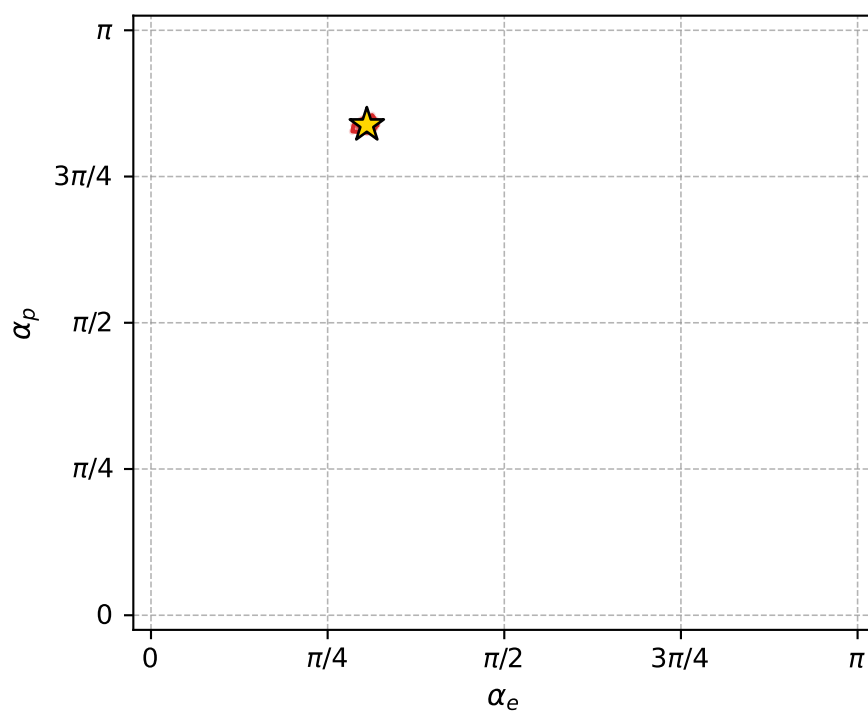
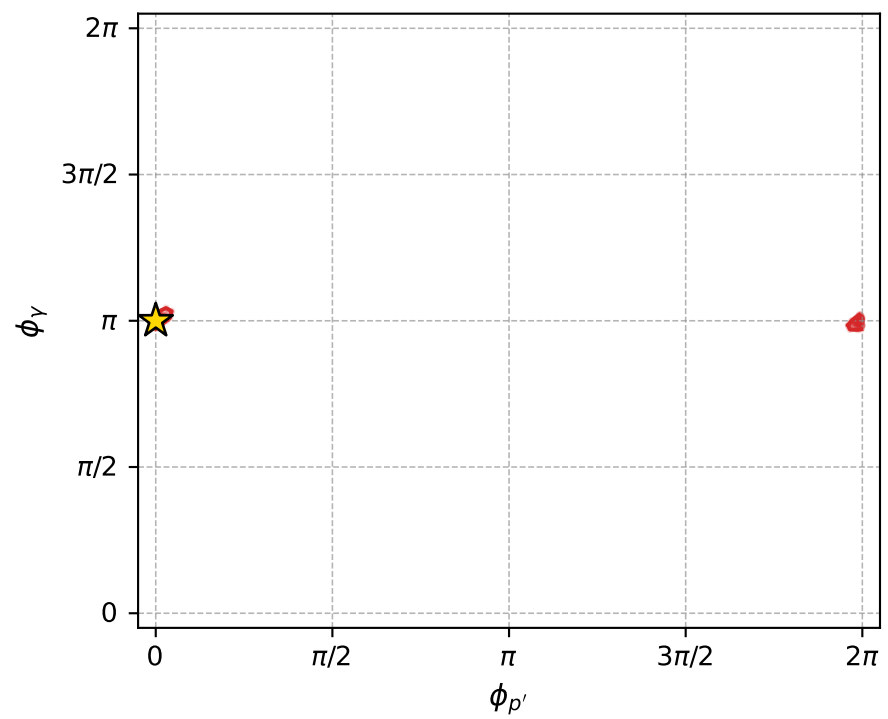
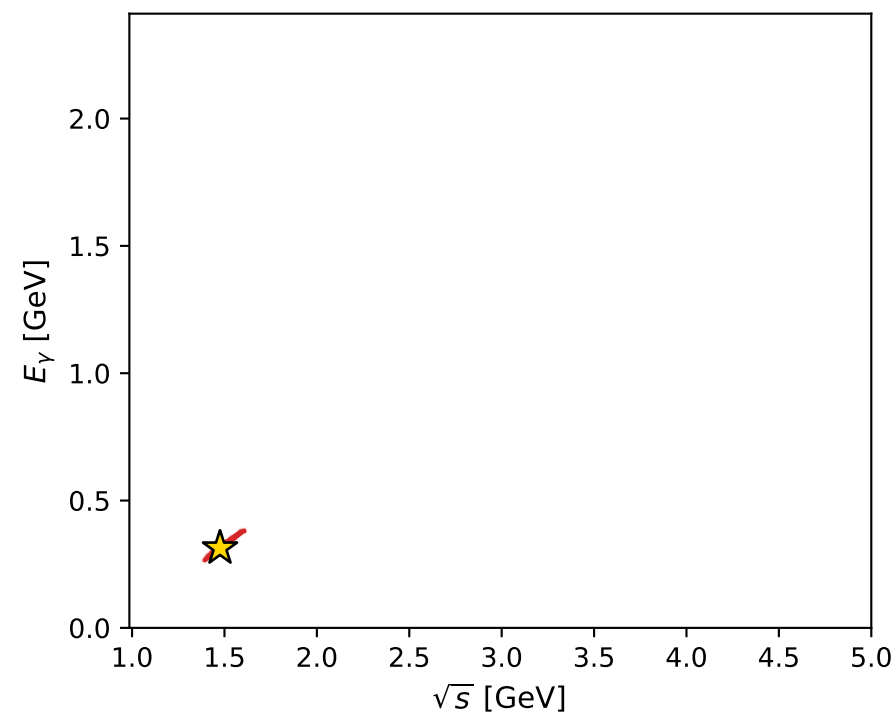
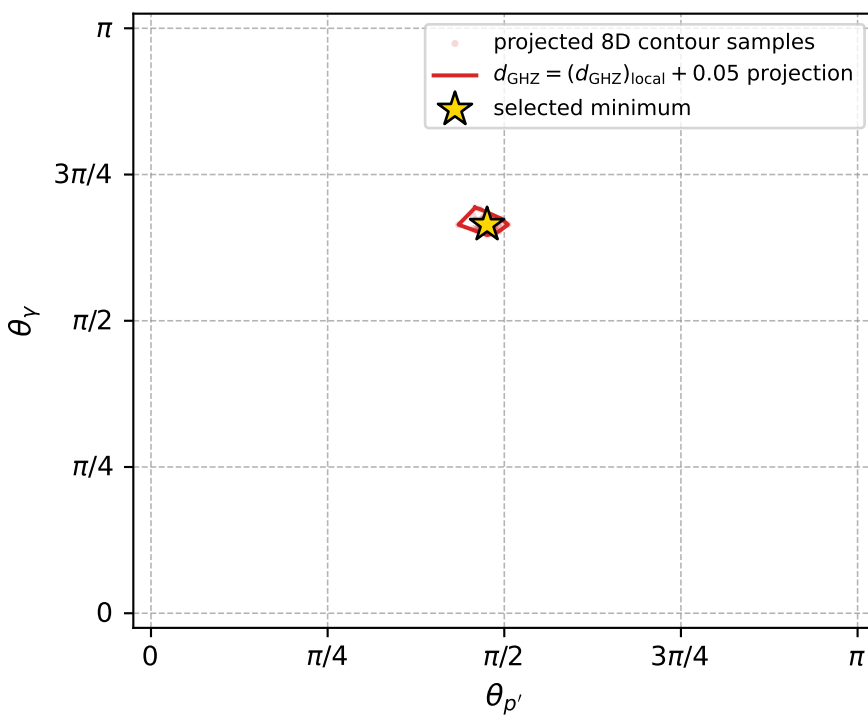
dghz_mixing_angles_polarization_02_local_minimum_271 $d_{\{\text{rm GHZ}\}}=0.000201774$
 region: polarization_cluster_02_local_minimum_271
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=0.95958649$ rad
 incoming lepton: $0.573859|+\rangle +0.818954|-\rangle$
 proton mixing angle: $\alpha_p=2.633207$ rad
 incoming proton: $-0.873531|+\rangle +0.486768|-\rangle$

$s=2.1783$, $\sqrt{s}=1.47591$, $|\vec{P}|=0.439884$, $|\vec{P}'|=0.110924$
 $\theta_{P'}=1.49429$, $\theta_Y=2.08684$, $\phi_{P'}=1.33256\text{e-}05$, $\phi_Y=3.14162$
 $E_Y=0.31329$, $\theta_{\ell'}=0.836587$, $\phi_{\ell'}=3.91526\text{e-}05$
 $Q^2=0.320405$, $x_B=0.328577$, $t=-0.189973$, $W^2=1.53457$, $y=0.750992$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.4399$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.4399$
 P $E= 1.0360$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.4399$
 ℓ' $E= 0.2181$ $p_x= 0.1619$ $p_y= 0.0000$ $p_z= 0.1461$
 P' $E= 0.9445$ $p_x= 0.1106$ $p_y= 0.0000$ $p_z= 0.0085$
 q_Y $E= 0.3133$ $p_x= -0.2725$ $p_y= -0.0000$ $p_z= -0.1546$



electron: polarization cluster P2, local minimum 271; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 0.00020177386$
 $d_{\text{GHZ}} \text{ contour} = 0.050201774$
 above global minimum = 0.00020175157
 8D contour samples = 256

selected phase-space point:

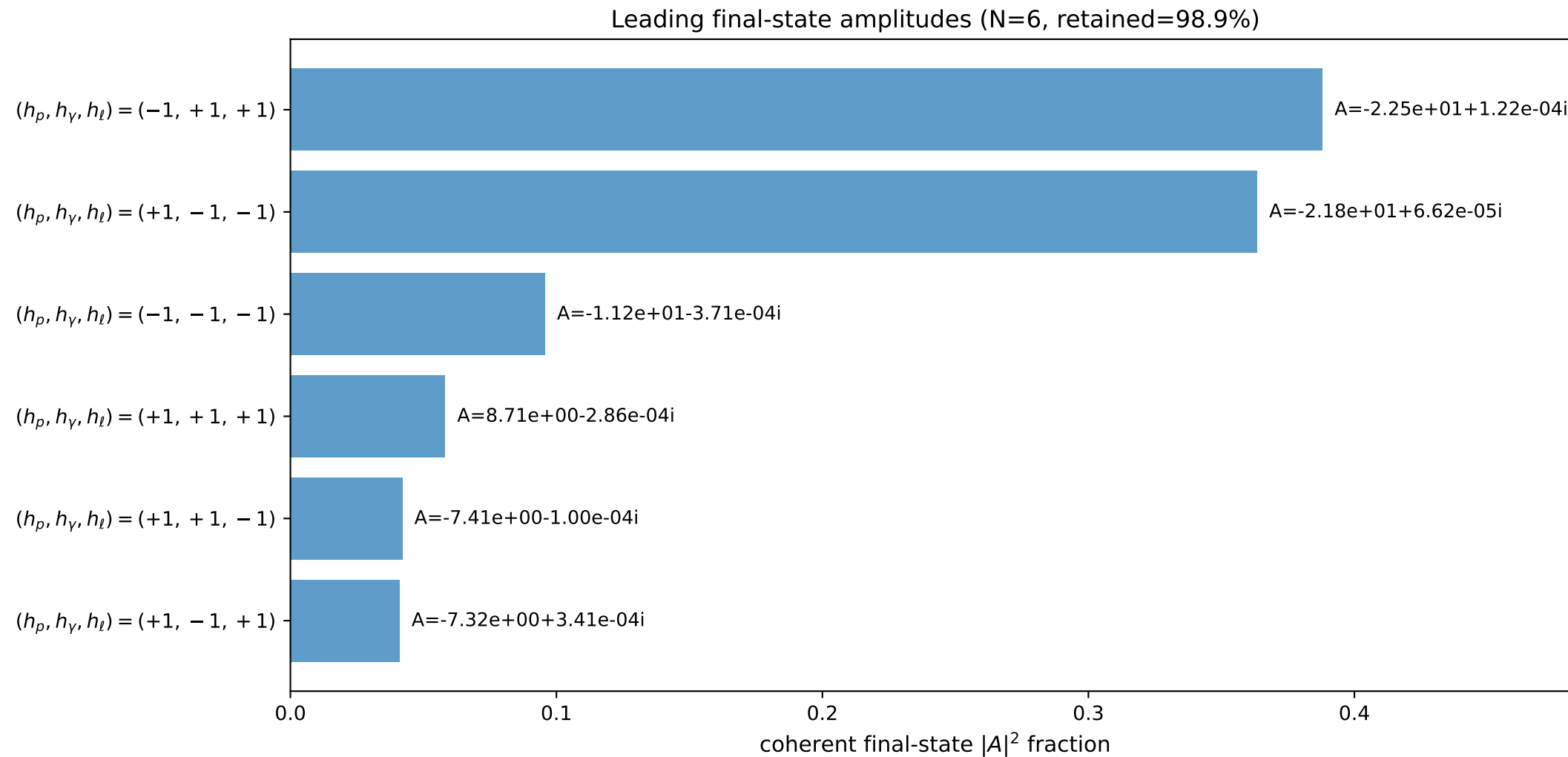
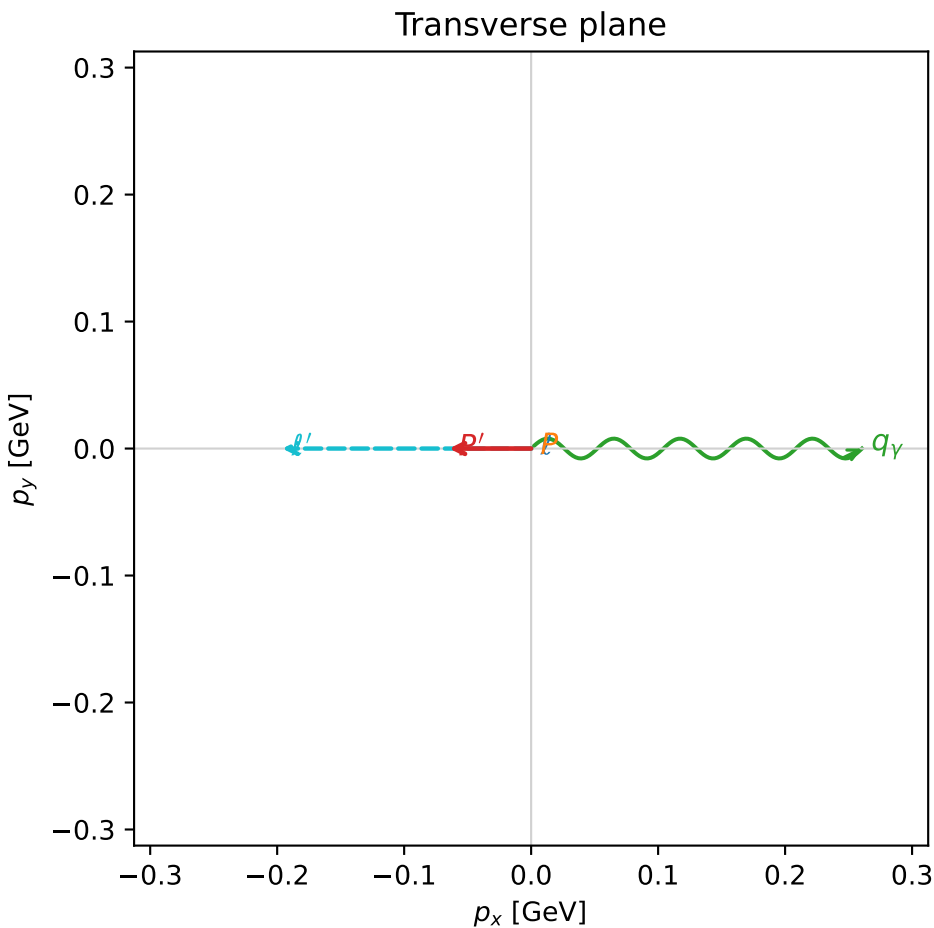
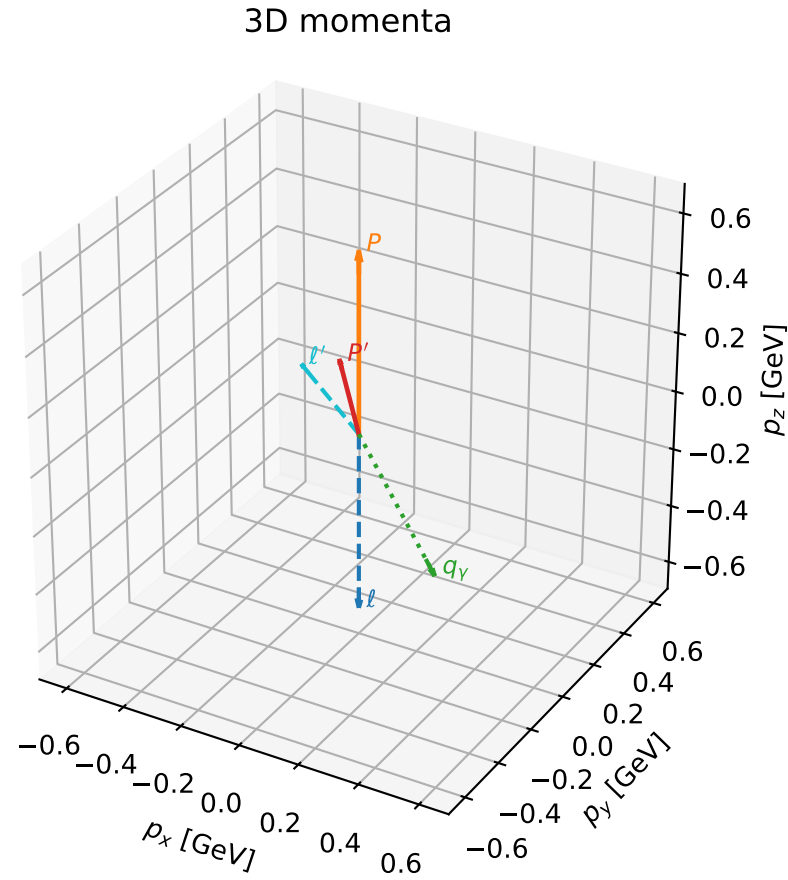
$\text{sqrt_s} = 1.475906$
 $\text{theta_p_out} = 1.494288$
 $\text{theta_gamma_out} = 2.086842$
 $\text{q0out} = 0.3132905$
 $\text{phi_p_out} = 1.332562\text{e-}05$
 $\text{phi_gamma_out} = 3.141621$
 $\text{alpha_e} = 0.9595865$
 $\text{alpha_p} = 2.633207$

coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

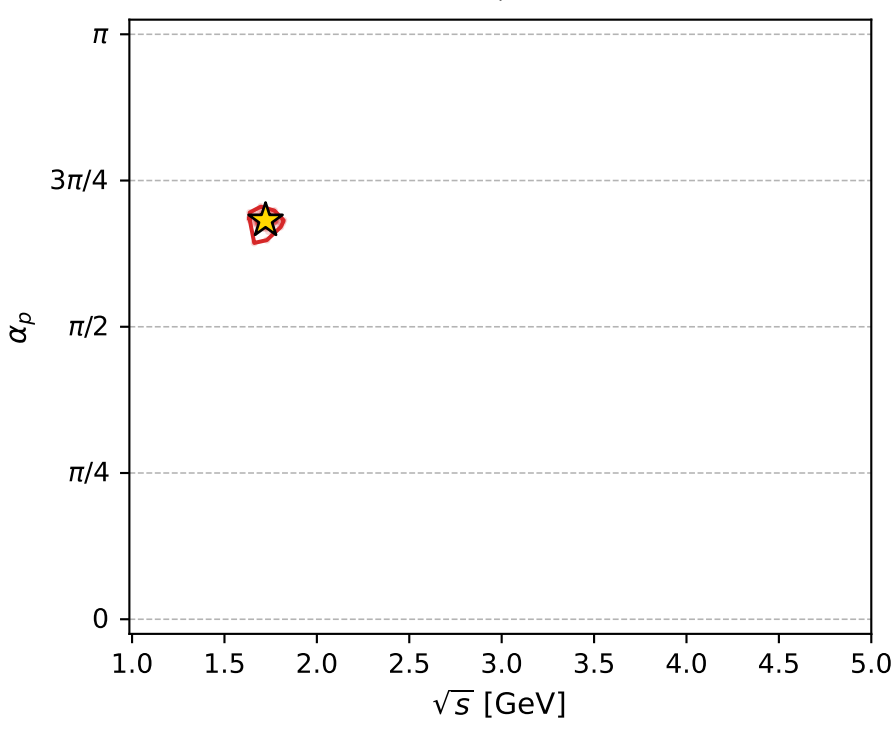
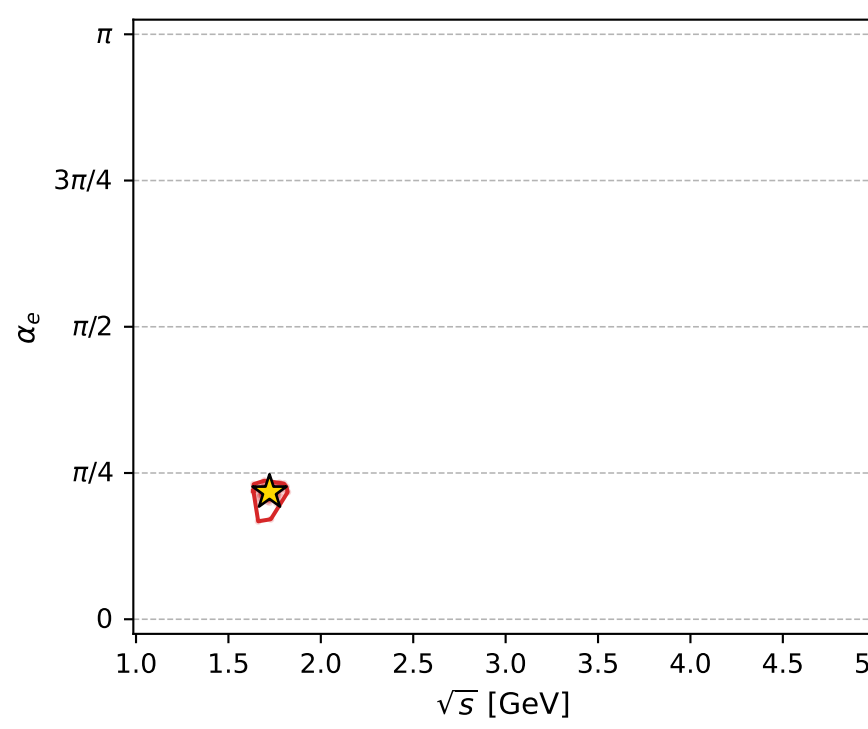
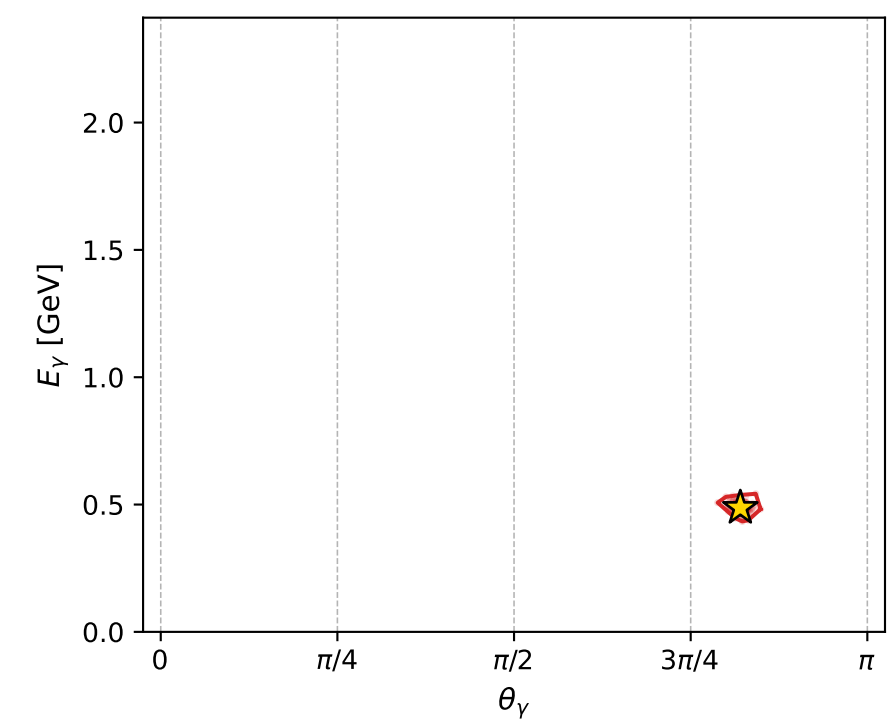
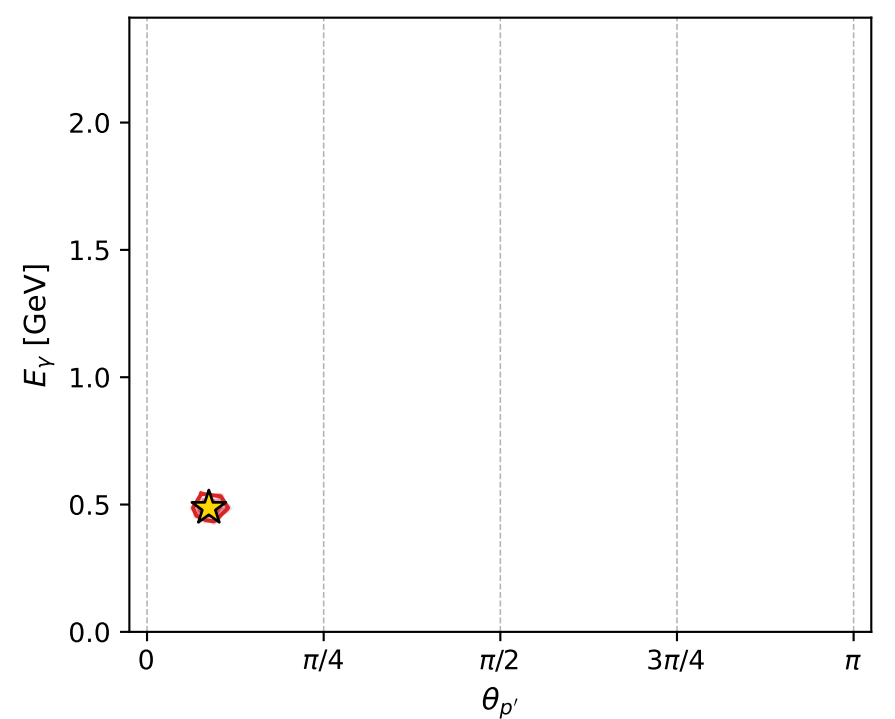
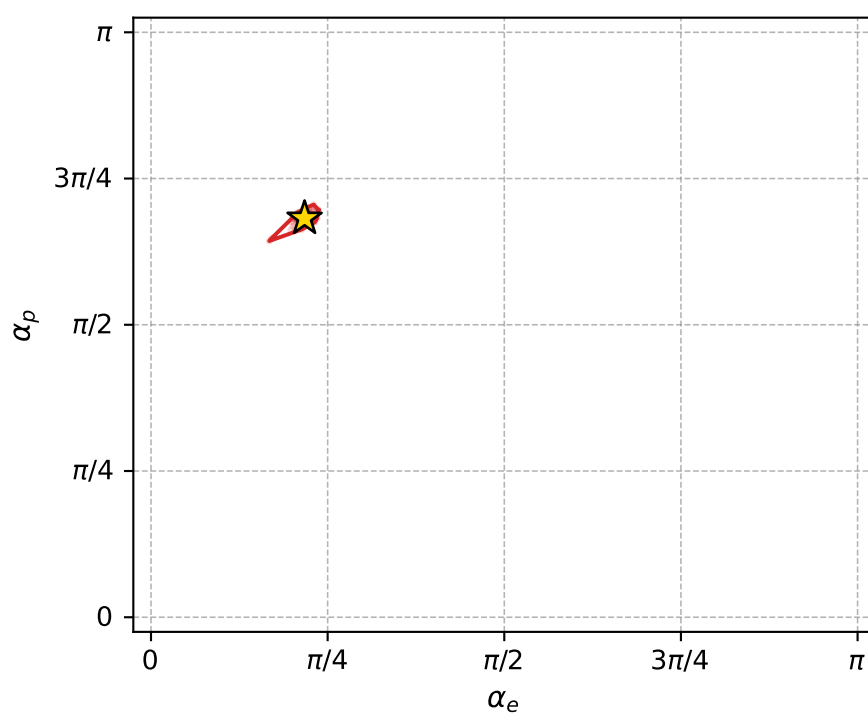
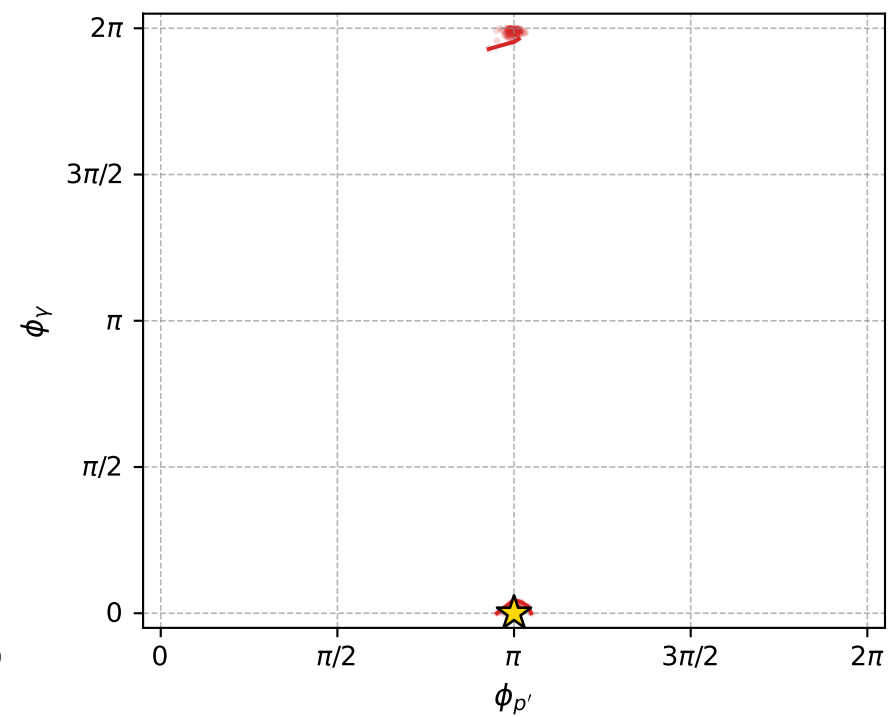
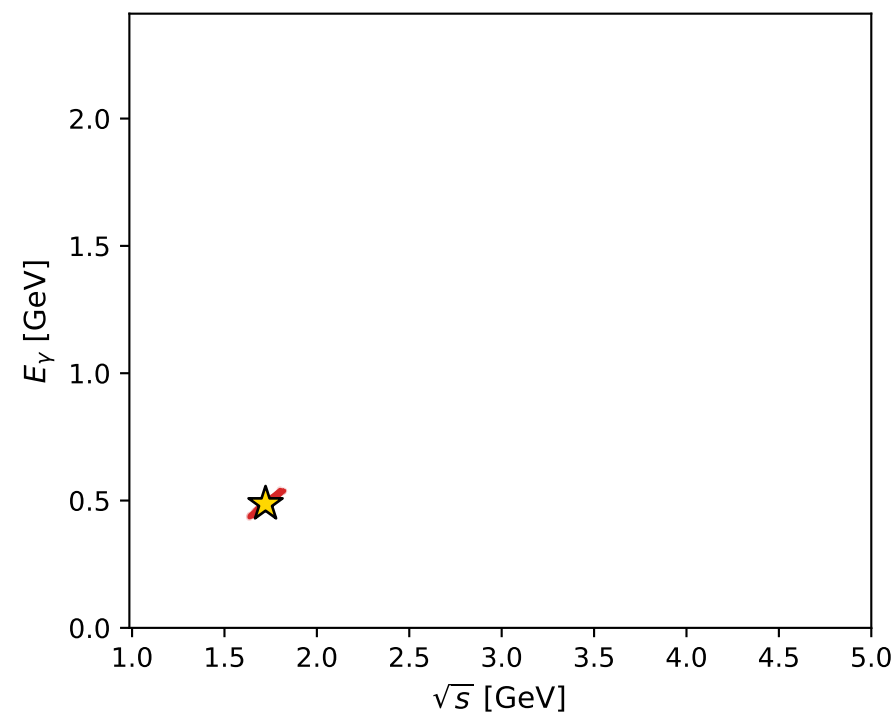
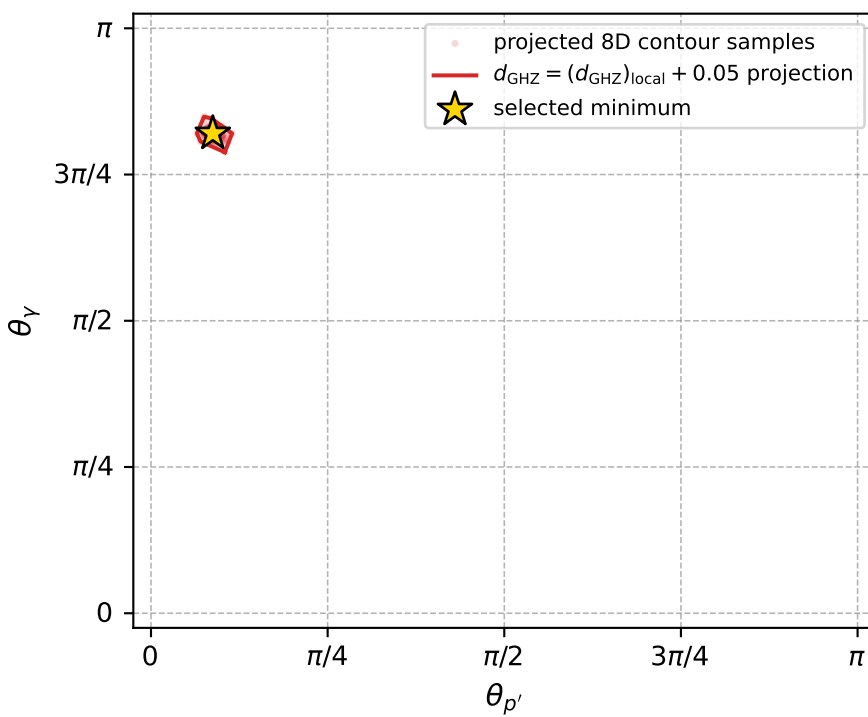
dghz_mixing_angles_polarization_02_local_minimum_273 $d_{\{\text{rm GHZ}\}}=0.000208012$
 region: polarization_cluster_02_local_minimum_273
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=0.68280243$ rad
 incoming lepton: $0.775808|+\rangle +0.63097|-\rangle$
 proton mixing angle: $\alpha_p=2.1422022$ rad
 incoming proton: $-0.540815|+\rangle +0.841141|-\rangle$

$s=2.96633$, $\sqrt{s}=1.7223$, $|\vec{P}|=0.605726$, $|\vec{P}'|=0.23788$
 $\theta_{p'}=0.274838$, $\theta_\gamma=2.57689$, $\phi_{p'}=3.14163$, $\phi_\gamma=3.12445\text{e-}05$
 $E_\gamma=0.486894$, $\theta_{\ell'}=0.821486$, $\phi_{\ell'}=3.14162$
 $Q^2=0.545234$, $x_B=0.318936$, $t=-0.12396$, $W^2=2.04415$, $y=0.81934$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.6057$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.6057$
 P $E= 1.1166$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.6057$
 ℓ' $E= 0.2677$ $p_x= -0.1960$ $p_y= -0.0000$ $p_z= 0.1824$
 P' $E= 0.9677$ $p_x= -0.0646$ $p_y= -0.0000$ $p_z= 0.2290$
 q_γ $E= 0.4869$ $p_x= 0.2606$ $p_y= 0.0000$ $p_z= -0.4113$



electron: polarization cluster P2, local minimum 273; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 0.0002080116$
 $d_{\text{GHZ}} \text{ contour} = 0.050208012$
 above global minimum = 0.00020798932
 8D contour samples = 256

selected phase-space point:

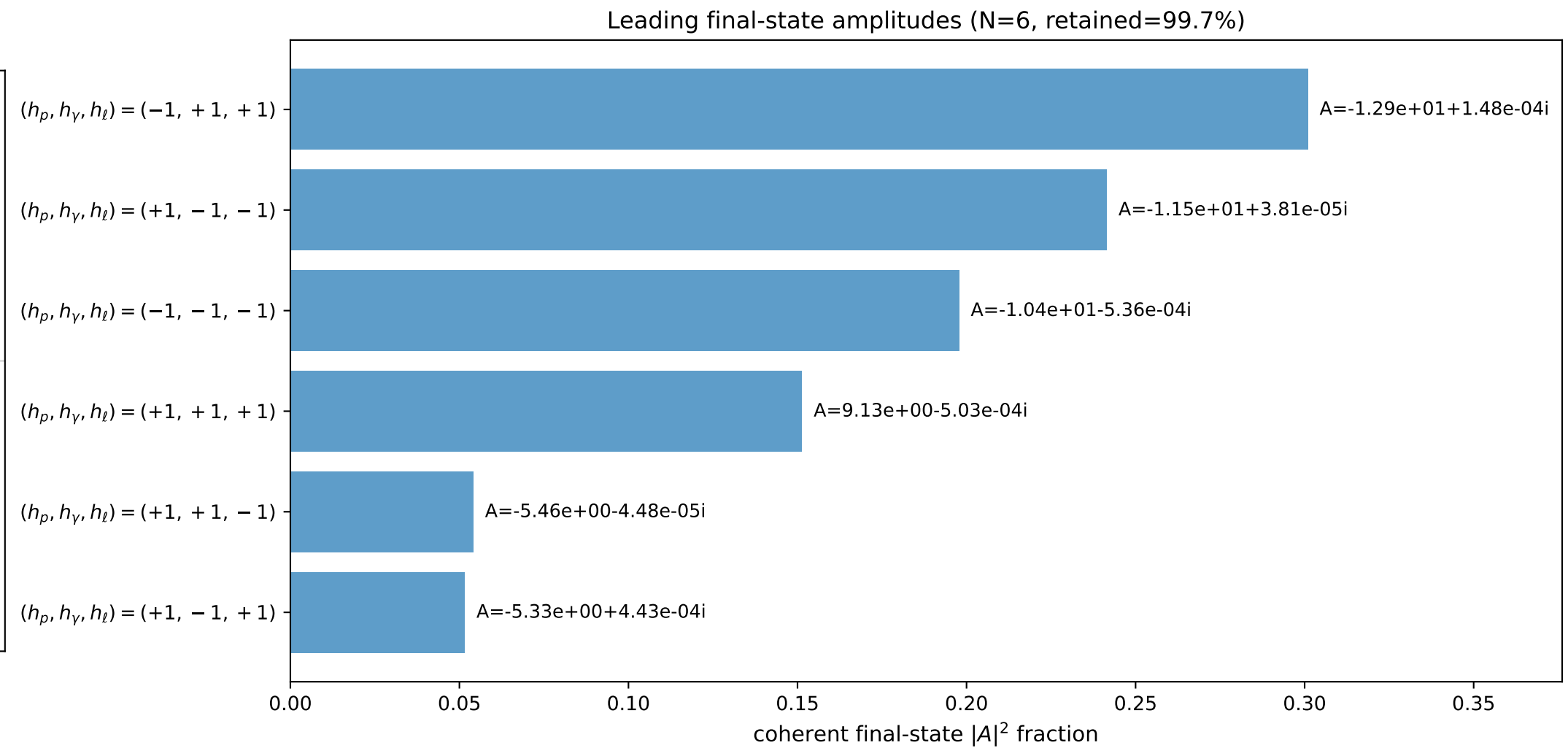
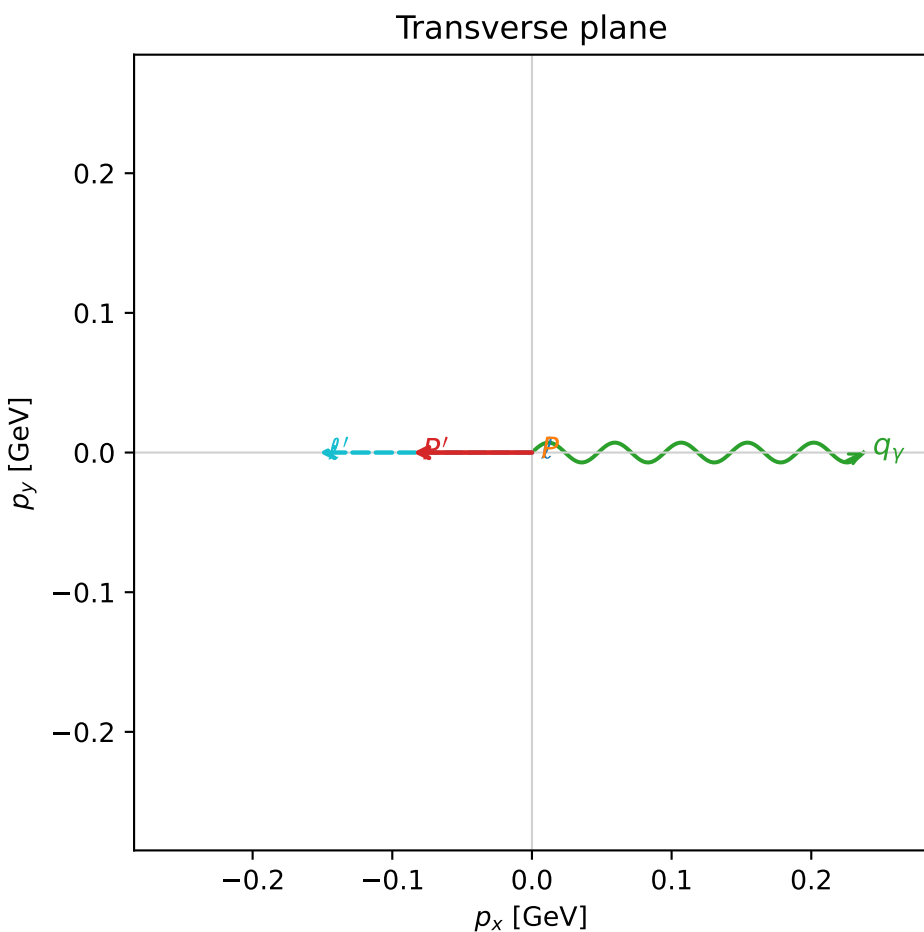
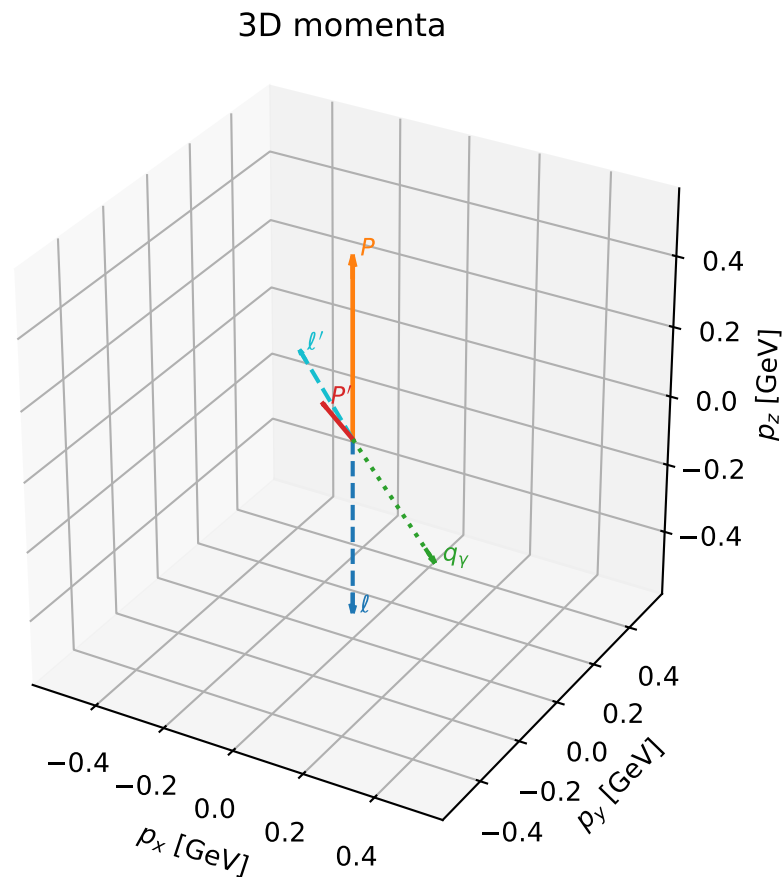
$\text{sqrt_s} = 1.722304$
 $\text{theta_p_out} = 0.2748376$
 $\text{theta_gamma_out} = 2.576889$
 $\text{q0out} = 0.486894$
 $\text{phi_p_out} = 3.141626$
 $\text{phi_gamma_out} = 3.124446\text{e-}05$
 $\text{alpha_e} = 0.6828024$
 $\text{alpha_p} = 2.142202$

coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

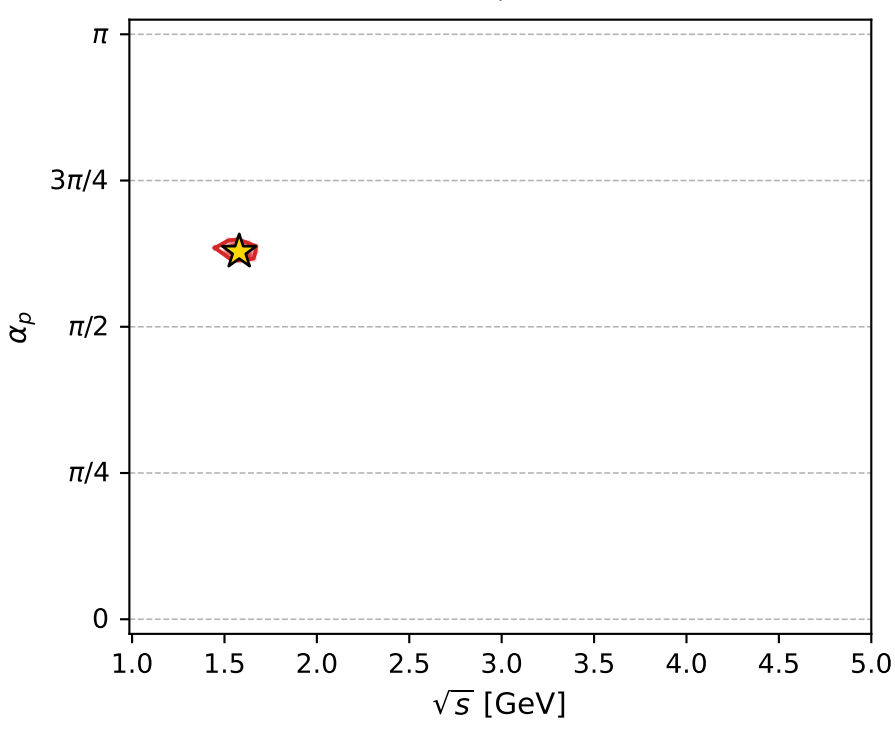
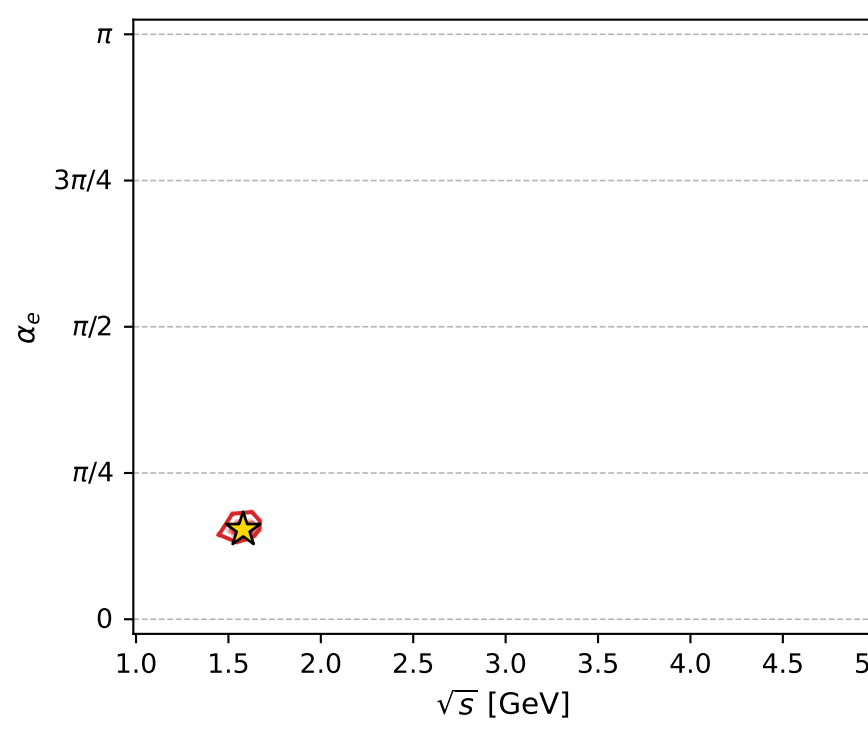
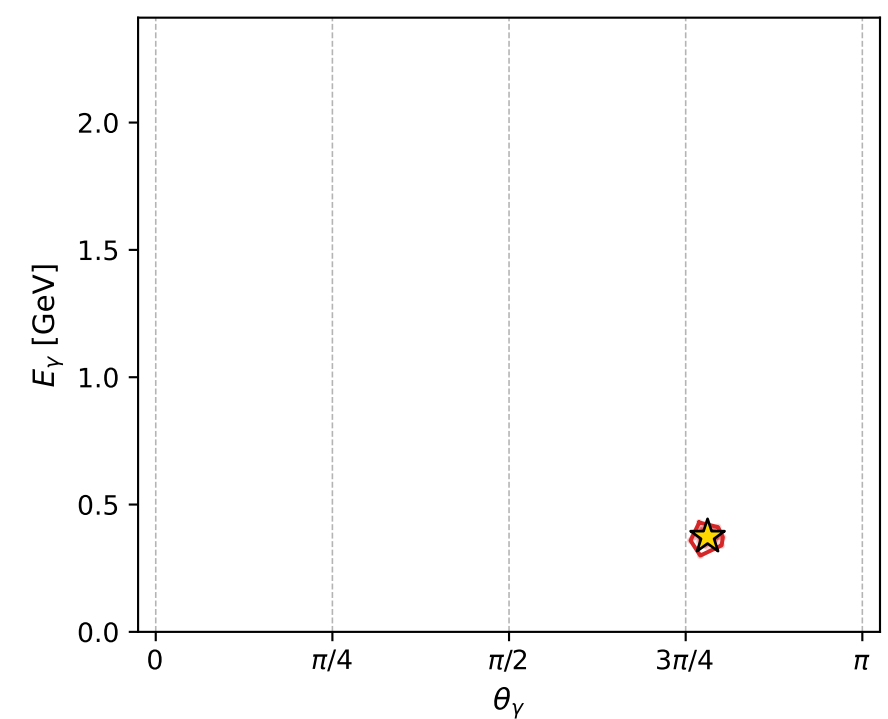
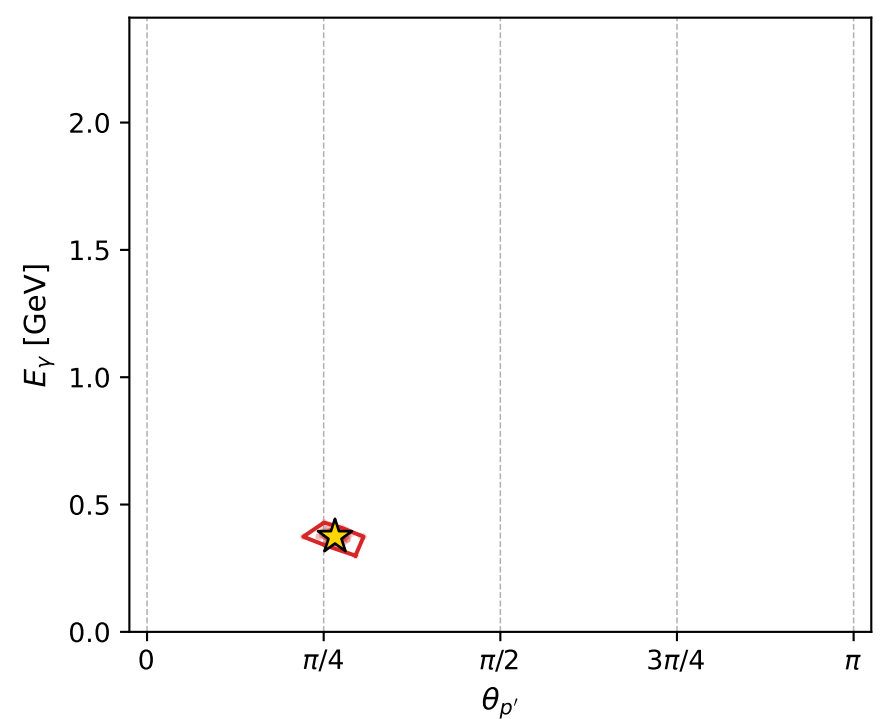
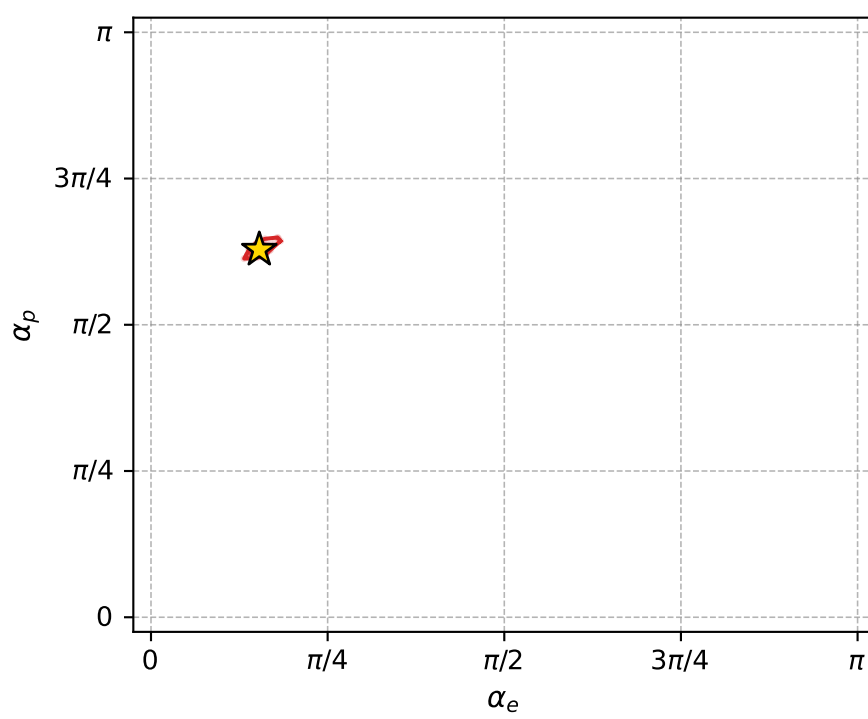
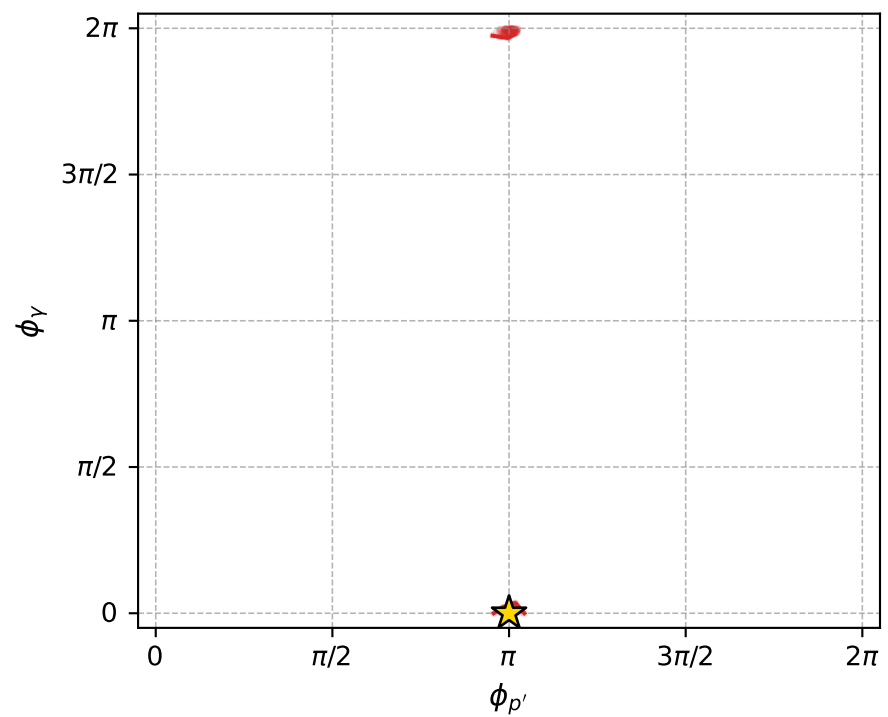
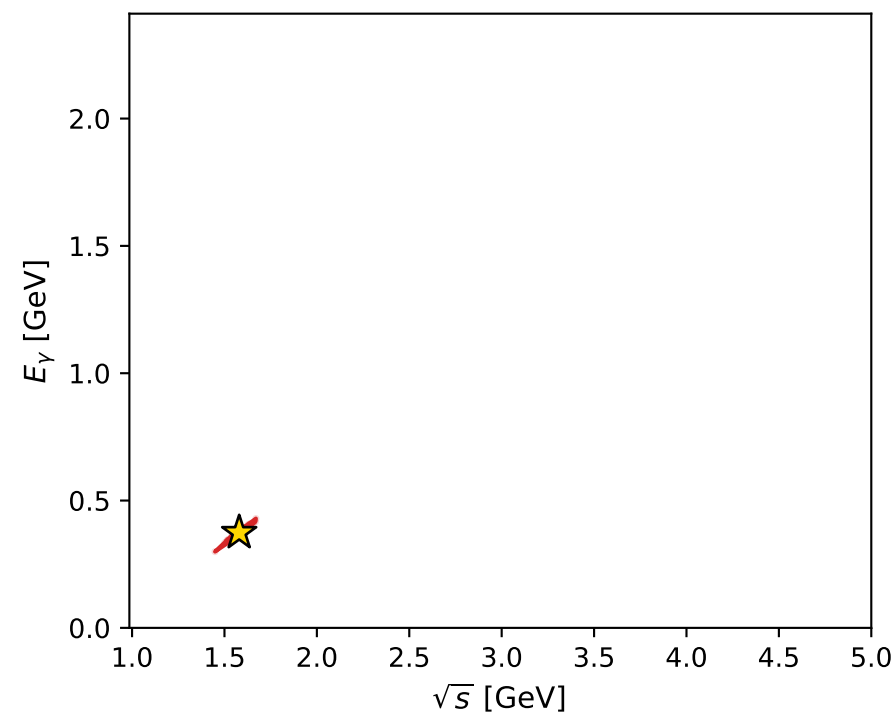
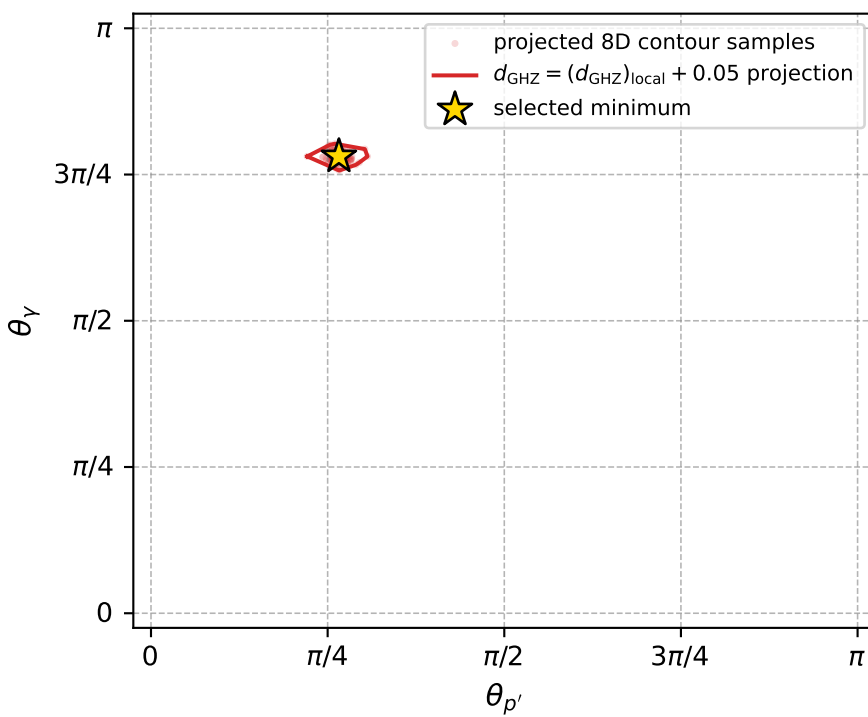
dghz_mixing_angles_polarization_02_local_minimum_274 d_{\rm GHZ}=0.000218394
region: polarization_cluster_02_local_minimum_274
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: $\alpha_e=0.48149496$ rad
incoming lepton: $0.886304|+\rangle +0.463105|-\rangle$
proton mixing angle: $\alpha_p=1.9738458$ rad
incoming proton: $-0.392225|+\rangle +0.919869|-\rangle$

$s=2.49541$, $\sqrt{s}=1.57969$, $|\vec{P}|=0.511356$, $|\vec{P}'|=0.1147$
 $\theta_{P'}=0.83558$, $\theta_Y=2.45347$, $\phi_{P'}=3.14165$, $\phi_Y=4.56851\text{e-}05$
 $E_Y=0.373805$, $\theta_{\ell'}=0.623483$, $\phi_{\ell'}=3.14163$
 $Q^2=0.483438$, $x_B=0.379244$, $t=-0.180746$, $W^2=1.67115$, $y=0.789036$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.5114$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.5114$
 P $E= 1.0683$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.5114$
 ℓ' $E= 0.2609$ $p_x= -0.1523$ $p_y= -0.0000$ $p_z= 0.2118$
 P' $E= 0.9450$ $p_x= -0.0851$ $p_y= -0.0000$ $p_z= 0.0769$
 q_Y $E= 0.3738$ $p_x= 0.2374$ $p_y= 0.0000$ $p_z= -0.2887$



electron: polarization cluster P2, local minimum 274; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 0.00021839418$
 $d_{\text{GHZ}} \text{ contour} = 0.050218394$
 above global minimum = 0.0002183719
 8D contour samples = 256

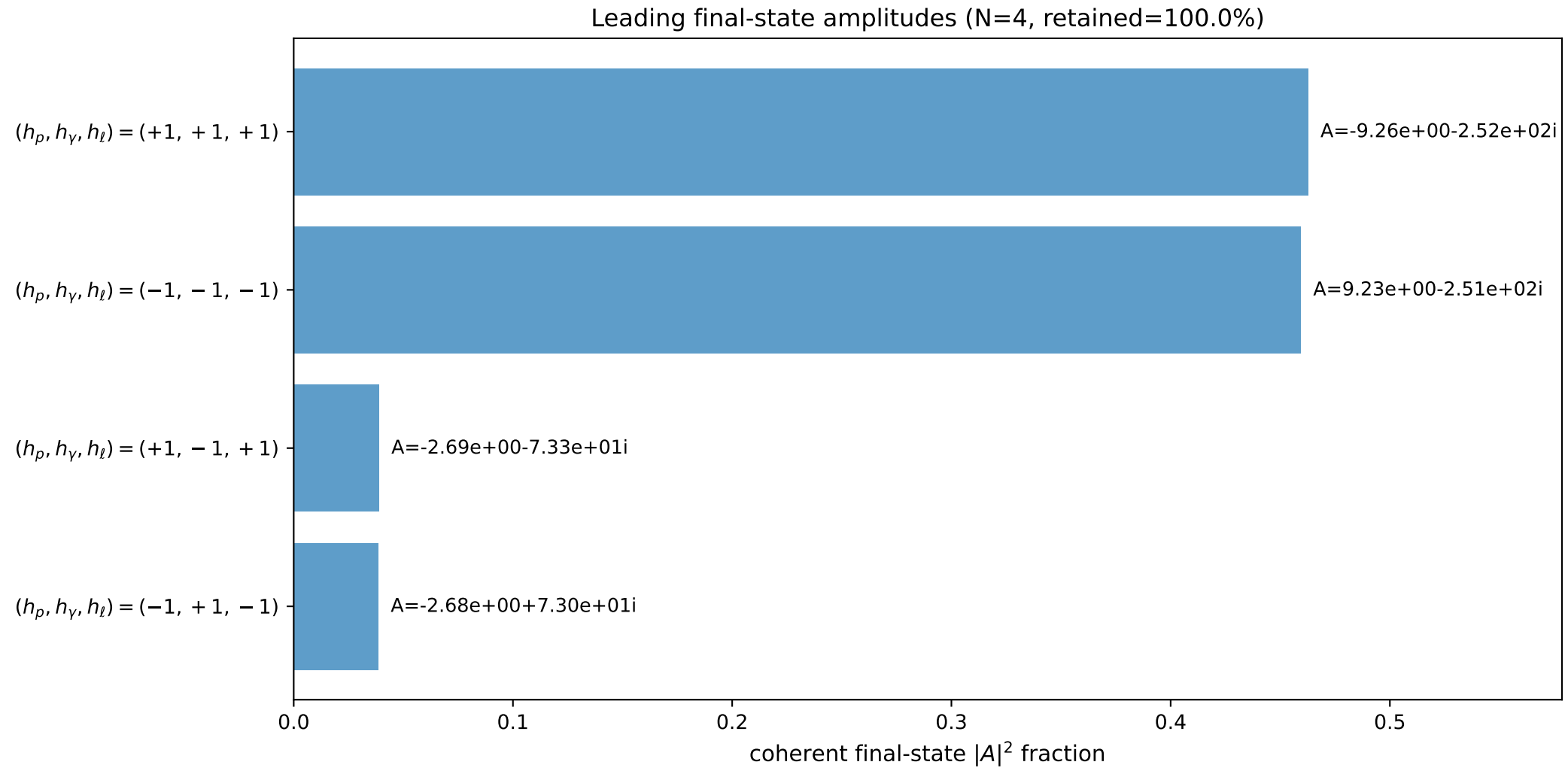
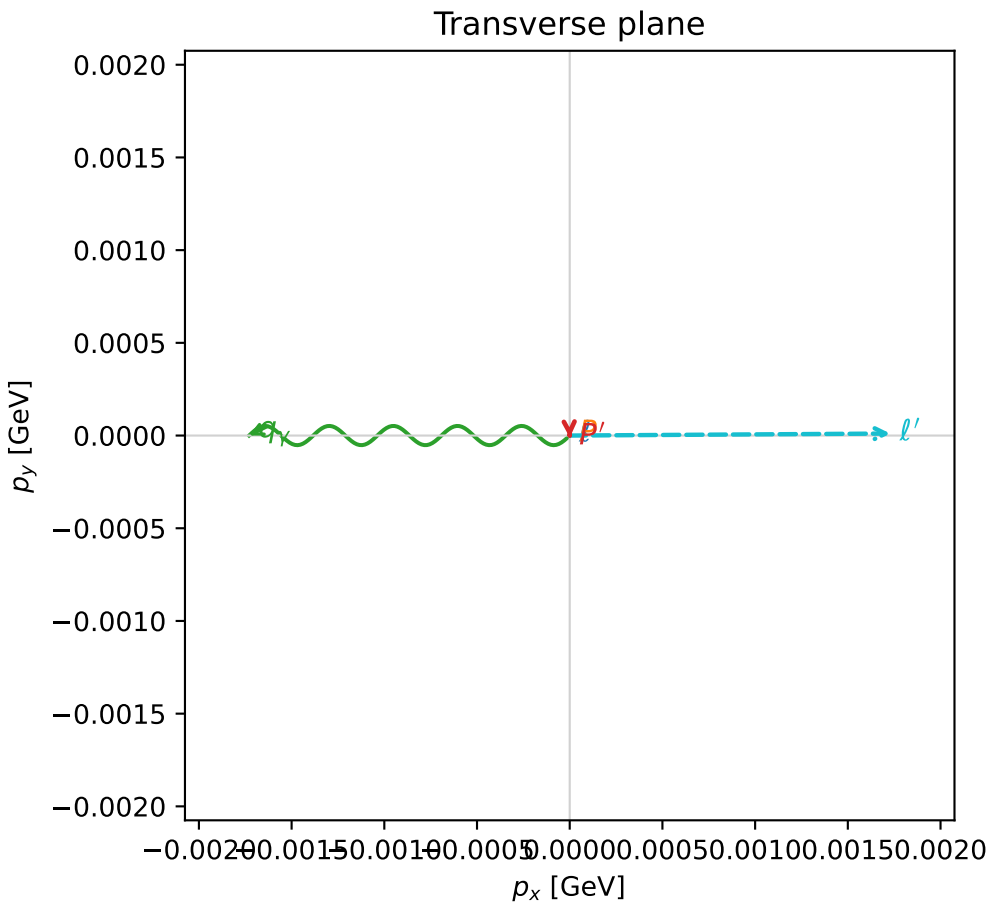
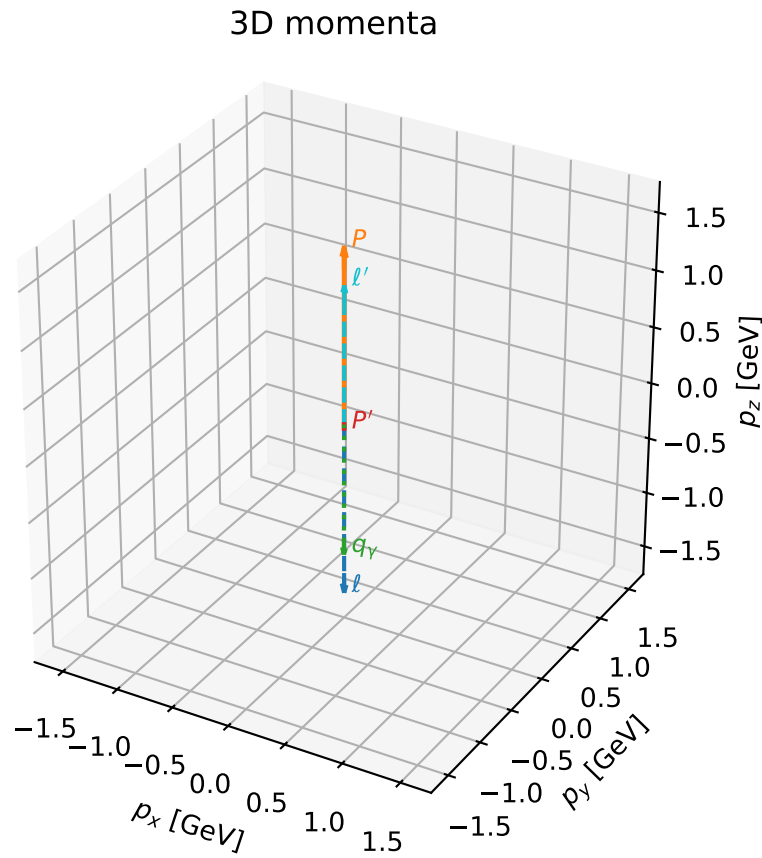
selected phase-space point:
 $\text{sqrt_s} = 1.579687$
 $\text{theta_p_out} = 0.8355805$
 $\text{theta_gamma_out} = 2.45347$
 $\text{q0out} = 0.3738051$
 $\text{phi_p_out} = 3.141645$
 $\text{phi_gamma_out} = 4.568508\text{e-}05$
 $\text{alpha_e} = 0.481495$
 $\text{alpha_p} = 1.973846$

coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

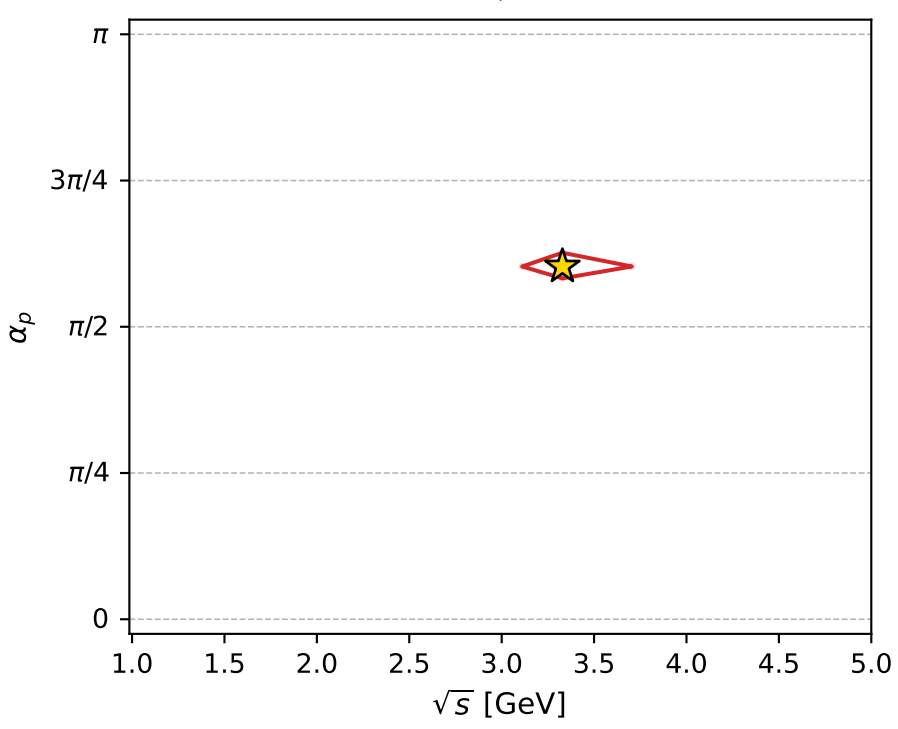
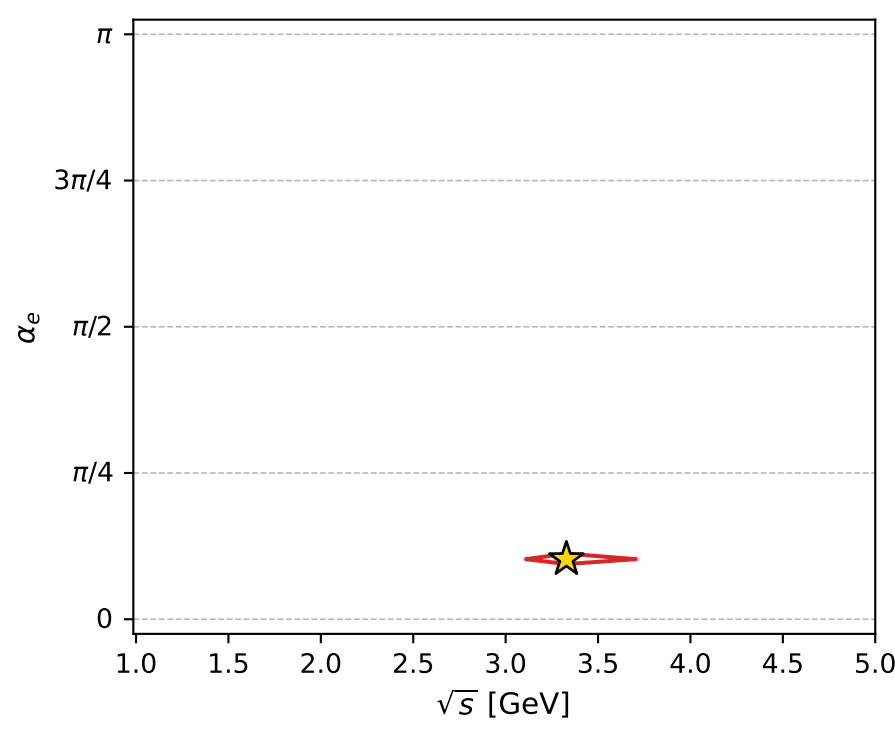
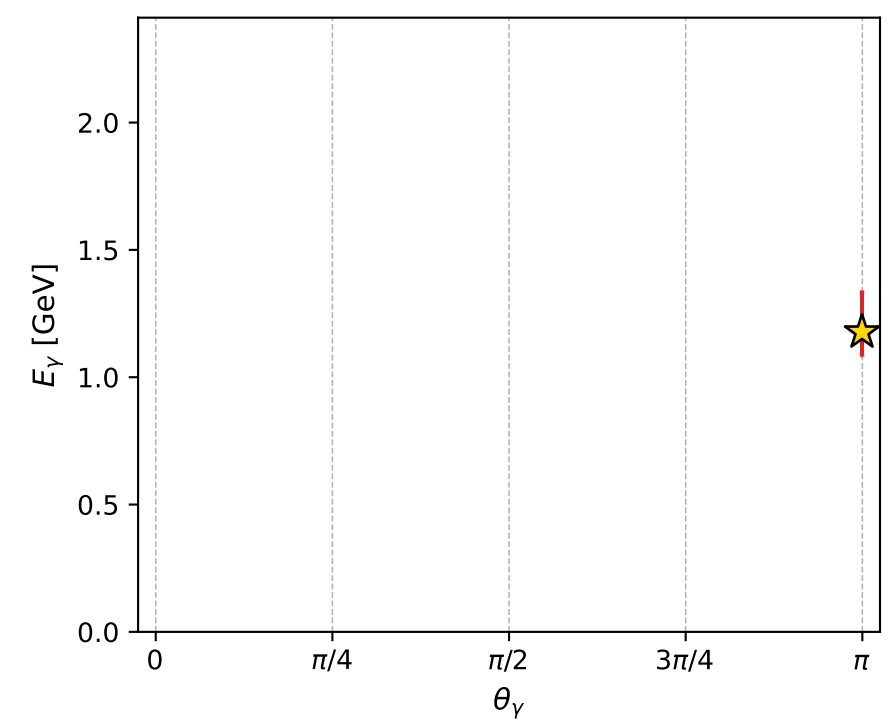
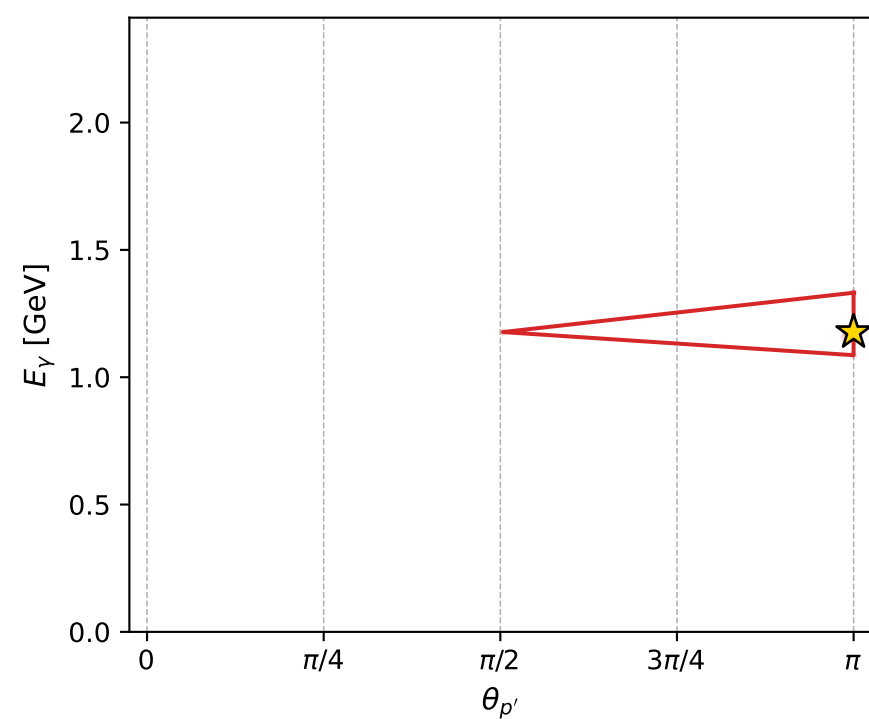
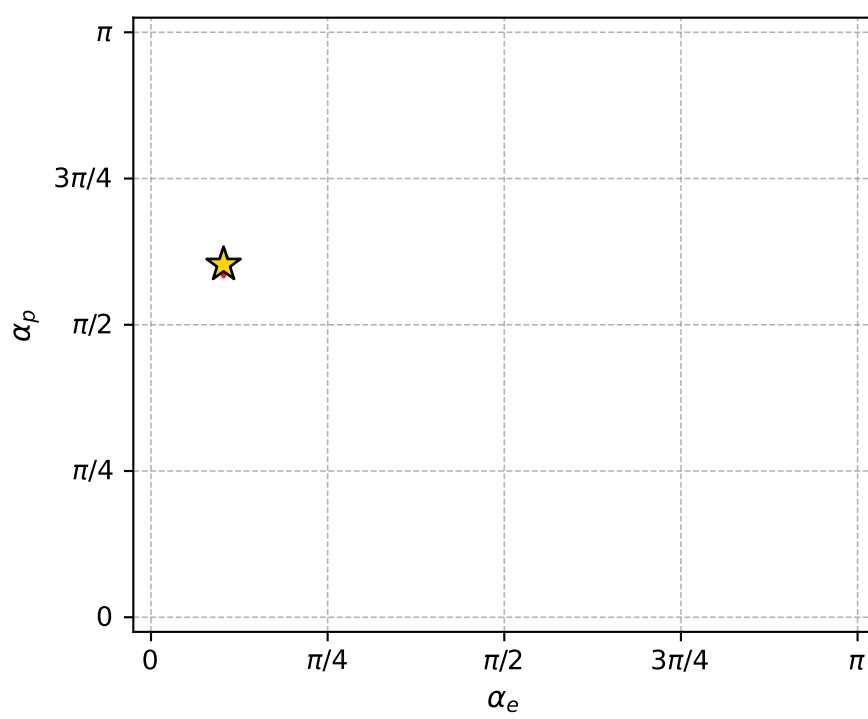
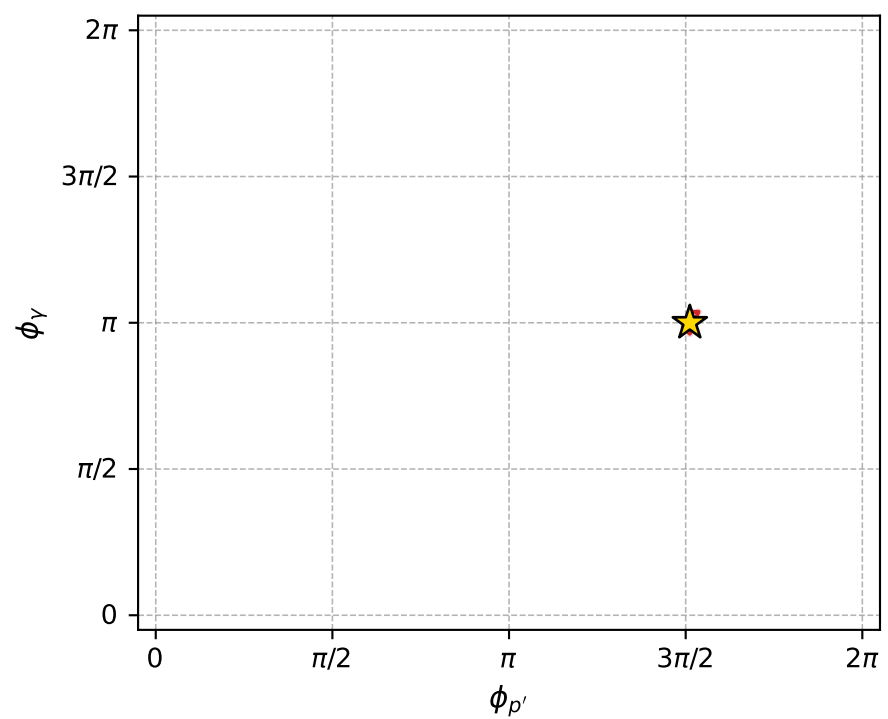
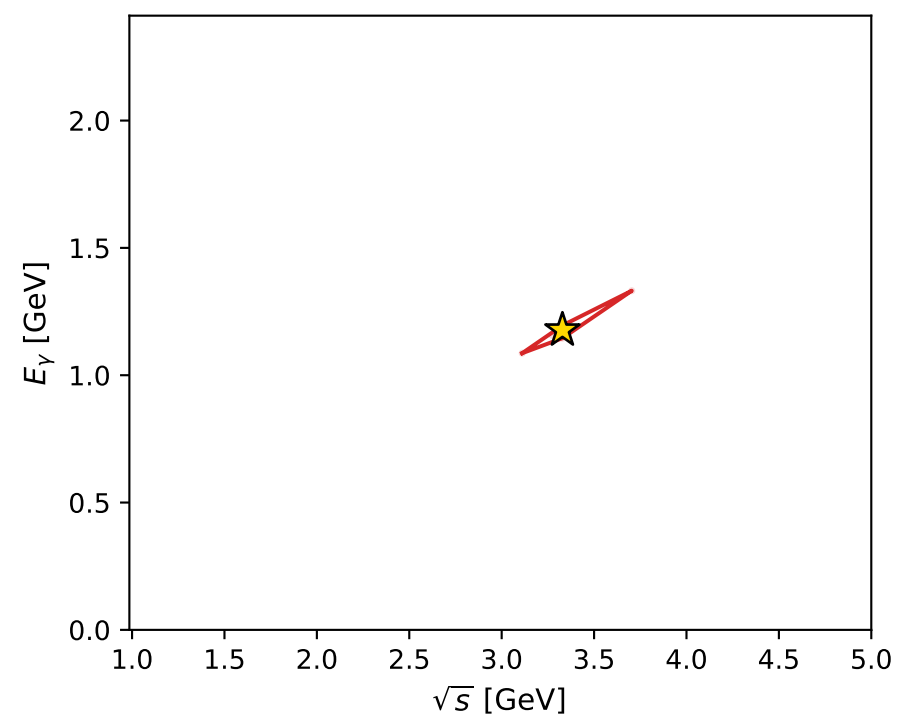
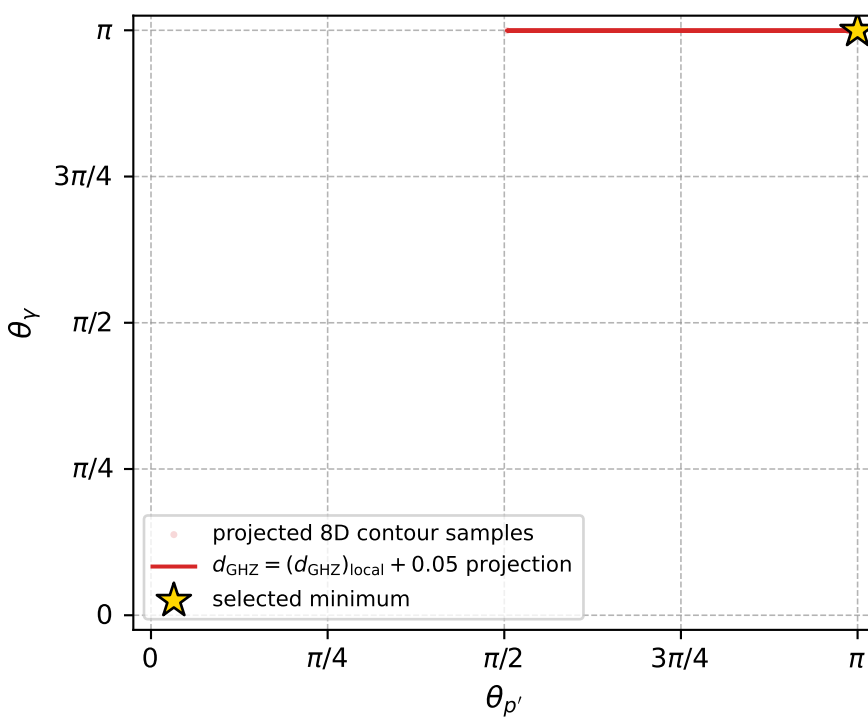
dghz_mixing_angles_polarization_02_local_minimum_275 d_{\rm GHZ}=0.000220156
region: polarization_cluster_02_local_minimum_275
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: $\alpha_e=0.32264262$ rad
incoming lepton: $0.948401|+\rangle +0.317074|-\rangle$
proton mixing angle: $\alpha_p=1.8945201$ rad
incoming proton: $-0.318099|+\rangle +0.948057|-\rangle$

$s=11.0801$, $\sqrt{s}=3.32869$, $|\vec{P}|=1.53218$, $|\vec{P}'|=0.0354715$
 $\theta_{p'}=3.14129$, $\theta_Y=3.14012$, $\phi_{p'}=4.74907$, $\phi_Y=3.14159$
 $E_Y=1.17727$, $\theta_{\ell'}=0.0014257$, $\phi_{\ell'}=0.00628435$
 $Q^2=7.43257$, $x_B=0.777531$, $t=-1.72166$, $W^2=3.00647$, $y=0.937148$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 1.5322$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.5322$
 P $E= 1.7965$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.5322$
 ℓ' $E= 1.2127$ $p_x= 0.0017$ $p_y= 0.0000$ $p_z= 1.2127$
 P' $E= 0.9387$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= -0.0355$
 q_Y $E= 1.1773$ $p_x= -0.0017$ $p_y= 0.0000$ $p_z= -1.1773$



electron: polarization cluster P2, local minimum 275; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 0.00022015596$
 $d_{\text{GHZ}} \text{ contour} = 0.050220156$
 above global minimum = 0.00022013368
 8D contour samples = 225

selected phase-space point:

$\text{sqrt}_s = 3.328686$
 $\theta_{p_{\text{out}}} = 3.141286$
 $\theta_{\gamma_{\text{out}}} = 3.140124$
 $q_{\text{out}} = 1.177272$
 $\phi_{p_{\text{out}}} = 4.749068$
 $\phi_{\gamma_{\text{out}}} = 3.141587$
 $\alpha_e = 0.3226426$
 $\alpha_p = 1.89452$

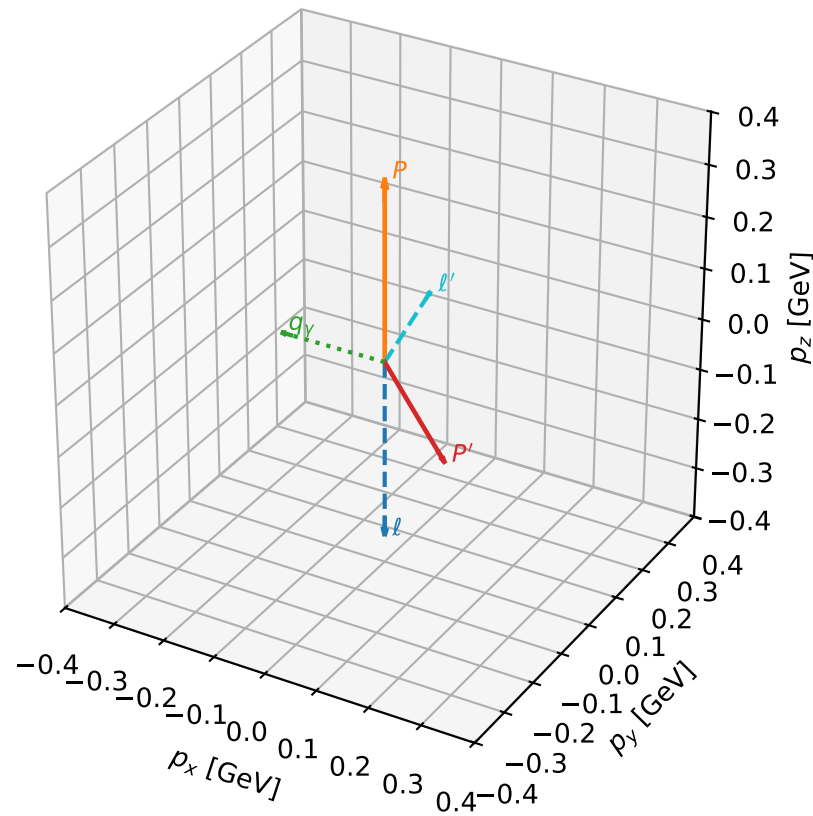
coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

dghz_mixing_angles_polarization_02_local_minimum_283 $d_{\{\text{rm GHZ}\}}=0.000236633$
region: polarization_cluster_02_local_minimum_283
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: $\alpha_e=1.2517711$ rad
incoming lepton: $0.313641|+\rangle +0.949542|-\rangle$
proton mixing angle: $\alpha_p=2.8822466$ rad
incoming proton: $-0.966558|+\rangle +0.256449|-\rangle$

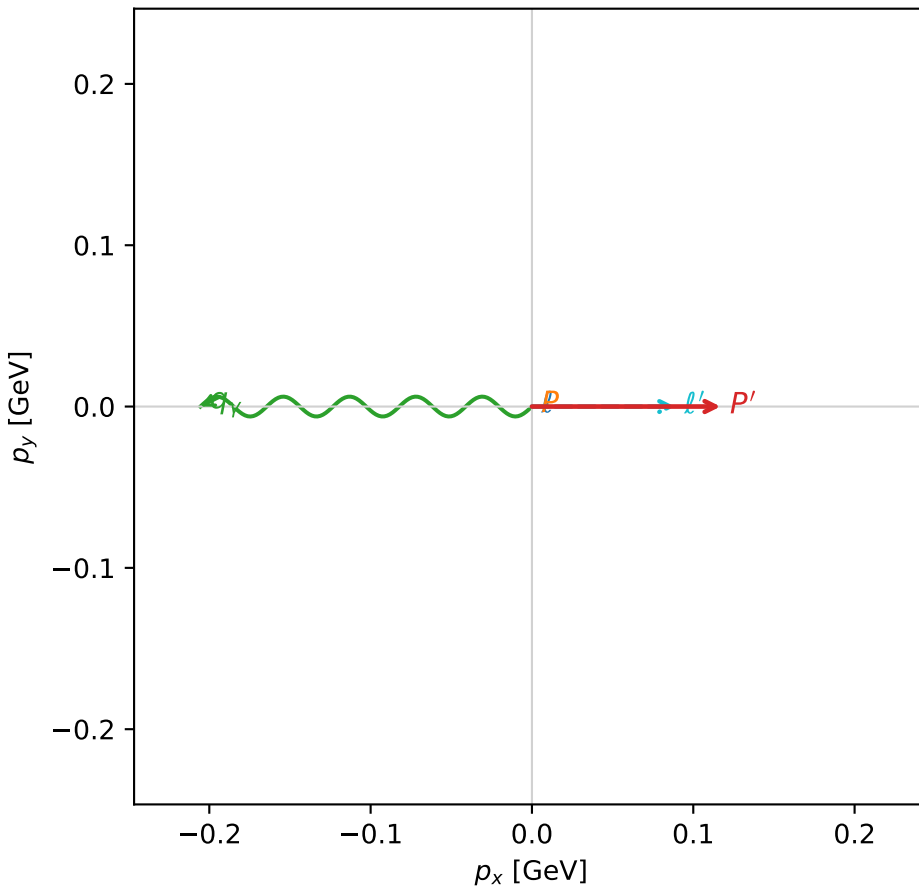
$s=1.82014$, $\sqrt{s}=1.34912$, $|\vec{P}|=0.348482$, $|\vec{P}'|=0.199624$
 $\theta_{P'}=2.51593$, $\theta_Y=1.5714$, $\phi_{P'}=0.000104733$, $\phi_Y=3.14168$
 $E_Y=0.205523$, $\theta_{\ell'}=0.500713$, $\phi_{\ell'}=5.80778\text{e-}05$
 $Q^2=0.241518$, $x_B=0.353237$, $t=-0.272332$, $W^2=1.32205$, $y=0.727143$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.3485$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.3485$
 P $E= 1.0006$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.3485$
 ℓ' $E= 0.1846$ $p_x= 0.0886$ $p_y= 0.0000$ $p_z= 0.1619$
 P' $E= 0.9590$ $p_x= 0.1169$ $p_y= 0.0000$ $p_z= -0.1618$
 q_Y $E= 0.2055$ $p_x= -0.2055$ $p_y= -0.0000$ $p_z= -0.0001$

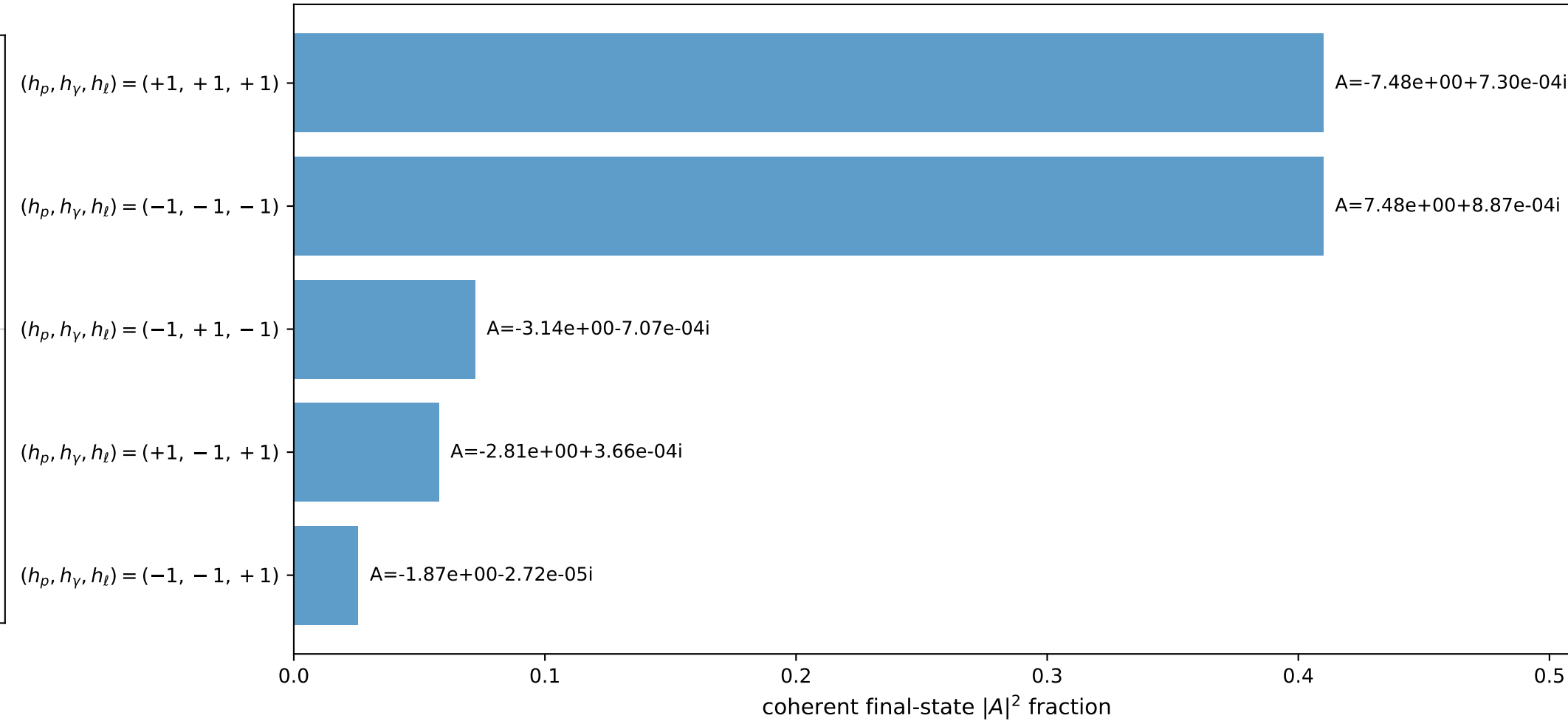
3D momenta



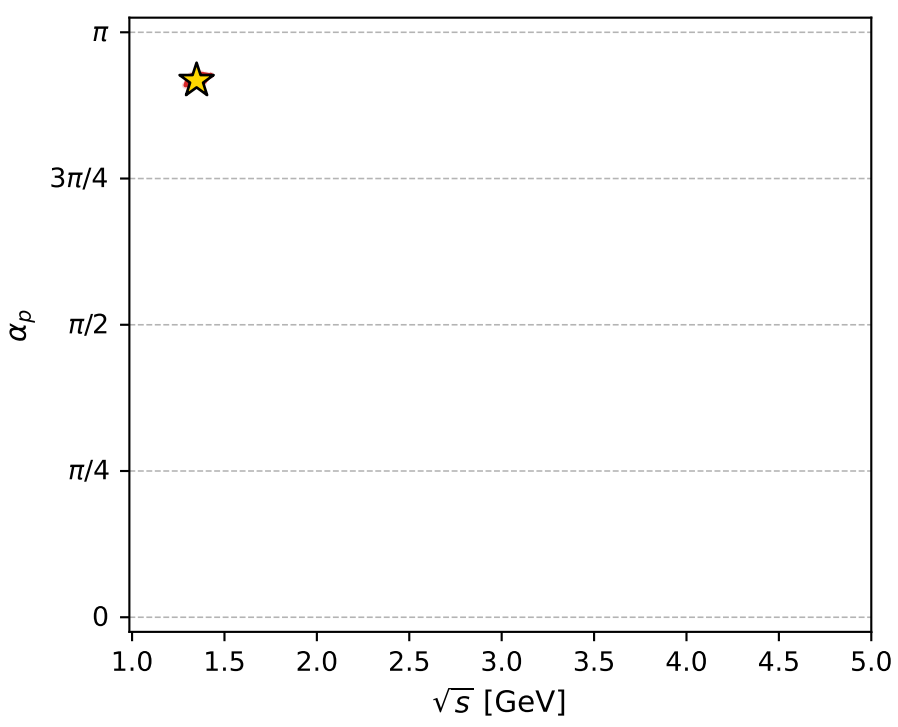
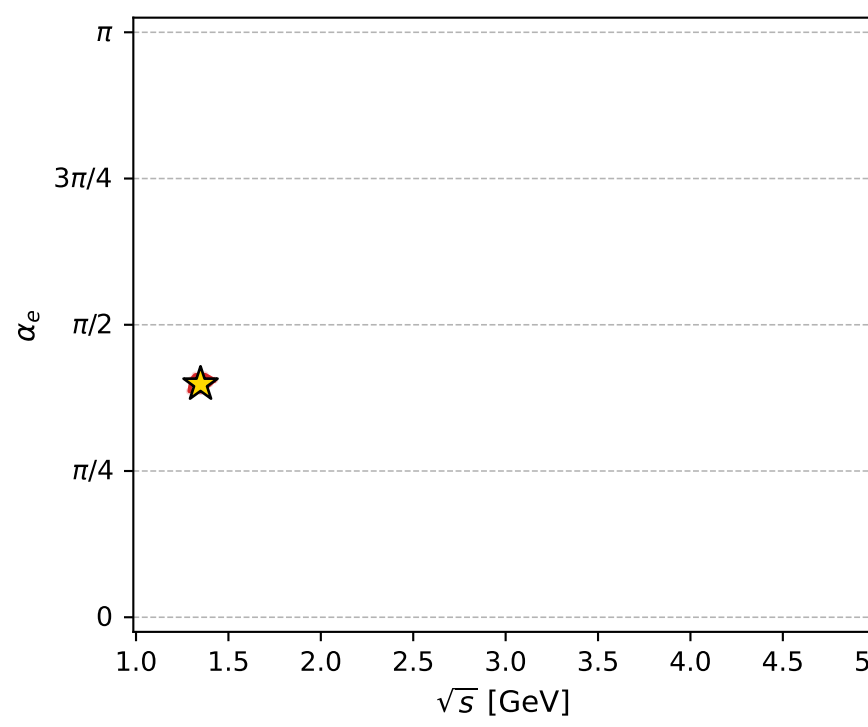
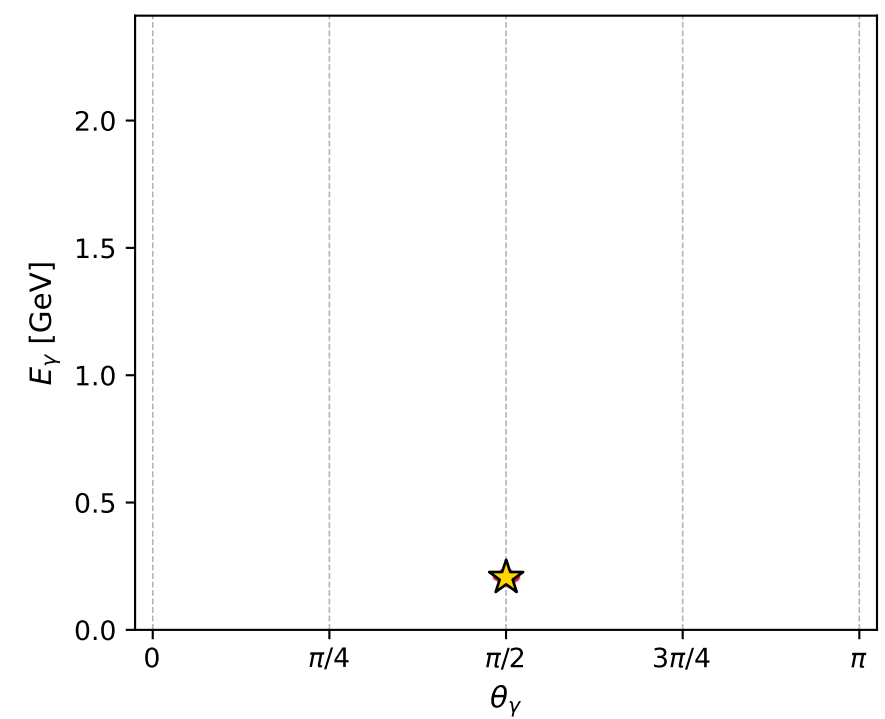
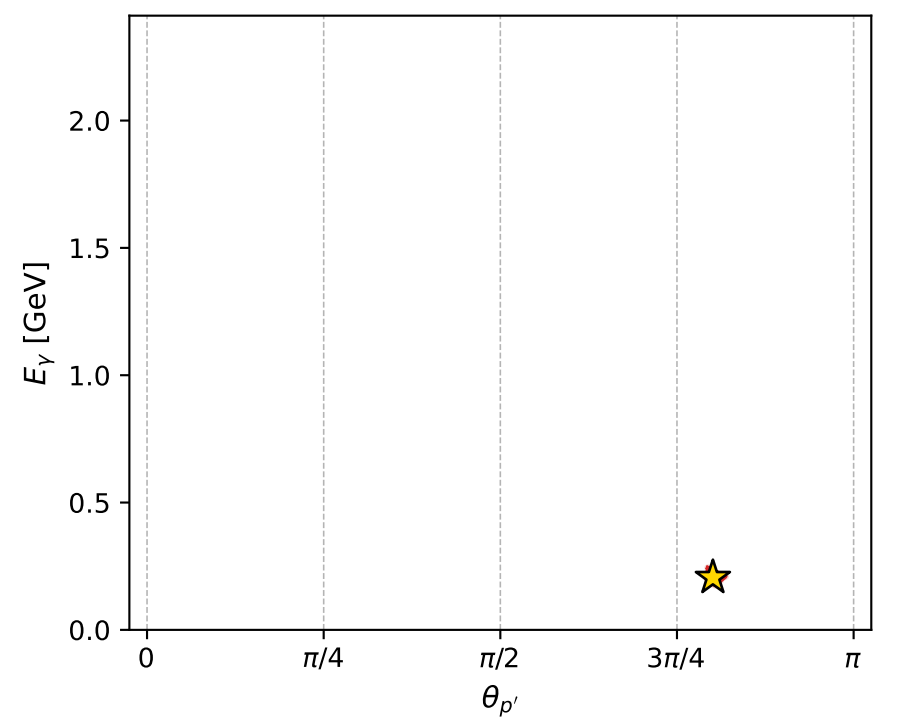
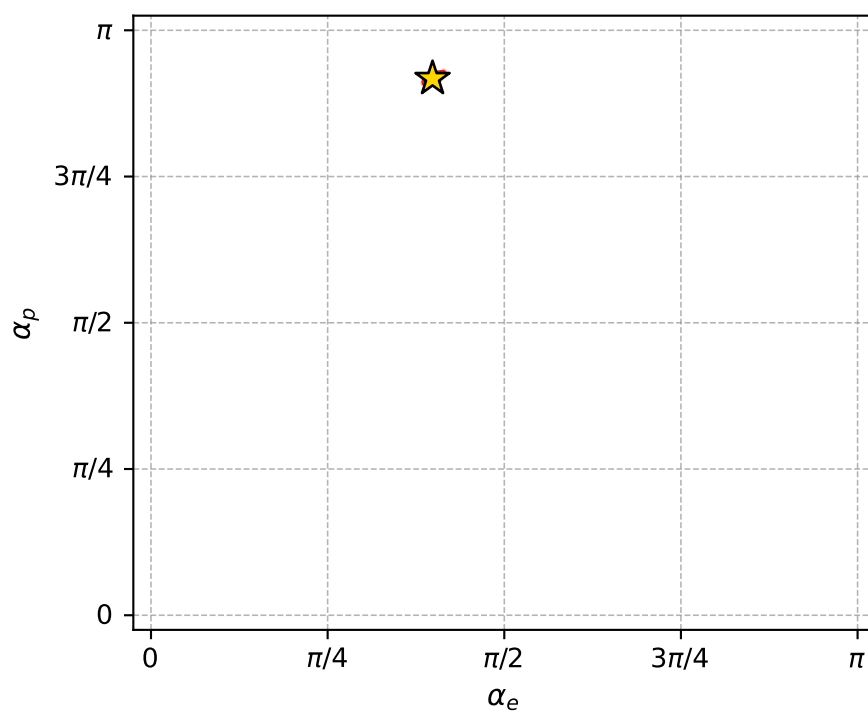
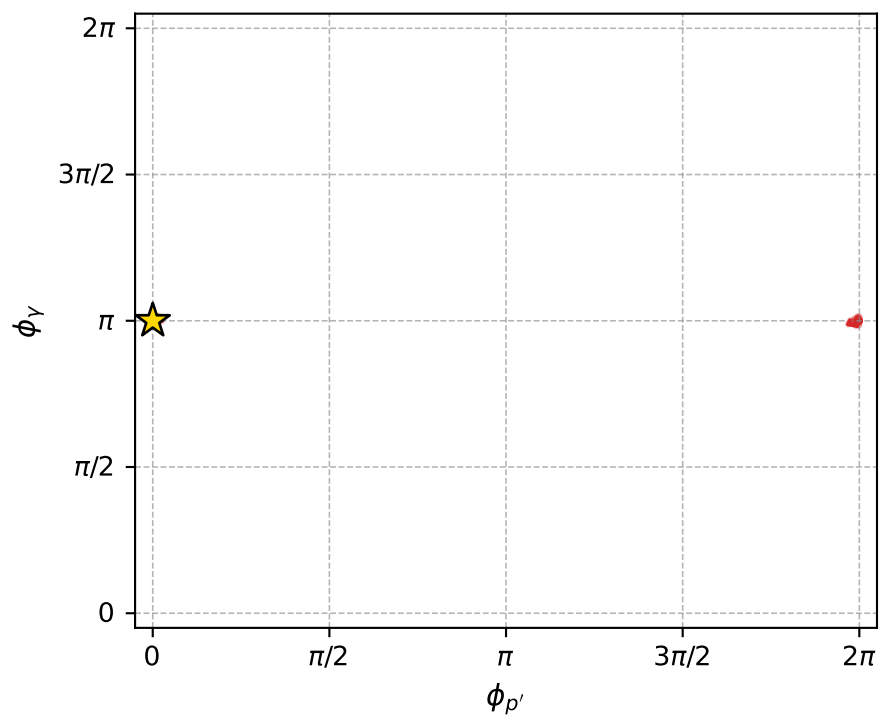
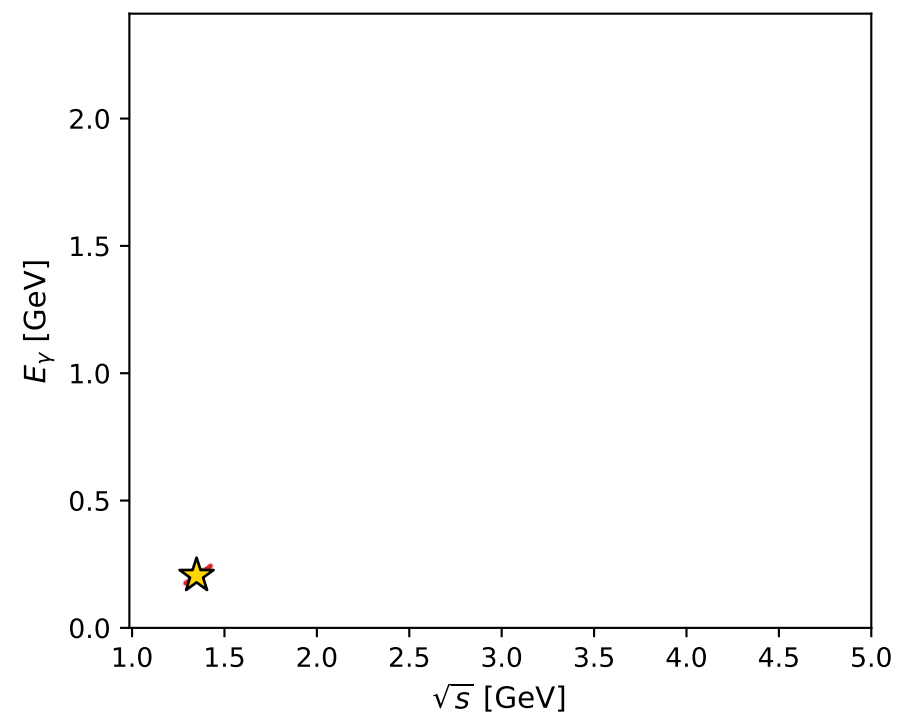
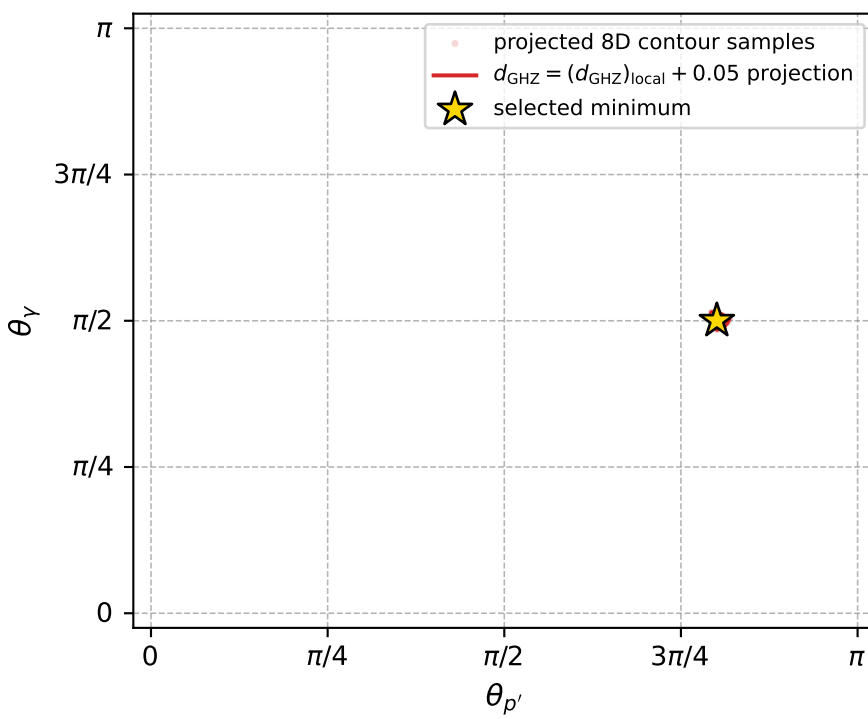
Transverse plane



Leading final-state amplitudes (N=5, retained=97.5%)



electron: polarization cluster P2, local minimum 283; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 0.00023663293$
 $d_{\text{GHZ}} \text{ contour} = 0.050236633$
 above global minimum = 0.00023661064
 8D contour samples = 256

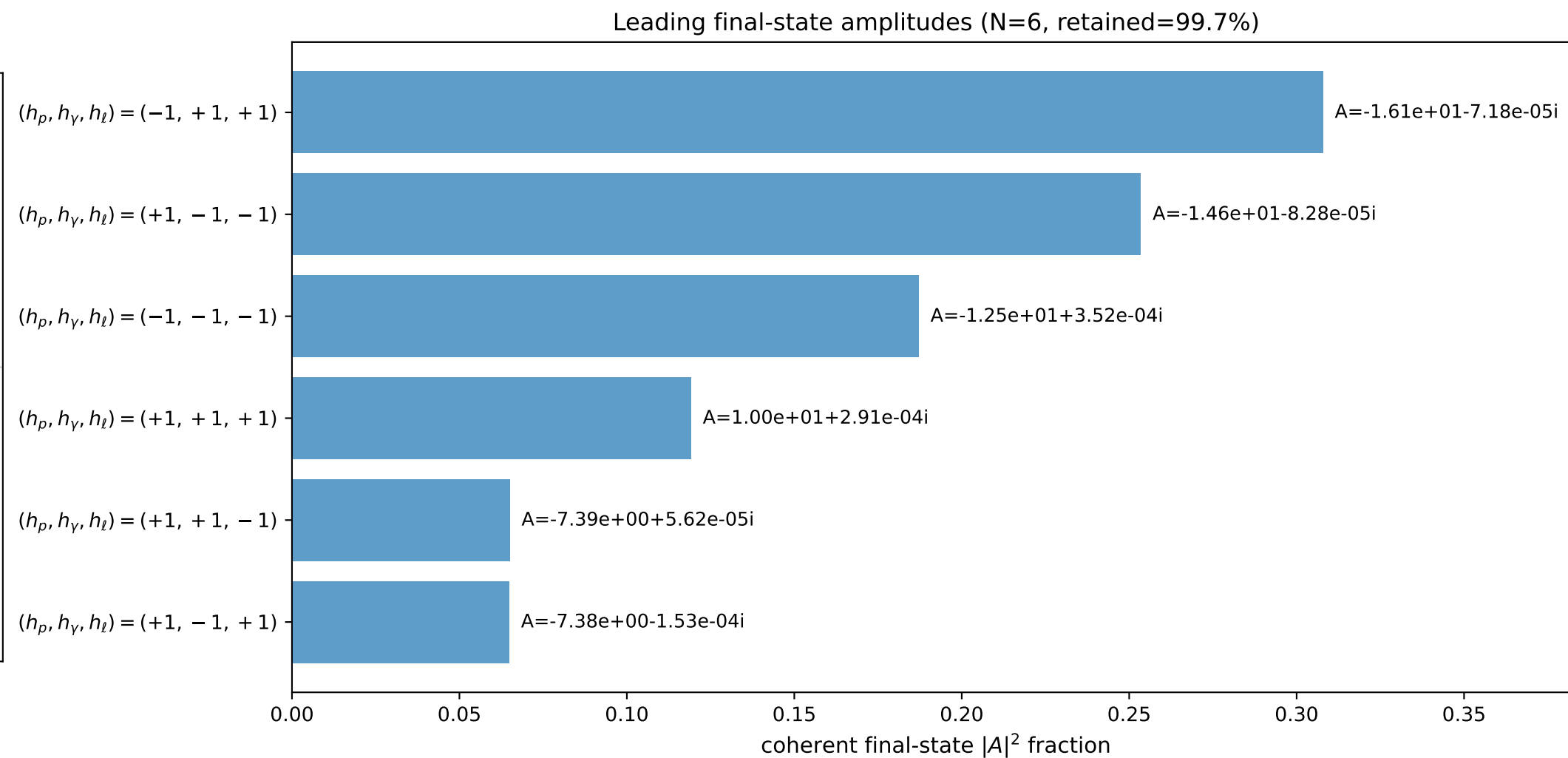
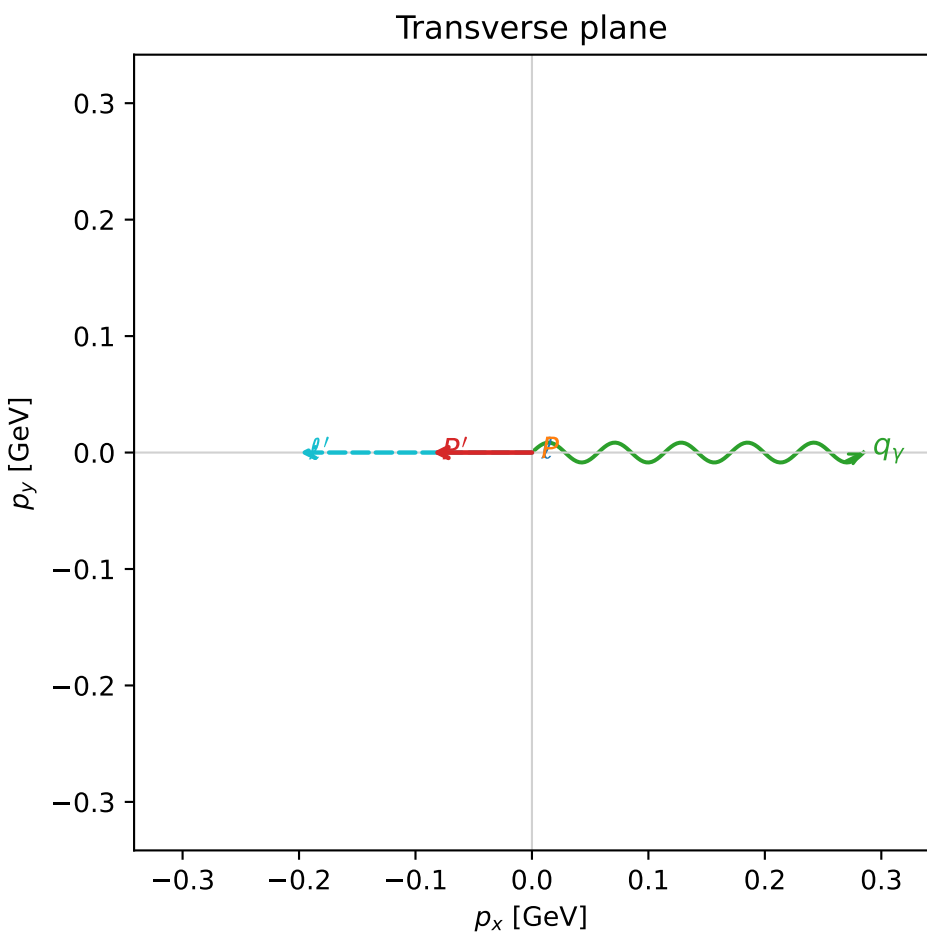
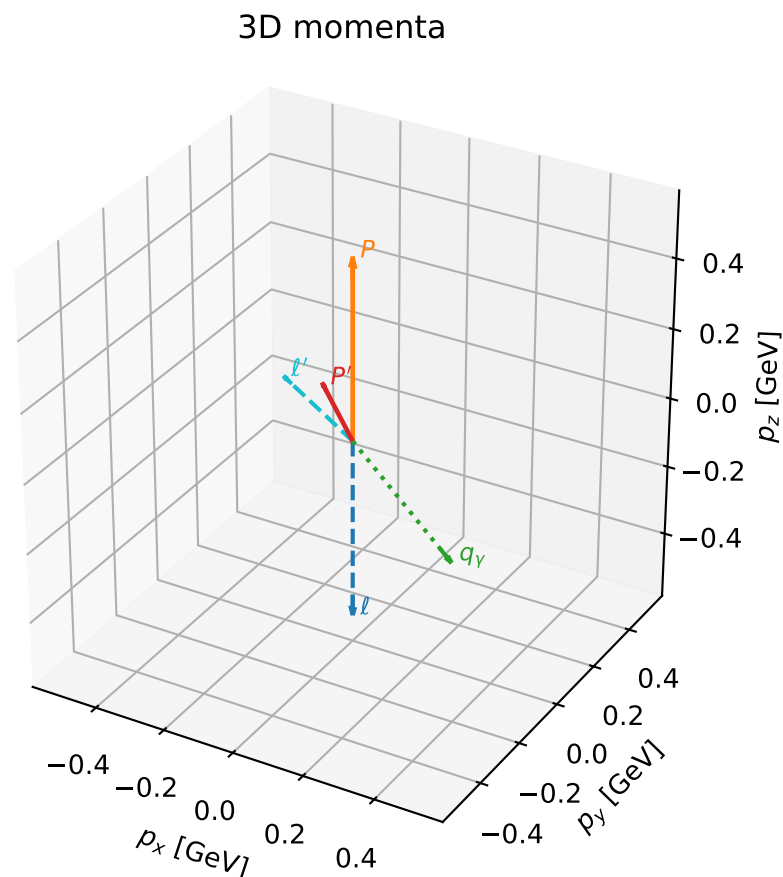
selected phase-space point:
 $\text{sqrt_s} = 1.349124$
 $\text{theta_p_out} = 2.515925$
 $\text{theta_gamma_out} = 1.571398$
 $\text{q0ut} = 0.2055225$
 $\text{phi_p_out} = 0.0001047327$
 $\text{phi_gamma_out} = 3.141677$
 $\text{alpha_e} = 1.251771$
 $\text{alpha_p} = 2.882247$

coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

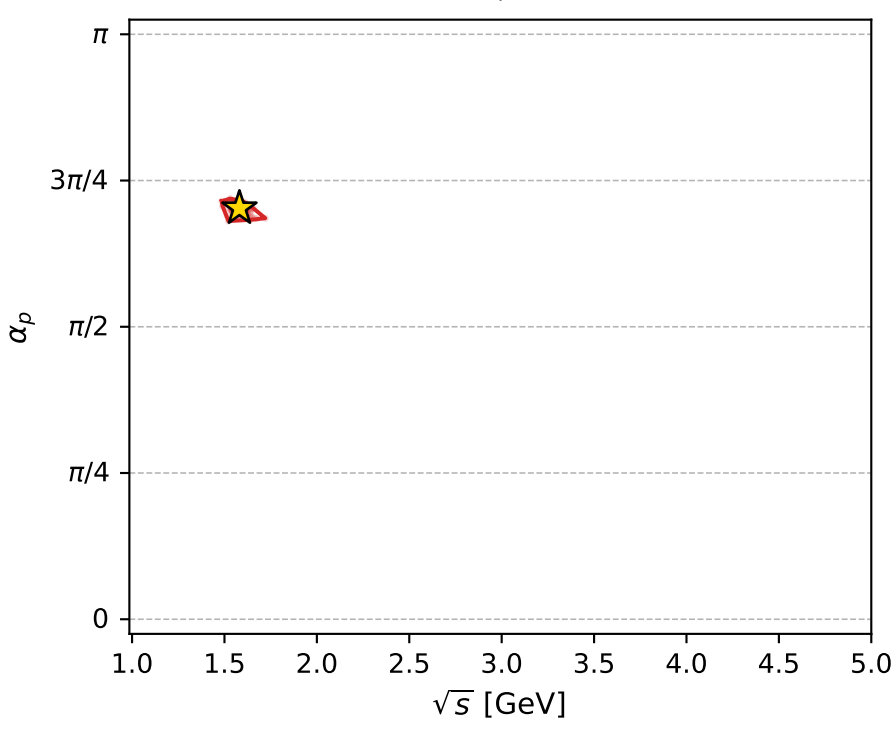
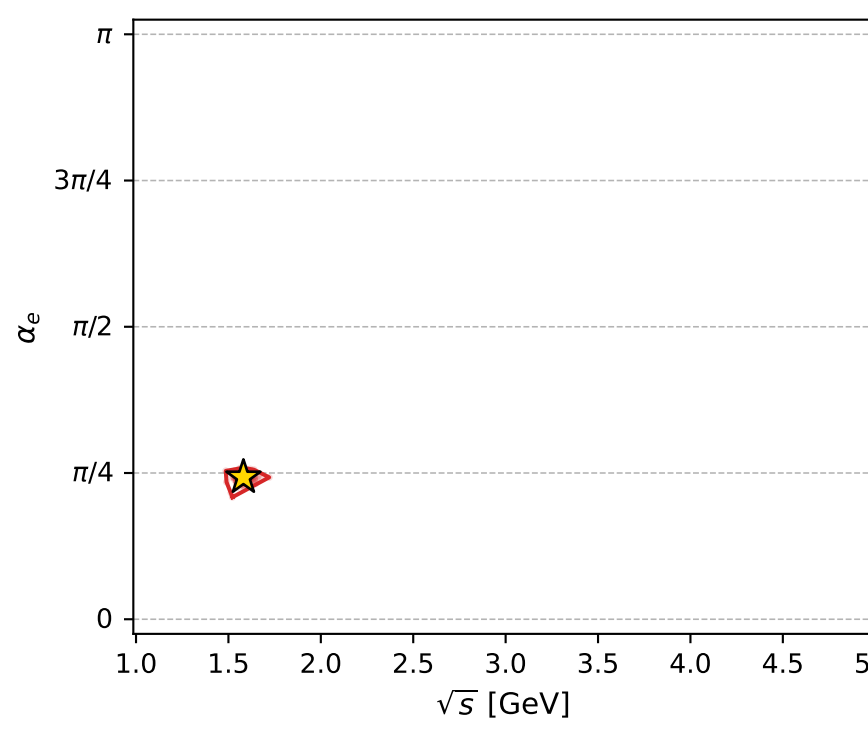
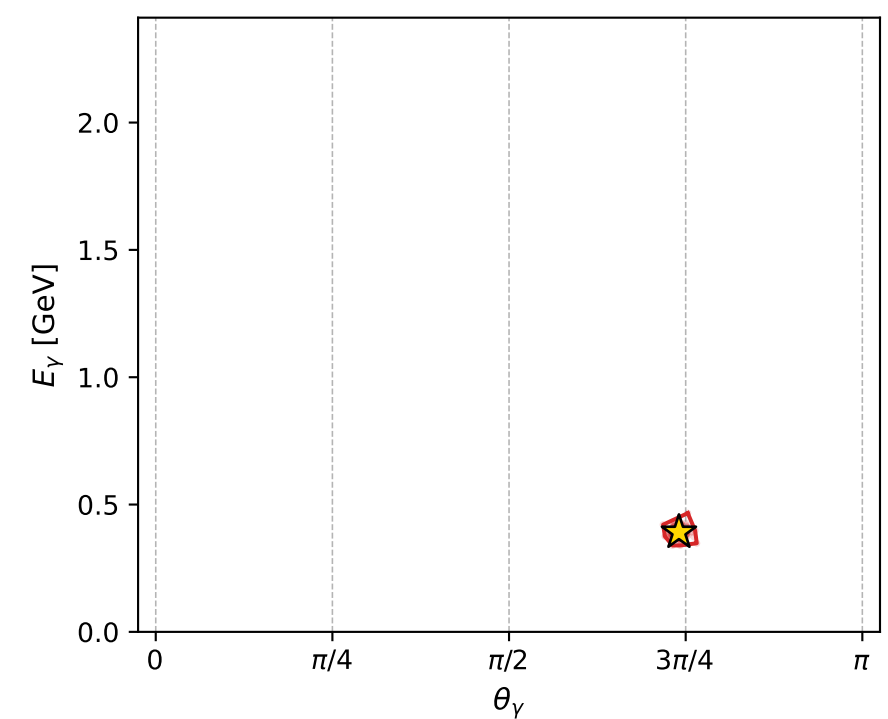
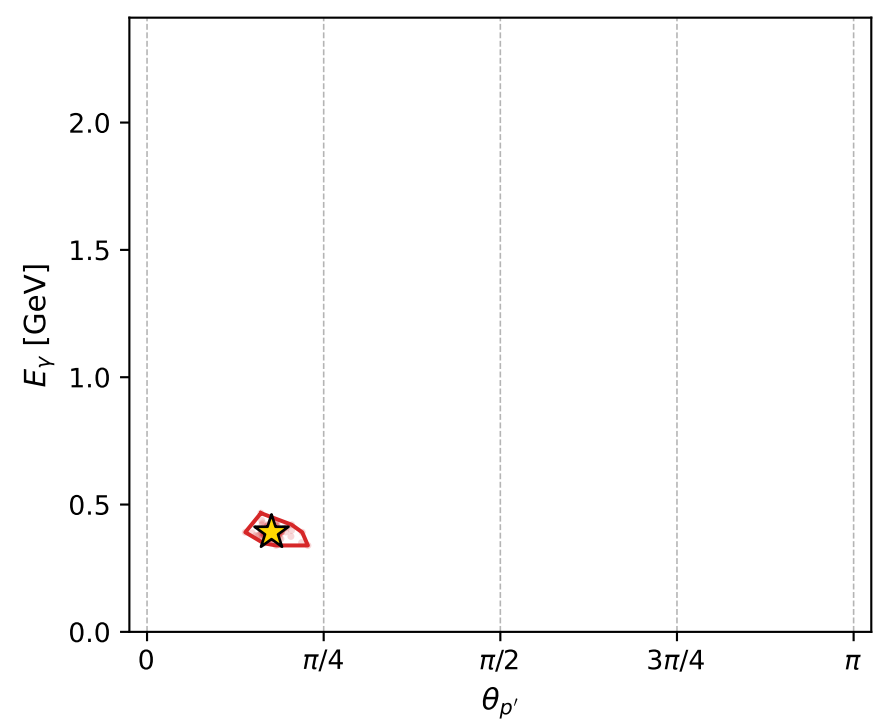
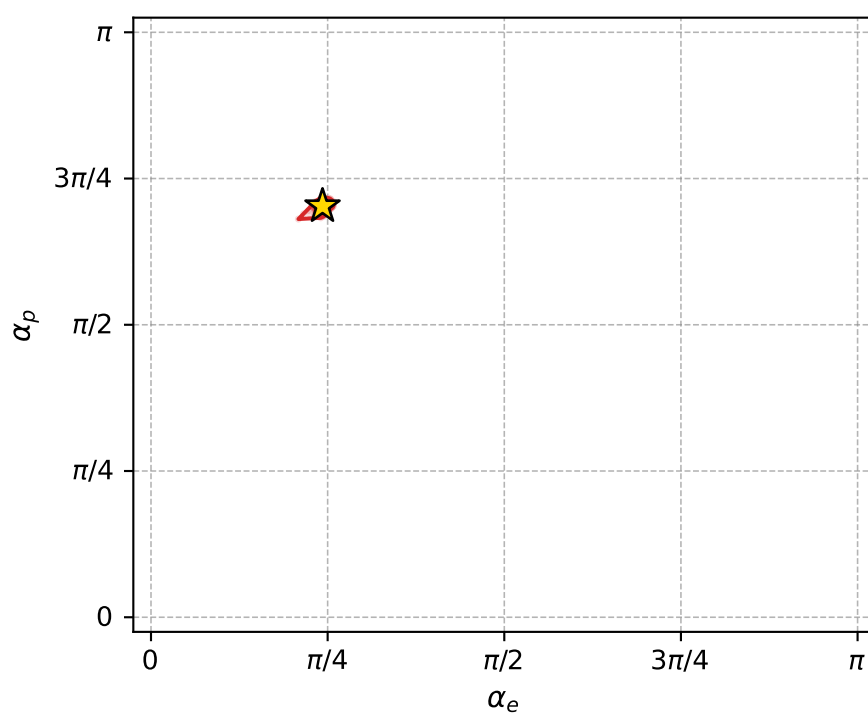
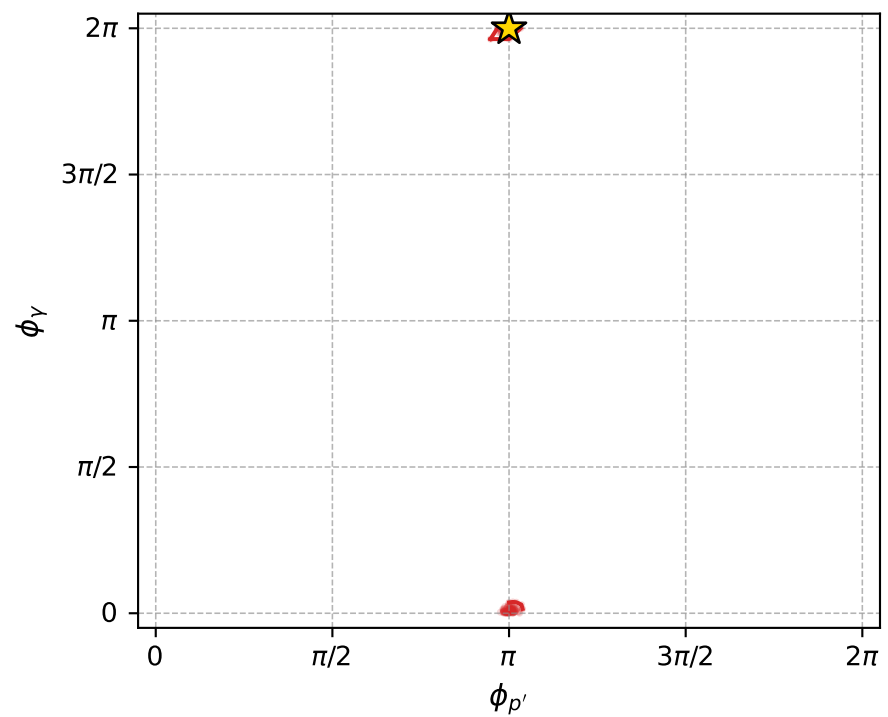
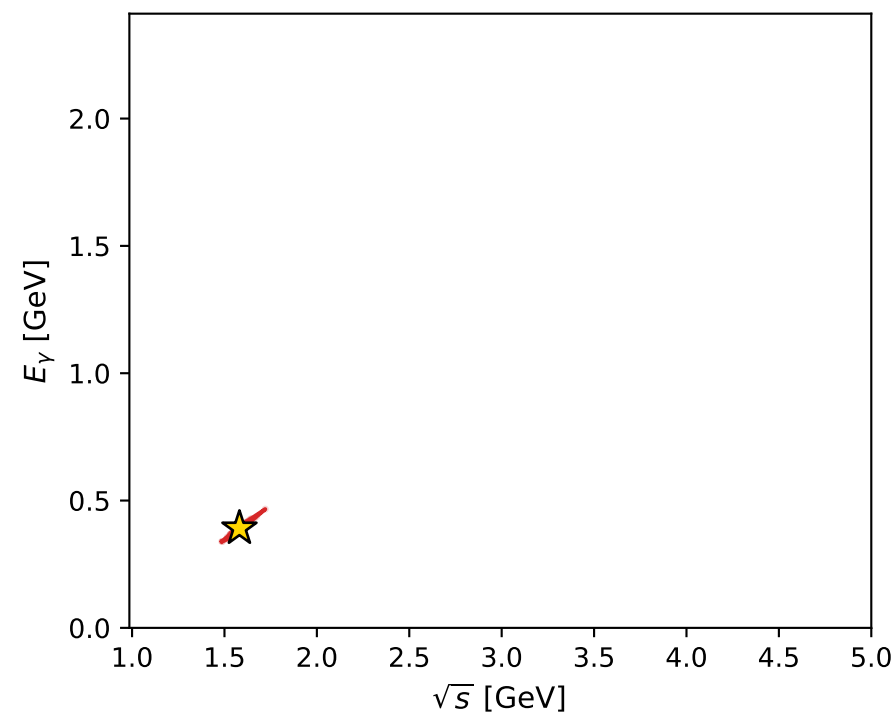
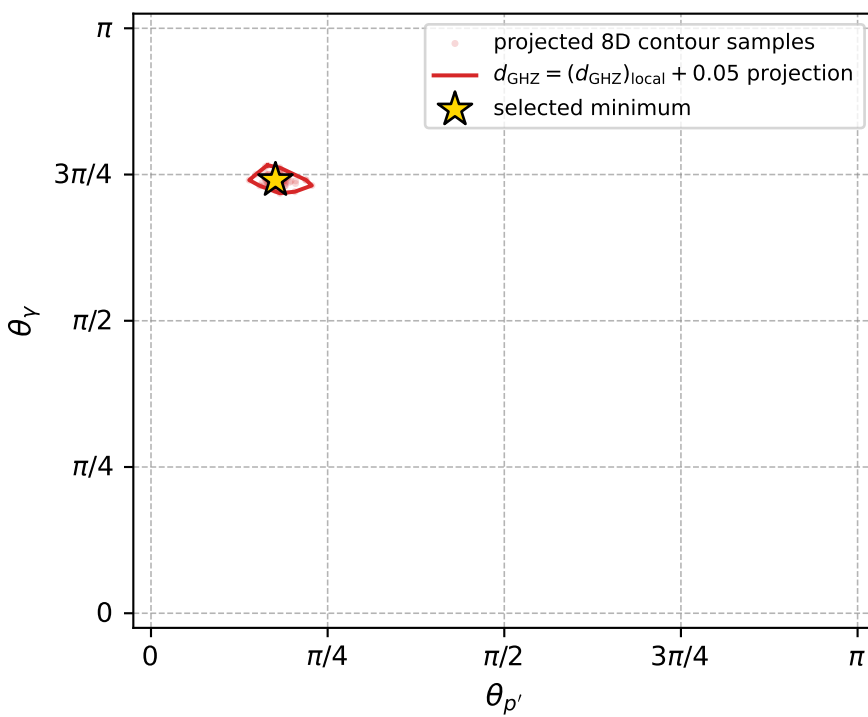
dghz_mixing_angles_polarization_02_local_minimum_285 $d_{\{\text{rm GHZ}\}}=0.000252337$
 region: polarization_cluster_02_local_minimum_285
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=0.76263008$ rad
 incoming lepton: $0.723022|+\rangle +0.690825|-\rangle$
 proton mixing angle: $\alpha_p=2.2074955$ rad
 incoming proton: $-0.594545|+\rangle +0.804063|-\rangle$

$s=2.49969$, $\sqrt{s}=1.58104$, $|\vec{P}|=0.512271$, $|\vec{P}'|=0.162407$
 $\theta_{P'}=0.553457$, $\theta_Y=2.32625$, $\phi_{P'}=3.14156$, $\phi_Y=6.28317$
 $E_Y=0.391105$, $\theta_{\ell'}=0.992955$, $\phi_{\ell'}=3.14158$
 $Q^2=0.376998$, $x_B=0.302972$, $t=-0.1336$, $W^2=1.74718$, $y=0.76818$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.5123$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.5123$
 P $E= 1.0688$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.5123$
 ℓ' $E= 0.2380$ $p_x= -0.1993$ $p_y= 0.0000$ $p_z= 0.1300$
 P' $E= 0.9520$ $p_x= -0.0854$ $p_y= 0.0000$ $p_z= 0.1382$
 q_Y $E= 0.3911$ $p_x= 0.2847$ $p_y= -0.0000$ $p_z= -0.2682$



electron: polarization cluster P2, local minimum 285; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 0.00025233733$
 $d_{\text{GHZ}} \text{ contour} = 0.050252337$
 above global minimum = 0.00025231505
 8D contour samples = 256

selected phase-space point:

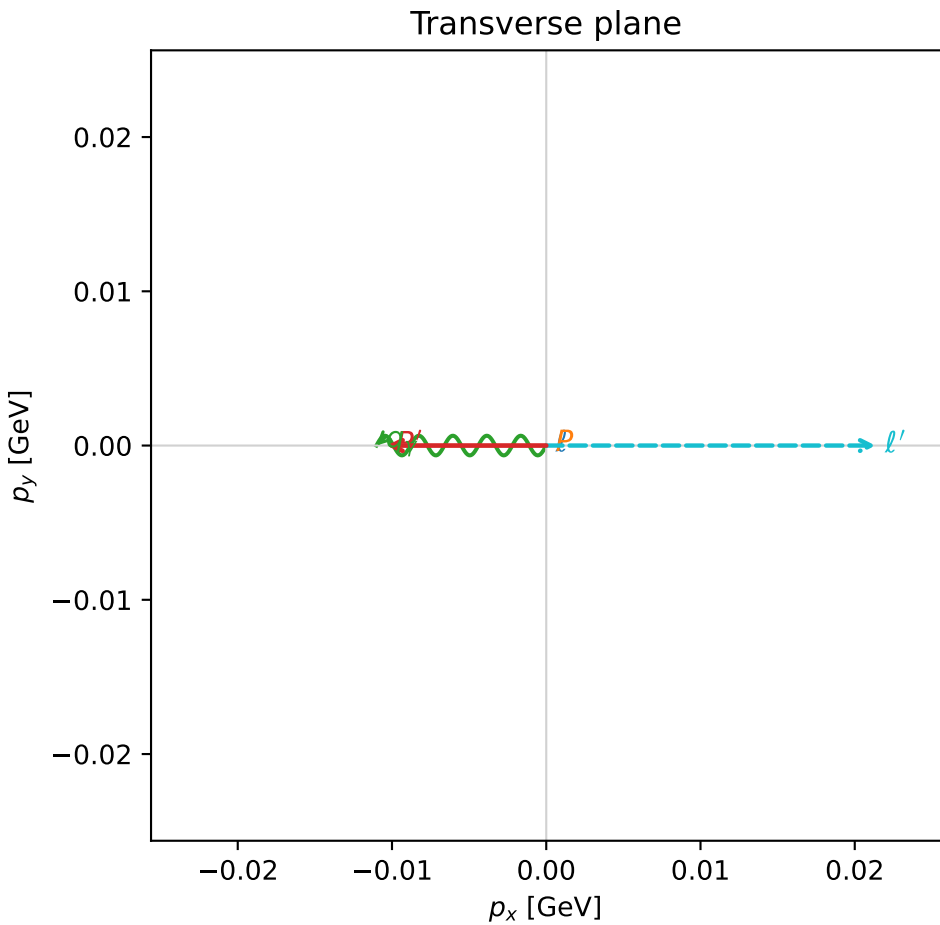
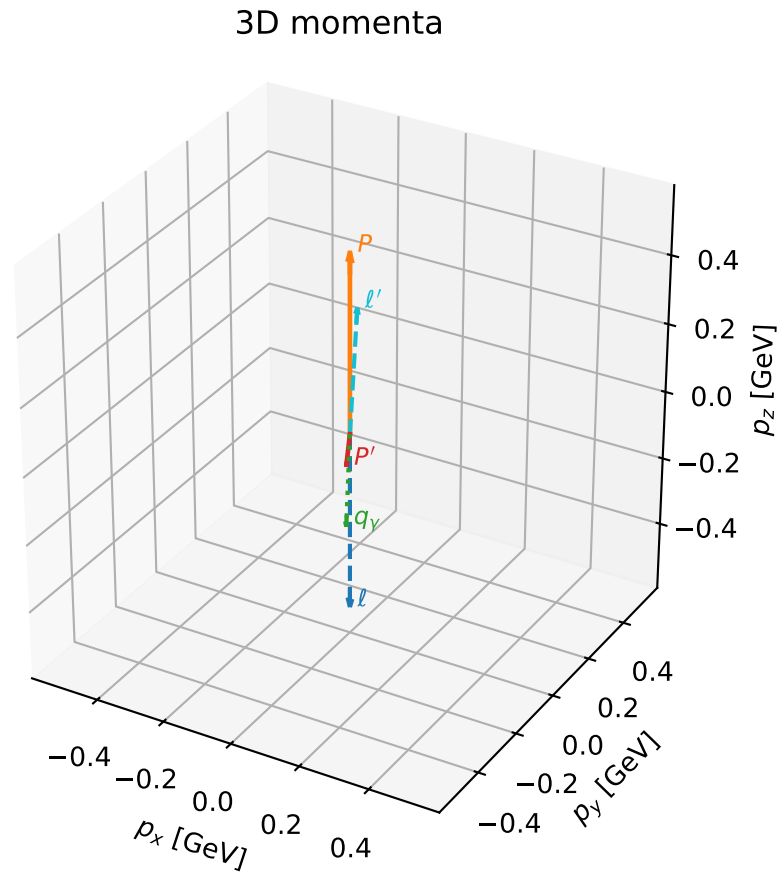
$\text{sqrt_s} = 1.58104$
 $\text{theta_p_out} = 0.5534566$
 $\text{theta_gamma_out} = 2.326254$
 $\text{q0ut} = 0.3911048$
 $\text{phi_p_out} = 3.141563$
 $\text{phi_gamma_out} = 6.283171$
 $\text{alpha_e} = 0.7626301$
 $\text{alpha_p} = 2.207496$

coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

dghz_mixing_angles_polarization_02_local_minimum_286 d_{\rm GHZ}=0.000253567
region: polarization_cluster_02_local_minimum_286
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: $\alpha_e=0.023844319$ rad
incoming lepton: $0.999716|+\rangle +0.0238421|-\rangle$
proton mixing angle: $\alpha_p=1.5427631$ rad
incoming proton: $0.0280295|+\rangle +0.999607|-\rangle$

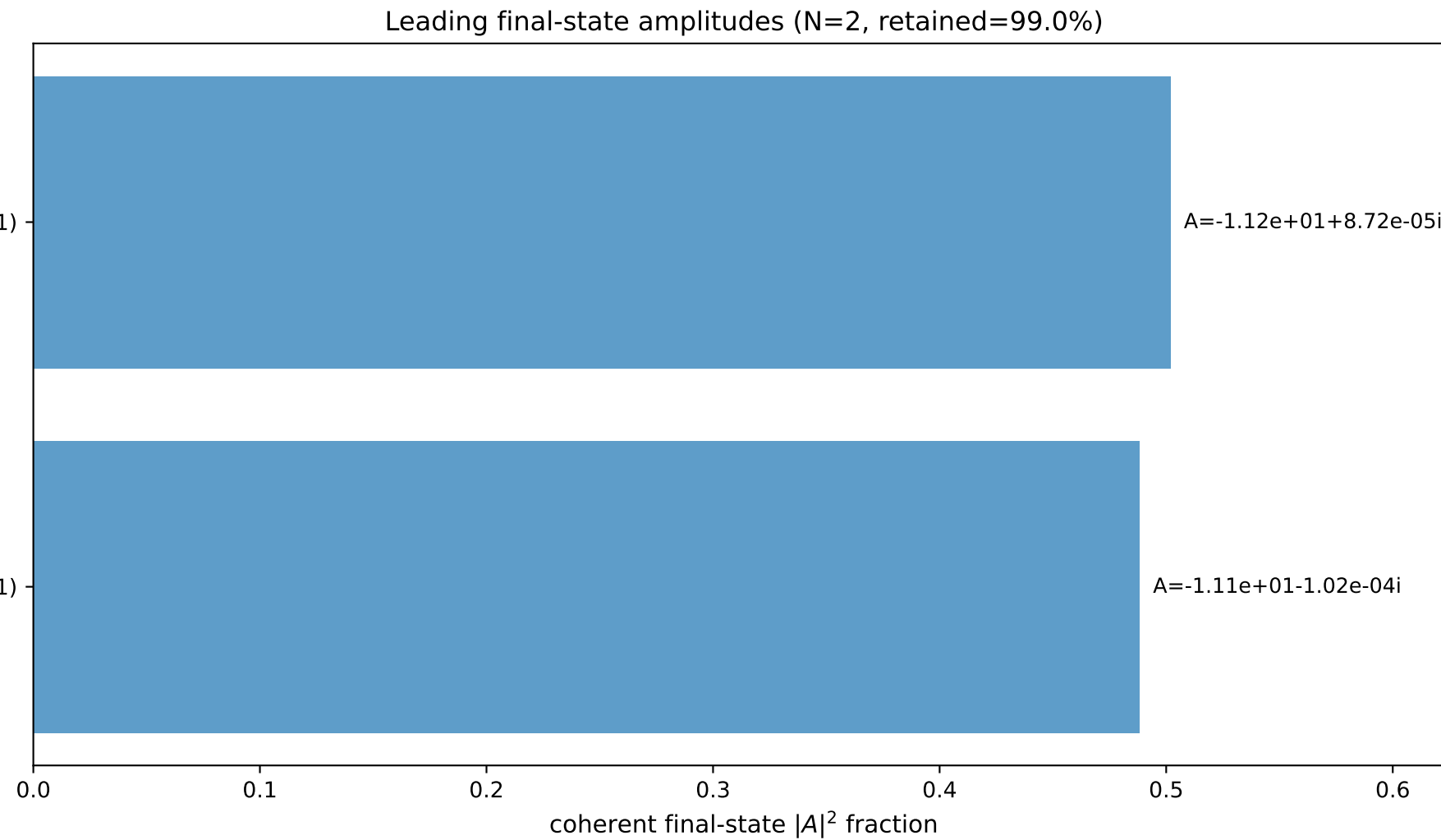
$s=2.54271$, $\sqrt{s}=1.59459$, $|\vec{P}|=0.52141$, $|\vec{P}'|=0.0940246$
 $\theta_{P'}=3.03139$, $\theta_V=3.10212$, $\phi_{P'}=3.1416$, $\phi_V=3.14162$
 $E_V=0.27902$, $\theta_{\ell'}=0.0572926$, $\phi_{\ell'}=1.62633\text{e-}05$
 $Q^2=0.777031$, $x_B=0.621249$, $t=-0.36114$, $W^2=1.35357$, $y=0.752168$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.5214$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.5214$
 P $E= 1.0732$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.5214$
 ℓ' $E= 0.3729$ $p_x= 0.0214$ $p_y= 0.0000$ $p_z= 0.3723$
 P' $E= 0.9427$ $p_x= -0.0103$ $p_y= -0.0000$ $p_z= -0.0935$
 q_Y $E= 0.2790$ $p_x= -0.0110$ $p_y= -0.0000$ $p_z= -0.2788$

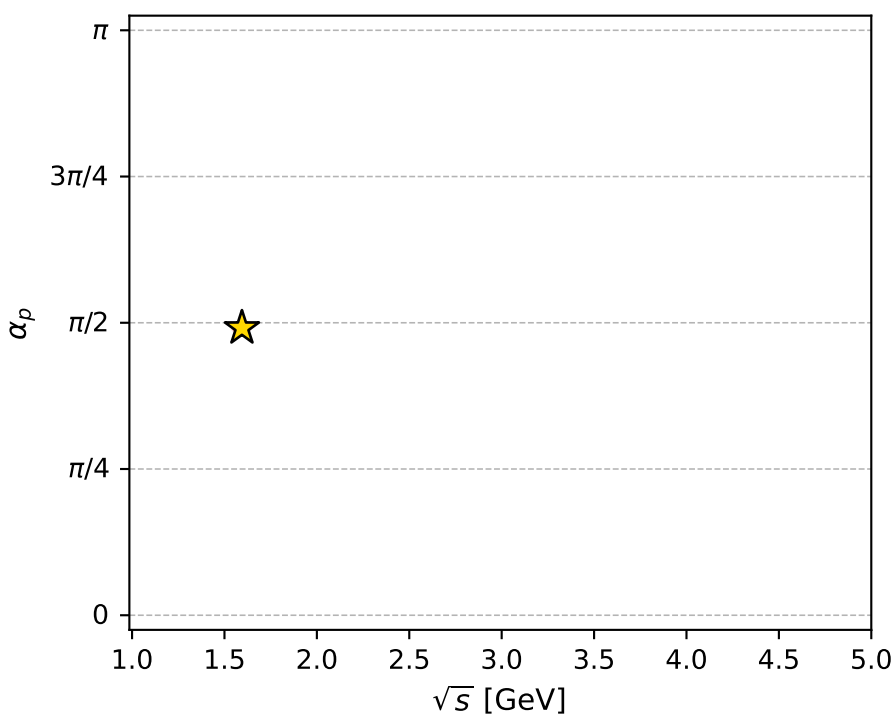
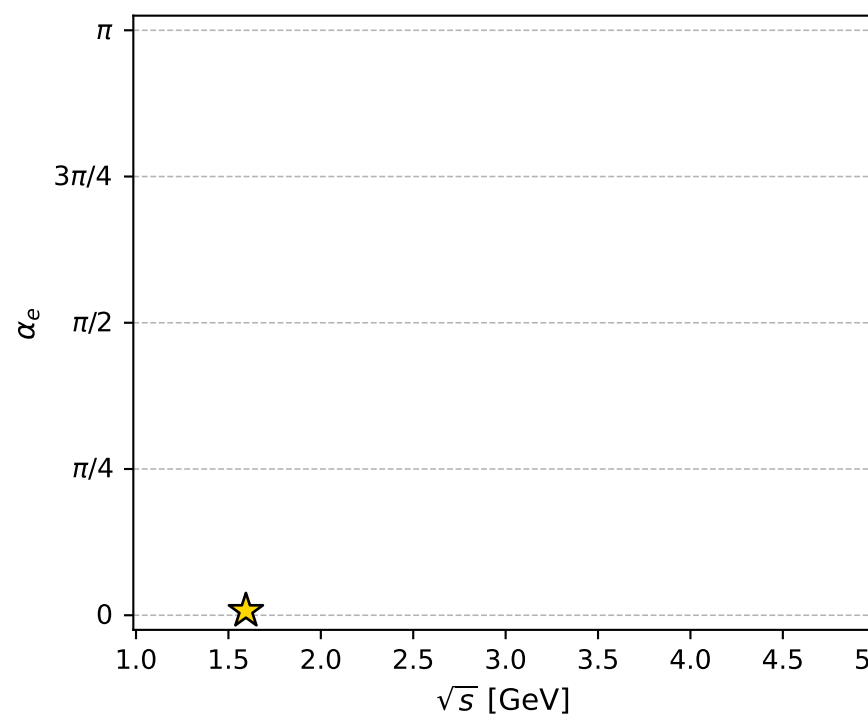
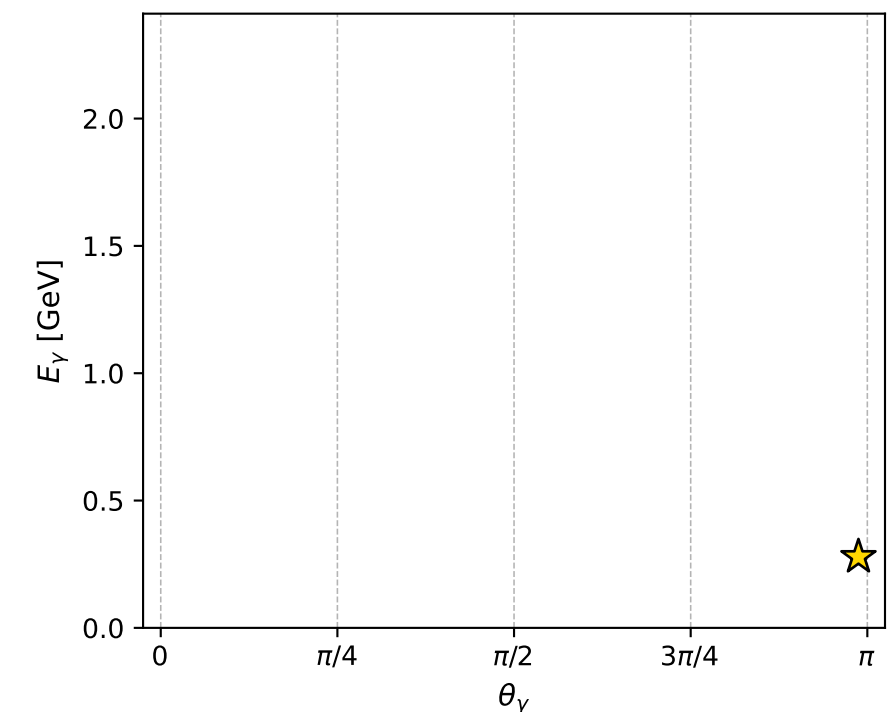
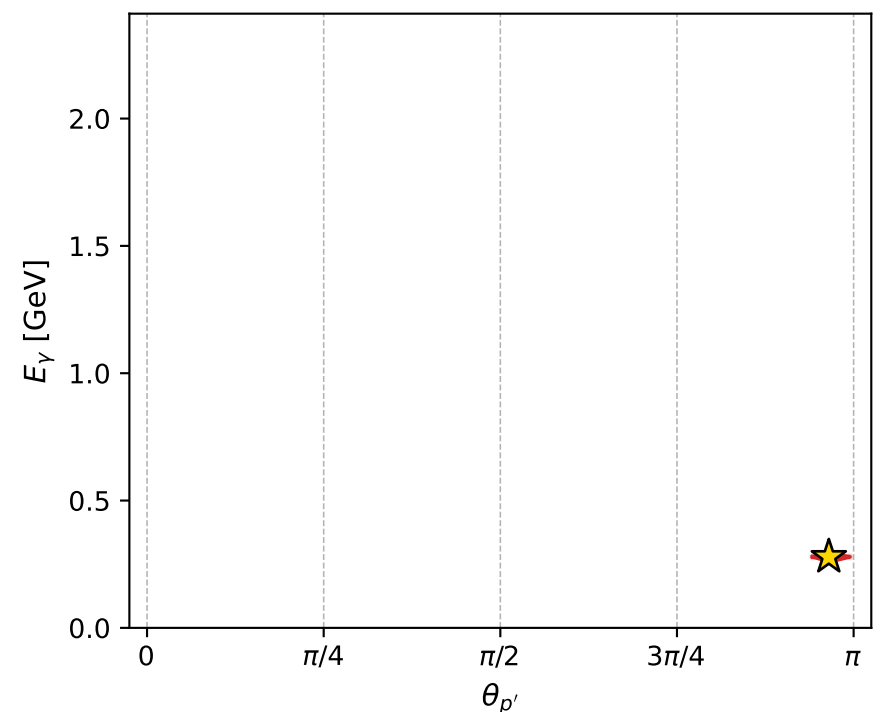
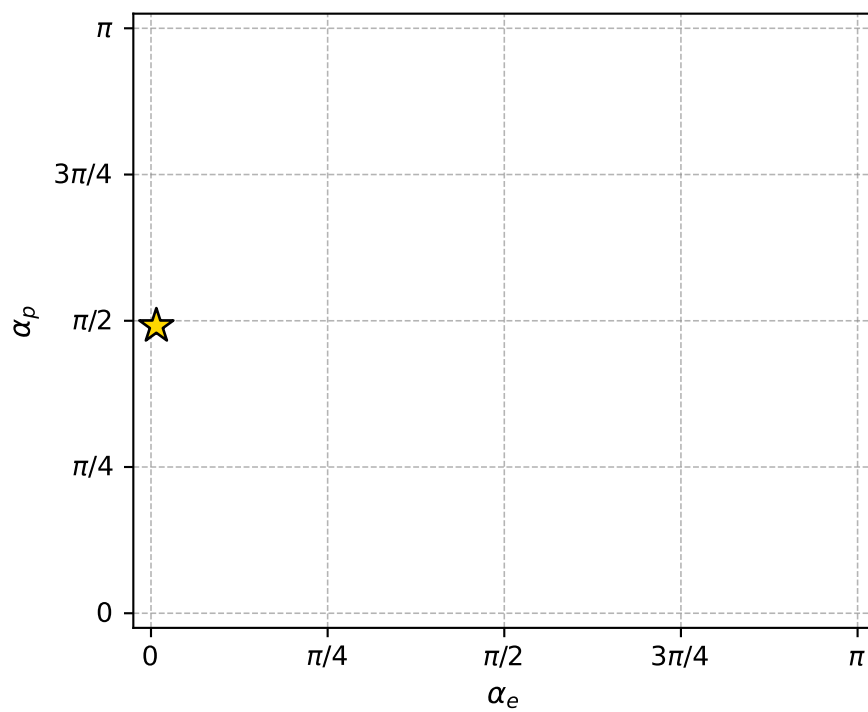
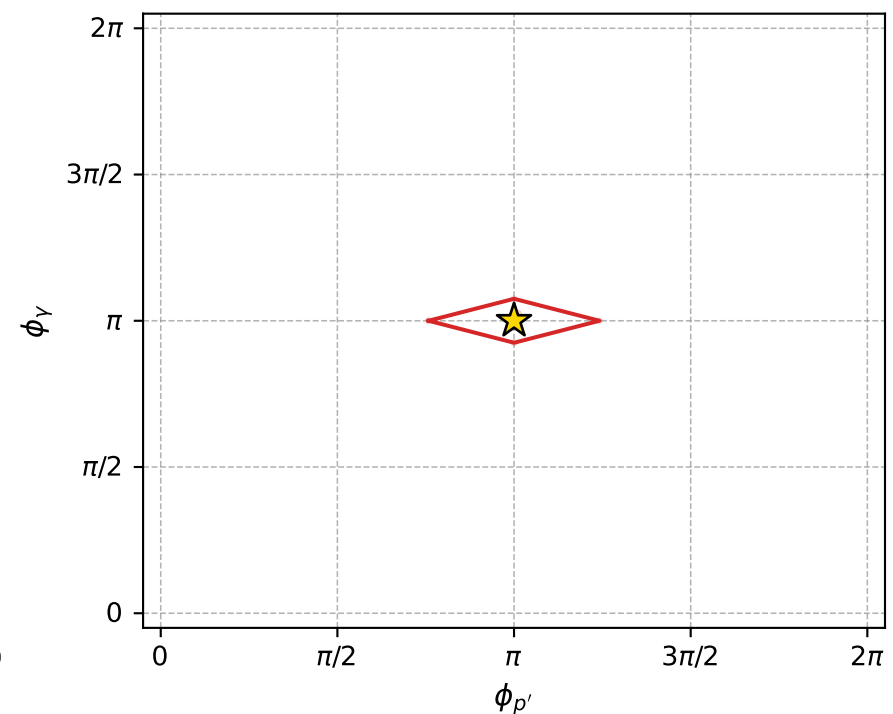
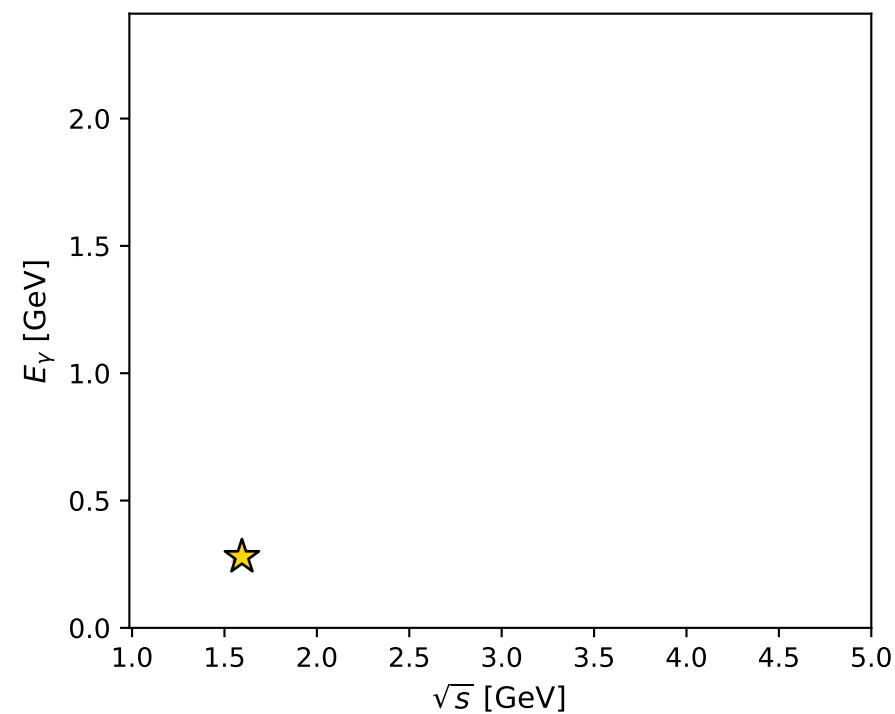
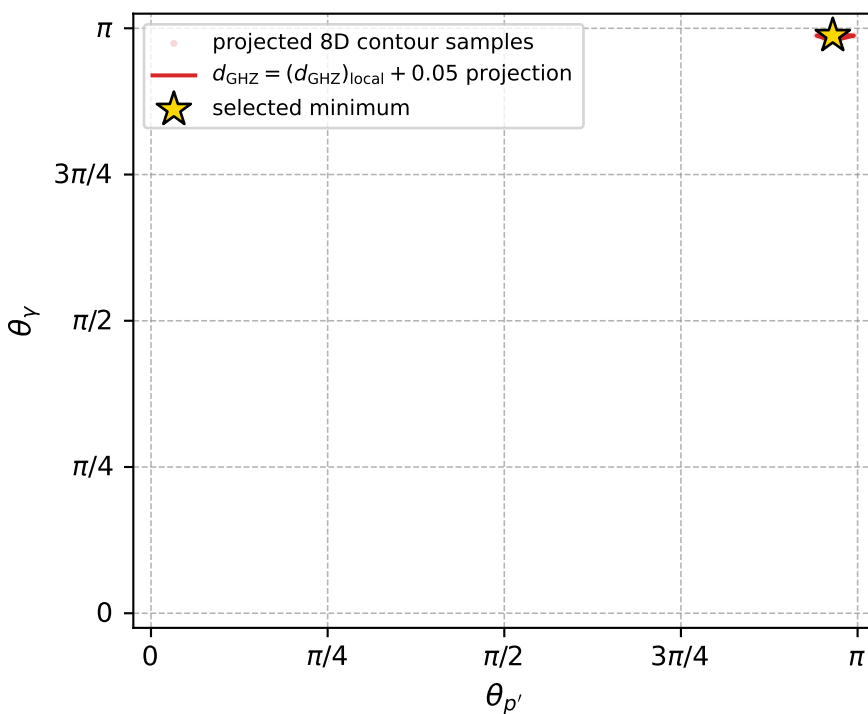


$(h_p, h_V, h_\ell) = (+1, +1, +1)$

$(h_p, h_V, h_\ell) = (-1, -1, -1)$



electron: polarization cluster P2, local minimum 286; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 0.00025356695$
 $d_{\text{GHZ}} \text{ contour} = 0.050253567$
 above global minimum = 0.00025354467
 8D contour samples = 256

selected phase-space point:

$\text{sqrt_s} = 1.594589$
 $\text{theta_p_out} = 3.031387$
 $\text{theta_gamma_out} = 3.102123$
 $\text{q0out} = 0.2790195$
 $\text{phi_p_out} = 3.141602$
 $\text{phi_gamma_out} = 3.141616$
 $\text{alpha_e} = 0.02384432$
 $\text{alpha_p} = 1.542763$

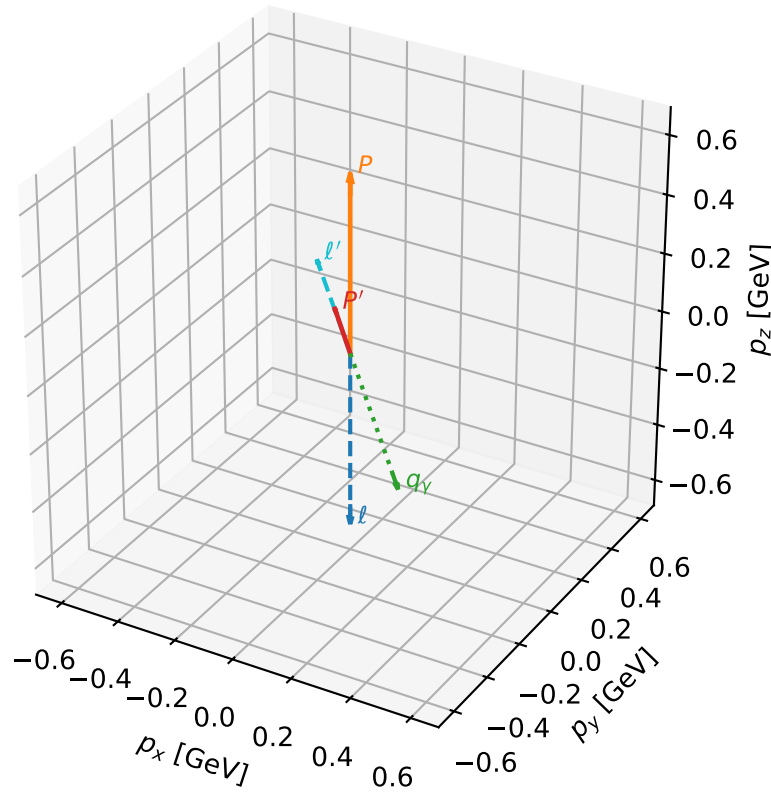
coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

dghz_mixing_angles_polarization_02_local_minimum_288 $d_{\{\text{rm GHZ}\}}=0.00025975$
 region: polarization_cluster_02_local_minimum_288
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=0.30245936$ rad
 incoming lepton: $0.954607|+\rangle +0.297869|-\rangle$
 proton mixing angle: $\alpha_p=1.8172517$ rad
 incoming proton: $-0.243968|+\rangle +0.969783|-\rangle$

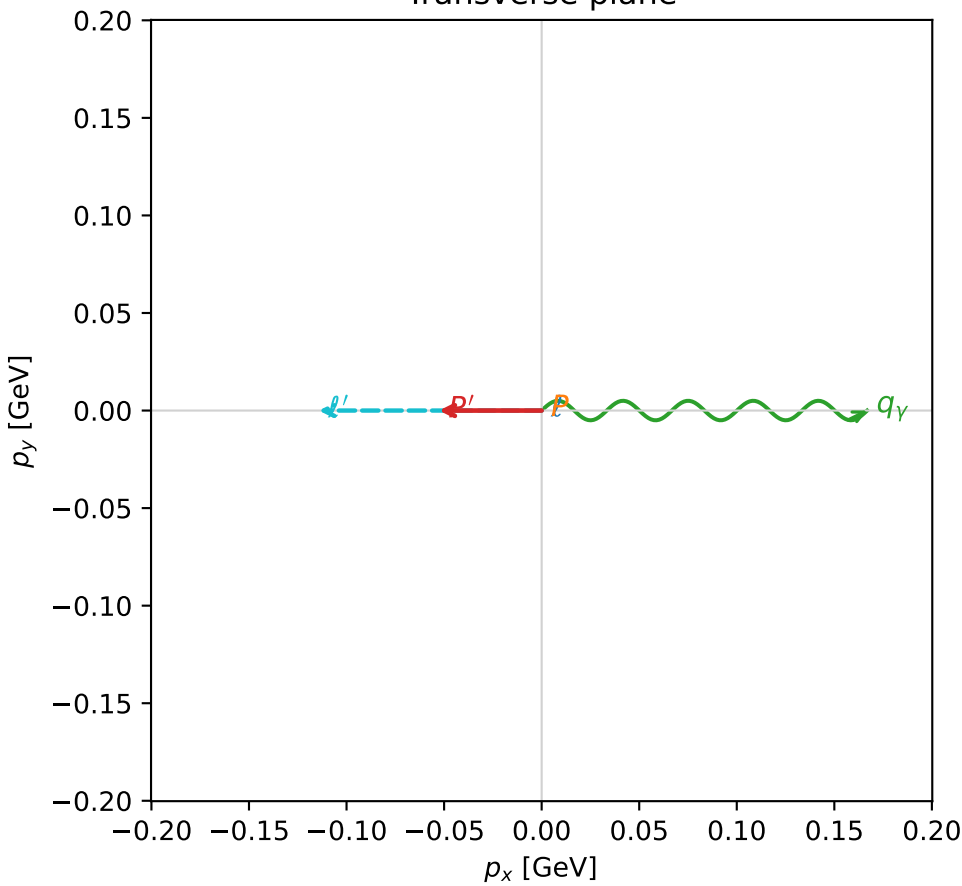
$s=2.94552$, $\sqrt{s}=1.71625$, $|\vec{P}|=0.601798$, $|\vec{P}'|=0.147268$
 $\theta_{P'}=0.365298$, $\theta_Y=2.7679$, $\phi_{P'}=3.14166$, $\phi_Y=3.79046\text{e-}05$
 $E_Y=0.45701$, $\theta_{\ell'}=0.377679$, $\phi_{\ell'}=3.14162$
 $Q^2=0.719354$, $x_B=0.41779$, $t=-0.191081$, $W^2=1.8823$, $y=0.833533$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.6018$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.6018$
 P $E= 1.1145$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.6018$
 ℓ' $E= 0.3098$ $p_x= -0.1142$ $p_y= -0.0000$ $p_z= 0.2879$
 P' $E= 0.9495$ $p_x= -0.0526$ $p_y= -0.0000$ $p_z= 0.1376$
 q_Y $E= 0.4570$ $p_x= 0.1668$ $p_y= 0.0000$ $p_z= -0.4255$

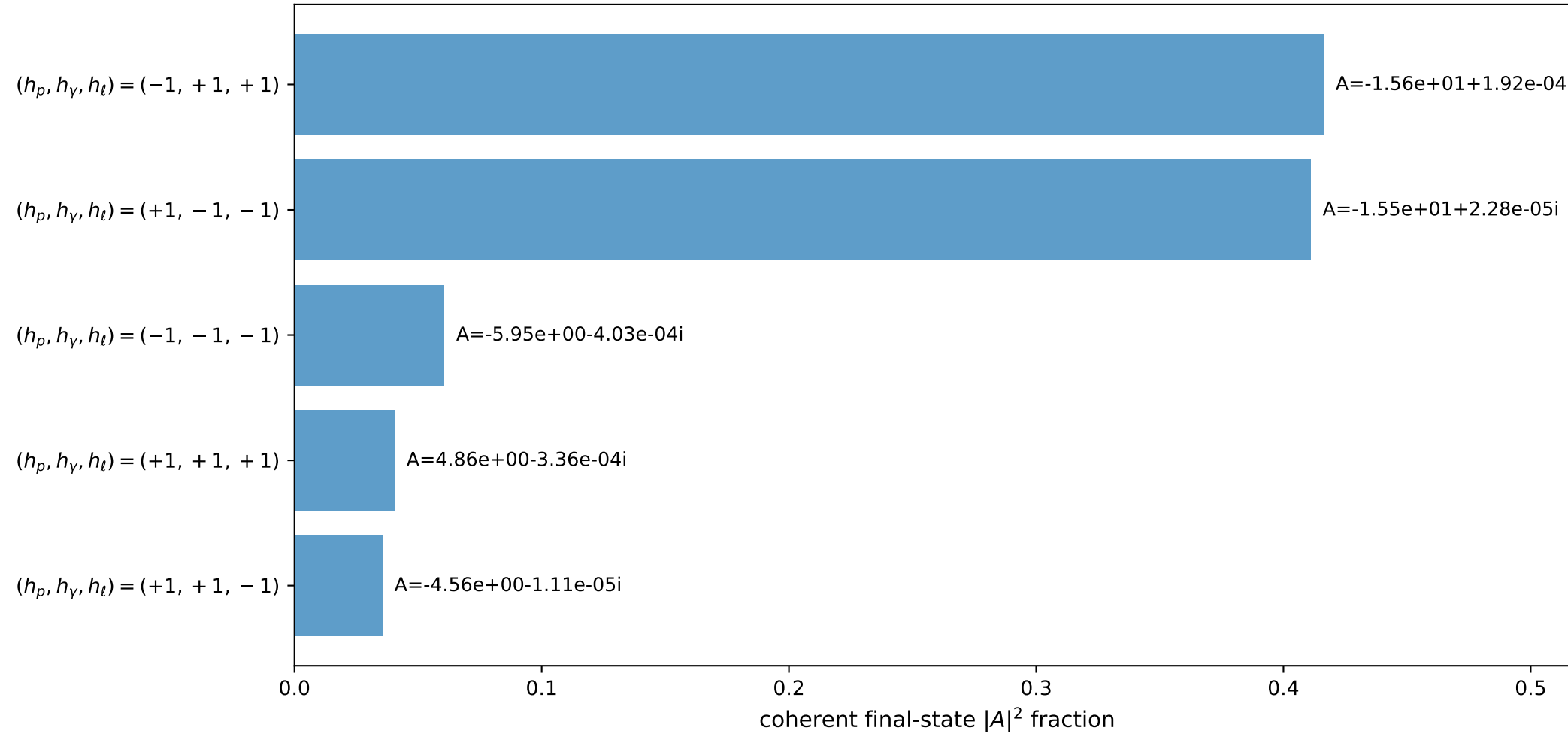
3D momenta



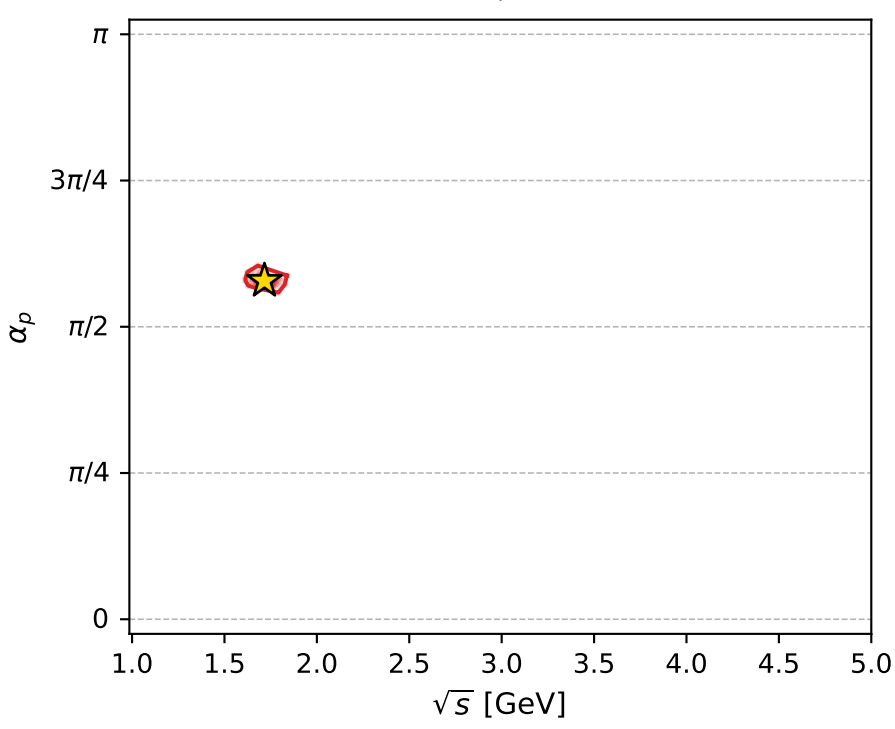
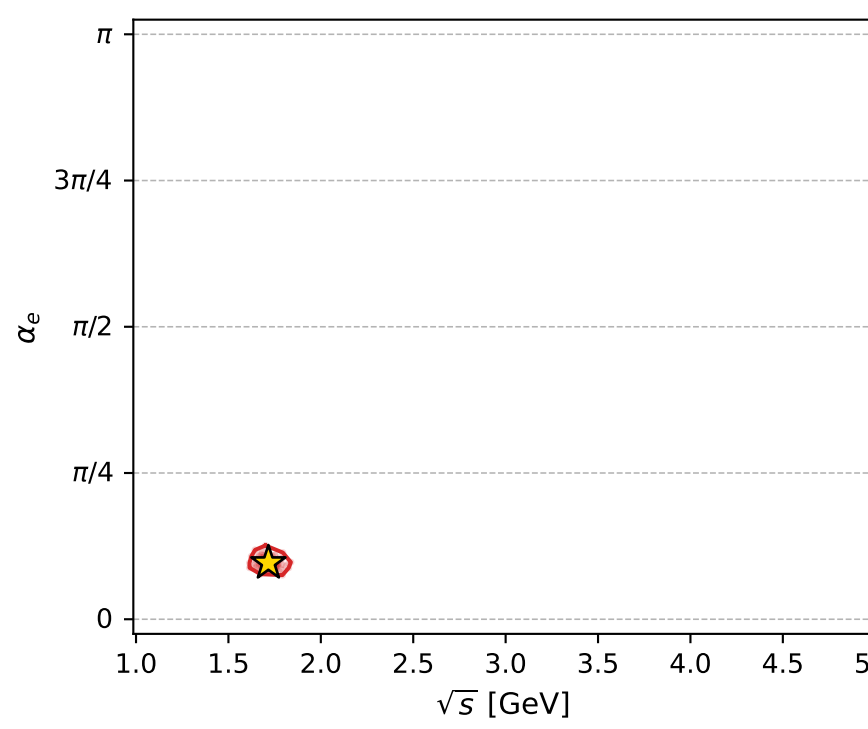
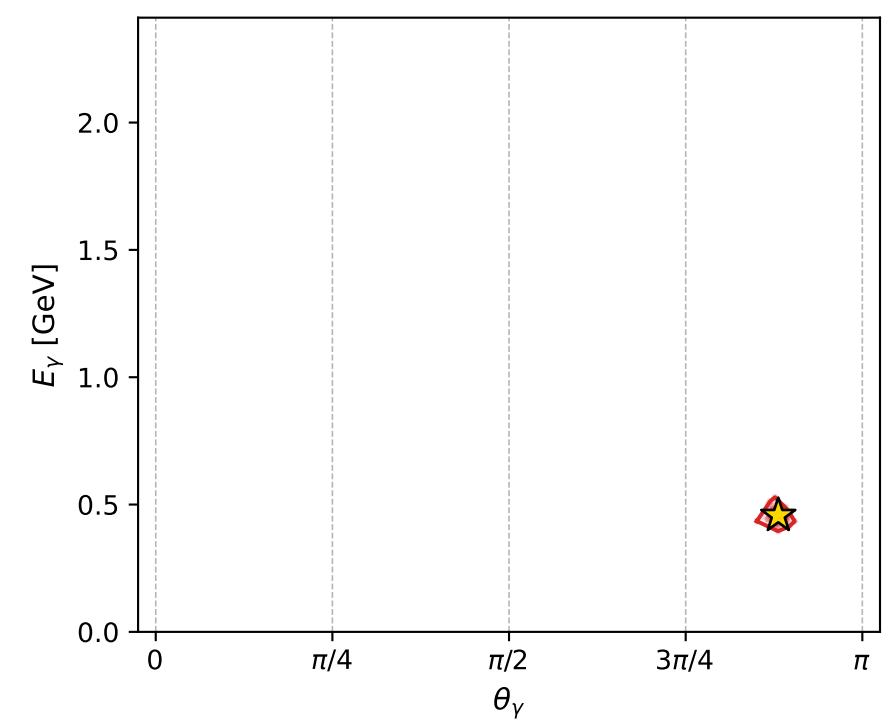
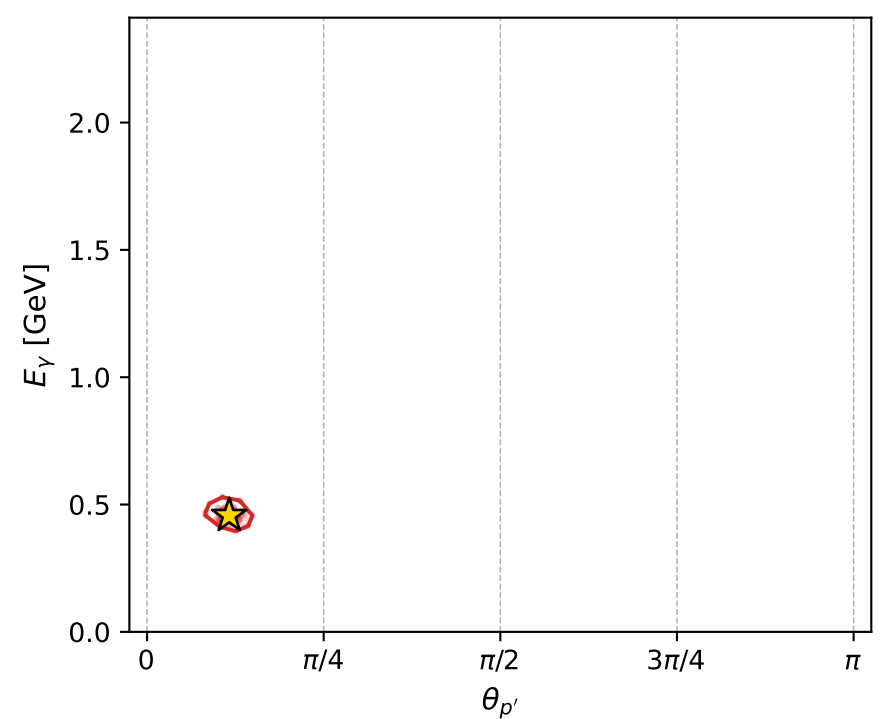
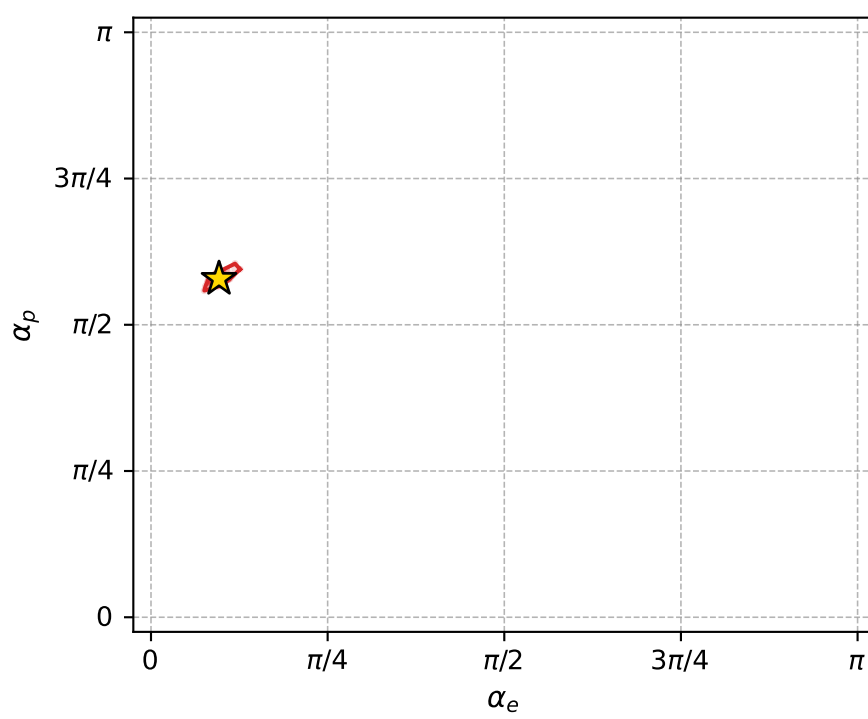
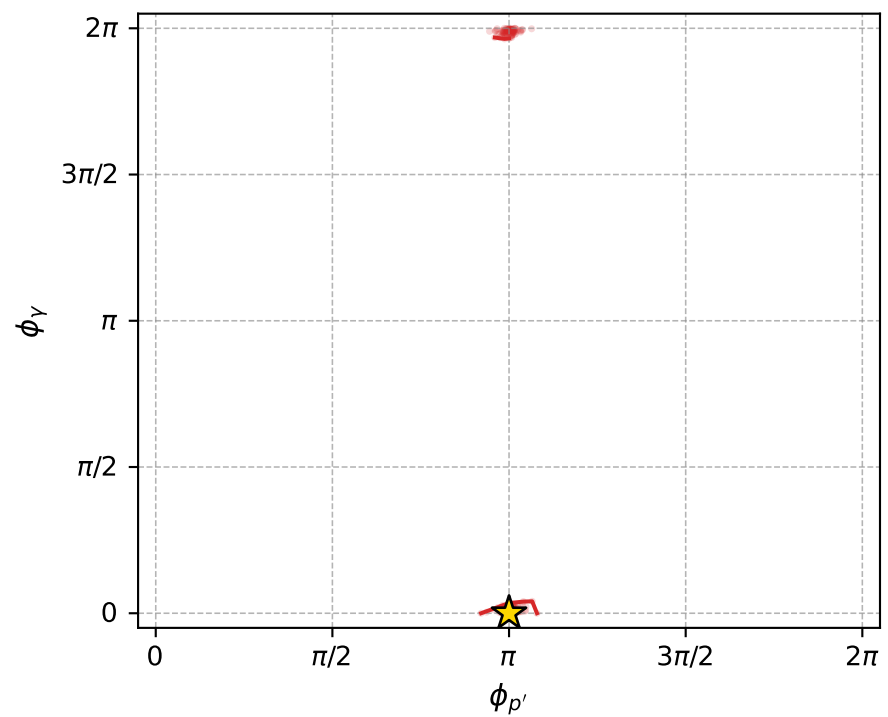
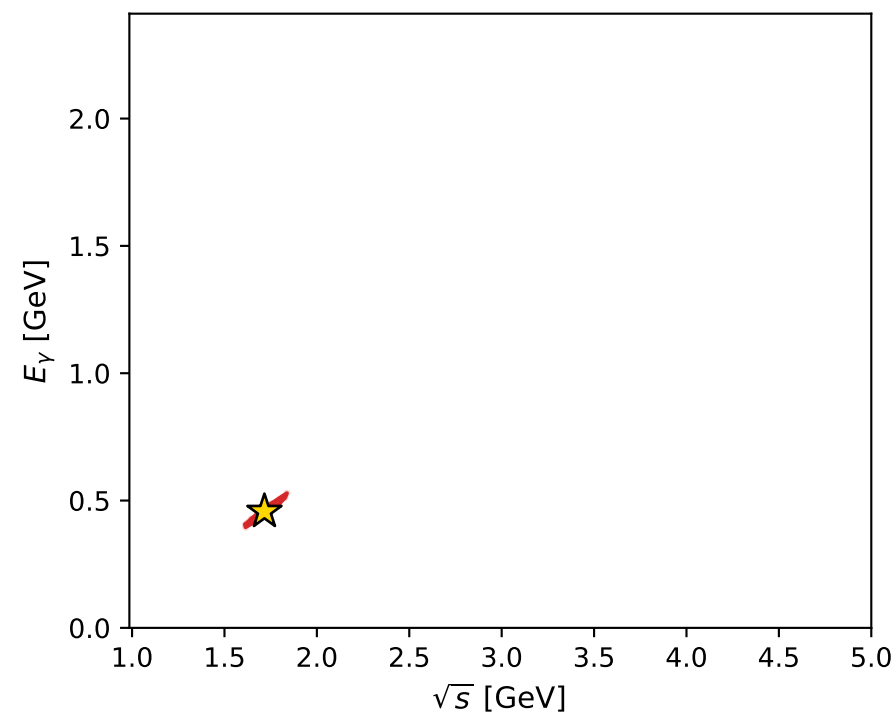
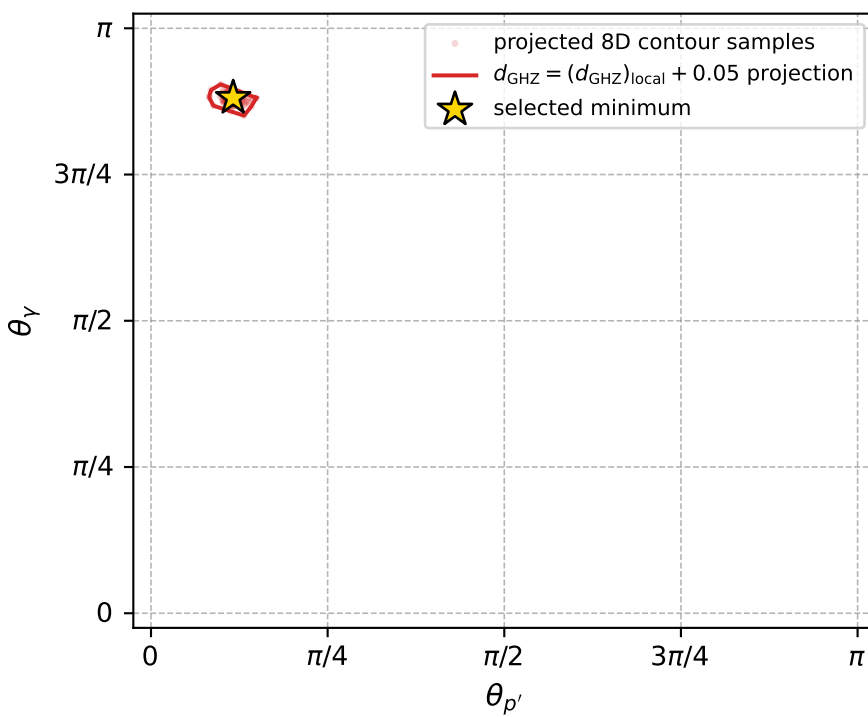
Transverse plane



Leading final-state amplitudes (N=5, retained=96.4%)



electron: polarization cluster P2, local minimum 288; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 0.00025974999$
 $d_{\text{GHZ}} \text{ contour} = 0.05025975$
 above global minimum = 0.00025972771
 8D contour samples = 256

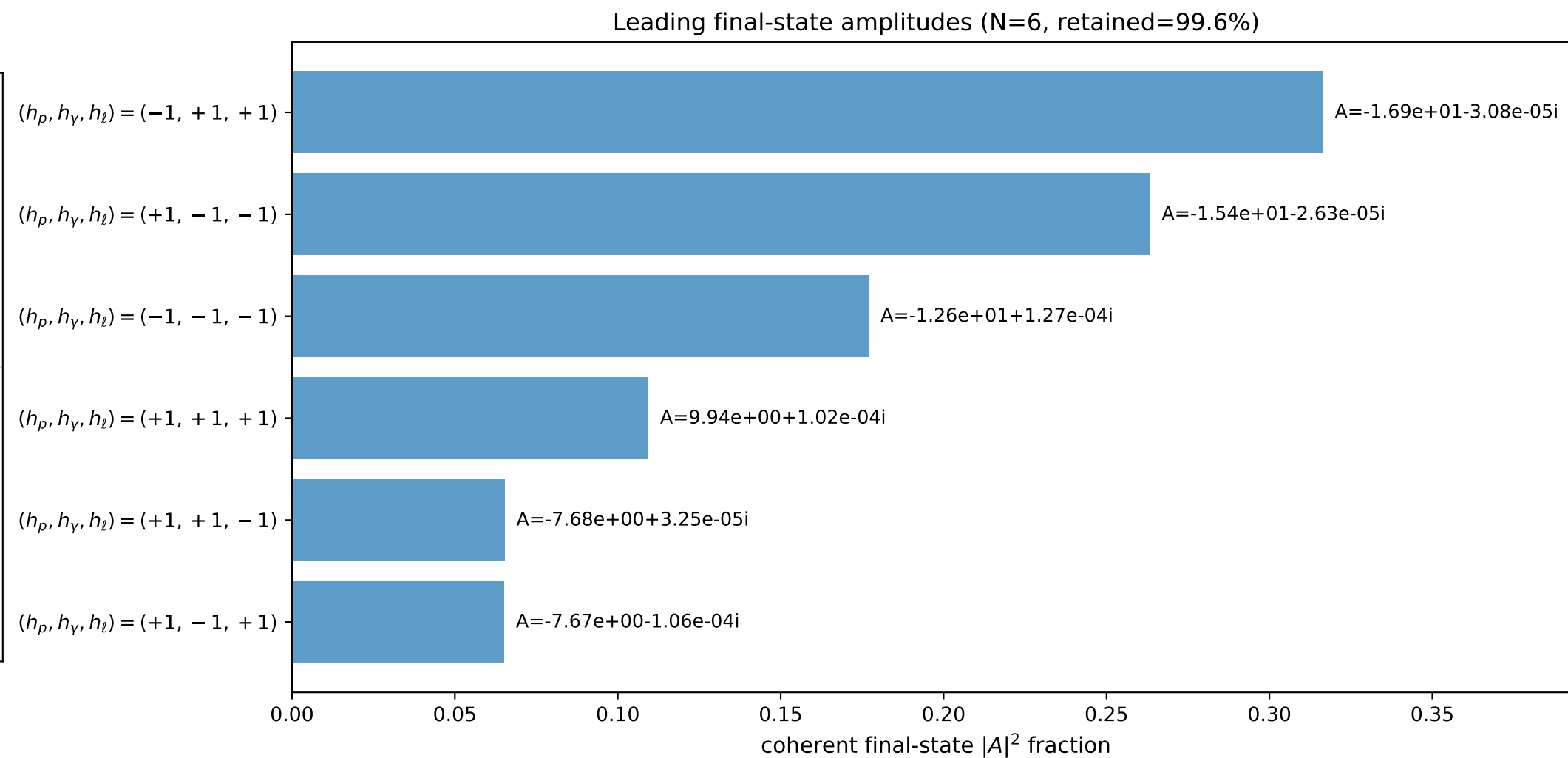
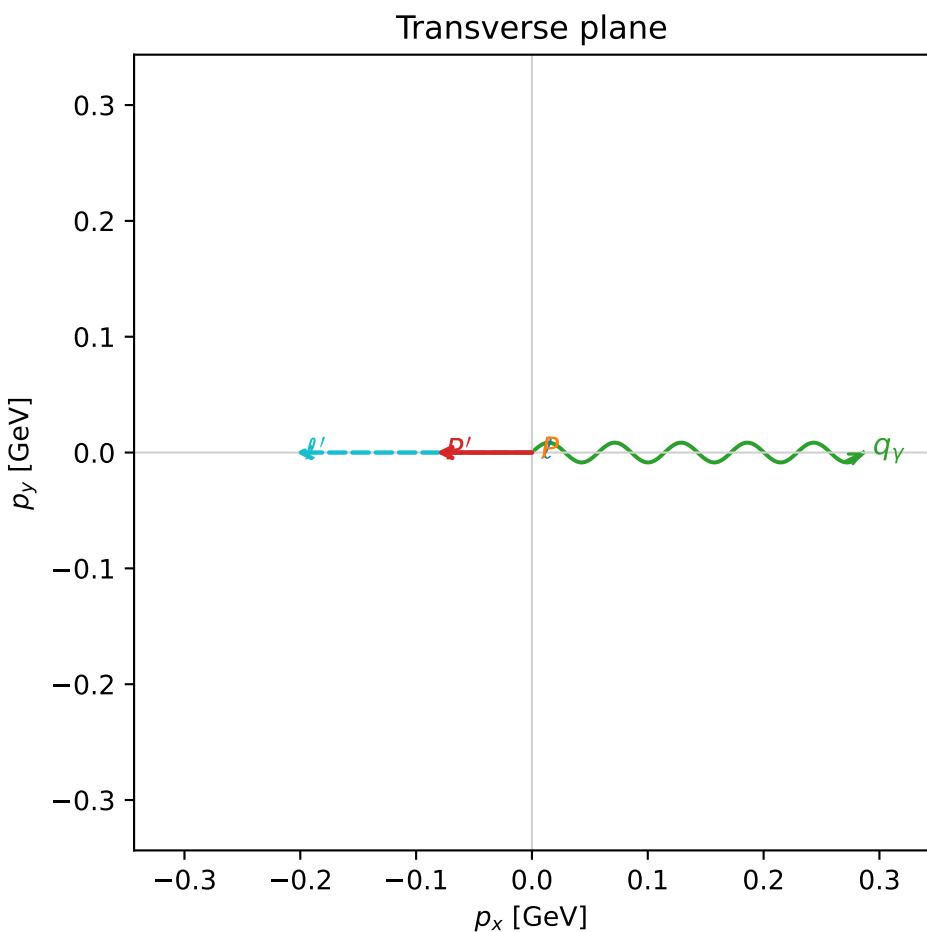
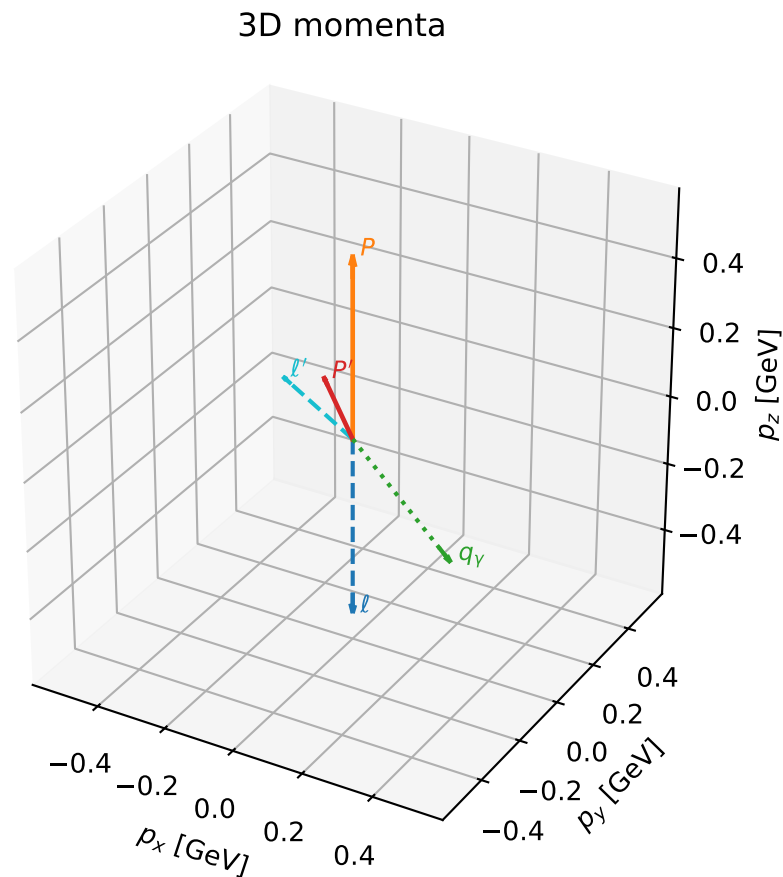
selected phase-space point:
 $\text{sqrt_s} = 1.716251$
 $\text{theta_p_out} = 0.3652981$
 $\text{theta_gamma_out} = 2.767903$
 $\text{q0ut} = 0.4570104$
 $\text{phi_p_out} = 3.141661$
 $\text{phi_gamma_out} = 3.790458\text{e-}05$
 $\text{alpha_e} = 0.3024594$
 $\text{alpha_p} = 1.817252$

coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

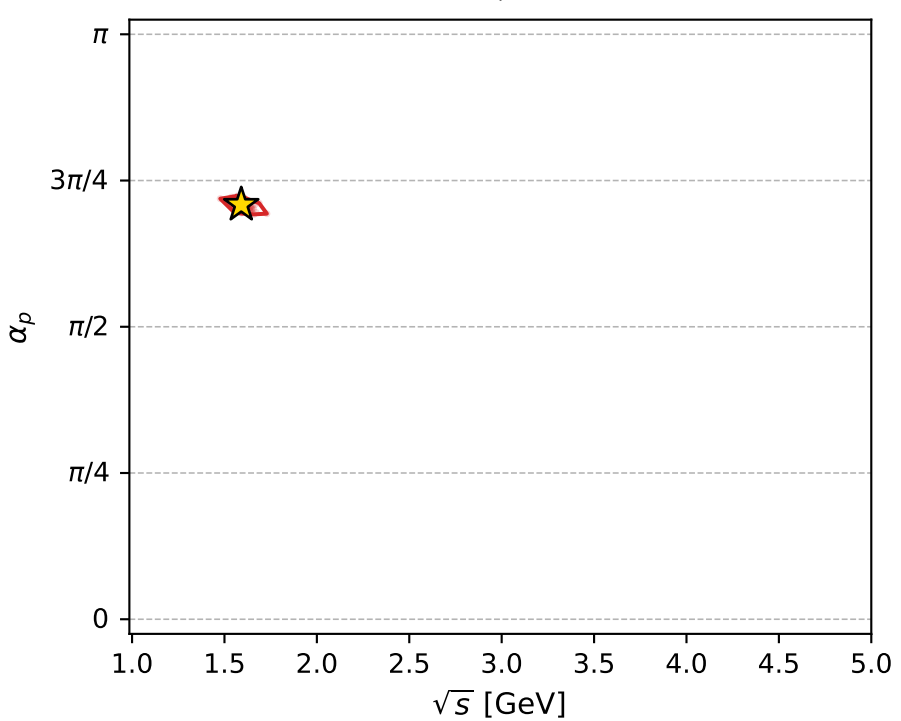
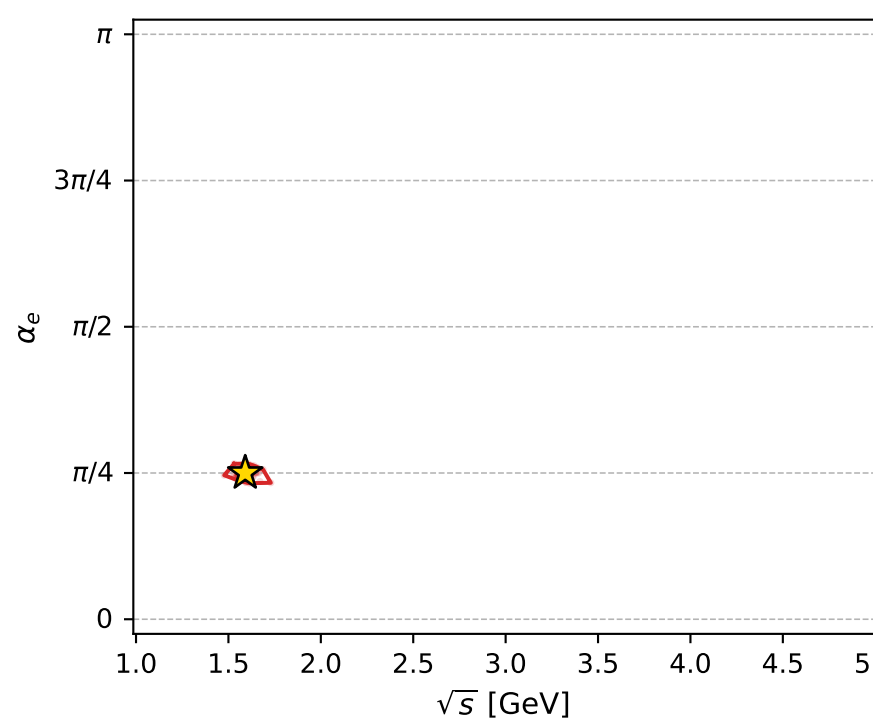
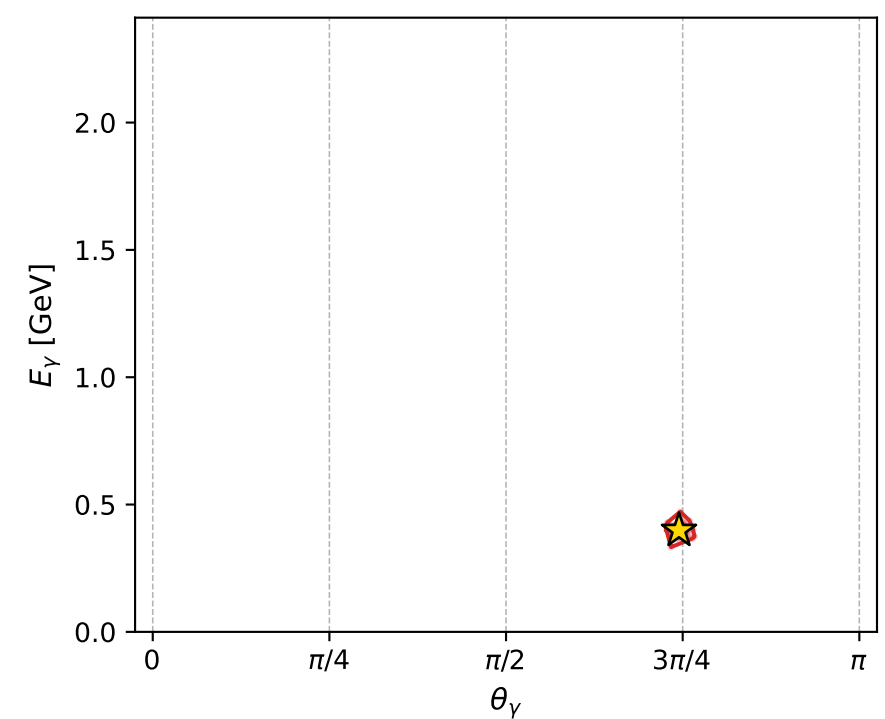
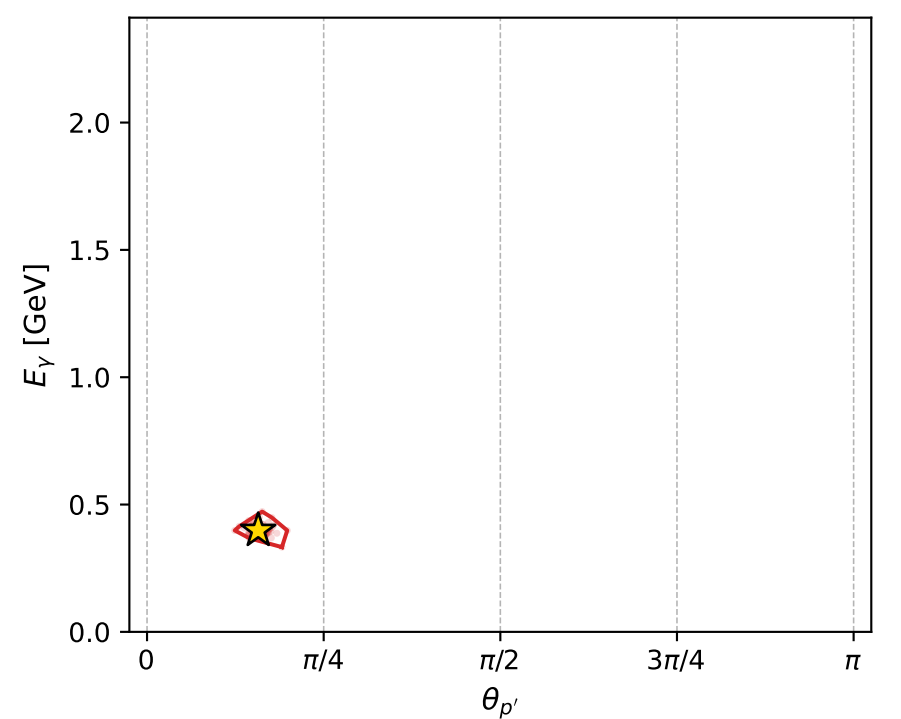
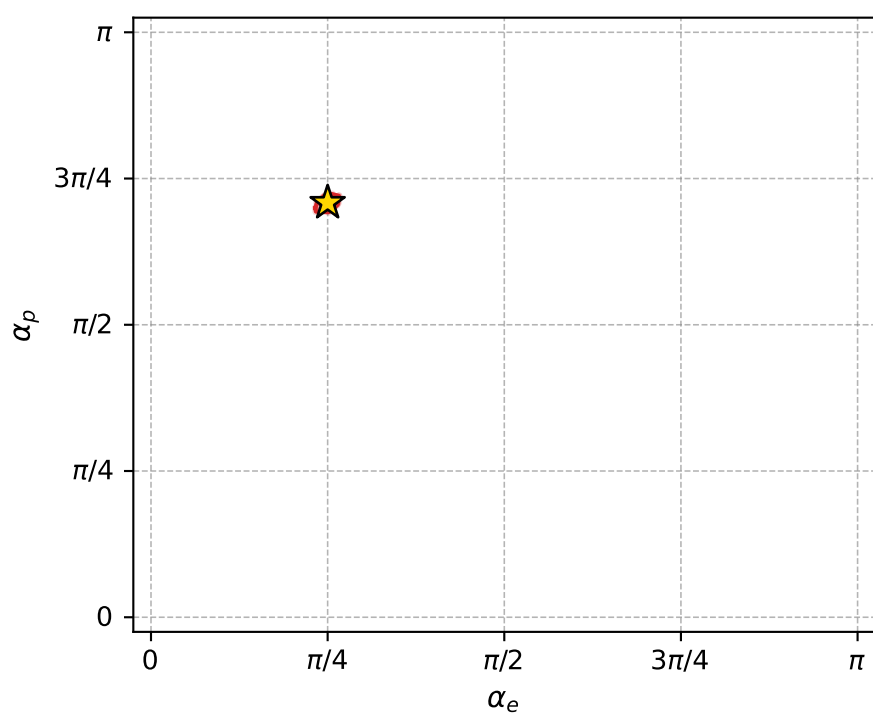
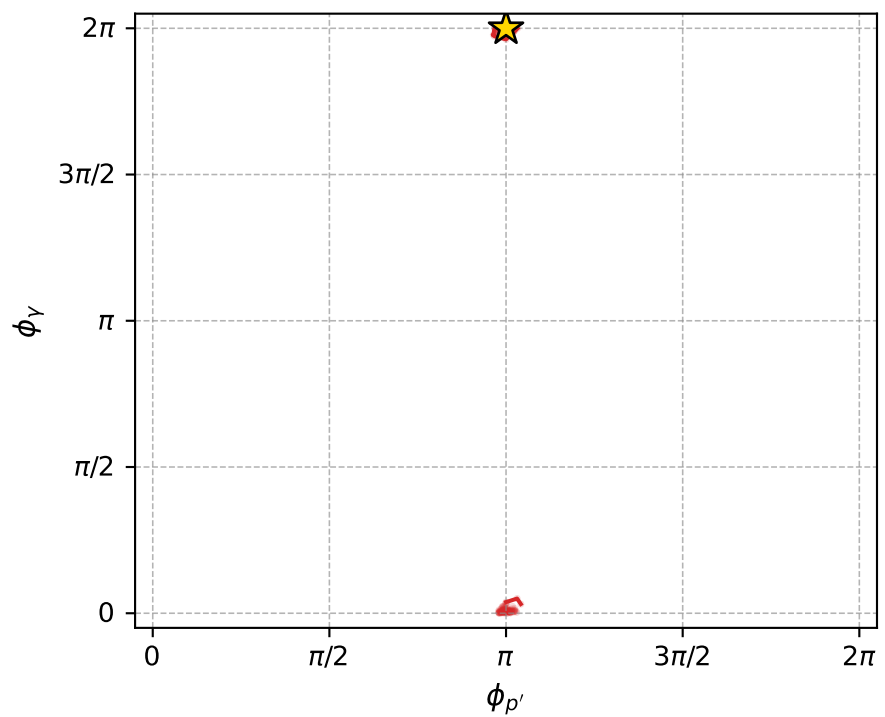
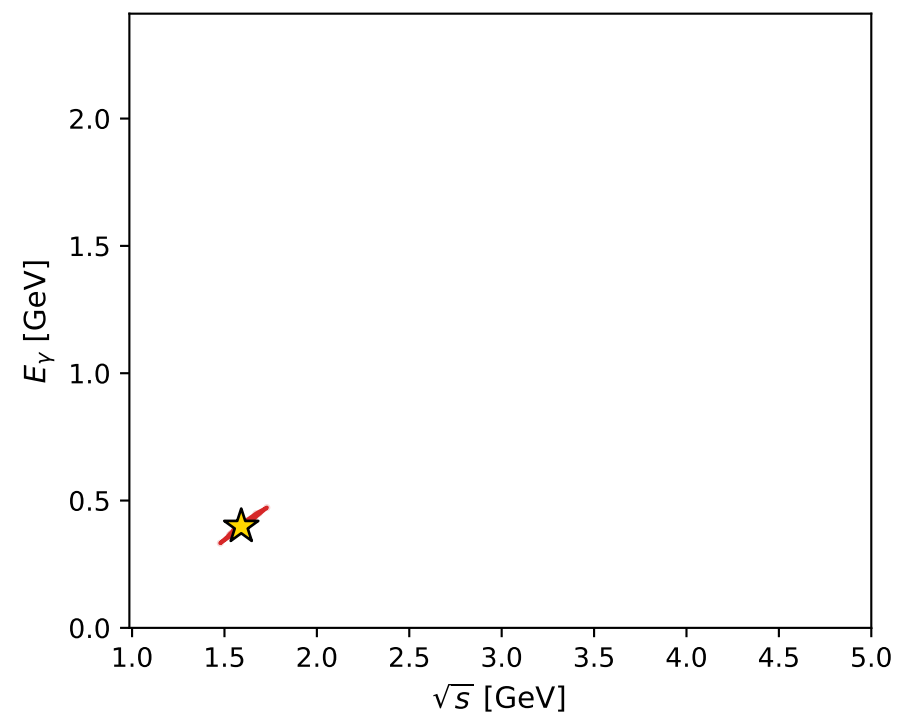
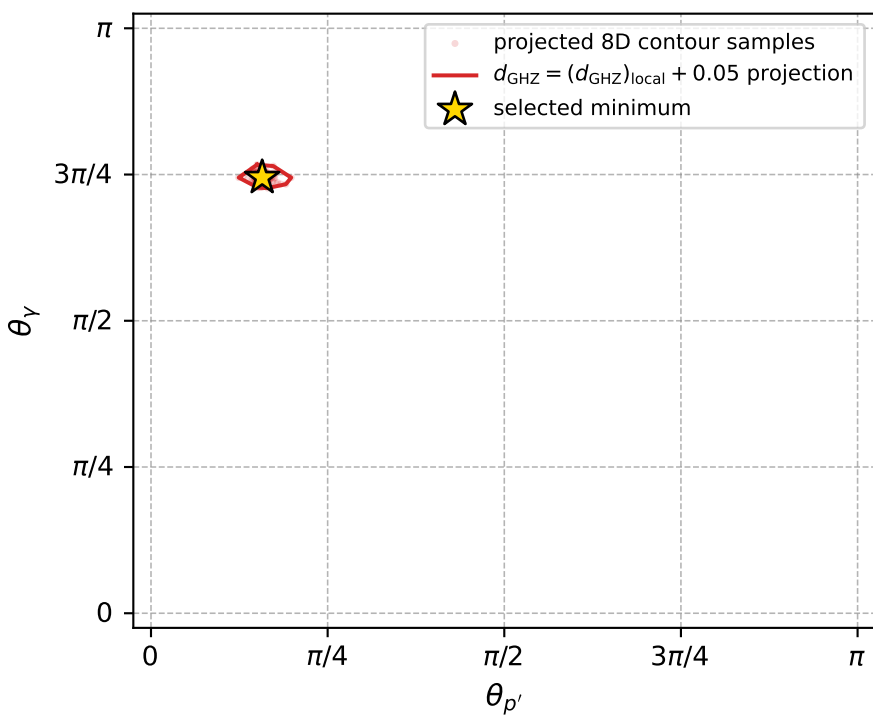
dghz_mixing_angles_polarization_02_local_minimum_292 $d_{\{\text{rm GHZ}\}}=0.000291151$
 region: polarization_cluster_02_local_minimum_292
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=0.78541057$ rad
 incoming lepton: $0.707098|+\rangle +0.707116|-\rangle$
 proton mixing angle: $\alpha_p=2.2259889$ rad
 incoming proton: $-0.609312|+\rangle +0.792931|-\rangle$

$s=2.53062$, $\sqrt{s}=1.59079$, $|\vec{P}|=0.518854$, $|\vec{P}'|=0.173893$
 $\theta_{P'}=0.494305$, $\theta_Y=2.33978$, $\phi_{P'}=3.14158$, $\phi_Y=6.28318$
 $E_Y=0.398332$, $\theta_{\ell'}=1.02436$, $\phi_{\ell'}=3.14158$
 $Q^2=0.376067$, $x_B=0.296558$, $t=-0.126684$, $W^2=1.77188$, $y=0.768187$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.5189$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.5189$
 P $E= 1.0719$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.5189$
 ℓ' $E= 0.2385$ $p_x= -0.2038$ $p_y= 0.0000$ $p_z= 0.1239$
 P' $E= 0.9540$ $p_x= -0.0825$ $p_y= 0.0000$ $p_z= 0.1531$
 q_Y $E= 0.3983$ $p_x= 0.2862$ $p_y= -0.0000$ $p_z= -0.2770$



electron: polarization cluster P2, local minimum 292; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 0.00029115057$
 $d_{\text{GHZ}} \text{ contour} = 0.050291151$
 above global minimum = 0.00029112829
 8D contour samples = 256

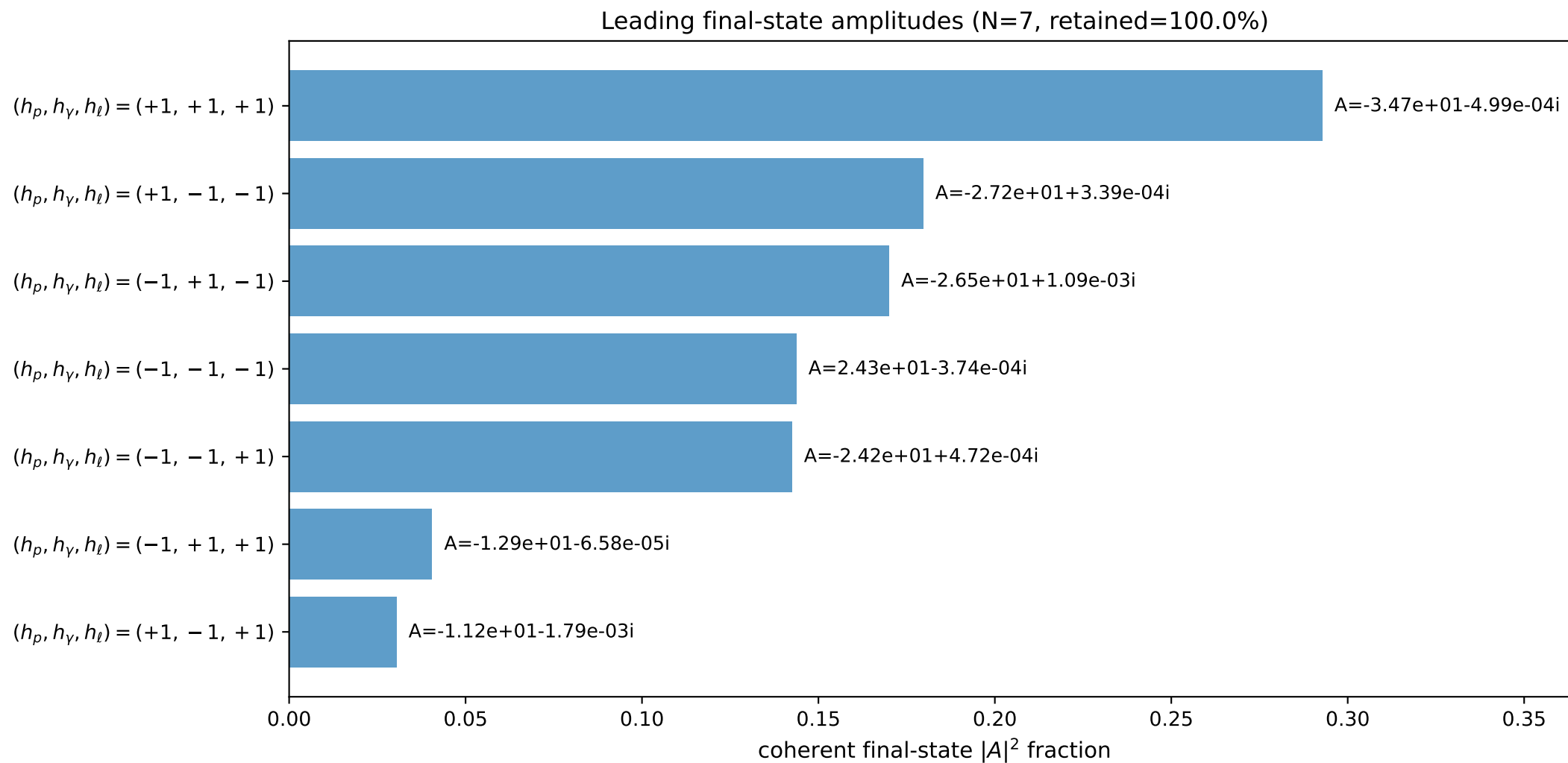
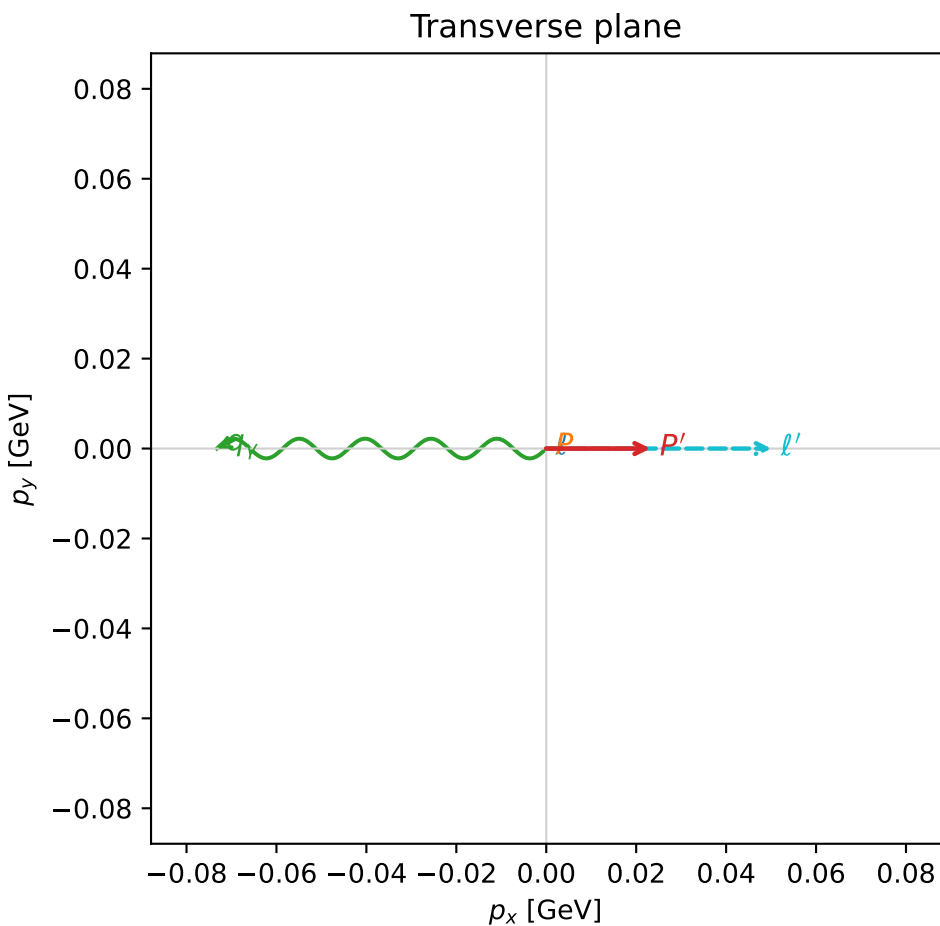
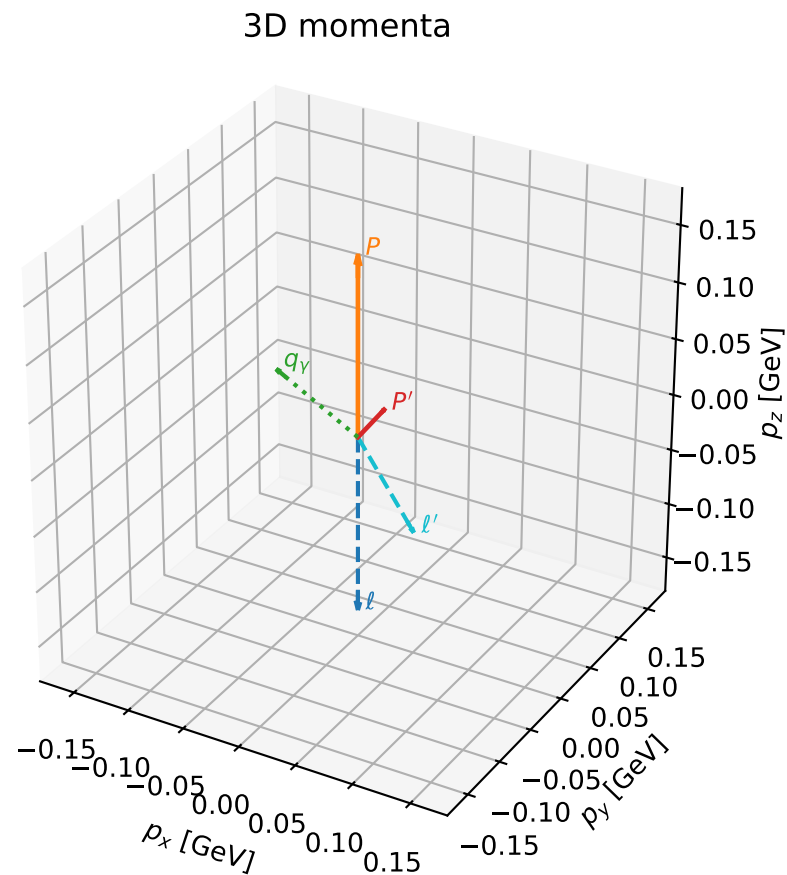
selected phase-space point:
 $\text{sqrt_s} = 1.590793$
 $\text{theta_p_out} = 0.4943051$
 $\text{theta_gamma_out} = 2.339778$
 $\text{q0out} = 0.3983323$
 $\text{phi_p_out} = 3.141582$
 $\text{phi_gamma_out} = 6.283176$
 $\text{alpha_e} = 0.7854106$
 $\text{alpha_p} = 2.225989$

coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

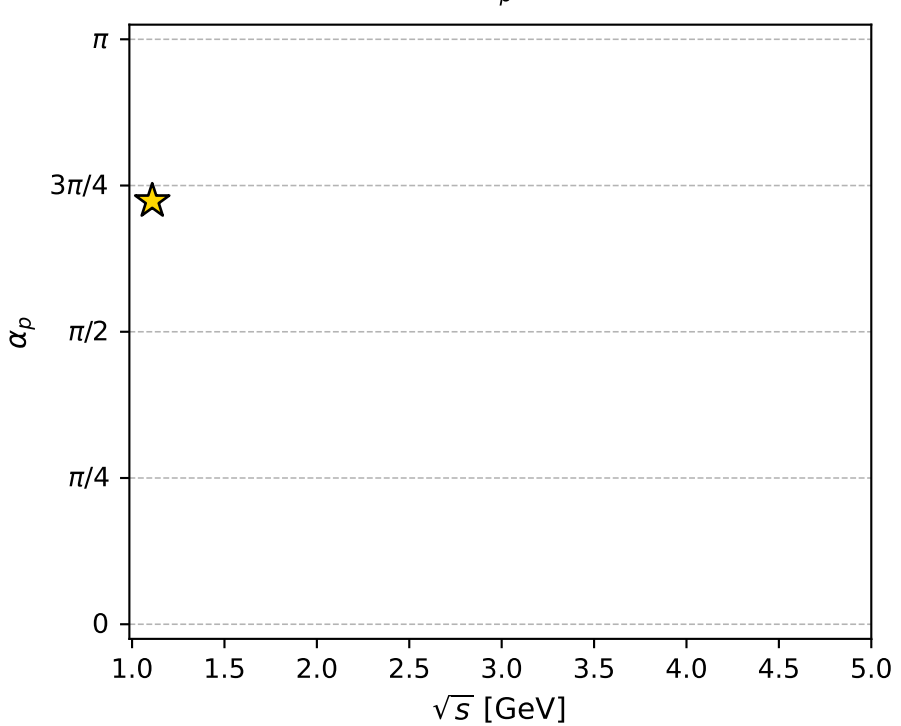
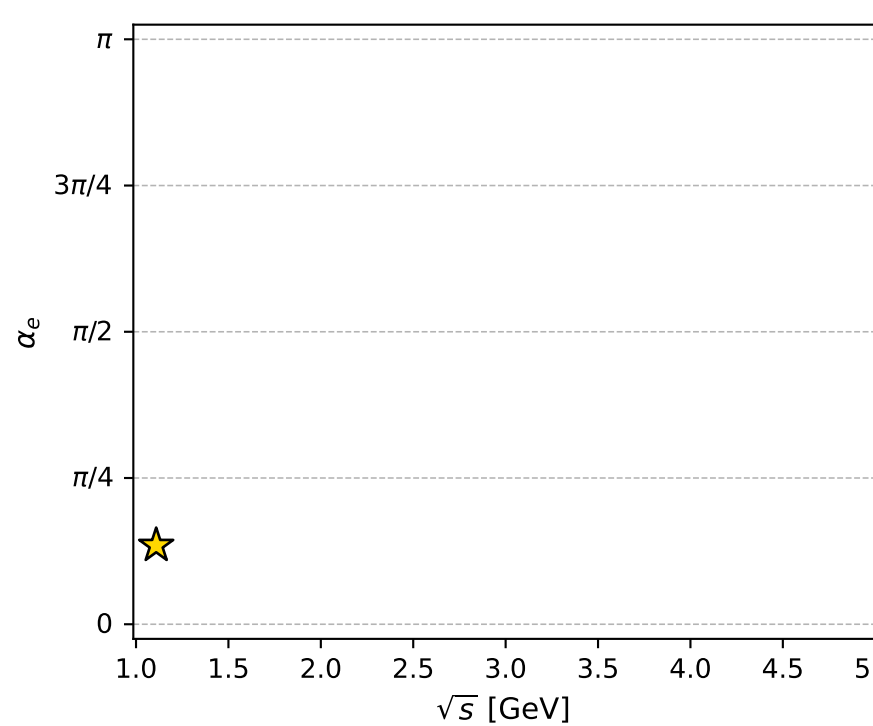
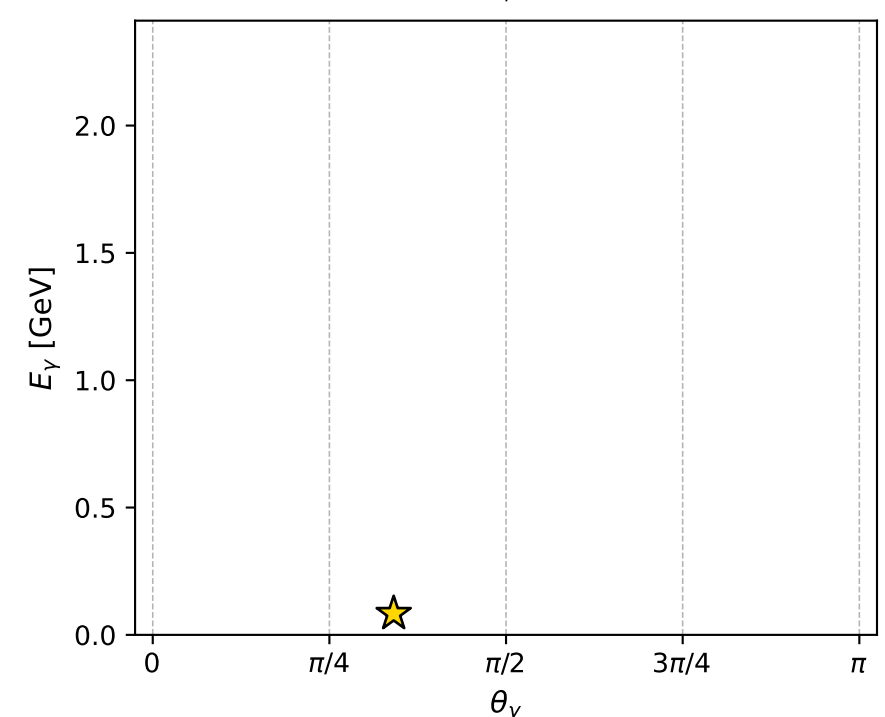
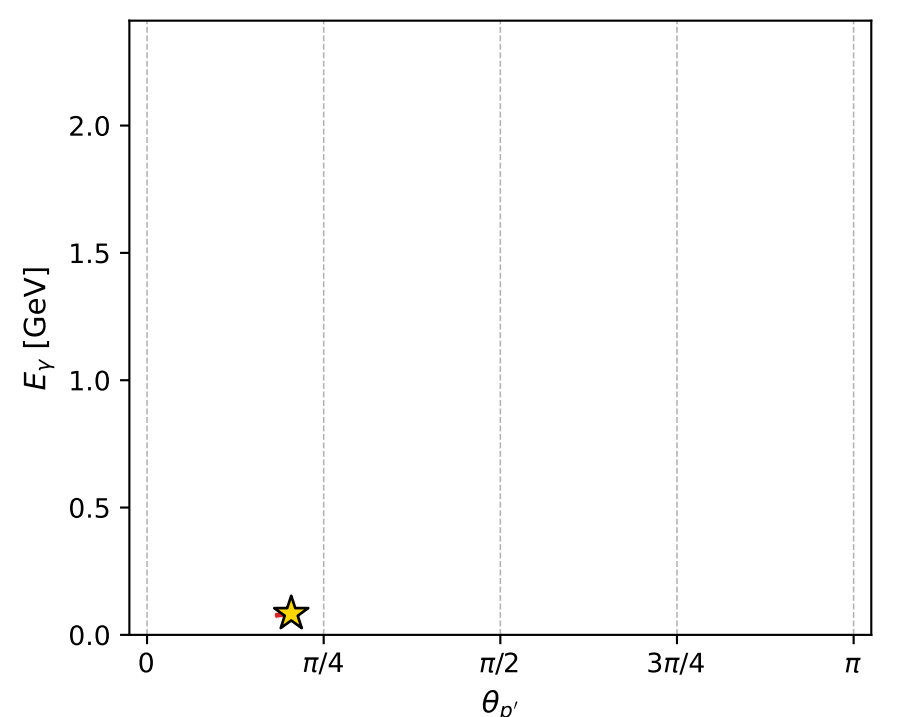
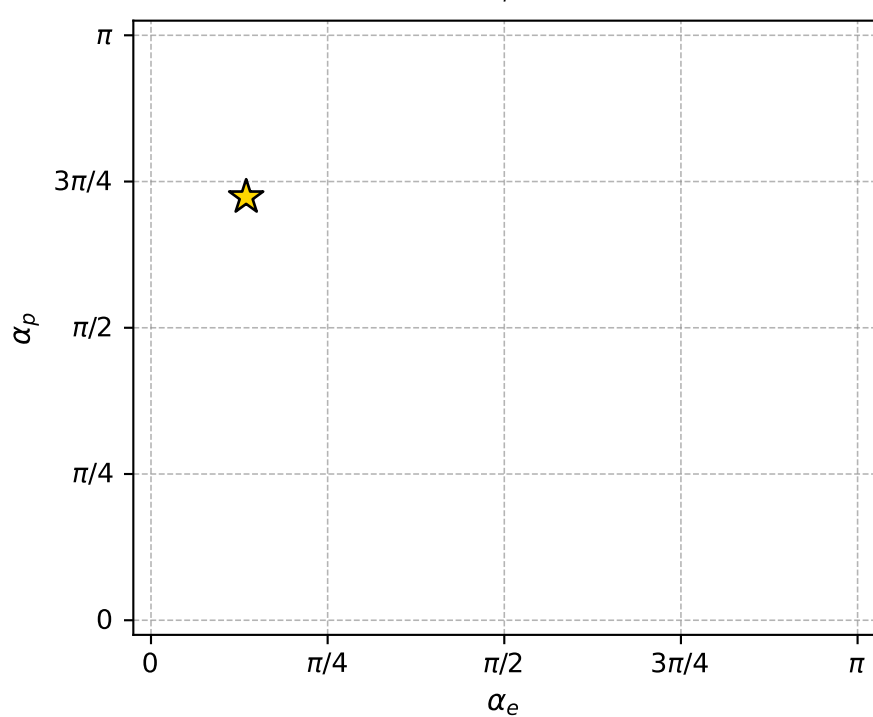
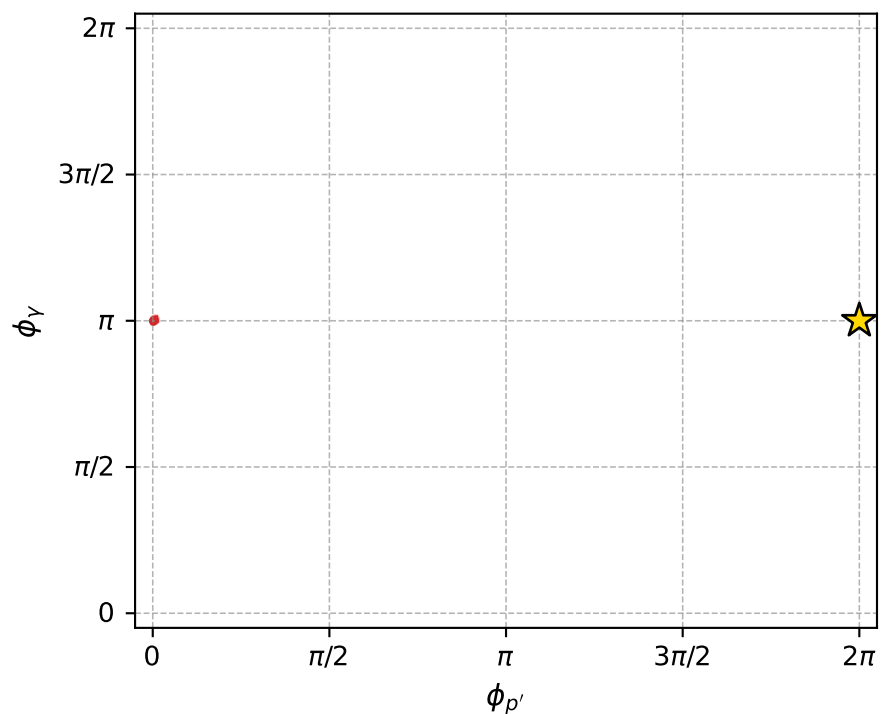
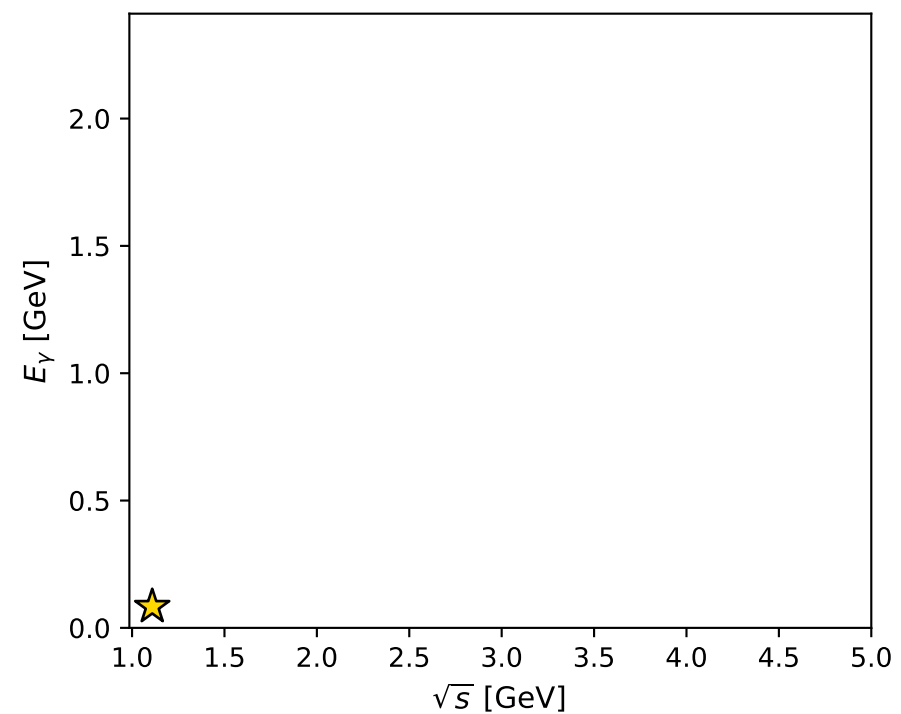
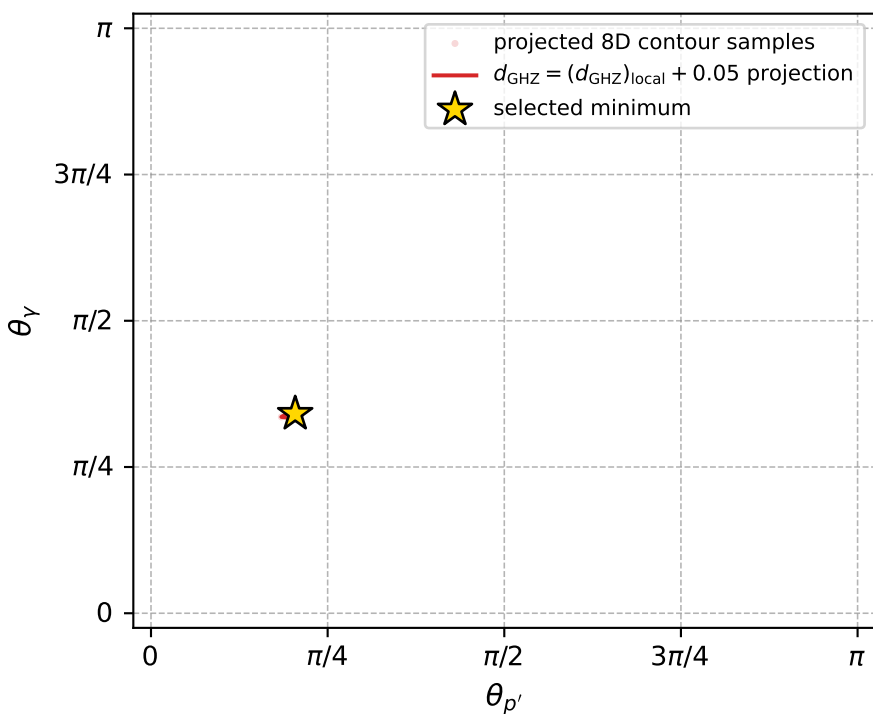
dghz_mixing_angles_polarization_02_local_minimum_294 $d_{\{\text{rm GHZ}\}}=0.000296925$
 region: polarization_cluster_02_local_minimum_294
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=0.42318044$ rad
 incoming lepton: $0.911787|+\rangle +0.410662|-\rangle$
 proton mixing angle: $\alpha_p=2.2727733$ rad
 incoming proton: $-0.645728|+\rangle +0.763567|-\rangle$

$s=1.23018$, $\sqrt{s}=1.10913$, $|\vec{P}|=0.157931$, $|\vec{P}'|=0.0387458$
 $\theta_{P'}=0.641094$, $\theta_V=1.07124$, $\phi_{P'}=6.28318$, $\phi_V=3.14156$
 $E_V=0.0834371$, $\theta_{\ell'}=2.52754$, $\phi_{\ell'}=6.28314$
 $Q^2=0.00501413$, $x_B=0.0308395$, $t=-0.0164813$, $W^2=1.03742$, $y=0.464092$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.1579$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1579$
 P $E= 0.9512$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1579$
 ℓ' $E= 0.0869$ $p_x= 0.0501$ $p_y= -0.0000$ $p_z= -0.0710$
 P' $E= 0.9388$ $p_x= 0.0232$ $p_y= -0.0000$ $p_z= 0.0311$
 q_V $E= 0.0834$ $p_x= -0.0732$ $p_y= 0.0000$ $p_z= 0.0400$



electron: polarization cluster P2, local minimum 294; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 0.00029692515$
 $d_{\text{GHZ}} \text{ contour} = 0.050296925$
 above global minimum = 0.00029690286
 8D contour samples = 256

selected phase-space point:

$\text{sqrt_s} = 1.109134$
 $\text{theta_p_out} = 0.6410941$
 $\text{theta_gamma_out} = 1.071242$
 $\text{q0out} = 0.08343712$
 $\text{phi_p_out} = 6.283182$
 $\text{phi_gamma_out} = 3.14156$
 $\text{alpha_e} = 0.4231804$
 $\text{alpha_p} = 2.272773$

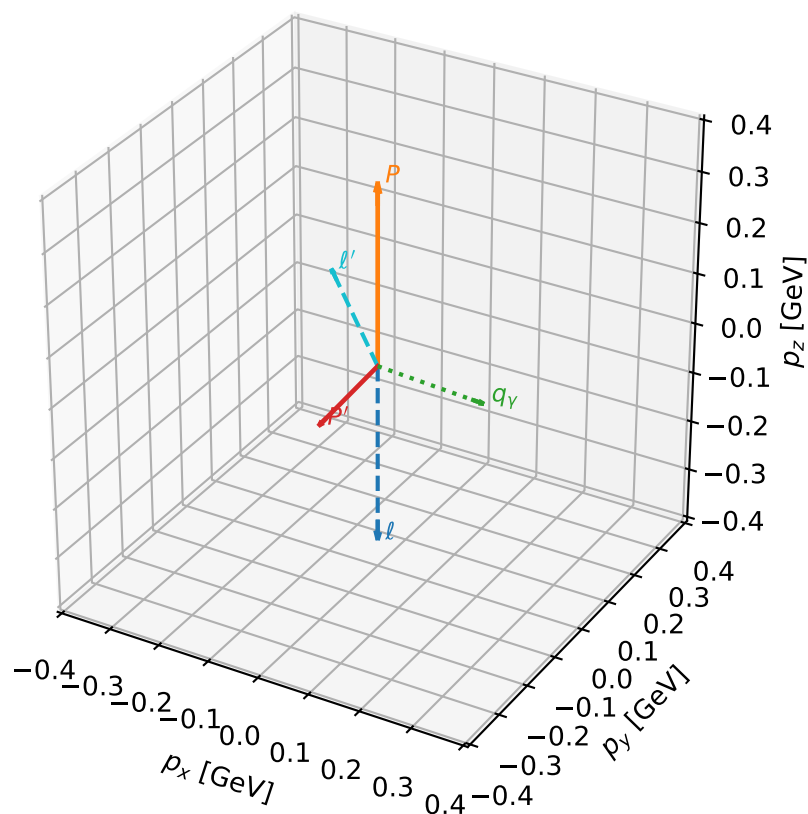
coherent mixing angles: minimum d_{GHz} configuration (gradient_local_minimum)

dghz_mixing_angles_polarization_02_local_minimum_295 $d_{\{\text{rm GHz}\}}=0.000302413$
 region: polarization_cluster_02_local_minimum_295
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=0.32449539$ rad
 incoming lepton: $0.947812|+\rangle +0.318831|-\rangle$
 proton mixing angle: $\alpha_p=1.8351053$ rad
 incoming proton: $-0.261242|+\rangle +0.965273|-\rangle$

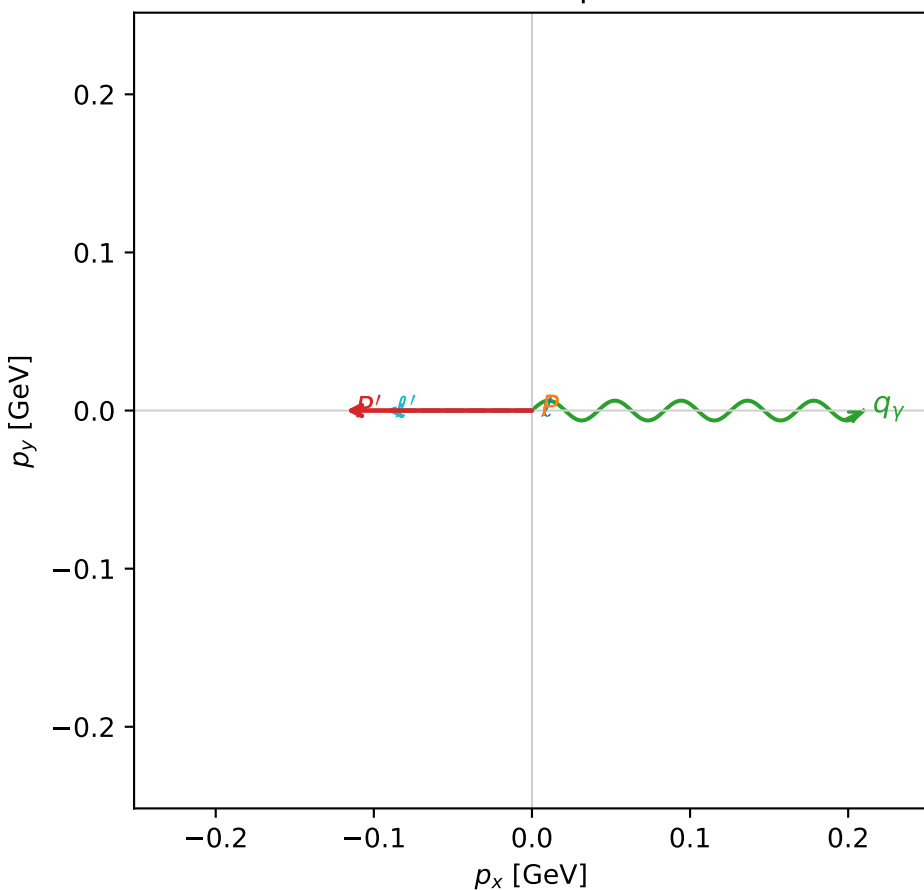
$s=1.84508$, $\sqrt{s}=1.35834$, $|\vec{P}|=0.3553$, $|\vec{P}'|=0.192777$
 $\theta_{P'}=2.48564$, $\theta_Y=1.6377$, $\phi_{P'}=3.14148$, $\phi_Y=6.28311$
 $E_Y=0.210167$, $\theta_{\ell'}=0.504539$, $\phi_{\ell'}=3.14157$
 $Q^2=0.253957$, $x_B=0.362025$, $t=-0.269895$, $W^2=1.32738$, $y=0.726755$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.3553$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.3553$
 P $E= 1.0030$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.3553$
 ℓ' $E= 0.1906$ $p_x= -0.0921$ $p_y= 0.0000$ $p_z= 0.1668$
 P' $E= 0.9576$ $p_x= -0.1176$ $p_y= 0.0000$ $p_z= -0.1528$
 q_Y $E= 0.2102$ $p_x= 0.2097$ $p_y= -0.0000$ $p_z= -0.0140$

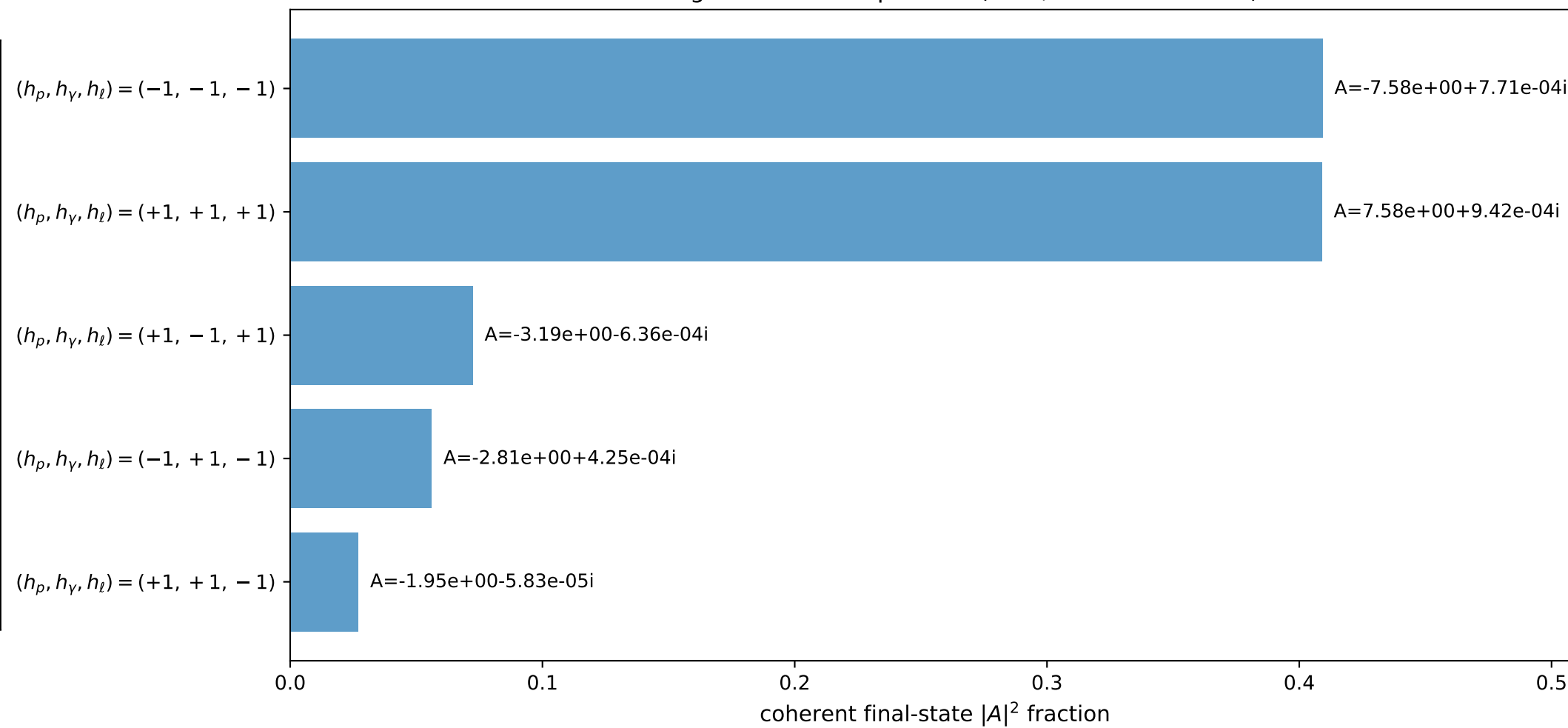
3D momenta



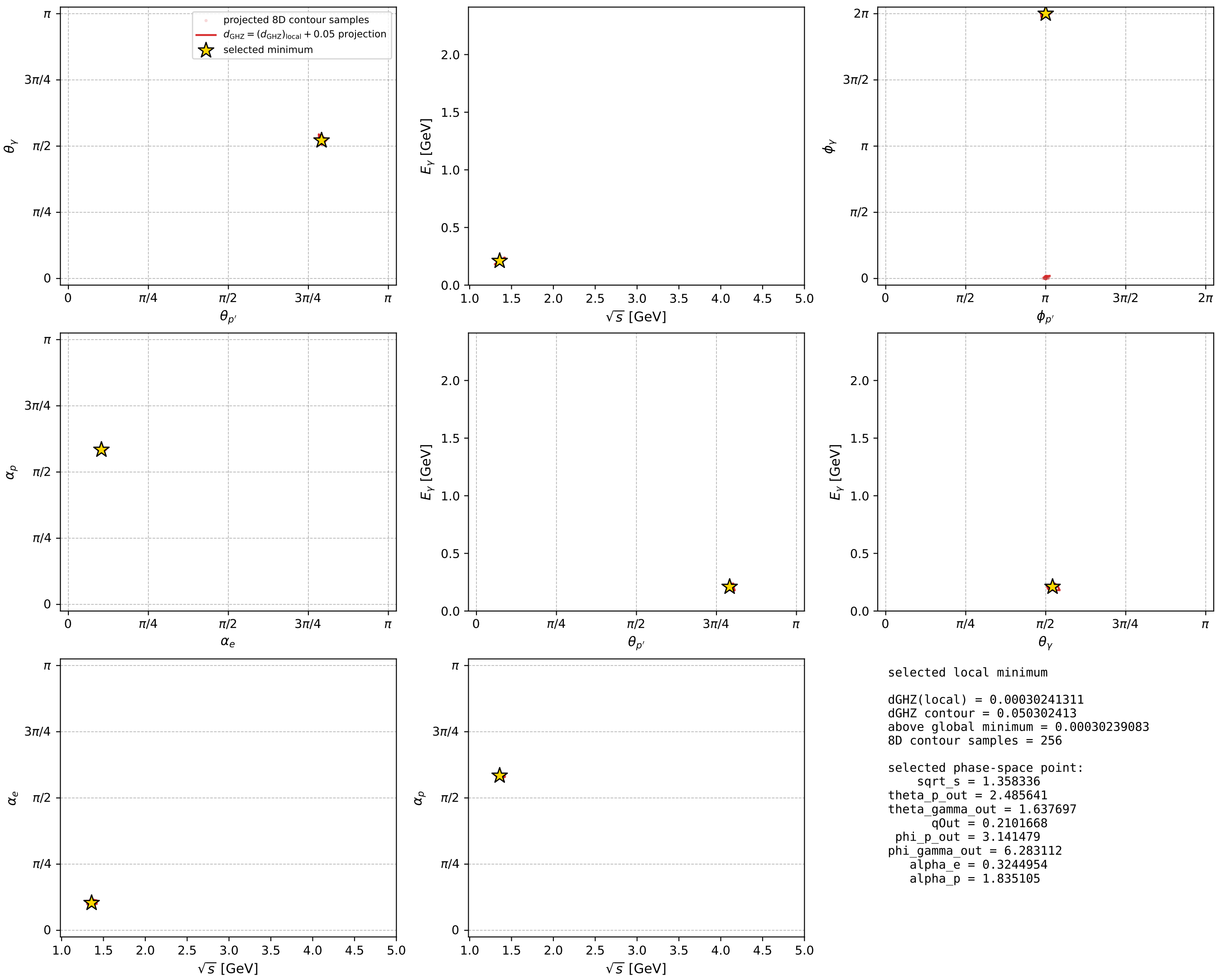
Transverse plane



Leading final-state amplitudes (N=5, retained=97.3%)



electron: polarization cluster P2, local minimum 295; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



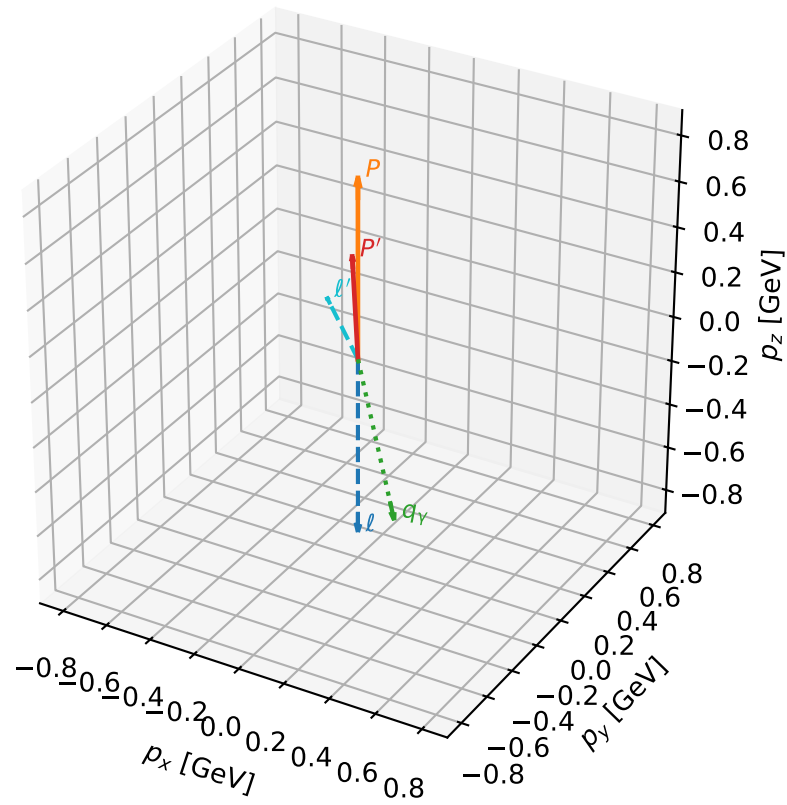
coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

dghz_mixing_angles_polarization_02_local_minimum_296 $d_{\{\text{rm GHZ}\}}=0.000302995$
 region: polarization_cluster_02_local_minimum_296
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=0.55513734$ rad
 incoming lepton: $0.849828|+\rangle +0.52706|-\rangle$
 proton mixing angle: $\alpha_p=2.048015$ rad
 incoming proton: $-0.45931|+\rangle +0.888276|-\rangle$

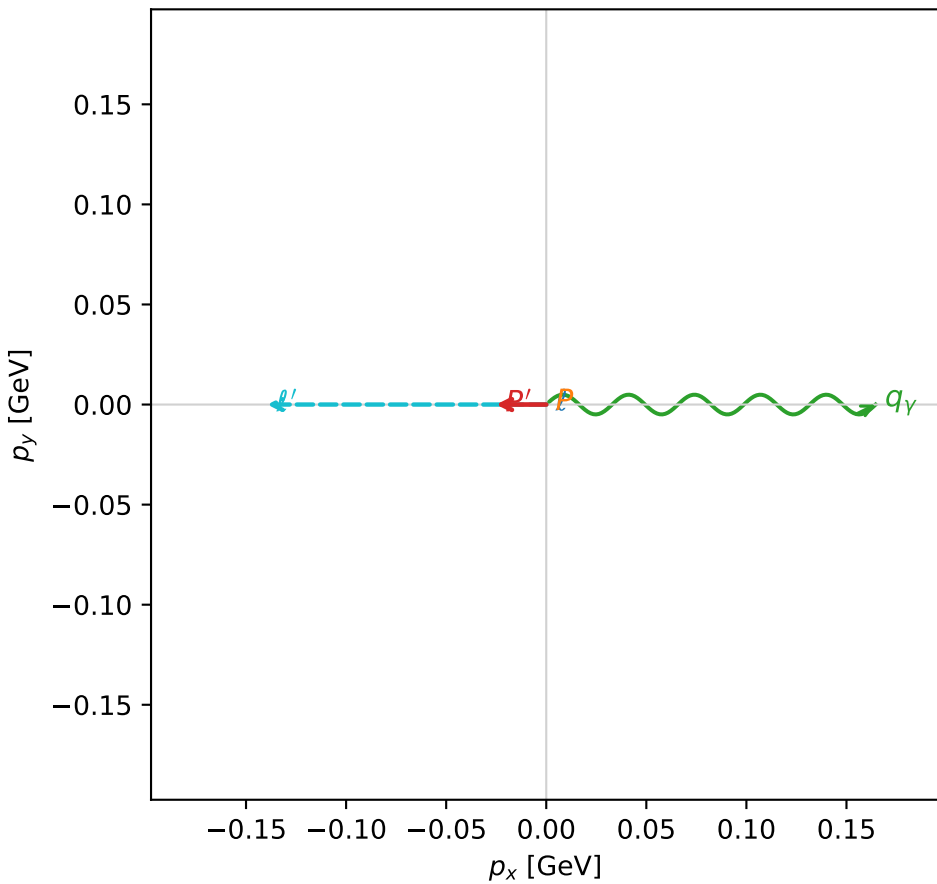
$s=4.06514$, $\sqrt{s}=2.01622$, $|\vec{P}|=0.789919$, $|\vec{P}'|=0.446256$
 $\theta_{P'}=0.0562449$, $\theta_Y=2.9051$, $\phi_{P'}=3.14149$, $\phi_Y=6.28313$
 $E_Y=0.702308$, $\theta_{\ell'}=0.531494$, $\phi_{\ell'}=3.14154$
 $Q^2=0.809472$, $x_B=0.280563$, $t=-0.0840407$, $W^2=2.95554$, $y=0.905777$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.7899$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.7899$
 P $E= 1.2263$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.7899$
 ℓ' $E= 0.2752$ $p_x= -0.1395$ $p_y= 0.0000$ $p_z= 0.2372$
 P' $E= 1.0387$ $p_x= -0.0251$ $p_y= 0.0000$ $p_z= 0.4456$
 q_Y $E= 0.7023$ $p_x= 0.1645$ $p_y= -0.0000$ $p_z= -0.6828$

3D momenta



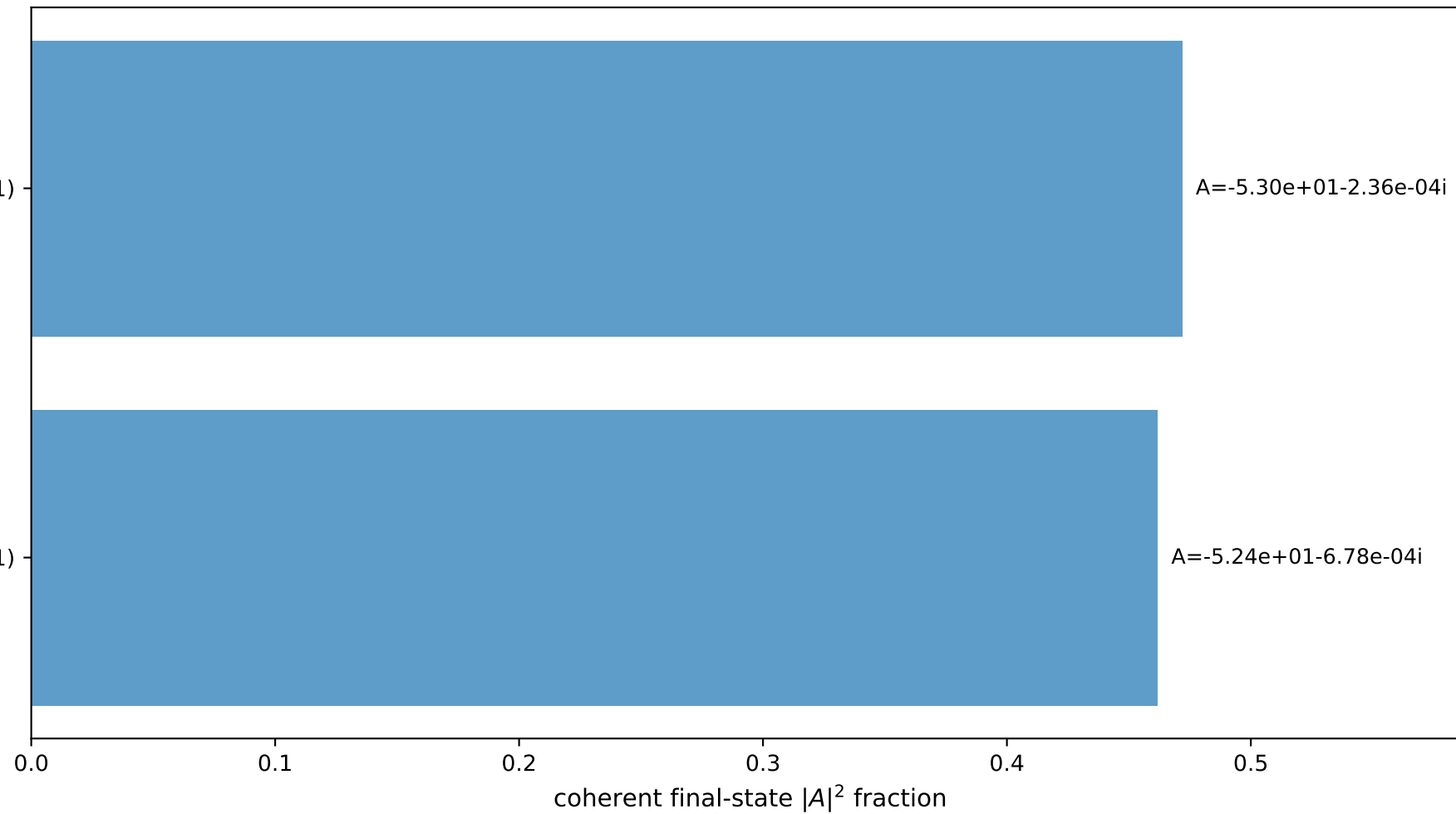
Transverse plane



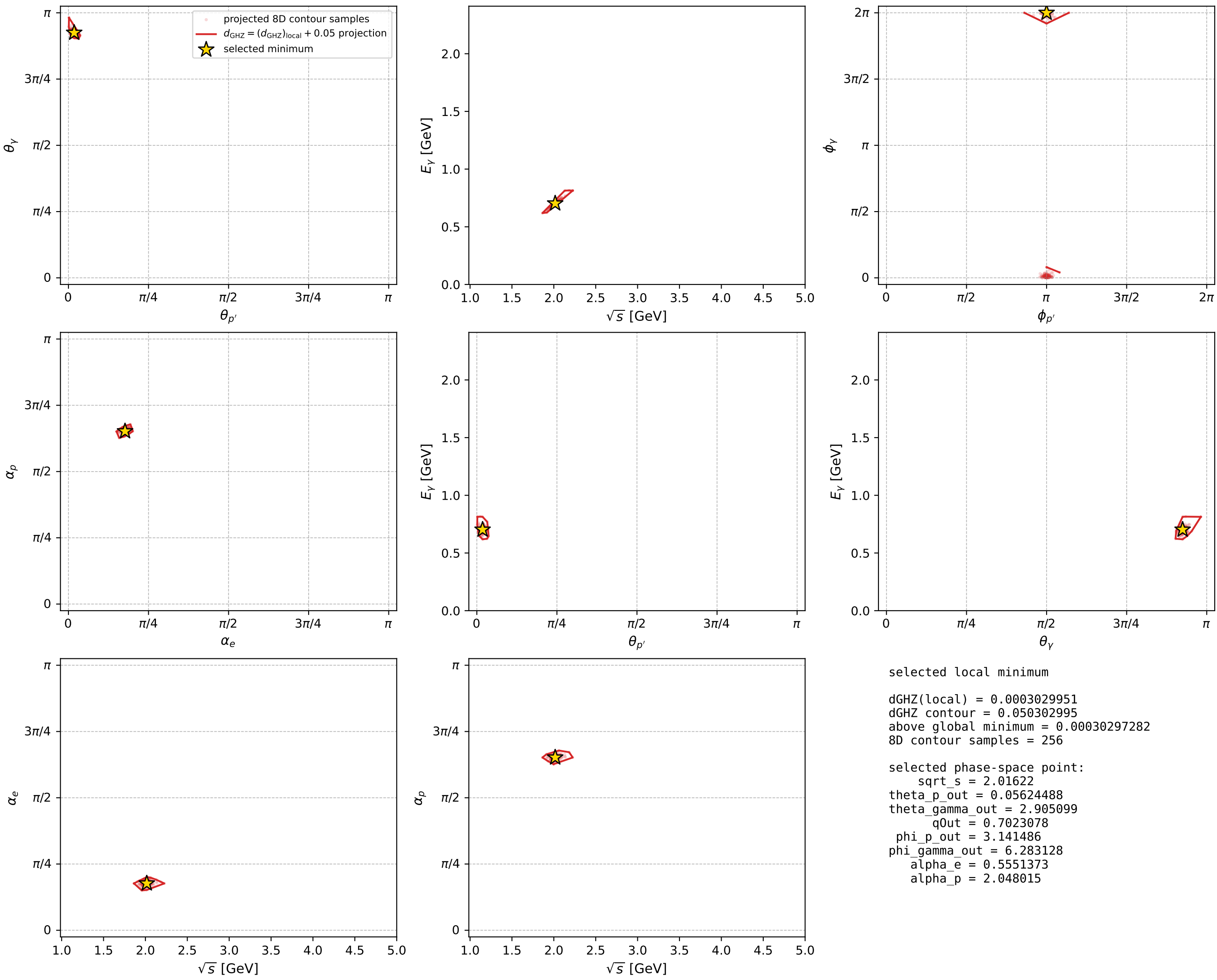
$(h_p, h_Y, h_\ell) = (+1, -1, -1)$

$(h_p, h_Y, h_\ell) = (-1, +1, +1)$

Leading final-state amplitudes (N=2, retained=93.4%)



electron: polarization cluster P2, local minimum 296; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



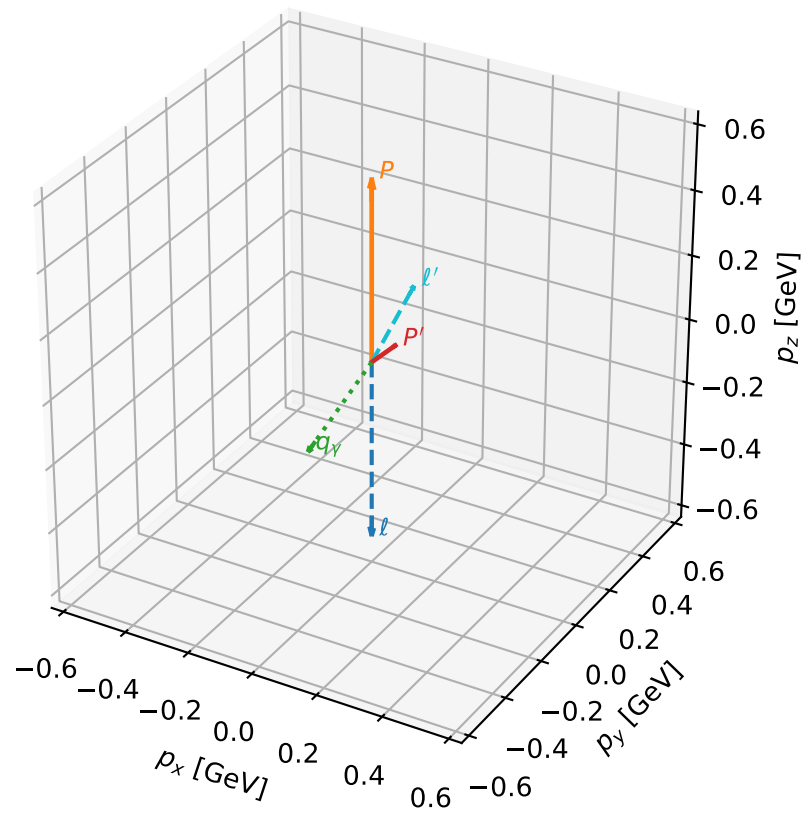
coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

dghz_mixing_angles_polarization_02_local_minimum_303 $d_{\{\text{rm GHZ}\}}=0.000329821$
 region: polarization_cluster_02_local_minimum_303
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=1.2321147$ rad
 incoming lepton: $0.332244|+\rangle +0.943194|-\rangle$
 proton mixing angle: $\alpha_p=2.8672676$ rad
 incoming proton: $-0.962608|+\rangle +0.270897|-\rangle$

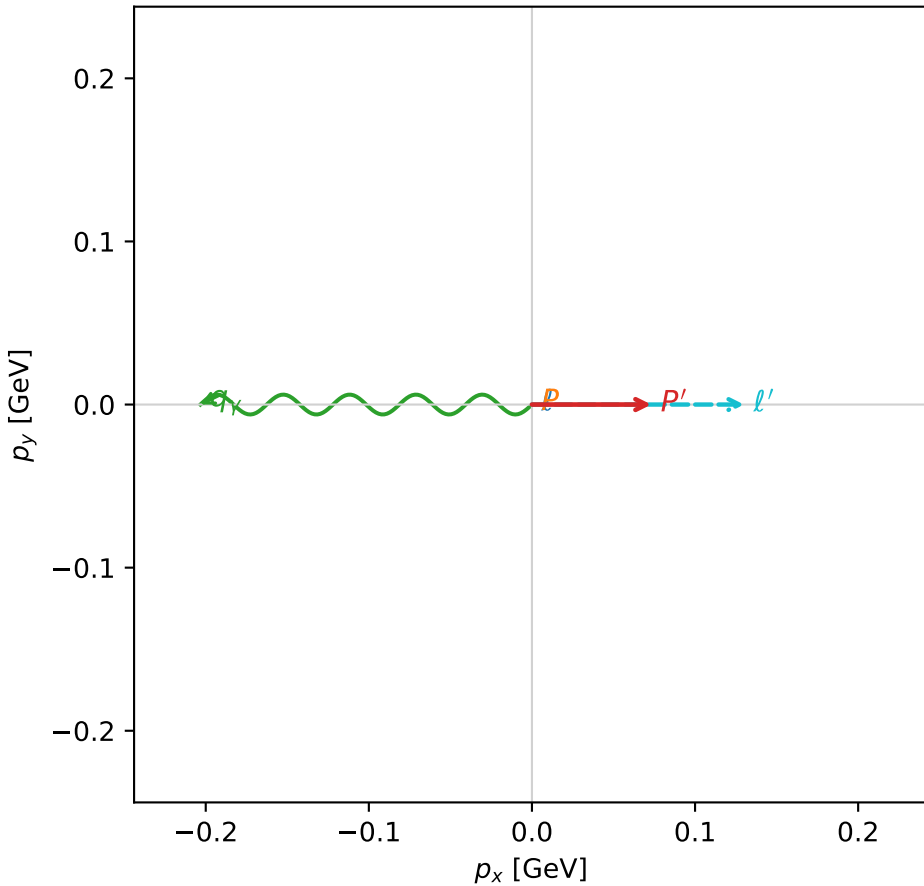
$s=2.70642$, $\sqrt{s}=1.64512$, $|\vec{P}|=0.555149$, $|\vec{P}'|=0.103183$
 $\theta_{P'}=0.788531$, $\theta_Y=2.60862$, $\phi_{P'}=7.73895\text{e-}06$, $\phi_Y=3.1416$
 $E_Y=0.400081$, $\theta_{\ell'}=0.446312$, $\phi_{\ell'}=6.64397\text{e-}06$
 $Q^2=0.636465$, $x_B=0.43255$, $t=-0.216675$, $W^2=1.7148$, $y=0.805566$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.5551$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.5551$
 P $E= 1.0900$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.5551$
 ℓ' $E= 0.3014$ $p_x= 0.1301$ $p_y= 0.0000$ $p_z= 0.2719$
 P' $E= 0.9437$ $p_x= 0.0732$ $p_y= 0.0000$ $p_z= 0.0727$
 q_Y $E= 0.4001$ $p_x= -0.2033$ $p_y= -0.0000$ $p_z= -0.3446$

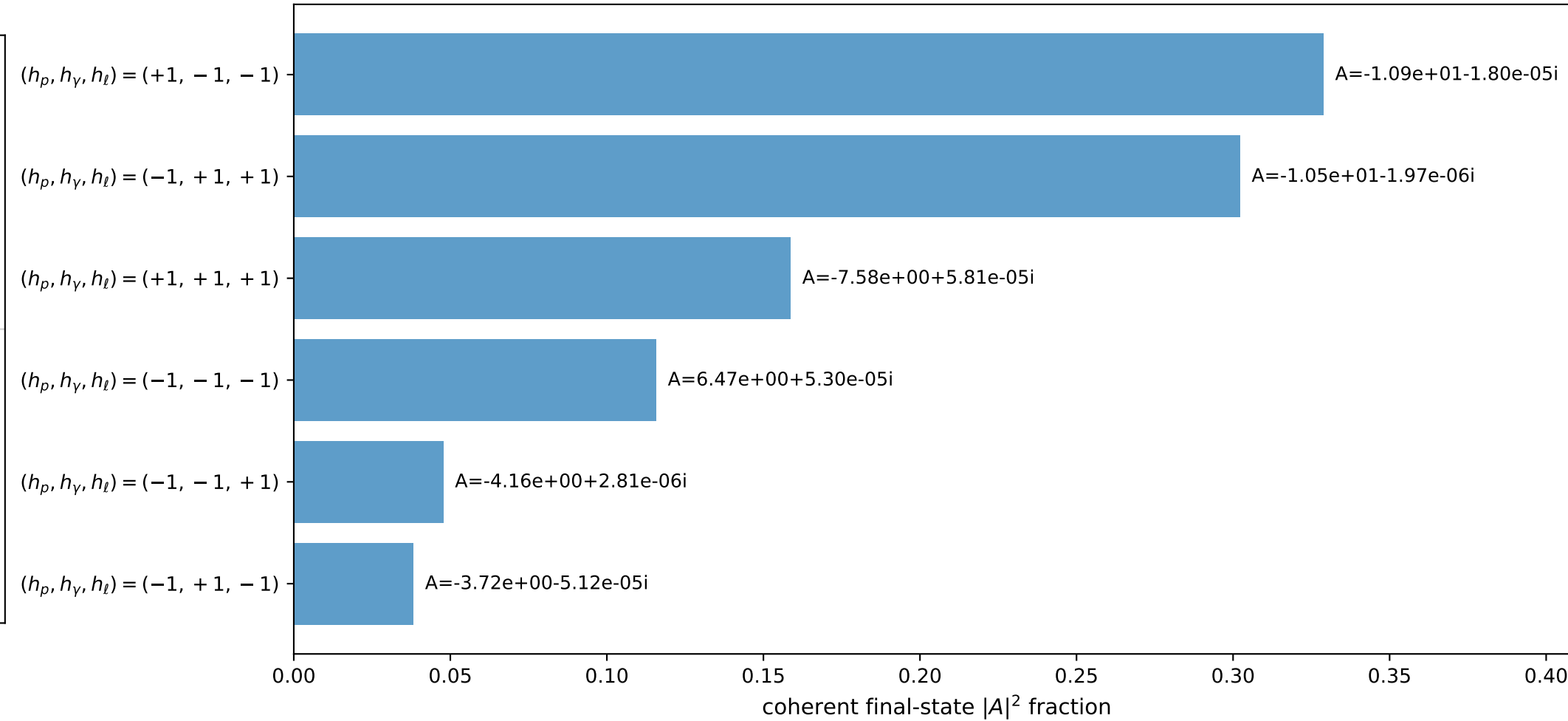
3D momenta



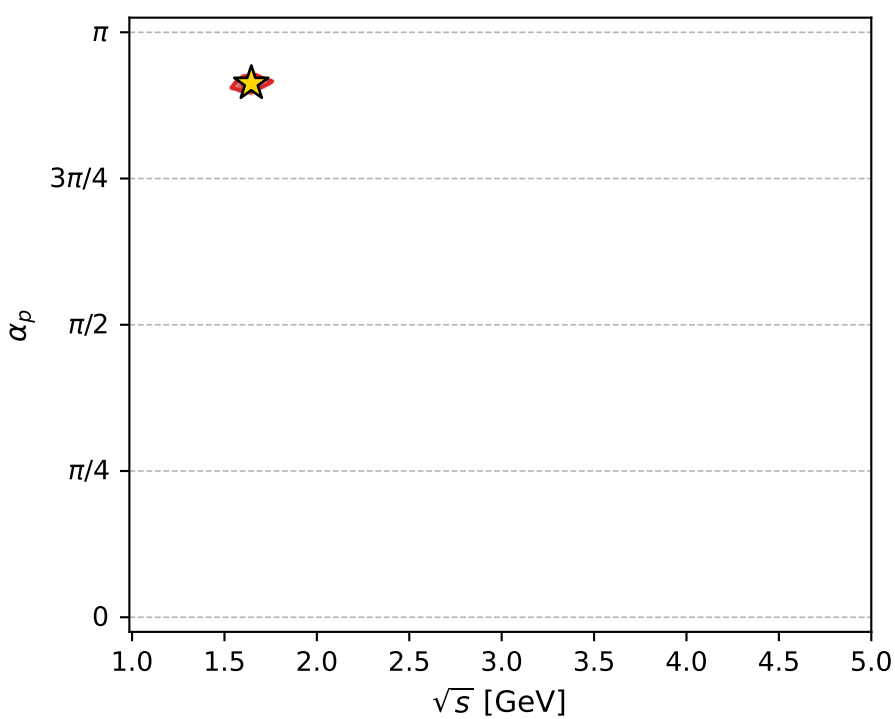
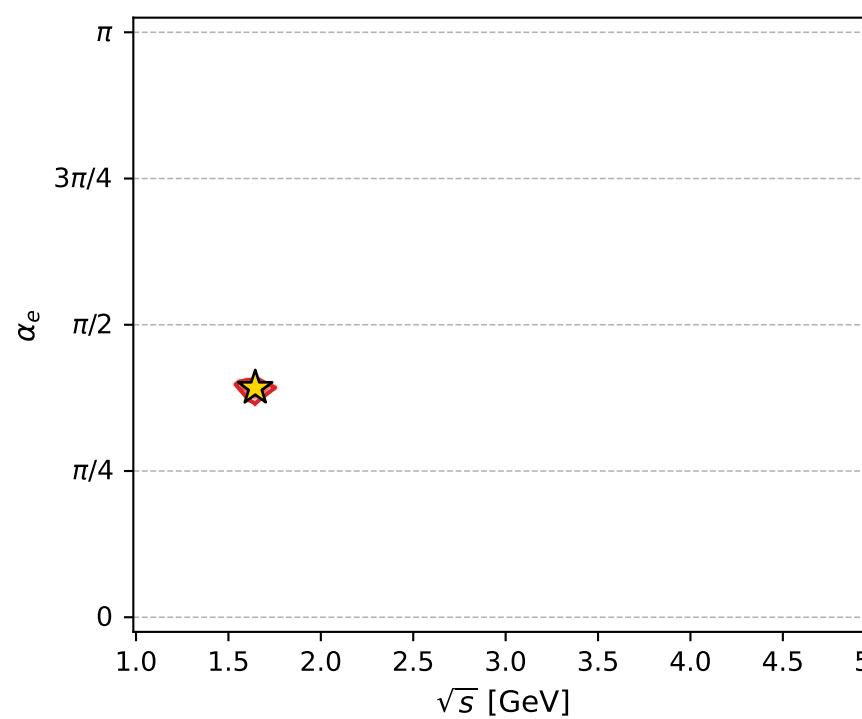
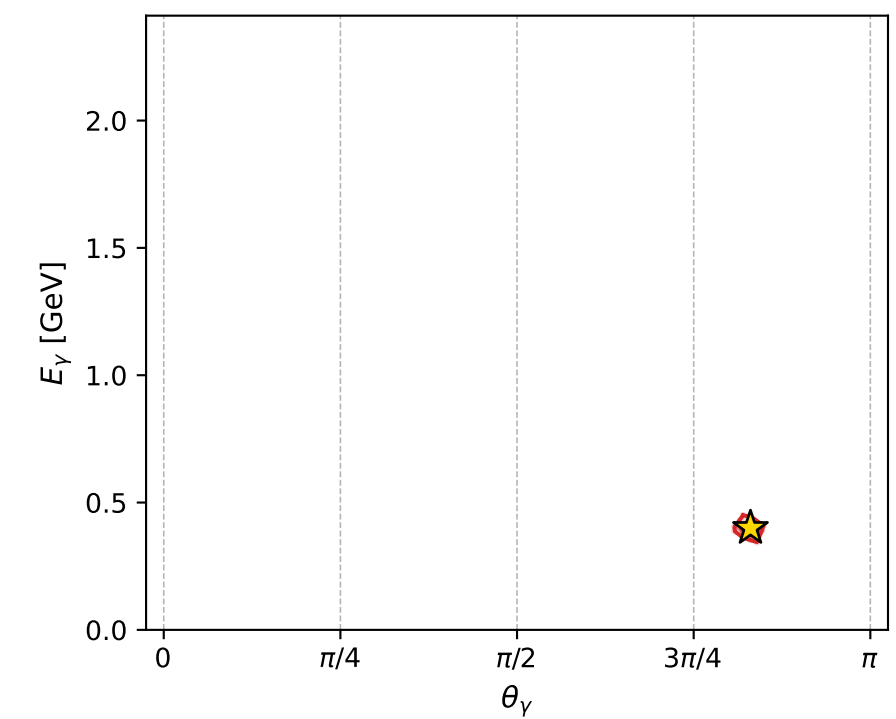
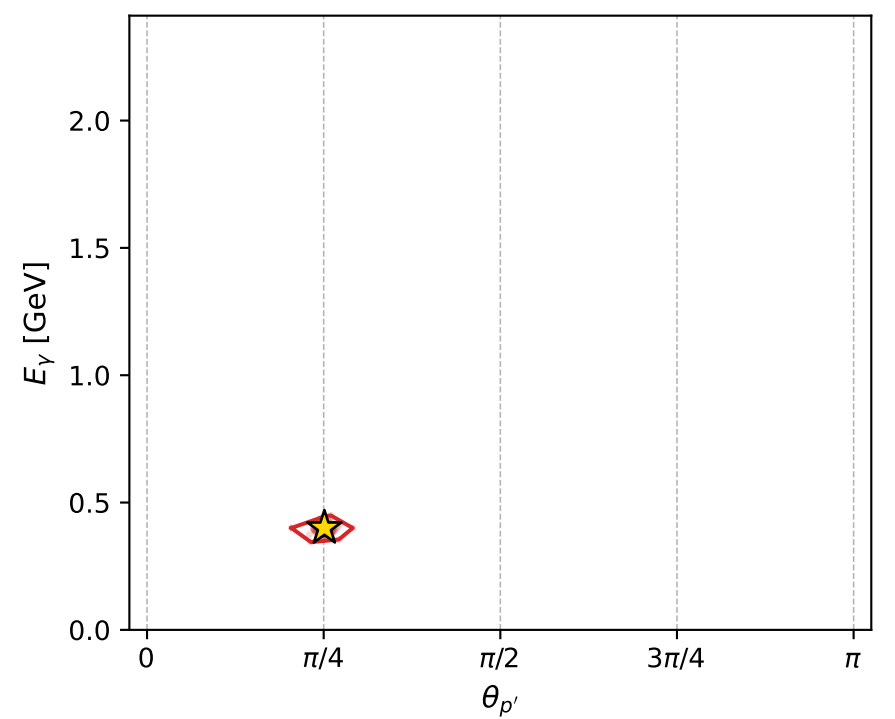
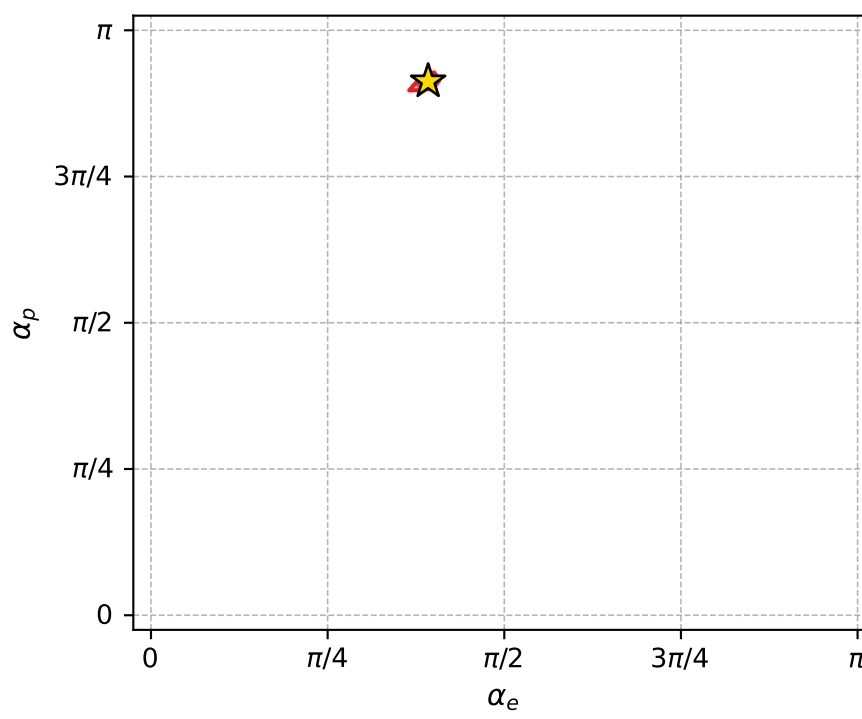
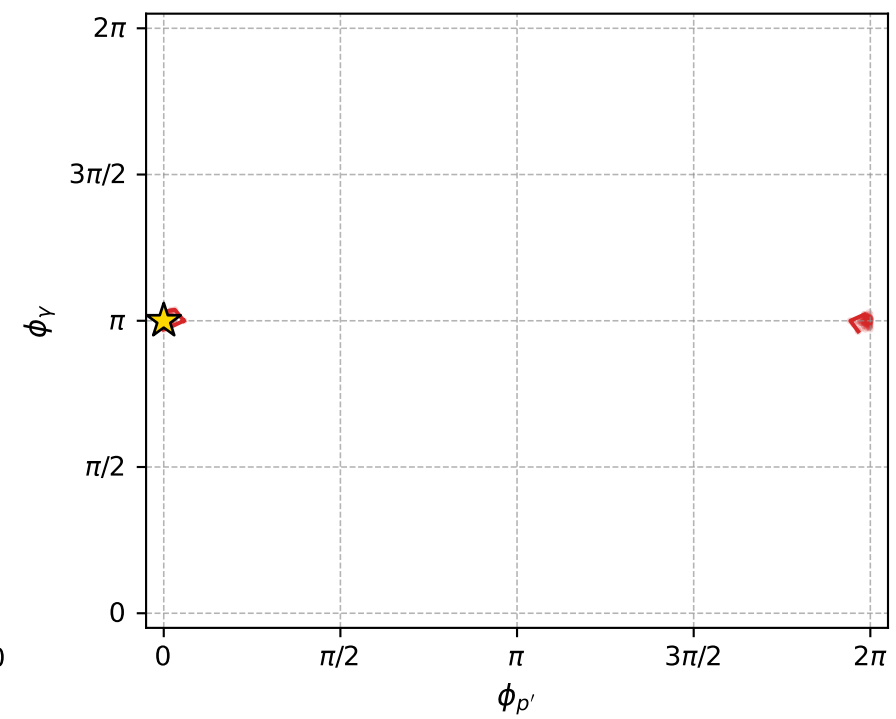
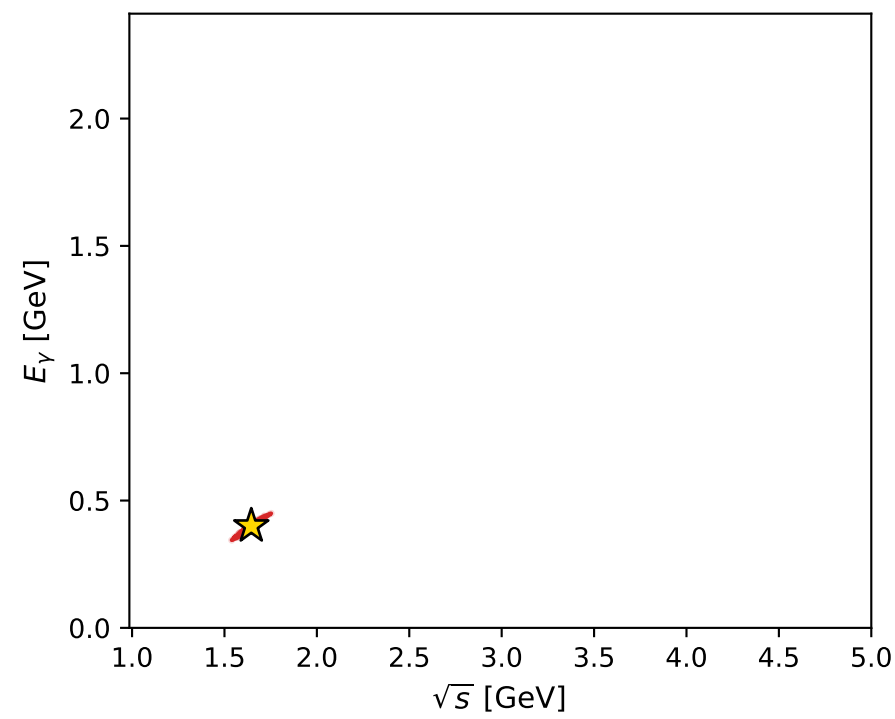
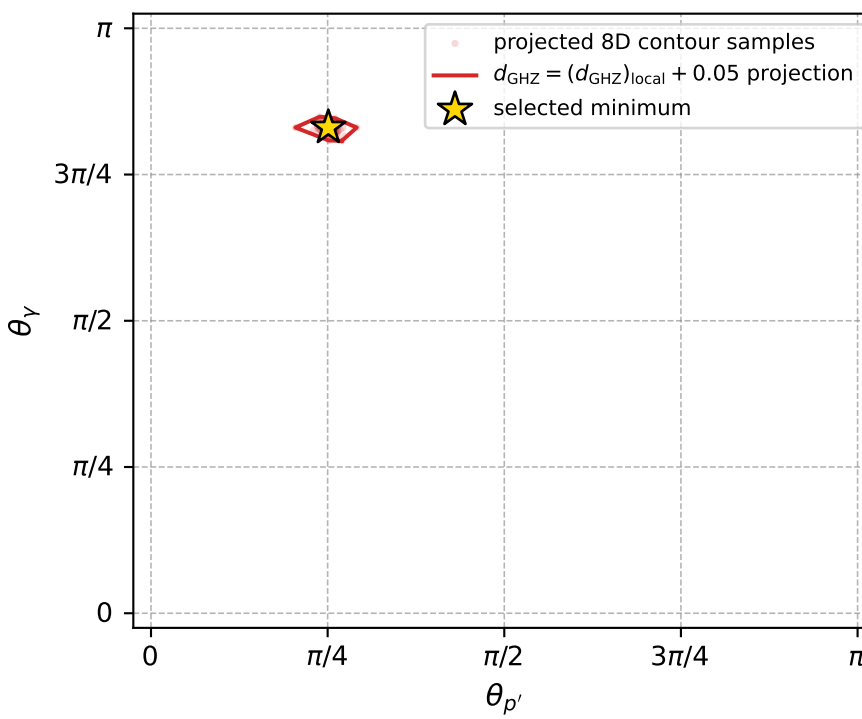
Transverse plane



Leading final-state amplitudes (N=6, retained=99.1%)



electron: polarization cluster P2, local minimum 303; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 0.00032982052$
 $d_{\text{GHZ}} \text{ contour} = 0.050329821$
 above global minimum = 0.00032979824
 8D contour samples = 256

selected phase-space point:

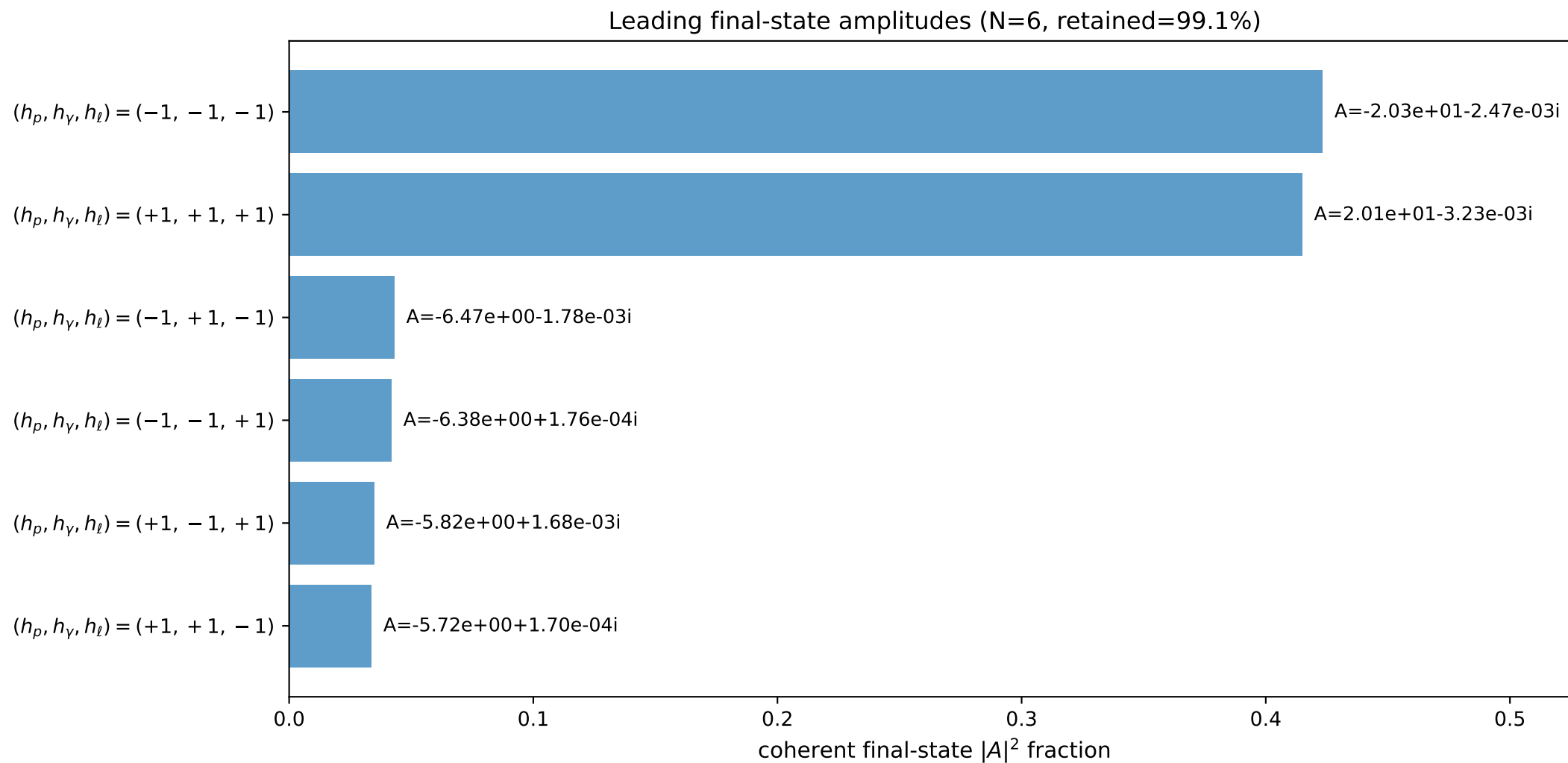
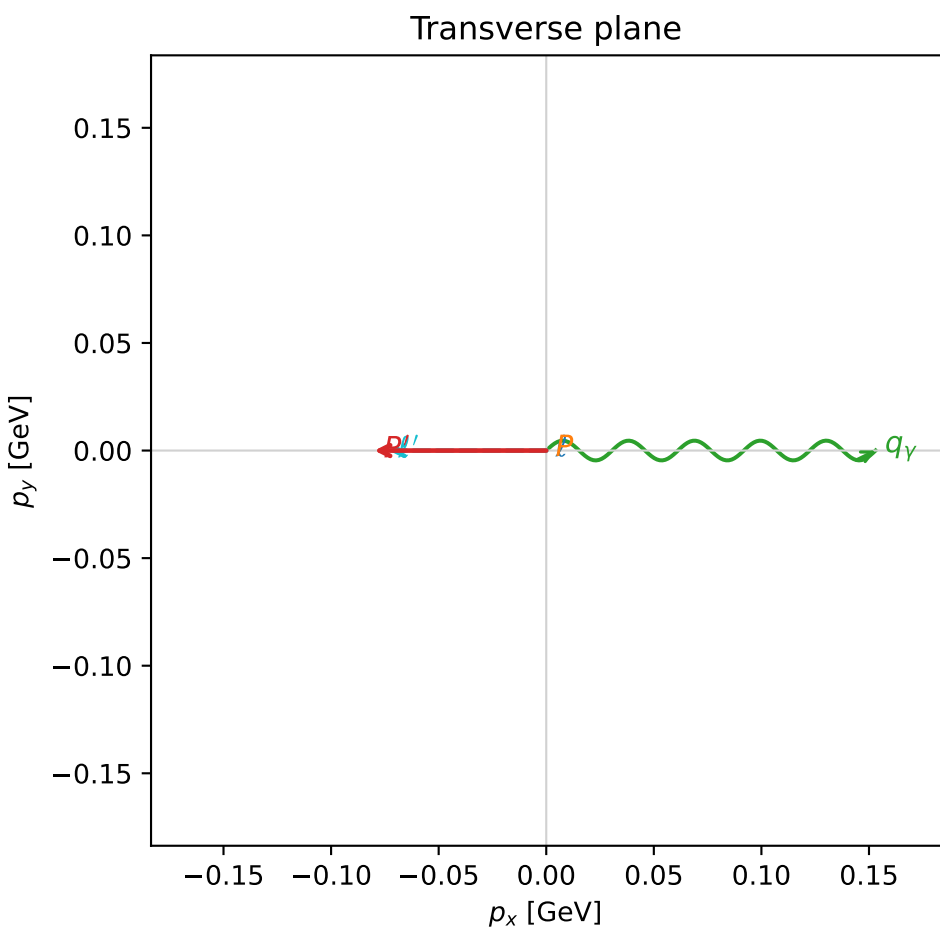
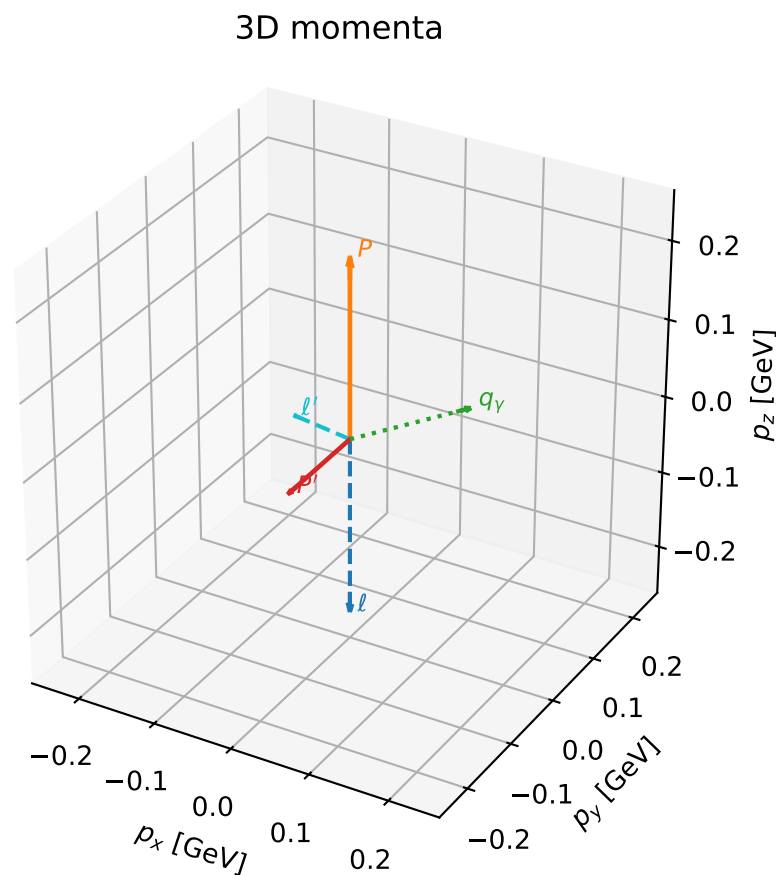
$\text{sqrt_s} = 1.645119$
 $\text{theta_p_out} = 0.7885315$
 $\text{theta_gamma_out} = 2.608625$
 $\text{q0out} = 0.4000807$
 $\text{phi_p_out} = 7.738947\text{e-}06$
 $\text{phi_gamma_out} = 3.1416$
 $\text{alpha_e} = 1.232115$
 $\text{alpha_p} = 2.867268$

coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

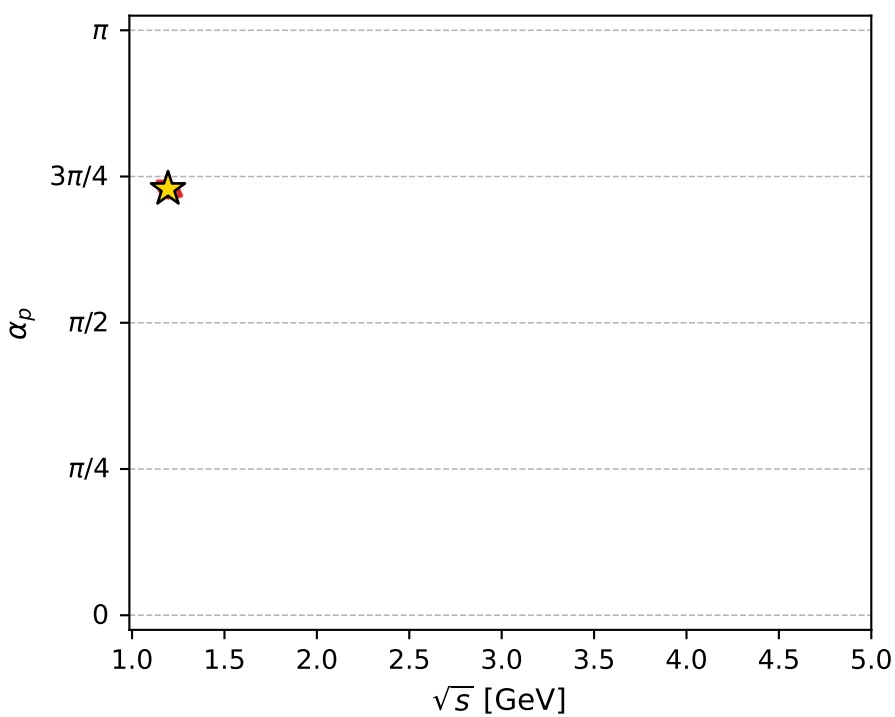
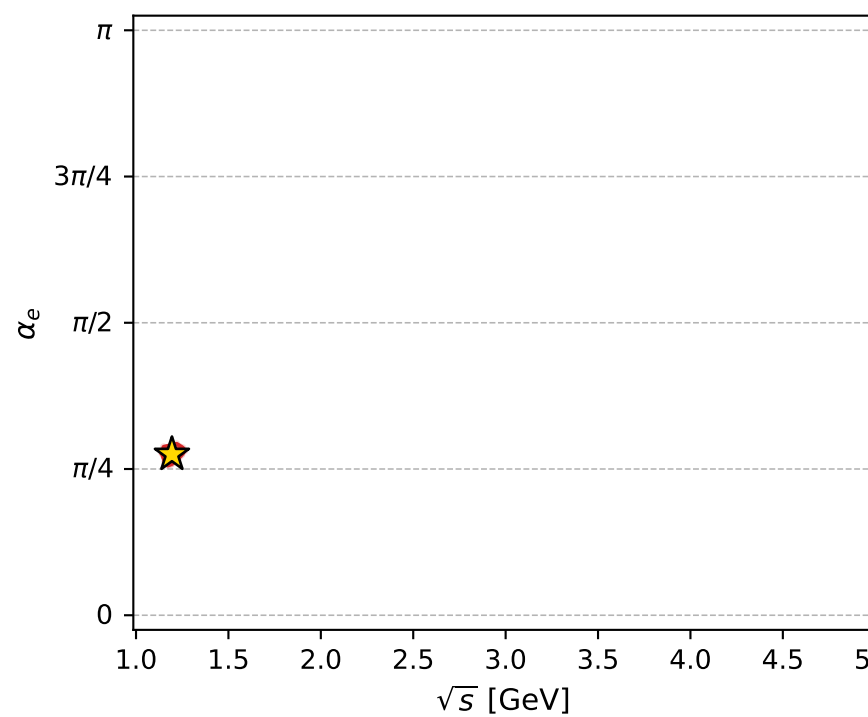
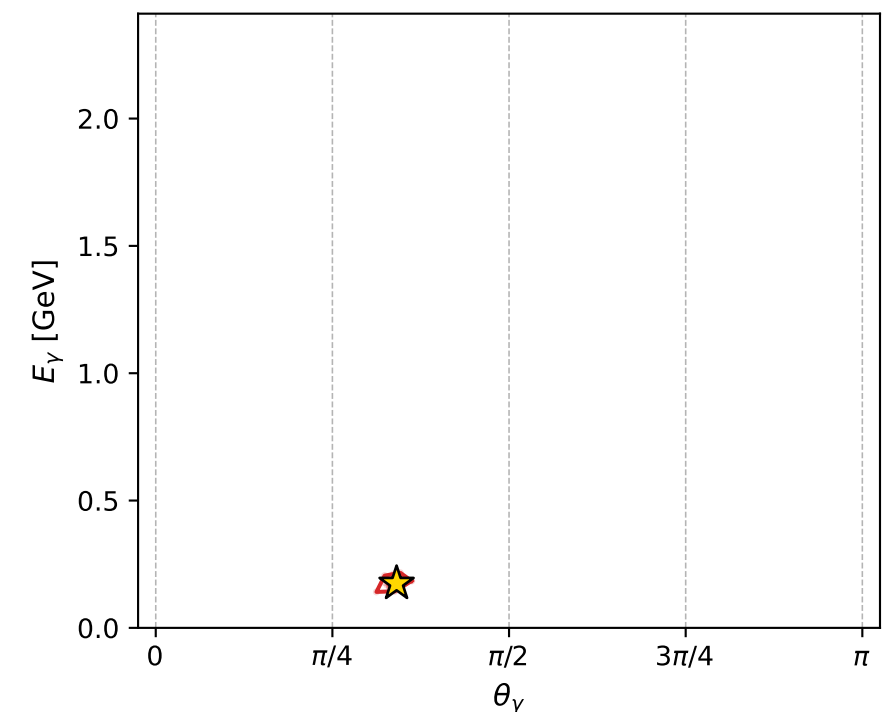
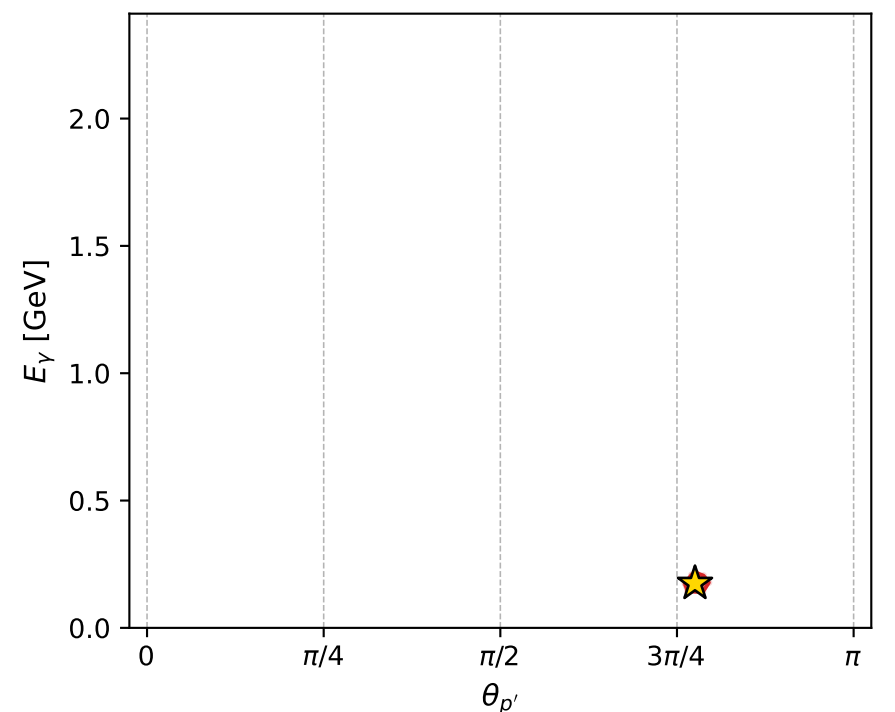
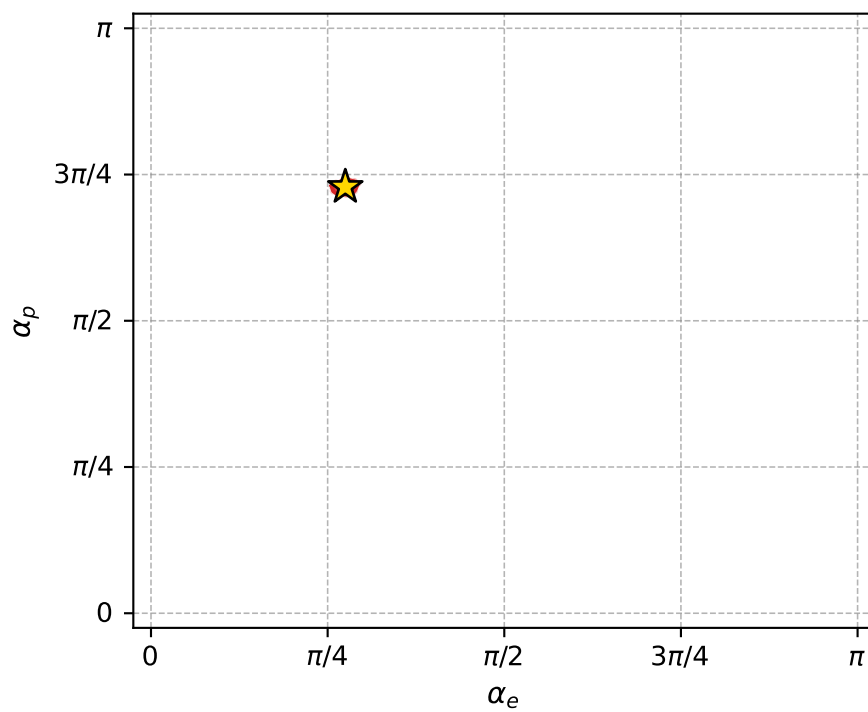
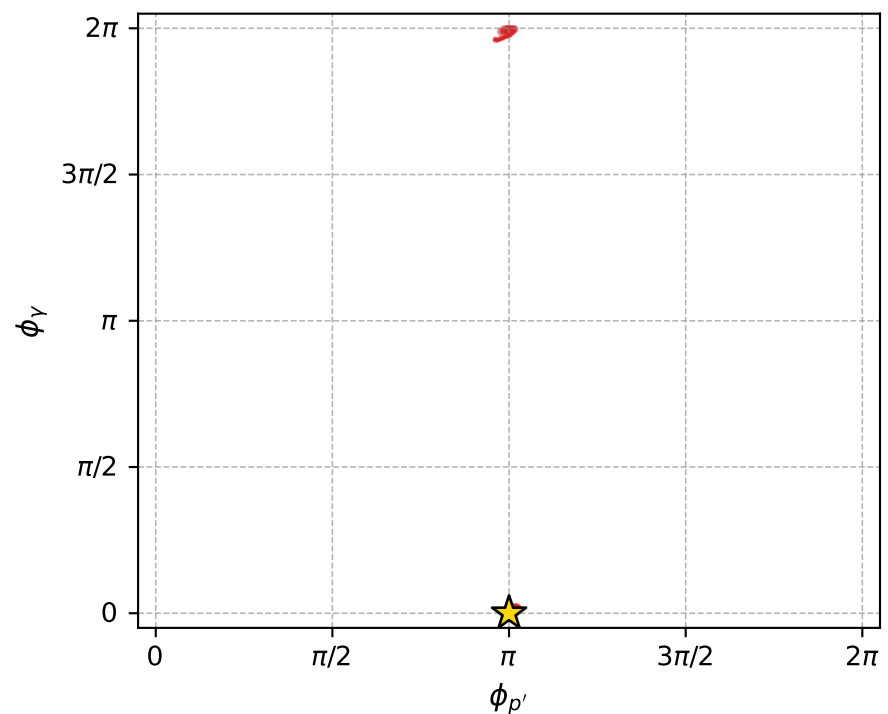
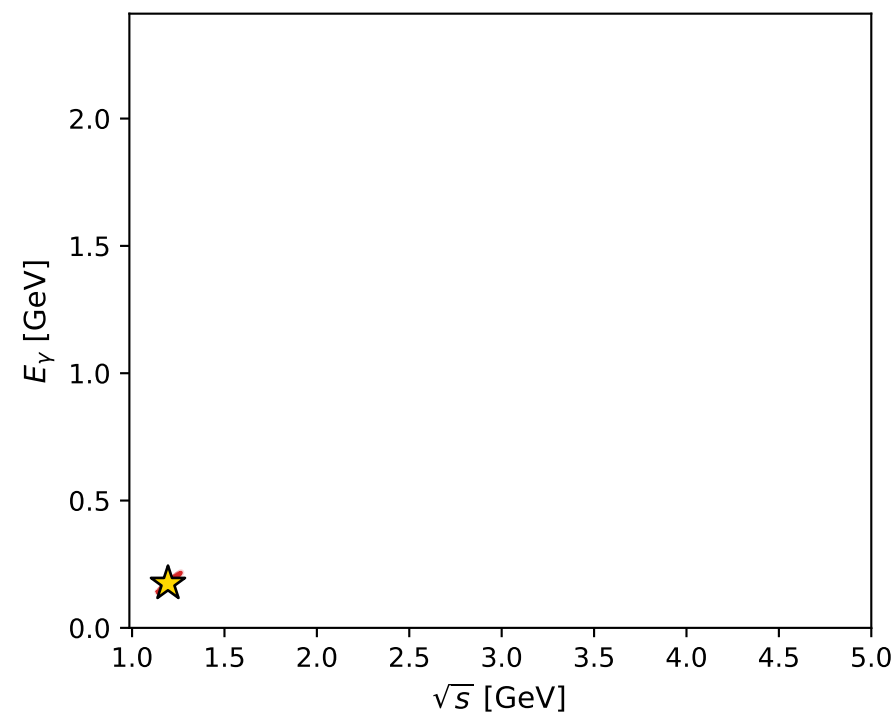
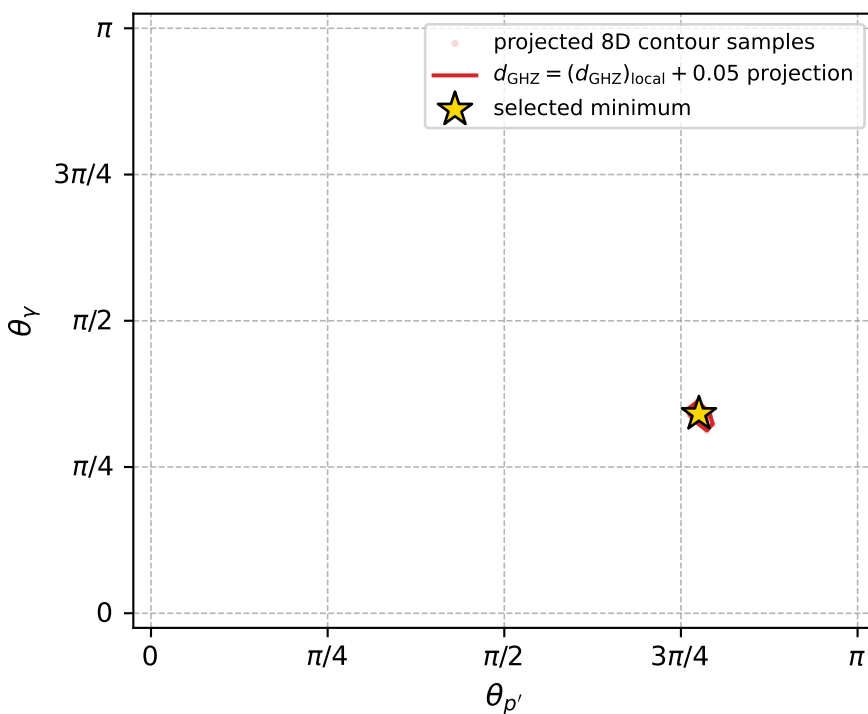
dghz_mixing_angles_polarization_02_local_minimum_305 $d_{\{\text{rm GHZ}\}}=0.000332496$
 region: polarization_cluster_02_local_minimum_305
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=0.86391292$ rad
 incoming lepton: $0.649467|+\rangle +0.76039|-\rangle$
 proton mixing angle: $\alpha_p=2.2894652$ rad
 incoming proton: $-0.658383|+\rangle +0.752683|-\rangle$

$s=1.42646$, $\sqrt{s}=1.19435$, $|\vec{P}|=0.228836$, $|\vec{P}'|=0.12329$
 $\theta_{P'}=2.43625$, $\theta_Y=1.07074$, $\phi_{P'}=3.14173$, $\phi_Y=0.000141885$
 $E_Y=0.174425$, $\theta_{\ell'}=1.4317$, $\phi_{\ell'}=3.14174$
 $Q^2=0.0384869$, $x_B=0.0941708$, $t=-0.110151$, $W^2=1.25005$, $y=0.747672$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.2288$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.2288$
 P $E= 0.9655$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.2288$
 ℓ' $E= 0.0739$ $p_x= -0.0731$ $p_y= -0.0000$ $p_z= 0.0102$
 P' $E= 0.9461$ $p_x= -0.0799$ $p_y= -0.0000$ $p_z= -0.0939$
 q_Y $E= 0.1744$ $p_x= 0.1531$ $p_y= 0.0000$ $p_z= 0.0836$



electron: polarization cluster P2, local minimum 305; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 0.0003324958$
 $d_{\text{GHZ}} \text{ contour} = 0.050332496$
 above global minimum = 0.00033247352
 8D contour samples = 256

selected phase-space point:

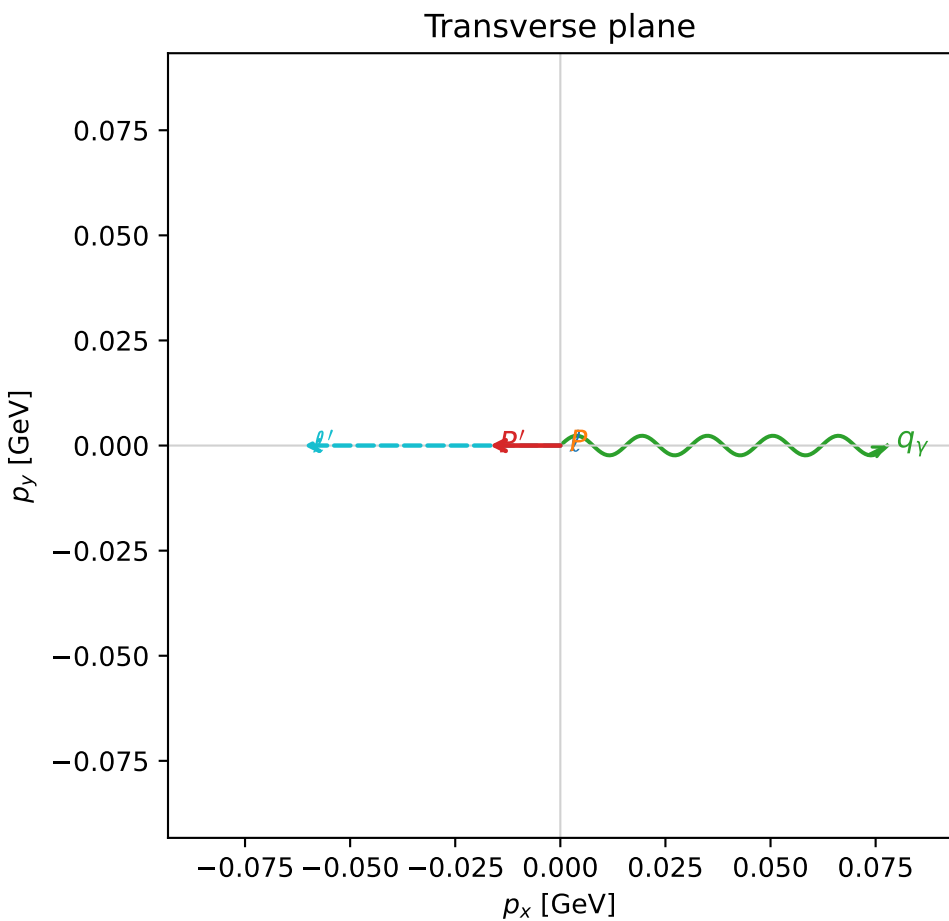
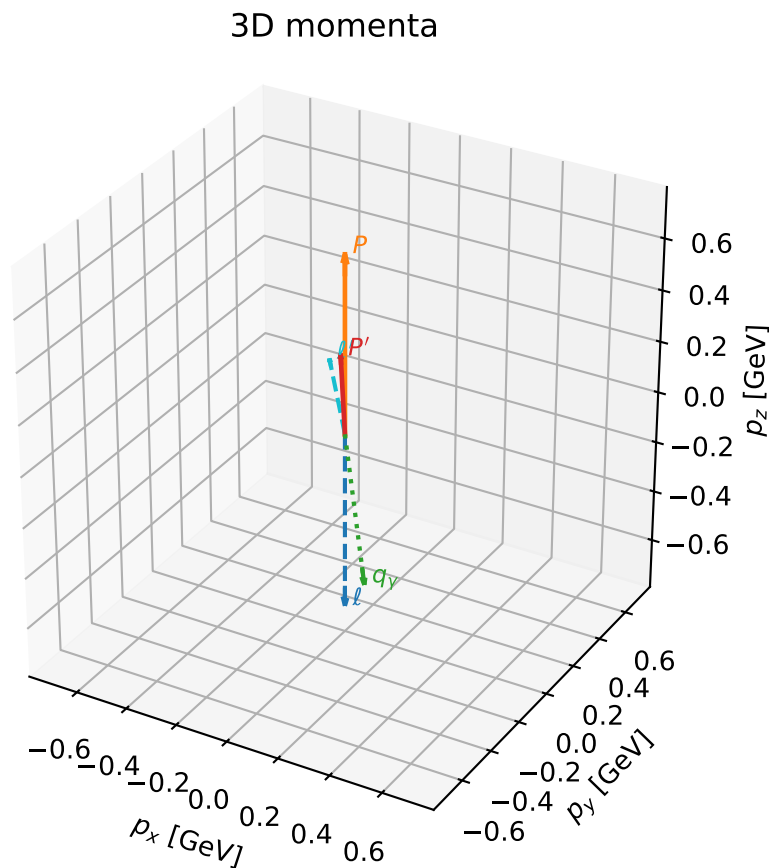
$\text{sqrt}_s = 1.194347$
 $\theta_{p_{\text{out}}} = 2.436246$
 $\theta_{\gamma_{\text{out}}} = 1.070743$
 $q_{\text{out}} = 0.1744249$
 $\phi_{p_{\text{out}}} = 3.141734$
 $\phi_{\gamma_{\text{out}}} = 0.0001418849$
 $\alpha_e = 0.8639129$
 $\alpha_p = 2.289465$

coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

dghz_mixing_angles_polarization_02_local_minimum_308 $d_{\text{GHZ}}=0.000352127$
region: polarization_cluster_02_local_minimum_308
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: $\alpha_e=0.20616489$ rad
incoming lepton: $0.978823|+\rangle +0.204708|-\rangle$
proton mixing angle: $\alpha_p=1.7420637$ rad
incoming proton: $-0.170431|+\rangle +0.98537|-\rangle$

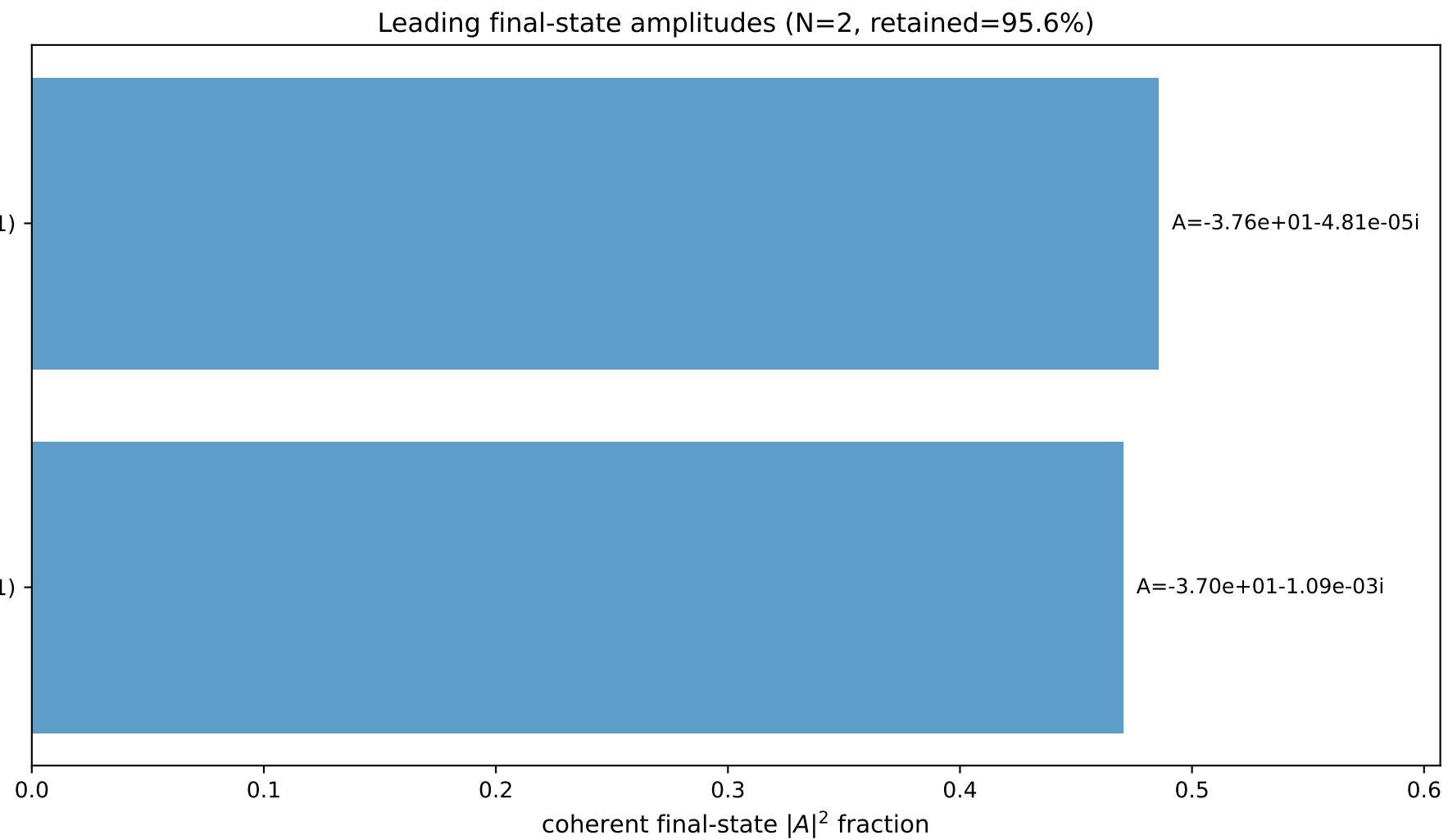
$s=3.44654$, $\sqrt{s}=1.85649$, $|\vec{P}|=0.691278$, $|\vec{P}'|=0.302862$
 $\theta_{p'}=0.0554148$, $\theta_Y=3.00846$, $\phi_{p'}=3.1414$, $\phi_Y=6.28312$
 $E_Y=0.58589$, $\theta_{\ell'}=0.215754$, $\phi_{\ell'}=3.14156$
 $Q^2=0.778686$, $x_B=0.340408$, $t=-0.11928$, $W^2=2.38866$, $y=0.891225$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.6913$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.6913$
 P $E= 1.1652$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.6913$
 ℓ' $E= 0.2849$ $p_x= -0.0610$ $p_y= 0.0000$ $p_z= 0.2783$
 P' $E= 0.9857$ $p_x= -0.0168$ $p_y= 0.0000$ $p_z= 0.3024$
 q_Y $E= 0.5859$ $p_x= 0.0778$ $p_y= -0.0000$ $p_z= -0.5807$

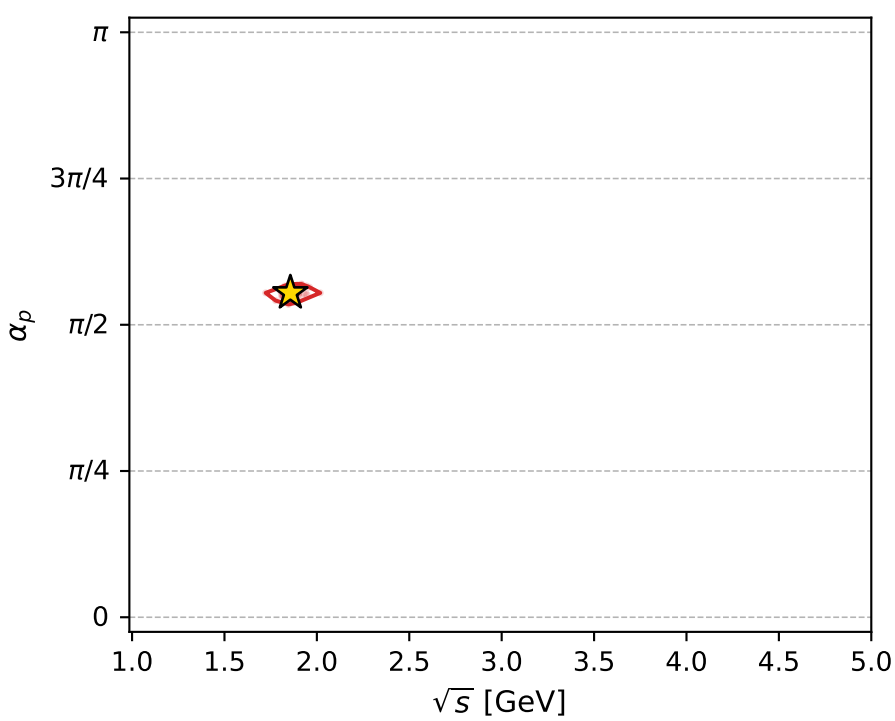
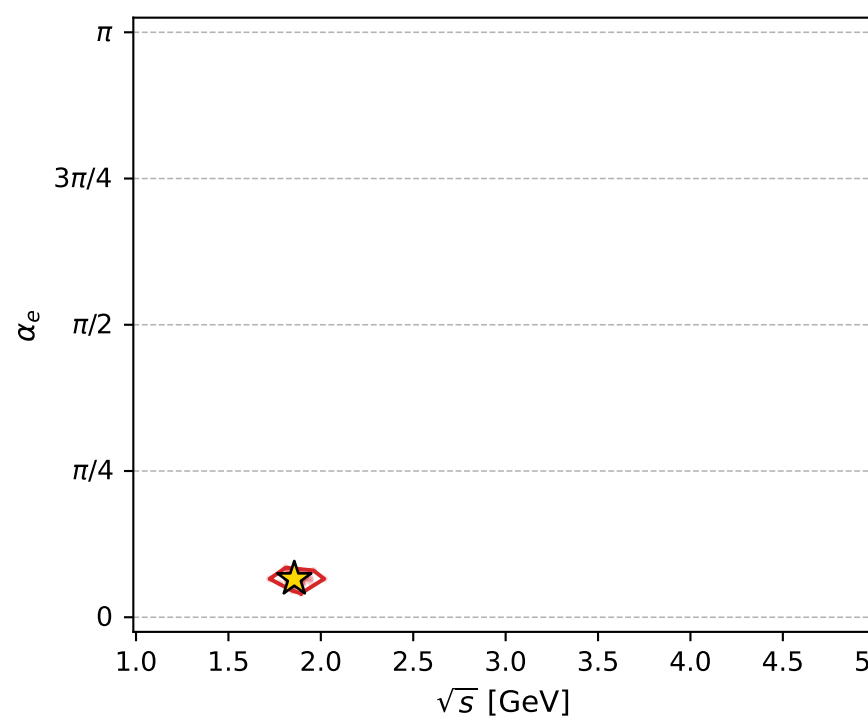
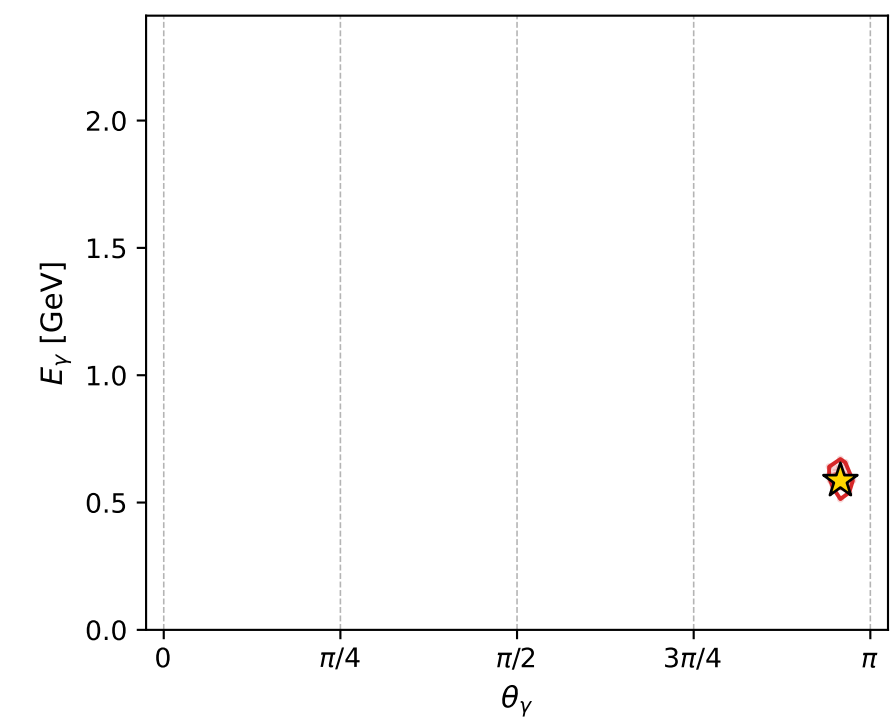
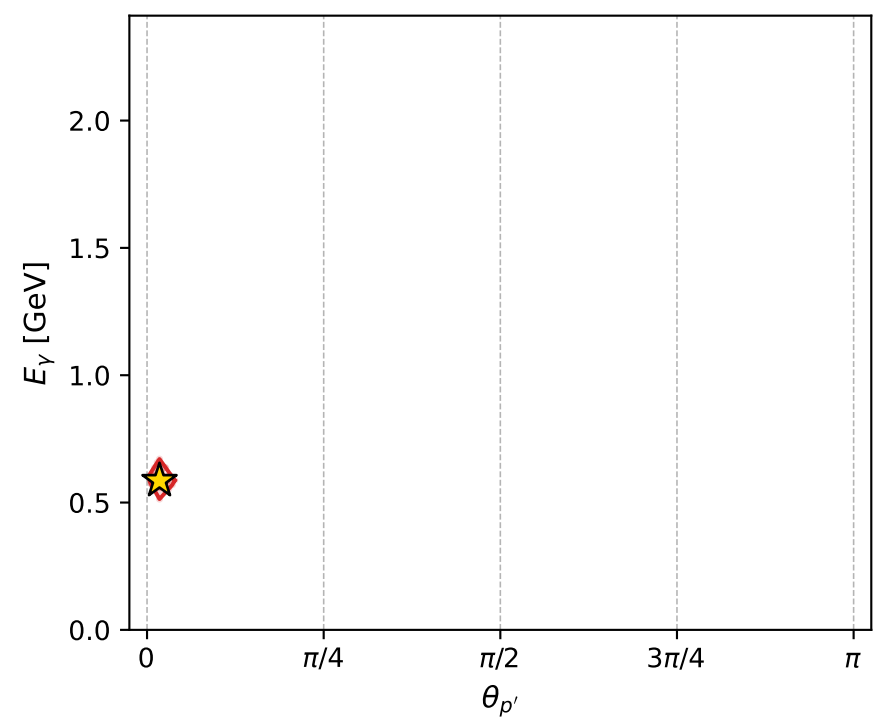
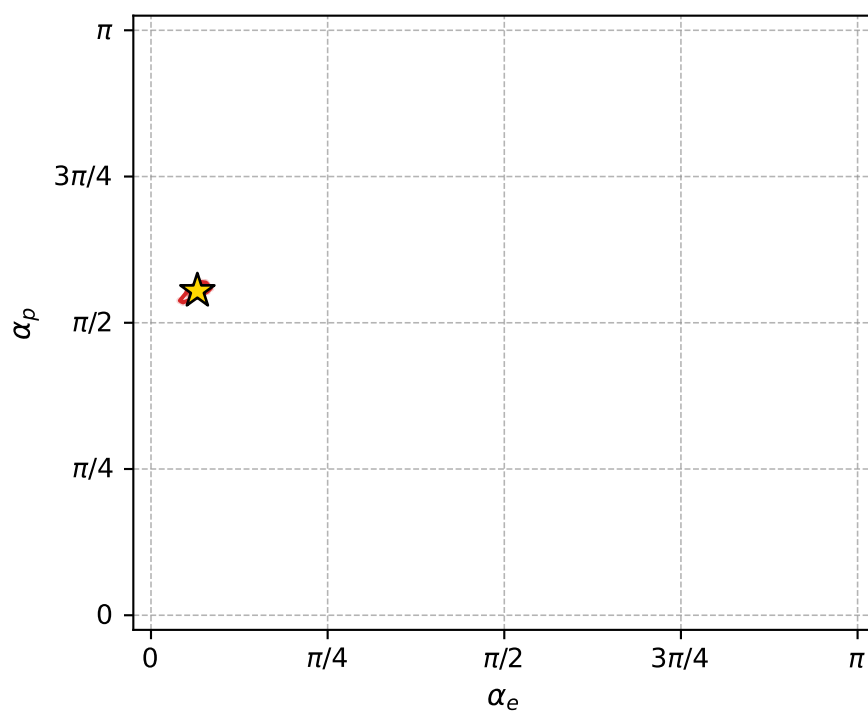
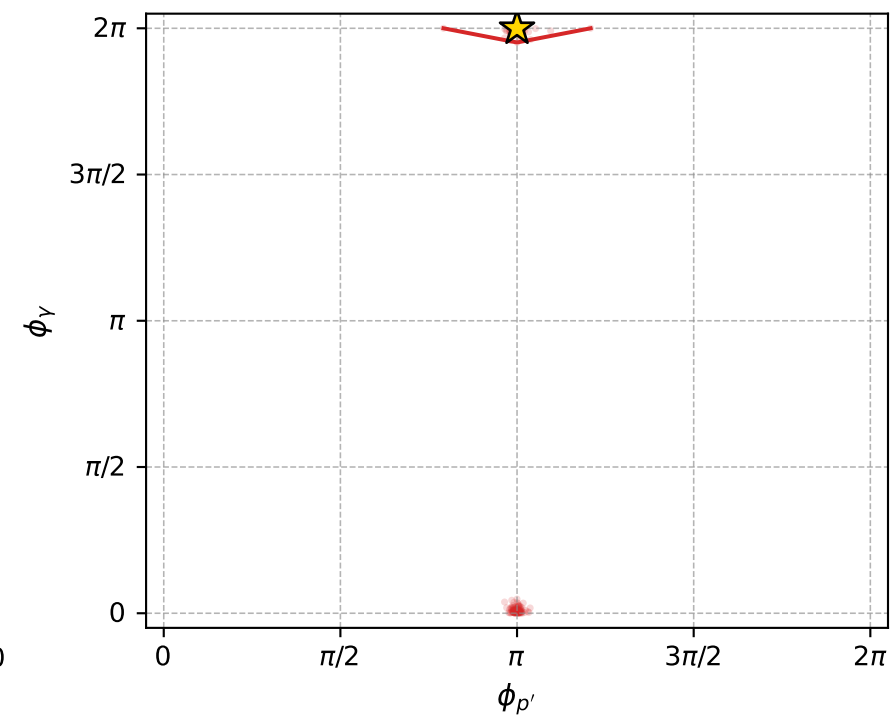
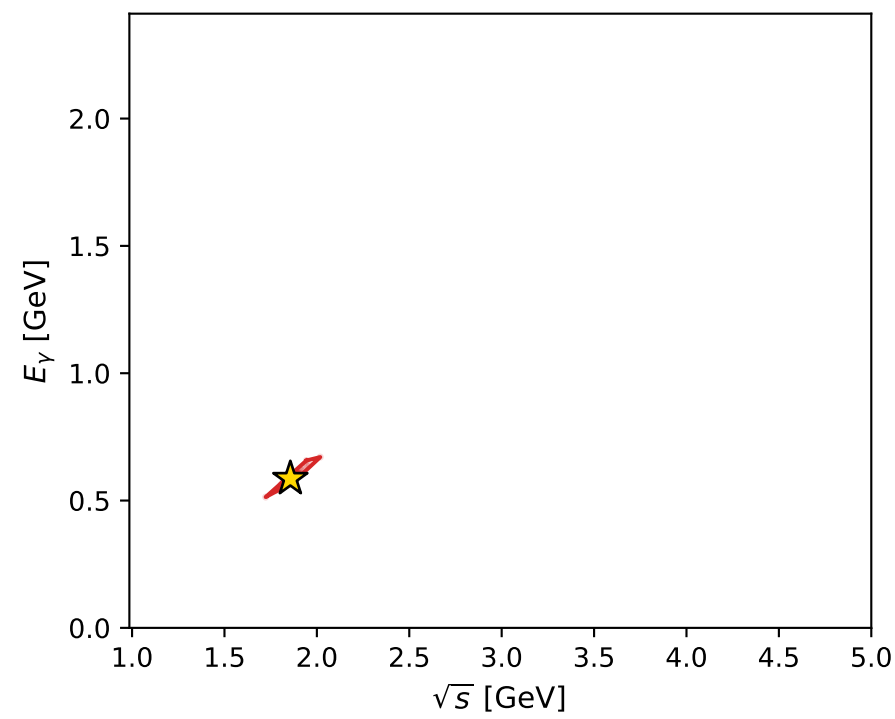
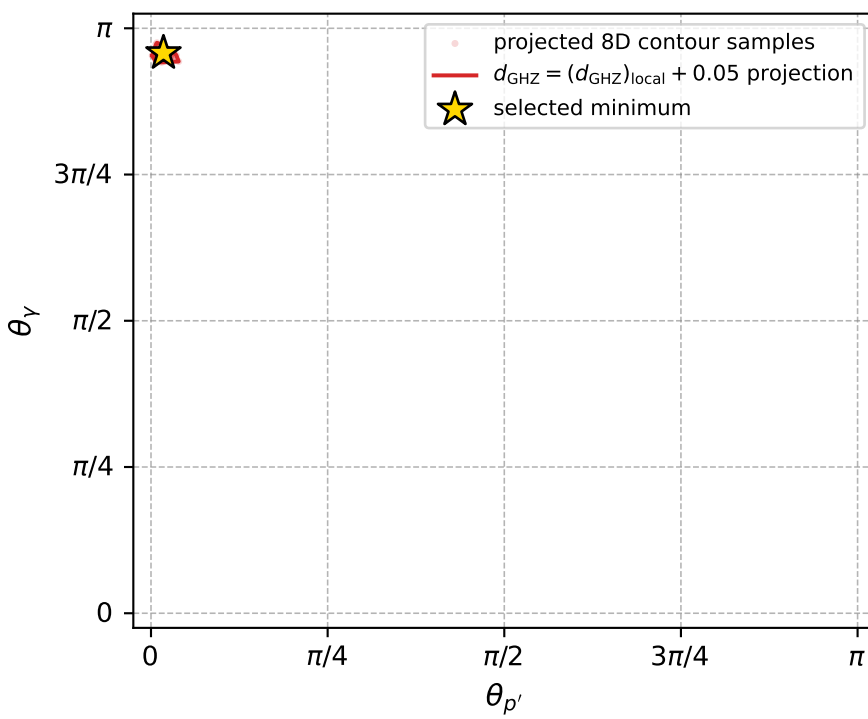


$(h_p, h_Y, h_\ell) = (+1, -1, -1)$

$(h_p, h_Y, h_\ell) = (-1, +1, +1)$



electron: polarization cluster P2, local minimum 308; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 0.00035212686$
 $d_{\text{GHZ}} \text{ contour} = 0.050352127$
 above global minimum = 0.00035210457
 8D contour samples = 254

selected phase-space point:

$\text{sqrt_s} = 1.856486$
 $\text{theta_p_out} = 0.05541481$
 $\text{theta_gamma_out} = 3.008462$
 $\text{q0ut} = 0.58589$
 $\text{phi_p_out} = 3.141402$
 $\text{phi_gamma_out} = 6.283122$
 $\text{alpha_e} = 0.2061649$
 $\text{alpha_p} = 1.742064$

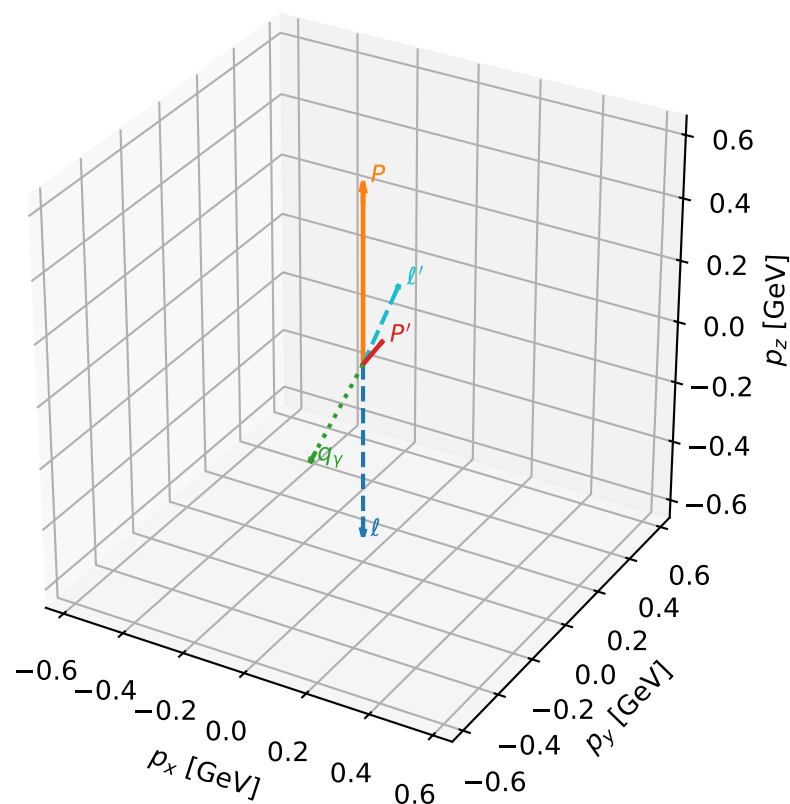
coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

dghz_mixing_angles_polarization_02_local_minimum_309 $d_{\{\text{rm GHZ}\}}=0.000358045$
 region: polarization_cluster_02_local_minimum_309
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=1.2781351$ rad
 incoming lepton: $0.288501|+\rangle + 0.95748|-\rangle$
 proton mixing angle: $\alpha_p=2.9053609$ rad
 incoming proton: $-0.972227|+\rangle + 0.234041|-\rangle$

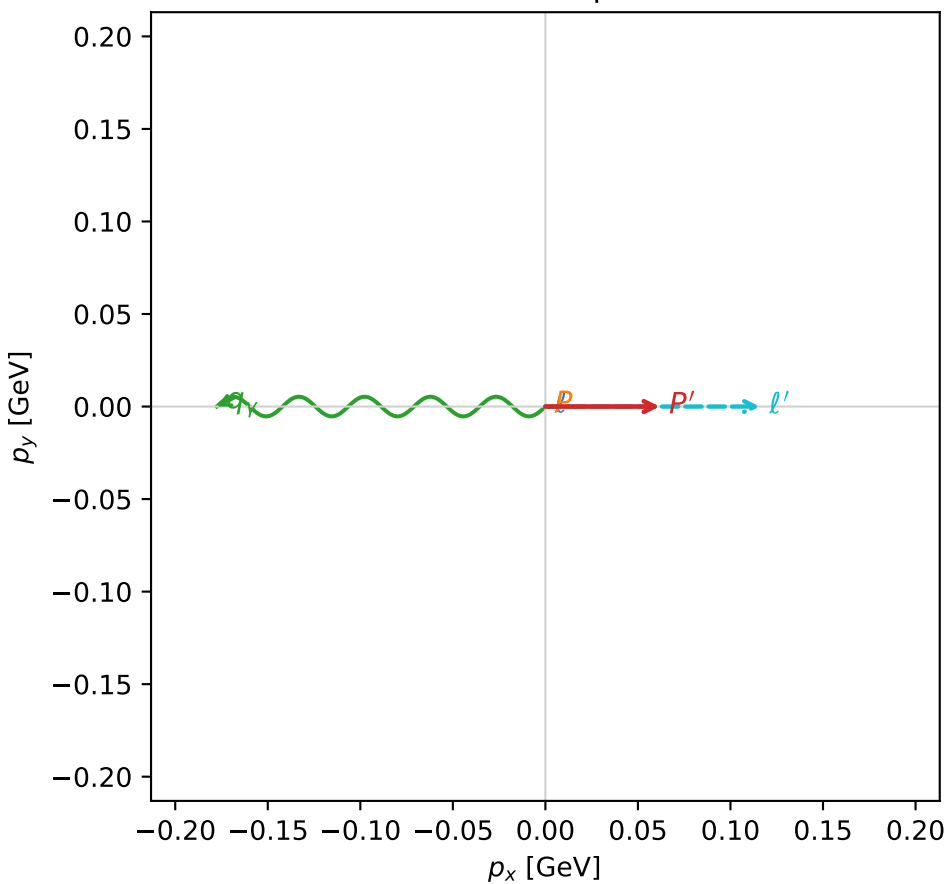
$s=2.80265$, $\sqrt{s}=1.67411$, $|\vec{P}|=0.574275$, $|\vec{P}'|=0.109471$
 $\theta_{P'}=0.599129$, $\theta_Y=2.70358$, $\phi_{P'}=0.000127264$, $\phi_Y=3.14165$
 $E_Y=0.41865$, $\theta_{\ell'}=0.381526$, $\phi_{\ell'}=2.71705e-05$
 $Q^2=0.688922$, $x_B=0.438772$, $t=-0.213772$, $W^2=1.76104$, $y=0.816576$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.5743$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.5743$
 P $E= 1.0998$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.5743$
 ℓ' $E= 0.3111$ $p_x= 0.1158$ $p_y= 0.0000$ $p_z= 0.2887$
 P' $E= 0.9444$ $p_x= 0.0617$ $p_y= 0.0000$ $p_z= 0.0904$
 q_Y $E= 0.4187$ $p_x= -0.1776$ $p_y= -0.0000$ $p_z= -0.3791$

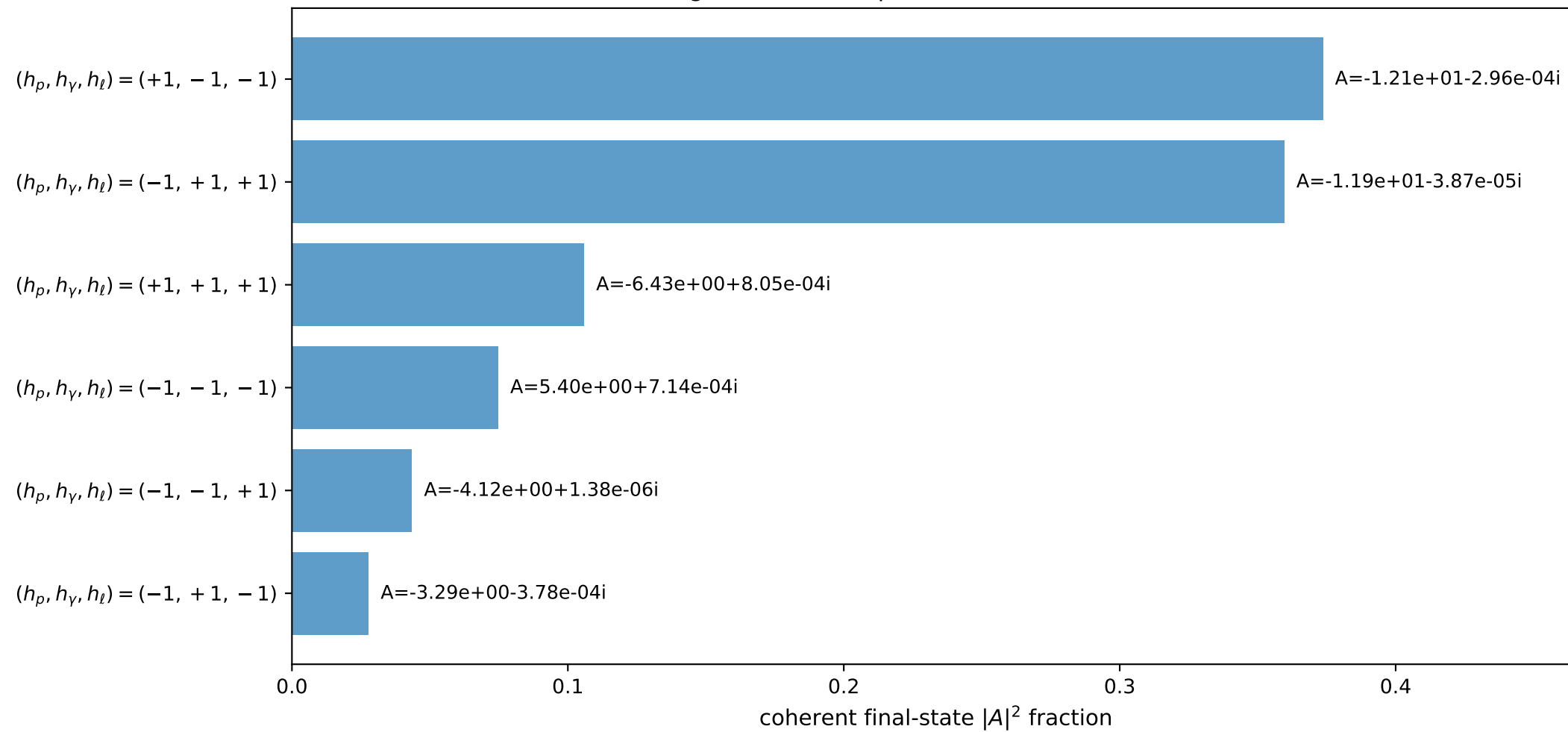
3D momenta



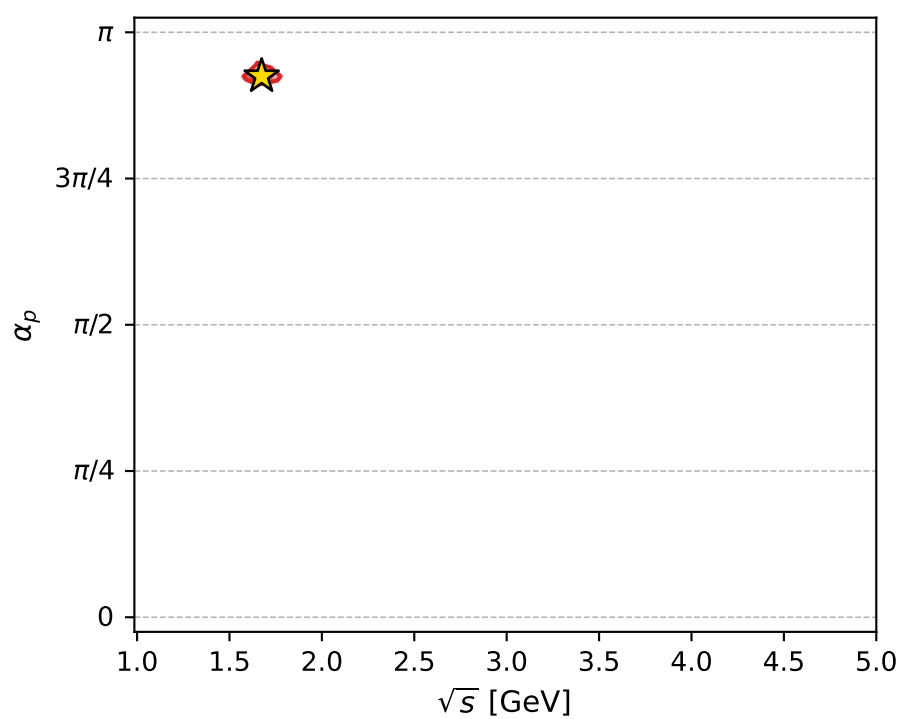
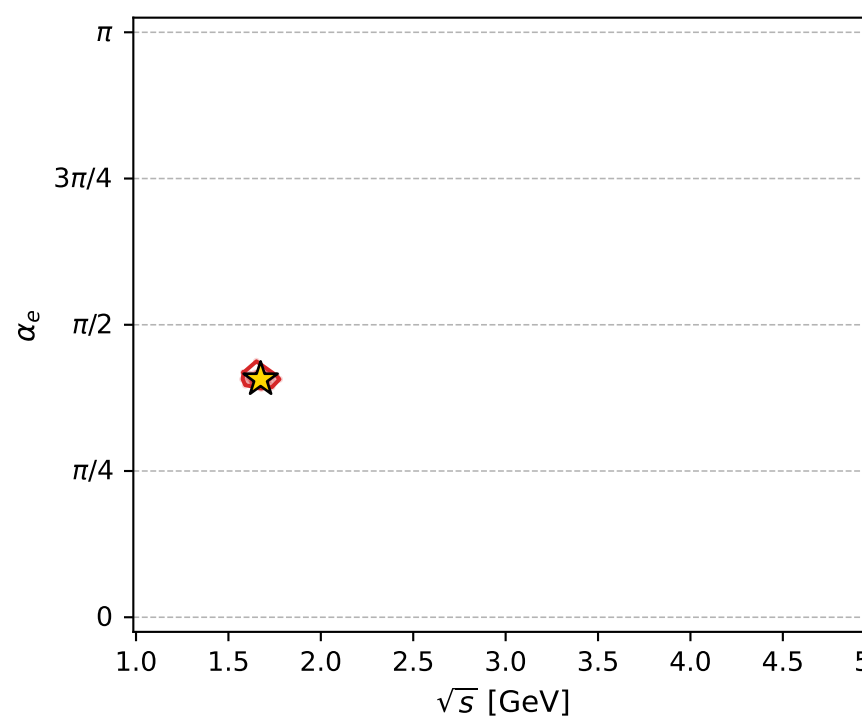
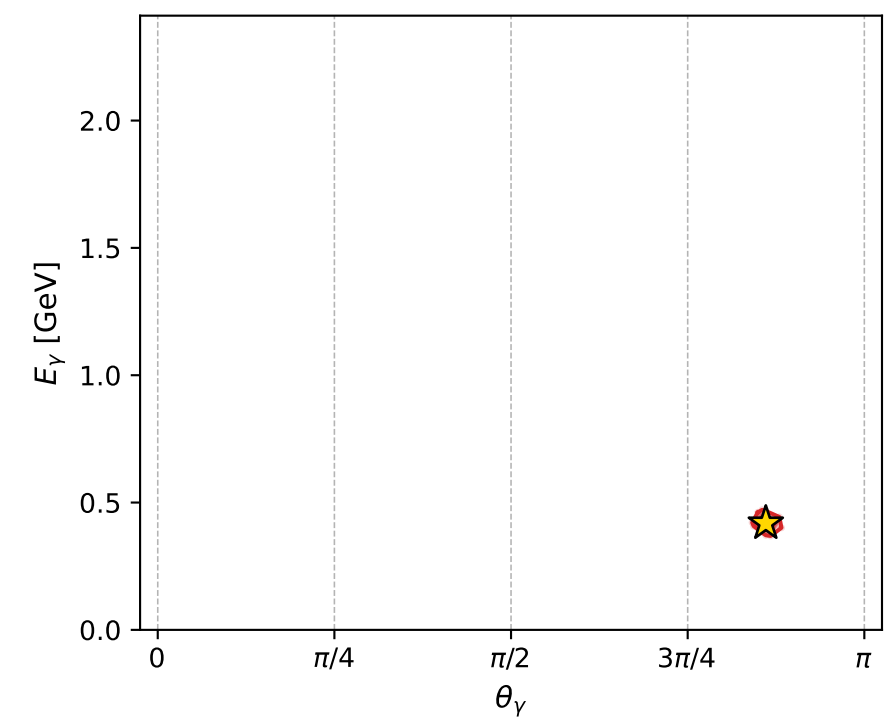
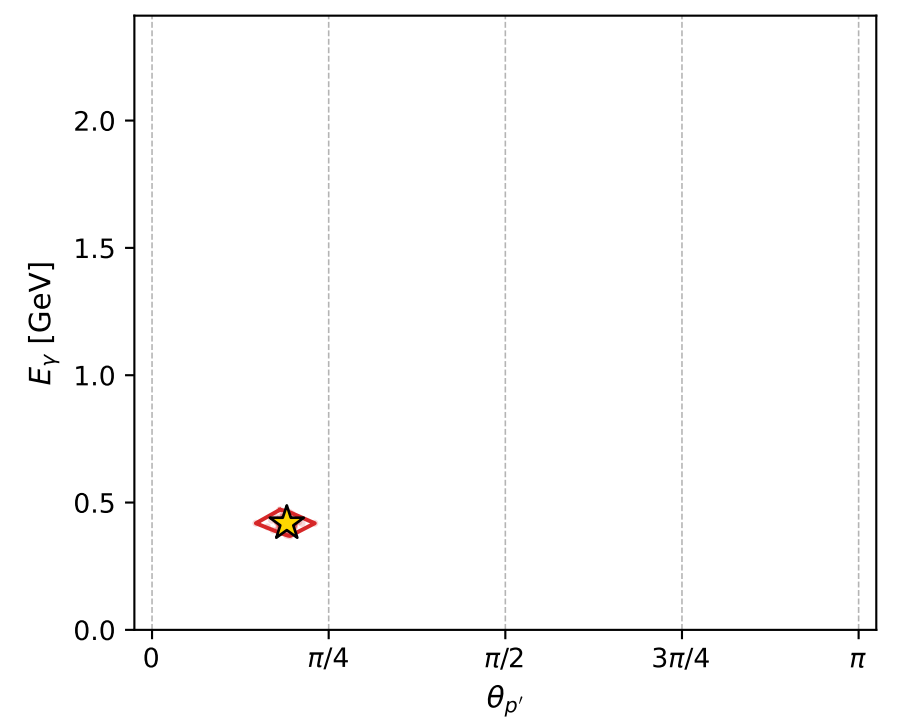
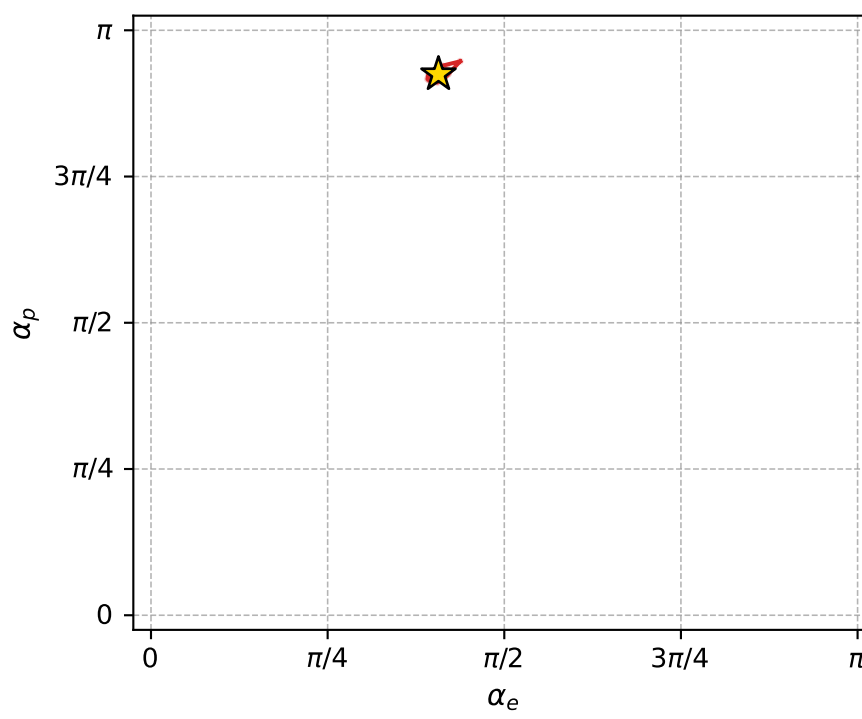
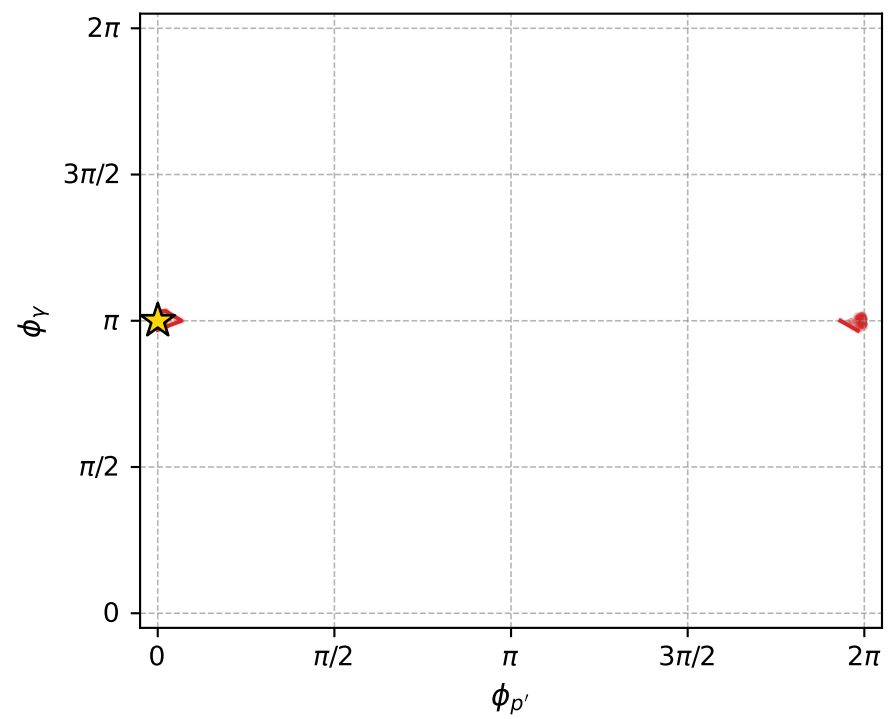
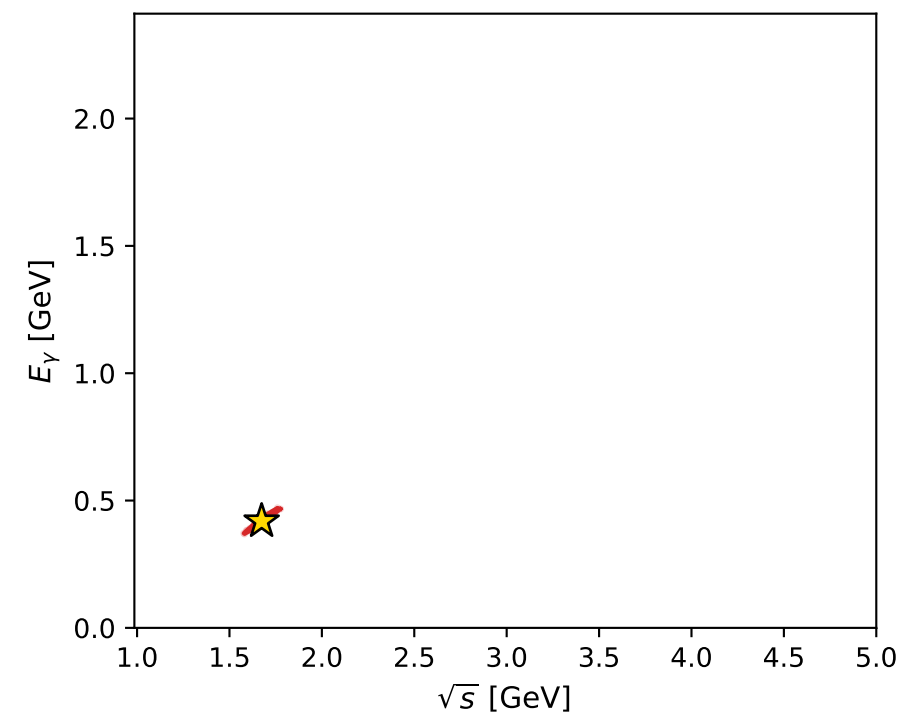
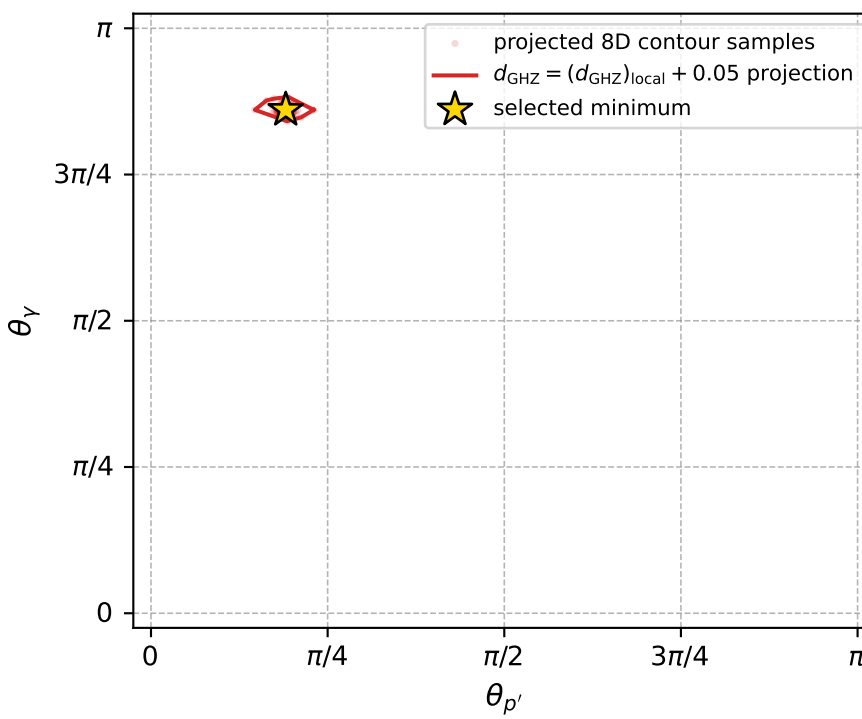
Transverse plane



Leading final-state amplitudes (N=6, retained=98.5%)



electron: polarization cluster P2, local minimum 309; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 0.0003580453$
 $d_{\text{GHZ}} \text{ contour} = 0.050358045$
 above global minimum = 0.00035802302
 8D contour samples = 256

selected phase-space point:

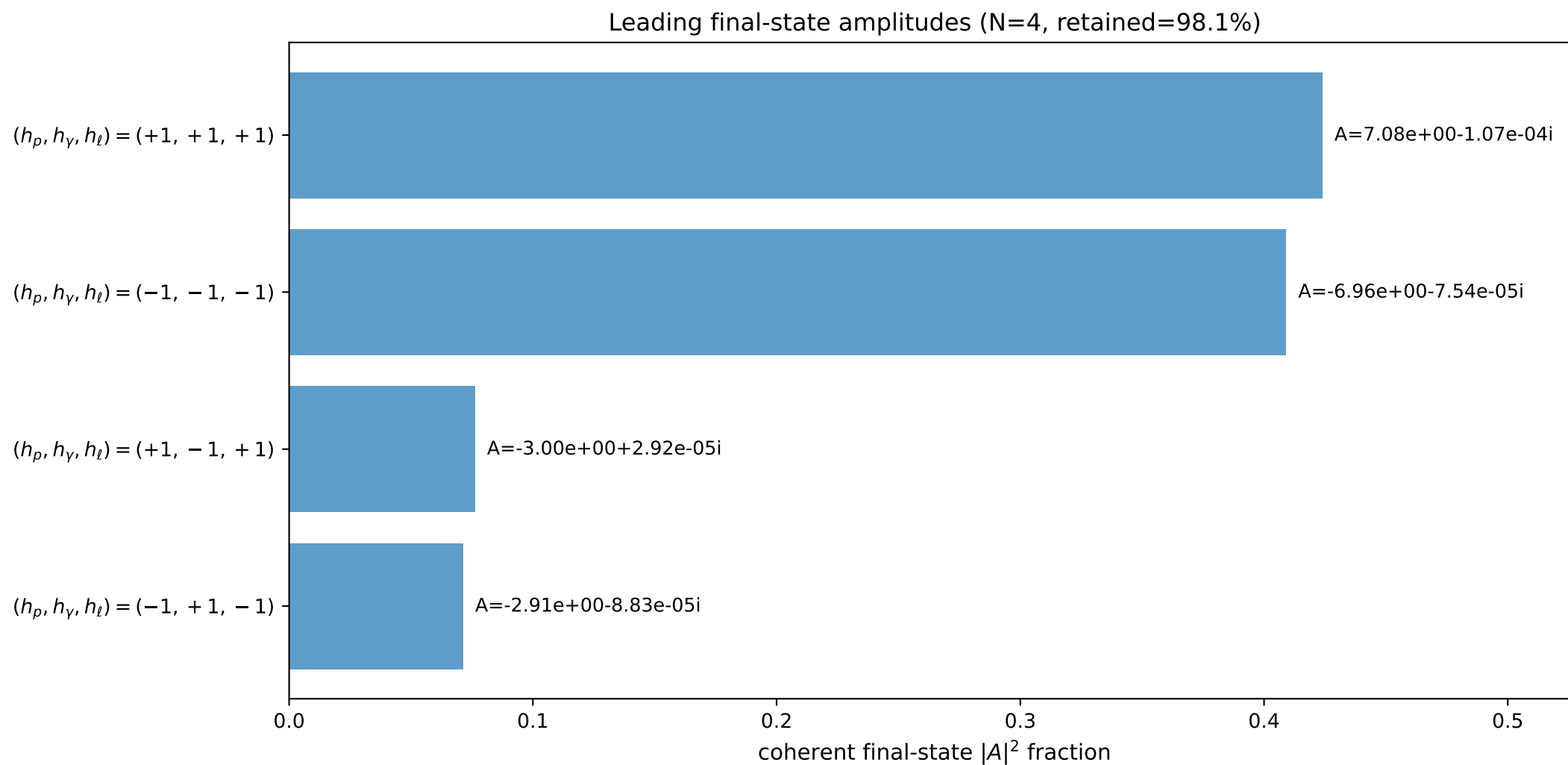
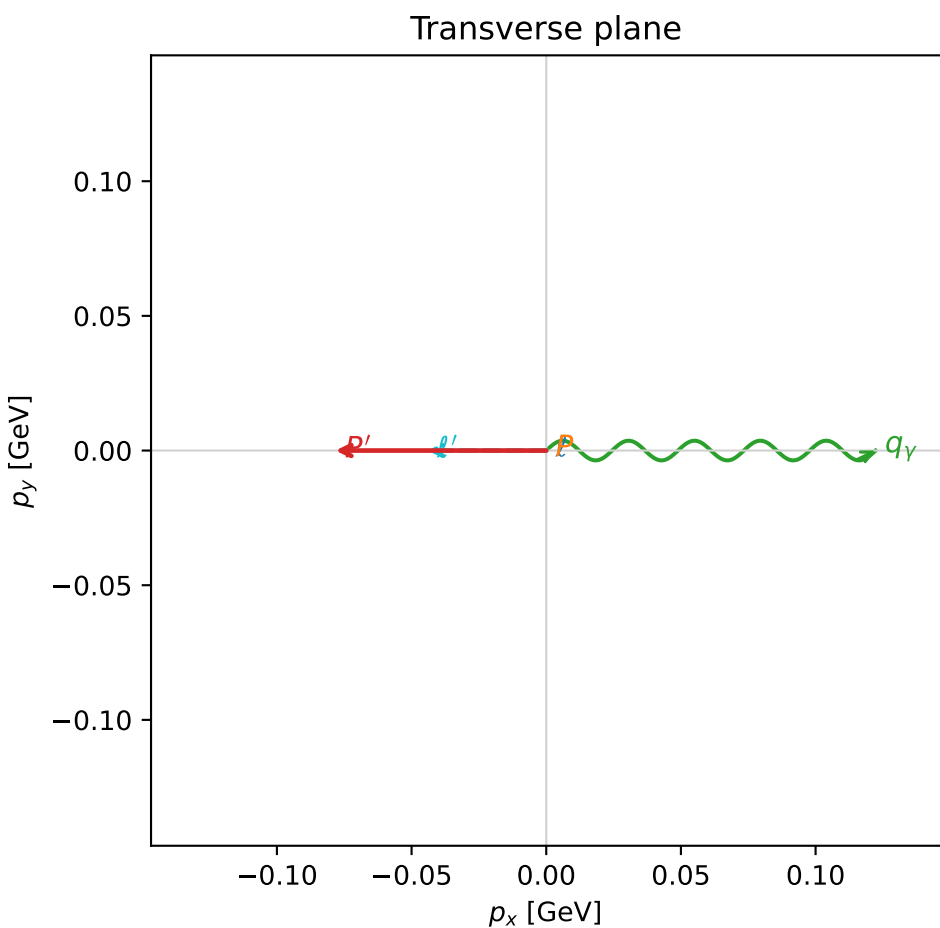
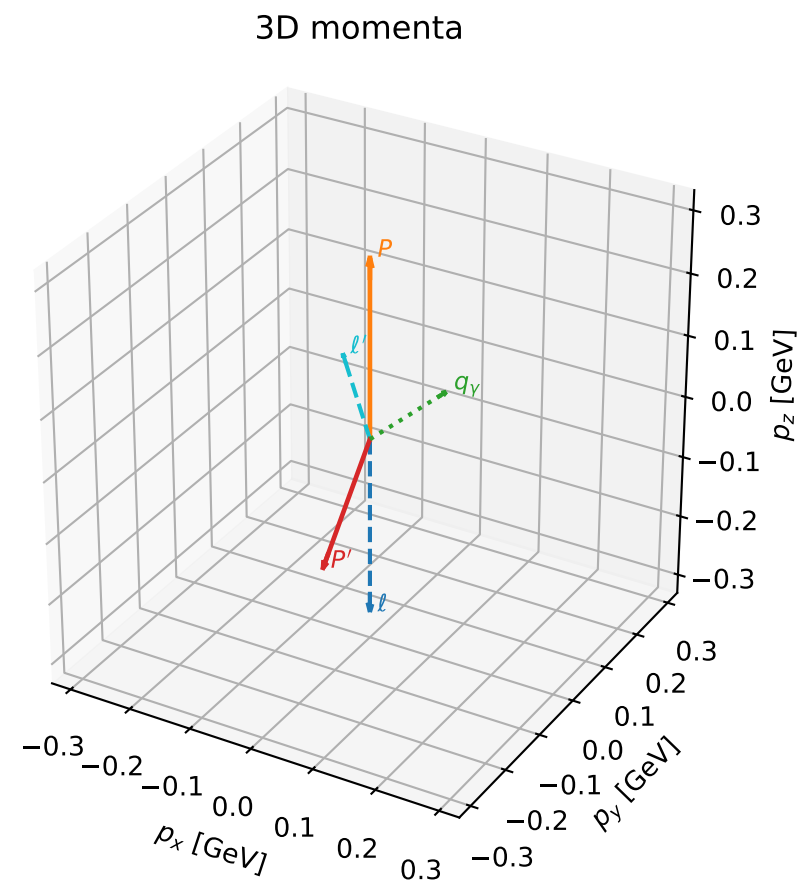
$\text{sqrt_s} = 1.67411$
 $\text{theta_p_out} = 0.5991289$
 $\text{theta_gamma_out} = 2.703584$
 $\text{qOut} = 0.4186502$
 $\text{phi_p_out} = 0.0001272637$
 $\text{phi_gamma_out} = 3.141655$
 $\text{alpha_e} = 1.278135$
 $\text{alpha_p} = 2.905361$

coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

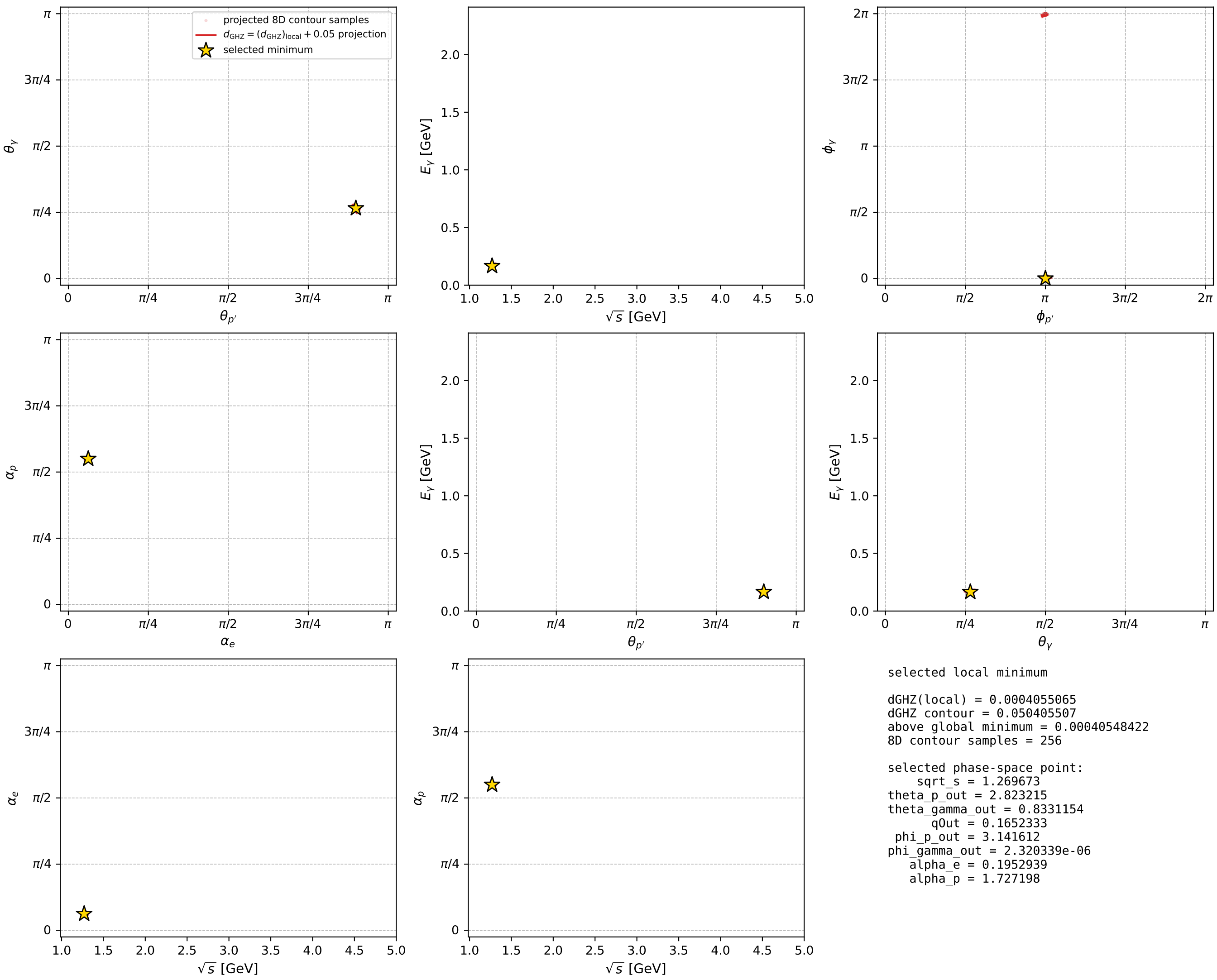
dghz_mixing_angles_polarization_02_local_minimum_317 $d_{\{\text{rm GHZ}\}}=0.000405507$
 region: polarization_cluster_02_local_minimum_317
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=0.19529395$ rad
 incoming lepton: $0.980991|+\rangle + 0.194055|-\rangle$
 proton mixing angle: $\alpha_p=1.7271979$ rad
 incoming proton: $-0.155765|+\rangle + 0.987794|-\rangle$

$s=1.61207$, $\sqrt{s}=1.26967$, $|\vec{P}|=0.288352$, $|\vec{P}'|=0.249942$
 $\theta_{P'}=2.82321$, $\theta_Y=0.833115$, $\phi_{P'}=3.14161$, $\phi_Y=2.32034e-06$
 $E_Y=0.165233$, $\theta_{\ell'}=0.33563$, $\phi_{\ell'}=3.14156$
 $Q^2=0.14992$, $x_B=0.276295$, $t=-0.282404$, $W^2=1.27253$, $y=0.741039$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.2884$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.2884$
 P $E= 0.9813$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.2884$
 ℓ' $E= 0.1337$ $p_x= -0.0440$ $p_y= 0.0000$ $p_z= 0.1262$
 P' $E= 0.9707$ $p_x= -0.0782$ $p_y= -0.0000$ $p_z= -0.2374$
 q_Y $E= 0.1652$ $p_x= 0.1223$ $p_y= 0.0000$ $p_z= 0.1111$



electron: polarization cluster P2, local minimum 317; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



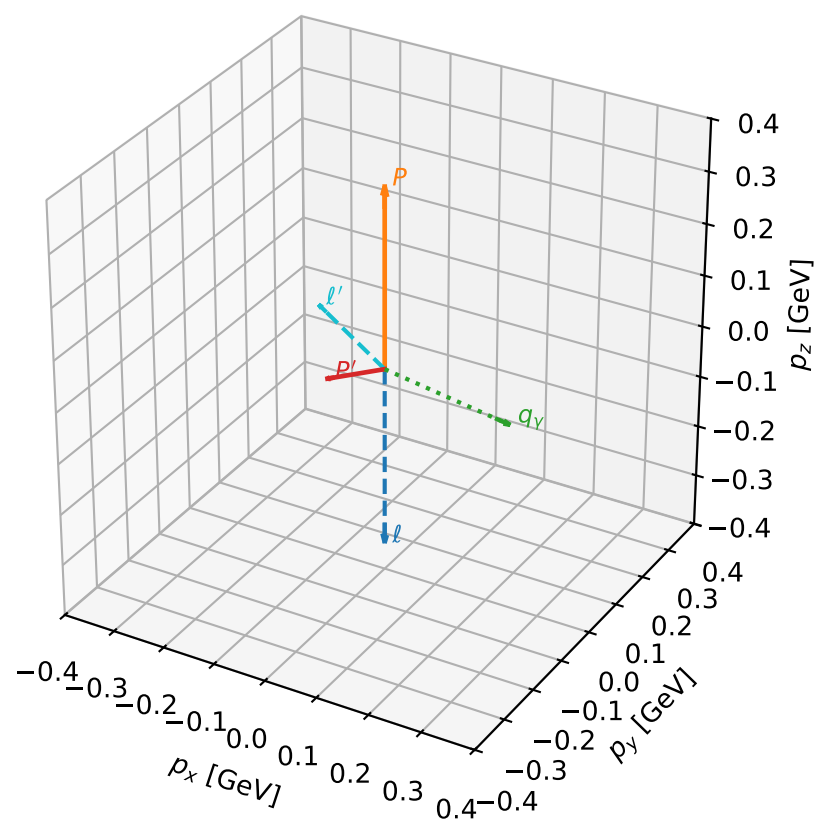
coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

dghz_mixing_angles_polarization_02_local_minimum_318 $d_{\{\text{rm GHZ}\}}=0.00040744$
 region: polarization_cluster_02_local_minimum_318
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=0.67085353$ rad
 incoming lepton: $0.783291|+\rangle +0.621655|-\rangle$
 proton mixing angle: $\alpha_p=2.1407506$ rad
 incoming proton: $-0.539594|+\rangle +0.841926|-\rangle$

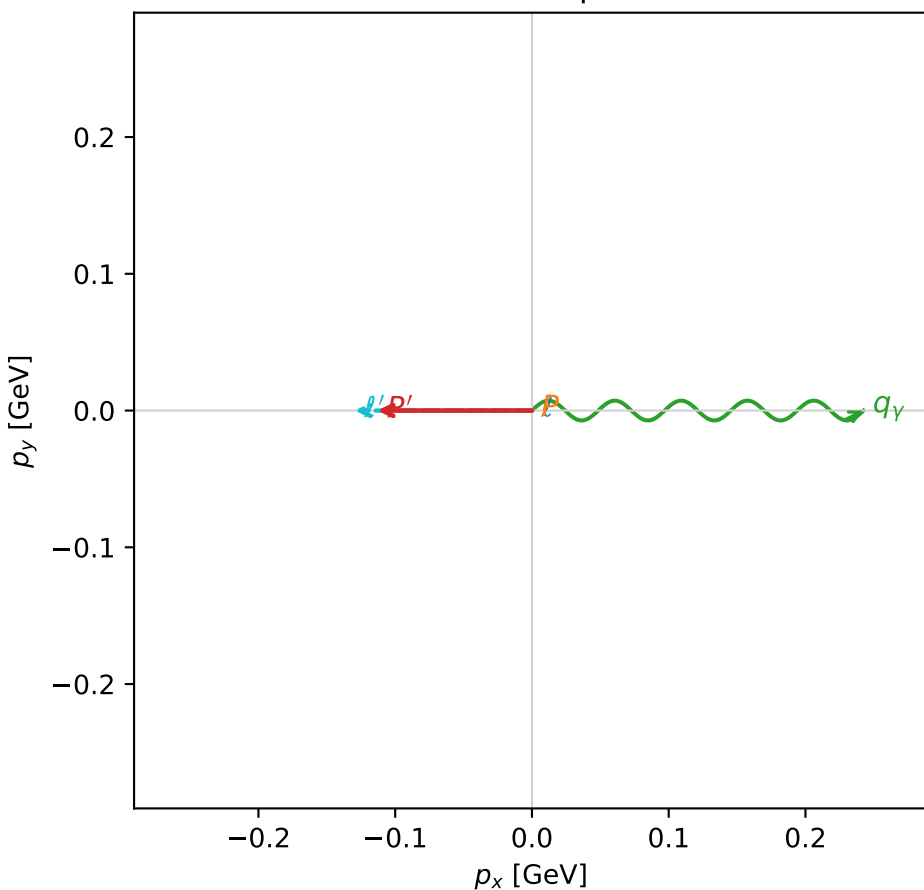
$s=1.81929$, $\sqrt{s}=1.34881$, $|\vec{P}|=0.34825$, $|\vec{P}'|=0.124119$
 $\theta_{P'}=1.99762$, $\theta_Y=1.72571$, $\phi_{P'}=3.14155$, $\phi_Y=6.28317$
 $E_Y=0.245388$, $\theta_{\ell'}=0.967264$, $\phi_{\ell'}=3.1416$
 $Q^2=0.171681$, $x_B=0.249922$, $t=-0.169514$, $W^2=1.3951$, $y=0.731215$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.3483$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.3482$
 P $E= 1.0006$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.3482$
 ℓ' $E= 0.1572$ $p_x= -0.1295$ $p_y= -0.0000$ $p_z= 0.0892$
 P' $E= 0.9462$ $p_x= -0.1130$ $p_y= 0.0000$ $p_z= -0.0514$
 q_Y $E= 0.2454$ $p_x= 0.2424$ $p_y= -0.0000$ $p_z= -0.0379$

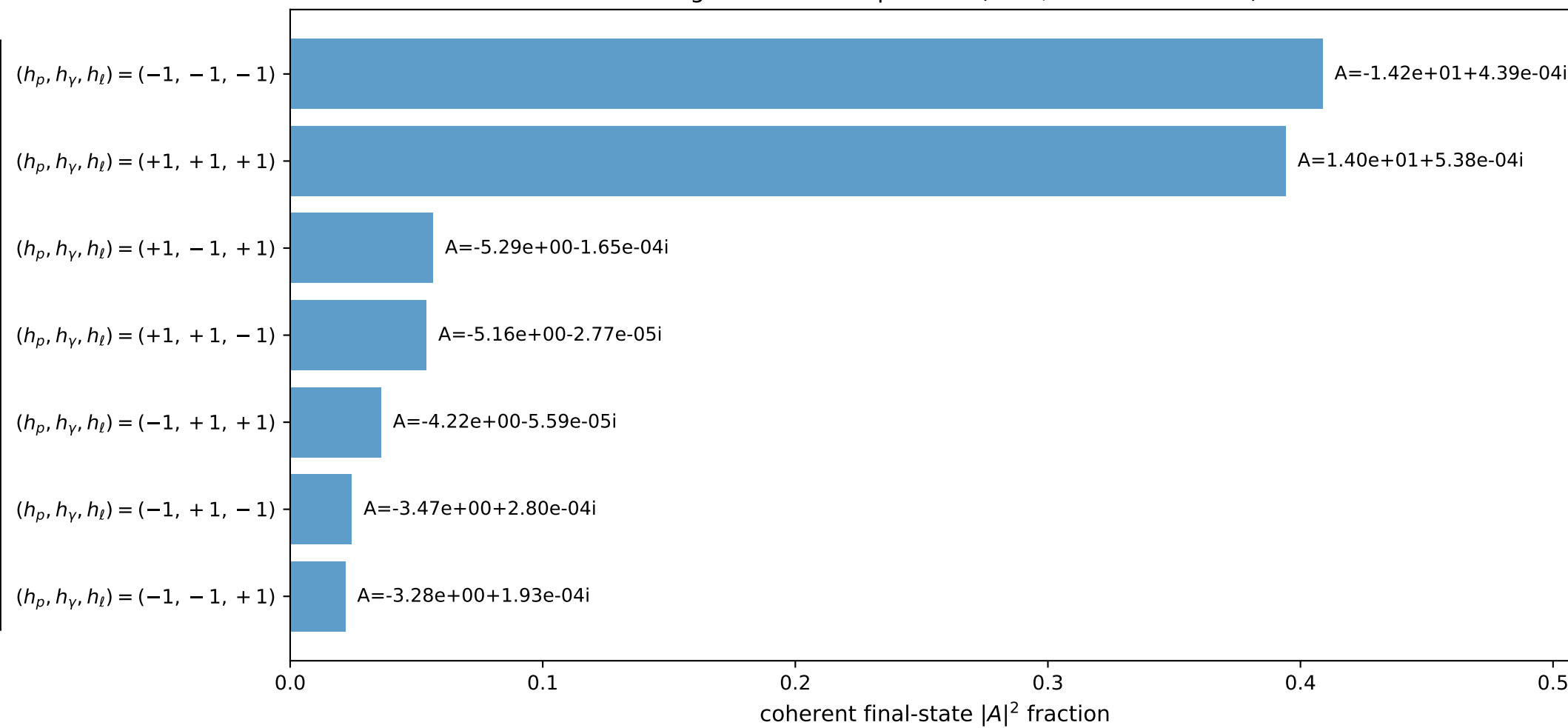
3D momenta



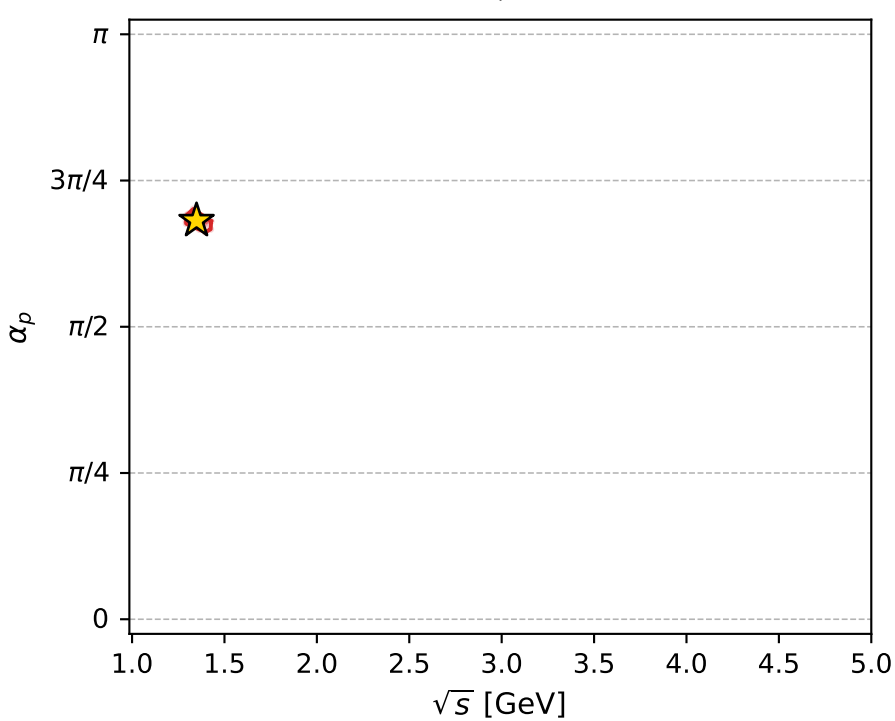
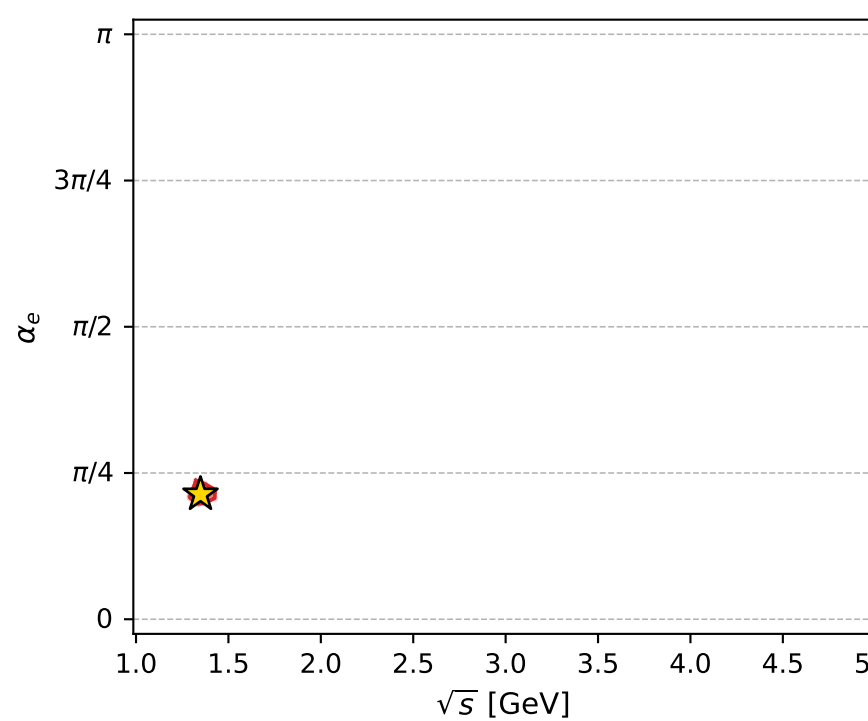
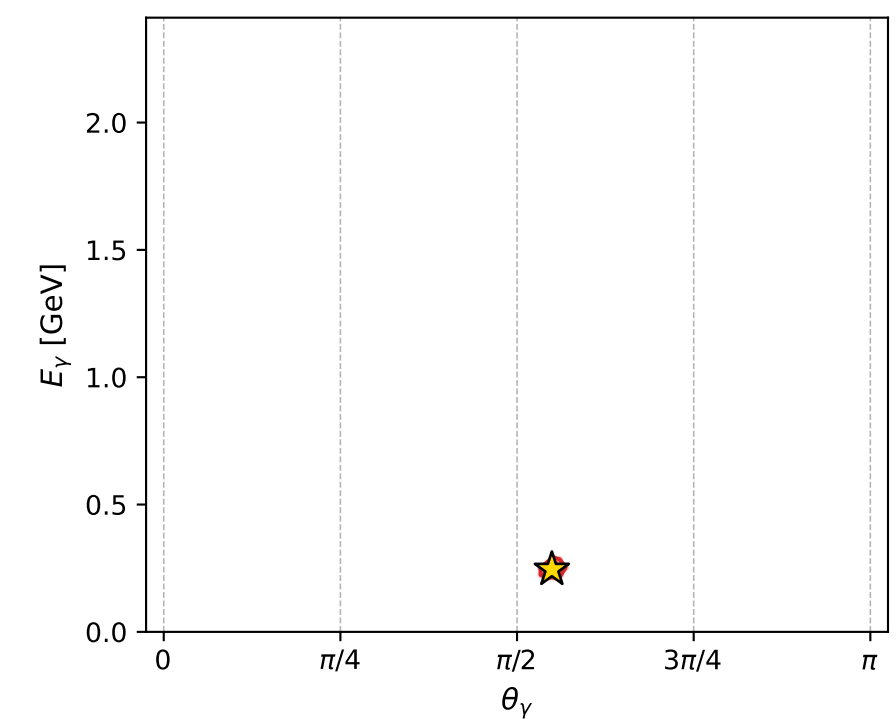
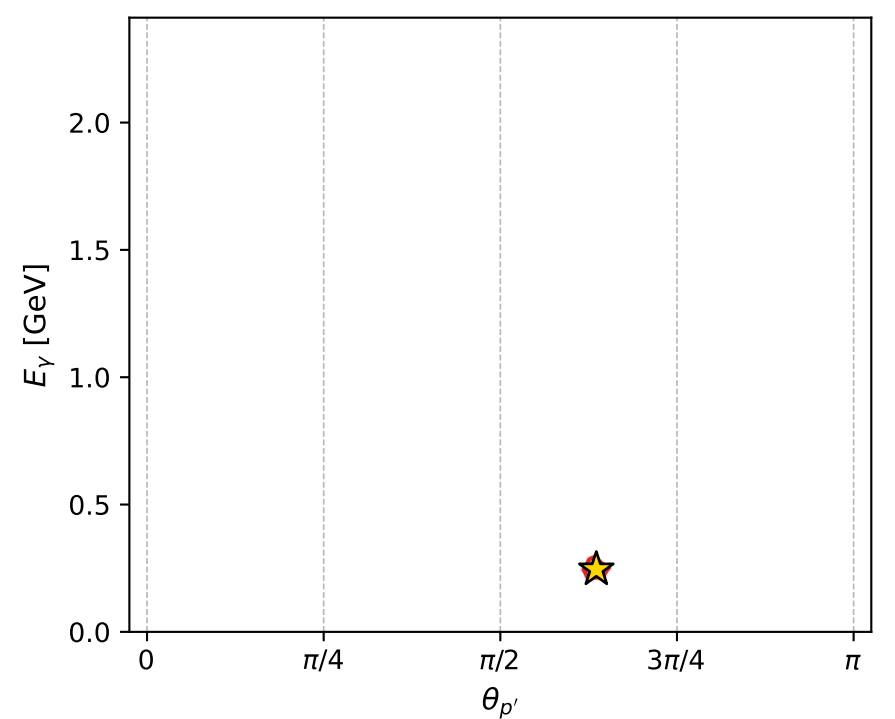
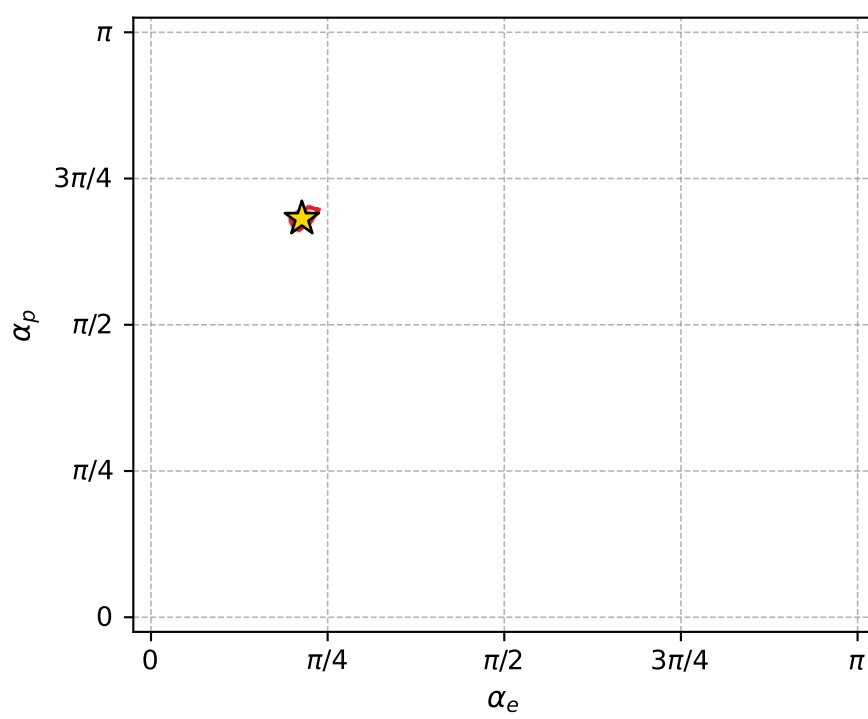
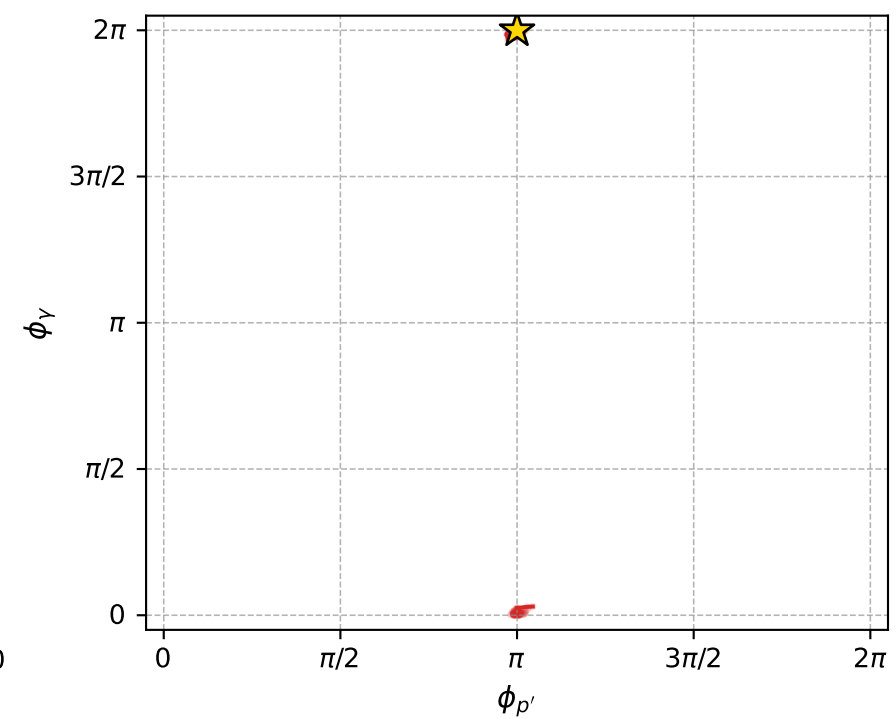
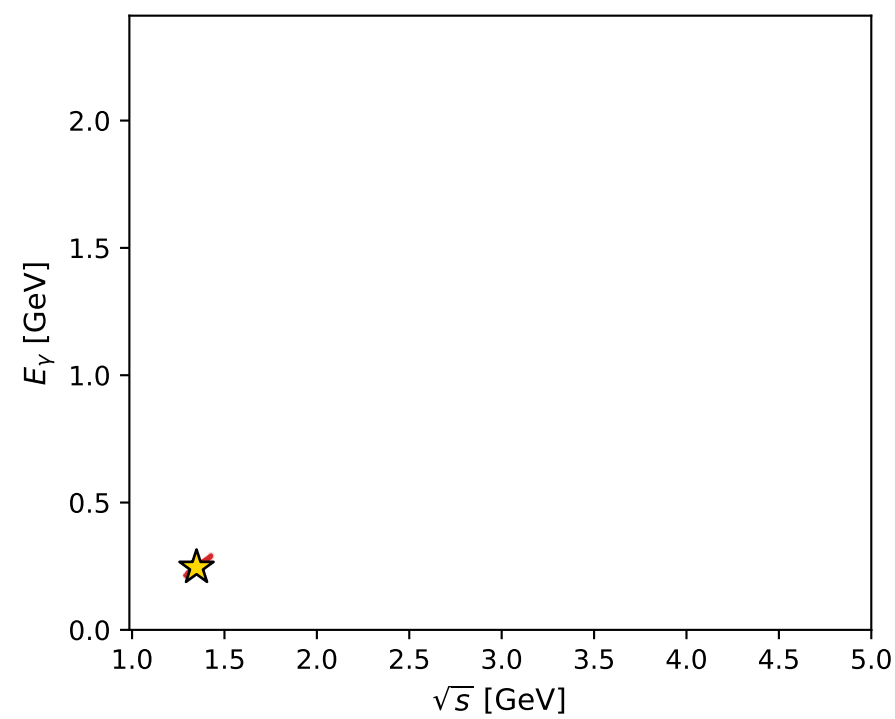
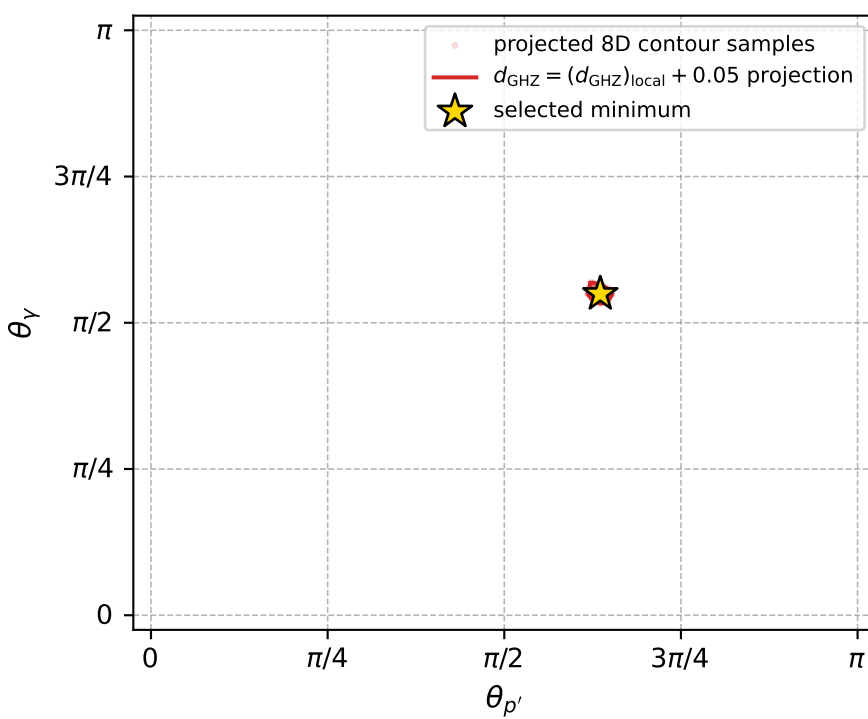
Transverse plane



Leading final-state amplitudes (N=7, retained=99.5%)



electron: polarization cluster P2, local minimum 318; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 0.00040744$
 $d_{\text{GHZ}} \text{ contour} = 0.05040744$
 above global minimum = 0.00040741771
 8D contour samples = 256

selected phase-space point:

$\text{sqrt_s} = 1.348811$
 $\text{theta_p_out} = 1.99762$
 $\text{theta_gamma_out} = 1.725709$
 $\text{q0out} = 0.2453884$
 $\text{phi_p_out} = 3.141552$
 $\text{phi_gamma_out} = 6.28317$
 $\text{alpha_e} = 0.6708535$
 $\text{alpha_p} = 2.140751$

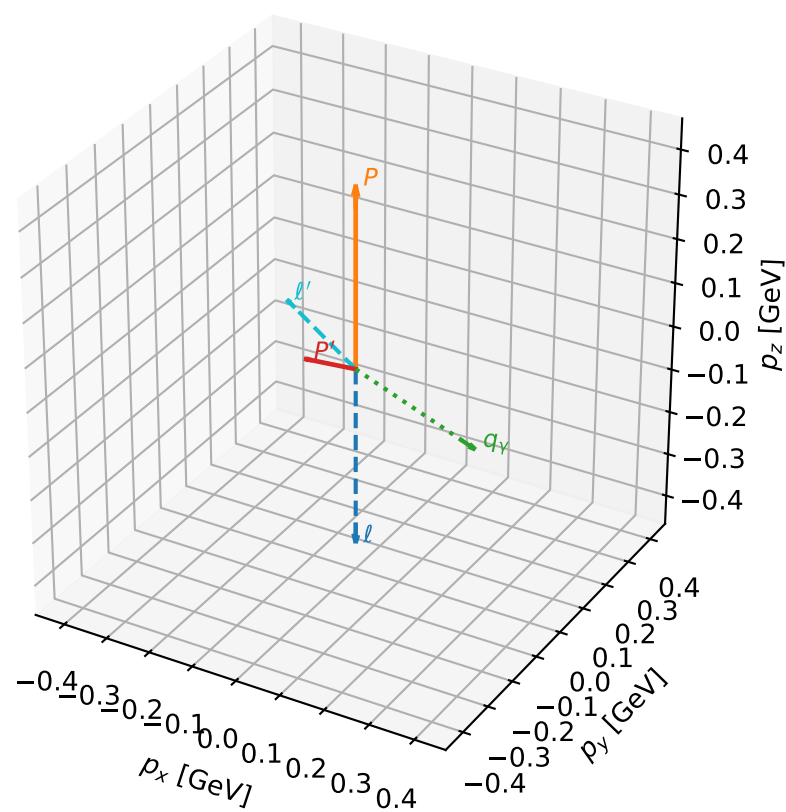
coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

dghz_mixing_angles_polarization_02_local_minimum_321 $d_{\{\text{rm GHZ}\}}=0.000422393$
 region: polarization_cluster_02_local_minimum_321
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=0.67740665$ rad
 incoming lepton: $0.779201|+\rangle +0.626774|-\rangle$
 proton mixing angle: $\alpha_p=2.1351082$ rad
 incoming proton: $-0.534834|+\rangle +0.844957|-\rangle$

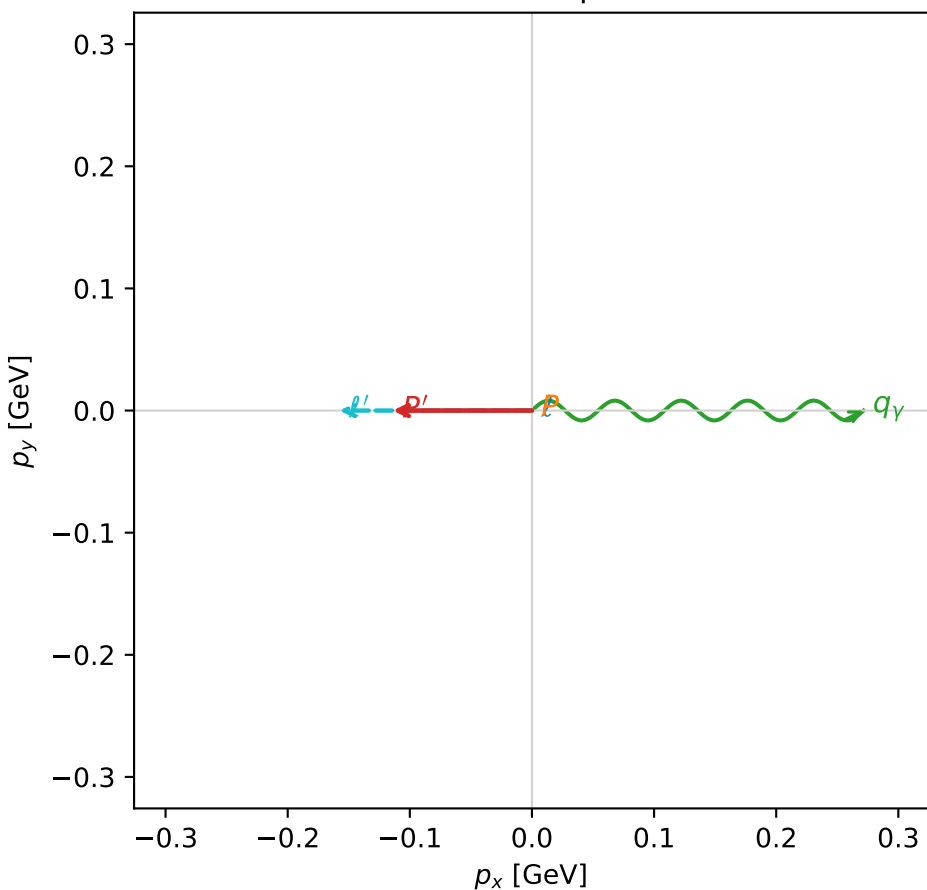
$s=2.04476$, $\sqrt{s}=1.42995$, $|\vec{P}|=0.407326$, $|\vec{P}'|=0.114295$
 $\theta_{P'}=1.65636$, $\theta_Y=1.93633$, $\phi_{P'}=3.14173$, $\phi_Y=0.000141849$
 $E_Y=0.290688$, $\theta_{\ell'}=0.945948$, $\phi_{\ell'}=3.14174$
 $Q^2=0.250911$, $x_B=0.291731$, $t=-0.1809$, $W^2=1.48901$, $y=0.738319$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.4073$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.4073$
 P $E= 1.0226$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.4073$
 ℓ' $E= 0.1943$ $p_x= -0.1576$ $p_y= -0.0000$ $p_z= 0.1137$
 P' $E= 0.9449$ $p_x= -0.1139$ $p_y= -0.0000$ $p_z= -0.0098$
 q_Y $E= 0.2907$ $p_x= 0.2715$ $p_y= 0.0000$ $p_z= -0.1039$

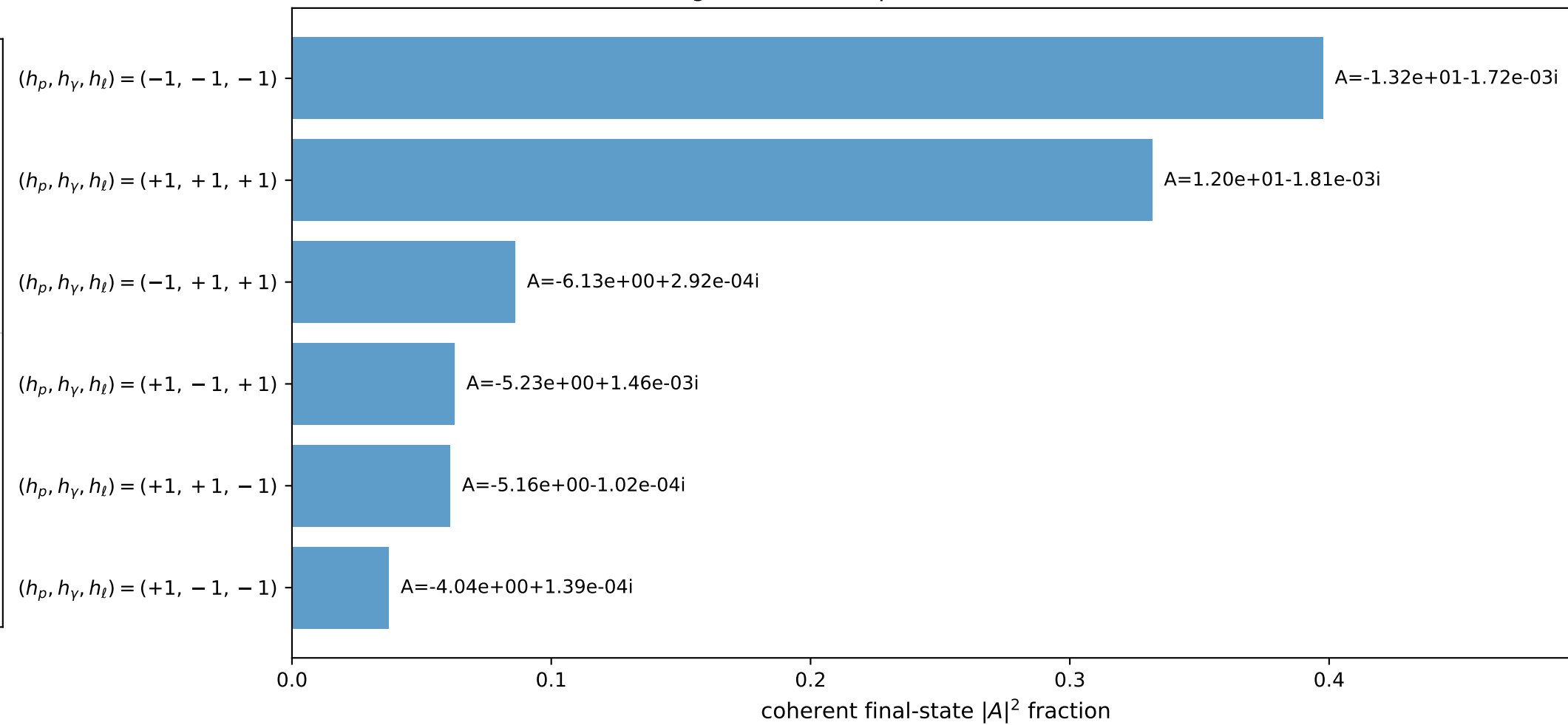
3D momenta



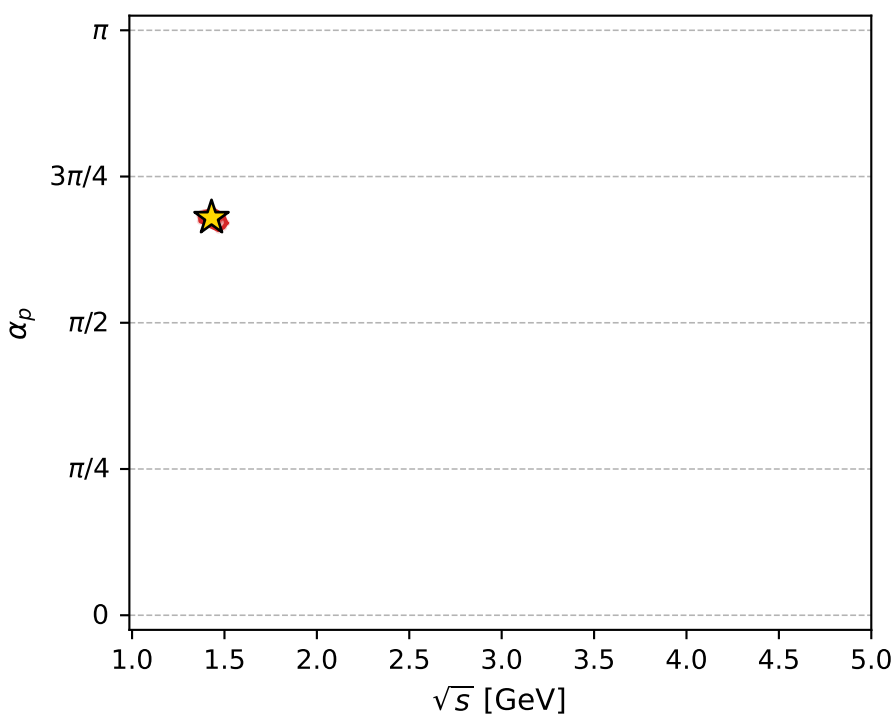
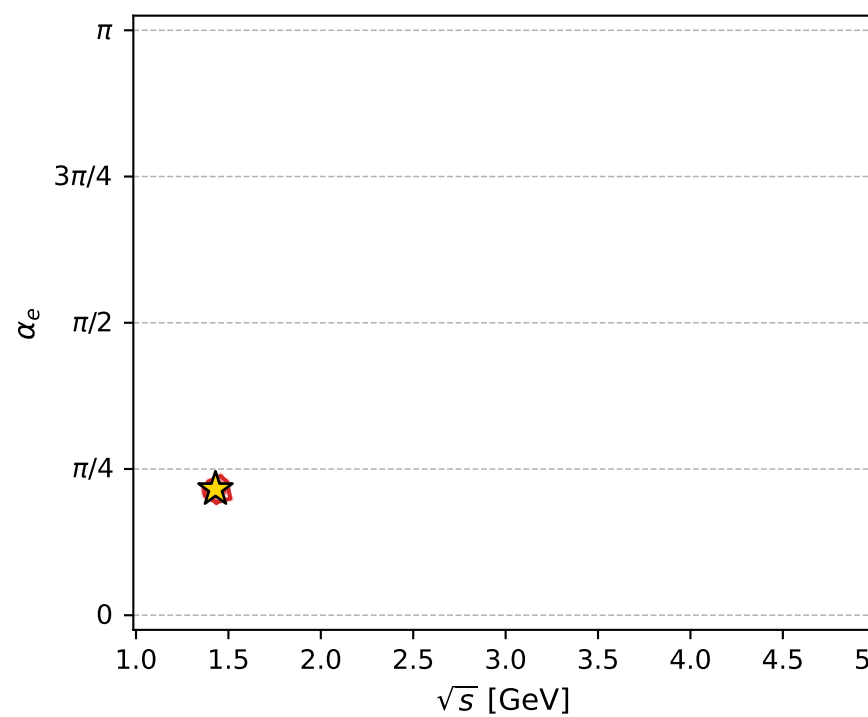
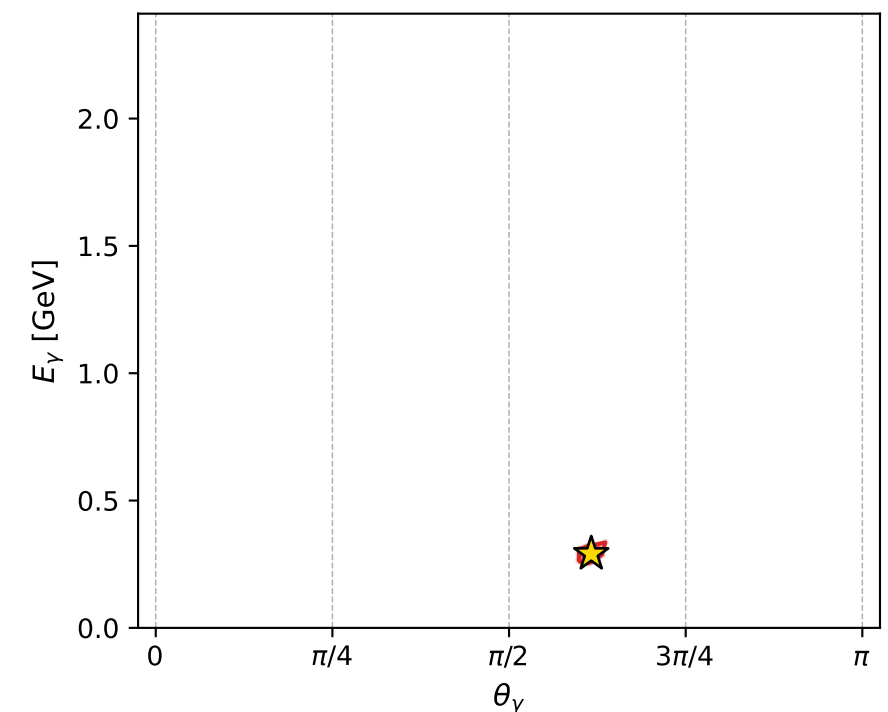
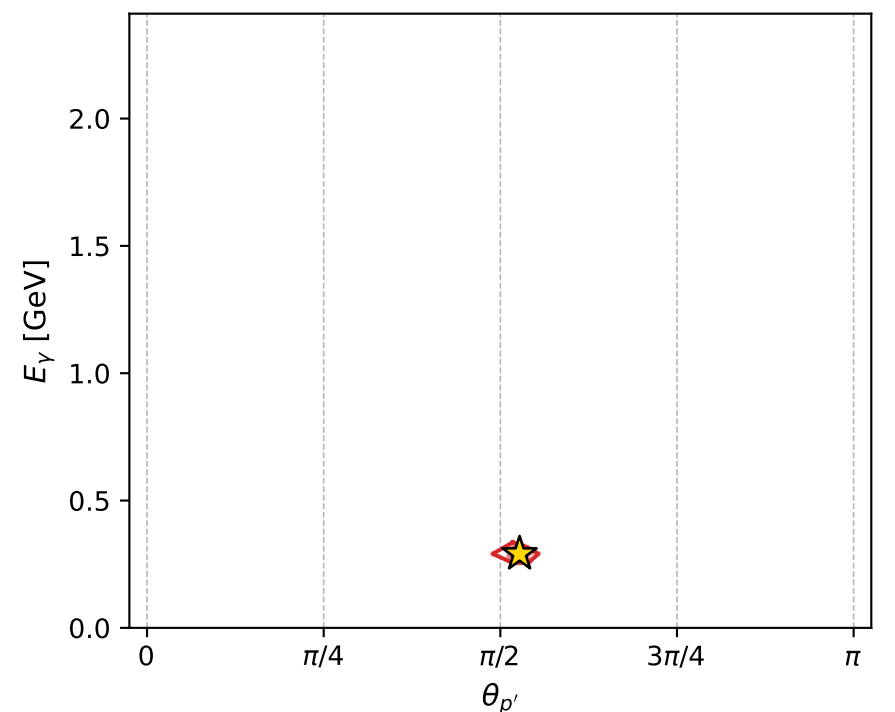
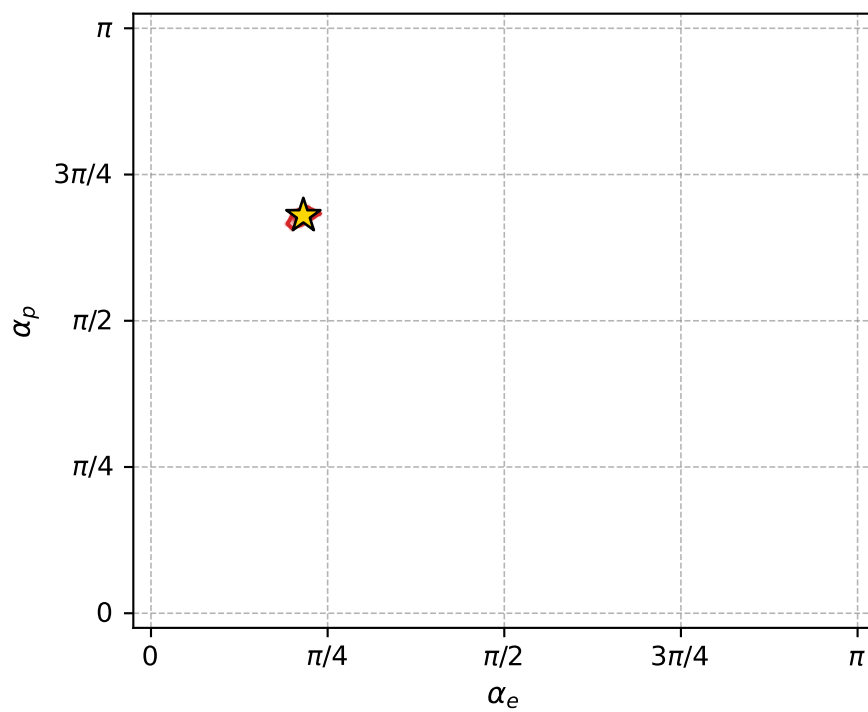
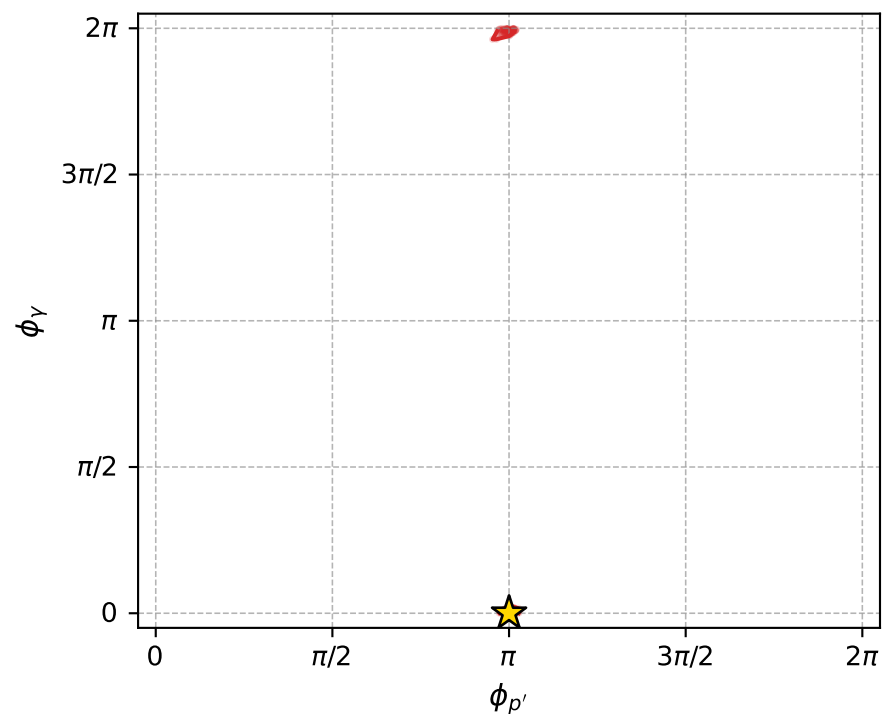
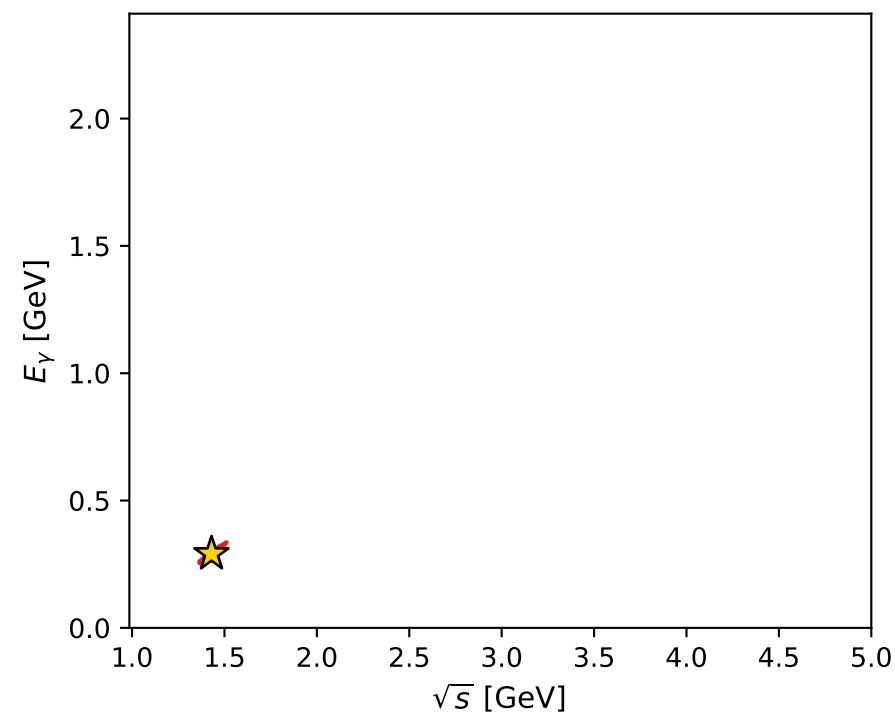
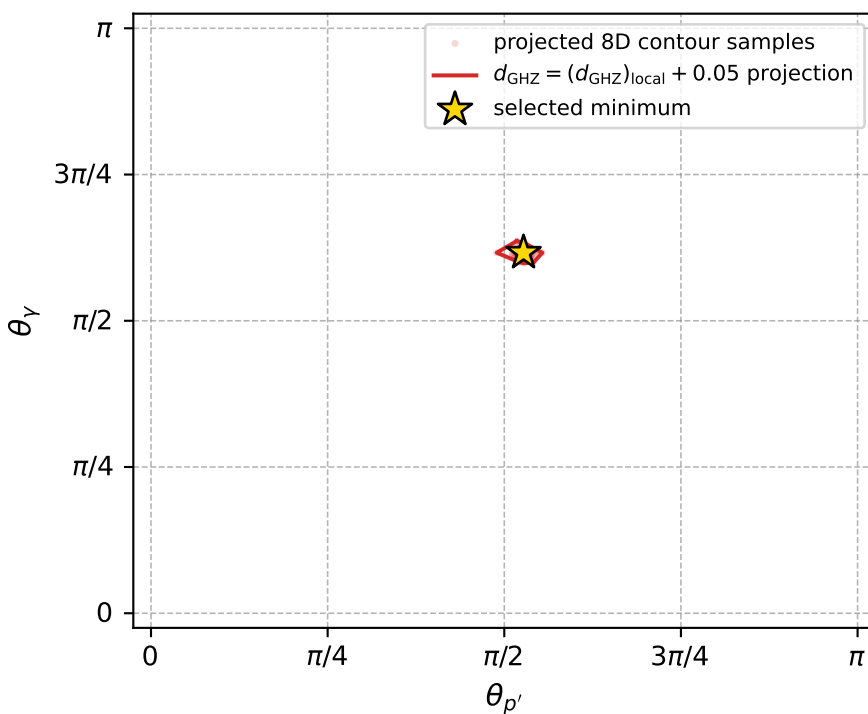
Transverse plane



Leading final-state amplitudes (N=6, retained=97.6%)



electron: polarization cluster P2, local minimum 321; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 0.00042239287$
 $d_{\text{GHZ}} \text{ contour} = 0.050422393$
 above global minimum = 0.00042237058
 8D contour samples = 256

selected phase-space point:

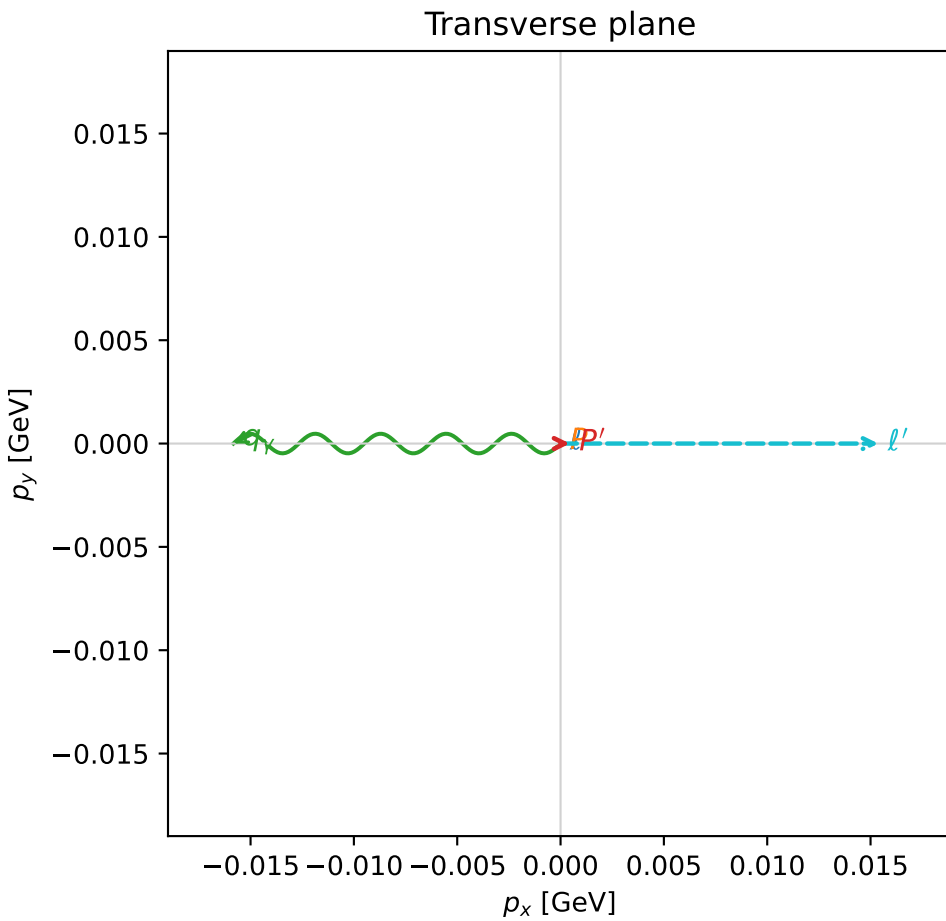
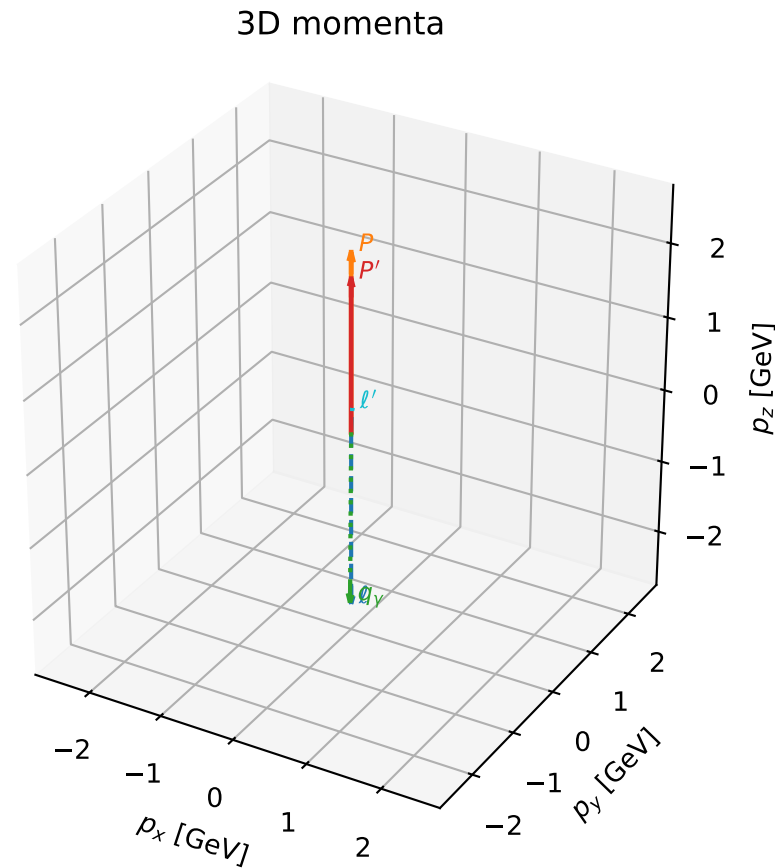
$\text{sqrt_s} = 1.42995$
 $\text{theta_p_out} = 1.656364$
 $\text{theta_gamma_out} = 1.936331$
 $\text{q0out} = 0.290688$
 $\text{phi_p_out} = 3.141728$
 $\text{phi_gamma_out} = 0.0001418489$
 $\text{alpha_e} = 0.6774067$
 $\text{alpha_p} = 2.135108$

coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

dghz_mixing_angles_polarization_02_local_minimum_323 $d_{\text{GHZ}}=0.000435487$
region: polarization_cluster_02_local_minimum_323
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: $\alpha_e=0.69364314$ rad
incoming lepton: $0.768922|+\rangle + 0.639343|-\rangle$
proton mixing angle: $\alpha_p=2.2770928$ rad
incoming proton: $-0.649021|+\rangle + 0.760771|-\rangle$

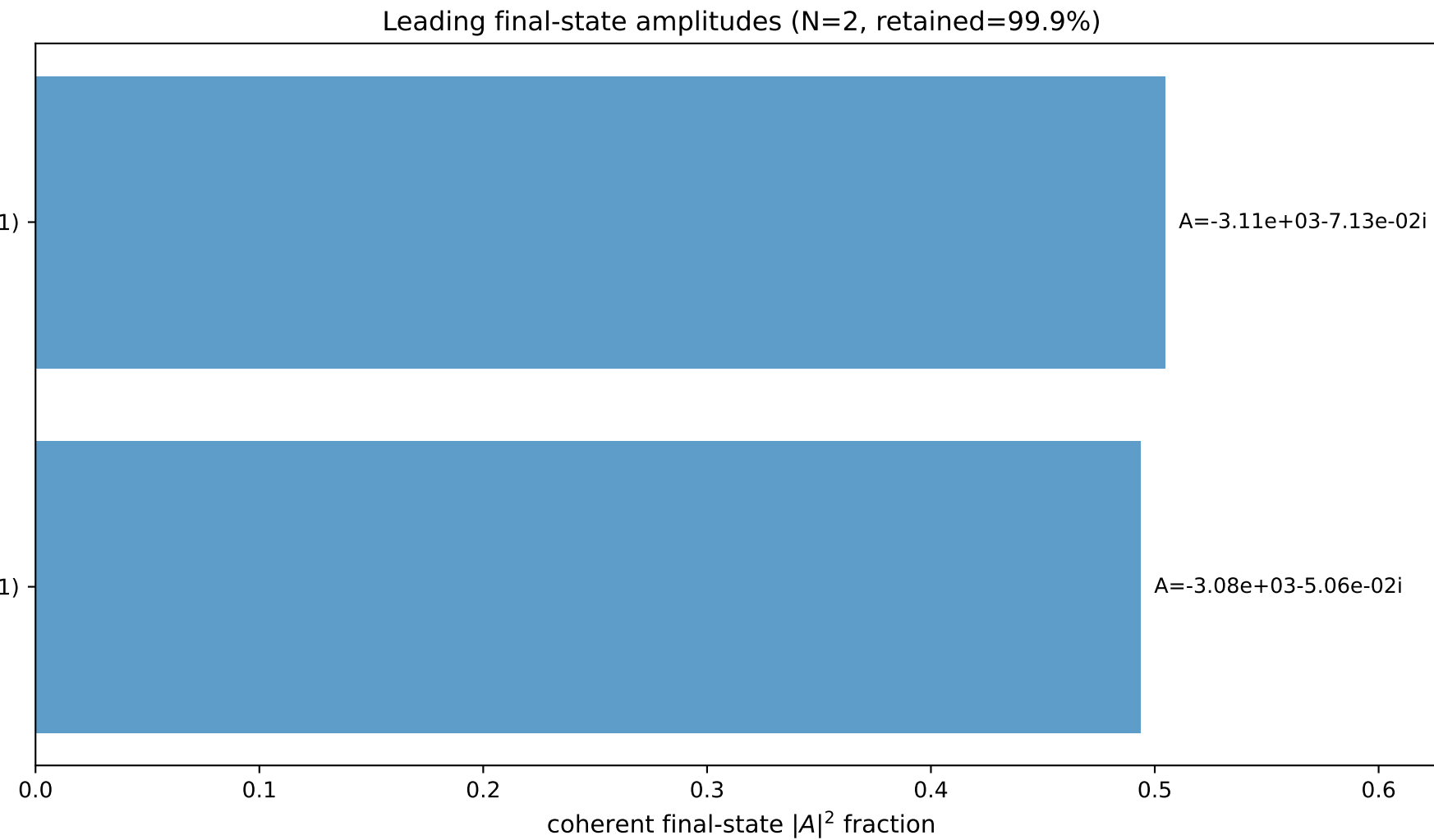
$s=25$, $\sqrt{s}=5$, $|\vec{P}|=2.41202$, $|\vec{P}'|=2.06443$
 $\theta_{p'}=0.000218156$, $\theta_Y=3.13499$, $\phi_{p'}=0.00267229$, $\phi_Y=3.14155$
 $E_Y=2.3983$, $\theta_{\ell'}=0.0460371$, $\phi_{\ell'}=6.28306$
 $Q^2=3.22234$, $x_B=0.134259$, $t=-0.0181278$, $W^2=21.6583$, $y=0.995053$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4120$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 P $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4120$
 ℓ' $E= 0.3342$ $p_x= 0.0154$ $p_y= -0.0000$ $p_z= 0.3338$
 P' $E= 2.2675$ $p_x= 0.0005$ $p_y= 0.0000$ $p_z= 2.0644$
 q_Y $E= 2.3983$ $p_x= -0.0158$ $p_y= 0.0000$ $p_z= -2.3982$

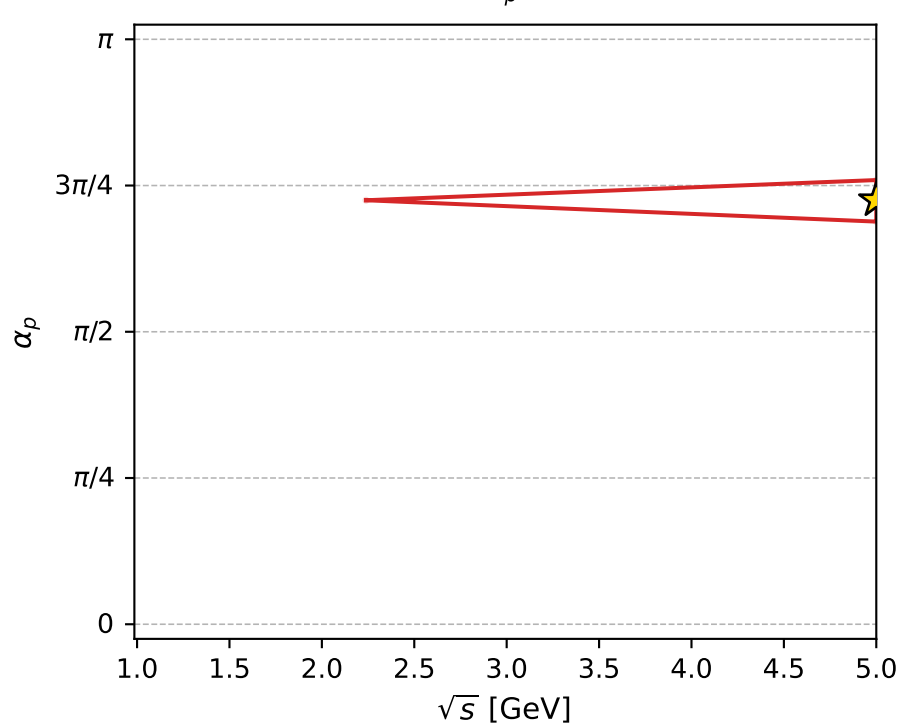
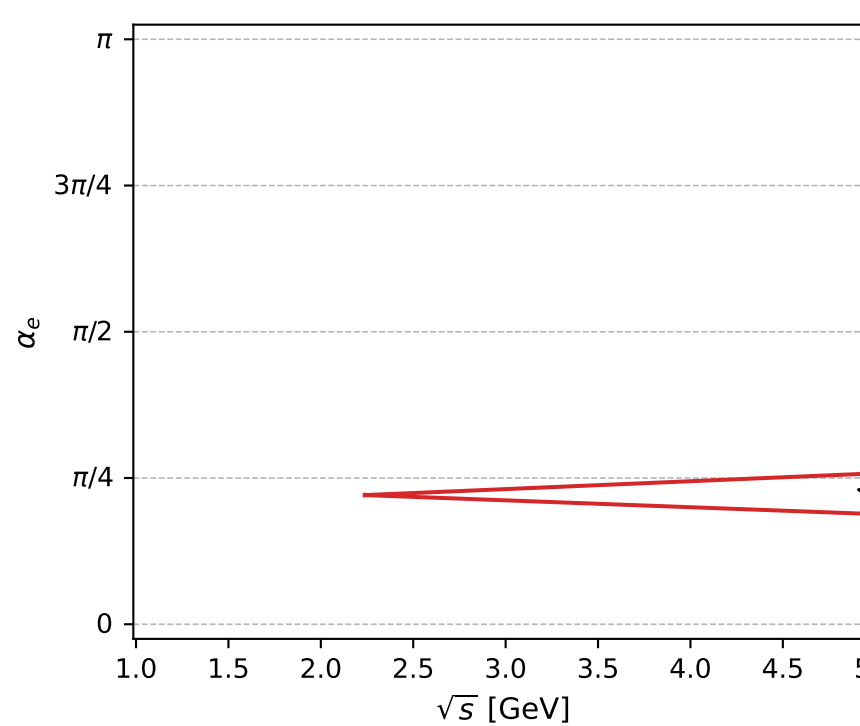
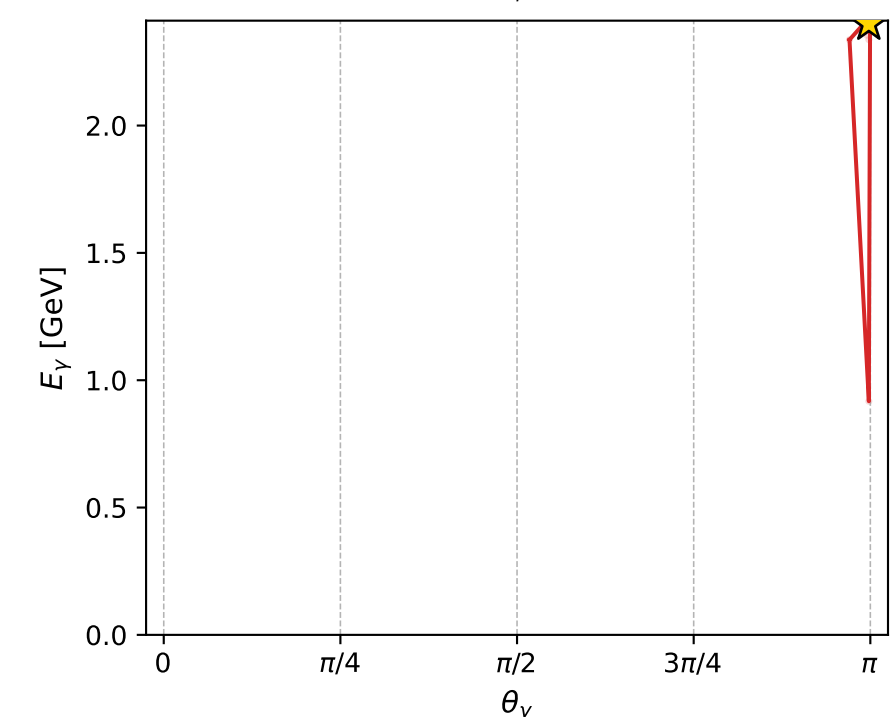
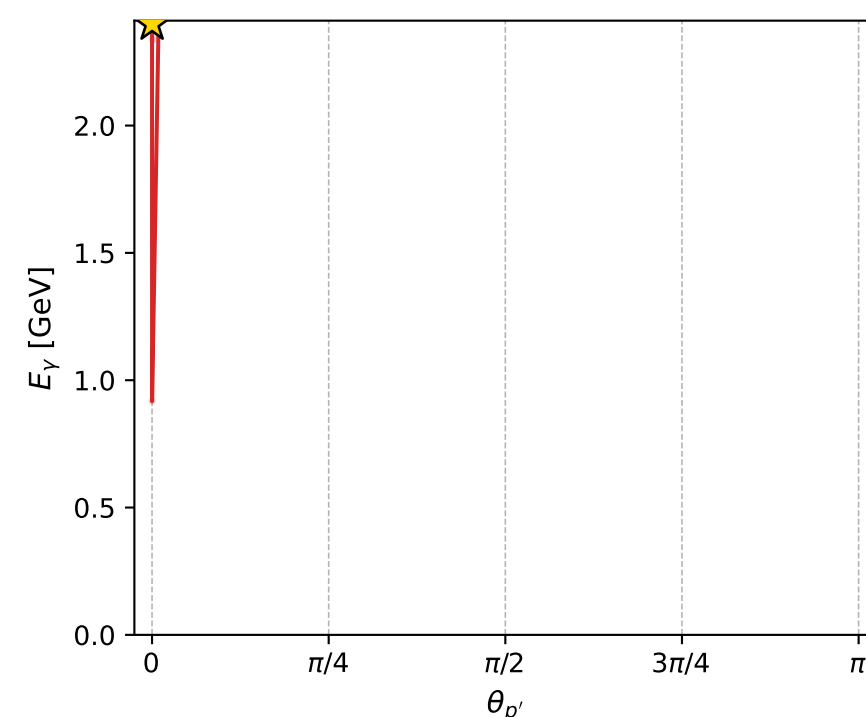
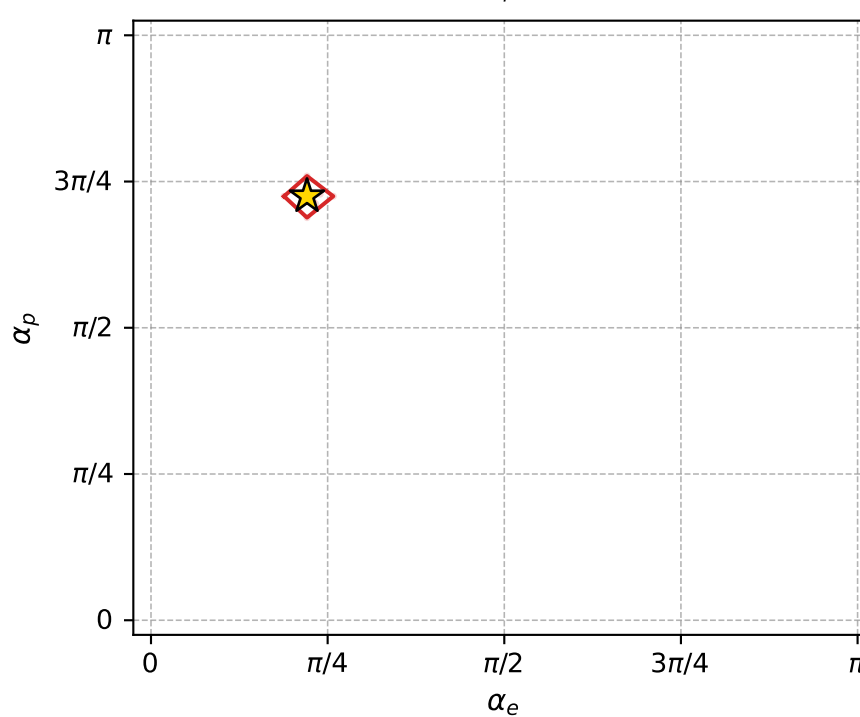
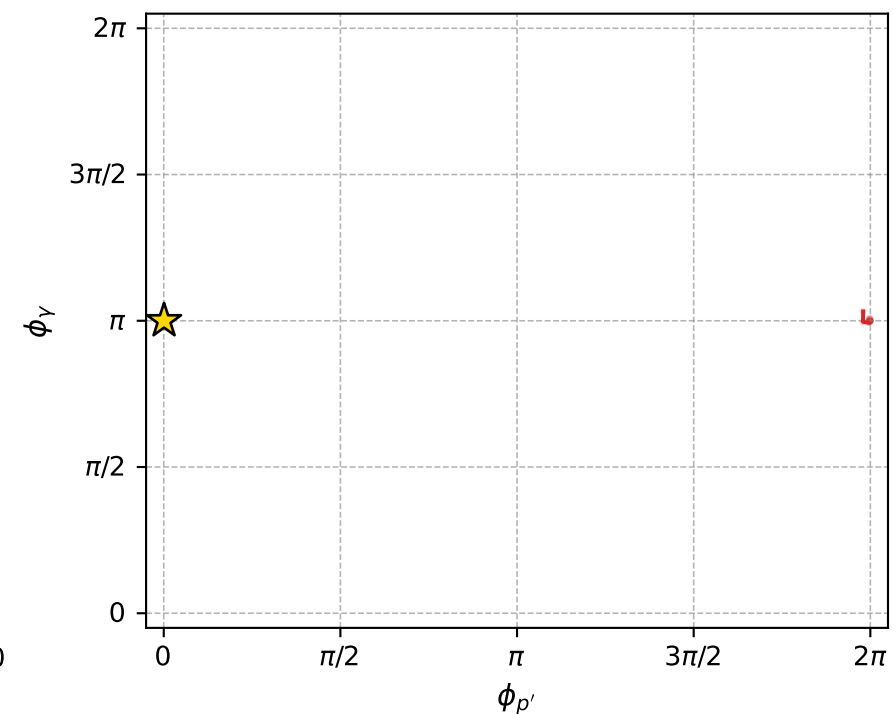
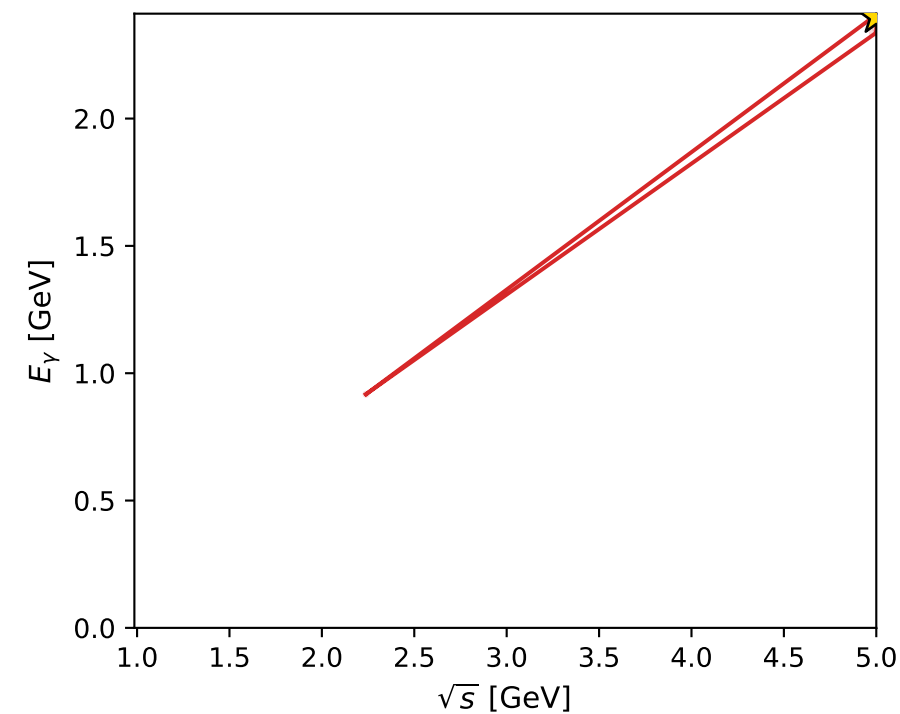
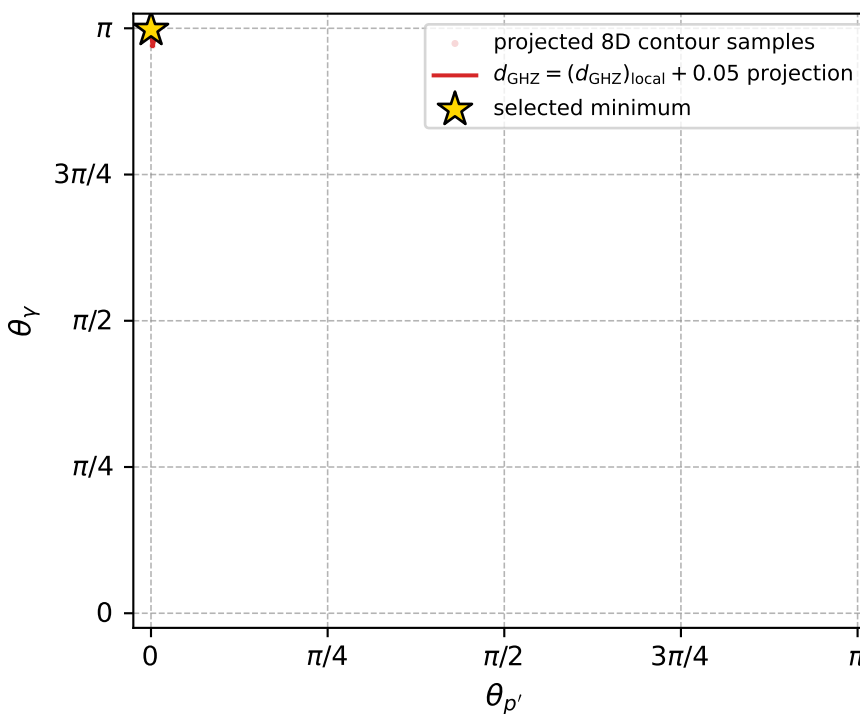


$(h_p, h_Y, h_{\ell}) = (-1, +1, +1)$

$(h_p, h_Y, h_{\ell}) = (+1, -1, -1)$



electron: polarization cluster P2, local minimum 323; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 0.00043548689$
 $d_{\text{GHZ}} \text{ contour} = 0.050435487$
 above global minimum = 0.00043546461
 8D contour samples = 78

selected phase-space point:

$\text{sqrt_s} = 5$
 $\text{theta_p_out} = 0.0002181565$
 $\text{theta_gamma_out} = 3.134993$
 $\text{q0out} = 2.398296$
 $\text{phi_p_out} = 0.002672285$
 $\text{phi_gamma_out} = 3.141548$
 $\text{alpha_e} = 0.6936431$
 $\text{alpha_p} = 2.277093$

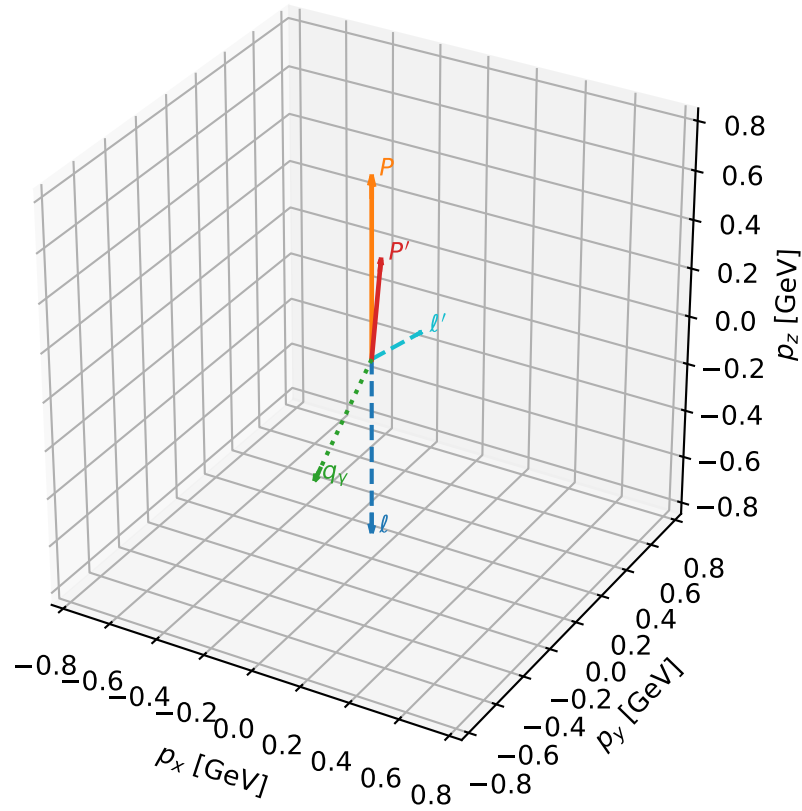
coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

dghz_mixing_angles_polarization_02_local_minimum_326 $d_{\{\text{rm GHZ}\}}=0.000464566$
 region: polarization_cluster_02_local_minimum_326
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=0.80010297$ rad
 incoming lepton: $0.696633|+\rangle +0.717428|-\rangle$
 proton mixing angle: $\alpha_p=2.4935623$ rad
 incoming proton: $-0.797274|+\rangle +0.603617|-\rangle$

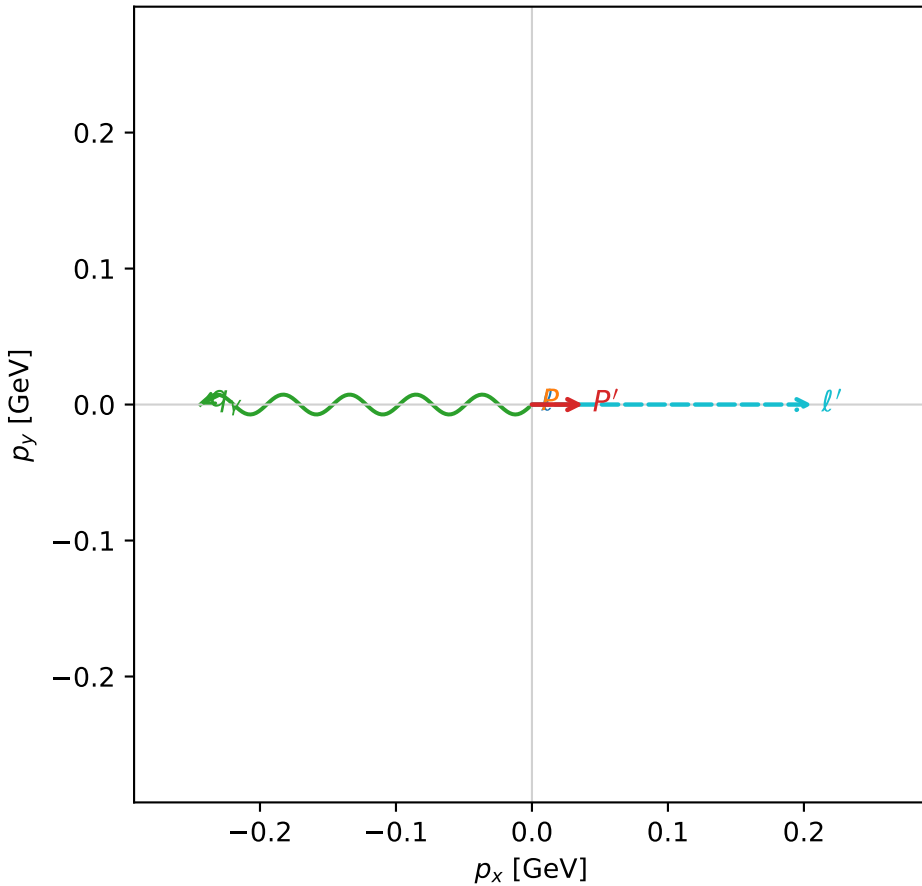
$s=3.73644$, $\sqrt{s}=1.93299$, $|\vec{P}|=0.738907$, $|\vec{P}'|=0.419049$
 $\theta_{P'}=0.0902752$, $\theta_Y=2.7491$, $\phi_{P'}=6.28315$, $\phi_Y=3.14158$
 $E_Y=0.637463$, $\theta_{\ell'}=0.876216$, $\phi_{\ell'}=6.28318$
 $Q^2=0.649975$, $x_B=0.263168$, $t=-0.0770314$, $W^2=2.69968$, $y=0.864601$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.7389$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.7389$
 P $E= 1.1941$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.7389$
 ℓ' $E= 0.2682$ $p_x= 0.2060$ $p_y= -0.0000$ $p_z= 0.1716$
 P' $E= 1.0273$ $p_x= 0.0378$ $p_y= -0.0000$ $p_z= 0.4173$
 q_Y $E= 0.6375$ $p_x= -0.2438$ $p_y= 0.0000$ $p_z= -0.5890$

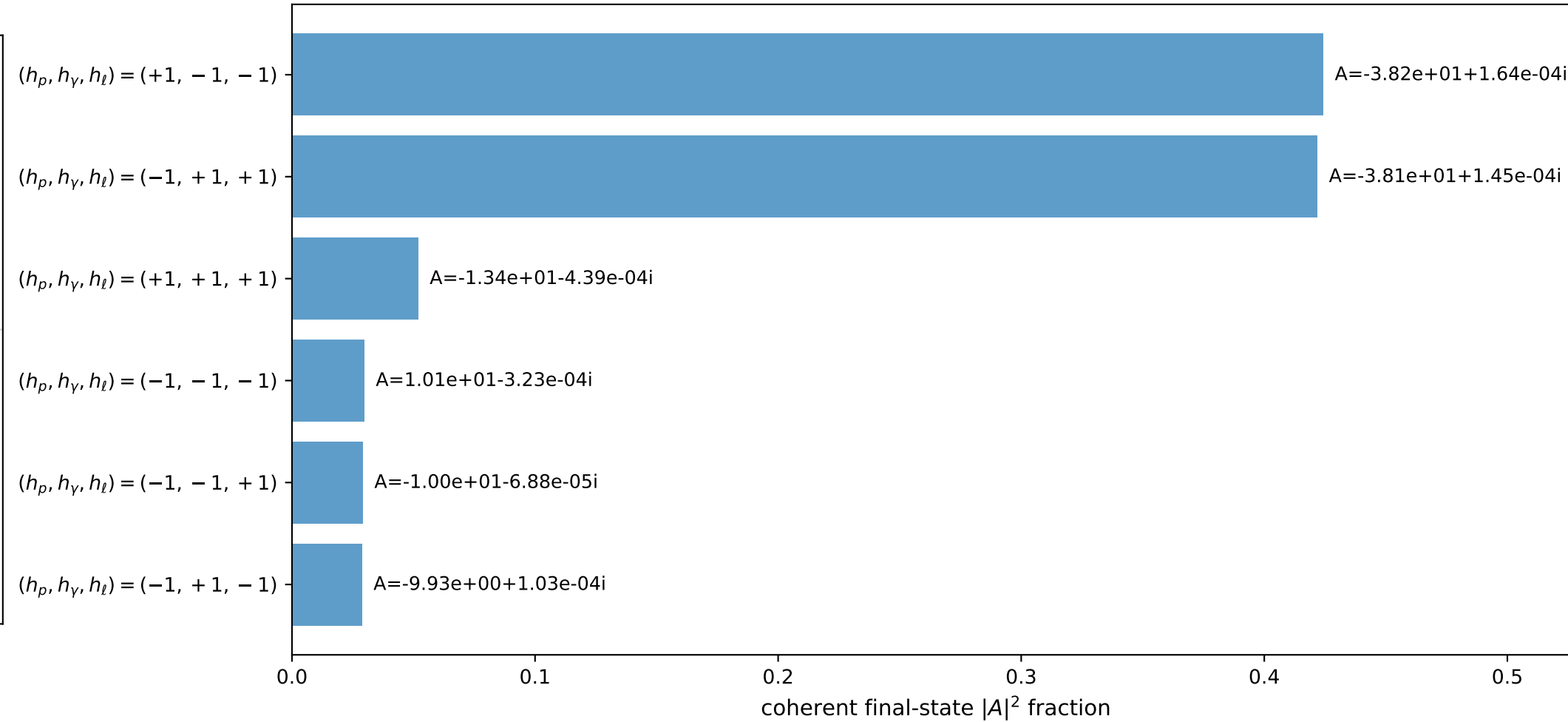
3D momenta



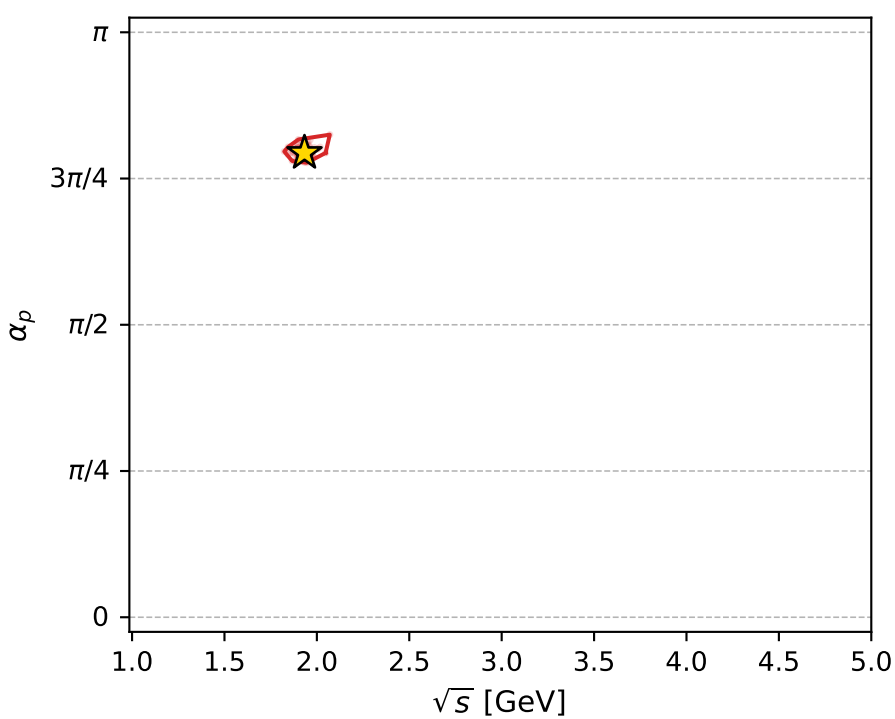
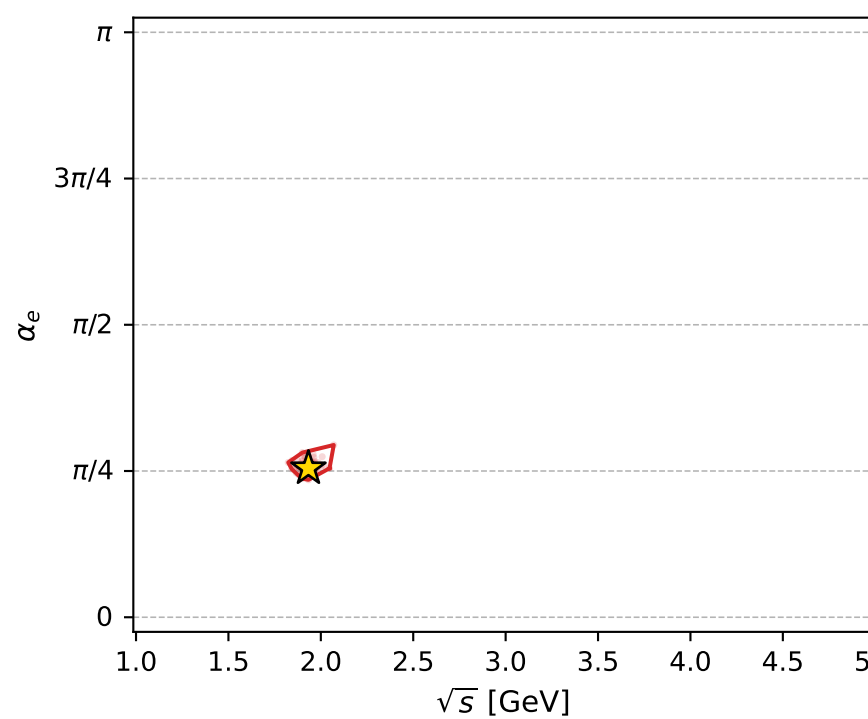
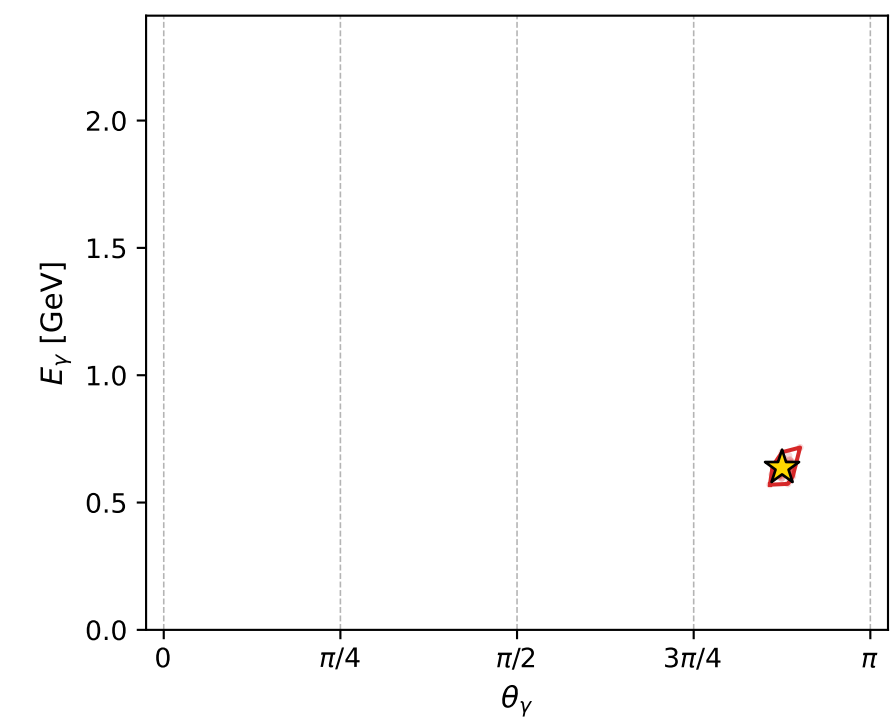
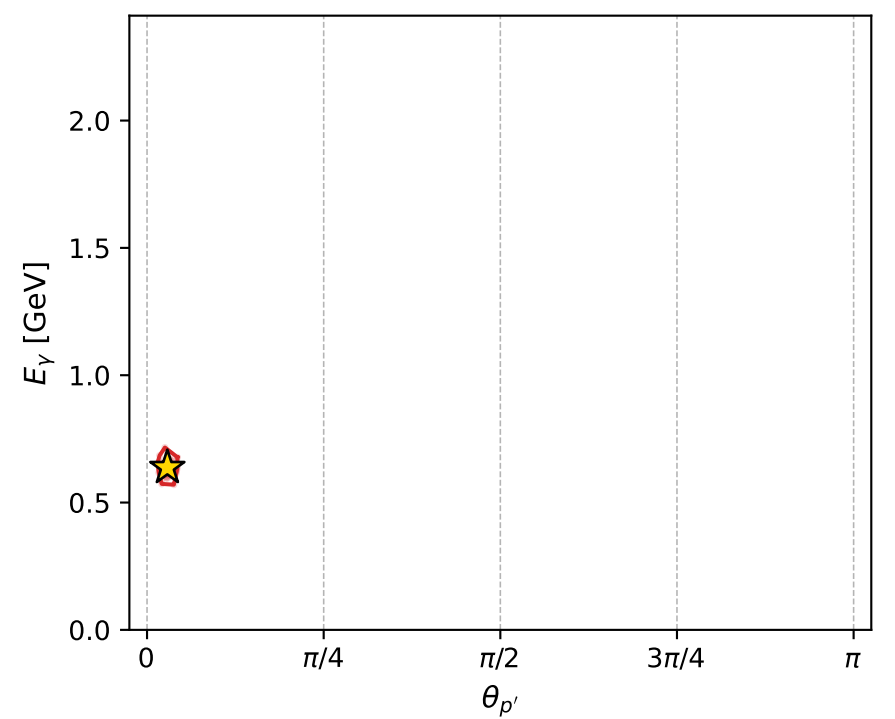
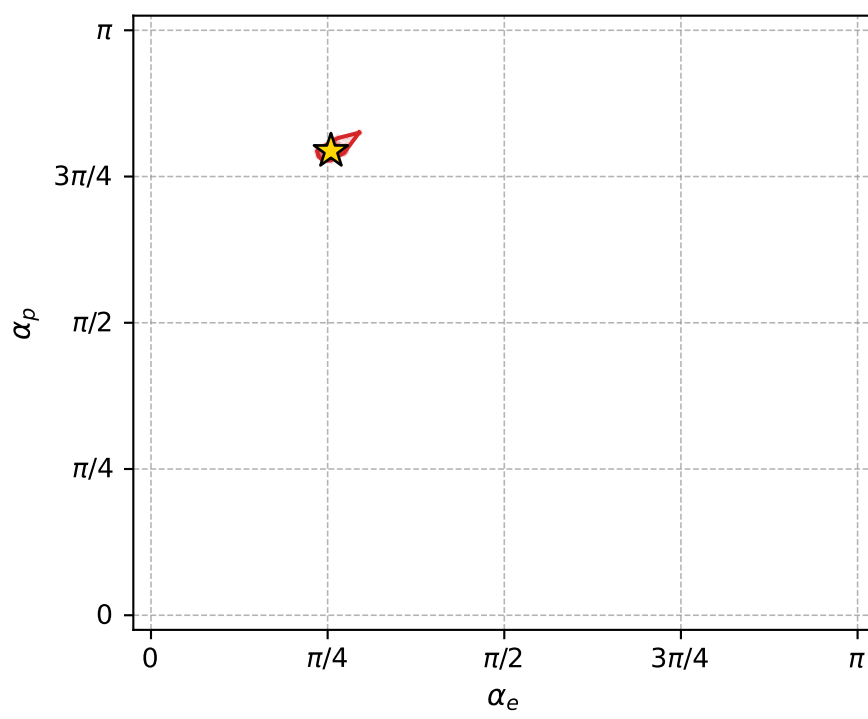
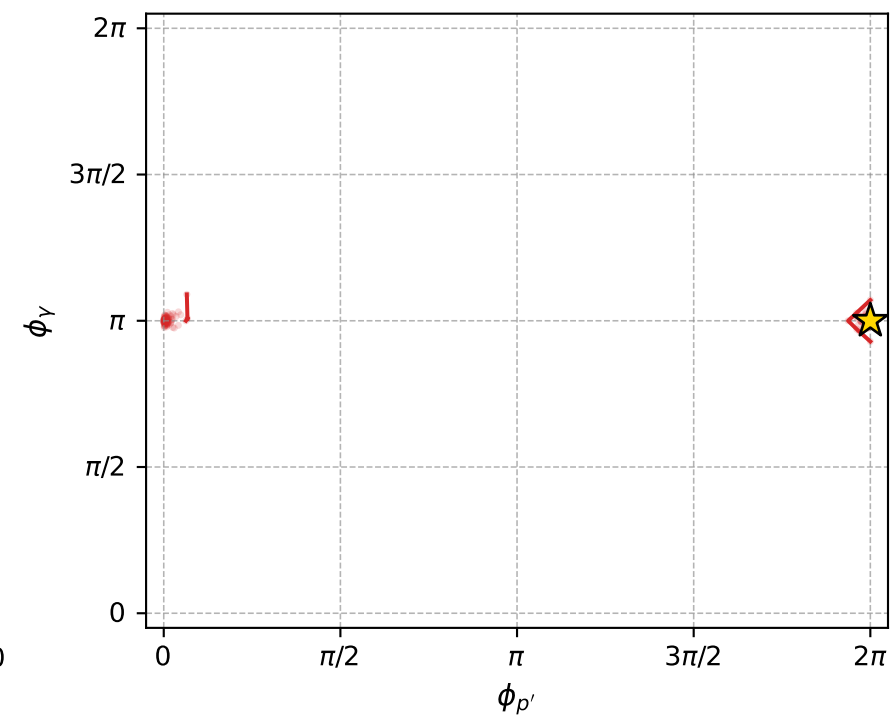
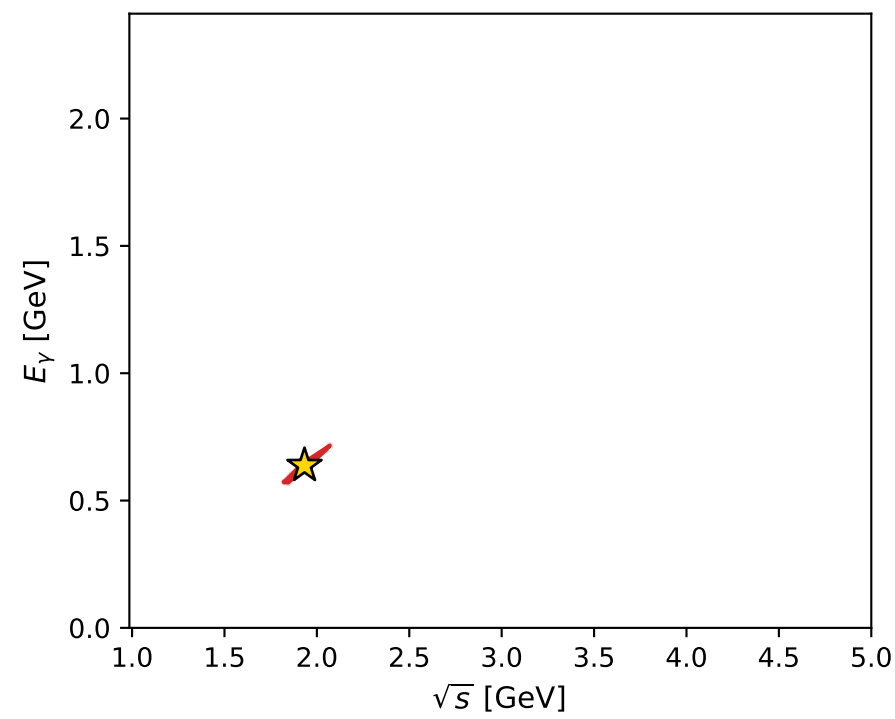
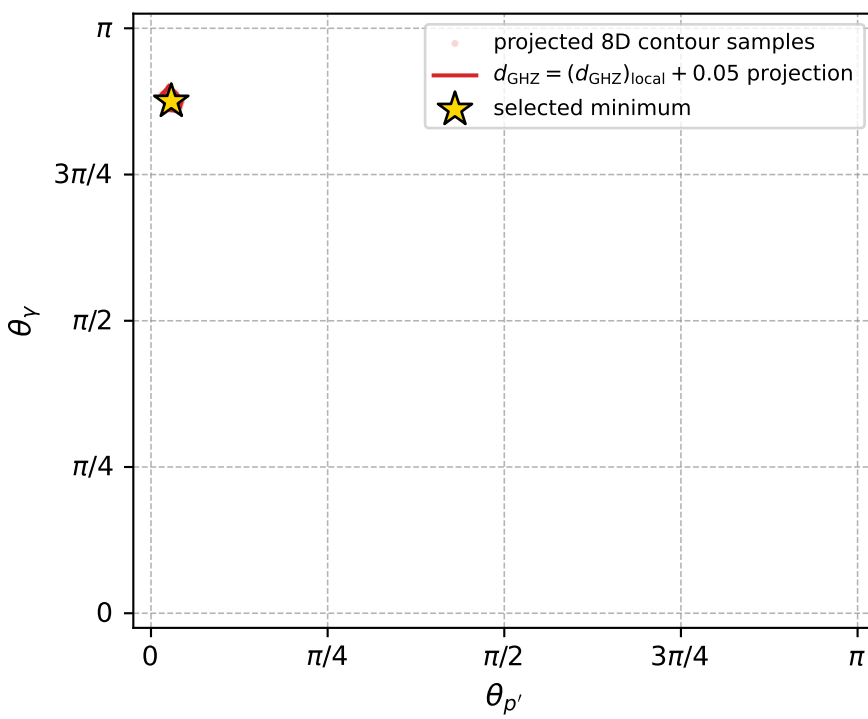
Transverse plane



Leading final-state amplitudes (N=6, retained=98.5%)



electron: polarization cluster P2, local minimum 326; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 0.00046456599$
 $d_{\text{GHZ}} \text{ contour} = 0.050464566$
 above global minimum = 0.0004645437
 8D contour samples = 256

selected phase-space point:
 $\text{sqrt_s} = 1.932987$
 $\text{theta_p_out} = 0.09027522$
 $\text{theta_gamma_out} = 2.749105$
 $\text{q0ut} = 0.6374634$
 $\text{phi_p_out} = 6.283152$
 $\text{phi_gamma_out} = 3.141584$
 $\text{alpha_e} = 0.800103$
 $\text{alpha_p} = 2.493562$

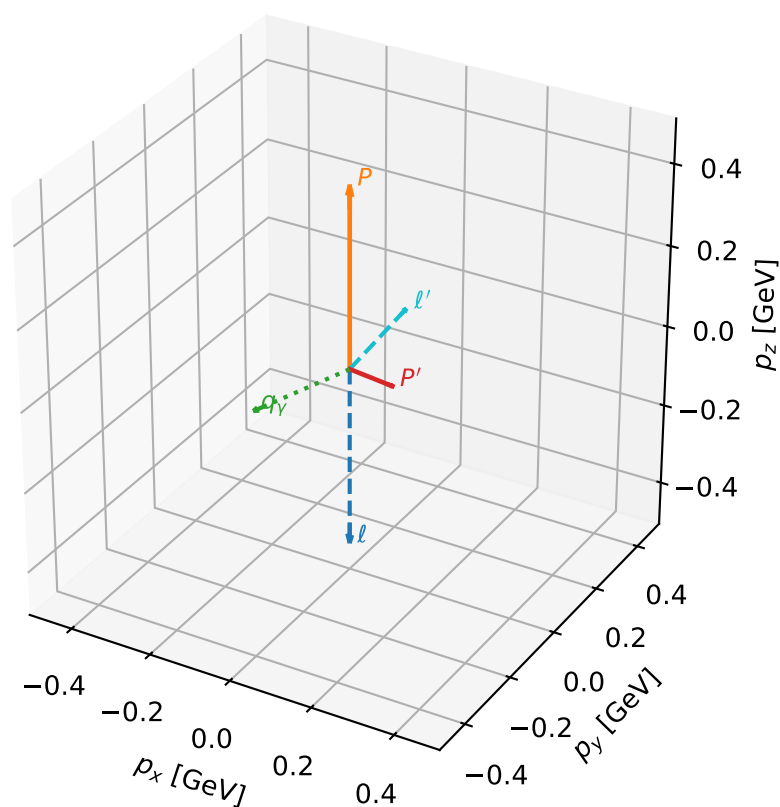
coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

dghz_mixing_angles_polarization_02_local_minimum_327 $d_{\{\text{rm GHZ}\}}=0.000465962$
 region: polarization_cluster_02_local_minimum_327
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=1.1114655$ rad
 incoming lepton: $0.443348|+\rangle +0.896349|-\rangle$
 proton mixing angle: $\alpha_p=2.7577672$ rad
 incoming proton: $-0.927239|+\rangle +0.37447|-\rangle$

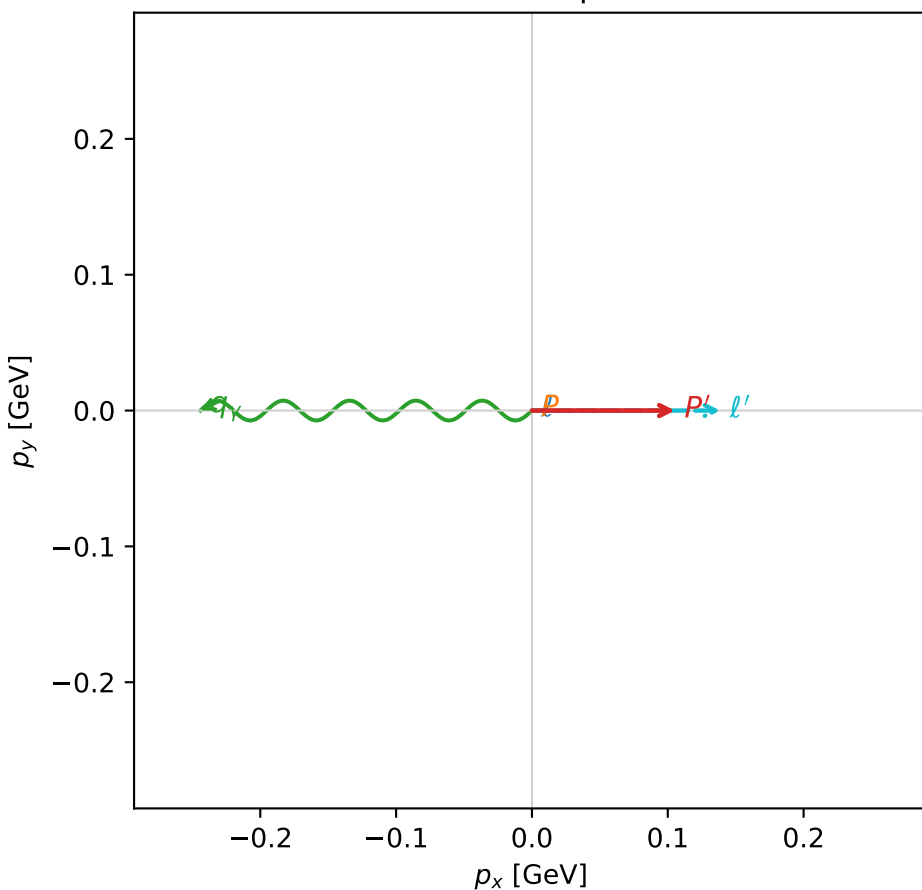
$s=2.18782$, $\sqrt{s}=1.47913$, $|\vec{P}|=0.442145$, $|\vec{P}'|=0.106014$
 $\theta_{P'}=1.67961$, $\theta_Y=2.19744$, $\phi_{P'}=6.28313$, $\phi_Y=3.14156$
 $E_Y=0.301344$, $\theta_{\ell'}=0.635045$, $\phi_{\ell'}=6.28317$
 $Q^2=0.373209$, $x_B=0.377167$, $t=-0.208261$, $W^2=1.49614$, $y=0.756516$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.4421$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.4421$
 P $E= 1.0370$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.4421$
 ℓ' $E= 0.2338$ $p_x= 0.1387$ $p_y= -0.0000$ $p_z= 0.1882$
 P' $E= 0.9440$ $p_x= 0.1054$ $p_y= -0.0000$ $p_z= -0.0115$
 q_Y $E= 0.3013$ $p_x= -0.2441$ $p_y= 0.0000$ $p_z= -0.1767$

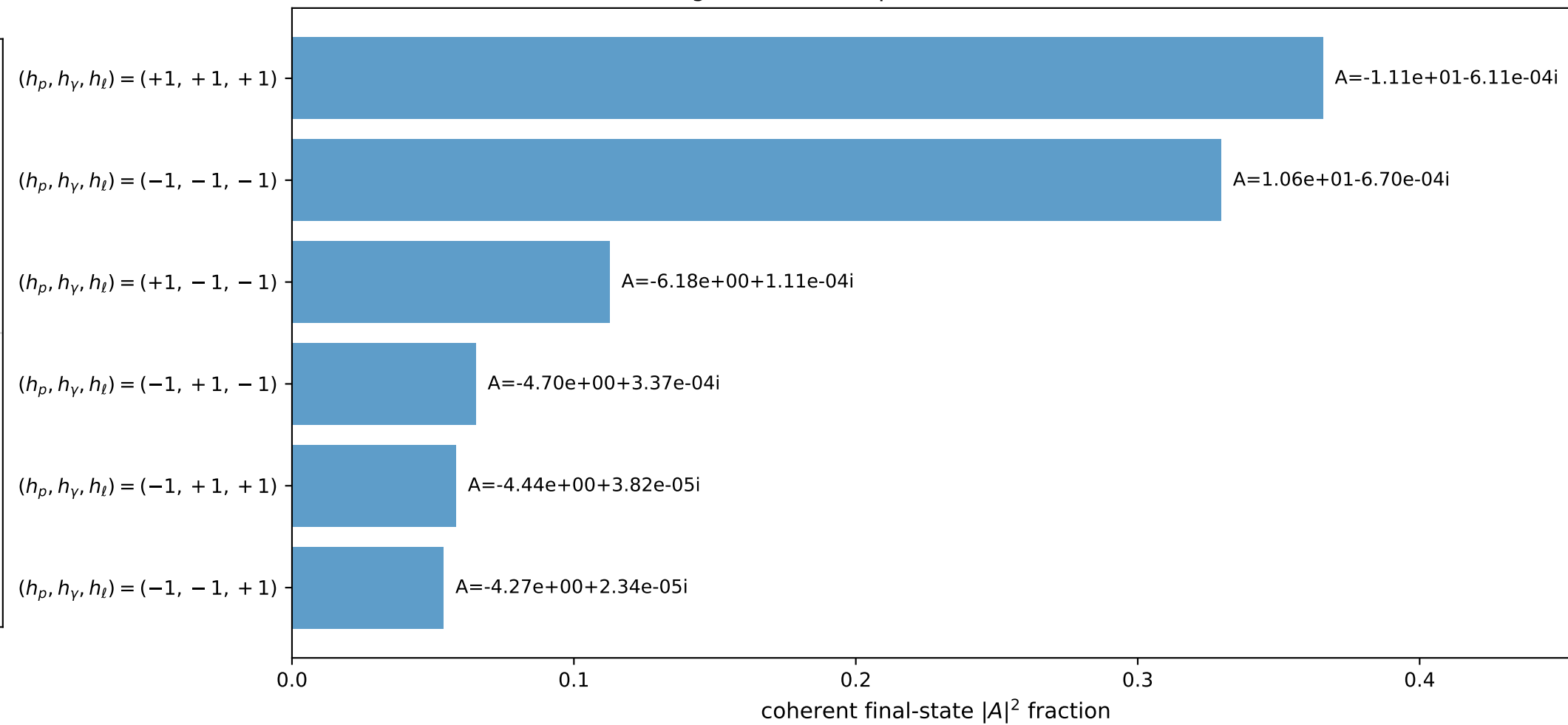
3D momenta



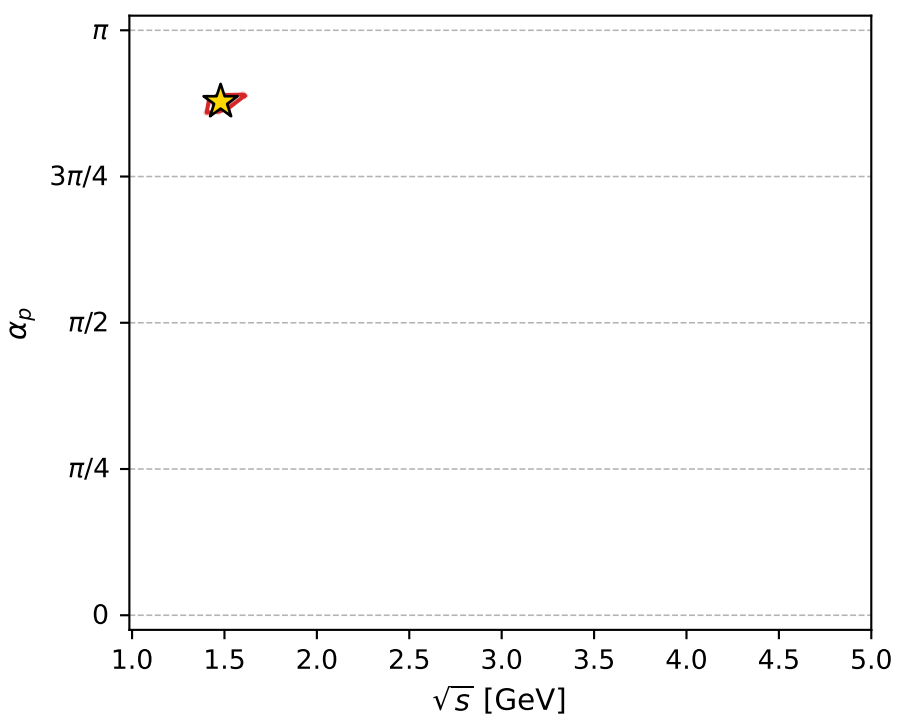
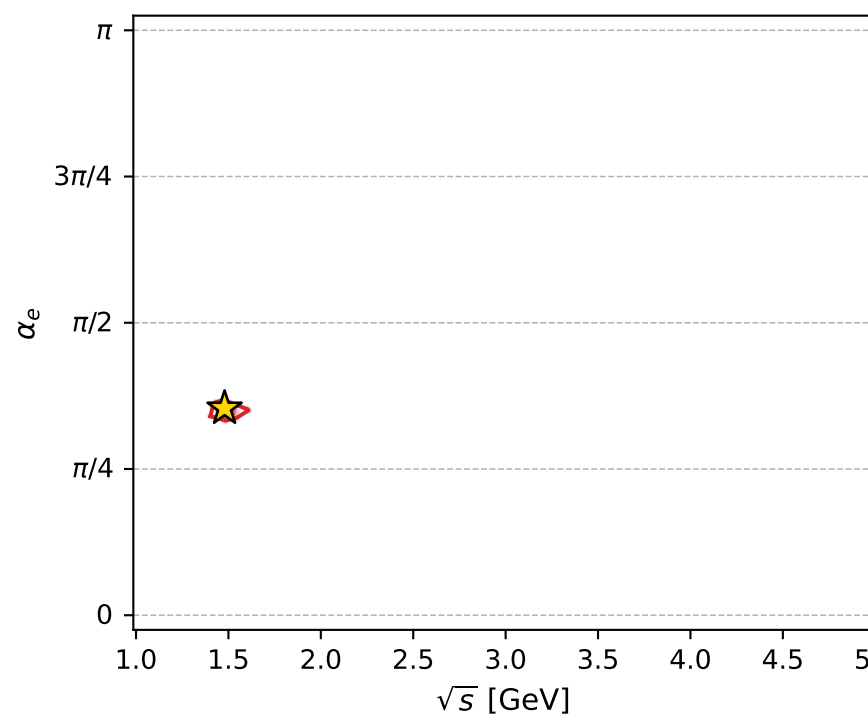
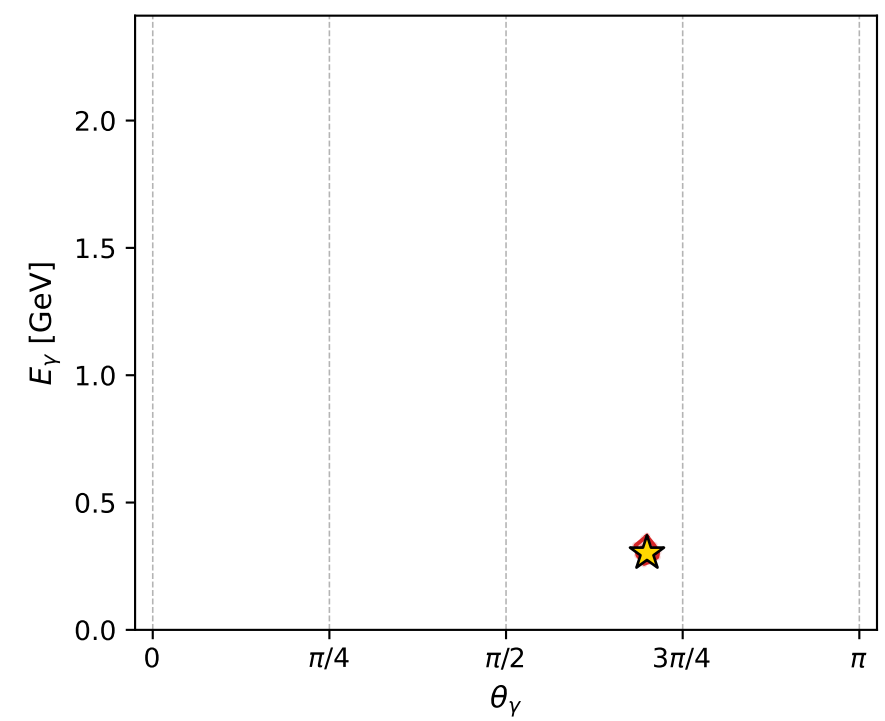
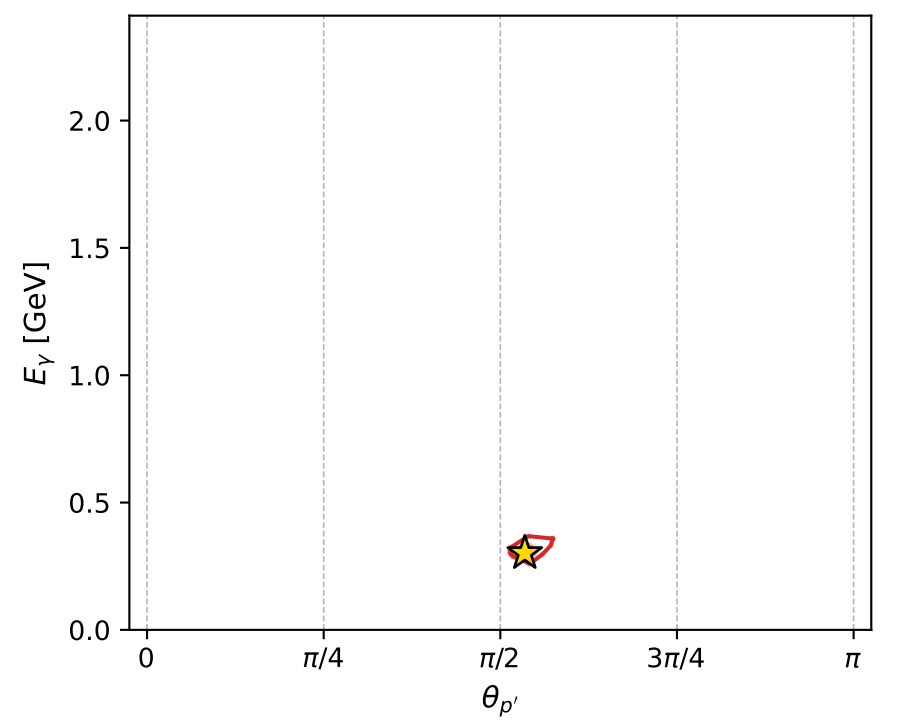
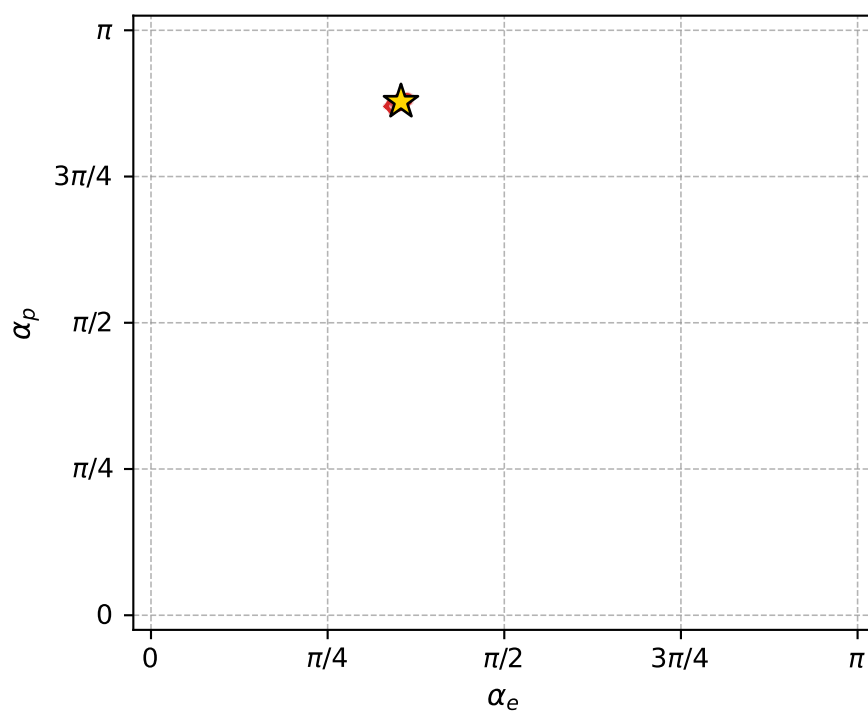
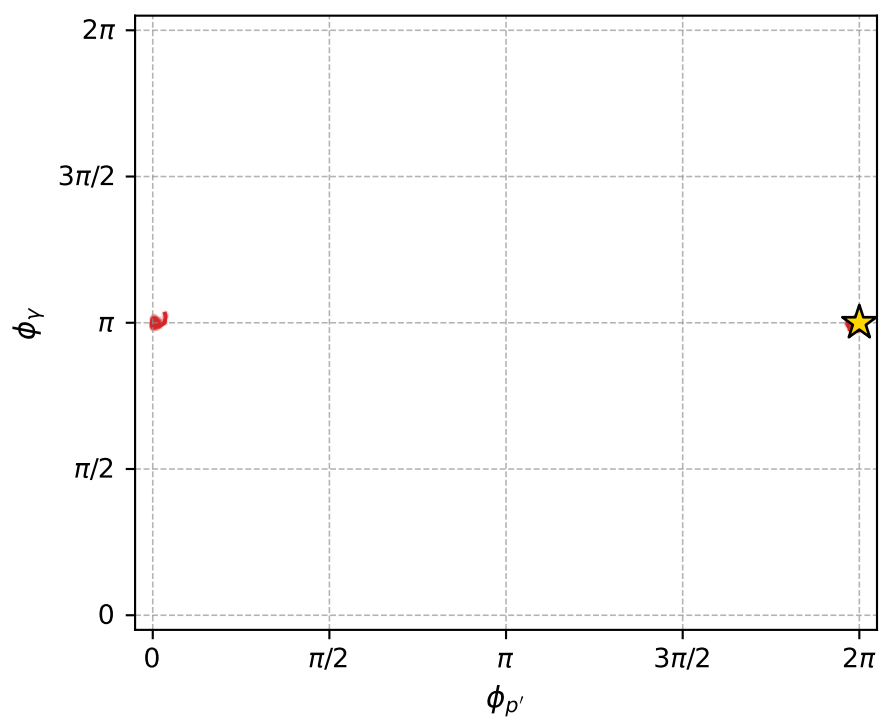
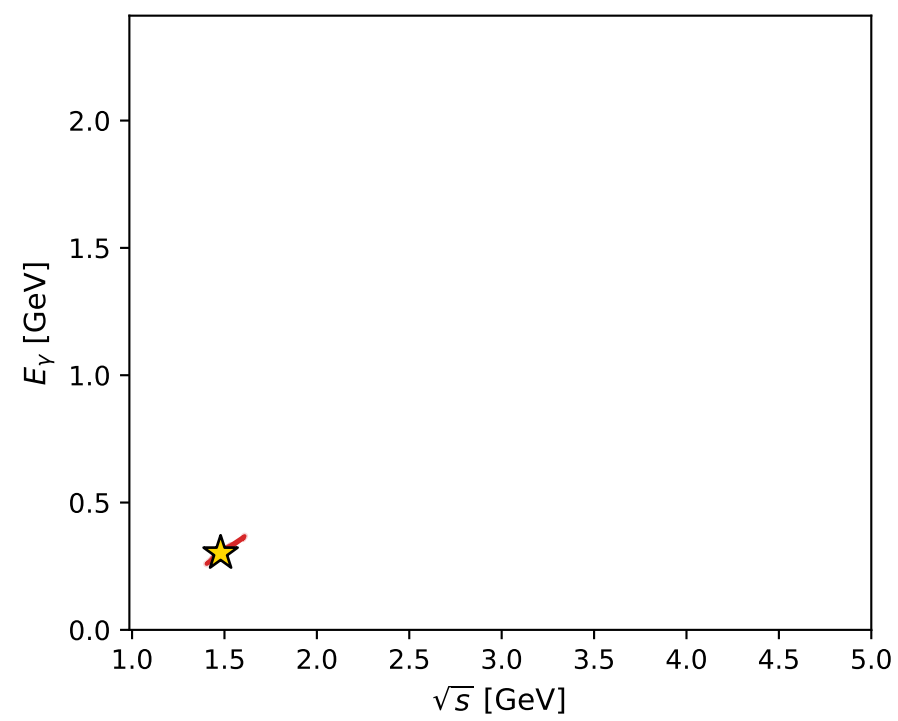
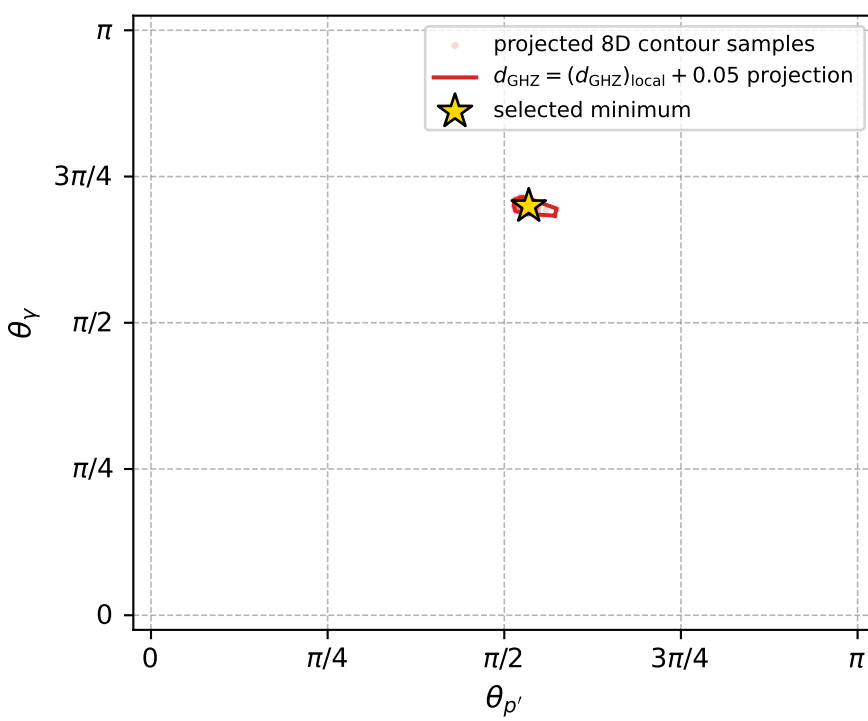
Transverse plane



Leading final-state amplitudes (N=6, retained=98.5%)



electron: polarization cluster P2, local minimum 327; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 0.00046596177$
 $d_{\text{GHZ}} \text{ contour} = 0.050465962$
 above global minimum = 0.00046593948
 8D contour samples = 256

selected phase-space point:

$\text{sqrt}_s = 1.47913$
 $\text{theta}_p_{\text{out}} = 1.679613$
 $\text{theta}_\gamma_{\text{out}} = 2.197441$
 $q_{\text{out}} = 0.3013438$
 $\text{phi}_p_{\text{out}} = 6.283126$
 $\text{phi}_\gamma_{\text{out}} = 3.14156$
 $\alpha_e = 1.111465$
 $\alpha_p = 2.757767$

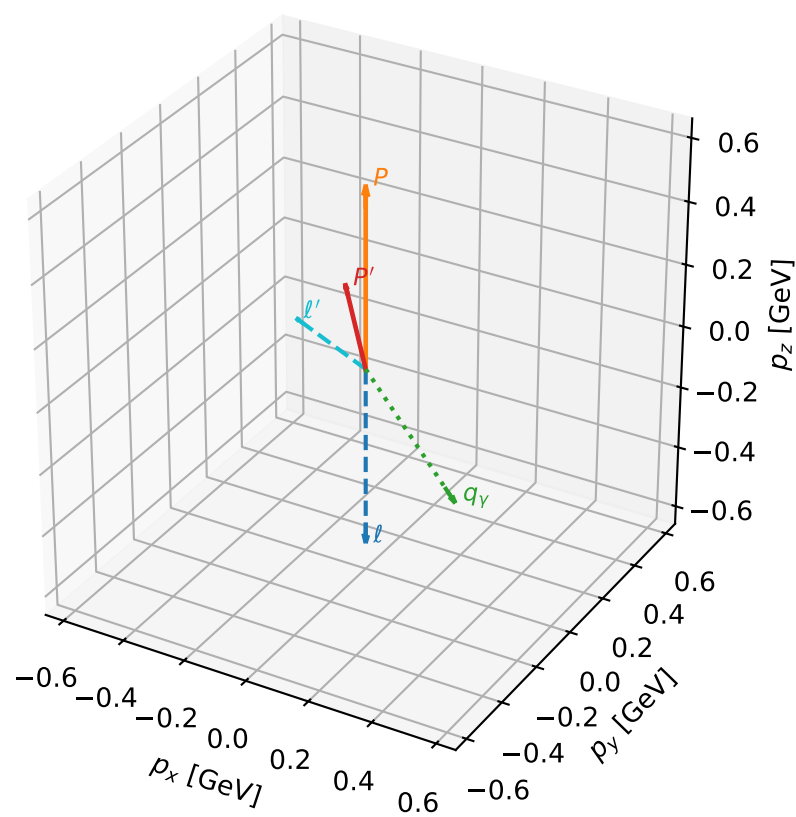
coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

dghz_mixing_angles_polarization_02_local_minimum_330 $d_{\{\text{rm GHZ}\}}=0.000492443$
region: polarization_cluster_02_local_minimum_330
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: $\alpha_e=0.87274305$ rad
incoming lepton: $0.642728|+\rangle +0.766095|-\rangle$
proton mixing angle: $\alpha_p=2.2937138$ rad
incoming proton: $-0.661575|+\rangle +0.749879|-\rangle$

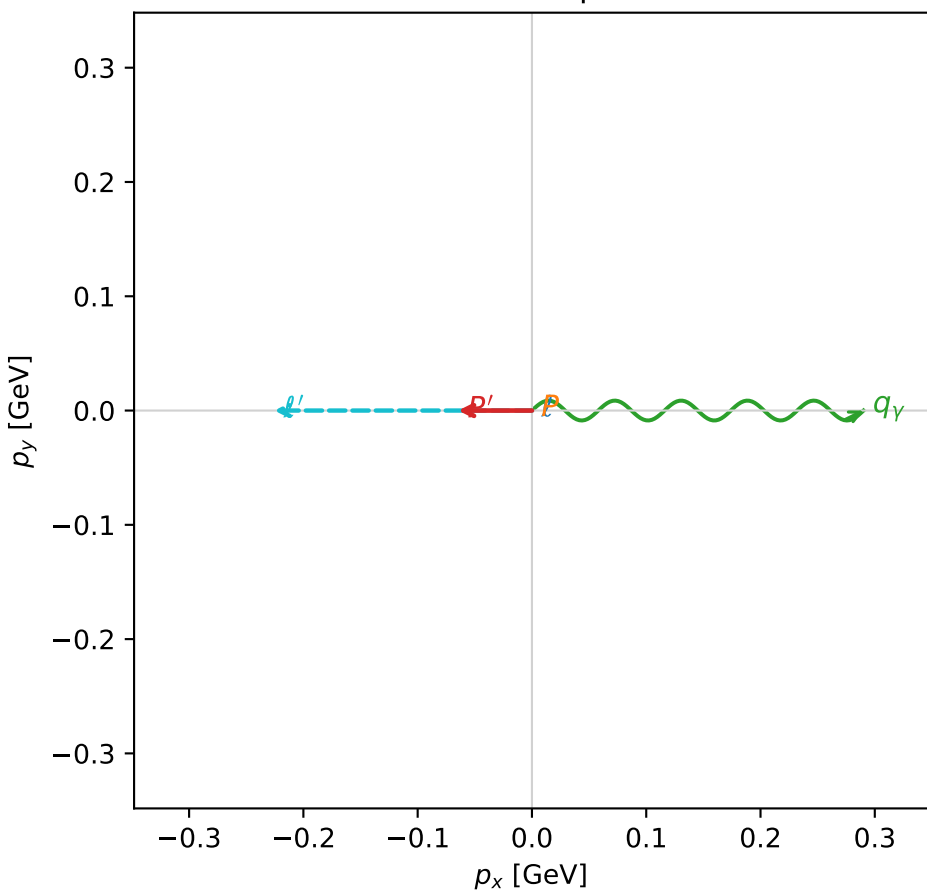
$s=2.8016$, $\sqrt{s}=1.6738$, $|\vec{P}|=0.57407$, $|\vec{P}'|=0.25755$
 $\theta_{P'}=0.253398$, $\theta_Y=2.44883$, $\phi_{P'}=3.14162$, $\phi_Y=1.52106\text{e-}05$
 $E_Y=0.454263$, $\theta_{\ell'}=1.15265$, $\phi_{\ell'}=3.1416$
 $Q^2=0.398454$, $x_B=0.266711$, $t=-0.0934962$, $W^2=1.97535$, $y=0.777393$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.5741$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.5741$
 P $E= 1.0997$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.5741$
 ℓ' $E= 0.2468$ $p_x= -0.2256$ $p_y= -0.0000$ $p_z= 0.1002$
 P' $E= 0.9727$ $p_x= -0.0646$ $p_y= -0.0000$ $p_z= 0.2493$
 q_Y $E= 0.4543$ $p_x= 0.2901$ $p_y= 0.0000$ $p_z= -0.3495$

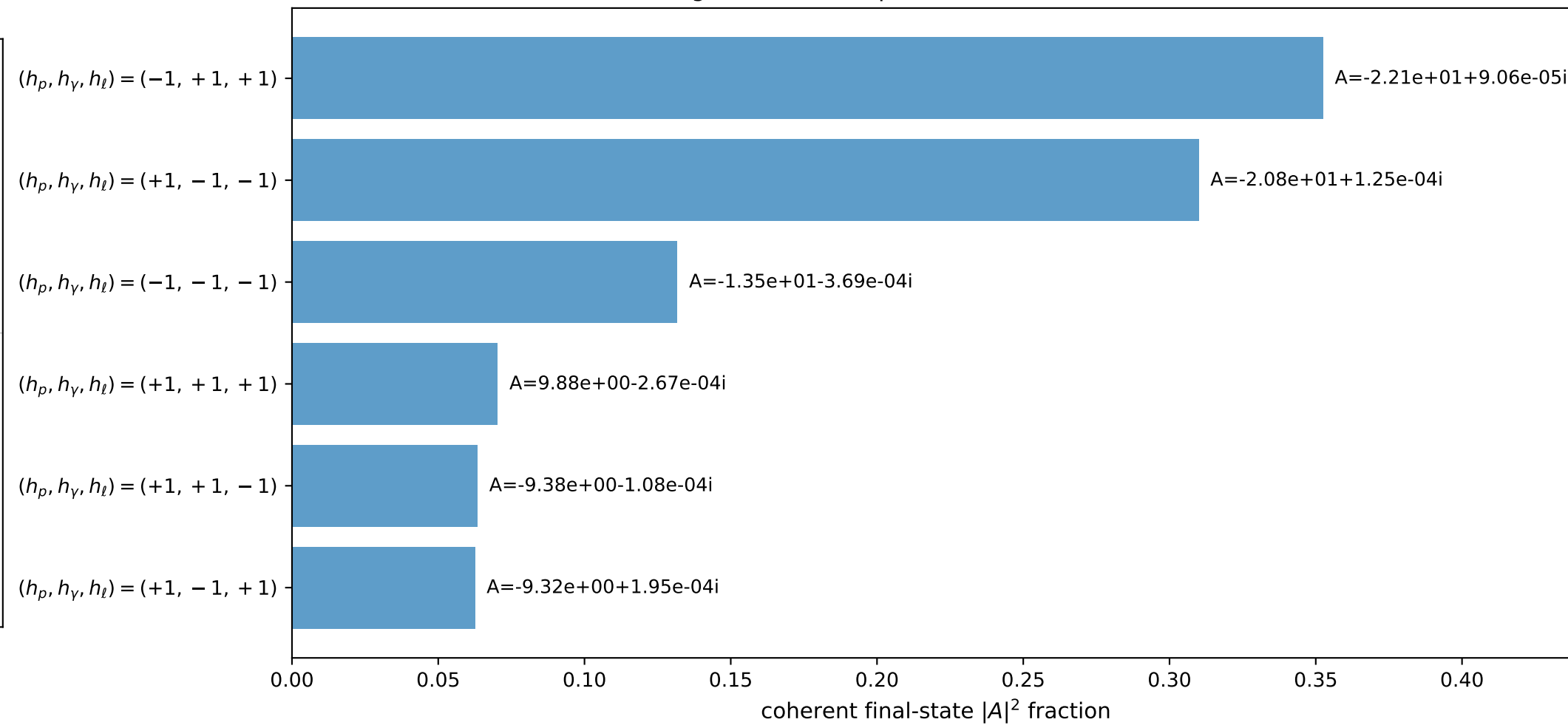
3D momenta



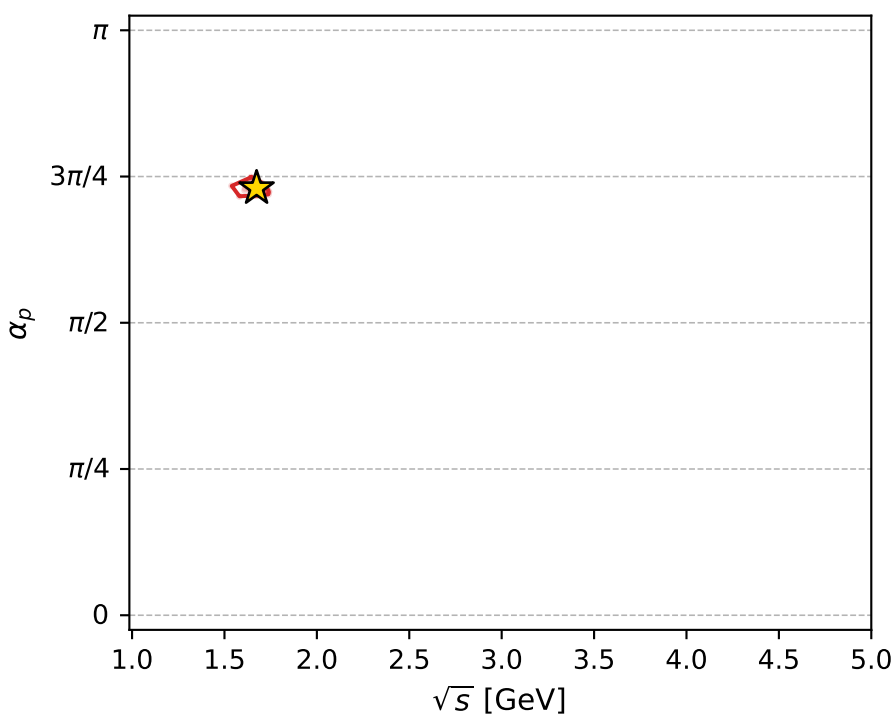
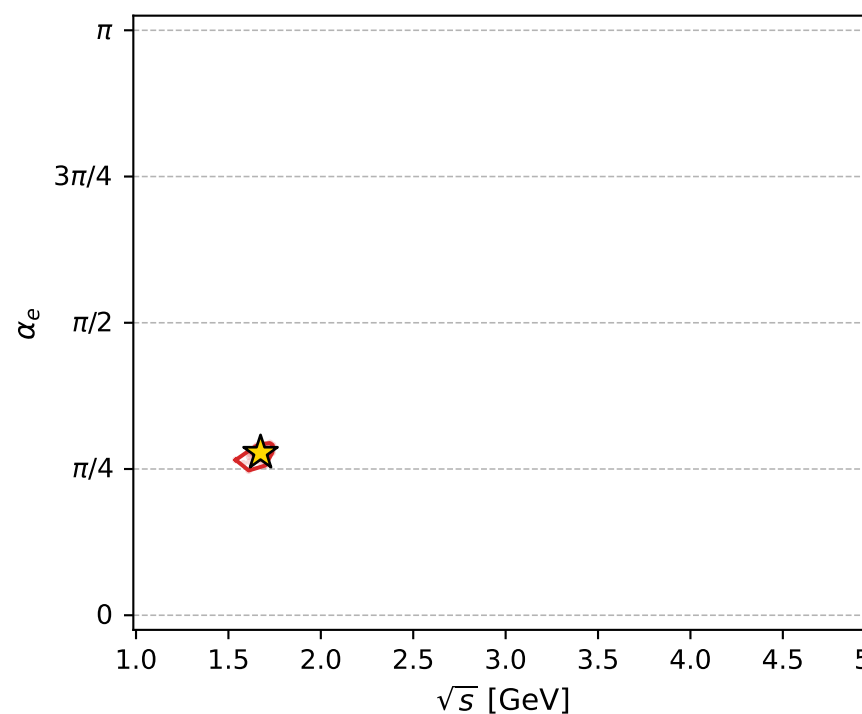
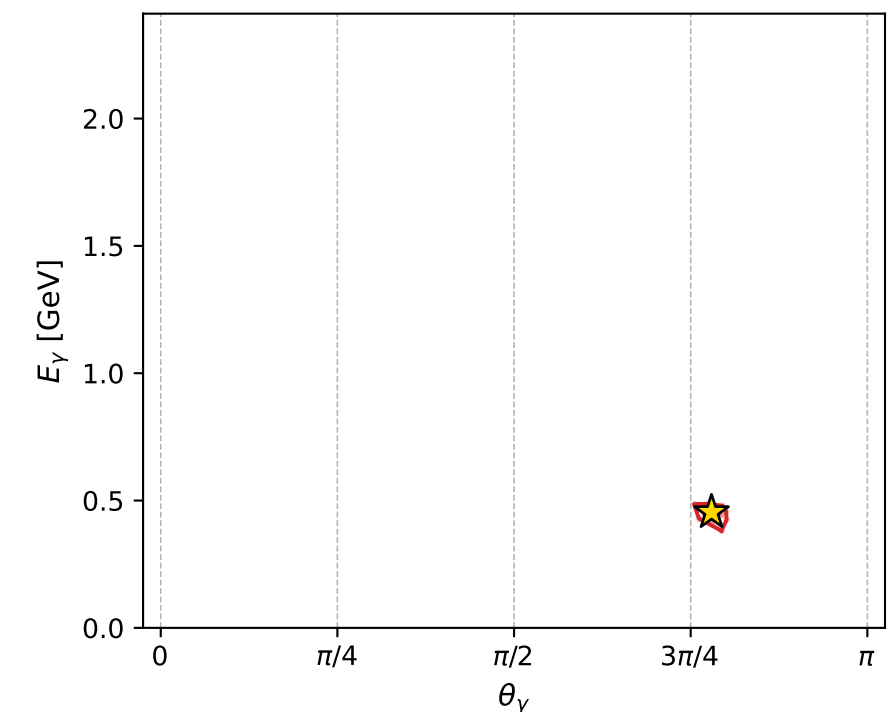
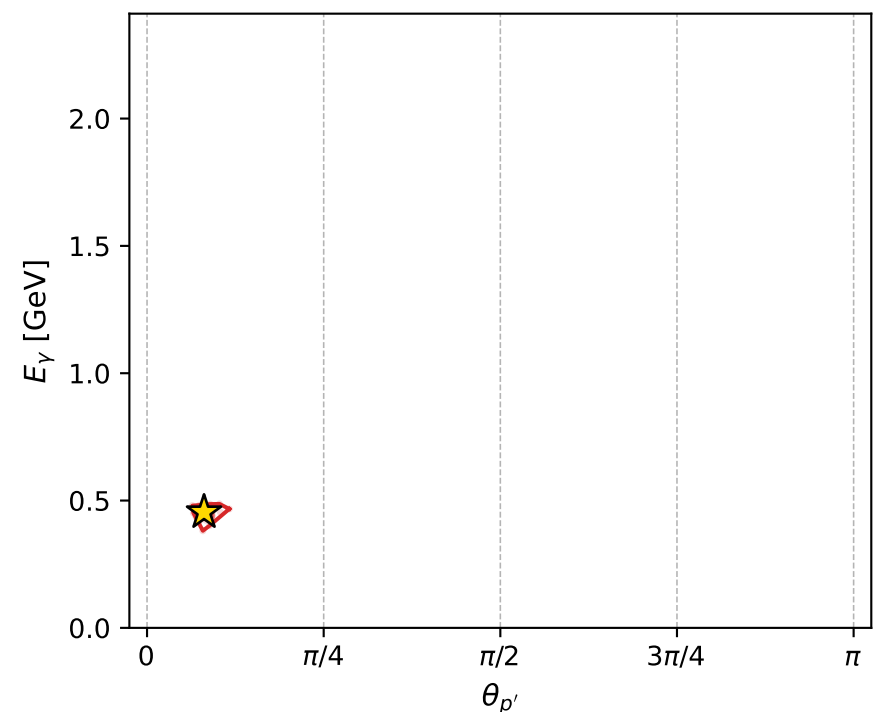
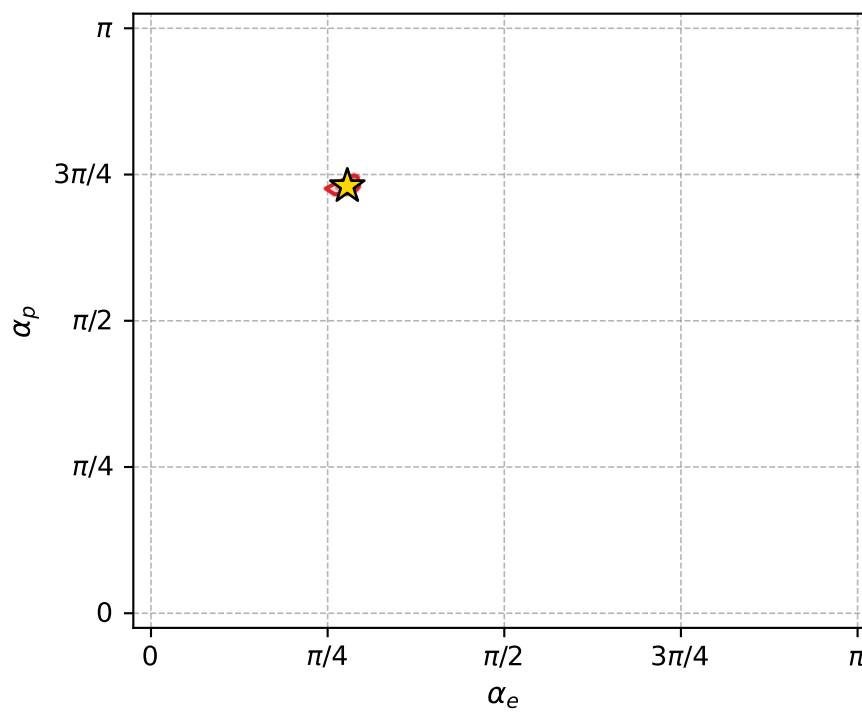
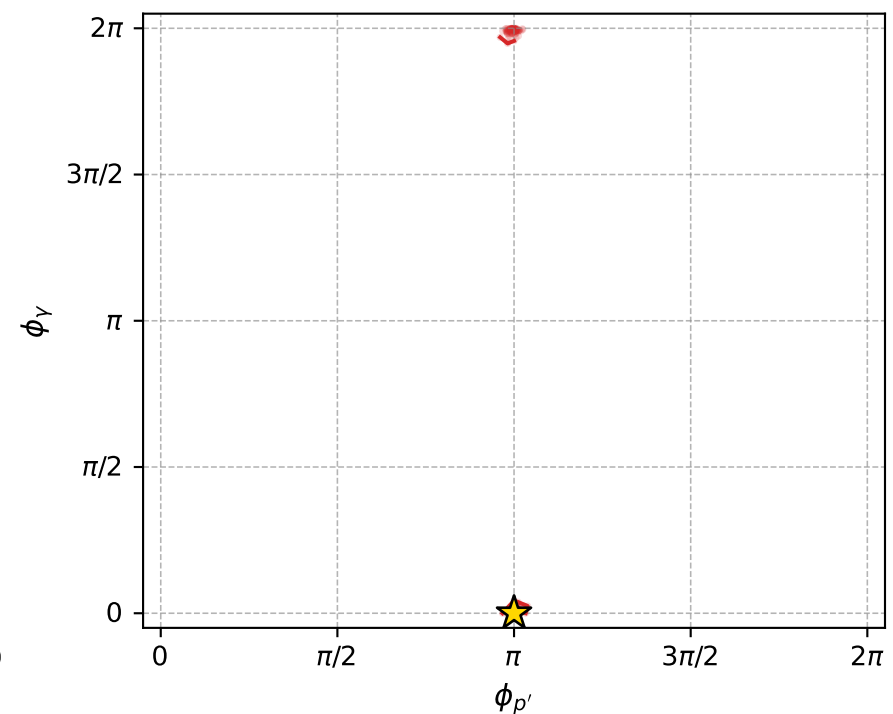
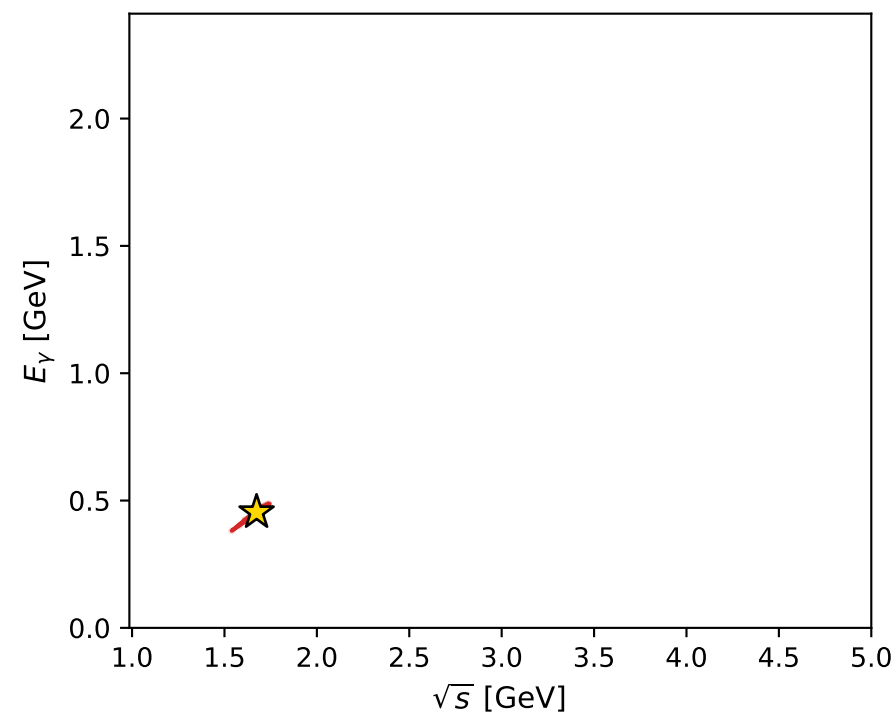
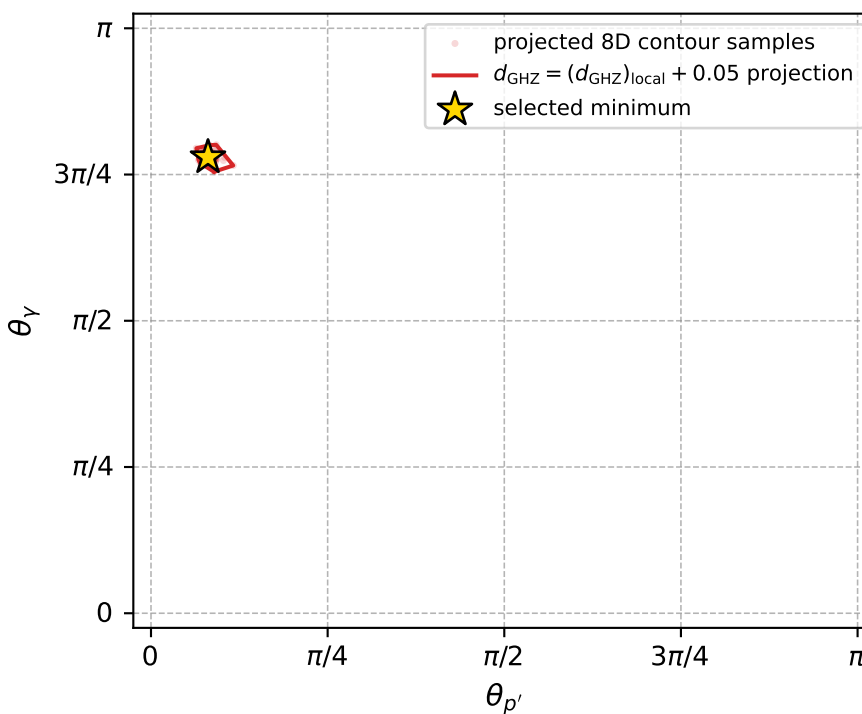
Transverse plane



Leading final-state amplitudes (N=6, retained=99.0%)



electron: polarization cluster P2, local minimum 330; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 0.00049244305$
 $d_{\text{GHZ}} \text{ contour} = 0.050492443$
 above global minimum = 0.00049242076
 8D contour samples = 256

selected phase-space point:

$\text{sqrt_s} = 1.673798$
 $\text{theta_p_out} = 0.2533982$
 $\text{theta_gamma_out} = 2.448834$
 $\text{qOut} = 0.4542627$
 $\text{phi_p_out} = 3.14162$
 $\text{phi_gamma_out} = 1.521056\text{e-}05$
 $\text{alpha_e} = 0.872743$
 $\text{alpha_p} = 2.293714$

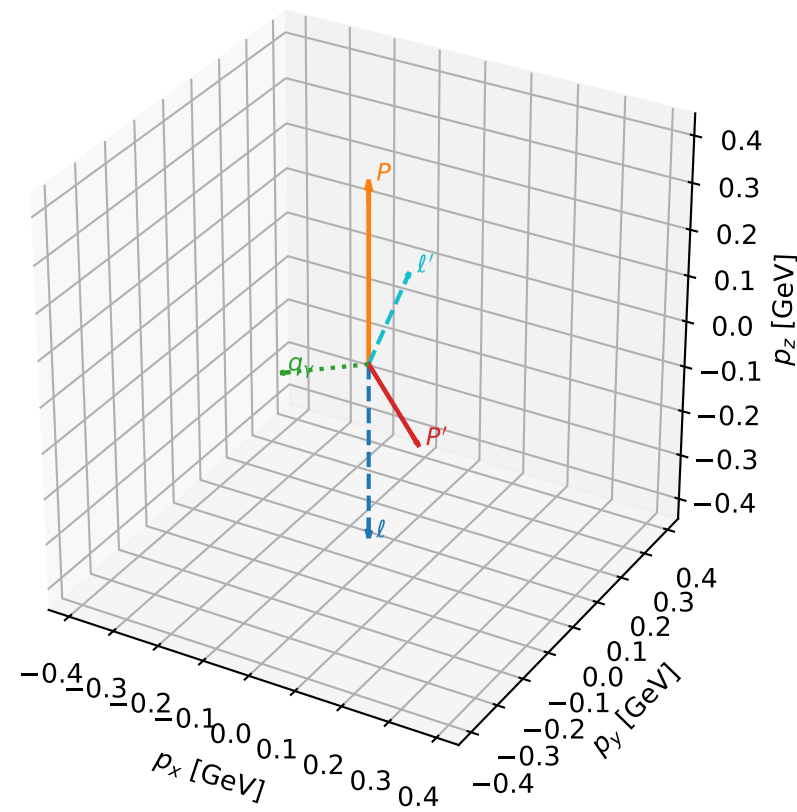
coherent mixing angles: minimum d_{GHz} configuration (gradient_local_minimum)

dghz_mixing_angles_polarization_02_local_minimum_332 $d_{\{\text{rm GHz}\}}=0.000506501$
region: polarization_cluster_02_local_minimum_332
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: $\alpha_e=1.3225702$ rad
incoming lepton: $0.245685|+\rangle +0.96935|-\rangle$
proton mixing angle: $\alpha_p=2.942323$ rad
incoming proton: $-0.980211|+\rangle +0.197953|-\rangle$

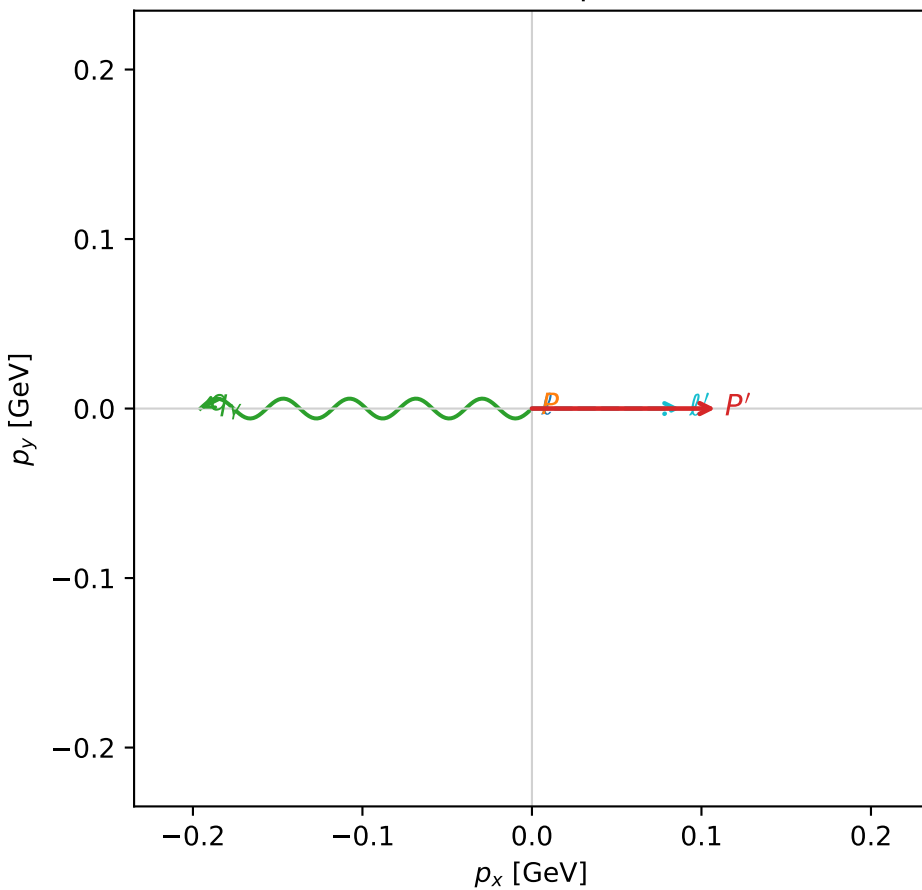
$s=1.96925$, $\sqrt{s}=1.4033$, $|\vec{P}|=0.388159$, $|\vec{P}'|=0.181221$
 $\theta_{P'}=2.50119$, $\theta_Y=1.94206$, $\phi_{P'}=2.8152\text{e-}05$, $\phi_Y=3.14163$
 $E_Y=0.2099$, $\theta_{\ell'}=0.375554$, $\phi_{\ell'}=5.56642\text{e-}05$
 $Q^2=0.356733$, $x_B=0.458516$, $t=-0.292742$, $W^2=1.30113$, $y=0.714163$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.3882$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.3882$
 P $E= 1.0151$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.3882$
 ℓ' $E= 0.2381$ $p_x= 0.0873$ $p_y= 0.0000$ $p_z= 0.2215$
 P' $E= 0.9553$ $p_x= 0.1083$ $p_y= 0.0000$ $p_z= -0.1453$
 q_Y $E= 0.2099$ $p_x= -0.1956$ $p_y= -0.0000$ $p_z= -0.0762$

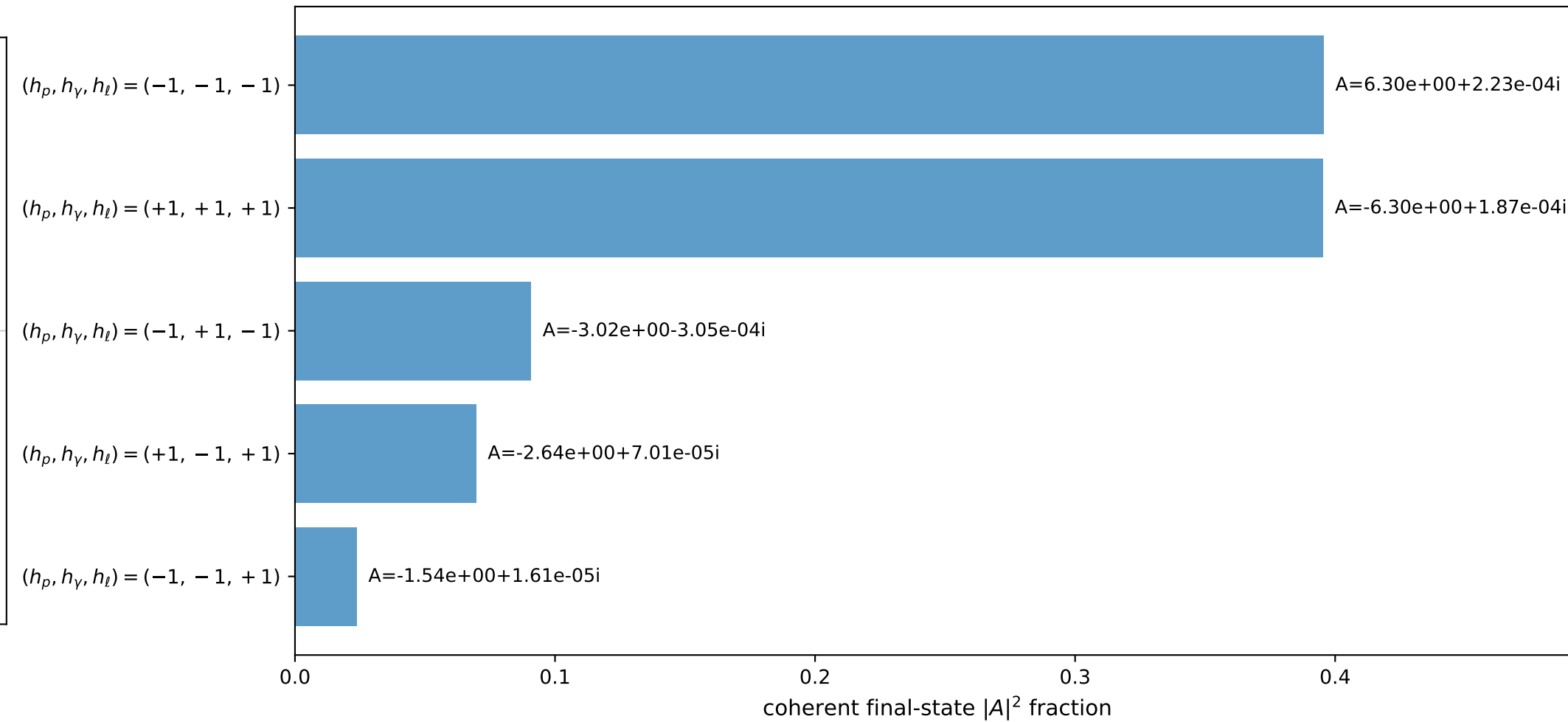
3D momenta



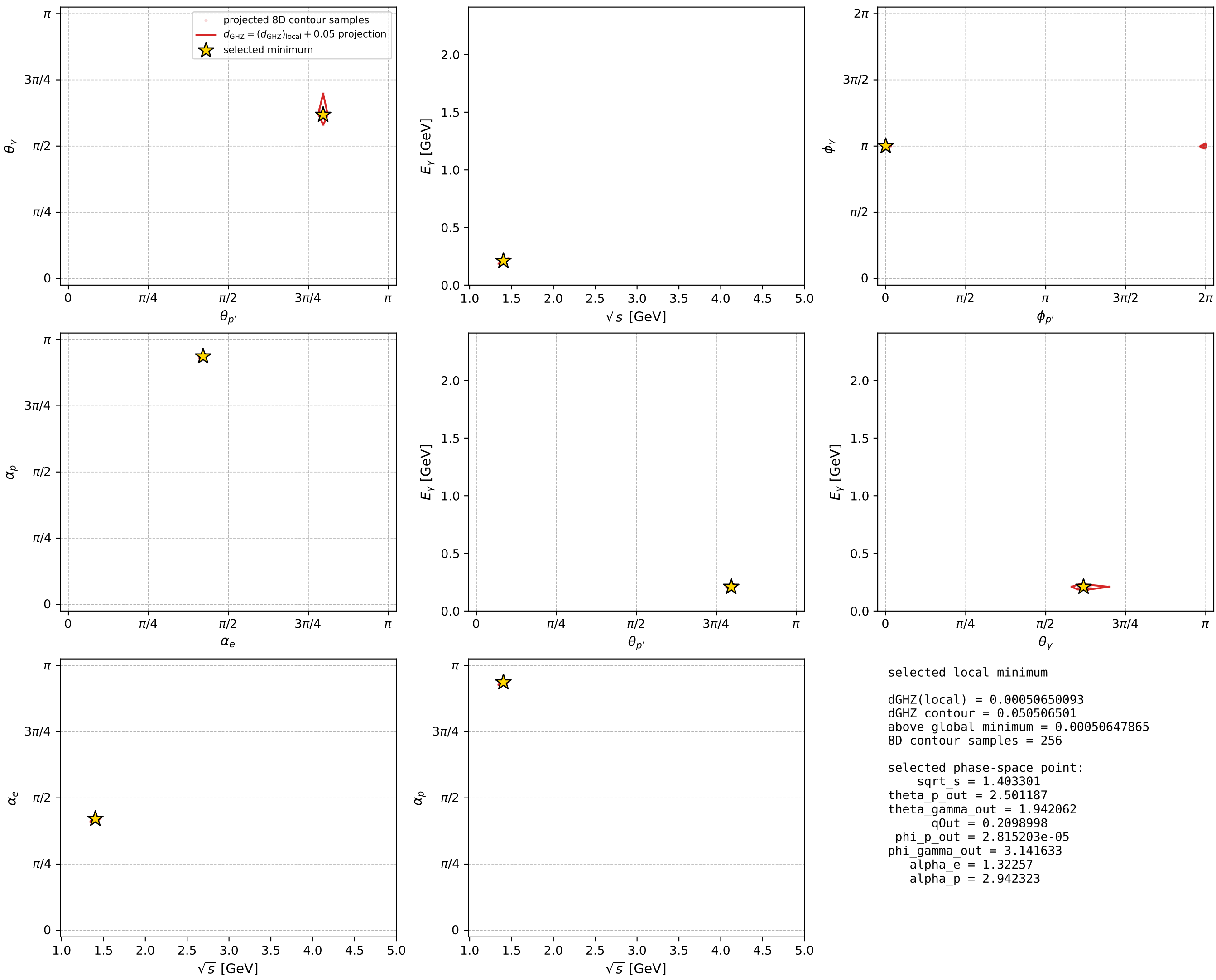
Transverse plane



Leading final-state amplitudes (N=5, retained=97.5%)



electron: polarization cluster P2, local minimum 332; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



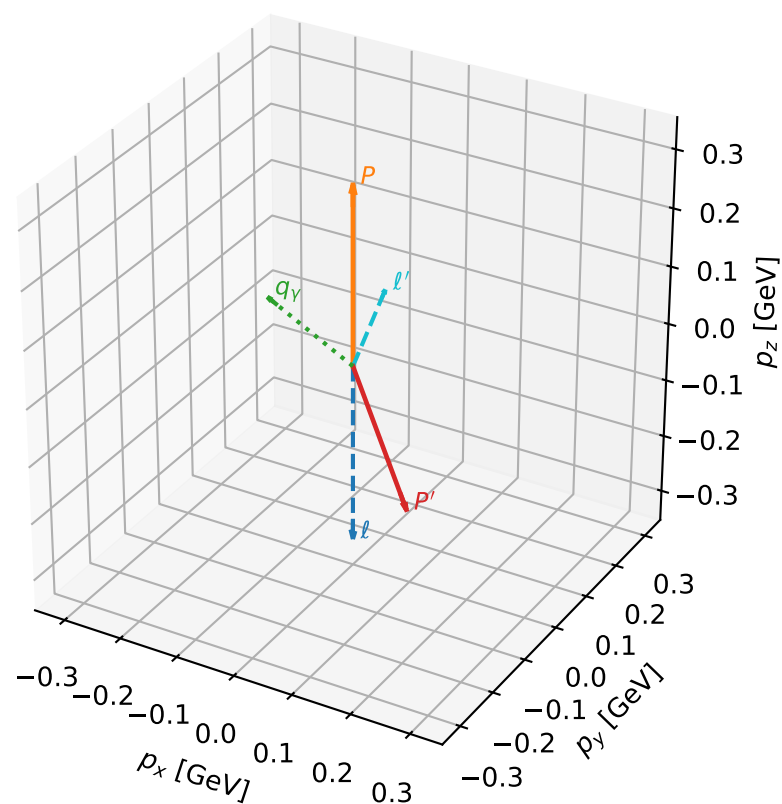
coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

dghz_mixing_angles_polarization_02_local_minimum_334 $d_{\{\text{rm GHZ}\}}=0.000509327$
 region: polarization_cluster_02_local_minimum_334
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=1.3534387$ rad
 incoming lepton: $0.21565|+\rangle +0.976471|-\rangle$
 proton mixing angle: $\alpha_p=2.9675287$ rad
 incoming proton: $-0.984889|+\rangle +0.173186|-\rangle$

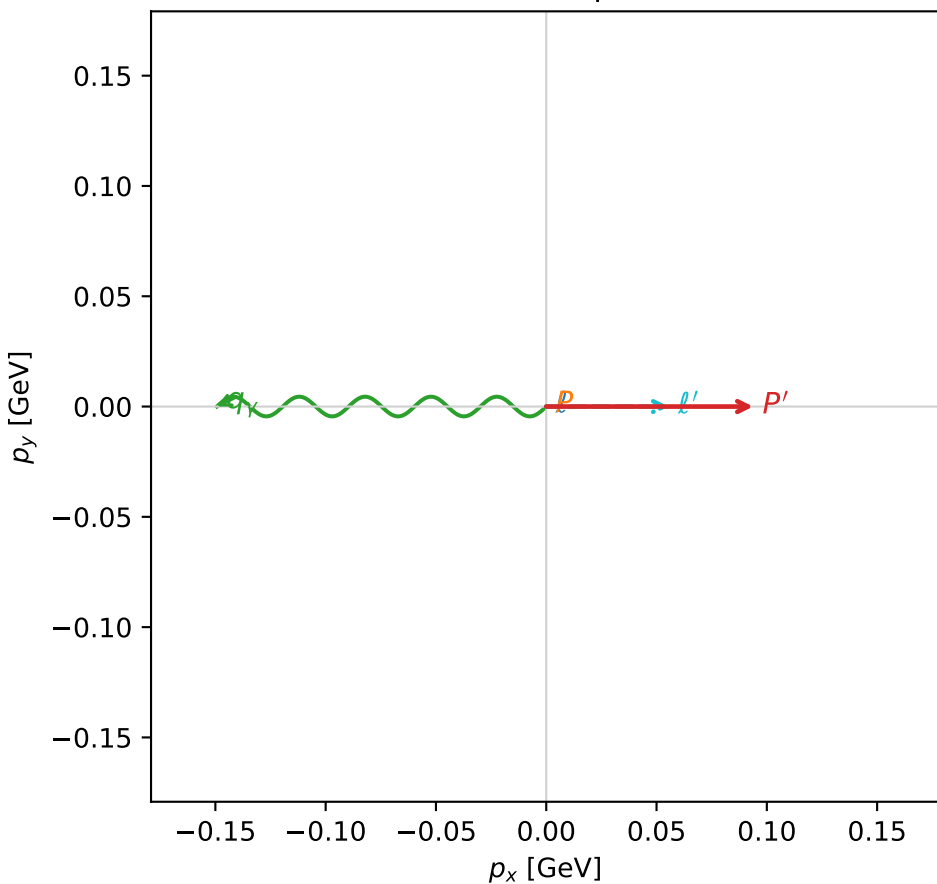
$s=1.67626$, $\sqrt{s}=1.2947$, $|\vec{P}|=0.307566$, $|\vec{P}'|=0.243641$
 $\theta_{P'}=2.74642$, $\theta_V=1.09075$, $\phi_{P'}=6.2831$, $\phi_V=3.14155$
 $E_V=0.168328$, $\theta_{\ell'}=0.360774$, $\phi_{\ell'}=2.52e-05$
 $Q^2=0.187231$, $x_B=0.324794$, $t=-0.291954$, $W^2=1.26907$, $y=0.723821$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.3076$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.3076$
 P $E= 0.9871$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.3076$
 ℓ' $E= 0.1573$ $p_x= 0.0555$ $p_y= 0.0000$ $p_z= 0.1471$
 P' $E= 0.9691$ $p_x= 0.0938$ $p_y= -0.0000$ $p_z= -0.2249$
 q_V $E= 0.1683$ $p_x= -0.1493$ $p_y= 0.0000$ $p_z= 0.0777$

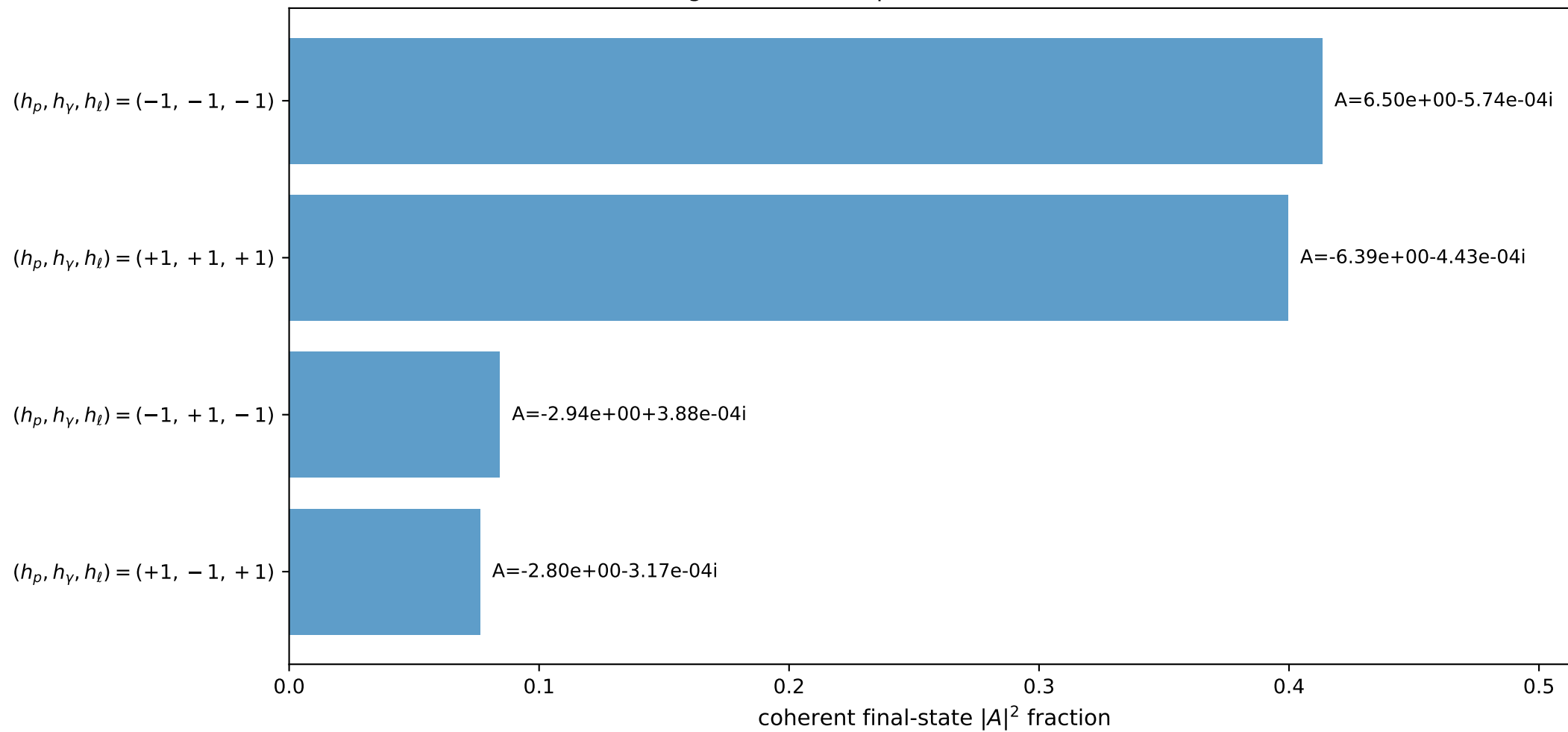
3D momenta



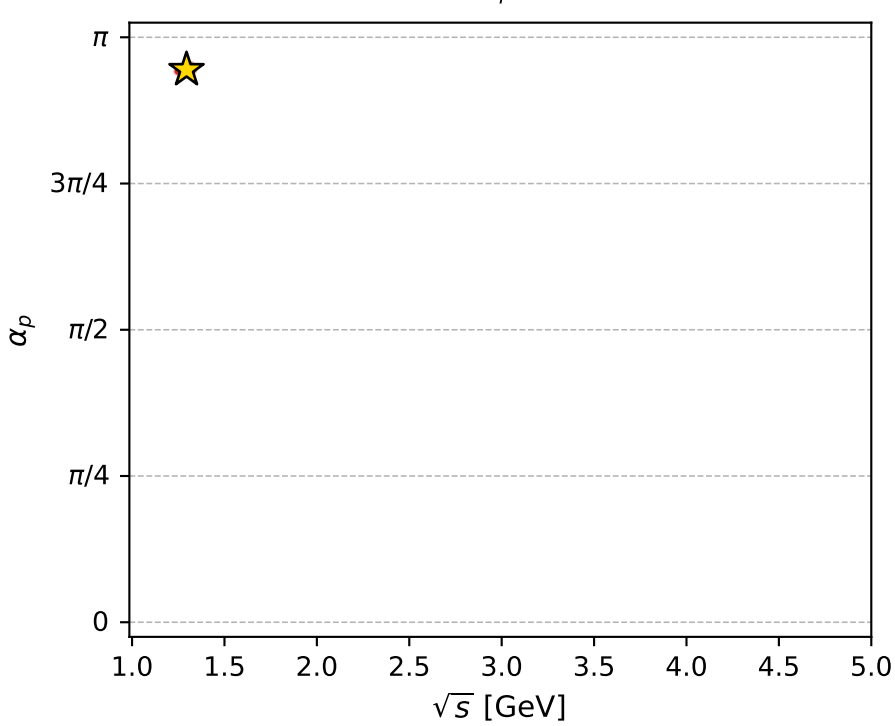
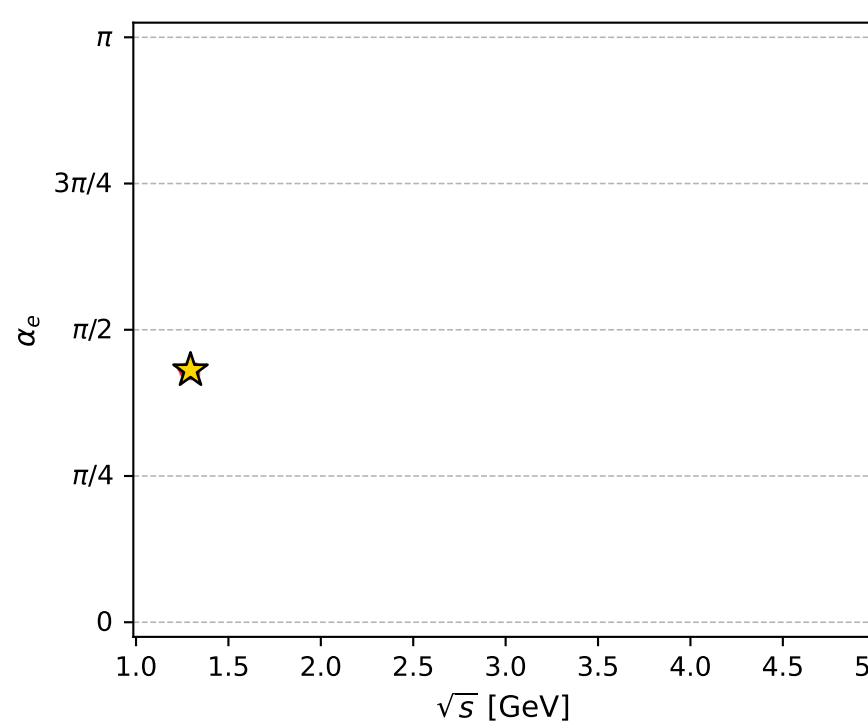
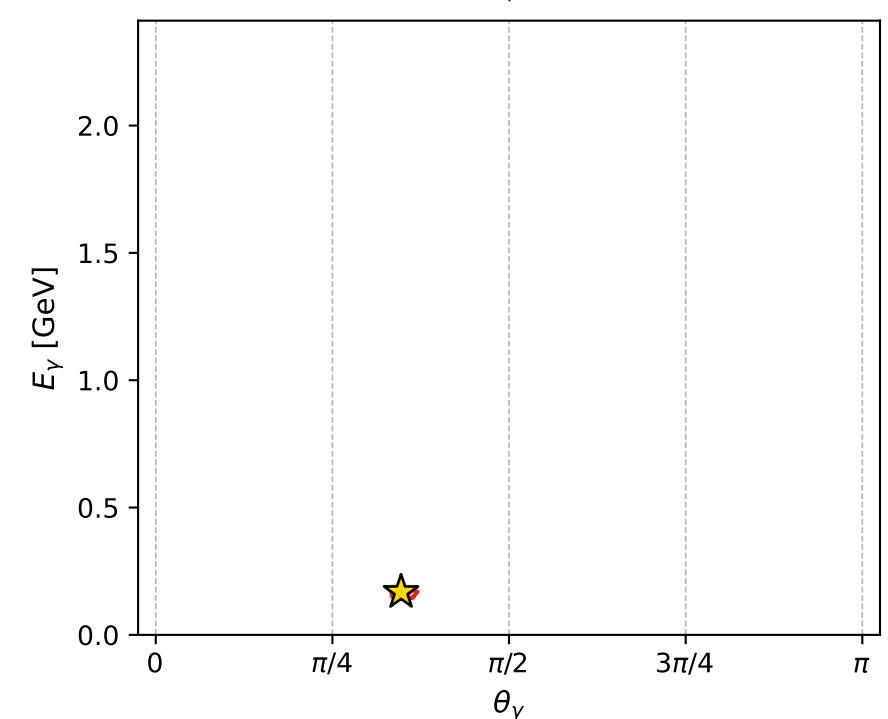
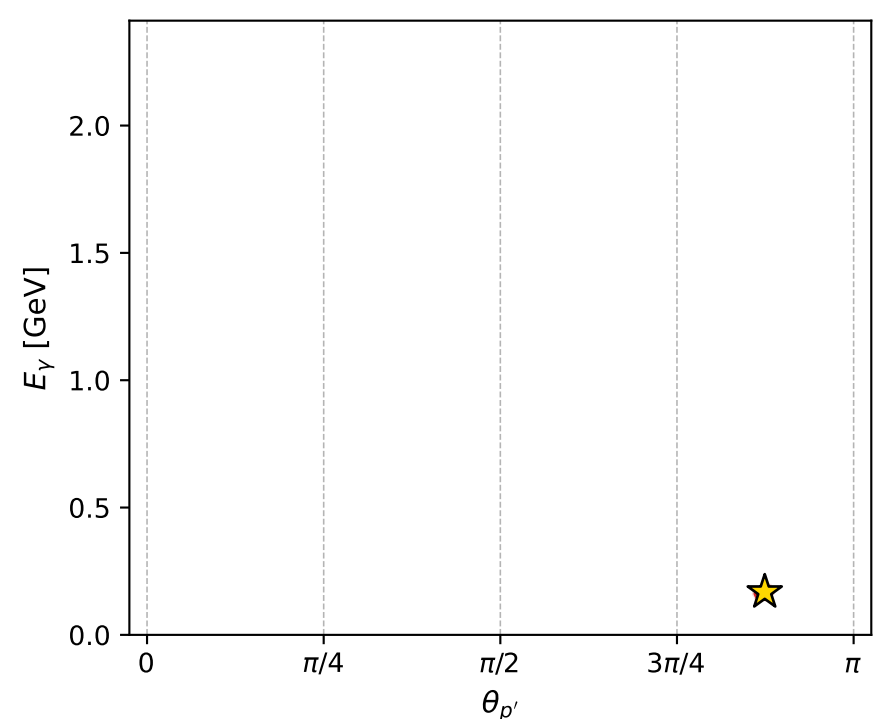
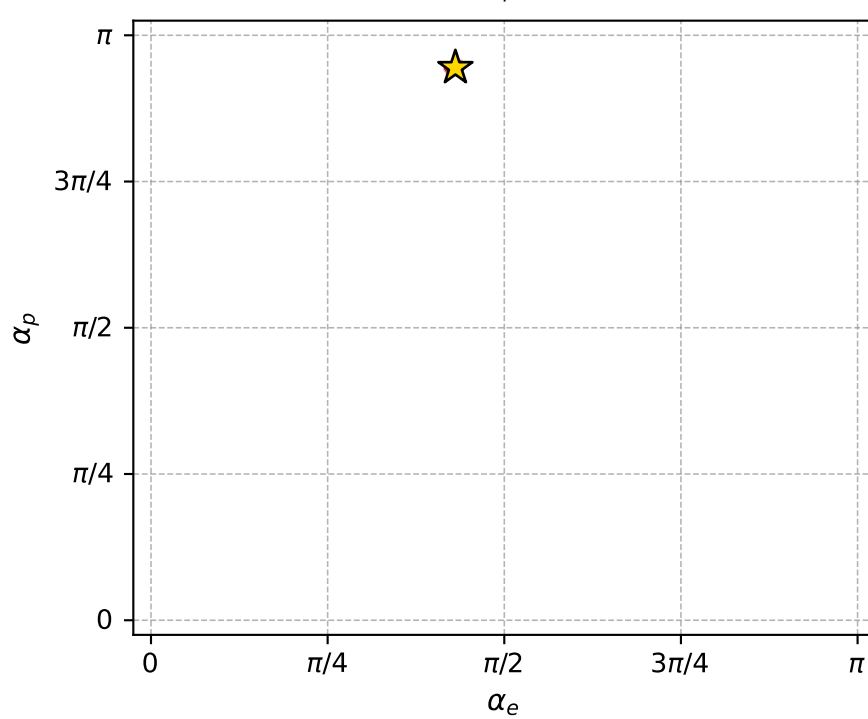
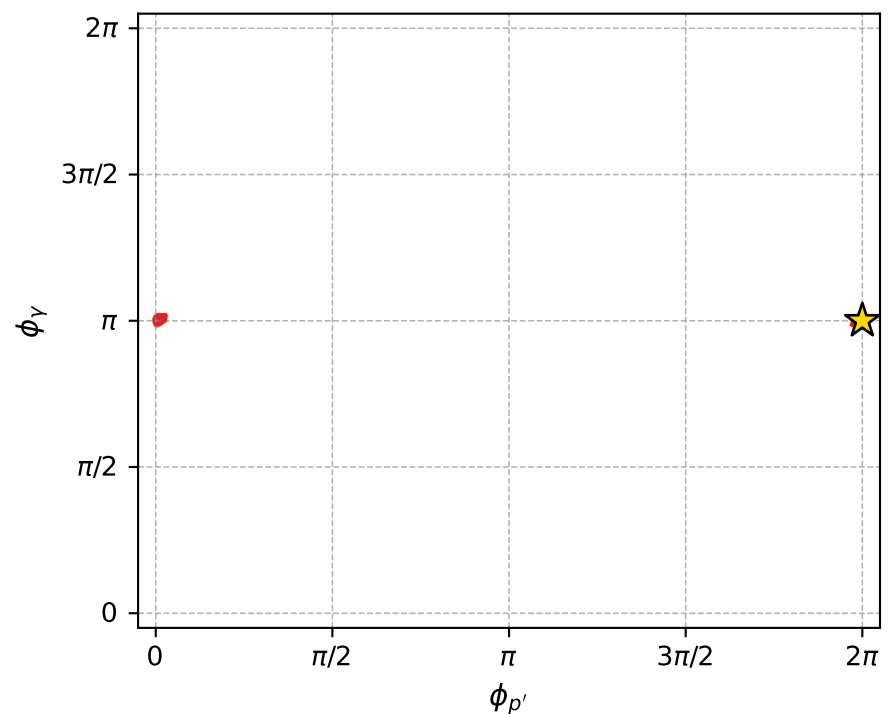
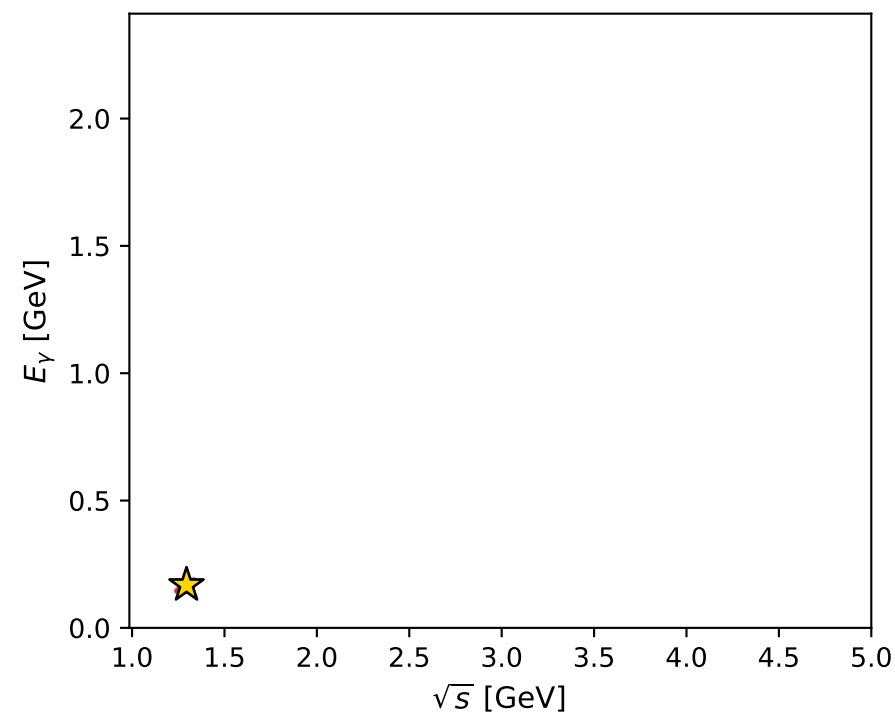
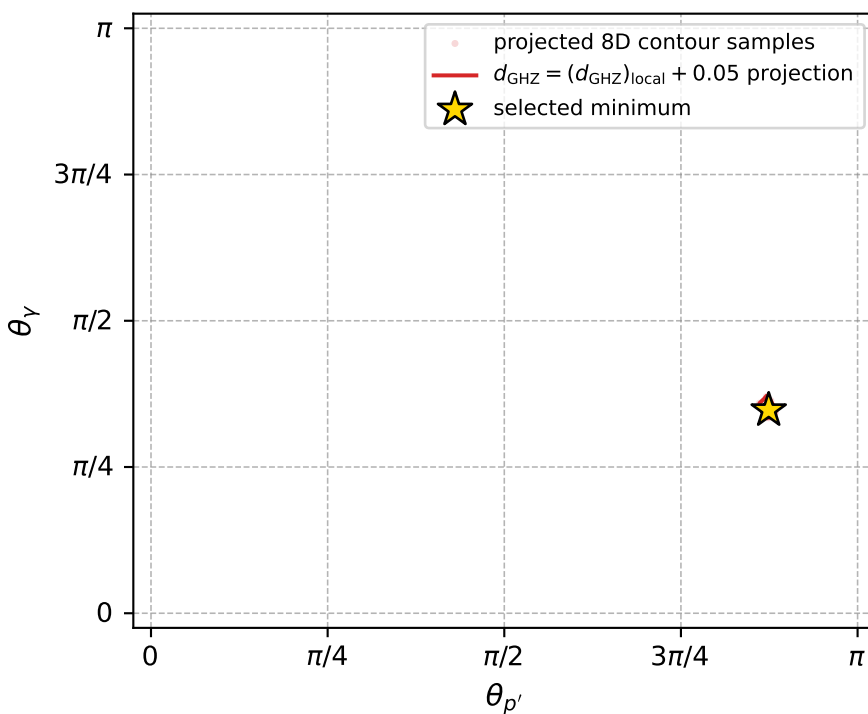
Transverse plane



Leading final-state amplitudes (N=4, retained=97.4%)



electron: polarization cluster P2, local minimum 334; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 0.00050932685$
 $d_{\text{GHZ}} \text{ contour} = 0.050509327$
 above global minimum = 0.00050930457
 8D contour samples = 256

selected phase-space point:

$\text{sqrt_s} = 1.294704$
 $\text{theta_p_out} = 2.74642$
 $\text{theta_gamma_out} = 1.09075$
 $\text{qOut} = 0.1683281$
 $\text{phi_p_out} = 6.283099$
 $\text{phi_gamma_out} = 3.141548$
 $\text{alpha_e} = 1.353439$
 $\text{alpha_p} = 2.967529$

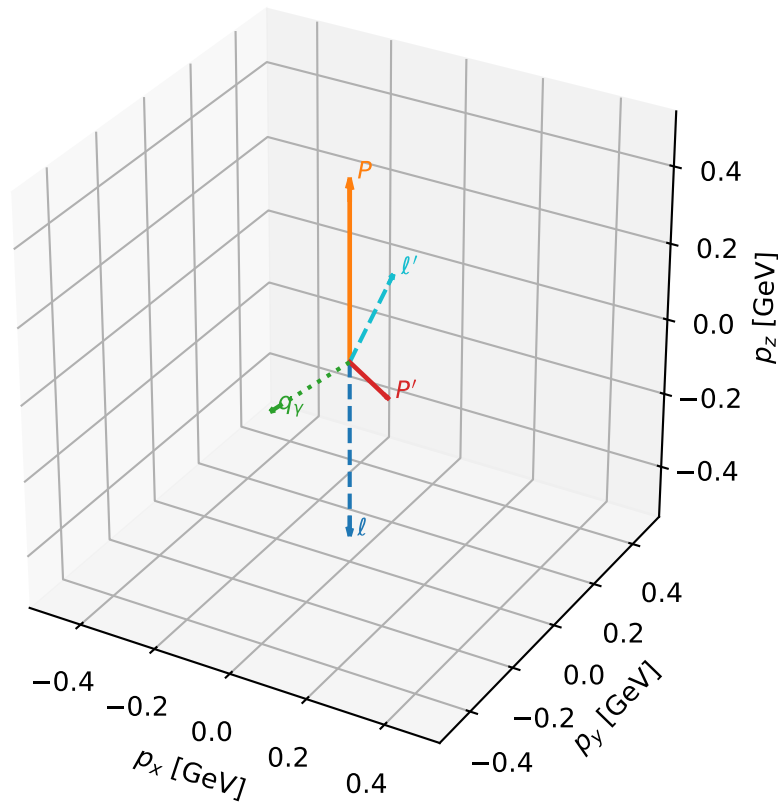
coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

dghz_mixing_angles_polarization_02_local_minimum_337 $d_{\{\text{rm GHZ}\}}=0.000559789$
 region: polarization_cluster_02_local_minimum_337
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=1.2898319$ rad
 incoming lepton: $0.277282|+\rangle +0.960788|-\rangle$
 proton mixing angle: $\alpha_p=2.9191798$ rad
 incoming proton: $-0.975368|+\rangle +0.220584|-\rangle$

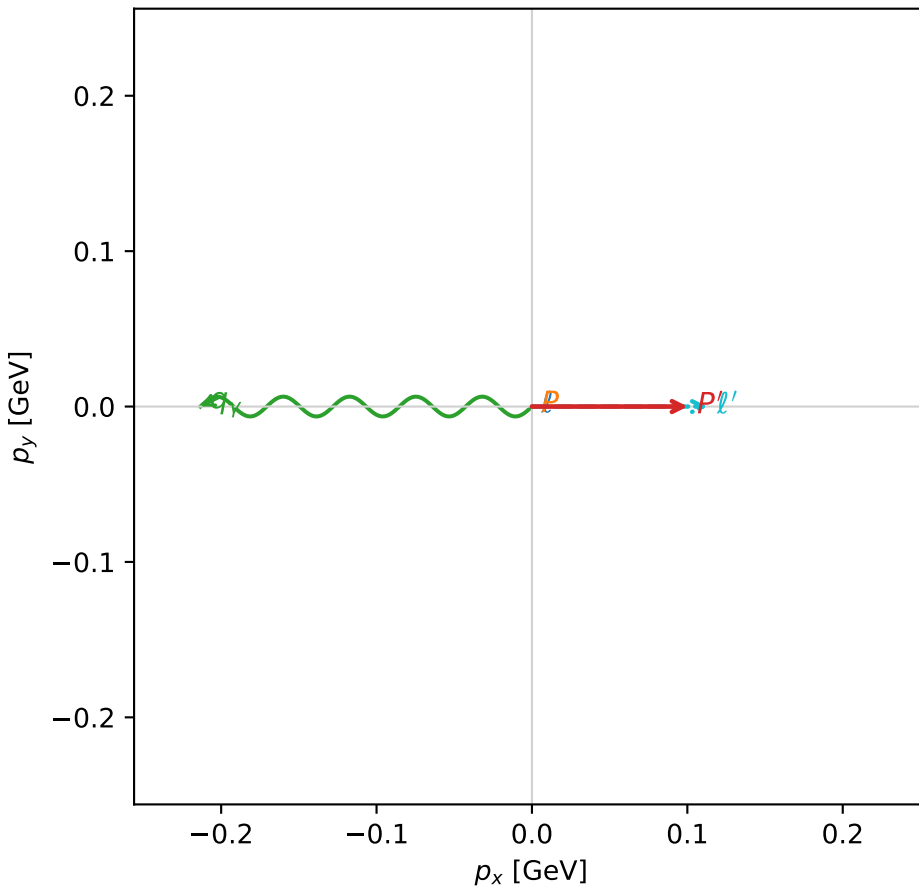
$s=2.30455$, $\sqrt{s}=1.51808$, $|\vec{P}|=0.469249$, $|\vec{P}'|=0.119857$
 $\theta_{P'}=2.15051$, $\theta_Y=2.3101$, $\phi_{P'}=1.28443\text{e-}05$, $\phi_Y=3.1416$
 $E_Y=0.288738$, $\theta_{\ell'}=0.409983$, $\phi_{\ell'}=9.88772\text{e-}06$
 $Q^2=0.510458$, $x_B=0.475385$, $t=-0.285527$, $W^2=1.44316$, $y=0.753682$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.4692$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.4692$
 P $E= 1.0488$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.4692$
 ℓ' $E= 0.2837$ $p_x= 0.1131$ $p_y= 0.0000$ $p_z= 0.2602$
 P' $E= 0.9456$ $p_x= 0.1003$ $p_y= 0.0000$ $p_z= -0.0657$
 q_Y $E= 0.2887$ $p_x= -0.2134$ $p_y= -0.0000$ $p_z= -0.1945$

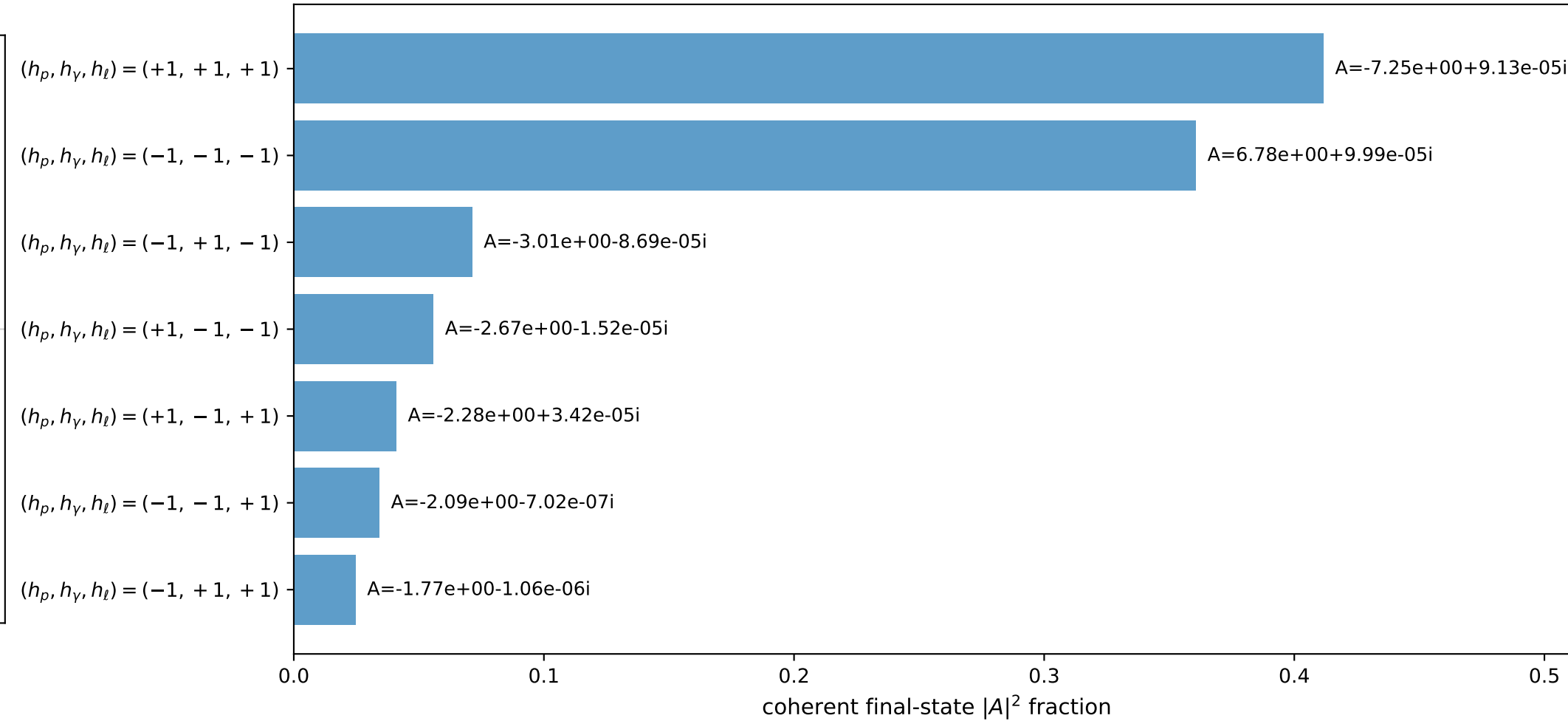
3D momenta



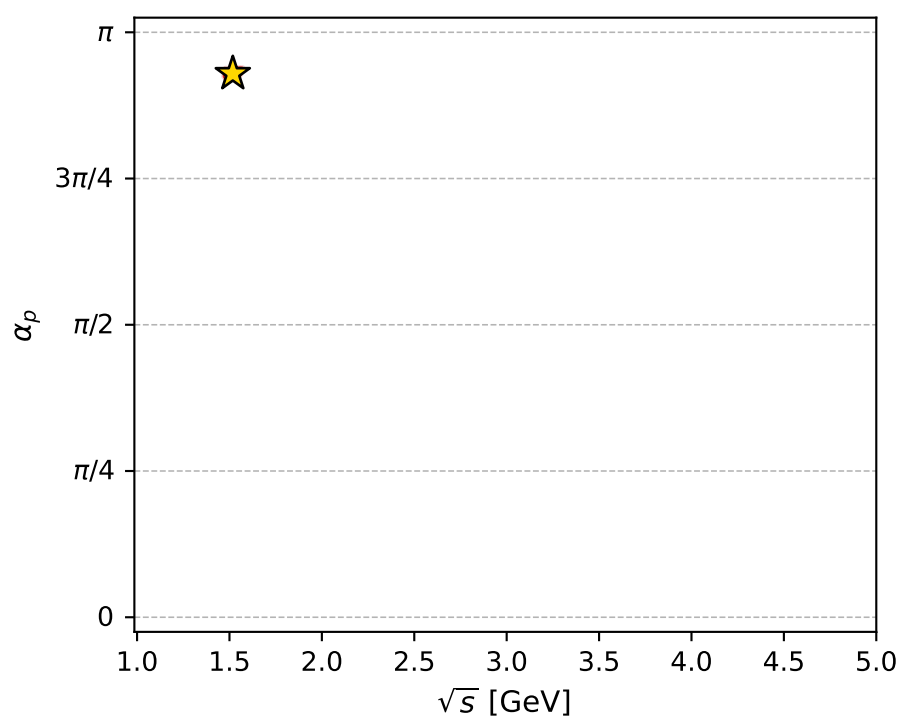
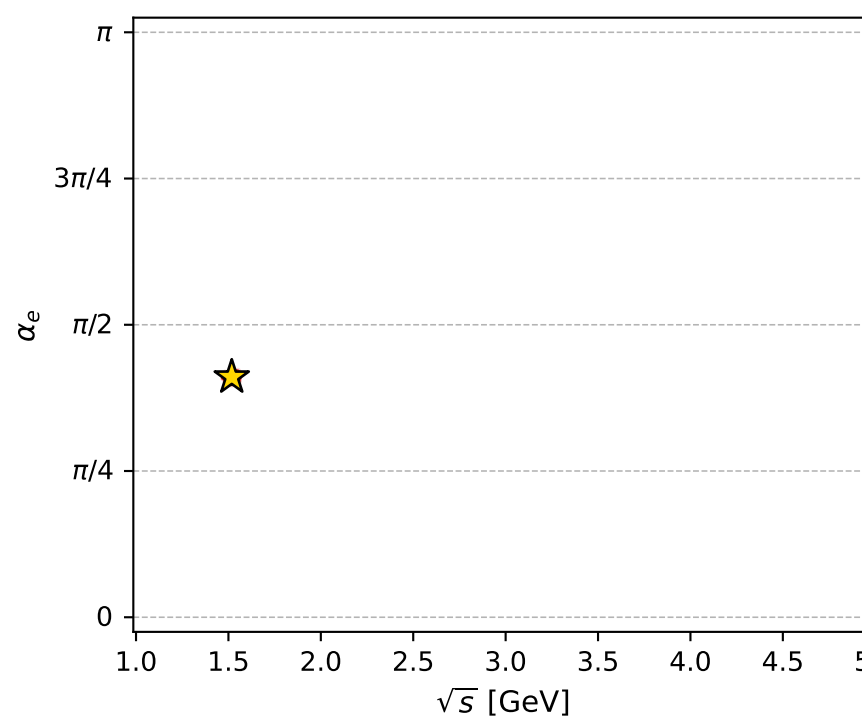
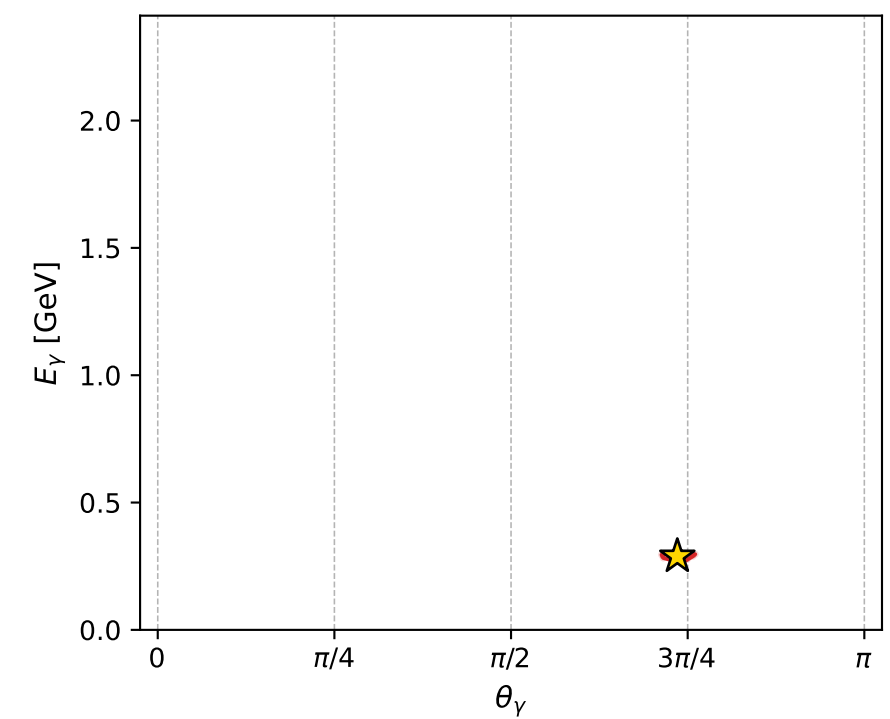
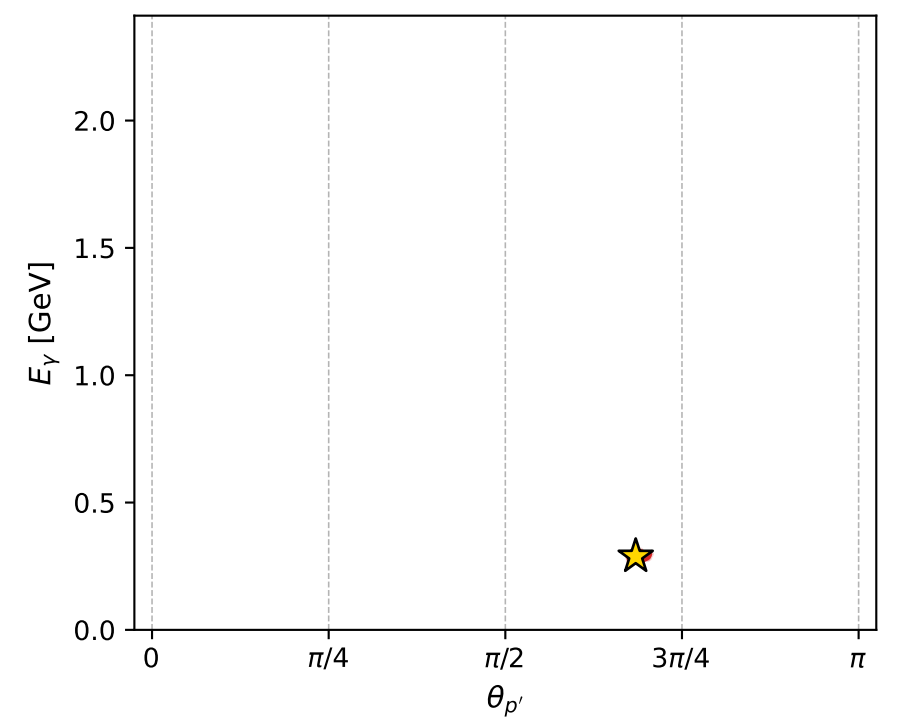
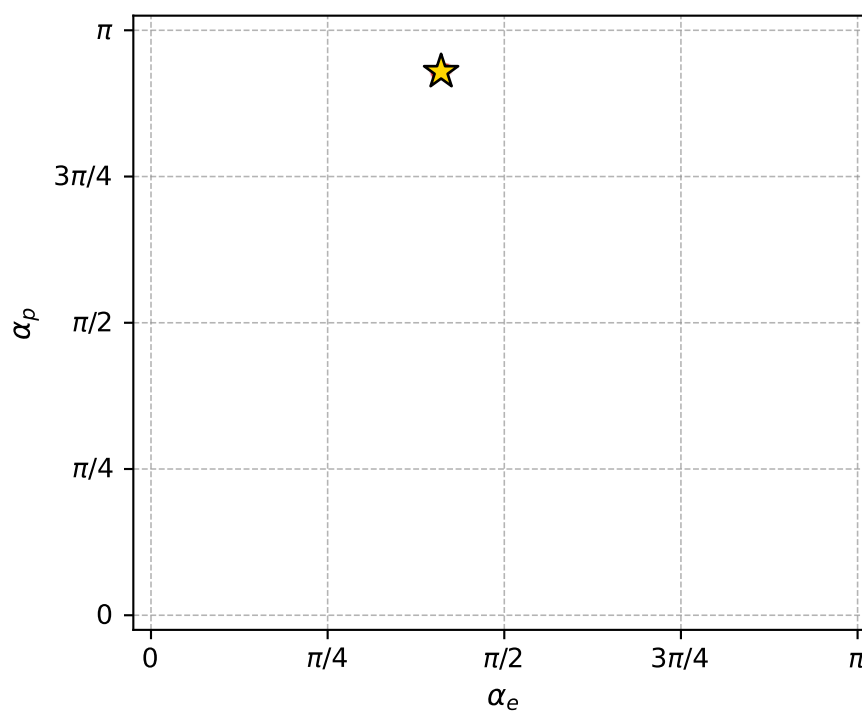
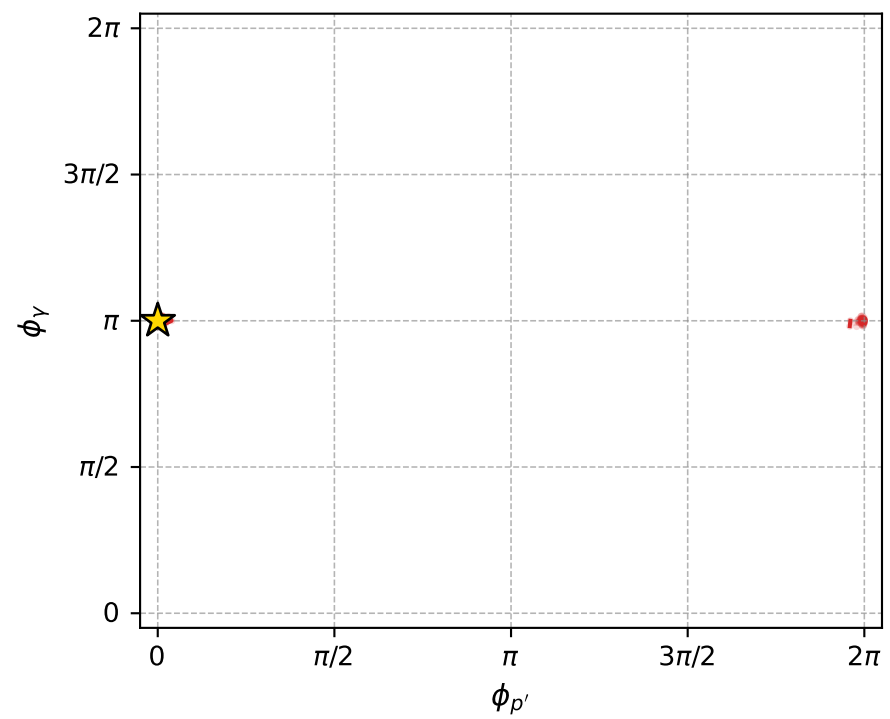
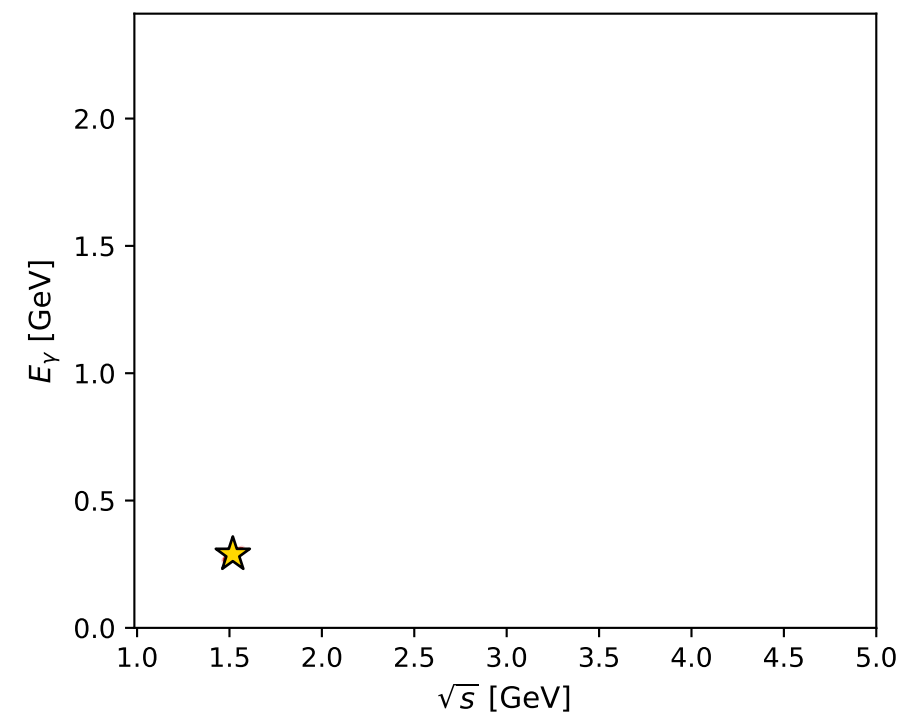
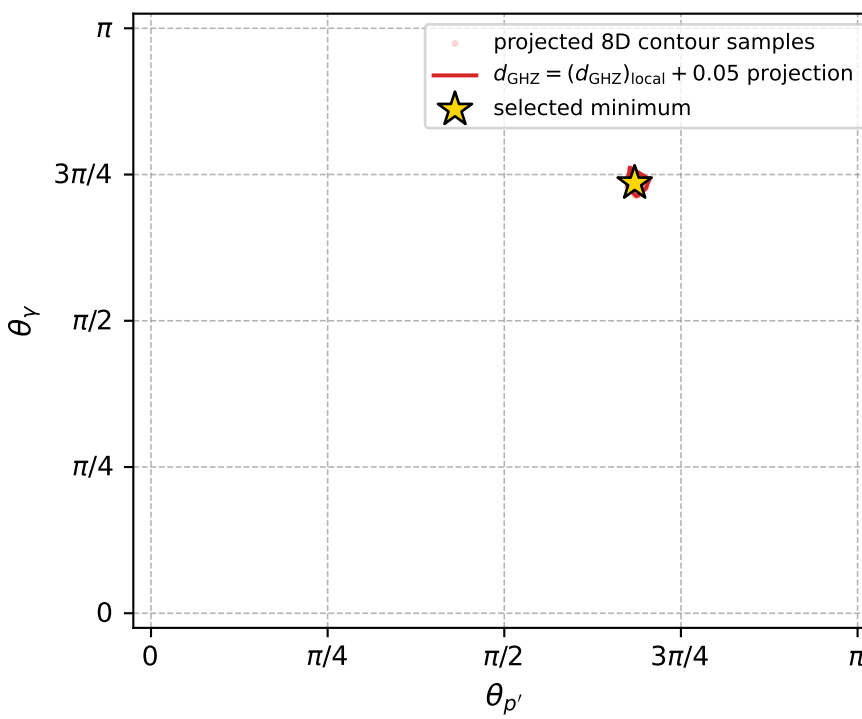
Transverse plane



Leading final-state amplitudes (N=7, retained=99.8%)



electron: polarization cluster P2, local minimum 337; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 0.00055978879$
 $d_{\text{GHZ}} \text{ contour} = 0.050559789$
 above global minimum = 0.00055976651
 8D contour samples = 256

selected phase-space point:

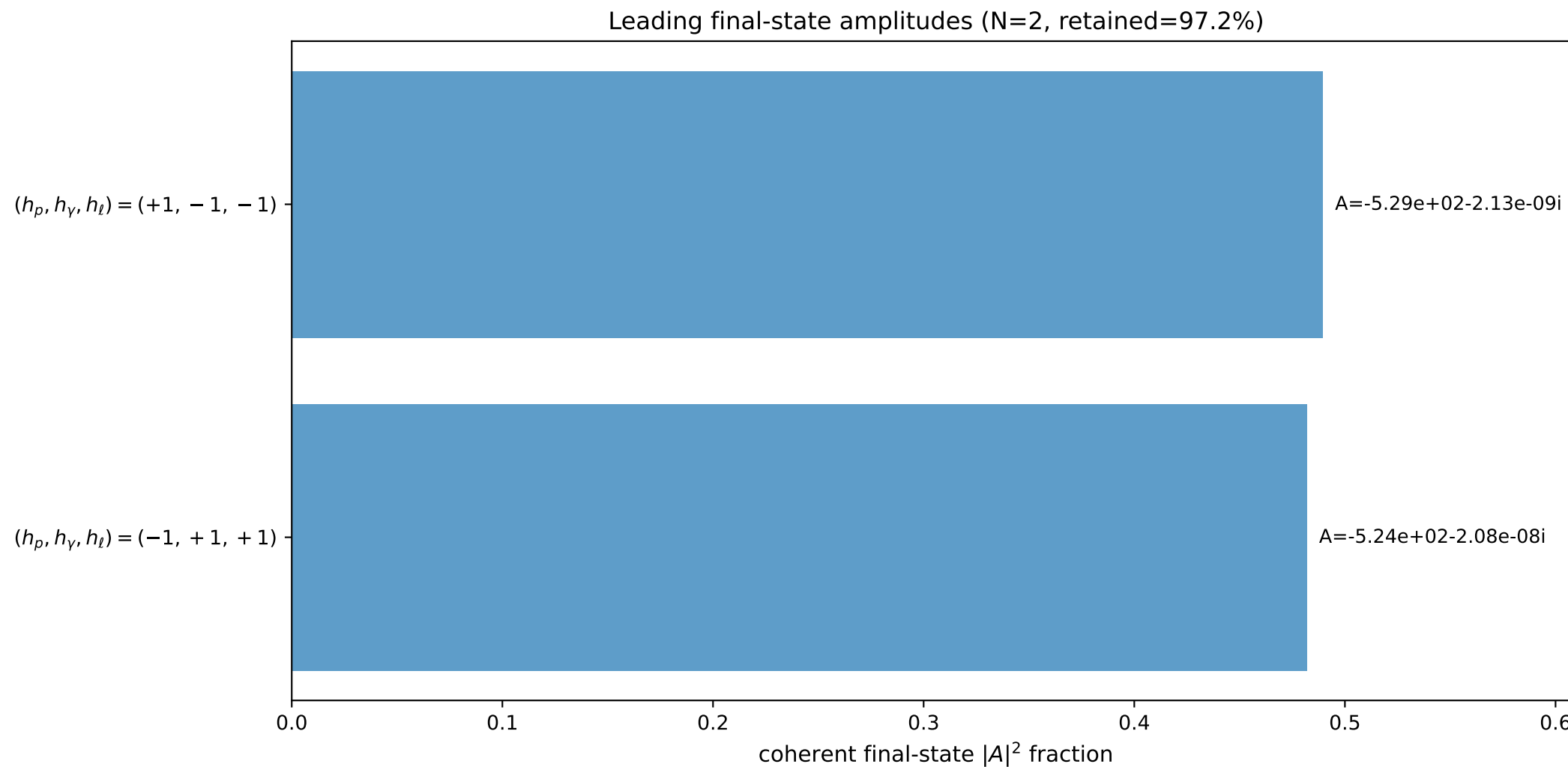
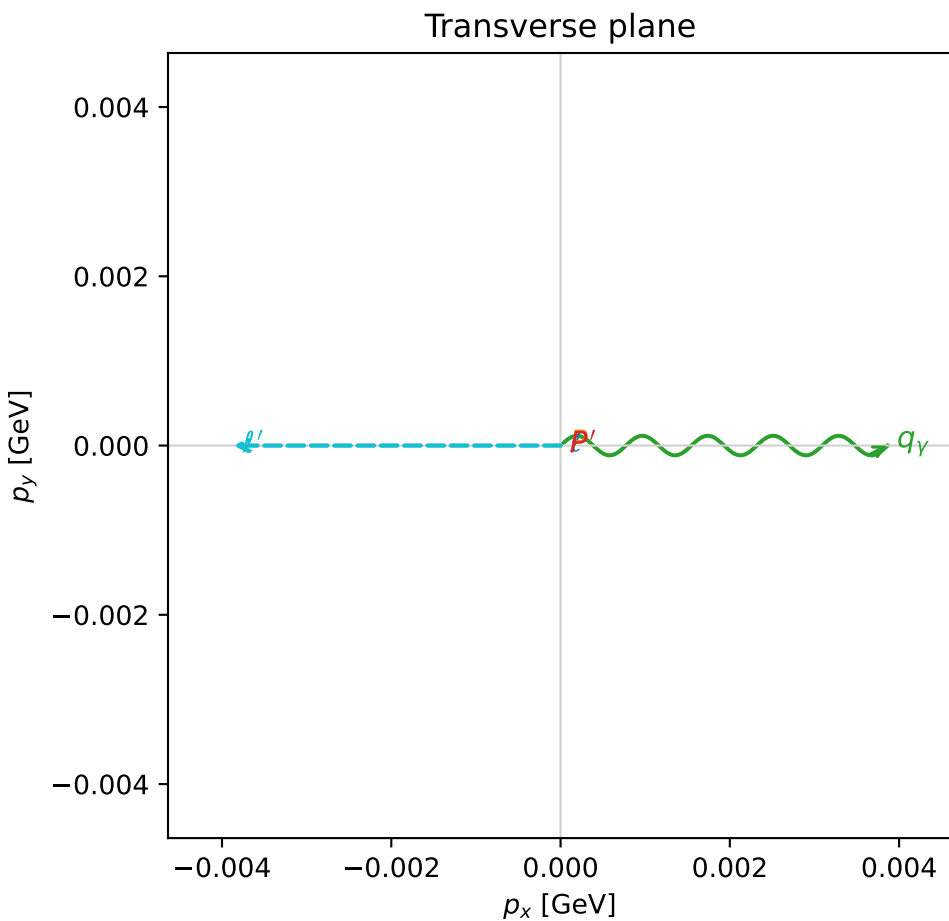
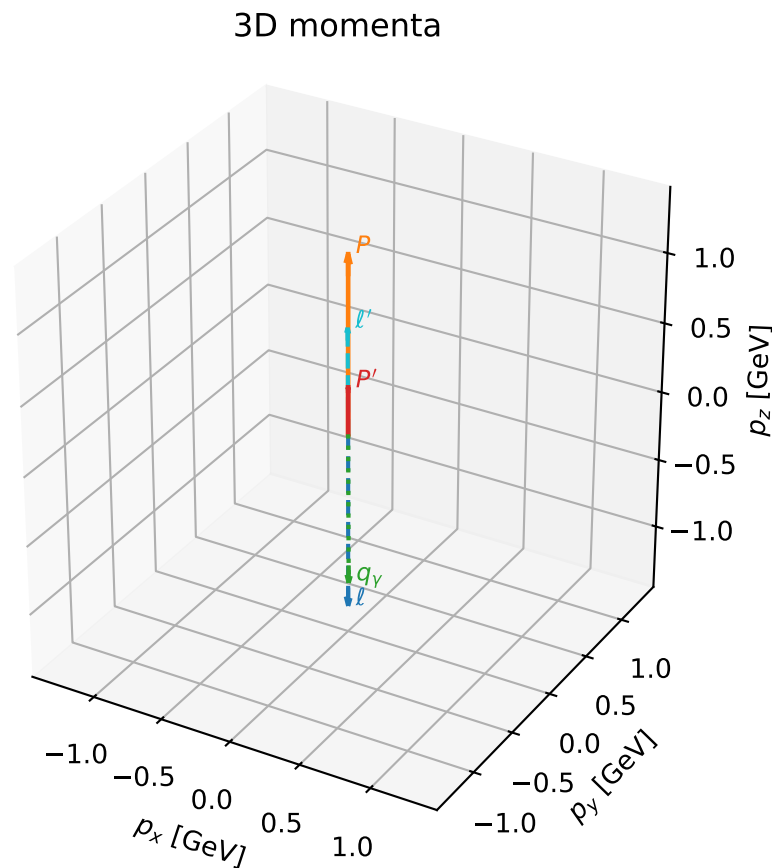
$\text{sqrt}_s = 1.518076$
 $\text{theta}_p_{\text{out}} = 2.150508$
 $\text{theta}_\gamma_{\text{out}} = 2.310096$
 $q_{\text{out}} = 0.2887381$
 $\text{phi}_p_{\text{out}} = 1.284431\text{e-}05$
 $\text{phi}_\gamma_{\text{out}} = 3.141604$
 $\alpha_e = 1.289832$
 $\alpha_p = 2.91918$

coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

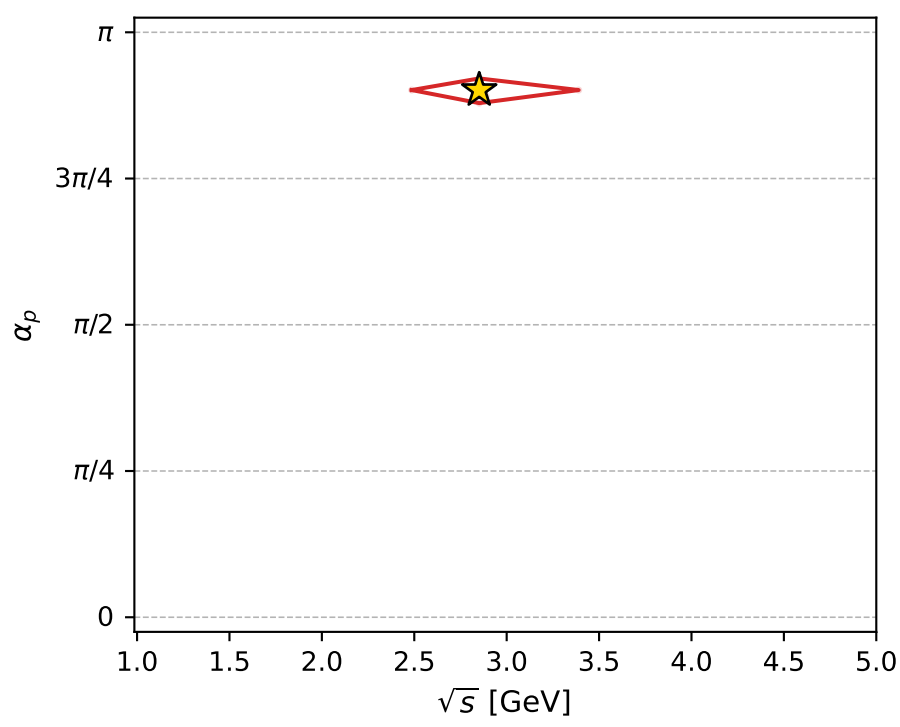
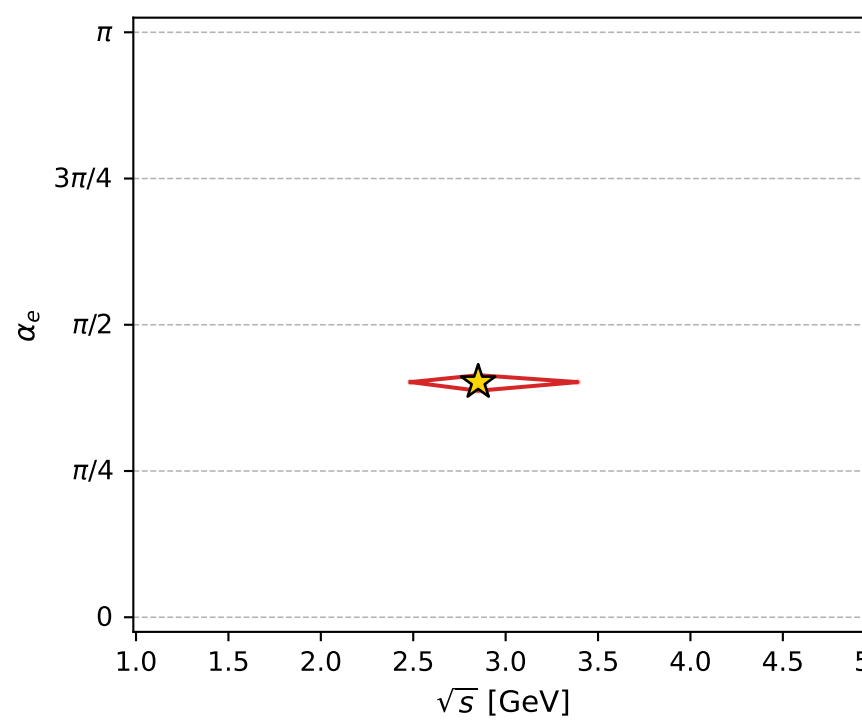
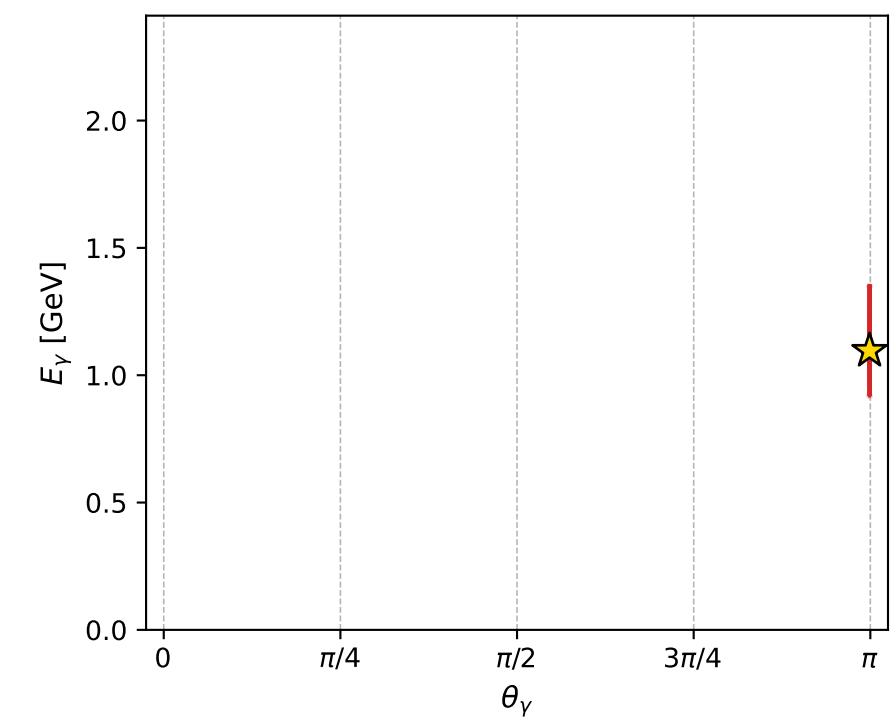
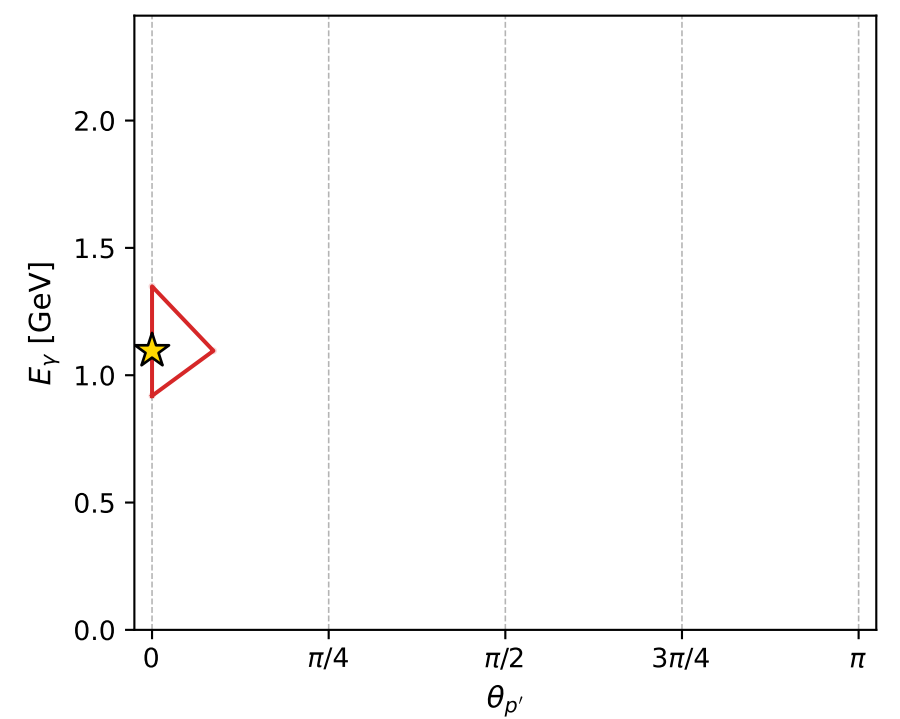
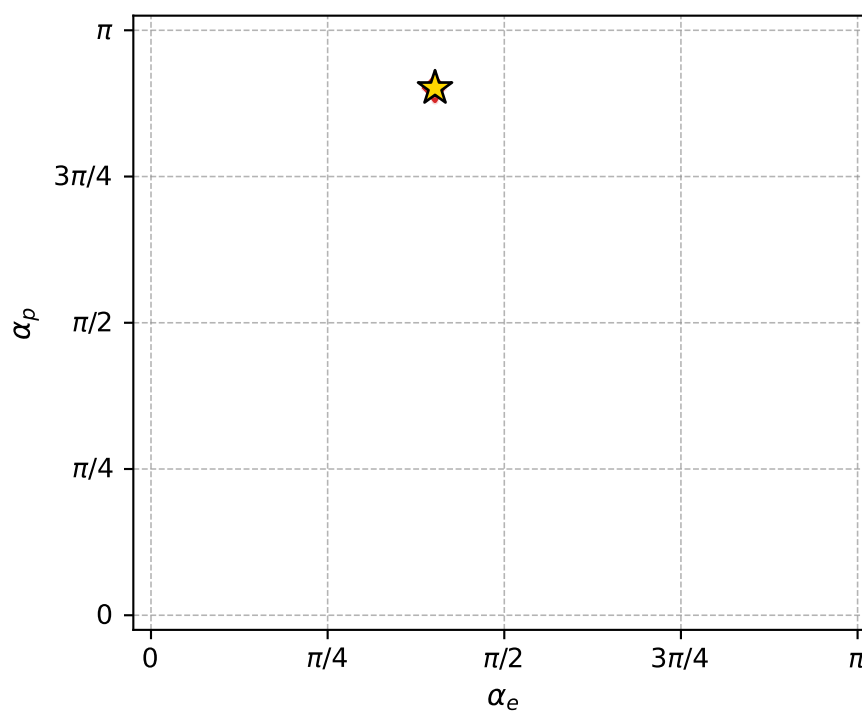
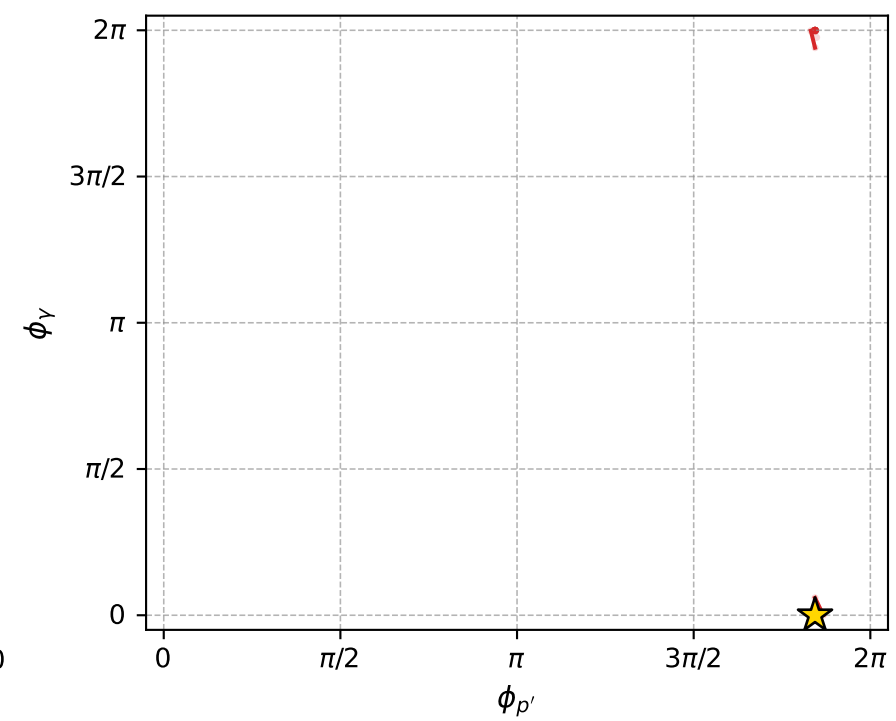
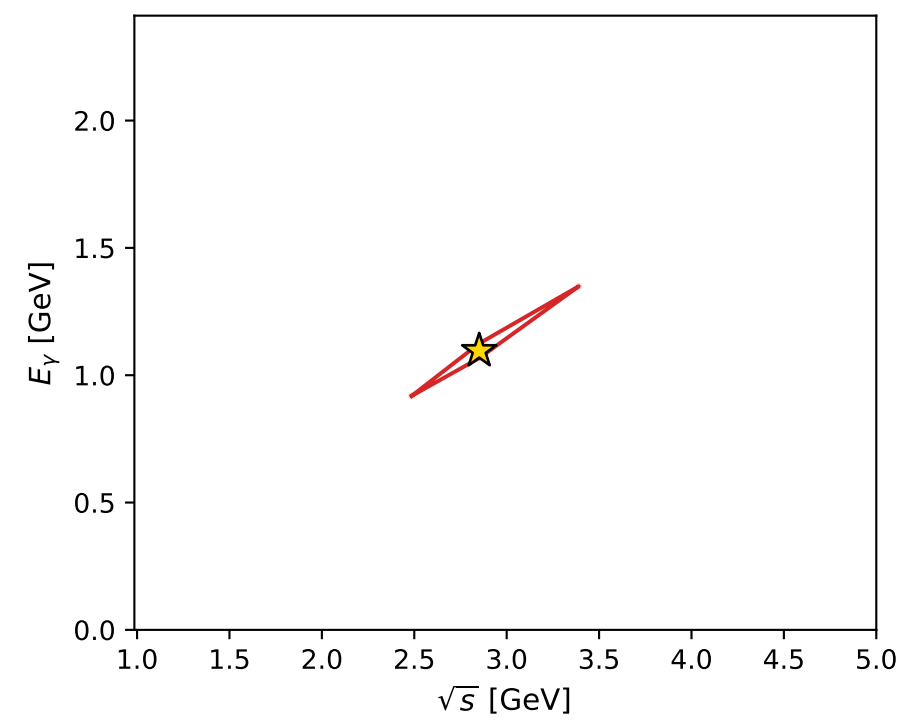
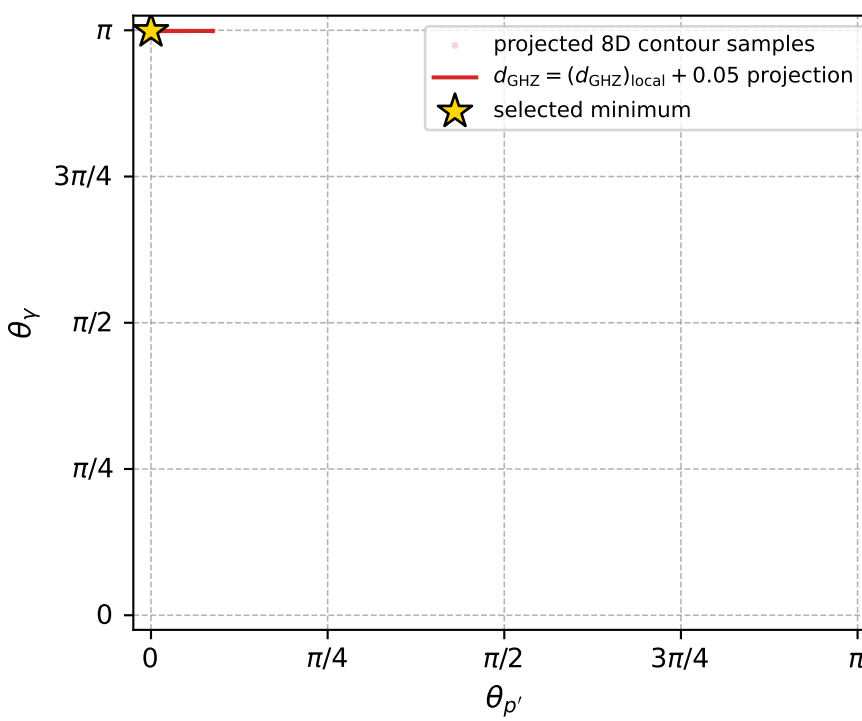
dghz_mixing_angles_polarization_02_local_minimum_339 $d_{\text{GHZ}}=0.000560516$
 region: polarization_cluster_02_local_minimum_339
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=1.2628387$ rad
 incoming lepton: $0.303113|+\rangle +0.952955|-\rangle$
 proton mixing angle: $\alpha_p=2.8313726$ rad
 incoming proton: $-0.952266|+\rangle +0.305268|-\rangle$

$s=8.13048$, $\sqrt{s}=2.8514$, $|\vec{P}|=1.27142$, $|\vec{P}'|=0.337119$
 $\theta_{p'}=0$, $\theta_Y=3.13807$, $\phi_{p'}=5.79128$, $\phi_Y=2.46636e-05$
 $E_Y=1.09589$, $\theta_{\ell'}=0.005094$, $\phi_{\ell'}=3.14162$
 $Q^2=3.85883$, $x_B=0.568952$, $t=-0.532742$, $W^2=3.80336$, $y=0.935414$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 1.2714$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.2714$
 P $E= 1.5800$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.2714$
 ℓ' $E= 0.7588$ $p_x= -0.0039$ $p_y= -0.0000$ $p_z= 0.7588$
 P' $E= 0.9967$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 0.3371$
 q_Y $E= 1.0959$ $p_x= 0.0039$ $p_y= 0.0000$ $p_z= -1.0959$



electron: polarization cluster P2, local minimum 339; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 0.00056051553$
 $d_{\text{GHZ}} \text{ contour} = 0.050560516$
 above global minimum = 0.00056049325
 8D contour samples = 132

selected phase-space point:

$\text{sqrt_s} = 2.851399$
 $\text{theta_p_out} = 0$
 $\text{theta_gamma_out} = 3.138066$
 $\text{q0ut} = 1.095887$
 $\text{phi_p_out} = 5.791285$
 $\text{phi_gamma_out} = 2.466362\text{e-}05$
 $\text{alpha_e} = 1.262839$
 $\text{alpha_p} = 2.831373$

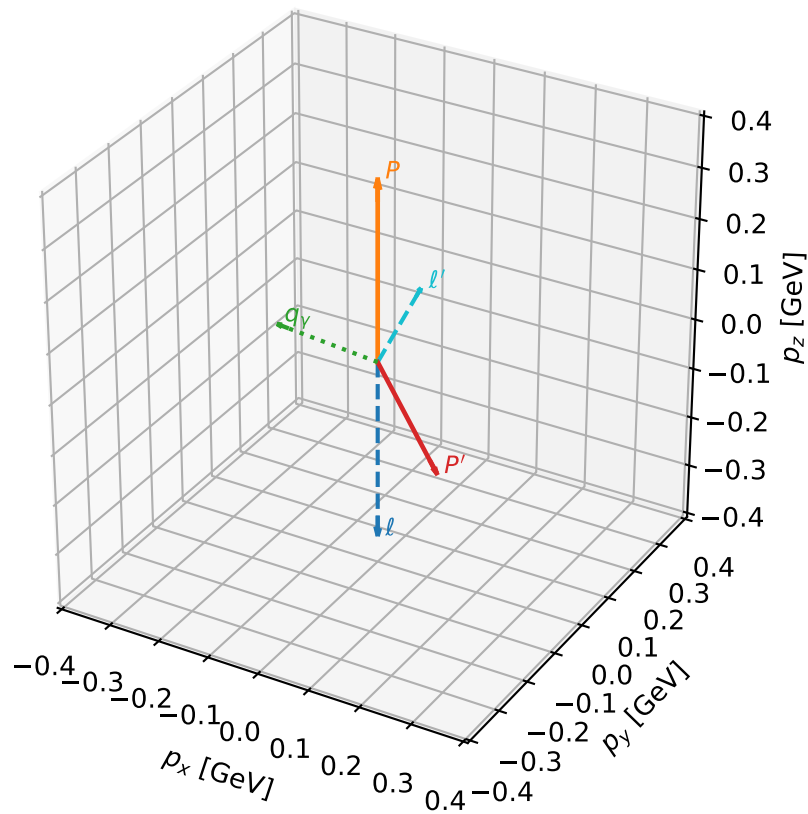
coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

dghz_mixing_angles_polarization_02_local_minimum_344 $d_{\{\text{rm GHZ}\}}=0.000631579$
 region: polarization_cluster_02_local_minimum_344
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=1.2828643$ rad
 incoming lepton: $0.28397|+\rangle + 0.958833|-\rangle$
 proton mixing angle: $\alpha_p=2.9141102$ rad
 incoming proton: $-0.974237|+\rangle + 0.225526|-\rangle$

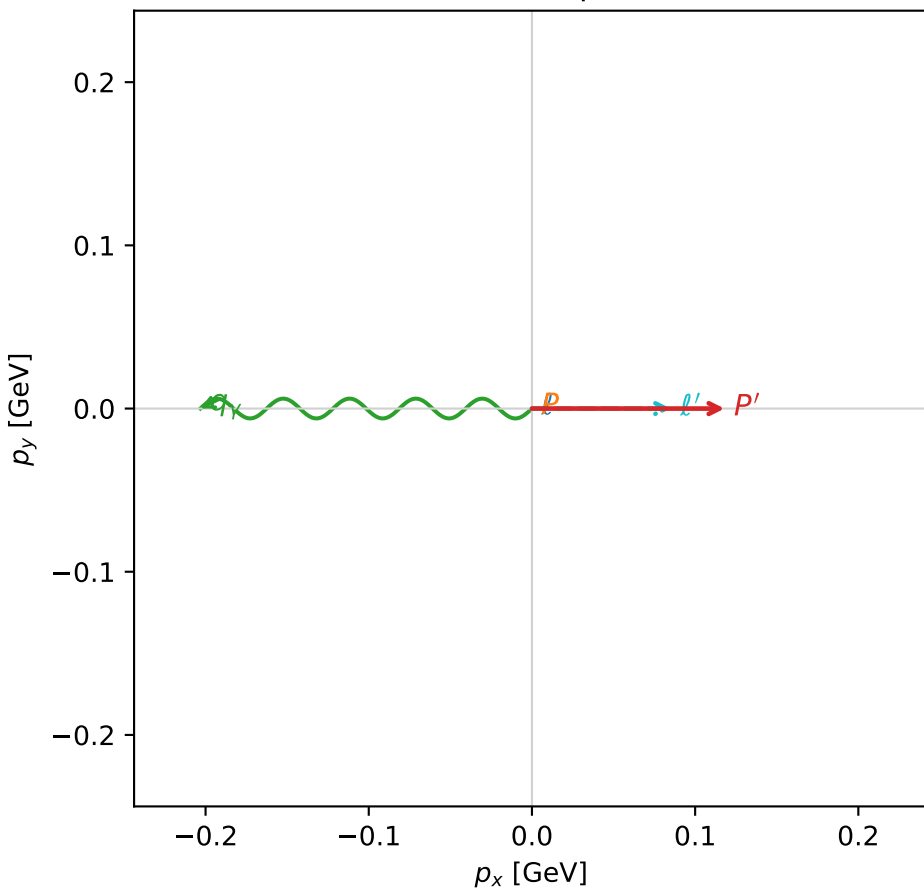
$s=1.84464$, $\sqrt{s}=1.35818$, $|\vec{P}|=0.355181$, $|\vec{P}'|=0.222347$
 $\theta_{P'}=2.58194$, $\theta_Y=1.48068$, $\phi_{P'}=6.28302$, $\phi_Y=3.14149$
 $E_Y=0.203999$, $\theta_{\ell'}=0.464096$, $\phi_{\ell'}=6.28317$
 $Q^2=0.255908$, $x_B=0.363454$, $t=-0.307921$, $W^2=1.32803$, $y=0.72979$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.3552$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.3552$
 P $E= 1.0030$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.3552$
 ℓ' $E= 0.1902$ $p_x= 0.0851$ $p_y= -0.0000$ $p_z= 0.1701$
 P' $E= 0.9640$ $p_x= 0.1180$ $p_y= -0.0000$ $p_z= -0.1884$
 q_Y $E= 0.2040$ $p_x= -0.2032$ $p_y= 0.0000$ $p_z= 0.0184$

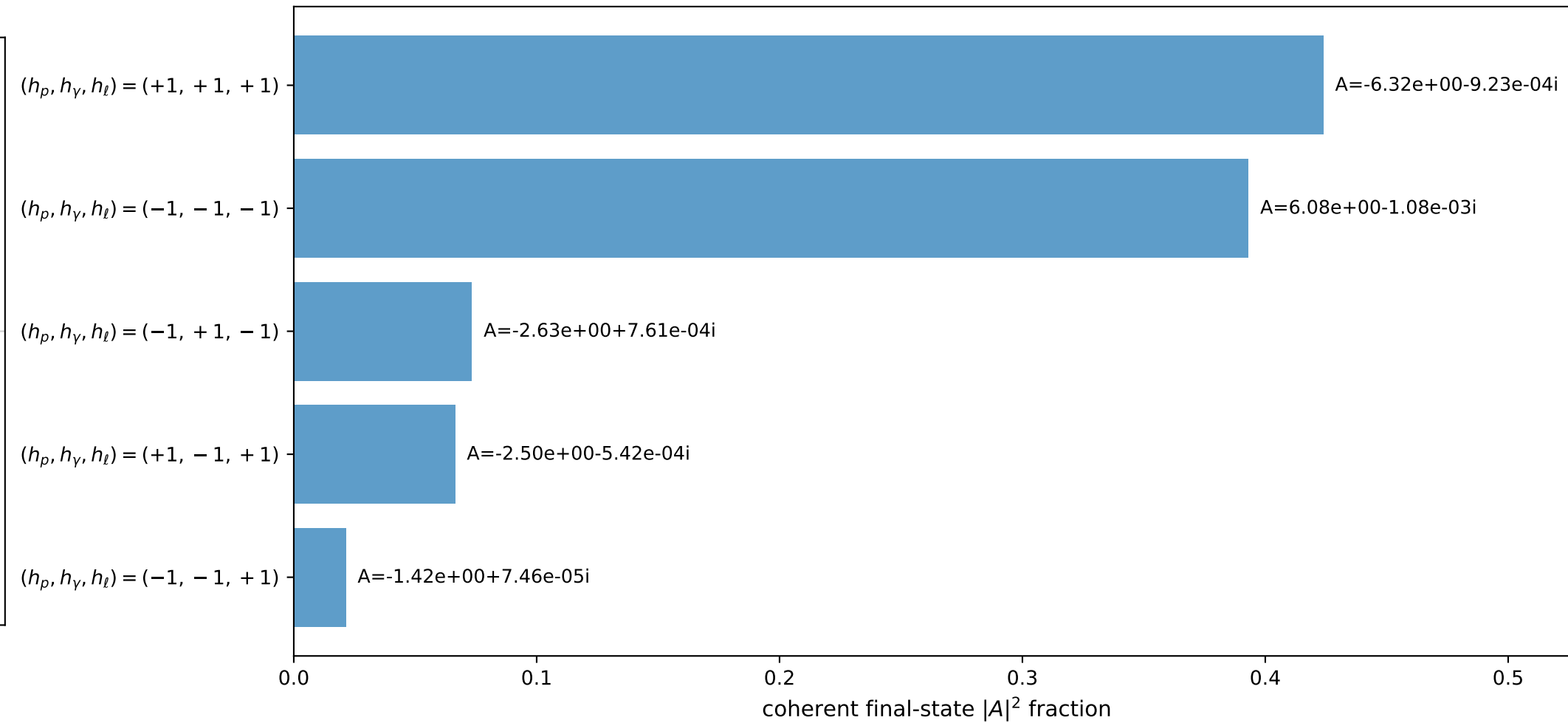
3D momenta



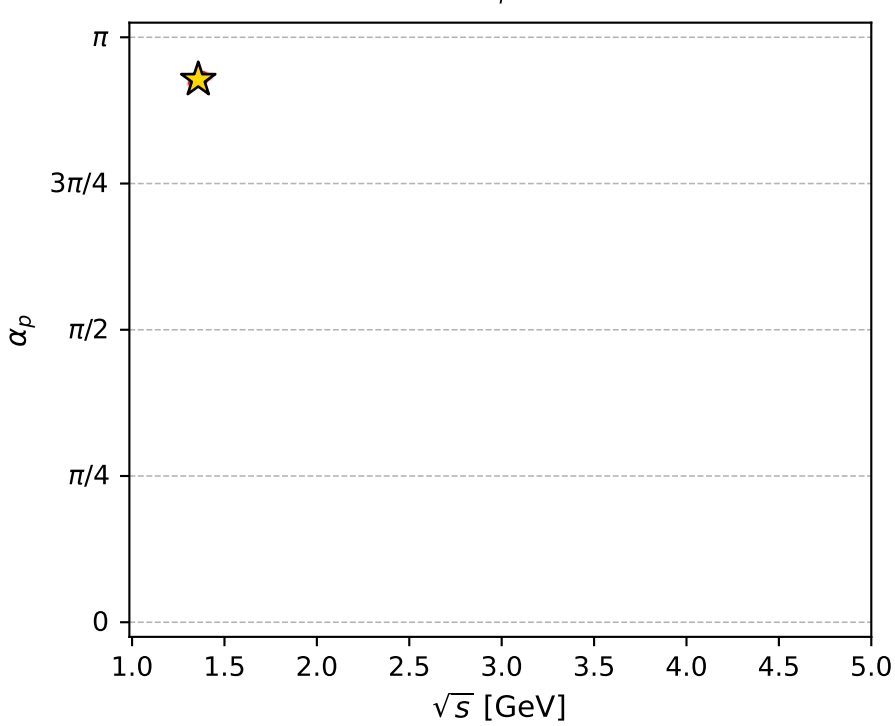
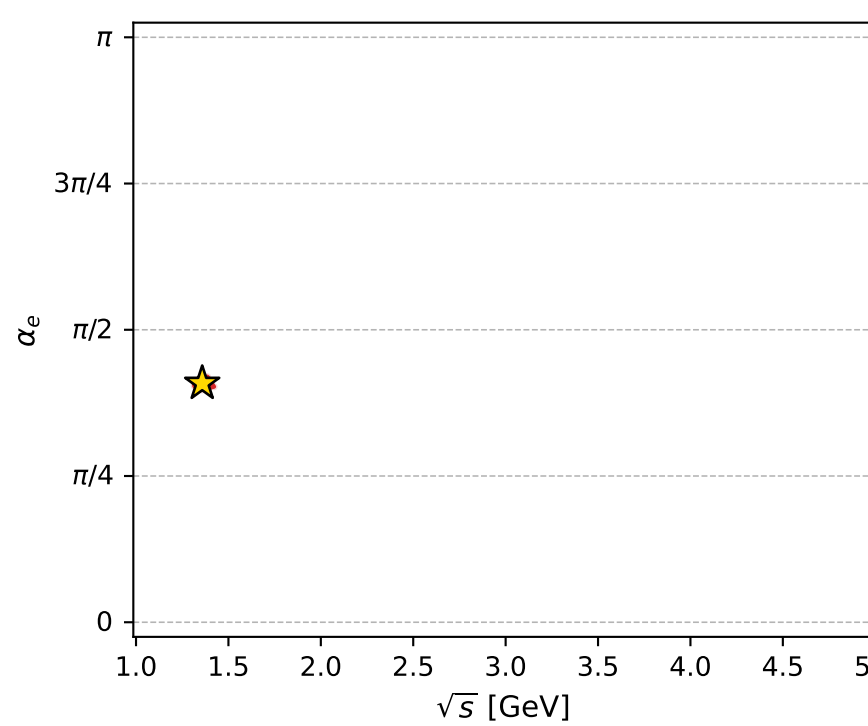
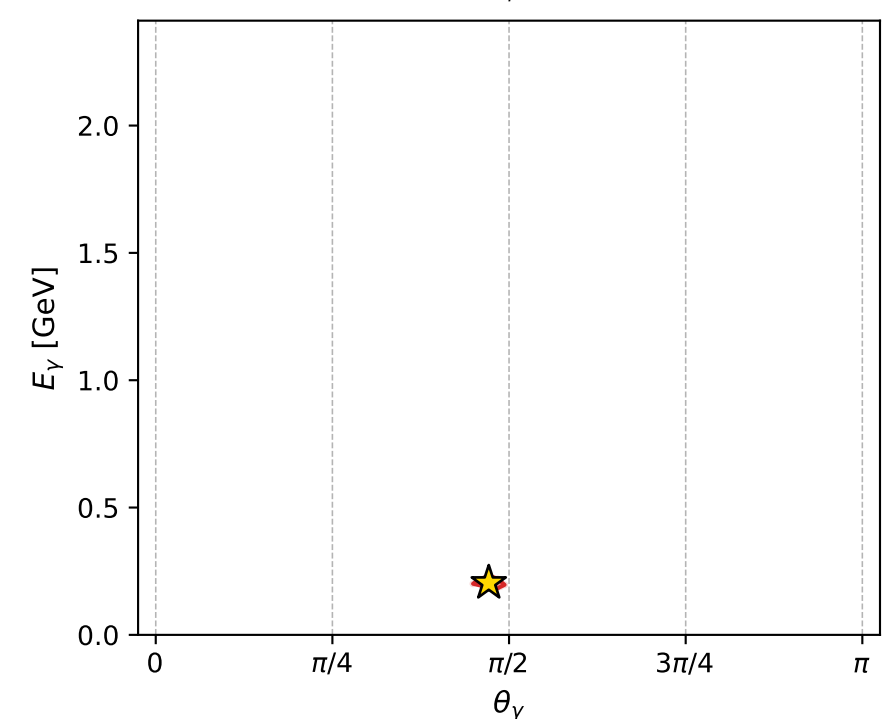
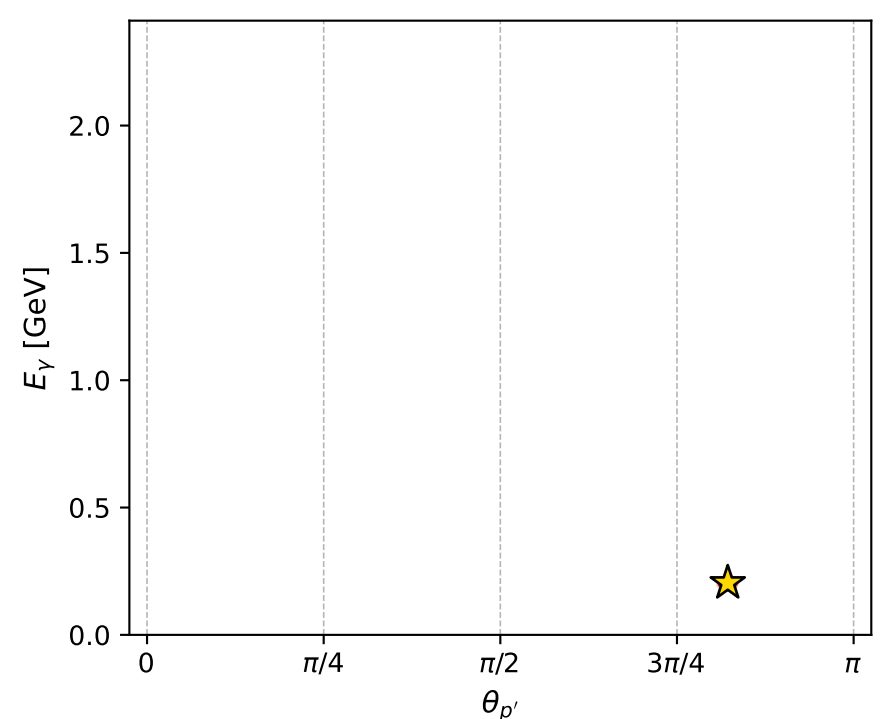
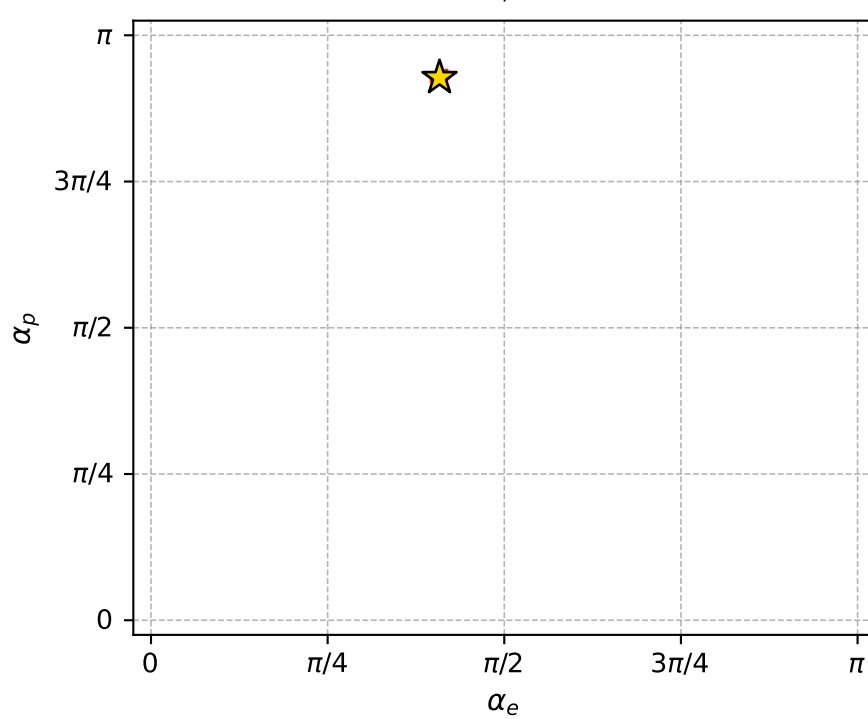
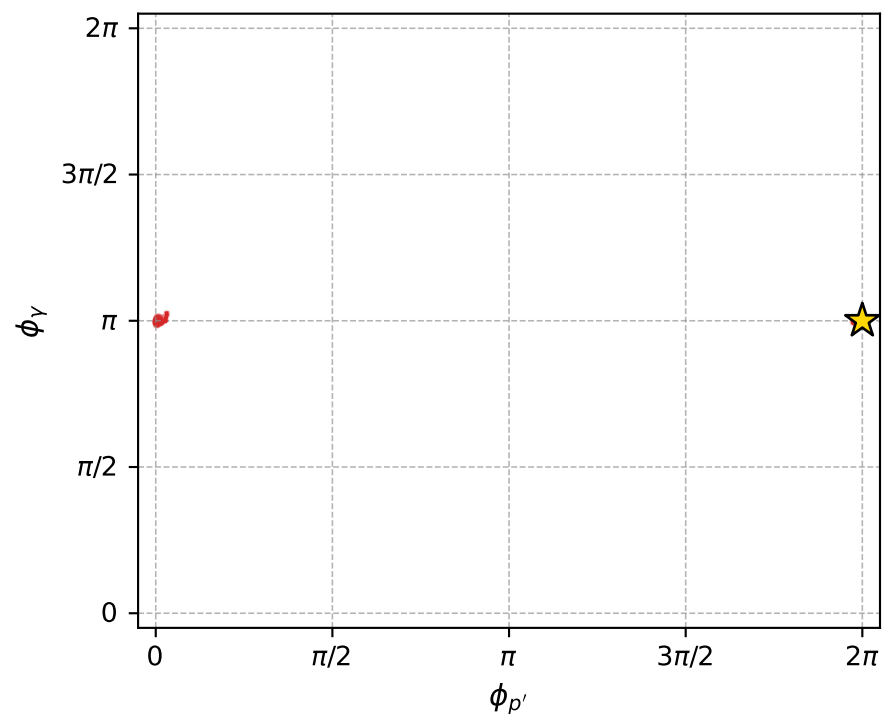
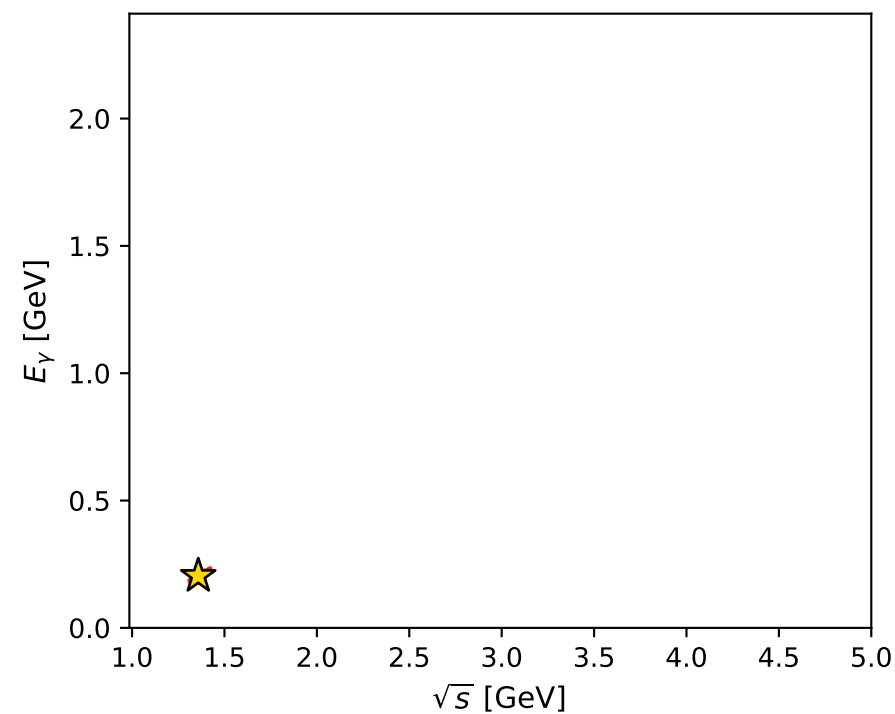
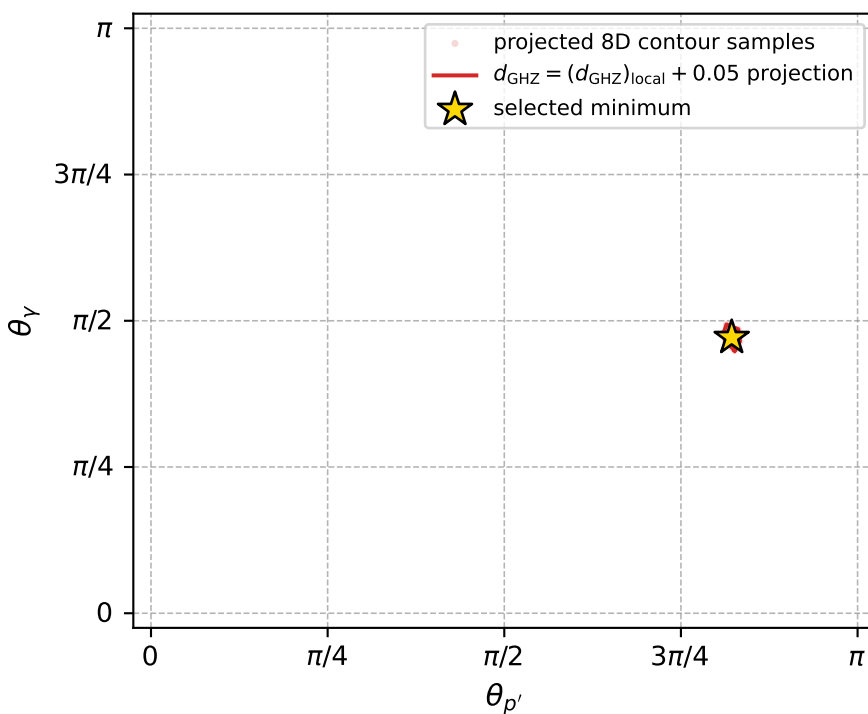
Transverse plane



Leading final-state amplitudes (N=5, retained=97.8%)



electron: polarization cluster P2, local minimum 344; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 0.00063157888$
 $d_{\text{GHZ}} \text{ contour} = 0.050631579$
 above global minimum = 0.0006315566
 8D contour samples = 256

selected phase-space point:

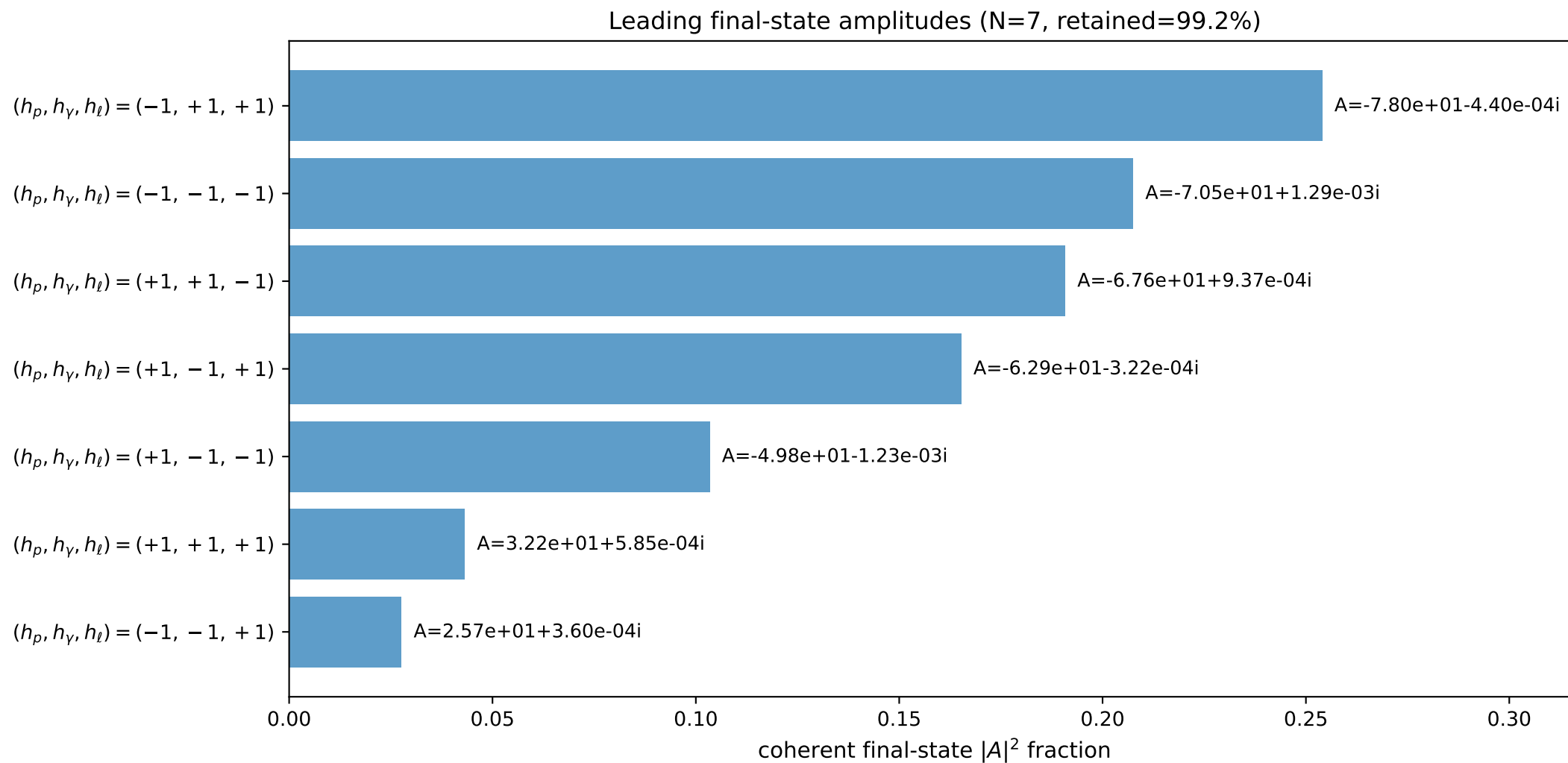
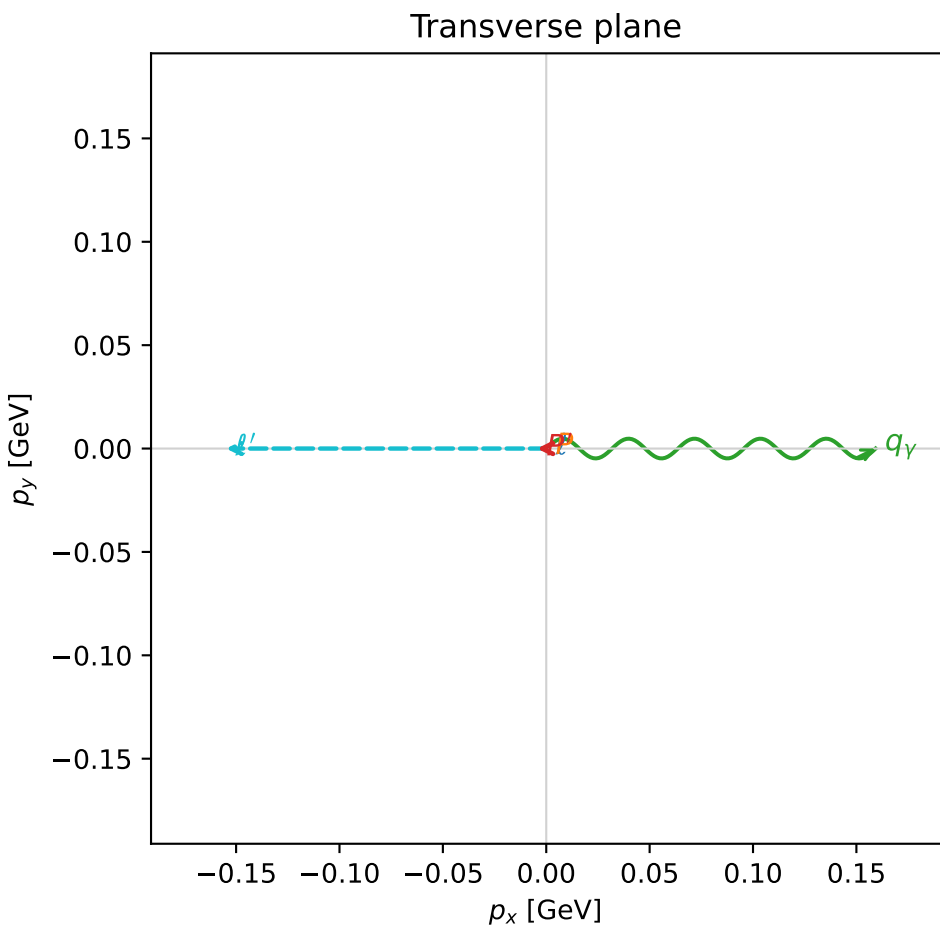
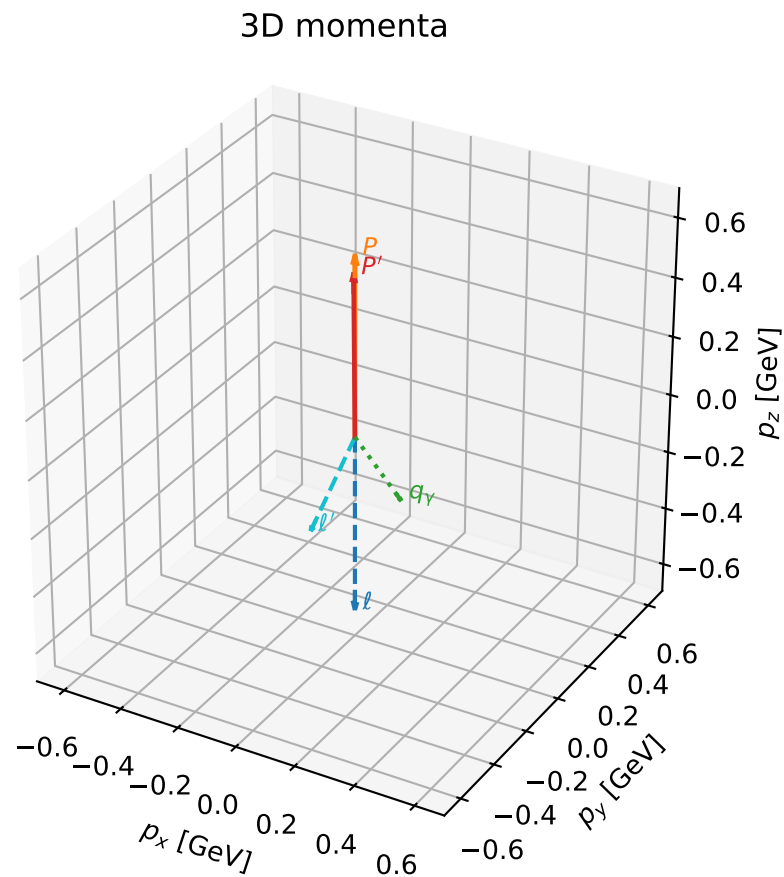
$\text{sqrt_s} = 1.358175$
 $\text{theta_p_out} = 2.581941$
 $\text{theta_gamma_out} = 1.480677$
 $\text{q0ut} = 0.2039987$
 $\text{phi_p_out} = 6.283019$
 $\text{phi_gamma_out} = 3.14149$
 $\text{alpha_e} = 1.282864$
 $\text{alpha_p} = 2.91411$

coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

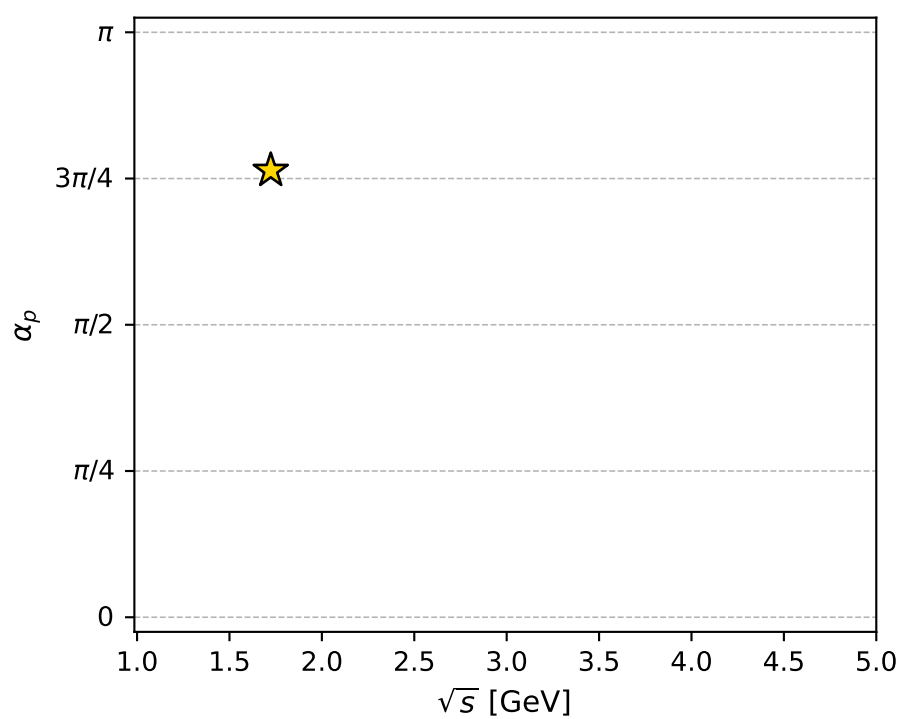
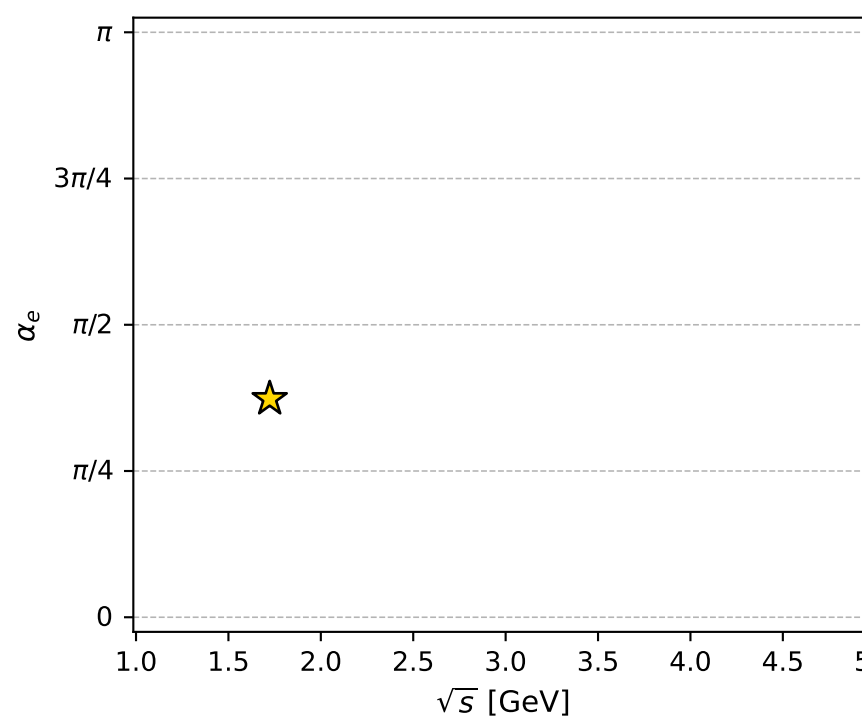
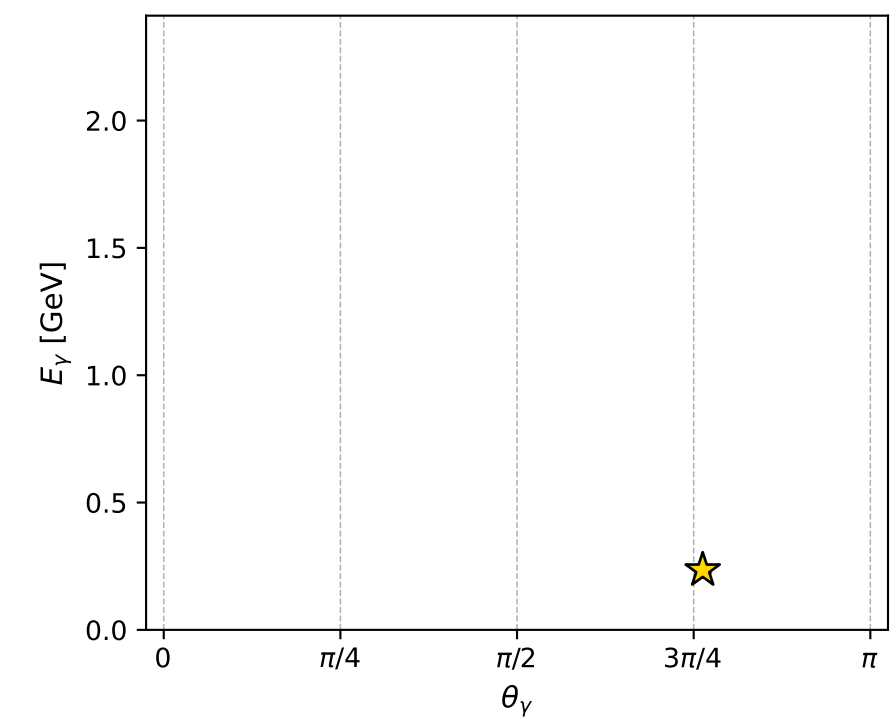
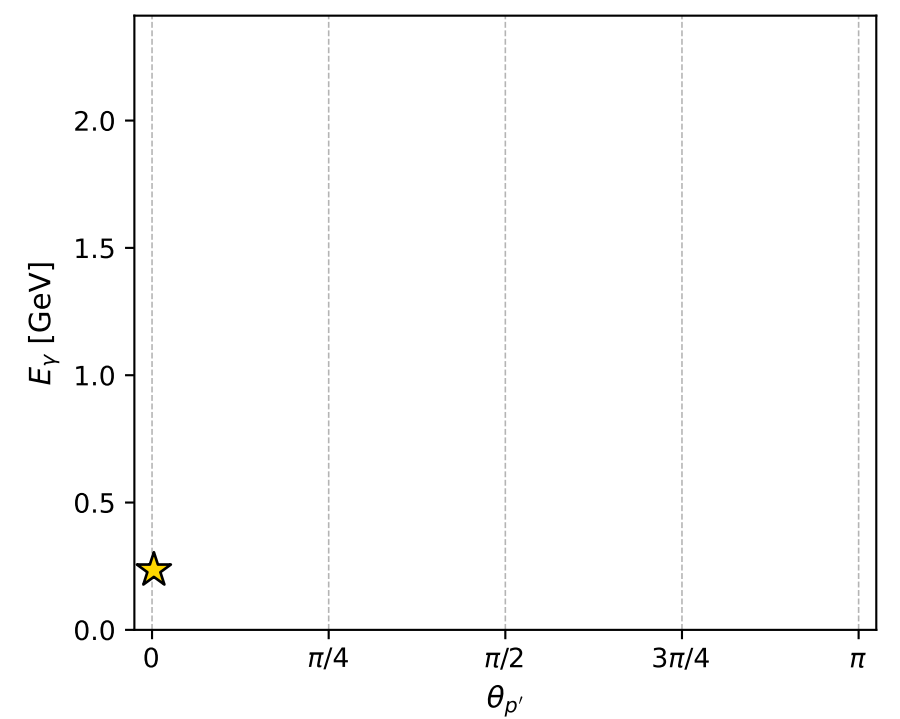
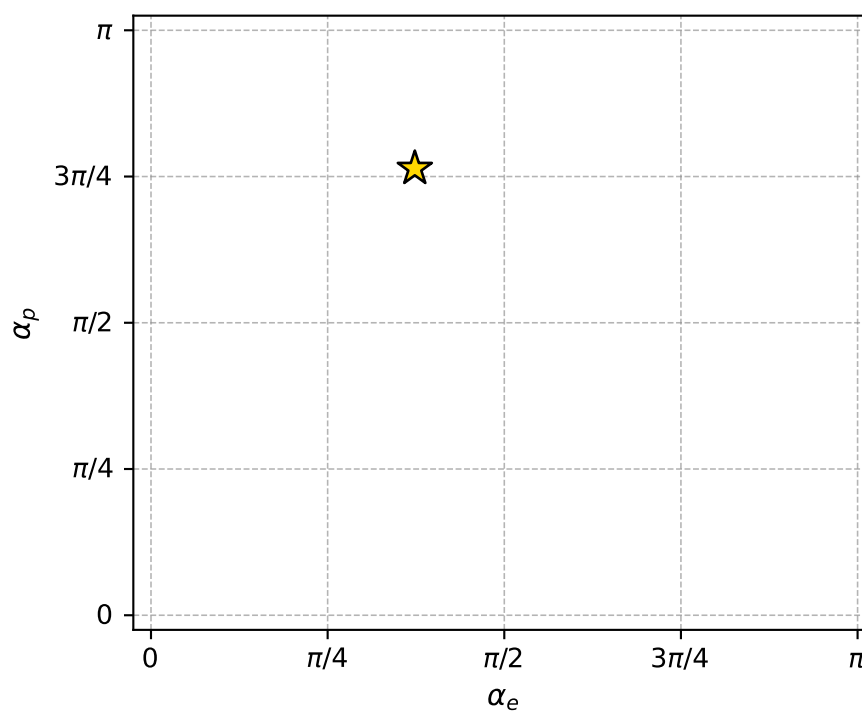
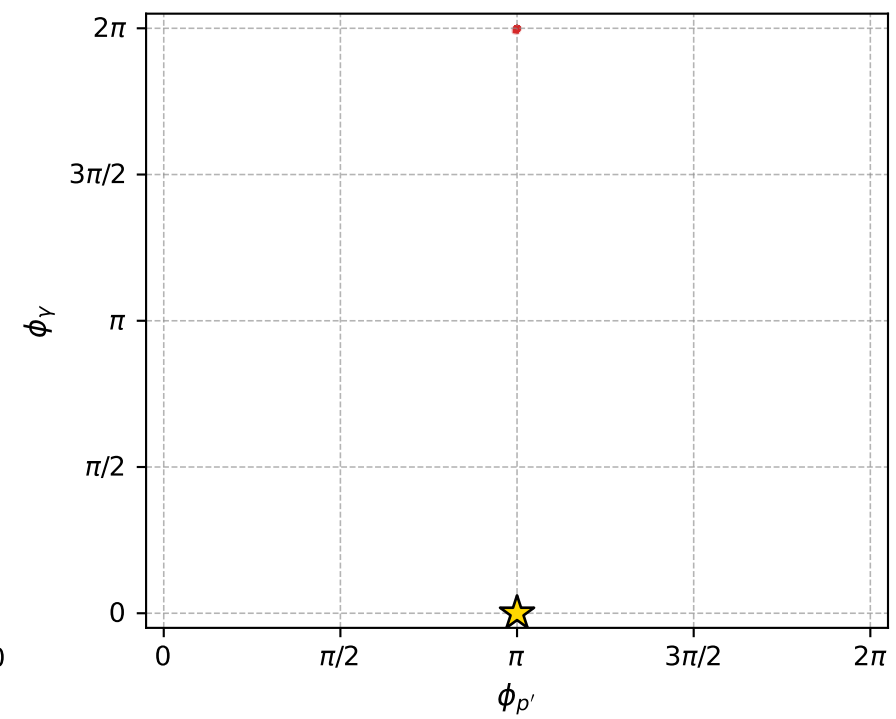
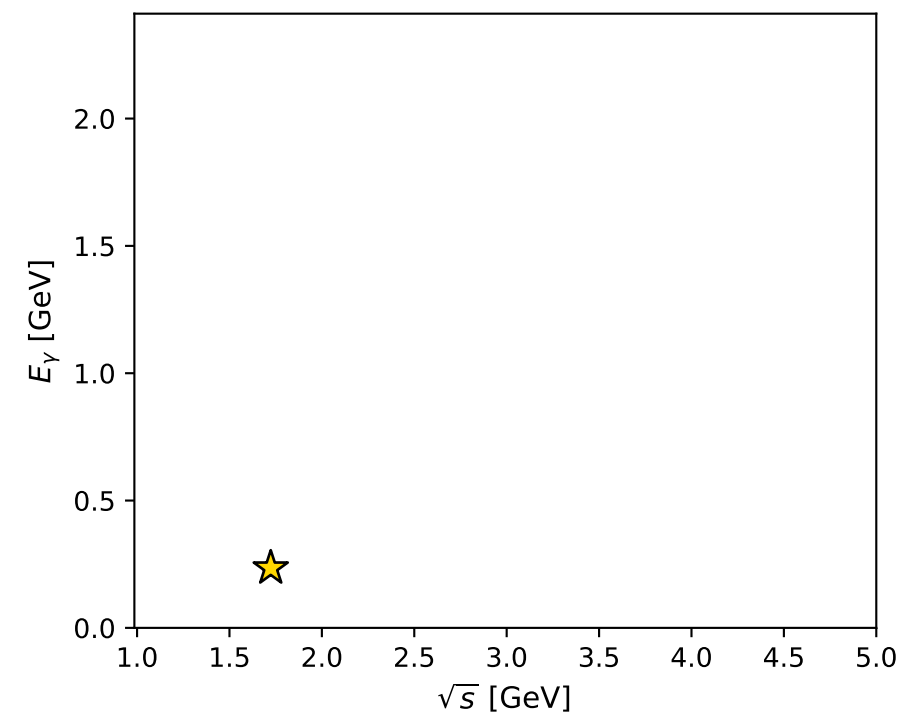
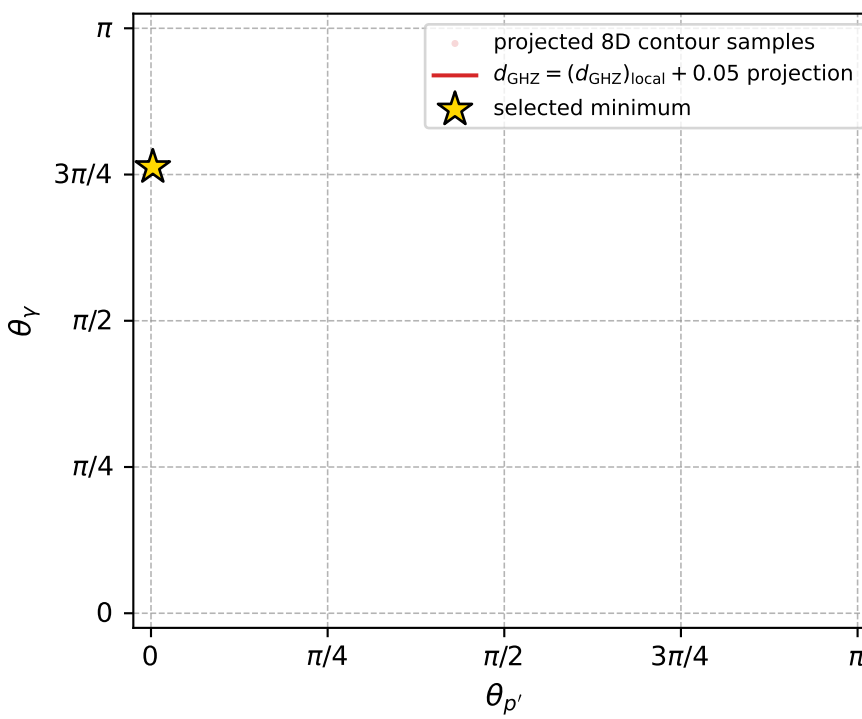
dghz_mixing_angles_polarization_02_local_minimum_347 d_{\rm GHZ}=0.00072087
 region: polarization_cluster_02_local_minimum_347
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=1.1729992$ rad
 incoming lepton: $0.387388|+\rangle + 0.921917|-\rangle$
 proton mixing angle: $\alpha_p=2.399453$ rad
 incoming proton: $-0.737024|+\rangle + 0.675866|-\rangle$

$s=2.97001$, $\sqrt{s}=1.72337$, $|\vec{P}|=0.606417$, $|\vec{P}'|=0.545266$
 $\theta_{P'}=0.00814588$, $\theta_Y=2.39633$, $\phi_{P'}=3.14157$, $\phi_Y=0$
 $E_Y=0.234874$, $\theta_{\ell'}=2.74777$, $\phi_{\ell'}=3.14159$
 $Q^2=0.0374651$, $x_B=0.0508503$, $t=-0.00273845$, $W^2=1.57915$, $y=0.352495$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.6064$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.6064$
 P $E= 1.1170$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.6064$
 ℓ' $E= 0.4035$ $p_x= -0.1548$ $p_y= -0.0000$ $p_z= -0.3726$
 P' $E= 1.0850$ $p_x= -0.0044$ $p_y= 0.0000$ $p_z= 0.5452$
 q_Y $E= 0.2349$ $p_x= 0.1593$ $p_y= 0.0000$ $p_z= -0.1726$



electron: polarization cluster P2, local minimum 347; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 0.0007208696$
 $d_{\text{GHZ}} \text{ contour} = 0.05072087$
 above global minimum = 0.00072084731
 8D contour samples = 256

selected phase-space point:

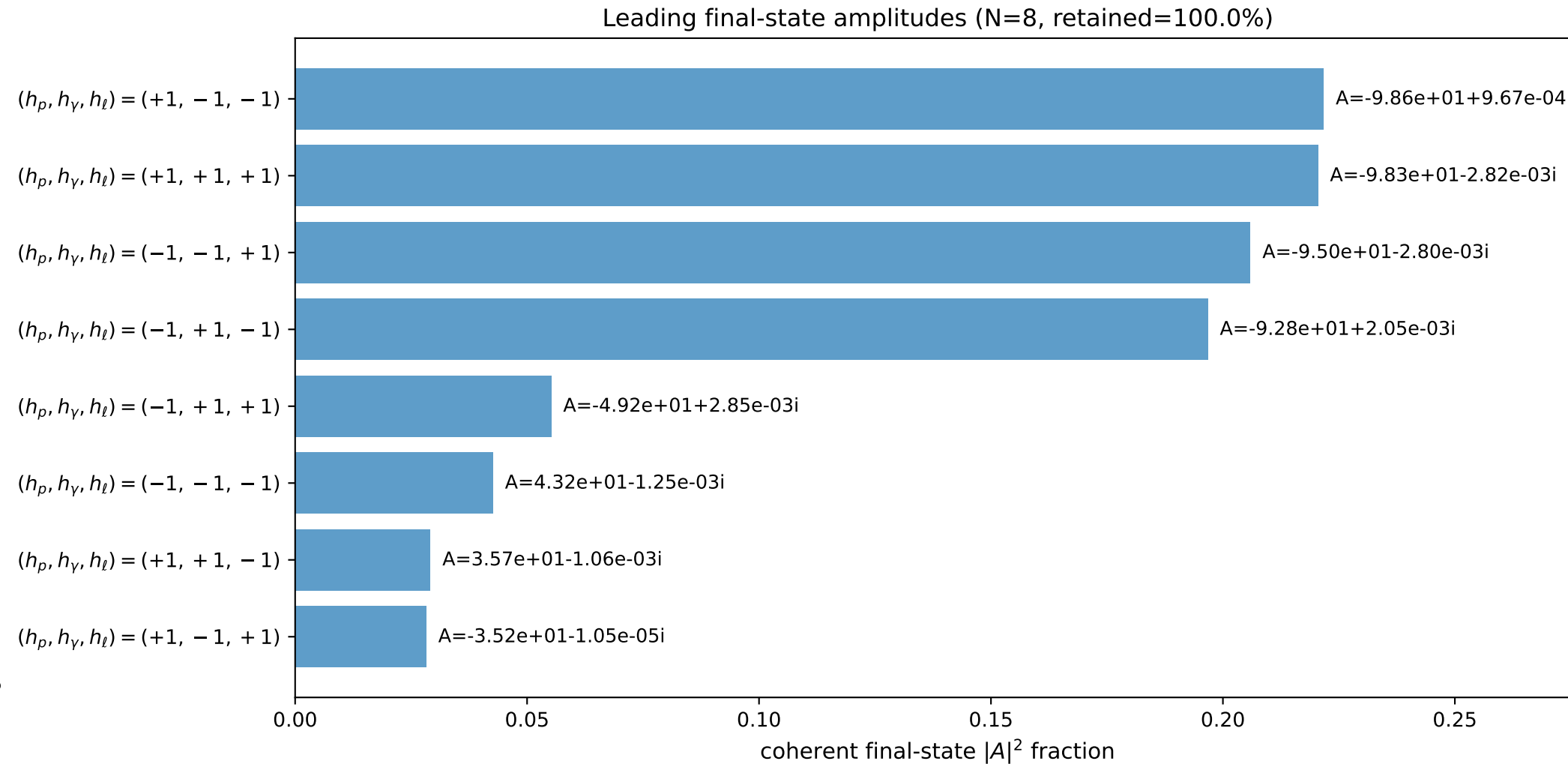
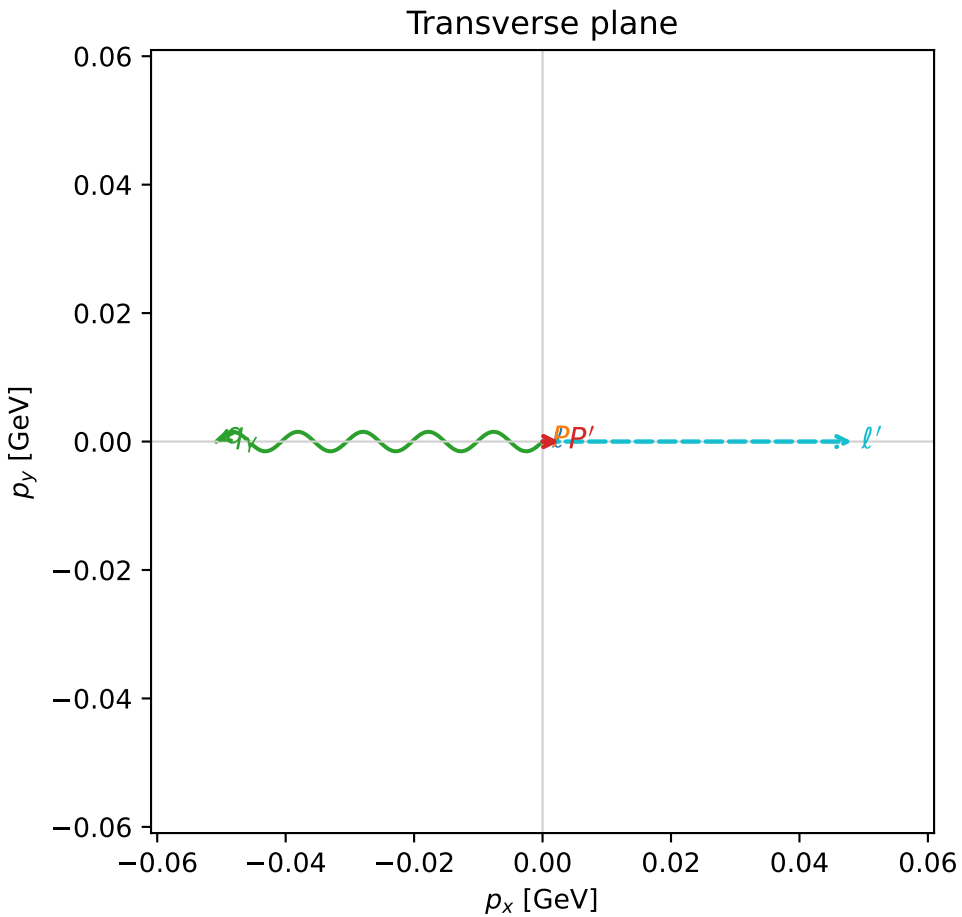
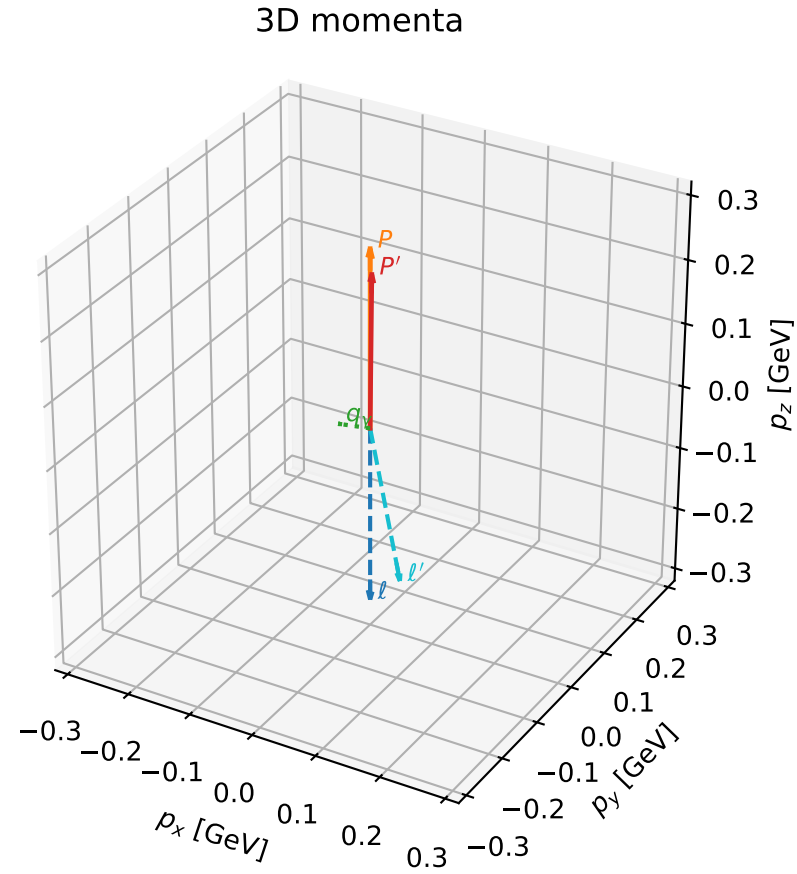
$\text{sqrt_s} = 1.723372$
 $\text{theta_p_out} = 0.008145883$
 $\text{theta_gamma_out} = 2.396325$
 $\text{q0out} = 0.2348737$
 $\text{phi_p_out} = 3.141574$
 $\text{phi_gamma_out} = 0$
 $\text{alpha_e} = 1.172999$
 $\text{alpha_p} = 2.399453$

coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

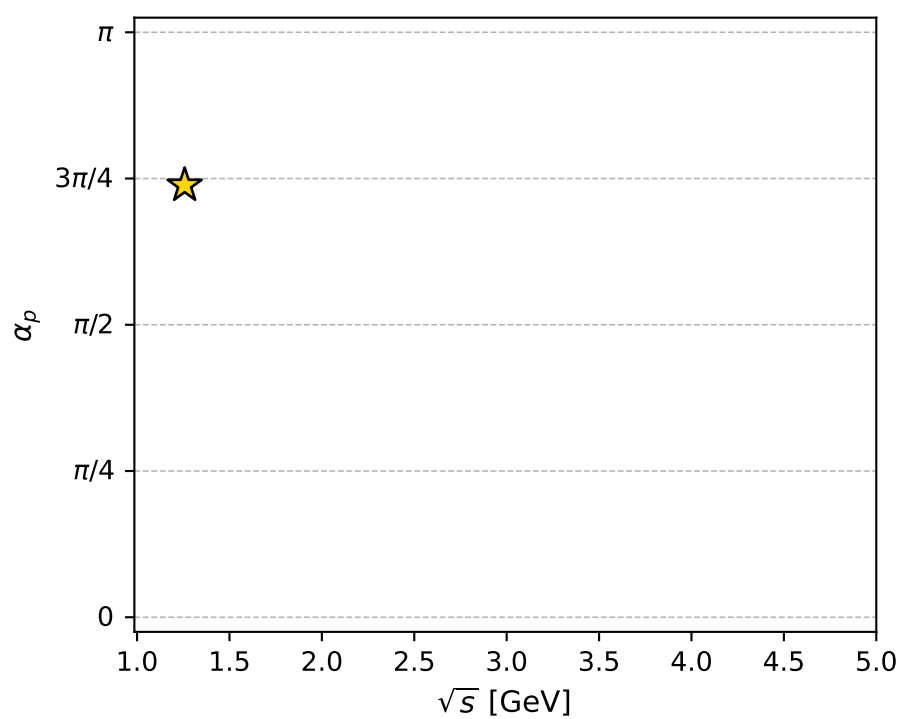
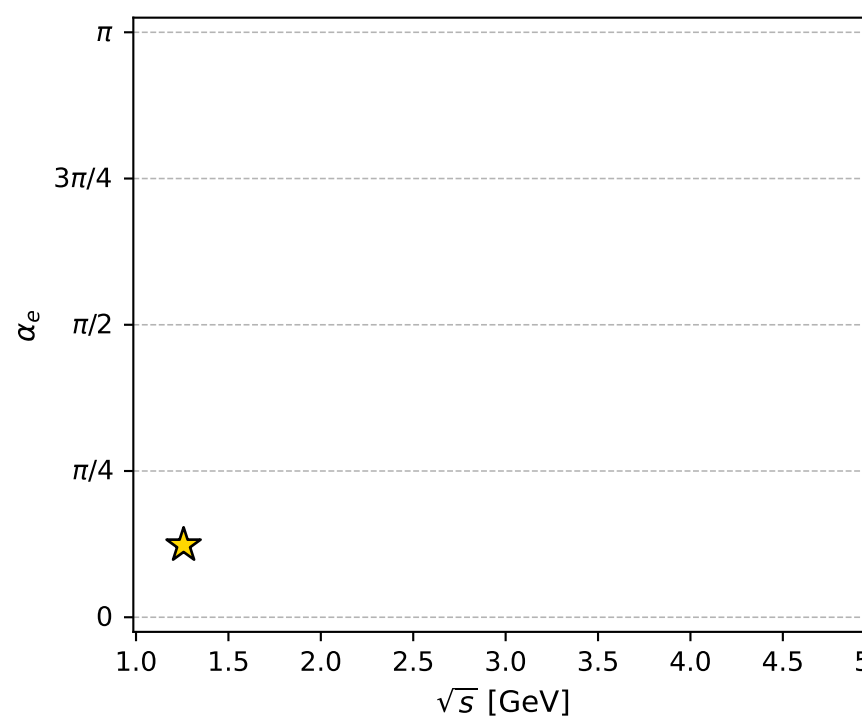
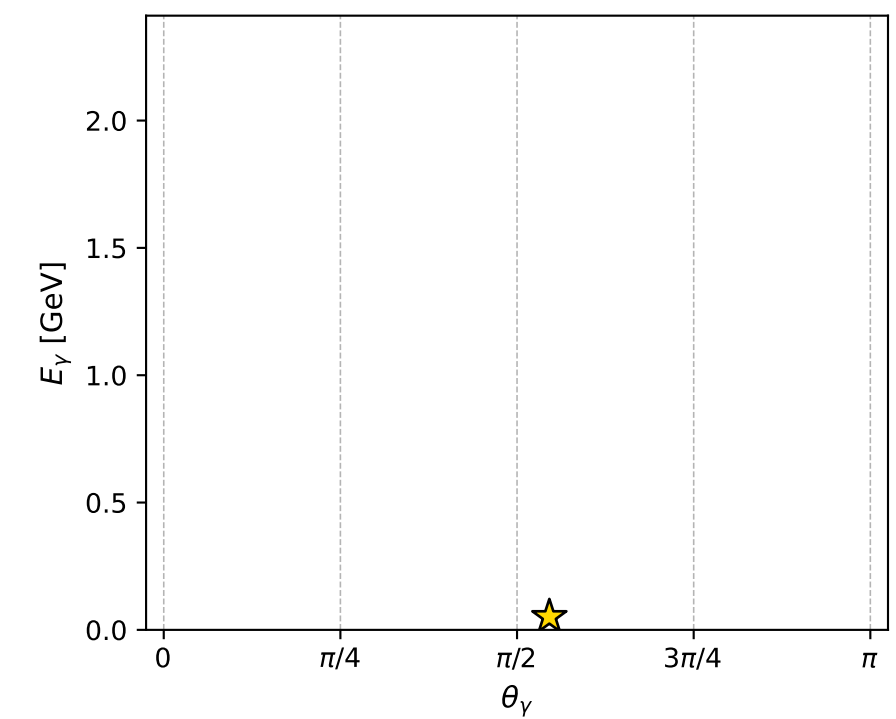
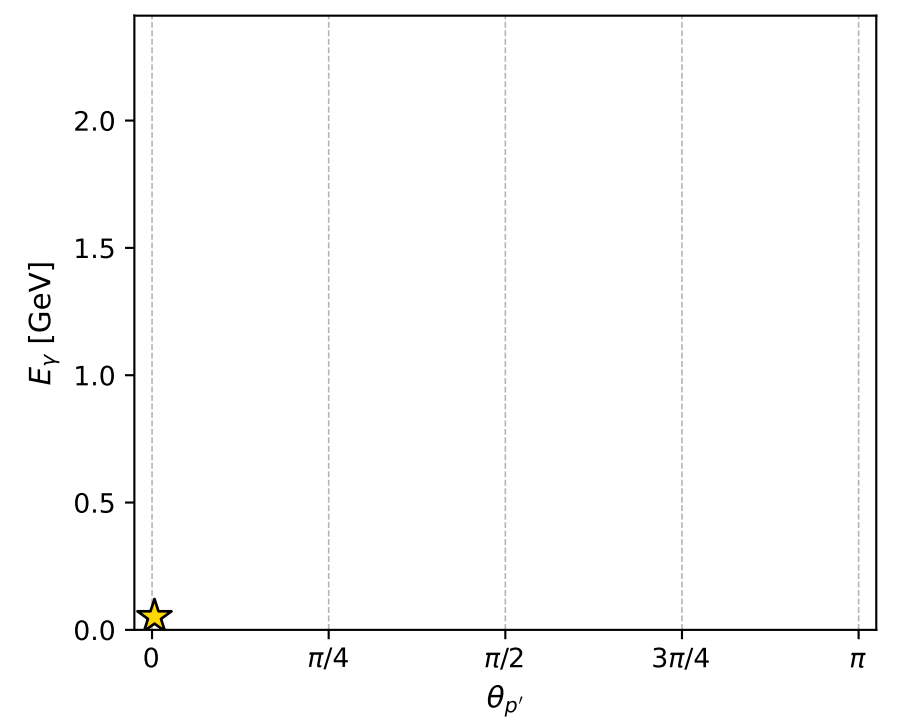
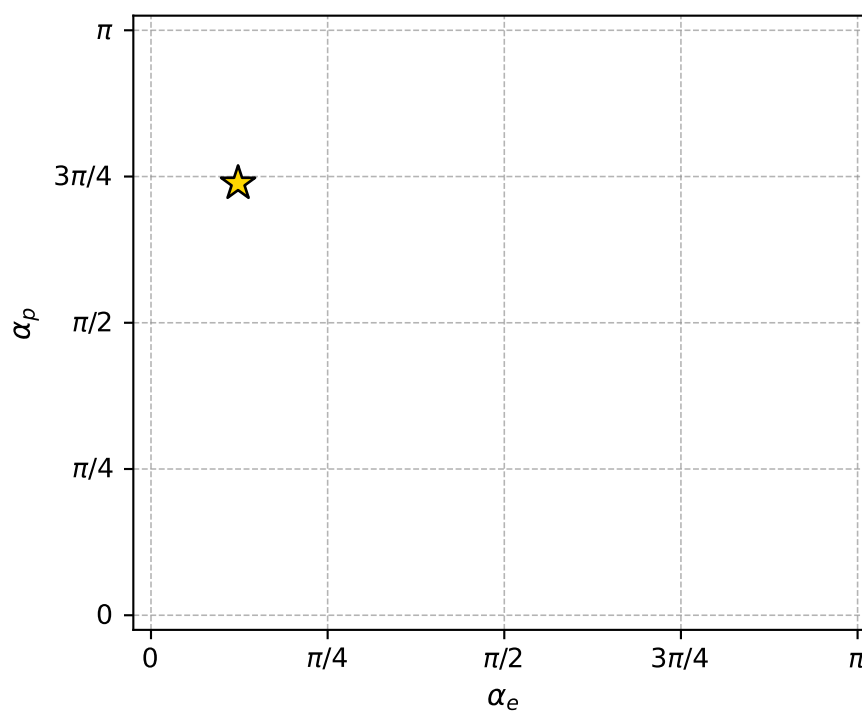
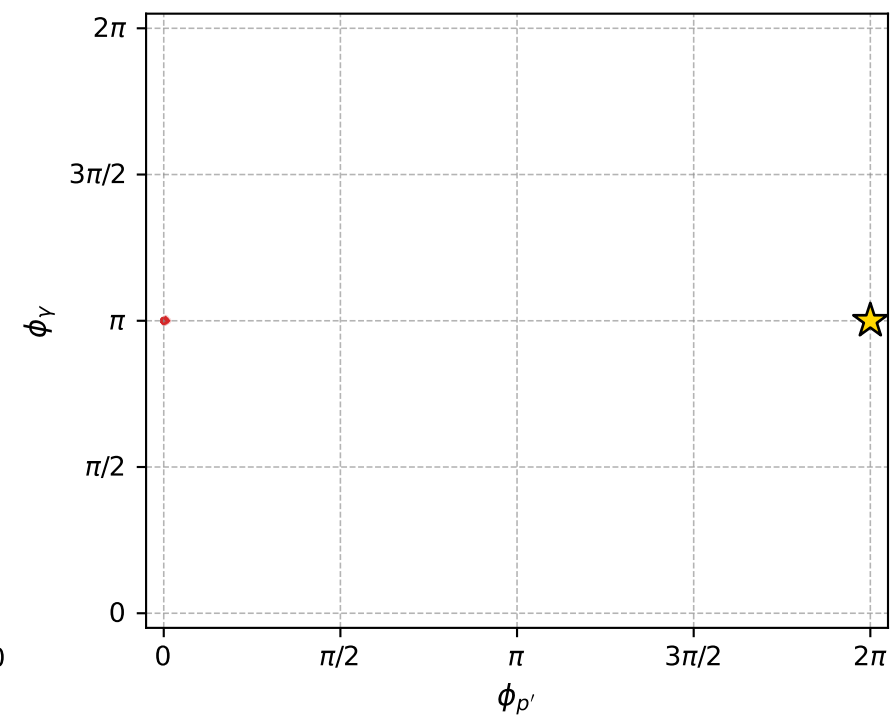
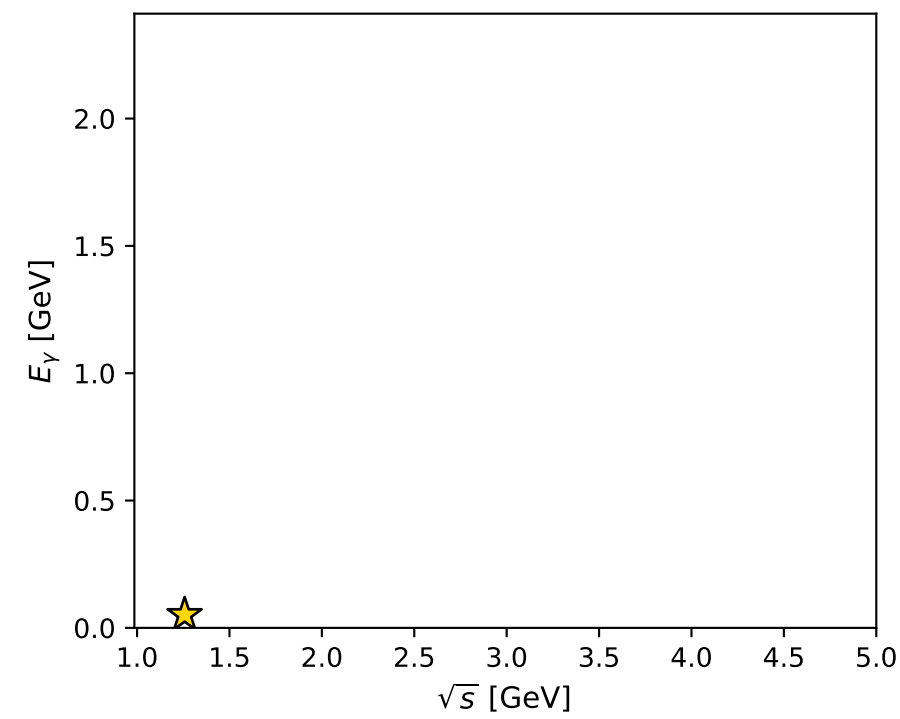
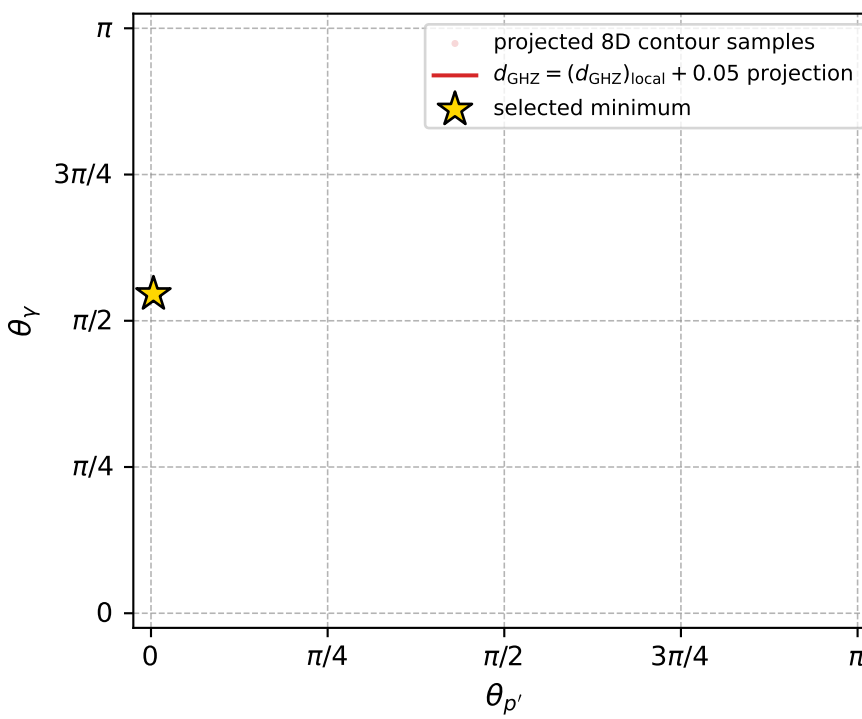
dghz_mixing_angles_polarization_02_local_minimum_349 $d_{\{\text{rm GHZ}\}}=0.000746926$
 region: polarization_cluster_02_local_minimum_349
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=0.38713594$ rad
 incoming lepton: $0.925994|+\rangle +0.377538|-\rangle$
 proton mixing angle: $\alpha_p=2.3198196$ rad
 incoming proton: $-0.680924|+\rangle +0.732354|-\rangle$

$s=1.58119$, $\sqrt{s}=1.25746$, $|\vec{P}|=0.278876$, $|\vec{P}'|=0.240271$
 $\theta_{P'}=0.010862$, $\theta_Y=1.71448$, $\phi_{P'}=6.28316$, $\phi_Y=3.14158$
 $E_Y=0.0513309$, $\theta_{\ell'}=2.93756$, $\phi_{\ell'}=6.28318$
 $Q^2=0.00275175$, $x_B=0.025971$, $t=-0.00139233$, $W^2=0.983047$, $y=0.151072$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.2789$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.2789$
 P $E= 0.9786$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.2789$
 ℓ' $E= 0.2378$ $p_x= 0.0482$ $p_y= -0.0000$ $p_z= -0.2329$
 P' $E= 0.9683$ $p_x= 0.0026$ $p_y= -0.0000$ $p_z= 0.2403$
 q_Y $E= 0.0513$ $p_x= -0.0508$ $p_y= 0.0000$ $p_z= -0.0074$



electron: polarization cluster P2, local minimum 349; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 0.00074692575$
 $d_{\text{GHZ}} \text{ contour} = 0.050746926$
 above global minimum = 0.00074690347
 8D contour samples = 256

selected phase-space point:

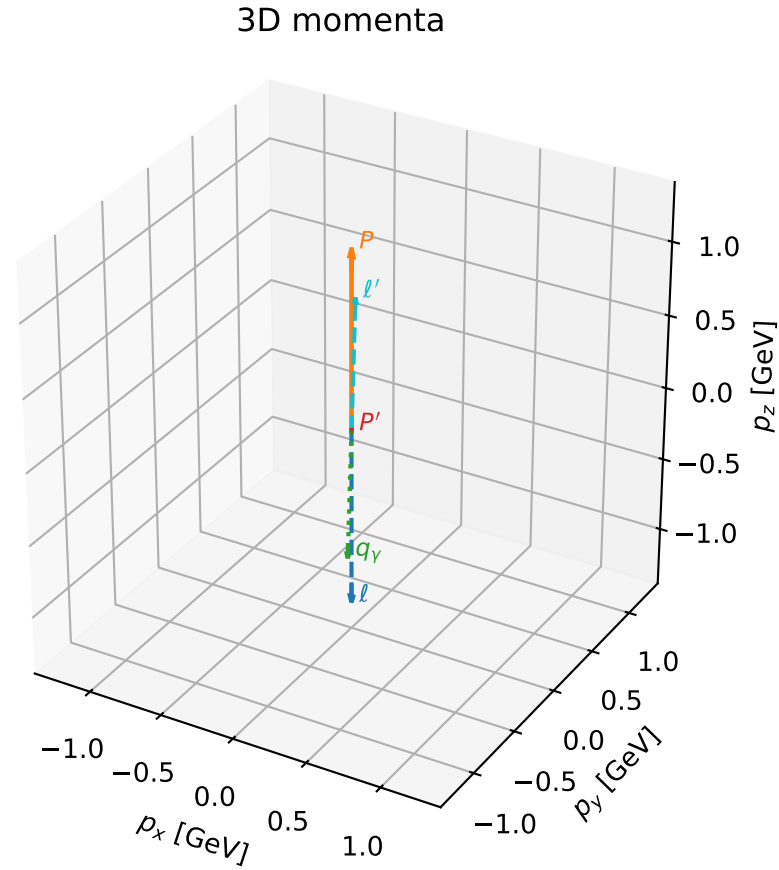
$\text{sqrt_s} = 1.257455$
 $\text{theta_p_out} = 0.01086199$
 $\text{theta_gamma_out} = 1.714482$
 $\text{q0ut} = 0.05133087$
 $\text{phi_p_out} = 6.283156$
 $\text{phi_gamma_out} = 3.141583$
 $\text{alpha_e} = 0.3871359$
 $\text{alpha_p} = 2.31982$

coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

dghz_mixing_angles_polarization_02_local_minimum_354 $d_{\{\text{rm GHZ}\}}=0.0010861$
 region: polarization_cluster_02_local_minimum_354
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=0.027969497$ rad
 incoming lepton: $0.999609|+\rangle +0.0279659|-\rangle$
 proton mixing angle: $\alpha_p=1.5433927$ rad
 incoming proton: $0.0274002|+\rangle +0.999625|-\rangle$

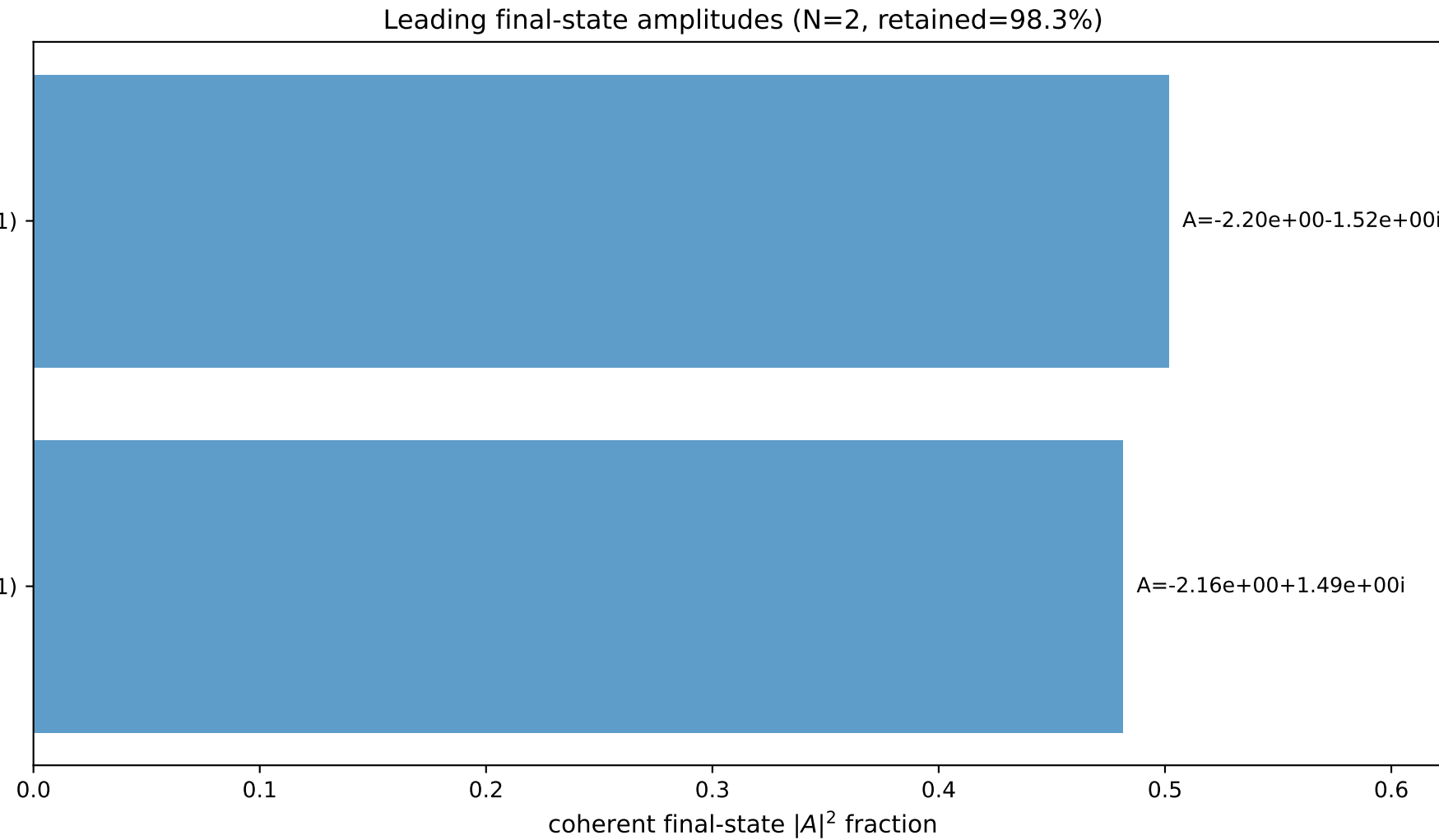
$s=7.55541$, $\sqrt{s}=2.74871$, $|\vec{P}|=1.21431$, $|\vec{P}'|=4.5944\text{e-}06$
 $\theta_{P'}=3.14154$, $\theta_V=3.11242$, $\phi_{P'}=3.74475$, $\phi_V=3.14156$
 $E_V=0.905353$, $\theta_{l'}=0.0291745$, $\phi_{l'}=6.28315$
 $Q^2=4.3966$, $x_B=0.721341$, $t=-1.11886$, $W^2=2.57828$, $y=0.913036$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 1.2143$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.2143$
 P $E= 1.5344$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.2143$
 l' $E= 0.9054$ $p_x= 0.0264$ $p_y= -0.0000$ $p_z= 0.9050$
 P' $E= 0.9380$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= -0.0000$
 q_Y $E= 0.9054$ $p_x= -0.0264$ $p_y= 0.0000$ $p_z= -0.9050$

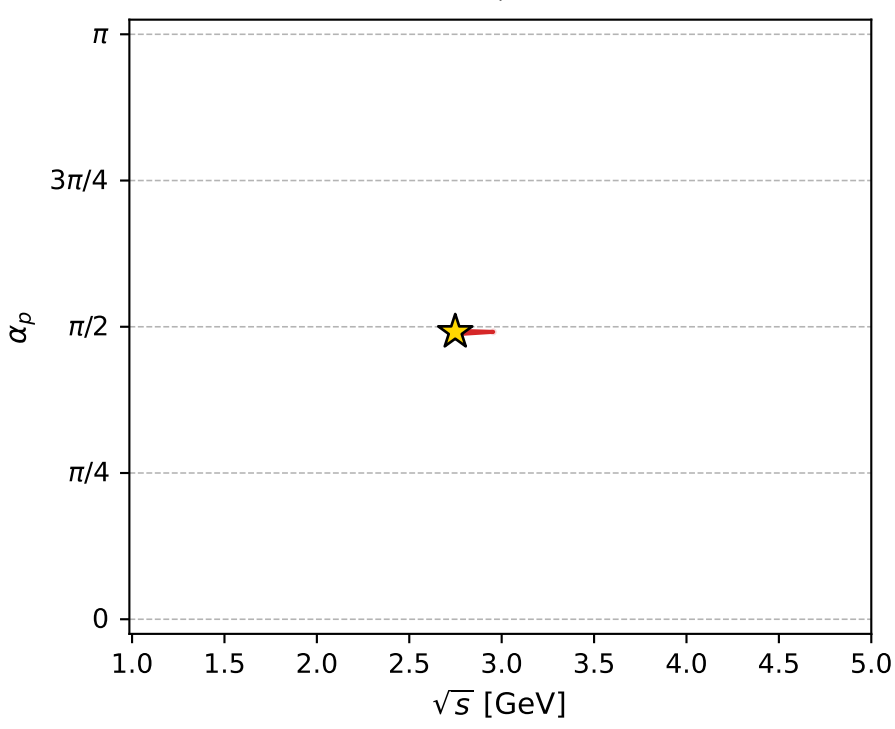
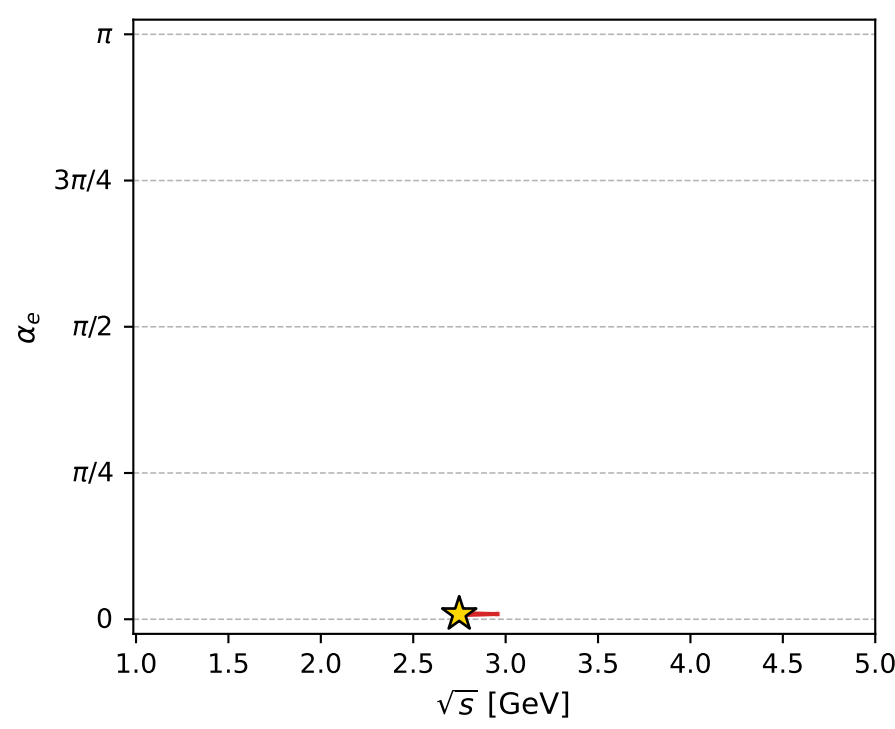
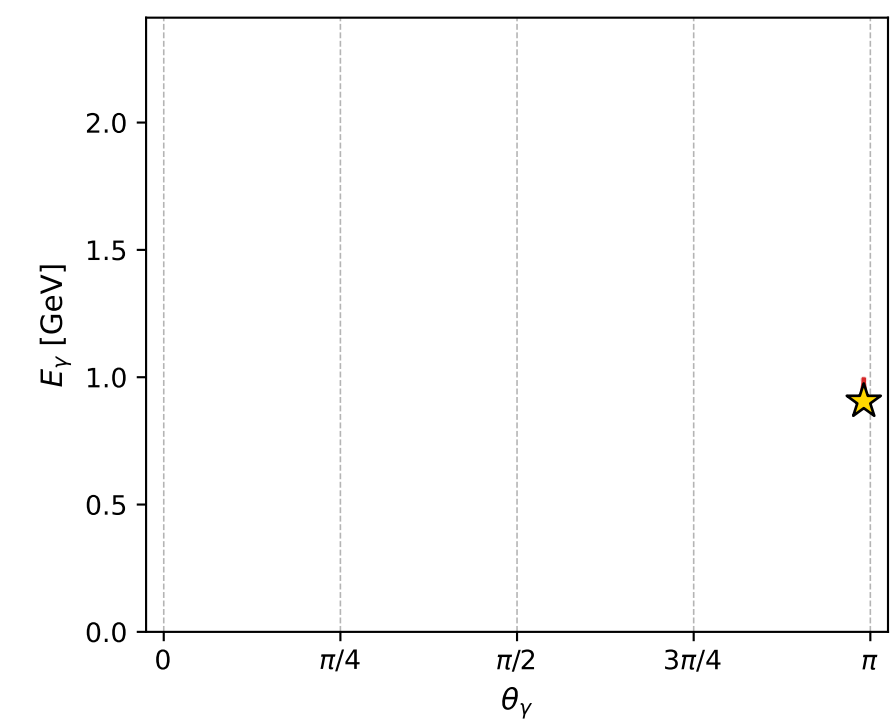
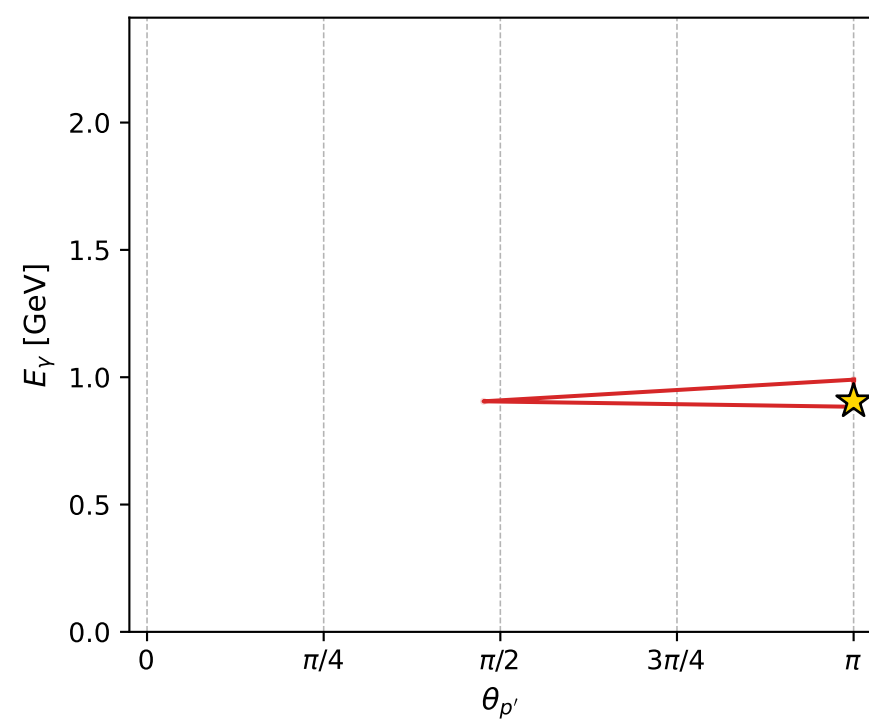
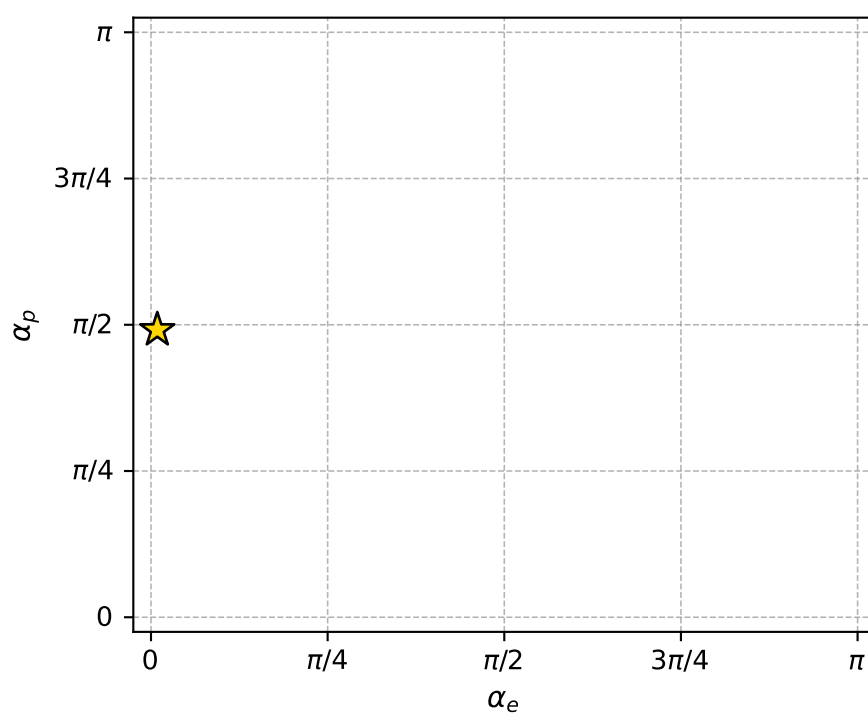
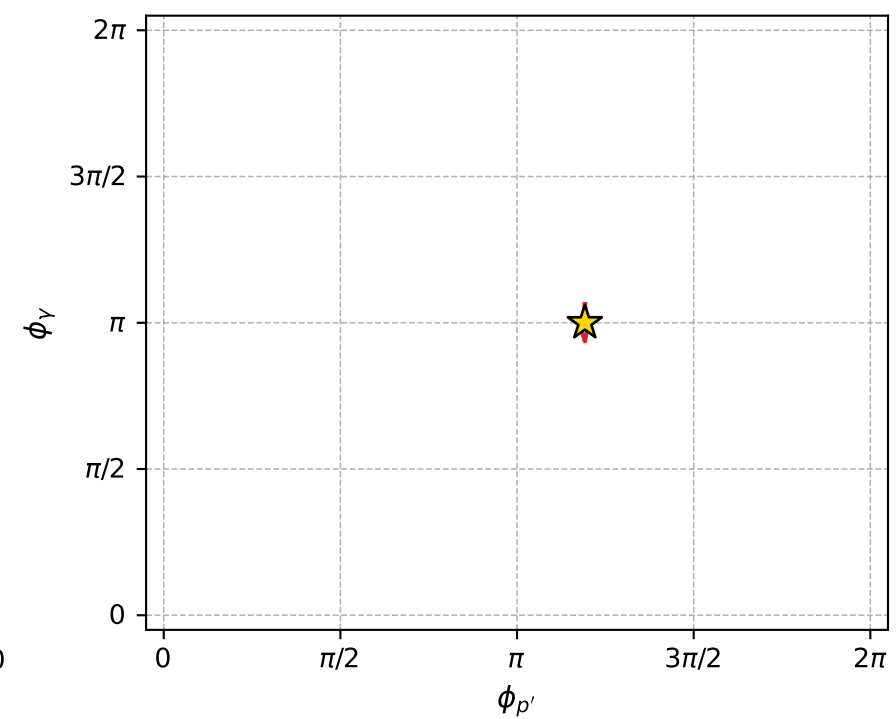
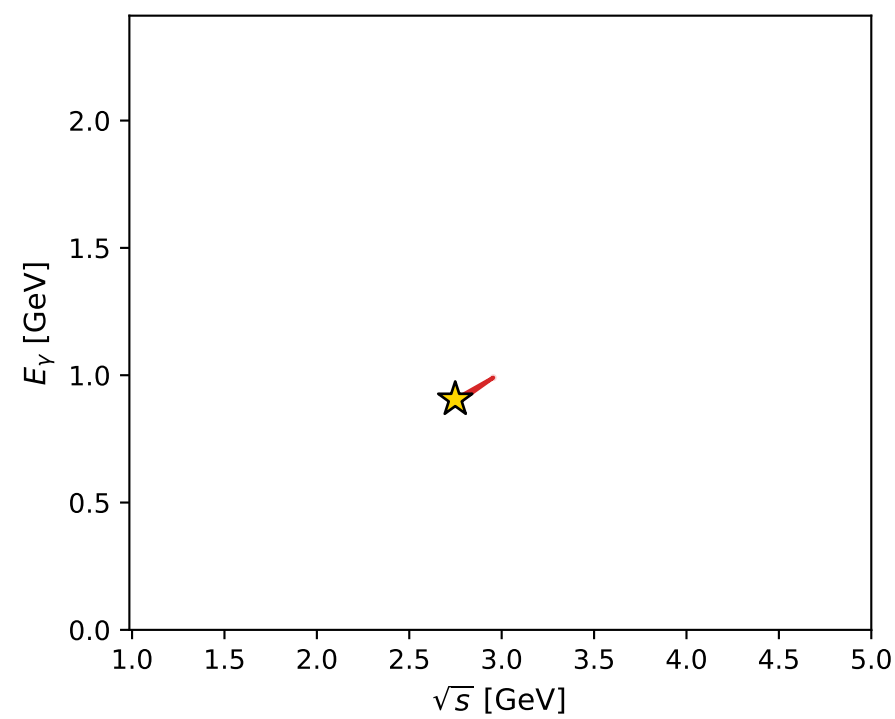
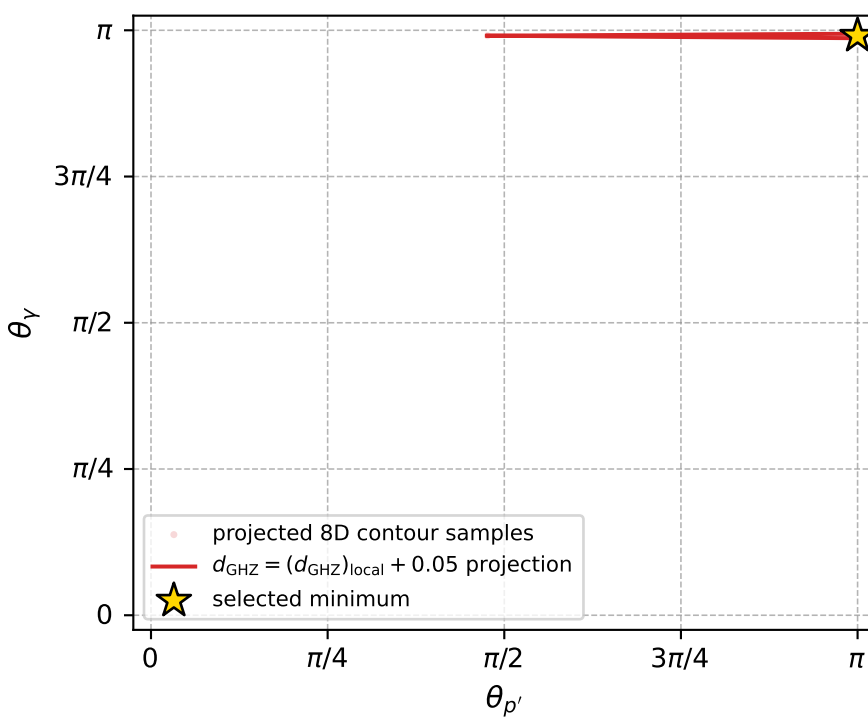


$(h_p, h_Y, h_l) = (-1, -1, -1)$

$(h_p, h_Y, h_l) = (+1, +1, +1)$



electron: polarization cluster P2, local minimum 354; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 0.0010860979$
 $d_{\text{GHZ}} \text{ contour} = 0.051086098$
 above global minimum = 0.0010860756
 8D contour samples = 186

selected phase-space point:

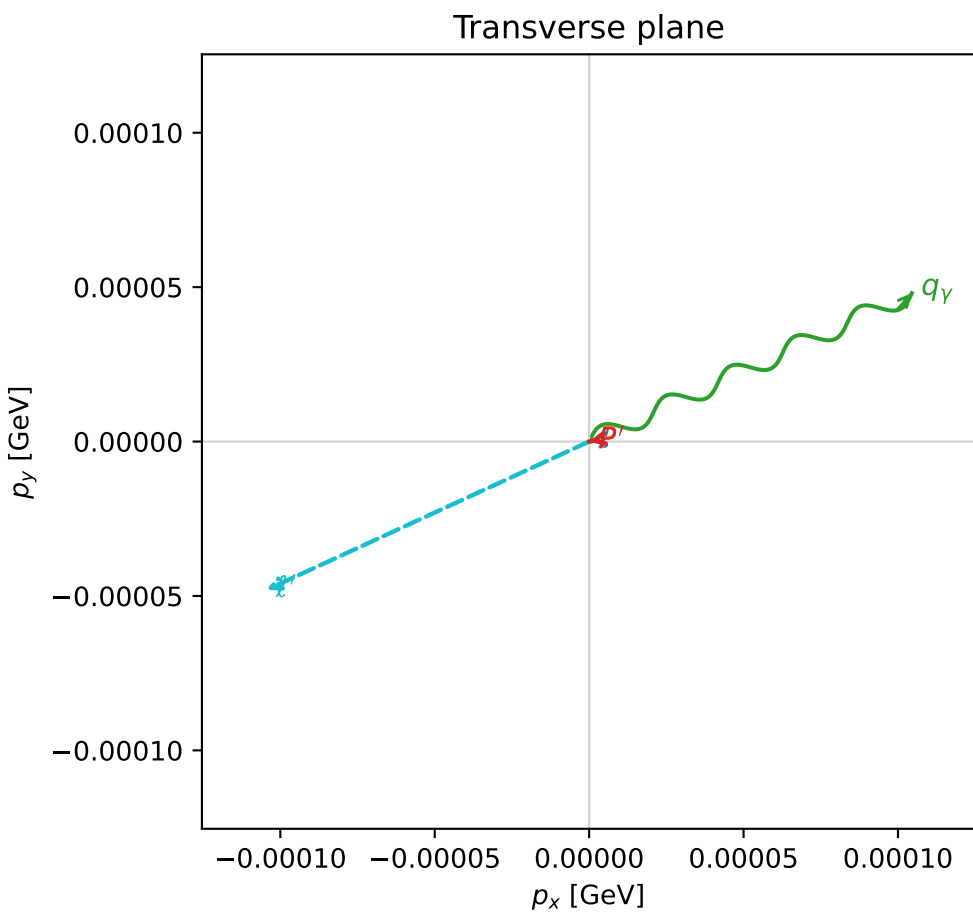
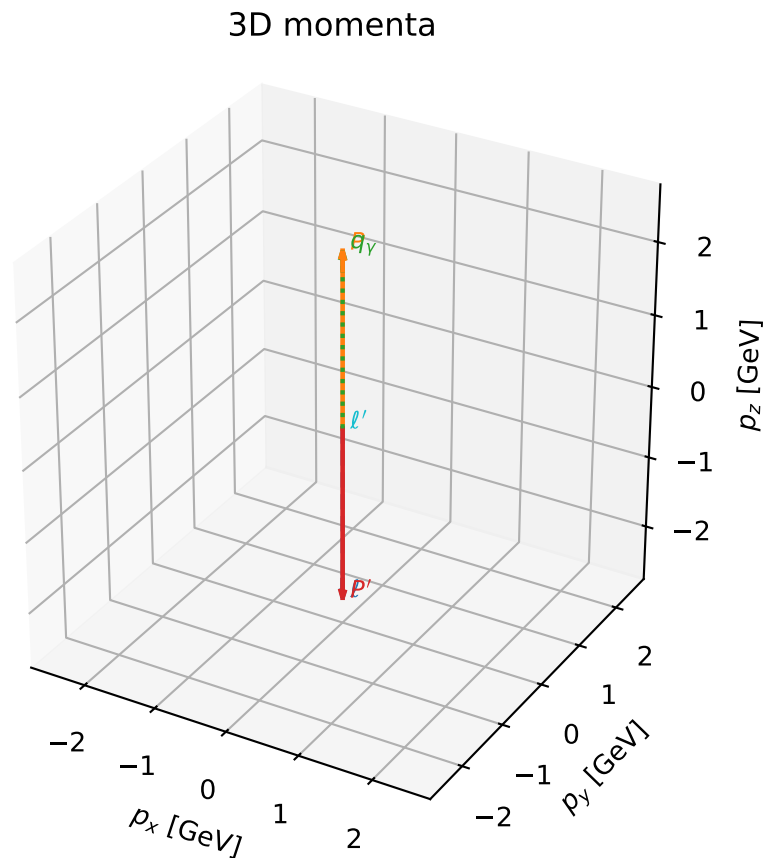
$\text{sqrt_s} = 2.74871$
 $\text{theta_p_out} = 3.141543$
 $\text{theta_gamma_out} = 3.112418$
 $\text{q0ut} = 0.9053526$
 $\text{phi_p_out} = 3.744754$
 $\text{phi_gamma_out} = 3.141562$
 $\text{alpha_e} = 0.0279695$
 $\text{alpha_p} = 1.543393$

coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

dghz_mixing_angles_polarization_02_local_minimum_355 d_{\rm GHZ}=0.00117646
region: polarization_cluster_02_local_minimum_355
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: $\alpha_e=0.78572109$ rad
incoming lepton: $0.706878|+\rangle +0.707335|-\rangle$
proton mixing angle: $\alpha_p=2.3570726$ rad
incoming proton: $-0.707727|+\rangle +0.706486|-\rangle$

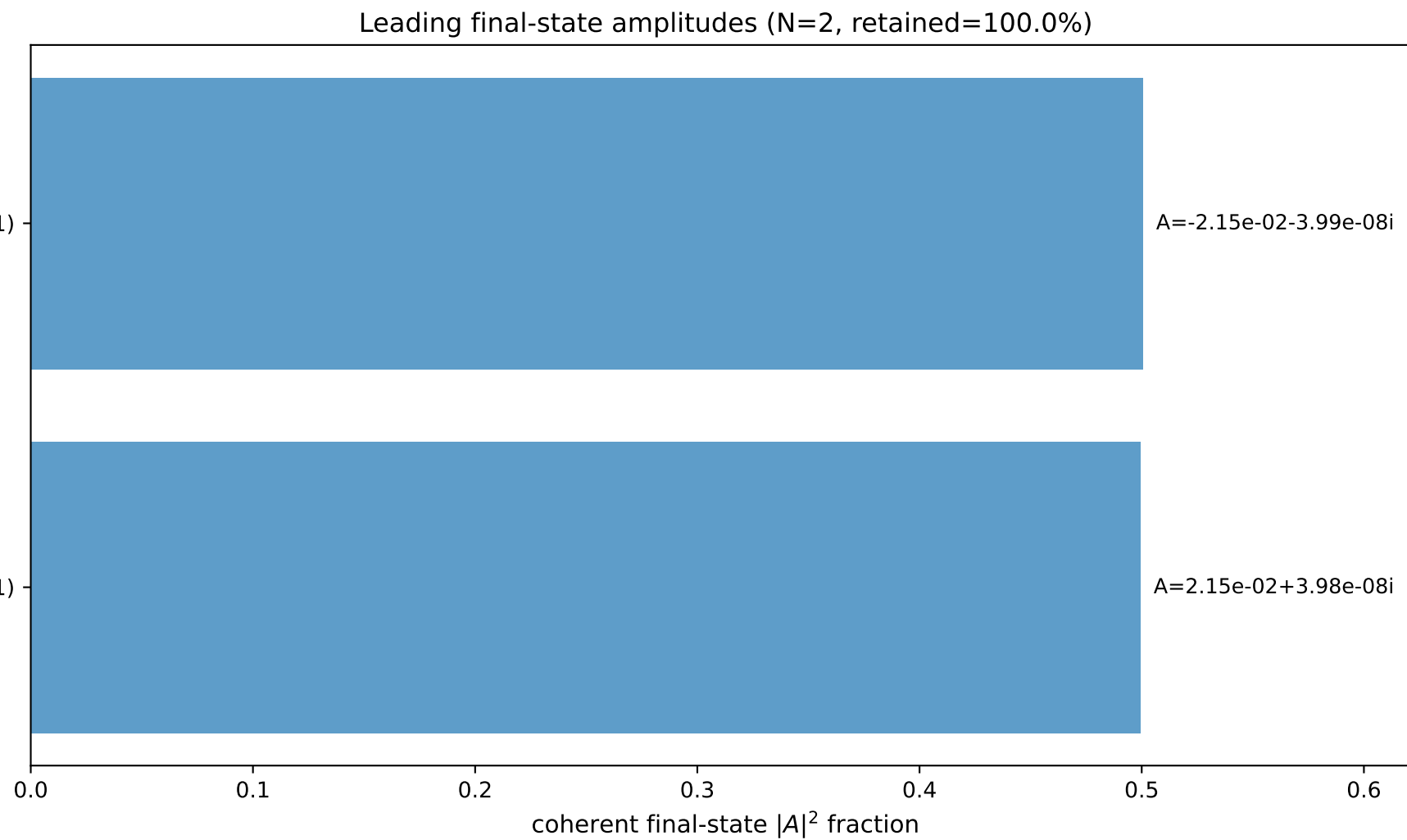
$s=25$, $\sqrt{s}=5$, $|\vec{P}|=2.41202$, $|\vec{P}'|=2.41055$
 $\theta_{p'}=3.14159$, $\theta_V=4.76846\text{e-}05$, $\phi_{p'}=3.33831$, $\phi_V=0.431012$
 $E_V=2.41192$, $\theta_{l'}=3.05787$, $\phi_{l'}=3.5726$
 $Q^2=0.000465873$, $x_B=1.9326\text{e-}05$, $t=-23.2571$, $W^2=24.9853$, $y=0.999411$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 2.4120$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 P $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4120$
 l' $E= 0.0015$ $p_x= -0.0001$ $p_y= -0.0000$ $p_z= -0.0014$
 P' $E= 2.5866$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= -2.4105$
 q_V $E= 2.4119$ $p_x= 0.0001$ $p_y= 0.0000$ $p_z= 2.4119$

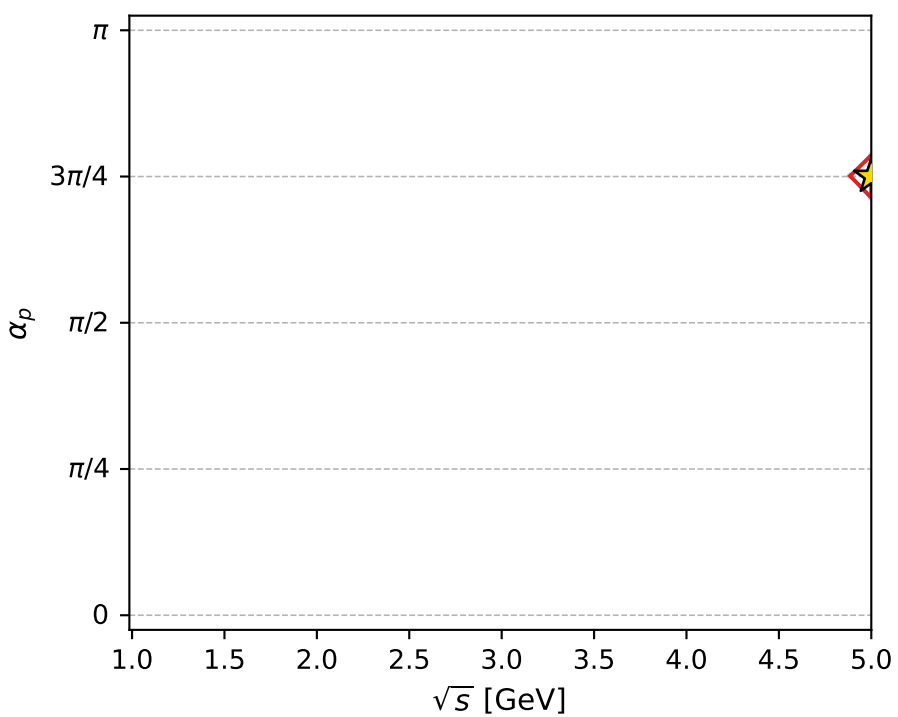
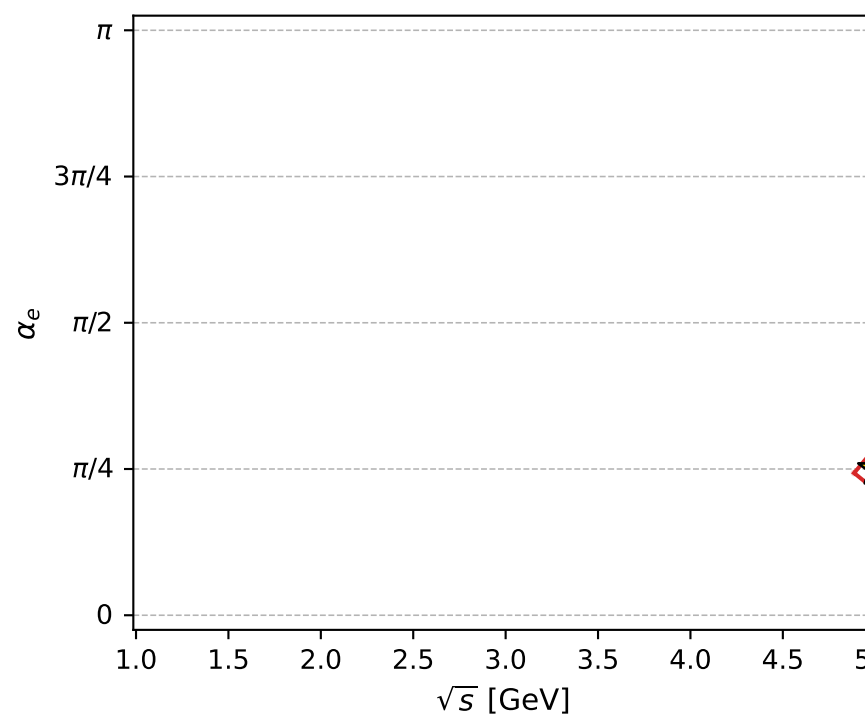
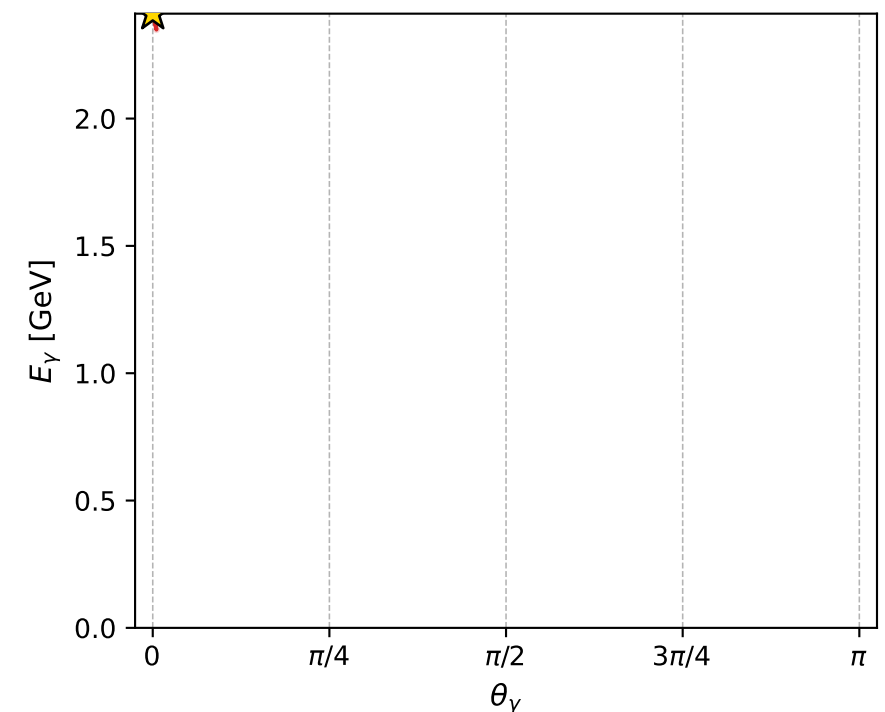
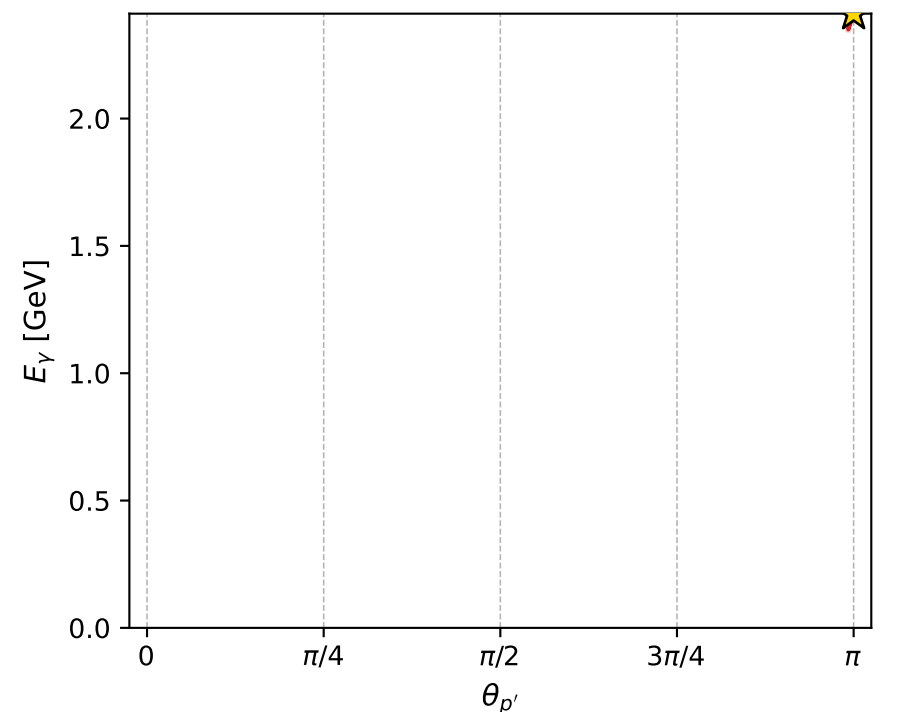
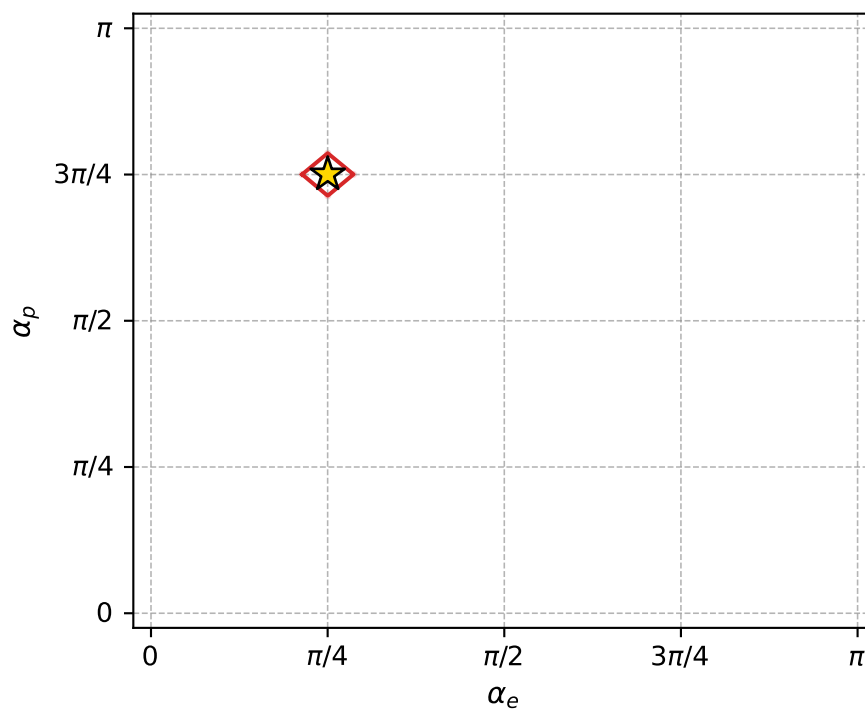
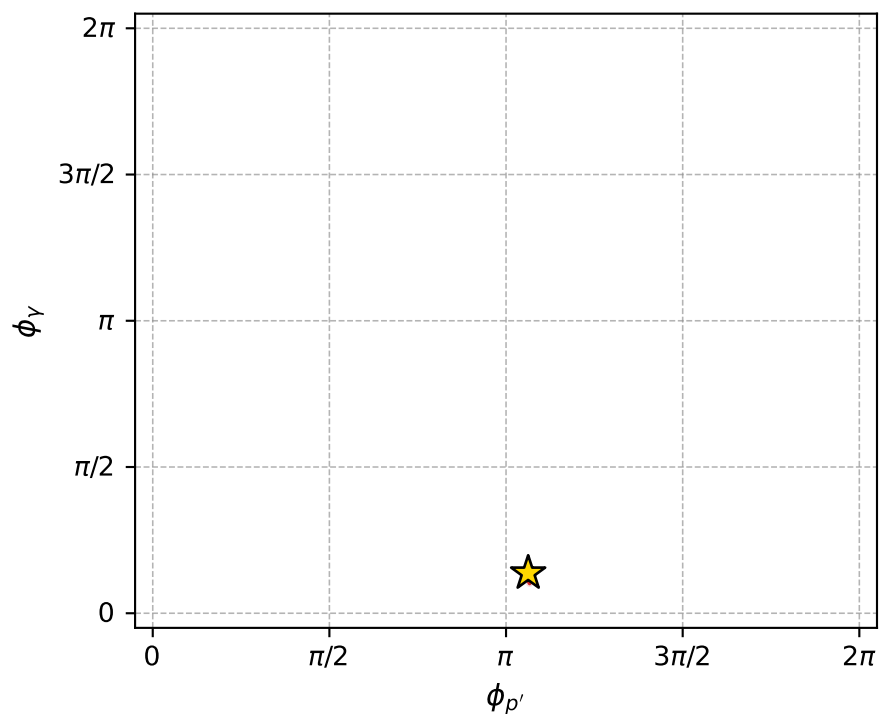
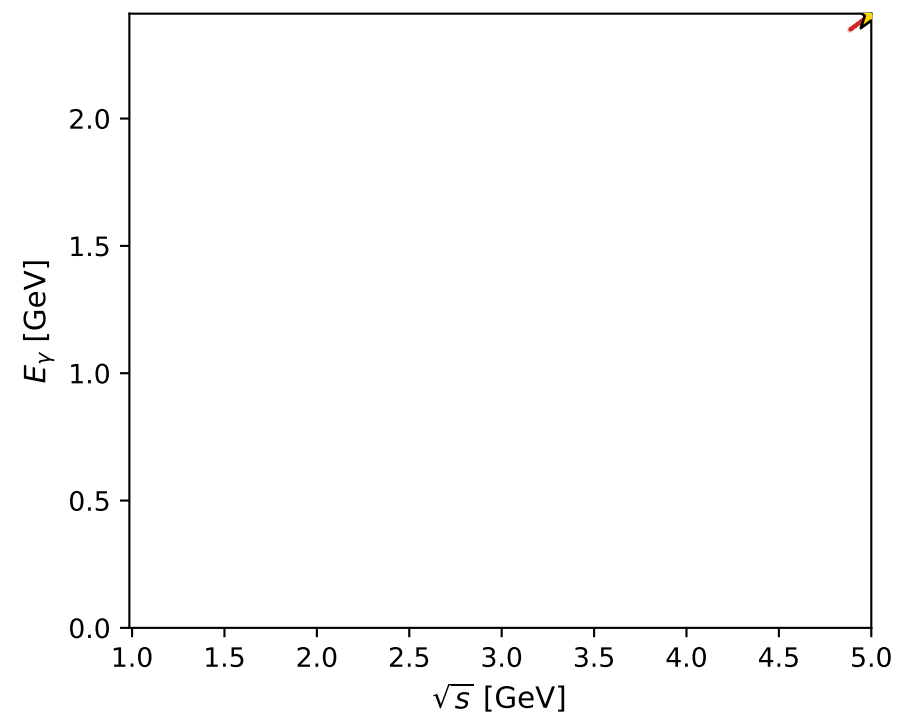
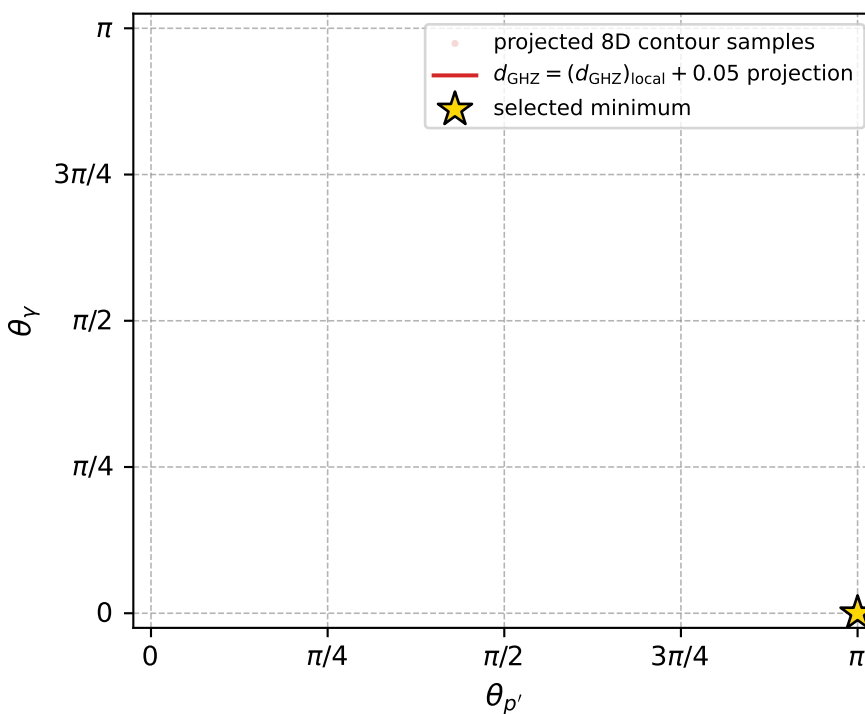


$(h_p, h_V, h_l) = (+1, +1, +1)$

$(h_p, h_V, h_l) = (-1, -1, -1)$



electron: polarization cluster P2, local minimum 355; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 0.0011764601$
 $d_{\text{GHZ}} \text{ contour} = 0.05117646$
 above global minimum = 0.0011764378
 8D contour samples = 30

selected phase-space point:

$\text{sqrt_s} = 5$
 $\text{theta_p_out} = 3.141593$
 $\text{theta_gamma_out} = 4.76846\text{e-}05$
 $\text{q0ut} = 2.411917$
 $\text{phi_p_out} = 3.33831$
 $\text{phi_gamma_out} = 0.4310123$
 $\text{alpha_e} = 0.7857211$
 $\text{alpha_p} = 2.357073$

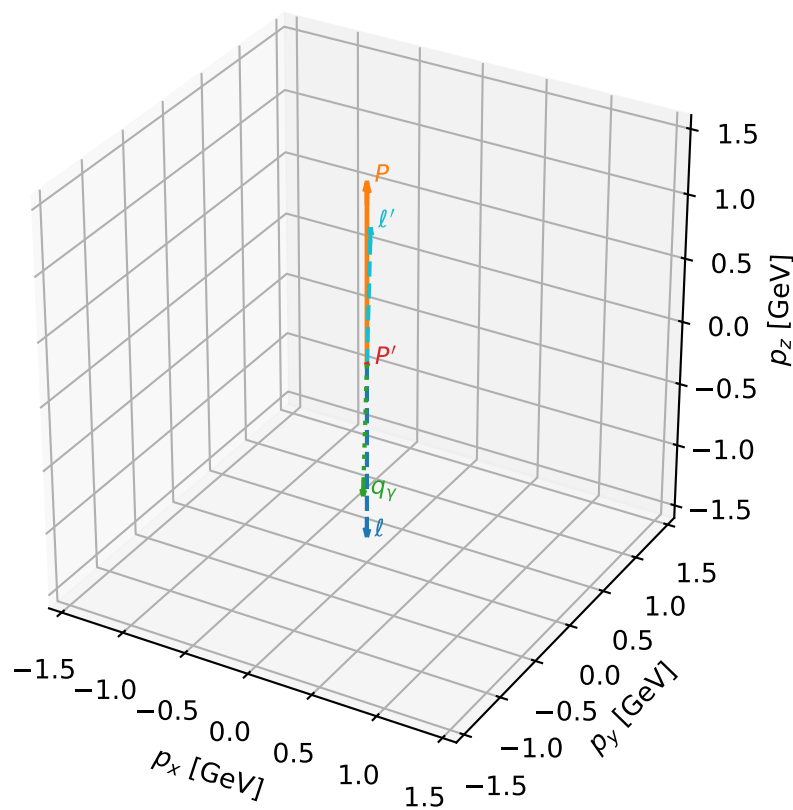
coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

dghz_mixing_angles_polarization_02_local_minimum_363 $d_{\{\text{rm GHZ}\}}=0.0023918$
region: polarization_cluster_02_local_minimum_363
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: $\alpha_e=0.02677023$ rad
incoming lepton: $0.999642|+\rangle +0.026767|-\rangle$
proton mixing angle: $\alpha_p=1.5442388$ rad
incoming proton: $0.0265544|+\rangle +0.999647|-\rangle$

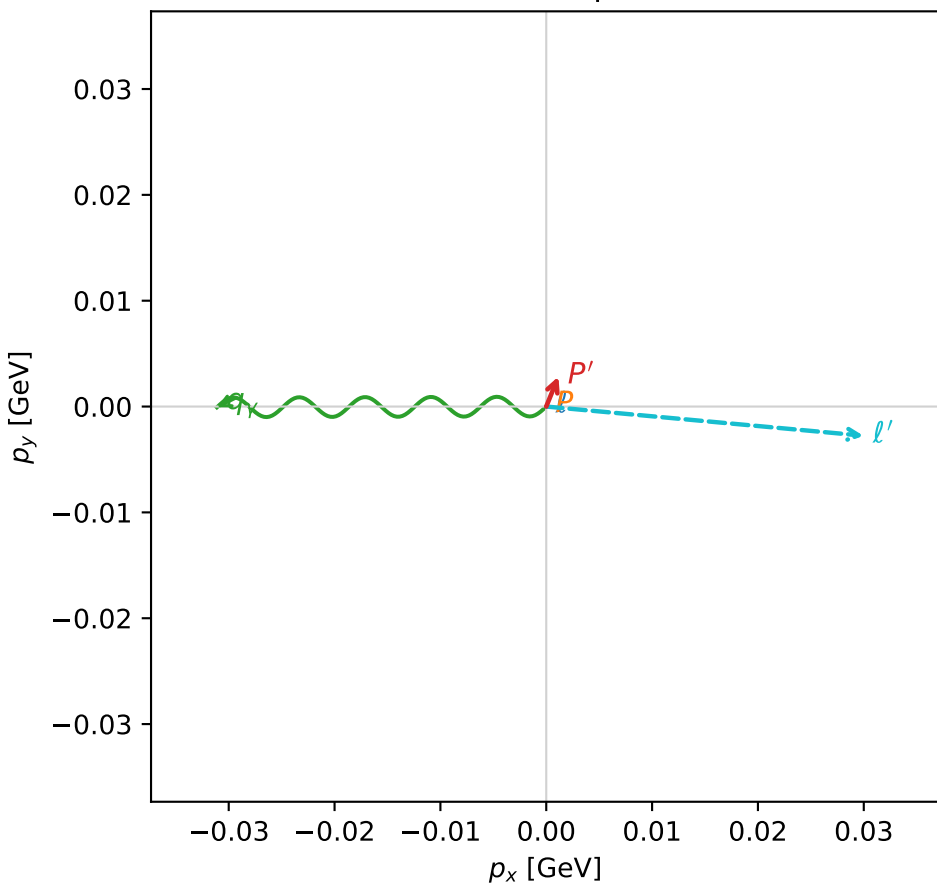
$s=9.46503$, $\sqrt{s}=3.07653$, $|\vec{P}|=1.39527$, $|\vec{P}'|=0.00305387$
 $\theta_{P'}=1.57555$, $\theta_V=3.11249$, $\phi_{P'}=1.18523$, $\phi_V=3.14395$
 $E_V=1.06927$, $\theta_{\ell'}=0.0281486$, $\phi_{\ell'}=6.19146$
 $Q^2=5.96642$, $x_B=0.748382$, $t=-1.39441$, $W^2=2.88584$, $y=0.928625$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 1.3953$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.3953$
 P $E= 1.6813$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.3953$
 ℓ' $E= 1.0693$ $p_x= 0.0300$ $p_y= -0.0028$ $p_z= 1.0688$
 P' $E= 0.9380$ $p_x= 0.0011$ $p_y= 0.0028$ $p_z= -0.0000$
 q_Y $E= 1.0693$ $p_x= -0.0311$ $p_y= -0.0001$ $p_z= -1.0688$

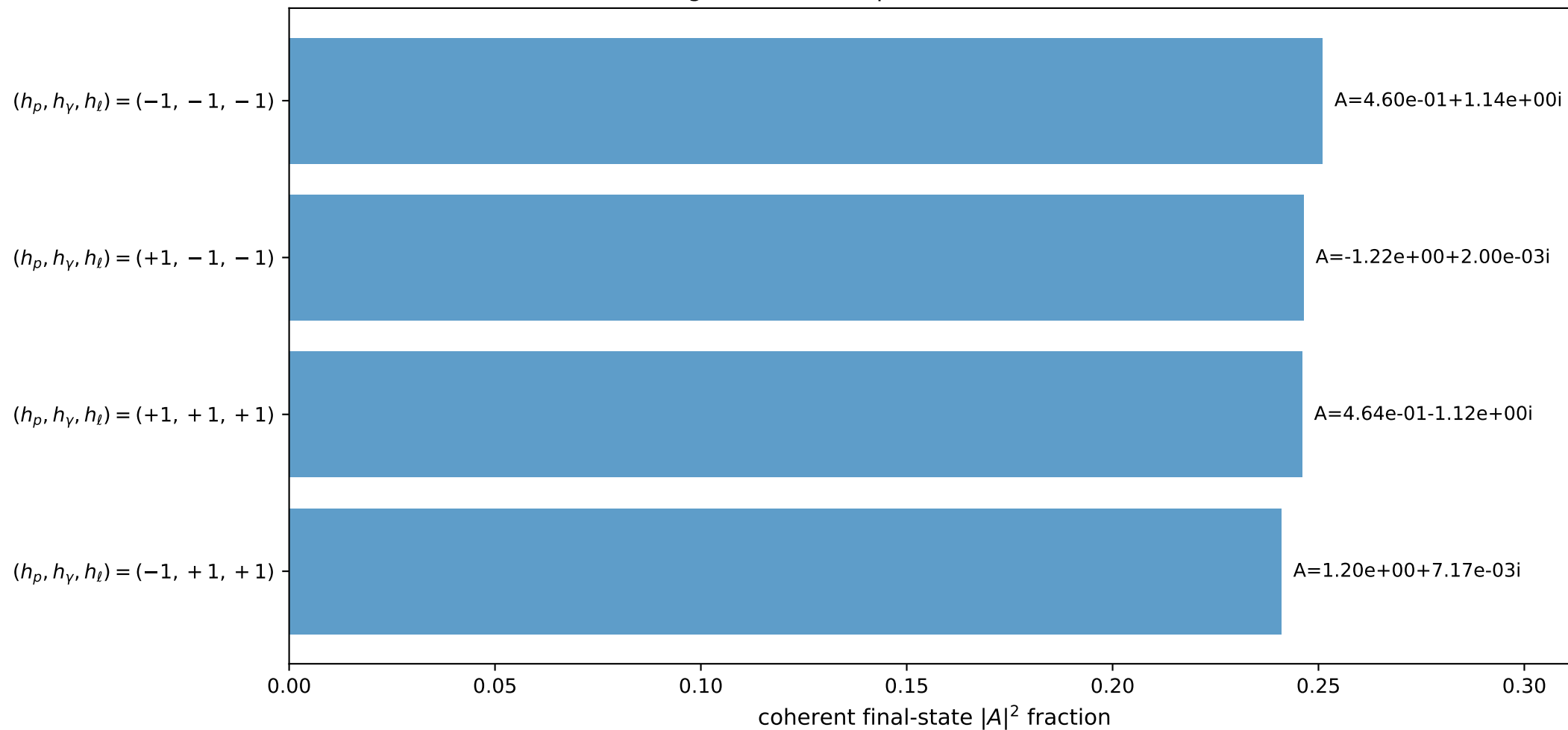
3D momenta



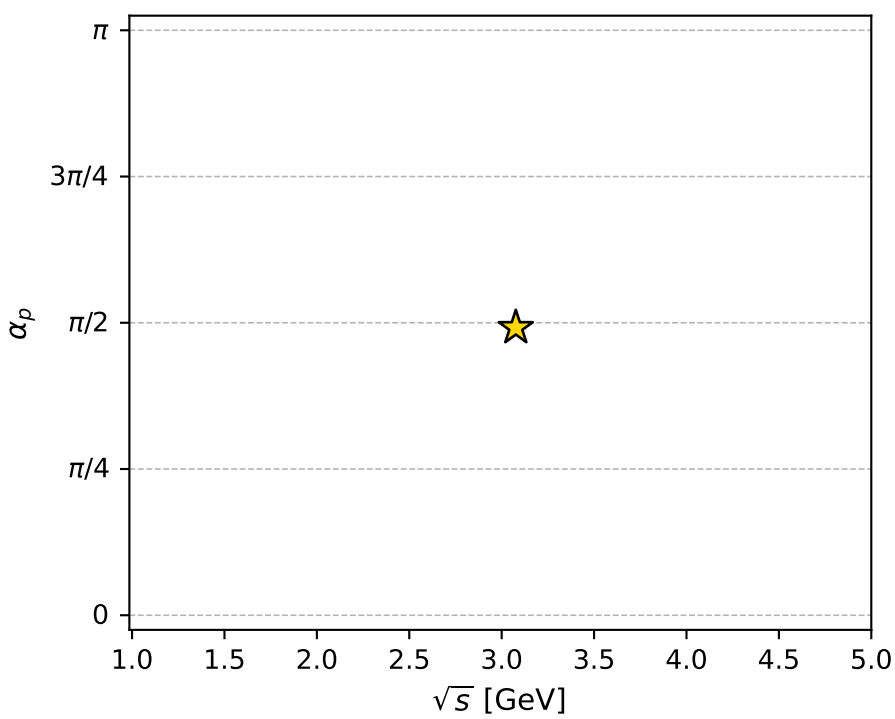
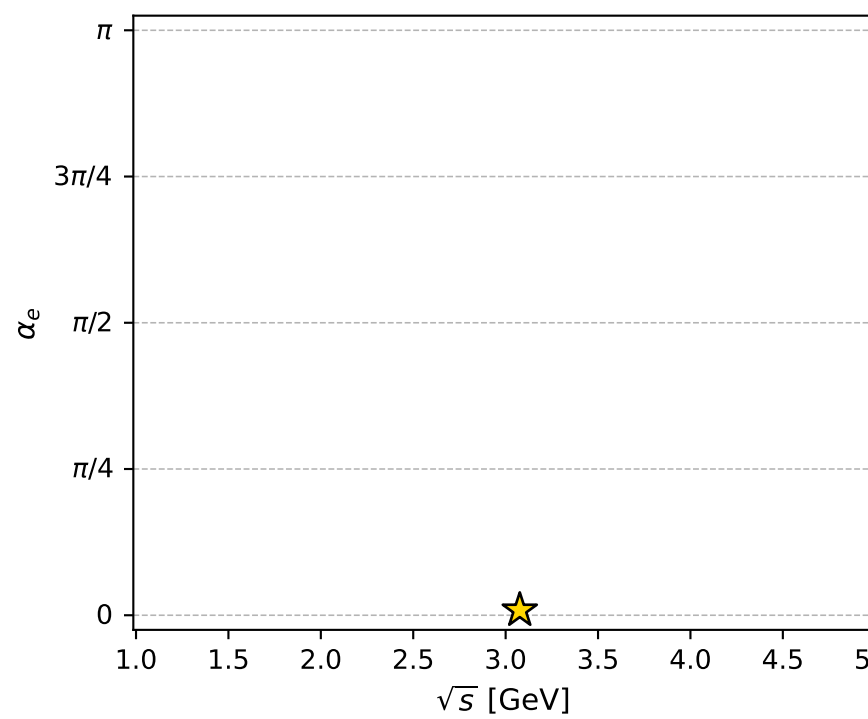
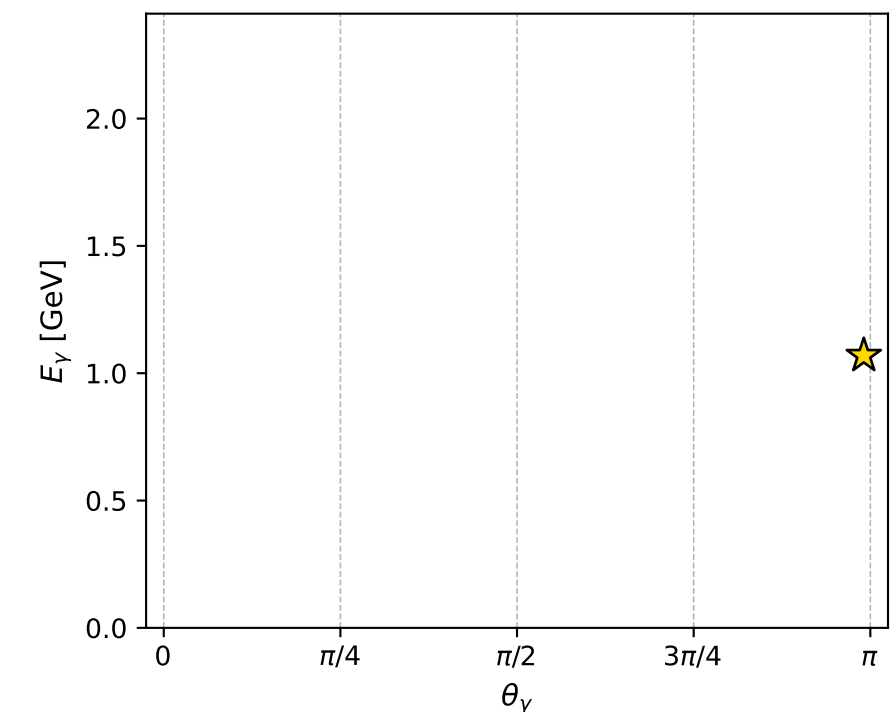
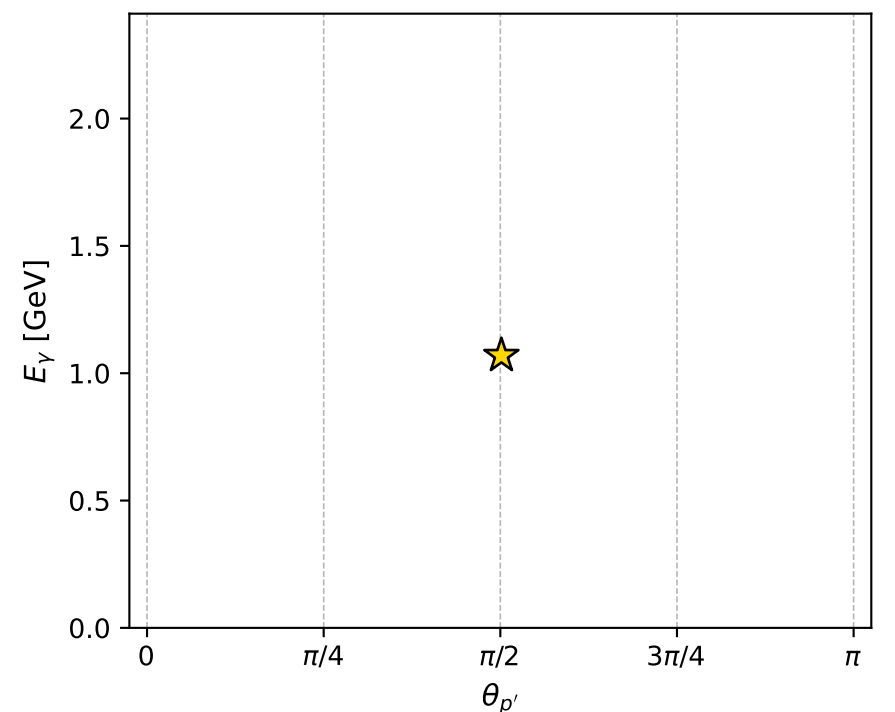
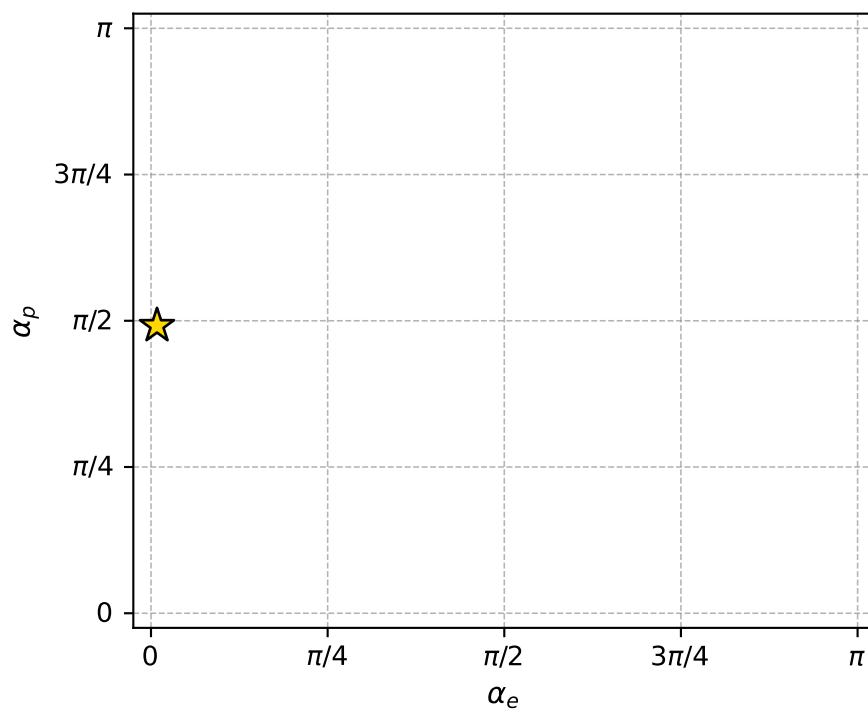
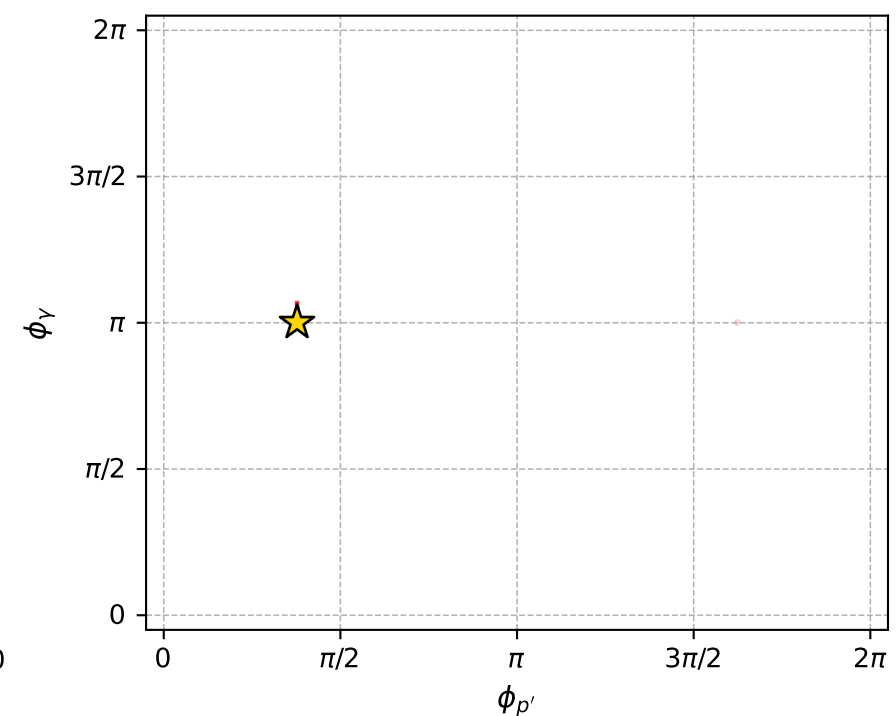
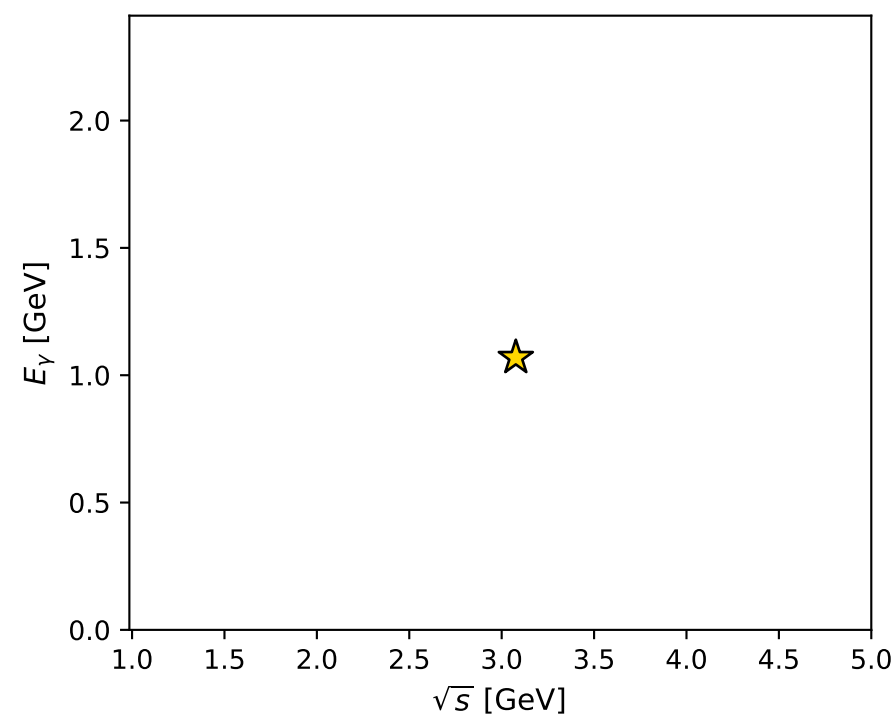
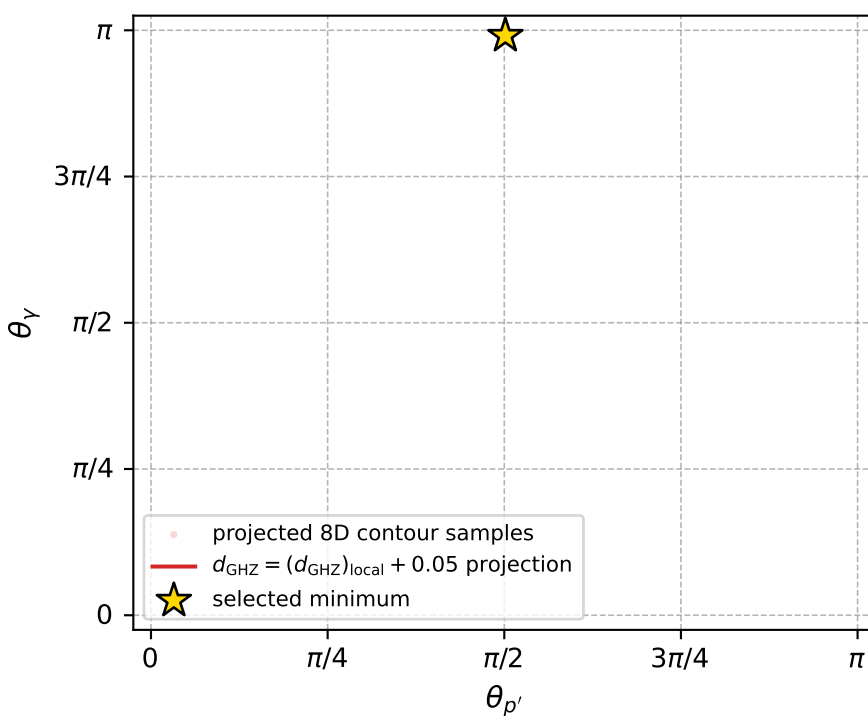
Transverse plane



Leading final-state amplitudes (N=4, retained=98.4%)



electron: polarization cluster P2, local minimum 363; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 0.0023918027$
 $d_{\text{GHZ}} \text{ contour} = 0.052391803$
 above global minimum = 0.0023917804
 8D contour samples = 255

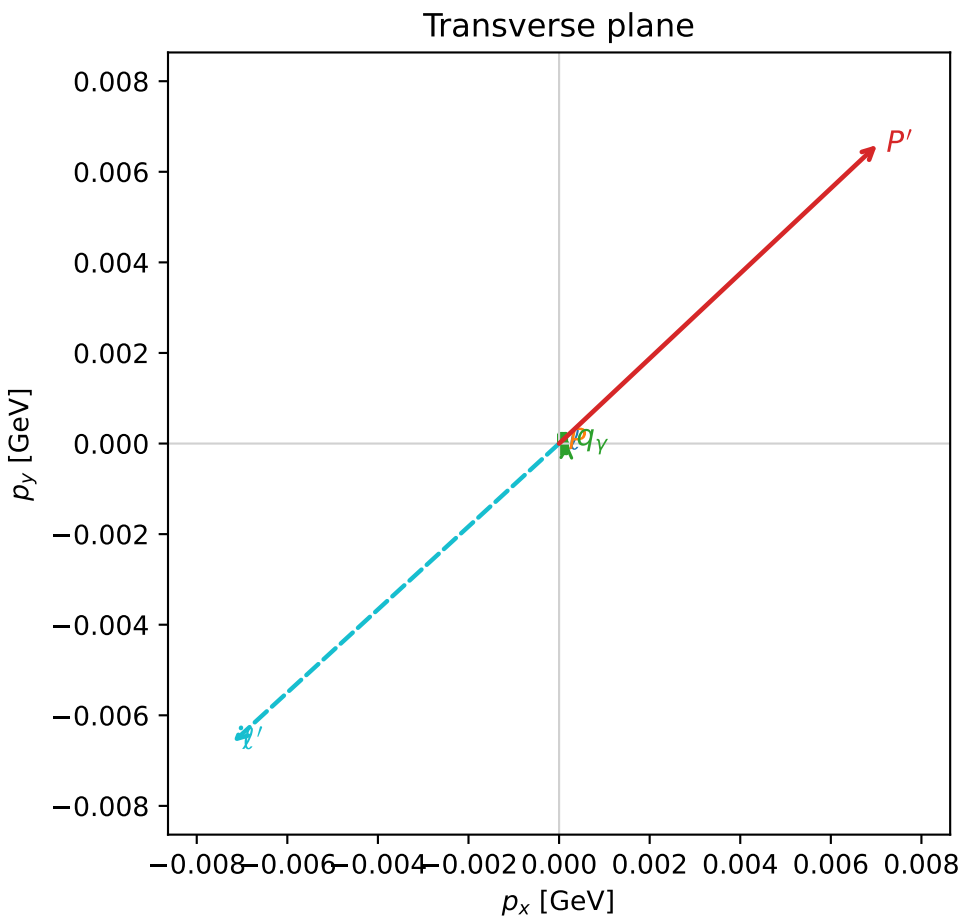
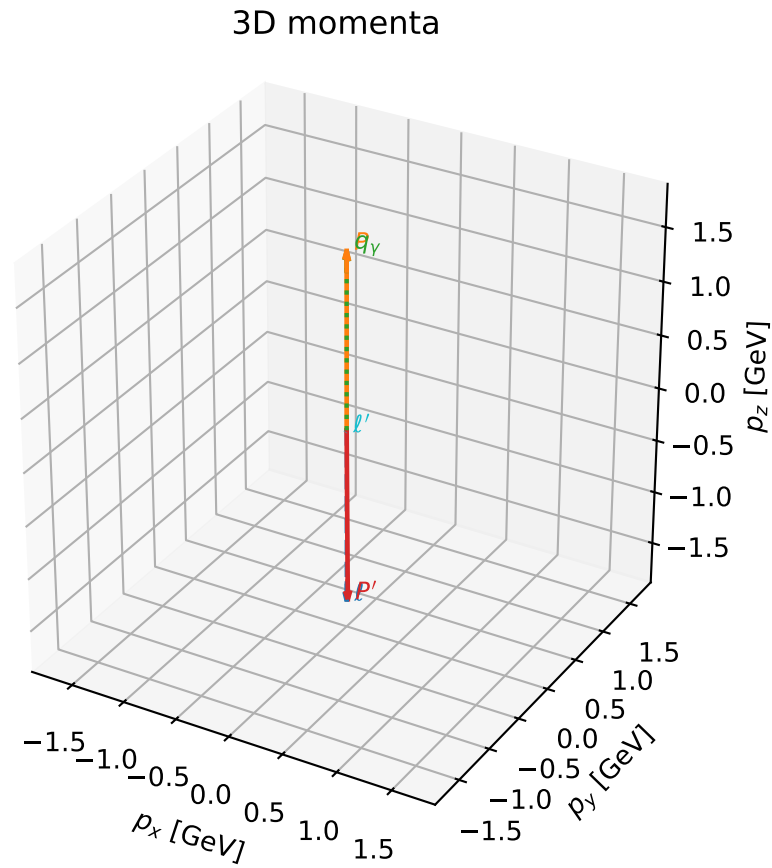
selected phase-space point:
 $\text{sqrt_s} = 3.076529$
 $\text{theta_p_out} = 1.575553$
 $\text{theta_gamma_out} = 3.112488$
 $\text{q0out} = 1.069269$
 $\text{phi_p_out} = 1.185235$
 $\text{phi_gamma_out} = 3.143946$
 $\text{alpha_e} = 0.02677023$
 $\text{alpha_p} = 1.544239$

coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

dghz_mixing_angles_polarization_02_local_minimum_370 d_{\rm GHZ}=0.0225057
 region: polarization_cluster_02_local_minimum_370
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=0.78532657$ rad
 incoming lepton: $0.707157|+\rangle +0.707056|-\rangle$
 proton mixing angle: $\alpha_p=2.3500697$ rad
 incoming proton: $-0.702763|+\rangle +0.711424|-\rangle$

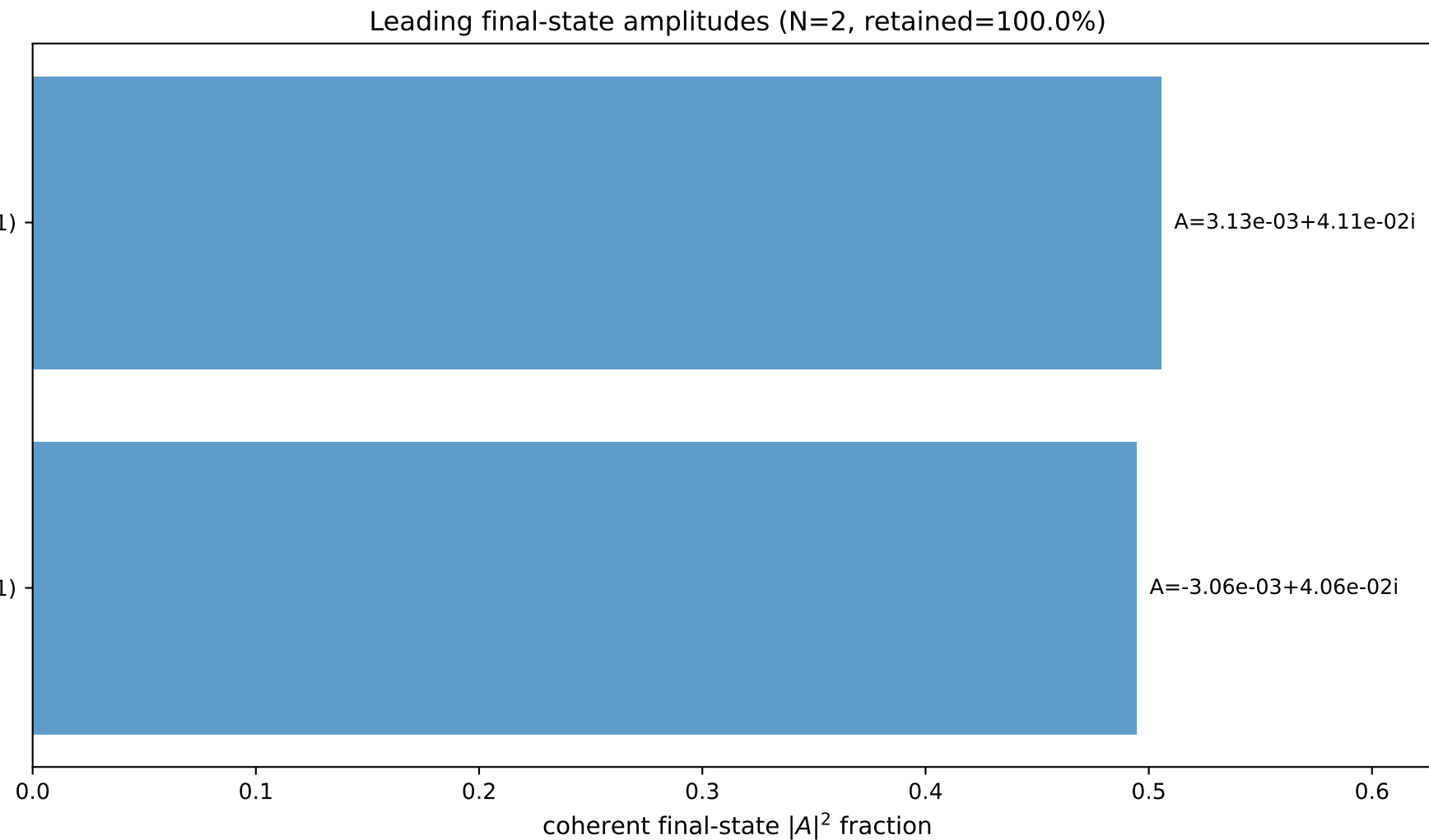
$s=12.52$, $\sqrt{s}=3.53836$, $|\vec{P}|=1.64485$, $|\vec{P}'|=1.62509$
 $\theta_{p'}=3.13567$, $\theta_V=0.000106839$, $\phi_{p'}=0.753952$, $\phi_V=0.000369628$
 $E_V=1.64223$, $\theta_{l'}=2.62489$, $\phi_{l'}=3.88324$
 $Q^2=0.00850448$, $x_B=0.000738953$, $t=-10.6921$, $W^2=12.3802$, $y=0.988717$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 1.6449$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.6449$
 P $E= 1.8935$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.6449$
 l' $E= 0.0198$ $p_x= -0.0072$ $p_y= -0.0066$ $p_z= -0.0172$
 P' $E= 1.8764$ $p_x= 0.0070$ $p_y= 0.0066$ $p_z= -1.6251$
 q_V $E= 1.6422$ $p_x= 0.0002$ $p_y= 0.0000$ $p_z= 1.6422$

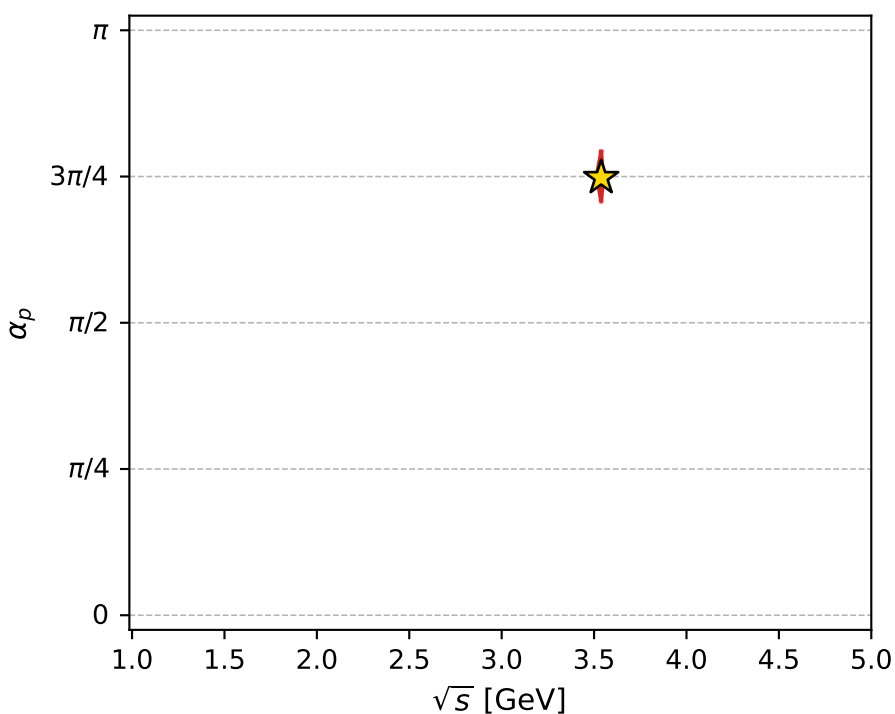
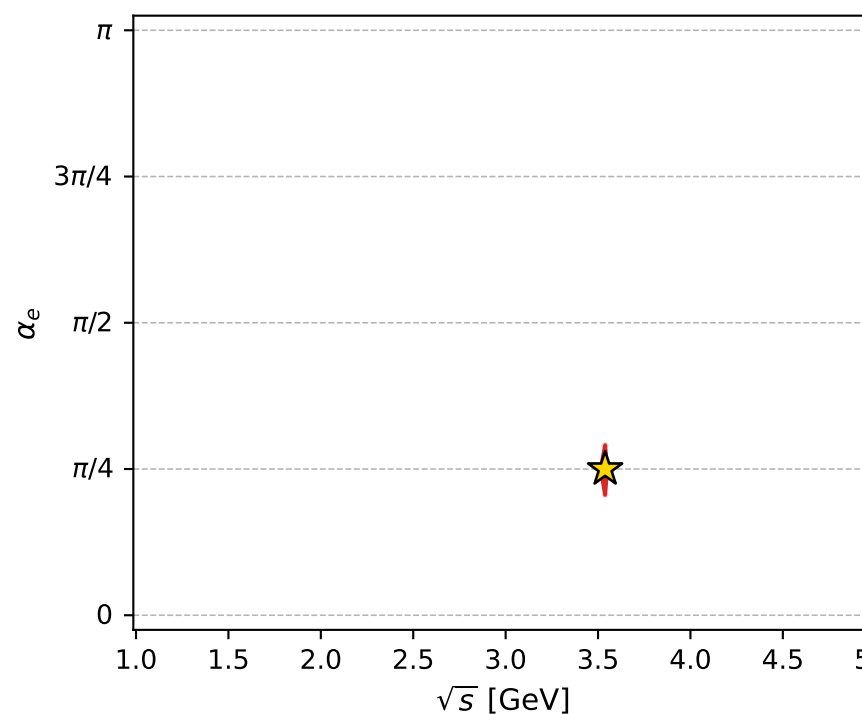
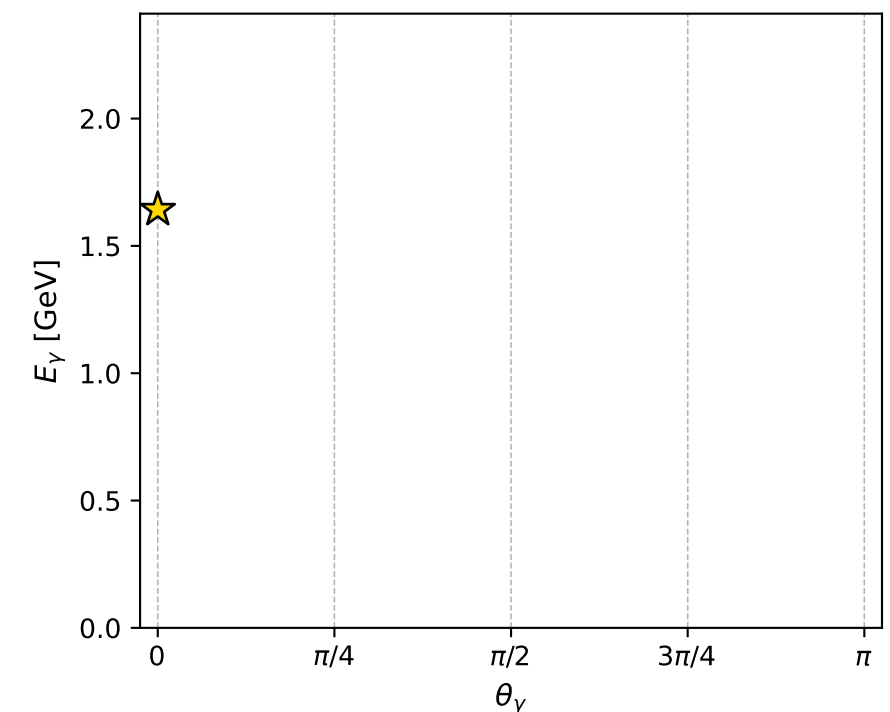
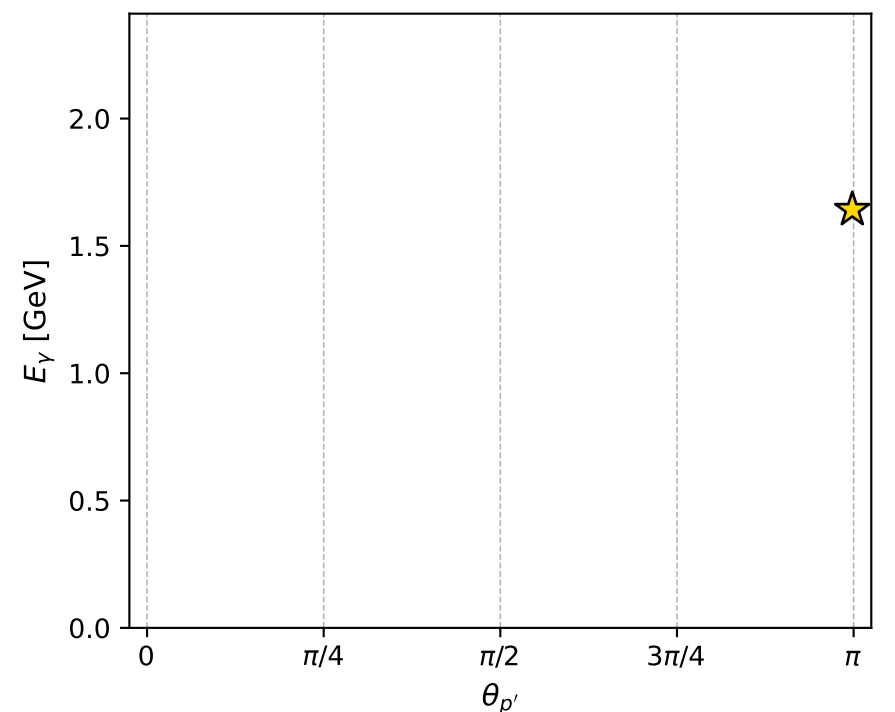
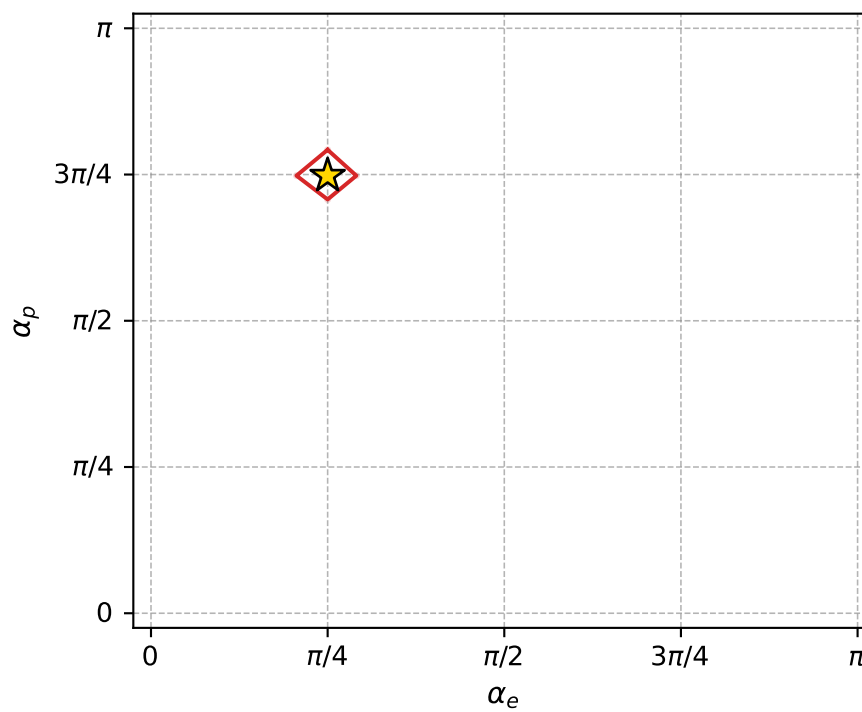
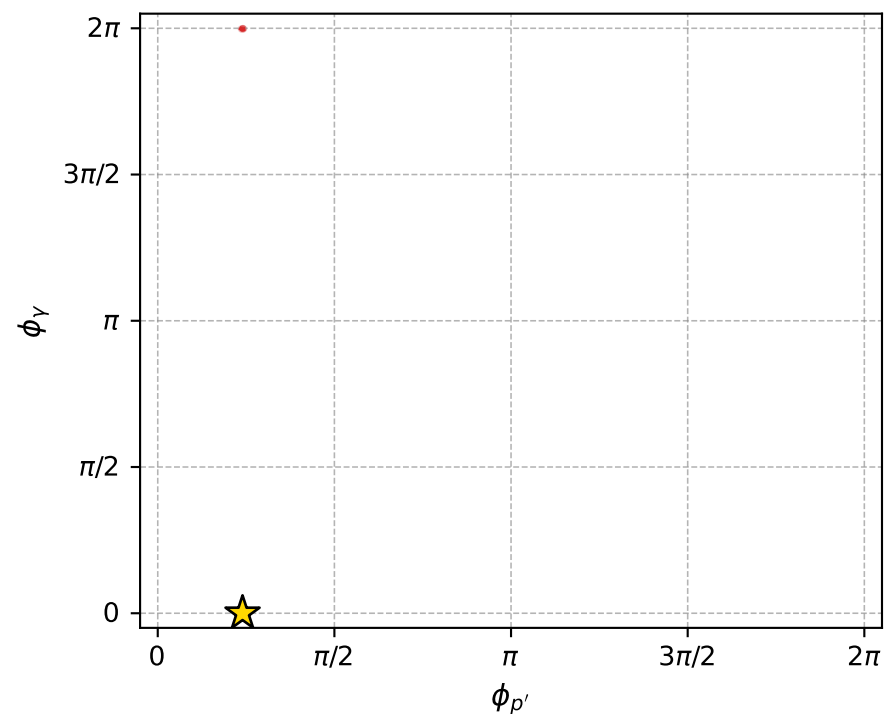
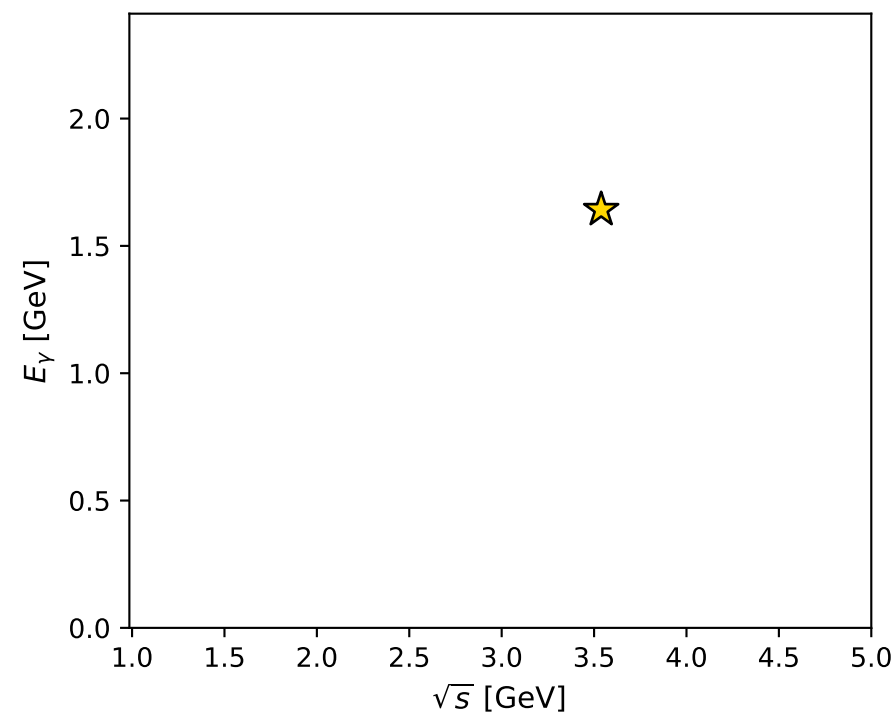
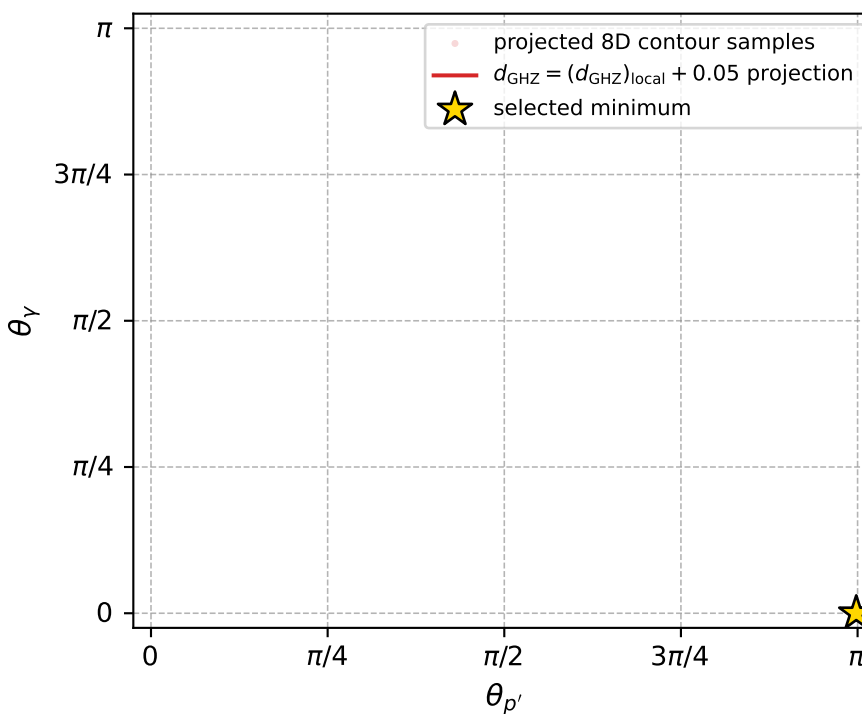


$(h_p, h_V, h_l) = (-1, -1, -1)$

$(h_p, h_V, h_l) = (+1, +1, +1)$



electron: polarization cluster P2, local minimum 370; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 0.022505684$
 $d_{\text{GHZ}} \text{ contour} = 0.072505684$
 above global minimum = 0.022505661
 8D contour samples = 155

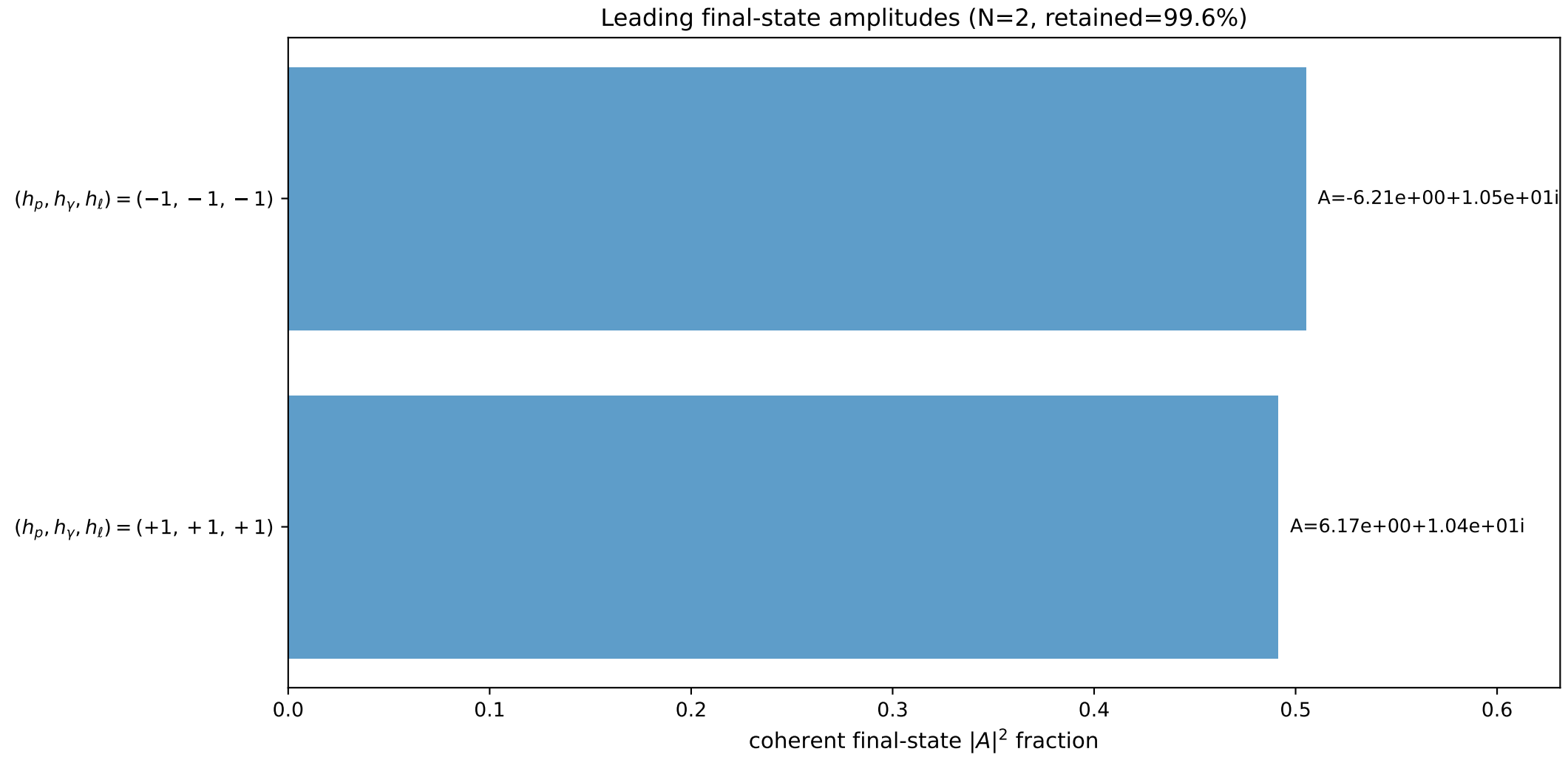
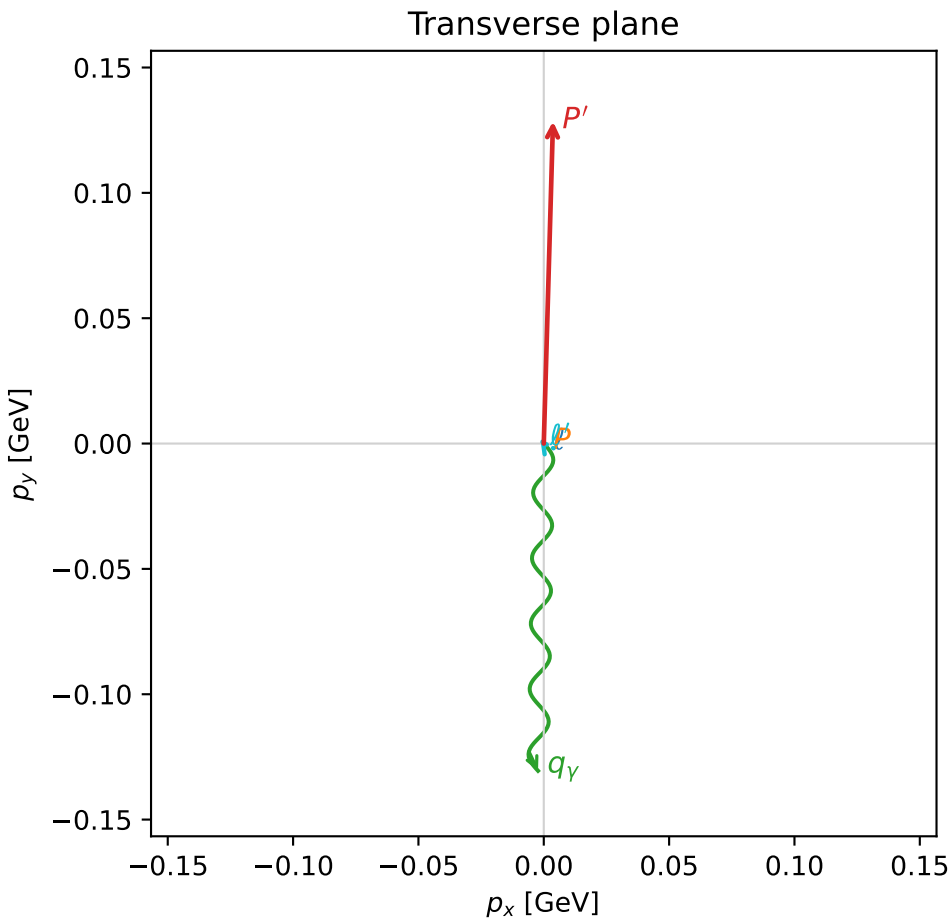
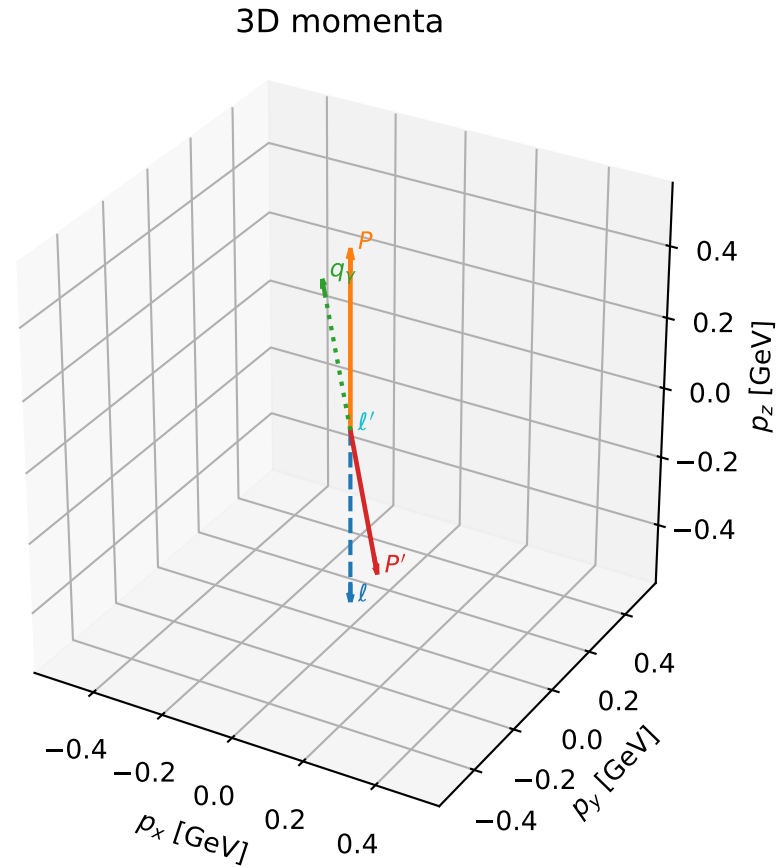
selected phase-space point:
 $\text{sqrt_s} = 3.538361$
 $\text{theta_p_out} = 3.135667$
 $\text{theta_gamma_out} = 0.000106839$
 $\text{q0out} = 1.642234$
 $\text{phi_p_out} = 0.7539521$
 $\text{phi_gamma_out} = 0.000369628$
 $\text{alpha_e} = 0.7853266$
 $\text{alpha_p} = 2.35007$

coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

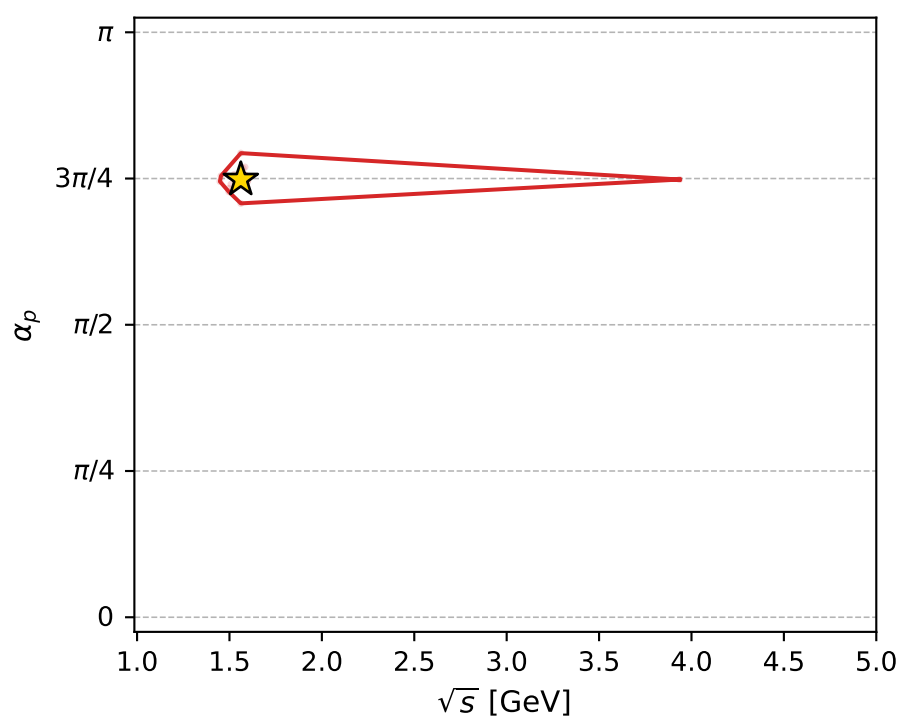
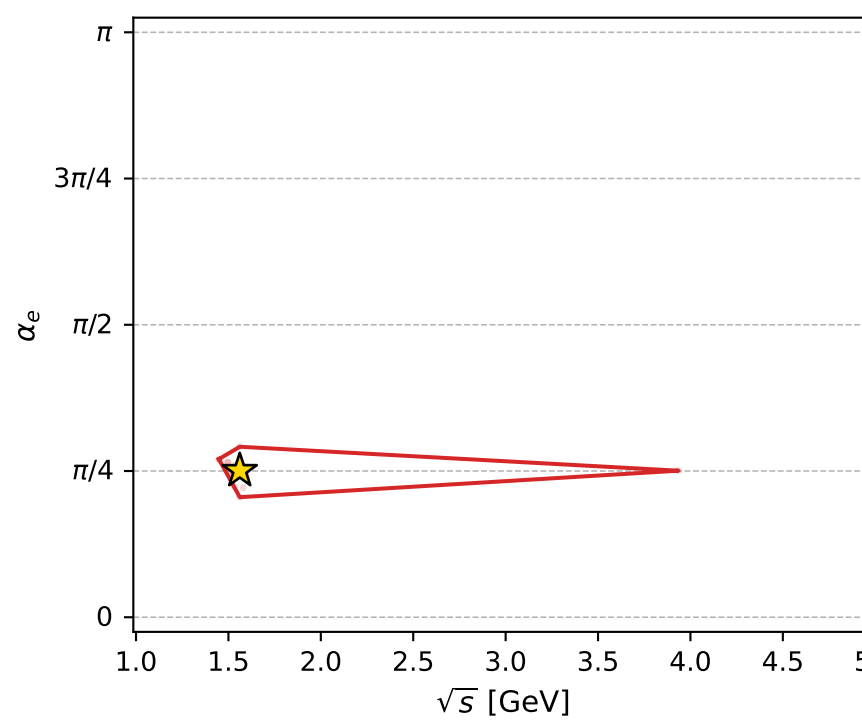
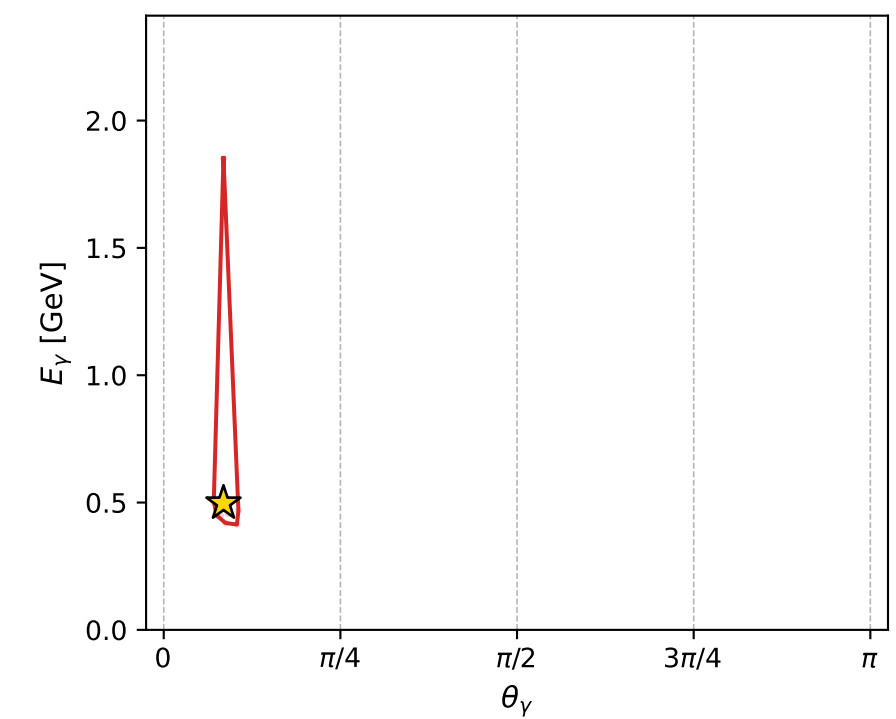
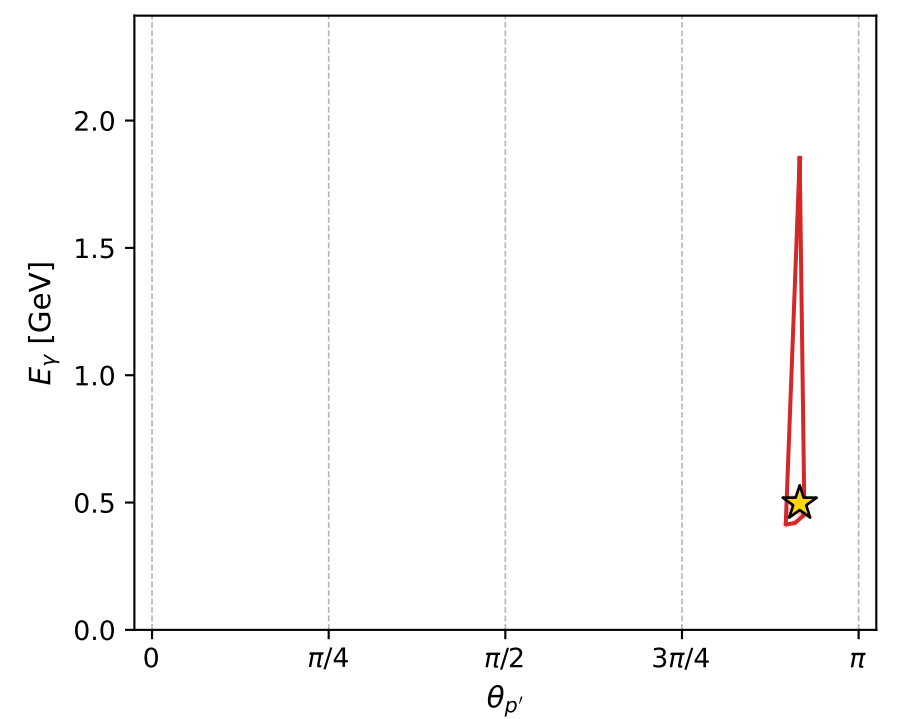
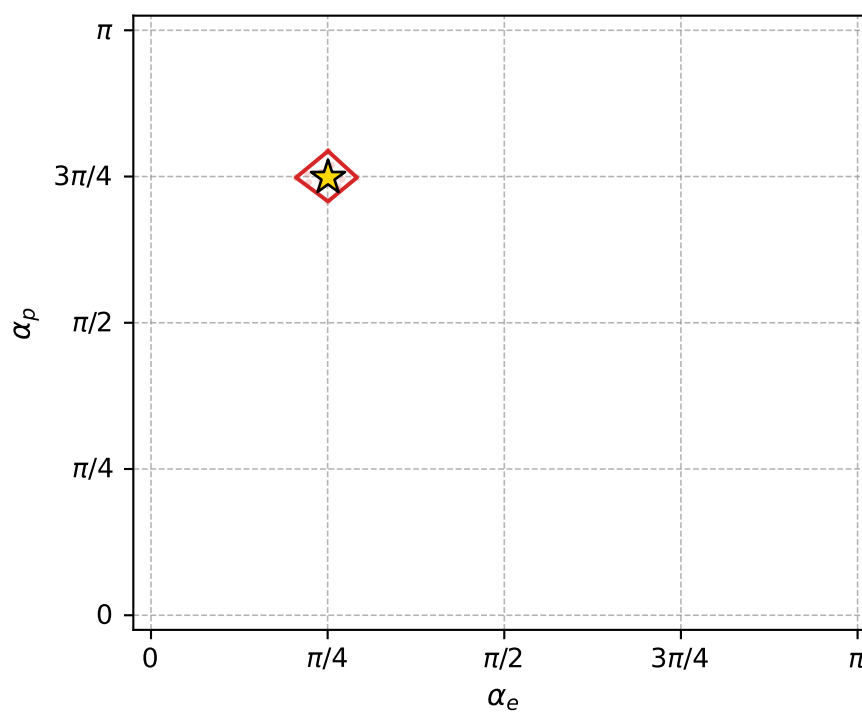
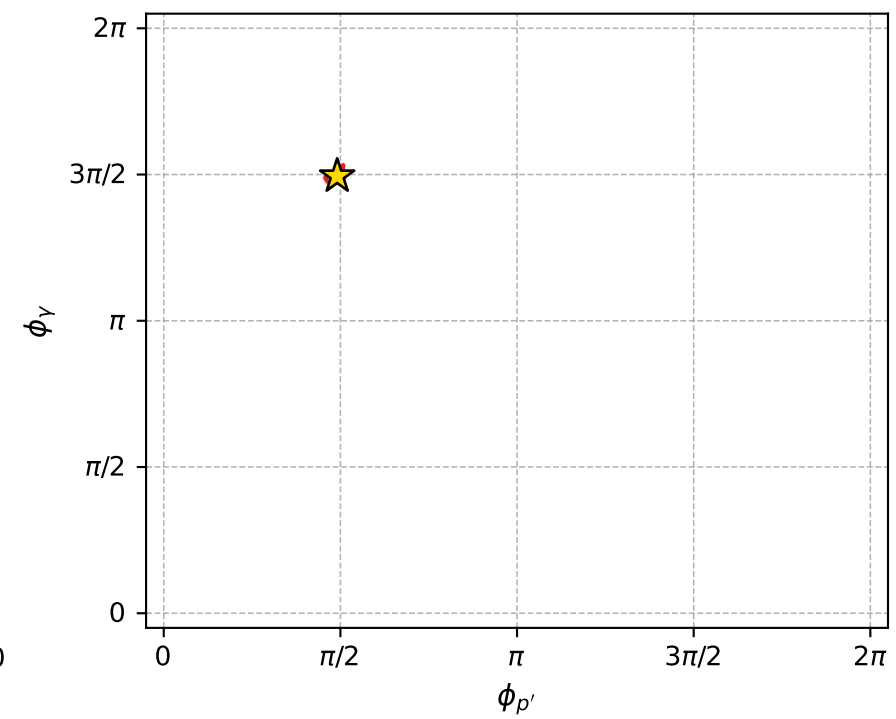
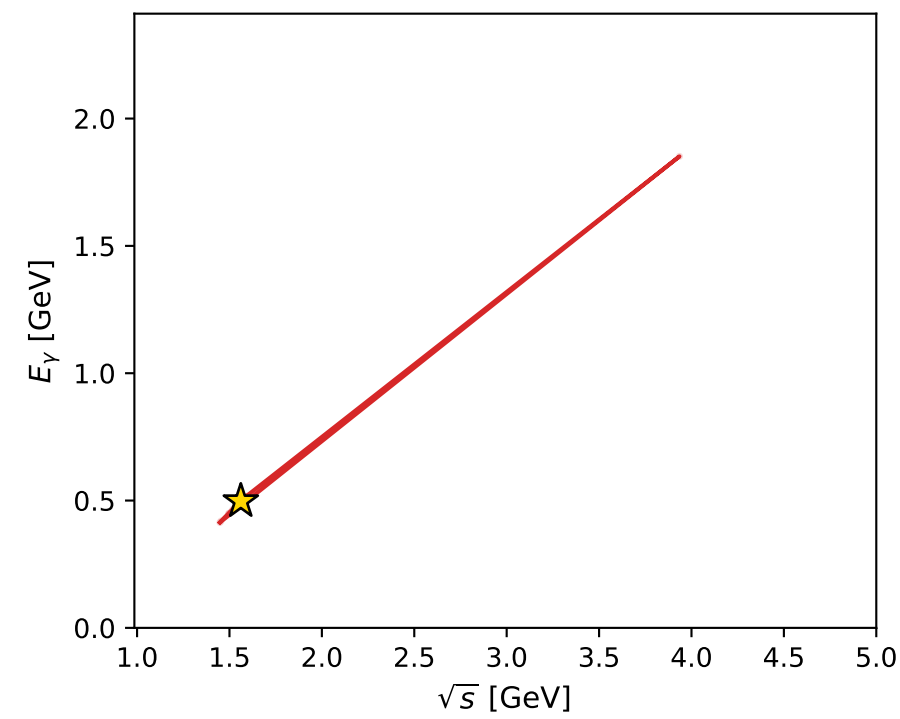
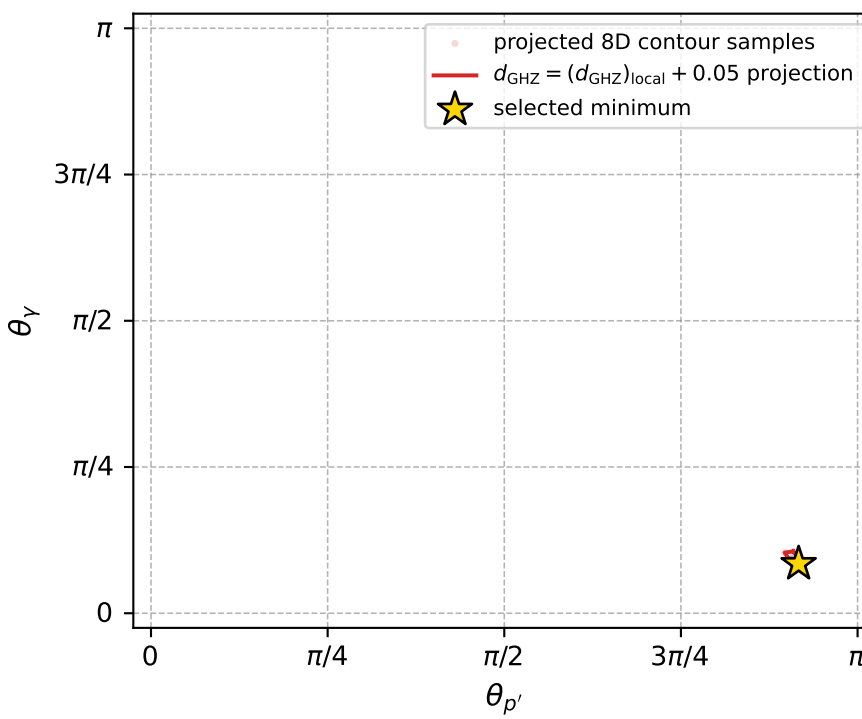
dghz_mixing_angles_polarization_02_local_minimum_371 $d_{\{\text{rm GHZ}\}}=0.0256597$
 region: polarization_cluster_02_local_minimum_371
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=0.78708844$ rad
 incoming lepton: $0.705911|+\rangle +0.708301|-\rangle$
 proton mixing angle: $\alpha_p=2.3508161$ rad
 incoming proton: $-0.703293|+\rangle +0.7109|-\rangle$

$s=2.43798$, $\sqrt{s}=1.5614$, $|\vec{P}|=0.498955$, $|\vec{P}'|=0.496309$
 $\theta_{p'}=2.87957$, $\theta_Y=0.26554$, $\phi_{p'}=1.54217$, $\phi_Y=4.69489$
 $E_Y=0.497557$, $\theta_{l'}=1.86425$, $\phi_{l'}=2.16912$
 $Q^2=0.0018846$, $x_B=0.00121447$, $t=-0.973645$, $W^2=2.42975$, $y=0.995924$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 0.4990$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.4990$
 P $E= 1.0624$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.4990$
 l' $E= 0.0026$ $p_x= -0.0014$ $p_y= 0.0020$ $p_z= -0.0007$
 P' $E= 1.0612$ $p_x= 0.0037$ $p_y= 0.1285$ $p_z= -0.4794$
 q_Y $E= 0.4976$ $p_x= -0.0023$ $p_y= -0.1306$ $p_z= 0.4801$



electron: polarization cluster P2, local minimum 371; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 0.025659696$
 $d_{\text{GHZ}} \text{ contour} = 0.075659696$
 above global minimum = 0.025659674
 8D contour samples = 256

selected phase-space point:

$\text{sqrt_s} = 1.561405$
 $\text{theta_p_out} = 2.879573$
 $\text{theta_gamma_out} = 0.26554$
 $\text{q0ut} = 0.4975571$
 $\text{phi_p_out} = 1.542169$
 $\text{phi_gamma_out} = 4.694891$
 $\alpha_e = 0.7870884$
 $\alpha_p = 2.350816$

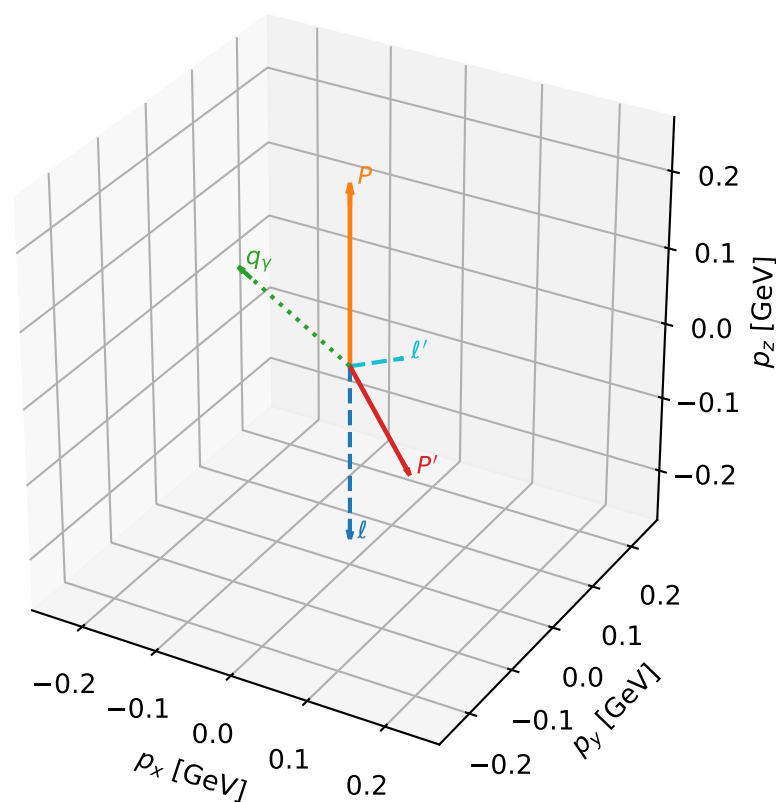
coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

dghz_mixing_angles_polarization_02_local_minimum_375 $d_{\{\text{rm GHZ}\}}=0.0355705$
 region: polarization_cluster_02_local_minimum_375
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=0.86247053$ rad
 incoming lepton: $0.650563|+\rangle +0.759452|-\rangle$
 proton mixing angle: $\alpha_p=2.5441438$ rad
 incoming proton: $-0.826773|+\rangle +0.562535|-\rangle$

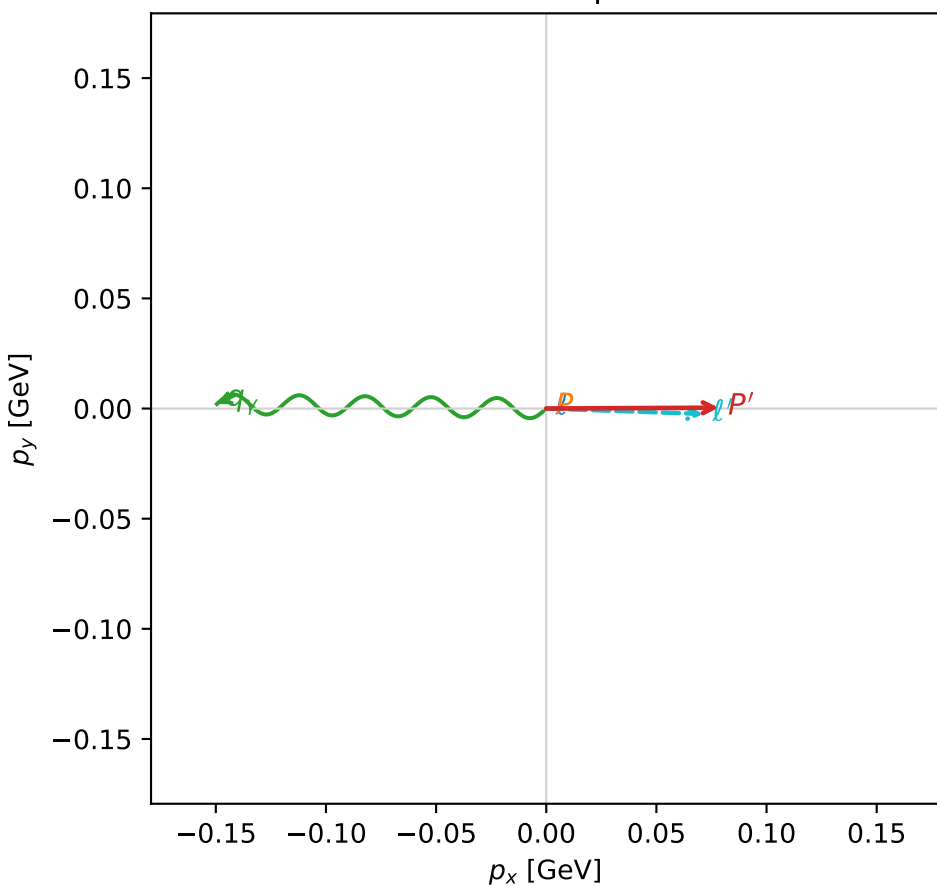
$s=1.44143$, $\sqrt{s}=1.20059$, $|\vec{P}|=0.233876$, $|\vec{P}'|=0.143424$
 $\theta_{P'}=2.56423$, $\theta_V=1.03848$, $\phi_{P'}=0.00499112$, $\phi_V=3.12772$
 $E_V=0.173528$, $\theta_{\ell'}=1.14752$, $\phi_{\ell'}=6.24859$
 $Q^2=0.0515786$, $x_B=0.121226$, $t=-0.131164$, $W^2=1.25374$, $y=0.757635$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.2339$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.2339$
 P $E= 0.9667$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.2339$
 ℓ' $E= 0.0782$ $p_x= 0.0712$ $p_y= -0.0025$ $p_z= 0.0321$
 P' $E= 0.9489$ $p_x= 0.0783$ $p_y= 0.0004$ $p_z= -0.1202$
 q_V $E= 0.1735$ $p_x= -0.1495$ $p_y= 0.0021$ $p_z= 0.0881$

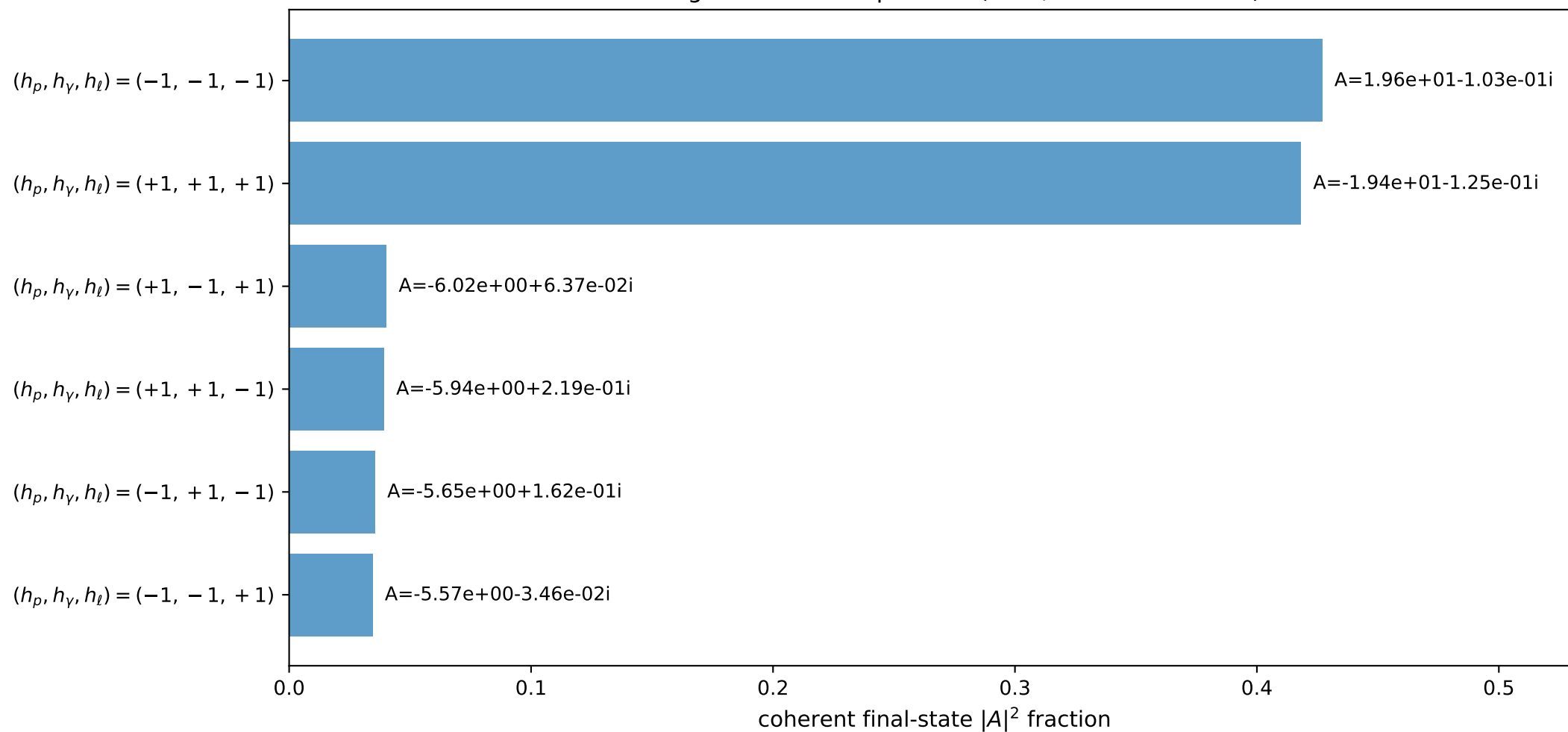
3D momenta



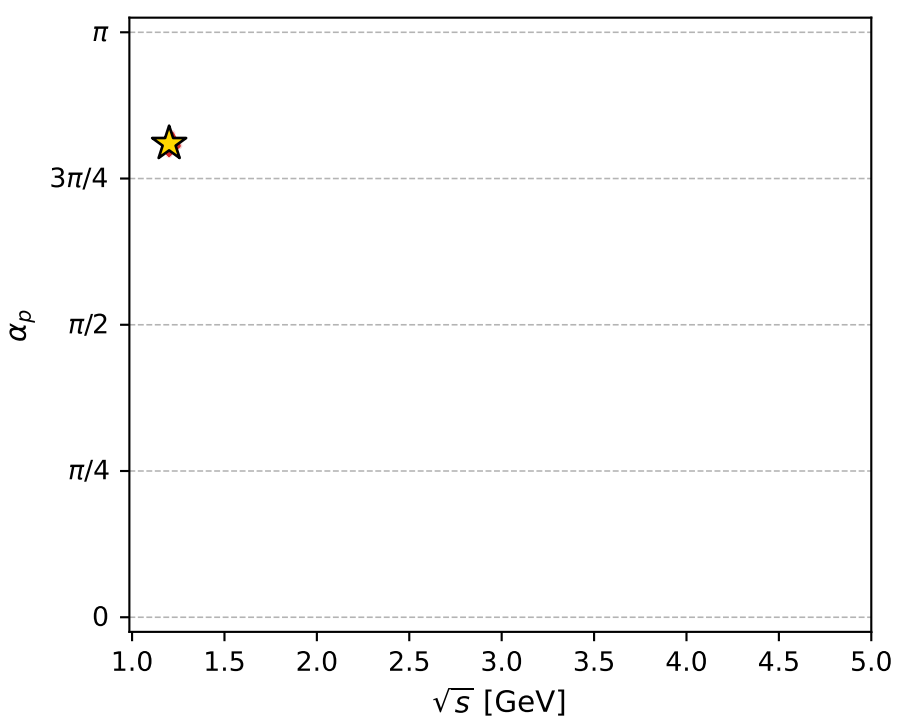
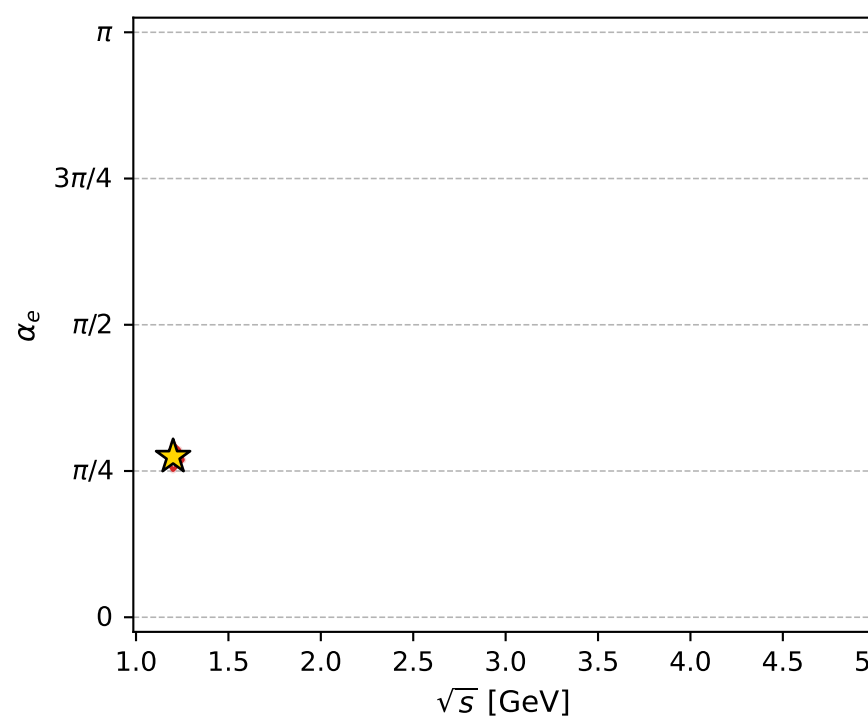
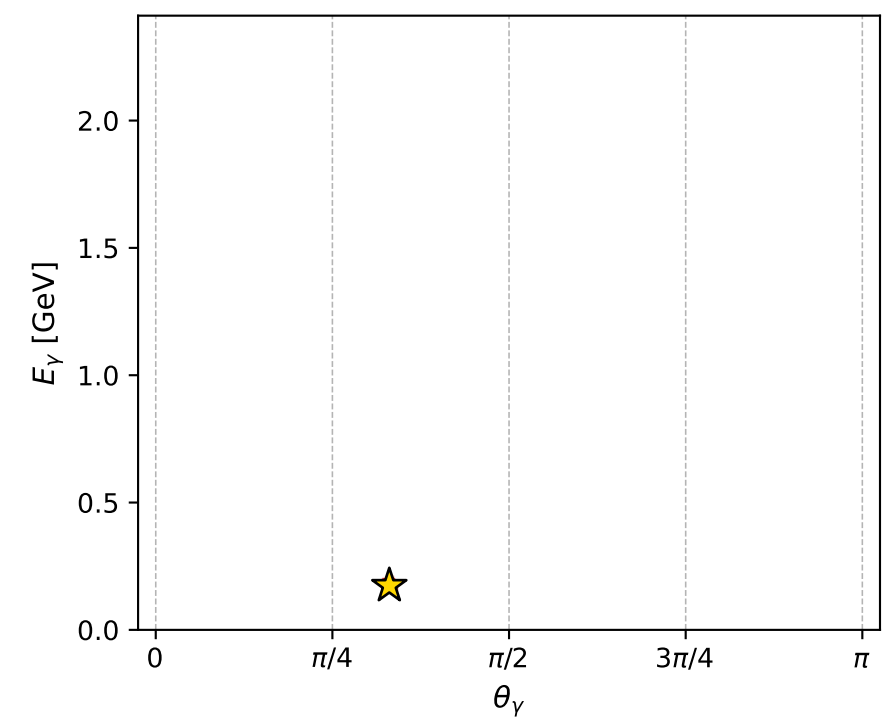
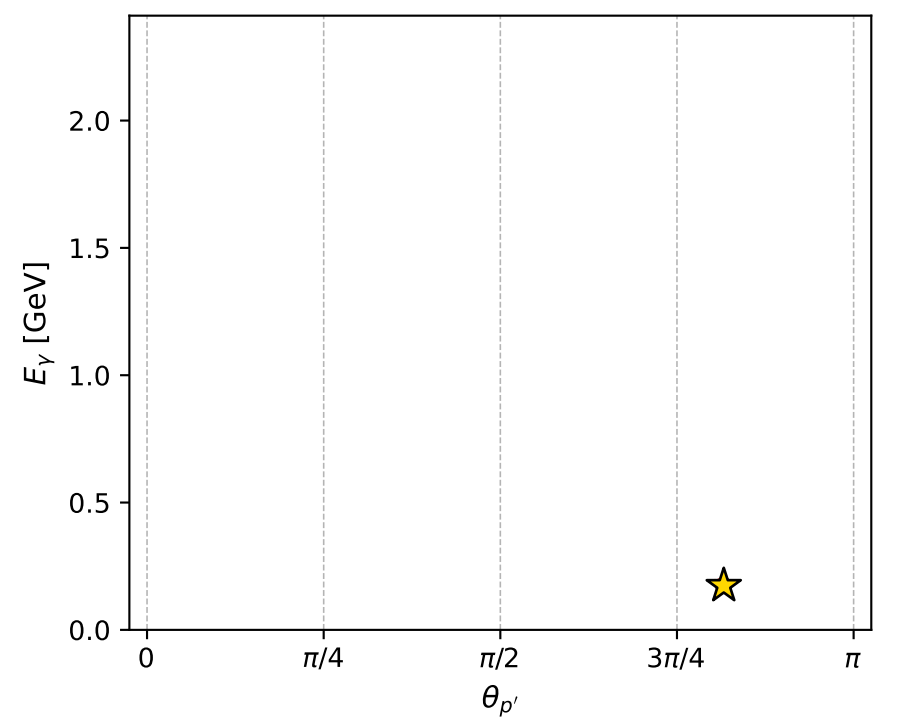
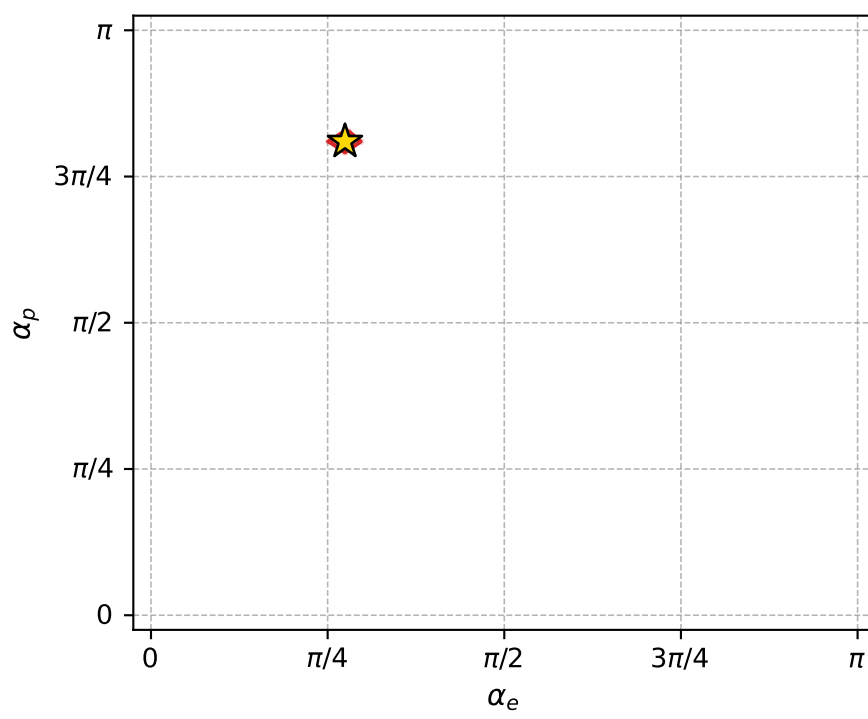
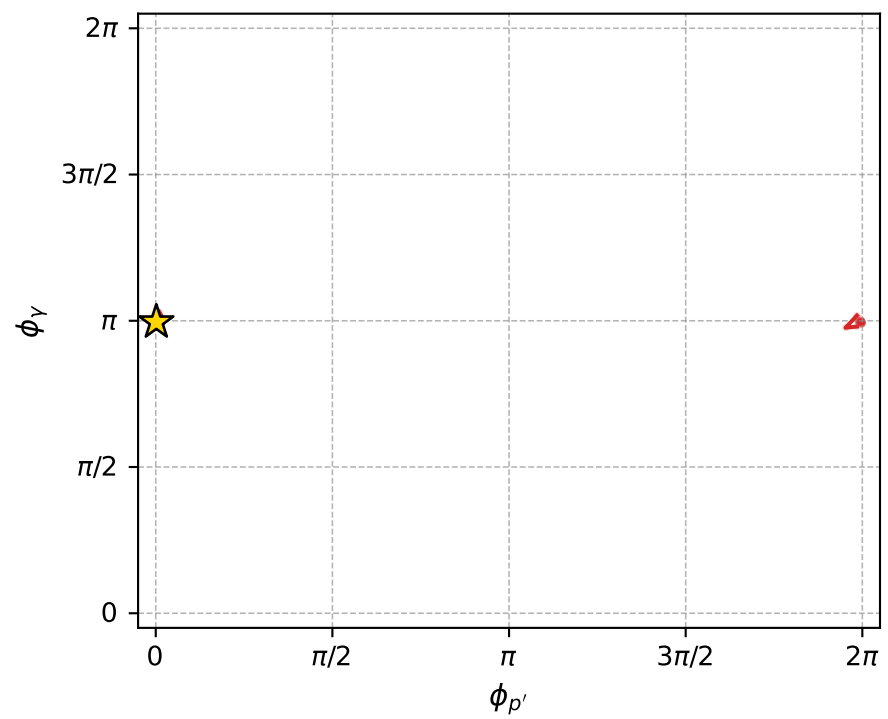
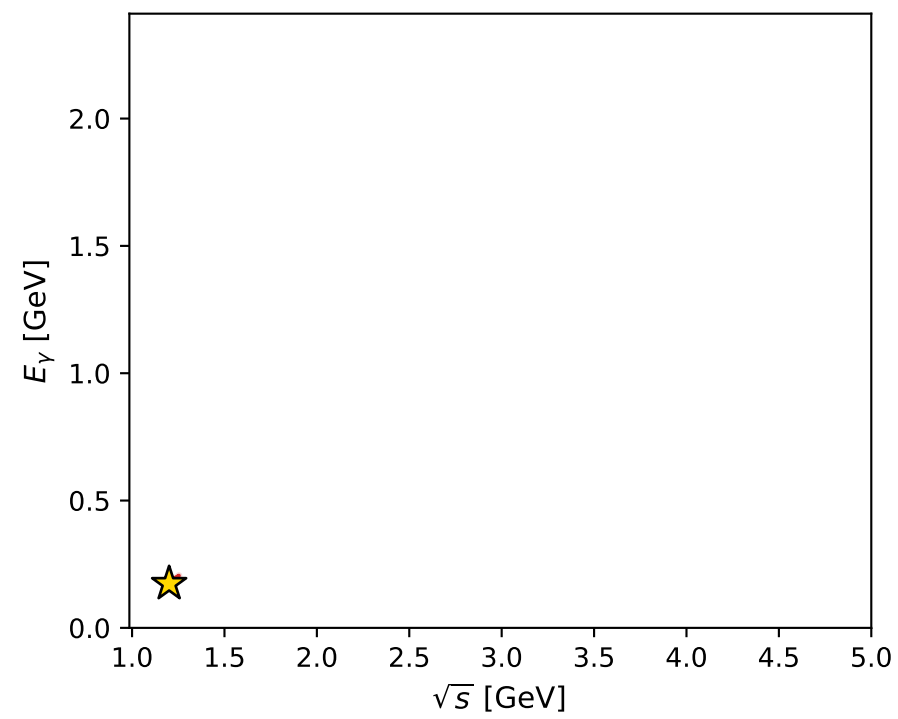
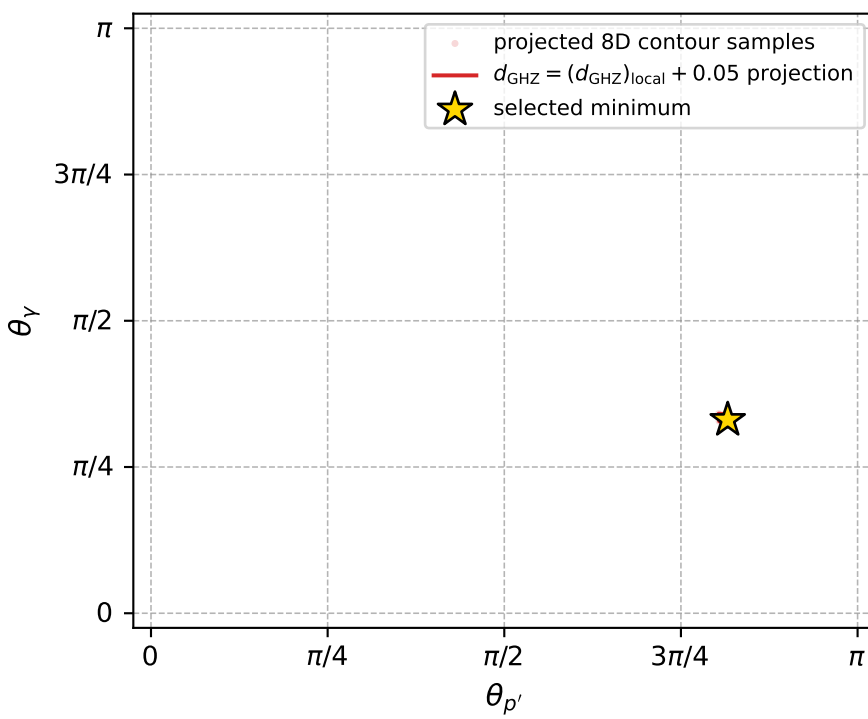
Transverse plane



Leading final-state amplitudes (N=6, retained=99.5%)



electron: polarization cluster P2, local minimum 375; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 0.035570468$
 $d_{\text{GHZ}} \text{ contour} = 0.085570468$
 above global minimum = 0.035570446
 8D contour samples = 256

selected phase-space point:

$\text{sqrt_s} = 1.200594$
 $\text{theta_p_out} = 2.564232$
 $\text{theta_gamma_out} = 1.038484$
 $\text{q0out} = 0.1735283$
 $\text{phi_p_out} = 0.004991124$
 $\text{phi_gamma_out} = 3.127718$
 $\text{alpha_e} = 0.8624705$
 $\text{alpha_p} = 2.544144$